

Purpose

Septoplasty is performed to correct a crooked (deviated) or dislocated septum, often as part of **plastic surgery** of the nose (**rhinoplasty**). The nasal septum has three functions: to support the nose, regulate air flow, and support the mucous membranes (mucosa) of the nose. Septoplasty is done to correct the shape of the nose caused by a deformed septum or correct deregulated air flow caused by a **deviated septum**. Septoplasty is often needed when the patient is having an operation to reduce the size of the nose (reductive rhinoplasty), because this operation usually reduces the amount of breathing space in the nose.

Septoplasty may also be done as a follow-up procedure following facial trauma, as the nose is frequently broken or dislocated by blows to the face resulting from automobile accidents, criminal assaults, or **sports injuries**.

Precautions

Septoplasty is ordinarily not performed within six months of a traumatic injury to the nose.

Description

Septoplasties are performed in the hospital with a combination of local and intravenous anesthesia. In some cases, hypnosis has been successfully used as anesthesia. After the patient is anesthetized, the surgeon makes a cut (incision) in the mucous tissue that covers the part of the septum that is made of cartilage. The tissue is lifted, exposing the cartilage and bony part of the septum. Usually, one side of the mucous tissue is left intact to provide support during healing. Cartilage is cut away as needed.

As the surgeon cuts away the cartilage, deformities tend to straighten themselves out, reducing the amount of cartilage that must be cut. Once the cartilage is cut, bony deformities can be corrected. For most patients, this is the extent of the surgery required to improve breathing through the nose and correct deformities. Some patients have bony obstructions at the base of the nasal chamber and require further surgery. These obstructions include bony spurs and ridges that contribute to drying, ulceration, or bleeding of the mucous tissue that covers the inside of the nasal passages. In these cases, the extent of the surgery depends on the nature of the deformities that need correcting.

During surgery, the patient's own cartilage that has been removed can be reused to provide support for

KEY TERMS

Cartilage—A tough, elastic connective tissue found in the joints, outer ear, nose, larynx, and other parts of the body.

Rhinoplasty—Plastic surgery of the nose.

Septum (plural, septa)—The dividing partition in the nose that separates the two nostrils. It is composed of bone and cartilage.

Splint—A thin piece of rigid material that is sometimes used during nasal surgery to hold certain structures in place until healing is underway.

the nose if needed. External septum supports are not usually needed. Splints may be needed occasionally to support cartilage when extensive cutting has been done. External splints can be used to support the cartilage for the first few days of healing. Tefla gauze is inserted in the nostril to support the flaps and cartilage and to absorb any bleeding or mucus.

A newer option for closing perforations in the septum is a button made of Silastic, a compound of silicone and rubber.

Preparation

Before performing a septoplasty, the surgeon will evaluate the difference in airflow between the two nostrils. In children, this assessment can be done very simply by asking the child to breathe out slowly on a small mirror held in front of the nose.

As with any other operation under **general anesthesia**, patients are evaluated for any physical conditions that might complicate surgery and for any medications that might affect blood clotting time.

Aftercare

Patients with septoplasties are usually sent home from the hospital later the same day or the morning after the surgery. All dressings inside the nose are removed before the patient leaves. Aftercare includes a list of detailed instructions for the patient that focus on preventing trauma to the nose.

Risks

The risks from a septoplasty are similar to those from other operations on the face: postoperative **pain** with some bleeding, swelling, bruising, or

discoloration. A few patients may have allergic reactions to the anesthetics. The operation in itself, however, is relatively low-risk in that it does not involve major blood vessels or vital organs. Infection is unlikely if proper surgical technique is observed.

Normal results

Normal results include improved breathing and airflow through the nostrils, and an acceptable outward shape of the nose.

Resources

BOOKS

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PERIODICALS

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Wain, H. J. "Reflections on Hypnotizability and Its Impact on Successful Surgical Hypnosis: A Sole Anesthetic for Septoplasty." *American Journal of Clinical Hypnosis* 46 (April 2004): 313–321.

ORGANIZATIONS

American Academy of Facial Plastic and Reconstructive Surgery (AAFPRS). 310 South Henry Street, Alexandria, VA 22314. (703) 299-9291. <<http://www.facemd.org>>.

American Society of Plastic Surgeons (ASPS). 444 East Algonquin Road, Arlington Heights, IL 60005. (847) 228-9900. <<http://www.plasticsurgery.org>>.

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Septum perforation see **Perforated septum**

Serenoa repens see **Saw palmetto**

Serotonin boosters see **Selective serotonin reuptake inhibitors**

Serum albumin test see **Protein components test**

Serum globulin test see **Protein components test**

Serum hepatitis see **Hepatitis B**

Serum protein electrophoresis see **Protein electrophoresis**

Serum sickness

Definition

Serum sickness is a type of delayed allergic response, appearing four to 10 days after exposure to some **antibiotics** or antiserum, the portion of serum that contains antibodies, such as gamma globulin, which may be given to provide immunization against some diseases.

Description

Serum sickness is very similar to an allergic reaction. The patient's immune system recognizes the proteins in the drug or antiserum as foreign proteins, and produces its own antibodies to protect against the foreign proteins. The newly formed antibodies bind with the foreign protein to form immune complexes. These immune complexes may enter the walls of blood vessels where they set off an inflammatory reaction.

While other types of allergic reactions may produce a rapid response, the serum sickness reaction is delayed because it takes time for the body to produce antibodies to the new protein.

Causes and symptoms

The usual symptoms are severe skin reactions, often on the palms of the hands and soles of the feet. **Fever**, sometimes as high as 104° F, is always present and usually appears before the skin rash.

Joint **pain** may be reported in up to 50% of cases. This is usually seen in the larger joints, but occasionally the finger and toe joints may also be involved.

Swelling of lymph nodes, particularly around the site of the injection, is seen in 10–20% of cases. There may also be swelling of the head and neck.

Urine analysis may show traces of blood and protein in the urine.

Other symptoms may involve the heart and central nervous system. These may include changes in vision, and difficulty in movement. Breathing difficulty may occur.

Traditionally, antitoxins were the most common cause of serum sickness, but those reports date from a time when most antitoxins were made from horse serum. As many as 16% of the people who received antirabies serum derived from horses developed serum sickness. The risk of a reaction to antitoxins has dropped dramatically since manufacturers have

started using human serum instead of horse serum to make their products.

Although antitoxins are the most common cause of serum sickness, a number of drugs have been reported to cause a serum sickness reaction. The following list is not complete, but indicates some of the drugs that have been associated with this type of reaction:

- allopurinol (Zyloprim)
- barbiturates
- captopril (Capoten)
- cephalosporin antibiotics
- griseofulvin (Fulvicin, Grifulvin)
- penicillins
- pehnytoin (Dilantin)
- procainamide (Procan SR, Procanbid, Pronestyl-SR)
- quinidine (Quinaglute, Quinidex, Quinora)
- streptokinase (Streptase, Kabikinase)
- sulfonamide antibacterial drugs

Of cases of serum sickness reported to the United States Food and Drug Administration, the drugs most commonly associated with the reaction have been the cephalosporin antibiotics, including cefaclor (Ceclor) and cefalexin (Keflex) and the sulfonamide combination trimethoprim-sulfamethoxazole (Bactrim, Septra.) This does not mean that these are high-risk drugs, since these drugs are very widely used, so that there are many people exposed to them.

In addition to these substances, allergenic extracts used for testing and immunization, hormones, and vaccines have been known to cause serum sickness.

Diagnosis

Diagnosis is made by observing the symptoms and reviewing the patient's medical and medication history. Although the symptoms of serum sickness may be similar to other conditions, patients who present with symptoms of serum sickness and who have a recent history of exposure to a drug or other product which may cause this type of reaction should be suspected of having serum sickness.

Treatment

The first step in treatment of serum sickness is always to discontinue the drug or other substance which is suspected of causing the reaction. After that, all treatment is symptomatic. Antihistamines, pain

KEY TERMS

Allergy—Altered body reaction, usually hypersensitivity, as a response to exposure to a specific substance.

Antibody—Any of a large number of proteins that are produced after stimulation by an antigen and act specifically against the antigen in an immune response.

Antihistamine—A drug that inhibits the actions of histamine. Histamine causes dilatation of capillaries, contraction of smooth muscle, and stimulation of gastric acid secretion.

Antitoxin—An antibody that is capable of neutralizing the specific toxin (a specific cause of disease) that stimulated its production in the body and is produced in animals for medical purposes by injection of a toxin or toxoid with the resulting serum being used to counteract the toxin in other individuals.

Serum—The clear yellowish fluid that remains from blood plasma after fibrinogen, prothrombin, and other clotting factors have been removed by clot formation—called also blood serum.

Sulfonamide—A sulfa drug, one of a large group of drugs used to treat bacterial infections.

relievers, and **corticosteroids** may be given to relieve the symptoms. The choice of treatment depends on the severity of the reaction.

Prognosis

Most serum sickness reactions are mild, and disappear on their own after one or two weeks as long as the cause is removed. Sometimes, symptoms of pain and discomfort may continue for several weeks, even after all the observable reactions such as skin rash and protein in the urine have disappeared. In very rare cases, however, there can be severe reactions and permanent damage. In very rare but extreme cases, serum sickness can lead to **shock**, permanent kidney damage, and even **death**.

Prevention

The most effective method of prevention is simple avoidance of antitoxins that may cause serum sickness. If patients have had a reaction in the past, particularly if the reaction was to a commonly used drug,

they should be made aware of it, and be advised to alert physicians and hospitals in the future. Patients who have had particularly severe reactions may be advised to wear identification bracelets, or use other means to alert health care providers.

When it is necessary to administer an antitoxin, skin tests may be used to identify people who are at risk of a reaction. If the situation does not allow enough time for skin testing, the antitoxin should be given along with an intravenous antihistamine. Other drugs, such as epinephrine, which may be needed for an emergency, should be available.

Resources

BOOKS

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PERIODICALS

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Fielding, Jonathan. "Our Health; Drug Reactions Differ From Side Effects." *The Los Angeles Times* February 7, 2000.

"VA Hospitals Test Smart Cards for Patient Information." *Computerworld* May 14, 2001.

ORGANIZATIONS

Action Against Allergy (AAA). PO Box 278, Twickenham Middlesex, Greater London TW1 4QQ, England.
American Allergy Association (AAA). 3104 E Camelback, Ste. 459 Phoenix, AZ 85016.

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Serum therapy see **Gammaglobulin**

Severe acute respiratory syndrome (SARS)

Definition

Severe acute respiratory syndrome (SARS) is the first emergent and highly transmissible viral disease to appear during the twenty-first century.

Description

Patients with SARS develop flu-like **fever**, **headache**, malaise, dry **cough** and other breathing difficulties. Many patients develop **pneumonia**, and in 5–10% of cases, the pneumonia and other complications are

severe enough to cause **death**. SARS is caused by a virus that is transmitted usually from person to person—predominantly by the aerosolized droplets of virus infected material.

The first known case of SARS was traced to a November 2002 case in Guangdong province, China. By mid-February 2003, Chinese health officials tracked more than 300 cases, including five deaths in Guangdong province from what was at the time described as an acute respiratory syndrome. Many flu-causing viruses have previously originated from Guangdong province because of cultural and exotic cuisine practices that bring animals, animal parts, and humans into close proximity. In such an environment, pathogens can more easily genetically mutate and make the leap from animal hosts to humans. The first cases of SARS showed high rates among Guangdong food handlers and chefs.

Chinese health officials initially remained silent about the outbreak, and no special precautions were taken to limit travel or prevent the spread of the disease. The world health community, therefore, had no chance to institute testing, **isolation**, and quarantine measures that might have prevented the subsequent global spread of the disease.

On February 21, Liu Jianlun, a 64-year-old Chinese physician from Zhongshan hospital (later determined to have been "super-spreader," a person capable of infecting unusually high numbers of contacts) traveled to Hong Kong to attend a family wedding despite the fact that he had a fever. Epidemiologists subsequently determined that, Jianlun passed on the SARS virus to other guests at the Metropole Hotel where he stayed—including an American businessman en route to Hanoi, three women from Singapore, two Canadians, and a Hong Kong resident. Jianlun's travel to Hong Kong and the subsequent travel of those he infected allowed SARS to spread from China to the infected travelers' destinations.

Johnny Chen, the American businessman, grew ill in Hanoi, Vietnam, and was admitted to a local hospital. Chen infected 20 health care workers at the hospital including noted Italian epidemiologist Carlo Urbani who worked at the Hanoi World Health Organization (WHO) office. Urbani provided medical care for Chen and first formally identified SARS as a unique disease on February 28, 2003. By early March, 22 hospital workers in Hanoi were ill with SARS.

Unaware of the problems in China, Urbani's report drew increased attention among epidemiologists when coupled with news reports in mid-March

that Hong Kong health officials had also discovered an outbreak of an acute respiratory syndrome among health care workers. Unsuspecting hospital workers admitted the Hong Kong man infected by Jianlun to a general ward at the Prince of Wales Hospital because it was assumed he had a typical severe pneumonia—a fairly routine admission. The first notice that clinicians were dealing with an unusual illness came—not from health notices from China of increasing illnesses and deaths due to SARS—but from the observation that hospital staff, along with those subsequently determined to have been in close proximity to the infected persons, began to show signs of illness. Eventually, 138 people, including 34 nurses, 20 doctors, 16 medical students, and 15 other health care workers, contracted pneumonia.

One of the most intriguing aspects of the early Hong Kong cases was a cluster of more than 250 SARS cases that occurred in a cluster of high-rise apartment buildings—many housing health care workers—that provided evidence of a high rate of secondary transmission. Epidemiologists conducted extensive investigations to rule out the hypothesis that the illnesses were related to some form of local contamination (e.g., sewage, bacteria on the ventilation system, etc.). Rumors began that the illness was due to cockroaches or rodents, but no scientific evidence supported the hypothesis that the disease pathogen was carried by insects or animals.

Hong Kong authorities then decided that those suffering the flu-like symptoms would be given the option of self-isolation, with family members allowed to remain confined at home or in special camps. Compliance checks were conducted by police.

One of the Canadians infected in Hong Kong, Kwan Sui-Chu, return to Toronto, Ontario, and died in a Toronto hospital on March 5. As in Hong Kong, because there were no alert from China about the SARS outbreak, Canadian officials did not initially suspect that Sui-Chu had been infected with a highly contagious virus, until Sui-Chu's son and five health care workers showed similar symptoms. By mid-April, Canada reported more than 130 SARS cases and 15 fatalities.

Increasingly faced with reports that provided evidence of global dissemination, on March 15, 2003, the World Health Organization (WHO) took the unusual step of issuing a travel warning that described SARS is a “worldwide health threat.” WHO officials announced that SARS cases, and potential cases, had been tracked from China to Singapore, Thailand, Vietnam, Indonesia, Philippines, and Canada.

Although the exact cause of the “acute respiratory syndrome” had not, at that time, been determined, WHO officials issuance of the precautionary warning to travelers bound for Southeast Asia about the potential SARS risk served as notice to public health officials about the potential dangers of SARS.

Within days of the first WHO warning, SARS cases were reported in United Kingdom, Spain, Slovenia, Germany, and in the United States.

WHO officials were initially encouraged that isolation procedures and alerts were working to stem the spread of SARS, as some countries reporting small numbers of cases experienced no further dissemination to hospital staff or others in contact with SARS victims. However, in some countries, including Canada, where SARS cases occurred before WHO alerts, SARS continued to spread beyond the bounds of isolated patients.

WHO officials responded by recommending increased screening and quarantine measures that included mandatory screening of persons returning from visits to the most severely affected areas in China, Southeast Asia, and Hong Kong.

In early April 2003, WHO took the controversial additional step of recommending against non-essential travel to Hong Kong and the Guangdong province of China. The recommendation, sought by infectious disease specialists, was not controversial within the medical community, but caused immediate concern regarding the potentially widespread economic impacts.

Mounting reports of SARS showed a increasing global dissemination of the virus. By April 9, the first confirmed reports of SARS cases in Africa reached WHO headquarters, and eight days later, a confirmed case was discovered in India.

Causes and symptoms

In mid-April 2003, Canadian scientists at the British Columbia **Cancer** Agency in Vancouver announced that they that sequenced the genome of the coronavirus most likely to be the cause of SARS. Within days, scientists at the Centers for Disease Control (CDC) in Atlanta, Georgia, offered a genomic map that confirmed more than 99% of the Canadian findings.

Both genetic maps were generated from studies of viruses isolated from SARS cases. The particular coronavirus mapped had a genomic sequence of 29,727 nucleotides—average for the family of coronavirus that typically contain between 29,000-31,000 nucleotides.

Proof that the coronavirus mapped was the specific virus responsible for SARS would eventually come from animal testing. Rhesus monkeys were exposed to the virus via injection and inhalation and then monitored to determine whether SARS like symptoms developed, and then if sick animals exhibited a histological pathology (i.e., an examination of the tissue and cellular level pathology) similar to findings in human patients. Other tests, including polymerase chain reaction (PCR) testing helped positively match the specific coronavirus present in the lung tissue, blood, and feces of infected animals to the exposure virus.

Identification of a specific pathogen can be a complex process, and positive identification requires thousands of tests. All testing is conducted with regard to testing Koch's postulates—the four conditions that must be met for an organism to be determined to the cause of a disease. First, the organism must be present in every case of the disease. Second, the organism must be able to be isolated from the host and grown in laboratory conditions. Third, the disease must be reproduced when the isolated organism is introduced into another, healthy host. The fourth postulate stipulates that the same organism must be able to be recovered and purified from the host that was experimentally infected.

Early data indicates that SARS has an incubation period range of two to 10 days, with an average incubation of about four days. Much of the inoculation period allows the virus to be both transported and spread by an asymptomatic carrier. With air travel, asymptomatic carriers can travel to anywhere in the world. The initial symptoms are non-specific and common to the flu. Infected cases then typically spike a high fever 100.4°F (38°C) as they develop a cough, **shortness of breath**, and difficulty breathing. SARS often fulminates (reaches its maximum progression) in a severe pneumonia that can cause **respiratory failure** and death in about 10% of its victims.

Diagnosis

Currently, initial tests include blood cultures, Gram stain, chest radiograph, and tests for other viral respiratory pathogens such as **influenza A** and **B**. Other serologic techniques are used, and if SARS is suspected, samples are forwarded to state/local public health departments and/or the CDC for coronavirus antibody testing.

Treatment

As of May 1, 2003, no therapy was demonstrated to have clinical effectiveness against the virus that

causes SARS, and physicians could offer only supportive therapy (e.g. administration of fluids, oxygen, ventilation, etc.).

Prognosis

By late April/early May 2003, WHO officials had confirmed reports of more than 3,000 cases of SARS from 18 different countries with 111 deaths attributed to the disease (about a 5–10% death rate). United States health officials reported 193 cases with no deaths. Significantly, all but 20 of the U.S. cases were linked to travel to infected areas, and the other 20 cases were accounted for by secondary transmission from infected patients to family members and health care workers.

Information on countries reporting SARS and the cumulative total of cases and deaths is updated each day on the WHO SARS web site at <<http://www.who.int/csr/sarscountry/en/>> .

Prevention

Until a vaccine is developed, isolation and quarantine remain potent tools in the modern public health arsenal. Both procedures seek to control exposure to infected individuals or materials. Isolation procedures are used with patients with a confirmed illness. Quarantine rules and procedures apply to individuals who are not currently ill, but are known to have been exposed to the illness (e.g., been in the company of an infected person or come in contact with infected materials).

Isolation and quarantine both act to restrict movement and to slow or stop the spread of disease within a community. Depending on the illness, patients placed in isolation may be cared for in hospitals, specialized health care facilities, or in less severe cases, at home. Isolation is a standard procedure for TB patients. In most cases, isolation is voluntary; however, isolation can be compelled by federal, state, and some local law.

States governments within the United States have a general authority to set and enforce quarantine conditions. At the federal level, the Centers for Disease Control and Prevention's (CDC) Division of Global Migration and Quarantine is empowered to detain, examine, or conditionally release (release with restrictions on movement or with a required treatment protocol) individuals suspected of carrying certain listed communicable diseases.

As of April 27, 2003, the CDC in Atlanta recommended SARS patients be voluntarily isolated, but

had not recommended enforced isolation or quarantine. Regardless, CDC and other public health officials, including the Surgeon General, sought and secured increased powers to deal with SARS. On April 4, 2003, U.S. President George W. Bush signed Presidential Executive Order 13295 that added SARS to a list of quarantinable communicable diseases. The order provided health officials with the broader powers to seek "...apprehension, detention, or conditional release of individuals to prevent the introduction, transmission, or spread of suspected communicable diseases..."

Travel advisories issued by WHO should be reviewed and people who must travel to areas with SARS outbreaks should follow such preventative measures as frequent hand washing and avoidance of large crowds. Likewise, family members caring for suspected and/or confirmed SARS patients should wash hands frequently, avoid direct contact with the patient's bodily fluids, and monitor their own possible development of symptoms closely.

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Severe combined immunodeficiency

Definition

Severe combined **immunodeficiency** (SCID) is the most serious human immunodeficiency disorder(s). It is a group of congenital disorders in which both the humoral part of the patient's immune system and the cells involved in immune responses fail to work properly. Children with SCID are vulnerable to recurrent severe infections, retarded growth, and early **death**.

Description

SCID is thought to affect between one in every 100,000 persons, and one in every 500,000 infants. Several different immune system disorders are currently grouped under SCID:

- Swiss-type agammaglobulinemia. This was the first type of SCID discovered, in Switzerland in the 1950s.
- Adenosine deaminase deficiency (ADA). About 50% of SCID cases are of this type. ADA deficiency

leads to low levels of B and T cells in the child's immune system.

- Autosomal recessive. About 40% of SCID cases are inherited from the parents in an autosomal recessive pattern.
- Bare lymphocyte syndrome. In this form of SCID, the white blood cells (lymphocytes) in the baby's blood are missing certain proteins. Without these proteins, the lymphocytes cannot activate the T cells in the immune system.
- SCID with leukopenia. Children with this form of SCID are lacking a type of white blood cell called a granulocyte.

In order to understand why SCID is considered the most severe immunodeficiency disorder, it is helpful to have an outline of the parts of the human immune system. It has three parts: cellular, humoral, and nonspecific. The cellular and humoral parts of the system are both needed to fight infections—they recognize disease agents and attack them. The cellular system is composed of many classes of T-lymphocytes (white blood cells that detect foreign invaders called antigens). The humoral system is made up of B cells, which are the only cells in the body that make antibodies. In SCID, neither the cellular nor the humoral part of the immune system is working properly.

Causes and symptoms

SCID is an inherited disorder. There are two ways in which a developing fetus' immune system can fail to develop normally. In the first type of genetic problem, both B and T cells are defective. In the second type, only the T cells are abnormal, but their defect affects the functioning of the B cells.

For the first few months of life, a child with SCID is protected by antibodies in the mother's blood. As early as three months of age, however, the SCID child begins to suffer from mouth infections (thrush), chronic **diarrhea**, otitis media and pulmonary infections, including pneumocystis pneumonia. The child loses weight, becomes very weak, and eventually dies from an opportunistic infection.

Diagnosis

SCID is diagnosed by the typing of T and B cells in the child's blood. B cells can be detected by immunofluorescence tests for surface markers (unique proteins) on the cells. T cells can be identified in tissue sections (samples) using enzyme-labeled antibodies.

KEY TERMS

Adenosine deaminase (ADA)—An enzyme that is lacking in a specific type of SCID. Children with an ADA deficiency have low levels of both B and T cells.

Antigens—A substance that usually causes the formation of an antibody. A foreign invaders in the body.

Autosomal recessive inheritance—A pattern of inheritance of a recessive gene where, among other things, both parents may not show symptoms.

B cell—A type of lymphocyte or white blood cell that is derived from precursor cells in the bone marrow.

Congenital—Present at the time of birth. Most forms of SCID are hereditary as well as congenital.

Gene therapy—An experimental treatment for SCID that consists of implanting a gene for ADA into an activated virus and merging it with some of the patient's own T cells. The corrected T cells are infused back into the patient every few months.

Humoral—Pertaining to or derived from a body fluid. The humoral part of the immune system includes antibodies and immunoglobulins in blood serum.

Lymphocyte—A type of white blood cell that is important in the formation of antibodies.

Orphan drug—A drug that is known to be useful in treatment but lacks sufficient funding for further research and development.

PEG-ADA—An orphan drug that is useful in treating SCID related to ADA deficiency.

T cells—Lymphocytes that originate in the thymus gland. T cells regulate the immune system's response to infections. The thymus gland is small or underdeveloped in children with SCID.

Thrush—A disease of the mouth caused by a yeast, *Candida albicans*.

Treatment

Patients with SCID can be treated with **antibiotics** and immune serum to protect them from infections, but these treatments cannot cure the disorder. Bone marrow transplants are currently regarded as one of the few effective standard treatments for SCID.

Investigational treatments

In 1990, the Food and Drug Administration (FDA) approved PEG-ADA, an orphan drug (not available in US but available elsewhere), for the treatment of SCID. PEG-ADA, which is also called pegademase bovine, works by replacing the ADA deficiency in children with this form of SCID. Children who receive weekly injections of PEG-ADA appear to have normal immune functions restored. Another treatment that is still in the experimental stage is **gene therapy**. In gene therapy, the children receive periodic infusions of their own T cells corrected with a gene for ADA that has been implanted in an activated virus.

Prognosis

Currently, there is no cure for SCID. Most untreated patients die before age two.

Prevention

Genetic counseling is recommended for parents of a child with SCID.

Resources

ORGANIZATIONS

Immune Deficiency Foundation. 25 W. Chesapeake Ave., Suite 206, Towson, MD 21204. (800) 296-4433.
< <http://www.primaryimmune.org> >.

National Organization for Rare Disorders. P.O. Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673.
< <http://www.rarediseases.org> >.

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Sex hormones tests

Definition

Sex hormones tests measure levels of the sex hormones, including estrogen, progesterone, and testosterone.

Purpose

The sex hormone tests are ordered to determine if secretion of these hormones is normal. Estrogen fraction test is done to evaluate sexual maturity, menstrual problems, and fertility problems in females. This test may also be used to test for tumors that excrete

estrogen. In pregnant women it aids in determining fetal-placental health. Estrogen fraction is also used to evaluate males who have enlargement of one or both breasts (gynecomastia), or who have feminization syndromes, where they display female sex characteristics.

Progesterone assay test is ordered to evaluate women who are having difficulty becoming pregnant or maintaining a **pregnancy**, and to monitor high-risk pregnancies.

Testosterone levels are ordered to evaluate:

- ambiguous sex characteristics
- **precocious puberty**
- virilizing syndromes in the female
- infertility in the male
- rare tumors of the ovary and testicle

Description

The sex hormones control the development of primary and secondary sexual characteristics. They regulate the sex-related functions of the body, such as the menstrual cycle or the production of eggs or sperm. There are three main types of sex hormones:

- the female sex hormones (called the estrogen hormones)
- the progesterone hormones (which help the body prepare for and maintain pregnancy)
- the male sex hormones, or the androgen hormones

Female sex hormones are responsible for normal menstruation and the development of secondary female characteristics. Testosterone is a hormone that induces **puberty** in the male and maintains male secondary sex characteristics. In females, the adrenal glands and the ovaries secrete small amounts of testosterone.

Estrogen

Estrogen is tested to evaluate menstrual status, sexual maturity, and **gynecomastia** (or feminization syndromes). It is a tumor marker for patients with certain ovarian tumors. E1, a type of estrogen, is the most active estrogen in the nonpregnant female.

E3 (estriol) is the major estrogen in the pregnant female. It is produced in the placenta. Excretion of estriol increases around the eighth week of gestation and continues to rise until shortly before delivery. Serial urine and blood studies of this hormone are used to assess placental function and fetal normality in high-risk pregnancies. Falling values during

pregnancy suggest fetoplacental deterioration and require prompt reassessment of the pregnancy, including the possibility of early delivery.

Progesterone

Progesterone is essential for the healthy functioning of the female reproductive system. Produced in the ovaries during the second half of the menstrual cycle, and by the placenta during pregnancy, small amounts of progesterone are also produced in the adrenal glands and testes.

After ovulation, an increase of progesterone causes the uterine lining to thicken in preparation for the implantation of a fertilized egg. If this event does not take place, progesterone and estrogen levels fall, resulting in shedding of the uterine lining.

Progesterone is essential during pregnancy, not only ensuring normal functioning of the placenta, but passing into the developing baby's circulation, where it is converted in the adrenal glands to corticosteroid hormones.

Testosterone

Testosterone is the most important of the male sex hormones. It is responsible for stimulating bone and muscle growth, and sexual development. It is produced by the testes and in very small amounts by the ovaries. Most testosterone tests measure total testosterone.

Testosterone stimulates sperm production (spermatogenesis), and influences the development of male secondary sex characteristics. Overproduction of testosterone caused by testicular, adrenal, or **pituitary tumors** in the young male may result in precocious puberty.

Overproduction of testosterone in females, caused by ovarian and adrenal tumors, can result in masculinization, the symptoms of which include cessation of the menstrual cycle (amenorrhea) and excessive growth of body hair (**hirsutism**).

When reduced levels of testosterone in the male indicate underactivity of the testes (hypogonadism), testosterone stimulation tests may be ordered.

Preparation

The progesterone and testosterone tests require a blood sample; it is not necessary for the patient to restrict food or fluids before the test. Testosterone specimens should be drawn in the morning, as testosterone levels are highest in the early morning hours.

The estrogen fraction test can be performed on blood and/or urine. It is not necessary for the patient to restrict food or fluids for either test. If a 24-hour urine test has been requested, the patient should call the laboratory for instructions.

Risks

Risks for these blood tests are minimal, but may include slight bleeding from the puncture site, **fainting** or feeling lightheaded after having blood drawn, or blood accumulating under the puncture site (hematoma).

Normal results

Estrogen levels vary in women, ranging from 24–149 picograms per ml of blood. In men, the normal range is between 12–34 picograms per ml of blood.

Progesterone levels vary from less than 150 nanograms per deciliter (ng/dL) of blood to 2,000 nanograms in menstruating women. During pregnancy, progesterone levels range from 1,500–20,000 ng/dL of blood.

Testosterone values vary from laboratory to laboratory, but can generally be found within the following levels:

- Men. 300–1,200 ng/dL
- Women. 30–95 ng/dL
- Prepubertal children. Less than 100 ng/dL (boys), less than 40 ng/dL (girls).

Abnormal results

Increased levels of estrogen are seen in feminization syndromes:

- when a male begins to develop female secondary sex characteristics
- during precocious puberty
- when children develop secondary sexual characteristics at an abnormally early age
- because of ovarian, testicular, or adrenal tumor
- During normal pregnancy, **cirrhosis**, and increased thyroid levels (hyperthyroidism)

Decreased levels of estrogen are found in the following conditions:

- a failing pregnancy
- during **menopause**
- anorexia nervosa

- primary and secondary hypogonadism
- turner's syndrome, seen in females with one missing X chromosome

Increased levels of progesterone are seen:

- during ovulation and pregnancy
- with certain types of ovarian cysts
- with a tumor of the ovary known as a choriocarcinoma

Decreased levels of progesterone are seen:

- in toxemia of pregnancy
- with a threatened abortion
- during placental failure
- after fetal **death**
- with amenorrhea
- due to ovarian dysfunction

Increased levels (male) of testosterone are found in:

- sexual precocity
- the viral infection of encephalitis
- tumors involving the adrenal glands
- testicular tumors
- excessive thyroid production (hyperthyroidism)
- testosterone resistance syndromes.

Decreased levels (male) of testosterone are seen in:

- Klinefelter syndrome
- a chromosomal deficiency
- primary and secondary hypogonadism
- down syndrome
- surgical removal of the testicles
- cirrhosis.

Increased levels (females) of testosterone are found in ovarian and adrenal tumors and in the presence of excessive hair growth of unknown cause (hirsutism).

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Sex reassignment surgery see **Gender reassignment surgery**

Sex therapy

Definition

Sex therapy is the treatment of **sexual dysfunction**.

Purpose

Sex therapy utilizes various techniques in order to relieve sexual dysfunction commonly caused by **premature ejaculation** or sexual **anxiety** and to improve the sexual health of the patient.

Precautions

Sexual dysfunction conjures up feelings of guilt, anger, insecurity, frustration, and rejection. Therapy is slow and requires open communication and understanding between sexual partners. Therapy may inadvertently address interpersonal communication problems.

Description

Sex therapy is conducted by a trained therapist, doctor, or psychologist. The initial sessions should cover a complete history not only of the sexual problem but of the entire relationship and each individual's background and personality. The sexual relationship should be discussed in the context of the entire relationship. In fact, sexual counseling may de-emphasize sex until other aspects of the relationship are better understood and communicated.

There are several techniques that combat sexual dysfunction and are used in sex therapy. They include:

- Semans' technique: which is used to help combat premature ejaculation with a "start-stop" approach to penis stimulation. By stimulating the man up to the point of ejaculation and then stopping, the man will become more aware of his response. More awareness leads to greater control, and open stimulation of both partners leads to greater communication and less anxiety. The start-stop technique is conducted four times until the man is allowed to ejaculate.
- Sensate focus therapy, the practice of nongenital and genital touching between partners in order to decrease sexual anxiety and build communication. First, partners explore each other's bodies without touching the genitals or breasts. Once the couple is comfortable with nongenital touching, they can expand to genital stimulation. Intercourse is prohibited in order to allow the partners to expand their intimacy and communication.

- Squeeze technique, which is used to treat premature ejaculation. When the man feels the urge to ejaculate, his partner squeezes his penis just below the head. This stops ejaculation and gives the man more control over his response.

Aftercare

Habits change slowly. All the techniques must be practiced faithfully for long periods of time to relearn behaviors. Communication is imperative.

Resources

BOOKS

Masters, William H., Virginia E. Johnson, and Robert C. Kolodny. *Heterosexuality*. New York: Harper Collins Publishers, Inc., 1994.

J. Ricker Polsdorfer, MD

Sexual abuse see **Rape and sexual assault**

Sexual arousal disorders see **Sexual dysfunction**

Sexual desire disorders see **Sexual dysfunction**

Sexual dysfunction

Definition

Sexual dysfunction is broadly defined as the inability to fully enjoy sexual intercourse. Specifically, sexual dysfunctions are disorders that interfere with a full sexual response cycle. These disorders make it difficult for a person to enjoy or to have sexual intercourse. While sexual dysfunction rarely threatens physical health, it can take a heavy psychological toll, bringing on depression, **anxiety**, and debilitating feelings of inadequacy.

Description

Sexual dysfunction takes different forms in men and women. A dysfunction can be life-long and always present, acquired, situational, or generalized, occurring despite the situation. A man may have a sexual problem if he:

- ejaculates before he or his partner desires
- does not ejaculate, or experiences delayed ejaculation

- is unable to have an erection sufficient for pleasurable intercourse
- feels **pain** during intercourse
- lacks or loses sexual desire

A woman may have a sexual problem if she:

- lacks or loses sexual desire
- has difficulty achieving orgasm
- feels anxiety during intercourse
- feels pain during intercourse
- feels vaginal or other muscles contract involuntarily before or during sex
- has inadequate lubrication

The most common sexual dysfunctions in men include:

- **Erectile dysfunction:** an impairment of the erectile reflex. The man is unable to have or maintain an erection that is firm enough for coitus or intercourse.
- **Premature ejaculation:** rapid ejaculation with minimal sexual stimulation before, on, or shortly after penetration and before the person wishes it.
- **Ejaculatory incompetence:** the inability to ejaculate within the vagina despite a firm erection and relatively high levels of sexual arousal.
- **Retarded ejaculation:** a condition in which the bladder neck does not close off properly during orgasm so that the semen spurts backward into the bladder.

Until recently, it was presumed that women were less sexual than men. In the past two decades, traditional views of female sexuality were all but demolished, and women's sexual needs became accepted as legitimate in their own right.

Female sexual dysfunctions include:

- **Sexual arousal disorder:** the inhibition of the general arousal aspect of sexual response. A woman with this disorder does not lubricate, her vagina does not swell, and the muscle that surrounds the outer third of the vagina does not tighten—a series of changes that normally prepare the body for orgasm (“the orgasmic platform”). Also, in this disorder, the woman typically does not feel erotic sensations.
- **Orgasmic disorder:** the impairment of the orgasmic component of the female sexual response. The woman may be sexually aroused but never reach orgasm. Orgasmic capacity is less than would be reasonable for her age, sexual experience, and the adequacy of sexual stimulation she receives.

- **Vaginismus:** a condition in which the muscles around the outer third of the vagina have involuntary spasms in response to attempts at vaginal penetration.
- **Painful intercourse:** a condition that can occur at any age. Pain can appear at the start of intercourse, mid-way through coital activities, at the time of orgasm, or after intercourse is completed. The pain can be felt as burning, sharp searing, or cramping; it can be external, within the vagina, or deep in the pelvic region or abdomen.

Causes and symptoms

Many factors, of both physical and psychological natures, can affect sexual response and performance. Injuries, ailments, and drugs are among the physical influences; in addition, there is increasing evidence that chemicals and other environmental pollutants depress sexual function. As for psychological factors, sexual dysfunction may have roots in traumatic events such as **rape** or incest, guilt feelings, a poor self-image, depression, chronic **fatigue**, certain religious beliefs, or marital problems. Dysfunction is often associated with anxiety. If a man operates under the misconception that all sexual activity must lead to intercourse and to orgasm by his partner, and if the expectation is not met, he may consider the act a failure.

Men

With premature ejaculation, physical causes are rare, although the problem is sometimes linked to a neurological disorder, prostate infection, or **urethritis**. Possible psychological causes include anxiety (mainly performance anxiety), guilt feelings about sex, and ambivalence toward women. However, research has failed to show a direct link between premature ejaculation and anxiety. Rather, premature ejaculation seems more related to sexual inexperience in learning to modulate arousal.

When men experience painful intercourse, the cause is usually physical; an infection of the prostate, urethra, or testes, or an allergic reaction to spermicide or condoms. Painful erections may be caused by **Peyronie's disease**, fibrous plaques on the upper side of the penis that often produce a bend during erection. **Cancer** of the penis or testes and arthritis of the lower back can also cause pain.

Retrograde ejaculation occurs in men who have had prostate or urethral surgery, take medication that keeps the bladder open, or suffer from diabetes, a disease that can injure the nerves that normally close the bladder during ejaculation.

Erectile dysfunction is more likely than other dysfunctions to have a physical cause. Drugs, diabetes (the most common physical cause), Parkinson's disease, **multiple sclerosis**, and spinal cord lesions can all be causes of erectile dysfunction. When physical causes are ruled out, anxiety is the most likely psychological cause of erectile dysfunction.

Women

Dysfunctions of arousal and orgasm in women also may be physical or psychological in origin. Among the most common causes are day-to-day discord with one's partner and inadequate stimulation by the partner. Finally, sexual desire can wane as one ages, although this varies greatly from person to person.

Pain during intercourse can occur for any number of reasons, and location is sometimes a clue to the cause. Pain in the vaginal area may be due to infection, such as urethritis; also, vaginal tissues may become thinner and more sensitive during breast-feeding and after **menopause**. Deeper pain may have a pelvic source, such as **endometriosis**, pelvic **adhesions**, or uterine abnormalities. Pain can also have a psychological cause, such as fear of injury, guilt feelings about sex, fear of **pregnancy** or injury to the fetus during pregnancy, or recollection of a previous painful experience.

Vaginismus may be provoked by these psychological causes as well, or it may begin as a response to pain, and continue after the pain is gone. Both partners should understand that the vaginal contraction is an involuntary response, outside the woman's control.

Similarly, insufficient lubrication is involuntary, and may be part of a complex cycle. Low sexual response may lead to inadequate lubrication, which may lead to discomfort, and so on.

Diagnosis

In deciding when a sexual dysfunction is present, it is necessary to remember that while some people may be interested in sex at almost any time, others have low or seemingly nonexistent levels of sexual interest. Only when it is a source of personal or relationship distress, instead of voluntary choice, is it classified as a sexual dysfunction.

The first step in diagnosing a sexual dysfunction is usually discussing the problem with a doctor, who will need to ask further questions in an attempt to differentiate among the types of sexual dysfunction. The physician may also perform a physical exam of the genitals, and may order further medical tests, including measurement of hormone levels in the blood. Men

may be referred to a specialist in diseases of the urinary and genital organs (urologist), and primary care physicians may refer women to a gynecologist.

Treatment

Treatments break down into two main kinds: behavioral psychotherapy and physical. **Sex therapy**, which is ideally provided by a member of the American Association of Sexual Educators, Counselors, and Therapists (AASECT), universally emphasizes correcting sexual misinformation, the importance of improved partner communication and honesty, anxiety reduction, sensual experience and pleasure, and interpersonal tolerance and acceptance. Sex therapists believe that many sexual disorders are rooted in learned patterns and values. These are termed psychogenic. An underlying assumption of sex therapy is that relatively short-term outpatient therapy can alleviate learned patterns, restrict symptoms, and allow a greater satisfaction with sexual experiences.

In some cases, a specific technique may be used during intercourse to correct a dysfunction. One of the most common is the "squeeze technique" to prevent premature ejaculation. When a man feels that an orgasm is imminent, he withdraws from his partner. Then, the man or his partner gently squeezes the head of the penis to halt the orgasm. After 20–30 seconds, the couple may resume intercourse. The couple may do this several times before the man proceeds to ejaculation.

In cases where significant sexual dysfunction is linked to a broader emotional problem, such as depression or **substance abuse**, intensive psychotherapy and/or pharmaceutical intervention may be appropriate.

In many cases, doctors may prescribe medications to treat an underlying physical cause or sexual dysfunction. Possible medical treatments include:

- clomipramine and fluoxetine for premature ejaculation
- papaverine and prostaglandin for erectile difficulties
- hormone replacement therapy for female dysfunctions
- Viagra, a pill approved in 1998 as a treatment for impotence

Alternative treatment

A variety of alternative therapies can be useful in the treatment of sexual dysfunction. Counseling or

KEY TERMS

Ejaculatory incompetence—The inability to ejaculate within the vagina.

Erectile dysfunction—Difficulty achieving or maintaining an erect penis.

Orgasmic disorder—The impairment of the ability to reach sexual climax.

Painful intercourse (dyspareunia)—Generally thought of as a female dysfunction but also affects males. Pain can occur anywhere.

Premature ejaculation—Rapid ejaculation before the person wishes it, usually in less than one to two minutes after beginning intercourse.

Retrograde ejaculation—A condition in which the semen spurts backward into the bladder.

Sexual arousal disorder—The inhibition of the general arousal aspect of sexual response.

Vaginismus—Muscles around the outer third of the vagina have involuntary spasms in response to attempts at vaginal penetration, not allowing for penetration.

psychotherapy is highly recommended to address any emotional or mental components of the disorder. Botanical medicine, either western, Chinese, or ayurvedic, as well as nutritional supplementation, can help resolve biochemical causes of sexual dysfunction. **Acupuncture** and homeopathic treatment can be helpful by focusing on the energetic aspects of the disorder.

Some problems with sexual function are normal. For example, women starting a new or first relationship may feel sore or bruised after intercourse and find that an over-the-counter lubricant makes sex more pleasurable. Simple techniques, such as soaking in a warm bath, may relax a person before intercourse and improve the experience. **Yoga** and **meditation** provide needed mental and physical relaxation for several conditions, such as vaginismus. Relaxation therapy eases and relieves anxiety about dysfunction. Massage is extremely effective at reducing **stress**, especially if performed by the partner.

Prognosis

There is no single cure for sexual dysfunctions, but almost all can be controlled. Most people who have a sexual dysfunction fare well once they get into a treatment program. For example, a high percentage of men

with premature ejaculation can be successfully treated in two to three months. Furthermore, the gains made in sex therapy tend to be long-lasting rather than short-lived.

Resources

ORGANIZATIONS

American Academy of Clinical Sexologists. 1929 18th St. NW, Suite 1166, Washington, DC 20009. (202) 462-2122.

American Association for Marriage and Family Therapy. 1133 15th St., NW Suite 300, Washington, DC 20005-2710. (202) 452-0109. <<http://www.aamft.org>>.

David James Doermann

Sexual perversions

Definition

Sexual perversions are conditions in which sexual excitement or orgasm is associated with acts or imagery that are considered unusual within the culture. To avoid problems associated with the stigmatization of labels, the neutral term paraphilia, derived from Greek roots meaning “alongside of” and “love,” is used to describe what used to be called sexual perversions. A paraphilia is a condition in which a person’s sexual arousal and gratification depend on a fantasy theme of an unusual situation or object that becomes the principal focus of sexual behavior.

Description

Paraphilias can revolve around a particular sexual object or a particular act. They are defined by *DSM-IV* as “sexual impulse disorders characterized by intensely arousing, recurrent sexual fantasies, urges and behaviors considered deviant with respect to cultural norms and that produce clinically significant distress or impairment in social, occupational or other important areas of psychosocial functioning.” The nature of a paraphilia is generally specific and unchanging, and most of the paraphilias are far more common in men than in women.

Paraphilias differ from what some people might consider “normal” sexual activity in that these behaviors cause significant distress or impairment in areas of life functioning. They do not refer to the normal use of sexual fantasy, activity, or objects to heighten sexual excitement where there is no distress or

impairment. The most common signs of sexual activity that can be classified as paraphilia include: the inability to resist an impulse for the sexual act, the requirement of participation by non-consenting or under-aged individuals, legal consequences, resulting sexual dysfunction, and interference with normal social relationships.

Paraphilias include fantasies, behaviors, and/or urges which:

- involve nonhuman sexual objects, such as shoes or undergarments
- require the suffering or humiliation of oneself or partner
- involve children or other non-consenting partners

The most common paraphilias are:

- exhibitionism, or exposure of the genitals
- fetishism, or the use of nonliving objects
- frotteurism, or touching and rubbing against a non-consenting person
- pedophilia, or the focus on prepubescent children
- sexual masochism, or the receiving of humiliation or suffering
- sexual sadism, or the inflicting of humiliation or suffering
- transvestic fetishism, or cross-dressing
- voyeurism, or watching others engage in undressing or sexual activity

A paraphiliac often has more than one paraphilia. Paraphilias often result in a variety of associated problems, such as guilt, depression, shame, isolation, and impairment in the capacity for normal social and sexual relationships. A paraphilia can, and often does, become highly idiosyncratic and ritualized.

Causes and symptoms

There is very little certainty about what causes a paraphilia. Psychoanalysts generally theorize that these conditions represent a regression to or a fixation at an earlier level of psychosexual development resulting in a repetitive pattern of sexual behavior that is not mature in its application and expression. In other words, an individual repeats or reverts to a sexual habit arising early in life. Another psychoanalytic theory holds that these conditions are all expressions of hostility in which sexual fantasies or unusual sexual acts become a means of obtaining revenge for a childhood trauma. The persistent, repetitive nature of the paraphilia is caused by an inability to erase the

underlying trauma completely. Indeed, a history of childhood sexual **abuse** is sometimes seen in individuals with paraphilias.

However, behaviorists suggest, instead, that the paraphilia begins via a process of conditioning. Nonsexual objects can become sexually arousing if they are frequently and repeatedly associated with a pleasurable sexual activity. The development of a paraphilia is not usually a matter of conditioning alone; there must usually be some predisposing factor, such as difficulty forming person-to-person sexual relationships or poor self-esteem.

The following are situations or causes that might lead someone in a paraphiliac direction:

- parents who humiliate and punish a small boy for strutting around with an erect penis
- a young boy who is sexually abused
- an individual who is dressed in a woman's clothes as a form of parental punishment
- fear of sexual performance or intimacy
- inadequate counseling
- excessive alcohol intake
- physiological problems
- sociocultural factors
- psychosexual trauma

Diagnosis

Whatever the cause, paraphiliacs apparently rarely seek treatment unless they are induced into it by an arrest or discovery by a family member. This makes diagnosis before a confrontation very difficult.

Paraphiliacs may select an occupation, or develop a hobby or volunteer work, that puts them in contact with the desired erotic stimuli, for example, selling women's shoes or lingerie in fetishism, or working with children in pedophilia. Other coexistent problems may be alcohol or drug abuse, intimacy problems, and personality disturbances, especially emotional immaturity. Additionally, there may be sexual dysfunctions. **Erectile dysfunction** and an inability to ejaculate may be common in attempts at sexual activity without the paraphiliac theme.

Paraphilias may be mild, moderate, or severe. An individual with mild paraphilia is markedly distressed by the recurrent paraphiliac urges but has never acted on them. The moderate has occasionally acted on the paraphilic urge. A severe paraphiliac has repeatedly acted on the urge.

Treatment

The literature describing treatment is fragmentary and incomplete. Traditional **psychoanalysis** has not been particularly effective with paraphilia and generally requires several years of treatment. Therapy with hypnosis has also had poor results. Current interests focus primarily on several behavioral techniques that include the following:

- Aversion imagery involves the pairing of a sexually arousing paraphilic stimulus with an unpleasant image, such as being arrested or having one's name appear in the newspaper.
- Desensitization procedures neutralize the anxiety-provoking aspects of nonparaphilic sexual situations and behavior by a process of gradual exposure. For example, a man afraid of having sexual contact with women his own age might be led through a series of relaxation procedures aimed at reducing his **anxiety**.
- Social skills training is used with either of the other approaches and is aimed at improving a person's ability to form interpersonal relationships.
- Orgasmic reconditioning may instruct a person to masturbate using his paraphilia fantasy and to switch to a more appropriate fantasy just at the moment of orgasm.

In addition to these therapies, drugs are sometimes prescribed to treat paraphilic behaviors. Drugs that drastically lower testosterone temporarily (anti-androgens) have been used for the control of repetitive deviant sexual behaviors and have been prescribed for paraphilia-related disorders as well. Cyproterone acetate inhibits testosterone directly at androgen receptor sites. In its oral form, the usual prescribed dosage range is 50–200 mg per day.

Serotonergics (drugs that boost levels of the brain chemical serotonin) are prescribed for anxious and depressive symptoms. Of the serotonergic agents reported, fluoxetine has received the most attention, although lithium, clomipramine, buspirone, and sertraline are reported as effective in case reports and open clinical trials with outpatients. Other alternative augmentation strategies that may be effective include adding a low dose of a secondary amine tricyclic antidepressant to the primary serotonergics, but these reports are only anecdotal.

Prognosis

Despite more than a decade of experience with psychotherapeutic treatment programs, most workers

KEY TERMS

Exhibitionism—Obtaining sexual arousal by exposing genitals to an unsuspecting stranger.

Fetishism—Obtaining sexual arousal using or thinking about an inanimate object or part of the body.

Frotteurism—Obtaining sexual arousal and gratification by rubbing one's genitals against others in public places.

Masochism—Sexual arousal by having pain and/or humiliation inflicted upon oneself.

Pedophilia—Sex or sexual activity with children who have not reached puberty.

Sadism—Sexual arousal through inflicting pain on another person.

Transvestitism—Sexual arousal from dressing in the clothes of the opposite sex.

Voyeurism—Sexual arousal by observing nude individuals without their knowledge.

in the field are not convinced that they have a high degree of success. Furthermore, because some cases involve severe abuse, many in the general public would prefer to “lock up” the sex offender than to have him out in the community in a treatment program or on parole after the treatment program has been completed.

Paraphilia and paraphilia-related disorders are more prevalent than most clinicians suspect. Since these disorders are cloaked in shame and guilt, the presence of these conditions may not be adequately revealed until a therapeutic alliance is firmly established. Once a diagnosis is established, appropriate education about possible behavioral therapies and appropriate use of psychopharmacological agents can improve the prognosis for these conditions.

Resources

ORGANIZATIONS

American Academy of Clinical Sexologists. 1929 18th St., N.W., Suite 1166, Washington, DC 20009. (202) 462-2122.

American Association for Marriage and Family Therapy. 1133 15th St., NW Suite 300, Washington, DC 20005-2710. (202) 452-0109. <<http://www.aamft.org>>.

David James Doermann

Sexually transmitted diseases

Definition

Sexually transmitted disease (STD) is a term used to describe more than 20 different infections that are transmitted through exchange of semen, blood, and other body fluids; or by direct contact with the affected body areas of people with STDs. Sexually transmitted diseases are also called venereal diseases.

Description

The Centers for Disease Control and Prevention (CDC) has reported that 85% of the most prevalent infectious diseases in the United States are sexually transmitted. The rate of STDs in this country is 50 to 100 times higher than that of any other industrialized nation. One in four sexually active Americans will be affected by an STD at some time in his or her life.

About 12 million new STD infections occur in the United States each year. One in four occurs in someone between the ages of 16 and 19. Almost 65% of all STD infections affect people under the age of 25.

Types of STDs

STDs can have very painful long-term consequences as well as immediate health problems. They can cause:

- birth defects
- blindness
- bone deformities
- brain damage
- cancer
- heart disease
- infertility and other abnormalities of the reproductive system
- mental retardation
- **death**

Some of the most common and potentially serious STDs in the United States include:

- **Chlamydia.** This STD is caused by the bacterium *Chlamydia trachomatis*, a microscopic organism that lives as a parasite inside human cells. Although over 526,000 cases of chlamydia were reported in the United States in 1997, the CDC estimates that nearly three million cases occur annually because 75% of women and 50% of men show no symptoms of the disease after infection. Approximately 40% of

women will develop **pelvic inflammatory disease** (PID) as a result of chlamydia infection, a leading cause of infertility.

- **Human papillomavirus (HPV).** HPV causes **genital warts** and is the single most important risk factor for **cervical cancer** in women. Over 100 types of HPV exist, but only about 30 of them can cause genital warts and are spread through sexual contact. In some instances, warts are passed from mother to child during **childbirth**, leading to a potentially life-threatening condition for newborns in which warts develop in the throat (laryngeal papillomatosis).
- **Genital herpes.** Herpes is an incurable viral infection thought to be one of the most common STDs in this country. It is caused by one of two types of herpes simplex viruses: HSV-1 (commonly causing oral herpes) or HSV-2 (usually causing genital herpes). The CDC estimates that 45 million Americans (one out of every five individuals 12 years of age or older) are infected with HSV-2; this number has increased 30% since the 1970s. HSV-2 infection is more common in women (one out of every four women) than men (one out of every five men) and in African Americans (45.9%) than Caucasians (17.6%).
- **Gonorrhea.** The bacterium *Neisseria gonorrhoeae* is the causative agent of gonorrhea and can be spread by vaginal, oral, or anal contact. The CDC reports that approximately 650,000 individuals are infected with gonorrhea each year in the United States, with 132.2 infections per 100,000 individuals occurring in 1999. Approximately 75% of American gonorrhea infections occur in persons aged 15 to 29 years old. In 1999, 75% of reported gonorrhea cases occurred among African Americans.
- **Syphilis.** Syphilis is a potentially life-threatening infection that increases the likelihood of acquiring or transmitting HIV. In 1998, the CDC reported approximately 38,000 cases of syphilis in the United States; this included 800 cases of congenital syphilis. Congenital syphilis causes irreversible health problems or death in as many as 40% of all live babies born to women with untreated syphilis.
- **Human immunodeficiencyvirus (HIV) infection.** In 2000, the CDC reported that 120,223 people in the United States are HIV-positive and 426,350 are living with **AIDS**. In addition, approximately 1,000-2,000 children are born each year with HIV infection. It is also estimated that 33 million adults and 1.3 million children worldwide were living with HIV/AIDS as of 1999 with 5.4 million being newly infected that year. There is no cure for this STD.

Drugs Used To Treat STDs

Brand Name (Generic Name)	Possible Common Side Effects Include:
Achromycin V (tetracycline hydrochloride)	Blurred vision, headache, dizziness, rash, hives, appetite loss, nausea and vomiting
Amoxil (amoxicillin)	Behavioral changes, diarrhea, hives, nausea and vomiting
Ceftin (cefuroxime axetil)	Nausea and vomiting, diarrhea, irritated skin
Doryx (doxycycline hyclate)	Itching (genital and/or rectal), nausea and vomiting, appetite loss, diarrhea, swelling
E.E.S., E-Mycin, ERYC, Ery-Tab, Erythrocine, Ilosone (erythromycin)	Diarrhea, nausea and vomiting, appetite loss, abdominal pain
Flagyl (metronidazole)	Numbness, tingling sensation in extremities, seizures
Floxin (ofloxacin)	Genital itching, nausea and vomiting, headache, diarrhea, dizziness
Minocin (minocycline hydrochloride)	Blurred vision, anemia, hives, rash, throat irritation
Noroxin (norfloxacin)	Headache, nausea, dizziness
Omnipen (ampicillin)	Itching, rash, hives, peeling skin, nausea and vomiting
Penetrex (enoxacin)	Nausea and vomiting
Zithromax (azithromycin)	Nausea and vomiting, diarrhea, abdominal pain
Zovirax (acyclovir)	Fluid retention, headache, rash, tingling sensation

Social groups and STDs

STDs affect certain population groups more severely than others. Women, young people, and members of minority groups are particularly affected. Women in any age bracket are more likely than men to develop medical complications related to STDs. With respect to racial and ethnic categories, the incidence of syphilis is 60 times higher among African Americans than among Caucasians, and four times higher in Hispanics than in Anglos. According to the CDC, in 1999 African Americans accounted for 77% of the total number of gonorrhea cases and nearly 46% of all genital herpes cases.

Causes and symptoms

The symptoms of STDs vary somewhat according to the disease agent (virus or bacterium), the sex of the patient, and the body systems affected. The symptoms of some STDs are easy to identify; others produce infections that may either go unnoticed for some time or are easy to confuse with other diseases.

Syphilis in particular can be confused with disorders ranging from **infectious mononucleosis** to allergic reactions to prescription medications. In addition, the incubation period of STDs varies. Some produce symptoms close enough to the time of sexual contact—often less than 48 hours later—for the patient to recognize the connection between the behavior and the symptoms. Others have a longer incubation period, so that the patient may not recognize the early symptoms as those of a sexually transmitted infection.

Some symptoms of STDs affect the genitals and reproductive organs:

- A woman who has an STD may bleed when she is not menstruating or has abnormal vaginal discharge. Vaginal burning, **itching**, and odor are common, and she may experience **pain** in her pelvic area while having sex
- A discharge from the tip of the penis may be a sign that a man has an STD. Males may also have painful or burning sensations when they urinate.
- There may be swelling of the lymph nodes near the groin area.
- Both men and women may develop skin **rashes**, sores, bumps, or blisters near the mouth or genitals. Homosexual men frequently develop these symptoms in the area around the anus.

Other symptoms of STDs are systemic, which means that they affect the body as a whole. These symptoms may include:

- fever, chills, and similar flu-like symptoms
- skin rashes over large parts of the body
- arthritis-like pains or aching in the joints
- throat swelling and redness that lasts for three weeks or longer

Diagnosis

A sexually active person who has symptoms of an STD or who has had an STD or symptoms of infection should be examined without delay by one of the following health care professionals:

- a specialist in **women's health** (gynecologist)
- a specialist in disorders of the urinary tract and the male sexual organs (urologist)
- a family physician
- a nurse practitioner
- a specialist in skin disorders (dermatologist).



Cultures on agar plates. (Photograph by T. McCarthy, Custom Medical Stock Photo. Reproduced by permission.)

The diagnostic process begins with a thorough **physical examination** and a detailed medical history that documents the patient's sexual history and assesses the risk of infection.

The doctor or other healthcare professional will:

- Describe the testing process. This includes all blood tests and other tests that may be relevant to the specific infection.
- Explain the meaning of the test results.
- Provide the patient with information regarding high-risk behaviors and any necessary treatments or procedures.

The doctor may suggest that a patient diagnosed with one STD be tested for others, as it's possible to have more than one STD at a time. One infection may hide the symptoms of another or create a climate that fosters its growth. At present, it is particularly important that persons who are HIV-positive be tested for syphilis as well.

Notification

The law in most parts of the United States requires public health officials to trace and contact the partners of persons with STDs. Minors, however, can get treatment without their parents' permission. Public health departments in most states can provide information about STD clinic locations; Planned Parenthood facilities provide testing and counseling. These agencies can also help with or assume the responsibility of notifying sexual partners who must be tested and may require treatment.

Treatment

Although self-care can relieve some of the pain of genital herpes or genital warts that has recurred after

being diagnosed and treated by a physician, other STD symptoms require immediate medical attention.

Antibiotics are prescribed to treat gonorrhea, chlamydia, syphilis, and other STDs caused by bacteria. Although prompt diagnosis and early treatment almost always cures these STDs, new infections can develop if exposure continues or is renewed. Viral infections can be treated symptomatically with antiviral medications.

Prognosis

The prognosis for recovery from STDs varies among the different diseases. The prognosis for recovery from gonorrhea, syphilis, and other STDs caused by bacteria is generally good, provided that the disease is diagnosed early and treated promptly. Untreated syphilis in particular can lead to long-term complications and disability. Viral STDs (genital herpes, genital warts, HIV) cannot be cured but must be treated on a long-term basis to relieve symptoms and prevent life-threatening complications.

Prevention

Vaccines

Vaccines for the prevention of **hepatitis A** and **hepatitis B** are currently recommended for gay and bisexual men, users of illegal drugs, health care workers, and others at risk of contracting these diseases. Vaccines to prevent other STDs are being tested and may be available within several years.

Lifestyle choices

The risk of becoming infected with an STD can be reduced or eliminated by changing certain personal behaviors. Abstaining from sexual relations or maintaining a mutually monogamous relationship with a partner are legitimate options. It is also wise to avoid sexual contact with partners who are known to be infected with an STD, whose health status is unknown, who **abuse** drugs, or who are involved in prostitution.

Use of condoms and other contraceptives

Men or women who have sex with a partner of known (or unsure) infection should make sure a new **condom** is used every time they have genital, oral, or anal contact. Used correctly and consistently, male condoms provide good protection against HIV and other STDs such as gonorrhea, chlamydia, and syphilis. Female condoms (lubricated sheaths inserted into

KEY TERMS

Chlamydia—A microorganism that resembles certain types of bacteria and causes several sexually transmitted diseases in humans.

Condom—A thin sheath worn over the penis during sexual intercourse to prevent pregnancy or the transmission of STDs. There are also female condoms.

Diaphragm—A dome-shaped device used to cover the back of a woman's vagina during intercourse in order to prevent pregnancy.

Pelvic inflammatory disease (PID)—An inflammation of the tubes leading from a woman's ovaries to the uterus (the Fallopian tubes), caused by a bacterial infection. PID is a leading cause of fertility problems in women.

Venereal disease—Another term for sexually transmitted disease.

the vagina) have also been shown to be effective in preventing HIV and other STDs. Condoms provide a measure of protection against genital herpes, genital warts, and hepatitis B.

Spermicides and diaphragms can decrease the risk of transmission of some STDs. They do not protect women from contracting HIV. Birth-control pills, patches, or injections do not prevent STDs. Neither do surgical sterilization or **hysterectomy**.

Hygienic measures

Urinating and washing the genital area with soap and water immediately after having sex may eliminate some germs before they cause infection. Douching, however, can spread infection deeper into the womb. It may also increase a woman's risk of developing pelvic inflammatory disease (PID).

Resources

ORGANIZATIONS

National STD Hotline. (800)227-8922.

Planned Parenthood Federation of America. (800)230-7526.
< <http://www.plannedparenthood.org> >.

OTHER

Sexually Transmitted Diseases. March 24, 2001.

< <http://www.cdc.gov/nchstp/dstd/dstdp.html> >.

Maureen Haggerty

Sexually transmitted diseases cultures

Definition

Sexually transmitted diseases are infections spread from person to person through sexual contact. A culture is a test in which a laboratory attempts to grow and identify the microorganism causing an infection.

Purpose

Sexually transmitted diseases (STDs) produce symptoms such as genital discharge, **pain** during urination, bleeding, pelvic pain, skin ulcers, or **urethritis**. Often, however, they produce no immediate symptoms. Therefore, the decision to test for these diseases must be based not only the presence of symptoms, but on whether or not a person is at risk of having one or more of the diseases. Activities, such as drug use and sex with more than one partner, put a person at high risk for these diseases.

STD cultures are necessary to diagnose certain types of STDs. Only after the infection is diagnosed can it be treated and further spread of the infection prevented. Left untreated, consequences of these diseases range from discomfort to **infertility** to **death**. In addition, these diseases, if present in a pregnant woman, can be passed from mother to fetus.

Description

Gonorrhea, **syphilis**, chlamydia, **chancroid**, herpes, human papillomavirus, human **immunodeficiency virus (HIV)**, and mycoplasma are common sexually transmitted diseases. Not all are diagnosed with a culture. For those that are, a sample of material is taken from the infection site, placed in a sterile container, and sent to the laboratory.

Bacterial cultures

In the laboratory, a portion of material from the infection site is spread over the surface of several different types of culture plates and placed in an incubator at body temperature for one to two days. Bacteria present in the sample will multiply and appear on the plates as visible colonies. They are identified by the appearance of their colonies and by the results of biochemical tests and a gram stain. The Gram stain is done by smearing part of a colony onto a microscope slide. After it dries, the slide is stained with purple and red stains, then examined under a microscope. The color of stain picked up by

the bacteria (purple or red), the shape (such as round or rectangle), and the size provide valuable clues as to the identity and which **antibiotics** might work best. Bacteria that stain purple are called Gram-positive; those that stain red are called gram-negative.

The result of the gram stain is available the same day or in less than an hour if requested by the physician. An early report, known as a preliminary report, is usually available after one day. This report will tell if any microorganisms have been found yet, and if so, their Gram stain appearance—for example, a Gram-negative rod or a gram-positive cocci. The final report, usually available in one to seven days, includes complete identification and an estimate of the quantity of the microorganisms isolated.

A sensitivity test, also called antibiotic susceptibility test, commonly done on bacteria isolated from an infection site, is not always done on bacteria isolated from a sexually transmitted disease. These bacteria often are treated using antibiotics that are part of a standard treatment protocol.

GONORRHEA. *Neisseria gonorrhoeae*, also called gonococcus or GC, causes gonorrhea. It infects the surfaces of the genitourinary tract, primarily the urethra in males and the cervix in females. On a gram stain done on material taken from an infection site, the bacteria appear as small gram-negative diplococci (pairs of round bacteria) inside white blood cells. *Neisseria gonorrhoeae* grows on a special culture plate called Thayer-Martin (TM) media in an environment with low levels of oxygen and high levels of carbon dioxide.

The best specimen from which to culture *Neisseria gonorrhoeae* is a swab of the urethra in a male or the cervix in a female. Other possible specimens include vagina, body fluid discharge, swab of genital lesion, or the first urine of the day. Final results usually are available after two days. Rapid nonculture tests are available to test for GC and provide results on the same or following day.

CHANCROID. Chancroid is caused by *Haemophilus ducreyi*. It is characterized by genital ulcers with nearby swollen lymph nodes. The specimen is collected by swabbing one of these pus-filled ulcers. The gram stain may not be helpful as this bacteria looks just like other *Haemophilus* bacteria. This bacteria only grows on special culture plates, so the physician must request a specific culture for a person who has symptoms of chancroid. Even using special culture plates, *Haemophilus ducreyi* is isolated from less than 80% of the ulcers it infects. If a culture is negative, the physician must diagnose chancroid based on the person's

symptoms and by ruling out other possible causes of these symptoms, such as syphilis.

MYCOPLASMA. Three types of mycoplasma organisms cause sexually transmitted urethritis in males and **pelvic inflammatory disease** and **cervicitis** in females: *Mycoplasma hominis*, *Mycoplasma genitalium*, and *Ureaplasma urealyticum*. These organisms require special culture plates and may take up to six days to grow. Samples are collected from the cervix in a female, the urethra or semen in a male, or urine.

SYPHILIS. Syphilis is caused by *Treponema pallidum*, one in a group of bacteria called spirochetes. It causes ulcers or chancres at the site of infection. The organism does not grow in culture. Using special techniques and stains, it is identified by looking at a sample of the ulcer or chancre under the microscope. Various blood tests also may be done to detect the treponema organism.

CHLAMYDIA. Chlamydia is caused by the gram-negative bacterium *Chlamydia trachomatis*. It is one of the most common STDs in the United States and generally appears in sexually active adolescents and young adults. While chlamydia often does not have any initial symptoms, it can, if left untreated, lead to pelvic inflammatory disease and sterility. Samples are collected from one or more of these infection sites: cervix in a female, urethra in a male, or the rectum. A portion of specimen is combined with a specific type of cell and allowed to incubate. Special stains are performed on the cultured cells, looking for evidence of the chlamydia organism within the cells. A swab can also be taken from the woman's vulva. Men and women can now be screened for Chlamydia with a urine sample. Urine-based screening has increased screening significantly, especially among men.

Viral cultures

To culture or grow a virus in the laboratory, a portion of specimen is mixed with commercially prepared animal cells in a test tube. Characteristic changes to the cells caused by the growing virus help identify the virus. The time to complete a viral culture varies with the type of virus. It may take several days or up to several weeks.

HERPES VIRUS. Herpes simplex virus type 2 is the cause of **genital herpes**. Diagnosis is usually made based on the person's symptoms. If a diagnosis needs confirmation, a viral culture is performed using material taken from an ulcer. A Tzanck smear is a microscope test that can rapidly detect signs of herpes infection in cells taken from an ulcer. The culture

takes up to 14 days. In 2004, the FDA approved a blood test to detect the antibodies to herpes virus.

HUMAN PAPILLOMAVIRUS. Human papillomavirus causes **genital warts**. This virus will not grow in culture; the diagnosis is based on the appearance of the **warts** and the person's symptoms. In late 2003, the U.S. Food and Drug Administration (FDA) approved a human papillomavirus (HPV) DNA test with a Pap smear for screening women age 30 and older. The combined test would help physicians determine which women were at extremely low risk for **cervical cancer** and which should be more closely monitored.

HIV. Human immunodeficiency virus (HIV) is usually diagnosed with a blood test. Cultures for HIV are possible, but rarely needed for diagnosis. However, newer rapid tests were developed in 2003 and approved by the FDA in 2004. These tests are cheaper and can deliver results in as little as three minutes. The FDA also approved an HIV test in 2004 that can detect HIV in saliva.

Preparation

Generally, the type of specimen depends on the type of infection. Cultures always should be collected before the person begins taking antibiotics. After collection of these specimens, each is placed into a sterile tube containing a liquid in which the organism can survive while in route to the laboratory. The new rapid HIV tests rely on blood samples collected from a finger stick or vein or on saliva collected from the mouth. Initial results are not sent to a lab but are processed onsite.

Urethral specimen

Men should not urinate one hour before collection of a urethral specimen. The physician inserts a sterile, cotton-tipped swab into the urethra.

Cervical specimen

Women should not douche or take a bath within 24 hours of collection of a cervical or vaginal culture. The physician inserts a moistened, nonlubricated vaginal speculum. After the cervix is exposed, the physician removes the cervical mucus using a cotton ball. Next, he or she inserts a sterile cotton-tipped swab into the endocervical canal and rotates the swab with firm pressure for about 30 seconds.

Vaginal specimen

Women should not douche or take a bath within 24 hours of collection of a cervical or vaginal culture.

KEY TERMS

Culture—A laboratory test done to grow and identify microorganisms causing infection.

Gram stain—Microscopic examination of a portion of a bacterial colony or sample from an infection site after it has been stained by special stains. Certain bacteria pick up the purple stain; these bacteria are called gram positive. Other bacteria pick up the red stain; these bacteria are called gram negative. The color of the bacteria, in addition to their size and shape, provide clues as to the identity of the bacteria.

Sensitivity test—A test that determines which antibiotics will kill the bacteria isolated from a culture.

Vulva—The external part of the woman's genital organs, including the vaginal vestibule.

The physician inserts a sterile, cotton-tipped swab into the vagina.

Anal specimen

The physician inserts a sterile, cotton-tipped swab about 1 inch into the anus and rotates the swab for 30 seconds. Stool must not contaminate the swab.

Oropharynx (throat) specimen

The person's tongue is held down with a tongue depressor, as a healthcare worker moves a sterile, cotton-tipped swab across the back of the throat and tonsil region.

Urine specimen

To collect a "clean-catch" urine, the person first washes the perineum, and the penis or labia and vulva. He or she begins urinating, letting the first portion pass into the toilet, then collecting the remainder into a sterile container.

Normal results

These microorganisms are not found in a normal culture. Many types of microorganisms, normally found on a person's skin and in the genitourinary tract, may contaminate the culture. If a mixture of these microorganisms grow in the culture, they are reported as normal flora.

Abnormal results

If a person has a positive culture for one or more of these microorganisms, treatment is started and his or her sexual partners should be notified and tested. Certain laws govern reporting and partner notification of various STDs. After treatment is completed, the person's physician may want a follow-up culture to confirm the infection is gone.

Resources

PERIODICALS

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- "Approval Sought for HIV-1 Test that Detects Antibodies in Oral Fluid or Plasma." *AIDS Weekly* October 27, 2003: 23.
- Boschert, Sherry. "Chlamydia Urine Test: Males Still Underscreened: Noninvasive Screening Test." *Pediatric News* August 2004: 10-12.
- "FDA Approves DNAwithPap for Screening Women (Greater than or Equal to) Age 30)." *Contemporary OB/Gyn* October 2003: 105.
- "FDA Approves OraQuick HIV-1/2 Test to Detect HIV-2 in Oral Fluid." *Biotech Week* July 21, 2004: 401.
- Kaye, Donald. "FDA Approves Herpes Antibody Test." *Clinical Infectious Diseases* September 15, 2004: 1.
- "New HIV Rapid Test Is 100 Percent Accurate." *Health & Medicine Week* September 15, 2003: 194.
- "New Three-minute Rapid HIV Test Launched in the United States." *Medical Devices & Surgical Technology Week* September 12, 2004: 102.
- "One-step HIV Test May Be Cheaper, Faster, Less Wasteful." *Medical Letter on the CDC & FDA* October 5, 2003: 5.
- St. Lawrence, Janet S., et al. "STD Screening, Testing, Case Reporting, and Clinical and Partner Notification Practices: A National Survey of U.S. Physicians." *The American Journal of Public Health* November 2002: 1784.

ORGANIZATIONS

- American Social Health Association. PO Box 13827, Research Triangle Park, NC 27709. (800) 227-8922. <<http://sunsite.unc.edu/ASHA>>.
- Centers for Disease Control and Prevention. National Center for HIV, STD, and TB Prevention. 1600 Clifton Road NE, Atlanta, GA 30333. (404) 639-8000. <<http://www.cdc.gov/nchstp/od/nchstp.html>>.

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SGOT see **Aspartate aminotransferase test**

Shaken baby syndrome

Definition

Shaken baby syndrome (SBS) is a collective term for the internal head injuries a baby or young child sustains from being violently shaken.

Description

Shaken baby syndrome was first described in medical literature in 1972. Physicians earlier labeled these injuries as accidental, but as more about child abuse became known, more cases of this syndrome were properly diagnosed.

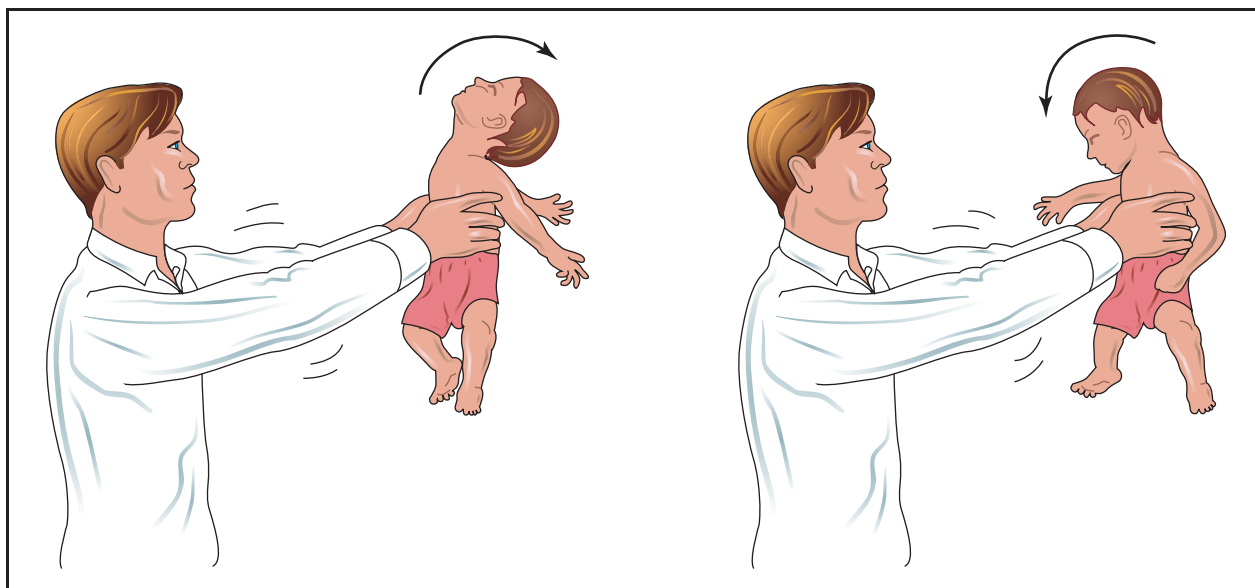
Every year, nearly 50,000 children in the United States are forcefully shaken by their caretakers. More than 60% of these children are boys. The victims are on average six to eight months old, but may be as old as five years or as young as a few days.

Men are more likely than women to shake a child; typically, these men are in their early 20s and are the baby's father or the mother's boyfriend. Women who inflict SBS are more likely to be babysitters or child care providers than the baby's mother. The shaking may occur as a response of frustration to the baby's inconsolable crying or as an action of routine **abuse**.

Causes and symptoms

Infants and small children are especially vulnerable to SBS because their neck muscles are still too weak to adequately support their disproportionately large heads, and their young brain tissue and blood vessels are extremely fragile. When an infant is vigorously shaken by the arms, legs, shoulders, or chest, the **whiplash** motion repeatedly jars the baby's brain with tremendous force, causing internal damage and bleeding. While there may be no obvious external signs of injury following shaking, the child may suffer internally from brain bleeding and bruising (called subdural hemorrhage and hematoma); brain swelling and damage (called cerebral **edema**); mental retardation; blindness, **hearing loss**, **paralysis**, speech impairment, and learning disabilities; and **death**. Nearly 2,000 children die every year as a result of being shaken.

Physicians may have difficulty initially diagnosing SBS because there are usually few witnesses to give a reliable account of the events leading to the trauma, few if any external injuries, and, upon close examination, the physical findings may not agree with the account given. A shaken baby may present one or



Shaken baby syndrome is a collective term for the internal head injuries a baby or young child sustains from being violently shaken. Because of the fragile state of an infant's brain tissue and blood vessels, when a baby is vigorously shaken by the chest, as shown in the illustration above, the whiplash motion repeatedly jars the baby's brain with extreme force, causing serious internal damage and bleeding. Nearly 2,000 American children die annually from this condition. (Illustration by Electronic Illustrators Group.)

more signs, including **vomiting**; difficulty breathing, sucking, swallowing, or making sounds; seizures; and altered consciousness.

Diagnosis

To diagnose SBS, physicians look for at least one of three classic conditions: bleeding at the back of one or both eyes (**retinal hemorrhage**), subdural hematoma, and cerebral edema. The diagnosis is confirmed by the results of either a computed tomography scan (CT scan) or magnetic resonance imaging (MRI).

Treatment

Appropriate treatment is determined by the type and severity of the trauma. Physicians may medically manage both internal and external injuries. Behavioral and educational impairments as a result of the injuries require the attention of additional specialists. Children with SBS may need physical therapy, speech therapy, vision therapy, and special education services.

Alternative treatment

There is no alternative to prompt medical treatment. An unresponsive child should never be put

KEY TERMS

Cerebral edema—Fluid collecting in the brain, causing tissue to swell.

Hematoma—A localized accumulation of blood in tissues as a result of hemorrhaging.

Hemorrhage—A condition of bleeding, usually severe.

Retinal hemorrhage—Bleeding of the retina, a key structure in vision located at the back of the eye.

Subdural hematoma—A localized accumulation of blood, sometimes mixed with spinal fluid, in the space of the brain beneath the membrane covering called the dura mater.

to bed, but must be taken to a hospital for immediate care.

Prognosis

Sadly, children who receive violent shaking have a poor prognosis for complete recovery. Those who do not die may experience permanent blindness, **mental retardation**, seizure disorders, or loss of motor control.

Prevention

Shaken baby syndrome is preventable with public education. Adults must be actively taught that shaking a child is never acceptable and can cause severe injury or death.

When the frustration from an incessantly crying baby becomes too much, caregivers should have a strategy for coping that does not harm the baby. The first step is to place the baby in a crib or playpen and leave the room in order to calm down. Counting to 10 and taking deep breaths may help. A friend or relative may be called to come over and assist. A calm adult may then resume trying to comfort the baby. A warm bottle, a dry diaper, soft music, a bath, or a ride in a swing, stroller, or car may be offered to soothe a crying child. Crying may also indicate **pain** or illness, such as from abdominal cramps or an earache. If the crying persists, the child should be seen by a physician.

Resources

ORGANIZATIONS

- American Humane Association, Children's Division. 63 Inverness Drive East, Englewood, CO 80112-5117. (800) 227-4645. <www.americanhumane.org>.
- Child Abuse Prevention Center of Utah. 2955 Harrison Boulevard, #102, Ogden, UT 84403. (888) 273-0071.
- National Center on Shaken Baby Syndrome. 2955 Harrison Blvd., #102, Ogden, UT 84403. (801) 627-3399. <<http://www.dontshake.com>>.

Bethany Thivierge

Shiatsu

Definition

Shiatsu is a manipulative therapy developed in Japan and incorporating techniques of *anma* (Japanese traditional massage), **acupressure**, stretching, and Western massage. Shiatsu involves applying pressure to special points or areas on the body in order to maintain physical and mental well being, treat disease, or alleviate discomfort. This therapy is considered holistic because it attempts to treat the whole person instead of a specific medical complaint. All types of acupressure generally focus on the same pressure points and so-called energy pathways, but may differ in terms of massage technique. Shiatsu, which can be translated as finger pressure, has been described as needle-free **acupuncture**.

Purpose

Shiatsu has a strong reputation for reducing **stress** and relieving **nausea and vomiting**. Shiatsu is also believed to improve circulation and boost the immune system. Some people use it to treat **diarrhea**, **indigestion**, **constipation**, and other disorders of the gastrointestinal tract; menstrual and menopausal problems; chronic **pain**; migraine; arthritis; **toothache**; **anxiety**; and depression. Shiatsu can be used to relieve muscular pain or tension, especially neck and back pain. It also appears to have sedative effects and may alleviate **insomnia**. In a broader sense, shiatsu is believed to enhance physical vitality and emotional well being.

Description

Origins

Shiatsu is an offshoot of *anma* that developed during the period after the Meiji Restoration in 1868. Traditional massage (*anma*) used during the age of shoguns was being criticized, and practitioners of *koho anma* (ancient way) displeased with it introduced new practices and new names for their therapies.

During the twentieth century, shiatsu distinguished itself from *anma* through the merging of Western knowledge of anatomy, *koho anma*, *ampuku* (abdominal massage), acupressure, *Do-In* (breathing practices), and Buddhism. Based on the work of Tamai Tempaku, shiatsu established itself in Japan and worldwide. The Shiatsu Therapists Association was founded in 1925 and clinics and schools followed. Students of Tempaku began teaching their own brand of shiatsu, creating branch disciplines. By 1955, the Japanese Ministry of Health and Welfare acknowledged shiatsu as a beneficial treatment, and licensing was established for practitioners.

Shiatsu and other forms of Japanese acupressure are based on the concept of *ki*, the Japanese term for the all-pervading energy that flows through everything in the universe. (This notion is borrowed from the Chinese, who refer to the omnipresent energy as *qi* or *chi*.) *Ki* tends to flow through the body along special energy pathways called meridians, each of which is associated with a vital organ. In Asian systems of traditional medicine, diseases are often believed to occur due to disruptions in the flow this energy through the body. These disruptions may stem from emotional factors, climate, or a host of other causes including stress, the presence of impurities in the body, and physical trauma.

The aim of shiatsu is to restore the proper flow of bodily energy by massaging the surface of the skin

along the meridian lines. Pressure may also be applied to any of the 600 or so acupoints. Acupoints, which are supposedly located just under the skin along the meridians, are tiny energy structures that affect the flow of ki through the body. When ki either stagnates and becomes deflected or accumulates in excess along one of these channels, stimulation to the acupoints, which are sensitive to pressure, can unblock and regulate the ki flow through toning or sedating treatment.

Western medicine has not proven the existence of meridians and acupoints. However, in one study, two French medical doctors conducted an experiment at Necker Hospital in Paris to test validity of the theory that energy is being transported along acupuncture meridians. They injected and traced isotopes with gamma-camera imaging. The meridians may actually correspond to nerve transmission lines. In this view, shiatsu and other forms of healing massage may trigger the emission of naturally occurring chemicals called neurotransmitters. Release of these chemical messengers may be responsible for some of the therapeutic effects associated with shiatsu, such as pain relief.

Preparations

People usually receive shiatsu therapy while lying on a floor mat or massage table or sitting up. The massage is performed through the clothing—preferably a thin garment made from natural fibers—and disrobing is not required. Pressure is often applied using the thumbs, though various other parts of the body may be employed, including fingertips, palms, knuckles, elbows, and knees—some therapists even use their feet. Shiatsu typically consists of sustained pressure (lasting up to 10 seconds at a time), squeezing, and stretching exercises. It may also involve gentle holding as well as rocking motions. A treatment session lasts anywhere from 30 to 90 minutes.

Before shiatsu treatment begins, the therapist usually performs a general health assessment. This involves taking a family medical history and discussing the physical and emotional health of the person seeking therapy. Typically, the practitioner also conducts a diagnostic examination by palpating the abdomen or back for any energy imbalances present in other parts of the body.

Precautions

While shiatsu is generally considered safe, there are a few precautions to consider. Because it may increase blood flow, this type of therapy is not recommended in people with bleeding problems, heart disease, or **cancer**.

KEY TERMS

Acupressure—An ancient form of Asian healing massage that involves applying pressure to special points or areas on the body in order to maintain good health, cure disease, and restore vitality.

Analgesic—Pain reliever.

Osteoporosis—A disease of the bones due to deficiency of bone matrix, occurring most frequently in postmenopausal women.

Palpate—Feel.

Massage therapy should always be used with caution in those with **osteoporosis**, fresh **wounds** or scar tissue, bone fractures, or inflammation.

Applying pressure to areas of the head is not recommended in people with epilepsy or high blood pressure, according to some practitioners of shiatsu.

Shiatsu is not considered effective in the treatment of fever, burns, and infectious diseases.

Shiatsu should not be performed right after a meal.

Side effects

When performed properly, shiatsu is not associated with any significant side effects. Some people may experience mild discomfort, which usually disappears during the course of the treatment session.

Research and general acceptance

Like many forms of massage, shiatsu is widely believed to have a relaxing effect on the body. There is also a significant amount of research suggesting that acupressure techniques can relieve **nausea** and **vomiting** associated with a variety of causes, including **pregnancy** and anesthetics and other drugs. In one study, acupressure was shown to significantly reduce the effects of nausea in 12 of 16 women suffering from morning sickness. Five days of this therapy also appeared to reduce anxiety and improve mood. Another investigation, published in 1999, studied the effects of acupressure on nausea resulting from the use of anesthetics. Pressure applied to an acupoint on the inside of the wrist appeared to alleviate nausea in patients who received anesthetics during the course of laparoscopic surgery.

Shiatsu may also produce sedative and analgesic effects. The sedative powers of acupressure were

investigated in a study published in the *Journals of Gerontology* 1999, which involved over 80 elderly people who suffered from sleeping difficulties. Compared to the people in the control groups, the 28 participants who received acupressure were able to sleep better. They slept for longer periods of time and were less likely to wake up during the night. The researchers concluded that acupressure may improve the quality of sleep in older adults. The use of acupressure in postoperative pain was investigated in a study published in 1996. In this study, which involved 40 knee surgery patients, one group received acupressure (15 acupoints were stimulated) while the control group received sham acupressure. Within an hour of treatment, members of the acupressure group reported less pain than those in the control group. The pain-relieving effects associated with acupressure lasted for 24 hours.

Shiatsu may benefit **stroke** victims. The results of at least one study (which did not include a control group) suggest that shiatsu may be useful during stroke rehabilitation when combined with other treatments.

Resources

BOOKS

Cook, Allan R. *Alternative Medicine Source book*. Detroit: Omnigraphics, 1999.

PERIODICALS

Chen, M.L., L.C. Lin, S.C. Wu, et al. "The effectiveness of Acupressure in Improving the Quality of Sleep of Institutionalized Residents." *J Gerontol A Biol Sci Med Sci* 1999; M389-94.

Harmon, D., J. Gardiner, R. Harrison, et al. "Acupressure and the Prevention of nausea and vomiting after laparoscopy." *Br J Anaesth* 1999; 387-390

ORGANIZATIONS

Acupressure Institute. 1533 Shattuck Avenue, Berkeley, CA 94709.

American Massage Therapy Association. 820 Davis Street, Suite 100, Evanston, IL. <<http://www.amtamassage.org>>.

American Oriental Bodywork Therapy Association. 50 Maple Place, Manhasset, NY 11030.

International School of Shiatsu. 10 South Clinton Street, Doylestown, PA 18901.

National Certification Board for Therapeutic Massage and Bodywork. 8201 Greensboro Drive, Suite 300, McLean, VA 22102.

OTHER

International School of Shiatsu. <<http://www.shiatsubo.com>>.

MEDLINE. <<http://igm.nlm.nih.gov>>.

Greg Annussek

Shigellosis

Definition

Shigellosis is an infection of the intestinal tract by a group of bacteria called *Shigella*. The bacteria is named in honor of Shiga, a Japanese researcher, who discovered the organism in 1897. The major symptoms are **diarrhea**, abdominal cramps, **fever**, and severe fluid loss (**dehydration**). Four different groups of *Shigella* can affect humans; of these, *S. dysenteriae* generally produces the most severe attacks, and *S. sonnei* the mildest.

Description

Shigellosis is a well-known cause of **traveler's diarrhea** and illness throughout the world. *Shigella* are extremely infectious bacteria, and ingestion of just 10 organisms is enough to cause severe diarrhea and dehydration. *Shigella* accounts for 10-20% of all cases of diarrhea worldwide, and in any given year infects over 140 million persons and kills 600,000, mostly children and the elderly. The most serious form of the disease is called **dysentery**, which is characterized by severe watery (and often blood- and mucus-streaked) diarrhea, abdominal cramping, rectal **pain**, and fever. *Shigella* is only one of several organisms that can cause dysentery, but the term bacillary dysentery is usually another name for shigellosis.

Most deaths are in less-developed or developing countries, but even in the United States, shigellosis can be a dangerous and potentially deadly disease. Poor hygiene, overcrowding, and improper storage of food are leading causes of infection. The following statistics show the marked difference in the frequency of cases between developed and less-developed countries; in the United States, about 30,000 individuals are hit by the disease each year or about 10 cases/100,000 population. On the other hand, infection in some areas of South America is 1,000 times more frequent. Shigellosis is most common in children below age five, and occurs less often in adults over 20.

Causes and symptoms

Shigella share several of the characteristics of a group of bacteria that inhabit the intestinal tract. *E coli*, another cause of food-borne illness, can be mistaken for *Shigella* both by physicians and the laboratory. Careful testing is needed to assure proper diagnosis and treatment.

Shigella are very resistant to the acid produced by the stomach, and this allows them to easily pass

through the gastrointestinal tract and infect the colon (large intestine). The result is a colitis that produces multiple ulcers, which can bleed. *Shigella* also produce a number of toxins (Shiga toxin and others) that increase the amount of fluid secretion by the intestinal tract. This fluid secretion is a major cause of the diarrhea symptoms.

Shigella infection spreads through food or water contaminated by human waste. Sources of transmission are:

- contaminated milk, ice cream, vegetables and other foods which often cause epidemics
- household contacts (40% of adults and 20% of children will develop infection from such a source)
- poor hygiene and overcrowded living conditions
- day care centers
- sexual practices which lead to oral-anal contact, directly or indirectly

Symptoms can be limited to only mild diarrhea or go on to full-blown dysentery. Dehydration results from the large fluid losses due to diarrhea, **vomiting**, and fever. Inability to eat or drink worsens the situation.

In developed countries, most infections are of the less severe type, and are often due to *S. sonnei*. The period between infection and symptoms (incubation period) varies from one to seven days. Shigellosis can last from a few days to several weeks, with an average of seven days.

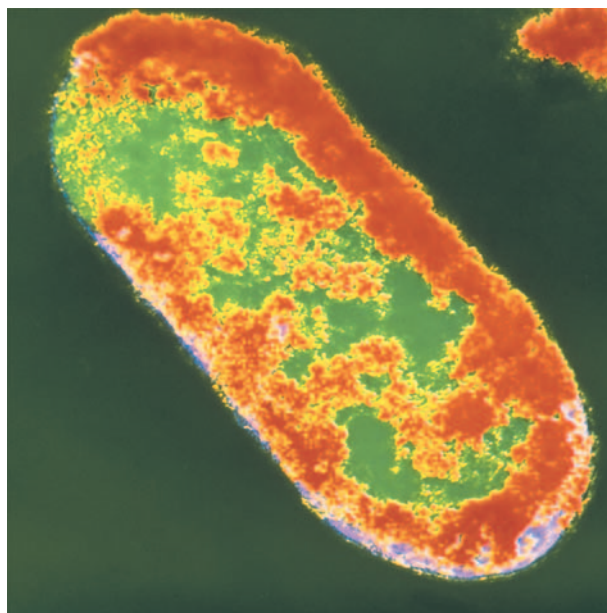
Complications

Areas outside the intestine can be involved, including:

- nervous system (irritation of the meninges or meningitis, encephalitis, and seizures)
- kidneys (producing hemolytic uremic syndrome or HUS which leads to kidney failure)
- joints (leading to an unusual form of arthritis called Reiter's syndrome)
- skin (rash)

One of the most serious complications of this disease is HUS, which involves the kidney. The main findings are kidney failure and damage to red blood cells. As many as 15% of patients die from this complication, and half the survivors develop chronic kidney failure, requiring dialysis.

Another life-threatening condition is toxic megacolon. Severe inflammation causes the colon to dilate or stretch, and the thin colon wall may eventually tear. Certain medications (particularly those that diminish intestinal contractions) may increase this risk, but this



A transmission electron microscopy (TEM) scan of *Shigella*, a genus of aerobic bacteria that causes dysentery in humans and animals. (Custom Medical Stock Photo. Reproduced by permission.)

interaction is unclear. Clues to this diagnosis include sudden decrease in diarrhea, swelling of the abdomen, and worsening abdominal pain.

Diagnosis

Shigellosis is one of the many causes of acute diarrhea. Culture (growing the bacteria in the laboratory) of freshly obtained diarrhea fluid is the only way to be certain of the diagnosis. But even this is not always positive, especially if the patient is already on **antibiotics**. *Shigella* are identified by a combination of their appearance under the microscope and various chemical tests. These studies take several days, so quicker means to recognize the bacteria and its toxins are being developed.

Treatment

The first aim of treatment is to keep up **nutrition** and avoid dehydration. Ideally, a physician should be consulted before starting any treatment. Antibiotics may not be necessary, except for the more severe infections. Many cases resolve before the diagnosis is established by culture. Medications that control diarrhea by slowing intestinal contractions can cause problems and should be avoided by patients with bloody diarrhea or fever, especially if antibiotics have not been started.

KEY TERMS

Dysentery—A disease marked by frequent watery bowel movements, often with blood and mucus, and characterized by pain, urgency to have a bowel movement, fever, and dehydration.

Traveler's diarrhea—An illness due to infection from a bacteria or parasite that occurs in persons traveling to areas where there is a high frequency of the illness. The disease is usually spread by contaminated food or water.

Oral Rehydration Solution(ORS)—A liquid preparation developed by the World Health Organization that can decrease fluid loss in persons with diarrhea. Originally developed to be prepared with materials available in the home, commercial preparations have recently come into use.

Antibiotic—A medication that is designed to kill or weaken bacteria.

Anti-motility medications—Medications such as loperamide (Imodium), dephenoxylate (Lomotil), or medications containing codeine or narcotics which decrease the ability of the intestine to contract. These may worsen the condition of a patient with dysentery or colitis.

Food-borne illness—A disease that is transmitted by eating or handling contaminated food.

Fluoroquinolones—A relatively new group of antibiotics that have had good success in treating infections with many gram-negative bacteria, such as *Shigella*. One drawback is that they should not be used in children under 17 years of age, because of possible effect on bone or cartilage growth.

Dialysis—A form of treatment for patients with kidneys that do not function properly. The treatment removes toxic wastes from the body that are normally removed by the kidneys.

Colitis—Inflammation of the colon or large bowel which has several causes. The lining of the colon becomes swollen, and ulcers often develop. The ability of the colon to absorb fluids is also affected, and diarrhea often results.

Carrier state—The continued presence of an organism (bacteria, virus, or parasite) in the body that does not cause symptoms, but is able to be transmitted and infect other persons.

Stool—Passage of fecal material; a bowel movement.

Meninges—Outer covering of the spinal cord and brain. Infection is called meningitis, which can lead to damage to the brain or spinal cord and lead to death.

Rehydration

The World Health Organization (WHO) has developed guidelines for a standard solution taken by mouth, and prepared from ingredients readily available at home. This Oral Rehydration Solution (ORS) includes salt, baking powder, sugar, orange juice, and water. Commercial preparations, such as Pedialyte, are also available. In many patients with mild symptoms, this is the only treatment needed. Severe dehydration usually requires intravenous fluid replacement.

Antibiotics

In the early and mid-1990s, researchers began to realize that not all cases of bacterial dysentery needed antibiotic treatment. Many patients improve without such therapy, and therefore these drugs are indicated only for treatment of moderate or severe disease, as found in the tropics. Choice of antibiotic is based on the type of bacteria found in the geographical area and on laboratory results. Recommendations as

of 1997 include ampicillin, sulfa derivatives such as Trimethoprim-Sulfamethoxazole (TMP-SMX) sold as Bactrim, or **fluoroquinolones** (such as Ciprofloxacin which is not FDA approved for use in children).

Prognosis

Many patients with mild infections need no specific treatment and recover completely. In those with severe infections, antibiotics will decrease the length of symptoms and the number of days bacteria appear in the feces. In rare cases, an individual may fail to clear the bacteria from the intestinal tract; the result is a persistent carrier state. This may be more frequent in **AIDS** (Acquired Immune Deficiency Syndrome) patients. Antibiotics are about 90% effective in eliminating these chronic infections.

In patients who have suffered particularly severe attacks, some degree of cramping and diarrhea can last for several weeks. This is usually due to damage

to the intestinal tract, which requires some time to heal. Since antibiotics can also produce a form of colitis, this must be considered as a possible cause of persistent or recurrent symptoms.

Prevention

Shigellosis is an extremely contagious disease; good hand washing techniques and proper precautions in food handling will help in avoiding spread of infection. Children in day care centers need to be reminded about hand washing during an outbreak to minimize spread. *Shigellosis* in schools or day care settings almost always disappears when holiday breaks occur, which sever the chain of transmission.

Traveler's diarrhea (TD)

Shigella accounts for about 10% of diarrhea illness in travelers to Mexico, South America, and the tropics. Most cases of TD are more of a nuisance than a life-threatening disease. However, bloody diarrhea is an indication that *Shigella* may be responsible.

In some cases though, aside from ruining a well deserved vacation, these infections can interrupt business conference schedules and, in the worst instances, lead to a life-threatening illness. Therefore, researchers have tried to find a safe, yet effective, way of preventing TD. Of course the best prevention is to follow closely the rules outlined by the WHO and other groups regarding eating fresh fruits, vegetables, and other foods.

One safe and effective method of preventing TD is the use of large doses of Pepto Bismol. Tablets are now available which are easier for travel; usage must start a few days before departure. Patients should be aware that Bismuth will turn bowel movements black.

Antibiotics have also proven to be highly effective in preventing TD. They can also produce significant side effects, and therefore a physician should be consulted before use. Like Pepto Bismol, antibiotics need to be started before beginning travel.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

David Kaminstein, MD

Shin splints

Definition

Shin splints refer to the sharp pains that occur down the front of the lower leg. They are a common complaint, particularly among runners and other athletes.

Description

Shin splints may refer to a number of lower leg complaints and injuries. In most cases, shin splints refer to the **pain** that results from overload on the tissues that connect muscles to the shin bone (tibia). They also may come from the small bone of the lower leg and ankle, called the fibula. The medical term for shin splints is medial tibial **stress** syndrome.

Next to ankle **sprains**, shin splints are probably the most common complaint of injury to the lower body. Most shin splints occur in the front (anterior) portion of the tibia; some also occur in the inside of the leg along the tibia. Runners probably suffer shin splints more than other people, but they also occur in people who play basketball and tennis and those who walk long distances, particularly on treadmills.

Causes and symptoms

The most common cause of shin splints is overdoing activities that constantly pound on the legs and feet. This may include sports with many stops and starts, running down hills or other tilted surfaces, or repeated walking. Simply training too long or too hard, especially without proper stretching and warm-up, can cause shin splints. People with flat feet, high arches, or feet that turn outward may be more prone to shin splints. Shoes that are worn or don't provide proper foot support also add to the problem.

Diagnosis

The physician will check the leg for tenderness. If the pain is in a single area of the tibia and hurts to the touch, the cause may be a stress fracture. The physician may order an x ray to rule out a stress fracture, but shin splints often can be diagnosed without x rays.

Treatment

Physicians usually recommend a period of rest for people with shin splints to let the area heal.

KEY TERMS

Podiatrist—A physician who specializes in the medical care and treatment of the human foot.

Stress fracture—A hairline fracture (narrow crack along the surface of a bone) that is caused by repeated stress to the bone, such as from jogging, rather than from a single heavy blow.

Usually, about three to four weeks is recommended, though the time varies depending on the patient and injury severity. Shin splints may be treated in phases, beginning with absolute rest and gradual return to activity. Ice and elevation of the foot may be used to help relieve pain and swelling in the first phase. If the person needs to keep in shape, stretching and water exercises that keep the foot from bearing weight may be allowed after initial treatment. As the patient returns to normal function, orthotic footwear and braces may be added to prevent re-injury.

Alternative treatment

Various massage techniques may help speed up recovery. Homeopathic physicians may recommend *Rhus tox*. Those using alternative remedies should ensure they are certified practitioners and should coordinate care with allopathic providers.

Prognosis

With proper rest, management, and prevention, people with shin splints can return to normal activity in a few weeks or more. However, continuing to perform the activity that caused the shin splints can lead to stress **fractures** of the tibia.

Prevention

Re-injury is most common in the first month after return to normal activity, and patients who have had shin splints should return to previous activities cautiously. The following can help prevent shin splints from occurring in people who run and perform stop and start physical activities:

- Warming up and stretching calf muscles before running or jogging. A podiatrist specializing in sports medicine or other sports medicine specialist may recommend specific stretching exercises.

- Strengthening muscles in the front lower leg (anterior tibialis) with resistance exercises or by walking on the heels three times daily for about 30 yards.
- Wearing quality shoes with arch supports. Runners should purchase new shoes about every 400 miles. A podiatrist can design special arch supports or orthotics for people with flat feet.
- Runs should be started at a slow pace and gradually increased.
- Athletes can cross-train in a sport that does not impact the feet and lower legs as much, such as swimming or riding a bicycle.

Resources

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OTHER

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Teresa G. Odle

Shingles

Definition

Shingles, also called herpes zoster or zona, gets its name from both the Latin and French words for belt or girdle and refers to girdle-like skin eruptions that may occur on the trunk of the body. The virus that causes **chickenpox**, the varicella zoster virus (VSV), can become dormant in nerve cells after an episode of chickenpox and later reemerge as shingles. Initially, red patches of rash develop into blisters. Because the virus travels along the nerve to the skin, it can damage the nerve and cause it to become inflamed. This condition can be very painful. If the **pain** persists long after

the rash disappears, it is known as postherpetic neuralgia.

Description

Any person who has had chickenpox can develop shingles. Approximately 500,000 cases of shingles occur every year in the United States, according to the National Institute of Allergy and Infectious Diseases (NIAID). Overall, approximately 20% of those who had chickenpox as children develop shingles at some time in their lives. People of all ages, even children, can be affected, but the incidence increases with age. Newborn infants, bone marrow and other transplant recipients, as well as individuals with immune systems weakened by disease or drugs are also at increased risk. However, most individuals who develop shingles do not have any underlying malignancy or other immunosuppressive condition.

Causes and symptoms

Shingles erupts along the course of the affected nerve, producing lesions anywhere on the body and may cause severe nerve pain. The most common areas to be affected are the face and trunk, which correspond to the areas where the chickenpox rash is most concentrated. The disease is caused by a reactivation of the chickenpox virus that has lain dormant in certain nerves following an episode of chickenpox. Exactly how or why this reactivation occurs is not clear; however, it is believed that the reactivation is triggered when the immune system becomes weakened, either as a result of **stress**, **fatigue**, certain medications, **chemotherapy**, or diseases, such as **cancer** or HIV. Further, it can be an early sign in persons with HIV that the immune system has deteriorated.

In some cases, the virus appears to be reactivated by mechanical irritation or minor surgical procedures. In one instance, the patient had an attack of shingles following **liposuction**.

Early signs of shingles are often vague and can easily be mistaken for other illnesses. The condition may begin with **fever** and malaise (a vague feeling of weakness or discomfort). Within two to four days, severe pain, **itching**, and numbness/tingling (paresthesia) or extreme sensitivity to touch (hyperesthesia) can develop, usually on the trunk and occasionally on the arms and legs. Pain may be continuous or intermittent, usually lasting from one to four weeks. It may occur at the time of the eruption, but can precede the eruption by days, occasionally making the diagnosis difficult. Signs and symptoms may include the following:

- itching, **tingling**, or severe burning pain
- red patches that develop into blisters
- grouped, dense, deep, small blisters that ooze and crust
- swollen lymph nodes

Diagnosis

Diagnosis is usually not possible until the skin lesions develop. Once they develop, however, the pattern and location of the blisters and the type of cell damage displayed are characteristic of the disease, allowing an accurate diagnosis primarily based upon the **physical examination**.

Although tests are rarely necessary, they may include the following:

- viral culture of skin lesion
- microscopic examination using a **Tzanck preparation**. This involves staining a smear obtained from a blister. Cells infected with the herpes virus will appear very large and contain many dark cell centers or nuclei.
- complete **blood count** (CBC) may show an elevated white blood cell count (WBC), a nonspecific sign of infection
- Rise in antibody to the virus
- Polymerase chain reaction (PCR) analysis. PCR testing has been found to be much faster and significantly more accurate than culturing the virus.

Treatment

Shingles almost always resolves spontaneously and may not require any treatment except for the relief of symptoms. In most people, the condition clears on its own in one or two weeks and seldom recurs.

Cool, wet compresses may help reduce pain. If there are blisters or crusting, applying compresses made with diluted vinegar will make the patient more comfortable. Mix one-quarter cup of white vinegar in two quarts of lukewarm water. Use the compress twice each day for 10 minutes. Stop using the compresses when the blisters have dried up.

Soothing baths and lotions such as colloidal oatmeal baths, starch baths or lotions, and calamine lotion may help to relieve itching and discomfort. Keep the skin clean, and do not re-use contaminated items. While the lesions continue to ooze, the person should be isolated to prevent infecting other susceptible individuals.



Shingles, or herpes zoster, on patient's buttocks and thigh.
(Custom Medical Stock Photo. Reproduced by permission.)

Later, when the crusts and scabs are separating, the skin may become dry, tight, and cracked. If that happens, rub on a small amount of plain petroleum jelly three or four times a day.

The **antiviral drugs** acyclovir, valacyclovir, and famciclovir can be used to treat shingles. These drugs may shorten the course of the illness. Their use results in more rapid healing of the blisters when drug therapy is started within 72 hours of the onset of the rash. In fact, the earlier the drugs are administered, the better, because early cases can sometimes be stopped. If taken later, these drugs are less effective but may still lessen the pain. Antiviral drug treatment does not seem to reduce the incidence of postherpetic neuralgia, but recent studies suggest famciclovir may cut the duration of postherpetic neuralgia in half. Side effects of typical oral doses of these antiviral drugs are minor with **headache** and nausea reported by 8–20 % of patients. Severely immunocompromised individuals, such as those with **AIDS**, may require intravenous administration of antiviral drugs.

Corticosteroids, such as prednisone, may be used to reduce inflammation but they do interfere with the functioning of the immune system. Corticosteroids, in combination with antiviral therapy, also are used to treat severe infections, such as those affecting the eyes, and to reduce severe pain.

Once the blisters are healed, some people continue to experience pain for months or even years

(postherpetic neuralgia). This pain can be excruciating. Consequently, the doctor may prescribe tranquilizers, sedatives, or antidepressants to be taken at night. As noted above attempts to treat postherpetic neuralgia with the antiviral drug famciclovir have shown some promising results. When all else fails, severe pain may require a permanent nerve block.

A newer medication to treat postherpetic neuralgia is pregabalin, to be marketed in the United States under the trade name Lyrica. Pregabalin was approved by the Food and Drug Administration in September 2004 for the treatment of **diabetic neuropathy** as well as postherpetic neuralgia. The drug has been shown to improve patients' sleep and overall quality of life as well as relieve pain. Its most common side effects are drowsiness, headache, **dry mouth**, and **dizziness**.

Alternative treatment

There are nonmedical methods of prevention and treatment that may speed recovery. For example, getting lots of rest, eating a healthy diet, exercising regularly, and minimizing stress are always helpful in preventing disease. Supplementation with vitamin B₁₂ during the first one to two days and continued supplementation with vitamin B complex, high levels of vitamin C with bioflavonoids, and calcium, are recommended to boost the immune system. Herbal antivirals such as **echinacea** can be effective in fighting infection and boosting the immune system.

Although no single alternative approach, technique, or remedy has yet been proven to reduce the pain, there are a few options which may be helpful. For example, topical applications of lemon balm (*Melissa officinalis*) or licorice (*Glycyrrhiza glabra*) and peppermint (*Mentha piperita*) may reduce pain and blistering. Homeopathic remedies include *Rhus toxicodendron* for blisters, *Mezereum* and *Arsenicum album* for pain, and *Ranunculus* for itching. Practitioners of Eastern medicine recommend self-hypnosis, **acupressure**, and **acupuncture** to alleviate pain.

Prognosis

Shingles usually clears up in two to three weeks and rarely recurs. Involvement of the nerves that cause movement may cause a temporary or permanent nerve **paralysis** and/or **tremors**. The elderly or debilitated patient may have a prolonged and difficult course. For them, the eruption is typically more extensive and inflammatory, occasionally resulting in blisters that bleed, areas where the skin actually dies,

KEY TERMS

Acyclovir—An antiviral drug that is available under the trade name Zovirax, in oral, intravenous, and topical forms. The drug blocks the replication of the varicella zoster virus.

Antibody—A specific protein produced by the immune system in response to a specific foreign protein or particle called an antigen.

Corticosteroid—A steroid that has similar properties to the steroid hormone produced by the adrenal cortex. It is used to alter immune responses to shingles.

Famciclovir—An oral antiviral drug that is available under the trade name Famvir. The drug blocks the replication of the varicella zoster virus.

Immunocompromised—A state in which the immune system is suppressed or not functioning properly.

Postherpetic neuralgia (PHN)—The term used to describe the pain after the rash associated with herpes zoster is gone.

Tzanck preparation—Procedure in which skin cells from a blister are stained and examined under the microscope. Visualization of large skin cells with many cell centers or nuclei indicates a positive diagnosis of herpes zoster when combined with results from a physical examination.

Valacyclovir—An oral antiviral drug that is available under the trade name Valtrex. The drug blocks the replication of the varicella zoster virus.

secondary bacterial infection, or extensive and permanent scarring.

Similarly, an immunocompromised patient usually has a more severe course that is frequently prolonged for weeks to months. They develop shingles frequently and the infection can spread to the skin, lungs, liver, gastrointestinal tract, brain, or other vital organs. Cases of chronic shingles have been reported in patients infected with AIDS, especially when they have a decreased number of one particular kind of immune cell, called CD4 lymphocytes. Depletion of CD4 lymphocytes is associated with more severe, chronic, and recurrent varicella-zoster virus infections. These lesions are typical at the onset but may turn into ulcers that do not heal.

Potentially serious complications can result from herpes zoster. Many individuals continue to

experience persistent pain long after the blisters heal. This pain, called postherpetic neuralgia or PHN, can be severe and debilitating. Postherpetic neuralgia can persist for months or years after the lesions have disappeared. The incidence of postherpetic neuralgia increases with age, and episodes in older individuals tend to be of longer duration. Most patients under 30 years of age experience no persistent pain. By age 40, the risk of prolonged pain lasting longer than one month increases to 33%. By age 70, the risk increases to 74%. The pain can adversely affect quality of life, but it does usually diminish over time. Another risk factor for PHN is female sex.

Other complications include a secondary bacterial infection, and rarely, potentially fatal inflammation of the brain (**encephalitis**) and the spread of an infection throughout the body. These rare, but extremely serious, complications are more likely to occur in those individuals who have weakened immune systems (immunocompromised).

Prevention

Strengthening the immune system by making lifestyle changes is thought to help prevent the development of shingles. A lifestyle designed to strengthen the immune system and maintain good overall health includes eating a well-balanced diet rich in essential **vitamins** and **minerals**, getting enough sleep, exercising regularly, and reducing stress.

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American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

National Institute of Allergy and Infectious Diseases (NIAID). Office of Communications and Public Liaison, 6610 Rockledge Drive, MSC 6612, Bethesda, MD 20892-6612. <<http://www.niaid.nih.gov>>.

David James Doermann
Rebecca J. Frey, PhD

Shock

Definition

Shock is a medical emergency in which the organs and tissues of the body are not receiving an adequate flow of blood. This deprives the organs and tissues of oxygen (carried in the blood) and allows the buildup of waste products. Shock can result in serious damage or even **death**.

Description

There are three stages of shock: Stage I (also called compensated, or nonprogressive), Stage II (also called decompensated or progressive), and Stage III (also called irreversible).

In Stage I of shock, when low blood flow (perfusion) is first detected, a number of systems are activated in order to maintain/restore perfusion. The result is that the heart beats faster, the blood vessels throughout the body become slightly smaller in diameter, and the kidney works to retain fluid in the circulatory system. All this serves to maximize blood flow to the most important organs and systems in the body. The patient in this stage of shock has very few symptoms, and treatment can completely halt any progression.

In Stage II of shock, these methods of compensation begin to fail. The systems of the body are unable to improve perfusion any longer, and the patient's symptoms reflect that fact. Oxygen deprivation in the brain causes the patient to become confused and

disoriented, while oxygen deprivation in the heart may cause chest **pain**. With quick and appropriate treatment, this stage of shock can be reversed.

In Stage III of shock, the length of time that poor perfusion has existed begins to take a permanent toll on the body's organs and tissues. The heart's functioning continues to spiral downward, and the kidneys usually shut down completely. Cells in organs and tissues throughout the body are injured and dying. The endpoint of Stage III shock is the patient's death.

Causes and symptoms

Shock is caused by three major categories of problems: cardiogenic (meaning problems associated with the heart's functioning); hypovolemic (meaning that the total volume of blood available to circulate is low); and **septic shock** (caused by overwhelming infection, usually by bacteria).

Cardiogenic shock can be caused by any disease, or event, which prevents the heart muscle from pumping strongly and consistently enough to circulate the blood normally. **Heart attack**, conditions which cause inflammation of the heart muscle (myocarditis), disturbances of the electrical rhythm of the heart, any kind of mass or fluid accumulation and/or blood clot which interferes with flow out of the heart can all significantly affect the heart's ability to adequately pump a normal quantity of blood.

Hypovolemic shock occurs when the total volume of blood in the body falls well below normal. This can occur when there is excess fluid loss, as in **dehydration** due to severe **vomiting** or diarrhea, diseases which cause excess urination (diabetes insipidus, diabetes mellitus, and kidney failure), extensive **burns**, blockage in the intestine, inflammation of the pancreas (**pancreatitis**), or severe bleeding of any kind.

Septic shock can occur when an untreated or inadequately treated infection (usually bacterial) is allowed to progress. Bacteria often produce poisonous chemicals (toxins) which can cause injury throughout the body. When large quantities of these bacteria, and their toxins, begin circulating in the bloodstream, every organ and tissue in the body is at risk of their damaging effects. The most damaging consequences of these bacteria and toxins include poor functioning of the heart muscle; widening of the diameter of the blood vessels; a drop in blood pressure; activation of the blood clotting system, causing **blood clots**, followed by a risk of uncontrollable bleeding; damage to the lungs, causing acute respiratory distress syndrome; liver failure; kidney failure; and **coma**.

KEY TERMS

Cardiogenic—Originating with the heart.

Deprivation—A condition of having too little of something.

Hypovolemic—Having a low volume.

Perfusion—Blood flow through an organ or tissue.

Sepsis—An overwhelming infection throughout the body, usually caused by bacteria in the bloodstream.

Initial symptoms of shock include cold, clammy hands and feet; pale or blue-tinged skin tone; weak, fast pulse rate; fast rate of breathing; low blood pressure. A variety of other symptoms may be present, but they are dependent on the underlying cause of shock.

Diagnosis

Diagnosis of shock is based on the patient's symptoms, as well as criteria including a significant drop in blood pressure, extremely low urine output, and blood tests that reveal overly acidic blood with a low circulating concentration of carbon dioxide. Other tests are performed, as appropriate, to try to determine the underlying condition responsible for the patient's state of shock.

Treatment

The most important goals in the treatment of shock include: quickly diagnosing the patient's state of shock; quickly intervening to halt the underlying condition (stopping bleeding, re-starting the heart, giving **antibiotics** to combat an infection, etc.); treating the effects of shock (low oxygen, increased acid in the blood, activation of the blood clotting system); and supporting vital functions (blood pressure, urine flow, heart function).

Treatment includes keeping the patient warm, with legs raised and head down to improve blood flow to the brain, putting a needle in a vein in order to give fluids or blood transfusions, as necessary; giving the patient extra oxygen to breathe and medications to improve the heart's functioning; and treating the underlying condition which led to shock.

Prognosis

The prognosis of an individual patient in shock depends on the stage of shock when treatment was

begun, the underlying condition causing shock, and the general medical state of the patient.

Prevention

The most preventable type of shock is caused by dehydration during illnesses with severe vomiting or **diarrhea**. Shock can be avoided by recognizing that a patient who is unable to drink in order to replace lost fluids needs to be given fluids intravenously (through a needle in a vein). Other types of shock are only preventable insofar as one can prevent their underlying conditions, or can monitor and manage those conditions well enough so that they never progress to the point of shock.

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Rosalyn Carson-DeWitt, MD

Shock I see **Adult respiratory distress syndrome**

Shock therapy see **Electroconvulsive therapy**

Shortness of breath

Definition

Shortness of breath, or dyspnea, is a feeling of difficult or labored breathing that is out of proportion to the patient's level of physical activity. It is a symptom of a variety of different diseases or disorders and may be either acute or chronic.

Description

The experience of dyspnea depends on its severity and underlying causes. The feeling itself results from a combination of impulses relayed to the brain from nerve endings in the lungs, rib cage, chest muscles, or diaphragm, combined with the patient's perception and interpretation of the sensation. In some cases, the patient's sensation of breathlessness is intensified by **anxiety** about its cause. Patients describe dyspnea variously as unpleasant shortness of breath, a feeling of increased effort or tiredness in moving the chest muscles, a panicky feeling of being smothered, or a sense of tightness or cramping in the chest wall.

Causes and symptoms

ACUTE DYSPNEA. Acute dyspnea with sudden onset is a frequent cause of emergency room visits. Most cases of acute dyspnea involve pulmonary (lung and breathing) disorders, cardiovascular disease, or chest trauma.

PULMONARY DISORDERS. Pulmonary disorders that can cause dyspnea include airway obstruction by a foreign object, swelling due to infection, or anaphylactic **shock**; acute **pneumonia**; hemorrhage from the lungs; or severe bronchospasms associated with **asthma**.

CARDIOVASCULAR DISEASE. Acute dyspnea can be caused by disturbances of the heart rhythm, failure of the left ventricle, mitral valve (a heart valve) dysfunction, or an embolus (a clump of tissue, fat, or gas) that is blocking the pulmonary circulation. Most pulmonary emboli (**blood clots**) originate in the deep veins of the lower legs and eventually migrate to the pulmonary artery.

TRAUMA. Chest injuries, both closed injuries and penetrating **wounds**, can cause **pneumothorax** (the presence of air inside the chest cavity), **bruises**, or fractured ribs. **Pain** from these injuries results in dyspnea. The impact of the driver's chest against the steering wheel in auto accidents is a frequent cause of closed chest injuries.

OTHER CAUSES. Anxiety attacks sometimes cause acute dyspnea; they may or may not be associated with chest pain. Anxiety attacks are often accompanied by hyperventilation, which is a breathing pattern characterized by abnormally rapid and deep breaths. Hyperventilation raises the oxygen level in the blood, causing chest pain and **dizziness**.

Chronic dyspnea

PULMONARY DISORDERS. Chronic dyspnea can be caused by asthma, chronic obstructive pulmonary disease (COPD), **bronchitis**, **emphysema**, inflammation of the lungs, **pulmonary hypertension**, tumors, or disorders of the vocal cords.

HEART DISEASE. Disorders of the left side of the heart or inadequate supply of blood to the heart muscle can cause dyspnea. In some cases a tumor in the heart or inflammation of the membrane surrounding the heart may cause dyspnea.

NEUROMUSCULAR DISORDERS. Neuromuscular disorders cause dyspnea from progressive deterioration of the patient's chest muscles. They include

muscular dystrophy, **myasthenia gravis**, and **amyotrophic lateral sclerosis**.

OTHER CAUSES. Patients who are severely anemic may develop dyspnea if they **exercise** vigorously. **Hyperthyroidism** or **hypothyroidism** may cause shortness of breath, and so may gastroesophageal reflux disease (GERD). Both chronic **anxiety disorders** and of physical fitness can also cause episodes of dyspnea. Deformities of the chest or **obesity** can cause dyspnea by limiting the movement of the chest wall and the ability of the lungs to fill completely.

Diagnosis

Patient history

The patient's history provides the doctor with such necessary information as a history of gastroesophageal reflux disease (GERD), asthma, or other allergic conditions; the presence of chest pain as well as difficulty breathing; recent accidents or recent surgery; information about **smoking** habits; the patient's baseline level of physical activity and exercise habits; and a psychiatric history of panic attacks or anxiety disorders.

ASSESSMENT OF BODY POSITION. How a person's body position affects his/her dyspnea symptoms sometimes gives hints as to the underlying cause of the disorder. Dyspnea that is worse when the patient is sitting up is called **platypnea** and indicates the possibility of **liver disease**. Dyspnea that is worse when the patient is lying down is called **orthopnea**, and is associated with heart disease or **paralysis** of the diaphragm. Paroxysmal nocturnal dyspnea (PND) refers to dyspnea that occurs during sleep and forces the patient to awake gasping for breath. It is usually relieved if the patient sits up or stands. PND may point to dysfunction of the left ventricle of the heart, **hypertension**, or narrowing of the mitral valve.

Physical examination

The doctor will examine the patient's chest in order to determine the rate and depth of breathing, the effort required, the condition of the patient's breathing muscles, and any evidence of chest deformities or trauma. He or she will listen for **wheezing**, **stridor**, or signs of fluid in the lungs. If the patient has a **fever**, the doctor will look for other signs of pneumonia. The doctor will check the patient's heart functions, including blood pressure, pulse rate, and the presence of **heart murmurs** or other abnormal heart sounds. If the doctor suspects a blood clot in one of the large veins leading to the heart, he or she will examine the patient's legs for signs of swelling.

Diagnostic tests

BASIC DIAGNOSTIC TESTS. Patients who are seen in emergency rooms are given a **chest x ray** and electrocardiogram (ECG) to assist the doctor in evaluating abnormalities of the chest wall, also to determine the position of the diaphragm, possible rib **fractures** or pneumothorax, irregular heartbeat, or the adequacy of the supply of blood to the heart muscle. Also, the patient may be given a breathing test on an instrument called a spirometer to screen for airway disorders.

The doctor may order blood tests and arterial blood gas tests to rule out anemia, hyperventilation—from an anxiety attack —, or thyroid dysfunction. A **sputum culture** can be used to test for pneumonia.

SPECIALIZED TESTS. Specialized tests may be ordered for patients with normal results from basic diagnostic tests for dyspnea. High-resolution CT scans can be used for suspected airway obstruction or mild emphysema. Tissue biopsy performed with a bronchoscope can be used for patients with suspected lung disease.

If the doctor suspects a **pulmonary embolism**, he or she may order ventilation-perfusion scanning to inspect lung function, an angiogram of blood vessels, or ultrasound studies of the leg veins. **Echocardiography** can be used to test for pulmonary hypertension and heart disease.

Pulmonary function studies or **electromyography** (EMG) are used to assess neuromuscular diseases. Exercise testing is used to assess dyspnea related to COPD, anxiety attacks, poor physical fitness, and the severity of lung or heart disease. The level of acidity in the patient's esophagus may be monitored to rule out GERD.

Treatment

Treatment of dyspnea depends on its underlying cause.

Acute dyspnea

Patients with acute dyspnea are given oxygen in the emergency room, with the following treatments for specific conditions:

- Asthma. Treatment with Alupent, epinephrine, or aminophylline.
- Anaphylactic shock. Treatment with Benadryl, steroids, or aminophylline, with hydrocortisone if necessary.
- Congestive **heart failure**. Treatment with oxygen, **diuretics**, and placing patient in upright position.

- Pneumonia. Treatment with **antibiotics** and removal of lung secretions.
- Anxiety attacks. Immediate treatment includes antidepressant medications. If the patient is hyperventilating, he or she may be asked to breathe into a paper bag to normalize breathing rhythm and the oxygen level of the blood.
- Pneumothorax. Surgical placement of a chest tube.

Chronic dyspnea

The treatment of chronic dyspnea depends on the underlying disorder. Asthma can often be managed with a combination of medications to reduce airway spasms and removal of allergens from the patient's environment. COPD requires both medication, lifestyle changes, and long-term physical **rehabilitation**. Anxiety disorders are usually treated with a combination of medication and psychotherapy. GERD can usually be managed with **antacids**, other medications, and dietary changes. There are no permanent cures for myasthenia gravis or muscular dystrophy.

Tumors and certain types of chest deformities can be treated surgically.

Alternative treatment

The appropriate alternative therapy for shortness of breath depends on the underlying cause of the condition. When dyspnea is acute and severe, **oxygen therapy** is used either in the doctor's office or in the emergency room. For shortness of breath with an underlying physical cause like asthma, anaphylactic shock, or pneumonia, the physical condition should be treated. Botanical and homeopathic remedies can be used for acute dyspnea, if the proper remedies and formulas are prescribed. If the dyspnea has a psychological basis (especially if it is caused by anxiety), **acupuncture**, botanical medicine, and homeopathy can help the patient heal at a deep level.

Prognosis

The prognosis for recovery depends on the underlying cause of the dyspnea, its severity, and the type of treatment required.

Prevention

Dyspnea caused by asthma can be minimized or prevented by removing dust and other triggers from the patient's environment. Long-term prevention of chronic dyspnea includes such lifestyle choices as regular aerobic exercise and avoidance of smoking.

KEY TERMS

Anaphylactic shock—A severe systemic reaction to an allergen that occurs in hypersensitive individuals. It can cause spasms of the larynx that block the patient's airway and cause dyspnea.

Dyspnea—A sensation of difficult or labored breathing.

Electromyography—A technique for recording electric currents in an active muscle in order to measure its level of function.

Orthopnea—Difficulty in breathing that occurs while the patient is lying down.

Paroxysmal nocturnal dyspnea (PND)—A form of dyspnea characterized by the patient's waking from sleep unable to breathe.

Platypnea—Dyspnea that occurs when the patient is sitting up.

Pneumothorax—The presence of air or gas inside the chest cavity.

Spirometer—An instrument that is used to test lung capacity. It is used to screen patients with dyspnea.

Stridor—A harsh or crowing breath sound caused by partial blockage of the patient's upper airway.

Wheezing—A whistling or musical sound caused by tightening of the air passages inside the patient's chest. Wheezing is most commonly associated with asthma.

Resources

BOOKS

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Rebecca J. Frey, PhD

Shy-Drager syndrome

Definition

Shy-Drager syndrome (SDS) is a rare condition that causes progressive damage to the autonomic nervous system. The autonomic nervous system controls vital involuntary body functions such as heart rate, breathing, and intestinal, urinary, and sexual

functions. The autonomic nervous system also controls skin and body temperature, and how the body responds to **stress**. Shy-Drager syndrome leads to dizziness or **fainting** when standing up, urinary incontinence, **impotence**, and muscle **tremors**.

Description

SDS was named for neurologists Milton Shy, M.D., from the National Institutes of Health, and Glenn Drager, M.D., from the Baylor College of Medicine, who first described the condition in 1960. It typically affects those between ages 50–70. It affects more men than women. In severe cases, the person cannot even stand up. Symptoms can be mild as well. Sometimes, people with mild cases are misdiagnosed as having **anxiety** or **hypertension**.

Many nonprescription drugs, such as cold medicines and diet capsules, can trigger extremely high blood pressure spikes in patients with SDS, even in very low doses. Therefore, these patients are at risk for strokes and excessive bleeding (hemorrhage) if they take even the recommended dosage of these drugs.

Causes and symptoms

The cause of SDS is unknown. Symptoms develop because of degeneration of certain groups of nerve cells in the spinal cord.

Patients with SDS usually have problems with the function of the autonomic nervous system. Progressive degeneration may occur in other areas of the nervous system as well. The hallmark of the syndrome is **dizziness** and fainting when arising or after standing still for a long time (postural **hypotension**). This is caused by low blood pressure and inadequate blood flow to the brain. When this problem becomes severe (for example, a blood pressure below 70/40 mmHg), it can lead to a momentary loss of consciousness. When the person faints, the blood pressure returns to normal and the person wakes up.

Many patients also notice impotence, **urinary incontinence**, dry mouth and skin, and trouble regulating body temperature because of abnormal sweating. Since the autonomic nervous system also controls the narrowing and widening of the iris, some patients with SDS have vision problems, such as trouble focusing.

In later stages, problems in the autonomic nervous system lead to breathing difficulties such as **sleep apnea**, loud breathing, and **snoring**. In advanced stages of the disease, patients can die from irregular heartbeat.

Other symptoms of SDS do not involve the autonomic nervous system. These include parkinsonism (muscle tremor, rigidity, and slow movements), double vision, problems controlling emotions, and wasting of muscles in the hands and feet. Eventually, patients may have problems chewing, swallowing, speaking, and breathing. There may be a loss of color pigment in the iris.

Diagnosis

While no blood test can reveal the disorder, a careful assessment of symptoms should alert a neurologist to suspect SDS. A combination of parkinsonism and certain autonomic problems (especially impotence, incontinence, and postural hypotension) are clear indications of the syndrome.

Tests of the autonomic nervous system may help diagnose the condition. In normal patients, blood levels of norepinephrine rise when they stand up. This doesn't happen in people with SDS. Norepinephrine is a hormone that helps maintain blood pressure by triggering certain blood vessels to constrict when blood pressure falls below normal. Another test for the condition is the Valsalva maneuver. In this test, the patient holds his or her breath and strains down as if having a bowel movement while the doctor monitors blood pressure and heart rate for 10 seconds. Patients with SDS will not have the normal increase in blood pressure and heart rate.

A variety of other tests can identify a broad range of autonomic problems in patients with SDS. Brain scans, however, don't usually reveal any problems.

Treatment

Medication can relieve many of the symptoms, especially the parkinsonism and low blood pressure. However, typical antiparkinsonism drugs such as carbidopa-levodopa (Sinemet) should be used with caution, since they often worsen the postural low blood pressure and may cause fainting.

Because postural hypotension is the most troublesome of the symptoms in the early years, treatments center on relieving this problem. Patients are encouraged to eat a liberal salt diet and drink plenty of fluids. They are advised to wear waist-high elastic hosiery and to sleep with the head elevated at least 5 in (13 cm). Other drug treatment includes fludrocortisone, indomethacin, nonsteroidal anti-inflammatory drugs, beta blockers, central stimulants, and other medications.

Occasionally, a pacemaker, **gastrostomy**, or tracheostomy may be needed. A pacemaker is a device

KEY TERMS

Autonomic nervous system—The part of the nervous system that controls the involuntary (apparently automatic) activities of organs, blood vessels, glands, and many other body tissues.

Degenerative—Degenerative disorders involve progressive impairment of both the structure and function of part of the body.

Gastrostomy—An artificial opening into the stomach through the abdomen to enable a patient to be fed via a feeding tube. The procedure is given to patients with SDS who are unable to chew or swallow.

Norepinephrine—A hormone that helps maintain blood pressure by triggering certain blood vessels to constrict when blood pressure falls below normal.

Sleep apnea—A sleep disorder characterized by periods of breathing cessation lasting for 10 seconds or more.

Tracheostomy—An opening through the neck into the trachea through which a tube may be inserted to maintain an effective airway and help a patient breathe.

that delivers electrical impulses to the heart to keep it beating regularly. A gastrostomy creates an opening in the stomach to connect a feeding tube from outside the body. In a tracheostomy an opening is made in the windpipe and a tube is inserted to maintain breathing.

Prognosis

While the course of the disease varies, and some patients live for up to 20 years after the symptoms first appear, most patients become severely disabled within seven or eight years. It is unusual for someone to survive more than 15 years after diagnosis.

Symptoms (especially tremor) often get worse if the patient smokes, because of the nicotine.

Many patients develop swallowing problems which may lead to recurrent episodes of pneumonia, a frequent cause of **death**. Others experience Cheyne-Stokes (periodic breathing). One of the most common causes of death is pulmonary embolus. This is caused by a blood clot in the main artery in the lung.

Prevention

Since scientists don't know the cause of Shy-Drager syndrome, there is no way to prevent the condition.

Resources

ORGANIZATIONS

American Academy of Neurology. 1080 Montreal Ave., St. Paul, MN 55116. (612) 695-1940.
< <http://www.aan.com> >.

Association for Neuro-Metabolic Disorders. 5223 Brookfield Lane, Sylvania, OH 43560-1809. (419) 885-1497.

National Institute of Neurological Disorders and Stroke. P.O. Box 5801, Bethesda, MD 20824. (800) 352-9424.
< <http://www.ninds.nih.gov/index.htm> >.

National Organization for Rare Disorders. P.O. Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673.
< <http://www.rarediseases.org> >.

Shy-Drager Syndrome Support Group. 2004 Howard Lane, Austin, TX 78728. (800) 288-5582.
< <http://www.shy-drager.com> >.

Carol A. Turkington

Shyness

Definition

Shyness is a personality trait that produces behaviors ranging from feeling uncomfortable at a party to an extreme fear of being watched by others while talking on the telephone.

Description

Shyness affects people of all ages. A toddler might run from strangers and cling to her parents. While kindergarten is frightening for many children; some students are anxious about the first day of school until they graduate from college. Job interviews are stressful for people uncomfortable talking about themselves. For some people, feelings of self-worth are related to their careers. Retirement may bring feelings of lower self-esteem.

Shyness is linked to brain activity, how a person was raised and other experiences, and the person's reaction to those experiences.

Social phobia

Extreme shyness is sometimes referred to as a social phobia. Also known as social **anxiety** disorder, a social phobia is a psychiatric condition defined as a "marked and persistent fear" of some situations. The shy person continues to go on job interviews. Social phobia may cause a person to remain unemployed,

according to the National Mental Health Association (NMHA). True social phobia affects about 3% of people.

Introversion

The introvert enjoy being alone and intentionally avoids situations like a party. The shy person wants to be around people. However, shyness is stronger than the desire to be sociable. The shy person is afraid to go to the party and stays home alone.

Causes and symptoms

Temperament is related to the amygdala, the part of the brain related to emotions and new situations. The amygdala evaluates new situations based on memories of past experiences. If the new situation appears threatening, the amygdala sends a warning signal. The amygdala in a shy person is extremely sensitive and much more active than that of an outgoing person. The increased activity causes the person to withdraw either physically or emotionally. This withdrawal is known as inhibition.

The baby runs from strangers; the job applicant laughs nervously when talking about his accomplishments. Brain activity is one component of shyness. Environment also plays a role. If the inhibited child has outgoing, nurturing parents, she will probably imitate their behavior. If parents and teachers are mocking and critical, a child may have a lifelong fear of the first day of school. A person with that background may compare himself with others and feel they are more capable than he is. The person embarrassed in a job interview could become anxious in future interviews.

At the root of shyness is a feeling of self-consciousness. This may cause the person to blush, tense up, or start sweating. Those are some reactions caused when the brain signals its warning. The person may avoid eye contact, look down, become very quiet, or fumble over words.

Symptoms vary because there are degrees of shyness. A person might be very quiet when meeting new people, but then become talkative when she feels comfortable with them. The jobseeker may not be afraid of social gatherings.

Social Phobia

Social phobia causes an extreme fear of being humiliated or embarrassed in front of people, according to the according to the NMHA. It may be connected to low self-esteem or feelings of inferiority. The

phobic is not fearful in all situations and may feel comfortable around people in most of the time.

However, social **phobias** have caused people to drop out of school, avoid making friends, and keep away from other fear-provoking situations. Phobic fears range from speaking in public and dating to using public restrooms or writing when other people are present.

According to the NMHA, phobic may feel that everyone is looking at them. A trivial mistake is regarded as much more serious, and blushing is painfully embarrassing. Social phobia is frequently accompanied by depression or **substance abuse**.

Diagnosis

In many cases, adults realize they are shy. In a sense they have diagnosed themselves, and may take steps to overcome their shyness. Teen-agers may also try to remedy their situations.

Adults and youths may buy self-help books or take classes on subjects like overcoming shyness and assertiveness training. These classes may be taught by counselors, psychologists, or people with experience conquering shyness. Health-care providers often schedule these classes. They are also taught in settings ranging from adult schools to social service agencies. Costs will vary at these classes.

Children may not know there are treatment solutions for their shyness. Parents and educators should be alert for symptoms of shyness in younger children. Schools and family resource centers can provide referrals if it appears counseling want their child diagnosed.

Medical diagnosis

Based on the child's circumstances, parents may take the child to their health care provider. Some insurance plans require an appointment with a doctor before a referral to a counselor or a psychologist. The health professional conducts an assessment and then recommends treatment.

Children and adults may need medical treatment for social phobia. The adult's diagnosis also starts with a medical exam to determine if there is a physical cause for symptoms. If that has been ruled out, the patient undergoes a psychiatric evaluation.

Diagnostic fees and the time allocated for evaluation vary for both shy and phobic people. Diagnosis could span several hour-long sessions that cover an initial evaluation, personality tests, and a meeting to set therapy goals. Each session could cost around \$90. Insurance may cover part of the costs.

Treatment

Shyness treatment concentrates on changing behavior so the person feels more at ease in shyness-provoking situations. The person may be guided by a self-help book or participate in individual or **group therapy**.

Books and therapy generally focus on behavioral therapy and **cognitive-behavioral therapy**. One method of behavioral therapy is to expose the person to the situation that triggers fear. This could start with rehearsing a job interview with a friend or making eye contact with a store clerk. Over time, the person goes on interviews to get experience rather than to be hired. Another person might move from eye contact to attending an enjoyable event like a concert to become more at ease around strangers.

Therapy also focuses on developing skills to cope in new situation. These include taking deep breaths to relax and practicing small talk. Cognitive-therapy helps the person learn how thinking patterns contribute to symptoms, according to NMHA. The person is taught techniques to change those thoughts or stop symptoms. This association maintains this therapy is very effective for people with social phobias.

Treatment costs vary from the price of a self-help book to the fees for therapy. Therapy sessions may be led by a licensed marriage and family counselor, a psychologist or psychiatrist. The cost of group therapy is for is generally an hourly fee, with therapy planned for a set time. The therapist might charge \$80 an hour for a social phobia group that meets three hours a week for 16 weeks.

Treatment may include medication. Prescription drugs like Paxil (paroxetine) are generally only prescribed to people with social **anxiety disorders**. Paxil is prescribed for depression and other **mood disorders**. The patient takes one tablet daily. Costs will vary, and a 30-day order could be priced at \$74 to \$84.

Insurance may cover part of the costs of therapy and medicine.

Alternative treatment

Alternative treatments for shyness focus on symptoms like tension and **stress**. Relaxation tapes and CDs guide the listener through a series of actions to relieve tension. The activity starts with deep breathing and then the person progressively focuses on the head and different parts of the body. The **exercise** may start with the head, neck, shoulders, moving down to the one foot and then the other. Some techniques involve tightly tensing and then releasing each part. Another

method is to concentrate on relaxing each part or imagine that it becomes warm.

Another self-treatment is **aromatherapy**. Lavender is a relaxing scent and is available in liquid form as an essential oil. Stress can be relieved by adding oil to a bath. Some people carry the oil with them. If they become anxious, the people can dab the oil on a cotton pad. They breathe in the lavender and feel calmer.

Prognosis

Shyness may not be a permanent. Children often outgrow shyness. Behavioral changes and therapy can help people feel more at ease. Furthermore, some aspects of shyness are positive. Shy people are frequently good listeners and are empathetic, aware of others' feelings.

Prevention

Shyness is a personality trait related to a person's biology and experiences. The part of shyness related to the brain cannot be changed. However, parents can provide a nurturing environment that helps prevent shyness. This will provide the child with a healthy mental attitude that helps prevent shyness. When faced with situations that could cause self-defeating shyness, children will have coping skills.

According to the National Mental Health Association, the basics of good mental health for children include:

- A family that provides unconditional love not related on accomplishments.
- Nurturing self-confidence and high self-esteem by praising children. Methods include encouraging a child to learn a new game. The parents should set realistic goals, assure children, and smile frequently. Parents should avoid sarcastic remarks, set realistic goals and let children know that all people make mistakes.
- Playing with other children helps the young learn how to develop friendships and problem-solving skills.
- Emphasizing that school is fun. Parents can play school with their child to demonstrate that learning is enjoyable. Enrolling children in preschool or children's programs allows them to learn, be creative, and develop social skills.
- When disciplining, parents should criticize the behavior, rather than berating the child.

Shyness prevention and adults

For adults prone to shyness, the issue is related more to treatment than prevention. Shyness for these

people has probably been an issue, one that surfaces at various times in their lives. A move, a **death** in the family, job loss, and other unsettling changes could cause emotions that include the fear associated with shyness.

In some circumstances, the person must go through the grieving process. In other situations, the person needs to do things that build self-confidence. Like the child, the adult needs a support system. A network of friends helps with encouragement and listens to the person's concerns.

To combat the avoidance symptom caused by shyness, the person should look into enjoyable pursuits. Recreational activities like walking groups combine physical exercise with the opportunity to socialize. Enrolling in a class at an adult school or community college provides the opportunity to learn and make new friends. Class topics range from upholstery to mystery book discussions. Classes like these can boost confidence as a person learns a hands-on skill or discovers that other mystery readers value her or his opinion.

Resources

PERIODICALS

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ORGANIZATIONS

American Psychological Association. 750 First Street, NE, Washington, DC 20002-4242. 800-374-2721. < <http://www.apa.org> >.

National Mental Health Association. 2001 N. Beauregard Street, 12th Floor, Alexandria, VA 22311. 703-684-7722. < <http://www.nmha.org> >.

Shyness Research Institute. 4201 Grant Line Road, New Albany, IN 47150. 812-941-2295. < <http://homepages.ius.edu/Special/Shyness> >.

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Liz Swain

Sick sinus syndrome

Definition

Sick sinus syndrome is a disorder of the sinus node of the heart, which regulates heartbeat. With sick sinus syndrome, the sinus node fails to signal properly, resulting in changes in the heart rate.

Description

The sinus node in the heart functions as the heart's pacemaker, or beat regulator. In sick sinus syndrome, patients normally will experience bradycardia, or slowed heart rate. Also, it is not uncommon to see fluctuations between slow and rapid heart rate (tachycardia). This makes the diagnosis and treatment of sick sinus syndrome more complicated than most other cardiac arrhythmias (irregular heart beats). A sick sinus node may be responsible for starting beats too slowly, pausing too long between initiation of heartbeats, or not producing heartbeats at all.

Causes and symptoms

Sick sinus syndrome may be brought on by the use of certain drugs, but is most common in elderly patients. Cardiac **amyloidosis**, a condition in which amyloid, a kind of protein, builds up in heart tissue, may affect the sinus node. Other conditions, such as **sarcoidosis** (round bumps in the tissue surrounding the heart and other organs), **Chagas' disease** (resulting from the bite of a bloodsucking insect) or certain cardiac **myopathies** can cause fiber-like tissue to grow around the normal sinus node, causing the node to malfunction.

A patient may not show any symptoms of sick sinus syndrome. In general, however, the common symptoms are those associated with slow heart rate, such as light-headedness, or dizziness, **fatigue** and **fainting**. Patients may also experience confusion, heart **palpitations**, **angina** or heart failure.

Diagnosis

A slow pulse, especially one that is irregular, may be the first indication of sick sinus syndrome. **Electrocardiography** (ECGs) is a commonly used method of detecting sick sinus syndrome. ECG monitoring for 24 hours is most useful, since with this syndrome, heart rate may alternate between slow and fast, and the determination of this fact can help differentiate sick sinus syndrome from other **arrhythmias**.

KEY TERMS

Arrhythmia—Irregular heart beat.

Atria—Plural for atrium. The atria are the upper chambers of the heart.

Bradycardia—A heart rate slower than normal.

Electrocardiograph (ECG)—A test of a patient's heartbeat that involves placing leads, or detectors, on the patient's chest to record electrical impulses in the heart. This test will produce a strip, or picture record of the heart's electrical functioning.

Myopathy—Weakness of muscle.

Pacemaker—A device implanted under the skin, below the collarbone, to regulate heartbeat. Leads from the device to the heart stimulate the electrical functions of the heart. Pacemakers are often used to control bradycardia and are usually smaller than a silver dollar.

Treatment

If drugs are causing the problem, their withdrawal may effectively eliminate the disorder. However, the treatment of sick sinus syndrome is normally delayed until a patient shows symptoms. Once treatment is indicated, most patients will receive a pacemaker. This is a permanent treatment involving implantation of a small device under the skin below the collarbone. Small electrodes run from the device to the heart; they deliver and regulate the electrical signals that cause the heart to beat. Patients with sick sinus syndrome should generally receive dual chamber pacing systems to prevent atrial fibrillation (involuntary contraction of the muscles of the atria). Some drugs are used to treat sick sinus syndrome, but digitalis should be used with caution. Often the use of drugs to regulate the heartbeat should be implemented only after the pacemaker has been placed, since these drugs may further worsen the slow heart rate.

Alternative treatment

The reduction or elimination of certain foods and substances, such as alcohol or **caffeine**, may be advised to control heart rate. Stress reduction may also assist with changes in rate. Homeopathic treatment can work on a deep healing level, while **acupuncture** and botanical medicine can offer supportive treatment for symptoms.

Prognosis

Patients with sick sinus syndrome face relatively normal lives if the disorder is controlled by a pacemaker. However, in some patients, the pacemaker does not adequately control the fluctuations in heart rate. Left untreated, or in severe cases, the heart could stop beating.

Prevention

Elimination of a drug therapy which aggravates sick sinus syndrome is the first line of treatment for some patients. Other causes of the syndrome are not preventable. However, proper treatment of those underlying conditions which affect the tissues of the heart may intervene to prevent sick sinus syndrome from becoming a significant problem.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300.

< <http://www.americanheart.org> >.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222.

< <http://www.nhlbi.nih.gov> >.

Teresa Odle

Sickle cell disease

Definition

Sickle cell disease describes a group of inherited blood disorders characterized by chronic anemia, painful events, and various complications due to associated tissue and organ damage.

Because sickle cell diseases are characterized by the rapid loss of red blood cells as they enter the circulation, they are classified as hemolytic disorders, "hemolytic" referring to the destruction of the cell membrane of red blood cells resulting in the release of hemoglobin.

Description

The most common and best-known type of sickle cell disease is sickle cell anemia, which is also called meniscocytosis, sicklemia, or SS disease. All types of sickle cell disease are caused by a genetic change in hemoglobin, the oxygen-carrying protein inside the

red blood cells. The red blood cells of affected individuals contain a predominance of a structural variant of the usual adult hemoglobin. This variant hemoglobin, called sickle hemoglobin, has a tendency to polymerize into rod-like structures that alter the shape of the usually flexible red blood cells. The cells take on a shape that resembles the curved blade of the sickle, an agricultural tool. Sickle cells have a shorter life span than normally shaped red blood cells. This results in chronic anemia characterized by low levels of hemoglobin and decreased numbers of red blood cells. Sickle cells are also less flexible and stickier than normal red blood cells, and can become trapped in small blood vessels preventing blood flow. This compromises the delivery of oxygen, which can result in **pain** and damage to associated tissues and organs. Sickle cell disease presents with marked variability, even within families.

Carriers of the sickle cell gene are said to have sickle cell trait. Unlike sickle cell disease, sickle cell trait does not cause health problems. In fact, sickle cell trait is protective against **malaria**, a disease caused by blood-borne parasites transmitted through mosquito bites. According to a widely accepted theory, the genetic mutation associated with the sickle cell trait occurred thousands of years ago. Coincidentally, this mutation increased the likelihood that carriers would survive malaria infection. Survivors then passed the mutation on to their offspring, and the trait became established throughout areas where malaria was common. As populations migrated, so did the sickle cell trait. Today, approximately one in 12 African Americans has sickle cell trait.

Worldwide, it has been estimated that one in every 250,000 babies is born annually with sickle cell disease. Sickle cell disease primarily affects people of African, Mediterranean, Middle Eastern, and Asian Indian ancestry. In the United States, sickle cell disease is most often seen in African Americans, in whom the disease occurs in one out of every 400 births. The disease has been described in individuals from several different ethnic backgrounds and is also seen with increased frequency in Latino Americans—particularly those of Caribbean, Central American, and South American ancestry. Approximately one in every 1000-1400 Latino births are affected.

Causes and symptoms

Humans normally make several types of the oxygen-carrying protein hemoglobin. An individual's stage in development determines whether he or she makes primarily embryonic, fetal, or adult hemoglobins. All types of hemoglobin are made of three

components: heme, alpha (or alpha-like) globin, and beta (or beta-like) globin. Sickle hemoglobin is the result of a genetic change in the beta globin component of normal adult hemoglobin. The beta globin gene is located on chromosome 11. The sickle cell form of the beta globin gene results from the substitution of a single DNA nucleotide, or genetic building-block. The change from adenine to thymine at codon (position) 6 of the beta globin gene leads to insertion of the amino acid valine—instead of glutamic acid—at this same position in the beta globin protein. As a result of this change, sickle hemoglobin has unique properties in comparison to the usual type of adult hemoglobin.

Most individuals have two normal copies of the beta globin gene, which make normal beta globin that is incorporated into adult hemoglobin. Individuals who have sickle cell trait (called sickle cell carriers) have one normal beta globin gene and one sickle cell gene. These individuals make both the usual adult hemoglobin and sickle hemoglobin in roughly equal proportions, so they do not experience any health problems as a result of having the trait. Although traces of blood in the urine and difficulty in concentrating the urine can occur, neither represents a significant health problem as a result of sickle cell trait. Of the millions of people with sickle cell trait worldwide, a small handful of individuals have experienced acute symptoms. In these very rare cases, individuals were subject to very severe physical strain.

When both members of a couple are carriers of sickle cell trait, there is a 25% chance in each **pregnancy** for the baby to inherit two sickle cell genes and have sickle cell anemia, or SS disease. Correspondingly, there is a 50% chance the baby will have sickle cell trait and a 25% chance that the baby will have the usual type of hemoglobin. Other types of sickle cell disease include SC disease, SD disease, and S/beta **thalassemia**. These conditions are caused by the co-inheritance of the sickle cell gene and another altered beta globin gene. For example, one parent may have sickle cell trait and the other parent may have hemoglobin C trait (another hemoglobin trait that does not cause health problems). For this couple, there would be a 25% chance of SC disease in each pregnancy.

Normal adult hemoglobin transports oxygen from the lungs to tissues throughout the body. Sickle hemoglobin can also transport oxygen. However, once the oxygen is released, sickle hemoglobin tends to polymerize (line-up) into rigid rods that alter the shape of the red blood cell. Sickling of the red blood cell can be triggered by low oxygen, such as occurs in organs with slow blood flow. It can also be triggered by cold temperatures and **dehydration**.

Sickle cells have a decreased life span in comparison to normal red blood cells. Normal red blood cells survive for approximately 120 days in the bloodstream; sickle cells last only 10–12 days. As a result, the bloodstream is chronically short of red blood cells and hemoglobin, and the affected individual develops anemia.

Sickle cells can create other complications. Due to their shape, they do not fit well through small blood vessels. As an aggravating factor, the outside surfaces of sickle cells may have altered chemical properties that increase the cells' 'stickiness'. These sticky sickle cells are more likely to adhere to the inside surfaces of small blood vessels, as well as to other blood cells. As a result of the sickle cells' shape and stickiness, blockages form in small blood vessels. Such blockages prevent oxygenated blood from reaching areas where it is needed, causing pain as well as organ and tissue damage.

The severity of symptoms cannot be predicted based solely on the genetic inheritance. Some individuals with sickle cell disease develop health- or life-threatening problems in infancy, but others may have only mild symptoms throughout their lives. Individuals may experience varying degrees of health at different stages in the life cycle. For the most part, this clinical variability is unpredictable, and the reasons for the observed variability can not usually be determined. However, certain types of sickle cell disease (i.e. SC disease) tend to result in fewer and less severe symptoms on average than other types of sickle cell disease (i.e. SS disease). Some additional modifying factors are known. For example, elevated levels of fetal hemoglobin in a child or adult can decrease the quantity and severity of some symptoms and complications. Fetal hemoglobin is a normally occurring hemoglobin that usually decreases from over 90% of the total hemoglobin to under 1% during the first year of life. This change is genetically determined, although some individuals may experience elevated levels of fetal hemoglobin due to variation in the genes that control fetal hemoglobin production. Such individuals often experience a reduction in their symptoms and complications due to the ability of fetal hemoglobin to prevent the polymerization of sickle hemoglobin, which leads to sickling of the red blood cell.

There are several symptoms that warrant immediate medical attention, including the following:

- signs of infection (**fever** greater than $>101^{\circ}\text{F}$ or 38.3°C , coughs frequently or breathing trouble, unusual crankiness, feeding difficulties)
- signs of severe anemia (pale skin or lips, yellowing of the skin or eyes, very tired, very weak)
- signs indicating possible dehydration (**vomiting, diarrhea**, fewer wet diapers)

- other signs (pain or swelling in the abdomen, swollen hands or feet, screams when touched)

These can be signs of various complications that occur in sickle cell disease.

Infections and effects on the spleen

Children with sickle cell disease who are under age three are particularly prone to life-threatening bacterial infections. *Streptococcus pneumoniae* is the most common offending bacteria, and invasive infection from this organism leads to **death** in 15% of cases. The spleen, an organ that helps to fight bacterial infections, is particularly vulnerable to the effects of sickling. Sickle cells can impede blood flow through the spleen, causing organ damage, which usually results in loss of spleen function by late childhood. The spleen can also become enlarged due to blockages and/or increased activity of the spleen. Rapid enlargement of the spleen may be a sign of another complication called *splenic sequestration*, which occurs mostly in young children and can be life-threatening. Widespread sickling in the spleen prevents adequate blood flow from the organ, removing increasing volumes of blood from the circulation and leading to accompanying signs of severe anemia.

Painful events

Painful events, also known as *vaso-occlusive events*, are a hallmark symptom of sickle cell disease. The frequency and duration of the pain can vary tremendously from person to person and over an individual's life cycle. Painful events are the most common cause of hospitalizations in sickle cell disease. However, only a small portion of individuals with sickle cell disease experience frequent and severe painful events. Most painful events can be managed at home. Pain results when small blood vessel blockages prevent oxygen from reaching tissues. Pain can affect any area of the body, although the extremities, chest, abdomen, and bones are frequently affected sites. There is some evidence that cold temperatures or infection can trigger a painful event, but most events occur for unknown reasons. The hand-foot syndrome, or *dactylitis*, is a particular type of painful event. Most common in toddlers, dactylitis results in pain and swelling in the hands and feet, sometimes accompanied by a fever.

Anemia

Sickle cells have a high turnover rate leading to a deficit of red blood cells in the bloodstream. Common symptoms of anemia include **fatigue**, paleness, and a **shortness of breath**. A particularly severe form of

anemia—aplastic anemia—occurs following infection with parvovirus. Parvovirus causes extensive destruction of the bone marrow, bringing production of new red blood cells to a halt. Bone marrow production resumes after seven to 10 days; however, given the short lives of sickle cells, even a brief shut-down in red blood cell production can cause a rapid decline in hemoglobin concentrations.

Delayed growth

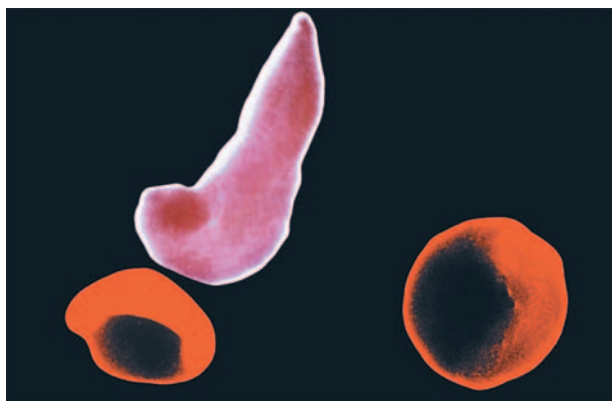
The energy demands of the bone marrow for red blood cell production compete with the demands of a growing body. Children with sickle cell anemia may have delayed growth and reach **puberty** at a later age than normal. By early adulthood, they catch up on growth and attain normal height; however, weight typically remains below average.

Stroke

Children with sickle cell disease have a significantly elevated risk of having a **stroke**, which can be one of the most concerning complications of sickle cell disease. Approximately 11% of individuals with sickle cell disease will have a recognizable stroke by the age of 20. **Magnetic resonance imaging** studies have found that 17% of children with sickle cell anemia have evidence of a previous stroke or clinically 'silent' stroke-like events called *transient ischemic events*. Stroke in sickle cell disease is usually caused by a blockage of a blood vessel, but about one fourth of the time may be caused by a hemorrhage (or rupture) of a blood vessel.

Strokes result in compromised delivery of oxygen to an area of the brain. The consequences of stroke can range from life-threatening, to severe physical or cognitive impairments, to apparent or subtle learning disabilities, to undetectable effects. Common stroke symptoms include weakness or **numbness** that affects one side of the body, sudden behavioral changes, loss of vision, confusion, loss of speech or the ability to understand spoken words, **dizziness**, **headache**, seizures, vomiting, or even **coma**.

Approximately two-thirds of the children who have a stroke will have at least one more. Transfusions have been shown to decrease the incidence of a second stroke. A recent study showed that children at highest risk to experience a first stroke were 10 times more likely to stroke if untreated when compared to high-risk children treated with chronic blood **transfusion** therapy. High-risk children were identified using transcranial doppler ultrasound technology to detect individuals with increased blood flow speeds due to constricted intracranial blood vessels.



A scanning electron microscopy (SEM) scan of red blood cells taken from a person with sickle cell anemia. The blood cells at the bottom are normal; the diseased, sickle-shaped cells appear at the top. (Photograph by Dr. Gopal Murti, Photo Researchers, Inc. Reproduced by permission.)

As of 2003, researchers are investigating various techniques for helping children with memory loss related to strokes caused by sickle cell disease.

Acute chest syndrome

Acute chest syndrome (ACS) is a leading cause of death for individuals with sickle cell disease, and recurrent attacks can lead to permanent lung damage. Therefore rapid diagnosis and treatment is of great importance. ACS can occur at any age and is similar but distinct from **pneumonia**. Affected persons may experience fever, **cough**, chest pain, and shortness of breath. ACS seems to have multiple causes including infection, sickling in the small blood vessels of the lungs, fat embolisms to the lungs, or a combination of factors.

Priapism

Males with sickle cell anemia may experience **priapism**, a condition characterized by a persistent and painful erection of the penis. Due to blood vessel blockage by sickle cells, blood is trapped in the tissue of the penis. Priapism may be short in duration or it may be prolonged. Priapism can be triggered by low oxygen (hypoxemia), alcohol consumption, or sexual intercourse. Since priapism can be extremely painful and result in damage to this tissue causing **impotence**, rapid treatment is essential.

Kidney disease

The environment in the kidney is particularly prone to damage from sickle cells. Signs of kidney damage can include blood in the urine, incontinence, and enlarged kidneys. Adults with sickle cell disease

often experience insufficient functioning of the kidneys, which can progress to kidney failure in a small percentage of adults with sickle cell disease.

Jaundice and gallstones

Jaundice is indicated by a yellow tone in the skin and eyes, and alone it is not a health concern. Jaundice may occur if bilirubin levels increase, which can occur with high levels of red blood cell destruction. Bilirubin is the final product of hemoglobin degradation, and is typically removed from the bloodstream by the liver. Therefore, jaundice can also be a sign of a poorly functioning liver, which may also be evidenced by an enlarged liver. Increased bilirubin also leads to increased chance for **gallstones** in children with sickle cell disease. Treatment, which may include removal of the gall bladder, may be selected if the gallstones start causing symptoms.

Retinopathy

The blood vessels that supply oxygen to the retina—the tissue at the back of the eye—may be blocked by sickle cells, leading to a condition called retinopathy. This is one of the only complications that is actually more common in SC disease as compared to SS disease. Retinopathy can be identified through regular ophthalmology evaluations and effectively treated in order to avoid damage to vision.

Joint problems

Avascular necrosis of the hip and shoulder joints, in which bone damage occurs due to compromised blood flow due to sickling, can occur later in childhood. This complication can affect an individual's physical abilities and result in substantial pain.

Diagnosis

Inheritance of sickle cell disease or trait cannot be prevented, but it may be predicted. Screening is recommended for individuals in high-risk populations. In the United States, African Americans and Latino Americans have the highest risk of having the disease or trait. Sickle cell is also common among individuals of Mediterranean, Middle Eastern, and Eastern Indian descent.

A complete **blood count** (CBC) will describe several aspects of an individual's blood cells. A person with sickle cell disease will have a lower than normal hemoglobin level, together with other characteristic red blood cell abnormalities. A *hemoglobin electrophoresis* is a test that can help identify the types and

quantities of hemoglobin made by an individual. This test uses an electric field applied across a slab of gel-like material. Hemoglobins migrate through this gel at various rates and to specific locations, depending on their size, shape, and electrical charge. Although sickle hemoglobin (Hb S) and regular adult hemoglobin (called Hb A) differ by only one amino acid, they can be clearly separated using **hemoglobin electrophoresis**. *Isoelectric focusing* and *high-performance liquid chromatography (HPLC)* use similar principles to separate hemoglobins and can be used instead of or in various combinations with hemoglobin electrophoresis to determine the types of hemoglobin present.

Another test called the sickledex can help confirm the presence of sickle hemoglobin, although this test cannot provide accurate or reliable diagnosis when used alone. When Hb S is present, but there is an absence or only a trace of Hb A, sickle cell anemia is a likely diagnosis. Additional beta globin *DNA testing*, which looks directly at the beta globin gene, can be performed to help confirm the diagnosis and establish the exact genetic type of sickle cell disease. CBC and hemoglobin electrophoresis are also typically used to diagnose sickle cell trait and various other types of beta globin traits.

Diagnosis of sickle cell disease can occur under various circumstances. If an individual has symptoms that are suggestive of this diagnosis, the above-described screening tests can be performed followed by DNA testing, if indicated. Screening at birth using HPLC or a related technique offers the opportunity for early intervention. More than 40 states include sickle cell screening as part of the usual battery of blood tests done for newborns. This allows for early identification and treatment. Hemoglobin trait screening is recommended for any individual of a high-risk ethnic background who may be considering having children. When both members of a couple are found to have sickle cell trait, or other related hemoglobin traits, they can receive **genetic counseling** regarding the risk of sickle cell disease in their future children and various testing options.

Sickle cell disease can be identified before birth through the use of prenatal diagnosis. *Chorionic villus sampling (CVS)* can be offered as early as 10 weeks of pregnancy and involves removing a sample of the placenta made by the baby and testing the cells. CVS carries a risk of causing a **miscarriage** that is between one-half to one percent.

Amniocentesis is generally offered between 15 and 22 weeks of pregnancy, but can sometimes be offered earlier. Two to three tablespoons of the fluid surrounding the baby is removed. This fluid contains

fetal cells that can be tested. This test carries a risk of causing a miscarriage, which is not greater than one percent. Pregnant woman and couples may choose prenatal testing in order to prepare for the birth of a baby that may have sickle cell disease. Alternately, knowing the diagnosis during pregnancy allows for the option of pregnancy termination.

Preimplantation genetic diagnosis (PGD) is a relatively new technique that involves in-vitro fertilization followed by **genetic testing** of one cell from each developing embryo. Only the embryos unaffected by sickle cell disease are transferred back into the uterus. PGD is currently available on a research basis only, and is relatively expensive.

Treatment

There are several practices intended to prevent some of the symptoms and complications of sickle cell disease. These include preventative **antibiotics**, good hydration, immunizations, and access to comprehensive care. Maintaining good health through adequate **nutrition**, avoiding stresses and infection, and getting proper rest is also important. Following these guidelines is intended to improve the health of individuals with sickle cell disease.

Penicillin

Infants are typically started on a course of penicillin that extends from infancy to age six. Use of this antibiotic is meant to ward off potentially fatal infections. Infections at any age are treated aggressively with antibiotics. Vaccines for common infections, such as *pneumococcal pneumonia*, are also recommended.

Pain management

Pain is one of the primary symptoms of sickle cell anemia, and controlling it is an important concern. The methods necessary for pain control are based on individual factors. Some people can gain adequate pain control through over-the-counter oral painkillers (**analgesics**). Other individuals, or painful events, may require stronger methods that can include administration of **narcotics**. Alternative therapies may be useful in avoiding or controlling pain, including relaxation, hydration, avoiding extremes of temperature, and the application of local warmth.

Blood transfusions

Blood transfusions are not usually given on a regular basis but are used to treat individuals with

frequent and severe painful events, severe anemia, and other emergencies. In some cases blood transfusions are given as a preventative measure, for example to treat spleen enlargement or prevent a second stroke (or a first stroke in an individual shown to be at high risk).

Regular blood transfusions have the potential to decrease formation of hemoglobin S, and reduce associated symptoms. However, there are limitations and risks associated with regular blood transfusions, including the risk of blood-borne infection and sensitization to proteins in the transfused blood that can make future transfusions very difficult. Most importantly, chronic blood transfusions can lead to iron overload. The body tends to store excess iron, such as that received through transfusions, in various organs. Over time, this iron storage can cause damage to various tissues and organs, such as the heart and endocrine organs.

Some of this damage can be prevented by the administration of a medication called *desferoxamine* that helps the body to eliminate excess iron through the urine. Alternately, some individuals receive a new, non-standard treatment called *erythrocytapheresis*. This involves the automated removal of sickle cells and is used in conjunction with a reduced number of regular transfusions. This treatment helps to reduce iron overload.

Hydroxyurea

Emphasis is being placed on developing drugs that treat sickle cell anemia directly. The most promising of these drugs in the late 1990s is hydroxyurea, a drug that was originally designed for anticancer treatment. Hydroxyurea has been shown to reduce the frequency of painful crises and acute chest syndrome in adults, and to lessen the need for blood transfusions. Hydroxyurea seems to work by inducing a higher production of fetal hemoglobin. The major side effects of the drug include decreased production of platelets, red blood cells, and certain white blood cells. The effects of long-term hydroxyurea treatment are unknown; however, a nine-year follow-up study of 299 adults with frequent painful crises reported in 2003 that taking hydroxyurea was associated with a 40% reduction in mortality.

Bone marrow transplantation

Bone marrow transplantation has been shown to cure sickle cell anemia in some cases. This treatment is reserved primarily for severely affected children with a healthy donor whose marrow proteins match those of the recipient, namely a brother or sister who has

inherited the same tissue type. Indications for a bone marrow transplant are stroke, recurrent acute chest syndrome, and chronic unrelieved pain.

Bone marrow transplantations tend to be the most successful in children; adults have a higher rate of transplant rejection and other complications. There is approximately a 10% fatality rate associated with bone marrow transplants done for sickle cell disease. Survivors face potential long-term complications, such as chronic graft-versus-host disease (an immune-mediated attack by the donor marrow against the recipient's tissues), **infertility**, and development of some forms of **cancer**. A relatively recent advance in transplantation involves the use of donor stem cells obtained from *cord blood*, the blood from the placenta that is otherwise discarded following the birth of a new baby. Cord blood cells, as opposed to fully mature bone marrow cells, appear to be less likely to result in graft-versus-host disease in the recipient. This increases the safety and efficacy of the transplant procedure.

Surgery

Certain surgical interventions are utilized in the treatment of specific sickle cell-related complications. Removal of a dysfunctional gallbladder or spleen can often lead to improvements in health. Investigations are currently underway to establish the efficacy of hip coring surgery, in which a portion of affected bone is removed to treat avascular necrosis of the hip. The hope is that this may provide an effective treatment to alleviate some pain and restore function in the affected hip.

Gene research

Replacing the gene that produces the defective hemoglobin in sickle cell disease patients with one that makes normal hemoglobin may be a possible treatment due to recent research. According to a 1998 report in *Science*, researchers studied the blood cells from people who carry the sickle cell gene. By using an enzyme called a ribosome, the study was able to alter sickle cells into normal cells. The ribosome cut out the mutated instructions in the cells' genetic pattern and replaced them with the correct instructions. Researchers hope that this will allow the cells to make normal hemoglobin—leading to the ultimate treatment for those with sickle cell disease.

In late 2001, genetic scientists reported that they had designed a gene that might lead to a future treatment of sickle cell anemia. Although the gene had not been tested in humans, early results showed that the injected gene protected cells from sickling. As of 2003,

KEY TERMS

Amino acid—Organic compounds that form the building blocks of protein. There are 20 types of amino acids (eight are “essential amino acids” which the body cannot make and must therefore be obtained from food).

Anemia—A blood condition in which the level of hemoglobin or the number of red blood cells falls below normal values. Common symptoms include paleness, fatigue, and shortness of breath.

Bilirubin—A yellow pigment that is the end result of hemoglobin breakdown. This pigment is metabolized in the liver and excreted from the body through the bile. Bloodstream levels are normally low; however, extensive red cell destruction leads to excessive bilirubin formation and jaundice.

Bone marrow—A spongy tissue located in the hollow centers of certain bones, such as the skull and hip bones. Bone marrow is the site of blood cell generation.

Bone marrow transplantation—A medical procedure used to treat some diseases that arise from defective blood cell formation in the bone marrow. Healthy bone marrow is extracted from a donor to replace the marrow in an ailing individual. Proteins on the surface of bone marrow cells must be identical or very closely matched between a donor and the recipient.

Globin—One of the component protein molecules found in hemoglobin. Normal adult hemoglobin has a pair each of alpha-globin and beta-globin molecules.

Heme—The iron-containing molecule in hemoglobin that serves as the site for oxygen binding.

Hemoglobin—Protein-iron compound in the blood that carries oxygen to the cells and carries carbon dioxide away from the cells.

Hemoglobin A—Normal adult hemoglobin that contains a heme molecule, two alpha-globin molecules, and two beta-globin molecules.

Hemoglobin electrophoresis—A laboratory test that separates molecules based on their size, shape, or electrical charge.

Hemoglobin S—Hemoglobin produced in association with the sickle cell trait; the beta-globin molecules of hemoglobin S are defective.

Hemolytic—Referring to the destruction of the cell membranes of red blood cells, resulting in the release of hemoglobin from the damaged cell.

Hydroxyurea—A drug that has been shown to induce production of fetal hemoglobin. Fetal hemoglobin has a pair of gamma-globin molecules in place of the typical beta-globins of adult hemoglobin. Higher-than-normal levels of fetal hemoglobin can prevent sickling from occurring.

Impotence—The inability to have a penile erection, which can be due to tissue damage resulting from sickling within the penis (priapism).

Iron overload—A side effect of frequent blood transfusions in which the body accumulates abnormally high levels of iron. Iron deposits can form in organs, particularly the heart, and cause life-threatening damage.

Jaundice—Yellowing of the skin or eyes due to excess of bilirubin in the blood.

Magnetic resonance imaging (MRI)—A technique that employs magnetic fields and radio waves to create detailed images of internal body structures and organs, including the brain.

Meniscocytosis—Another word for sickle cell disease.

Mutation—A permanent change in the genetic material that may alter a trait or characteristic of an individual, or manifest as disease, and can be transmitted to offspring.

Narcotics—Strong, prescription medication that can be effective in treating pain, but have the potential to be habit-forming if their use is not supervised correctly.

Nucleic acid—The cellular molecules DNA and RNA that act as coded instructions for the production of proteins and are copied for transmission of inherited traits.

Ophthalmology—The medical specialty of vision and the eye.

Placenta—The organ responsible for oxygen and nutrition exchange between a pregnant mother and her developing baby.

Red blood cell—Hemoglobin-containing blood cells that transport oxygen from the lungs to tissues. In the tissues, the red blood cells exchange their oxygen for carbon dioxide, which is brought back to the lungs to be exhaled.

Screening—Process through which carriers of a trait may be identified within a population.

Sickle cell—A red blood cell that has assumed an elongated shape due to the presence of hemoglobin S.

experiments in **gene therapy** for sickle cell disease have been carried out in mice, using lentiviral vectors to transfer the corrective gene into the mouse's stem cells. This technique, however, has not yet been attempted in human subjects as of late 2003.

Psychosocial support

As in any lifelong, chronic disease, comprehensive care is important. Assistance with the emotional, social, family-planning, economic, vocational, and other consequences of sickle cell disease can enable affected individuals to better access and benefit from their medical care.

Prognosis

Sickle cell disease is characteristically variable between and within affected individuals. Predicting the course of the disorder based solely on genes is not possible. Several factors aside from genetic inheritance determine the prognosis for affected individuals, including the frequency, severity, and nature of specific complications in any given individual. The availability and access of comprehensive medical care also plays an important role in preventing and treating serious, acute complications, which cause the majority of sickle cell-related deaths. For those individuals who do not experience such acute events, life expectancy is probably substantially greater than the average for all people with sickle cell disease. The impact of recent medical advances supports the hypothesis that current life expectancies may be significantly greater than those estimated in the early 1990s. At that time, individuals with SS disease lived to the early- to mid-40s, and those with SC disease lived into the upper 50s *on average*. As of 2003, the life expectancy of persons with SS is over 50. With early detection and comprehensive medical care, most people with sickle cell disease are in fairly good health most of the time. Most individuals can be expected to live well into adulthood, enjoying an improved quality of life including the ability to choose a variety of education, career, and family-planning options for themselves.

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ORGANIZATIONS

Mayo Foundation for Medical Education and Research. <<http://www.mayohealth.org>>.

Sickle Cell Disease Association of America. 200 Corporate Point, Suite 495, Culver City, CA 90230-7633. (310) 216-6363. (800) 421-8453. <<http://sicklecelldisease.org/>>.

Sickle Cell Disease Program, Division of Blood Diseases and Resources. National Heart, Lung, and Blood Institute. II Rockledge Centre, 6701 Rockledge Dr. MSC 7950, Bethesda, MD 20892-7950. (301) 435-0055.

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Sideroblastic anemia

Definition

Sideroblastic anemia is a term used to describe a group of rare blood disorders characterized by the

bone marrow's inability to manufacture normal red blood cells.

Description

Named for the Greek words for iron and germ, sideroblastic anemia is one of the principal types of iron-utilization anemia. Abnormal, iron-saturated red cells are present in the blood of people who have this disease. Although the iron circulates normally from the plasma to the bone marrow, where new red blood cells are created, it is not properly incorporated into new red blood cells.

Sideroblastic anemia can be inherited, but the disease is usually acquired as a result of illness or exposure to toxic substances.

Sideroblastic anemia is a disease of adults.

Causes and symptoms

The cause of sideroblastic anemia cannot always be identified. Drug toxicity, alcohol abuse, and **lead poisoning** are common causes of this condition.

Sideroblastic anemia is also associated with:

- leukemia
- lymphoma (**cancer** of the lymph glands)
- myeloma (cancer of the bone marrow)
- rheumatoid arthritis, and other inflammatory diseases

Symptoms of sideroblastic anemia are the same as symptoms of the disease that causes the condition, as well as anemia

Complications

Possible complications of sideroblastic anemia include:

- congestive **heart failure**
- diabetes mellitus
- enlargement of the liver and spleen
- formation of liver nodules and scar tissue
- irregular heartbeat
- recurring inflammation of the sac that surrounds the heart
- secondary **hypopituitarism** (dwarfism)
- skin darkening
- underactivity of the thyroid gland

Diagnosis

Blood tests are used to examine the appearance and other characteristics of red cells and to measure the amount of iron in the blood. **Bone marrow biopsy** is also used.

Treatment

Acquired sideroblastic anemia may be cured when the condition that causes it is treated or removed.

If the cause of a patient's anemia cannot be determined, blood transfusions may be necessary. Medications are prescribed to stimulate excretion or excess iron that accumulates as a result of these transfusions.

In rare instances, treatment with oral pyridoxine (a B-complex vitamin) benefits patients whose sideroblastic anemia was present at birth. This treatment improves the condition of some patients but does not cure the anemia.

Prognosis

Sideroblastic anemia of unknown origin may lead to leukemia. It may take as long as 10 years for this disease progression to take place.

Resources

ORGANIZATIONS

Leukemia Society of America, Inc. 600 Third Ave., New York, NY 10016. (800) 955-4572. <<http://www.leukemia.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

National Organization for Rare Disorders. P.O. Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673. <<http://www.rarediseases.org>>.

Maureen Haggerty

SIDS see **Sudden infant death syndrome**

Sigmoidoscopy

Definition

Sigmoidoscopy is a procedure by which a doctor inserts either a short and rigid or slightly longer and flexible fiber-optic tube into the rectum to examine the lower portion of the large intestine (or bowel).

Purpose

Sigmoidoscopy is used most often in screening for colorectal **cancer** or to determine the cause of rectal bleeding. It is also used for the diagnosis of inflammatory bowel disease and other benign diseases of the lower intestine.

Cancer of the rectum and colon is the second most common cancer in the United States, and claims the lives of approximately 56,000 people annually. As a result, The American Cancer Society recommends that people age 50 and over be screened for colorectal cancer every five years. The screening includes a flexible sigmoidoscopy. Screening at an earlier age should be done on patients who have a family history of colon or **rectal cancer**, or small growths in the colon (polyps).

Individuals with inflammatory bowel disease (Crohn's colitis or **ulcerative colitis**) are at increased risk for colorectal cancer and should begin their screenings at a younger age, and be screened more frequently. Many doctors screen such patients more often than every three to five years. Those with ulcerative colitis should be screened beginning 10 years after the onset of disease; those with Crohn's colitis beginning 15 years after the onset of disease.

Some doctors prefer to do this screening with a colonoscopy, which allows them to see the entire colon (certain patients, such as those with Crohn's colitis or ulcerative colitis, must be screened with a colonoscopy). However, compared with sigmoidoscopy, **colonoscopy** is a longer process, causes more discomfort, and is more costly.

Studies have indicated that about one-fourth of all precancerous or small cancerous growths in the colorectal region can be seen with a rigid sigmoidoscope. The longer, flexible version, which is the primary type of sigmoidoscope used in the screening process, can detect more than one-half of all growths in this region. This examination is usually performed in combination with a **fecal occult blood test**, in an effort to increase detection of polyps and cancers that lie beyond the scope's reach.

Precautions

Sigmoidoscopy can usually be conducted in a doctor's office or a health clinic. However, some individuals should have the procedure done in a hospital day surgery facility. These include patients with rectal bleeding, and patients whose blood does not clot well (possibly as a result of blood-thinning medications).

The exam is not always adequate. A 2004 study reported that among older patients and women,

sigmoidoscopy is not always effective, particularly because insertion depth is not adequate. For unknown reasons, this is almost twice as true for women as for men.

Description

Most sigmoidoscopy is done with a flexible fiber-optic tube. The tube contains a light source and a camera lens. The doctor moves the sigmoidoscope up beyond the rectum (the first 1 ft/30 cm of the colon), examining the interior walls of the rectum. If a 2 ft/60 cm scope is used, the next portion of the colon can also be examined for any irregularities.

The procedure takes 20 to 30 minutes, during which time the patient will remain awake. Light **sedation** may be given to some patients. There is some discomfort (usually bloating and cramping) because air is injected into the bowel to widen the passage for the sigmoidoscope. **Pain** is rare except in individuals with active inflammatory bowel disease.

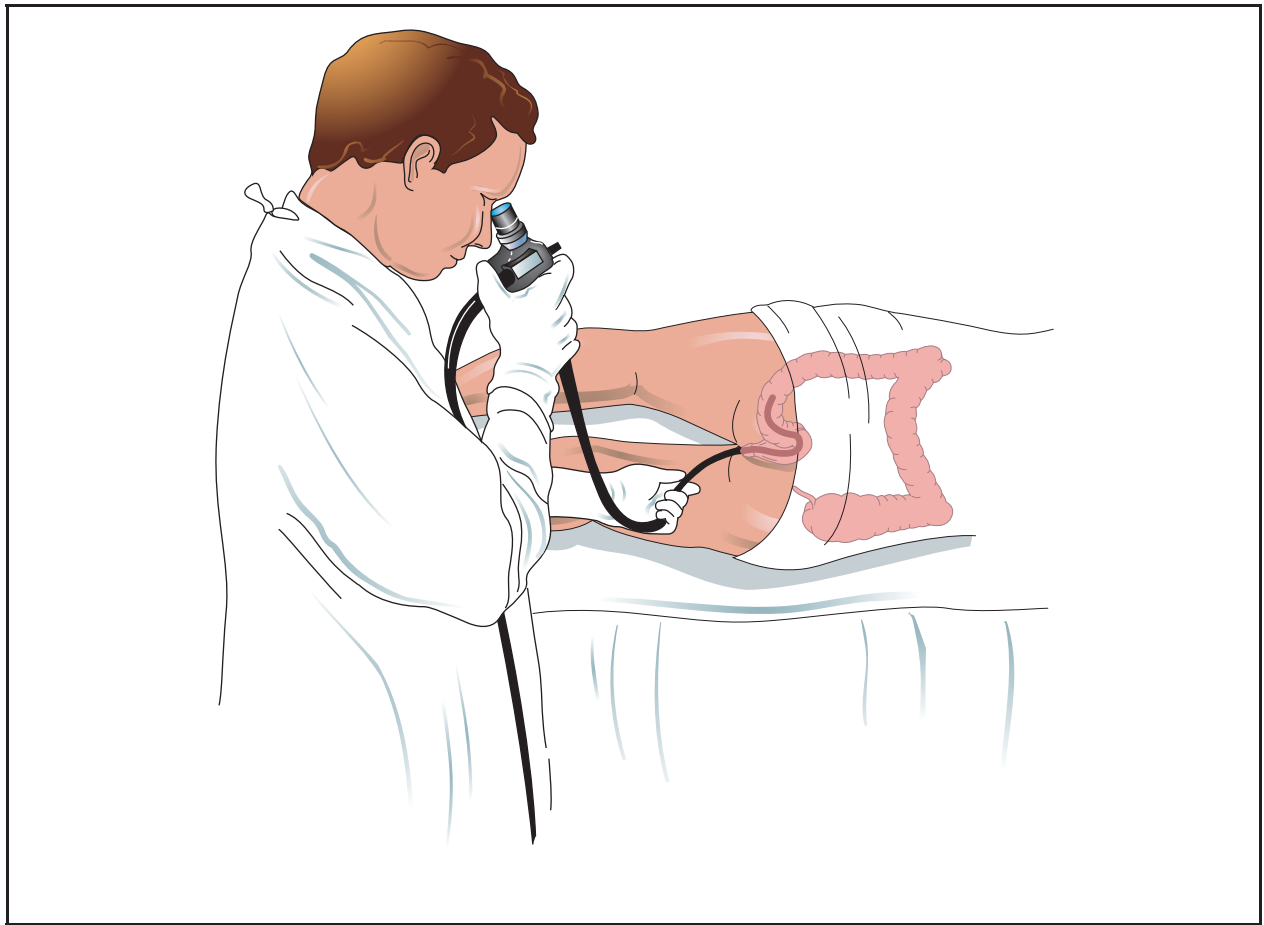
In a colorectal cancer screening, the doctor is looking for polyps or tumors. Studies have shown that over time, many polyps develop into cancerous lesions and tumors. Using instruments threaded through the fiber-optic tube, cancerous or precancerous polyps can either be removed or biopsied during the sigmoidoscopy. People who have cancerous polyps removed can be referred for full colonoscopy, or more frequent sigmoidoscopy, as necessary.

The doctor may also look for signs of ulcerative colitis, which include a loss of blood flow to the lining of the bowel, a thickening of the lining, and sometimes a discharge of blood and pus mixed with stool. The doctor can also look for **Crohn's disease**, which often appears as shallow or deep ulcerations, or erosions and fissures in the lining of the colon. In many cases, these signs appear in the first few centimeters of the colon above the rectum, and it is not necessary to do a full colonoscopic exam.

Private insurance plans often cover the cost of sigmoidoscopy for screening in healthy individuals over 50, or for diagnostic purposes. Medicare covers the cost for diagnostic exams, and may cover the costs for screening exams.

Preparation

The purpose of preparation for sigmoidoscopy is to clean the lower bowel of stool so that the doctor can see the lining. Many patients are required to consume only clear liquids on the day before the test, and to take two **enemas** on the morning of the procedure. The



Sigmoidoscopy is a procedure most often used in screening for colorectal cancer and as a test in diagnosis of possible inflammatory bowel disease. As illustrated above, the physician can view the rectum and colon through a sigmoidoscope, a 12 inch (30 cm) or 24 inch (60 cm) flexible fiber-optic tube which contains a light source and a lens. (*Illustration by Electronic Illustrators Group.*)

bowel is cleaner, however, if patients also take an oral laxative preparation of 1.5 oz phospho-soda the evening before the sigmoidoscopy.

Certain medications should be avoided for a week before having a sigmoidoscopy. These include:

- aspirin, or products containing aspirin
- ibuprofen products (Nuprin, Advil, or Motrin)
- iron or **vitamins** containing iron

Although most prescription medication can be taken as usual, patients should check with their doctor in advance.

Aftercare

Patients may feel mild cramping after the procedure that will improve after passing gas. Patients can resume their normal activities almost immediately.

Risks

There is a slight risk of bleeding from the procedure. This risk is heightened in individuals whose blood does not clot well, either due to disease or medication, and in those with active inflammatory bowel disease. The most serious complication of sigmoidoscopy is bowel perforation (tear). This complication is very rare, however, occurring only about once in every 7,500 procedures.

Normal results

A normal exam shows a smooth bowel wall with no evidence of inflammation, polyps or tumors.

Abnormal results

For a cancer screening sigmoidoscopy, an abnormal result involves one or more noncancerous or

KEY TERMS

Biopsy—A procedure where a piece of tissue is removed from a patient for diagnostic testing.

Colorectal cancer—Cancer of the large intestine, or colon, including the rectum (the last 16 in of the large intestine before the anus).

Inflammatory bowel disease—Ulcerative colitis or Crohn's colitis; chronic conditions characterized by periods of diarrhea, bloating, abdominal cramps, and pain, sometimes accompanied by weight loss and malnutrition because of the inability to absorb nutrients.

Polyp—A small growth that can be precancerous when it appears in the colon.

precancerous polyps or tumors. Patients showing polyps have an increased risk of developing colorectal cancer in the future.

Small polyps can be completely removed. Larger polyps or tumors usually require the doctor to remove a portion of the growth for diagnostic testing. Depending on the test results, the patient is then scheduled to have the growth removed surgically, either as an urgent matter if it is cancerous, or as an elective surgery within a few months if it is noncancerous.

In a diagnostic sigmoidoscopy, an abnormal result shows signs of active inflammatory bowel disease, either a thickening of the intestinal lining consistent with ulcerative colitis, or ulcerations or fissures consistent with Crohn's disease.

Sigmoidoscopy is a procedure most often used in screening for colorectal cancer and as a test in diagnosis of possible inflammatory bowel disease. As illustrated above, the physician can view the rectum and colon through a sigmoidoscope, a 12 inch (30 cm) or 24 inch (60 cm) flexible fiber-optic tube which contains a light source and a lens.

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Teresa G. Odle

Sildenafil citrate

Definition

Sildenafil citrate (Viagra) is a medication used to treat *erectile dysfunction* (ED), or **impotence**, in men.

Purpose

Labeled use

Viagra treats **erectile dysfunction**, the inability to achieve and/or maintain an erection of the penis that is adequate for sexual intercourse. Ten to fifteen million men in the United States suffer from ED, and by age 65, up to 25% of men have experienced impotence problems. Erectile dysfunction can be caused by a number of physical and psychological conditions, including diabetes, depression, **prostate cancer**, **spinal cord injury**, **multiple sclerosis**, arteriosclerosis, and heart disease. Injuries to the penis that cause nerve, tissue, or vascular damage can trigger impotence. It is also a common side effect of some prescription medications, including **antihistamines**, antidepressants, antihypertensives, antipsychotics, **beta blockers**, **diuretics**, tranquilizers, appetite suppressants, cimetidine (Tagamet), and **finasteride** (Propecia).

A study of African American and Hispanic men published in 2002 reported that Viagra appears to be equally safe and equally effective across different racial and ethnic groups in the United States.

Investigational uses

Although not approved for use in women, clinical studies have shown that sildenafil citrate may be

effective in relieving female **sexual dysfunction** for some women. In one study, both female and male study participants who suffered from sexual dysfunction related to their use of such psychotropic medications as **benzodiazepines** reported an increase in arousal and overall sexual satisfaction when they began taking Viagra. Several studies have also indicated the drug may be effective in improving libido and arousal in women taking selective serotonin uptake inhibitors (SSRIs).

Another possible use of sildenafil in women is the treatment of **infertility**. Women who have had repeated failures with **in vitro fertilization** (IVF) due to poor development of the tissue that lines the uterus may benefit from treatment with vaginal suppositories containing sildenafil. One study reported that 70% of patients had a significant thickening of the uterine lining, with 29% having a successful implantation of a new embryo, and 45% achieving ongoing pregnancies.

Another investigational study conducted by researchers at Johns Hopkins University School of Medicine in Baltimore, and published in the August 2000 issue of the *Journal of Clinical Investigation* found that Viagra may have additional clinical promise for people with diabetes beyond treating ED. In animal studies, Viagra was effective in relaxing the pyloric muscle of stomach, improving digestion and relieving the symptoms of gastroparesis. Up to 75% of people with diabetes suffer from gastroparesis, which causes bloating, **nausea**, loss of appetite, and **vomiting**. Further human studies are needed to evaluate Viagra's effectiveness in treating this common diabetic complication.

Because of its capacity to enhance nitric oxide production, sildenafil has been investigated as a possible treatment for other disorders that are caused by impaired nitric oxide production. One such disorder is esophageal motility dysfunction (**achalasia**), in which the smooth muscles of the esophagus and the cardiac sphincter remain constricted, causing difficulty in swallowing, regurgitation of food, and chest **pain** when eating. A study published in 2000 in the journal *Gastroenterology* found that sildenafil temporarily improved the condition in some patients by relaxing the lower esophageal muscles. An Italian study reported in 2002 that sildenafil shows genuine promise as a treatment for spastic **esophageal disorders**.

Precautions

Viagra is not labeled or approved for use by women or children, or by men without erectile dysfunction. The medication may also be contraindicated

(not recommended for use) in patients with certain medical conditions.

Because sexual activity can **stress** the heart, men who have heart problems should check with their physician to see if sexual activity is recommended. Viagra may trigger temporary **hypotension** (low blood pressure) and is known to increase cardiovascular nerve activity, so it is prescribed with caution in men with a history of **heart attack**, arteriosclerosis, **angina**, arrhythmia, and chronic low blood pressure problems. However, a study published in the March 15, 2001, *British Medical Journal* found no evidence that the drug causes a higher incidence of heart attack. A four-year update on the safety of Viagra published in September 2002 corroborated the findings of the British report, and stated that the only absolute contraindication for the use of sildenafil is the concurrent use of nitrates.

Anyone experiencing cardiovascular symptoms such as **dizziness**, chest or arm pain, and nausea when participating in sexual activity after taking Viagra should stop the encounter. They should also not take Viagra again until they have discussed the episode with their healthcare provider.

It is recommended that men with kidney or liver impairments, and men over age 65, start at the lowest possible dosage of Viagra (25 mg). Clinical studies have shown that the drug builds up in the plasma of these patients to a concentration that is three to eight times higher than normal. Caution is also recommended in prescribing the drug to individuals with *retinitis pigmentosa*, a rare genetic eye disorder. Viagra should not be taken more than once per day by anyone.

Viagra has not been studied for use on patients with stomach ulcers and bleeding disorders, and its safety in these individuals is unknown. Men who have either of these conditions should let their physician know before taking Viagra. It should also be used with caution in men with misshapen or deformed penises, such as those with Peyronie's disease, cavernosal fibrosis, or with angulation of the penis.

Men who take medications containing nitrates (e.g., nitroglycerin, isosorbide mononitrate, isosorbide dinitrate) should never take Viagra, as the interaction between the two drugs may cause a dramatic drop in blood pressure, and possibly trigger a heart attack or **stroke**. This includes illegal recreational drugs such as amyl nitrates (also known as poppers).

Viagra may also interact with other prescription and over-the-counter (OTC) medications, either magnifying or diluting the intended therapeutic effects of

one or both drugs. Some drugs that have a known interaction with Viagra include the protease inhibitor ritonavir and the antibiotic erythromycin. For this reason, it is critical that men who are prescribed Viagra let their healthcare providers know all the medications they are taking.

Other medications and therapies for erectile dysfunction, including vacuum or pump devices, drug injections (Caverject), and urethral suppositories (MUSE), should never be used in conjunction with Viagra.

Description

Sildenafil citrate was originally developed in 1991 as a treatment for angina, or chest pain. The drug, marketed under the name Viagra, received FDA market clearance as a treatment for impotence in March 1998, and since that time it has been prescribed for over 10 million men worldwide. It was the first oral medication approved for ED treatment. A newer drug, tadalafil, has been developed to treat men who do not respond to sildenafil. Tadalafil has gained preliminary approval in the European Union (EU), and is in the final stages of regulatory approval in Canada as of November 2002.

Viagra is a vasodilator, a drug that has the effect of dilating the blood vessels. It works by improving blood circulation to the penis, and by enhancing the effects of *nitric oxide*, the agent that relaxes the smooth muscle of the penis and regulates blood vessels during sexual stimulation, allowing the penis to become engorged and achieve an erection.

The average recommended dose of Viagra is 50 mg. For men that do not respond adequately to this amount, the dosage may be increased up to 100 mg or decreased to 25 mg. The medication is taken approximately one hour before sexual activity is planned, and may remain effective for up to four hours.

Viagra does not increase sexual desire. Sexual stimulation and arousal are required for the medication to be effective. Despite its widespread use as a recreational drug, it is not an aphrodisiac and there is no clinical evidence that it improves sexual performance in men who are not suffering from ED.

Many insurance plans provide coverage or reimbursement for sildenafil citrate provided it is prescribed to treat erectile dysfunction. A 1999 report issued by a health insurance consulting group indicated that almost half of the men taking Viagra at least once weekly receive insurance reimbursement for the drug. The pills cost approximately \$10 each,

and insurers may limit coverage to a specific number of pills each month.

Preparation

Viagra requires time to be absorbed by the body and become effective. The average recommended time frame for taking the drug is one hour before initiating sexual activity, although depending on an individual's response to the drug, this time can vary from four hours to 30 minutes.

Men should always consult with their physician before beginning treatment with sildenafil citrate. The medication is not for everyone, and a healthcare professional needs to evaluate medical history and perform a thorough medical examination before prescribing the drug. In addition, erectile dysfunction may be a symptom of an undiagnosed condition (i.e., diabetes) for which treatment is critical, and may actually reverse the impotence problem.

Risks

The most commonly reported side effects of Viagra are **headache**, flushing of the face, upset stomach, and nasal congestion.

Other less common side effects include, but are not limited to:

- vision problems, including sensitivity to light, blurred vision, and a color tinge to vision
- urinary tract infection
- diarrhea
- dizziness
- rash

Side effects may be reduced or eliminated through adjustments to dosage. Men who experience these symptoms should consult their physician.

Priapism, a painful and prolonged erection that lasts for two to six hours, is a rare but potentially serious side effect of Viagra. Because prolonged erection can permanently damage the tissues of the penis, anyone who experiences an erection lasting over four hours should call a healthcare professional immediately.

Men who are taking Viagra and inadvertently or intentionally take a medication containing nitrates may suffer from life-threatening hypotension—a severe drop in blood pressure.

The cardiovascular risks of sildenafil citrate are still under investigation. The drug is known to cause

KEY TERMS

Angina—Angina pectoris, or chest pain, caused by an insufficient supply of oxygen and decreased blood flow to the heart muscle. Angina is frequently the first sign of coronary artery disease.

Angulation of the penis—Abnormal bend or angle to the structure of the penis.

Antidepressants—Medications prescribed to relieve major depression. Classes of antidepressants include selective serotonin reuptake inhibitors (fluoxetine/Prozac, sertraline/Zoloft), tricyclics (amitriptyline/Elavil), MAOIs (phenelzine/Nardil), and heterocyclics (bupropion/Wellbutrin, trazodone/Desyrel).

Antihistamines—A drug used to treat allergic conditions that counteracts histamines — a substance in the body that causes itching, vascular changes, and mucus secretion when released by cells.

Antihypertensives—Medications used to treat high blood pressure.

Antipsychotics—A class of drugs used to control psychotic symptoms in patients with psychotic disorders such as schizophrenia and delusional disorder. Antipsychotics include risperidone (Risperdal), haloperidol (Haldol), and chlorpromazine (Thorazine).

Arrhythmia—Irregular heartbeat caused by erratic electrical signals or nerve impulses to the cardiac muscles.

Artherosclerosis—The cause of coronary artery disease, in which the walls of the coronary arteries thicken due to the accumulation of plaque in the blood vessels.

Beta blockers—Drugs that lower blood pressure and reduce stress to the heart by blocking the actions of beta receptors that control the speed and strength of heart muscle contractions and blood vessel dilation.

Cavernosal fibrosis—The formation of abnormal fibrous tissue in the erectile tissue of the penis.

Diuretics—Any substance that increases urine output.

Erectile dysfunction—Impotence; the inability of a man to achieve and/or maintain an erection of sufficient quality for sexual intercourse.

Gastroparesis—Nerve damage of the stomach that delays or stops stomach emptying, resulting in nausea, vomiting, bloating, discomfort, and weight loss.

Peyronie's disease—A disease which causes a hardening of the corpora cavernosa, the erectile tissue of the penis. The penis may become misshapen and/or curved as a result.

Placebo—An inactive substance with no pharmacological action that is administered to some patients in clinical trials to determine the relative effectiveness of another drug administered to a second group of patients.

Priapism—A painful, abnormally prolonged erection (i.e., four or more hours).

Protease inhibitor—A drug that inhibits the action of enzymes.

Retinitis pigmentosa—An inherited degenerative eye disease that impairs night vision and drastically narrows the field of vision.

Selective serotonin uptake inhibitors (SSRIs)—Drugs that regulate depression by blocking the reabsorption of serotonin in the brain consequently raising serotonin levels. SSRIs include fluoxetine (Prozac), sertraline (Zoloft), and paroxetine (Paxil).

Serotonin—One of three major neurotransmitters found in the brain that is linked to emotions.

dips in blood pressure and to boost cardiovascular nerve activity. Some cardiovascular-related deaths have been reported in men who use Viagra, but it is unclear whether the fatalities were due to the drug itself or to the underlying heart disease. Further complicating the picture is the fact that the stress of sexual activity may have triggered the fatal cardiac event with or without the use of Viagra. The *BMJ* study, and a report published in the April 18, 2001 issue of the *Journal of the American Medical Association (JAMA)* suggest that the drug does not increase the risk of heart attack. However, *JAMA* also notes

that further studies are necessary to confirm this finding.

Although it is a prescription drug, as of early 2001 there was still a thriving illicit market for Viagra via the Internet. Aside from the health risks recreational use of the drug poses to individuals with heart conditions and other contraindicated disorders, any adverse effects caused by Viagra cannot be tracked by regulatory authorities if it has been illegally obtained. In addition, the drug appears to be toxic in large doses. In November 2002, a group of French toxicologists

reported the case of a 56-year-old male who took a fatal overdose of Viagra.

Normal results

When used as directed, Viagra allows men with erectile dysfunction to achieve and maintain a penile erection when aroused during sexual activity. Double-blind, randomized clinical trials of sildenafil citrate have shown that the drug has an 63–82% efficacy rate in improving erectile activity among men with ED, depending on the dose administered (between 25 and 100 mg), compared to a 24% improvement in men receiving a placebo.

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ORGANIZATIONS

American Heart Association. American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-630 or (800) 242-8721. inquire@heart.org. <http://www.americanheart.org>.

U.S. Food and Drug Administration (FDA), Center for Drug Evaluation and Research. Viagra Information. <http://www.fda.gov/cder/consumerinfo/viagra/default.htm>.

OTHER

Pfizer, Inc. Viagra Information Site. <http://www.viagra.com/>.

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Silent thyroiditis see **Thyroiditis**

Silicosis

Definition

Silicosis is a progressive disease that belongs to a group of lung disorders called pneumoconioses. Silicosis is marked by the formation of lumps (nodules) and fibrous scar tissue in the lungs. It is the oldest known occupational lung disease, and is caused by exposure to inhaled particles of silica, mostly from quartz in rocks, sand, and similar substances.

Description

It is estimated that there are TWO million workers in the United States employed in occupations at risk for the development of silicosis. These include miners, foundry workers, stonecutters, potters and ceramics workers, sandblasters, tunnel workers, and rock drillers. Silicosis is mostly found in adults over 40. It has four forms:

- **Chronic.** Chronic silicosis may take 15 or more years of exposure to develop. There is only mild impairment of lung functioning. Chronic silicosis may progress to more advanced forms.
- **Complicated.** Patients with complicated silicosis have noticeable **shortness of breath**, weight loss, and

extensive formation of fibrous tissue (fibrosis) in the lungs. These patients are at risk for developing **tuberculosis** (TB).

- **Accelerated.** This form of silicosis appears after 5-10 years of intense exposure. The symptoms are similar to those of complicated silicosis. Patients in this group often develop rheumatoid arthritis and other autoimmune disorders.
- **Acute.** Acute silicosis develops within six months to two years of intense exposure to silica. The patient loses a great deal of weight and is constantly short of breath. These patients are at severe risk of TB.

Causes and symptoms

The precise mechanism that triggers the development of silicosis is still unclear. What is known is that particles of silica dust get trapped in the tiny sacs (alveoli) in the lungs where air exchange takes place. White blood cells called macrophages in the alveoli ingest the silica and die. The resulting inflammation attracts other macrophages to the region. The nodule forms when the immune system forms fibrous tissue to seal off the reactive area. The disease process may stop at this point, or speed up and destroy large areas of the lung. The fibrosis may continue even after the worker is no longer exposed to silica.

Early symptoms of silicosis include shortness of breath after exercising and a harsh, dry cough. Patients may have more trouble breathing and **cough** up blood as the disease progresses. Congestive **heart failure** can give their nails a bluish tint. Patients with advanced silicosis may have trouble sleeping and experience chest **pain**, hoarseness, and loss of appetite. Silicosis patients are at high risk for TB, and should be checked for the disease during the doctor's examination.

Diagnosis

Diagnosis of silicosis is based on:

- A detailed occupational history.
- Chest x rays will usually show small round opaque areas in chronic silicosis; the round areas are larger in complicated and accelerated silicosis.
- bronchoscopy
- lung function tests

It should be noted that the severity of the patient's symptoms does not always correlate with x-ray findings or lung function test results.

KEY TERMS

Fibrosis—The development of excess fibrous connective tissue in an organ. Fibrosis of the lungs is a symptom of silicosis.

Pneumoconiosis (plural, pneumoconioses)—Any chronic lung disease caused by inhaling particles of silica or similar substances that lead to loss of lung function.

Silica—A substance (silicon dioxide) occurring in quartz sand, flint, and agate. It is used in making glass, scouring and grinding powders, pottery, etc.

Treatment

Symptom management

There is no cure for silicosis. Therapy is intended to relieve symptoms, treat complications, and prevent respiratory infections. It includes careful monitoring for signs of TB. Respiratory symptoms may be treated with bronchodilators, increased fluid intake, steam inhalation, and physical therapy. Patients with severe breathing difficulties may be given **oxygen therapy** or placed on a mechanical ventilator. Acute silicosis may progress to complete **respiratory failure**. Heart-lung transplants are the only hope for some patients.

Patients with silicosis should call their doctor for any of the following symptoms:

- tiredness or mental confusion
- continued weight loss
- coughing up blood
- fever, chest pain, breathlessness, or new unexplained symptoms

Lifestyle changes

Patients with silicosis should be advised to quit smoking, prevent infections by avoiding crowds and persons with colds or similar infections, and receive vaccinations against **influenza** and **pneumonia**. They should be encouraged to increase their **exercise** capacity by keeping up regular activity, and to learn to pace themselves with their daily routine.

Prognosis

Silicosis is currently incurable. The prognosis for patients with chronic silicosis is generally good. Acute

silicosis, however, may progress rapidly to respiratory failure and death.

Prevention

Silicosis is a preventable disease. Preventive occupational safety measures include:

- controls to minimize workplace exposure to silica dust
- substitution of substances—especially in sandblasting—that are less hazardous than silica
- clear identification of dangerous areas in the workplace
- Informing workers about the dangers of overexposure to silica dust, training them in safety techniques, and giving them appropriate protective clothing and equipment.

Coworkers of anyone diagnosed with silicosis should be examined for symptoms of the disease. The state health department and the Occupational Safety and Health Administration (OSHA) or the Mine Safety and Health Administration (MSHA) must be notified whenever a diagnosis of silicosis is confirmed.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Maureen Haggerty

Silo-filler's disease see **Lung diseases due to gas or chemical exposure**

Simethicone see **Antigas agents**

Singer's nodules see **Vocal cord nodules and polyps**

Sinus endoscopy

Definition

An endoscope is a narrow flexible tube which contains an optical device like a telescope or magnifying lens with a bright light. In sinus endoscopy, the endoscope is inserted into the nose, and the interior of the nasal passages, sinuses, and throat is examined.

Purpose

Sinus endoscopy is used to help diagnose structural defects, infection or damage to the sinuses, or structures in the nose and throat. It may be used to view polyps and growths in the sinuses and to investigate causes of recurrent inflammation of the sinuses (**sinusitis**). During surgical procedures, an endoscope may be used to view the area to correct sinus-drainage problems or to remove polyps from the nose and throat.

Precautions

Insertion of the endoscope may cause a gag reflex and some discomfort, however, no special precautions are required to prepare for nasal endoscopy.

Description

This procedure can be done in a physician's office. The endoscope is inserted into a nostril and is threaded through the sinus passages to the throat. To make viewing of these areas easier, and to record the areas being examined, a camera, monitor, or other such viewing device is connected to the endoscope.

Preparation

For the procedure, the patient is usually awake and seated upright in a chair. A local anesthetic spray or liquid may be applied to the throat to make insertion of the endoscope less uncomfortable.

Aftercare

After the endoscope is removed, the patient can return to most normal activities. If an anesthetic was used, the patient may have to wait until the **numbness** wears off to be able to eat or drink.

Risks

The insertion and removal of the endoscope may stimulate a gag reflex, and can cause some discomfort. The procedure may also irritate the tissues of the nose and throat, causing a **nosebleed** or coughing.

Normal results

Under normal conditions, no polyps, or growths are found in the sinuses. There should also be no evidence of infection, swelling, injury, or any structural defect that would prevent normal draining of the sinuses.

Abnormal results

Polyps, growths, infections, or structural defects of the nasal passages are considered abnormal.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.
Ear Foundation. 1817 Patterson St., Nashville, TN 37203. (800) 545-4327. <<http://www.earfoundation.org>>.

OTHER

"Endoscopic Plastic Surgery." *Southern California Plastic Surgery Group* Page. <<http://www.face-doctor.com>>.

Altha Roberts Edgren

Sinus x ray see **Skull x rays**

Sinusitis

Definition

Sinusitis refers to an inflammation of the sinuses, airspaces within the bones of the face. Sinusitis is most often due to an infection within these spaces.

Description

The sinuses are paired air pockets located within the bones of the face. They are:

- the frontal sinuses; located above the eyes, in the center region of each eyebrow
- the maxillary sinuses; located within the cheekbones, just to either side of the nose
- the ethmoid sinuses; located between the eyes, just behind the bridge of the nose.
- the sphenoid sinuses; Located just behind the ethmoid sinuses, and behind the eyes.

The sinuses are connected with the nose. They are lined with the same kind of skin found elsewhere within the respiratory tract. This skin has tiny little hairs projecting from it, called cilia. The cilia beat constantly, to help move the mucus produced in the sinuses into the respiratory tract. The beating cilia sweeping the mucus along the respiratory tract helps to clear the respiratory tract of any debris, or any organisms which may be present. When the lining of the sinuses is at all swollen, the swelling interferes with

the normal flow of mucus. Trapped mucus can then fill the sinuses, causing an uncomfortable sensation of pressure and providing an excellent environment for the growth of infection-causing bacteria.

Causes and symptoms

Sinusitis is almost always due to an infection, although swelling from **allergies** can mimic the symptoms of pressure, **pain**, and congestion; and allergies can set the stage for a bacterial infection. Bacteria are the most common cause of sinus infection. *Streptococcus pneumoniae* causes about 33% of all cases, while *Haemophilus influenzae* causes about 25% of all cases. Sinusitis in children may be caused by *Moraxella catarrhalis* (20%). In people with weakened immune systems (including patients with diabetes; acquired **immunodeficiency** syndrome or **AIDS**; and patients who are taking medications which lower their immune resistance, such as **cancer** and transplant patients), sinusitis may be caused by fungi such as *Aspergillus*, *Candida*, or Mucorales.

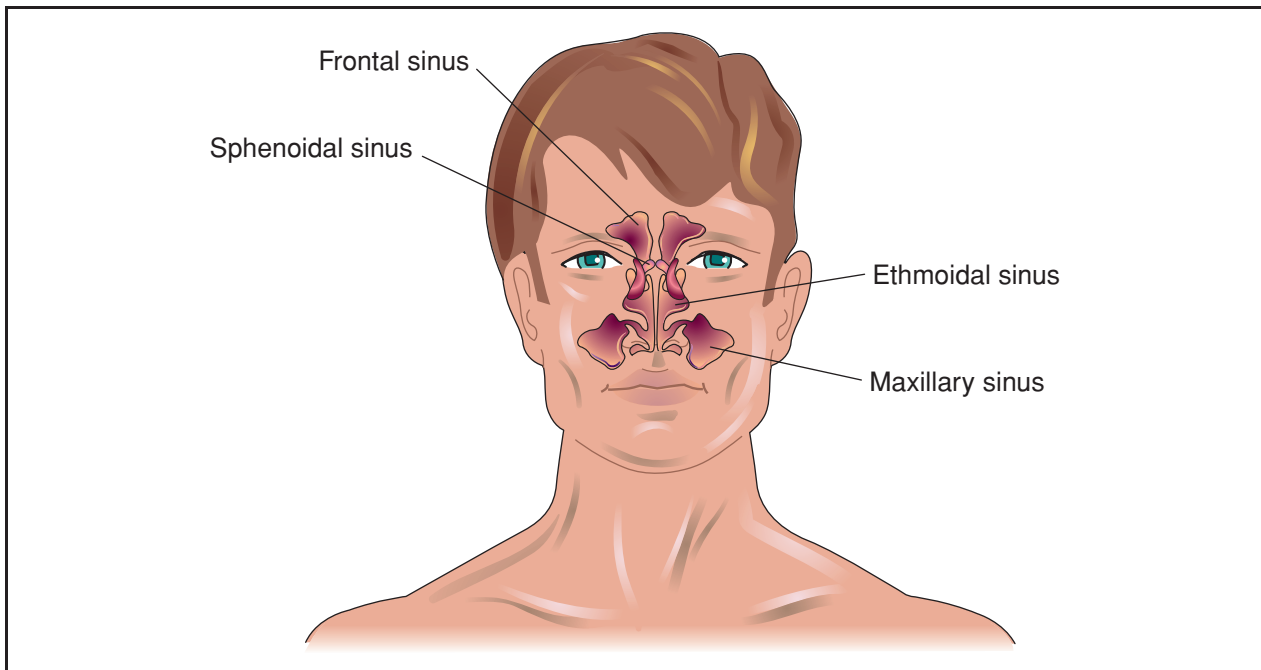
Acute sinusitis usually follows some type of upper respiratory tract infection or cold. Instead of ending, the cold seems to linger on, with constant or even worsening congestion. Drainage from the nose often changes from a clear color to a thicker, yellowish-green. There may be **fever**. **Headache** and pain over the affected sinuses may occur, as well as a feeling of pressure which may worsen when the patient bends over. There may be pain in the jaw or teeth. Some children, in particular, get upset stomachs from the infected drainage going down the back of their throats, and being swallowed into their stomachs. Some patients develop a **cough**.

Chronic sinusitis occurs when the problem has existed for at least three months. There is rarely a fever with chronic sinusitis. Sinus pain and pressure is frequent, as is nasal congestion. Because of the nature of the swelling in the sinuses, they may not be able to drain out the nose. Drainage, therefore, drips constantly down the back of the throat, resulting in a continuously sore throat and **bad breath**.

Diagnosis

Diagnosis is sometimes tricky, because the symptoms so often resemble those of an uncomplicated cold. However, sinusitis should be strongly suspected when a cold lingers beyond about a week's time.

Medical practitioners have differing levels of trust of certain basic examinations commonly conducted in the office. For example, tapping over the sinuses may



Sinusitis is the inflammation of the sinuses caused by a bacterial infection. Sometimes diagnosis may be problematic because the symptoms often mimic those of the common cold. Sinusitis is usually treated with antibiotics. (Illustration by Electronic Illustrators Group.)

cause pain in patients with sinusitis, but it may not. A procedure called “sinus transillumination” may, or may not, also be helpful. Using a flashlight pressed up against the skin of the cheek, the practitioner will look in the patient’s open mouth. When the sinuses are full of air (under normal conditions), the light will project through the sinus, and will be visible on the roof of the mouth as a lit-up, reddened area. When the sinuses are full of mucus, the light will be stopped. While this simple test can be helpful, it is certainly not a perfect way to diagnose or rule out the diagnosis of sinusitis.

X-ray pictures and CT scans of the sinuses are helpful for both acute and chronic sinusitis. People with chronic sinusitis should also be checked for allergies; and they may need a procedure with a scope to see if any kind of anatomic obstruction is causing their illness. For example, the septum (the cartilage which separates the two nasal cavities from each other) may be slightly displaced, called a **deviated septum**. This can result in chronic obstruction, setting the person up for the development of an infection.

Treatment

Antibiotic medications are used to treat acute sinusitis. Suitable **antibiotics** include sulfa drugs, amoxicillin, and a variety of **cephalosporins**. These medications are usually given for about two weeks, but may be given

for even longer periods of time. **Decongestants**, or the short-term use of decongestant nose sprays, can be useful. **Acetaminophen** and ibuprofen can decrease the pain and headache associated with sinusitis. Also, running a humidifier can prevent mucus within the nasal passages from drying out uncomfortably, and can help soothe any accompanying **sore throat** or cough.

Chronic sinusitis is often treated initially with antibiotics. Steroid nasal sprays may be used to decrease swelling in the nasal passages. If an anatomic reason is found for chronic sinusitis, it may need to be corrected with surgery. If a surgical procedure is necessary, samples are usually taken at the same time to allow identification of any organisms present which may be causing infection.

Fungal sinusitis will require surgery to clean out the sinuses. Then, a relatively long course of a very strong antifungal medication called amphotericin B is given through a needle in the vein (intravenously).

Alternative treatment

Chronic sinusitis is often associated with **food allergies**. An elimination/challenge diet is recommended to identify and eliminate allergenic foods. Irrigating the sinuses with a salt water solution is often recommended for sinusitis and allergies, in

KEY TERMS

Cilia—Tiny, hair-like projections from a cell. Within the respiratory tract, the cilia act to move mucus along, in an effort to continually flush out and clean the respiratory tract.

Sinus—An air-filled cavity.

order to clear the nasal passages of mucus. Another solution for nasal lavage (washing) utilizes powdered goldenseal (*Hydrastis canadensis*). Other herbal treatments, taken internally, include a mixture made of eyebright (*Euphrasia officinalis*), goldenseal, yarrow (*Achillea millefolium*), horseradish, and ephedra (*Ephedra sinica*), or, when infection is present, a mixture made of echinacea (*Echinacea* spp.), wild indigo, and poke root (*Phytolacca decandra-Americana*).

Homeopathic practitioners find a number of remedies useful for treating sinusitis. Among those they recommend are: *Arsenicum album*, *Kalium bichromium*, *Nux vomica*, *Mercurius iodatus*, and *Silica*.

Acupuncture has been used to treat sinusitis, as have a variety of dietary supplements, including **vitamins** A, C, and E, and the mineral zinc. Contrast **hydrotherapy** (hot and cold compresses, alternating 3 minutes hot, 30 seconds cold, repeated 3 times always ending with cold) applied directly over the sinuses can relieve pressure and enhance healing. A direct inhalation of essential oils (2 drops of oil to 2 cups of water) using thyme, rosemary, and lavender can help open the sinuses and kill bacteria that cause infection.

Prognosis

Prognosis for sinus infections is usually excellent, although some individuals may find that they are particularly prone to contracting such infections after a cold. Fungal sinusitis, however, has a relatively high **death** rate.

Prevention

Prevention involves the usual standards of good hygiene to cut down on the number of colds an individual catches. Avoiding exposure to cigarette smoke, identifying and treating allergies, and avoiding deep dives in swimming pools may help prevent sinus infections. During the winter, it is a good idea to use a humidifier. Dry nasal passages may crack, allowing bacteria to enter. When allergies are diagnosed, a number of nasal sprays are available to try to prevent

inflammation within the nasal passageways, thus allowing the normal flow of mucus.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.

Rosalyn Carson-DeWitt, MD

Situs inversus

Definition

Situs inversus is a condition in which the organs of the chest and abdomen are arranged in a perfect mirror image reversal of the normal positioning.

Description

Normal human development results in an asymmetrical arrangement of the organs within the chest and abdomen. Typically, the heart lies on the left side of the body (*levocardia*), the liver and spleen lie on the right, and the lung on the left has two lobes while the lung on the right has three lobes. This normal arrangement is known as *situs solitus*.

However, in about 1 in 8,500 people, the organs of the chest and abdomen are arranged in the exact opposite position: the heart is on the right (*dextrocardia*), as is the two-lobed lung, and the liver, spleen, and three-lobed lung are on the left. Yet because this arrangement, called *situs inversus*, is a perfect mirror image, the relationship between the organs is not changed, so functional problems rarely occur.

Causes and symptoms

Early in the normal development of an embryo, the tube-like structure that becomes the heart forms a loop toward the left, identifying the left/right axis along which the other organs should be positioned. Although the mechanism that causes the heart loop to go left is not fully understood, at least one gene has been identified to have a role in this process. However, it is thought that many factors may be involved in causing situs inversus. Rarely, situs inversus can run in families, but most often it is an isolated and accidental event occurring in an individual for the first time in the family.

Most people with situs inversus have no medical symptoms or complications resulting from the condition. Although only 3-5% of people with situs inversus have any type of functional heart defect, this is higher than the rate of heart defects in the general population, which is less than 1%.

It is estimated that about 25% of people with situs inversus have an underlying condition called primary ciliary dyskinesia (PCD). PCD, also known as Kartagener's syndrome, is characterized as situs inversus, chronic sinus infections, increased mucous secretions from the lungs, and increased susceptibility to respiratory infections. PCD is caused by a defect in the cilia that impairs their normal movements.

Diagnosis

Situs inversus should be detected by a thorough **physical examination**. It is often picked up when a physician, using a stethoscope, hears otherwise normal heart sounds on the right side of the body instead of the left. To confirm the suspected diagnosis of situs inversus, imaging studies such as MRI, CT, or ultrasound may be ordered, and a referral may be made to a cardiologist or internist for completeness. Imaging studies will also rule out the possibility of random arrangement of the organs, or heterotaxy, which has a much higher risk for serious medical complications.

Treatment

There is no treatment for situs inversus. In the unlikely case that a heart defect is present, it should be treated accordingly by a cardiologist.

Individuals who have situs inversus should be sure to inform all physicians involved in their medical care. In addition to preventing unnecessary confusion, this will reduce the risk of missing a crucial diagnosis that presents with location-specific symptoms (such as **appendicitis**).

Alternative treatment

Not applicable.

Prognosis

The prognosis for an individual with situs inversus is good, and in the absence of a heart defect or other underlying diagnosis, life expectancy is normal.

Prevention

There is no known method of preventing situs inversus.

KEY TERMS

Cilia—Tiny hairlike projections on certain cells within the body; cilia produce lashing or whipping movements to direct or cause motion of substances or fluids within the body.

Gene—A single unit of genetic information, providing the body with instruction for a specific biological task.

MRI—An imaging study that uses magnetic forces to produce an image of the body's internal structures.

CT—A special technique that uses a computer to create a cross-sectional image of the body from a series of x rays.

Ultrasound—An imaging study that uses high-frequency sound waves to form a visual image of the body's internal structures.

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American Heart Association. National Center. 7272 Greenville Avenue, Dallas, TX 75231-4596. (214) 373-6300. (800) 242-8721. inquire@heart.org.

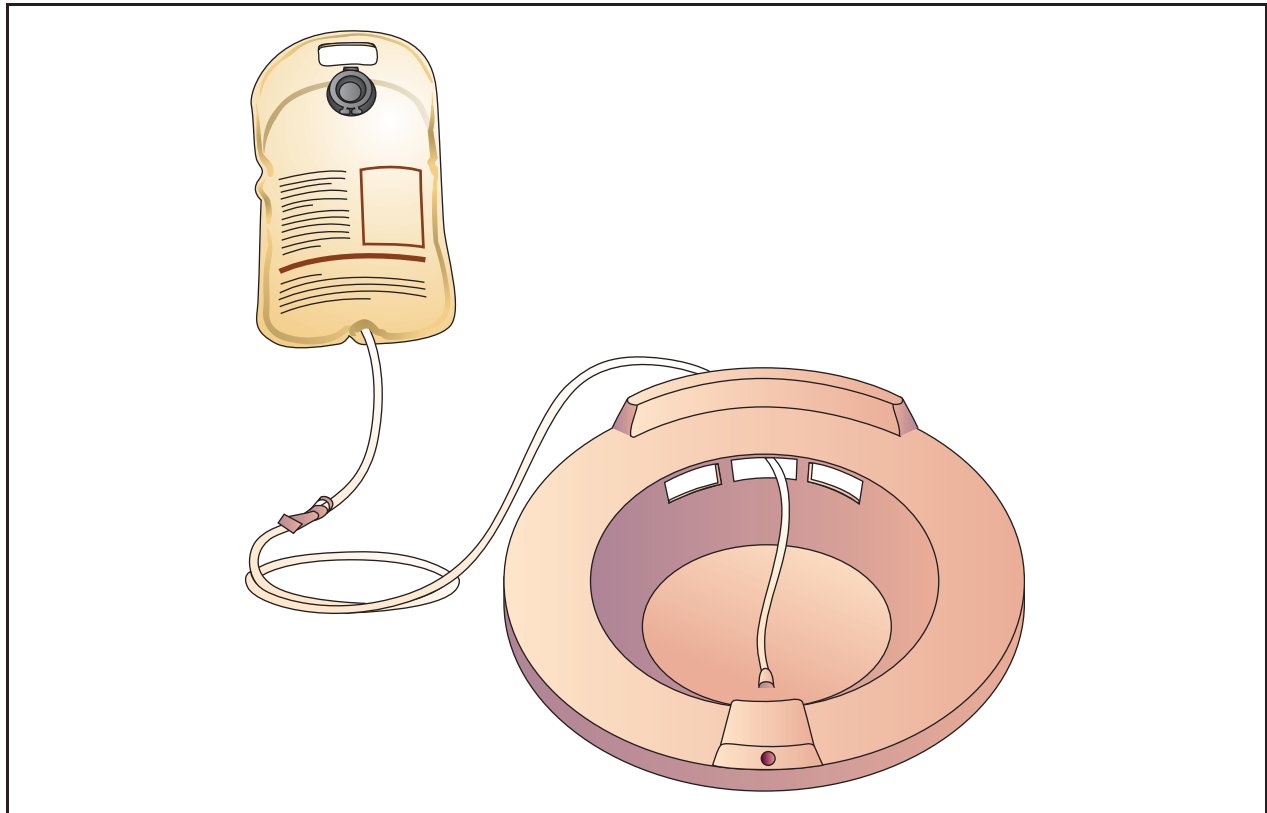
NIH/National Heart, Lung and Blood Institute Information Center. P.O. Box 30105. Bethesda, MD 20824-0105. (301) 592-8573.

Stefanie B. N. Dugan, M.S.

Sitz bath

Definition

A sitz bath (also called a hip bath) is a type of bath in which only the hips and buttocks are soaked in water or saline solution. Its name comes from the German verb "sitzen," meaning "to sit."



Equipment used for sitz baths. A sitz bath, in which only the hips and buttocks are soaked in water or saline solution, is used for patients who have had surgery in the rectal area or to ease discomfort from bladder, prostate, or vaginal infections. (Illustration by Electronic Illustrators Group.)

Purpose

A sitz bath is used for patients who have had surgery in the area of the rectum, or to ease the **pain** of **hemorrhoids**, uterine cramps, prostate infections, painful ovaries, and/or testicles. It is also used to ease discomfort from infections of the bladder, prostate, or vagina. Inflammatory bowel diseases are also treated with sitz baths.

Precautions

Some patients may become dizzy when standing up after sitting in hot water; it is best to have someone else present when doing a contrast sitz bath.

Description

The sitz bath is a European tradition in which only the pelvis and abdominal area are placed in water, with the upper body, arms, legs, and feet out of the water. The water can be warm or cool and one or two tubs may be used.

Warm sitz baths are one of the easiest and most effective ways to ease the pain of hemorrhoids. A warm bath is also effective in lessening the discomfort associated with **genital herpes**, uterine cramps, and other painful conditions in the pelvic area.

For prostate pain, patients should take two hot sitz baths a day, for about 15 minutes each.

To ease discomfort from a vaginal yeast infection, women should take a warm saline sitz bath. To prepare, fill the tub to hip height with warm water and add 1/2 c of salt (enough to make the water taste salty) and 1/2 vinegar. Sit in the bath for 20 minutes (or until the water gets cool). The vinegar will help bring the vaginal pH back to 4.5 (pH is a measurement of how acid or alkaline a fluid is).

A brief, cool sitz bath helps ease inflammation, **constipation**, and vaginal discharge. It can be used to tone the muscles in cases of bladder or bowel incontinence.

Other conditions respond to a “contrast bath” of both hot and cold. For this a patient should have a tub

KEY TERMS

pH—A standard laboratory test that measures how acidic or alkaline a solution is.

Saline solution—Another word for salt water.

of hot water (about 110°F/43°C) and one tub of ice water. The patient should sit in the hot water for three to four minutes and in the cold for 30–60 seconds. This is repeated three to five times, always ending with the cold water.

If two tubs are not handy, the patient may sit in a hot bath (up to the navel). Then the patient stands up in the water and pulls a cold towel between the legs and over the pelvis in front and back. The cold towel is held in place for up to 60 seconds. Then the patient should sit back into the hot bath, and repeat the process 3–5 times, ending with the cold towel.

Preparation

The bath should be filled with 3–4 in (8–10cm) of water. For most conditions, nothing else should be added (no bubble bath or oil).

Aftercare

The area should be carefully patted dry and, if necessary, clean dressings should be applied.

Risks

Sitz baths pose almost no risk. On rare occasions, patients can feel dizzy or experience rapid heart beat because of blood vessel dilation.

Normal results

Swelling goes down; discomfort is eased; healing is promoted.

Carol A. Turkington

Sjögren's syndrome

Definition

Sjögren's syndrome (SS) is a disorder in which the mouth and eyes become extremely dry. Sjögren's

syndrome is often associated with other autoimmune disorders. It is named for Henrik Sjögren, a Swedish ophthalmologist.

Description

Like other **autoimmune disorders**, Sjögren's syndrome occurs when the body's immune system mistakenly begins treating parts of the body as foreign invaders. While the immune cells should attack and kill invaders like bacteria, viruses, and fungi, these cells should not attack the body itself. In autoimmune disorders, however, cells called antibodies see tissues of the body as foreign, and help to start a chain of events that results in damage and destruction of those tissues.

There are three types of Sjögren's syndrome. Primary Sjögren's syndrome occurs by itself, with no other associated disorders. Secondary Sjögren's syndrome occurs along with other autoimmune disorders, like systemic lupus erythematosus, **rheumatoid arthritis**, **scleroderma**, **vasculitis**, or **polymyositis**. When the disorder is limited to involvement of the eyes, with no other organ or tissue involvement evident, it is called sicca complex.

Women are about nine times more likely to suffer from Sjögren's syndrome than are men. SS affects all age groups, although most patients are diagnosed when they are between 40 and 55 years old. Sjögren's syndrome is commonly associated with other autoimmune disorders. In fact, 30% of patients with certain autoimmune disorders will also have Sjögren's syndrome.

SS is found in all races and ethnic groups. It is thought to affect between 0.1% and 3% of the population in the United States; this range reflects the lack of a uniform set of diagnostic criteria. According to the American College of Rheumatology, between 1 million and 4 million Americans have Sjögren's syndrome.

Causes and symptoms

The cause of Sjögren's syndrome has not been clearly defined, but several causes are suspected. The syndrome sometimes runs in families. Other potential causes include hormonal factors (since there are more women than men with the disease) and viral factors. The viral theory suggests that the immune system is activated in response to a viral invader, but then fails to turn itself off. Some other immune malfunction then causes the overly active immune system to begin attacking the body's own tissues. In 2004 a group of Greek researchers presented evidence that a coxsackievirus may be the disease organism that triggers SS.

The main problem in Sjögren's syndrome is dryness. The salivary glands are often attacked and slowly destroyed, leaving the mouth extremely dry and sticky. Swallowing and talking become difficult. Normally, the saliva washes the teeth clean. Saliva cannot perform this function in Sjögren's syndrome, so the teeth develop many cavities and decay quickly. The parotid glands produce the majority of the mouth's saliva. They are located lying over the jaw bones behind the area of the cheeks and in front of the ears, and may become significantly enlarged in Sjögren's syndrome.

The eyes also become extremely dry as the tear glands (called glands of lacrimation) are slowly destroyed. Eye symptoms include **itching**, burning, redness, increased sensitivity to light, and thick secretions gathering at the eye corners closest to the nose. The cornea may have small irritated pits in its surface (ulcerations).

Destruction of glands in other areas of the body may cause a variety of symptoms. In the nose, dryness may result in nosebleeds. In the rest of the respiratory tract, the rates of ear infection, hoarseness, **bronchitis**, and **pneumonia** may increase. Vaginal dryness can be quite uncomfortable. Rarely, the pancreas may slow production of enzymes important for digestion. The kidney may malfunction. About 33% of all patients with Sjögren's syndrome have other symptoms unrelated to gland destruction. These symptoms include fatigue, decreased energy, fevers, muscle aches and pains, and joint **pain**.

Many patients with SS also develop a variety of skin problems that include dry patches, vasculitis, and cutaneous B-cell lymphoma. These and other dermatologic disorders are more common in SS than was previously thought.

Patients who also have other autoimmune diseases will suffer from the symptoms specific to those conditions.

In addition to physical symptoms, patients with SS appear to be at increased risk for depression and other **mood disorders**.

Diagnosis

Diagnosis of Sjögren's syndrome is based on the patient having at least three consecutive months of bothersome eye and/or mouth dryness. A variety of tests can then be done to determine the quantity of tears produced, the quantity of saliva produced, and the presence or absence of antibodies that could be involved in the destruction of glands.

KEY TERMS

Autoimmune disorder—A disorder in which the body's immune cells mistake the body's own tissues as foreign invaders; the immune cells then work to destroy tissues in the body.

Cornea—A transparent structure of the eye over the iris and pupil; light must pass through the cornea to make vision possible.

Coxsackievirus—Any of a group of enteroviruses that produce a disease in humans characterized by fever and rash. Coxsackieviruses are named for the town in upstate New York where they were first identified.

Immune system—The complex network of organs and blood cells that protect the body from foreign invaders, like bacteria, viruses, and fungi.

Treatment

There is no cure for Sjögren's syndrome. Instead, treatment usually attempts to reduce the discomfort and complications associated with dryness of the eyes and mouth (and other areas). Artificial tears are available, and may need to be used up to every 30 minutes. By using these types of products, the patient is more comfortable and avoids the complications associated with eyes that are overly dry. **Dry mouth** is treated by sipping fluids slowly but constantly throughout the day. Sugarless chewing gum can also be helpful. An artificial saliva is available for use as a mouthwash. Patients may also be given such drugs as pilocarpine (Salagen) or cevimeline (Evoxac) to increase saliva and tear secretions. Careful dental hygiene is important in order to avoid **tooth decay**, and it is wise for patients to decrease sugar intake. Vaginal dryness can be treated with certain gel preparations. Steroid medications may be required when other symptoms of autoimmune disorders complicate Sjögren's syndrome. However, these medications should be avoided when possible because they may make the cornea thin and even more susceptible to injury.

Prognosis

The prognosis for patients with primary Sjögren's syndrome is particularly good; these patients have a normal life expectancy. Although the condition is quite annoying, serious complications rarely occur. The prognosis for patients with secondary Sjögren's syndrome varies since it depends on the prognosis for the accompanying autoimmune disorder.

Prevention

Since the cause of Sjögren's syndrome is unknown as of 2004, there are no known ways to prevent this syndrome.

Resources

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ORGANIZATIONS

American College of Rheumatology. 1800 Century Place, Suite 250, Atlanta, GA 30345-4300. (404) 633-3777. Fax: (404) 633-1870. <<http://www.rheumatology.org>>.

Sjögren's Syndrome Foundation, Inc. 8120 Woodmont Avenue, Bethesda, MD 20814. (800) 475-6473. Fax: (301) 718-0322. <<http://www.sjogrens.org>>.

OTHER

American College of Rheumatology Fact Sheet. "Sjögren's Syndrome." <http://www.rheumatology.org/public/factsheets/sjogrens_new.asp?aud=pat>.

Rosalyn Carson-DeWitt, MD
Rebecca J. Frey, PhD

Skeletal traction see **Traction;**
Immobilization

Skin abrasion see **Skin resurfacing**

Skin allergy test see **Allergy tests**

Skin biopsy

Definition

A skin biopsy is a procedure in which a small piece of living skin is removed from the body for examination, usually under a microscope, to establish a precise diagnosis. Skin biopsies are usually brief, straightforward procedures performed by a skin specialist (dermatologist) or family physician.

Purpose

The word *biopsy* is taken from Greek words that mean "to view life." The term describes what a specialist in identifying diseases (pathologist) does with tissue obtained from a skin biopsy. The pathologist *visually* examines the tissue under a microscope.

A skin biopsy is used to make a diagnosis of many skin disorders. Information from the biopsy also helps the doctor choose the best treatment for the patient.

Doctors perform skin biopsies to:

- make a diagnosis
- confirm a diagnosis made from the patient's medical history and a **physical examination**
- check whether a treatment prescribed for a previously diagnosed condition is working
- check the edges of tissue removed with a tumor to make certain it contains all the diseased tissue

Skin biopsies also can serve a therapeutic purpose. Many skin abnormalities (lesions) can be removed completely during the biopsy procedure.

Precautions

A patient taking **aspirin** or another blood thinner (anticoagulant) may be asked to stop taking them a week or more before the skin biopsy. This adjustment in medication will prevent excessive bleeding during the procedure and allow for normal blood clotting.

Some patients are allergic to lidocaine, the numbing agent most frequently used during a skin biopsy. The doctor can usually substitute another anesthetic agent.

Description

The first part of the skin biopsy test is obtaining a sample of tissue that best represents the lesion being evaluated. Many biopsy techniques are available. The choice of technique and precise location from which to take the biopsy material are determined by factors such as the type and shape of the lesion. Biopsies can be classified as excisional or incisional. In excisional biopsy, the lesion is completely removed; in incisional biopsy, a portion of the lesion is removed.

The most common biopsy techniques are:

- **Shave biopsy.** A scalpel or razor blade is used to shave off a thin layer of the lesion parallel to the skin.
- **Punch biopsy.** A small cylindrical punch is screwed into the lesion through the full thickness of the skin and a plug of tissue is removed. A stitch or two may be needed to close the wound.
- **Scalpel biopsy.** A scalpel is used to make a standard surgical incision or excision to remove tissue. This technique is most often used for large or deep lesions. The wound is closed with stitches.
- **Scissors biopsy.** Scissors are used to snip off surface (superficial) skin growths and lesions that grow from a stem or column of tissue. Such growths are sometimes seen on the eyelids or neck.

After the biopsy tissue is removed, bleeding may be controlled by applying pressure or by burning with electricity or chemicals. **Antibiotics** often are applied to the wound to prevent infection. Stitches may be placed in the wound, or the wound may be bandaged and allowed to heal on its own.

The second part of the skin biopsy test is handling and examining the tissue sample. Drying and structural damage to the tissue sample must be prevented, so it should be placed immediately in an appropriate preservative, such as formaldehyde.

The pathologist can use a variety of laboratory techniques to process the biopsy tissue. Tissue stains and several different kinds of microscopes are used. Because there are many skin disorders (broadly called dermatosis and **dermatitis**), the pathologist has extensive training in their accurate identification. Cases of melanoma, the most malignant kind of skin **cancer**, have almost tripled in the past 30 years. Because melanoma grows very rapidly in the skin, quick and accurate diagnosis is important.

Preparation

The area of the biopsy is cleansed thoroughly with alcohol or a disinfectant containing iodine. Sterile

KEY TERMS

Benign—Noncancerous.

Dermatitis—A skin disorder that causes inflammation, that is, redness, swelling, heat, and pain.

Dermatologist—A doctor who specializes in skin care and treatment.

Dermatosis—A noninflammatory skin disorder.

Lesion—An area of abnormal or injured skin.

Malignant—Cancerous.

Pathologist—A person who specializes in studying diseases. In particular, this person examines the structural and functional changes in the tissues and organs of the body that are caused by disease or that cause disease themselves.

cloths (drapes) may be positioned, and a local anesthetic, usually lidocaine, is injected into the skin near the lesion. Sometimes the anesthetic contains epinephrine, a drug that helps reduce bleeding during the biopsy. Sterile gloves and surgical instruments are always used to reduce the risk of infection.

Aftercare

If stitches have been placed, they should be kept clean and dry until removed. Stitches are usually removed five to 10 days after the biopsy. Sometimes the patient is instructed to put protective ointment on the stitches before showering. **Wounds** that have not been stitched should be cleaned with soap and water daily until they heal. Adhesive strips should be left in place for two to three weeks. **Pain** medications usually are not necessary.

Risks

Infection and bleeding occur rarely after skin biopsy. If the skin biopsy may leave a scar, the patient usually is asked to give informed consent before the test.

Normal results

The biopsy reveals normal skin layers.

Abnormal results

The biopsy reveals a noncancerous (benign) or cancerous (malignant) lesion. Benign lesions may require treatment.

Resources

ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050.
<<http://www.aad.org>>.

Collette L. Placek

Skin cancer see **Melanoma**

Skin cancer, non-melanoma

Definition

Non-melanoma skin **cancer** is a malignant growth of the external surface or epithelial layer of the skin.

Description

Skin cancer is the growth of abnormal cells capable of invading and destroying other associated skin cells. Skin cancer is often subdivided into either melanoma or non-melanoma. Melanoma is a dark-pigmented, usually malignant tumor arising from a skin cell capable of making the pigment melanin (a melanocyte). Non-melanoma skin cancer most often originates from the external skin surface as a squamous cell carcinoma or a basal cell carcinoma.

The cells of a cancerous growth originate from a single cell that reproduces uncontrollably, resulting in the formation of a tumor. Exposure to sunlight is documented as the main cause of almost 800,000 cases of non-melanoma skin cancer diagnosed each year in the United States. The incidence increases for those living where direct sunshine is plentiful, such as near the equator.

Basal cell carcinoma affects the skin's basal layer and has the potential to grow progressively larger in size, although it rarely spreads to distant areas (metastasizes). Basal cell carcinoma accounts for 80% of skin cancers (excluding melanoma), whereas squamous cell cancer makes up about 20%. Squamous cell carcinoma is a malignant growth of the external surface of the skin. Squamous cell cancers metastasize at a rate of 2–6%, with up to 10% of lesions affecting the ear and lip.

Causes and symptoms

Cumulative sun exposure is considered a significant risk factor for non-melanoma skin cancer. There

is evidence suggesting that early, intense exposure causing blistering **sunburn** in childhood may also play an important role in the cause of non-melanoma skin cancer. Basal cell carcinoma most frequently affects the skin of the face, with the next most common sites being the ears, the backs of the hands, the shoulders, and the arms. It is prevalent in both sexes and most commonly occurs in people over 40.

About 1–2% of all skin cancers develop within burn **scars**; squamous cell carcinomas account for about 95% of these cancers, with 3% being basal cell carcinomas and the remainder malignant melanomas.

Basal cell carcinomas usually appear as small **skin lesions** that persist for at least three weeks. This form of non-melanomatous skin cancer looks flat and waxy, with the edges of the lesion translucent and rounded. The edges also contain small fresh blood vessels. An ulcer in the center of the lesion gives it a dimpled appearance. Basal cell carcinoma lesions vary from 4–6mm in size, but can slowly grow larger if untreated.

Squamous cell carcinoma also involves skin exposed to the sun, such as the face, ears, hands, or arms. This form of non-melanoma is also most common among people over 40. Squamous cell carcinoma presents itself as a small, scaling, raised bump on the skin with a crusting ulcer in the center, but without **pain** and **itching**.

Basal cell and squamous cell carcinomas can grow more easily when people have a suppressed immune system because they are taking immunosuppressive drugs or are exposed to radiation. Some people must take immunosuppressive drugs to prevent the rejection of a transplanted organ or because they have a disease in which the immune system attacks the body's own tissues (autoimmune illnesses); others may need **radiation therapy** to treat another form of cancer. Because of this, everyone taking these immunosuppressive drugs or receiving radiation treatments should undergo complete skin examination at regular intervals. If proper treatment is delayed and the tumor continues to grow, the tumor cells can spread (metastasize) to muscle, bone, nerves, and possibly the brain.

Diagnosis

To diagnose skin cancer, doctors must carefully examine the lesion and ask the patient about how long it has been there, whether it itches or bleeds, and other questions about the patient's medical history. If skin cancer cannot be ruled out, a sample of the tissue is removed and examined under a microscope (a biopsy). A definitive diagnosis of squamous or basal cell cancer

can only be made with microscopic examination of the tumor cells. Once skin cancer has been diagnosed, the stage of the disease's development is determined. The information from the biopsy and staging allows the physician and patient to plan for treatment and possible surgical intervention.

Treatment

A variety of treatment options are available for those diagnosed with non-melanoma skin cancer. Some carcinomas can be removed by cryosurgery, the process of freezing with liquid nitrogen. Uncomplicated and previously untreated basal cell carcinoma of the trunk and arms is often treated with curettage and electrodesiccation, which is the scraping of the lesion and the destruction of any remaining malignant cells with an electrical current. Removal of a lesion layer-by-layer down to normal margins (Mohs' surgery) is an effective treatment for both basal and squamous cell carcinoma. Radiation therapy is best reserved for older, debilitated patients or when the tumor is considered inoperable. Laser therapy is sometimes useful in specific cases; however, this form of treatment is not widely used to treat skin cancer.

A newer type of radiation treatment for non-melanoma skin cancers consists of low-energy x rays delivered through the tip of a portable needle-like probe at a high dosage rate. This method allows the radiologist to treat the cancer while sparing the surrounding normal skin. The device was effective in treating 83% of skin lesions in a group of patients diagnosed with non-melanomatous skin cancers during a Phase I trial for the Food and Drug Administration (FDA). There were no cases of damage to surrounding tissues.

Some topical (applied onto the skin) creams and ointments may be used to treat certain types of non-melanoma skin cancers. For example, imiquimod cream, a topical immune stimulator, has been shown effective in treating superficial basal cell carcinoma. The company who marketed the drug was seeking FDA approval.

Alternative treatment

Alternative medicine aims to prevent rather than treat skin cancer. **Vitamins** have been shown to prevent sunburn and, possibly, skin cancer. Some dermatologists have suggested that taking vitamins E and C may help prevent sunburn. In one particular study, men and women took these vitamins for eight days prior to being exposed to ultraviolet light. The researchers found that those who consumed vitamins



A close up image of a precancerous mole that could develop into a melanoma. Melanomas arise from pigment-producing cells, while non-melanoma skin cancer arises from squamous cells or basal cells. (Custom Medical Stock Photo. Reproduced by permission.)

required about 20% more ultraviolet light to induce sunburn than did people who didn't take vitamins. This is the first study that indicates the oral use of vitamins E and C increases resistance to sunburn. These antioxidants are thought to reduce the risk of skin cancer, and are expected to provide protection from the sun even in lower doses. Other antioxidant nutrients, including beta carotene, selenium, zinc, and the bioflavonoid quercetin have been suggested as possibly preventing skin cancer. However, a 2003 study reported that selenium was not effective in preventing basal cell carcinoma and may even increase risk of squamous cell carcinoma and total non-melanoma skin cancer. Antioxidant herbs such as bilberry (*Vaccinium myrtillus*), hawthorn (*Crataegus laevigata*), tumeric (*Curcuma longa*), and ginkgo (*Ginkgo biloba*) also have been presented as helpful in preventing skin cancers.

KEY TERMS

Autoimmune—Pertaining to an immune response by the body against one of its own tissues or types of cells.

Curettage—The removal of tissue or growths by scraping with a curette.

Dermatologist—A physician specializing in the branch of medicine concerned with skin.

Electrodesiccation—To make dry, dull, or lifeless with the use of electrical current.

Lesion—A patch of skin that has been infected or diseased.

Topical—Referring to a medication or other preparation applied to the skin or the outside of the body.

A team of researchers at Duke University reported in 2003 that topical application of a combination of 15% vitamin C and 1% vitamin E over a four-day period offered significant protection against sunburn. The researchers suggest that this combination may protect skin against **aging** caused by sunlight as well.

Another antioxidant that appears to counter the effects of severe sun exposure is superoxide dismutase, or SOD. SOD must be given in injectable form, however, because it is destroyed in the digestive tract.

As of 2003, researchers were also looking at botanical compounds that could be added to skin care products applied externally to lower the risk of skin cancer. Several botanical compounds had been tested on animals and found to be effective in preventing skin cancer, but further research needs to be done in human subjects.

Prognosis

Both squamous and basal cell carcinoma are curable with appropriate treatment, although basal cell carcinomas have about a 5% rate of recurrence. Early detection remains critical for a positive prognosis. Although it is rare for basal cell carcinomas to metastasize, their metastases can rapidly lead to **death** if they invade the eyes, ears, mouth, or the membranes covering the brain.

Prevention

Avoiding exposure to the sun reduces the incidence of non-melanoma skin cancer. Sunscreen with

a sun-protective factor of 15 or higher is helpful in prevention, along with a hat and clothing to shield the skin from sun damage. People should examine their skin monthly for unusual lesions, especially if previous skin cancers have been experienced.

Advances in photographic technique have now made it easier to track the development of **moles** with the help of whole-body photographs. A growing number of hospitals are offering these photographs as part of outpatient mole-monitoring services.

Resources

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ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>>.

Centers for Disease Control and Prevention (CDC). Cancer Prevention and Control Program. 4770 Buford Highway, NE, MS K64, Atlanta, GA 30341. (888) 842-6355. <<http://www.cdc.gov/cancer/comments.htm>>.

National Cancer Institute (NCI). NCI Public Inquiries Office, Suite 3036A, 6116 Executive Boulevard, MSC8332, Bethesda, MD 20892-8322. (800) 4-CANCER or (800) 332-8615 (TTY). <<http://www.nci.nih.gov>>.

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Skin culture

Definition

A skin culture is a test that is done to identify the microorganism (bacteria, fungus, or virus) causing a skin infection and to determine the antibiotic or other treatment that will effectively treat the infection.

Purpose

Microorganisms can infect healthy skin, but more often they infect skin already damaged by an injury or abrasion. Skin infections are contagious and, if left untreated, can lead to serious complications. A culture enables a physician to diagnose and treat a skin infection.

Description

Several groups of microorganisms cause skin infections: bacteria, fungi (molds and yeast), and viruses. Based on the appearance of the infection, the physician determines what group of microorganisms is likely causing the infection, then he or she collects a

specimen for one or more types of cultures. A sample of material—such as skin cells, pus, or fluid—is taken from the infection site, placed in a sterile container, and sent to the laboratory. In the laboratory, each type of culture is handled differently.

Bacterial infections are the most common. Bacteria cause lesions, ulcers, **cellulitis**, and **boils**. Pyoderma are pus-containing skin infections, such as **impetigo**, caused by *Staphylococcus* or group A *Streptococcus* bacteria. To culture bacteria, a portion of material from the infection site is spread over the surface of a culture plate and placed in an incubator at body temperature for one to two days. Bacteria in the skin sample multiply and appear on the plates as visible colonies. They are identified by noting the appearance of their colonies, and by performing biochemical tests and a Gram's stain.

The Gram's stain is done by smearing part of a colony onto a microscope slide. After it dries, the slide is colored with purple and red stains, then examined under a microscope. The color of stain picked up and retained by the bacteria (purple or red), their shape (such as round or rectangle), and their size provide valuable clues as to their identity.

A sensitivity test, also called antibiotic susceptibility test, is also done. The bacteria are tested against different **antibiotics** to determine which will effectively treat the infection by killing the bacteria.

Fungal cultures are done less frequently. A group of fungi called dermatophytes cause a skin infection called **ringworm**. Yeast causes an infection called thrush. These infections are usually diagnosed using a method other than culture, such as the KOH test. A culture is done only when specific identification of the mold or yeast is necessary. The specimen is spread on a culture plate designed to grow fungi, then incubated. Several different biochemical tests and stains are used to identify molds and yeasts.

Viruses, such as herpes, can also cause skin infections. Specimens for viral cultures are mixed with commercially-prepared animal cells in a test tube. Characteristic changes to the cells caused by the growing virus help identify the virus.

Results for bacterial cultures are usually available in one to three days. Cultures for fungi and viruses may take longer—up to three weeks. Cultures are covered by insurance.

Preparation

After cleaning the infected area with sterile saline and alcohol, the physician collects skin cells, pus, or

KEY TERMS

Pyoderma—A pus-containing skin infection, such as impetigo, caused by *Staphylococcus* or group A *Streptococcus* bacteria.

Sensitivity test—A test that determines which antibiotics will treat an infection by killing the bacteria.

fluid using a needle or swab. If necessary, the physician will open a lesion to collect the specimen. To collect a specimen for a fungal culture, the physician uses a scalpel to scrape skin cells into a sterile container.

Normal results

Many types of microorganisms are normally found on a person's skin. Presence of these microorganisms is noted on a skin culture report as "normal flora."

Abnormal results

A microorganism is considered to be a cause of the infection if it is either the only or predominant microorganism that grew, if it grew in large numbers, or if it is known to produce infection.

Resources

PERIODICALS

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Nancy J. Nordenson

Skin grafting

Definition

Skin grafting is a surgical procedure by which skin or skin substitute is placed over a burn or non-healing wound to permanently replace damaged or missing skin or provide a temporary wound covering.

Purpose

Wounds such as third-degree **burns** must be covered as quickly as possible to prevent infection or loss of fluid. Wounds that are left to heal on their own can

contract, often resulting in serious scarring; if the wound is large enough, the scar can actually prevent movement of limbs. Non-healing wounds, such as diabetic ulcers, venous ulcers, or pressure sores, can be treated with skin grafts to prevent infection and further progression of the wounded area.

Precautions

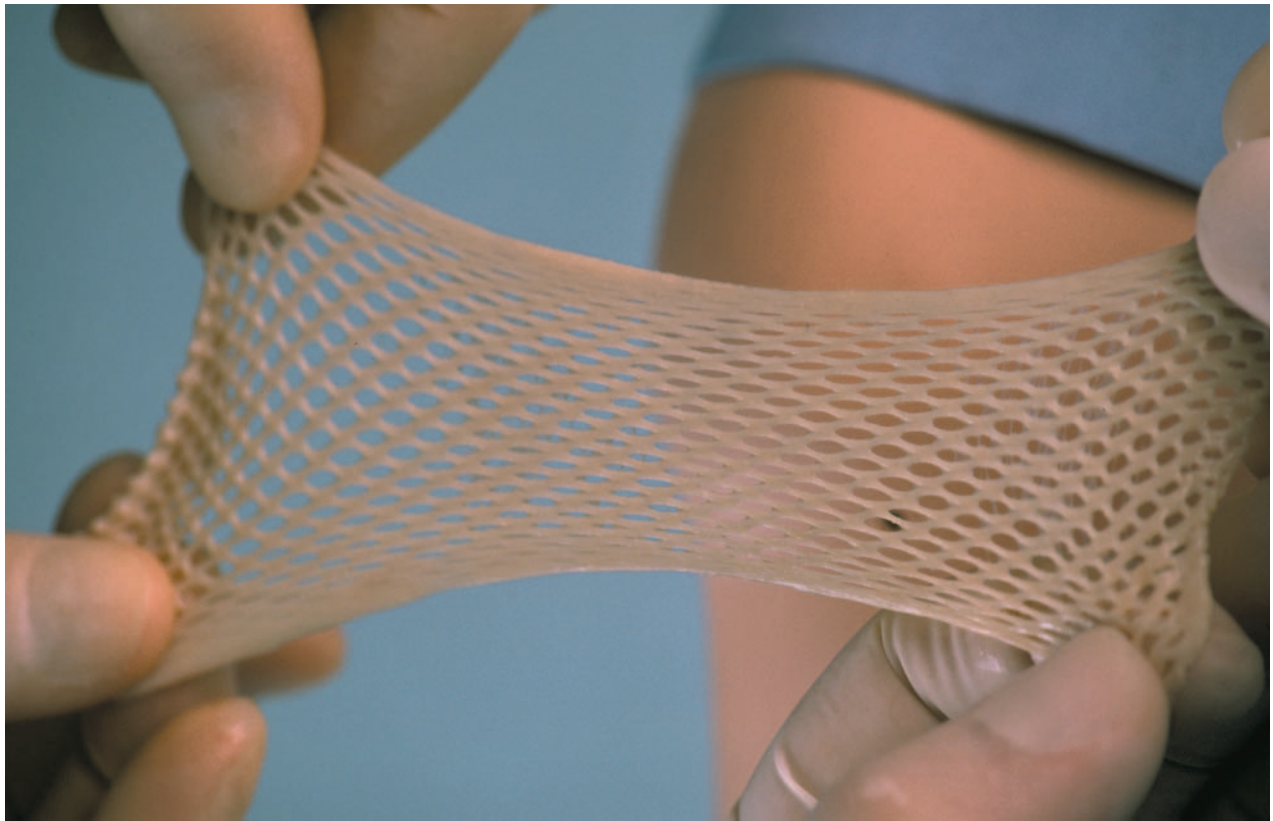
Skin grafting is generally not used for first- or second-degree burns, which generally heal with little or no scarring. Also, the tissue for grafting and the recipient site must be as sterile as possible to prevent later infection that could result in failure of the graft.

Description

The skin is the largest organ of the human body. It consists of two main layers: the epidermis is the outer layer, sitting on and nourished by the thicker dermis. These two layers are approximately 0.04–0.08 in (1–2 mm) thick. The epidermis consists of an outer layer of dead cells, which provides a tough, protective coating, and several layers of rapidly dividing cells called keratinocytes. The dermis contains the blood vessels, nerves, sweat glands, hair follicles, and oil glands. The dermis consists mainly of connective tissue, primarily the protein collagen, which gives the skin its flexibility and provides structural support. Fibroblasts, which make collagen, are the main cell type in the dermis.

Skin protects the body from fluid loss, aids in temperature regulation, and helps prevent disease-causing bacteria or viruses from entering the body. Skin that is damaged extensively by burns or non-healing wounds can compromise the health and well-being of the patient. More than 50,000 people are hospitalized for burn treatment each year in the United States, and 5,500 die. Approximately 4 million people suffer from non-healing wounds, including 1.5 million with venous ulcers and 800,000 with diabetic ulcers, which result in 55,000 amputations per year in the United States.

Skin for grafting can be obtained from another area of the patient's body, called an autograft, if there is enough undamaged skin available, and if the patient is healthy enough to undergo the additional surgery required. Alternatively, skin can be obtained from another person (donor skin from cadavers is frozen, stored, and available for use), called an allograft, or from an animal (usually a pig), called a xenograft. Allografts and xenografts provide only temporary covering—they are rejected by the patient's immune system within seven to 10 days and must be replaced with an autograft.



This skin graft is ready for application. (Photograph by Ted Horowitz, *The Stock Market*. Reproduced by permission.)

A split-thickness skin graft takes mainly the epidermis and a little of the dermis, and usually heals within several days. The wound must not be too deep if a split-thickness graft is going to be successful, since the blood vessels that will nourish the grafted tissue must come from the dermis of the wound itself.

A full-thickness graft involves both layers of the skin. Full-thickness autografts provide better contour, more natural color, and less contraction at the grafted site. The main disadvantage of full-thickness skin grafts is that the wound at the donor site is larger and requires more careful management; often a split-thickness graft must be used to cover the donor site.

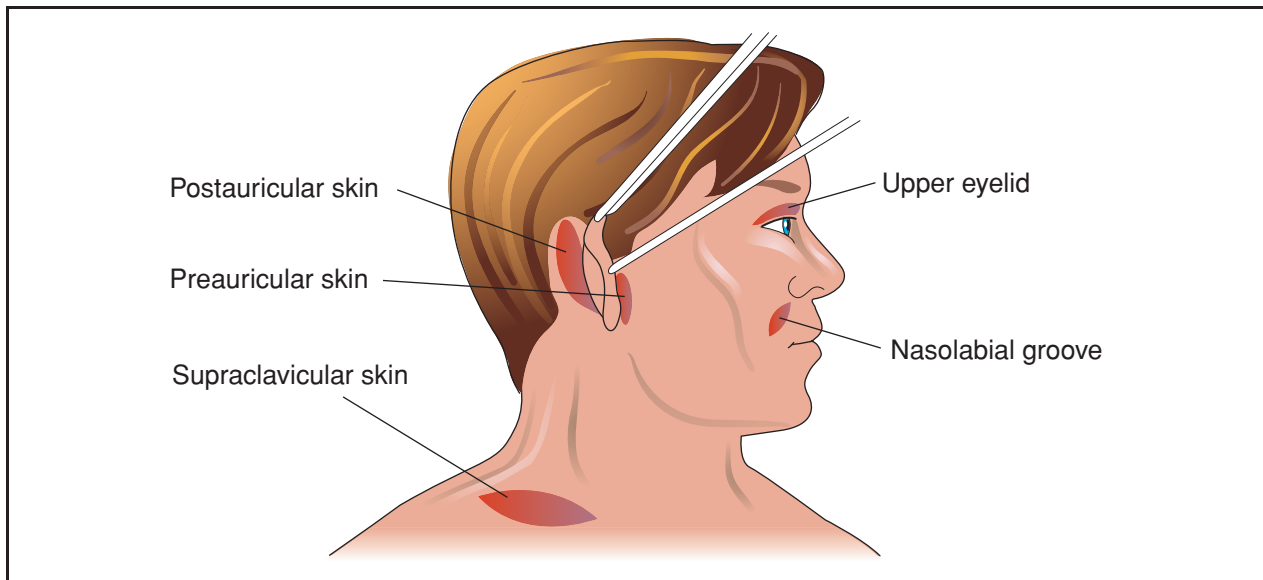
A composite skin graft is sometime used, consisting of combinations of skin and fat, skin and cartilage, or dermis and fat. Composite grafts are used where three-dimensional reconstruction is necessary. For example, a wedge of ear containing skin and cartilage can be used to repair the nose.

Several artificial skin products are available for burns or non-healing wounds. Unlike allografts and

xenografts, these products are not rejected by the patient's body and actually encourage the generation of new tissue. Artificial skin usually consists of a synthetic epidermis and a collagen-based dermis. This artificial dermis, the fibers of which are arranged in a lattice, acts as a template for the formation of new tissue. Fibroblasts, blood vessels, nerve fibers, and lymph vessels from surrounding healthy tissue cross into the collagen lattice, which eventually degrades as these cells and structures build a new dermis. The synthetic epidermis, which acts as a temporary barrier during this process, is eventually replaced with a split-thickness autograft or with an epidermis cultured in the laboratory from the patient's own epithelial cells. The cost for the synthetic products is about \$1,000 for a 40-in (100-cm) square piece of artificial skin, in addition to the costs of the surgery. This procedure is covered by insurance.

Aftercare

Once a skin graft has been put in place, even after it has healed, it must be maintained carefully. Patients



Skin grafting is a surgical procedure by which skin or a skin substitute is placed over a burn or non-healing wound to replace the damaged skin or provide a temporary wound covering. Skin for grafting can be obtained from another area of the patient's body, such as the face and neck, as shown in the illustration above. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Allograft—Tissue that is taken from one person's body and grafted to another person.

Autograft—Tissue that is taken from one part of a person's body and transplanted to a different part of the same person.

Collagen—A protein that provides structural support; the main component of connective tissue.

Dermis—The underlayer of skin, containing blood vessels, nerves, hair follicles, and oil and sweat glands.

Epidermis—The outer layer of skin, consisting of a layer of dead cells that perform a protective function and a second layer of dividing cells.

Fibroblasts—A type of cell found in connective tissue; produces collagen.

Keratinocytes—Cells found in the epidermis. The keratinocytes at the outer surface of the epidermis are dead and form a tough protective layer. The cells underneath divide to replenish the supply.

Xenograft—Tissue that is transplanted from one species to another (e.g., pigs to humans).

who have grafts on their legs should remain in bed for seven to 10 days, with their legs elevated. For several

months, the patient should support the graft with an Ace bandage or Jobst stocking. Grafts in other areas of the body should be similarly supported after healing to decrease the amount of contracture.

Grafted skin does not contain sweat or oil glands, and should be lubricated daily for two to three months with a bland oil (e.g., mineral oil) to prevent drying and cracking.

Risks

The risks of skin grafting include those inherent in any surgical procedure that involves anesthesia. These include reactions to the medications, problems breathing, bleeding, and infection. In addition, the risks of an allograft procedure include transmission of infectious disease.

Normal results

A skin graft should provide significant improvement in the quality of the wound site, and may prevent the serious complications associated with burns or non-healing wounds.

Abnormal results

Failure of a graft can result from poor blood flow, swelling, or infection.

Resources

ORGANIZATIONS

American Burn Association. 625 N. Michigan Ave., Suite 1530, Chicago, IL 60611. (800) 548-2876.
< <http://www.ameriburn.org> >.

American Diabetes Association. 1701 North Beauregard Street, Alexandria, VA 22311. (800) 342-2383.
< <http://www.diabetes.org> >.

Lisa Christenson, PhD

Skin lesion removal

Definition

Skin lesion removal employs a variety of techniques, from relatively simple biopsies to more complex surgical excisions, to remove lesions that range from benign growths to **malignant melanoma**.

Purpose

Sometimes the purpose of skin lesion removal is to excise an unsightly mole or other cosmetically unattractive skin growth. Other times, physicians will remove a skin lesion to make certain it is not cancerous, and, if it proves cancerous, to prevent its spread to other parts of the body.

Precautions

Most skin lesion removal procedures require few precautions. The area to be treated is cleaned before the procedure with alcohol or another antibacterial preparation, but generally it is not necessary to use a sterile operating room. Most procedures are performed on an outpatient basis, using a local anesthetic. Some of the more complex procedures may require specialized equipment available only in an outpatient surgery center. Most of the procedures are not highly invasive and, frequently, can be well-tolerated by young and old patients, as well as those with other medical conditions.

Description

A variety of techniques are used to remove **skin lesions**. The particular technique selected will depend on such factors as the seriousness of the lesion, its location, and the patient's ability to tolerate the procedure. Some of the simpler techniques, such as a

biopsy or cryosurgery, can be performed by a primary care physician. Some of the more complex techniques, such as excision with a scalpel, electrosurgery, or laser surgery, are typically performed by a dermatologic surgeon, plastic surgeon, or other surgical specialist. Often, the technique selected will depend on how familiar the physician is with the procedure and how comfortable he or she is with performing it.

Biopsy

In this procedure, the physician commonly injects a local anesthetic at the site of the skin lesion, then removes a sample of the lesion, so that a definite diagnosis can be made. The sample is sent to a pathology laboratory, where it is examined under a microscope. Certain characteristic skin cells, and their arrangement in the skin, offer clues to the type of skin lesion, and whether it is cancerous or otherwise poses danger. Depending on the results of the microscopic examination, additional surgery may be scheduled.

A variety of methods are used to obtain a **skin biopsy**. The physician may use a scalpel to cut a piece or remove all of the lesion for examination. Lesions that are confined to the surface may be sampled with a shave biopsy, where the physician holds a scalpel blade parallel to the surface of the skin and slides the blade across the base of the lesion, removing a sample. Some physicians use a single-edge razor blade for this, instead of a scalpel. A physician may also perform a punch biopsy, in which a small circular punch removes a plug of skin.

Excision

When excising a lesion, the physician attempts to remove it completely by using a scalpel to cut the shape of an ellipse around the lesion. Leaving an elliptical wound, rather than a circular wound, makes it easier to insert stitches. If a lesion is suspected to be cancerous, the physician will not cut directly around the lesion, but will attempt to also remove a healthy margin of tissue surrounding it. This is to ensure that no cancerous cells remain, which would allow the tumor to reappear. To prevent recurrence of basal and squamous cell skin cancers, experts recommend a margin of 0.08–0.16 in (2–4 mm) for malignant melanoma, the margin may be 1.2 in (3 cm) or more.

Destruction

Not all lesions need to be excised. A physician may simply seek to destroy the lesion using a number of destructive techniques. These techniques do not

leave sufficient material to be examined by a pathologist, however, and are best used in cases where a visual diagnosis is certain.

- **Cryosurgery.** This technique employs an extremely cold liquid or instrument to freeze and destroy abnormal skin cells that require removal. Liquid nitrogen is the most commonly used cryogen. It is typically sprayed on the lesion in several freeze-thaw cycles to ensure adequate destruction of the lesion.
- **Curettage.** In this procedure, an instrument with a circular cutting loop at the end is drawn across the lesion, starting at the middle and moving outward. With successive strokes, the physician scrapes portions of the lesion away. Sometimes a physician will use the curet to reduce the size of the lesion before turning to another technique to finish removing it.
- **Electrosurgery.** This utilizes an alternating current to selectively destroy skin tissue. Depending on the type of current and device used, physicians may use electrosurgical equipment to dry up surface lesions (electrodesiccation), to burn off the lesion (electrocoagulation), or to cut the lesion (electrosection). One advantage of electrosurgery is that it minimizes bleeding.

Mohs' micrographic surgery

The real extent of some lesions may not be readily apparent to the eye, making it difficult for the surgeon to decide where to make incisions. If some cancer cells are left behind, for example, the **cancer** may reappear or spread. In a technique called Mohs' micrographic surgery, surgeons begin by removing a lesion and examining its margins under a microscope for evidence of cancer. If cancerous cells are found, the surgeon then removes another ring of tissue and examines the margins again. The process is repeated until the margins appear clear of cancerous cells. The technique is considered ideal for aggressive tumors in areas such as the nose or upper lip, where an excision with wide margins may be difficult to repair, and may leave a cosmetically poor appearance.

Lasers

Laser surgery is now applied to a variety of skin lesions, ranging from spider veins to more extensive blood vessel lesions called hemangiomas. Until recently, CO₂ lasers were among the more common laser devices used by physicians, primarily to destroy skin lesions. Other lasers, such as the Nd:YAG and flashlamp-pumped pulse dye laser have been developed to achieve more selective results when used to

treat vascular lesions, such as hemangiomas, or pigmented lesions, such as café-au-lait spots.

Preparation

No extensive preparation is required for skin lesion removal. Most procedures can be performed on an outpatient basis with a local anesthetic. The lesion and surrounding area is cleaned with an antibacterial compound before the procedure. A sterile operating room is not required.

Aftercare

The amount of aftercare will vary, depending on the skin lesion removal technique. For biopsy, curettage, cryosurgery, and electrosurgery procedures, the patient is told to keep the wound clean and dry. Healing will take at least several weeks, and may take longer, depending on the size of the wound and other factors. Healing times will also vary with excisions and with Mohs' micrographic surgery, particularly if a skin graft or skin flap is needed to repair the resulting wound. Laser surgery may produce changes in skin coloration that often resolve in time. **Pain** is usually minimal following most outpatient procedures, so pain medicines are not routinely prescribed. Some areas of the body, such as the scalp and fingers, can be more painful than others, however, and a pain medicine may be required.

Risks

All surgical procedures present risk of infection. Keeping the wound clean and dry can minimize the risk. **Antibiotics** are not routinely given to prevent infection in skin surgery, but some doctors believe they have a role. Other potential complications include:

- bleeding below the skin, which may create a hematoma and sometimes requires the wound to be reopened and drained,
- temporary or permanent nerve damage resulting from excision in an area with extensive and shallow nerve branches,
- wounds that may reopen after they have been stitched closed, increasing the risk of infection and scarring.

Normal results

Depending on the complexity of the skin lesion removal procedure, patients can frequently resume

KEY TERMS

Curet—A surgical instrument with a circular cutting loop at one end. The curet is pulled over the skin lesion in repeated strokes to remove one portion of the lesion at a time.

Mohs' micrographic surgery—A surgical technique in which successive rings of skin tissue are removed and examined under a microscope to ensure that no cancer is left.

Shave biopsy—A method of removing a sample of skin lesion so it can be examined by a pathologist. A scalpel or razor blade is held parallel to the skin's surface and is used to slice the lesion at its base.

their normal routine the day of surgery. Healing frequently will take place within weeks. Some excisions will require later reconstructive procedures to improve the appearance left by the original procedure.

Abnormal results

In addition to the complications outlined above, it is always possible that the skin lesion will reappear, requiring further surgery.

Resources

ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

American Society for Dermatologic Surgery. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-9830. <<http://www.asds-net.org>>.

American Society of Plastic and Reconstructive Surgeons. 44 E. Algonquin Rd., Arlington Heights, IL 60005. (847) 228-9900. <<http://www.plasticsurgery.org>>.

Richard H. Camer

Skin lesions

Definition

A skin lesion is a superficial growth or patch of the skin that does not resemble the area surrounding it.

Description

Skin lesions can be grouped into two categories: primary and secondary. Primary skin lesions are variations in color or texture that may be present at birth, such as **moles** or **birthmarks**, or that may be acquired during a person's lifetime, such as those associated with infectious diseases (e.g. **warts**, **acne**, or **psoriasis**), allergic reactions (e.g. **hives** or **contact dermatitis**), or environmental agents (e.g. **sunburn**, pressure, or temperature extremes). Secondary skin lesions are those changes in the skin that result from primary skin lesions, either as a natural progression or as a result of a person manipulating (e.g. scratching or picking at) a primary lesion.

The major types of primary lesions are:

- **Macule.** A small, circular, flat spot less than $\frac{2}{5}$ in (1 cm) in diameter. The color of a macule is not the same as that of nearby skin. Macules come in a variety of shapes and are usually brown, white, or red. Examples of macules include freckles and flat moles. A macule more than $\frac{2}{5}$ in (1 cm) in diameter is called a patch.
- **Vesicle.** A raised lesion less than $\frac{1}{5}$ in (5 mm) across and filled with a clear fluid. Vesicles that are more than $\frac{1}{5}$ in (5 mm) across are called bullae or blisters. These lesions may be the result of sunburns, insect bites, chemical irritation, or certain viral infections, such as herpes.
- **Pustule.** A raised lesion filled with pus. A pustule is usually the result of an infection, such as acne, impetigo, or **boils**.
- **Papule.** A solid, raised lesion less than $\frac{2}{5}$ in (1 cm) across. A patch of closely grouped papules more than $\frac{2}{5}$ in (1 cm) across is called a plaque. Papules and plaques can be rough in texture and red, pink, or brown in color. Papules are associated with such conditions as warts, **syphilis**, psoriasis, seborrheic and actinic keratoses, **lichen planus**, and skin **cancer**.
- **Nodule.** A solid lesion that has distinct edges and that is usually more deeply rooted than a papule. Doctors often describe a nodule as "palpable," meaning that, when examined by touch, it can be felt as a hard mass distinct from the tissue surrounding it. A nodule more than 2 cm in diameter is called a tumor. Nodules are associated with, among other conditions, keratinous cysts, lipomas, fibromas, and some types of lymphomas.
- **Wheal.** A skin elevation caused by swelling that can be itchy and usually disappears soon after erupting. Wheals are generally associated with an allergic reaction, such as to a drug or an insect bite.

- **Telangiectasia.** Small, dilated blood vessels that appear close to the surface of the skin. Telangiectasia is often a symptom of such diseases as **rosacea** or **scleroderma**.

The major types of secondary skin lesions are:

- **Ulcer.** Lesion that involves loss of the upper portion of the skin (epidermis) and part of the lower portion (dermis). Ulcers can result from acute conditions such as bacterial infection or trauma, or from more chronic conditions, such as scleroderma or disorders involving peripheral veins and arteries. An ulcer that appears as a deep crack that extends to the dermis is called a fissure.
- **Scale.** A dry, horny build-up of dead skin cells that often flakes off the surface of the skin. Diseases that promote scale include fungal infections, psoriasis, and **seborrheic dermatitis**.
- **Crust.** A dried collection of blood, serum, or pus. Also called a scab, a crust is often part of the normal healing process of many infectious lesions.
- **Erosion.** Lesion that involves loss of the epidermis.
- **Excoriation.** A hollow, crusted area caused by scratching or picking at a primary lesion.
- **Scar.** Discolored, fibrous tissue that permanently replaces normal skin after destruction of the dermis. A very thick and raised scar is called a keloid.
- **Lichenification.** Rough, thick epidermis with exaggerated skin lines. This is often a characteristic of scratch **dermatitis** and **atopic dermatitis**.
- **Atrophy.** An area of skin that has become very thin and wrinkled. Normally seen in older individuals and people who are using very strong topical corticosteroid medication.

Causes and symptoms

Skin lesions can be caused by a wide variety of conditions and diseases. A tendency toward developing moles, freckles, or birthmarks may be inherited. Infection of the skin itself by bacteria, viruses, fungi, or parasites is the most common cause of skin lesions. Acne, **athlete's foot** (tinea pedis), warts, and **scabies** are examples of skin infections that cause lesions. Allergic reactions and sensitivity to outside environmental factors can also lead to the formation of skin lesions. Underlying conditions can also precipitate the appearance of skin lesions. For example, the decreased sensitivity and poor circulation that accompanies **diabetes mellitus** can contribute to the formation of extensive ulcers on extremities such as the feet. Infections of body's entire system can cause the sudden onset of skin

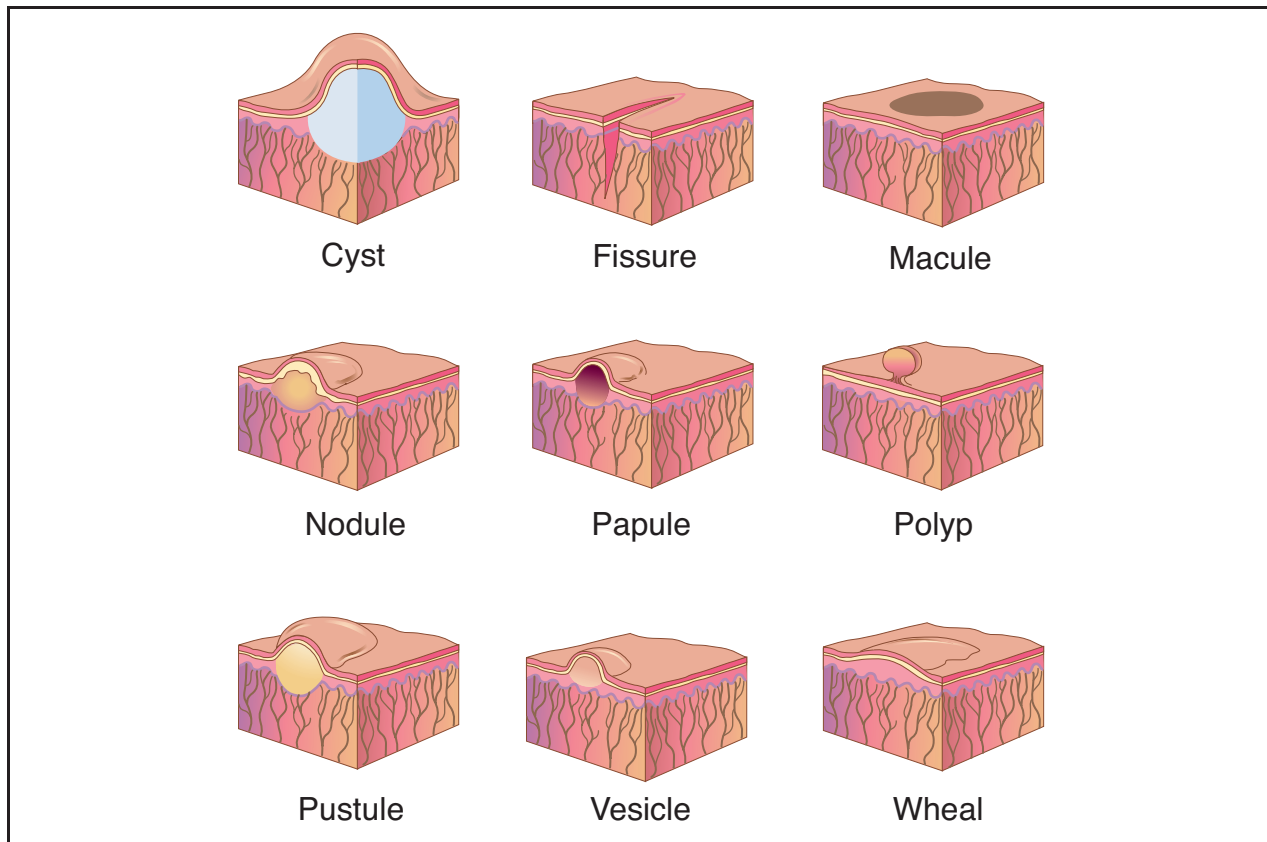
lesions. For example, skin lesions are a hallmark symptom of such diseases as chicken pox, herpes, and small pox. Cancers affecting the skin, including basal cell carcinoma, squamous cell carcinoma, **malignant melanoma**, and **Kaposi's sarcoma**, are recognized by their lesions.

Diagnosis

Diagnosis of the underlying cause of skin lesions is usually based on patient history, characteristics of the lesion, and where and how it appears on the patient's body (e.g. pustules confined to the face, neck and upper back can indicate acne, while scales appearing on the scalp and face may indicate seborrheic dermatitis). To determine the cause of an infection, doctors may also take scrapings or swab samples from lesions for examination under a microscope or for use in bacterial, fungal, or viral cultures. In cases where a fungal infection is suspected, a doctor may examine a patient's skin under ultraviolet light using a filter device called a **Woods light**—under these conditions, certain species will taken on specific fluorescent colors. Dermatologists may also use contrast lighting and subdued lighting to detect variations in the skin. When involvement of the immune system is suspected, doctors may order a immunofluorescence test, which detects antibodies to specific antigens using a fluorescent chemical. In cases of contact dermatitis, a condition in which a allergic reaction to something irritates the skin, doctors may use patch tests, in which samples of specific antigens are introduced into the skin via a scratch or a needle prick, to determine what substances are provoking the reaction.

The vast majority of skin lesions are noncancerous. However, doctors will determine whether or not a particular lesion or lesions are cancerous based on observation and the results of an excisional or punch biopsy, in which a tissue sample is excised for microscopic analysis. Since early detection is a key to successful treatment, individuals should examine their skin on a monthly basis for changes to existing moles, the presence of new moles, or a change in a certain area of skin. When examining moles, factors to look for include:

- **Asymmetry.** A normal mole is round, whereas a suspicious mole is uneven.
- **Border.** A normal mole has a clear-cut border with the surrounding skin, whereas the edges of a suspect mole may be irregular.
- **Color.** Normal moles are uniformly tan or brown, but cancerous moles may appear as mixtures of red, white, blue, brown, purple, or black.



A skin lesion is an abnormal growth or an area of skin that does not resemble the skin surrounding it. The illustrations above feature some of the different types of skin lesions. (Illustration by Electronic Illustrators Group.)

- Diameter. Normal moles are usually less than $\frac{1}{5}$ in (5 mm) in diameter, a skin lesion greater than this may be suspected as cancerous.

Treatment

Treatment of skin lesions depends upon the underlying cause, what type of lesions they are, and the patient's overall health. If the cause of the lesions is an allergic reaction, removing the allergen from the patient's environment is the most effective treatment. Topical preparations can also be used to clean and protect irritated skin as well as to remove dead skin cells and scales. These may come in a variety of forms, including ointments, creams, lotions, and solutions. **Topical antibiotics**, fungicides, pediculicides (agents that kill lice), and scabicides (agents that kill the scabies parasite) can be applied to treat appropriate skin infections. Oral medications may be taken to address systemic infections or conditions. Deeply infected lesions may require minor surgery to lance and drain pus. Topical agents to sooth irritated skin and reduce inflammation may also be applied. **Corticosteroids** are

particularly effective in reducing inflammation and **itching** (puritis). Oatmeal baths, baking soda mixtures, and calamine lotion are also recommended for the relief of these symptoms. A type of corticosteroid may be used to reduce the appearance of keloid **scars**. Absorbent powders may also be used to reduce moisture and prevent the spread of infection. In cases of ulcers that are slow to heal, pressure dressings may be used. At times, surgical removal of a lesion may be recommended—this is the usual course of therapy for skin cancer. Surgical removal usually involves a simple excision under local anesthetic, but it may also be accomplished through freezing (**cryotherapy**) or **laser surgery**.

Prognosis

Skin lesions such as moles, freckles, and birthmarks are a normal part of skin and will not disappear unless deliberately removed by a surgical procedure. Lesions due to an allergic reaction often subside soon after the offending agent is removed. Healing of lesions due to infections or disorders depends upon

KEY TERMS

Corticosteroid—A type of steroid medication that helps relieve itching (pruritis) and reduce inflammation.

Fibroma—A usually benign tumor consisting of fibrous tissue.

Lesion—A possibly abnormal change or difference in a tissue or structure, such as the skin.

Lipoma—A usually benign tumor of fatty tissue.

Patch test—Test in which different antigens (substances that cause an allergic reaction) are introduced into a patient's skin via a needle prick or scratch and then observed for evidence of an allergic reaction to one or more of them. Also known as a scratch test.

Woods light—Device that allows only ultraviolet light to pass through it.

the type of infection or disorder and the overall health of the individual. Prognosis for skin cancer primarily depends upon whether or not the lesion is localized and whether or not it has spread to other areas of the body, such as the lymph nodes. In cases where the lesion is localized and has not spread to other parts of the body, the cure rate is 95-100%.

Prevention

Not all skin lesions are preventable; moles and freckles, for example, are benign growths that are common and unavoidable. However others can be avoided or minimized by taking certain precautions. Skin lesions caused by an allergic reaction can be avoided by determining what the offending agent is and removing it from the home or workplace, or, if this is impossible, developing strategies for safely handling it, such as with gloves and protective clothing. Keeping the skin, nails, and scalp clean and moisturized can help reduce or prevent the incidence of infectious skin diseases, as can not sharing personal care items such as combs and make-up with others. Skin lesions associated with **sexually transmitted diseases** can be prevented by the use of condoms. Scratching or picking at existing lesions should be avoided since this usually serves only to spread infection and may result in scarring. Individuals who have systemic conditions, such as diabetes mellitus or poor circulation, that could lead to serious skin lesions should inspect their bodies regularly for changes in their skin's condition. Regular visual inspection of

the skin is also a key to preventing or minimizing the occurrence of skin cancer, as is the regular use of sun screens with an SPF of 15 or more.

Resources

BOOKS

Rosen, Theodore, Marilyn B. Lanning, and Marcia J. Hill. *The Nurse's Atlas of Dermatology*. Boston: Little Brown & Co., 1983.

Bridget Travers

Skin pigmentation disorders

Definition

Skin pigmentation disorders are conditions that cause the skin to appear lighter or darker than normal, or blotchy and discolored.

Description

People of all races have skin pigmentation disorders. Some disorders, like **albinism** (which affects one out of every 17,000 people) are rare. Others, such as age spots, are very common.

Skin pigmentation disorders occur because the body produces either too much or too little melanin, a pigment that creates hair, skin, and eye color. Melanin protects the body by absorbing ultraviolet light.

In hypopigmentation means the body does not produce enough melanin. Albinism, for example, is an inherited condition that causes a lack of pigment. So people with albinism typically have light skin, white or pale yellow hair, and light blue or gray eyes. Another condition called vitiligo, creates smooth, depigmented white spots on the skin. Vitiligo affects nearly 2% of the population, but it strikes people between 10 and 30 years old more often, and is more evident in people with darker skin.

In **hyperpigmentation**, the body produces too much melanin, causing skin to become darker than usual. Lichen simplex chronicus is a skin disorder with severe **itching** that causes thick, dark patches of skin to develop. Lamellar **ichthyosis** (fish scale disease) is an inherited disease that also is characterized by darkened, scaly, dry patches of skin.

Hyperpigmentation also occurs in melasma, a dark mask-like discoloration that covers the cheeks and bridge of the nose. Melasma can occur during

the end of **pregnancy**. People with the autoimmune disease (when immune cells, which attack invaders, become abnormally programmed to kill self cells inside the body) systemic lupus also may develop a similar butterfly-shaped mask on their faces. In addition, many people have moles, freckles, age spots, and **birthmarks**, ranging from red or brown to bluish, black, covering various parts of their bodies.

Causes and symptoms

Scientists are still studying the reasons why skin pigmentation disorders occur. In some cases there are tangible causes, such as sun exposure, drug reactions or genetic inheritance. In other cases, it is not as clear.

Albinism is an inherited recessive trait. Albinism has many different forms, but most people who have this condition have pale skin, hair, and eyes. Melanin also creates eye color, and serves as a filter that prevents too much light from entering the eye. Since they lack melanin in their eyes, many people with albinism also have **visual impairment**. With little skin pigmentation, they also **sunburn** easily and are more prone to skin **cancer**.

The hypopigmentation spots associated with vitiligo sometimes form where a person has been cut or injured. Research has shown that the light patches associated with vitiligo do not contain melanocytes, the type of skin cells that create melanin. Some scientists believe vitiligo may be caused by an autoimmune disorder. It also has been linked to other conditions such as **hyperthyroidism** (too much thyroid hormone) and Addison's Disease, which affects the adrenal gland.

Hyperpigmentation can be caused by many factors, from too much sunbathing to drug reactions or poor **nutrition**. Wounds and **scars** also can develop darker patches of skin. A psychological syndrome gives people with lichen simplex chronicus to develop a compulsive need to scratch, which causes dark, leathery skin to form. This can lead to permanent scarring and infection if untreated. Scientists believe lamellar ichthyosis is caused by genetics.

The mask caused by melasma may be related to pregnancy hormones, and usually disappears after a woman gives birth. Birthmarks, **moles**, and **aging** spots usually are harmless. Some moles, however, can change in size, color, texture, or start bleeding, which could indicate possible skin cancer.

Diagnosis

Diagnostic tests vary for different types of skin pigmentation disorders. Physicians usually can

KEY TERMS

Melanin—A pigment that creates hair, skin and eye color. Melanin also protects the body by absorbing ultraviolet light.

Melanocytes— The type of skin cells that create melanin.

Albinism— An inherited condition that causes a lack of pigment. People with albinism typically have light skin, white or pale yellow hair and light blue or gray eyes

Hypopigmentation—A skin condition that occurs when the body has too little melanin, or pigment.

Hyperpigmentation— A skin condition that occurs when the body has too much melanin, or pigment.

Lichen simplex chronicus—A skin disorder with severe itching that causes thick, dark patches of skin to develop.

Vitiligo—A skin disorder that creates smooth, depigmented white spots on the skin.

Lamellar ichthyosis— Also called fish scale disease, this inherited condition is characterized by darkened, scaly, dry patches of skin.

Melasma—A dark mask-like discoloration that covers the cheeks and bridge of the nose. Also called "the mask of pregnancy."

diagnose albinism by looking carefully at a person's hair, skin, and eyes. They may order blood tests and eye exams as well. A visual exam also is enough to diagnose vitiligo.

For most hyperpigmentation disorders, doctors can make a diagnosis by looking at a person's appearance. To detect conditions like **lichen simplex chronicus** or lamellar ichthyosis, or skin cancer, they may also do a biopsy to remove some of the affected skin for further study under a microscope. Some physicians also use a wood's lamp, or black light test, to diagnose skin conditions. Affected areas would absorb the ultraviolet light and stand out with fluorescent colors in the darkened room.

Treatment

For albinism, healthcare providers advise people to cover up, use sunscreen and avoid excess sunlight to prevent skin cancer. People with albinism also must wear protective sunglasses

and, in some cases, prescription corrective lenses. Surgery may be necessary to correct visual impairments.

To treat vitiligo, physicians may prescribe a combination of photo-sensitive medications like trimethylpsoralen and ultraviolet light therapy to darken the spots. If the person has depigmented patches covering more than 50% of the body, doctors also may be able to use skin bleaching agents like monobenzone to give the skin a lighter, more uniform appearance. Other options include cosmetic concealers and **skin grafting**.

Skin-lightening creams are available for hyperpigmentation disorders. Doctors also advise staying out of the sun. Counseling with a dietitian may help in cases caused by poor nutrition. For lichen simplex chronicus, doctors could prescribe antihistamines and topical steroid creams to stop the itching. If a mole or birthmark appears suspicious, physicians often will surgically remove it to prevent skin cancer.

Prognosis

Most skin pigmentation disorders do not affect a person's health, only the outward appearance.

Prevention

In most cases, doctors will recommend using sunscreen and avoiding too much sun exposure.

Resources

PERIODICALS

Wilson, Tracy. "The Paler Side of Beauty." *Heart and Soul* 6, no. 1 (February 1999): 30-33.

ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. <<http://www.aad.org>>.

National Organization for Albinism and Hypopigmentation (NOAH), 1530 Locust St., #29, Philadelphia, PA, 19102-4415. (800) 473-2310. <<http://www.albinism.org>>.

OTHER

MelanomaNet. <<http://www.skincarephysicians.com/melanomanet/index.html>>.

National Weather Service. "Ultraviolet Light Index." <<http://www.nws.noaa.gov/pa/secnews/uv/index.html>>.

Melissa Knopper

Skin resurfacing

Definition

Skin resurfacing employs a variety of techniques to change the surface texture and appearance of the skin. Common skin resurfacing techniques include chemical peels, dermabrasion, and laser resurfacing.

Purpose

Skin resurfacing procedures may be performed for cosmetic reasons, such as diminishing the appearance of wrinkles around the mouth or eyes. They may also be used as a medical treatment, such as removing large numbers of certain precancerous lesions called actinic keratoses. Physicians sometimes combine techniques, using dermabrasion or laser resurfacing on some areas of the face, while performing a chemical peel on other areas.

Precautions

As the popularity of skin resurfacing techniques has increased, many unqualified or inexperienced providers have entered the field. Patients should choose their provider with the same degree of care they take for any other medical procedure. Complications of skin resurfacing techniques can be serious, including severe infection and scarring.

Patient's with active herpesvirus infections are not good candidates for resurfacing procedures. Persons who tend to scar easily may also experience poor results. Patients who have recently used the oral **acne** medication isotretinoin (Accutane) may be at higher risk of scarring following skin resurfacing.

Description

Chemical peel

Chemical peels employ a variety of caustic chemicals to selectively destroy several layers of skin. The peeling solutions are "painted on," area-by-area, to ensure that the entire face is treated. After the skin heals, discoloration, wrinkles, and other surface irregularities are often eliminated.

Chemical peels are divided into three types: superficial, medium-depth, and deep. The type of peel depends on the strength of the chemical used, and on how deeply it penetrates. Superficial peels are used for fine wrinkles, sun damage, acne, and **rosacea**. The medium-depth peel is used for more obvious wrinkles and sun damage, as well as for precancerous lesions

like actinic keratoses. Deep peels are used for the most severe wrinkling and sun damage.

Dermabrasion

Dermabrasion uses an abrasive tool to selectively remove layers of skin. Some physicians use a hand-held motorized tool with a small wire brush or diamond-impregnated grinding wheel at the end. Other physicians prefer to abrade the skin by hand with an abrasive pad or other instrument. Acne scarring is one of the prime uses for dermabrasion. It also can be used to treat wrinkling, remove surgical **scars**, and obliterate **tattoos**.

Laser resurfacing

Laser resurfacing is the most recently developed technique for skin resurfacing. Specially designed, pulsed CO₂ lasers can vaporize skin layer-by-layer, causing minimal damage to other skin tissue. Special scanning devices move the laser light across the skin in predetermined patterns, ensuring proper exposure. Wrinkling around the eyes, mouth, and cheeks are the primary uses for laser resurfacing. Smile lines or those associated with other facial muscles tend to reappear after laser resurfacing. Laser resurfacing appears to achieve its best results as a spot treatment; patients expecting complete elimination of their wrinkles will not be satisfied.

Preparation

Chemical peel

Preparation for the chemical peel begins several weeks before the actual procedure. To promote turnover of skin cells, patients use a mild glycolic acid lotion or cream in the morning, and the acne cream tretinoin in the evening. They also use hydroquinone cream, a bleaching product that helps prevent later discoloration. To prevent reappearance of a herpes simplex virus infection, antiviral medicine is started a few days before the procedure and continues until the skin has healed.

Patients arrive for the procedure wearing no makeup. The physician “degreases” the patient’s face using alcohol or another cleanser. Some degree of **pain** accompanies all types of peels. For a superficial peel, use of a hand held fan to cool the face during the procedure is often sufficient. For medium-depth peels, the patient may take a sedative or **aspirin**. During the procedure, cold compresses and a hand-held fan can also reduce pain. Deep peels can be extremely painful. Some physicians prefer **general**

anesthesia, but local anesthetics combined with intravenous sedatives are frequently sufficient to control pain.

Dermabrasion

Dermabrasion does not require much preparation. It is usually performed under **local anesthesia**, although some physicians use intravenous **sedation** or general anesthesia. The physician begins by marking the areas to be treated and then chilling them with ice packs. In order to stiffen the skin, a spray refrigerant is applied to the area, which also helps control pain. Some physicians prefer to inject the area with a solution of saline and local anesthetic, which also leaves the skin’s surface more solid. Since dermabrasion can cause quite a bit of bleeding, physicians and their assistants will wear gloves, gowns, and masks to protect themselves from possible blood-transmitted infection.

Laser resurfacing

Antiviral medications should be started several days before the procedure. Laser resurfacing is performed under local anesthesia. An oral sedative may also be taken. The patient’s eyes must be shielded, and the area surrounding the face should be shielded with wet drapes or crumpled foil to catch stray beams of laser light. The physician will mark the areas to be treated before beginning the procedure.

Aftercare

Chemical peel

Within a day or so following a superficial peel, the skin will turn faint pink or brown. Over the next few days, dead skin will peel away. Patients will be instructed to wash their skin frequently with a mild cleanser and cool water, then apply an ointment to the skin to keep it moist. After a medium-depth peel, the skin turns deep red or brown, and crusts may form. Care is similar to that following a superficial peel. Redness may persist for a week or more. Deep-peeled skin will turn brown and crusty. There may also be swelling and some oozing of fluid. Frequent washing and ointments are favored over dressings. The skin typically heals in about two weeks, but redness may persist.

Dermabrasion

Following the procedure, an ointment may be applied, and the wound will be covered with a dressing and mask. Patients with a history of herpesvirus

KEY TERMS

Actinic keratosis—A crusty, scaly skin lesion, caused by exposure to the sun, which can transform into skin cancer.

Herpesviruses—A family of viruses responsible for cold sores, chicken pox, and genital herpes.

Isotretinoin—A powerful vitamin A derivative used in the treatment of acne. It can promote scarring after skin resurfacing procedures.

infections will begin taking an antiviral medication to prevent a recurrence. After 24 hours, the dressing is removed, and ointment is reapplied to keep the wound moist. Patients are encouraged to wash their face with plain water and reapply ointment every few hours. This relieves **itching** and pain and helps remove oozing fluid and other matter. Patients may require a pain medication. A steroid medication may be taken during the first few days to reduce swelling. The skin will take a week or more to heal, but may remain very red.

Laser resurfacing

The skin should be kept moist following laser resurfacing. This promotes more rapid healing and reduces the risk of infection. Some physicians favor application of ointments only to the skin; others prefer the use of dressings. In either case, care of the skin is similar to that given following a chemical peel. The face is washed with plain water to remove ooze, and an ointment is reapplied. Healing will take approximately two weeks. Pain medications and a steroid to reduce swelling may also be taken.

Risks

All resurfacing procedures can lead to infection and scarring. It is also possible that skin coloration will be altered, or that redness of the skin will be prolonged for many months. Some of the peeling agents used in deep chemical peels can affect the function of the heart.

Normal results

Depending on the resurfacing techniques selected, it is possible to improve the appearance of skin damaged by sun, age, or disease in many people. Skin resurfacing techniques address only the surface of the skin; procedures such as face-lift surgery or

blepharoplasty may be needed to repair other age-related skin changes. All resurfacing procedures are accompanied by some pain, redness, and skin color changes. These may persist for several months following the procedure, but they usually resolve over time.

Abnormal results

As noted above, resurfacing procedures can reactivate herpesvirus infections or lead to new, sometimes serious infections. All resurfacing techniques intentionally create skin **wounds**, creating the possibility for scarring. Abnormal results such as these can be minimized with use of antiviral medications prior to the procedure and good wound care afterward. Selection of an experienced, reputable provider also is key.

Resources

ORGANIZATIONS

- American Society for Dermatologic Surgery. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-9830. <<http://www.asds-net.org>>.
- American Society for Laser Medicine and Surgery. 2404 Stewart Square, Wausau, WI 54401. (715) 845-9283. <<http://www.aslms.org>>.
- American Society of Plastic and Reconstructive Surgeons. 44 E. Algonquin Rd., Arlington Heights, IL 60005. (847) 228-9900. <<http://www.plasticsurgery.org>>.

Richard H. Camer

Skin traction see **Traction; Immobilization**

Skull x rays

Definition

Skull x rays are performed to examine the nose, sinuses, and facial bones. These studies may also be referred to as sinus x rays. X-ray studies produce films, also known as radiographs, by aiming x rays at soft bones and tissues of the body. X-ray beams are similar to light waves, except their shorter wavelength allows them to penetrate dense substances, producing images and shadows on film.

Purpose

Doctors may order skull x rays to aid in the diagnosis of a variety of diseases or injuries.

Sinusitis

Sinus x rays may be ordered to confirm a diagnosis of **sinusitis**, or sinus infection.

Fractures

A skull x ray may detect bone **fractures** resulting from injury or disease. The skull x ray should clearly show the skull cap, jaw bones, and facial bones.

Tumors

Skull radiographs may indicate tumors in facial bones, tissues, or the sinuses. Tumors may be benign (not cancerous) or malignant (cancerous).

Other

Birth defects (referred to as congenital anomalies) may be detected on a skull x ray by changes in bone structure. Abnormal tissues or glands resulting from various conditions or diseases may also be shown on a skull radiograph.

Precautions

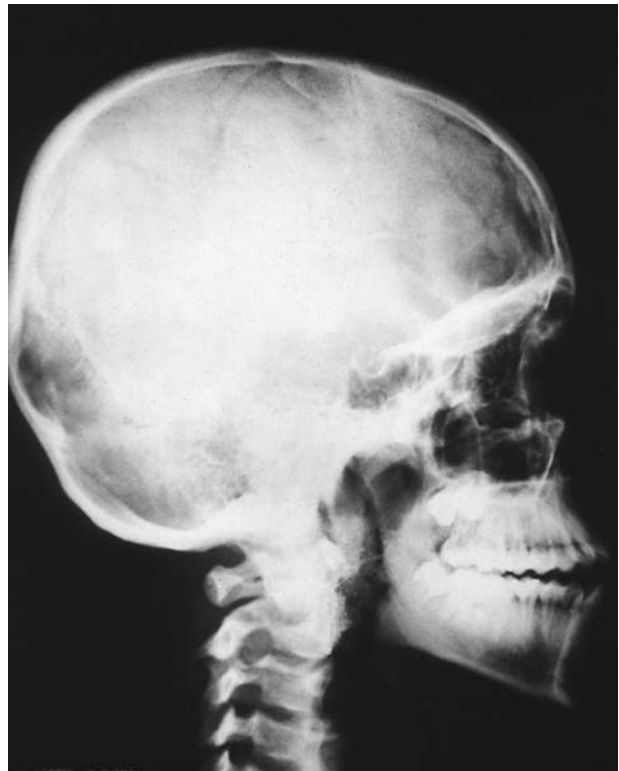
As with any x-ray procedure, women who may be pregnant are advised against having a skull x ray if it is not absolutely necessary. However, a lead apron may be worn across the abdomen during the procedure to protect the fetus. Children are also more sensitive to x-ray exposure. Children of both sexes should wear a protective covering (a lead apron) in the genital/reproductive area. In general, skull x-ray exposure is minimal and x-ray equipment and procedures are monitored to ensure radiation safety.

Description

Skull or sinus x rays may be performed in a doctor's office that has x-ray equipment and a technologist available. The exam may also be performed in an outpatient radiology facility or a hospital radiology department.

In many instances, particularly for sinus views, the patient will sit upright in a chair, perhaps with the head held stable by a foam vise. A film cassette is located behind the patient. The x-ray tube is in front of the patient and may be moved to allow for different positions and views. A patient may also be asked to move his or her head at various angles and positions.

In some cases, technologists will ask the patient to lie on a table and will place the head and neck at various angles. In routine skull x rays, as many as



A skull x ray. (Photo Researchers. Reproduced by permission.)

five different views may be taken to allow a clear picture of various bones and tissues. The length of the test will vary depending on the number of views taken, but in general, it should last about 10 minutes. The technologist will usually ask a patient to wait while the films are being developed to ensure that they are clear before going to the radiologist.

Preparation

There is no preparation for the patient prior to arriving at the radiology facility. Patients will be asked to remove jewelry, dentures, or other metal objects that may produce artifacts on the film. The referring doctor or x-ray technologist can answer any questions regarding the procedure. Any woman who is, or may be, pregnant should tell the technologist.

Aftercare

There is no aftercare required following skull or sinus x-ray procedures.

Risks

There are no common side effects from skull or sinus x ray. The patient may feel some discomfort in

KEY TERMS

Radiograph—The actual picture or film produced by an x-ray study.

X ray—A form of electromagnetic radiation with shorter wavelengths than normal light. X rays can penetrate most structures.

the positioning of the head and neck, but will have no complications. Any x-ray procedure carries minimal radiation risk, and children and pregnant women should be protected from radiation exposure to the abdominal or genital areas.

Normal results

Normal results should indicate sinuses, bones, tissues, and other observed areas are of normal size, shape, and thickness for the patient's age and medical history. Results, whether normal or abnormal, will be provided to the referring doctor in a written report.

Abnormal results

Abnormal results may include:

Sinusitis

Air in sinuses will show up on a radiograph as black, but fluid will be cloudy or white (opaque). This helps the radiologist to identify trapped fluids in the sinuses. In chronic sinusitis, the radiologist may also note thickening or hardening of the bony wall of an infected sinus.

Fractures

Radiologists may recognize even tiny facial bone fractures as a line of defect.

Tumors

Tumors may be visible if the bony sinus wall is distorted or destroyed. Abnormal findings may result in follow-up imaging studies.

Other

Skull x rays may also detect disorders that show up as changes in bone structure, such as Paget's disease of the bone or acromegaly (a disorder associated with excess growth hormones from the pituitary gland). Areas of calcification, or gathering of calcium

deposits, may indicate a condition such as an infection of bone or bone marrow (**osteomyelitis**).

Resources

ORGANIZATIONS

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. <<http://www.nci.nih.gov>>.

National Head Injury Foundation, Inc. (888) 222-5287. <<http://www.nhif.org/home.html>>.

Radiological Society of North America. 820 Jorie Boulevard, Oak Brook, IL 60523-2251. (630) 571-2670. <<http://www.rsna.org>>.

Teresa Odle

SLE see **Systemic lupus erythematosus**

Sleep apnea

Definition

Sleep apnea is a condition in which breathing stops for more than ten seconds during sleep. Sleep apnea is a major, though often unrecognized, cause of daytime sleepiness. It can have serious negative effects on a person's quality of life, and is thought to be considerably underdiagnosed in the United States.

Description

A sleeping person normally breathes continuously and uninterrupted throughout the night. A person with sleep apnea, however, has frequent episodes (up to 400–500 per night) in which he or she stops breathing. This interruption of breathing is called “apnea.” Breathing usually stops for about 30 seconds; then the person usually startles awake with a loud snort and begins to breathe again, gradually falling back to sleep.

There are two forms of sleep apnea. In obstructive sleep apnea (OSA), breathing stops because tissue in the throat closes off the airway. In central sleep apnea, (CSA), the brain centers responsible for breathing fail to send messages to the breathing muscles. OSA is much more common than CSA. It is thought that about 1–10% of adults are affected by OSA; only about one tenth of that number have CSA. OSA can affect people of any age and of either sex, but it is most common in middle-aged, somewhat overweight men, especially those who use alcohol.

Causes and symptoms

Obstructive sleep apnea

Obstructive sleep apnea occurs when part of the airway is closed off (usually at the back of the throat) while a person is trying to inhale during sleep. People whose airways are slightly narrower than average are more likely to be affected by OSA. **Obesity**, especially obesity in the neck, can increase the risk of developing OSA, because the fat tissue tends to narrow the airway. In some people, the airway is blocked by enlarged tonsils, an enlarged tongue, jaw deformities, or growths in the neck that compress the airway. Blocked nasal passages may also play a part in some people.

When a person begins to inhale, the expansion of the lungs lowers the air pressure inside the airway. If the muscles that keep the airway open are not working hard enough, the airway narrows and may collapse, shutting off the supply of air to the lungs. OSA occurs during sleep because the neck muscles that keep the airway open are not as active then. Congestion in the nose can make collapse more likely, since the extra effort needed to inhale will lower the pressure in the airway even more. Drinking alcohol or taking tranquilizers in the evening worsens this situation, because these cause the neck muscles to relax. (These drugs also lower the “respiratory drive” in the nervous system, reducing breathing rate and strength.)

People with OSA almost always snore heavily, because the same narrowing of the airway that causes **snoring** can also cause OSA. Snoring may actually help cause OSA as well, because the vibration of the throat tissues can cause them to swell. However, most people who snore do not go on to develop OSA.

Other risk factors for developing OSA include male sex; **pregnancy**; a family history of the disorder; and **smoking**. With regard to gender, it has been found that male sex hormones sometimes cause changes in the size or structure of the upper airway. The weight gain that accompanies pregnancy can affect a woman’s breathing patterns during sleep, particularly during the third trimester. With regard to family history, OSA is known to run in families even though no gene or genes associated with the disorder have been identified as of 2002. Smoking increases the risk of developing OSA because it causes inflammation, swelling, and narrowing of the upper airway.

Some patients being treated for **head and neck cancer** develop OSA as a result of physical changes in the muscles and other tissues of the neck and throat. Doctors recommend prompt treatment of the OSA to improve the patient’s quality of life.

Central sleep apnea

In central sleep apnea, the airway remains open, but the nerve signals controlling the respiratory muscles are not regulated properly. This can cause wide fluctuations in the level of carbon dioxide (CO₂) in the blood. Normal activity in the body produces CO₂, which is brought by the blood to the lungs for exhalation. When the blood level of CO₂ rises, brain centers respond by increasing the rate of respiration, clearing the CO₂. As blood levels fall again, respiration slows down. Normally, this interaction of CO₂ and breathing rate maintains the CO₂ level within very narrow limits. CSA can occur when the regulation system becomes insensitive to CO₂ levels, allowing wide fluctuations in both CO₂ levels and breathing rates. High CO₂ levels cause very rapid breathing (hyperventilation), which then lowers CO₂ so much that breathing becomes very slow or even stops. CSA occurs during sleep because when a person is awake, breathing is usually stimulated by other signals, including conscious awareness of breathing rate.

A combination of the two forms is also possible, and is called mixed sleep apnea. Mixed sleep apnea episodes usually begin with a reduced central respiratory drive, followed by obstruction.

OSA and CSA cause similar symptoms. The most common symptoms are:

- daytime sleepiness
- morning headaches
- a feeling that sleep is not restful
- disorientation upon waking
- poor judgment
- personality changes

Sleepiness is caused not only by the frequent interruption of sleep, but by the inability to enter long periods of deep sleep, during which the body performs numerous restorative functions. OSA is one of the leading causes of daytime sleepiness, and is a major risk factor for motor vehicle accidents. Headaches and disorientation are caused by low oxygen levels during sleep, from the lack of regular breathing.

Other symptoms of sleep apnea may include **sexual dysfunction**, loss of concentration, memory loss, intellectual impairment, and behavioral changes including **anxiety** and depression.

Sleep apnea is also associated with night sweats and nocturia, or increased frequency of urination at night. Bedwetting in children is also linked to sleep apnea.

Sleep apnea can also cause serious changes in the cardiovascular system. Daytime **hypertension** (high blood pressure) is common. An increase in the number of red blood cells (polycythemia) is possible, as is an enlarged left ventricle of the heart (**cor pulmonale**), and left ventricular failure. In some people, sleep apnea causes life-threatening changes in the rhythm of the heart, including heartbeat slowing (bradycardia), racing (tachycardia), and other types of “arrhythmias.” Sudden **death** may occur from such **arrhythmias**. Patients with the **Pickwickian syndrome** (named after a Charles Dickens character) are obese and sleepy, with right **heart failure**, **pulmonary hypertension**, and chronic daytime low blood oxygen (hypoxemia) and increased blood CO₂ (hypercapnia).

Diagnosis

Excessive daytime sleepiness is the complaint that usually brings a person to see the doctor. A careful medical history will include questions about alcohol or tranquilizer use, snoring (often reported by the person's partner), and morning headaches or disorientation. A physical exam will include examination of the throat to look for narrowing or obstruction. Blood pressure is also measured. Measuring heart rate or blood levels of oxygen and CO₂ during the daytime will not usually be done, since these are abnormal only at night in most patients.

In some cases the person's dentist may suggest the diagnosis of OSA on the basis of a dental checkup or evaluation of the patient for oral surgery.

Confirmation of the diagnosis usually requires making measurements while the person sleeps. These tests are called a **polysomnography** study, and are conducted during an overnight stay in a specialized sleep laboratory. Important parts of the polysomnography study include measurements of:

- heart rate
- airflow at the mouth and nose
- respiratory effort
- sleep stage (light sleep, deep sleep, dream sleep, etc.)
- oxygen level in the blood, using a noninvasive probe (ear oximetry)

Simplified studies done overnight at home are also possible, and may be appropriate for people whose profile strongly suggests the presence of obstructive sleep apnea; that is, middle-aged, somewhat overweight men, who snore and have high blood pressure. The home-based study usually includes ear oximetry and cardiac measurements. If these measurements

support the diagnosis of OSA, initial treatment is usually suggested without polysomnography. Home-based measurements are not used to rule out OSA, however, and if the measurements do not support the OSA diagnosis, polysomnography may be needed to define the problem further.

Both types of studies are usually covered by insurance with the appropriate referral from a physician. Without insurance, lab-based polysomnography cost approximately \$1,500 in 1997, while overnight home monitoring cost between \$500 and \$1,000.

Treatment

Behavioral changes

Treatment of obstructive sleep apnea begins with reducing the use of alcohol or tranquilizers in the evening, if these have been contributing to the problem. Weight loss is also effective, but if the weight returns, as it often does, so does the apnea. Changing sleeping position may be effective; snoring and sleep apnea are both most common when a person sleeps on his back. Turning to sleep on the side may be enough to clear up the symptoms. Raising the head of the bed may also help. Opening of the nasal passages can provide some relief. There are a variety of nasal devices such as clips, tapes, or holders which may help, though discomfort may limit their use. Nasal **decongestants** may be useful, but should not be taken for sleep apnea without the consent of the treating physician.

Oxygen and drug therapy

Supplemental nighttime oxygen can be useful for some people with either central and obstructive sleep apnea. Tricyclic **antidepressant drugs** such as protriptyline (Vivactil) may help by increasing the muscle tone of the upper airway muscles, but their side effects may severely limit their usefulness.

Mechanical ventilation

For moderate to severe sleep apnea, the most successful treatment is nighttime use of a ventilator, called a CPAP machine. CPAP (continuous positive airway pressure) blows air into the airway continuously, preventing its collapse. CPAP requires the use of a nasal mask. The appropriate pressure setting for the CPAP machine is determined by polysomnography in the sleep lab. Its effects are dramatic; daytime sleepiness usually disappears within one to two days after treatment begins. CPAP is used to treat both obstructive and central sleep apnea.

CPAP is tolerated well by about two-thirds of patients who try it. Bilevel positive airway pressure (BiPAP), is an alternative form of ventilation. With BiPAP, the ventilator reduces the air pressure when the person exhales. This is more comfortable for some.

Surgery

Surgery can be used to correct obstructions in the airways. The most common surgery is called UPPP, for uvulopalatopharyngoplasty. This surgery removes tissue from the rear of the mouth and top of the throat. The tissues removed include parts of the uvula (the flap of tissue that hangs down at the back of the mouth), the soft palate, and the pharynx. Tonsils and adenoids are usually removed in this operation. This operation significantly improves sleep apnea in slightly more than half of all cases.

Reconstructive surgery is possible for those whose OSA is due to constriction of the airway by lower jaw deformities. Genioplasty, which is a procedure that plastic surgeons usually perform to reshape a patient's chin to improve his or her appearance, is now being done to reshape the upper airway in patients with OSA.

When other forms of treatment are not successful, obstructive sleep apnea may be treated by a tracheostomy. In this procedure, an opening is made into the trachea (windpipe) below the obstruction, and a tube inserted to maintain an air passage. A tracheostomy requires a great deal of care to prevent infection of the tracheostomy site. In addition, since air is no longer being filtered and moistened by the nasal passages before entering the lungs, the lower airways can become dry and susceptible to infection as well. Tracheostomy is usually reserved for those whose apnea has led to life-threatening heart arrhythmias, and who have not been treated successfully with other treatments.

Oral appliances

Another approach to treating OSA involves the use of oral appliances intended to improve breathing either by holding the tongue in place or by pushing the lower jaw forward during sleep to increase the air volume in the upper airway. The first type of oral appliance is known as a tongue retaining device or TRD. The second type is variously called an oral protrusive device (OPD) or mandibular advancement splint (MAS), because it holds the mandible, or lower jaw, forward during sleep. These oral devices appear to work best for patients with mild-to-moderate OSA, and in some cases can postpone or prevent the need for surgery. Their rate of patient compliance is about

KEY TERMS

Continuous positive airway pressure (CPAP)—A ventilation system that blows a gentle stream of air into the nose to keep the airway open.

Genioplasty—An operation performed to reshape the chin. Genioplasties are often done to treat OSA because the procedure changes the structure of the patient's upper airway.

Mandible—The medical term for the lower jaw. One type of oral appliance used to treat OSA pushes the mandible forward in order to ease breathing during sleep.

Nocturia—Excessive need to urinate at night. Nocturia is a symptom of OSA and often increases the patient's daytime sleepiness.

Polysomnography—A group of tests administered to analyze heart, blood, and breathing patterns during sleep.

Tracheotomy—A surgical procedure in which a small hole is cut into the trachea, or windpipe, below the level of the vocal cords.

Uvulopalatopharyngoplasty (UPPP)—An operation to remove excess tissue at the back of the throat to prevent it from closing off the airway during sleep.

50%; most patients who stop using oral appliances do so because their teeth are in poor condition. TRDs and OPDs can be fitted by dentists; however, most dentists work together with the patient's physician following a polysomnogram rather than prescribing the device by themselves.

Prognosis

The combination of behavioral changes, ventilation assistance, drug therapy, and surgery allow most people with sleep apnea to be treated successfully, although it may take some time to determine the most effective and least intrusive treatment. Polysomnography testing is usually required after beginning a treatment to determine how effective it has been.

Prevention

For people who snore frequently, weight control, avoidance of evening alcohol or tranquilizers, and adjustment of sleeping position may help reduce the risk of developing obstructive sleep apnea.

Resources

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American Dental Association. 211 East Chicago Avenue, Chicago, IL 60611. (312) 440-2500. <www.ada.org>.

American Sleep Apnea Association. 1424 K Street NW, Suite 302, Washington, DC 20005. (202) 293-3650. Fax: (202) 293-3656. <www.sleepapnea.org>.

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Sleep disorders

Definition

Sleep disorders are a group of syndromes characterized by disturbance in the patient's amount of sleep, quality or timing of sleep, or in behaviors or physiological conditions associated with sleep. There are about 70 different sleep disorders. To qualify for the diagnosis of sleep disorder, the condition must be a persistent problem, cause the patient significant emotional distress, and interfere with his or her social or occupational functioning.

Although sleep is a basic behavior in animals as well as humans, researchers still do not completely understand all of its functions in maintaining health. In the past 30 years, however, laboratory studies on human volunteers have yielded new information about the different types of sleep. Researchers have

learned about the cyclical patterns of different types of sleep and their relationships to breathing, heart rate, brain waves, and other physical functions. These measurements are obtained by a technique called **polysomnography**.

There are five stages of human sleep. Four stages have non-rapid eye movement (NREM) sleep, with unique brain wave patterns and physical changes occurring. Dreaming occurs in the fifth stage, during rapid eye movement (REM) sleep.

- Stage 1 NREM sleep. This stage occurs while a person is falling asleep. It represents about 5% of a normal adult's sleep time.
- Stage 2 NREM sleep. In this stage, (the beginning of "true" sleep), the person's electroencephalogram (EEG) will show distinctive wave forms called sleep spindles and K complexes. About 50% of sleep time is stage 2 REM sleep.
- Stages 3 and 4 NREM sleep. Also called delta or slow wave sleep, these are the deepest levels of human sleep and represent 10–20% of sleep time. They usually occur during the first 30–50% of the sleeping period.
- REM sleep. REM sleep accounts for 20–25% of total sleep time. It usually begins about 90 minutes after the person falls asleep, an important measure called REM latency. It alternates with NREM sleep about every hour and a half throughout the night. REM periods increase in length over the course of the night.

Sleep cycles vary with a person's age. Children and adolescents have longer periods of stage 3 and stage 4 NREM sleep than do middle aged or elderly adults. Because of this difference, the doctor will need to take a patient's age into account when evaluating a sleep disorder. Total REM sleep also declines with age.

The average length of nighttime sleep varies among people. Most people sleep between seven and nine hours a night. This population average appears to be constant throughout the world. In temperate climates, however, people often notice that sleep time varies with the seasons. It is not unusual for people in North America and Europe to sleep about 40 minutes longer per night during the winter.

Description

Sleep disorders are classified based on what causes them. Primary sleep disorders are distinguished from those that are not caused by other mental disorders,

prescription medications, **substance abuse**, or medical conditions. The two major categories of primary sleep disorders are the dyssomnias and the parasomnias.

Dyssomnias

Dyssomnias are primary sleep disorders in which the patient suffers from changes in the amount, restfulness, and timing of sleep. The most important dyssomnia is primary **insomnia**, which is defined as difficulty in falling asleep or remaining asleep that lasts for at least one month. It is estimated that 35% of adults in the United States experience insomnia during any given year, but the number of these adults who are experiencing true primary insomnia is unknown. Primary insomnia can be caused by a traumatic event related to sleep or bedtime, and it is often associated with increased physical or psychological arousal at night. People who experience primary insomnia are often anxious about not being able to sleep. The person may then associate all sleep-related things (their bed, bedtime, etc.) with frustration, making the problem worse. The person then becomes more stressed about not sleeping. Primary insomnia usually begins when the person is a young adult or in middle age.

Hypersomnia is a condition marked by excessive sleepiness during normal waking hours. The patient has either lengthy episodes of daytime sleep or episodes of daytime sleep on a daily basis even though he or she is sleeping normally at night. In some cases, patients with primary hypersomnia have difficulty waking in the morning and may appear confused or angry. This condition is sometimes called sleep drunkenness and is more common in males. The number of people with primary hypersomnia is unknown, although 5–10% of patients in sleep disorder clinics have the disorder. Primary hypersomnia usually affects young adults between the ages of 15 and 30.

Nocturnal myoclonus and **restless legs syndrome** (RLS) can cause either insomnia or hypersomnia in adults. Patients with nocturnal myoclonus wake up because of cramps or twitches in the calves. These patients feel sleepy the next day. Nocturnal myoclonus is sometimes called periodic limb movement disorder (PLMD). RLS patients have a crawly or aching feeling in their calves that can be relieved by moving or rubbing the legs. RLS often prevents the patient from falling asleep until the early hours of the morning, when the condition is less intense.

Kleine-Levin syndrome is a recurrent form of hypersomnia that affects a person three or four times a year. Doctors do not know the cause of this

syndrome. It is marked by two to three days of sleeping 18–20 hours per day, hypersexual behavior, compulsive eating, and irritability. Men are three times more likely than women to have the syndrome. Currently, there is no cure for this disorder.

Narcolepsy is a dyssomnia characterized by recurrent “sleep attacks” that the patient cannot fight. The sleep attacks are about 10–20 minutes long. The patient feels refreshed by the sleep, but typically feels sleepy again several hours later. Narcolepsy has three major symptoms in addition to sleep attacks: cataplexy, **hallucinations**, and sleep **paralysis**. Cataplexy is the sudden loss of muscle tone and stability (“drop attacks”). Hallucinations may occur just before falling asleep (hypnagogic) or right after waking up (hypnopompic) and are associated with an episode of REM sleep. Sleep paralysis occurs during the transition from being asleep to waking up. About 40% of patients with narcolepsy have or have had another mental disorder. Although narcolepsy is often regarded as an adult disorder, it has been reported in children as young as three years old. Almost 18% of patients with narcolepsy are 10 years old or younger. It is estimated that 0.02–0.16% of the general population suffer from narcolepsy. Men and women are equally affected.

Breathing-related sleep disorders are syndromes in which the patient’s sleep is interrupted by problems with his or her breathing. There are three types of breathing-related sleep disorders:

- **Obstructive sleep apnea syndrome.** This is the most common form of breathing-related sleep disorder, marked by episodes of blockage in the upper airway during sleep. It is found primarily in obese people. Patients with this disorder typically alternate between periods of **snoring** or gasping (when their airway is partly open) and periods of silence (when their airway is blocked). Very loud snoring is a clue to this disorder.
- **Central sleep apnea syndrome.** This disorder is primarily found in elderly patients with heart or neurological conditions that affect their ability to breathe properly. It is not associated with airway blockage and may be related to brain disease.
- **Central alveolar hypoventilation syndrome.** This disorder is found most often in extremely obese people. The patient’s airway is not blocked, but his or her blood oxygen level is too low.
- **Mixed-type sleep apnea syndrome.** This disorder combines symptoms of both obstructive and central sleep apnea.

Circadian rhythm sleep disorders are dyssomnias resulting from a discrepancy between the person’s

daily sleep/wake patterns and demands of social activities, shift work, or travel. The term circadian comes from a Latin word meaning daily. There are three circadian rhythm sleep disorders. Delayed sleep phase type is characterized by going to bed and arising later than most people. **Jet lag** type is caused by travel to a new time zone. Shift work type is caused by the schedule of a person’s job. People who are ordinarily early risers appear to be more vulnerable to jet lag and shift work-related circadian rhythm disorders than people who are “night owls.” There are some patients who do not fit the pattern of these three disorders and appear to be the opposite of the delayed sleep phase type. These patients have an advanced sleep phase pattern and cannot stay awake in the evening, but wake up on their own in the early morning.

PARASOMNIAS. Parasomnias are primary sleep disorders in which the patient’s behavior is affected by specific sleep stages or transitions between sleeping and waking. They are sometimes described as disorders of physiological arousal during sleep.

Nightmare disorder is a parasomnia in which the patient is repeatedly awakened from sleep by frightening dreams and is fully alert on awakening. The actual rate of occurrence of nightmare disorder is unknown. Approximately 10–50% of children between three and five years old have nightmares. They occur during REM sleep, usually in the second half of the night. The child is usually able to remember the content of the nightmare and may be afraid to go back to sleep. More females than males have this disorder, but it is not known whether the sex difference reflects a difference in occurrence or a difference in reporting. Nightmare disorder is most likely to occur in children or adults under severe or traumatic **stress**.

Sleep terror disorder is a parasomnia in which the patient awakens screaming or crying. The patient also has physical signs of arousal, like sweating, shaking, etc. It is sometimes referred to as pavor nocturnus. Unlike nightmares, sleep terrors typically occur in stage 3 or stage 4 NREM sleep during the first third of the night. The patient may be confused or disoriented for several minutes and cannot recall the content of the dream. He or she may fall asleep again and not remember the episode the next morning. Sleep terror disorder is most common in children four to 12 years old and is outgrown in adolescence. It affects about 3% of children. Fewer than 1% of adults have the disorder. In adults, it usually begins between the ages of 20 and 30. In children, more males than females have the disorder. In adults, men and women are equally affected.

Sleepwalking disorder, which is sometimes called somnambulism, occurs when the patient is capable of complex movements during sleep, including walking. Like sleep terror disorder, sleepwalking occurs during stage 3 and stage 4 NREM sleep during the first part of the night. If the patient is awakened during a sleepwalking episode, he or she may be disoriented and have no memory of the behavior. In addition to walking around, patients with sleepwalking disorder have been reported to eat, use the bathroom, unlock doors, or talk to others. It is estimated that 10–30% of children have at least one episode of sleepwalking. However, only 1–5% meet the criteria for sleepwalking disorder. The disorder is most common in children eight to 12 years old. It is unusual for sleepwalking to occur for the first time in adults.

Unlike sleepwalking, REM sleep behavior disorder occurs later in the night and the patient can remember what they were dreaming. The physical activities of the patient are often violent.

Sleep disorders related to other conditions

In addition to the primary sleep disorders, there are three categories of sleep disorders that are caused by or related to substance use or other physical or mental disorders.

SLEEP DISORDERS RELATED TO MENTAL DISORDERS. Many mental disorders, especially depression or one of the **anxiety disorders**, can cause sleep disturbances. Psychiatric disorders are the most common cause of chronic insomnia.

SLEEP DISORDERS DUE TO MEDICAL CONDITIONS. Some patients with chronic neurological conditions like Parkinson's disease or Huntington's disease may develop sleep disorders. Sleep disorders have also been associated with viral **encephalitis**, brain disease, and hypo- or **hyperthyroidism**.

SUBSTANCE-INDUCED SLEEP DISORDERS. The use of drugs, alcohol, and **caffeine** frequently produces disturbances in sleep patterns. Alcohol **abuse** is associated with insomnia. The person may initially feel sleepy after drinking, but wakes up or sleeps fitfully during the second half of the night. Alcohol can also increase the severity of breathing-related sleep disorders. With amphetamines or **cocaine**, the patient typically suffers from insomnia during drug use and hypersomnia during drug withdrawal. Opioids usually make short-term users sleepy. However, long-term users develop tolerance and may suffer from insomnia.

In addition to alcohol and drugs that are abused, a variety of prescription medications can affect sleep patterns. These medications include **antihistamines**,

corticosteroids, **asthma** medicines, and drugs that affect the central nervous system.

Sleep disorders in children and adolescents

Pediatricians estimate that 20–30% of children have difficulties with sleep that are serious enough to disturb their families. Although sleepwalking and night terror disorder occur more frequently in children than in adults, children can also suffer from narcolepsy and sleep apnea syndrome.

Causes and symptoms

The causes of sleep disorders have already been discussed with respect to the classification of these disorders.

The most important symptoms of sleep disorders are insomnia and sleepiness during waking hours. Insomnia is by far the more common of the two symptoms. It covers a number of different patterns of sleep disturbance. These patterns include inability to fall asleep at bedtime, repeated awakening during the night, and/or inability to go back to sleep once awakened.

Diagnosis

Diagnosis of sleep disorders usually requires a psychological history as well as a medical history. With the exception of sleep apnea syndromes, physical examinations are not usually revealing. The patient's sex and age are useful starting points in assessing the problem. The doctor may also talk to other family members in order to obtain information about the patient's symptoms. The family's observations are particularly important to evaluate sleepwalking, kicking in bed, snoring loudly, or other behaviors that the patient cannot remember.

Sleep logs

Many doctors ask patients to keep a sleep diary or sleep log for a minimum of one to two weeks in order to evaluate the severity and characteristics of the sleep disturbance. The patient records medications taken as well as the length of time spent in bed, the quality of the sleep, and similar information. Some sleep logs are designed to indicate circadian sleep patterns as well as simple duration or restfulness of sleep.

Psychological testing

The doctor may use **psychological tests** or inventories to evaluate insomnia because it is frequently



A patient suffering from acute sleep apnea is hooked up to monitors in preparation for a night's sleep at a Stanford University sleep lab. (Photograph by Russell D. Curtis, Photo Researchers, Inc. Reproduced by permission.)

associated with mood or affective disorders. The **Minnesota Multiphasic Personality Inventory** (MMPI), the Millon Clinical Multiaxial Inventory (MCMI), the Beck Depression Inventory, and the Zung Depression Scale are the tests most commonly used in evaluating this symptom.

SELF-REPORT TESTS. The Epworth Sleepiness Scale, a self-rating form recently developed in Australia, consists of eight questions used to assess daytime sleepiness. Scores range from 0–24, with scores higher than 16 indicating severe daytime sleepiness.

Laboratory studies

If the doctor is considering breathing-related sleep disorders, myoclonus, or narcolepsy as possible diagnoses, he or she may ask the patient to be tested in a sleep laboratory or at home with portable instruments.

POLYSOMNOGRAPHY. Polysomnography can be used to help diagnose sleep disorders as well as conduct research into sleep. In some cases the patient is

tested in a special sleep laboratory. The advantage of this testing is the availability and expertise of trained technologists, but it is expensive. As of 2001, however, portable equipment is available for home recording of certain specific physiological functions.

MULTIPLE SLEEP LATENCY TEST (MSLT). The multiple sleep latency test (MSLT) is frequently used to measure the severity of the patient's daytime sleepiness. The test measures sleep latency (the speed with which the patient falls asleep) during a series of planned naps during the day. The test also measures the amount of REM sleep that occurs. Two or more episodes of REM sleep under these conditions indicates narcolepsy. This test can also be used to help diagnose primary hypersomnia.

REPEATED TEST OF SUSTAINED WAKEFULNESS (RTSW). The repeated test of sustained wakefulness (RTSW) is a test that measures sleep latency by challenging the patient's ability to stay awake. In the RTSW, the patient is placed in a quiet room with dim lighting and is asked to stay awake. As with the MSLT, the testing pattern is repeated at intervals during the day.

Treatment

Treatment for a sleep disorder depends on what is causing the disorder. For example, if major depression is the cause of insomnia, then treatment of the depression with antidepressants should resolve the insomnia.

Medications

Sedative or hypnotic medications are generally recommended only for insomnia related to a temporary stress (like surgery or grief) because of the potential for **addiction** or overdose. Trazodone, a sedating antidepressant, is often used for chronic insomnia that does not respond to other treatments. Sleep medications may also cause problems for elderly patients because of possible interactions with their other prescription medications. Among the safer hypnotic agents are lorazepam, temazepam, and zolpidem. Chloral hydrate is often preferred for short-term treatment in elderly patients because of its mildness. Short-term treatment is recommended because this drug may be habit forming.

Narcolepsy is treated with stimulants such as dextroamphetamine sulfate or methylphenidate. Nocturnal myoclonus has been successfully treated with clonazepam.

Children with sleep terror disorder or sleepwalking are usually treated with **benzodiazepines** because this type of medication suppresses stage 3 and stage 4 NREM sleep.

Psychotherapy

Psychotherapy is recommended for patients with sleep disorders associated with other mental disorders. In many cases the patient's scores on the Beck or Zung inventories will suggest the appropriate direction of treatment.

Sleep education

"Sleep hygiene" or sleep education for sleep disorders often includes instructing the patient in methods to enhance sleep. Patients are advised to:

- wait until he or she is sleepy before going to bed
- avoid using the bedroom for work, reading, or watching television
- get up at the same time every morning no matter how much or how little he or she slept
- avoid **smoking** and avoid drinking liquids with caffeine

- get some physical **exercise** early in the day every day
- limit fluid intake after dinner; in particular, avoid alcohol because it frequently causes interrupted sleep
- learn to meditate or practice relaxation techniques
- avoid tossing and turning in bed; instead, he or she should get up and listen to relaxing music or read

Lifestyle changes

Patients with sleep apnea or hypopnea are encouraged to stop smoking, avoid alcohol or drugs of abuse, and lose weight in order to improve the stability of the upper airway.

In some cases, patients with sleep disorders related to jet lag or shift work may need to change employment or travel patterns. Patients may need to avoid rapid changes in shifts at work.

Children with nightmare disorder may benefit from limits on television or movies. Violent scenes or frightening science fiction stories appear to influence the frequency and intensity of children's nightmares.

Surgery

Although making a surgical opening into the windpipe (a tracheostomy) for sleep apnea or hypopnea in adults is a treatment of last resort, it is occasionally performed if the patient's disorder is life threatening and cannot be treated by other methods. In children and adolescents, surgical removal of the tonsils and adenoids is a fairly common and successful treatment for sleep apnea. Most sleep apnea patients are treated with continuous positive airway pressure (CPAP). Sometimes an oral prosthesis is used for mild sleep apnea.

Alternative treatment

Some alternative approaches may be effective in treating insomnia caused by **anxiety** or emotional stress. **Meditation** practice, breathing exercises, and **yoga** can break the vicious cycle of sleeplessness, worry about inability to sleep, and further sleeplessness for some people. Yoga can help some people to relax muscular tension in a direct fashion. The breathing exercises and meditation can keep some patients from obsessing about sleep.

Homeopathic practitioners recommend that people with chronic insomnia see a professional homeopath. They do, however, prescribe specific remedies for at-home treatment of temporary insomnia: *Nux vomica* for alcohol or substance-related insomnia, *Ignatia* for insomnia caused by

KEY TERMS

Apnea—The temporary absence of breathing. Sleep apnea consists of repeated episodes of temporary suspension of breathing during sleep.

Cataplexy—Sudden loss of muscle tone (often causing a person to fall), usually triggered by intense emotion. It is regarded as a diagnostic sign of narcolepsy.

Circadian rhythm—Any body rhythm that recurs in 24-hour cycles. The sleep-wake cycle is an example of a circadian rhythm.

Dyssomnia—A primary sleep disorder in which the patient suffers from changes in the quantity, quality, or timing of sleep.

Electroencephalogram (EEG)—The record obtained by a device that measures electrical impulses in the brain.

Hypersomnia—An abnormal increase of 25% or more in time spent sleeping. Patients usually have excessive daytime sleepiness.

Hypnotic—A medication that makes a person sleep.

Hypopnea—Shallow or excessively slow breathing usually caused by partial closure of the upper airway during sleep, leading to disruption of sleep.

Insomnia—Difficulty in falling asleep or remaining asleep.

Jet lag—A temporary disruption of the body's sleep-wake rhythm following high-speed air travel across several time zones. Jet lag is most severe in people who have crossed eight or more time zones in 24 hours.

Kleine-Levin syndrome—A disorder that occurs primarily in young males, three or four times a year. The syndrome is marked by episodes of hypersomnia, hypersexual behavior, and excessive eating.

Narcolepsy—A life-long sleep disorder marked by four symptoms: sudden brief sleep attacks, cataplexy, temporary paralysis, and hallucinations. The hallucinations are associated with falling asleep or the transition from sleeping to waking.

Nocturnal myoclonus—A disorder in which the patient is awakened repeatedly during the night by

cramps or twitches in the calf muscles. Nocturnal myoclonus is sometimes called periodic limb movement disorder (PLMD).

Non-rapid eye movement (NREM) sleep—A type of sleep that differs from rapid eye movement (REM) sleep. The four stages of NREM sleep account for 75–80% of total sleeping time.

Parasomnia—A primary sleep disorder in which the person's physiology or behaviors are affected by sleep, the sleep stage, or the transition from sleeping to waking.

Pavor nocturnus—Another term for sleep terror disorder.

Polysomnography—Laboratory measurement of a patient's basic physiological processes during sleep. Polysomnography usually measures eye movement, brain waves, and muscular tension.

Primary sleep disorder—A sleep disorder that cannot be attributed to a medical condition, another mental disorder, or prescription medications or other substances.

Rapid eye movement (REM) sleep—A phase of sleep during which the person's eyes move rapidly beneath the lids. It accounts for 20–25% of sleep time. Dreaming occurs during REM sleep.

REM latency—After a person falls asleep, the amount of time it takes for the first onset of REM sleep.

Restless legs syndrome (RLS)—A disorder in which the patient experiences crawling, aching, or other disagreeable sensations in the calves that can be relieved by movement. RLS is a frequent cause of difficulty falling asleep at night.

Sedative—A medication given to calm agitated patients; sometimes used as a synonym for hypnotic.

Sleep latency—The amount of time that it takes to fall asleep. Sleep latency is measured in minutes and is important in diagnosing depression.

Somnambulism—Another term for sleepwalking.

grief, *Arsenicum* for insomnia caused by fear or anxiety, and *Passiflora* for insomnia related to mental stress.

Melatonin has also been used as an alternative treatment for sleep disorders. Melatonin is produced in the body by the pineal gland at the base of the brain.

This substance is thought to be related to the body's circadian rhythms.

Practitioners of Chinese medicine usually treat insomnia as a symptom of excess yang energy. Cinnabar is recommended for chronic nightmares. Either magnetic magnetite or “dragon bones” is recommended

for insomnia associated with **hysteria** or fear. If the insomnia appears to be associated with excess yang energy arising from the liver, the practitioner will give the patient oyster shells. **Acupuncture** treatments can help bring about balance and facilitate sleep.

Dietary changes like eliminating stimulant foods (coffee, cola, chocolate) and late-night meals or snacks can be effective in treating some sleep disorders. Nutritional supplementation with magnesium, as well as botanical medicines that calm the nervous system, can also be helpful. Among the botanical remedies that may be effective for sleep disorders are valerian (*Valeriana officinalis*), passionflower (*Passiflora incarnata*), and skullcap (*Scutellaria lateriflora*).

Prognosis

The prognosis depends on the specific disorder. Children usually outgrow sleep disorders. Patients with Kleine-Levin syndrome usually get better around age 40. Narcolepsy is a life-long disorder. The prognosis for sleep disorders related to other conditions depends on successful treatment of the substance abuse, medical condition, or other mental disorder. The prognosis for primary sleep disorders is affected by many things, including the patient's age, sex, occupation, personality characteristics, family circumstances, neighborhood environment, and similar factors.

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Sleep study see **Polysomnography**

Sleeping drugs see **Anti-insomnia drugs**

Sleeping sickness

Definition

Sleeping sickness (also called trypanosomiasis) is an infection caused by *Trypanosoma* protozoa; it is passed to humans through the bite of the tsetse fly. If left untreated, the infection progresses to **death** within months or years.

DAVID BRUCE (1855–1931)

David Bruce was born in Melbourne, Australia, on May 29, 1855, to Scottish immigrants. Bruce's family moved back to Scotland when he was five years old. Bruce attended the University of Edinburgh where he studied natural history and medicine. Following his graduation, he accepted a position working with a doctor. In 1883, Bruce married Mary Elizabeth Steele who would help him with his work throughout his life.

In 1884, Bruce began to study the disease "Malta, Mediterranean" when he and Mary were stationed in Malta with the Army Medical Service. Using a microscope, Bruce discovered that the disease was caused by a "microoccus" that grew in the individual's spleen. The organism responsible for this disease was ultimately isolated by Bernhard L. F. Bang. In 1905, Bruce led a scientific team that discovered that the soldiers who contracted the disease had ingested the milk of infected goats. The disease disappeared when the soldiers quit drinking goat's milk. Many physicians began calling the disease "brucellosis" in honor of Bruce's discoveries. Bruce also conducted research in Africa where he found that the tsetse fly could infect humans, as well as animals, with the "nagana" disease. Ultimately, his work would prove that sleeping sickness was caused by the tsetse fly.

In 1903, Bruce became the director of the Royal Society's Sleeping Sickness Commission and, in 1908, he was knighted. He served as commandant of the Royal Army Medical College after he and his wife returned to England. Bruce died on November 20, 1931.

Description

Protozoa are single-celled organisms considered to be the simplest life form in the animal kingdom. The protozoa responsible for sleeping sickness are a variety that bear numerous flagella (hair-like projections from the cell that help the cell to move). These protozoa exist only on the continent of Africa. The type of protozoa causing sleeping sickness in humans is referred to as the *Trypanosoma brucei* complex, which can be divided further into Rhodesian (Central and East African) and Gambian (Central and West African) subspecies.

The Rhodesian variety live within antelopes in savanna and woodland areas, and they cause no problems with the antelope's health. The protozoa are then acquired by tsetse flies when they bite and suck the blood of an infected antelope or cow.

Within the tsetse fly, the protozoa cycle through several different life forms; ultimately they migrate to

the salivary glands of the tsetse fly. Once the protozoa are harbored in the salivary glands, they are ready to be deposited into the bloodstream of the fly's next source of a blood meal.

Humans most likely to become infected by Rhodesian trypanosomes are people such as game wardens and visitors to game parks in East Africa, who may be bitten by a tsetse fly that has fed on game (antelope) carrying the protozoa. The Rhodesian variety of sleeping sickness causes a much more severe illness, with even greater likelihood of eventual death than the Gambian form.

The Gambian variety of *Trypanosoma* thrives in tropical rain forests throughout Central and West Africa; it does not infect game or cattle, and is primarily a threat to people dwelling in such areas, rarely infecting visitors.

Causes and symptoms

The first sign of infection with the trypanosome may be a sore appearing at the site of the tsetse fly bite about two to three days after having been bitten. Redness, **pain**, and swelling occur, but are often ignored by the patient.

Stage I illness

Two to three weeks later, Stage I disease develops as a result of the protozoa being carried through the blood and lymph circulation of the host. This systemic (meaning that symptoms affect the whole body) phase of the illness is characterized by a **fever** that rises quite high, then falls to normal, then respikes (rises rapidly). A rash with intense **itching** may be present, and **head-ache** and mental confusion may occur. The Gambian form, in particular, includes extreme swelling of lymph tissue, with enlargement of both the spleen and liver, and greatly swollen lymph nodes. "Winterbottom's sign" is classic of Gambian sleeping sickness, and consists of a visibly swollen area of lymph nodes located behind the ear and just above the base of the neck. During this stage, the heart may be affected by a severe inflammatory reaction, particularly when the infection is caused by the Rhodesian variety of trypanosomiasis.

Many of the symptoms of sleeping sickness are actually the result of attempts by the patient's immune system to get rid of the invading organism. The heightened activity of the cells of the immune system result in damage to the patient's own organs, anemia, and leaky blood vessels. These leaks in the blood vessels end up helping to further spread the protozoa throughout the afflicted person's body.

One reason for the intense reaction of the immune system to the presence of the trypanosomes is also the reason why the trypanosomes survive so well despite the efforts of the immune system to eradicate them. The protozoa causing sleeping sickness are able to rapidly change specific markers (unique proteins) on their outer coats. These kinds of markers usually serve to stimulate the host's immune system to produce immune cells that will specifically target the marker, allowing quick destruction of those cells bearing the markers. Trypanosomes, however, are able to express new markers at such a high rate of change that the host's immune system is constantly trying to catch up.

Stage II illness

Stage II sleeping sickness involves the nervous system. Gambian sleeping sickness, in particular, has a clearly delineated phase in which the predominant symptoms involve the brain. The patient's speech becomes slurred, mental processes slow, and the patient sits and stares for long periods of time, or sleeps. Other symptoms resemble Parkinson's disease, including imbalance when walking, slow and shuffling gait, trembling of the limbs, involuntary movements, muscle tightness, and increasing mental confusion. Untreated, these symptoms eventually lead to **coma** and then to death.

Diagnosis

Diagnosis of sleeping sickness can be made by microscopic examination of fluid from the original sore at the site of the tsetse fly bite. Trypanosomes will be present in the fluid for a short period of time following the bite. If the sore has already resolved, fluid can be obtained from swollen lymph nodes for examination. Other methods of trypanosome diagnosis involve culturing blood, lymph node fluid, bone marrow, or spinal fluid. These cultures are then injected into rats, which develop blood-borne protozoa infection that can be detected in blood smears within one to two weeks. However, this last method is effective only for the Rhodesian variety of sleeping sickness.

Treatment

Without treatment, sleeping sickness will lead to death. Unfortunately, however, those medications effective against the *Trypanosoma brucei* complex protozoa all have significant potential side effects for the patient. Suramin, eflornithine, pentamidine, and

KEY TERMS

Immune system—That network of tissues and cells throughout the body that is responsible for ridding the body of any invaders, such as viruses, bacteria, protozoa, etc.

Protozoa—Single-celled organisms considered to be the simplest life form in the animal kingdom.

several drugs that contain arsenic (a chemical which in higher doses is highly poisonous to humans), are all effective anti-trypanosomal agents. Each of these drugs, however, requires careful monitoring to ensure that the drugs themselves do not cause serious complications such as fatal hypersensitivity (allergic) reaction, kidney or liver damage, or inflammation of the brain.

Prevention

Prevention of sleeping sickness requires avoiding contact with the tsetse fly. Insect repellents and clothing that covers the limbs to the wrists and ankles are advisable. Public health measures have included drug treatment of humans who are infected with one of the *Trypanosoma brucei* complex. There are currently no immunizations available to prevent the acquisition of sleeping sickness.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Sleepwalking see **Sleep disorders**

Slipped disk see **Herniated disk**

Slit lamp examination see **Eye examination**

Small-for-gestational-age infant see
Intrauterine growth retardation

Small bowel follow-through (SBFT) see
Upper GI exam

Small cell lung cancer see **Lung cancer, small cell**

Small intestine biopsy

Definition

A biopsy is a diagnostic procedure in which tissue or cells are removed from a part of the body and specially prepared for examination under a microscope. When the tissue involved is part of the small intestine, the procedure is called a small-intestine (or small-bowel) biopsy.

Purpose

The small-bowel biopsy is used to diagnose and confirm disease of the intestinal mucosa (the lining of the small intestine). The test is most commonly done to test for tumors of the small bowel or malabsorption syndromes.

Precautions

Due to the slight risk of bleeding during or after this procedure, **aspirin**, aspirin-containing medications, **nonsteroidal anti-inflammatory drugs**, and anticoagulants and **antiplatelet drugs** should be withheld for at least five days before the test.

Description

The small intestine is approximately 21 ft (6.4m). It has three sections: the duodenum (a short, curved segment fixed to the back wall of the abdomen), the jejunum, and the ileum (two larger, coiled, and mobile segments). Some digestion occurs in the stomach, but the small intestine is mainly responsible for digestion and absorption of foods.

Malabsorption syndromes occur when certain conditions result in impaired absorption of nutrients, **vitamins**, or **minerals** from the diet by the lining of the small intestine. For example, injury to the intestinal lining can interfere with absorption, as can infections, intestinal parasites, some drugs, blockage of the lymphatic vessels, poor blood supply to the intestine, or diseases like sprue.

Malabsorption is suspected when a patient not only loses weight, but has **diarrhea** and nutritional deficiencies despite eating well (weight loss alone can have other causes). Laboratory tests like fecal fat, a measurement of fat in stool samples collected over 72 hours, are the most reliable tests for diagnosing fat malabsorption, but abnormalities of the small intestine itself are diagnosed by small-intestine biopsy.

Several different methods are used to detect abnormalities of the small intestine. A tissue specimen can be obtained by using an endoscope (a flexible viewing tube), or by using a thin tube with a small cutting instrument at the end. This latter procedure is ordered when specimens larger than those provided by endoscopic biopsy are needed, because it allows removal of tissue from areas beyond the reach of an endoscope.

Several similar types of capsules are used for tissue collection. In each, a mercury-weighted bag is attached to one end of the capsule, while a thin polyethylene tube about 5 ft (1.5m) long is attached to the other end. Once the bag, capsule, and tube are in place in the small bowel, suction on the tube draws the tissue into the capsule and closes it, cutting off the piece of tissue within. This is an invasive procedure, but it causes little **pain** and complications are rare.

A newer method of obtaining diagnostic information about the small intestine was approved by the Food and Drug Administration (FDA) in 2001. Known as the M2A Imaging System, the device was developed by a company in Atlanta, Georgia. The M2A system consists of an imaging capsule, a portable belt-pack image receiver and recorder, and a specially modified computer. The patient swallows the capsule, which is the size of a large pill. A miniature lens in the capsule transmits images through an antenna/transmitter to the belt-pack receiver, which the patient wears under ordinary clothing as he or she goes about daily activities. The belt-pack recording device is returned after seven or eight hours to the doctor, who then examines the images recorded as a digital video. The capsule itself is simply allowed to pass through the digestive tract.

Preparation requires only **fasting** the night before the M2A examination and taking nothing but clear liquids for two hours after swallowing the capsule. After four hours the patient can eat food without interfering with the test. As of the early 2000s, the M2A system is used to evaluate gastrointestinal bleeding from unknown causes, inflammatory bowel disease, some malabsorption syndromes, and to monitor surgical patients following small-bowel transplantation. The system has shown good results in detecting **Crohn's disease** undiagnosed by conventional methods.

Small-intestine biopsy procedure

After application of a topical anesthetic to the back of the patient's throat, the capsule and the tube are introduced, and the patient is asked to swallow as the tube is advanced. The patient is then placed on the right side and the instrument tip is advanced another 20 in (51cm) or so. The tube's position is checked by

fluoroscopy or by instilling air through the tube and listening with a stethoscope for air to enter the stomach.

The tube is advanced 2–4 in (5.1–10 cm) at a time to pass the capsule through the stomach outlet (pylorus). When fluoroscopy confirms that the capsule has passed the pylorus, small samples of small intestine tissue are obtained by the instrument's cutting edge, after which the instrument and tube are withdrawn. The entire procedure may be completed in minutes.

Preparation

This procedure requires tissue specimens from the small intestine through means of a tube inserted into the stomach through the mouth. The patient is to withhold food and fluids for at least eight hours before the test.

Aftercare

The patient should not have anything to eat or drink until the topical anesthetic wears off (usually about one to two hours). If intravenous sedatives were administered during the procedure, the patient should not drive for the remainder of the day. Complications from this procedure are uncommon, but can occur. The patient is to note any abdominal pain or bleeding and report either immediately to the doctor.

Risks

Complications from this procedure are rare, but can include bleeding (hemorrhage), bacterial infection with **fever** and pain, and bowel puncture (perforation). The patient should immediately report any abdominal pain or bleeding to the physician in charge. Biopsy is contraindicated in uncooperative patients, those taking aspirin or anticoagulants, and in those with uncontrolled bleeding disorders.

Normal results

Normal results are no abnormalities seen on gross examination of the specimen(s) or under the microscope after tissue preparation.

Abnormal results

Small-intestine tissue exhibiting abnormalities may indicate Whipple's disease, a malabsorption disease; lymphoma, a group of cancers; and parasitic infections like **giardiasis**, strongyloidiasis, and coccidiosis. When biopsy indicates celiac sprue (a malabsorption disorder), infectious **gastroenteritis**

KEY TERMS

Sprue—A disorder of impaired absorption of nutrients from the diet by the small intestine (malabsorption), resulting in malnutrition. Two forms of sprue exist: tropical sprue, which occurs mainly in tropical regions; and celiac sprue, which occurs more widely and is due to sensitivity to the wheat protein gluten.

Whipple's disease—A disorder of impaired absorption of nutrients by the small intestine. Symptoms include diarrhea, abdominal pain, progressive weight loss, joint pain, swollen lymph nodes, abnormal skin pigmentation, anemia, and fever. The precise cause is unknown, but it is probably due to an unidentified bacterial infection.

Wireless capsule endoscopy—A newer method of examining the small bowel by means of a capsule swallowed by the patient. The capsule contains a miniaturized lens and an antenna that transmits information to a belt-pack recorder worn by the patient during the day.

(inflammation of the gastrointestinal tract), folate and B₁₂ deficiency, or **malnutrition**, confirmation studies are needed for conclusive diagnosis.

Resources

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Smallpox

Definition

Smallpox is an infection caused by the variola virus, a member of the poxvirus family. Throughout history, smallpox has been a greatly feared disease because it was responsible for huge epidemics worldwide that resulted in large numbers of deaths. In 1980, the World Health Organization (WHO) announced that an extensive program of **vaccination** against the disease had resulted in the complete eradication of the virus, with the exception of samples of stored virus in two laboratories.

Description

Smallpox is strictly an infection of human beings. Animals and insects can neither be infected by smallpox, nor carry the virus in any form. Most infections are caused by contact with a person who has already developed the characteristic **skin lesions** (pox) of the disease, although a person who has a less severe infection (not symptomatic or diagnosable in the usual way) can unwittingly spread the virus.

Causes and symptoms

Smallpox is a relatively contagious disease, which accounts for its ability to cause massive epidemics. The variola virus is acquired from direct contact with individuals infected with the disease, from contaminated air droplets, and even from objects used by another smallpox victim (books, blankets, utensils, etc.). The respiratory tract is the usual entry point for the variola virus into a human being.

After the virus enters the body, there is a 12–14 day incubation period during which the virus multiplies, although no symptoms are recognizable. After the incubation period, symptoms appear abruptly and include **fever**, chills, and muscle aches. Two to three days later, a bumpy rash begins appearing first on the face and forearms. The rash progresses, ultimately reaching the chest, abdomen, and back. Seven to ten days after the rash appears, the patient is most infectious. The individual bumps (papules) fill with clear fluid and eventually become pus-filled over the course of 10–12 days. These pox eventually scab over, each leaving a permanently scarred pock or pit when the scab drops off.

Initially, the smallpox symptoms and rash appear similar to **chickenpox**. However, unlike chickenpox, smallpox lesions develop at the same rate so that

they are all visible in the same stage. Another major difference is that smallpox occurs primarily on the face and extremities, whereas chickenpox tends to be concentrated on the face and trunk area.

Complications such as bacterial infection of the open skin lesions, **pneumonia**, or bone infections are the major causes of **death** from smallpox. A very severe and quickly fatal form called “sledgehammer smallpox,” occurs in 5-10% of patients and results in massive, uncontrollable bleeding (hemorrhage) from the skin lesions, as well as from the mouth, nose, and other areas of the body. This form is very infectious and usually fatal five to seven days after onset.

Fear of smallpox comes from both the epidemic nature of the disease, as well as from the fact that no therapies have ever been discovered to either treat the symptoms of smallpox, or shorten the course of the disease.

Diagnosis

In modern times, a diagnosis of smallpox is made using an electron microscope to identify virus in fluid from the papules, urine, or in the patient’s blood prior to the appearance of the papular rash.

Treatment

No treatments have been developed to halt the progression of the disease. Treatment for smallpox is only supportive, meaning that it is aimed at keeping a patient as comfortable as possible. **Antibiotics** are sometimes administered to prevent secondary bacterial infections.

Prognosis

Approximately one in three patients die from smallpox, with the more severe, hemorrhagic form nearly 100% fatal. Patients who survive smallpox infection nearly always have multiple areas of scarring where each pock has been.

Prevention

From about the tenth century in China, India, and the Americas, it is noted that individuals who had even a mild case of smallpox could not be infected again. Fascinating accounts appear in writings from all over the world of ways in which people tried to prevent smallpox. Material from people mildly ill with smallpox (fluid or pus from the papules, scabs over the pox) was scratched into the skin of people who had never had the illness, in an attempt to produce a mild



Smallpox pustules on the arm of an Asian Indian man. (Photograph by C. James Webb, Phototake NYC. Reproduced by permission.)

reaction and its accompanying protective effect. These efforts often resulted in full-fledged smallpox, and probably served only to help effectively spread the infection throughout a community. In fact, such crude smallpox “vaccinations” were against the law in Colonial America.

In 1798, Edward Jenner published a paper in which he discussed his important observation that milkmaids who contracted a mild infection of the hands (called cowpox, and caused by a relative of the variola virus) appeared to be immune to smallpox. Jenner created an immunization against smallpox using the pus found in the lesions of cowpox infection. Jenner’s paper led to much work in the area of vaccinations, and ultimately resulted in the creation of a very effective vaccination against smallpox that utilized the vaccinia virus, another close relative of variola. Indeed, the term vaccination is derived from *vacce*, Latin for cow and related to the

cowpox link. Later, the term was applied to other vaccinations.

In 1967, WHO began its attempt to eradicate the smallpox virus worldwide. The methods used in the program were simple:

- Careful surveillance for all smallpox infections worldwide, to allow for quick diagnosis and immediate quarantine of patients.
- Immediate vaccination of all contacts diagnosed with infection, in order to interrupt the virus' usual pattern of infection.

The WHO's program was extremely successful, and the virus was declared eradicated worldwide in May 1980.

Future concerns

Today, two laboratories (the Centers for Disease Control and Prevention in Atlanta, Georgia, and the Russian State Centre for Research on Virology and Biotechnology in Koltsovo, Novosibirsk Region) officially retain samples of the smallpox virus. These samples, as well as stockpiles of the smallpox vaccine, are stored because some level of concern exists that another poxvirus could undergo genetic changes (mutate) and cause human infection. There is also the remote chance that smallpox virus could somehow escape from the laboratories where it is stored. For these reasons, surveillance continues of various animal groups that continue to be infected with viruses related to the variola virus, and large quantities of vaccine are stored in different countries around the world, so that response to any future threat by the smallpox virus could be prompt.

Of greatest concern is the potential use of smallpox as a biological weapon. Since 1980, when the WHO announced smallpox had been eradicated, essentially no one has been vaccinated against the disease. Those individuals vaccinated prior to 1980 are believed to be susceptible as well because immunity only lasts 15-20 years. These circumstances coupled with the nature of smallpox to spread quickly from person to person could lead to devastating consequences.

The United States and Russia are the only two countries to officially house remaining samples of the virus. However, it is believed that other countries, such as Iraq, may have obtained samples of the smallpox virus during the Cold War through their association with Russia. It is also possible that scientists with access to the virus may have sold their services and knowledge to other governments.

KEY TERMS

Epidemic—A situation in which a particular infection is experienced by a very large percentage of the people in a given community within a given time frame.

Eradicate—To completely do away with something, eliminate it, end its existence.

Hemorrhage—Bleeding that is massive, uncontrollable, and often life-threatening.

Lesion—The tissue disruption or the loss of function caused by a particular disease process.

Papules—Firm bumps on the skin.

Pox—A pus-filled bump on the skin.

Vaccine—A preparation using a non-infectious element or relative of a particular virus or bacteria, and administered with the intention of halting the progress of an infection, or completely preventing it.

On June 22-23, 2001, four U.S. organizations (CSIS—Center for Strategic and International Studies, Johns Hopkins Center for Civilian Biodefense Studies, ANSER—Analytic Services Inc., and MIPT—Memorial Institute for the Prevention of Terrorism) presented a fictitious scenario of the United States' response to a deliberate introduction of smallpox titled *Dark Winter*. This **exercise** demonstrated that if such an event were to occur, the United States would be ill prepared on several fronts. The primary concern is an inadequate supply of vaccine, which is essential to preventing disease development in exposed persons. Between 1997 and 2001, two companies were contracted to produce additional smallpox vaccines for both military and civilian use. Through these contracts, an additional 40 million doses would be made available for civilian use by 2005. In the meantime, studies are underway to determine if the existing vaccines can be diluted in order to increase the number of doses available for immediate use. Results from a very small group of volunteers tested in 2000 found that at one-tenth strength, the existing smallpox vaccines are approximately 70% effective. In late 2001, a new study began evaluating the effectiveness of the vaccine at one-fifth strength.

In the event that smallpox is reintroduced into the current population, it will be imperative that doctors immediately recognize the symptoms and isolate the individual to prevent further spread of the disease. Prompt vaccination of any persons who had contact

with the patient is also necessary to prevent additional cases of smallpox from developing. Controlling and containing spread of this disease is critical for prevention of a world-wide epidemic that would have a devastating impact on current populations.

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Rosalyn Carson-DeWitt, MD

Smelling disorders

Definition

Smelling disorders are disturbances of the olfactory sense, which is known as the sense of smell. These nasal dysfunctions range from the total loss of smell (**anosmia**) to dysosmia, a distorted sense of smell.

Description

An awareness of how the olfactory system works is helpful for understanding how smelling disorders affect the sense of smell. People detect odors because sensory receptors located in the nose carry smell sensations to the brain. The receptors, which are nerve cell endings, are found in the mucous membrane in the

roof of the nose. This section of the nose called the olfactory area is located just below the brain's frontal lobes.

In the olfactory area are millions of tiny olfactory cells. Each cell contains about 12 cilia, tiny hairs that extend into a mucus layer. The mucus moistens the cilia. Mucus also catches odor molecules, while receptors in the cilia stimulate the molecules and send nerve impulses to the brain.

Olfactory nerve fibers carry the impulse to two olfactory bulbs located in the brain. Information is processed in the bulbs and then sent to the cerebral cortex. Once the transmission is inside the smell center of the brain, a person experiences the sense of smell.

A person with a normal sense of smell (normosmia) is able to distinguish 10,000 odors. The sense of smell stimulates salivary glands. As a result, smelling disorders often affect the sense of taste. The olfactory sense allows people to experience pleasurable odors like the scent of roses. And smell is thought to contribute to sexual attraction.

A smelling disorder that affects the sense of smell is generally not life-threatening. However, it can be dangerous. Without a sense of smell, a person might eat spoiled food. Lack of a sense of smell could pose a health risk if a person has little appetite and fails to eat enough. Furthermore, without a sense of smell, a person might not detect a gas leak or the smell of something burning. Loss of smell and the resulting loss of taste may lead to depression.

Types of smelling disorders

Smelling disorders differ in the way that the sense of smell is affected and how long a person has the disorder. For example, anosmia, the loss of the sense of smell, is often a temporary symptom of a cold or flu. However, a **head injury** could cause permanent anosmia. In addition, a head injury could produce dysosmia, the distorted sense of smell that could cause a person to hallucinate a foul odor.

Smelling disorders are categorized as:

- Anosmia, the loss of the sense of smell. It is the most common smelling disorder. This condition can be temporary or permanent.
- Dysosmia is a distorted sense of smell. A person senses non-existent unpleasant odors. It can be caused by medical and mental conditions.
- Hyperosmia is an increased sensitivity to smell. It can be a characteristic of someone with a neurotic or histrionic personality.

- Hyposmia is the diminished sense of smell. This is usually a temporary condition that a person may experience after a case of acute **influenza**. Sometimes this condition is referred to as partial anosmia.
- Presbyosmia refers to the lessening or loss of the olfactory sense that occurs when a person ages.

Smelling disorder demographics

Anosmia occurs in about 10% of head trauma injuries, and head trauma is a leading cause of anosmia in young adults. In older adults, the disorder is generally caused by viral infection. **Aging** may also bring a loss of the sense of smell. In rare cases, anosmia is inherited. It is a symptom of male **hypogonadism** (Kallmann's syndrome).

Olfactory **hallucinations** known as dysosmia are generally associated with psychological conditions. In some cases, people may believe they are the source of foul odors.

Causes and symptoms

Anosmia is the most common type of smelling disorder. Loss of the olfactory sense is generally caused by nasal congestion or obstruction. Temporary partial anosmia often occurs when a person has a cold, the flu, or some types of **rhinitis**, especially hay **fever (allergic rhinitis)**. During these conditions, nasal mucus membranes become inflamed. Other causes for anosmia are:

- **Nasal polyps** and other disorders that prevent air from getting to the area in the nose where the smell receptors are found. Hay fever or an allergy may cause one or more polyps to show up.
- Viral upper respiratory infection.
- Atrophic rhinitis. This condition causes mucus membrane to waste away. The person may experience some level of permanent anosmia. One symptom of this condition is that a person expels a foul-smelling discharge.
- Hypertrophic rhinitis. Mucous membrane thickens, covering the olfactory nerve endings. If not treated, hypertrophic rhinitis can lead to permanent anosmia. This discharge could overpower other odors.
- Cigarettes. **Smoking** aggravates the nose's membrane and intensifies nasal polyp symptoms.
- A crooked nose or a deviated septum.
- When the olfactory bulbs, tracts, or central connections are destroyed. This can occur in situations such as head trauma, infections or nasal or sinus surgery.
- Head injury. If both olfactory nerves are torn during a head injury, permanent anosmia results.
- Medications such as **antihistamines** and **decongestants**, especially prolonged use of decongestants.
- Drugs like amphetamines, estrogen, naphazoline, phenothiazines, and reserpine.
- The aging process may cause the sense to lessen. In most cases, there is no other obvious cause for the disorder.
- A tumor behind the nose or in the membranes surrounding the brain.
- Lead poisoning.
- Exposure to insecticides or other chemicals.
- Radiation therapy.
- Nervous disorders.
- Idiopathic loss, which means there is no diagnosable cause for the condition.

Anosmia symptoms

Most people with anosmia can distinguish salty, sweet, bitter, and sour tastes since the tongue senses these tastes. However, people with anosmia cannot sense other tastes. Since taste is largely based on the olfactory sense, people complain of losing the sense of taste (ageusia).

Dysosmia

Infected nasal sinuses and damage to the olfactory bulbs can cause dysosmia, the distorted sense of smell. Head trauma can cause this disorder. Poor **oral hygiene** can lead to dysosmia. In these cases, a person may also find that disagreeable odors are accompanied by the sensing of unpleasant tastes. In addition, brain-stem disease can cause smelling disorders. An epileptic seizure can include olfactory hallucinations.

Mental conditions such as depression and **schizophrenia** may be accompanied by dysosmia. In addition, when people who are person severely dependent on alcohol quit drinking, they may experience dysosmia.

Diagnosis

If a smelling disorder is a symptom of a mental condition such as schizophrenia, diagnosis should be part of treatment for that condition.

When the condition is caused by a medical condition such as **allergies** or a viral infection, a person may notice that the olfactory sense is impaired during that

condition. If the smelling disorder continues after the person is well, an appointment should be made with a primarily health care provider.

Diagnosis of smelling disorders begins with a health assessment to determine the cause of the olfactory impairment. The patient's primary care doctor will ask if the patient has a cold, allergies, **sinusitis**, or an upper respiratory infection.

Treatment of a head injury or follow-up medical appointment should address smelling disorders. In all cases, discussion of the symptoms covers issues such as when the smelling disorder started, if this has been an ongoing problem, and whether the disorder is becoming more intense. The assessment will include questions about whether the patient can taste food and if the disorder affects all odors or specific smells. The patient will also be asked about medications taken.

Physical examinations

The **physical examination** will include a thorough inspection of the nose, nasopharynx, and the examination of the upper respiratory tract. The examination could include sinus transillumination, placement of a light on the face to help determine if sinuses are full. **Skull x rays** may be required to determine the presence of tumors in the nose or brain.

The patient may be referred to a neurologist;—an ear, nose, and throat specialist;—or to a center that specializes in treatment of smelling disorders.

Other diagnostic tests could be required. These include:

- A CT scan (**computed tomography scans**) of the head. Also known as a CAT scan, this process provides a more detailed image than the x ray.
- Olfactory nerve testing.
- Nasal cytology, which involves the study of mucus under a microscope.
- Testing to determine the scope of smelling disorder. A basic smell test involves the patient trying to identify each one of a group of different odors. A variation of this is a scratch-and-sniff test. The patient may be asked to differentiate among concentrations of one odor. The alcohol sniff test that involves use of a material soaked in isopropyl alcohol. Patients close their eyes and the doctor moves around. Patients tell the doctor when they smell the alcohol.
- The patient may also take a taste test.

Medical costs

The costs for diagnosis and treatment vary because of the different types of smelling disorders, the range of causes for olfactory dysfunction, and the different types of treatment.

There are also differences in what health plans require in terms of patient co-pay. A health plan could cover treatments ranging from the initial appointment with a primary care provider to the surgery to remove brain tumors.

In addition, some health plans cover costs of treatment at specialized facilities like the Center for Smell and Taste Disorders at the University of Colorado Health Sciences Center in Denver. A series of tests including a taste-and-smell test cost \$250 in May of 2001.

Treatment

Treating a condition that causes a smelling disorder can sometimes restore the olfactory sense. Treatments for smelling disorders are as varied as the olfactory dysfunctions. Treatment for smelling disorders ranges from lifestyle changes to surgery. Treatment of mental conditions could affect the smelling disorder. In some cases, the disorder can't be treated, and the person must adjust to the loss of the sense of smell. Anosmia associated with aging is not treatable.

Basic treatments for anosmia

The sense of smell should return after a condition like a cold or the flu ends. Decongestants such as Sudafed help reduce congestion related to colds, allergies, and sinus conditions. Manufacturer's dosage recommendations should be followed. If anosmia is related to excessive use of nasal decongestants, a person should discontinue use of those medications.

Saline sprays can be used to clean the interior of the nose.

If smoking causes anosmia, a person should quit smoking.

The sense of smell may return after treatment of allergic or bacterial rhinitis and sinusitis. An over-the-counter antihistamine such as Actifed may provide relief.

If allergies cause anosmia, adjustments should be made to avoid allergens. If dust causes allergies, care should be taken to clean areas such as the bedroom.

Antibiotics may be prescribed for infections.

Other medications prescribed for smelling disorders include steroids such as Prednisone. It should only be used for a short time since longterm use could lessen resistance to infection.

Surgical treatment

Removal of nasal polyps and benign tumors may cause the sense of smell to return. Polyp removal is an uncomplicated surgery. Generally, only a local anesthetic is needed.

Septoplasty straightens the nasal passage. It is generally an outpatient surgery, with local or **general anesthesia** required. **Rhinoplasty** straightens the structure of the nose. This surgery could be combined with septoplasty.

Endoscopic sinus surgery opens sinus drainage channels. This outpatient surgery is an option after a person sees no improvement after trying treatments such as medications.

Surgical treatment may not be effective in conditions that result in the destruction of the olfactory nerve or its central passages. However, regeneration of those tissues may cause the sense to return.

Enhancing taste

Without a sense of smell, most people can still taste salt and flavors that are sweet, sour, and bitter. People with anosmia could distinguish other tastes by adding spices such as pepper to food. These spices stimulate facial nerves that also sense flavors.

Alternative treatment

Alternative treatments for smelling disorder center around the theory that zinc supplements help improve the sense of smell. The supplement is said to be effective when the olfactory sense is impaired by conditions such as a head injury or an upper respiratory infection. A person should take 50 mg of zinc picolinate each day after eating. This procedure might be effective in the case of head injury. However, it may be several months before results are seen. **Acupuncture** may also produce results.

If polyps cause a smelling disorder, a change in diet could be helpful. A person should avoid dairy products, take supplements such as garlic, and follow other recommendations from a health care practitioner. A daily dosage of 5,000–10,000 mg of vitamin C could cut back on the amount of polyps. **Vitamins** should not be taken all at once. A multi-vitamin and mineral complex could also help.

KEY TERMS

Histrionic—A behavior characterized by an excitable nature and the constant desire for stimulation.

Nasopharynx—The passage that connects the nasal cavity to the top of the throat.

Neurotic—Behavior characterized by neurosis, mental functional disorders with symptoms such as anxiety, depression, compulsions, and phobias.

Polyp—A benign growth in areas such as the nasal passage.

Rhinitis—The inflammation of the mucous membrane in the nose.

Septum—A sheet of cartilage and bone that separates the nostrils.

Sinusitis—Inflammation of the paranasal sinuses because of allergic reactions or viral, bacterial, or functional infections.

Prognosis

In cases where smelling disorders are treatable, the outcome is positive because the olfactory sense is restored. In those cases where the sense of smell is lost, the person must make adjustments to adapt to life without that sense. Those adjustments include using spices like pepper to stimulate tastebuds.

Since a person with anosmia can no longer smell food to determine whether it is safe to eat, care should be taken. The person who lives with other people can ask them if food smells fresh. People who live alone should discard food if there is a chance that it has spoiled. Other home safety measures include installing smoke alarms and gas detectors. Cooking on an electric stove is preferable to a gas stove.

Furthermore, people with smelling disorders can find support groups. These are often associated with smell and taste clinics. In addition, there are on-line bulletin board where people can share experiences. One site contains descriptions of how things smell. Those words provide a connection to a missing sense in the same way that sign language allows the hearing-impaired to understand the spoken word.

Prevention

Not all causes of smelling disorders can be prevented. However, people with a disorder should not smoke and should ask those around them not to

smoke. Those with smelling disorders related to allergies should be taken to avoid allergens. Since head trauma injuries can lead to smelling disorders, people should wear protective helmets when bicycling or participating in sports like football.

Resources

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Watson, Lyall. *Jacob's Organ and the Remarkable Sense of Smell*. New York: W. W. Norton & Company, 2000.

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery. One Prince St., Alexandria, VA 22314-3357. (703) 836-4444. <<http://www.ent.org>>.

Center for Taste and Smell Disorders. University of Colorado Health Sciences Center, 4200 E. Ninth Ave., Denver, CO 80262. (303) 315-5660. <<http://www.hsc.colorado.edu>>.

U.S. Department of Health and Human Services. 200 Independence Avenue, SW, Washington, DC 20201. (877) 696-6775. <<http://www.hhs.gov>>.

Liz Swain

Smoke inhalation

Definition

Smoke inhalation is breathing in the harmful gases, vapors, and particulate matter contained in smoke.

Description

Smoke inhalation typically occurs in victims or firefighters caught in structural fires. However, cigarette **smoking** also causes similar damage on a smaller scale over a longer period of time. People who are trapped in fires may suffer from smoke inhalation independent of receiving skin **burns**; however, the incidence of smoke inhalation increases with the percentage of total body surface area burned. Smoke inhalation contributes to the total number of fire-related deaths each year for several reasons: the damage is serious; its diagnosis is not always easy and there are no sensitive diagnostic tests; and patients may not show symptoms until 24–48 hours after the event. Children under age 11 and adults over age 70 are most vulnerable to the effects of smoke inhalation.

Causes and symptoms

The harmful materials given off by combustion injure the airways and lungs in three ways: heat damage, tissue irritation, and oxygen **starvation** of tissues (asphyxiation). Signs of heat damage are singed nasal hairs, burns around and inside the nose and mouth, and internal swelling of the throat. Tissue irritation of the throat and lungs may appear as noisy breathing, coughing, hoarseness, black or gray spittle, and fluid in the lungs. Asphyxiation is apparent from **shortness of breath** and blue-gray or cherry-red skin color. In some cases, the patient may not be conscious or breathing.

Diagnosis

In addition to looking for the signs of heat damage, tissue irritation, and asphyxiation, the physician will assess the patient's breathing by the respiratory rate (number of breaths per minute) and motion of the chest as the lungs inflate and deflate. The patient's circulation is also evaluated by the pulse rate (number of heartbeats per minute) and blood pressure. Blood tests will indicate the levels of oxygen and byproducts of poisonous gases. Chest x rays are too insensitive to show damage to delicate respiratory tissues, but can show fluid in the lungs (**pulmonary edema**).

The physician may perform a **bronchoscopy**, a visual examination in which the airways and lungs are seen through a fiber optic tube inserted down the patient's windpipe (trachea). Other pulmonary function tests may be performed to measure how efficiently the lungs are working.

Treatment

Treatment will vary with the severity of the damage caused by smoke inhalation. The primary focus of treatment is to maintain an open airway and provide an adequate level of oxygen. If the airway is open and stable, the patient may be given high-flow humidified 100% oxygen by mask. If swelling of the airway tissues is closing off the airway, the patient may require the insertion of an endotracheal tube to artificially maintain an open airway.

Oxygen is often the only medication necessary. However, patients who have a **cough** with **wheezing** (bronchospasm), indicating that the bronchial airways are narrowed or blocked, may be given a bronchodilator to relax the muscles and increase ventilation. There are also antidotes for specific poisonous gases in the blood; dosage is dependent upon the level indicated by blood tests. **Antibiotics** are not given until

KEY TERMS

Asphyxiation—Oxygen starvation of tissues. Chemicals such as carbon monoxide prevent the blood from carrying sufficient oxygen to the brain and other organs. As a result, the person may lose consciousness, stop breathing, and die without artificial respiration (assisted breathing) and other means of elevating the blood oxygen level.

Hyperbaric oxygen therapy—Pure oxygen is administered to the patient in a special chamber at three times the normal atmospheric pressure. The patient gets more oxygen faster to overcome severe asphyxiation.

Pulmonary—Pertaining to the lungs.

Pulmonary edema—The filling of the lungs with fluid as the body's response to injury or infection.

sputum and blood cultures confirm the presence of a bacterial infection.

In institutions where it is available, hyperbaric **oxygen therapy** may be used to treat smoke inhalation resulting in severe carbon monoxide or cyanide **poisoning**. This treatment requires a special chamber in which the patient receives pure oxygen at three times the normal atmospheric pressure, thus receiving more oxygen faster to overcome loss of consciousness, altered mental state, cardiovascular dysfunction, pulmonary **edema**, and severe neurological damage.

Alternative treatment

Botanical medicine can help to maintain open airways and heal damaged mucous membranes. It can also help support the entire respiratory system. **Acupuncture** and homeopathic treatment can provide support to the whole person who has suffered a traumatic injury such as smoke inhalation.

Prognosis

Although the outcome depends of the severity of the smoke inhalation and the severity of any accompanying burns or other injuries, with prompt medical treatment, the prognosis for recovery is good. However, some patients may experience chronic pulmonary problems following smoke inhalation, and those with **asthma** or other chronic respiratory conditions prior to smoke inhalation may find their original conditions have been aggravated by the inhalation injury.

Prevention

Smoke inhalation is best avoided by preventing structural fires. This includes inspection of wiring, safe use and storage of flammable liquids, and maintenance of clean, well-ventilated chimneys, wood stoves, and space heaters. Properly placed and working smoke detectors in combination with rapid evacuation plans will minimize a person's exposure to smoke in the event of a fire. When escaping a burning building, a person should move close to the floor where there is more cool, clear air to breathe because hot air rises, carrying gases and particulate matter upward. Finally, firefighters should wear proper protective gear.

Resources

OTHER

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Bethany Thivierge

Smoking-cessation drugs

Definition

Smoking cessation drugs are medicines that help people stop smoking cigarettes or using other forms of tobacco.

Purpose

People who smoke cigarettes or use other forms of tobacco often have a difficult time when they try to stop. The difficulty is partly psychological; they get in the habit of using tobacco at certain times of day or while they are doing certain things, such as having a cup of coffee or reading the newspaper. But the habit is also hard to break for physical reasons. Tobacco contains nicotine, a drug that is as addictive as **cocaine** or heroin. Of those who have ever tried even a single cigarette, about a third will become nicotine-dependent. A person who is addicted to nicotine has withdrawal symptoms, such as irritability, **anxiety**, difficulty concentrating, and craving for tobacco when he or she stops using tobacco.

Some people can stop smoking through will-power alone, but most do better if they have support from friends, family, a physician or pharmacist, or a formal stop-smoking program. Heavy tobacco users



The nicotine patch brand Nicoderm. (Fieldmark Publications. Reproduced by permission.)

may find that smoking cessation products also help by easing their withdrawal symptoms. Most smoking cessation products contain nicotine, but the nicotine is delivered in small, steady doses spread out over many hours. In contrast, when a person inhales a cigarette, nicotine enters the lungs and then travels to the brain within seconds, delivering the “rush” that smokers come to crave. Another difference is that smoking cessation products do not contain the tar, carbon monoxide, and other toxins that make cigarettes so harmful to people’s health. According to one Canadian study, tobacco smoke contains over 40 different chemicals known to cause **cancer**.

The importance of smoking cessation is reflected in legal penalties against the tobacco industry for its long-standing denial of the harm caused by tobacco

products. Recent legal findings against the tobacco industry have led to legislation in three states concerning lawsuits against the industry. In Florida, state agencies can sue on behalf of Medicaid recipients for repayment of benefits. Maryland allows the use of statistical analysis in lawsuits against tobacco companies. In Vermont, the state can bring direct lawsuits against tobacco manufacturers to recover Medicaid benefits for tobacco-related illnesses paid after April 1998. Nineteen states (Alabama, Alaska, Connecticut, Florida, Hawaii, Louisiana, Maryland, Massachusetts, Minnesota, Mississippi, Montana, New Hampshire, New York, Rhode Island, Texas, Vermont, Virginia, Washington, and Wisconsin) have set aside as of 2001 a portion of their money from tobacco settlements to smoking prevention programs.

Description

Nicotine replacement products

Smoking cessation drugs that contain nicotine are also called nicotine substitution products or nicotine replacement therapy. There are four forms approved by the Food and Drug Administration (FDA) as of 2001—chewing gum, skin patch, nasal spray, and inhaler. The nasal spray and inhaler are available only with a prescription, but the gum and some brands of the patch can be bought over the counter (without a prescription). People who buy the nonprescription products should check with a physician before starting to use them. The patches are sold under the brand names Nicotrol, Nicoderm CQ, and Habitrol (prescription only). The gum is sold under the brand name Nicorette. The nasal spray and inhaler are marketed as Nicotrol NS and Nicotrol respectively. The costs of these products are about \$30 for a box of 48 pieces of the gum and about \$30 per week for the patches.

Other medications

Another type of smoking cessation drug, bupropion (Zyban), also reduces craving and withdrawal symptoms, although it is not a nicotine replacement product. Bupropion is an antidepressant medication that is thought to help people stop smoking by mimicking some of the effects of tobacco on brain tissue. Bupropion can be used together with nicotine replacement products; several studies indicate that the combination helps more smokers quit than either method by itself.

Buspirone (BuSpar) is a tranquilizer that appears to be effective in helping smokers deal with feelings of anxiety resulting from tobacco withdrawal.

Alternative approaches

Other approaches that have been used to help smokers quit include hypnosis and **acupuncture**. The evidence for the usefulness of hypnosis is largely anecdotal; it appears to be most helpful when used in combination with nicotine replacement products or bupropion. Although acupuncture has been used in Western countries since the 1970s to help people quit smoking, it does not appear to be particularly effective in this regard. A British study that was published in 1999 found that smokers who received acupuncture did not have a higher quit rate than those who received only sham acupuncture.

Recommended dosage

The recommended dosage of nicotine replacement products depends on the method of administration.

Each form of this medicine comes with detailed instructions for its use. Following directions exactly is very important. For example, nicotine gum should not be chewed like regular chewing gum. It must be chewed very slowly until it has a slight taste or causes a slight **tingling** sensation in the mouth; then “parked” between the cheek and gum until the taste and tingling goes away; then chewed and parked in the same way for about 30 minutes. Nicotine patches and other products also must be used correctly to be effective. Some patches are meant to be worn only during the day and removed at night; others are worn 24 hours a day.

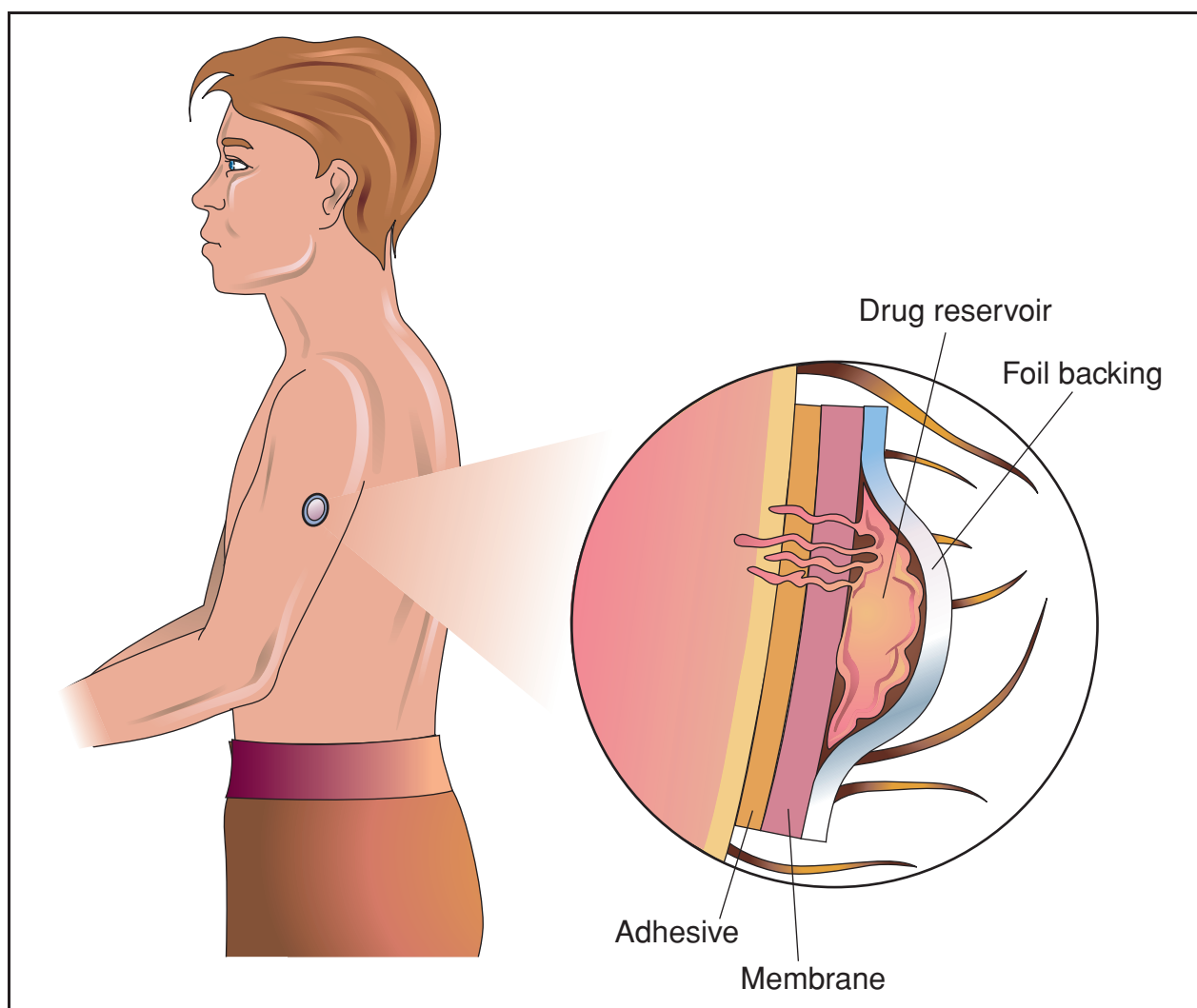
Smokers who are heavily dependent on nicotine may want to ask their doctors about using a combination of nicotine replacement products. Studies done between 1995 and 2000 indicate that combining the transdermal patch with either the gum or the nasal spray helps more smokers quit than any of the three products by themselves. It is thought that the higher success rate is due to the different rates of speed at which these products deliver nicotine to the body. The nasal spray delivers nicotine very rapidly, and can be used to relieve intense cravings at times of the day when the smoker is accustomed to having a cigarette, while the patch delivers a smaller dosage of nicotine to the body at a steadier rate.

Precautions

Seeing a physician regularly while using smoking cessation drugs is important. The physician will check to make sure the medicine is working as it should and will watch for unwanted side effects.

Seeing a physician regularly while using smoking cessation drugs is important. The physician will check to make sure the medicine is working as it should and will watch for unwanted side effects.

- nausea
- vomiting
- severe **pain** in the stomach or abdomen
- severe diarrhea
- severe dizziness
- fainting
- convulsions (seizures)
- low blood pressure
- fast, weak, or irregular heartbeat
- hearing or vision problems
- severe breathing problems
- severe watering of the mouth or drooling



The nicotine patch is a type of transepidermal patch designed to deliver nicotine, the addictive substance contained in cigarettes, directly through the skin and into the blood stream. The patch contains a drug reservoir sandwiched between a nonpermeable back layer and a permeable adhesive layer that attaches to the skin. The drug leeches slowly out of the reservoir, releasing small amounts of the drug at a constant rate for up to 24 hours. (Illustration by Electronic Illustrators Group.)

- cold sweat
- severe headache
- confusion
- severe weakness

Keep these drugs, including thrown-away patches and gum—out of the reach of children and pets. Even a small amount of nicotine can seriously harm a child or animal.

Nicotine in any form should not be used during **pregnancy**, as it may harm the fetus or cause **miscarriage**. Women who may become pregnant should use effective birth control while taking smoking cessation drugs. Women who become pregnant while taking this

medicine should stop taking it immediately and check with their physicians.

Nicotine passes into breast milk and may cause problems for nursing babies. Women who are breastfeeding and want to use smoking cessation drugs may need to stop breastfeeding during treatment.

Anyone who has had unusual reactions to nicotine in the past should let his or her physician know before using a smoking cessation drug. The physician should also be told about any **allergies** to foods, dyes, preservatives, or other substances. People who have had a rash or irritation from adhesive bandages should check with a physician before using a nicotine patch.



The wheals on the arm of this patient was caused by an allergic reaction to nicotine patches used to help subdue the urge to smoke. (Custom Medical Stock Photo. Reproduced by permission.)

Smoking cessation patches, gum, and other products may make certain medical problems worse. Before using a smoking cessation drug, people with any of these medical problems should make sure their physicians are aware of their conditions:

- heart or blood vessel disease
- high blood pressure
- diabetes
- overactive thyroid
- skin rash or irritation
- stomach ulcer
- pheochromocytoma (pcc) (a tumor of the adrenal medulla)
- dental problems or mouth sores
- sore throat
- jaw pain or temporomandibular joint disorder (TMJ)

There are also precautions to take with bupropion and buspirone. Bupropion should not be taken by patients with a history of seizures, high blood pressure, anorexia, or **bulimia nervosa**. People taking buspirone should be careful about driving or operating heavy machinery until they can tell whether the drug makes them drowsy as a side effect. Although buspirone does not interact with alcohol as intensely as most

KEY TERMS

Acupuncture—A Chinese medical practice that treats illness or addictions by the insertion of very thin steel needles at specified points along the body's energy channels.

Bupropion—An antidepressant medication given to smokers for nicotine withdrawal symptoms. It is sold under the trade name Zyban.

Buspirone—An antianxiety medication that is also given for withdrawal symptoms. It is sold under the trade name BuSpar.

Nicotine—A colorless, oily chemical found in tobacco that makes people physically dependent on smoking. It is poisonous in large doses.

Withdrawal symptoms—A group of physical or mental symptoms that may occur when a person suddenly stops using a drug on which he or she has become dependent.

tranquilizers do, patients should still use alcohol cautiously if they are taking buspirone.

Side effects

Each type of smoking cessation product may cause minor side effects that usually go away as the body adjusts to the drug. These usually do not need medical attention unless they continue or they interfere with normal activities. For example, nicotine gum may cause belching, jaw aches, or sore mouth or throat. Nicotine patches may cause redness, **itching**, or burning where the patch is applied. The nasal spray may irritate the nose and sinuses, while the inhaler may cause throat irritation or coughing.

If nicotine gum injures the mouth, teeth, or dental work, check with a physician as soon as possible. Other side effects are possible. Anyone who has unusual symptoms while using smoking cessation drugs should get in touch with his or her physician.

The side effects of bupropion include **dry mouth** and difficulty sleeping. The possible side effects of buspirone include headaches and drowsiness.

Interactions

People taking certain drugs may need to change their doses when they stop smoking. Anyone who uses a smoking cessation drug should let the physician know all other medicines he or she is taking and

should ask whether the doses need to be changed. Examples of drugs that may be affected when a person stops smoking are:

- insulin
- airway opening drugs (**bronchodilators**) such as aminophylline (Somophyllin), oxtriphylline (Choledyl) or theophylline (Somophyllin-T)
- opioid (narcotic) pain relievers such as propoxyphene (Darvon)
- the beta blocker propranolol (Inderal)

Other drugs may also interact with smoking cessation drugs. Be sure to check with a physician or pharmacist before combining smoking cessation drugs with any other prescription or nonprescription (over-the-counter) medicine.

Bupropion should not be used by patients who are also taking monoamine oxidase inhibitor (MAOI) medications. These include such drugs as furazolidone, isocarboxazid, and phenelzine. Bupropion may also interact with phenytoin, carbamazepine, and levodopa. Buspirone also interacts with MAOIs, as well as with trazadone and haloperidol.

Resources

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American Association for Respiratory Care. 11030 Ables Lane, Dallas, TX 75229. <<http://www.aarc.org>>.

American Cancer Society (ACS). 1599 Clifton Road, NE, Atlanta, GA 30329. (404) 320-3333 or (800) ACS-2345. Fax: (404) 329-7530. <<http://www.cancer.org>>.

American Lung Association. 1740 Broadway, 14th Floor, New York, NY 10019. (212) 315-8700 or (800) 586-4872 (LUNG USA).

Office on Smoking and Health. Centers for Disease Control and Prevention. Mailstop K-50, 4770 Buford Highway

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Rebecca J. Frey, PhD

Smoking

Definition

Smoking is the inhalation of the smoke of burning tobacco encased in cigarettes, pipes, and cigars. Casual smoking is the act of smoking only occasionally, usually in a social situation or to relieve **stress**. A smoking habit is a physical **addiction** to tobacco products. Many health experts now regard habitual smoking as a psychological addiction, too, and one with serious health consequences.

Description

The U.S. Food and Drug Administration has asserted that cigarettes and smokeless tobacco should be considered nicotine delivery devices. Nicotine, the active ingredient in tobacco, is inhaled into the lungs, where most of it stays. The rest passes into the bloodstream, reaching the brain in about 10 seconds and dispersing throughout the body in about 20 seconds.

Depending on the circumstances and the amount consumed, nicotine can act as either a stimulant or tranquilizer. This can explain why some people report that smoking gives them energy and stimulates their mental activity, while others note that smoking relieves **anxiety** and relaxes them. The initial "kick" results in part from the drug's stimulation of the adrenal glands and resulting release of epinephrine into the blood. Epinephrine causes several physiological changes—it temporarily narrows the arteries, raises the blood pressure, raises the levels of fat in the blood, and increases the heart rate and flow of blood

from the heart. Some researchers think epinephrine contributes to smokers' increased risk of high blood pressure.

Nicotine, by itself, increases the risk of heart disease. However, when a person smokes, he or she is ingesting a lot more than nicotine. Smoke from a cigarette, pipe, or cigar is made up of many additional toxic chemicals, including tar and carbon monoxide. Tar is a sticky substance that forms into deposits in the lungs, causing lung **cancer** and respiratory distress. Carbon monoxide limits the amount of oxygen that the red blood cells can convey throughout your body. Also, it may damage the inner walls of the arteries, which allows fat to build up in them.

Besides tar, nicotine, and carbon monoxide, tobacco smoke contains 4,000 different chemicals. More than 200 of these chemicals are known to be toxic. Nonsmokers who are exposed to tobacco smoke also take in these toxic chemicals. They inhale the smoke exhaled by the smoker as well as the more toxic *sidestream smoke*—the smoke from the end of the burning cigarette, cigar, or pipe.

Here's why sidestream smoke is more toxic than exhaled smoke: When a person smokes, the smoke he or she inhales and then breathes out leaves harmful deposits inside the body. But because lungs partially cleanse the smoke, exhaled smoke contains fewer poisonous chemicals. That's why exposure to tobacco smoke is dangerous even for a nonsmoker.

Causes and symptoms

No one starts smoking to become addicted to nicotine. It isn't known how much nicotine may be consumed before the body becomes addicted. However, once smoking becomes a habit, the smoker faces a lifetime of health risks associated with one of the strongest addictions known to man.

About 70% of smokers in the United States would like to quit; in any given year, however, only about 3.6% of the country's 47 million smokers quit successfully.

Although specific genes have not yet been identified as of 2003, researchers think that genetic factors contribute substantially to developing a smoking habit. Several twin studies have led to estimates of 46–84% heritability for smoking. It is thought that some genetic variations affect the speed of nicotine metabolism in the body and the activity level of nicotinic receptors in the brain.

Symptoms That Occur After Quitting Smoking

Symptom	Cause	Duration	Relief
Craving for cigarette	nicotine craving	first week can linger for months	distract yourself with other activity
Irritability, impatience	nicotine craving	2 to 4 weeks	Exercise, relaxation techniques, avoid caffeine
Insomnia	nicotine craving temporarily reduces deep sleep	2 to 4 weeks	Avoid caffeine after 6 PM relaxation techniques; exercise
Fatigue	lack of nicotine stimulation	2 to 4 weeks	Nap
Lack of concentration	lack of nicotine stimulation	A few weeks	Reduce workload; avoid stress
Hunger	cigarettes craving confused hunger pangs	Up to several weeks	Drink water or low calorie drinks; eat low-calorie snacks
Coughing, dry throat, nasal drip	Body ridding itself of mucus in lungs and airways	Several weeks	Drink plenty of fluids; use cough drops
Constipation, gas	Intestinal movement decreases with lack of nicotine	1 to 2 weeks	Drink plenty of fluids; add fiber to diet; exercise

Smoking risks

Smoking is recognized as the leading preventable cause of **death**, causing or contributing to the deaths of approximately 430,700 Americans each year. Anyone with a smoking habit has an increased chance of lung, cervical, and other types of cancer; respiratory diseases such as **emphysema**, **asthma**, and chronic **bronchitis**; and cardiovascular disease, such as **heart attack**, high blood pressure, **stroke**, and **atherosclerosis** (narrowing and hardening of the arteries). The risk of stroke is especially high in women who take birth control pills.

Smoking can damage fertility, making it harder to conceive, and it can interfere with the growth of the fetus during **pregnancy**. It accounts for an estimated 14% of premature births and 10% of infant deaths. There is some evidence that smoking may cause **impotence** in some men.

Because smoking affects so many of the body's systems, smokers often have vitamin deficiencies and suffer oxidative damage caused by free radicals. Free radicals are molecules that steal electrons from other

molecules, turning the other molecules into free radicals and destabilizing the molecules in the body's cells.

Smoking is recognized as one of several factors that might be related to a higher risk of hip **fractures** in older adults.

Studies reveal that the more a person smokes, the more likely he is to sustain illnesses such as cancer, chronic bronchitis, and emphysema. But even smokers who indulge in the habit only occasionally are more prone to these diseases.

Some brands of cigarettes are advertised as “low tar,” but no cigarette is truly safe. If a smoker switches to a low-tar cigarette, he is likely to inhale longer and more deeply to get the chemicals his body craves. A smoker has to quit the habit entirely in order to improve his health and decrease the chance of disease.

Though some people believe chewing tobacco is safer, it also carries health risks. People who chew tobacco have an increased risk of heart disease and mouth and throat cancer. Pipe and cigar smokers have increased health risks as well, even though these smokers generally do not inhale as deeply as cigarette smokers do. These groups haven't been studied as extensively as cigarette smokers, but there is evidence that they may be at a slightly lower risk of cardiovascular problems but a higher risk of cancer and various types of circulatory conditions.

Recent research reveals that passive smokers, or those who unavoidably breathe in second-hand tobacco smoke, have an increased chance of many health problems such as lung cancer and asthma, and in children, **sudden infant death syndrome**.

Smokers' symptoms

Smokers are likely to exhibit a variety of symptoms that reveal the damage caused by smoking. A nagging morning **cough** may be one sign of a tobacco habit. Other symptoms include **shortness of breath**, **wheezing**, and frequent occurrences of respiratory illness, such as bronchitis. Smoking also increases **fatigue** and decreases the smoker's sense of smell and taste. Smokers are more likely to develop poor circulation, with cold hands and feet and premature wrinkles.

Sometimes the illnesses that result from smoking come on silently with little warning. For instance, **coronary artery disease** may exhibit few or no symptoms. At other times, there will be warning signs, such as bloody discharge from a woman's vagina, a sign of cancer of the cervix. Another warning sign is a hacking cough, worse than the usual smoker's cough, that brings up phlegm or blood—a sign of lung cancer.

Withdrawal symptoms

A smoker who tries to quit may expect one or more of these withdrawal symptoms: **nausea**, **constipation** or **diarrhea**, drowsiness, loss of concentration, **insomnia**, **headache**, nausea, and irritability.

Diagnosis

It's not easy to quit smoking. That's why it may be wise for a smoker to turn to his physician for help. For the greatest success in quitting and to help with the withdrawal symptoms, the smoker should talk over a treatment plan with his doctor or alternative practitioner. He should have a general **physical examination** to gauge his general health and uncover any deficiencies. He should also have a thorough evaluation for some of the serious diseases that smoking can cause.

Treatment

Research shows that most smokers who want to quit benefit from the support of other people. It helps to quit with a friend or to join a group such as those organized by the American Cancer Society. These groups provide support and teach behavior modification methods that can help the smoker quit. The smoker's physician can often refer him to such groups.

Other alternatives to help with the withdrawal symptoms of kicking the habit include nicotine replacement therapy in the form of gum, patches, nasal sprays, and oral inhalers. These are available by prescription or over the counter. A physician can provide advice on how to use them. They slowly release a small amount of nicotine into the bloodstream, satisfying the smoker's physical craving. Over time, the amount of gum the smoker chews is decreased and the amount of time between applying the patches is increased. This helps wean the smoker from nicotine slowly, eventually beating his addiction to the drug. But there's one important caution: If the smoker lights up while taking a nicotine replacement, a nicotine overdose may cause serious health problems.

The prescription drug Zyban (bupropion hydrochloride) has shown some success in helping smokers quit. This drug contains no nicotine, and was originally developed as an antidepressant. It isn't known exactly how bupropion works to suppress the desire for nicotine. A five-year study of bupropion reported in 2003 that the drug has a very good record for safety and effectiveness in treating tobacco dependence. Its most common side effect is insomnia, which can also result from nicotine withdrawal.

Researchers are investigating two new types of drugs as possible treatments for tobacco dependence as of 2003. The first is an alkaloid known as 18-methoxycoronaridine (18-MC), which selectively blocks the nicotinic receptors in brain tissue. Another approach involves developing drugs that inhibit the activity of cytochrome P450 2A6 (CYP2A6), which controls the metabolism of nicotine.

Expected results

Research on smoking shows that most smokers desire to quit. But smoking is so addictive that fewer than 20% of the people who try ever successfully kick the habit. Still, many people attempt to quit smoking over and over again, despite the difficulties—the cravings and withdrawal symptoms, such as irritability and restlessness.

For those who do quit, the benefits to health are well worth the effort. The good news is that once a smoker quits the health effects are immediate and dramatic. After the first day, oxygen and carbon monoxide levels in the blood return to normal. At two days, nerve endings begin to grow back and the senses of taste and smell revive. Within two weeks to three months, circulation and breathing improve. After one year of not smoking, the risk of heart disease is reduced by 50%. After 15 years of abstinence, the risks of health problems from smoking virtually vanish. A smoker who quits for good often feels a lot better too, with less fatigue and fewer respiratory illnesses.

Alternative treatment

There are a wide range of alternative treatments that can help a smoker quit the habit, including **hypnotherapy**, herbs, **acupuncture**, and **meditation**. For example, a controlled trial demonstrated that self-massage can help smokers crave less intensely, smoke fewer cigarettes, and in some cases completely give them up.

Hypnotherapy

Hypnotherapy helps the smoker achieve a trance-like state, during which the deepest levels of the mind are accessed. A session with a hypnotherapist may begin with a discussion of whether the smoker really wants to and truly has the motivation to stop smoking. The therapist will explain how hypnosis can reduce the stress-related symptoms that sometimes come with kicking the habit.

Often the therapist will discuss the dangers of smoking with the patient and begin to “reframe”

the patient’s thinking about smoking. Many smokers are convinced they can’t quit, and the therapist can help persuade them that they can change this behavior. These suggestions are then repeated while the smoker is under hypnosis. The therapist may also suggest while the smoker is under hypnosis that his feelings of worry, anxiety, and irritability will decrease.

In a review of 17 studies of the effectiveness of hypnotherapy, the percentage of people treated by hypnosis who still were not smoking after six months ranged from 4–8%. In programs that included several hours of treatment, intense interpersonal interaction, individualized suggestions, and follow-up treatment, success rates were above 50%.

Aromatherapy

One study demonstrated that inhaling the vapor from black pepper extract can reduce symptoms associated with smoking withdrawal. Other essential oils can be used for relieving the anxiety a smoker often experiences while quitting.

Herbs

A variety of herbs can help smokers reduce their cravings for nicotine, calm their irritability, and even reverse the oxidative cellular damage done by smoking. Lobelia, sometimes called Indian tobacco, has historically been used as a substitute for tobacco. It contains a substance called lobeline, which decreases the craving for nicotine by bolstering the nervous system and calming the smoker. In high doses, lobelia can cause **vomiting**, but the average dose—about 10 drops per day—should pose no problems.

Herbs that can help relax a smoker during withdrawal include wild oats and kava kava.

To reduce the oral fixation supplied by a nicotine habit, a smoker can chew on licorice root—the plant, not the candy. Licorice is good for the liver, which is a major player in the body’s **detoxification** process. Licorice also acts as a tonic for the adrenal system, which helps reduce stress. And there’s an added benefit: If a smoker tries to light up after chewing on licorice root, the cigarette tastes like burned cardboard.

Other botanicals that can help repair free-radical damage to the lungs and cardiovascular system are those high in flavonoids, such as hawthorn, ginkgo biloba, and bilberry, as well as antioxidants such as vitamin A, vitamin C, zinc, and selenium.

Acupuncture

This ancient Chinese method of healing is used commonly to help beat addictions, including smoking. The acupuncturist will use hair-thin needles to stimulate the body's *qi*, or healthy energy. Acupuncture is a sophisticated treatment system based on revitalizing *qi*, which supposedly flows through the body in defined pathways called meridians. During an addiction like smoking, *qi* isn't flowing smoothly or gets stuck, the theory goes.

Points in the ear and feet are stimulated to help the smoker overcome his addiction. Often the acupuncturist will recommend keeping the needles in for five to seven days to calm the smoker and keep him balanced.

Vitamins

Smoking seriously depletes vitamin C in the body and leaves it more susceptible to infections. Vitamin C can prevent or reduce free-radical damage by acting as an antioxidant in the lungs. Smokers need additional C, in higher dosage than nonsmokers. Fish in the diet supplies **Omega-3 fatty acids**, which are associated with a reduced risk of chronic obstructive pulmonary disease (emphysema or chronic bronchitis) in smokers. Omega-3 fats also provide cardiovascular benefits as well as an anti-depressive effect. Vitamin therapy doesn't reduce craving but it can help beat some of the damage created by smoking. Vitamin B₁₂ and **folic acid** may help protect against smoking-induced cancer.

Prevention

How do you give up your cigarettes for good and never go back to them again?

Here are a few tips from the experts:

- Have a plan and set a definite quit date.
- Get rid of all the cigarettes and ashtrays at home or in your desk at work.
- Don't allow others to smoke in your house.
- Tell your friends and neighbors that you're quitting. Doing so helps make quitting a matter of pride.
- Chew sugarless gum or eat sugar-free hard candy to redirect the oral fixation that comes with smoking. This will prevent weight gain, too.
- Eat as much as you want, but only low-calorie foods and drinks. Drink plenty of water. This may help with the feelings of tension and restlessness that quitting can bring. After eight weeks, you'll lose your craving for tobacco, so it's safe then to return to your usual eating habits.

KEY TERMS

Antioxidant—Any substance that reduces the damage caused by oxidation, such as the harm caused by free radicals.

Chronic bronchitis—A smoking-related respiratory illness in which the membranes that line the bronchi, or the lung's air passages, narrow over time. Symptoms include a morning cough that brings up phlegm, breathlessness, and wheezing.

Cytochrome—A substance that contains iron and acts as a hydrogen carrier for the eventual release of energy in aerobic respiration.

Emphysema—An incurable, smoking-related disease, in which the air sacs at the end of the lung's bronchi become weak and inefficient. People with emphysema often first notice shortness of breath, repeated wheezing and coughing that brings up phlegm.

Epinephrine—A nervous system hormone stimulated by the nicotine in tobacco. It increases heart rate and may raise smokers' blood pressure.

Flavonoid—A food chemical that helps to limit oxidative damage to the body's cells, and protects against heart disease and cancer.

Free radical—An unstable molecule that causes oxidative damage by stealing electrons from surrounding molecules, thereby disrupting activity in the body's cells.

Nicotine—The addictive ingredient of tobacco, it acts on the nervous system and is both stimulating and calming.

Nicotine replacement therapy—A method of weaning a smoker away from both nicotine and the oral fixation that accompanies a smoking habit by giving the smoker smaller and smaller doses of nicotine in the form of a patch or gum.

Sidestream smoke—The smoke that is emitted from the burning end of a cigarette or cigar, or that comes from the end of a pipe. Along with exhaled smoke, it is a constituent of second-hand smoke.

- Stay away from social situations that prompt you to smoke. Dine in the nonsmoking section of restaurants.
- Spend the money you save not smoking on an occasional treat for yourself.

Resources

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ORGANIZATIONS

- American Association of Oriental Medicine. 5530 Wisconsin Avenue, Suite 1210, Chevy Chase, MD 20815. (301) 941-1064 or (888) 500-7999. <<http://www.aaom.org>>.
- American Cancer Society. Contact the local organization or call (800) 227-2345. <<http://www.cancer.org>>.
- American Lung Association. 1740 Broadway, New York, NY 10019. (800) 586-4872 or (212) 315-8700. <<http://www.lungusa.org>>.
- Herb Research Foundation. 1007 Pearl St., Suite 200, Boulder CO 80302. (303) 449-2265. <<http://www.herbs.org>>.
- National Heart, Lung, and Blood Institute (NHLBI). Building 31, Room 5A52, 31 Center Drive, MSC 2486, Bethesda, MD 20892. (301) 592-8573. <<http://www.nhlbi.nih.gov>>.

Smoking, Tobacco, and Health Information Line. Centers for Disease Control and Prevention. Mailstop K-50, 4770 Buford Highway NE, Atlanta, GA 30341-3724. (800) 232-1311. <<http://www.cdc.gov/tobacco>>.

OTHER

Virtual Office of the Surgeon General: Tobacco Cessation Guideline. <<http://www.surgeongeneral.gov/tobacco>>.

Barbara Boughton

Snoring

Definition

Snoring is a sound generated during sleep by vibration of loose tissue in the upper airway.

Description

Snoring is one symptom of a group of disorders known as sleep disordered breathing. It occurs when the soft palate, uvula, tongue, tonsils, and/or muscles in the back of the throat rub against each other and generate a vibrating sound during sleep. Twenty percent of all adults are chronic snorers, and 45% of normal adults snore occasionally. As people grow older, their chance of snoring increases. Approximately half of all individuals over 60 snore regularly.

In some cases, snoring is a symptom of a more serious disorder called obstructed **sleep apnea** (OSA). OSA occurs when part of the airway is closed off (usually at the back of the throat) while a person is trying to inhale during sleep, and breathing stops for more than 10 seconds before resuming again. These breathless episodes can occur as many as several hundred times a night.

People with OSA almost always snore heavily, because the same narrowing of the airway that causes snoring can also cause OSA. Snoring may actually attribute to OSA as well, because the vibration of the throat tissues which occurs in snoring can cause the tissue to swell.

Snoring is associated with physical problems as well as social **stress**. People who do not suffer from OSA may be diagnosed with socially unacceptable snoring (SUS), which refers to snoring that is loud enough to prevent the sleeper's bed partner or roommate from sleeping. SUS is a factor in the breakup of some marriages and other long-term relationships.

Moreover, a study published in 2002 indicates that people who snore are at increased risk of developing type 2 diabetes. Snoring appears to be a risk factor that is independent of body weight or a family history of diabetes.

Causes and symptoms

There are several major causes of snoring, including:

- Excessively relaxed throat muscles. Alcohol, drugs, and sedatives can cause the throat muscles to become lax, and/or the tongue to pull back into the airway.
- Large uvula. The piece of tissue that hangs from the back of the throat is called the uvula. Individuals with a large or longer than average uvula can suffer from snoring when the uvula vibrates in the airway.
- Large tonsils and/or adenoids. The tonsils (tissue at the back of either side of the throat) can also vibrate if they are larger than normal, as can the adenoids.
- Excessive weight. Overweight people are more likely to snore. This is frequently caused by the extra throat and neck tissue they are carrying around.
- Nasal congestion. Colds and **allergies** can plug the nose, creating a vacuum in the throat that results in snoring as airflow increases.
- Cysts and tumors. Cysts and/or tumors of the throat can trigger snoring.
- Structural problems of the nose. A **deviated septum** or other nasal problems can also cause snoring.

Diagnosis

A patient interview, and possibly an interview with the patient's spouse or anyone else in the household who has witnessed the snoring, is usually enough for a diagnosis of snoring. A medical history that includes questions about alcohol or tranquilizer use; past ear, nose, and throat problems; and the pattern and degree of snoring will be completed, and a physical exam will be performed to determine the cause of the problem. This will typically include examination of the throat to look for narrowing, obstruction, or malformations. If the snoring is suspected to be a symptom of a more serious disorder such as obstructive sleep apnea, the patient will require further testing. This testing is called a **polysomnography** study, and is conducted during an overnight stay in a specialized sleep laboratory. The polysomnography study include measurements of heart rate, airflow at the mouth and nose, respiratory effort, sleep stage (light sleep, deep sleep, dream sleep, etc.), and oxygen level in the blood.

In some cases the patient may be referred to a dentist or orthodontist for evaluation of the jaw structure and dentition.

In addition, the patient may be examined by sleep endoscopy. In this procedure, the patient is given a medication (midazolam) to induce sleep. His or her throat and nasal passages are then examined with a flexible laryngoscope. In many cases, sleep endoscopy reveals obstructions that are not apparent during a standard **physical examination** of the throat. Many patients are found to have obstructions at more than one level in their breathing passages.

Treatment

Several surgical procedures are available for treating chronic snoring. These include:

- Uvulopalatopharyngoplasty (UPPP), a surgical procedure which involves removing excess throat tissues (e.g., tonsils, parts of the soft palate) to expand the airway.
- Laser-assisted uvulopalatoplasty (LAUP) uses a surgical laser to remove part of the uvula and palate.
- Palatal stiffening is a minimally-invasive surgical technique where a laser or a cauterizer is used to produce scar tissue in the soft palate in order to stop the vibrations that produce snoring.
- Radiofrequency ablation is another technique which uses scarring to shrink the uvula and/or soft palate. A needle electrode is used to shrink and scar the mouth and throat tissues.

Alternative treatment

There are a number of remedies for snoring, but few are proven clinically effective. Popular treatments include:

- Mechanical devices. Many splints, braces, and other devices are available which reposition the nose, jaw, and/or mouth in order to clear the airways. Other devices are designed to wake an individual when snoring occurs. Patients should consult a dentist or orthodontist about these devices, as most require custom fitting. In addition, persons with certain types of gum disease or dental problems should not be fitted with oral appliances to stop snoring.
- Nasal strips. Nasal strips that attach like an adhesive bandage to the bridge of the nose are available at most drugstores, and can help stop snoring in some individuals by opening the nasal passages.

- Continuous positive airway pressure (CPAP). Some chronic snorers find relief by sleeping with a nasal mask which provides air pressure to the throat.
- **Decongestants.** Snoring caused by nasal congestion may be successfully treated with decongestants. Some effective herbal remedies that clear the nasal passages include golden rod (*Solidago virgaurea*) and golden seal (*Hydrastis canadensis*). Steam inhalation of essential oils of eucalyptus blue gum (*Eucalyptus globulus*) or peppermint (*Mentha x piperata*) can also relieve congestion.
- Weight loss. Snoring thought to be caused by excessive weight may be curtailed by a sensible weight loss and **exercise** program.
- Sleep position. Snoring usually worsens when an individual sleeps on his or her back, so sleeping on one's side may alleviate the problem. Those who have difficulty staying in a side sleeping position may find sleeping with pillows behind them helps them maintain the position longer. Other devices include a new vest designed to prevent the sleeper from lying on his or her back.
- Bed adjustments. For some people, raising the head of the bed solves their snoring problem. A slight incline can prevent the tongue from retracting into the back of the throat. Bricks, wooden blocks, or specially designed wedges can be used to elevate the head of the bed approximately 4–16 in (10–41 cm).

Alternative treatments that have been reported to be effective for patients whose snoring is caused by colds or allergies include **acupuncture**, homeopathy, and **aromatherapy** treatments. Aromatherapy treatments for snoring typically make use of marjoram oil, which is thought to be particularly effective in clearing the nasal passages.

Prevention

Adults with a history of snoring may be able to prevent snoring episodes with the following measures:

- avoid alcohol and sedatives before bedtime
- remove allergens from the bedroom
- use a decongestant before bed
- sleep on the side, not the back

Resources

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Beers, Mark H., MD, and Robert Berkow, MD., editors. "Disorders of the Oral Region." Section 9, Chapter 105 In *The Merck Manual of Diagnosis and Therapy*.

KEY TERMS

Ablation—The removal of abnormal tissue growths by surgery.

Cauterize—To seal tissue or blood vessels using a heat or electrical source.

Continuous positive airway pressure (CPAP)—A ventilation device that blows a gentle stream of air into the nose during sleep to keep the airway open.

Deviated septum—A hole or perforation in the septum, the wall that divides the two nasal cavities.

Endoscope—A slender optical instrument that allows a doctor to examine the inside of the throat or other hollow organ. Sleep endoscopy is a technique that allows the doctor to detect previously unsuspected obstructions in the patient's nose and throat.

Obstructive sleep apnea (OSA)—A potentially life-threatening condition characterized by episodes of breathing cessation during sleep alternating with snoring or disordered breathing. The low levels of oxygen in the blood of patients with OSA may eventually cause heart problems or stroke.

Polysomnography—A technique for diagnosing sleep disorders with the use of a machine that records the pulse, breathing rate and other variables while the patient sleeps.

Soft palate—The structure at the roof of the mouth that separates the mouth and the pharynx.

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Sodium imbalance see **Hypernatremia; Hyponatremia**

Somatization disorder see **Somatoform disorders**

Somatoform disorders

Definition

The somatoform disorders are a group of mental disturbances placed in a common category on the basis of their external symptoms. These disorders are characterized by physical complaints that appear to be medical in origin but that cannot be explained in terms of a physical disease, the results of **substance abuse**, or by another mental disorder. In order to meet the criteria for a somatoform disorder, the physical symptoms must be serious enough to interfere with the patient's employment or relationships, and must be symptoms that are not under the patient's voluntary control.

It is helpful to understand that the present classification of these disorders reflects recent historical changes in the practice of medicine and psychiatry. When psychiatry first became a separate branch of medicine at the end of the nineteenth century, the term *hysteria* was commonly used to describe mental disorders characterized by altered states of consciousness (for example, sleepwalking or trance states) or physical symptoms (for example, a "paralyzed" arm or leg with no neurologic cause) that could not be fully explained by a medical disease. The term *dissociation* was used for the psychological mechanism that allows the mind to split off uncomfortable feelings, memories, or ideas so that they are lost to conscious recall. Sigmund Freud and other pioneering psychoanalysts thought that the hysterical patient's symptoms resulted from dissociated thoughts or memories re-emerging through bodily functions or trance states. Prior to the categorization all mental disorders that were considered to be forms of **hysteria** were grouped together on the basis of this theory about their cause. Since 1980, however, the somatoform disorders and the so-called **dissociative disorders** have been placed in separate categories on the basis of their chief

symptoms. In general, the somatoform disorders are characterized by disturbances in the patient's physical sensations or ability to move the limbs or walk, while the dissociative disorders are marked by disturbances in the patient's sense of identity or memory.

Description

As a group, the somatoform disorders are difficult to recognize and treat because patients often have long histories of medical or surgical treatment with several different doctors. In addition, the physical symptoms are not under the patient's conscious control, so that he or she is not intentionally trying to confuse the doctor or complicate the process of diagnosis. Somatoform disorders are, however, a significant problem for the health care system because patients with these disturbances overuse medical services and resources.

Somatization disorder (Briquet's syndrome)

Somatization disorder was formerly called Briquet's syndrome, after the French physician who first recognized it. The distinguishing characteristic of this disorder is a group or pattern of symptoms in several different organ systems of the patient's body that cannot be accounted for by medical illness. The criteria for this disorder require four symptoms of **pain**, two symptoms in the digestive tract, one symptom involving the sexual organs, and one symptom related to the nervous system. Somatization disorder usually begins before the age of 30. It is estimated that 0.2% of the United States population will develop this disorder in the course of their lives. Another researcher estimates that 1% of all women in the United States have symptoms of this disorder. The female-to-male ratio is estimated to range between 5:1 and 20:1.

Somatization disorder is considered to be a chronic disturbance that tends to persist throughout the patient's life. It is also likely to run in families. Some psychiatrists think that the high female-to-male ratio in this disorder reflects the cultural pressures on women in North American society and the social "permission" given to women to be physically weak or sickly.

Conversion disorder

Conversion disorder is a condition in which the patient's senses or ability to walk or move are impaired without a recognized medical or neurological disease or cause and in which psychological factors

(such as **stress** or trauma) are judged to be temporarily related to onset or exacerbation. The disorder gets its name from the notion that the patient is converting a psychological conflict or problem into an inability to move specific parts of the body or to use the senses normally. An example of a conversion reaction would be a patient who loses his or her voice in a situation in which he or she is afraid to speak. The symptom simultaneously contains the **anxiety** and serves to get the patient out of the threatening situation. The resolution of the emotion that underlies the physical symptom is called the patient's *primary gain*, and the change in the patient's social, occupational, or family situation that results from the symptom is called a *secondary gain*. Doctors sometimes use these terms when they discuss the aftereffects of conversion disorder or of other somatoform disorders on the patient's emotional adjustment and lifestyle.

The specific physical symptoms of conversion disorder may include a loss of balance or **paralysis** of an arm or leg; the inability to swallow or speak; the loss of touch or pain sensation; going blind or deaf; seeing double; or having **hallucinations**, seizures, or convulsions.

Unlike somatization disorder, conversion disorder may begin at any age, and it does not appear to run in families. It is estimated that as many as 34% of the population experiences conversion symptoms over a lifetime, but that the disorder is more likely to occur among less educated or sophisticated people. Conversion disorder is not usually a chronic disturbance; 90% of patients recover within a month, and most do not have recurrences. The female-to-male ratio is between 2:1 and 5:1. Male patients are likely to develop conversion disorders in occupational settings or military service.

Pain disorder

Pain disorder is marked by the presence of severe pain as the focus of the patient's concern. This category of somatoform disorder covers a range of patients with a variety of ailments, including chronic headaches, back problems, arthritis, muscle aches and cramps, or pelvic pain. In some cases the patient's pain appears to be largely due to psychological factors, but in other cases the pain is derived from a medical condition as well as the patient's psychology.

Pain disorder is relatively common in the general population, partly because of the frequency of work-related injuries in the United States. This disorder appears to be more common in older adults, and the sex ratio is nearly equal, with a female-to-male ratio of 2:1.

Hypochondriasis

Hypochondriasis is a somatoform disorder marked by excessive fear of or preoccupation with having a serious illness that persists in spite of medical testing and reassurance. It was formerly called hypochondriacal neurosis.

Although hypochondriasis is usually considered a disorder of young adults, it is now increasingly recognized in children and adolescents. It may also develop in elderly people without previous histories of health-related fears. The disorder accounts for about 5% of psychiatric patients, and is equally common in men and women. Hypochondriasis may persist over a number of years but usually occurs as a series of episodes rather than continuous treatment-seeking. The flare-ups of the disorder are often correlated with stressful events in the patient's life.

Body dysmorphic disorder

Body dysmorphic disorder is a new category of somatoform disorders. It is defined as a preoccupation with an imagined or exaggerated defect in appearance. Most cases involve features on the patient's face or head, but other body parts—especially those associated with sexual attractiveness, such as the breasts or genitals—may also be the focus of concern.

Body dysmorphic disorder is regarded as a chronic condition that usually begins in the patient's late teens and fluctuates over the course of time. It was initially considered to be a relatively unusual disorder, but may be more common than was formerly thought. It appears to affect men and women with equal frequency. Patients with body dysmorphic disorder frequently have histories of seeking or obtaining **plastic surgery** or other procedures to repair or treat the supposed defect. Some may even meet the criteria for a delusional disorder of the somatic type.

Somatoform disorders in children and adolescents

The most common somatoform disorders in children and adolescents are conversion disorders, although body dysmorphic disorders are being reported more frequently. Conversion reactions in this age group usually reflect stress in the family or problems with school rather than long-term psychiatric disturbances. Some psychiatrists speculate that adolescents with conversion disorders frequently have overprotective or overinvolved parents with a subconscious need to see their child as sick; in many cases the son or daughter's symptoms become the center of family attention. The rise in body dysmorphic disorders in adolescents is thought to reflect

the increased influence of media preoccupation with physical perfection.

Causes and symptoms

Because groups the somatoform disorders are categorized on the basis of symptom patterns, their causes as presently understood include several different factors.

Family stress

Family stress is believed to be one of the most common causes of somatoform disorders in children and adolescents. Conversion disorders in this age group may also be connected with physical or sexual **abuse** within the family of origin.

Parental modeling

Somatization disorder and hypochondriasis may result in part from the patient's unconscious reflection or imitation of parental behaviors. This "copycat" behavior is particularly likely if the patient's parent derived considerable secondary gain from his or her symptoms.

Cultural influences

Cultural influences appear to affect the gender ratios and body locations of somatoform disorders, as well as their frequency in a specific population. Some cultures (for example, Greek and Puerto Rican) report higher rates of somatization disorder among men than is the case for the United States. In addition, researchers found lower levels of somatization disorder among people with higher levels of education. People in Asia and Africa are more likely to report certain types of physical sensations (for example, burning hands or feet, or the feeling of ants crawling under the skin) than are Westerners.

Biological factors

Genetic or biological factors may also play a role. For example, people who suffer from somatization disorder may also differ in how they perceive and process pain.

Diagnosis

Accurate diagnosis of somatoform disorders is important to prevent unnecessary surgery, laboratory tests, or other treatments or procedures. Because somatoform disorders are associated with physical symptoms, patients are often diagnosed by primary

care physicians as well as by psychiatrists. In many cases the diagnosis is made in a general medical clinic. Children and adolescents with somatoform disorders are most likely to be diagnosed by pediatricians. Diagnosis of somatoform disorders requires a thorough physical workup to exclude medical and neurological conditions, or to assess their severity in patients with pain disorder. A detailed examination is especially necessary when conversion disorder is a possible diagnosis, because some neurological conditions—including **multiple sclerosis** and myasthenia gravis—have on occasion been misdiagnosed as conversion disorder. Some patients who receive a diagnosis of somatoform disorder ultimately go on to develop neurological disorders.

In addition to ruling out medical causes for the patient's symptoms, a doctor who is evaluating a patient for a somatization disorder will consider the possibility of other psychiatric diagnoses or of overlapping psychiatric disorders. Somatoform disorders often coexist with **personality disorders** because of the chicken-and-egg relationship between physical illness and certain types of character structure or personality traits. At one time, the influence of Freud's theory of hysteria led doctors to assume that the patient's hidden emotional needs "cause" the illness. But in many instances, the patient's personality may have changed over time due to the stresses of adjusting to a chronic disease. This gradual transformation is particularly likely in patients with pain disorder. Patients with somatization disorder often develop panic attacks or **agoraphobia** together with their physical symptoms. In addition to anxiety or personality disorders, the doctor will usually consider major depression as a possible diagnosis when evaluating a patient with symptoms of a somatoform disorder. Pain disorders may be associated with depression, and body dysmorphic disorder may be associated with obsessive-compulsive disease.

Treatment

Relationship with primary care practitioner

Because patients with somatoform disorders often have lengthy medical histories, a long-term relationship with a trusted primary care practitioner (PCP) is a safeguard against unnecessary treatments as well as a comfort to the patient. Many PCPs prefer to schedule brief appointments on a regular basis with the patient and keep referrals to specialists to a minimum. This practice also allows them to monitor the patient for any new physical symptoms or diseases. However, some PCPs work with a psychiatric consultant.

Medications

Patients with somatoform disorders are sometimes given **anxiety drugs** or **antidepressant drugs** if they have been diagnosed with a coexisting mood or anxiety disorder. In general, however, it is considered better practice to avoid prescribing medications for these patients since they are likely to become psychologically dependent on them. However, body dysmorphic disorder has been successfully treated with **selective serotonin reuptake inhibitors (SSRI)** antidepressants.

Psychotherapy

Patients with somatoform disorders are not considered good candidates for **psychoanalysis** and other forms of insight-oriented psychotherapy. They can benefit, however, from supportive approaches to treatment that are aimed at symptom reduction and stabilization of the patient's personality. Some patients with pain disorder benefit from **group therapy** or support groups, particularly if their social network has been limited by their pain symptoms. **Cognitive-behavioral therapy** is also used sometimes to treat pain disorder.

Family therapy is usually recommended for children or adolescents with somatoform disorders, particularly if the parents seem to be using the child as a focus to divert attention from other difficulties. Working with families of chronic pain patients also helps avoid reinforcing dependency within the family setting.

Hypnosis is a technique that is sometimes used as part of a general psychotherapeutic approach to conversion disorder because it may allow patients to recover memories or thoughts connected with the onset of the physical symptoms.

Alternative treatment

Patients with somatization disorder or pain disorder may be helped by a variety of alternative therapies including **acupuncture**, **hydrotherapy**, therapeutic massage, **meditation**, botanical medicine, and homeopathic treatment. Relief of symptoms, including pain, can occur on the physical level, as well as on the mental, emotional, and spiritual levels.

Prognosis

The prognosis for somatoform disorders depends, as a rule, on the patient's age and whether the disorder is chronic or episodic. In general, somatization

KEY TERMS

Briquet's syndrome—Another name for somatization disorder.

Conversion disorder—A somatoform disorder characterized by the transformation of a psychological feeling or impulse into a physical symptom. Conversion disorder was previously called hysterical neurosis, conversion type.

Dissociation—A psychological mechanism in which the mind splits off certain aspects of a traumatic event from conscious awareness. Dissociation can affect the patient's memory, sense of reality, and sense of identity.

Hysteria—The earliest term for a psychoneurotic disturbance marked by emotional outbursts and/or disturbances of movement and sense perception. Some forms of hysteria are now classified as somatoform disorders and others are grouped with the dissociative disorders.

Hysterical neurosis—An older term for conversion disorder or dissociative disorder.

Primary gain—The immediate relief from guilt, anxiety, or other unpleasant feelings that a patient derives from a symptom.

Repression—A unconscious psychological mechanism in which painful or unacceptable ideas, memories, or feelings are removed from conscious awareness or recall.

Secondary gain—The social, occupational, or interpersonal advantages that a patient derives from symptoms. A patient's being relieved of his or her share of household chores by other family members would be an example of secondary gain.

Somatoform disorder—A category of psychiatric disorder characterized by physical complaints that appear to be medical in origin but that cannot be explained in terms of a physical disease, the results of substance abuse, or by another mental disorder.

disorder and body dysmorphic disorder rarely resolve completely. Hypochondriasis and pain disorder may resolve if there are significant improvements in the patient's overall health and life circumstances, and people with both disorders may go through periods when symptoms become less severe (remissions) or become worse (exacerbations). Conversion disorder tends to be rapidly resolved, but may recur in about 25% of all cases.

Prevention

Generalizations regarding prevention of somatoform disorders are difficult because these syndromes affect different age groups, vary in their symptom patterns and persistence, and result from different problems of adjustment to the surrounding culture. In theory, allowing expression of emotional pain in children, rather than regarding it as "weak," might reduce the secondary gain of physical symptoms that draw the care or attention of parents.

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Somatotrophic hormone test see **Growth hormone tests**

Sore throat

Definition

Sore throat, also called pharyngitis, is a painful inflammation of the mucous membranes lining the pharynx. It is a symptom of many conditions, but most often is associated with colds or **influenza**. Sore throat may be caused by either viral or bacterial infections or environmental conditions. Most sore throats heal without complications, but they should not be ignored because some develop into serious illnesses.

Description

Almost everyone gets a sore throat at one time or another, although children in child care or grade school have them more often than adolescents and adults. Sore throats are most common during the winter months when upper respiratory infections (colds) are more frequent.

Sore throats can be either acute or chronic. Acute sore throats are the more common. They appear suddenly and last from three to about seven days. A chronic sore throat lasts much longer and is a symptom of an unresolved underlying condition or disease, such as a sinus infection.

Causes and symptoms

Sore throats have many different causes, and may or may not be accompanied by cold symptoms, **fever**, or swollen lymph glands. Proper treatment depends on understanding the cause of the sore throat.

Viral sore throat

Viruses cause 90–95% of all sore throats. Cold and flu viruses are the main culprits. These viruses cause an inflammation in the throat and occasionally the tonsils (**tonsillitis**). Cold symptoms almost always accompany a viral sore throat. These can include a runny nose, **cough**, congestion, hoarseness, **conjunctivitis**, and fever. The level of throat **pain** varies from uncomfortable to excruciating, when it is painful for the patient to eat, breathe, swallow, or speak.

Another group of viruses that cause sore throat are the adenoviruses. These may also cause infections of the lungs and ears. In addition to a sore throat, symptoms that accompany an adenovirus infection include cough, runny nose, white bumps on the tonsils and throat, mild **diarrhea**, **vomiting**, and a rash. The sore throat lasts about one week.

A third type of virus that can cause severe sore throat is the coxsackie virus. It can cause a disease called herpangina. Although anyone can get herpangina, it is most common in children up to age ten and is more prevalent in the summer or early autumn. Herpangina is sometimes called summer sore throat.

Three to six days after being exposed to the virus, an infected person develops a sudden sore throat that is accompanied by a substantial fever usually between 102–104°F (38.9–40°C). Tiny grayish-white blisters form on the throat and in the mouth. These fester and become small ulcers. Throat pain is often severe, interfering with swallowing. Children may become dehydrated if they are reluctant to eat or drink because of the pain. In addition, people with herpangina may vomit, have abdominal pain, and generally feel ill and miserable.

One other common cause of a viral sore throat is mononucleosis. Mononucleosis occurs when the Epstein-Barr virus infects one specific type of lymphocyte. The infection spreads to the lymphatic system, respiratory system, liver, spleen, and throat. Symptoms appear 30–50 days after exposure.

Mononucleosis, sometimes called the kissing disease, is extremely common. It is estimated that by the age of 35–40, 80–95% of Americans will have had mononucleosis. Often, symptoms are mild, especially in young children, and are diagnosed as a cold. Since symptoms are more severe in adolescents and adults,

more cases are diagnosed as mononucleosis in this age group. One of the main symptoms of mononucleosis is a severe sore throat.

Although a runny nose and cough are much more likely to accompany a sore throat caused by a virus than one caused by a bacteria, there is no absolute way to tell what is causing the sore throat without a laboratory test. Viral sore throats are contagious and are passed directly from person to person by coughing and sneezing.

Bacterial sore throat

From 5–10% of sore throats are caused by bacteria. The most common bacterial sore throat results from an infection by group A *Streptococcus*. This type of infection is commonly called **strep throat**. Anyone can get strep throat, but it is most common in school age children.

Pharyngeal **gonorrhea**, a sexually transmitted bacterial disease, causes a severe sore throat. Gonorrhea in the throat is transmitted by having oral sex with an infected person.

Noninfectious sore throat

Not all sore throats are caused by infection. Postnasal drip can irritate the throat and make it sore. It can be caused by hay fever and other **allergies** that irritate the sinuses. Environmental and other conditions, such as heavy **smoking** or breathing second-hand smoke, heavy alcohol consumption, breathing polluted air or chemical fumes, or swallowing substances that burn or scratch the throat can also cause pharyngitis. Dry air, like that in airplanes or from forced hot air furnaces, can make the throat sore. People who breathe through their mouths at night because of nasal congestion often get sore throats that improve as the day progresses. Sore throat caused by environmental conditions is not contagious.

Diagnosis

It is easy for people to tell if they have a sore throat, but difficult to know what has caused it without laboratory tests. Most sore throats are minor and heal without any complications. A small number of bacterial sore throats do develop into serious diseases. Because of this, it is advisable to see a doctor if a sore throat lasts more than a few days or is accompanied by fever, **nausea**, or abdominal pain.

Diagnosis of a sore throat by a doctor begins with a **physical examination** of the throat and chest. The doctor will also look for signs of other illness, such as a



This young woman is having her sore throat examined by a medical practitioner using a fiber-optic tongue depressor.
(Custom Medical Stock Photo. Reproduced by permission.)

sinus infection or **bronchitis**. Since both bacterial and viral sore throat are contagious and pass easily from person to person, the doctor will seek information about whether the patient has been around other people with flu, sore throat, colds, or strep throat. If it appears that the patient may have strep throat, the doctor will do laboratory tests.

If mononucleosis is suspected, the doctor may do a mono spot test to look for antibodies indicating the presence of the Epstein-Barr virus. The test is inexpensive, takes only a few minutes, and can be done in a physician's office. An inexpensive blood test can also determine the presence of antibodies to the mononucleosis virus.

Treatment

Effective treatment varies depending on the cause of the sore throat. As frustrating as it may be to the patient, viral sore throat is best left to run its course without drug treatment. **Antibiotics** have no effect on a viral sore throat. They do not shorten the length of the illness, nor do they lessen the symptoms.

Sore throat caused by a streptococci or another bacteria must be treated with antibiotics. Penicillin is the preferred medication. Oral penicillin must be taken for 10 days. Patients need to take the entire amount of antibiotic prescribed, even after symptoms of the sore throat improve. Stopping the antibiotic

early can lead to a return of the sore throat. Occasionally a single injection of long-acting penicillin G is given instead of 10 days of oral treatment. These medications generally cost under \$15.

Because mononucleosis is caused by a virus, there is no specific drug treatment available. Rest, a healthy diet, plenty of fluids, limiting heavy **exercise** and competitive sports, and treatment of aches with **acetaminophen** (Datril, Tylenol, Panadol) or ibuprofen (Advil, Nuprin, Motrin, Medipren) will help the illness pass. Nearly 90% of mononucleosis infections are mild. The infected person does not normally get the disease again.

In the case of chronic sore throat, it is necessary to treat the underlying disease to heal the sore throat. If a sore throat caused by environmental factors, the aggravating stimulus should be eliminated from the sufferer's environment.

Home care for sore throat

Regardless of the cause of a sore throat, there are some home care steps that people can take to ease their discomfort. These include:

- taking acetaminophen or ibuprofen for pain; **aspirin** should not be given to children because of its association with increased risk for **Reye's Syndrome**, a serious disease
- gargling with warm double strength tea or warm salt water made by adding 1 tsp of salt to 8 oz (237 ml) of water
- drinking plenty of fluids, but avoiding acid juices like orange juice, which can irritate the throat (sucking on popsicles is a good way to get fluids into children)
- eating soft, nutritious foods like noodle soup and avoiding spicy foods
- refraining from smoking
- resting until the fever is gone, then resuming strenuous activities gradually
- a room humidifier may make sore throat sufferers more comfortable
- antiseptic lozenges and sprays may aggravate the sore throat rather than improve it

Alternative treatment

Alternative treatment focuses on easing the symptoms of sore throat using herbs and botanical medicines.

- Aromatherapists recommend inhaling the fragrances of essential oils of lavender (*Lavandula officinalis*),

thyme (*Thymus vulgaris*), eucalyptus (*Eucalyptus globulus*), sage (*Salvia officinalis*), and sandalwood.

- Ayurvedic practitioners suggest gargling with a mixture of water, salt, and tumeric (*Curcuma longa*) powder or astringents such as alum, sumac, sage, and bayberry (*Myrica* spp.).
- Herbalists recommend taking osha root (*Ligusticum porteri*) internally for infection or drinking ginger (*Zingiber officinale*) or slippery elm (*Ulmus fulva*) tea for pain.
- Homeopaths may treat sore throats with superdilute solutions *Lachesis*, *Belladonna*, *Phytolacca*, yellow jasmine (*Gelsemium*), or mercury.
- Nutritional recommendations include zinc lozenges every two hours along with vitamin C with bioflavonoids, vitamin A, and beta-carotene supplements.

Prognosis

Sore throat caused by a viral infection generally clears up on its own within one week with no complications. The exception is mononucleosis. Ninety percent of cases of mononucleosis clear up without medical intervention or complications, so long as **dehydration** does not occur. In young children the symptoms may last only a week, but in adolescents the symptoms last longer. Adults over age 30 have the most severe and long lasting symptoms. Adults may take up to six months to recover. In all age groups **fatigue** and weakness may continue for up to six weeks after other symptoms disappear.

In rare cases of mononucleosis, breathing may be obstructed because of swollen tonsils, adenoids, and lymph glands. If this happens, the patient should immediately seek emergency medical care.

Patients with bacterial sore throat begin feeling better about 24 hours after starting antibiotics. Untreated strep throat has the potential to cause **scarlet fever**, kidney damage, or **rheumatic fever**. Scarlet fever causes a rash, and can cause high fever and convulsions. Rheumatic fever causes inflammation of the heart and damage to the heart valves. Taking antibiotics within the first week of a strep infection will prevent these complications. People with strep throat remain contagious until after they have been taking antibiotics for 24 hours.

Prevention

There is no way to prevent a sore throat; however, the risk of getting one or passing one on to another person can be minimized by:

KEY TERMS

Antigen—A foreign protein to which the body reacts by making antibodies

Conjunctivitis—An inflammation of the membrane surrounding the eye; also known as pinkeye.

Lymphocyte—A type of white blood cell. Lymphocytes play an important role in fighting disease.

Pharynx—The pharynx is the part of the throat that lies between the mouth and the larynx or voice box.

Toxin—A poison. In the case of scarlet fever, the toxin is secreted as a byproduct of the growth of the streptococcus bacteria and causes a rash.

- washing hands well and frequently
- avoiding close contact with someone who has a sore throat
- not sharing food and eating utensils with anyone
- not smoking
- staying out of polluted air

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Sotalol see **Antiarrhythmic drugs**

Sound therapy see **Music therapy**

South American blastomycosis

Definition

South American **blastomycosis** is a potentially fatal, chronic fungus infection that occurs more often in men. The infection may affect different parts of the body, including the lungs or the skin, and may cause ulcers of the mouth, voicebox, and nose.

KEY TERMS

Amphotericin B—A drug used to treat fungal infections.

Sulfonamide drugs—A group of antibacterial drugs used to treat infections of the lungs and skin, among other things.

Description

South American blastomycosis occurs primarily in Brazil, although cases crop up in Mexico, Central America, or other parts of South America. It affects men between ages 20 and 50 about 10 times more often than women.

The disease is far more serious than its North American variant (North American blastomycosis), which is endemic to the eastern United States, southern Canada, and the midwest.

South American blastomycosis is known medically as paracoccidioidal granuloma, or paracoccidioidomycosis. The infection has a very long incubation period (at least five years).

Causes and symptoms

South American blastomycosis is caused by the yeast-like fungus *Paracoccidioides brasiliensis* that is acquired by breathing in the spores of the fungus, which is commonly found in old wood and soil. It may appear very similar to **tuberculosis**; in fact, both diseases may infect a patient at the same time.

Symptoms include ulcers in the mouth, larynx and nose, in addition to large, draining lymph nodes, **cough**, chest **pain**, swollen lymph glands, weight loss, and lesions on the skin, genitals, and intestines. There may also be lesions in the liver, spleen, intestines, and adrenal glands.

Diagnosis

A physician can diagnose the condition by microscopic examination of a smear prepared from a lesion or sputum (spit). Biopsy specimens may also reveal the infection. While blood tests are helpful, they can't determine the difference between past and active infection.

Treatment

The primary goal of treatment is to control the infection. The best treatment has been amphotericin

B. Sulfonamide drugs have been used and can stop the progress of the infection, but they don't kill the fungus.

Scientists are studying new treatments for the fungal infection, including ketoconazole, fluconazole, and itraconazole, which appear to be equally effective as amphotericin B, according to research.

Prognosis

The disease is chronic and often fatal. Because blastomycosis may be recurrent, patients should continue follow-up care for several years.

Prevention

There is no way to prevent the disease.

Resources

ORGANIZATIONS

National Institute of Allergy and Infectious Disease.
Building 31, Room 7A-50, 31 Center Drive MSC 2520,
Bethesda, MD 20892-2520. (301) 496-5717. <<http://www.niaid.nih.gov/default.htm>>.
National Organization for Rare Disorders. P.O. Box 8923,
New Fairfield, CT 06812-8923. (800) 999-6673.
<<http://www.rarediseases.org>>.

Carol A. Turkington

Space medicine see **Aviation medicine**

Spanish flu see **Influenza**

Spastic colitis see **Irritable bowel syndrome**

Spastic colon see **Irritable bowel syndrome**

Speech disorders

Definition

According to the American Speech-Language-Hearing Association (ASHA), a language disorder is an impairment in comprehension use of the spoken, written, or other symbol system.

Description

Speech disorders affect the language and mechanics, the content of speech, or the function of language in communication. Because speech disorders affect a person's ability to communicate effectively,

every aspect of the person's life can be affected, for example, the person's ability to make friends, and to communicate at school or at work.

Amyotrophic lateral sclerosis (ALS)

Amyotrophic lateral sclerosis (ALS), also known as Lou Gehrig's disease, is a neurological disease that attacks the nerve cells in the brain that control voluntary muscles. ALS causes motor neurons to die so that the brain and spinal cord are unable to send messages to the muscles telling them to move. Because the muscles are not functioning, they begin to atrophy. Muscles in the face and jaw can be affected, and thereby affecting a person's speech.

Aphasia

Aphasia results from damage to the language centers of the brain, which affects a person's ability to communicate through speaking, listening, and writing.

Persons with aphasia have trouble with expressive language, what is said, or receptive language, what is understood. Not only are speech and understanding speech affected, but also reading and writing is affected. The severity of aphasia varies from person to person, but in the most severe cases, a person may not be able to understand speech at all. Persons with mild aphasia may only become confused when speech becomes lengthy and complicated.

Developmental apraxia of speech

Developmental **apraxia** is a disorder that affects the nervous system and affects a person's ability to sequence and say sounds, syllables, and words. The brain does not send the correct messages to the mouth and jaw so that the person can say what he or she wants to say.

Children who are suffering from this disorder don't babble as an infant and first words are delayed. Older children may have more difficulty with longer phrases, and may appear to be searching for words to express a thought. Listeners will likely have a difficult time understanding the child.

Laryngeal cancer

Laryngeal cancer is characterized by a malignant growth in the larynx, or the voice box, which sometimes requires removal of the larynx or part of it.

Cancer anywhere in the throat affects speech, swallowing, and chewing. Depending on the size of the growth, a person may have trouble moving the

mouth and lips. Therefore, speech sounds and eating will be affected and a person will have trouble communicating.

Orofacial myofunctional disorders

Orofacial myofunctional disorder (OMD) causes the tongue to move forward in an exaggerated manner while a person is speaking or swallowing. The tongue also may protrude when resting in the mouth.

Because heredity contributes to the size and shape of a person's mouth, there may be genetic reasons for the disorder. **Allergies** also affect the mouth and face muscles, which make it difficult to breathe because of nasal congestion. Because a person may sleep with the tongue protruding, lip muscles weaken. Enlarged tonsils also can block airways, creating the same breathing problems. Additionally, thumb-sucking, nail-biting, and teeth-clenching and grinding also can contribute to the disorder.

Stuttering

Stuttering is a disorder of speech fluency that frequently interrupts the flow of speech.

Because children typically stumble and confuse their words as speech develops, stuttering is not immediately evident. It is usually when children become older and continue to stumble that stuttering becomes evident.

Causes and symptoms

Amyotrophic lateral sclerosis (ALS)

Initial symptoms include weakness in any part of the body, and appendages begin to tire easily. Occasionally the disease affects only one appendage rather than both at the same time. Persons with ALS may have trouble maintaining balance and may stumble or have difficulty with tasks that require manual dexterity, such as buttoning a shirt or tying a shoe.

Eventually, the diaphragm and chest wall become so weak that a person cannot breathe on his or her own and needs the help of a ventilator. Because of the lack of muscle strength, a person with ALS will experience difficulty speaking loudly and clearly until the person is unable to speak at all using the vocal cords. The person will have difficulty pronouncing words and have difficulty completing lengthy sentences.

Along with the difficulty in speaking also comes difficulty in chewing and swallowing. Food can be broken down and pureed to make it easier to chew and swallow. However, a person eventually will have

difficulty chewing and swallowing foods that are broken down or pureed. When ability to eat is affected, proper **nutrition** and body weight also are affected, and medical professionals may decide that it is best to put in a feeding tube.

Aphasia

Stroke is the most common cause of aphasia, although other injuries, such as a **brain tumor** or gunshot wound, also can cause aphasia.

Developmental apraxia of speech

Developmental apraxia is a disorder that affects the nervous system and affects a person's ability to sequence and say sounds, syllables, and words. The brain does not send the correct messages to the mouth and jaw so that the person can say what he or she wants to say.

Children who are suffering from this disorder don't babble as an infant and first words are delayed. Older children may have more difficulty with longer phrases, and may appear to be searching for words to express a thought. Listeners will likely have a difficult time understanding the child.

There is no known cause for developmental apraxia of speech. Symptoms include weakness of the jaw, tongue, and lips, and delayed speech development. Persons with the disorder also may have trouble identifying an object in the mouth using the sense of touch, which is known as oral-sensory perception.

Laryngeal cancer

Any kind of **smoking** of cigarettes, cigars, or tobacco and alcohol **abuse** contribute to oral cancer, including smokeless tobacco. Persons with laryngeal cancer or another type of oral cancer may have a red or white patch or lump in the mouth. Symptoms also include difficulty chewing, swallowing, or chewing.

Stuttering

There is no known cause for stuttering, although poor muscle coordination and the rate of language development are believed to contribute to it.

Stuttering is characterized by repetition of sounds, syllables, portions of a word, words, and complete phrases; stretching the sounds and syllables; hesitation between words; words spoken in spurts; tense muscles in the jaw and mouth; and a feeling of loss of control.

Diagnosis

Amyotrophic lateral sclerosis (ALS)

About 20,000 people in the United States have ALS at any given time with 5,000 new cases diagnosed every year. ALS is in the same family of disorders as **multiple sclerosis**, Parkinson's disease, and **muscular dystrophy**. Persons of all races and ethnic groups are afflicted by the disease, although men are more likely to have it than women.

Aphasia

About 700,000 persons in the United States have strokes every year, and one million are estimated to have aphasia.

Developmental apraxia of speech

A child suspected to have apraxia should first have his or her hearing tested to determine if the child has any deafness. Muscle development in the face and jaw should be evaluated and speech exercises tested. Articulation of words should be tested as well as the person's expressive and receptive language skills.

Laryngeal cancer

It is likely that a dentist or physician will first detect signs of possible cancer. Oral cancer makes up about 2–5% of all cancers, and about 30,000 cases are diagnosed each year. Twice as many men than women are diagnosed with cancer typically between the ages of 50 and 70.

Orofacial myofunctional disorders

The diagnosis of orofacial myofunctional disorder affects speech sounds because of weak tongue tip muscles, although a person's speech may not be affected at all.

Stuttering

Stuttering is a problem that most likely will manifest itself during childhood rather than adulthood.

Treatment

Amyotrophic lateral sclerosis (ALS)

In addition to treatments such as a feeding tube, a person with ALS would likely enlist the help of a speech therapist to help him or her determine ways in which he or she can maintain vocal control. A person also may enlist the help of an occupational therapist, a

medical professional trained to help persons who have trouble with activities of daily living such as dressing, bathing, and eating.

Aphasia

A speech-language pathologist can perform drills and exercises with a person that include practice in naming objects and following directions to try to improve skills. The person learns the best way to express himself or herself. **Group therapy** also is an option, which focuses on structured discussions.

Developmental apraxia of speech

Treatment should focus on the coordination of motor movements necessary during speech production, which includes controlling breathing. A speech-language pathologist teaches exercises to a person with apraxia that will strengthen the jaws, lips, and tongue to improve coordination during speech. The therapist uses tactile, auditory, and visual feedback to direct the brain to move the muscles used during speech.

Laryngeal cancer

Depending on when the cancer is first detected, and depending on the size of the cancer, the entire larynx may not need to be removed. Radiation, **chemotherapy**, or partial removal can be done in lieu of complete removal. In these cases, the voice may be preserved although the quality likely will be affected.

Orofacial myofunctional disorders

In cases where speech is affected, a speech pathologist should be consulted to help control breathing problems and work on speech articulation. The lip, palate, tongue, and facial muscles should be evaluated so that errors in speech can be detected. Therapy includes increasing awareness of the mouth and facial muscles, as well as the posture of the mouth and tongue. Muscle **exercise** can be done to increase strength and control.

Stuttering

A treatment plan by a speech therapist includes improving fluency and ease with which a person speaks. Strategies include reducing the rate of speech and using slower speech movements; articulating lightly; and starting air flow for speech before any other muscle movement.

Alternative treatment

Developmental apraxia of speech

Some persons with apraxia may decide to use alternative communication systems, such as a computer that transcribes and “speaks” what a person is directing it to say. These augmentative systems should only be used when a person is so severely impaired that effective speech or communication isn’t possible.

Laryngeal cancer

In cases of a full **laryngectomy**, a hole is made in the neck and, rather than using the mouth and nose to talk and breath, the person must use the hole.

Once the larynx is removed, the person needs to develop a new speech system without a voice. A speech pathologist should follow one of three plans: esophageal speech, artificial larynx, or tracheoesophageal puncture (TEP).

- **Esophageal speech.** Without a larynx, a person is no longer able to exhale air from the lungs through the mouth to speak. Using esophageal speech, the person inhales and traps the air in the throat, causing the esophagus to vibrate and create sound.
- **Artificial larynx.** A mechanical instrument can be used that produces sound for some speech. These devices can be held against the neck or used by inserting a tube in the mouth.
- **Tracheoesophageal puncture.** This is a popular method in restoring speech production. During surgery, a hole is made between the trachea and esophagus and a valve is inserted into the hole. The person breathes air into the lungs and then covers the hole in the throat. During exhalation, the esophagus vibrates and creates speech.

Stuttering

A person suffering from stuttering may employ distraction strategies to help him or her stop stuttering. Typically, a person stuttering becomes frustrated and embarrassed; subsequently, encouraging the person to think of something or do something else may break the stuttering cycle.

Prognosis

Amyotrophic lateral sclerosis (ALS)

ALS patients often die of **respiratory failure** within three to five years of being diagnosed, although some persons have been known to survive as many as 10 years or longer.

KEY TERMS

Neurons—Nerve cells in the brain, brain stem, and spinal cord that connect the nervous system and the muscles.

Aphasia

Persons with aphasia can improve and eventually function in more typical public settings, and possibly return to school or work.

Developmental apraxia of speech

With proper treatment, apraxia can be brought under control and the person will be able to function normally as an adult.

Laryngeal cancer

Full removal of the larynx removes the risk of a cancer relapse, although other parts of the throat and mouth can be affected.

Orofacial myofunctional disorders

A person can learn to control this disorder with proper treatment and maintain normal speech and breathing patterns.

Stuttering

With proper speech therapy, stuttering can be controlled or eliminated.

Prevention

Laryngeal cancer

Persons should not engage in smoking or drug abuse to decrease the risk of oral cancer.

Orofacial myofunctional disorders

In cases where the cause is evident, such as allergies or enlarged tonsils, a person should first remedy that problem; perhaps have the tonsils removed and treat allergies with medication.

Resources

BOOKS

Paul, Rhea. *Language Disorders from Infancy through Adolescence*. 2nd ed. St. Louis: Mosby, Inc., 2001.

ORGANIZATIONS

American Speech-Language-Hearing Association. 1801 Rockville Pike, Rockville, MD 20852. (800) 638-8255. < <http://www.asha.org> >.

Meghan Gourley

Speech disturbance see **Aphasia**

Speech therapy see **Rehabilitation**

Sperm count see **Semen analysis**

Spina bifida

Definition

Spina bifida is a serious birth abnormality in which the spinal cord is malformed and lacks its usual protective skeletal and soft tissue coverings.

Description

Spina bifida may appear in the body midline anywhere from the neck to the buttocks. In its most severe form, termed spinal rachischisis, the entire spinal canal is open, exposing the spinal cord and nerves. More commonly, the abnormality appears as a localized mass on the back that is covered by skin or by the meninges, the three-layered membrane that envelopes the spina cord. Spina bifida is usually readily apparent at birth because of the malformation of the back and **paralysis** below the level of the abnormality.

Various forms of spina bifida are known as meningocele, myelomeningocele, spina bifida aperta, open spina bifida, myelodysplasia, spinal dysraphism, spinal rachischisis, myelocele, and meningocele. The term meningocele is used when the spine malformation contains only the protective covering (meninges) of the spinal cord. The other terms indicate involvement of the spinal cord and nerves in the malformation. A related term, spina bifida occulta, indicates that one or more of the bony bodies in the spine are incompletely hardened, but that there is no abnormality of the spinal cord itself.

Spina bifida occurs worldwide, but there has been a steady downward trend in occurrence rates over the past 50–70 years, particularly in regions of high prevalence. The highest prevalence rates, about one in 200 pregnancies, have been reported from certain northern provinces in China. Intermediate prevalence rates, about one in 1,000 pregnancies, have been found in

Central and South America. The lowest prevalence rates, less than one in 2,000 pregnancies, have been found in the European countries. The highest regional prevalence in the United States of about one in 500 pregnancies has occurred in the Southeast.

Causes and symptoms

Spina bifida may occur as an isolated abnormality or in the company of other malformations. As an isolated abnormality, spina bifida is caused by the combination of genetic factors and environmental influences that bring about malformation of the spine and spinal column. The specific genes and environmental influences that contribute to the many-faceted causes of spina bifida are not completely known. An insufficiency of **folic acid** is known to be one influential nutritional factor. Changes (mutations) in genes involving the metabolism of folic acid are believed to be significant genetic risk factors. The recurrence risk after the birth of an infant with isolated spina bifida is 3–5%. Recurrence may be for spina bifida or another type of spinal abnormality.

Spina bifida may arise because of chromosome abnormalities, single gene mutations, or specific environmental insults such as maternal **diabetes mellitus** or prenatal exposure to certain **anticonvulsant drugs**. The recurrence risk varies with each of these specific causes.

In most cases, spina bifida is obvious at birth because of malformation of the spine. The spine may be completely open, exposing the spinal cord and nerves. More commonly, the spine abnormality appears as a mass on the back covered by membrane (meninges) or skin. Spina bifida may occur anywhere from the base of the skull to the buttocks. About 75% of abnormalities occur in the lower back (lumbar) region. In rare instances, the spinal cord malformation may occur internally, sometimes with a connection to the gastrointestinal tract.

In spina bifida, many complications arise, dependent in part on the level and severity of the spine malformation. As a rule, the nerves below the level of the abnormality develop in a faulty manner and fail to function, resulting in paralysis and loss of sensation below the level of the spine malformation. Since most abnormalities occur in the lumbar region, the lower limbs are paralyzed and lack sensation. Furthermore, the bowel and bladder have inadequate nerve connections, causing an inability to control bowel and bladder function. Most infants also develop hydrocephaly, an accumulation of excess fluid in the four cavities of the brain. At least one of every seven cases develop

findings of Chiari II malformation, a condition in which the lower part of the brain is crowded and may be forced into the upper part of the spinal cavity.

There are a number of mild variant forms of spina bifida, including multiple vertebral abnormalities, skin dimples, tufts of hair, and localized areas of skin deficiency over the spine. Two variants, lipomeningocele and lipomyelomeningocele, typically occur in the lower back area (lumbar or sacral) of the spine. In these conditions, a tumor of fatty tissue becomes isolated among the nerves below the spinal cord, which may result in tethering of the spinal cord and complications similar to those with open spina bifida.

Diagnosis

Few disorders are to be confused with open spina bifida. The diagnosis is usually obvious based on the external findings at birth. Paralysis below the level of the abnormality and fluid on the brain (hydrocephaly) may contribute to the diagnosis. Other spine abnormalities such as congenital **scoliosis** and **kyphosis**, or soft tissue tumors overlying the spine, are not likely to have these accompanying findings. In cases in which there are no external findings, the diagnosis is more difficult and may not become evident until neurological abnormalities or hydrocephaly develop weeks, months, or years following birth.

Prenatal diagnosis may be made in most cases with ultrasound examination after 12–14 weeks of **pregnancy**. Many cases are also detected by the testing of the mother's blood for the level of alpha-fetoprotein at about 16 weeks of pregnancy. If the spine malformation is not skin covered, alpha-fetoprotein from the fetus' circulation may leak into the surrounding amniotic fluid, a small portion of which is absorbed into the mother's blood.

Treatment

Aggressive surgical and medical management have improved the survival and function of infants with spina bifida. Initial surgery may be carried out during the first days of life, providing protection against injury and infection. Subsequent surgery is often necessary to protect against excessive curvature of the spine, and in the presence of hydrocephaly, to place a mechanical shunt to decrease the pressure and amount of cerebrospinal fluid in the cavities of the brain. Because of weakness or paralysis below the level of the spine abnormality, most children will require physical therapy, bracing, and other orthopedic assistance to enable them to walk. A variety of



An infant with spina bifida. (Photograph by Biophoto Associates, Photo Researchers, Inc. Reproduced by permission.)

approaches including periodic bladder catheterization, surgical diversion of urine, and **antibiotics** are used to protect urinary function.

Although most individuals with spina bifida have normal intellectual function, learning disabilities or **mental retardation** occur in a minority. This may result, in part, from hydrocephaly and/or infections of the nervous system. Children so affected may benefit from early educational intervention, physical therapy, and occupational therapy. Counseling to improve self-image and lessen barriers to socialization becomes important in late childhood and adolescence.

Open fetal surgery has been performed for spina bifida during the last half of pregnancy. After direct closure of the spine malformation, the fetus is returned to the womb. By preventing chronic intrauterine exposure to mechanical and chemical trauma, **prenatal surgery** improves neurological function and leads to fewer complications after birth. Fetal surgery is considered experimental, and results have been mixed.

Prevention of isolated spina bifida and other spinal abnormalities has become possible during recent decades. The major prevention is through the

KEY TERMS

Chiari II anomaly—A structural abnormality of the lower portion of the brain (cerebellum and brain stem) associated with spina bifida. The lower structures of the brain are crowded and may be forced into the foramen magnum, the opening through which the brain and spinal cord are connected.

Fetus—The term used to describe a developing human infant from approximately the third month of pregnancy until delivery. The term embryo is used prior to the third month.

Hydrocephalus—The excess accumulation of cerebrospinal fluid around the brain, often causing enlargement of the head.

use of a B vitamin, folic acid, for several months prior to and following conception. The Centers for Disease Control and Prevention recommend the intake of 400 micrograms of synthetic folic acid every day for all women of childbearing years.

Prognosis

More than 80% of infants born with spina bifida survive with surgical and medical management. Although complications from paralysis, hydrocephaly, Chiari II malformation, and urinary tract deterioration threaten the well-being of the survivors, the outlook for normal intellectual function is good.

Resources

ORGANIZATIONS

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Ave., White Plains, NY 10605. (888) 663-4637. <<http://www.modimes.org>>.

National Birth Defects Prevention Network. Atlanta, GA (770) 488-3550. <<http://www.nbdpn.org>>.

Shriners Hospitals for Children. International Shrine Headquarters, 2900 Rocky Point Dr., Tampa, FL 33607-1460. (813) 281-0300. <<http://www.shrinershq.org>>.

Spina Bifida Association of America. 4590 MacArthur Blvd. NW, Suite 250, Washington, DC 20007-4226. (800) 621-3141 or (202) 944-3285. Fax: (202) 944-3295.

Roger E. Stevenson

Spina bifida occulta see **Spina bifida**

Spinal cord injury

Definition

Spinal cord injury is damage to the spinal cord that causes loss of sensation and motor control.

Description

Approximately 10,000 new spinal cord injuries (SCIs) occur each year in the United States. About 250,000 people are currently affected. Spinal cord injuries can happen to anyone at any time of life. The typical patient, however, is a man between the ages of 19 and 26, injured in a motor vehicle accident (about 50% of all SCIs), a fall (20%), an act of violence (15%), or a sporting accident (14%). Alcohol or other drug **abuse** plays an important role in a large percentage of all spinal cord injuries. Six percent of people who receive injuries to the lower spine die within a year, and 40% of people who receive the more frequent higher injuries die within a year.

Short-term costs for hospitalization, equipment, and home modifications are approximately \$140,000

for an SCI patient capable of independent living. Lifetime costs may exceed one million dollars. Costs may be three to four times higher for the SCI patient who needs long-term institutional care. Overall costs to the American economy in direct payments and lost productivity are more than \$10 billion per year.

Causes and symptoms

Causes

The spinal cord is about as big around as the index finger. It descends from the brain down the back through hollow channels of the backbone. The spinal cord is made of nerve cells (neurons). The nerve cells carry sensory data from the areas outside the spinal cord (periphery) to the brain, and they carry motor commands from brain to periphery. Peripheral neurons are bundled together to make up the 31 pairs of peripheral nerve roots. The peripheral nerve roots enter and exit the spinal cord by passing through the spaces between the stacked vertebrae. Each pair of nerves is named for the vertebra from which it exits. These are known as:

- C1-8. These nerves enter from the eight cervical or neck vertebrae.
- T1-12. These nerves enter from the thoracic or chest vertebrae.
- L1-5. These nerves enter from the lumbar vertebrae of the lower back.
- S1-5. These nerves enter through the sacral or pelvic vertebrae.
- Coccygeal. These nerves enter through the coccyx or tailbone.

Peripheral nerves carry motor commands to the muscles and internal organs, and they carry sensations from these areas and from the body's surface. (Sensory data from the head, including sight, sound, smell, and taste, do not pass through the spinal cord and are not affected by most SCIs.) Damage to the spinal cord interrupts these signals. The interruption damages motor functions that allow the muscles to move, sensory functions such as feeling heat and cold, and autonomic functions such as urination, sexual function, sweating, and blood pressure.

Spinal cord injuries most often occur where the spine is most flexible, in the regions of C5-C7 of the neck, and T10-L2 at the base of the rib cage. Several physically distinct types of damage are recognized. Sudden and violent jolts to nearby tissues can jar the cord. This jarring causes a temporary spinal **concussion**. Concussion symptoms usually disappear

completely within several hours. A spinal contusion or bruise is bleeding within the spinal column. The pressure from the excess fluid may kill spinal cord neurons. Spinal compression is caused by some object, such as a tumor, pressing on the cord. Lacerations or tears cause direct damage to cord neurons. Lacerations can be caused by bone fragments or missiles such as bullets. Spinal transection describes the complete severing of the cord. Most spinal cord injuries involve two or more of these types of damage.

Symptoms

PARALYSIS AND LOSS OF SENSATION. The extent to which movement and sensation are damaged depends on the level of the spinal cord injury. Nerves leaving the spinal cord at different levels control sensation and movement in different parts of the body. The distribution is roughly as follows:

- C1-C4: head and neck.
- C3-C5: diaphragm (chest and breathing).
- C5-T1: shoulders, arms and hands.
- T2-T12: chest and abdomen (excluding internal organs).
- L1-L4: abdomen (excluding internal organs), buttocks, genitals, and upper legs.
- L4-S1: legs.
- S2-S4: genitals and muscles of the perineum.

Damage below T1, which lies at the base of the rib cage, causes **paralysis** and loss of sensation in the legs and trunk below the injury. Injury at this level usually does no damage to the arms and hands. Paralysis of the legs is called paraplegia. Damage above T1 involves the arms as well as the legs. Paralysis of all four limbs is called quadriplegia or tetraplegia. Cervical or neck injuries not only cause quadriplegia but also may cause difficulty in breathing. Damage in the lower part of the neck may leave enough diaphragm control to allow unassisted breathing. Patients with damage at C3 or above, just below the base of the skull, require mechanical assistance to breathe.

Symptoms also depend on the extent of spinal cord injury. A completely severed cord causes paralysis and loss of sensation below the wound. If the cord is only partially severed, some function will remain below the injury. Damage limited to the front portion of the cord causes paralysis and loss of sensations of **pain** and temperature. Other sensation may be preserved. Damage to the center of the cord may spare the legs but paralyze the arms. Damage to the right or

left half causes loss of position sense, paralysis on the side of the injury, and loss of pain and temperature sensation on the opposite side.

DEEP VENOUS THROMBOSIS. Blood does not flow normally to a paralyzed limb that is inactive for long periods. The blood pools in the deep veins and forms clots, a condition known as **deep vein thrombosis**. A clot or thrombus can break free and lodge in smaller arteries in the brain, causing a **stroke**, or in the lungs, causing **pulmonary embolism**.

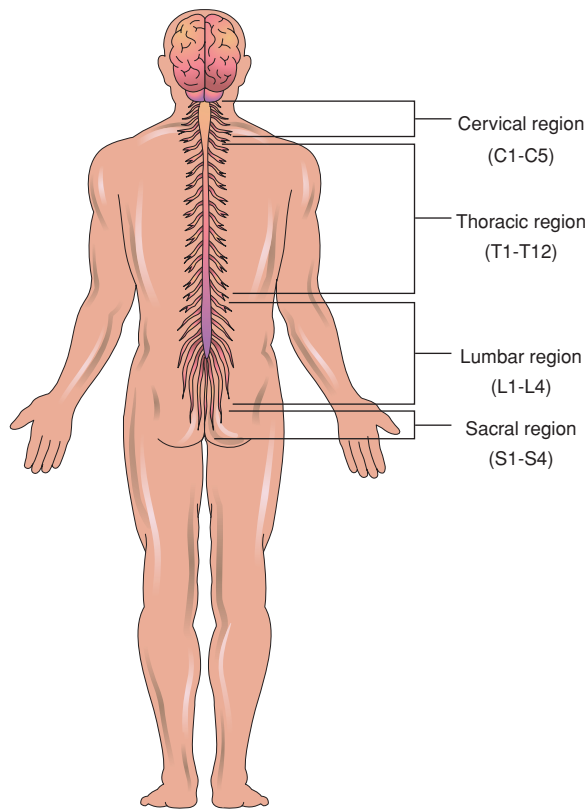
PRESSURE ULCERS. Inability to move also leads to pressure ulcers or bed sores. Pressure ulcers form where skin remains in contact with a bed or chair for a long time. The most common sites of pressure ulcers are the buttocks, hips, and heels.

SPASTICITY AND CONTRACTURE. A paralyzed limb is incapable of active movement, but the muscle still has tone, a constant low level of contraction. Normal muscle tone requires communication between the muscle and the brain. Spinal cord injury prevents the brain from telling the muscle to relax. The result is prolonged muscle contraction or spasticity. Because the muscles that extend and those that bend a joint are not usually equal in strength, the involved joint is bent, often severely. This constant pressure causes deformity. As the muscle remains in the shortened position over several weeks or months, the tendons remodel and cause permanent muscle shortening or contracture. When muscles have permanently shortened, the inner surfaces of joints, such as armpits or palms, cannot be cleaned and the skin breaks down in that area.

HETEROTOPIC OSSIFICATION. Heterotopic ossification is an abnormal deposit of bone in muscles and tendons that may occur after injury. It is most common in the hips and knees. Initially heterotopic ossification causes localized swelling, warmth, redness, and stiffness of the muscle. It usually begins one to four months after the injury and is rare after one year.

AUTONOMIC DYSREFLEXIA. Body organs that regulate themselves, such as the heart, gastrointestinal tract, and glands, are controlled by groups of nerves called autonomic nerves. Autonomic nerves emerge from three different places: above the spinal column, in the lower back from vertebrae T1-L4, and from the lowest regions of the sacrum at the base of the spine. In general, these three groups of autonomic nerves operate in balance. Spinal cord injury can disrupt this balance, a condition called autonomic dysreflexia or autonomic hyperreflexia. Patients with injuries at T6 or above are at greatest risk.

In autonomic dysreflexia, irritation of the skin, bowel, or bladder causes a highly exaggerated



The extent of sensory and motor loss resulting from a spinal cord injury depends on the level of the injury because nerves at different levels control sensation and movement in different parts of the body. The distribution is as follows: C1-C4: head and neck; C3-C5: diaphragm; C5-T1: shoulders, arms, and hands; T2-T12: chest and abdomen (excluding internal organs); L1-L4: abdomen (excluding internal organs), buttocks, genitals, upper legs; L4-S3: legs; S2-S4: genitals, muscles of the perineum. (Illustration by Electronic Illustrators Group.)

response from autonomic nerves. This response is caused by the uncontrolled release of norepinephrine, a hormone similar to adrenaline. Uncontrolled release of norepinephrine causes a rapid rise in blood pressure and a slowing of the heart rate. These symptoms are accompanied by throbbing **headache**, **nausea**, **anxiety**, sweating, and goose bumps below the level of the injury. The elevated blood pressure can rapidly cause loss of consciousness, seizures, cerebral hemorrhage, and **death**. Autonomic dysreflexia is most often caused by an over-full bladder or bladder infection, impaction or hard impassable fecal mass in the bowel, or skin irritation from tight clothing, **sunburn**, or other irritant. Inability to sense these irritants before the autonomic reaction begins is a major cause of dysreflexia.

LOSS OF BLADDER AND BOWEL CONTROL. Bladder and bowel control require both motor nerves and the autonomic nervous system. Both of these systems may be damaged by SCI. When the autonomic nervous system triggers an urge to urinate or defecate, continence is maintained by contracting the anal or urethral sphincters. A sphincter is a ring of muscle that contracts to close off a passage or opening in the body. When the neural connections to these muscles are severed, conscious control is lost. In addition, loss of feeling may prevent sensations of fullness from reaching the brain. To compensate, the patient may help empty the bowel or bladder by using physical maneuvers that stimulate autonomic contractions before they would otherwise begin. However, the patient may not be able to relax the sphincters. If the sphincters cannot be relaxed, the patient will retain urine or feces.

Retention of urine may cause muscular changes in the bladder and urethral sphincter that make the problem worse. Urinary tract infection is common. Retention of feces can cause impaction. Symptoms of impaction include loss of appetite and nausea. Untreated impaction may cause perforation of the large intestine and rapid overwhelming infection.

SEXUAL DYSFUNCTION. Men who have sustained SCI may be unable to achieve an erection or ejaculate. Sperm formation may be abnormal too, reducing fertility. Fertility and the ability to achieve orgasm are less impaired for women. Women may still be able to become pregnant and deliver vaginally with proper medical care.

Diagnosis

The location and extent of spinal cord injury is determined with **computed tomography scans** (CT scans), **magnetic resonance imaging** (MRI) scans, and x rays. X rays may be enhanced with an injected contrast dye.

Treatment

A person who may have a spinal cord injury should not be moved. Treatment of SCI begins with **immobilization**. This strategy prevents partial injuries of the cord from severing it completely. Use of splints to completely immobilize suspected SCI at the scene of the injury has helped reduce the severity of spinal cord injuries in the last two decades. Intravenous methylprednisone, a steroidal anti-inflammatory drug, is given during the first 24 hours to reduce inflammation and tissue destruction.

Rehabilitation after spinal cord injury seeks to prevent complications, promote recovery, and make the most of remaining function. Rehabilitation is a complex and long-term process. It requires a team of professionals, including a neurologist, physiatrist or rehabilitation specialist, physical therapist, and occupational therapist. Other specialists who may be needed include a respiratory therapist, vocational rehabilitation counselor, social worker, speech-language pathologist, nutritionist, special education teacher, recreation therapist, and clinical psychologist. Support groups provide a critical source of information, advice, and support for SCI patients.

Paralysis and loss of sensation

Some limited mobility and sensation may be recovered, but the extent and speed of this recovery cannot be predicted. Experimental electrical stimulation has been shown to allow some control of muscle contraction in paraplegia. This experimental technique offers the possibility of unaided walking. Further development of current control systems will be needed before useful movement is possible outside the laboratory.

The physical therapist focuses on mobility, to maintain range of motion of affected limbs and reduce contracture and deformity. Physical therapy helps compensate for lost skills by using those muscles that are still functional. It also helps to increase any residual strength and control in affected muscles. A physical therapist suggests adaptive equipment such as braces, canes, or wheelchairs.

An occupational therapist works to restore ability to perform the activities of daily living, such as eating and grooming, with tools and new techniques. The occupational therapist also designs modifications of the home and workplace to match the individual impairment.

A pulmonologist or respiratory therapist promotes airway hygiene through instruction in assisted coughing techniques and postural drainage. The respiratory professional also prescribes and provides instruction in the use of ventilators, facial or nasal masks, and tracheostomy equipment where necessary.

Pressure ulcers

Pressure ulcers are prevented by turning in bed at least every two hours. The patient should be turned more frequently when redness begins to develop in sensitive areas. Special mattresses and chair cushions can distribute weight more evenly to reduce pressure.

Electrical stimulation is sometimes used to promote muscle movement to prevent pressure ulcers.

Spasticity and contracture

Range of motion (ROM) exercises help to prevent contracture. Chemicals can be used to prevent **contractures** from becoming fixed when ROM **exercise** is inadequate. Phenol or alcohol can be injected onto the nerve or botulinum toxin directly into the muscle. Botulinum toxin is associated with fewer complications, but it is more expensive than phenol and alcohol. Contractures can be released by cutting the shortened tendon or transferring it surgically to a different site on the bone where its pull will not cause as much deformity. Such tendon transfers may also be used to increase strength in partially functional extremities.

Heterotopic ossification

Etidronate disodium (Didronel), a drug that regulates the body's use of calcium, is used to prevent heterotopic ossification. Treatment begins three weeks after the injury and continues for 12 weeks. Surgical removal of ossified tissue is possible.

Autonomic dysreflexia

Autonomic dysreflexia is prevented by bowel and bladder care and attention to potential irritants. It is treated by prompt removal of the irritant. Drugs to lower blood pressure are used when necessary. People with SCI should educate friends and family members about the symptoms and treatment of dysreflexia, because immediate attention is necessary.

Loss of bladder and bowel control

Normal bowel function is promoted through adequate fluid intake and a diet rich in fiber. Evacuation is stimulated by deliberately increasing the abdominal pressure, either voluntarily or by using an abdominal binder.

Bladder care involves continual or intermittent catheterization. The full bladder may be detected by feeling its bulge against the abdominal wall. Urinary tract infection is a significant complication of catheterization and requires frequent monitoring.

Sexual dysfunction

Counseling can help in adjusting to changes in sexual function after spinal cord injury. Erection may be enhanced through the same means used to treat **erectile dysfunction** in the general population.

KEY TERMS

Autonomic nervous system—The part of the nervous system that controls involuntary functions such as sweating and blood pressure.

Botulinum toxin—Any of a group of potent bacterial toxins or poisons produced by different strains of the bacterium *Clostridium botulinum*.

Computed tomography (CT)—An imaging technique in which cross-sectional x rays of the body are compiled to create a three-dimensional image of the body's internal structures.

Magnetic resonance imaging (MRI)—An imaging technique that uses a large circular magnet and radio waves to generate signals from atoms in the body. These signals are used to construct images of internal structures.

Motor—Of or pertaining to motion, the body apparatus involved in movement, or the brain functions that direct purposeful activity.

Motor nerve—Motor or efferent nerve cells carry impulses from the brain to muscle or organ tissue.

Peripheral nervous system—The part of the nervous system that is outside the brain and spinal cord. Sensory, motor, and autonomic nerves are included.

Postural drainage—The use of positioning to drain secretions from the bronchial tubes and lungs into the trachea or windpipe.

Range of motion (ROM)—The range of motion of a joint from full extension to full flexion (bending) measured in degrees like a circle.

Sensory nerves—Sensory or afferent nerves carry impulses of sensation from the periphery or outward parts of the body to the brain. Sensations include feelings, impressions, and awareness of the state of the body.

Voluntary—An action or thought undertaken or controlled by a person's free will or choice.

Prognosis

The prognosis of SCI depends on the location and extent of injury. Injuries of the neck above C4 with significant involvement of the diaphragm hold the gravest prognosis. Respiratory infection is one of the leading causes of death in long-term SCI. Overall, 85% of SCI patients who survive the

first 24 hours are alive 10 years after their injuries. Recovery of function is impossible to predict. Partial recovery is more likely after an incomplete wound than after the spinal cord has been completely severed.

Prevention

Risk of spinal cord injury can be reduced through prevention of the accidents that lead to it. Chances of injury from automobile accidents, the major cause of SCIs, can be significantly reduced by driving at safe speeds, avoiding alcohol while driving, and using seat belts.

Resources

ORGANIZATIONS

National Spinal Cord Injury Association. 8300 Colesville Road, Silver Spring, Maryland 20910. (301) 588-6959. <<http://www.erols.com/nsCIA>>.

Richard Robinson

Spinal cord tumors

Definition

A spinal cord tumor is a benign or cancerous growth in the spinal cord, between the membranes covering the spinal cord, or in the spinal canal. A tumor in this location can compress the spinal cord or its nerve roots; therefore, even a noncancerous growth can be disabling unless properly treated.

Description

The spinal cord contains bundles of nerves that carry messages between the brain and the body. Because the spinal cord is rigidly encased in bone, any tumor that grows on or near it can compress the nerves, and interfere in this communication. About 10,000 Americans develop spinal cord tumors each year, and about 40% of these are cancerous. Similar to brain tumors, spinal cord growths are rare.

Newly formed tumors that begin within the spinal cord are unusual, especially among children and the elderly. More typically, tumors originate elsewhere in the body and move through the bloodstream (metastasize) to the spinal cord.

Causes and symptoms

Scientists don't know what causes these tumors, although the noncancerous growths may be hereditary or present since birth.

When the tumor presses on the spinal cord, it causes symptoms including;

- back **pain**
- severe or burning pain in other parts of the body
- numbness or cold
- progressive loss of muscle strength or sensation in the legs
- loss of bladder or bowel control

A tumor in the top of the spinal column can cause pain radiating from the arms or neck; a tumor in the lower spine may cause leg or back pain. If there are several tumors in different areas of the spinal cord at the same time, it may cause symptoms in a variety of spots on the body.

Diagnosis

Suspected spinal cord compression, by tumor, is a medical emergency. Prompt intervention may prevent **paralysis**.

If a neurological exam and review of symptoms suggest a spinal cord tumor, the doctor may order some of these additional tests:

- MRI or CT scan
- myelography
- blood and spinal fluid studies
- x rays of the spine
- biopsy
- radionuclide bone scan

Treatment

If the tumor is malignant and has metastasized, treatment depends on the type of the primary **cancer**. Surgery is usually the first step in treating benign and malignant tumors outside the spinal cord. Tumors inside the spinal cord may not be able to be completely removed with surgery. If they can not be, radiation and **chemotherapy** treatments may be effective. Treatment also may include pain relievers and cortisone drugs to lessen swelling around the tumor, and ease pressure on the spinal cord.

KEY TERMS

Computed tomography scans (CT Scan)—The CT scan combines an X-ray with a computer to create a detailed picture of the spinal cord. It may help to determine the type of tumor, locate swelling or bleeding, and check results of treatment.

Magnetic resonance imaging (MRI) — MRI is an imaging technique that uses a magnetic field to scan the body's tissues and structures. It gives a better picture of tumors located near bone than does a CT scan, without the risk of radiation, and can provide a three-dimensional image of the tumor.

Myelogram—A myelogram is an x ray exam of the spinal cord, nerves and other tissues within the spinal cord that are highlighted by injected contrast dye.

Prognosis

Early diagnosis and treatment can produce a higher success rate. Long-term survival also depends on the tumor's type, location, and size. Surgery to remove the bone around the cord can ease pressure on the spinal nerves and nerve pathways, which will usually ease pain and other symptoms; however, it may make walking more difficult. Physical therapy and **rehabilitation** may help.

Prevention

Since spinal cord tumors usually are the result of a cancer that has first appeared elsewhere in the body, early detection of cancer in other organs may prevent spinal cord tumors. Lifestyle changes, as stopping **smoking**, to lower the risk of the development of other types of cancer, may also help.

Resources

ORGANIZATIONS

National Institute of Neurological Disorders and Stroke.
P.O. Box 5801, Bethesda, MD 20824. (800) 352-9424.
< <http://www.ninds.nih.gov/index.htm> >.

Carol A. Turkington

Spinal fluid analysis see **Cerebrospinal fluid (CSF) analysis**

Spinal fusion see **Disk removal**

Spinal instrumentation

Definition

Spinal instrumentation is a method of straightening and stabilizing the spine after spinal fusion, by surgically attaching hooks, rods, and wire to the spine in a way that redistributes the stresses on the bones and keeps them in proper alignment.

Purpose

Spinal instrumentation is used to treat instability and deformity of the spine. Instability occurs when the spine no longer maintains its normal shape during movement. Such instability results in nerve damage, spinal deformities, and disabling **pain**. Spinal deformities may be caused by:

- birth defects
- fractures
- marfan syndrome
- neurofibromatosis
- neuromuscular diseases
- severe injuries
- tumors

Curvature of the spine (**scoliosis**) is usually treated with spinal fusion and spinal instrumentation. Scoliosis is a disorder of unknown origin. It causes bending and twisting of the spine that eventually results in distortion of the chest and back. About 85% of cases occur in girls between the ages of 12 and 15, who are experiencing adolescent growth spurt.

Spinal instrumentation serves three purposes. It provides a stable, rigid column that encourages bones to fuse after spinal-fusion surgery. Second, it redirects the stresses over a wider area. Third, it restores the spine to its proper alignment.

Different types of spinal instrumentation are used to treat different spinal problems. Several common types of spinal instrumentation are explained below. Although the details of the insertion of rods, wires, and hooks varies, the purpose of all spinal instrumentation is the same—to correct and stabilize the backbone.

Harrington rod

The Harrington Rod is one of the oldest and most proven forms of spinal instrumentation. It is used to

straighten and stabilize the spine when curvature is greater than 60 degrees. It is an appropriate treatment for scoliosis.

Advantages of the Harrington rod are its relative simplicity of installation, the low rate of complications, and a proven record of reducing curvature of the spine. The main disadvantage is that the patient must remain in a body cast for about six months, then wear a brace for another three to six months while the bone fusion solidifies.

Luque rod

Luque rods are custom contoured metal rods that are fixed to each segment (vertebra) in the affected part of the spine. The main advantage is that the patient may not need to wear a cast or brace after the procedure. The main disadvantage is that the risk of injury to the nerves and spinal cord is higher than with some other forms of instrumentation. This is because wires must be threaded through each vertebra near the spinal column, increasing the risk of such damage. Luque rods are sometimes used to treat scoliosis.

Drummond instrumentation

Drummond instrumentation, also called Harrington-Drummond instrumentation, uses a Harrington rod on the concave side of the spine and a Luque rod on the convex side. The advantage is that each vertebra segment is fixed, with the risk of nerve injury decreased over Luque rod instrumentation. The disadvantage is that, like Harrington rod instrumentation, the patient must wear a cast and a brace after surgery.

Cotrel-Dubousset instrumentation

Cotrel-Dubousset instrumentation uses hooks and rods in a cross-linked pattern to realign the spine and redistribute the biomechanical **stress**. The main advantage of Cotrel-Dubousset instrumentation is that, because of the extensive cross-linking, the patient may have to wear a cast or brace after surgery. The disadvantage is the complexity of the operation and the number of hooks and cross-links that may fail.

Zeilke instrumentation

Zeilke instrumentation is similar to Cotrel-Dubousset instrumentation, but is used to treat double curvature of the spine. It requires wearing a brace for many months after surgery.

Other forms of instrumentation

The Kaneda device is used to treat fractured thoracic or lumbar vertebrae when it is suspected that bone fragments are present in the spinal canal. Variations on the basic forms of spinal instrumentation, such as Wisconsin instrumentation, are being refined as technology improves. A physician chooses the proper type of instrumentation based on the type of disorder, the age and health of the patient, and on the physician's experience.

Precautions

Since the hooks and rods of spinal instrumentation are anchored in the bones of the back, spinal instrumentation should not be performed on people with serious **osteoporosis**. To overcome this limitation, techniques are being explored that help anchor instrumentation in fragile bones.

Description

Spinal instrumentation is performed by a neuro and/or orthopedic surgical team with special experience in spinal operations. The surgery is done in a hospital under **general anesthesia**. It is done at the same time as spinal fusion.

The surgeon strips the muscles away from the area to be fused. The surface of the bone is peeled away. A piece of bone is removed from the hip and placed along side the area to be fused. The stripping of the bone helps the bone graft to fuse.

After the fusion site is prepared, the rods, hooks, and wires are inserted. There is some variation in how this is done based on the spinal instrumentation chosen. In general, Harrington rods are the simplest instrumentation to install, and Cotrel-Dubousset instrumentation is the most complex and risky. Once the rods are in place, the incision is closed.

Preparation

Spinal fusion with spinal instrumentation is major surgery. The patient will undergo many tests to determine that nature and exact location of the back problem. These tests are likely to include x rays, **magnetic resonance imaging (MRI)**, **computed tomography scans (CT scans)**, and myelograms. In addition, the patient will undergo a battery of blood and urine tests, and possibly an electrocardiogram to provide the surgeon and anesthesiologist with information that will allow the operation to be performed safely. In Harrington rod instrumentation, the patient may be placed in **traction** or an upper body cast to stretch contracted muscles before surgery.

KEY TERMS

Lumbar vertebrae—The vertebrae of the lower back below the level of the ribs.

Marfan syndrome—A rare hereditary defect that affects the connective tissue.

Neurofibromatosis—A rare hereditary disease that involves the growth of lesions that may affect the spinal cord.

Osteoporosis—A bone disorder, usually seen in the elderly, in which the bones become increasingly less dense and more brittle.

Spinal fusion—An operation in which the bones of the lower spine are permanently joined together using a bone graft obtained usually from the hip.

Thoracic vertebrae—The vertebrae in the chest region to which the ribs attach.

Aftercare

After surgery, the patient will be confined to bed. A catheter is inserted so that the patient can urinate without getting up. Vital signs are monitored, and the patient's position is changed frequently so that **bedsores** do not develop.

Recovery from spinal instrumentation can be a long, arduous process. Movement is severely limited for a period of time. In certain types of instrumentation, the patient is put in a cast to allow the realigned bones to stay in position until healing takes place. This can be as long as six to eight months. Many patients will need to wear a brace after the cast is removed.

During the recovery period, the patient is taught respiratory exercises to help maintain respiratory function during the time of limited mobility. Physical therapists assist the patient in learning self-care and in performing strengthening and range of motion exercises. Length of hospital stay depends on the age and health of the patient, as well as the specific problem that was corrected. The patient can expect to remain under a physician's care for many months.

Risks

Spinal instrumentation carries a significant risk of nerve damage and **paralysis**. The skill of the surgeon can affect the outcome of the operation, so patients should look for a hospital and surgical team that has a lot of experience doing spinal procedures.

After surgery there is a risk of infection or an inflammatory reaction due to the presence of the foreign material in the body. Serious infection of the membranes covering the spinal cord and brain can occur. In the long-term, the instrumentation may move or break, causing nerve damage and requiring a second surgery. Some bone grafts do not heal well, lengthening the time the patient must spend in a cast or brace, or necessitating additional surgery. Casting and wearing a brace may take an emotional toll, especially on young people. Patients who have had spinal instrumentation must avoid contact sports, and, for the rest of their lives, eliminate situations that will abnormally put stress on their spines.

Normal results

Many young people with scoliosis heal with significantly improved alignment of the spine. Results of spinal instrumentation done for other conditions vary widely.

Resources

ORGANIZATIONS

National Scoliosis Foundation. 5 Cabot Place, Stoughton, MA 02074. (800) 673-6922. <<http://www.scoliosis.org>>.

OTHER

Orthogate. <<http://owl.orthogate.org/>>.

Tish Davidson, A.M.

Spinal meningitis see **Meningitis**

Spinal stenosis

Definition

Spinal stenosis is any narrowing of the spinal canal that causes compression of the spinal nerve cord. Spinal stenosis causes **pain** and may cause loss of some body functions.

Description

Spinal stenosis is a progressive narrowing of the opening in the spinal canal. The spine is a long series of bones called vertebrae. Between each pair of vertebra is a fibrous intervertebral disk. Collectively, the vertebrae and disks are called the backbone. Each vertebra has a hole through it. These holes line up to form the spinal canal. A large bundle of nerves called the spinal cord

runs through the spinal canal. This bundle of 31 nerves carries messages between the brain and the various parts of the body. At each vertebra, some smaller nerves branch out from these nerve roots to serve the muscles and tissue in the immediate area. When the spinal canal narrows, nerve roots in the spinal cord are squeezed. Pressure on the nerve roots causes chronic pain and loss of control over some functions because communication with the brain is interrupted. The lower back and legs are most affected by spinal stenosis. The nerve roots that supply the legs are near the bottom of the spinal cord. The pain gets worse after standing for a long time and after some forms of **exercise**. The posture required by these physical activities increases the **stress** on the nerve roots. Spinal stenosis usually affects people over 50 years of age. Women have the condition more frequently than men do.

Cervical spinal stenosis is a narrowing of the vertebrae of the neck (cervical vertebrae). The disease and its effects are similar to stenosis in the lower spine. A narrower opening in the cervical vertebrae can also put pressure on arteries entering the spinal column, cutting off the blood supply to the remainder of the spinal cord.

Causes and symptoms

Spinal stenosis causes pain in the buttocks, thigh, and calf and increasing weakness in the legs. The patient may also have difficulty controlling bladder and bowel functions. The pain of spinal stenosis seems more severe when the patient walks downhill. Spinal stenosis can be congenital, acquired, or a combination. Congenital spinal stenosis is a birth defect. Acquired spinal stenosis develops after birth. It is usually a consequence of tissue destruction (degeneration) caused by an infectious disease or a disease in which the immune system attacks the body's own cells (autoimmune disease). The two most common causes of spinal stenosis are birth defect and progressive degeneration of the tissue of the joints (**osteoarthritis**). Other causes include improper alignment of the vertebrae as in spondylolisthesis, destruction of bone tissue as in Paget's disease, or an overgrowth of bone tissue as in diffuse idiopathic skeletal hyperostosis. The spinal canal is usually more than 0.5 in (12 mm) in diameter. A smaller diameter indicates stenosis. The diameter of the cervical spine ranges is 0.6–1 in (15–12 mm). Any opening under 0.5 in (13 mm) in diameter is considered evidence of stenosis. Acquired spinal stenosis usually begins with degeneration of the intervertebral disks or the surfaces of the vertebrae or both. In trying to heal this degeneration, the body builds up the spinal column. In the process, the spinal canal can become narrower.

Diagnosis

The physician must determine that the symptoms are caused by spinal stenosis. Conditions that can cause similar symptoms include a slipped (herniated) intervertebral disk, spinal tumors, and disorders of the blood flow (circulatory disorders). Spinal stenosis causes back and leg pain. The leg pain is usually worse when the patient is standing or walking. Some forms of spinal stenosis are less painful when the patient is riding an exercise bike because the forward tilt of the body changes the pressure in the spinal column. Doppler scanning can trace the flow of blood to determine whether the pain is caused by circulatory problems. X-ray images, **computed tomography scans** (CT scans), and **magnetic resonance imaging** (MRI) scans can reveal any narrowing of the spinal canal. **Electromyography**, nerve conduction velocity, or **evoked potential studies** can locate problems in the muscles indicating areas of spinal cord compression.

Treatment

Mild cases of spinal stenosis may be treated with rest, **nonsteroidal anti-inflammatory drugs** (such as **aspirin**), and **muscle relaxants**. Spinal stenosis can be a progressive disease, however, and the source of pressure may have to be surgically removed (surgical decompression) if the patient is losing control over bladder and bowel functions. The surgical procedure removes bone and other tissues that have entered the spinal canal or put pressure on the spinal cord. Two vertebrae may be fused, to eliminate improper alignment, such as that caused by spondylolisthesis. For surgery, patients lie on their sides or in a modified kneeling position. This position reduces bleeding and places the spine in proper alignment. Alignment is especially important if vertebrae are to be fused. Surgical decompression can eliminate leg pain and restore control of the legs, bladder, and bowels, but usually does not eliminate lower back pain. Physical therapy and massage can help reduce the symptoms of spinal stenosis. An exercise program should be developed to increase flexibility and mobility. A brace or corset may be worn to improve posture. Activities that place stress on the lower back muscles should be avoided.

Prognosis

Surgical decompression does not stop the degenerative processes that cause spinal stenosis, and the condition can develop again. Nevertheless, most

KEY TERMS

Computed tomography (CT) Scans—An imaging technique in which cross-sectional x rays of the body are compiled to create a three-dimensional image of the body's internal structures.

Congenital—Present before birth. The term is used to describe disorders that developed in the fetal stage.

Doppler scanning—A procedure in which ultrasound images are used to watch a moving structure such as the flow of blood or the beating of the heart.

Electromyography—A test that uses electrodes to record the electrical activity of muscle. The information gathered is used to find disorders of the nerves that serve the muscles.

Evoked potential—A test of nerve response that uses electrodes placed on the scalp to measure brain reaction to a stimulus such as a touch.

Magnetic resonance imaging (MRI)—An imaging technique that uses a large circular magnet and radio waves to generate signals from atoms in the body. These signals are used to construct images of internal structures.

Nerve conduction velocity test—A test that measures the time it takes a nerve impulse to travel a specific distance over the nerve after electronic stimulation.

Stenosis—The narrowing or constriction of a channel or opening.

patients achieve good results with surgical decompression. The patient will probably continue to have lower back pain after the surgical procedure.

Resources

BOOKS

Berkow, Robert, editor. *Merck Manual of Medical Information*. Whitehouse Station, NJ: Merck Research Laboratories, 2004.

John T. Lohr, PhD

Spinal tap see **Cerebrospinal fluid (CSF) analysis**

Spirometry see **Pulmonary function test**

Spleen, enlarged see **Hypersplenism**

Spleen removal see **Splenectomy**

Splenectomy

Definition

Splenectomy is the surgical removal of the spleen, which is an organ that is part of the lymphatic system. The spleen is a dark-purple, bean-shaped organ located in the upper left side of the abdomen, just behind the bottom of the rib cage. In adults, the spleen is about 4.8 X 2.8 X 1.6 in (12 X 7 X 4 cm) in size, and weighs about 4–5 oz (113–14 zg). Its functions include a role in the immune system; filtering foreign substances from the blood; removing worn-out blood cells from the blood; regulating blood flow to the liver; and sometimes storing blood cells. The storage of blood cells is called sequestration. In healthy adults, about 30% of blood platelets are sequestered in the spleen.

Purpose

Splenectomies are performed for a variety of different reasons and with different degrees of urgency. Most splenectomies are done after the patient has been diagnosed with **hypersplenism**. Hypersplenism is not a specific disease but a group of symptoms, or syndrome, that can be produced by a number of different disorders. It is characterized by enlargement of the spleen (splenomegaly), defects in the blood cells, and an abnormally high turnover of blood cells. It is almost always associated with splenomegaly caused by specific disorders such as **cirrhosis** of the liver or certain cancers. The decision to perform a splenectomy depends on the severity and prognosis of the disease that is causing the hypersplenism.

Splenectomy always necessary

There are two diseases for which splenectomy is the only treatment—primary cancers of the spleen and a blood disorder called hereditary spherocytosis (HS). In HS, the absence of a specific protein in the red blood cell membrane leads to the formation of relatively fragile cells that are easily damaged when they pass through the spleen. The cell destruction does not occur elsewhere in the body and ends when the spleen is removed. HS can appear at any age, even in newborns, although doctors prefer to put off removing the spleen until the child is five or six years old.

Splenectomy usually necessary

There are some disorders in which splenectomy is usually recommended. They include:

- Immune (idiopathic) thrombocytopenic purpura (ITP). ITP is a disease involving platelet destruction.

Splenectomy has been regarded as the definitive treatment for this disease and is effective in about 70% of chronic ITP cases. More recently, however, the introduction of new drugs in the treatment of ITP has reopened the question as to whether splenectomy is always the best treatment option.

- Trauma. The spleen can be ruptured by blunt as well as penetrating injuries to the chest or abdomen. Car accidents are the most common cause of blunt traumatic injury to the spleen.
- Abscesses in the spleen. These are relatively uncommon but have a high mortality rate.
- Rupture of the splenic artery. Rupture sometimes occurs as a complication of pregnancy.
- Hereditary elliptocytosis. This is a relatively rare disorder. It is similar to HS in that it is characterized by red blood cells with defective membranes that are destroyed by the spleen.

Splenectomy sometimes necessary

In other disorders, the spleen may or may not be removed.

- Hodgkin's disease, a serious form of **cancer** that causes lymph nodes to enlarge. Splenectomy is often performed in order to find out how far the disease has progressed.
- Thrombotic thrombocytopenic purpura (TTP). TTP is a rare disorder marked by **fever**, kidney failure, and an abnormal decrease in the number of platelets. Splenectomy is one part of treatment for TTP.
- Autoimmune hemolytic disorders. These disorders may appear in patients of any age but are most common in patients over 50. The red blood cells are destroyed by antibodies produced by the patient's own body (autoantibodies).
- **Myelofibrosis**. Myelofibrosis is a disorder in which bone marrow is replaced by fibrous tissue. It produces severe and painful splenomegaly. Splenectomy does not cure myelofibrosis but may be performed to relieve **pain** caused by the swollen spleen.
- **Thalassemia**. Thalassemia is a hereditary form of anemia that is most common in people of Mediterranean origin. Splenectomy is sometimes performed if the patient's spleen has become painfully enlarged.

Precautions

Patients should be carefully assessed regarding the need for a splenectomy. Because of the spleen's role in

protecting people against infection, it should not be removed unless necessary. The operation is relatively safe for young and middle-aged adults. Older adults, especially those with cardiac or pulmonary disease, are more vulnerable to post-surgical infections. Thromboembolism following splenectomy is another complication for this patient group, which has about 10% mortality following the surgery. Splenectomies are performed in children only when the benefits outweigh the risks.

The most important part of the assessment is the measurement of splenomegaly. The normal spleen cannot be felt when the doctor examines the patient's abdomen. A spleen that is large enough to be felt indicates splenomegaly. In some cases the doctor will hear a dull sound when he or she thumps (percusses) the patient's abdomen near the ribs on the left side. Imaging studies that can be used to demonstrate splenomegaly include ultrasound tests, technetium-99m sulfur colloid imaging, and CT scans. The rate of platelet or red blood cell destruction by the spleen can be measured by tagging blood cells with radioactive chromium or platelets with radioactive indium.

Description

Complete splenectomy

REMOVAL OF ENLARGED SPLEEN. Splenectomy is performed under **general anesthesia**. The most common technique is used to remove greatly enlarged spleens. After the surgeon makes a cut (incision) in the abdomen, the artery to the spleen is tied to prevent blood loss and reduce the spleen's size. It also helps prevent further sequestration of blood cells. The surgeon detaches the ligaments holding the spleen in place and removes it. In many cases, tissue samples will be sent to a laboratory for analysis.

REMOVAL OF RUPTURED SPLEEN. When the spleen has been ruptured by trauma, the surgeon approaches the organ from its underside and fastens the splenic artery.

In some cases, the doctor may prefer conservative (nonsurgical) management of a ruptured spleen, most often when the patient's blood pressure is stable and there are no signs of other abdominal injuries. In the case of multiple abdominal trauma, however, the spleen is usually removed.

Partial splenectomy

In some cases the surgeon removes only part of the spleen. This procedure is considered by some to be

a useful compromise that reduces pain from an enlarged spleen while leaving the patient less vulnerable to infection. Long-term follow-up of the results of partial splenectomies has not yet been done.

Laparoscopic splenectomy

Laparoscopic splenectomy, or removal of the spleen through several small incisions, has been more frequently used in recent years. Laparoscopic surgery involves the use of surgical instruments, with the assistance of a tiny camera and video monitor. Laparoscopic procedures reduce the length of hospital stay, the level of post-operative pain, and the risk of infection. They also leave smaller **scars**. Laparoscopic splenectomy is not, however, the best option for many patients.

Laparoscopic splenectomy is gaining increased acceptance in the early 2000s as an alternative to open splenectomy for a wide variety of disorders, although splenomegaly still presents an obstacle to laparoscopic splenectomy; massive splenomegaly has been considered a contraindication. In patients with enlarged spleens, however, laparoscopic splenectomy is associated with less morbidity, decreased **transfusion** rates, and shorter hospital stays than when the open approach is used. Patients with enlarged spleens usually have more severe hematologic diseases related to greater morbidity; therefore, laparoscopic splenectomy has potential advantages.

The most frequent serious complication following laparoscopic splenectomy is damage to the pancreas. Application of a hydrogel sealant to the pancreas during surgery, however, appears to significantly reduce the risk of leakage from the pancreas.

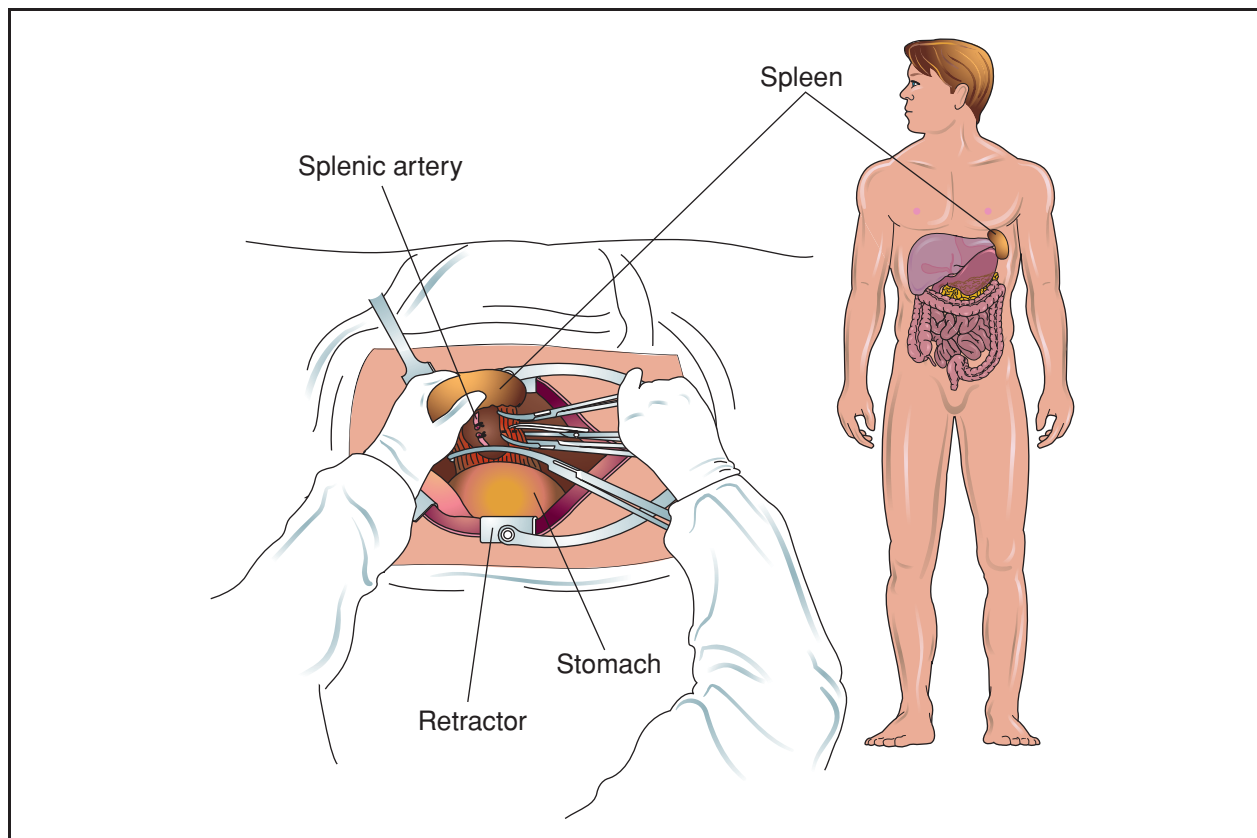
Splenic embolization

Splenic embolization is an alternative to splenectomy that is used in some patients who are poor surgical risks. Embolization involves plugging or blocking the splenic artery to shrink the size of the spleen. The substances that are injected during this procedure include polyvinyl alcohol foam, polystyrene, and silicone. Embolization is a technique that needs further study and refinement.

Preparation

Preoperative preparation for nonemergency splenectomy includes:

- Correction of abnormalities of blood clotting and the number of red blood cells.



Splenectomy is the surgical removal of the spleen. This procedure is performed as a last result in most diseases involving the spleen. In some cases, however, splenectomy does not address the underlying causes of splenomegaly or other conditions affecting the spleen. (Illustration by Electronic Illustrators Group.)

- Treatment of any infections.
- Control of immune reactions. Patients are usually given protective vaccinations about a month before surgery. The most common vaccines used are Pneumovax or Pnu-Imune 23 (against pneumococcal infections) and Menomune-A/C/Y/W-135 (against meningococcal infections).

Aftercare

Immediately following surgery, patients should follow the physician's instructions and take all medications intended to prevent infection. Blood transfusions may be indicated for some patients to replace defective blood cells. The most important part of aftercare, however, is long-term caution regarding vulnerability to infection. Patients should see their doctor at once if they have a fever or any other sign of infection, and avoid travel to areas where exposure to **malaria** or similar diseases is likely. Children with splenectomies may be kept on antibiotic therapy until they are 16 years old. All patients can be given a

booster dose of pneumococcal vaccine five to 10 years after splenectomy.

Risks

The chief risk following splenectomy is overwhelming bacterial infection, or postsplenectomy **sepsis**. This vulnerability results from the body's decreased ability to clear bacteria from the blood, and lowered levels of a protein in blood plasma that helps to fight viruses (immunoglobulin M). The risk of dying from infection after splenectomy is highest in children, especially in the first two years after surgery. The risk of postsplenectomy sepsis can be reduced by vaccinations before the operation. Some doctors also recommend a two-year course of penicillin following splenectomy or long-term treatment with ampicillin.

Other risks following splenectomy include inflammation of the pancreas and collapse of the lungs. In some cases, splenectomy does not address the underlying causes of splenomegaly or other conditions. Excessive bleeding after the operation is an additional

KEY TERMS

Embolization—An alternative to splenectomy that involves injecting silicone or similar substances into the splenic artery to shrink the size of the spleen.

Hereditary spherocytosis (HS)—A blood disorder in which the red blood cells are relatively fragile and are damaged or destroyed when they pass through the spleen. Splenectomy is the only treatment for HS.

Hypersplenism—A syndrome marked by enlargement of the spleen, defects in one or more types of blood cells, and a high turnover of blood cells.

Immune or idiopathic thrombocytopenic purpura (ITP)—A blood disease that results in destruction of platelets, which are blood cells involved in clotting.

Laparoscope—An instrument used to view the abdominal cavity through a small incision and perform surgery on a small area, such as the spleen.

Pneumovax—A vaccine that is given to splenectomy patients to protect them against bacterial infections. Other vaccines include Pnu-Imune and Menomune.

Sepsis—A generalized infection of the body, most often caused by bacteria.

Sequestration—A process in which the spleen withdraws some normal blood cells from circulation and holds them in case the body needs extra blood in an emergency. In hypersplenism, the spleen sequesters too many blood cells.

Splenomegaly—Abnormal enlargement of the spleen.

Thromboembolism—A clot in the blood that forms and blocks a blood vessel. It can lead to infarction, or death of the surrounding tissue due to lack of blood supply.

possible complication, particularly for ITP patients. Infection immediately following surgery may also occur.

Normal results

Results depend on the reason for the operation. In blood disorders, the splenectomy will remove the cause of the blood cell destruction. Normal results for patients with an enlarged spleen are relief of pain and of the complications of splenomegaly. It is not always possible, however, to predict which patients will respond well or to what degree.

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Splenic trauma

Definition

Splenic trauma is physical injury to the spleen, the lymphatic organ located in the upper left side of the abdomen just under the rib cage. The spleen weighs between 75 and 150 grams (between 0.16 and 0.33 pounds) in adults.

Description

The spleen is an organ that produces white blood cells, filters the blood (10–15% of the total blood supply every minute), stores red blood cells and platelets, and destroys those that are **aging**. It is located near the stomach on the left side of the abdomen. A direct blow to the abdomen may bruise, tear or shatter the spleen. Trauma to the spleen can cause varying degrees of damage, the major problem associated with internal bleeding. Mild splenic subcapsular hematomas are injuries in which bleeding is limited to small areas on and immediately around the spleen. Splenic contusions refer to bruising and bleeding on and around larger areas of the spleen. Lacerations (tears) are the most common splenic trauma injuries. Tears tend to occur on the areas between the three main blood vessels of the spleen. Because of the abundant blood supply, splenic trauma may cause serious internal bleeding. Most injuries to the spleen in children heal spontaneously. Severe trauma can cause the spleen or its blood vessels to rupture or fragment.

Splenic trauma is more common in children than in adults. In general, children are prone to abdominal injuries due to accidents and falls and because their abdominal organs are less protected by bone, muscle and fat. Abdominal injuries including splenic trauma are the most common cause of preventable deaths in children.

Causes and symptoms

The most common cause of injury to the spleen is blunt abdominal trauma. Blunt trauma is often caused by a direct blow to the belly, car and motorcycle accidents, falls, sports mishaps, and fights. The spleen is the most commonly injured organ in blunt abdominal trauma; splenic injury occurs in nearly 25% of injuries of this type. Penetrating injuries such as those from stabbing, gunshot **wounds**, and accidental impaling also account for cases of splenic trauma, although far less frequently than blunt trauma.

In adults, ruptured spleens may have been preceded by conditions causing rapid splenic

enlargement, such as infections, particularly those caused by the Epstein-Barr virus (EBV); **cancer**; immune system disorders; diseases of the spleen; or circulatory problems. In a very few cases the spleen may be injured by a spell of violent coughing. This type of rupture is known as an atraumatic rupture.

A spleen that has become enlarged and fragile from disease is sometimes ruptured by a doctor or medical student in the course of palpating (feeling) the patient's abdomen, or damaged by a surgeon in the course of an operation on other abdominal organs.

Damage to the spleen may cause localized or general abdominal **pain**, tenderness, and swelling. Fractured ribs may be present. Splenic trauma may cause mild or severe internal bleeding, leading to **shock** and for which symptoms include rapid heartbeat, **shortness of breath**, thirst, pale or clammy skin, weak pulse, low blood pressure, **dizziness**, **fainting**, sweating. **Vomiting** blood, blood in the stools or urine, deterioration of vital signs, and loss of consciousness are other symptoms.

Diagnosis

The goal of diagnosis of all abdominal traumas is to detect and treat life-threatening injuries as quickly as possible. The physician will determine the extent of organ damage and whether surgery will be necessary while providing appropriate emergency care. Initial diagnosis consists of detailing all circumstances of the injury from the patient and bystanders as well as the close **physical examination** of the patient and measurement of vital signs. Blood tests, **urinalysis**, stool samples and x rays of the chest and abdomen are usually performed. Plain x rays may show abdominal air pockets that indicate internal ruptures, but are rarely helpful because they do not show splenic and intra-abdominal damage.

Several other diagnostic tests may be used for the noninvasive and accurate assessment of splenic damage: **computed tomography scans** (CT), of **magnetic resonance imaging** (MRI), radionuclide scanning, and ultrasonography. Ultrasonography—particularly focused abdominal sonographic technique (FAST)—has now become a standard bedside technique in many hospitals to check for bleeding in the abdomen. Imaging tests allow doctors to determine the necessity and type of surgery required. The CT scan has been shown to be the most available and accurate test for abdominal trauma. MRI tests are accurate but costly and less available in some hospitals, while radionuclide scanning requires more time and patient stability. Peritoneal lavage is another diagnostic technique in

which the abdominal cavity is entered and flushed to check for bleeding. When patients exhibit shock, infection, or prolonged internal bleeding, exploratory **laparoscopy** is used for emergency diagnosis.

Treatment

Not long ago nearly all cases of splenic trauma were treated by laparoscopy, opening the abdomen, and by **splenectomy**, the surgical removal of the spleen. This approach resulted from the difficulty in assessing the severity of the injury, the potential dangers of shock and **death**, and the beliefs that the spleen healed poorly and that it was not an important organ. Nowadays, improved techniques of diagnosis and monitoring (particularly the introduction of CT scans), as well as understanding that removal of the spleen creates future risk of a lowered capacity to fight infection has modified treatment approaches. Research over the past two decades has shown that the spleen has high healing potential, and confirmed that children are more susceptible to infection after splenectomy (post splenectomy **sepsis**, PSS). PSS has a mortality rate of over 50% and standard procedure now avoids splenectomy as much as possible. Adult splenic trauma is treated by splenectomy more often than children's; for unknown reasons, the adult spleen more frequently spontaneously ruptures after injury. Adults are also less susceptible to PSS.

Nonoperative treatment

In nonoperative therapy, splenic trauma patients are monitored closely, often in intensive care units for several days. Fluid and blood levels are observed and maintained by intravenous fluid and possible blood transfusions. Follow-up scans may be used to observe the healing process.

Operative treatment

Splenic trauma patients require surgery when nonoperative treatment fails, when major or prolonged internal bleeding exists and for gunshot and many stab wounds. Whenever possible, surgeons try to preserve at least part of the spleen and try to repair its blood vessels.

Prognosis

The ample blood supply to the spleen can promote rapid healing. Studies have shown that intra-abdominal bleeding associated with splenic trauma stops without surgical intervention in up to two out of three cases in children. When trauma patients stabilize

KEY TERMS

Computed tomography (CT) scan—Computer-aided x-ray examination that allows cross-sectional views of organs and tissues.

Laparoscope—An optical or fiberoptic instrument that is inserted by incision in the abdominal wall and is used to view the interior of the peritoneal cavity.

Laparoscopy—Procedure using a laparoscope to view organs, obtain tissue samples and perform surgery.

Magnetic resonance imaging (MRI)—Imaging technique using magnets and radio waves to provide internal pictures of the body.

Radionuclide scanning—Diagnostic test in which a radioactive dye is injected into the bloodstream and photographed to display internal vessels, organs and tissues.

Ultrasonography—Imaging test using sound waves to view internal organs and tissues.

during nonoperative therapy, chances are high that surgery will be avoided and that spleen injuries will heal themselves. Splenic trauma patients undergoing diagnostic tests such as CT and MRI scans have improved chances of avoiding splenectomy and retaining whole or partial spleens.

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American Association for the Surgery of Trauma home page. < <http://www.aast.org> >.

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Split personality see **Multiple personality disorder**

Spontaneous abortion see **Miscarriage**

Sporothrix schenckii infection see **Sporotrichosis**

Sporotrichosis

Definition

Sporotrichosis is a chronic infection caused by the microscopic fungus *Sporothrix schenckii*. The disease causes ulcers on the skin that are painless but do not heal, as well as nodules or knots in the lymph channels near the surface of the body. Infrequently, sporotrichosis affects the lungs, joints, or central nervous system and can cause serious illness.

Description

The fungus that causes sporotrichosis is found in spagnum moss, soil, and rotting vegetation. Anyone can get sporotrichosis, but it is most common among nursery workers, farm laborers, and gardeners handling spagnum moss, roses, or barberry bushes. Cases have also been reported in workers whose jobs took them under houses into crawl spaces contaminated with the fungus. Children who played on baled hay have also gotten the disease. Sporotrichosis is sometimes called spagnum moss disease or alcoholic rose gardener's disease.

Causes and symptoms

The fungus causing sporotrichosis enters the body through scratches or cuts in the skin.

Therefore, people who handle plants with sharp thorns or needles, like roses, barberry, or pines, are more likely to get sporotrichosis. Sporotrichosis is not passed directly from person to person, so it is not possible to catch sporotrichosis from another person who has it.

The first signs of sporotrichosis are painless pink, red, or purple bumps usually on the finger, hand, or arm where the fungus entered the body. These bumps may appear anywhere from one to 12 weeks after infection, but usually appear within three weeks. Unlike many other fungal infections sporotrichosis does not cause **fever** or any feelings of general ill health.

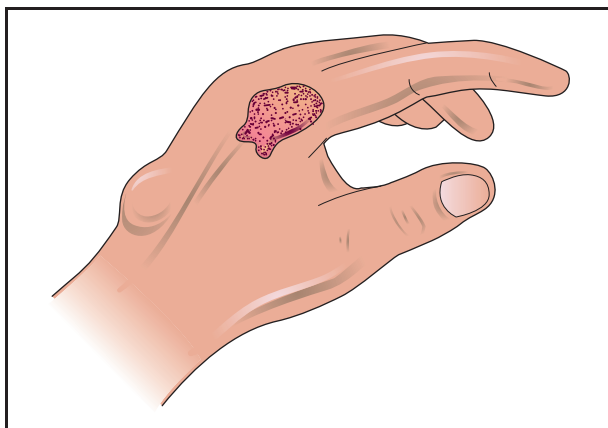
The reddish bumps eventually expand and fester, creating skin ulcers that do not heal. In addition, the infection often moves to nearby lymph nodes. Although most cases of sporotrichosis are limited to the skin and lymph channels, occasionally the joints, lungs, and central nervous system become infected. In rare cases, **death** may result.

People who have weakened immune systems, either from a disease such as acquired immune deficiency syndrome (**AIDS**) or leukemia, or as the result of medications they take (**corticosteroids**, **chemotherapy** drugs), are more likely to get sporotrichosis and are more at risk for the disease to spread to the internal organs. Alcoholics and people with **diabetes mellitus** or a pre-existing lung disease are also more likely to become infected. Although sporotrichosis is painless, it is important for people with symptoms to see a doctor and receive treatment.

Diagnosis

The preferred way to diagnose sporotrichosis is for a doctor to obtain a sample of fluid from a freshly opened sore and send it to a laboratory to be cultured. The procedure is fast and painless. It is possible to confirm the presence of advanced sporotrichosis through a blood test or a biopsy. Doctors may also take a blood sample to perform tests that rule out other fungal infections or diseases such as **tuberculosis** or bacterial **osteomyelitis**.

Dermatologists and doctors who work with AIDS patients are more likely to have experience in diagnosing sporotrichosis. In at least one state, New York, the laboratory test to confirm this disease is provided free through the state health department. In other cases, diagnosis should be covered by health insurance at the same level as other diagnostic laboratory tests.



Sporotrichosis is a chronic infection caused by the microscopic fungus *Sporothrix schenckii*. It produces ulcers on the skin that are painless but do not heal, and nodules or knots in the lymph channels near the surface of the body. (Illustration by Electronic Illustrators Group.)

Treatment

When sporotrichosis is limited to the skin and lymph system, it is usually treated with a saturated solution of potassium iodine that the patient dilutes with water or juice and drinks several times a day. The iodine solution can only be prescribed by a physician. This treatment must be continued for many weeks. Skin ulcers should be treated like any open wound and covered with a clean bandage to prevent a secondary bacterial infection. The drug itraconazol (Sporanox), taken orally, is also available to treat sporotrichosis.

In serious cases of sporotrichosis, when the internal organs are infected, the preferred treatment is the drug amphotericin B. Amphotericin B is a strong anti-fungal drug with potentially severe toxic side effects. It is given intravenously, so hospitalization is required for treatment. The patient may also receive other drugs to minimize the side effects of the amphotericin B.

Alternative treatment

Alternative treatment for fungal infections focuses on maintaining general good health and eating a diet low in dairy products, sugars, including honey and fruit juice, and foods, such as beer, that contain yeast. This is complemented by a diet high in raw food. Supplements of and vitamins C, E, and A, B complex, and pantothenic acid may also be added to the diet, as may *Lactobacillus acidophilus*, bifidobacteria, and garlic capsules.

Fungicidal herbs such as myrrh (*Commiphora molmol*), tea tree oil (*Melaleuca* spp.), citrus seed

KEY TERMS

Acidophilus—The bacteria *Lactobacillus acidophilus*, usually found in yogurt.

Bacterial osteomyelitis—An infection of the bone or bone marrow that is caused by a bacterium.

Bifidobacteria—A group of bacteria normally present in the intestine. Commercial supplements are available.

Corticosteroids—A group of hormones produced naturally by the adrenal gland or manufactured synthetically. They are often used to treat inflammation. Examples include cortisone and prednisone.

Lymph channels—The vessels that transport lymph throughout the body. Lymph is a clear fluid that contains cells important in forming antibodies that fight infection.

extract, pau d'arco tea, and garlic (*Allium sativum*) may also be applied directly to the infected skin.

Prognosis

Most cases of sporotrichosis are confined to the skin and lymph system. With treatment, skin sores begin healing in one to two months, but complete recovery often takes six months or more. People who have AIDS are also more likely to have the fungus spread throughout the body, causing a life-threatening infection. In people whose bones and joints are infected or who have pulmonary lesions, surgery may be necessary.

Prevention

Since an opening in the skin is necessary for the sporotrichosis fungus to enter the body, the best way to prevent the disease is to avoid accidental scrapes and cuts on the hands and arms by wearing gloves and long sleeves while gardening. Washing hands and arms well after working with roses, barberry, spagnum moss, and other potential sources of the fungus may also provide some protection.

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Sports injuries

Definition

Sports injuries result from acute trauma or repetitive **stress** associated with athletic activities. Sports injuries can affect bones or soft tissue (ligaments, muscles, tendons).

Professional dancers are increasingly recognized as performing athletes, and many of the treatments and preventive measures utilized in sports medicine are now applied to dance-related injuries.

It is also important to remember that many types of injuries that affect athletes may also occur in workers in certain occupations; for example, many people in the building trades develop **tennis elbow** or golfer's elbow. The principles of sports medicine can be applied in the treatment of most common musculoskeletal injuries.

Description

Adults are less likely to suffer sports injuries than children, whose vulnerability is heightened by immature reflexes, an inability to recognize and evaluate risks, and underdeveloped coordination.

In 2002, about 20.3 million Americans suffered a sports injury. Of those, 53% were minor enough to be self-treated or left untreated. However, about 10 million Americans annually receive medical attention for their sports-related injuries. That equates to almost 26 per 1,000 people. The highest rate is among children age five to 14 years old (59.3 per 1,000 people). As many as 20% of children who play sports get hurt, and about 25% of their injuries are classified as serious. Boys aged 12 to 17 are the highest risk group. More than 775,000 boys and girls under age 14 are treated in hospital emergency rooms for sports-related injuries.

Injury rates are highest for athletes who participate in contact sports, but the most serious injuries are associated with individual activities. Between one-half and two-thirds of childhood sports injuries occur during practice, or in the course of unorganized athletic activity.

Baseball and softball are the leading causes of sports-related facial trauma in the United States, with 68% of these injuries caused by contact with the ball rather than player-player collision or being hit by a swung bat.

Types of sports injuries

About 95% of sports injuries are minor soft tissue traumas.

The most common sports injury is a bruise (contusion). It is caused when blood collects at the site of an injury and discolors the skin.

Sprains account for one-third of all sports injuries. A sprain is a partial or complete tear of a ligament, a strong band of tissue that connects bones to one another and stabilizes joints.

A strain is a partial or complete tear of:

- muscle (tissue composed of cells that enable the body to move)
- tendon (strong connective tissue that links muscles to bones)

Inflammation of a tendon (**tendinitis**) and inflammation of one of the fluid-filled sacs that allow tendons to move easily over bones (**bursitis**) usually result from minor stresses that repeatedly aggravate the same part of the body. These conditions often occur at the same time.

SKELETAL INJURIES. Fractures account for 5–6% of all sports injuries. The bones of the arms and legs are most apt to be broken. Sports activities rarely involve fractures of the spine or skull. The bones of the legs and feet are most susceptible to stress fractures, which occur when muscle **strains** or contractions make bones bend. Stress fractures are especially common in ballet dancers, long-distance runners, and in people whose bones are thin.

Shin splints are characterized by soreness and slight swelling of the front, inside, and back of the lower leg, and by sharp **pain** that develops while exercising and gradually intensifies. Shin splints are caused by overuse or by stress fractures that result from the repeated foot pounding associated with activities such as aerobics, long-distance running, basketball, and volleyball.

A compartment syndrome is a potentially debilitating condition in which the muscles of the lower leg grow too large to be contained within membranes that enclose them. This condition is characterized by **numbness and tingling**. Untreated compartment syndrome can result in long-term loss of function.

BRAIN INJURIES. Brain injury is the primary cause of fatal sports-related injuries. **Concussion**, which is also called mild traumatic brain injury or MTBI, can result from even minor blows to the head. A concussion can cause loss of consciousness and may affect:

- balance
- comprehension
- coordination
- hearing



Chauncy Billups, a guard for the Denver Nuggets, grimaces after spraining his ankle during a game. (AP/Wide World Photos. Reproduced by permission.)

- memory
- vision

Causes and symptoms

Common causes of sports injuries include:

- athletic equipment that malfunctions or is used incorrectly
- falls
- forceful high-speed collisions between players
- wear and tear on areas of the body that are continually subjected to stress

Symptoms include:

- instability or obvious dislocation of a joint
- pain

- swelling
- weakness

Diagnosis

Symptoms that persist, intensify, or reduce the athlete's ability to play without pain should be evaluated by an orthopedic surgeon. Prompt diagnosis often can prevent minor injuries from becoming major problems, or causing long-term damage.

An orthopedic surgeon should examine anyone:

- who is prevented from playing by severe pain associated with acute injury
- whose ability to play has declined due to chronic or long-term consequences of an injury
- whose injury has caused visible deformities in an arm or leg.

The physician will perform a **physical examination**, ask how the injury occurred, and what symptoms the patient has experienced. X rays and other imaging studies of bones and soft tissues may be ordered.

Anyone who has suffered a blow to the head should be examined immediately, and at five-minute intervals until normal comprehension has returned. The initial examination measures the athlete's:

- awareness
- concentration
- short-term memory

Subsequent evaluations of concussion assess:

- dizziness
- headache
- nausea
- visual disturbances

Treatment

Treatment for minor soft tissue injuries generally consists of:

- compressing the injured area with an elastic bandage
- elevation
- ice
- rest.

Anti-inflammatories, taken by mouth or injected into the swelling, may be used to treat bursitis. Anti-inflammatory medications and exercises to correct muscle imbalances usually are used to treat tendinitis. If the athlete keeps stressing inflamed tendons, they may rupture, and casting or surgery is sometimes necessary to correct this condition.

Orthopedic surgery may be required to repair serious **sprains and strains**.

Controlling inflammation as well as restoring normal use and mobility are the goals of treatment for overuse injuries.

Athletes who have been injured are usually advised to limit their activities until their injuries are healed. The physician may suggest special exercises or behavior modifications for athletes who have had several injuries. Athletes who have been severely injured may be advised to stop playing altogether.

Prevention

Every child who plans to participate in organized athletic activity should have a pre-season sports physical. This special examination is performed by a pediatrician or family physician who:

- carefully evaluates the site of any previous injury
- may recommend special stretching and strengthening exercises to help growing athletes create and preserve proper muscle and joint interaction
- pays special attention to the cardiovascular and skeletal systems.

Telling the physician which sport the athlete plays will help that physician determine which parts of the body will be subjected to the most stress. The physician then will be able to suggest to the athlete steps to take to minimize the chance of getting hurt.

Other injury-reducing game plans include:

- being in shape
- knowing and obeying the rules that regulate the activity
- not playing when tired, ill, or in pain
- not using steroids, which can improve athletic performance but cause life-threatening problems
- taking good care of athletic equipment and using it properly
- wearing appropriate protective equipment

On a larger scale, sports injuries are becoming a public health concern in America. Prevention efforts include wearing protective devices (such as bicycle helmets and pads when skating or skateboarding), and educating both children and adults about safety. Other preventive efforts include changes in the rules of the game or sport to minimize injuries. For example, wearing goggles will be mandatory in women's lacrosse as of 2005 in order to reverse the rising rate of eye and other facial injuries in that sport. Research also continues on improving equipment. For example, thick rubber insoles can help prevent against repetitive injuries from running, but scientists recently observed that they can add to injuries in sports such as soccer, where athletes need to make quick changes of direction. On the other hand, recent improvements in the design and construction of football helmets have been credited with a significant decline in the frequency and severity of head injuries among football players.

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- American Academy of Otolaryngology—Head and Neck Surgery. One Prince Street, Alexandria, VA 22314-3357. (703) 836-4444. <<http://www.entnet.org>>.
- American College of Sports Medicine (ACSM). 401 West Michigan Street, Indianapolis, IN 46202-3233. (317) 637-9200. Fax: (317) 634-7817. <<http://www.acsm.org>>.
- Institute for Preventative Sports Medicine. P.O. Box 7032, Ann Arbor, MI 48107 (313) 434-3390. <<http://www.ipism.org>>.

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Sports vision see **Vision training**

Spouse abuse see **Abuse**

Sprains and strains

Definition

Sprain refers to damage or tearing of ligaments or a joint capsule. Strain refers to damage or tearing of a muscle.

Description

When excessive force is applied to a joint, the ligaments that hold the bones together may be torn or damaged. This results in a sprain, and its seriousness depends on how badly the ligaments are torn. Any joint can be sprained, but the most frequently injured joints are the ankle, knee, and finger.

Strains are tears in the muscle. Sometimes called pulled muscles, they usually occur because of overexertion or improper lifting techniques. Sprains and strains are common. Anyone can have them.

Children under age eight are less likely to have sprains than are older people. Children’s ligaments are tighter, and their bones are more apt to break before a ligament tears. People who are active in sports suffer more strains and sprains than less active people. Repeated sprains in the same joint make the joint less stable and more prone to future sprains.

Causes and symptoms

There are three grades of sprains. Grade I sprains are mild injuries where there is no tearing of the ligament, and no joint function is lost, although there may be tenderness and slight swelling.

Grade II sprains are caused by a partial tear in the ligament. These sprains are characterized by obvious swelling, extensive bruising, **pain**, difficulty bearing weight, and reduced function of the joint.

Grade III, or third degree, sprains are caused by complete tearing of the ligament where there is severe pain, loss of joint function, widespread swelling and bruising, and the inability to bear weight. These symptoms are similar to those of bone **fractures**.

Strains can range from mild muscle stiffness to great soreness. Strains result from overuse of muscles, improper use of the muscles, or as the result of injury in another part of the body when the body compensates for pain by altering the way it moves.

Diagnosis

Grade I sprains and mild strains are usually self-diagnosed. Grade II and III sprains are often seen by a

KEY TERMS

Ligament—Tough, fibrous connective tissue that holds bones together at joints.

physician, who x rays the area to differentiate between a sprain and a fracture.

Treatment

Grade I sprains and mild strains can be treated at home. Basic first aid for sprains consists of RICE: Rest, Ice for 48 hours, Compression (wrapping in an elastic bandage), and Elevation of the sprain above the level of the heart. Over-the-counter pain medication such as **acetaminophen** (Tylenol) or ibuprofen (Motrin) can be taken for pain.

In addition to RICE, people with grade II and grade III sprains in the ankle or knee usually need to use crutches until the sprains have healed enough to bear weight. Sometimes, physical therapy or home exercises are needed to restore the strength and flexibility of the joint.

Grade III sprains are usually immobilized in a cast for several weeks to see if the sprain heals. Pain medication is prescribed. Surgery may be necessary to relieve pain and restore function. Athletic people under age 40 are the most likely candidates for surgery, especially with grade III knee sprains. For complete healing, physical therapy usually will follow surgery.

Alternative treatment

Alternative practitioners endorse RICE and conventional treatments. In addition, nutritional therapists recommend vitamin C and bioflavonoids to supplement a diet high in whole grains, fresh fruits, and vegetables. Anti-inflammatories, such as bromelain (a proteolytic enzyme from pineapples) and tumeric (*Curcuma longa*), may also be helpful. The homeopathic remedy arnica (*Arnica montana*) may be used initially for a few days, followed by ruta (*Ruta graveolens*) for joint-related injuries or *Rhus toxicodendron* for muscle-related injuries. If surgery is needed, alternative practitioners can recommend pre- and post-surgical therapies that will enhance healing.

Prognosis

Moderate sprains heal within two to four weeks, but it can take months to recover from severe ligament

tears. Until recently, tearing the ligaments of the knee meant the end to an athlete's career. Improved surgical and rehabilitative techniques now offer the possibility of complete recovery. However, once a joint has been sprained, it will never be as strong as it was before.

Prevention

Sprains and strains can be prevented by warming-up before exercising, using proper lifting techniques, wearing properly fitting shoes, and taping or bracing the joint.

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Tish Davidson, A.M.

Sputum culture

Definition

Sputum is material coughed up from the lungs and expectorated (spit out) through the mouth. A sputum culture is done to find and identify the microorganism causing an infection of the lower respiratory tract such as **pneumonia** (an infection of the lung). If a microorganism is found, more testing is done to determine which **antibiotics** will be effective in treating the infection.

Purpose

A person with a **fever** and a continuing **cough** that produces pus-like material and/or blood may have an infection of the lower respiratory tract. Infections of the lungs and bronchial tubes are caused by several types of microorganisms, including bacteria, fungi (molds and yeast), and viruses. A **chest x ray** provides visual evidence of an infection; a culture can grow the microorganism causing the infection. The microorganism is grown in the laboratory so it can be identified, and tested for its response to medications, such as antifungals and antibiotics.

Description

Based on the clinical condition of the patient, the physician determines what group of microorganism is

likely to be causing the infection, and then orders one or more specific types of cultures: bacterial, viral, or fungal (for yeast and molds). For all culture types, the sputum must be collected into a sterile container. The sputum specimen must be collected carefully, so that bacteria that normally live in the mouth and saliva don't contaminate the sputum and complicate the process of identifying the cause of the infectious agent. Once in the laboratory, each culture type is handled differently.

Bacterial culture

A portion of the sputum is smeared on a microscope slide for a Gram stain. Another portion is spread over the surface of several different types of culture plates, and placed in an incubator at body temperature for one to two days.

A Gram stain is done by staining the slide with purple and red stains, then examining it under a microscope. Gram staining checks that the specimen does not contain saliva or material from the mouth. If many epithelial (skin) cells and few white blood cells are seen, the specimen is not pure sputum and is not adequate for culture. Depending on laboratory policy, the specimen may be rejected and a new specimen requested. If many white blood cells and bacteria of one type are seen, this is an early confirmation of infection. The color of stain picked up by the bacteria (purple or red), their shape (such as round or rectangular), and their size provide valuable clues as to their identity and helps the physician predict what antibiotics might work best before the entire test is completed. Bacteria that stain purple are called gram-positive; those that stain red are called gram-negative.

During incubation, bacteria present in the sputum sample multiply and will appear on the plates as visible colonies. The bacteria are identified by the appearance of their colonies, by the results of biochemical tests, and through a Gram stain of part of a colony.

A sensitivity test, also called antibiotic susceptibility test, is also done. The bacteria are tested against different antibiotics to determine which will treat the infection by killing the bacteria.

The initial result of the Gram stain is available the same day, or in less than an hour if requested by the physician. An early report, known as a preliminary report, is usually available after one day. This report will tell if any bacteria have been found yet, and if so, their Gram stain appearance—for example, a gram-negative rod, or a gram-positive cocci. The final report, usually available in one to three days, includes complete identification and an estimate of the quantity

of the bacteria and a list of the antibiotics to which they are sensitive.

Fungal culture

To look for mold or yeast, a fungal culture is done. The sputum sample is spread on special culture plates that will encourage the growth of mold and yeast. Different biochemical tests and stains are used to identify molds and yeast. Cultures for fungi may take several weeks.

Viral culture

Viruses are a common cause of pneumonia. For a viral culture, sputum is mixed with commercially-prepared animal cells in a test tube. Characteristic changes to the cells caused by the growing virus help identify the virus. The time to complete a viral culture varies with the type of virus. It may take from several days to several weeks.

Special procedures

Tuberculosis is caused by a slow-growing bacteria called *Mycobacterium tuberculosis*. Because it does not easily grow using routine culture methods, special procedures are used to grow and identify this bacteria. When a sputum sample for tuberculosis first comes into the laboratory, a small portion of the sputum is smeared on a microscope slide and stained with a special stain, called an acid-fast stain. The stained sputum is examined under a microscope for tuberculosis organisms, which pick-up the stain, making them visible. This smear is a rapid screen for the organism, and allows the physician to receive a preliminary report within 24 hours.

To culture for tuberculosis, portions of the sputum are spread on and placed into special culture plates and tubes of broth that promote the growth of the organism. Growth in broth is faster than growth on culture plates. Instruments are available that can detect growth in broth, speeding the process even further. Growth and identification may take two to four weeks.

Other microorganisms that cause various types of lower respiratory tract infections also require special culture procedures to grow and identify. *Mycoplasma pneumonia* causes a mild to moderate form of pneumonia, commonly called walking pneumonia; *Bordetella pertussis* causes **whooping cough**; *Legionella pneumophila*, Legionnaire's disease; *Chlamydia pneumoniae*, an atypical pneumonia; and *Chlamydia psittaci*, **parrot fever**.

Pneumocystis carinii causes pneumonia in people with weakened immune systems, such as people with **AIDS**. This organism does not grow in culture. Special stains are done on sputum when pneumonia caused by this organism is suspected. The diagnosis is based on the results of these stains, the patient's symptoms, and medical history.

Sputum culture is also called sputum culture and sensitivity.

It is possible that sputum cultures will eventually be replaced in the diagnosis of tuberculosis by newer molecular techniques. These advanced methods speed the diagnostic process as well as improve its accuracy. As of late 2002, four molecular techniques are increasingly used in laboratories around the world to diagnose TB. They include polymerase chain reaction to detect mycobacterial DNA in patient specimens; nucleic acid probes to identify mycobacteria in culture; restriction fragment length polymorphism analysis to compare different strains of TB for epidemiological studies; and genetic-based susceptibility testing to identify drug-resistant strains of mycobacteria.

Preparation

The specimen for culture should be collected before antibiotics are begun. Antibiotics in the person's system may prevent microorganisms present in the sputum from growing in culture.

The best time to collect a sputum sample is early in the morning, before having anything to eat or drink. The patient should first rinse his or her mouth with water to decrease mouth bacteria and dilute saliva. Through a deep cough, the patient must cough up sputum from within the chest. Taking deep breaths and lowering the head helps bring up the sputum. Sputum must not be held in the mouth but immediately spat into a sterile container. For tuberculosis, the physician may want the patient to collect sputum samples on three consecutive mornings.

If coughing up sputum is difficult, a health care worker can have the patient breathe in sterile saline produced by a nebulizer. This nebulized saline coats the respiratory tract, loosening the sputum, and making it easier to cough up. Sputum may also be collected by a physician during a **bronchoscopy** procedure. Bronchoscopy, however, is not regarded as a cost-effective way of obtaining a useful sample.

If tuberculosis is suspected, collection of sputum should be carried out in an **isolation** room, with all attending healthcare workers wearing masks.

KEY TERMS

Acid-fast stain—A special stain done to microscopically identify the bacteria that cause tuberculosis.

Culture—A laboratory test done to grow and identify microorganisms causing infection.

Gram stain—Microscopic examination of a portion of a bacterial colony or sample from an infection site after it has been stained by special stains. Certain bacteria pick up and retain the purple stain; these bacteria are called gram-positive. Other bacteria lose the purple stain and retain the red stain; these bacteria are called gram-negative. The color of the bacteria, in addition to their size and shape, provide clues as to the identity of the bacteria.

Normal flora—The mixture of bacteria normally found at specific body sites.

Pneumonia—An infection of the lungs.

Sputum—Material coughed up from the lower respiratory tract and expectorated through the mouth.

Sensitivity test—A test that determines which antibiotics will kill the bacteria that has been isolated from a culture.

In addition to special precautions in collecting sputum when tuberculosis is suspected, workers in hospital laboratories must take extra care to inactivate unstained smear preparations that may contain *M. tuberculosis*. As of 2002, the most effective deactivation technique is the use of a solution of 5% phenol in ethanol.

Normal results

Sputum from a healthy person would have no growth on culture. A mixture of microorganisms, however, normally found in a person's mouth and saliva often contaminate the culture. If these microorganisms grow in the culture, they may be reported as normal flora contamination.

Abnormal results

The presence of bacteria and white blood cells on the Gram stain and the isolation of a microorganism from culture, other than normal flora contamination, is evidence of a lower respiratory tract infection.

Microorganisms commonly isolated from sputum include: *Streptococcus pneumoniae*, *Haemophilus*

influenzae, *Staphylococcus aureus*, *Legionella pneumophila*, *Mycoplasma pneumonia*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Bordetella pertussis*, and *Escherichia coli*.

Resources

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ORGANIZATIONS

- American Lung Association. 432 Park Avenue South, New York, NY 10016. (800) LUNG-USA.
< www.lungusa.org >.
- National Heart, Lung, and Blood Institute (NHLBI). P. O. Box 30105, Bethesda, MD 20824-0105. (301) 592-8573.
< www.nhlbi.nih.gov >.

Nancy J. Nordenson
Rebecca J. Frey, PhD

Squam see **Skin cancer, non-melanoma**

Squint see **Strabismus**

SSPE see **Subacute sclerosing panencephalitis**

SSRIs see **Selective serotonin reuptake inhibitors**

SSSS see **Staphylococcal scalded skin syndrome**

Stanford-Binet intelligence scales

Definition

The Stanford-Binet intelligence scale is a standardized test that assesses intelligence and cognitive abilities in children and adults aged two to 23.

Purpose

The Stanford-Binet intelligence scale is used as a tool in school placement, in determining the presence of a learning disability or a developmental delay, and in tracking intellectual development. In addition, it is sometimes included in neuropsychological testing to assess the brain function of individuals with neurological impairments.

Precautions

Although the Stanford-Binet was developed for children as young as two, examiners should be cautious in using the test to screen very young children for developmental delays or disabilities. The test cannot be used to diagnose **mental retardation** in children aged three and under, and the scoring design may not detect developmental problems in preschool-age children.

Intelligence testing requires a clinically trained examiner. The Stanford-Binet intelligence scale should be administered and interpreted by a trained professional, preferably a psychologist.

Description

The Stanford-Binet intelligence scale is a direct descendent of the Binet-Simon scale, the first intelligence scale created in 1905 by psychologist Alfred Binet and Dr. Theophilus Simon. This revised edition, released in 1986, was designed with a larger, more diverse, representative sample to minimize the gender and racial inequities that had been criticized in earlier versions of the test.

The Stanford-Binet scale tests intelligence across four areas: verbal reasoning, quantitative reasoning, abstract/visual reasoning, and short-term memory. The areas are covered by 15 subtests, including vocabulary, comprehension, verbal absurdities, pattern analysis, matrices, paper folding and cutting, copying, quantitative, number series, equation building, memory for sentences, memory for digits, memory for objects, and bead memory.



The Stanford-Binet intelligence scale. (Photo Researchers Inc. Reproduced by permission.)

All test subjects take an initial vocabulary test, which along with the subject's age, determines the number and level of subtests to be administered. Total testing time is 45–90 minutes, depending on the subject's age and the number of subtests given. Raw scores are based on the number of items answered, and are converted into a standard age score corresponding to age group, similar to an IQ measure.

The 1997 Medicare reimbursement rate for psychological and neuropsychological testing, including intelligence testing, is \$58.35 an hour. Billing time typically includes test administration, scoring and interpretation, and reporting. Many insurance plans cover all or a portion of diagnostic psychological testing.

Normal results

The Stanford-Binet is a standardized test, meaning that norms were established during the design phase of the test by administering the test to a large, representative sample of the test population. The test has a mean, or average, standard score of 100 and a standard deviation of 16 (subtests have a mean of 50 and a standard deviation of 8). The standard deviation indicates how far above or below the norm the subject's score is. For example, an eight-year-old is assessed with the Stanford-Binet scale and achieves a standard age score of 116. The mean score of 100 is the average level at which all eight-year-olds in the representative sample performed. This child's score would be one standard deviation above that norm.

While standard age scores provide a reference point for evaluation, they represent an average of a variety of skill areas. A trained psychologist will evaluate and interpret an individual's performance on the

KEY TERMS

Norms—Normative or mean score for a particular age group.

Representative sample—A random sample of people that adequately represents the test-taking population in age, gender, race, and socioeconomic standing.

Standard deviation—A measure of the distribution of scores around the average (mean). In a normal distribution, two standard deviations above and below the mean includes about 95% of all samples.

Standardization—The process of determining established norms and procedures for a test to act as a standard reference point for future test results. The Stanford-Binet test was standardized on a national representative sample of 5,000 subjects.

scale's subtests to discover strengths and weaknesses and offer recommendations based upon these findings.

Resources

ORGANIZATIONS

American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700.
< <http://www.apa.org> >.

Paula Anne Ford-Martin

Stapedectomy

Definition

Stapedectomy is a surgical procedure in which the innermost bone (stapes) of the three bones (the stapes, the incus, and the malleus) of the middle ear is removed, and replaced with a small plastic tube of stainless-steel wire (a prosthesis) to improve the movement of sound to the inner ear.

Purpose

A stapedectomy is used to treat progressive **hearing loss** caused by **otosclerosis**, a condition in which spongy bone hardens around the base of the stapes. This condition fixes the stapes to the opening of the inner ear, so that the stapes no longer vibrates

properly; therefore, the transmission of sound to the inner ear is disrupted. Untreated otosclerosis eventually results in total deafness, usually in both ears.

Description

With the patient under local or **general anesthesia**, the surgeon opens the ear canal and folds the eardrum forward. Using an operating microscope, the surgeon is able to see the structures in detail, and evaluates the bones of hearing (ossicles) to confirm the diagnosis of otosclerosis.

Next, the surgeon separates the stapes from the incus; freed from the stapes, the incus and malleus bones can now move when pressed. A laser (or other tiny instrument) vaporizes the tendon and arch of the stapes bone, which is then removed from the middle ear.

The surgeon then opens the window that joins the middle ear to the inner ear and acts as the platform for the stapes bone. The surgeon directs the laser's beam at the window to make a tiny opening, and gently clips the prosthesis to the incus bone. A piece of tissue is taken from a small incision behind the ear lobe and used to help seal the hole in the window and around the prosthesis. The eardrum is then gently replaced and repaired, and held there by absorbable packing ointment or a gelatin sponge. The procedure usually takes about an hour and a half.

Good candidates for the surgery are those who have a fixed stapes from otosclerosis, and a conductive hearing loss at least 20 dB. Patients with a severe hearing loss might still benefit from a stapedectomy, if only to improve their hearing to the point where a hearing aid can be of help. The procedure can improve hearing in more than 90% of cases.

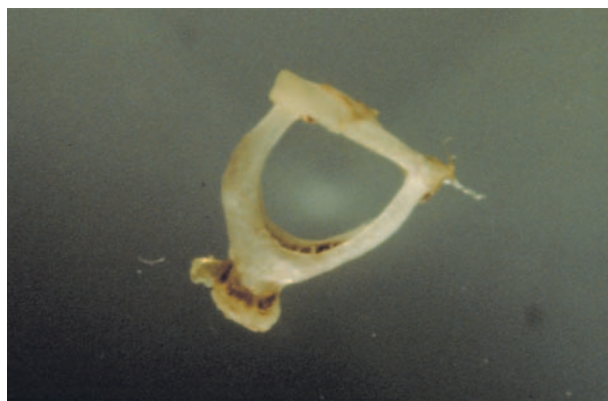
Preparation

Prior to admission to the hospital, the patient will be given a hearing test to measure the degree of deafness, and a full ear, nose, and throat exam.

Most surgeons prefer to use general anesthesia; in this case, an injection will be given to the patient before surgery.

Aftercare

The patient is usually discharged the morning after surgery. **Antibiotics** are given up to five days after surgery to prevent infection; packing and sutures are removed about a week after surgery.



A human stapes bone (located in middle ear) extracted during a stapedectomy. (Custom Medical Stock Photo. Reproduced by permission.)

It is important that the patient not put pressure on the ear for a few days after surgery. Blowing one's nose, lifting heavy objects, swimming underwater, descending rapidly in high-rise elevators, or taking an airplane flight should be avoided.

Right after surgery, the ear is usually quite sensitive, so the patient should avoid loud noises until the ear retrains itself to hear sounds properly.

It is extremely important that the patient avoid getting the ear wet until it has completely healed. Water in the ear could cause an infection; most seriously, water could enter the middle ear and cause an infection within the inner ear, which could then lead to a complete hearing loss. When taking a shower, and washing the hair, the patient should plug the ear with a cotton ball or lamb's wool ball, soaked in Vaseline. The surgeon should give specific instructions about when and how this can be done.

Usually, the patient may return to work and normal activities about a week after leaving the hospital, although if the patient's job involves heavy lifting, three weeks of home rest is recommend. Three days after surgery, the patient may fly in pressurized aircraft.

Risks

The most serious risk is an increased hearing loss, which occurs in about one percent of patients. Because of this risk, a stapedectomy is usually performed on only one ear at a time.

Less common complications include:

- temporary change in taste (due to nerve damage) or lack of taste

KEY TERMS

Cochlea—The hearing part of the inner ear. This snail-shaped structure contains fluid and thousands of microscopic hair cells tuned to various frequencies, in addition to the organ of Corti (the receptor for hearing).

Conductive hearing loss—A type of medically treatable hearing loss in which the inner ear is usually normal, but there are specific problems in the middle or outer ears that prevent sound from getting to the inner ear in a normal way.

Incus—The middle of the three bones of the middle ear. It is also known as the “anvil.”

Malleus—One of the three bones of the middle ear. It is also known as the “hammer.”

Ossicles—The three small bones of the middle ear: the malleus (hammer), the incus (anvil) and the stapes (stirrup). These bones help carry sound from the eardrum to the inner ear.

Vertigo—A feeling of dizziness together with a sensation of movement and a feeling of rotating in space.

- perforated eardrum
- vertigo that may persist and require surgery
- damage to the chain of three small bones attached to the eardrum
- temporary facial nerve **paralysis**
- ringing in the ears

Severe **dizziness** or vertigo may be a signal that there has been an incomplete seal between the fluids of the middle and inner ear. If this is the case, the patient needs immediate bed rest, an exam by the ear surgeon, and (rarely) an operation to reopen the eardrum to check the prosthesis.

Normal results

Most patients are slightly dizzy for the first day or two after surgery, and may have a slight **headache**. Hearing improves once the swelling subsides, the slight bleeding behind the ear drum dries up, and the packing is absorbed or removed, usually within two weeks. Hearing continues to get better over the next three months.

About 90% of patients will have a completely successful surgery, with markedly improved hearing. In 8% of cases, hearing improves, but not quite as

patients usually expect. About half the patients who had ringing in the ears (**tinnitus**) before surgery will have significant relief within six weeks after the procedure.

Resources

ORGANIZATIONS

American Academy of Otolaryngology-Head and Neck Surgery, Inc. One Prince St., Alexandria VA 22314-3357. (703) 836-4444. < <http://www.entnet.org> >.

Better Hearing Institute. 515 King Street, Suite 420, Alexandria, VA 22314. (703) 684-3391.

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Staphylococcal infections

Definition

Staphylococcal (staph) infections are communicable conditions caused by certain bacteria and generally characterized by the formation of abscesses. They are the leading cause of primary infections originating in hospitals (nosocomial infections) in the United States.

Description

Classified since the early twentieth century as among the deadliest of all disease-causing organisms, staph exists on the skin or inside the nostrils of 20–30% of healthy people. It is sometimes found in breast tissue, the mouth, and the genital, urinary, and upper respiratory tracts.

Although staph bacteria are usually harmless, when injury or a break in the skin enables the organisms to invade the body and overcome the body's natural defenses, consequences can range from minor discomfort to **death**. Infection is most apt to occur in:

- newborns
- women who are breastfeeding
- individuals whose immune systems have been undermined by radiation treatments, **chemotherapy**, or medication
- intravenous drug users
- those with surgical incisions, skin disorders, and serious illness like **cancer**, diabetes, and lung disease

Types of infections

Staph infections produce pus-filled pockets (abscesses) located just beneath the surface of the skin or deep within the body. Risk of infection is greatest among the very young and the very old.

A localized staph infection is confined to a ring of dead and dying white blood cells and bacteria. The skin above it feels warm to the touch. Most of these abscesses eventually burst, and pus that leaks onto the skin can cause new infections.

A small fraction of localized staph infections enter the bloodstream and spread through the body. In children, these systemic (affecting the whole body) or disseminated infections frequently affect the ends of the long bones of the arms or legs, causing a bone infection called **osteomyelitis**. When adults develop invasive staph infections, bacteria are most apt to cause abscesses of the brain, heart, kidneys, liver, lungs, or spleen.

Staphylococcus aureus

Named for the golden color of the bacteria grown under laboratory conditions, *S. aureus* is a hardy organism that can survive in extreme temperatures or other inhospitable circumstances. About 70–90% of the population carry this strain of staph in the nostrils at some time. Although present on the skin of only 5–20% of healthy people, as many as 40% carry it elsewhere, such as in the throat, vagina, or rectum, for varying periods of time, from hours to years, without developing symptoms or becoming ill.

S. aureus flourishes in hospitals, where it infects healthcare personnel and patients who have had surgery; who have acute **dermatitis**, insulin-dependent diabetes, or dialysis-dependent **kidney disease**; or who receive frequent allergy-desensitization injections. Staph bacteria can also contaminate bedclothes, catheters, and other objects.

S. aureus causes a variety of infections. **Boils** and inflammation of the skin surrounding a hair shaft (**folliculitis**) are the most common. Toxic **shock** (TSS) and scalded skin syndrome (SSSS) are among the most serious.

TOXIC SHOCK. Toxic shock syndrome is a life-threatening infection characterized by severe **head-ache**, **sore throat**, **fever** as high as 105°F, and a sun-burn-like rash that spreads from the face to the rest of the body. Symptoms appear suddenly; they also include **dehydration** and watery **diarrhea**.

Inadequate blood flow to peripheral parts of the body (shock) and loss of consciousness occur within

the first 48 hours. Between the third and seventh day of illness, skin peels from the palms of the hands, soles of the feet, and other parts of the body. Kidney, liver, and muscle damage often occur.

SCALDED SKIN SYNDROME. Rare in adults and most common in newborns and other children under the age of five, scalded skin syndrome originates with a localized skin infection. A mild fever and/or an increase in the number of infection-fighting white blood cells may occur.

A bright red rash spreads from the face to other parts of the body and eventually forms scales. Large, soft blisters develop at the site of infection and elsewhere. When they burst, they expose inflamed skin that looks as if it had been burned.

MISCELLANEOUS INFECTIONS. *S. aureus* can also cause:

- arthritis
- bacteria in the bloodstream (**bacteremia**)
- pockets of infection and pus under the skin (carbuncles)
- tissue inflammation that spreads below the skin, causing **pain** and swelling (cellulitis)
- inflammation of the valves and walls of the heart (**endocarditis**)
- inflammation of tissue that enclosed and protects the spinal cord and brain (**meningitis**)
- inflammation of bone and bone marrow (osteomyelitis)
- pneumonia

Other strains of staph

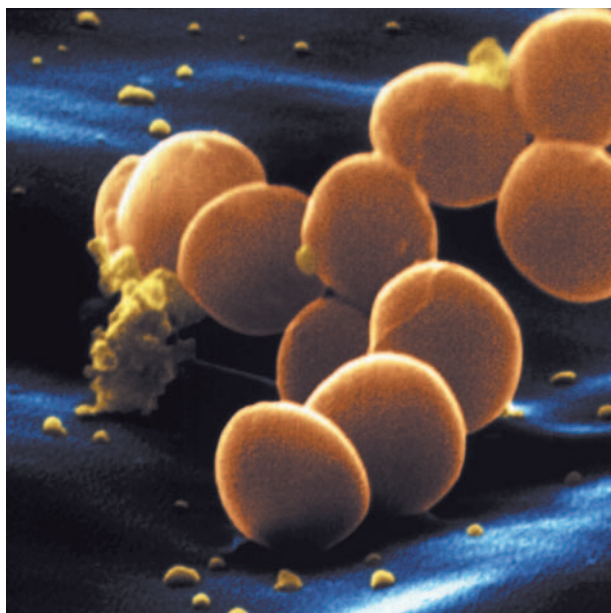
S. EPIDERMIDIS. Capable of clinging to tubing (as in that used for intravenous feeding, etc.), prosthetic devices, and other non-living surfaces, *S. epidermidis* is the organism that most often contaminates devices that provide direct access to the bloodstream.

The primary cause of bacteremia in hospital patients, this strain of staph is most likely to infect cancer patients, whose immune systems have been compromised, and high-risk newborns receiving intravenous supplements.

S. epidermidis also accounts for two of every five cases of prosthetic valve endocarditis. Prosthetic valve endocarditis is endocarditis as a complication of the implantation of an artificial valve in the heart. Although contamination usually occurs during surgery, symptoms of infection may not become evident until a year after the operation. More than



A close-up of a woman's finger and nail cuticle infected with *Staphylococcus aureus*. (Custom Medical Stock Photo. Reproduced by permission.)



A micrographic image of *Staphylococcus aureus*. (Photograph by Oliver Meckes, Photo Researchers, Inc. Reproduced by permission.)

half of the patients who develop prosthetic valve endocarditis die.

STAPHYLOCOCCUS SAPROPHYTICUS. Existing within and around the tube-like structure that carries urine from the bladder (urethra) of about 5% of healthy males and females, *S. saprophyticus* is the second most common cause of unobstructed urinary tract infections (UTIs) in sexually active young women. This strain of staph is responsible for 10–20% of infections affecting healthy outpatients.

Causes and symptoms

Staph bacteria can spread through the air, but infection is almost always the result of direct contact with open sores or body fluids contaminated by these organisms.

Staph bacteria often enter the body through inflamed hair follicles or oil glands. Or they penetrate skin damaged by **burns**, cuts and scrapes, infection, insect bites, or **wounds**.

Multiplying beneath the skin, bacteria infect and destroy tissue in the area where they entered the body. Staph infection of the blood (staphylococcal bacteremia) develops when bacteria from a local infection infiltrate the lymph glands and bloodstream. These infections, which can usually be traced to contaminated catheters or intravenous devices, usually cause persistent high fever. They may cause shock. They also can cause death within a short time.

Warning signs

Common symptoms of staph infection include:

- pain or swelling around a cut, or an area of skin that has been scraped
- boils or other skin abscesses
- blistering, peeling, or scaling of the skin; this is most common in infants and young children
- enlarged lymph nodes in the neck, armpits, or groin

A family physician should be notified whenever:

- Lymph nodes in the neck, armpits, or groin become swollen or tender.
- An area of skin that has been cut or scraped becomes painful or swollen, feels hot, or produces pus. These symptoms may mean the infection has spread to the bloodstream.
- A boil or carbuncle appears on any part of the face or spine. Staph infections affecting these areas can spread to the brain or spinal cord.
- A boil becomes very sore. Usually a sign that infection has spread, this condition may be accompanied by fever, chills, and red streaks radiating from the site of the original infection.
- Boils that develop repeatedly. This type of recurrent infection could be a symptom of diabetes.

Diagnosis

Blood tests that show unusually high concentrations of white blood cells can suggest staph infection, but diagnosis is based on laboratory analysis of

material removed from pus-filled sores, and on analysis of normally uninfected body fluids, such as, blood and urine. Also, x rays can enable doctors to locate internal abscesses and estimate the severity of infection. Needle biopsy (removing tissue with a needle, then examining it under a microscope) may be used to assess bone involvement.

Treatment

Superficial staph infections can generally be cured by keeping the area clean, using soaps that leave a germ-killing film on the skin, and applying warm, moist compresses to the affected area for 20–30 minutes three or four times a day.

Severe or recurrent infections may require a seven to 10 day course of treatment with penicillin or other oral **antibiotics**. The location of the infection and the identity of the causal bacteria determines which of several effective medications should be prescribed.

In case of a more serious infection, antibiotics may be administered intravenously for as long as six weeks. Intravenous antibiotics are also used to treat staph infections around the eyes or on other parts of the face.

Surgery may be required to drain or remove abscesses that form on internal organs, or on shunts or other devices implanted inside the body.

Alternative treatment

Alternative therapies for staph infection are meant to strengthen the immune system and prevent recurrences. Among the therapies believed to be helpful for the person with a staph infection are **yoga** (to stimulate the immune system and promote relaxation), **acupuncture** (to draw heat away from the infection), and herbal remedies. Herbs that may help the body overcome, or withstand, staph infection include:

- **Garlic** (*Allium sativum*). This herb is believed to have antibacterial properties. Herbalists recommend consuming three garlic cloves or three garlic oil capsules a day, starting when symptoms of infection first appear.
- **Cleavers** (*Galium aparine*). This anti-inflammatory herb is believed to support the lymphatic system. It may be taken internally to help heal staph abscesses and reduce swelling of the lymph nodes. A cleavers compress can also be applied directly to a skin infection.
- **Goldenseal** (*Hydrastis canadensis*). Another herb believed to fight infection and reduce inflammation,

KEY TERMS

Abscess—A cavity containing pus surrounded by inflamed tissue.

Endocarditis—Inflammation of the lining of the heart, and/or the heart valves, caused by infection.

Nosocomial infections—Infections that were not present before the patient came to a hospital, but were acquired by a patient while in the hospital.

goldenseal may be taken internally when symptoms of infection first appear. Skin infections can be treated by making a paste of water and powdered goldenseal root and applying it directly to the affected area. The preparation should be covered with a clean bandage and left in place overnight.

- **Echinacea** (*Echinacea* spp.). Taken internally, this herb is believed to have antibiotic properties and is also thought to strengthen the immune system.
- **Thyme** (*Thymus vulgaris*), lavender (*Lavandula officinalis*), or bergamot (*Citrus bergamot*) oils. These oils are believed to have antibacterial properties and may help to prevent the scarring that may result from skin infections. A few drops of these oils are added to water and then a compress soaked in the water is applied to the affected area.
- **Tea tree oil** (*Melaleuca* spp.). Another infection-fighting herb, this oil can be applied directly to a boil or other skin infection.

Prognosis

Most healthy people who develop staph infections recover fully within a short time. Others develop repeated infections. Some become seriously ill, requiring long-term therapy or emergency care. A small percentage die.

Prevention

Healthcare providers and patients should always wash their hands thoroughly with warm water and soap after treating a staph infection or touching an open wound or the pus it produces. Pus that oozes onto the skin from the site of an infection should be removed immediately. This affected area should then be cleansed with antiseptic or with antibacterial soap.

To prevent infection from spreading from one part of the body to another, it is important to shower

rather than bathe during the healing process. Because staph infection is easily transmitted from one member of a household to others, towels, washcloths, and bed linens used by someone with a staph infection should not be used by anyone else. They should be changed daily until symptoms disappear, and laundered separately in hot water with bleach.

Children should frequently be reminded not to share:

- brushes, combs, or hair accessories
- caps
- clothing
- sleeping bags
- sports equipment
- other personal items

A diet rich in green, yellow, and orange vegetables can bolster natural immunity. A doctor or nutritionist may recommend **vitamins** or mineral supplements to compensate for specific dietary deficiencies. Drinking eight to 10 glasses of water a day can help flush disease-causing organisms from the body.

Because some strains of staph bacteria are known to contaminate artificial limbs, prosthetic devices implanted within the body, and tubes used to administer medication or drain fluids from the body, catheters and other devices should be removed on a regular basis, if possible, and examined for microscopic signs of staph. Symptoms may not become evident until many months after contamination has occurred, so this practice should be followed even with patients who show no sign of infection.

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Maureen Haggerty

Staphylococcal scalded skin syndrome

Definition

Staphylococcal scalded skin syndrome (SSSS) is a disease, caused by a type of bacteria, in which large sheets of skin may peel away.

Description

SSSS primarily strikes children under the age of five, particularly infants. Clusters of SSSS cases (epidemics) can occur in newborn nurseries, when staff in those nurseries accidentally pass the causative bacteria between patients. It can also strike other age groups who have weakened immune systems. Such immunocompromised patients include those with **kidney disease**, people undergoing **cancer chemotherapy**, organ transplant patients, and individuals with acquired **immunodeficiency syndrome (AIDS)**.

Causes and symptoms

SSSS is caused by a type of bacteria called *Staphylococcus aureus*. This bacteria produces a chemical called an epidermolytic toxin ("epiderm," deriving from the Greek words *epi*, meaning on, and *derma*, meaning skin, refers to the top layer of skin; "-lytic," deriving from the Greek word *lysis*, which literally denotes the act of undoing, means breaking or destroying; a toxin is a poison). While the bacteria itself is not spread throughout the body, it affects all of the skin by sending this toxin through the bloodstream.

SSSS begins with a small area of infection. In newborn babies, this may appear as a crusted area around the umbilicus, or in the diaper area. In children between the ages of one and six, a small, red, crusty bump appears near the nose or ear. The child may have no energy, and may have a **fever**. The skin becomes sensitive and uncomfortable even before the rash is fully visible. The rash starts out as bright red patches around the original area of crusting. Blisters may appear, and the skin may look wrinkled. When the blisters pop, they leave pitted areas. Even gently touching these red patches of skin may cause them to peel away in jagged sheets. The skin below is shiny, moist, and bright pink. Within a day or two, the top layer of skin all over the body is peeling off in large sheets.

The dangers of this illness include the chance that a different kind of bacteria will invade through the open areas in the skin and cause a serious systemic infection (**sepsis**). A lot of body fluid is lost as the skin peels away, and the layer underneath dries. **Dehydration** is a danger at this point.

Diagnosis

Although good patient care includes taking specimens of blister fluid and smears from the nose or throat, no bacteria are usually demonstrated. SSSS is usually diagnosed on the basis of the typical progression of symptoms in a child of this age, prone to this

KEY TERMS

Epidermis—The top layer of skin.

Epidermolytic—Damaging to the top layer of skin.

Sepsis—An overwhelming infection affecting all the systems of the body.

disorder. A sample of skin (**skin biopsy**) should be taken, prepared, and examined under a microscope. If the patient's disease is truly SSSS, the biopsy will show a characteristic appearance. There will be no accumulation of those cells usually present in the case of a bacterial infection. Instead, there will be evidence of disruption of only the top layer of skin (epidermis).

Treatment

Treatment involves careful attention to avoid the development of dehydration. A variety of lotions and creams are available to apply to areas where the epidermis has peeled away. This both soothes the sensitive areas, and protects against drying and further moisture loss.

Prognosis

Most patients heal from SSSS within about 10–14 days. Healing occurs without scarring in the majority of patients. **Death** may occur if severe dehydration or sepsis complicate the illness. About 3% of children die of these complications; about 50% of immunocompromised adults die of these complications.

Prevention

As always, good hygiene can prevent the passage of the causative bacteria between people. In the event of an outbreak in a newborn nursery, members of the staff should have nasal smears taken to identify an adult who may be unknowingly carrying the bacteria and passing it on to the babies.

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Rosalyn Carson-DeWitt, MD

Staphylococcal food poisoning see **Food poisoning**

Starvation

Definition

Starvation is the result of a severe or total lack of nutrients needed for the maintenance of life.

Description

Adequate **nutrition** has two components, necessary nutrients and energy in the form of calories. It is possible to ingest enough energy without a well-balanced selection of individual nutrients and produce diseases that are noticeably different from those resulting from an overall insufficiency of nutrients and energy. Although all foods are a source of energy for the human body, it is possible to consume a seemingly adequate amount of food without getting the required minimum of energy (calories). For example, marasmus is the result of a diet that is deficient mainly in energy. Children who get enough calories, but not enough protein have kwashiorkor. This is typical in cultures with a limited variety of foods that eat mostly a single staple carbohydrate like maize or rice. These conditions overlap and are associated with multiple vitamin and mineral deficits, most of which have specific names and set of problems associated with them.

- Marasmus produces a very skinny child with stunted growth.
- Children with kwashiorkor have body fat, an enlarged liver, and edema—swelling from excess water in the tissues. They also have growth retardation.
- Niacin deficiency produces **pellagra** characterized by **diarrhea**, skin **rashes**, brain dysfunction, tongue, mouth and vaginal irritation, and trouble swallowing.
- Thiamine (Vitamin B₁) deficiency causes **beriberi**, which can appear as **heart failure** and **edema**, a brain and nerve disease, or both.
- Riboflavin deficiency causes a sore mouth and throat, a skin rash, and anemia.
- Lack of vitamin C (ascorbic acid)—scurvy—causes hair damage, bleeding under the skin, in muscles and joints, gum disease, poor wound healing, and in severe cases convulsions, **fever**, loss of blood pressure, and **death**.
- Vitamin B₁₂ is needed to keep the nervous system working properly. It and pyridoxine (vitamin B₆) are both necessary for blood formation.
- Vitamin A deficiency causes at first loss of night vision and eventually blindness from destruction of the cornea, a disease called keratomalacia.

- Vitamin K is necessary for blood clotting.
- Vitamin D regulates calcium balance. Without it, children get **rickets** and adults get osteomalacia.

Causes and symptoms

Starvation may result from a number of factors. They include:

- anorexia nervosa, which is an eating disorder characterized by extreme calorie restriction
- intentional fasting
- coma
- stroke
- inability to obtain food (famine; **child abuse**; aftermath of war or other disaster; being lost in wilderness or desert areas)
- severe gastrointestinal disease

Since the body will combat **malnutrition** by breaking down its own fat and eventually its own tissue, a whole host of symptoms can appear. The body's structure, as well as its functions, are affected. Starved adults may lose as much as 50% of their normal body weight.

Characteristic symptoms of starvation include:

- shrinkage of such vital organs as the heart, lungs, ovaries, or testes, and gradual loss of their functions
- chronic diarrhea
- anemia
- reduction in muscle mass and consequent weakness
- lowered body temperature combined with extreme sensitivity to cold
- decreased ability to digest food because of lack of digestive acid production
- irritability and difficulty with mental concentration
- immune deficiency
- swelling from fluid under the skin
- decreased sex drive

Complete starvation in adults leads to death within eight to 12 weeks. In the final stages of starvation, adult humans experience a variety of neurological and psychiatric symptoms, including **hallucinations** and convulsions, as well as severe muscle **pain** and disturbances in heart rhythm.

In children, chronic malnutrition is marked by growth retardation. Anemia is the first sign to appear in an adult. Swelling of the legs is next, due to a drop in the protein content of the blood. Loss of resistance to

KEY TERMS

Anemia—Not enough red blood cells in the blood.

Anorexia nervosa—Eating disorder marked by malnutrition and weight loss commonly occurring in young women.

Cornea—The clear part of the front of the eye that admits light.

Kwashiorkor—Severe malnutrition in children caused by mainly by a protein-poor diet, characterized by growth retardation.

Marasmus—Severe malnutrition in children caused by a diet lacking mainly in calories. Can also be caused by disease and parasitic infection.

infection follows next, along with poor wound healing. There is also progressive weakness and difficulty swallowing, which may lead to inhaling food. At the same time, the signs of specific nutrient deficiencies may appear.

Treatment

If the degree of malnutrition is severe, the intestines may not tolerate a fully balanced diet. They may, in fact, not be able to absorb adequate nutrition at all. Carefully prepared elemental **diets** or intravenous feeding must begin the treatment. A formula consisting of 42% dried skim milk, 32% edible oil, and 25% sucrose plus electrolyte, mineral, and vitamin supplements is recommended for the first phase of refeeding. The treatment back to health is long and first begins with liquids. Gradually, solid foods are introduced and a daily diet providing 5,000 calories or more is instituted.

Prognosis

People can recover from severe degrees of starvation to a normal stature and function. Children, however, may suffer from permanent **mental retardation** or growth defects if their deprivation was long and extreme.

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J. Ricker Polsdorfer, MD
Rebecca J. Frey, PhD

Stasis dermatitis see **Dermatitis**

Static encephalopathy see **Cerebral palsy**

STDs see **Sexually transmitted diseases**

Steatosis see **Fatty liver**

Steele-Richardson-Olszewski syndrome see **Progressive supranuclear palsy**

Stein-Leventhal syndrome see **Polycystic ovary syndrome**

Steinert's disease see **Myotonic dystrophy**

Stem cell therapy see **Bone marrow transplantation**

Stem cell transplantation

Definition

Stem cells are basic human cells that reproduce (replicate) easily, providing a continuous source of new, sometimes different types of cells. A stem cell transplant is a procedure that replaces unhealthy stem cells with healthy ones.

Purpose

Physicians use stem cell transplants to treat many diseases that damage or destroy bone marrow, found in the soft fatty tissue inside the bones. Examples of these diseases are leukemia and **multiple myeloma**. Some patients lack bone marrow because of aggressive **cancer** treatments or diseases such as **aplastic anemia**, which causes abnormal blood cell production.

Recent advances in stem cell research have made it a treatment possibility for patients with certain types of lymphomas, genetic disorders, and **autoimmune disorders** as well. Researchers are hoping to eventually

harvest stem cells to treat diseases such as Parkinson's disease, type 1 diabetes, Alzheimer's disease, **liver disease**, arthritis, and spinal cord injuries.

Precautions

The physician will weigh many factors when determining if a patient is a candidate for stem cell transplantation, including overall health and function of many vital organ systems. Stem cell transplantation is an aggressive treatment and may not be recommended for some patients, including those with heart, kidney, or lung disorders. If the patient has an aggressive cancer that has spread throughout the body, he or she may not be considered for a stem cell transplant. It once was thought that stem cell transplants were not safe in patients over age 60, but new research shows that elderly patients can safely receive stem cells from donors.

Many ethical and legal factors are impacting the research and development into stem cell transplantation. Much debate surrounds scientific advances. For example, human embryos, fetal tissues and umbilical cords are sources of stem cells that may be transplanted or used for disease research. Some people have ethical problems with the use of embryos in fertility clinics for stem cell research or transplantation. Some link stem cell transplantation for disease with cloning and want to stop funding for stem cell research over fear of human cloning. A study released in 2005 stated that 63% of Americans back embryonic stem cell research and 70% support federal legislation to promote more research. Meanwhile, scientists continue to develop new and exciting possibilities for transplanting stem cells into the human body that may one day lead to new treatments for previously incurable diseases. Many do so with private funding.

Description

Stem cell transplants sometimes are called bone marrow transplants. Nearly 100 years ago, physicians tried to give patients with leukemia and anemia bone marrow by mouth. These treatments were not successful, but led to experiments showing healthy bone marrow transfused into the blood stream could restore damaged bone marrow.

Today, two types of stem cell transplants are performed most often. When a patient's own stem cells are collected (harvested) before they are destroyed by high-dose **chemotherapy** or **radiation therapy** treatments, then returned to the same patient's body, it is called an autologous transplant. Using stem cells from another person, or a donor, is called allogenic

transplant. In many cases, donor cells come from a relative, such as a brother or sister. However, the likelihood that a sibling will match the patient is only about 25%. Stem cells may need to come from a person not related to the recipient.

To find out if a patient could receive stem cells from a donor, physicians developed human leukocyte antigen (HLA) testing to match tissue types. The next challenge became finding donors. Throughout the 1980s and 1990s, private individuals, hospitals, foundations, and states worked to set up a nationwide registry of bone marrow donors. The National Marrow Donor Program (NMDP) now has the largest stem cell donor registry in the world. In July 2003, more than five million volunteer donors and more than 28,000 units of umbilical cord blood were listed on the NMDP registry. More than 16,000 bone marrow transplants have been performed since the NMDP was founded in 1986.

Stem cell transplants normally take place at specialized centers. Donor cells are taken from the donor in an operating room while the patient is unconscious and under **general anesthesia**. Bone marrow normally is harvested from the top of the hip bone. The marrow usually is filtered, treated, and either transplanted immediately or frozen for later use. Stem cells are transfused through an intravenous (IV) catheter that physicians insert in the patient's neck or chest. Physicians refer to this step as the "rescue process." The stem cells replace old cells. For example, the transplanted cells travel to the bone cavities and begin replacing old bone marrow.

Preparation

Standard preparation involves getting rid of the remaining disease and damaged cells. The exact process depends on the patient. In many cases, the patient will receive chemotherapy. Some also receive radiation therapy. Another goal of preparation is to depress the immune system. This makes it less likely that the patient's body will reject the new stem cells. New advances have been made that allow some of the patient's diseased cells to remain and mix with the new cells. Immediately before transplantation, the treating physician and staff will give the patient special instructions and precautions, depending on his or her disease and exact procedure.

Aftercare

Stem cells take up to three weeks to begin producing new cells or bone marrow, a process called

KEY TERMS

Catheter—A medical device shaped like a tube that physicians can insert into vessels, canals, or passageways to more easily inject or withdraw fluids.

Embryo—A developing human from the time of conception to the end of the eighth week after conception.

Engraftment—The process of transplanted stem cells reproducing new cells.

engraftment. Until engraftment is complete, patients may bleed easily and are at risk for infections. They may be required to stay in the hospital at least one week following surgery until blood cell counts reach a safe level. Once home, patients usually must be closely monitored, be careful not to risk infection, and may be anemic and sleep most of the time. They may receive **antibiotics** to prevent infection and will receive specific instructions for care from the transplant center staff. Most stem cell transplant patients cannot return to work or normal activities for up to six months following the transplant.

Risks

In addition to the risk of a life-threatening infection following a stem cell transplant, patients receiving stem cells from donors risk serious complications from graft-versus-host disease (GVHD). GVHD is caused when the donor's cells react against the patient's (recipient's) tissue. Sometimes, the patient's body simply rejects the new cells. Researchers continue to explore ways to lessen risks of complications following stem cell transplants. For example, recent studies recommend a **measles vaccination** after stem cell transplantation to prevent measles in recipients, who are at risk because of their low immunity.

Normal results

When **bone marrow transplantation** is successful, it prolongs the life of a person who might have otherwise died from the disease that caused damage to the bone marrow. People with more siblings have a higher chance of finding a compatible donor, though successful stem cell transplants also occur with donors who are not related to the recipients. Patients usually can return to normal activities within six months to one year after the transplant.

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- “Patients Should Receive Measles Vaccine After Stem Cell Transplantation.” *Life Science Weekly* (Nov. 23, 2004):883.

ORGANIZATIONS

- International Myeloma Foundation. 12650 Riverside Drive, Suite 206. North Hollywood, CA 91607-3421. 800-452-CURE. <http://www.myeloma.org>.
- National Marrow Donor Program. Suite 500, 3001 Broadway St. NE, Minneapolis, MN 55413-1753. 800-627-7692. <http://www.marrow.org>.

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- Advances in Stem Cell Transplants: A Primer for Patients.* Web page. National Marrow Donor Program, 2005. http://www.marrow.org/PATIENT/advances_stem_cell_transplants_primer_patients.html.

Teresa G. Odle

Sterilization see **Tubal ligation; Vasectomy**

Stillbirth

Definition

A stillbirth is defined as the **death** of a fetus at any time after the twentieth week of **pregnancy**. Stillbirth is also referred to as intrauterine fetal death (IUFD).

Description

It is important to distinguish between a stillbirth and other words that describe the unintentional end of a pregnancy. A pregnancy that ends before the twentieth week is called a **miscarriage** rather than a stillbirth, even though the death of the fetus is a common cause of miscarriage. After the twentieth week, the unintended end of a pregnancy is called a stillbirth if the infant is dead at birth and premature delivery if it is born alive.

Factors that increase a mother's risk of stillbirth include: age over 35, **malnutrition**, inadequate prenatal care, **smoking**, and alcohol or drug **abuse**.

Causes and symptoms

Causes

A number of different disorders can cause stillbirth. They include:

- Pre-eclampsia and **eclampsia**. These are disorders of late pregnancy characterized by high blood pressure, fluid retention, and protein in the urine.
- Diabetes in the mother.
- Hemorrhage.
- Abnormalities in the fetus caused by infectious diseases, including **syphilis**, **toxoplasmosis**, German measles (**rubella**), and **influenza**.
- Severe **birth defects**, including **spina bifida**. Birth defects are responsible for about 20% of stillbirths.
- Postmaturity. Postmaturity is a condition in which the pregnancy has lasted 41 weeks or longer.
- Unknown causes. These account for about one-third of stillbirths.

Symptoms

In most cases the only symptom of stillbirth is that the mother notices that the baby has stopped moving. In some cases, the first sign of fetal death is **premature labor**. Premature labor is marked by a rush of fluid from the vagina, caused by the tearing of the membrane around the baby; and by abdominal cramps or contractions.

Diagnosis

When the mother notices that fetal movement has stopped, the doctor can use several techniques to evaluate whether the baby has died. The doctor can listen for the fetal heartbeat with a stethoscope, use Doppler ultrasound to detect the heartbeat, or give the mother an electronic fetal nonstress test. In this test, the mother lies on her back with electronic monitors attached to her abdomen. The monitors record the baby's heart rate, movements, and contractions of the uterus.

Treatment

Medical

In most cases of intrauterine death, the mother will go into labor within two weeks of the baby's death. If the mother does not go into labor, the doctor will bring on (induce) labor in order to prevent the risk of hemorrhage. Labor is usually induced by giving the mother a drug (oxytocin) that cause the uterus to contract.

KEY TERMS

Alpha-fetoprotein analysis—A blood test that can be done after the sixteenth week of pregnancy to evaluate the possibility of spina bifida and other birth defects in the fetus.

Electronic fetal nonstress test—A test in which electronic monitors attached to the mother's abdomen to detect contractions of the uterus as well as the baby's heartbeat and movements.

Miscarriage—The spontaneous end of a pregnancy before the twentieth week. The death of the fetus is a common cause of miscarriage.

Oxytocin—A drug that is given to induce labor in some cases of stillbirth.

Pre-eclampsia and eclampsia—Disorders of late pregnancy associated with high blood pressure, fluid retention, and protein in the urine. They can cause stillbirth.

Premature delivery—The birth of a live baby when a pregnancy ends spontaneously after the twentieth week.

Follow-up therapy

Emotional support from family and friends, self-help groups, and counseling by a mental health professional can help bereaved parents cope with their loss.

Prognosis

With the exception of women with diabetes, women who have a stillbirth have as good a chance of carrying a future pregnancy to term as women who are pregnant for the first time.

Prevention

The risk of stillbirth can be lowered to some extent by good prenatal care and the mother's avoidance of exposure to infectious diseases, smoking, alcohol abuse, or drug consumption. Tests before delivery (**antepartum testing**), such as ultrasound, the alpha-fetoprotein blood test, and the electronic fetal nonstress test, can be used to evaluate the health of the fetus before there is a stillbirth.

Resources

BOOKS

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ORGANIZATIONS

Compassionate Friends. P.O. Box 3696, Oak Brook, IL 60522. (877) 969-0010. <<http://www.compassionatefriends.org>>.

GriefNet. P.O. Box 3272, Ann Arbor, MI 48106. <<http://rivendell.org>>.

Hannah's Prayer. P.O. Box 5016, Auburn CA 95604. (775) 852-9202. <<http://www.hannah.org>>.

M.E.N.D. (Mommies Enduring Neonatal Death). P.O. Box 1007, Coppell, TX 75067. (972) 459-2396; (888) 695-6363. <http://www.mend.org/home_index.asp>.

Pregnancy and Infant Loss Support (SHARE). St. Joseph Health Center, 300 First Capitol Dr., St. Charles, MO 63301. (800) 821-6819. <<http://www.nationalshareoffice.com/index.html>>.

Carol A. Turkington

Stings see **Bites and stings**

Stockholm syndrome

Definition

Stockholm syndrome refers to a group of psychological symptoms that occur in some persons in a captive or hostage situation. It has received considerable media publicity in recent years because it has been used to explain the behavior of such well-known kidnapping victims as Patty Hearst (1974) and Elizabeth Smart (2002). The term takes its name from a bank robbery in Stockholm, Sweden, in August 1973. The robber took four employees of the bank (three women and one man) into the vault with him and kept them hostage for 131 hours. After the employees were finally released, they appeared to have formed a paradoxical emotional bond with their captor; they told reporters that they saw the police as their enemy rather than the bank robber, and that they had positive feelings toward the criminal. The syndrome was first named by Nils Bejerot (1921–1988), a medical professor who specialized in **addiction** research and served as a psychiatric consultant to the Swedish police during the standoff at the bank. Stockholm syndrome is also known as Survival Identification Syndrome.

Description

Stockholm syndrome is considered a complex reaction to a frightening situation, and experts do not agree completely on all of its characteristic features or on the factors that make some people more

susceptible than others to developing it. One reason for the disagreement is that it would be unethical to test theories about the syndrome by experimenting on human beings. The data for understanding the syndrome are derived from actual hostage situations since 1973 that differ considerably from one another in terms of location, number of people involved, and time frame. Another source of disagreement concerns the extent to which the syndrome can be used to explain other historical phenomena or more commonplace types of abusive relationships. Many researchers believe that Stockholm syndrome helps to explain certain behaviors of survivors of World War II concentration camps; members of religious cults; battered wives; incest survivors; and physically or emotionally abused children as well as persons taken hostage by criminals or terrorists.

Most experts, however, agree that Stockholm syndrome has three central characteristics:

- The hostages have negative feelings about the police or other authorities.
- The hostages have positive feelings toward their captor(s).
- The captors develop positive feelings toward the hostages.

Causes & symptoms

Stockholm syndrome does not affect all hostages (or persons in comparable situations); in fact, a Federal Bureau of Investigation (FBI) study of over 1200 hostage-taking incidents found that 92% of the hostages did *not* develop Stockholm syndrome. FBI researchers then interviewed flight attendants who had been taken hostage during airplane hijackings, and concluded that three factors are necessary for the syndrome to develop:

- The crisis situation lasts for several days or longer.
- The hostage takers remain in contact with the hostages; that is, the hostages are not placed in a separate room.
- The hostage takers show some kindness toward the hostages or at least refrain from harming them. Hostages abused by captors typically feel anger toward them and do not usually develop the syndrome.

In addition, people who often feel helpless in other stressful life situations or are willing to do anything in order to survive seem to be more susceptible to developing Stockholm syndrome if they are taken hostage.

People with Stockholm syndrome report the same symptoms as those diagnosed with posttraumatic

KEY TERMS

Coping—In psychology, a term that refers to a person's patterns of response to stress. Some patterns of coping may lower a person's risk of developing Stockholm syndrome in a hostage situation.

Flashback—The re-emergence of a traumatic memory as a vivid recollection of sounds, images, and sensations associated with the trauma. The person having the flashback typically feels as if they are reliving the event. Flashbacks were first described by doctors treating combat veterans of World War I (1914–1918).

Identification with an aggressor—In psychology, an unconscious process in which a person adopts the perspective or behavior patterns of a captor or abuser. Some researchers consider it a partial explanation of Stockholm syndrome.

Regression—In psychology, a return to earlier, usually childish or infantile, patterns of thought or behavior.

Syndrome—A set of symptoms that occur together.

stress disorder (PTSD): **insomnia**, nightmares, general irritability, difficulty concentrating, being easily startled, feelings of unreality or confusion, inability to enjoy previously pleasurable experiences, increased distrust of others, and flashbacks.

Diagnosis

Stockholm syndrome is a descriptive term for a pattern of coping with a traumatic situation rather than a diagnostic category. Most psychiatrists would use the diagnostic criteria for **acute stress disorder** or posttraumatic stress disorder when evaluating a person with Stockholm syndrome.

Treatment

Treatment of Stockholm syndrome is the same as for PTSD, most commonly a combination of medications for short-term sleep disturbances and psychotherapy for the longer-term symptoms.

Prognosis

The prognosis for recovery from Stockholm syndrome is generally good, but the length of treatment needed depends on several variables. These include the

nature of the hostage situation; the length of time the crisis lasted, and the individual patient's general coping style and previous experience(s) of trauma.

Prevention

Prevention of Stockholm syndrome at the level of the larger society includes further development of crisis intervention skills on the part of law enforcement as well as strategies to prevent kidnapping or hostage-taking incidents in the first place. Prevention at the individual level is difficult as of the early 2000s because researchers have not been able to identify all the factors that may place some persons at greater risk than others; in addition, they disagree on the specific psychological mechanisms involved in Stockholm syndrome. Some regard the syndrome as a form of regression (return to childish patterns of thought or action) while others explain it in terms of emotional **paralysis** ("frozen fright") or identification with the aggressor.

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Rebecca Frey, PhD

Stomach acid determination see **Gastric acid determination**

Stomach cancer

Definition

Stomach **cancer** (also known as gastric cancer) is a disease in which the cells forming the inner lining of the stomach become abnormal and start to divide uncontrollably, forming a mass called a tumor.

Description

The stomach is a J-shaped organ that lies in the left and central portion of the abdomen. The stomach produces many digestive juices and acids that mix with food and aid in the process of digestion. There are five regions of the stomach that doctors refer to when determining the origin of stomach cancer. These are:

- the cardia, area surrounding the cardiac sphincter which controls movement of food from the esophagus into the stomach,
- the fundus, upper expanded area adjacent to the cardiac region,
- the antrum, lower region of the stomach where it begins to narrow,
- the prepyloric, region just before or nearest the pylorus,
- and the pylorus, the terminal region where the stomach joins the small intestine

Cancer can develop in any of the five sections of the stomach. Symptoms and outcomes of the disease will vary depending on the location of the cancer.

Based on previous data from the National Cancer Institute and the United States Census, the American Cancer Society estimates that 21,700 Americans will be diagnosed with stomach cancer during 2001 and approximately 13,000 deaths will result from the disease. In most areas, men are affected by stomach cancer nearly twice as often as women. Most cases of stomach cancer are diagnosed between the ages of 50

and 70, but in families with a hereditary risk of stomach cancer, younger cases are more frequently seen.

Stomach cancer is one of the leading causes of cancer deaths in several areas of the world, most notably Japan and other Asian countries. In Japan it appears almost ten times as frequently as in the United States. The number of new stomach cancer cases is decreasing in some areas, however, especially in developed countries. In the United States, incidence rates have dropped from 30 individuals per 100,000 in the 1930s, to only 8 in 100,000 individuals developing stomach cancer by the 1980s. The use of refrigerated foods and increased consumption of fresh fruits and vegetables, instead of preserved foods with high salt content, may be a reason for the decline.

Causes and symptoms

While the exact cause for stomach cancer has not been identified, several potential factors have lead to increased numbers of individuals developing the disease and therefore, significant risk has been associated. Diet, work environment, exposure to the bacterium *Helicobacter pylori*, and a history of stomach disorders such as ulcers or polyps are some of these believed causes.

Studies have shown that eating foods with high quantities of salt and nitrites increases the risk of stomach cancer. The diet in a specific region can have a great impact on its residents. Making changes to the types of foods consumed has been shown to decrease likelihood of disease, even for individuals from countries with higher risk. For example, Japanese people who move to the United States or Europe and change the types of foods they eat have a far lower chance of developing the disease than do Japanese people who remain in Japan and do not change their dietary habits. Eating recommended amounts of fruit and vegetables may lower a person's chances of developing this cancer.

A high risk for developing stomach cancers has been linked to certain industries as well. The best proven association is between stomach cancer and persons who work in coal mining and those who work processing timber, nickel, and rubber. An unusually large number of these workers have been diagnosed with this form of cancer.

Several studies have identified a bacterium (*Helicobacter pylori*) that causes stomach ulcers (inflammation in the inner lining of the stomach). Chronic (long-term) infection of the stomach with these bacteria may lead to a particular type of cancer

(lymphomas or mucosa-associated lymphoid tissue [MALT]) in the stomach.

Another risk factor is the development of polyps, benign growths in the lining of the stomach. Although polyps are not cancerous, some may have the potential to turn cancerous. People in blood group A are also at elevated risk for this cancer for unknown reasons. Other speculative causes of stomach cancer include previous stomach surgery for ulcers or other conditions, or a form of anemia known as **pernicious anemia**.

Stomach cancer is a slow-growing cancer. It may be years before the tumor grows very large and produces distinct symptoms. In the early stages of the disease, the patient may only have mild discomfort, **indigestion**, **heartburn**, a bloated feeling after eating, and mild **nausea**. In the advanced stages, a patient will have loss of appetite and resultant weight loss, stomach pains, **vomiting**, difficulty in swallowing, and blood in the stool. Stomach cancer often spreads (metastasizes) to adjoining organs such as the esophagus, adjacent lymph nodes, liver, or colon.

Diagnosis

Unfortunately, many patients diagnosed with stomach cancer experience **pain** for two or three years before informing a doctor of their symptoms. When a doctor suspects stomach cancer from the symptoms described by the patient, a complete medical history will be taken to check for any risk factors. A thorough **physical examination** will be conducted to assess all the symptoms. Laboratory tests may be ordered to check for blood in the stool (**fecal occult blood test**) and anemia (low red blood cell count), which often accompany gastric cancer.

In some countries, such as Japan, it is appropriate for patients to be given routine screening examinations for stomach cancer, as the risk of developing cancer in that society is very high. Such screening might be useful for all high-risk populations. Due to the low prevalence of stomach cancer in the United States, routine screening is usually not recommended unless a family history of the disease exists.

Whether as a screening test or because a doctor suspects a patient may have symptoms of stomach cancer, endoscopy or barium x rays are used in diagnosing stomach cancer. For a barium x ray of the upper gastrointestinal tract, the patient is given a chalky, white solution of barium sulfate to drink. This solution coats the esophagus, the stomach, and the small intestine. Air may be pumped into the stomach after the barium solution in order to get a clearer

picture. Multiple x rays are then taken. The barium coating helps to identify any abnormalities in the lining of the stomach.

In another more frequently used test, known as upper gastrointestinal endoscopy, a thin, flexible, lighted tube (endoscope) is passed down the patient's throat and into the stomach. The doctor can view the lining of the esophagus and the stomach through the tube. Sometimes, a small ultrasound probe is attached at the end of the endoscope. This probe sends high frequency sound waves that bounce off the stomach wall. A computer creates an image of the stomach wall by translating the pattern of echoes generated by the reflected sound waves. This procedure is known as an endoscopic ultrasound or EUS.

Endoscopy has several advantages, in that the physician is able to see any abnormalities directly. In addition, if any suspicious-looking patches are seen, biopsy forceps can be passed painlessly through the tube to collect some tissue for microscopic examination. This is known as a biopsy. EUS is beneficial because it can provide valuable information on depth of tumor invasion.

After stomach cancer has been diagnosed and before treatment starts, another type of x-ray scan is taken. Computed tomography (CT) is an imaging procedure that produces a three-dimensional picture of organs or structures inside the body. CT scans are used to obtain additional information in regard to how large the tumor is and what parts of the stomach it borders; whether the cancer has spread to the lymph nodes; and whether it has spread to distant parts of the body (metastasized), such as the liver, lung, or bone. A CT scan of the chest, abdomen, and pelvis is taken. If the tumor has gone through the wall of the stomach and extends to the liver, pancreas, or spleen, the CT will often show this. Although a CT scan is an effective way of evaluating whether cancer has spread to some of the lymph nodes, it is less effective than EUS in evaluating whether the nodes closest to the stomach are free of cancer. However, CT scans, like barium x-rays, have the advantage of being less invasive than upper endoscopy.

Laparoscopy is another procedure used to stage some patients with stomach cancer. This involves a medical device similar to an endoscope. A laparoscopy is a minimally invasive surgery technique with one or a few small incisions, which can be performed on an outpatient basis, followed by rapid recovery. Patients who may receive **radiation therapy** or **chemotherapy** before surgery may undergo a laparoscopic procedure to determine the precise stage of cancer. The patient

with bone pain or with certain laboratory results should be given a bone scan.

Benign gastric neoplasms are tumors of the stomach that cause no major harm. One of the most common is called a submucosal leiomyoma. If a leiomyoma starts to bleed, surgery should be performed to remove it. However, many leiomyomas require no treatment. Diagnosis of stomach cancers should be conducted carefully so that if the tumor does not require treatment the patient is not subjected to a surgical operation.

Treatment

More than 95% of stomach cancers are caused by adenocarcinomas, malignant cancers that originate in glandular tissues. The remaining 5% of stomach cancers include lymphomas and other types of cancers. It is important that gastric lymphomas be accurately diagnosed because these cancers have a much better prognosis than stomach adenocarcinomas. Approximately half of the people with gastric lymphomas survive five years after diagnosis. Treatment for gastric lymphoma involves surgery combined with chemotherapy and radiation therapy.

Staging of stomach cancer is based on how deep the growth has penetrated the stomach lining; to what extent (if any) it has invaded surrounding lymph nodes; and to what extent (if any) it has spread to distant parts of the body (metastasized). The more confined the cancer, the better the chance for a cure.

One important factor in the staging of adenocarcinoma of the stomach is whether or not the tumor has invaded the surrounding tissue and, if it has, how deep it has penetrated. If invasion is limited, prognosis is favorable. Disease tissue that is more localized improves the outcome of surgical procedures performed to remove the diseased area of the stomach. This is called a resection of the stomach.

Several distinct ways of classifying stomach cancer according to cell type have been proposed. The Lauren classification is encountered most frequently. According to this classification system, gastric adenocarcinomas are either called intestinal or diffuse. Intestinal cancers are much like a type of intestinal cancer called intestinal carcinoma. Intestinal tumors are more frequently found in males and in older patients. The prognosis for these tumors is better than that for diffuse tumors. Diffuse tumors are more likely to infiltrate, that is, to move into another organ of the body.



An excised section of a human stomach showing a cancerous tumor (center, triangular shape). (Custom Medical Stock Photo. Reproduced by permission.)

Because symptoms of stomach cancer are so mild, treatment often does not commence until the disease is well advanced. The three standard modes of treatment for stomach cancer include surgery, radiation therapy, and chemotherapy. While deciding on the patient's treatment plan, the doctor takes into account many factors. The location of the cancer and its stage are important considerations. In addition, the patient's age, general health status, and personal preferences are also taken into account.

Surgery

In the early stages of stomach cancer, surgery may be used to remove the cancer. Surgical removal of adenocarcinoma is the only treatment capable of eliminating the disease. Laparoscopy is often used before surgery to investigate whether or not the tumor can be removed surgically. If the cancer is widespread and

cannot be removed with surgery, an attempt will be made to remove blockage and control symptoms such as pain or bleeding. Depending on the location of the cancer, a portion of the stomach may be removed, a procedure called a partial **gastrectomy**. In a surgical procedure known as total gastrectomy, the entire stomach may be removed. However, doctors prefer to leave at least part of the stomach if possible. Patients who have been given a partial gastrectomy achieve a better quality of life than those having a total gastrectomy and typically lead normal lives. Even when the entire stomach is removed, the patients quickly adjust to a different eating schedule. This involves eating small quantities of food more frequently. High-protein foods are generally recommended.

Partial or total gastrectomy is often accompanied by other surgical procedures. Lymph nodes are frequently removed and nearby organs, or parts of these organs, may be removed if cancer has spread to them. Such organs may include the pancreas, colon, or spleen.

Preliminary studies suggest that patients who have tumors that cannot be removed by surgery at the start of therapy may become candidates for surgery later. Combinations of chemotherapy and radiation therapy are sometimes able to reduce disease for which surgery is not initially appropriate. Preliminary studies are being performed to determine if some of these patients can become candidates for surgical procedures after such therapies are applied.

Chemotherapy

Whether or not patients undergoing surgery for stomach cancer should receive chemotherapy is a controversial issue. Chemotherapy involves administering anti-cancer drugs either intravenously (through a vein in the arm) or orally (in the form of pills). This can either be used as the primary mode of treatment or after surgery to destroy any cancerous cells that may have migrated to distant sites. Most cancers of the gastrointestinal tract do not respond well to chemotherapy, however, adenocarcinoma of the stomach and advanced stages of cancer are exceptions.

Chemotherapy medicines such as doxorubicin, mitomycin C, and 5-fluorouracil, used alone, provide benefit to at least one in five patients. Combinations of agents may provide even more benefit, although it is not certain that this includes longer survival. For example, some doctors use what is called the FAM regimen, which combines 5-fluorouracil, doxorubicin, and mitomycin. Some doctors prefer using 5-fluorouracil alone to FAM since side effects are more moderate. Another combination some doctors are using involve high doses

of the medications methotrexate, 5- fluorouracil, and doxorubicin. Other combinations that have shown benefit include the ELF regimen, a combination of leucovorin, 5- fluorouracil, and etoposide. The EAP regimen, a combination of etoposide, doxorubicin, and cisplatin is also used.

Although chemotherapy using a single medicine is sometimes used, the best response rates are often achieved with combinations of medicines. Therefore, in addition to studies exploring the effectiveness of new medicines there are other studies attempting to evaluate how to best combine existing forms of chemotherapy to bring the greatest degree of help to patients.

Radiation therapy

Radiation therapy is often used after surgery to destroy the cancer cells that may not have been completely removed during surgery. To treat stomach cancer, external beam radiation therapy is generally used. In this procedure, high-energy rays from a machine that is outside of the body are concentrated on the area of the tumor. In the advanced stages of gastric cancer, radiation therapy is used to ease symptoms such as pain and bleeding. However, studies of radiation treatment for stomach cancer have shown that the way it has been used it has been ineffective for many patients.

Researchers are actively assessing the role of chemotherapy and radiation therapy used before a surgical procedure is conducted. They are searching for ways to use both chemotherapy and radiation therapy so that they increase the length of survival of patients more effectively than current methods are able to do.

Prognosis

Overall, approximately 20% of patients with stomach cancer live at least five years following diagnosis. Patients diagnosed with stomach cancer in its early stages have a far better prognosis than those for whom it is in the later stages. In the early stages, the tumor is small, lymph nodes are unaffected, and the cancer has not migrated to the lungs or the liver. Unfortunately, only about 20% of patients with stomach cancer are diagnosed before the cancer had spread to the lymph nodes or formed a distant metastasis.

It is important to remember that statistics on prognosis may be misleading. Newer therapies are being developed rapidly and five-year survival has not yet been measured with these. Also, the largest

KEY TERMS

Adenocarcinoma—Malignant cancers that originate in the tissues of glands or that form glandular structures.

Anemia—A condition in which iron levels in the blood are low.

Barium x ray (upper GI)—An x-ray test of the upper part of the gastrointestinal (GI) tract (including the esophagus, stomach, and a small portion of the small intestine) after the patient is given a white, chalky barium sulfate solution to drink. This substance coats the upper GI and the x rays reveal any abnormality in the lining of the stomach and the upper GI.

Biopsy—Removal of a tissue sample for examination under the microscope to check for cancer cells.

Chemotherapy—Treatment of cancer with synthetic drugs that destroy the tumor either by inhibiting the growth of the cancerous cells or by killing the cancer cells.

Endoscopic ultrasound (EUS)—A medical procedure in which sound waves are sent to the stomach wall by an ultrasound probe attached to the end of an endoscope. The pattern of echoes generated by the reflected sound waves are translated into an image of the stomach wall by a computer.

External radiation therapy—Radiation therapy that focuses high-energy rays from a machine on the area of the tumor.

Infiltrate—A tumor that moves into another organ of the body.

Polyp—An abnormal growth that develops on the inside of a hollow organ such as the colon, stomach, or nose.

Radiation therapy—Treatment using high-energy radiation from x-ray machines, cobalt, radium, or other sources.

Total gastrectomy—Surgical removal (excision) of the entire stomach.

Upper endoscopy—A medical procedure in which a thin, lighted, flexible tube (endoscope) is inserted down the patient's throat. Through this tube the doctor can view the lining of the esophagus, stomach, and the upper part of the small intestine.

group of people diagnosed with stomach cancer are between 60 and 70 years of age, suggesting that some of these patients die not from cancer but from other age-related diseases. As a result, some patients with stomach cancer may be expected to have longer survival than did patients just ten years ago.

Prevention

Avoiding many of the risk factors associated with stomach cancer may prevent its development. Excessive amounts of salted, smoked, and pickled foods should be avoided, as should foods high in nitrates. A diet that includes recommended amounts of fruits and vegetables is believed to lower the risk of several cancers, including stomach cancer. The American Cancer Society recommends eating at least five servings of fruits and vegetables daily and choosing six servings of food from other plant sources, such as grains, pasta, beans, cereals, and whole grain bread.

Abstaining from tobacco and excessive amounts of alcohol will reduce the risk for many cancers. In countries where stomach cancer is common, such as Japan, early detection is important for successful treatment.

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Stomach flu see **Gastroenteritis**

Stomach flushing

Definition

Stomach flushing is the repeated introduction of fluids into the stomach through a nasogastric tube, and their subsequent withdrawal by **nasogastric suction**.

Purpose

Stomach flushing is performed to aid in controlling gastrointestinal bleeding or to cleanse the stomach of poisons.

Controlling stomach bleeding

Bleeding from the esophagus due to ruptured veins or bleeding from the stomach due to ulcers is a medical emergency. In an attempt to stop the bleeding, the stomach is flushed with large quantities of body-temperature saline solution or ice water. This procedure is called stomach flushing or gastric lavage.

Stomach flushing to control bleeding is not uniformly accepted, and some experts believe it is of little benefit and exposes the patient to unnecessary risks. It is usually done in conjunction with the administration of drugs to constrict the blood vessels.

Stomach flushing to remove poisons

At one time, stomach flushing was common practice to remove certain poisons. Recent thinking by the American Academy of Clinical Toxicology is that stomach flushing should not be used routinely with poisoned patients. It is useful only if the patient has swallowed a life-threatening quantity of poison, and when the flushing can be done within 60 minutes of having swallowed the poison.

Precautions

In **poisoning** cases, stomach flushing should not be used if the poison is a strong corrosive acid (hydrochloric acid, sulfuric acid), alkali (lye, ammonia), or a volatile hydrocarbon such as gasoline. Stomach flushing should also not be done on patients who are having convulsions. Patients who are losing or have lost consciousness must have their airways intubated before a nasogastric tube is inserted.

Description

Stomach flushing is performed in a hospital emergency room or intensive care unit by an emergency

KEY TERMS

Electrolytes—Salts and minerals that ionize in body fluids. Common human electrolytes are sodium chloride, potassium, calcium, and sodium bicarbonate. Electrolytes control the fluid balance of the body and are important in muscle contraction, energy generation, and almost any major biochemical reaction in the body.

Saline—A salt water solution that mimics the concentration of electrolytes in the blood.

room physician or gastroenterologist. A nasogastric tube is inserted, and small amounts of saline or ice water are introduced into the stomach and withdrawn. The procedure is repeated until the withdrawn fluid is clear.

Preparation

Little preparation is necessary for this procedure other than educating the patient as to what will happen. The patient should remove dental appliances before the nasogastric tube is inserted.

Aftercare

After stomach flushing, the patient's vital signs will be monitored. Checks will be made for fluid and electrolyte imbalances. If necessary, additional treatment to prevent gastrointestinal bleeding or poisoning will be done.

Risks

In poisoning cases, stomach flushing delays the administration of **activated charcoal**, which may be more beneficial to treating the patient than flushing the stomach. In addition, stomach flushing may stimulate bleeding from the esophagus or stomach. The patient may inhale some of the stomach contents, causing aspiration, **pneumonia**, or infection in the lungs. Fluid and electrolyte imbalances are more likely to occur in older, sicker patients. Mechanical damage to the throat is more likely in patients who are uncooperative.

Normal results

Stomach flushing is usually tolerated by patients and is a temporary treatment, performed in conjunction with other therapies.

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Tish Davidson, A.M.

Stomach removal see **Gastrectomy**

Stomach resection see **Gastrectomy**

Stomatitis

Definition

Inflammation of the mucous lining of any of the structures in the mouth, which may involve the cheeks, gums, tongue, lips, and roof or floor of the mouth. The word "stomatitis" literally means inflammation of the mouth. The inflammation can be caused by conditions in the mouth itself, such as poor **oral hygiene**, poorly fitted dentures, or from mouth **burns** from hot food or drinks, or by conditions that affect the entire body, such as medications, allergic reactions, or infections.

Description

Stomatitis is an inflammation of the lining of any of the soft-tissue structures of the mouth. Stomatitis is usually a painful condition, associated with redness, swelling, and occasional bleeding from the affected area. **Bad breath** (halitosis) may also accompany the condition. Stomatitis affects all age groups, from the infant to the elderly.

Causes and symptoms

A number of factors can cause stomatitis; it is a fairly common problem in the general adult population in North America. Poorly fitted oral appliances, cheek biting, or jagged teeth can persistently irritate the oral structures. Chronic mouth breathing due to plugged nasal airways can cause dryness of the mouth tissues, which in turn leads to irritation. Drinking beverages that are too hot can burn the mouth, leading to irritation and **pain**. Diseases, such as herpetic infections (the **common cold sore**), **gonorrhea**, **measles**, leukemia, **AIDS**, and lack of vitamin C can present with oral signs. Other systemic diseases associated with stomatitis include inflammatory bowel disease (IBD)

and Behçet's syndrome, an inflammatory multisystem disorder of unknown cause.

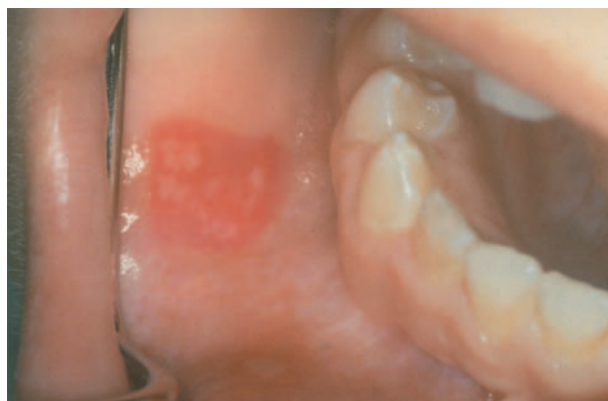
Aphthous stomatitis, also known as recurrent aphthous ulcers (RAU) or **canker sores**, is a specific type of stomatitis that presents with shallow, painful ulcers that are usually located on the lips, cheeks, gums, or roof or floor of the mouth. These ulcers can range from pinpoint size to up to 1 in (2.5 cm) or more in diameter. Though the causes of canker sores are unknown, nutritional deficiencies, especially of vitamin B₁₂, folate, or iron is suspected. Generalized or contact stomatitis can result from excessive use of alcohol, spices, hot food, or tobacco products. Sensitivity to mouthwashes, toothpastes, and lipstick can irritate the lining of the mouth. Exposure to heavy metals, such as mercury, lead, or bismuth can cause stomatitis. Thrush, a fungal infection, is a type of stomatitis.

Diagnosis

Diagnosis of stomatitis can be difficult. A patient's history may disclose a dietary deficiency, a systemic disease, or contact with materials causing an allergic reaction. A **physical examination** is done to evaluate the oral lesions and other skin problems. Blood tests may be done to determine if any infection is present. Scrapings of the lining of the mouth may be sent to the laboratory for microscopic evaluation, or cultures of the mouth may be done to determine if an infectious agent may be the cause of the problem.

Treatment

The treatment of stomatitis is based on the problem causing it. Local cleansing and good oral hygiene are fundamental. Sharp-edged foods such as peanuts, tacos, and potato chips should be avoided. A soft-bristled toothbrush should be used, and the teeth and gums should be brushed carefully; the patient should avoid banging the toothbrush into the gums. Local factors, such as ill-fitting dental appliances or sharp teeth, can be corrected by a dentist. An infectious cause can usually be treated with medication. Systemic problems, such as AIDS, leukemia, and anemia are treated by the appropriate medical specialist. Minor mouth burns from hot beverages or hot foods will usually resolve on their own in a week or so. Chronic problems with aphthous stomatitis are treated by first correcting any vitamin B₁₂, iron, or folate deficiencies. If those therapies are unsuccessful, medication can be prescribed which can be applied to each aphthous ulcer with a cotton-tipped applicator. This therapy is successful with a limited number of



This patient is afflicted with stomatitis, a common inflammatory disease of the mouth. (Photograph by Edward H. Gill, Custom Medical Stock Photo. Reproduced by permission.)

patients. More recently, low-power treatment with a carbon dioxide laser has been found to relieve the discomfort of recurrent aphthae. Major outbreaks of aphthous stomatitis can be treated with tetracycline **antibiotics** or **corticosteroids**. Valacyclovir has been shown to be effective in treating stomatitis caused by herpesviruses.

Patients may also be given topical anesthetics (usually a 2% lidocaine gel) to relieve pain and a protective paste (Orabase) or a coating agent like Kaopectate to protect eroded areas from further irritation from dentures, braces, or teeth.

Alternative treatment

Alternate treatment of stomatitis mainly involves prevention of the problem. Patients with such dental appliances as dentures should visit their dentist on a regular basis. Patients with systemic diseases or chronic medical problems need to ask their health care provider what types of oral problems they can expect from their particular disease. These patients must also contact their medical clinic at the first sign of problems. Common sense needs to be exercised when consuming hot foods or drinks. Use of tobacco products should be discouraged. Alcohol should be used in moderation. Mouthwashes and toothpastes known to the patient to cause problems should be avoided.

Botanical medicine can assist in resolving stomatitis. One herb, calendula (*Calendula officinalis*), in tincture form (an alcohol-based herbal extract) and diluted for a mouth rinse, can be quite effective in treating aphthous stomatitis and other manifestations of stomatitis.

KEY TERMS

Aphthous stomatitis—A specific type of stomatitis presenting with shallow, painful ulcers. Also known as *canker sores*.

Stomatitis—Inflammation of the lining of the mouth, gums, or tongue.

Thrush—A form of stomatitis caused by *Candida* fungi and characterized by cream-colored or bluish patches on the tongue, mouth, or pharynx.

More recently, a group of researchers in Brazil have reported that an extract made from the leaves of *Trichilia glabra*, a plant found in South America, is effective in killing several viruses that cause stomatitis.

Prognosis

The prognosis for the resolution of stomatitis is based on the cause of the problem. Many local factors can be modified, treated, or avoided. Infectious causes of stomatitis can usually be managed with medication, or, if the problem is being caused by a certain drug, by changing the offending agent.

Prevention

Stomatitis caused by local irritants can be prevented by good oral hygiene, regular dental checkups, and good dietary habits. Problems with stomatitis caused by systemic disease can be minimized by good oral hygiene and closely following the medical therapy prescribed by the patient's health care provider.

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American Medical Association. 515 N. State St., Chicago, IL 60612. (312) 464-5000. < <http://www.ama-assn.org> >.

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Stone removal see **Gallstone removal**

Stool culture

Definition

Stool culture is a test to identify bacteria in patients with a suspected infection of the digestive tract. A sample of the patient's feces is placed in a special medium where bacteria is then grown. The bacteria that grow in the culture are identified using a microscope and biochemical tests.

Purpose

Stool culture is performed to identify bacteria or other organisms in persons with symptoms of gastrointestinal infection, most commonly **diarrhea**. Identification of the organism is necessary to determine the treatment of the patient's infection or to trace the cause of an outbreak or epidemic of certain types of diarrhea.

According to the Centers for Disease Control and Prevention (CDC), doctors are most likely to order a stool culture for patients with any of the following characteristics: **AIDS**, bloody stools, diarrhea lasting

longer than three days, high **fever**, history of recent travel abroad, or severe **dehydration**.

Precautions

Stool culture is performed only if an infection of the digestive tract is suspected. The test has no harmful effects.

Description

Stool culture may also be called fecal culture. To obtain a specimen for culture, the patient is asked to collect a stool sample into a special sterile container. In some cases, the container may contain a transport solution. Specimens may need to be collected on three consecutive days. It is important to return the specimen to the doctor's office or the laboratory in the time specified by the physician or nurse. Laboratories do not accept stool specimens contaminated with water, urine, or other materials.

The culture test involves placing a sample of the stool on a special substance, called a medium, that provides nutrients for certain organisms to grow and reproduce. The medium is usually a thick gel-like substance. The culture is done in a test tube—or on a flat round culture plate—which is incubated at the proper temperature for growth of the bacteria. After a colony of bacteria grows in the medium, the type of bacteria is identified by observing the colony's growth, its physical characteristics, and its microscopic features. The bacteria may be dyed with special stains that make it easier to identify features specific to particular bacteria.

The length of time needed to perform a stool culture depends on the laboratory where it is done and the culture methods used. Stool culture usually takes 72 hours or longer to complete. Some organisms may take several weeks to grow in a culture.

An antibiotic sensitivity test may be done after a specific bacterium is identified. This test shows which **antibiotics** will be most effective for treating the infection.

Although most intestinal infections are caused by bacteria, in some cases a fungal or viral culture may be necessary. The most common bacterial infections of the digestive tract are caused by *Shigella*, *Salmonella*, *Campylobacter*, and *Yersinia*. Patients taking certain antibiotics may be susceptible to infection with *Clostridium difficile*. In some cases, as with *Clostridium difficile*, the stool culture is used to detect the toxin (poison or harmful chemical) produced by the bacteria.

Patients with AIDS, or other immune system diseases, may also have gastrointestinal infections caused by such fungi as *Candida*, or viral organisms including cytomegalovirus.

Several intestinal parasites may cause gastrointestinal infection and diarrhea. Parasites are not cultured, but are identified microscopically in a test called "Stool Ova and Parasites."

Insurance coverage for stool culture may vary among different insurance plans. This common test usually is covered if ordered by a physician approved by the patient's insurance plan, and if it is done at an approved laboratory.

Alternative methods

Newer methods of testing stool samples for specific disease organisms include various forms of polymerase chain reaction (PCR) assays. One type that has been used to test for several different types of intestinal viruses at the same time is the RT-PCR, which stands for reverse transcriptase polymerase chain reaction. This assay measures changes in an organism's messenger RNA. RT-PCR assays have several advantages over standard stool cultures: they require only very small samples of material; they can be performed much more rapidly; and they can be used to test environmental water for virus contamination as well as human stool samples.

Preparation

The physician or other healthcare provider will ask the patient for a complete medical history and perform a **physical examination** to determine possible causes of the gastrointestinal problem. Information about the patient's diet, any medications taken, and recent travel may provide clues to the identity of possible infectious organisms.

Stool culture normally doesn't require any special preparation. Patients do not need to change their diet before collecting the specimen. Intake of some substances can contaminate the stool specimen and should not be taken the day before collection. These substances include castor oil, bismuth, and laxative preparations containing psyllium hydrophilic mucilloid.

Normal results

Bacteria that are normally found in the intestines include *Pseudomonas* and *Escherichia coli*. These enteric bacteria (bacteria of the gastrointestinal

KEY TERMS

Bismuth—A substance used in medicines to treat diarrhea, nausea, and indigestion.

Enteric—Pertaining to the intestine.

Enterotoxigenic—Refers to an organism that produces toxins in the gastrointestinal tract that cause such things as vomiting, diarrhea, and other symptoms of food poisoning.

Feces—Material excreted by the intestines.

Flora—Refers to normal bacteria found in a healthy person.

Gastrointestinal—Referring to the digestive tract; the stomach and intestines.

Psyllium hydrophilic mucilloid—A plant material contained in some laxatives.

Sterile—Free of microorganisms.

Toxin—A poison; usually refers to protein poisons produced by bacteria, animals, and some plants.

system) are considered normal flora and usually do not cause infection in the digestive tract.

Abnormal results

Bacteria that do not normally inhabit the digestive tract, and that are known to cause gastrointestinal infection include *Shigella*, *Salmonella*, *Campylobacter*, and *Yersinia*. *Clostridium difficile* produces a toxin that can cause severe diarrhea. Other bacteria that produce toxins are *Staphylococcus aureus*, *Bacillus cereus*, and enterotoxigenic (producing disease in the digestive system) *Escherichia coli*. Although *Escherichia coli* is a normal bacteria found in the intestines, the enterotoxigenic type of this bacteria can be acquired from eating contaminated meat, juice, or fruits. It produces a toxin that causes severe inflammation and bleeding of the colon.

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ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Toni Rizzo
Rebecca J. Frey, PhD

Stool fat test

Definition

Stool fats, also known as fecal fats, or fecal lipids, are fats that are excreted in the feces. When secretions from the pancreas and liver are adequate, emulsified dietary fats are almost completely absorbed in the small intestine. When a malabsorption disorder or other cause disrupts this process, excretion of fat in the stool increases.

Purpose

This test evaluates digestion of fats by determining excessive excretion of lipids in patients exhibiting signs of malabsorption, such as weight loss, abdominal distention, and scaly skin.

Precautions

Drugs that may increase fecal fat levels include **enemas** and **laxatives**, especially mineral oil. Drugs that

may decrease fecal fat include Metamucil and barium. Other substances that can affect test results include alcohol, potassium chloride, calcium carbonate, neomycin, kanamycin, and other broad-spectrum **antibiotics**.

Description

Excessive excretion of fecal fat is called steatorrhea, a condition that is suspected when the patient has large, “greasy,” and foul-smelling stools. Both digestive and absorptive disorders can cause steatorrhea. Digestive disorders affect the production and release of the enzyme lipase from the pancreas, or bile from the liver, which are substances that aid digestion of fats; absorptive disorders disturb the absorptive and enzyme functions of the intestine. Any condition that causes malabsorption or maldigestion is also associated with increased fecal fat. As an example, children with **cystic fibrosis** have mucous plugs that block the pancreatic ducts. The absence or significant decrease of the pancreatic enzymes, amylase, lipase, trypsin, and chymotrypsin limits fat protein and carbohydrate digestion, resulting in steatorrhea due to fat malabsorption.

Both qualitative and quantitative tests are used to identify excessive fecal fat. The qualitative test involves staining a specimen of stool with a special dye, then examining it microscopically for evidence of malabsorption, such as undigested muscle fiber and various fats. The quantitative test involves drying and weighing a 72-hour stool specimen, then using an extraction technique to separate the fats, which are subsequently evaporated and weighed. This measurement of the total output of fecal fat per 24 hours in a three-day specimen is the most reliable test for steatorrhea.

Preparation

This test requires a 72-hour stool collection. The patient should abstain from alcohol during this time and maintain a high-fat diet (100 g/day) for three days before the test, and during the collection period. The patient should call the laboratory for instructions on how to collect the specimen.

Normal results

Reference values vary from laboratory to laboratory, but are generally found within the range of 5–7 g/24 hr.

It should be noted that children, especially infants, cannot ingest the 100 g/day of fat that is suggested for the test. Therefore, a fat retention coefficient is

determined by measuring the difference between ingested fat and fecal fat, and expressing that difference as a percentage. The figure, called the fat retention coefficient, is 95% or greater in healthy children and adults. A low value is indicative of steatorrhea.

Abnormal results

Increased fecal fat levels are found in cystic fibrosis, malabsorption secondary to other conditions like Whipple’s disease or **Crohn’s disease**, maldigestion secondary to pancreatic or bile duct obstruction, and “short-gut” syndrome secondary to surgical resection, bypass, or congenital anomaly.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby’s Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Stool O & P test

Definition

The stool O & P test is the stool ova and parasites test. In this test, a stool sample is examined for the presence of intestinal parasites and their eggs, which are called ova.

Purpose

The ova and parasites test is performed to look for and identify intestinal parasites and their eggs in persons with symptoms of gastrointestinal infection. Patients may have no symptoms, or experience **diarrhea**, blood in the stools, and other gastrointestinal distress. Identification of a particular parasite indicates the cause of the patient’s disease and determines the medication needed to treat it.

Precautions

Stool O & P is performed if an infection of the digestive tract is suspected. The test has no harmful effects.

Description

Examination of the stool for ova and parasites is done to diagnose parasitic infection of the intestines.

The test may be done in the doctor's office or a laboratory. The patient collects a stool sample in one or more sterile containers containing special chemical fixatives. The feces should be collected directly into the container. It must not be contaminated with urine, water, or other materials. Three specimens are often needed—collected every other day, or every third day. However, as many as six specimens may be needed to diagnose the amoeba *Entamoeba histolytica*. The specimen does not need to be refrigerated. It should be delivered to the doctor's office or laboratory within 12 hours.

In the laboratory, the stool sample is observed for signs of parasites and their eggs. Some parasites are large enough to be seen without a microscope. For others, microscope slides are prepared with fresh unstained stool, and with stool dyed with special stains. These preparations are observed with a microscope for the presence of parasites or their eggs.

An unstained stool examination for ova and parasites normally only takes a few minutes. If specimen staining and other preparation is done, the test may take longer. When the specimen is sent to a laboratory, the results may take eight to 24 hours to be reported.

The most common intestinal parasites in North America that cause infections are:

- roundworms: *Ascaris lumbricoides*
- hookworms: *Necator americanus*
- pinworms: *Enterobius follicularis*
- tapeworms: *Diphyllobothrium latum*, *Taenia saginata*, and *Taenia solium*
- protozoa: *Entamoeba histolytica* (an amoeba), and *Giardia lamblia* (a flagellate)

Numerous other parasites are found in other parts of the world. These may be contracted by travelers to other countries. Patients with acquired immune deficiency syndrome (AIDS) or other immune system disorders are commonly infected with the parasites in the *Microsporidia* family, *Cryptosporidium*, and *Isospora belli*.

Insurance coverage for stool ova and parasites may vary among different insurance plans. This test usually is covered if ordered by a physician approved by the patient's insurance plan, and if it is done at an approved laboratory.

Preparation

The physician, or other healthcare provider, will ask the patient for a complete medical history, and perform a **physical examination** to determine possible

KEY TERMS

Amoeba—A type of protozoa (one-celled animal) that can move or change its shape by extending projections of its cytoplasm.

Bismuth—A substance used in medicines to treat diarrhea, nausea, and indigestion.

Cryptosporidium—A type of parasitic protozoa.

Feces—Material excreted by the intestines.

Flagellate—A microorganism that uses flagella (hair-like projections) to move.

Gastrointestinal—Referring to the digestive tract; the stomach and intestines.

Isospora belli—A type of parasitic protozoa.

Microsporidia—A type of parasitic protozoa.

Ova—Eggs.

Parasite—An organism that lives on or inside another living organism (host), causing damage to the host.

Pathogenic—Disease-causing.

Protozoa—One-celled eukaryotic organisms belonging to the kingdom Protista.

Sterile—Free of microorganisms.

causes of the gastrointestinal symptoms. Information about the patient's diet, any medications taken, and recent travel may provide clues to the identity of possible infectious parasites.

Collecting a stool sample for ova and parasite detection normally doesn't require any special preparation. Patients do not need to change their diet before collecting the specimen. Patients should avoid taking any medications or treatments containing mineral oil, castor oil, or bismuth, magnesium or other antidiarrheal medicines, or **antibiotics** for seven to 10 days before collecting the specimen.

Normal results

Normally, parasites and eggs should not be found in stools. Some parasites are not pathogenic, which means they do not cause disease. If these are found, no treatment is necessary.

Abnormal results

The presence of any pathogenic parasite indicates an intestinal parasitic infection. Depending on the

parasite identified, other tests may need to be performed to determine if the parasite has invaded other parts of the body. Some parasites travel from the intestines to other parts of the body and may already have caused damage to other tissues by the time a diagnosis is made. For example, the roundworm, *Ascaris* penetrates the intestinal wall and can cause inflammation in the abdomen. It can also migrate to the lungs and cause **pneumonia**. This kind of injury can occur weeks before the roundworm eggs show up in the stool.

Other types of damage caused by intestinal parasites include anemia due to hemorrhage caused by hookworms, and anemia caused by depletion of vitamin B₁₂ through the action of tapeworms.

When a parasite is identified, the patient can be treated with the appropriate medications to eliminate the parasite.

Resources

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Toni Rizzo

Stool occult blood test see **Fecal occult blood test**

Stool ova and parasites test see **Stool O & P test**

Strabismus

Definition

Strabismus is a condition in which the eyes do not point in the same direction. It can also be referred to as a tropia or squint.

Description

Strabismus occurs in 2–5% of all children. About half are born with the condition, which causes one or both eyes to turn:

- inward (esotropia or “crossed eyes”)
- outward (exotropia or “wall eyes”)
- upward (hypertropia)
- downward (hypotropia)

Strabismus is equally common in boys and girls. It sometimes runs in families.

Types of strabismus

Esotropia is the most common type of strabismus in infants. Accommodative esotropia develops in children under age two who cross their eyes when focusing on objects nearby. This usually occurs in children who are moderately to highly farsighted (hyperopic).

Another common form of strabismus, exotropia, may only be noticeable when a child looks at far-away objects, daydreams, or is tired or sick.

Sometimes the eye turn is always in the same eye; however sometimes the turn alternates from one eye to the other’.

Most children with strabismus have comitant strabismus. No matter where they look, the degree of deviation does not change. In incomitant strabismus, the amount of misalignment depends upon which direction the eyes are pointed.

False strabismus (pseudostabismus)

A child may appear to have a turned eye, however this appearance may actually be due to:

- extra skin that covers the inner corner of the eye
- a broad, flat nose
- eyes set unusually close together or far apart

This condition, false strabismus, usually disappears as the child’s face grows. An eye doctor needs to determine whether the eyeturn is true or pseudostabismus.

With normal vision, both eyes send the brain the same message. This binocular fixation (both eyes looking directly at the same object) is necessary to see three-dimensionally and to aid in depth perception. When an eye is misaligned, the brain receives two different images. Young children learn to ignore distorted messages from a misaligned eye, but adults with strabismus often develop double vision (diplopia).

A baby’s eyes should be straight and parallel by three or four months of age. A child who develops strabismus after the age of eight or nine years is said to have adult-onset strabismus.

Causes and symptoms

Strabismus can be caused by a defect in muscles or the part of the brain that controls eye movement. It is especially common in children who have:

- brain tumors
- cerebral palsy
- Down syndrome
- hydrocephalus
- other disorders that affect the brain

Diseases that cause partial or total blindness can cause strabismus. So can extreme farsightedness, **cataracts**, eye injury, or having much better vision in one eye than the other.

In adults, strabismus is usually caused by:

- diabetes
- head trauma
- stroke
- brain tumor
- other diseases affecting nerves that control eye muscles

The most obvious symptom of strabismus is an eye that isn't always straight. The deviation can vary from day to day or during the day. People who have strabismus often squint in bright sunlight or tilt their heads to focus their eyes.

Diagnosis

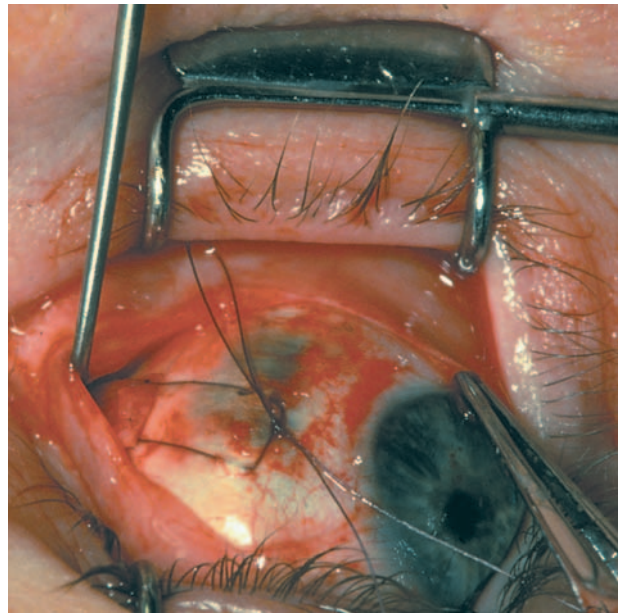
Every baby's eyes should be examined by the age of six months. A baby whose eyes have not straightened by the age of four months should be examined to rule out serious disease.

A pediatrician, family doctor, ophthalmologist, or optometrist licensed to use diagnostic drugs uses drops that dilate the pupils and temporarily paralyze eye-focusing muscles to evaluate visual status and ocular health. Early diagnosis is important. Some eye turns may be a result of a tumor. Untreated strabismus can damage vision in the unused eye and possibly result in lazy eye (**amblyopia**).

Treatment

Preserving or restoring vision and improving appearance may involve one or more of the following:

- glasses to aid in focusing and straighten the eye(s)
- patching to force infants and young children to use and straighten the weaker eye
- eye drops or ointments as a substitute for patching or glasses, or to make glasses more effective
- surgery to tighten, relax, or reposition eye muscles



A close-up of ophthalmic surgery being performed to correct strabismus. (Photograph by Michael English, M.D. Custom Medical Stock Photo. Reproduced by permission.)

- medication injected into an overactive eye muscle to allow the opposite muscle to straighten the eye
- vision training (also called eye exercises)

Prognosis

Early consistent treatment usually improves vision and appearance. The most satisfactory results are achieved if the condition is corrected before the age of seven years old.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424. <<http://www.eyenet.org>>.

American Academy of Pediatric Ophthalmology and Strabismus (AAPOS). <<http://med-aapos.bu.edu>>.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

Maureen Haggerty

Strawberry marks see **Birthmarks**

Strengthening exercises see **Exercise**

Strep culture see **Throat culture**

Strep test see **Streptococcal antibody tests**

Strep throat

Definition

Streptococcal **sore throat**, or strep throat as it is more commonly called, is an infection of the mucous membranes lining the pharynx. Sometimes the tonsils are also infected (**tonsillitis**). The disease is caused by group A *Streptococcus* bacteria. Untreated strep throat may develop into **rheumatic fever** or other serious conditions.

Description

Strep throat accounts for between 5–10% of all sore throats. Although anyone can get strep throat, it is most common in school-age children. People who smoke, who are fatigued, run down, or who live in damp, crowded conditions are also more likely to become infected. Children under age two and adults who are not around children are less likely to get the disease.

Strep throat occurs most frequently from November to April. The disease passes directly from person to person by coughing, sneezing, and close contact. Very occasionally the disease is passed through food, when a food handler infected with strep throat accidentally contaminates food by coughing or sneezing. Statistically, if someone in the household is infected, one out of every four other household members may get strep throat within two to seven days.

Causes and symptoms

A person with strep throat suddenly develops a painful sore throat one to five days after being exposed to the streptococcus bacteria. The **pain** is indistinguishable from sore throats caused by other diseases.

The infected person usually feels tired and has a **fever**, sometimes accompanied by chills, **headache**, muscle aches, swollen lymph glands, and **nausea**. Young children may complain of abdominal pain. The tonsils look swollen and are bright red, with white or yellow patches of pus on them. Sometimes the roof of the mouth is red or has small red spots. Often a person with strep throat has **bad breath**.

Despite these common symptoms, strep throat can be deceptive. It is possible to have the disease and not show any of these symptoms. Many young children complain only of a headache and stomachache, without the characteristic sore throat.

Occasionally, within a few days of developing the sore throat, an individual may develop a fine, rough, sunburn-like rash over the face and upper body, and

have a fever of 101–104°F (38.3–40°C). The tongue becomes bright red, with a flecked, strawberry-like appearance. When a rash develops, this form of strep throat is called **scarlet fever**. The rash is a reaction to toxins released by the streptococcus bacteria. Scarlet fever is no more dangerous than strep throat, and is treated the same way. The rash disappears in about five days. One to three weeks later, patches of skin may peel off, as might occur with a **sunburn**, especially on the fingers and toes.

Untreated strep throat can cause rheumatic fever. This is a serious illness, although it occurs rarely. The most recent outbreak appeared in the United States in the mid-1980s. Rheumatic fever occurs most often in children between the ages of five and 15, and may have a genetic component, since it seems to run in families. Although the strep throat that causes rheumatic fever is contagious, rheumatic fever itself is not.

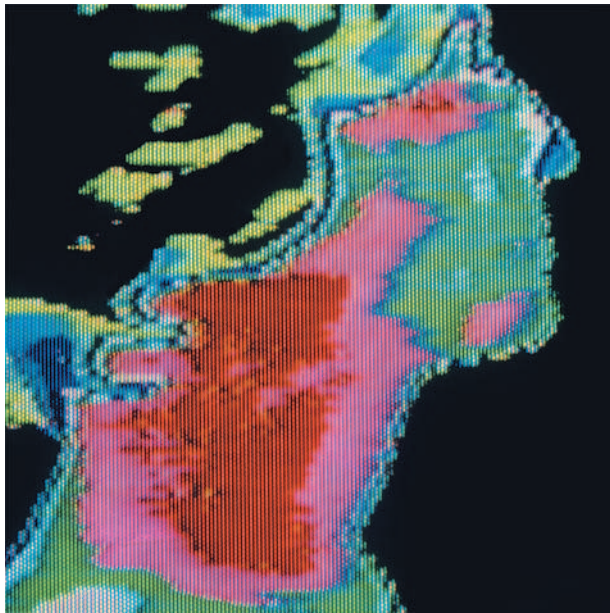
Rheumatic fever begins one to six weeks after an untreated streptococcal infection. The joints, especially the wrists, elbows, knees, and ankles become red, sore, and swollen. The infected person develops a high fever, and possibly a rapid heartbeat when lying down, paleness, **shortness of breath**, and fluid retention. A red rash over the trunk may come and go for weeks or months. An acute attack of rheumatic fever lasts about three months.

Rheumatic fever can cause permanent damage to the heart and heart valves. It can be prevented by promptly treating **streptococcal infections** with **antibiotics**. It does not occur if all the streptococcus bacteria are killed within the first 10–12 days after infection.

In the 1990s, outbreaks of a virulent strain of group A *Streptococcus* were reported to cause a toxic-shock-like illness and a severe invasive infection called necrotizing fasciitis, which destroys skin and muscle tissue. Although these diseases are caused by group A *Streptococci*, they rarely begin with strep throat. Usually the streptococcus bacteria enters the body through a skin wound. These complications are rare. However, since the **death** rate in necrotizing fasciitis is 30–50%, it is wise to seek prompt treatment for any streptococcal infection.

Diagnosis

Diagnosis of a strep throat by a doctor begins with a **physical examination** of the throat and chest. The doctor will also look for signs of other illness, such as a sinus infection or **bronchitis**, and seek information about whether the patient has been around other people with strep throat. If it appears that the patient may have strep throat, the doctor will do laboratory tests.



A thermographic image showing a streptococcal sore throat, or strep throat. (Photograph by Howard Sochurek, *The Stock Market*. Reproduced by permission.)

There are two types of tests to determine if a person has strep throat. A rapid strep test can only determine the presence of streptococcal bacteria, but will not tell if the sore throat is caused by another kind of bacteria. The results are available in about 20 minutes. The advantage of this test is the speed with which a diagnosis can be made.

The rapid strep test has a false negative rate of about 20%. In other words, in about 20% of cases where no strep is detected by the rapid strep test, the patient actually does have strep throat. Because of this, when a rapid strep test is negative, the doctor often does a **throat culture**.

For a rapid strep test or a throat culture, a nurse will use a sterile swab to reach down into the throat and obtain a sample of material from the sore area. The procedure takes only a few seconds, but may cause gagging.

For a throat culture a sample of swabbed material is cultured, or grown, in the laboratory on a medium that allows technicians to determine what kind of bacteria are present. Results take 24–48 hours. The test is very accurate and will show the presence of other kinds of bacteria besides *Streptococci*. It is important not to take any leftover antibiotics before visiting the doctor and having a throat culture. Even small amounts of antibiotics can suppress the bacteria and mask its presence in the throat culture.

In the event that rheumatic fever is suspected, the doctor does a blood test. This test, called an antistreptolysin-O test, will tell the doctor whether the person has recently been infected with strep bacteria. This helps the doctor distinguish between rheumatic fever and **rheumatoid arthritis**.

Treatment

Strep throat is treated with antibiotics. Penicillin is the preferred medication. Oral penicillin must be taken for 10-days. Patients need to take the entire amount of antibiotic prescribed and not discontinue taking the medication when they feel better. Stopping the antibiotic early can lead to a return of the strep infection. Occasionally, a single injection of long-acting penicillin (Bicillin) is given instead of 10 days of oral treatment.

About 10% of the time, penicillin is not effective against the strep bacteria. When this happens a doctor may prescribe other antibiotics such as amoxicillin (Amoxil, Pentamox, Sumox, Trimox), clindamycin (Cleocin), or a cephalosporin (Keflex, Durocef, Ceclor). Erythromycin (Eryzole, Pediazole, Ilosone), another inexpensive antibiotic, is given to people who are allergic to penicillin. Scarlet fever is treated with the same antibiotics as strep throat.

Without treatment, the symptoms of strep throat begin subsiding in four or five days. However, because of the possibility of getting rheumatic fever, it is important to treat strep throat promptly with antibiotics. If rheumatic fever does occur, it is also treated with antibiotics. Anti-inflammatory drugs, such as steroids, are used to treat joint swelling. **Diuretics** are used to reduce water retention. Once the rheumatic fever becomes inactive, children may continue on low doses of antibiotics to prevent a reoccurrence. Necrotizing fasciitis is treated with intravenous antibiotics.

Home care for strep throat

There are home care steps that people can take to ease the discomfort of their strep symptoms.

- Take **acetaminophen** or ibuprofen for pain. **Aspirin** should not be given to children because of its association with an increase in **Reye's Syndrome**, a serious disease.
- Gargle with warm double strength tea or warm salt water, made by adding one teaspoon of salt to eight ounces of water, to relieve sore throat pain.
- Drink plenty of fluids, but avoid acidic juices like orange juice because they irritate the throat.

- Eat soft, nutritious foods like noodle soup. Avoid spicy foods.
- Avoid smoke and smoking.
- Rest until the fever is gone, then resume strenuous activities gradually.
- Use a room humidifier, as it may make sore throat sufferers more comfortable.
- Be aware that antiseptic lozenges and sprays may aggravate the sore throat rather than improve it.

Alternative treatment

Alternative treatment focuses on easing the symptoms of strep throat through herbs and botanical medicines. Some practitioners suggest using these treatments in addition to antibiotics, since they primarily address the comfort of the patient and not the underlying infection. Many practitioners recommend *Lactobacillus acidophilus* to offset the suppressive effects of antibiotics on the beneficial bacteria of the intestines.

Some suggested treatments include:

- Inhaling fragrances of the essential oils of lavender (*Lavandula officinalis*), thyme (*Thymus vulgaris*), eucalyptus (*Eucalyptus globulus*), sage (*Salvia officinalis*), and sandalwood (Aromatherapy).
- Gargling with a mixture of water, salt, and tumeric (*Curcuma longa*) powder or astringents, such as alum, sumac, sage, and bayberry (Ayurvedic medicine).
- Taking osha root (*Ligusticum porteri*) internally for infection or drinking tea made of sage, **echinacea** (*Echinacea* spp.) and cleavers (*Gallium aparine*) Osha root has an unpleasant taste many children will not accept (Botanical medicine).

Prognosis

Patients with strep throat begin feeling better about 24 hours after starting antibiotics. Symptoms rarely last longer than five days.

People remain contagious until after they have been taking antibiotics for 24 hours. Children should not return to school or childcare until they are no longer contagious. Food handlers should not work for the first 24 hours after antibiotic treatment, because strep infections are occasionally passed through contaminated food. People who are not treated with antibiotics can continue to spread strep bacteria for several months.

KEY TERMS

Lactobacillus acidophilus—A bacteria found in yogurt that changes the balance of the bacteria in the intestine in a beneficial way.

About 10% of strep throat cases do not respond to penicillin. People who have even a mild sore throat after a 10 day treatment with antibiotic should return to their doctor. An explanation for this may be that the person is just a carrier of strep, and that something else is causing the sore throat.

Taking antibiotics within the first week of a strep infection will prevent rheumatic fever and other complications. If rheumatic fever does occur, the outcomes vary considerably. Some cases may be cured. In others there may be permanent damage to the heart and heart valves. In rare cases, rheumatic fever can be fatal.

Necrotizing fasciitis has a death rate of 30–50%. Patients who survive often suffer a great deal of tissue and muscle loss. Fortunately, this complication of a streptococcus infection is very rare.

Prevention

There is no way to prevent getting a strep throat. However, the risk of getting one or passing one on to another person can be minimized by:

- washing hands well and frequently, especially after nose blowing or sneezing and before food handling
- disposing of used tissues properly
- avoiding close contact with someone who has a strep throat
- not sharing food and eating utensils with anyone
- not smoking

Resources

OTHER

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Tish Davidson, A.M.

Streptobacillary rat-bite fever see **Rat-bite fever**

Streptococcal antibody tests

Definition

Streptococcal infections are caused by a micro-organism called *Streptococcus*. Three streptococcal antibody tests are available: the antistreptolysin O titer (ASO), the antideoxyribonuclease-B titer (anti-DNase-B, or ADB), and the streptozyme test.

Purpose

The antistreptolysin O titer, or ASO, is ordered primarily to determine whether a previous group A *Streptococcus* infection has caused a poststreptococcal disease, such as **scarlet fever**, **rheumatic fever**, or a **kidney disease** called **glomerulonephritis**.

The anti-DNase-B (ADB) test is performed to determine a previous infection of a specific type of *Streptococcus*, group A beta-hemolytic *Streptococcus*. Identification of infections of this type are particularly important in suspected cases of acute rheumatic **fever** (ARF) or acute glomerulonephritis.

Streptozyme is a screening test used to detect antibodies to several streptococcal antigens. An antigen is a substance that can trigger an immune response, resulting in production of an antibody as part of the body's defense against infection and disease.

Precautions

For the ASO test, increased levels of fats, called beta lipoproteins, in the blood can neutralize streptolysin O and cause a false-positive ASO titer. **Antibiotics**, which reduce the number of streptococci and thereby suppress ASO production, may decrease ASO levels. Steroids, which suppress the immune system, consequently may also suppress ASO production. Also Group A streptococcal infections of the skin may not produce an ASO response. Antibiotics also may decrease anti-DNase-B (ADB) levels.

Description

Streptococcal infections are caused by bacteria known as *Streptococcus*. There are several disease-causing strains of streptococci (groups A, B, C, D, and G), which are identified by their behavior, chemistry, and appearance. Each group causes specific types of infections and symptoms. These antibody tests are useful for group A streptococci. Group A streptococci are the most virulent species for humans and are the cause of **strep throat**, **tonsillitis**, wound and skin

infections, blood infections (septicemia), scarlet fever, **pneumonia**, rheumatic fever, **Sydenham's chorea** (formerly called St. Vitus' dance), and glomerulonephritis.

Although symptoms may suggest a streptococcal infection, the diagnosis must be confirmed by tests. The best procedure, and one that is used for an acute infection, is to take a sample from the infected area for culture, a means of growing bacteria artificially in the laboratory. However, cultures are useless about two to three weeks after initial infection, so the ASO, anti-DNase-B, and streptozyme tests are used to determine if a streptococcal infection is present.

Antistreptolysin O titer (ASO)

The ASO titer is used to demonstrate the body's reaction to an infection caused by group A beta-hemolytic streptococci. Group A streptococci produce the enzyme streptolysin O, which can destroy (lyse) red blood cells. Because streptolysin O is antigenic (contains a protein foreign to the body), the body reacts by producing antistreptolysin O (ASO), which is a neutralizing antibody. ASO appears in the blood serum one week to one month after the onset of a strep infection. A high titer (high levels of ASO) is not specific for any type of poststreptococcal disease, but it does indicate if a streptococcal infection is or has been present.

Serial (several given in a row) ASO testing is often performed to determine the difference between an acute or convalescent blood sample. The diagnosis of a previous strep infection is confirmed when serial titers of ASO rise over a period of weeks, then fall slowly. ASO titers peak during the third week after the onset of acute symptoms of a streptococcal disease; at six months after onset, approximately 30% of patients exhibit abnormal titers.

Antideoxyribonuclease-B titer (anti-DNase B, or ADB)

Anti-DNase-B, or ADB, also detects antigens produced by group A strep, and is elevated in most patients with rheumatic fever and poststreptococcal glomerulonephritis. This test is often done concurrently with the ASO titer, and subsequent testing is usually performed to detect differences in the acute and convalescent blood samples. When ASO and ADB are performed concurrently, 95% of previous strep infections are detected. If both are repeatedly negative, the likelihood is that the patient's symptoms are not caused by a poststreptococcal disease.

When evaluating patients with acute rheumatic fever, the American Heart Association recommends the ASO titer rather than ADB. Even though the ADB

is more sensitive than ASO, its results are too variable. It also should be noted that, while ASO is the recommended test, when ASO and ADB are done together, the combination is better than either ASO or ADB alone.

Streptozyme

The streptozyme test is often used as a screening test for antibodies to the streptococcal antigens NADase, DNase, streptokinase, streptolysin O, and hyaluronidase. This test is most useful in evaluating suspected poststreptococcal disease following *Streptococcus pyogenes* infection, such as rheumatic fever.

Streptozyme has certain advantages over ASO and ADB. It can detect several antibodies in a single assay, it is technically quick and easy, and it is unaffected by factors that can produce false-positives in the ASO test. The disadvantages are that, while it detects different antibodies, it does not determine which one has been detected, and it is not as sensitive in children as in adults. In fact, borderline antibody elevations, which could be significant in children, may not be detected at all. As with the ASO and ADB, a serially rising titer is more significant than a single determination.

Preparation

These tests are performed on blood specimens drawn from the patient's vein. The patient does not need to fast before these tests.

Risks

The risks associated with these tests are minimal, but may include slight bleeding from the blood-drawing site, **fainting** or feeling lightheaded after the blood is drawn, or blood accumulating under the puncture site (hematoma).

Normal results

Antistreptolysin O titer:

- adult: 160 Todd units/ml
- child: six months to two years: 50 Todd units/ml; two to four years: 160 Todd units/ml; five to 12 years: 170–330 Todd units/ml
- newborn: similar to the mother's value

Antideoxyribonuclease-B titer:

- adult: 85 units
- child (preschool): 60 units
- child (school age): 170 units

KEY TERMS

Antibody—A protein manufactured by a type of white blood cells called lymphocytes, in response to the presence of an antigen, or foreign protein, in the body. Because bacteria, viruses, and other organisms commonly contain many antigens, antibodies are formed against these foreign proteins to neutralize or destroy the invaders.

Antigen—A substance that can trigger a defensive response in the body, resulting in production of an antibody as part of the body's defense against infection and disease. Many antigens are foreign proteins not found naturally in the body, and include bacteria, viruses, toxins, and tissues from another person used in organ transplantation.

Glomerulonephritis—An inflammation of the glomeruli, the filtering units of the kidney. Damage to these structures hampers removal of waste products, salt, and water from the bloodstream, which may cause serious complications. This disorder can be mild and cause no symptoms, or so severe enough to cause kidney failure.

Rheumatic fever—A disease that causes inflammation in various body tissues. Rare in most developed countries, but reported to be on the increase again in parts of the United States. Joint inflammation occurs, but more serious is the frequency with which the disease permanently damages the heart. The nervous system may also be affected, causing Sydenham's chorea.

Sydenham's chorea—A childhood disorder of the central nervous system. Once called St. Vitus' dance, the condition is characterized by involuntary, jerky movements that usually follow an attack of rheumatic fever. Rare in the United States today, but a common disorder in developing countries. Usually resolves in two to three months with no long-term adverse effects.

Streptozyme: less than 100 streptozyme units.

Abnormal results

Antistreptolysin O titer: Increased levels are seen after the second week of an untreated infection in acute streptococcal infection, and are also increased with acute rheumatic fever, acute glomerulonephritis (66% of patients will not have high ASO titers), and scarlet fever.

Antideoxyribonuclease-B titer: Increased titers are seen in cases of acute rheumatic fever and post-streptococcal glomerulonephritis.

Streptozyme: As this is a screening test for antibodies to streptococcal antigens, increased levels require more definitive tests to confirm diagnosis.

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Janis O. Flores

Streptococcal gangrene see **Gangrene**

Streptococcal infections

Definition

Streptococcal (strep) infections are communicable diseases that develop when bacteria normally found on the skin or in the intestines, mouth, nose, reproductive tract, or urinary tract invade other parts of the body and contaminate blood or tissue.

Some strep infections don't produce symptoms. Some are fatal.

Description

Most people have some form of strep bacteria in their body at some time. A person who hosts bacteria without showing signs of infection is considered a carrier.

Types of infection

Primary strep infections invade healthy tissue, and most often affect the throat. Secondary strep infections invade tissue already weakened by injury or illness. They frequently affect the bones, ears, eyes, joints, or intestines.

Both primary and secondary strep infections can travel from affected tissues to lymph glands, enter the bloodstream, and spread throughout the body.

Numerous strains of strep bacteria have been identified. Types A, B, C, D, and G are most likely to make people sick.

Group A

Group A strep (GAS) is the form of strep bacteria most apt to be associated with serious illness.

Between 10,000 and 15,000 GAS infections occur in the United States every year. Most are mild inflammations of the throat or skin, where the bacteria are normally found; however, GAS infections can be deadly.

Two of the most severe invasive GAS infections are necrotizing fasciitis or flesh-eating bacteria (destruction of muscle tissue and fat) and **toxic shock syndrome** (a rapidly progressive disorder that causes **shock** and damages internal organs).

GROUP B. Group B strep (GBS) most often affects pregnant women, infants, the elderly, and chronically ill adults.

Since first emerging in the 1970s, GBS has been the primary cause of life-threatening illness and **death** in newborns. GBS exists in the reproductive tract of 20-25% of all pregnant women. Although no more than 2% of these women develop invasive infection, 40-73% transmit bacteria to their babies during delivery.

About 12,000 of the 3.5 million babies born in the United States each year develop GBS disease in infancy. About 75% of them develop early-onset infection. Sometimes evident within a few hours of birth and always apparent within the first week of life, this condition causes inflammation of the membranes covering the brain and spinal cord (**meningitis**), **pneumonia**, blood infection (**sepsis**) and other problems.

Late-onset GBS develops between the ages of seven days and three months. It often causes meningitis. About half of all cases of this rare condition can be traced to mothers who are GBS carriers. The cause of the others is unknown.

GBS has also been linked to a history of **breast cancer**.

GROUP C. Group C strep (GCS) is a common source of infection in animals. It rarely causes human illness.

GROUP D. Group D strep (GDS) is a common cause of wound infections in hospital patients. GDS is also associated with:

- abnormal growth of tissue in the gastrointestinal tract
- urinary tract infection (UTI)
- womb infections in women who have just given birth

GROUP G. Normally present on the skin, in the mouth and throat, and in the intestines and genital tract, Group G strep (GGS) is most likely to lead to infection in alcoholics and in people who have **cancer**, **diabetes mellitus**, **rheumatoid arthritis**, and other conditions that suppress immune-system activity.

GGS can cause a variety of infections, including:

- bacteria in the bloodstream (bacteremia)
- inflammation of the connective tissue structure surrounding a joint (bursitis)
- endocarditis (a condition that affects the lining of the heart chambers and the heart valves)
- meningitis
- inflammation of bone and bone marrow (osteomyelitis)
- inflammation of the lining of the abdomen (peritonitis)

Causes and symptoms

Streptococcal infection occurs when bacteria contaminate cuts or open sores or otherwise penetrate the body's natural defenses.

GAS

GAS is transmitted by direct contact with saliva, nasal discharge, or open **wounds** of someone who has the infection. Chronic illness, **kidney disease** treated by dialysis, and steroid use increase vulnerability to infection.

About one of five people with GAS infection develops a sore, inflamed throat, and pus on the tonsils. The majority of those infected by GAS either have no symptoms or develop enlarged lymph nodes, **fever**, **headache**, **nausea**, **vomiting**, weakness, and a rapid heartbeat.

Flesh-eating bacteria is characterized by fever, extreme **pain**, and swelling and redness at a site where skin is broken.

Symptoms of toxic shock include abdominal pain, confusion, **dizziness**, and widespread red skin rash.

GBS

A pregnant woman who has GBS infection can develop infections of the bladder, blood, and urinary tract, and deliver a baby who is infected or stillborn. The risk of transmitting GBS infection during birth is highest in a woman whose labor begins before the 37th week of **pregnancy** or lasts more than 18 hours or who:



The scarlet fever rash on this person's arm was caused by a streptococcal infection. (Custom Medical Stock Photo. Reproduced by permission.)

- becomes a GBS carrier during the final stages of pregnancy
- has a GBS urinary-tract infection
- has already given birth to a baby infected with GBS
- develops a fever during labor

More than 13% of babies who develop GBS infection during birth or within the first few months of life develop neurologic disorders. An equal number of them die.

Among men, and in women who are not pregnant, the most common consequences of GBS infection are pneumonia and infections of blood, skin, and soft tissue.

Miscellaneous symptoms

Other symptoms associated with strep infection include:

- anemia
- elevated white blood cell counts
- inflammation of the epiglottis (epiglottitis)
- heart murmur
- high blood pressure
- infection of the heart muscle
- kidney inflammation (**nephritis**)
- swelling of the face and ankles

Diagnosis

Strep bacteria can be obtained by swabbing the back of the throat or the rectum with a piece of sterile cotton. Microscopic examination of the smear can identify which type of bacteria has been collected.

Treatment

Penicillin and other **antibiotics** are used to treat strep infections.

It takes less than 24 hours for antibiotics to eliminate an infected person's ability to transmit GAS.

Guidelines developed by the American Academy of Obstetrics and Gynecology (AAOG), the American Academy of Pediatrics (AAP), and the Centers for Disease Control and Prevention (CDC) recommend administering intravenous antibiotics to a woman at high risk of passing GBS infection on to her child, and offering the medication to any pregnant woman who wants it.

Initiating antibiotic therapy at least four hours before birth allows medication to become concentrated enough to protect the baby during passage through the birth canal.

Babies infected with GBS during or shortly after birth may die. Those who survive often require lengthy hospital stays and develop vision or **hearing loss** and other permanent disabilities.

Alternative treatment

Conventional medicine is very successful in treating strep infections. However, several alternative therapies, including homeopathy and botanical medicine, may help relieve symptoms or support the person with a strep infection. For example, several herbs, including garlic (*Allium sativum*), **echinacea** (*Echinacea* spp.), and goldenseal (*Hydrastis canadensis*), are believed to strengthen the immune system, thus helping the body fight a current infection, as well as helping prevent future infections.

Prognosis

GAS is responsible for more than 2,000 deaths a year. About 20% of people infected with flesh-eating bacteria die. So do three of every five who develop toxic shock syndrome.

Early-onset GBS kills 15% of the infants it affects. Late-onset disease claims the lives of 10% of babies who develop it.

GBS infections are fatal in about 20% of the men and non-pregnant women who develop them.

About 10–15% of non-GAS strep infections are fatal. Antibiotic therapy, begun when symptoms first appear, may increase a patient's chance of survival.

Prevention

Washing the hands frequently, especially before eating and after using the bathroom, and keeping wounds clean can help prevent strep infection. Exposure to infected people should be avoided, and a family physician should be notified by anyone who develops an extremely **sore throat** or pain, redness, swelling, or drainage at the site of a wound or break in the skin.

Until vaccines to prevent strep infection become available, 12 monthly doses of oral or injected antibiotics may prevent some types of recurrent infection.

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Maureen Haggerty

Streptococcal sore throat see **Strep throat**

Streptococcal toxic shock syndrome see **Toxic shock syndrome**

Streptomycin see **Aminoglycosides**

Streptozyme test see **Streptococcal antibody tests**

Stress

Definition

Stress is defined as an organism's total response to environmental demands or pressures. When stress was first studied in the 1950s, the term was used to denote both the causes and the experienced effects of these pressures. More recently, however, the word stressor has been used for the stimulus that provokes a stress response. One recurrent disagreement among researchers concerns the definition of stress in humans. Is it primarily an external response that can be measured by changes in glandular secretions, skin reactions, and other physical functions, or is it an internal interpretation of, or reaction to, a stressor; or is it both?

Description

Stress in humans results from interactions between persons and their environment that are perceived as straining or exceeding their adaptive

Top Ten Stressful Life Events

Death of spouse
 Divorce
 Marital separation
 Jail term or death of close family member
 Personal injury or illness
 Marriage
 Loss of job due to termination
 Marital reconciliation or retirement
 Pregnancy
 Change in financial state

capacities and threatening their well-being. The element of perception indicates that human stress responses reflect differences in personality, as well as differences in physical strength or general health.

Risk factors for stress-related illnesses are a mix of personal, interpersonal, and social variables. These factors include lack or loss of control over one's physical environment, and lack or loss of social support networks. People who are dependent on others (e.g., children or the elderly) or who are socially disadvantaged (because of race, gender, educational level, or similar factors) are at greater risk of developing stress-related illnesses. Other risk factors include feelings of helplessness, hopelessness, extreme fear or anger, and cynicism or distrust of others.

Causes and symptoms

Causes

The causes of stress can include any event or occurrence that a person considers a threat to his or her coping strategies or resources. Researchers generally agree that a certain degree of stress is a normal part of a living organism's response to the inevitable changes in its physical or social environment, and that positive, as well as negative, events can generate stress as well as negative occurrences. Stress-related disease, however, results from excessive and prolonged demands on an organism's coping resources. It is now believed that 80–90% of all disease is stress-related.

Recent research indicates that some vulnerability to stress is genetic. Scientists at the University of Wisconsin and King's College London discovered that people who inherited a short, or stress-sensitive, version of the serotonin transporter gene were almost three times as likely to experience depression following

a stressful event as people with the long version of the gene. Further research is likely to identify other genes that affect susceptibility to stress.

One cause of stress that has affected large sectors of the general population around the world since 2001 is terrorism. The events of September 11, 2001, the sniper shootings in Virginia and Maryland and the Bali nightclub bombing in 2002, the **suicide** bombings in the Middle East in 2003, have all been shown to cause short-term symptoms of stress in people who read about them or watch television news reports as well as those who witnessed the actual events. Stress related to terrorist attacks also appears to affect people in countries far from the location of the attack as well as those in the immediate vicinity. It is too soon to tell how stress related to episodes of terrorism will affect human health over long periods of time, but researchers are already beginning to investigate this question. In 2004 the Centers for Disease Control and Prevention (CDC) released a report on the aftereffects of the World Trade Center attacks on rescue and recovery workers and volunteers. The researchers found that over half the 11,700 people who were interviewed met threshold criteria for a mental health evaluation. A longer-term evaluation of these workers is underway.

A new condition that has been identified since 9/11 is childhood traumatic grief, or CTG. CTG refers to an intense stress reaction that may develop in children following the loss of a parent, sibling, or other loved one during a traumatic event. As defined by the National Child Traumatic Stress Network (NCTSN), "Children with childhood traumatic grief experience the cause of [the loved one's] **death** as horrifying or terrifying, whether the death was sudden and unexpected (for example, due to homicide, suicide, motor vehicle accident, drug overdose, natural disaster, war, terrorism, and so on) or due to natural causes (**cancer**, **heart attack**, and so forth). Even if the manner of death does not appear to others to be sudden, shocking, or frightening, children who perceive the death in this way may develop childhood traumatic grief. In this condition, even happy thoughts and memories of the deceased person remind children of the traumatic way in which the deceased died." More information on the identification and treatment of childhood traumatic grief can be obtained from the NCTSN web site, < http://www.nctsn.org/nccts/nav.do?pid=hom_main > .

Symptoms

The symptoms of stress can be either physical or psychological. Stress-related physical illnesses, such as **irritable bowel syndrome**, heart attacks, arthritis, and chronic headaches, result from long-term

overstimulation of a part of the nervous system that regulates the heart rate, blood pressure, and digestive system. Stress-related emotional illness results from inadequate or inappropriate responses to major changes in one's life situation, such as marriage, completing one's education, becoming a parent, losing a job, or retirement. Psychiatrists sometimes use the term adjustment disorder to describe this type of illness. In the workplace, stress-related illness often takes the form of burnout—a loss of interest in or ability to perform one's job due to long-term high stress levels. For example, palliative care nurses are at high risk of burnout due to their inability to prevent their patients from dying or even to relieve their physical suffering in some circumstances.

Diagnosis

When the doctor suspects that a patient's illness is connected to stress, he or she will take a careful history that includes stressors in the patient's life (family or employment problems, other illnesses, etc.). Many physicians will evaluate the patient's personality as well, in order to assess his or her coping resources and emotional response patterns. There are a number of personality inventories and **psychological tests** that doctors can use to help diagnose the amount of stress that the patient experiences and the coping strategies that he or she uses to deal with them. A variation on this theme is to identify what the patient perceives as threatening as well as stressful. Stress-related illness can be diagnosed by primary care doctors, as well as by those who specialize in psychiatry. The doctor will need to distinguish between **adjustment disorders** and **anxiety** or **mood disorders**, and between psychiatric disorders and physical illnesses (e.g., thyroid activity) that have psychological side effects.

Treatment

Recent advances in the understanding of the many complex connections between the human mind and body have produced a variety of mainstream approaches to stress-related illness. Present treatment regimens may include one or more of the following:

- **Medications.** These may include drugs to control blood pressure or other physical symptoms of stress, as well as drugs that affect the patient's mood (tranquilizers or antidepressants).
- **Stress management programs.** These may be either individual or group treatments, and usually involve analysis of the stressors in the patient's life. They often focus on job or workplace-related stress.

- **Behavioral approaches.** These strategies include relaxation techniques, breathing exercises, and physical **exercise** programs including walking.
- **Massage.** Therapeutic massage relieves stress by relaxing the large groups of muscles in the back, neck, arms, and legs.
- **Cognitive therapy.** These approaches teach patients to reframe or mentally reinterpret the stressors in their lives in order to modify the body's physical reactions.
- **Meditation and associated spiritual or religious practices.** Recent studies have found positive correlations between these practices and stress hardiness.

Alternative treatment

Treatment of stress is one area in which the boundaries between traditional and alternative therapies have changed in recent years, in part because some forms of physical exercise (**yoga**, **tai chi**, aikido) that were once associated with the counterculture have become widely accepted as useful parts of mainstream **stress reduction** programs. Other alternative therapies for stress that are occasionally recommended by mainstream medicine include **aromatherapy**, dance therapy, **biofeedback**, nutrition-based treatments (including dietary guidelines and **nutritional supplements**), **acupuncture**, homeopathy, and herbal medicine.

Prognosis

The prognosis for recovery from a stress-related illness is related to a wide variety of factors in a person's life, many of which are genetically determined (race, sex, illnesses that run in families) or beyond the individual's control (economic trends, cultural stereotypes and prejudices). It is possible, however, for humans to learn new responses to stress and, thus, change their experiences of it. A person's ability to remain healthy in stressful situations is sometimes referred to as stress hardiness. Stress-hardy people have a cluster of personality traits that strengthen their ability to cope. These traits include believing in the importance of what they are doing; believing that they have some power to influence their situation; and viewing life's changes as positive opportunities rather than as threats.

Prevention

Complete prevention of stress is neither possible nor desirable, because stress is an important stimulus

KEY TERMS

Adjustment disorder—A psychiatric disorder marked by inappropriate or inadequate responses to a change in life circumstances. Depression following retirement from work is an example of adjustment disorder.

Biofeedback—A technique in which patients learn to modify certain body functions, such as temperature or pulse rate, with the help of a monitoring machine.

Burnout—An emotional condition, marked by tiredness, loss of interest, or frustration, that interferes with job performance. Burnout is usually regarded as the result of prolonged stress.

Stress hardiness—A personality characteristic that enables persons to stay healthy in stressful circumstances. It includes belief in one's ability to influence the situation; being committed to or fully engaged in one's activities; and having a positive view of change.

Stress management—A category of popularized programs and techniques intended to help people deal more effectively with stress.

Stressor—A stimulus, or event, that provokes a stress response in an organism. Stressors can be categorized as acute or chronic, and as external or internal to the organism.

of human growth and creativity, as well as an inevitable part of life. In addition, specific strategies for stress prevention vary widely from person to person, depending on the nature and number of the stressors in an individual's life, and the amount of control he or she has over these factors. In general, however, a combination of attitudinal and behavioral changes works well for most patients. The best form of prevention appears to be parental modeling of healthy attitudes and behaviors within the family.

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- The American Institute of Stress. 124 Park Avenue, Yonkers, NY 10703 (914) 963-1200. Fax: (914) 965-6267. <<http://www.stress.org>>.
- Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.
- National Child Traumatic Stress Initiative. Center for Mental Health Services, Substance Abuse and Mental

Health Services Administration, Department of Health and Human Services, 5600 Fishers Lane, Parklawn Building, Room 17C-26, Rockville, MD 20857. (301) 443-2940. <http://www.ncatsnet.org/ncats/nav.do?pid=hom_main>.

National Institute of Mental Health (NIMH). 6001 Executive Boulevard, Room 8184, MSC 9663, Bethesda, MD 20892-9663. (301) 443-4513. <<http://www.nimh.nih.gov>>.

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Rebecca J. Frey, PhD

Stress reduction

Definition

Stress is the body's normal response to anything that disturbs its natural physical, emotional, or mental balance. Stress reduction refers to various strategies that counteract this response and produce a sense of relaxation and tranquility.

Purpose

Although stress is a natural phenomenon of living, stress that is not controlled and that continues for a long period of time can seriously compromise health. For this reason, stress must be understood, managed and appropriately reduced. Several very different strategies and therapies are available that help with relaxation and stress management.

Precautions

Stress reduction can only present a problem if an individual attributes an actual, serious condition or disease to being simply a stress-related response and avoids consulting a physician.

People who have undergone a severe trauma (criminal assault, combat, natural or transportation disaster, etc.) may experience symptoms of **post-traumatic stress disorder** (PTSD) or **acute stress**

disorder (ASD). These disorders are defined by their temporal connection to a traumatic event in the patient's life, and are characterized by a cluster of **anxiety** and dissociative symptoms. They interfere with the patient's normal level of functioning, and require some form of supportive therapy. People who experience a sense of detachment or unreality, emotional numbing, a general feeling of being dazed, **amnesia** for part of the traumatic event, or similar symptoms should consult a medical doctor in addition to using other approaches to stress reduction.

Description

Everyone encounters stress every day. Although most people think of it as something negative that happens to them, in fact stress itself is really neither good nor bad but is neutral or nonspecific. Stress may be internal (from within ourselves) or external (such as noise from the environment) and does not always result from something unpleasant. A certain amount of stress in our lives is actually essential to being sufficiently stimulated to meet the challenges of everyday life, but when stress is constant and acute, it can have dangerous consequences. Since stress is both natural and unavoidable, it is necessary to understand it and to learn how to deal with it, particularly how to reduce it.

The specific and immediate cause of stress is called the stressor. A stressor can be something dramatic or terrible, such as a violent experience or the **death** of a loved one, or it can be a positive and rewarding event, like marriage or a promotion. The stressor can be internal, such as feelings of guilt or anger felt in a relationship, or it can be external, such as a natural disaster or the ordinary rigors and frustrations of commuting. It can also have a physical source, like simple **exercise** or hard work, or it can be strictly mental, like worry. Our bodies react the same way physiologically no matter what the source and reasons for stress might be.

From a physical standpoint, the body reacts to stress in a standard and predictable manner. When stress occurs, the brain immediately receives nerve impulses. These impulses initiate an automatic sequence carried out by the body's sympathetic nervous system: it begins with stimulation of the brain's hypothalamus, which sends nerve impulses to both the adrenal and the pituitary glands. Also called the "fight or flight" response, this automatic physiological process is known to have evolved in humans and animals to enable them to cope with sudden life-threatening emergencies. When faced with a major stressor, the

body's biochemistry instantly hurtles into a ready mode that marshals all the possible resources necessary to either escape or do battle. Thus, the adrenal glands located on top of the kidneys provide an instant surge of adrenaline, the body's rocket fuel, quickening the heart rate and blood flow and providing every cell with extra oxygen. They also release cortisol or hydrocortisone, causing an increase in both amino acids (the building blocks of proteins) and blood sugar. These will be needed if tissue repair must take place. Finally, the pituitary gland at the base of the brain releases a variety of hormones, endorphins among them, that act as natural painkillers and permit the body to do things it ordinarily cannot do. Thus, at just about the same time a stressor is recognized by the body, the heart and breathing rate spikes, the pupils dilate to let in more light, perspiration increases and digestion slows, and the body is aroused, energized, and temporarily feels no **pain**. This sequence of events allows individuals to do whatever is required to save themselves, whether it is to flee from a predator or engage in combat and fend off an attack.

While these automatic physiological responses served early man well and were essential to survival of the species, today's men and women rarely must literally fight for their lives or dodge and elude a predator. Yet their bodies' automatic response to stress has remained unchanged in a radically changed, modern world. Whether caveman or corporate executive, when the fight-or-flight response kicks in, a three-stage process begins. Stage one is the alarm stage in which the body releases hormones and prepares for extreme physical action. Resistance is stage two in which the body attempts to resist yet adapt to the stress and to repair any damage done. The final stage, exhaustion, occurs if the stress remains constant. It is especially dangerous since stage one's physical response may begin all over again. The persistence of stress and stage three's exhaustion is the point at which disease can occur. The body may then experience severe debilitating conditions like migraine, heart irregularities, and mental illness. The body's functions may even shut down altogether.

Although different individuals may have different levels of tolerance to stress, chronic stress will eventually wear down even the strongest of people. Prolonged stress can cause biochemical imbalances that weaken the immune system and invite serious illness. Overall, stress that persists is known to interfere with digestion and, more seriously, alter brain chemistry, create hormonal imbalances, increase heart rate, raise blood pressure, and negatively affect both metabolic and immune function. It is also

important to recognize that although stress itself is not a disease, it can worsen any number of already serious physical conditions. Many physicians feel that chronic stress can so overtax an individual's physical resources and ways of coping that **cancer**, **stroke**, and heart disease can occur. While long-term stress can seriously affect one's quality of life and lead to major, sometimes fatal, diseases, prolonged stress also results in the everyday miseries of **headache** and allergy, digestive disorders and **fatigue**, irritable bladder and **impotence**, **insomnia**, anxiety, depression, and simple aches and pains. Researchers exploring the connection between stress and susceptibility to colds exposed stressed individuals (who had experienced a death in the family, become divorced, or had recently moved) to cold viruses and then tested for antibodies a month later. Results indicated that severely stressed individuals were four times more likely to become infected.

It follows that if stress can cause or contribute to illness, then reducing stress should have the opposite effect and perhaps even encourage healing. Probably the most important step toward reducing the stress in everyone's life is to understand the nature of stress and to learn how to condition ourselves to be able to gain some control over it. Being able to recognize that we are stressed is probably the first step toward understanding. Of the many signs and symptoms that alert us, some are obvious and require only common sense to recognize. Short-term noticeable effects of stress include sweaty palms and other types of perspiration, dilated pupils, and difficulty in swallowing ("a lump in the throat"). Tightness in the chest is another stress signal as are stomach problems and some skin conditions. Stress that is the result of prolonged anxiety (a sense of apprehension) often results in feelings of panic or actual trembling, fatigue, insomnia, and **shortness of breath**, heart **palpitations** and **dizziness**, and sometimes simple irritability. Although none of these symptoms is pleasant, they are relatively minor compared to the silent but much more serious internal effects that can lead to immune-related disorders and even cancer and heart disease.

Fortunately, stress and the negative effects it creates can be reduced by a wide variety of therapeutic approaches. When successfully applied, many of these therapies or strategies can both reduce stress and reverse its damage. Before selecting a particular therapy, it is important to be able to distinguish bad or unhealthy stress from the type that is not bad. Researchers have found that the most important variable among types of stress is an individual's sense of control in a given situation. The least harmful stress

scenario is one in which an individual has a sufficient degree of control or some idea of predictability. Put simply, predictable pain is less stressful because individuals know when to relax (gaining relief from pain as well as protecting themselves from its damaging effects). But when individuals have no warning of pain, they are in a state of constant stress. An example from daily life might be the difference between the stress experienced by top executives who are in control of their fate and their middle-level managers who are not. The former can pick and choose when to enter or engage a stressful situation or problem, but the latter have no control nor any ability to predict when such a situation will arise and are constantly on alert or in a state of anxiety.

For those with little control over situations that make them anxious, there are basically two ways to deal with their stress. One is to remove or at least reduce the stressor, and the other is to increase their resistance to it. Although there are many strategies to achieve each of these, all of them can be reduced to some variation of a single simple concept—relaxation. While there is no one single technique or therapy for everyone to use to manage and reduce stress, there is certainly some combination of lifestyle change, diet, exercise, and relaxation that will allow all types of individuals to better manage the stress in their lives. Although relaxation is at the core of most stress reduction methods, it is not something that everyone can fully achieve without assistance and guidance. Interestingly, our modern life experiences often do not provide us with the coping skills needed to deal with stressful stimuli, and increasingly, people find that simple relaxing is something that they must learn how to do.

Fortunately, there are a number of relaxation therapies that enable the willing individual to achieve deep, beneficial relaxation. In fact, there are almost too many from which to choose. A 1997 book on stress remedies cowritten by the editors of *Prevention* magazine and published by Rodale Press is organized alphabetically and lists fifty-nine separate stress-reducing techniques and subjects, from Acceptance to **Yoga**. These and many other methods of reducing stress can be grouped into the following general categories: mind-body therapies, body work and movement therapies, and herbal-based **diets** and natural regimens. Many of the specific techniques in these categories can be part of a self-help or self-care approach, although some require the help of an experienced practitioner.

Therapies that focus on the mind/body connection are based on the fact that thinking and emotions

can have physical effects on the body. These techniques encourage the individual to take control and learning how to cope with stressors rather than trying to eliminate them. Such therapies range from individual counseling and **meditation** or involvement with a support group to the mystery of **guided imagery** and the technology of **biofeedback**. They all have the common goal of evoking the physiological relaxation response, in which a person can achieve such beneficial internal results as lowering blood pressure and decreasing gastric acid secretion.

Body work and movement therapies include techniques ranging from dance therapy and the gentleness of massage to **reflexology** and the rigors of **rolfing**. Body work is based partly on the therapeutic power of human touch and can also include manipulation, realignment, and posture correction. Movement therapies are a particular form of physical exercise, although they attempt to do much more than simply get a person into shape. Most usually emphasize the mind/body connection and strive to put people in better touch with both their bodies and their feelings. Body work and movement therapies can be as vigorous as deep tissue manipulation or as simple and minimal as the Alexander technique's light posture corrections.

Herbal remedies for stress are usually part of a larger system of natural, **holistic medicine**. Whether Chinese traditional medicine, its counterpart from India, or the homeopathy of the West, all these systems of natural medicine have a holistic focus and emphasize the need for inner balance. All demonstrate how the individual's physical, emotional, mental, and spiritual states are connected and use natural substances as part of the treatment for reducing stress. Such therapies range from the occasional purging (cleansing) of **Ayurvedic medicine** to the sleep-inducing properties of chamomile tea. They also can include the use of cayenne to relieve pain, fragrant essential oils from flowers to evoke a pleasing response and relieve tension, or aloe vera to soothe burned skin.

A list of some of the more common therapies and techniques available for reducing stress includes:

- **Acupuncture.** Insertion of needles at certain spots under the skin for the purpose of attaining balance by either releasing blocked energy or draining off excess energy.
- **Alexander technique.** Improving the alignment of head, neck, and back claims to achieve efficient posture and movement.
- **Aromatherapy.** Massage with essential oils from flowers claims to affect mood and produce a sense of well-being.

- **Art therapy.** Creating something allows free expression and results in feelings of achievement and mood change.
 - **Autogenic training therapy.** A form of deep meditation or self-hypnosis.
 - **Autosuggestion therapy.** A form of verbal therapy involving repetition of a positive idea.
 - **Ayurvedic medicine.** A complete system of daily living based on awareness of one's particular constitution.
 - **Behavioral therapy.** A variety of psychotherapies that are based on changing ourselves by retraining.
 - **Bach flower therapy.** Herbal remedies that are prepared from flowers acting energetically to soothe the mind and body.
 - **Bioenergetics.** A practice that encourages sudden release of tensions by crying or kicking.
 - **Biofeedback.** Monitoring rates of body functions and using data to influence and gain control over autonomic functions.
 - **Breathing for relaxation.** Stylized breathing technique to control and lower body functions.
 - **Counseling.** Work with a therapist trained in talking-based therapy.
 - **Dance movement therapy.** Freedom of expression through movement.
 - **Feldenkrais method.** Slow, light movements alter habits and reeducate neuromuscular system.
 - **Flotation therapy.** Floating in a soundproof tank with no external stimulation.
 - **Guided imagery.** Creating a mental picture of what is desired. Also called Creative imagery or Visualization.
 - **Herbal medicine.** Uses substances derived from plants as treatment instead of synthetic drugs.
 - **Homeopathy.** Uses minute doses of plant, animal, and mineral substances to stimulate the body's natural healing.
 - **Hydrotherapy.** Use of water internally and externally for healing purposes.
 - **Hypnotherapy.** Hypnosis in order to identify and release patterns that keep an individual from a personal balance point.
 - **Kinesiology.** Uses muscle testing to correct imbalances in the body's "energy system." Also called Touch for Health.
 - **Massage.** Use of touch and manipulation to soothe. Can also employ vigorous deep tissue manipulation.
 - **Meditation.** Deep, relaxed, receptive, and focused concentration on a single object, sound, or word.
 - **Music therapy.** Playing or listening to music to create an emotional reaction.
 - **Naturopathy.** A complete health care system that uses a variety of natural healing therapies.
 - **Psychotherapy.** A talking-based therapy with a mental health professional to get at the root of a conflict, modify behavior and disruptive negative thought patterns.
 - **Reflexology.** Manipulation of zones of the feet that relate to the major organs, glands, and areas of the body.
 - **Rolfing.** Vigorous manipulation of the body's connective tissue to restore "balance."
 - **Shiatsu.** Traditional Japanese finger pressure massage therapy.
 - **Sound therapy.** Uses sound waves to slow the body's autonomic system.
 - **Tai chi chuan.** System of slow, continuous exercises based on rhythm and equilibrium.
 - **Yoga.** System of exercises that combines certain positions with deep breathing and meditation.
- These and many other techniques, systems, and therapies are available to the person searching for some way to reduce and manage the stress of everyday life. Some methods are very simple and can be easily learned, while others are high-tech and often involve a practitioner. A search for common elements among most of these stress-reducing systems reveals several obvious strategies that nearly everyone can employ on their own. However, it is important to know and recognize the signals of stress. Further, it is easier to resist the negative effects of stress by eating properly and getting sufficient sleep and exercise.
- Nearly all stress-reducing systems are geared to evoking some degree of beneficial mind/body relaxation, and most include some version of the following:
- mental time out
 - deep breathing
 - meditation and singular focus
 - gentle, repetitive exercise
- The best stress reduction system is the one that works for the individual. Whether stress can be relieved by laughter, mellow music, repetition of a

single word, self-massage, vigorous activity, or simply by doing everyday chores in a mindful state of heightened awareness, it is important that stress be recognized and managed every day. Studies have shown that regular relaxation eventually makes the body less responsive to its stress hormones and acts as a sort of natural tranquilizer. People can build their own immune defense against the stress response.

Many companies have introduced workplace stress management programs to improve their employees' health. These programs typically include instruction on emotional refocusing or restructuring, and have been shown to be beneficial in reducing the participants' blood pressure, heart rate, and other signs of emotional upset. In addition, stress management programs designed for persons in specific high-stress occupations (medicine, law enforcement, emergency response, etc.) have proved to be effective in reducing burnout and helping members of these professions cope with the specific stresses of their respective jobs.

An additional general strategy for handling stress in family life or the workplace is the cultivation of a group of character traits that has been termed "psychological hardiness." These traits include believing in the importance of what one is doing; believing that one has some power to influence the immediate situation; and viewing life's changes as positive opportunities rather than as threats. These qualities are sometimes referred to as the "3 Cs," which stand for commitment, control and challenge. Approaches to stress reduction that enhance these qualities are especially beneficial to people.

Newer trends in stress reduction

One trend in stress reduction in the early 2000s is the development of stress management programs or stress reduction strategies tailored to specific categories of people, often defined by their occupation or by a chronic health condition. For example, journalists who cover traumatic events are increasingly recognized as susceptible to developing posttraumatic stress disorder. With regard to specific diseases, stress management programs have been pioneered as of 2004 for patients with **asthma** or lupus erythematosus.

Another new trend in stress reduction is the development of programs designed for communities as well as individuals. After the events of September 11, 2001, many mental health professionals recognized that acts of terrorism or mass violence affect large groups of people, and that psychiatric interventions need to

address stress as a group experience as well as an individual one.

Risks

All relaxation-based therapies to reduce stress are virtually free of serious risk.

Normal results

Learning how to manage stress has the short-term benefits of giving people some sense of control in their lives, providing them with positive coping strategies, and making them more relaxed and healthier. The long-term benefits can be a stronger immune system, proper hormonal balance, and reduced susceptibility to such serious, life-threatening diseases as heart disease and cancer.

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KEY TERMS

Adrenal gland—A pair of glands that rest on the top of each kidney that produce steroids, such as sex hormones and those concerned with metabolic functions.

Amino acid—Organic acids that are the main components of proteins and are synthesized by living cells.

Antibody—A type of protein produced in the blood in response to a foreign substance that destroys the intruding substance; it is responsible for immunity.

Burnout—An emotional condition marked by tiredness, loss of interest, or frustration that interferes with job performance. Burnout is usually regarded as the result of prolonged stress.

Chronic—Long-term or frequently recurring.

Debilitating—Weakening, or reducing the strength of.

Dilate—To enlarge, open wide, or distend.

Endorphins—A group of proteins with powerful pain-killing properties that originate naturally in the brain.

Holistic—That which pertains to the entire person, involving the body, mind, and spirit.

Hydrocortisone—A steroid hormone produced by the adrenal glands that provides resistance to stress.

Hypothalamus—A part of the brain that controls some of the body's automatic regulatory functions.

Immune function—The state in which the body recognizes foreign materials and is able to neutralize them before they can do any harm.

Impotence—The inability of the male to engage in sexual intercourse because of insufficient erection.

Insomnia—Inability to sleep under normal conditions.

Metabolic function—Those processes necessary for the maintenance of a living organism.

Neuromuscular—Relating to nerve and muscle or their interaction.

Physiological—Dealing with the functions and processes of the body.

Pituitary gland—A gland at the base of the brain responsible for growth, maturation, and reproduction.

Sympathetic nervous system—That part of the autonomic nervous system that affects contraction of muscles and blood vessels. Stimulation of this system by a stressor triggers the production of hormones that prepare the body for fight or flight.

Therapeutic—Curative or healing.

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ORGANIZATIONS

American Institute of Stress. 124 Park Avenue. Yonkers, NY 10703. (914) 963-1200. <<http://www.stress.org>>.

American Psychiatric Association (APA). 1400 K Street, NW, Washington, DC 20005. (888) 357-7924. <<http://www.psych.org>>.

Center for Mindfulness. University of Massachusetts Medical Center, 55 Lake Avenue North, Worcester, MA 01655. (508) 856-2656. <<http://www.umassmed.edu/cfm>>.

National Institute of Mental Health (NIMH) Office of Communications. 6001 Executive Boulevard, Room 8184, MSC 9663, Bethesda, MD 20892-9663. (866) 615-NIMH or (301) 443-4513. <<http://www.nimh.nih.gov>>.

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National Institute of Mental Health (NIMH). *Anxiety Disorders*. NIH Publication No. 02-3879. Bethesda, MD: NIMH, 2002.

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Stress test

Definition

Used to evaluate heart function, a **stress** test requires that a patient exercises on a treadmill or **exercise** bicycle while his or her heart rate, breathing, blood pressure, electrocardiogram (ECG), and feeling of well being are monitored.

Purpose

When the body is active, it requires more oxygen than when it is at rest, and, therefore, the heart has to pump more blood. Because of the increased stress on the heart, exercise can reveal coronary problems that are not apparent when the body is at rest. This is why the stress test, though not perfect, remains the best initial, noninvasive, practical coronary test.

The stress test helps doctors determine how well the heart handles the increased demands imposed by physical activity. It is particularly useful for evaluating possible **coronary artery disease**, detecting inadequate supply of oxygen-rich blood to the tissues of the heart muscle (**ischemia**), and determining safe levels of exercise in people with existing heart disease.

Precautions

The exercise stress test carries a very slight risk (1 in 100,000) of causing a **heart attack**. For this reason, the exercise stress test should be attended by a health care professional with a defibrillator and other emergency equipment on standby.

The patient must be aware of the symptoms of a heart attack and stop the test if he or she develops any of the following symptoms:

- an unsteady gait
- confusion
- skin is grayish or cold and clammy
- **dizziness** or **fainting**
- a drop in blood pressure
- chest **pain** (**angina**)
- irregular heart beat (cardiac arrhythmias)

Description

The technician affixes electrodes to specific areas of the patient's chest, using special adhesive patches with a special gel that conducts electrical impulses. Typically, electrodes are placed under each collarbone

and each bottom rib, and six electrodes are placed across the chest in a rough outline of the heart. Then the technician attaches wires from the electrodes to an ECG, which records the electrical activity picked up by the electrodes.

The technician runs resting ECG tests while the patient is lying down, then standing up, and then breathing heavily for half a minute. These tests can later be compared with the ECG tests performed while the patient is exercising. The patient's blood pressure is taken and the blood pressure cuff is left in place, so that blood pressure can be measured periodically throughout the test.

The patient begins riding a stationary bicycle or walking on a treadmill. Gradually the intensity of the exercise is increased. For example, if the patient is walking on a treadmill, the speed of the treadmill increases and the treadmill is tilted upward to simulate an incline. If the patient is on an exercise bicycle, the resistance or "drag" is gradually increased. The patient continues exercising at increasing intensity until he or she reaches his or her target heart rate (generally set at a minimum of 85% of the maximal predicted heart rate based on the patient's age) or experiences severe **fatigue**, dizziness, or chest pain. During this time, the patient's heart rate, ECG pattern, and blood pressure are continually monitored.

In some cases, other tests, such as **echocardiography** or thallium scanning, are also used in conjunction with the exercise stress test. For instance, recent studies suggest that women have a high rate of false negatives (results showing no problem when one exists) and false positives (results showing a problem when one does not exist) with the stress test. They may benefit from another test, such as exercise echocardiography. People who are unable to exercise may be injected with drugs that mimic the effects of exercise on the heart and given a thallium scan, which can detect the same abnormalities that an exercise test can.

Preparation

Patients are usually instructed not to eat or smoke for several hours before the test. They should also tell the doctor about any medications they are taking. They should wear comfortable sneakers and exercise clothing.

Aftercare

After the test, the patient should rest until blood pressure and heart rate return to normal. If all goes

KEY TERMS

Angina—Chest pain from a poor blood supply to the heart muscle due to narrowing of the coronary arteries.

Cardiac arrhythmia—An irregular heart rate or rhythm.

Coronary arteries—Two arteries that branch off from the aorta and supply blood to the heart.

Defibrillator—A device that delivers an electric shock to the heart muscle through the chest wall in order to restore a normal heart rate.

False negative—Test results showing no problem when one exists.

False positive—Test results showing a problem when one does not exist.

Hypertrophy—The overgrowth of muscle.

Ischemia—Diminished supply of oxygen-rich blood to an organ or area of the body.

well, and there are no signs of distress, the patient may return to his or her normal daily activities.

Risks

There is a very slight risk of a heart attack from the exercise, as well as cardiac arrhythmia (irregular heart beats), angina, or cardiac arrest (about one in 100,000).

Normal results

A normal result of an exercise stress test shows normal electrocardiogram tracings and heart rate, blood pressure within the normal range, and no angina, unusual dizziness, or **shortness of breath**.

Abnormal results

A number of abnormalities may show up on an exercise stress test. An abnormal electrocardiogram (ECG) may indicate deprivation of oxygen-rich blood to the heart muscle (ST wave segment depression, for example), heart rhythm disturbances, or structural abnormalities of the heart, such as overgrowth of muscle (hypertrophy). If the blood pressure rises too high or the patient experiences distressing symptoms during the test, the heart may be unable to handle the increased workload. Stress test abnormalities usually require further evaluation and therapy.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>> .

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>> .

Robert Scott Dinsmoor

Stridor

Definition

Stridor is a term used to describe noisy breathing in general, and to refer specifically to a high-pitched crowing sound associated with **croup**, respiratory infection, and airway obstruction.

Description

Stridor occurs when erratic air currents attempt to force their way through breathing passages narrowed by:

- illness
- infection
- the presence of **foreign objects**
- throat abnormalities

Stridor can usually be heard from a distance but is sometimes audible only during deep breathing. Someone who has stridor may crow and wheeze when:

- inhaling
- exhaling
- inhaling and exhaling

Most common in young children, whose naturally small airways are easily obstructed, stridor can be a symptom of a life-threatening respiratory emergency.

Causes and symptoms

During childhood, stridor is usually caused by infection of the cartilage flap (epiglottitis) that covers the opening of the windpipe to prevent **choking** during swallowing. It can also be caused by a toy or other tiny object the child has tried to swallow.

Laryngomalacia is a common cause of a rapid, low-pitched form of stridor that may be heard when

a baby inhales. This harmless condition does not require medical attention. It usually disappears by the time the child is 18 months old.

The most common causes of stridor in adults are:

- abscess or swelling of the upper airway
- paralysis or malfunction of the vocal cords
- tumor.

Other common causes of stridor include:

- enlargement of the thyroid gland (**goiter**)
- swelling of the voice box (laryngeal **edema**)
- narrowing of the windpipe (tracheal stenosis)

When stridor is caused by a condition that slowly narrows the airway, crowing and **wheezing** may not develop until the obstruction has become severe.

Diagnosis

When stridor is present in a newborn, pediatricians and neonatologists look for evidence of:

- heart defects inherent at birth (congenital)
- neurological disorders
- General toxicity.

If examinations do not reveal the reasons for the baby's noisy breathing, the air passages are assumed to be the cause of the problem.

Listening to an older child or adult breathe usually enables pediatricians, family physicians, and pulmonary specialists to estimate where an airway obstruction is located. The extent of the obstruction can be calculated by assessing the patient's:

- complexion
- chest movements
- breathing rate
- level of consciousness

X rays and direct examination of the voice box (larynx) and breathing passages indicate the exact location of the obstruction or inflammation. Flow-volume loops and pulse oximetry are diagnostic tools used to measure how much air flows through the breathing passages, and how much oxygen those passages contain.

Pulmonary function tests may also be performed.

Treatment

The cause of this condition determines the way it is treated.

Life-threatening emergencies may require:

- the insertion of a breathing tube through the mouth and nose (tracheal intubation)
- the insertion of a breathing tube directly into the windpipe (tracheostomy)

Resources

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Maureen Haggerty

Stroke

Definition

A stroke is the sudden death of brain cells in a localized area due to inadequate blood flow.

Description

A stroke occurs when blood flow is interrupted to part of the brain. Without blood to supply oxygen and nutrients and to remove waste products, brain cells quickly begin to die. Depending on the region of the brain affected, a stroke may cause **paralysis**, speech impairment, loss of memory and reasoning ability, **coma**, or death. A stroke also is sometimes called a brain attack or a cerebrovascular accident (CVA).

Some important stroke statistics include:

- more than one-half million people in the United States experience a new or recurrent stroke each year
- stroke is the third leading cause of death in the United States and the leading cause of disability
- stroke kills about 160,000 Americans each year, or almost one out of three stroke victims
- three million Americans are currently permanently disabled from stroke
- in the United States, stroke costs about \$30 billion per year in direct costs and loss of productivity
- two-thirds of strokes occur in people over age 65 but they can occur at any age
- strokes affect men more often than women, although women are more likely to die from a stroke

- strokes affect blacks more often than whites, and are more likely to be fatal among blacks

Stroke is a medical emergency requiring immediate treatment. Prompt treatment improves the chances of survival and increases the degree of recovery that may be expected. A person who may have suffered a stroke should be seen in a hospital emergency room without delay. Treatment to break up a blood clot, the major cause of stroke, must begin within three hours of the stroke to be effective. Improved medical treatment of all types of stroke has resulted in a dramatic decline in death rates in recent decades. In 1950, nine in ten died from stroke, compared to slightly less than one in three in the twenty-first century. However, about two-thirds of stroke survivors will have disabilities ranging from moderate to severe.

Causes and symptoms

Causes

There are four main types of stroke. Cerebral thrombosis and cerebral **embolism** are caused by **blood clots** that block an artery supplying the brain, either in the brain itself or in the neck. These account for 70–80% of all strokes. **Subarachnoid hemorrhage** and intracerebral hemorrhage occur when a blood vessel bursts around or in the brain.

Cerebral thrombosis occurs when a blood clot, or thrombus, forms within the brain itself, blocking the flow of blood through the affected vessel. Clots most often form due to “hardening” (**atherosclerosis**) of brain arteries. Cerebral thrombosis occurs most often at night or early in the morning. Cerebral thrombosis is often preceded by a **transient ischemic attack**, or TIA, sometimes called a “mini-stroke.” In a TIA, blood flow is temporarily interrupted, causing short-lived stroke-like symptoms. Recognizing the occurrence of a TIA, and seeking immediate treatment, is an important step in stroke prevention.

Cerebral embolism occurs when a blood clot from elsewhere in the circulatory system breaks free. If it becomes lodged in an artery supplying the brain, either in the brain or in the neck, it can cause a stroke. The most common cause of cerebral embolism is atrial fibrillation, a disorder of the heart beat. In atrial fibrillation, the upper chambers (atria) of the heart beat weakly and rapidly, instead of slowly and steadily. Blood within the atria is not completely emptied. This stagnant blood may form clots within the atria, which can then break off and enter the circulation. Atrial fibrillation is a factor in about 15% of all strokes. The risk of a stroke from atrial fibrillation can be dramatically reduced with daily use of anticoagulant medication.

Hemorrhage, or bleeding, occurs when a blood vessel breaks, either from trauma or excess internal pressure. The vessels most likely to break are those with preexisting defects such as an aneurysm. An aneurysm is a “pouching out” of a blood vessel caused by a weak arterial wall. Brain aneurysms are surprisingly common. According to **autopsy** studies, about 6% of all Americans have them. Aneurysms rarely cause symptoms until they burst. Aneurysms are most likely to burst when blood pressure is highest, and controlling blood pressure is an important preventive strategy.

Intracerebral hemorrhage affects vessels within the brain itself, while subarachnoid hemorrhage affects arteries at the brain’s surface, just below the protective arachnoid membrane. Intracerebral hemorrhages represent about 10% of all strokes, while subarachnoid hemorrhages account for about 7%.

In addition to depriving affected tissues of blood supply, the accumulation of fluid within the inflexible skull creates excess pressure on brain tissue, which can quickly become fatal. Nonetheless, recovery may be more complete for a person who survives hemorrhage than for one who survives a clot, because the blood deprivation effects usually are not as severe.

Death of brain cells triggers a chain reaction in which toxic chemicals created by cell death affect other nearby cells. This is one reason why prompt treatment can have such a dramatic effect on final recovery.

Risk factors

Risk factors for stroke involve age, sex, heredity, predisposing diseases or other medical conditions, use of certain medications, and lifestyle choices:

- **Age and sex.** The risk of stroke increases with age, doubling for each decade after age 55. Men are more likely to have a stroke than women.
- **Heredity.** Blacks, Asians, and Hispanics have higher rates of stroke than do whites, related partly to higher blood pressure. People with a family history of stroke are at greater risk.
- **Diseases.** Stroke risk is increased for people with diabetes, heart disease (especially atrial fibrillation), high blood pressure, prior stroke, or TIA. Risk of stroke increases tenfold for someone with one or more TIAs.
- **Other medical conditions.** Stroke risk increases with **obesity**, high blood cholesterol level, or high red blood cell count.
- **Hormone replacement therapy.** In mid-2003, a large clinical trial called the Women’s Health Initiative was halted when researchers discovered several

potentially dangerous effects of combined hormone replacement therapy on postmenopausal women. In addition to increasing the risk of some cancers and **dementia**, combined estrogen and progesterone therapy increased risk of ischemic stroke by 31% among study participants.

- Lifestyle choices. Stroke risk increases with cigarette **smoking** (especially if combined with the use of **oral contraceptives**), low level of physical activity, alcohol consumption above two drinks per day, or use of **cocaine** or intravenous drugs.

Symptoms

Symptoms of an embolic stroke usually come on quite suddenly and are at their most intense right from the start, while symptoms of a thrombotic stroke come on more gradually. Symptoms may include:

- blurring or decreased vision in one or both eyes
- severe **headache**, often described as “the worst headache of my life”
- weakness, **numbness**, or paralysis of the face, arm, or leg, usually confined to one side of the body
- dizziness, loss of balance or coordination, especially when combined with other symptoms

Diagnosis

The diagnosis of stroke is begun with a careful medical history, especially concerning the onset and distribution of symptoms, presence of risk factors, and the exclusion of other possible causes. A brief neurological exam is performed to identify the degree and location of any deficits, such as weakness, incoordination, or visual losses.

Once stroke is suspected, a computed tomography scan (CT scan) or **magnetic resonance imaging** (MRI) scan is performed to distinguish a stroke caused by blood clot from one caused by hemorrhage, a critical distinction that guides therapy. Blood and urine tests are done routinely to look for possible abnormalities.

Other investigations that may be performed to guide treatment include an electrocardiogram, **angiography**, ultrasound, and electroencephalogram.

Treatment

Emergency treatment

Emergency treatment of stroke from a blood clot is aimed at dissolving the clot. This “thrombolytic therapy” currently is performed most often with tissue plasminogen activator, or t-PA. t-PA must be administered

within three hours of the stroke event. Therefore, patients who awaken with stroke symptoms are ineligible for t-PA therapy, as the time of onset cannot be accurately determined. t-PA therapy has been shown to improve recovery and decrease long-term disability in selected patients. t-PA therapy carries a 6.4% risk of inducing a cerebral hemorrhage, and is not appropriate for patients with bleeding disorders, very high blood pressure, known aneurysms, any evidence of intracranial hemorrhage, or incidence of stroke, head trauma, or intracranial surgery within the past three months. Patients with clot-related (thrombotic or embolic) stroke who are ineligible for t-PA treatment may be treated with heparin or other blood thinners, or with **aspirin** or other anti-clotting agents in some cases.

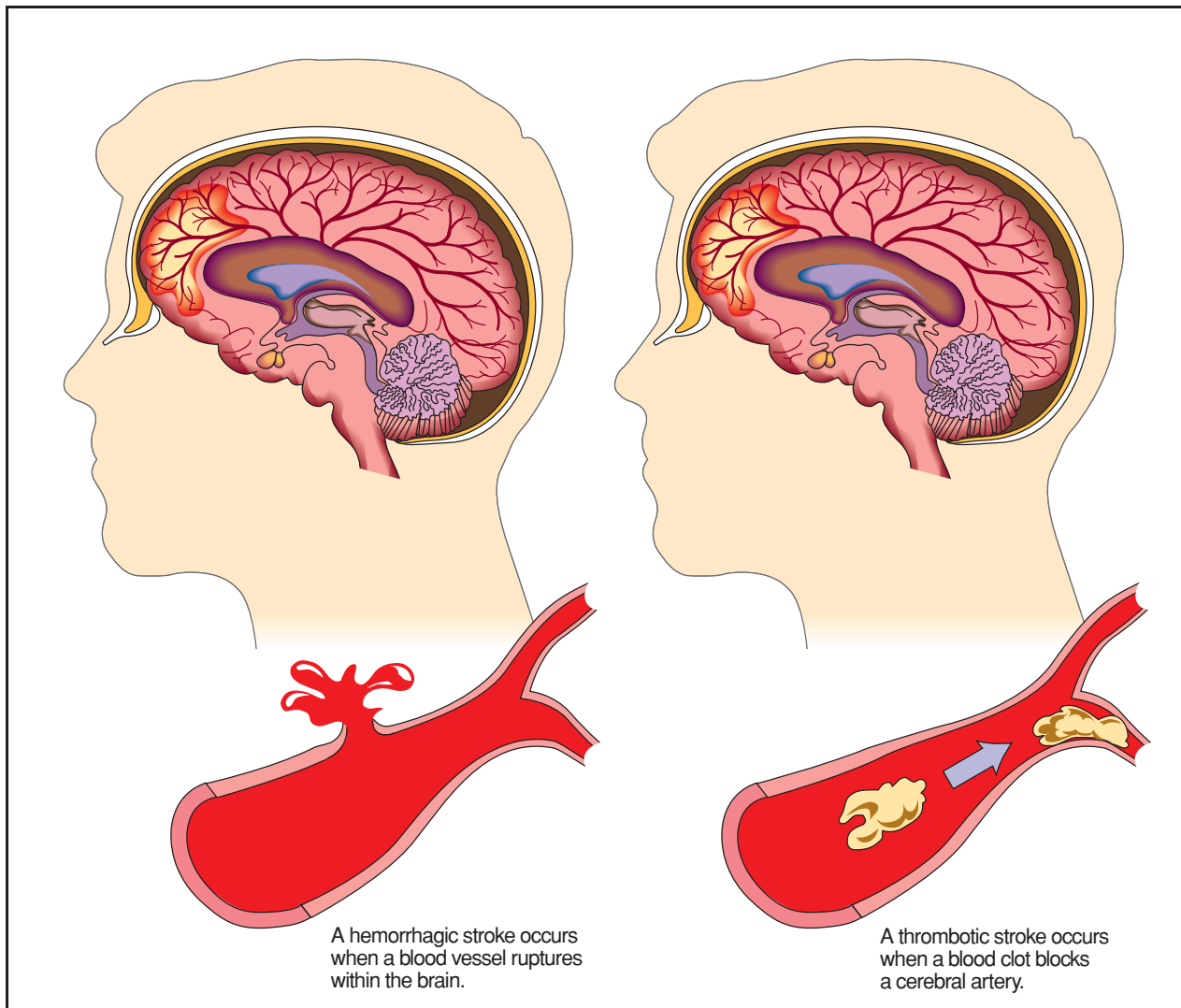
Emergency treatment of hemorrhagic stroke is aimed at controlling intracranial pressure. Intravenous urea or mannitol plus hyperventilation is the most common treatment. **Corticosteroids** also may be used. Patients with reversible bleeding disorders, such as those due to anticoagulant treatment, should have these bleeding disorders reversed, if possible.

Surgery for hemorrhage due to aneurysm may be performed if the aneurysm is close enough to the cranial surface to allow access. Ruptured vessels are closed off to prevent rebleeding. For aneurysms that are difficult to reach surgically, endovascular treatment may be used. In this procedure, a catheter is guided from a larger artery up into the brain to reach the aneurysm. Small coils of wire are discharged into the aneurysm, which plug it up and block off blood flow from the main artery.

Rehabilitation

Rehabilitation refers to a comprehensive program designed to regain function as much as possible and compensate for permanent losses. Approximately 10% of stroke survivors are without any significant disability and able to function independently. Another 10% are so severely affected that they must remain institutionalized for severe disability. The remaining 80% can return home with appropriate therapy, training, support, and care services.

Rehabilitation is coordinated by a team of medical professionals and may include the services of a neurologist, a physician who specializes in rehabilitation medicine (physiatrist), a physical therapist, an occupational therapist, a speech-language pathologist, a nutritionist, a mental health professional, and a social worker. Rehabilitation services may be provided in an acute care hospital, rehabilitation hospital, long-term care facility, outpatient clinic, or at home.



A hemorrhagic stroke (left) compared to a thrombotic stroke (right). (Illustration by Hans & Cassady, Inc.)

The rehabilitation program is based on the patient's individual deficits and strengths. Strokes on the left side of the brain primarily affect the right half of the body, and vice versa. In addition, in left brain dominant people, who constitute a significant majority of the population, left brain strokes usually lead to speech and language deficits, while right brain strokes may affect spatial perception. Patients with right brain strokes also may deny their illness, neglect the affected side of their body, and behave impulsively.

Rehabilitation may be complicated by cognitive losses, including diminished ability to understand and follow directions. Poor results are more likely in patients with significant or prolonged cognitive changes, sensory losses, language deficits, or incontinence.

PREVENTING COMPLICATIONS. Rehabilitation begins with prevention of stroke recurrence and other medical complications. The risk of stroke recurrence may be reduced with many of the same measures used to prevent stroke, including quitting smoking and controlling blood pressure.

One of the most common medical complications following stroke is deep venous thrombosis, in which a clot forms within a limb immobilized by paralysis. Clots that break free often become lodged in an artery feeding the lungs. This type of **pulmonary embolism** is a common cause of death in the weeks following a stroke. Resuming activity within a day or two after the stroke is an important preventive measure, along with use of elastic stockings on the lower limbs. Drugs that

prevent clotting may be given, including intravenous heparin and oral warfarin.

Weakness and loss of coordination of the swallowing muscles may impair swallowing (dysphagia), and allow food to enter the lower airway. This may lead to aspiration **pneumonia**, another common cause of death shortly after a stroke. Dysphagia may be treated with retraining exercises and temporary use of pureed foods.

Depression occurs in 30–60% of stroke patients. Antidepressants and psychotherapy may be used in combination.

Other medical complications include urinary tract infections, pressure ulcers, falls, and seizures.

TYPES OF REHABILITATIVE THERAPY. Brain tissue that dies in a stroke cannot regenerate. In some cases, the functions of that tissue may be performed by other brain regions after a training period. In other cases, compensatory actions may be developed to replace lost abilities.

Physical therapy is used to maintain and restore range of motion and strength in affected limbs, and to maximize mobility in walking, wheelchair use, and transferring (from wheelchair to toilet or from standing to sitting, for instance). The physical therapist advises on mobility aids such as wheelchairs, braces, and canes. In the recovery period, a stroke patient may develop muscle spasticity and **contractures**, or abnormal contractions. Contractures may be treated with a combination of stretching and splinting.

Occupational therapy improves self-care skills such as feeding, bathing, and dressing, and helps develop effective compensatory strategies and devices for activities of daily living. A speech-language pathologist focuses on communication and swallowing skills. When dysphagia is a problem, a nutritionist can advise alternative meals that provide adequate **nutrition**.

Mental health professionals may be involved in the treatment of depression or loss of thinking (cognitive) skills. A social worker may help coordinate services and ease the transition out of the hospital back into the home. Both social workers and mental health professionals may help counsel the patient and family during the difficult rehabilitation period. Caring for a person affected with stroke requires learning a new set of skills and adapting to new demands and limitations. Home caregivers may develop **stress**, **anxiety**, and depression. Caring for the caregiver is an important part of the overall stroke treatment program.

Support groups can provide an important source of information, advice, and comfort for stroke

patients and for caregivers. Joining a support group can be one of the most important steps in the rehabilitation process.

Prognosis

Stroke is fatal for about 27% of white males, 52% of black males, 23% of white females, and 40% of black females. Stroke survivors may be left with significant deficits. Emergency treatment and comprehensive rehabilitation can significantly improve both survival and recovery. A 2003 study found that treating people who have had a stroke with certain antidepressant medications, even if they were not depressed, could increase their chances of living longer. People who received the treatment were less likely to die from cardiovascular events than those who did not receive **antidepressant drugs**.

Prevention

Damage from stroke may be significantly reduced through emergency treatment. Knowing the symptoms of stroke is as important as knowing those of a **heart attack**. Patients with stroke symptoms should seek emergency treatment without delay, which may mean dialing 911 rather than their family physician.

The risk of stroke can be reduced through lifestyle changes:

- quitting smoking
- controlling blood pressure
- getting regular exercise
- keeping body weight down
- avoiding excessive alcohol consumption
- getting regular checkups and following the doctor's advice regarding diet and medicines, particularly hormone replacement therapy.

Treatment of atrial fibrillation may significantly reduce the risk of stroke. Preventive anticoagulant therapy may benefit those with untreated atrial fibrillation. Warfarin (Coumadin) has proven to be more effective than aspirin for those with higher risk. A new drug called ximelagatran (Exanta) with fewer side effects has been introduced in Europe. The drug's manufacturer was applying for FDA approval to market the drug for use in preventing stroke and other thromboembolic complications in early 2004.

In 2003, physicians at the Framingham Heart Study derived new risk scores to help physicians determine which patients with new onset of atrial fibrillation are at higher risk for stroke alone or for stroke or

KEY TERMS

Aneurysm—A pouchlike bulging of a blood vessel. Aneurysms can rupture, leading to stroke.

Atrial fibrillation—A disorder of the heart beat associated with a higher risk of stroke. In this disorder, the upper chambers (atria) of the heart do not completely empty when the heart beats, which can allow blood clots to form.

Cerebral embolism—A blockage of blood flow through a vessel in the brain by a blood clot that formed elsewhere in the body and traveled to the brain.

Cerebral thrombosis—A blockage of blood flow through a vessel in the brain by a blood clot that formed in the brain itself.

Intracerebral hemorrhage—A cause of some strokes in which vessels within the brain begin bleeding.

Subarachnoid hemorrhage—A cause of some strokes in which arteries on the surface of the brain begin bleeding.

Tissue plasminogen activator (tPA)—A substance that is sometimes given to patients within three hours of a stroke to dissolve blood clots within the brain.

death. Screening for aneurysms may be an effective preventive measure in those with a family history of aneurysms or autosomal **polycystic kidney disease**, which tends to be associated with aneurysms.

Resources

PERIODICALS

“HRT Increases Risk of Dementia and Stroke.”

Contemporary OB/GYN July 2003: 16–21.

“New Classification Scheme Helpful to Predict Risk of Stroke or Death.” *Heart Disease Weekly* September 14, 2003: 3.

“New Drug Application Submitted to FDA for Exanta.”

Heart Disease Weekly January 25, 2004: 79.

“New Stroke Prevention Drug.” *Chemist & Druggist*

September 13, 2003: 24.

“Post-stroke Antidepressant Treatment Appears to Reduce Death Rate.” *Heart Disease Weekly* October 26, 2003: 56.

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Stroke Association. 9707 E. Easter Lane, Englewood, Co. 80112. (800) 787-6537. <<http://www.stroke.org>>.

Richard Robinson
Teresa G. Odle

Strongyloidiasis see **Threadworm infection**

Structural integration see **Rolfing**

Stupor see **Coma**

Stuttering

Definition

Stuttering is a speech problem characterized by repetitions, pauses, or drawn out syllables, words, and phrases. Stutterers are different than people experiencing normal fluency problems because a stutterer's disfluency is more severe and consistent than that of people who do not stutter.

Description

Normal language development in a child can include a period of disfluency. Children might repeat syllables or words once or twice. Sometimes, children experiencing normal disfluencies hesitate during speech or use fillers, including “um,” with frequency. These developmental problems usually happen between one and five years of age. Often, parents are concerned about the disfluency they hear in their children. In fact, about 25% of all children experience speech disfluencies during development concern their parents because of their severity.

A child with mild stuttering, however, will repeat sounds more than twice. Parents and teachers often notice the child's facial muscles become tense and he or she might struggle to speak. The child's voice pitch might rise with repetitions, and some children experience occasional periods when airflow or voice stops for seconds at a time. Children with more severe stuttering stutter through more than 10% of their speech. This child exhibits considerable tension and tries to avoid stuttering by using different words. In these children, complete blocks of speech are more common than repetitions or prolongations, during which children lengthen syllables or words.

Stuttering usually begins in childhood when the child is developing language skills, and it rarely

develops in adulthood with only 1% of the population affected by the disorder. Stuttering does not affect intelligence. Teens often experience more noticeable problems with stuttering as they enter the dating scene and increase their social interactions. Stuttering can severely affect one's life. Often, adults who are concerned about stuttering choose their careers based on the disability.

The degree of stuttering is often inconsistent. Stutterers can be fluent in some situations. Many find that they stop stuttering when singing or doing other activities involving speech. Some have good and bad days when it comes to stuttering. On good days, a stutterer might be able to talk fluently using words that usually cause him to repeat, pause or prolongate sounds, syllables, parts of words, entire words, or phrases.

Causes and symptoms

There is no known cause of stuttering. Some believe that it has a physical cause and that it might be related to a breakdown in the neurological system. Stuttering starts early in life and often is inherited. Brain scan research has revealed that there might be abnormalities in the brains of stutterers, while they are stuttering. Myths about why stuttering occurs abound. Some cultures believe that stuttering is caused by emotional problems, tickling an infant too much or because a mother ate improperly during breastfeeding. None have been proven to be true. It is believed that some drugs might induce stuttering-like conditions. These include antidepressants, **antihistamines**, tranquilizers and **selective serotonin reuptake inhibitors**.

Diagnosis

Speech and language therapists diagnose stuttering by asking stutterers to read out loud, pronounce specific words, and talk. Some also order hearing tests. The tests will determine whether or not a person needs speech therapy.

Treatment

Researchers don't understand what causes stuttering. However, progress has been made into what contributes to the development of the disability and, therefore, in some cases it can be prevented in childhood with the help of therapy early on. Therapy can help people of all ages suffering from the speech disability. While not an overnight cure, therapy can offer positive results and more fluent speech patterns. The goals of therapy are to reduce stuttering frequency,

KEY TERMS

Antipsychotics—A class of drugs used to treat psychotic or neurotic behavior.

Disfluency—An interruption in speech flow.

Neuroleptics—Antipsychotic drugs that affect psychomotor activity.

decrease the tension and struggle of stuttering, become educated about stuttering, and learn to use effective communications skills, such as making eye contact, to further enhance speech. The therapy focuses on helping stutterers to discover easier and different ways of producing sounds and expressing thoughts. The success of therapy depends largely on the stutterer's willingness to work at getting better.

The duration of stuttering therapy needed varies among stutterers. Sometimes, it helps stutterers if they have therapy intermittently throughout their lives.

Parents, teachers and others can do things to help ease stuttering. These include: talking slowly, but normally, clearly, and in a relaxed manner to a stutterer; answering questions after a pause to encourage a relaxed transaction; trying not to make stuttering worse by getting annoyed by a person's stuttering; giving stutterers reassurance about their stuttering; and encourage the stutterer to talk about his or her stuttering.

Electronic fluency aids help some stutterers when used as an adjunct to therapy. Medications, such as antipsychotics and neuroleptics, have been used to treat stuttering with limited success.

Alternative treatment

Some use relaxation techniques to help their stuttering.

Prognosis

More than three million Americans stutter and four times more males are affected than females. Winston Churchill, Marilyn Monroe, Carly Simon, James Earl Jones and King George VI are among the many people who stuttered but went on to live successful professional lives. Decades of research have yielded no answers to the causes of stuttering; still much has been learned about what contributes to stuttering's development and how to prevent it in children. People who stutter can get better through therapy.

Prevention

New and exciting developments are occurring in researchers' understanding of the genetics of stuttering. Researchers are finding the locations of genes that predispose people to stuttering. While genetic factors will not explain all stuttering, genetics will help to uncover the disability's causes. Speech therapy, especially that performed at a young age, can stop the progression of stuttering.

Resources

ORGANIZATIONS

National Stuttering Foundation of America. 1-(800) 992-9392. <<http://www.stutteringhelp.org>>.

OTHER

The Stuttering Home Page. Minnesota State University, Mankato. <<http://www.mandato.msus.edu/deprt/comdis/kuster/stutter.html>>.

"Stuttering." The Nemours Foundation, KidsHealth.org. <<http://kidshealth.org>>.

"What is Stuttering?" Robert W. Quesal, PhD, Professor and Program Director. Communications Sciences and Disorders. Western Illinois University. <<http://www.wiu.edu>>.

Lisette Hilton

See also **Eyelid disorders**

Subacute sclerosing panencephalitis

Definition

Subacute sclerosing panencephalitis is a rare, progressive brain disorder caused by an abnormal immune response to the **measles** virus.

Description

This fatal condition is a complication of measles, and affects children and young adults before the age of 20. It usually occurs in boys more often than in girls, but is extremely rare, appearing in only one out of a million cases of measles.

Causes and symptoms

Experts believe this condition is a form of measles **encephalitis** (swelling of the brain), caused by an improper response by the immune system to the measles virus.

KEY TERMS

Measles encephalitis—A serious complication of measles occurring in about one out of every 1,000 cases, causing headache, drowsiness, and vomiting seven to ten days after the rash appears. Seizures and coma can follow, which may lead to retardation and death.

The condition begins with behavioral changes, memory loss, irritability, and problems with school work. As the neurological damage increases, the child experiences seizures, involuntary movements, and further neurological deterioration. Eventually, the child starts suffering from progressive **dementia**. The optic nerve begins to shrink and weaken (atrophy) and subsequently the child becomes blind.

Diagnosis

Blood tests and spinal fluid reveal high levels of antibodies to measles virus, and there is a characteristically abnormal electroencephalogram (EEG), or brain wave test. Typically, there is a history of measles infection two to ten years before symptoms begin.

Treatment

There is no standard treatment, and a number of **antiviral drugs** have been tested with little success. Treatment of symptoms, including the use of **anticonvulsant drugs**, can be helpful.

Prognosis

While there may be periodic remissions during the course of this disease, it is usually fatal (often from **pneumonia**) within one to three years after onset.

Resources

ORGANIZATIONS

National Institute of Allergy and Infectious Disease. Building 31, Room 7A-50, 31 Center Drive MSC 2520, Bethesda, MD 20892-2520. (301) 496-5717. <<http://www.niaid.nih.gov/default.htm>>.

National Organization for Rare Disorders. P.O. Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673. <<http://www.rarediseases.org>>.

Carol A. Turkington

Subacute spongiform encephalopathy see **Creutzfeldt-Jakob disease**

Subacute thyroiditis see **Thyroiditis**

Subarachnoid hemorrhage

Definition

A subarachnoid hemorrhage is an abnormal and very dangerous condition in which blood collects beneath the arachnoid mater, a membrane that covers the brain. This area, called the subarachnoid space, normally contains cerebrospinal fluid. The accumulation of blood in the subarachnoid space can lead to **stroke**, seizures, and other complications. Additionally, subarachnoid hemorrhages may cause permanent brain damage and a number of harmful biochemical events in the brain. A subarachnoid hemorrhage and the related problems are frequently fatal.

Description

Subarachnoid hemorrhages are classified into two general categories: traumatic and spontaneous. Traumatic refers to brain injury that might be sustained in an accident or a fall. Spontaneous subarachnoid hemorrhages occur with little or no warning and are frequently caused by ruptured aneurysms or blood vessel abnormalities in the brain.

Traumatic brain injury is a critical problem in the United States. According to annual figures compiled by the Brain Injury Association, approximately 373,000 people are hospitalized, more than 56,000 people die, and 99,000 survive with permanent disabilities due to traumatic brain injuries. The leading causes of injury are bicycle, motorcycle, and automobile accidents, with a significant minority due to accidental falls, and sports and recreation mishaps.

Exact statistics are not available on traumatic subarachnoid hemorrhages, but several large clinical studies have found an incidence of 23–39% in relation to severe **head injury**. Furthermore, subarachnoid hemorrhages have been described in the medical literature as the most common brain injury found during **autopsy** investigations of head trauma.

Spontaneous subarachnoid hemorrhages are often due to an aneurysm (a bulge or sac-like projection from a blood vessel) which bursts. **Arteriovenous malformations** (AVMs), which are abnormal interfaces between arteries and veins, may also rupture

and release blood into the subarachnoid space. Both aneurysms and AVMs are associated with weak spots in the walls of blood vessels and account for approximately 60% of all spontaneous subarachnoid hemorrhages. The rest may be attributed to other causes, such as **cancer** or infection, or are of unknown origin.

In industrialized countries, it is estimated that there are 6.5–26.4 cases of spontaneous subarachnoid hemorrhage per 100,000 people annually. Certain factors raise the risk of suffering a hemorrhage. Aneurysms are acquired over a person's lifetime and are rarely a factor in subarachnoid hemorrhage before age 20. Conversely, AVMs are present at birth. In some cases, there may be a genetic predisposition for aneurysms or AVMs. Other factors that have been implicated, but not definitively linked to spontaneous subarachnoid hemorrhages, include **atherosclerosis**, cigarette use, extreme alcohol consumption, and the use of illegal drugs, such as **cocaine**. The exact role of high blood pressure is somewhat unclear, but since it does seem linked to the formation of aneurysms, it may be considered an indirect risk factor.

The immediate danger due to subarachnoid hemorrhage, whether traumatic or spontaneous, is **ischemia**. Ischemia refers to tissue damage caused by restricted or blocked blood flow. The areas of the brain that do not receive adequate blood and oxygen can suffer irreparable injury, leading to permanent brain damage or **death**. An individual who survives the initial hemorrhage is susceptible to a number of complications in the following hours, days, and weeks.

The most common complications are intracranial **hypertension**, vasospasm, and **hydrocephalus**. Intracranial hypertension, or high pressure within the brain, can lead to further bleeding from damaged blood vessels; a complication associated with a 70% fatality rate. Vasospasm, or blood vessel constriction, is a principal cause of secondary ischemia. The blood vessels in the brain constrict in reaction to chemicals released by blood breaking down within the subarachnoid space. As the blood vessels become narrower, blood flow in the brain becomes increasingly restricted. Approximately one third of spontaneous subarachnoid hemorrhages and 30–60% of traumatic bleeds are followed by vasospasm. Hydrocephalus, an accumulation of fluid in the chambers of the brain (ventricles) due to restricted circulation of cerebrospinal fluid, follows approximately 15% of subarachnoid hemorrhages. Because cerebrospinal fluid cannot drain properly, pressure accumulates on the brain, possibly prompting further ischemic complications.

Causes and symptoms

Whether through trauma or disease, subarachnoid hemorrhages are caused by blood being released by a damaged blood vessel and accumulating in the subarachnoid space. Symptoms associated with traumatic subarachnoid hemorrhage may or may not resemble those associated with spontaneous hemorrhage, as trauma can involve multiple injuries with overlapping symptoms.

Typically, a spontaneous subarachnoid hemorrhage is indicated by a sudden, severe **headache**. **Nausea, vomiting, and dizziness** frequently accompany the **pain**. Loss of consciousness occurs in about half the cases of spontaneous hemorrhage. A **coma**, usually brief, may occur. A stiff neck, **fever**, and aversion to light may appear following the hemorrhage. Neurologic symptoms may include partial **paralysis**, loss of vision, seizures, and speech difficulties.

Spontaneous subarachnoid hemorrhages may be preceded by warning signs prior to the initial bleed. Sentinel, or warning, headaches may be present in the days or weeks before an aneurysm or AVM ruptures. These headaches can be accompanied by dizziness, nausea, and vomiting, and possibly neurologic symptoms. Approximately 50% of AVMs are discovered before they bleed significantly; however, most aneurysms are not diagnosed before they rupture.

Diagnosis

To make a diagnosis, a health-care provider takes a detailed history of the symptoms and does a **physical examination**. The symptoms may mimic other disorders and diagnosis can be complicated, especially if the individual is unconscious. The sudden, severe headache can fuel suspicion of a subarachnoid hemorrhage or similar event, and a computed tomography scan (CT scan) or **magnetic resonance imaging (MRI)** scan is considered essential to a quick diagnosis. The MRI is less sensitive than the CT in detecting acute subarachnoid bleeding, but more sensitive in diagnosing AVM or aneurysm.

A CT scan reveals blood that has escaped into the subarachnoid space. For the best results, the scan should be done within 12 hours of the hemorrhage. If this is not possible, lumbar puncture and examination of the cerebrospinal fluid is advised. Lumbar puncture is also done in cases in which the CT scan doesn't reveal a hemorrhage, but there is a high suspicion that one has occurred. In subarachnoid hemorrhage, cerebrospinal fluid shows red blood cells and/or xanthochromia, a yellowish tinge caused by blood

breakdown products. Xanthochromia first appears six to 12 hours after subarachnoid hemorrhage, making it advisable to delay lumbar puncture until at least 12 hours after the onset of symptoms for a more definite diagnosis.

Once a hemorrhage, AVM, or aneurysm has been diagnosed, further tests are done to pinpoint the damage. The CT scan may be useful in giving the general location, but cerebral **angiography** maps out the exact details. This procedure involves injecting a special dye into the blood stream. This dye makes blood vessels visible in x rays of the area.

Treatment

The initial course of treatment focuses on stabilizing the hemorrhage victim. Depending on the individual's condition, this may involve intubation and mechanical ventilation, supplemental oxygen, intravenous fluids, and close monitoring of vital signs. If the person suffers seizures, an anticonvulsant, such as phenytoin (Dilantin), is administered. Nimodipine, a calcium channel blocker, may be given to prevent vasospasm and its complications. Sedatives and medications for pain, nausea, and vomiting are administered as needed.

Once the individual is stabilized, cerebral angiography is done to locate the damaged blood vessel. This information and the individual's condition are considered before attempting surgical treatment. Surgery is necessary to remove the damaged area of the blood vessel and prevent a second hemorrhage. The specific neurosurgical procedures depend on the location and type of blood vessel damage. Typically, clip ligation is the preferred means of treating an aneurysm, and surgical excision, radiosurgery, or endovascular embolization are used to manage an AVM.

Prognosis

Individuals who are conscious and demonstrate few neurologic symptoms when they reach medical help have the best prognosis. However, the overall prospects for subarachnoid hemorrhage patients are generally not good. Of the individuals who suffer an aneurysmal hemorrhage, approximately 15% do not live long enough to get medical treatment. Another 20-40% will not survive the complications caused by the hemorrhage, and approximately 12% of the survivors will experience permanent neurologic disability. Neurologic disabilities may include partial paralysis, weakened or numbed areas of the body, cognitive or speech difficulties, and vision problems. Individuals

KEY TERMS

Aneurysm—A weak point in a blood vessel where the pressure of the blood causes the vessel wall to bulge outwards. An aneurysm may also appear as a sac-like projection from the blood vessel wall.

Arachnoid mater—One of three membranes that encase the brain and spinal cord. The arachnoid mater is the middle membrane.

Arteriovenous malformation—An abnormal tangle of arteries and veins in which the arteries feed directly into the veins without a normal intervening capillary bed.

Atherosclerosis—An abnormal condition in which lipids, or fats, form deposits on the inside walls of blood vessels.

Cerebral angiography—A medical test in which an x-ray visible dye is injected into blood vessels to allow them to be imaged on an x ray.

Cerebrospinal fluid—The clear, normally colorless fluid found within the subarachnoid space.

Computerized tomography (CT) scan—Cross-sectional x rays of the body compiled to create a three-dimensional image of the body's internal structures.

Hemorrhage—The escape of blood from blood vessels.

Hydrocephalus—Enlargement of the chambers in the brain (ventricles) caused by an accumulation of cerebrospinal fluid.

Intracranial hypertension—Abnormally high pressure within the brain.

Ischemia—A condition in which blood flow is cut off or restricted from a particular area. The tissue becomes starved of oxygen and nutrients, resulting in tissue death.

Ischemic—Referring to ischemia.

Lumbar puncture—A diagnostic procedure in which a needle is inserted into the lower spine to withdraw a small amount of cerebrospinal fluid. This fluid is examined to assess trauma to the brain.

Subarachnoid—Referring to the space underneath the arachnoid mater.

Vasospasm—The constriction or narrowing of blood vessels. In cases of hemorrhage, the constriction is prompted by chemical signals from the escaped blood as it breaks down.

whose subarachnoid hemorrhages occur as a result of AVMs have a slightly better prognosis, although the risk of death is approximately 10–15% for each hemorrhage.

Subarachnoid hemorrhage associated with traumatic brain injury has a poor prognosis. In clinical studies, 46–78% of head injury cases involving subarachnoid hemorrhage resulted in severe disability, vegetative survival, or death. Furthermore, it is possible that traumatic subarachnoid hemorrhages are accompanied by additional injuries, which would further diminish survival and recovery rates.

Prevention

Traumatic brain injury is the leading cause of subarachnoid hemorrhages, so it follows that efforts to prevent head injury would prevent these hemorrhages. Since accidents cannot always be prevented, measures to minimize potential damage are always advisable. Use of activity-appropriate protective gear, such as bicycle helmets, motorcycle helmets, and sports head gear, is strongly encouraged and promoted by medical associations, consumer organizations, advocacy

groups, and health-care professionals. These same groups also advise using seat belts in automobiles.

Spontaneous subarachnoid hemorrhages are more difficult to prevent. Since there may be a genetic component to aneurysms and AVMs, close relatives to individuals with these conditions may consider being screened to assess their own status. Quitting **smoking** and keeping blood pressure within normal limits may also reduce the risk of suffering a spontaneous subarachnoid hemorrhage.

Resources

ORGANIZATIONS

Brain Injury Association of America. 105 North Alfred St., Alexandria, VA 22314. (800) 444-6443. <<http://www.biausa.org>>.

National Stroke Association. 9707 E. Easter Lane, Englewood, Co. 80112. (800) 787-6537. <<http://www.stroke.org>>.

Julia Barrett

Subdural empyema see **Central nervous system infections**

Subdural hematoma

Definition

A subdural hematoma is a collection of blood in the space between the outer layer (dura) and middle layers of the covering of the brain (the meninges). It is most often caused by torn, bleeding veins on the inside of the dura as a result of a blow to the head.

Description

Subdural hematomas most often affect people who are prone to falling. Only a slight hit on the head or even a fall to the ground without hitting the head may be enough to tear veins in the brain, often without fracturing the skull. There may be no external evidence of the bruising on the brain's surface.

Small subdural hematomas may not be very serious, and the blood can be slowly absorbed over several weeks. Larger hematomas, however, can gradually enlarge over several weeks, even though the bleeding has stopped. This enlargement can compress the brain itself, possibly leading to **death** if the blood is not drained.

The time between the injury and the appearance of symptoms can vary from less than 48 hours to several weeks, or more. Symptoms appearing in less than 48 hours are due to an acute subdural hematoma. This type of bleeding is often fatal, and results from tearing of the venous sinus. If more than two weeks have passed before symptoms appear, the condition is called a chronic subdural hematoma, resulting from tearing of the smaller vein. The young and the old are most likely to experience a chronic condition. This chronic form is less risky, as pressure of the veins against the skull lessens the bleeding. Prompt medical care can reduce the probability of permanent brain damage.

Causes and symptoms

A subdural hematoma is caused by an injury to the head that tears blood vessels. In childhood, hematomas are a common complication of falls. A subdural hematoma also may be an indication of **child abuse**, as evidenced by **shaken baby syndrome**.

Symptoms tend to fluctuate, and include:

- **headache**
- episodes of confusion and drowsiness



Subdural hematoma present on autopsied body. (Custom Medical Stock Photo. Reproduced by permission.)

- one-sided weakness or **paralysis**
- lethargy
- enlarged or asymmetric pupils
- convulsions or loss of consciousness after **head injury**
- coma

A doctor should be contacted immediately if symptoms appear. Because these symptoms mimic the signs of a **stroke**, the patient should tell the doctor about any head injury within the previous few months.

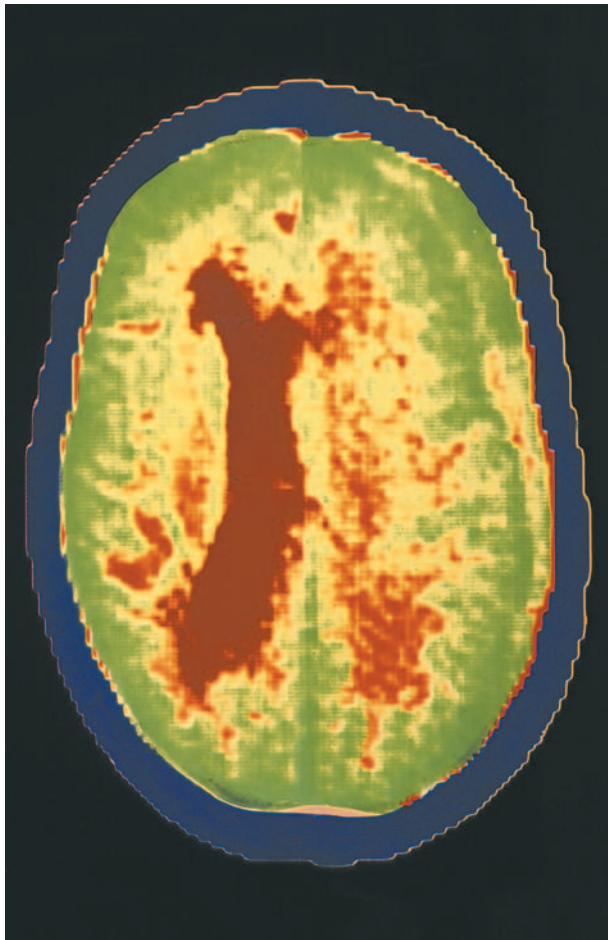
In an infant, symptoms may include increased pressure within the skull, growing head size, bulging fontanelle (one of two soft spots on a infant's skull), **vomiting**, irritability, lethargy, and seizures. In cases of **child abuse**, there may be **fractures** of the skull or other bones.

Diagnosis

A chronic subdural hematoma can be difficult to diagnose, but a slow loss of consciousness after a head injury is assumed to be a hematoma unless proven otherwise. The hematoma can be confirmed with **magnetic resonance imaging** (MRI), which is the preferred type of scan; a hematoma can be hard to detect on a computed tomography scan (CT scan), depending on how long after the hemorrhage the CT is done.

Treatment

Small hematomas that do not cause symptoms may not need to be treated. Otherwise, the hematoma should be surgically removed. Liquid blood can be drained from burr holes drilled into the skull. The surgeon may have to open a section of skull to remove a large hematoma or to tie off the bleeding vein.



CT scan indicating subdural hematoma highlighted as a red mass on the center left of the brain. (Photo Researchers. Reproduced by permission.)

KEY TERMS

Corticosteroids—A group of drugs similar to natural corticosteroid hormones produced by the adrenal glands. The drugs have a wide variety of applications, including use for inflammatory disorders and swelling.

Diuretics—A group of drugs that helps remove excess water from the body by increasing the amount lost by urination.

Fontanelle—One of the two soft areas on a baby's scalp; a membrane-covered gap between the bones of the skull.

Corticosteroids and **diuretics** can control brain swelling. After surgery, **anticonvulsant drugs** (such as phenytoin) may help control or prevent seizures, which can begin as late as two years after the head injury.

Prognosis

If treatment is provided soon enough, recovery is usually complete. Headache, **amnesia**, attention problems, **anxiety**, and giddiness may continue for some time after surgery. Most symptoms in adults usually disappear within six months, with further improvement over several years. Children tend to recover much faster.

Prevention

Because a subdural hematoma usually follows a head injury, preventing head injury can prevent a hematoma.

Resources

ORGANIZATIONS

American Academy of Neurology. 1080 Montreal Ave., St. Paul, MN 55116. (612) 695-1940. <<http://www.aan.com>>.

Brain Injury Association of America. 105 North Alfred St., Alexandria, VA 22314. (800) 444-6443. <<http://www.biausa.org>>.

Head Injury Hotline. P.O. Box 84151, Seattle WA 98124. (206) 621-8558. <<http://www.headinjury.com>>.

Head Trauma Support Project, Inc. 2500 Marconi Ave., Ste. 203, Sacramento, CA 95821. (916) 482-5770.

Carol A. Turkington

Subdural hemorrhage see **Subdural hematoma**

Subluxations see **Dislocations and subluxations**

Substance abuse and dependence

Definition

Substance **abuse** and dependence refer to any continued pathological use of a medication, non-medically indicated drug (called drugs of abuse), or toxin. They normally are distinguished as follows.

Substance abuse is any pattern of substance use that results in repeated adverse social consequences related to drug-taking—for example, interpersonal conflicts, failure to meet work, family, or school obligations, or legal problems. Substance dependence, commonly known as **addiction**, is characterized by physiological and behavioral symptoms related to substance use. These symptoms include the need for increasing amounts of the substance to maintain desired effects, withdrawal if drug-taking ceases, and a great deal of time spent in activities related to substance use.

Substance abuse is more likely to be diagnosed among those who have just begun taking drugs and is often an early symptom of substance dependence. However, substance dependence can appear without substance abuse, and substance abuse can persist for extended periods of time without a transition to substance dependence.

Description

Substance abuse and dependence are disorders that affect all population groups although specific patterns of abuse and dependence vary with age, gender, culture, and socioeconomic status. According to data from the National Longitudinal Alcohol Epidemiologic Survey, 13.3% of a survey group of Americans exhibited symptoms of alcohol dependence during their lifetime, and 4.4% exhibited symptoms of alcohol dependence during the past 12 months. According to the 1997 National Household Survey on Drug Abuse, 6.4% of those surveyed had used an illicit drug in the past month.

Although substance dependence can begin at any age, to people aged 18 to 24 have relatively high substance use rates, and dependence often arises sometime during the ages of 20 to 49. Gender proportions vary according to the class of drugs, but substance use disorders are in general more frequently seen in men. A 2004 report revealed that in a 2002 national survey, more than 2.6 million youths age 12 to 17 had used inhalants more than once.

In addition to being an individual health disorder, substance abuse and dependence may be viewed as a public health problem with far-ranging health, economic, and adverse social implications. Substance-related disorders are associated with teen **pregnancy** and the transmission of **sexually transmitted diseases** (STDs), as well as failure in school, unemployment, domestic violence, homelessness, and crimes such as

rape and sexual assault, aggravated assault, robbery, burglary, and larceny. According to the National Institute on Alcohol Abuse and **Alcoholism** (NIAAA), the estimated cost of alcohol-related disorders alone (including health care expenditures, lost productivity, and premature **death**) was \$166.5 billion in 1995.

The term substance, when discussed in the context of substance abuse and dependence, refers to medications, drugs of abuse, and toxins. These substances have an intoxicating effect, desired by the user, which can have either stimulating (speeding up) or depressive/sedating (slowing down) effects on the body. Substance dependence and/or abuse can involve any of the following 10 classes of substances:

- alcohol
- amphetamines (including “crystal meth,” some medications used in the treatment of attention deficit disorder [ADD], and amphetamine-like substances found in appetite suppressants)
- cannabis (including **marijuana** and hashish)
- cocaine (including “crack”)
- hallucinogens (including **LSD**, mescaline, and MDMA [“ecstasy”])
- inhalants (including compounds found in gasoline, glue, and paint thinners)
- nicotine (substance dependence only)
- opioids (including morphine, heroin, codeine, **methadone**, oxycodone [Oxycontin (TM)])
- phencyclidine (including PCP, angel dust, ketamine)
- sedative, hypnotic, and anxiolytic (anti-anxiety) substances (including **benzodiazepines** such as valium, **barbiturates**, prescription sleeping medications, and most prescription anti-anxiety medications)

Caffeine has been identified as a substance in this context, but as yet there is insufficient evidence to establish whether caffeine-related symptoms fall under substance abuse and dependence.

Substances of abuse may thus be illicit drugs, readily available substances such as alcohol or glue, over-the-counter drugs, or prescription medications. In many cases, a prescription medication that becomes a substance of abuse may have been a legal, medically indicated prescription for the user, but the pattern of use diverges from the use prescribed by the physician.

Frequency Of Substance Abuse By Gender And Age

Men	
Ages 18 to 29	17 to 24 percent
Ages 30 to 44	11 to 14 percent
Ages 45 to 64	6 to 8 percent
Over age 65	1 to 3 percent
Women	
Ages 18 to 29	4 to 10 percent
Ages 30 to 44	2 to 4 percent
Ages 45 to 64	1 to 2 percent
Over age 65	less than 1 percent

Causes and symptoms

Causes

The causes of substance dependence are not well established, but three factors are believed to contribute to substance-related disorders: genetic factors, psychopathology, and social learning. In genetic epidemiological studies of alcoholism, the probability of identical twins both exhibiting alcohol dependence was significantly greater than with fraternal twins, thus suggesting a genetic component in alcoholism. It is unclear, however, whether the genetic factor is related to alcoholism directly, or whether it is linked to other psychiatric disorders that are known to be associated with substance abuse. For example, there is evidence that alcoholic males from families with **depressive disorders** tend to have more severe courses of substance dependence than alcoholic men from families without such family histories.

These and other findings suggest substance use may be a way to relieve the symptoms of a psychological disorder. In this model, unless the underlying pathology is treated, attempts to permanently stop substance dependence are ineffective. Psychopathologies that are associated with substance dependence include antisocial personality disorder, **bipolar disorder**, depression, **anxiety disorder**, and **schizophrenia**.

A third factor related to substance dependence is social environment. In this model, drug-taking is essentially a socially learned behavior. Local social norms determine the likelihood that a person is exposed to the substance and whether continued use is reinforced. For example, individuals may, by observing family or peer role models, learn that substance use is a normal way to relieve daily stresses. External

penalties, such as legal or social sanctions, may reduce the likelihood of substance use.

At the level of neurobiology, it is believed that substances of abuse operate through similar pathways in the brain. The chemical changes induced by the stimulation of these pathways by initial use of the substance lead to the desire to continue substance use, and eventual substance dependence.

Symptoms

The DSM-IV-TR identifies seven criteria (symptoms), at least three of which must be met during a given 12-month period, for the diagnosis of substance dependence:

- Tolerance, as defined either by the need for increasing amounts of the substance to obtain the desired effect or by experiencing less effect with extended use of the same amount of the substance.
- Withdrawal, as exhibited either by experiencing unpleasant mental, physiological, and emotional changes when drug-taking ceases or by using the substance as a way to relieve or prevent withdrawal symptoms.
- Longer duration of taking substance or use in greater quantities than was originally intended.
- Persistent desire or repeated unsuccessful efforts to stop or lessen substance use.
- A relatively large amount of time spent in securing and using the substance, or in recovering from the effects of the substance.
- Important work and social activities reduced because of substance use.
- Continued substance use despite negative physical and psychological effects of use.

Although not explicitly listed in the DSM-IV-TR criteria, “craving,” or the overwhelming desire to use the substance regardless of countervailing forces, is a universally-reported symptom of substance dependence.

Symptoms of substance abuse, as specified by DSM-IV-TR, include one or more of the following occurring during a given 12-month period:

- Substance use resulting in a recurrent failure to fulfill work, school, or home obligations (work absences, substance-related school suspensions, neglect of children).
- Substance use in physically hazardous situations such as driving or operating machinery.

- Substance use resulting in legal problems such as drug-related arrests.
- Continued substance use despite negative social and relationship consequences of use.

In addition to the general symptoms, there are other physical signs and symptoms of substance abuse that are related to specific drug classes:

- Signs and symptoms of alcohol intoxication include such physical signs as slurred speech, lack of coordination, unsteady gait, memory impairment, and stupor, as well as behavior changes shortly after alcohol ingestion, including inappropriate aggressive behavior, mood volatility, and impaired functioning.
- Amphetamine users may exhibit rapid heartbeat, elevated or depressed blood pressure, dilated (enlarged) pupils, weight loss, as well as excessively high energy, inability to sleep, confusion, and occasional paranoid psychotic behavior.
- Cannabis users may exhibit red eyes with dilated pupils, increased appetite, **dry mouth**, and rapid pulse; they may also be sluggish and slow to react.
- Cocaine users may exhibit rapid heart rate, elevated or depressed blood pressure, dilated pupils, weight loss, in addition to wide variations in their energy level, severe mood disturbances, **psychosis**, and **paranoia**.
- Users of hallucinogens may exhibit anxiety or depression, paranoia, and unusual behavior in response to **hallucinations** (imagined sights, voices, sounds, or smells that appear real). Signs include dilated pupils, rapid heart rate, **tremors**, lack of coordination, and sweating. Flashbacks, or the re-experiencing of a hallucination long after stopping substance use, are also a symptom of hallucinogen use.
- Users of inhalants experience **dizziness**, spastic eye movements, lack of coordination, slurred speech, and slowed reflexes. Associated behaviors may include belligerence, tendency toward violence, apathy, and impaired judgment.
- Opioid drug users exhibit slurred speech, drowsiness, impaired memory, and constricted (small) pupils. They may appear slowed in their physical movements.
- Phencyclidine users exhibit spastic eye movements, rapid heartbeat, decreased sensitivity to **pain**, and lack of muscular coordination. They may show belligerence, predisposition to violence, impulsiveness, and agitation.

- Users of sedative, hypnotic, or anxiolytic drugs show slurred speech, unsteady gait, inattentiveness, and impaired memory. They may display inappropriate behavior, mood volatility, and impaired functioning.

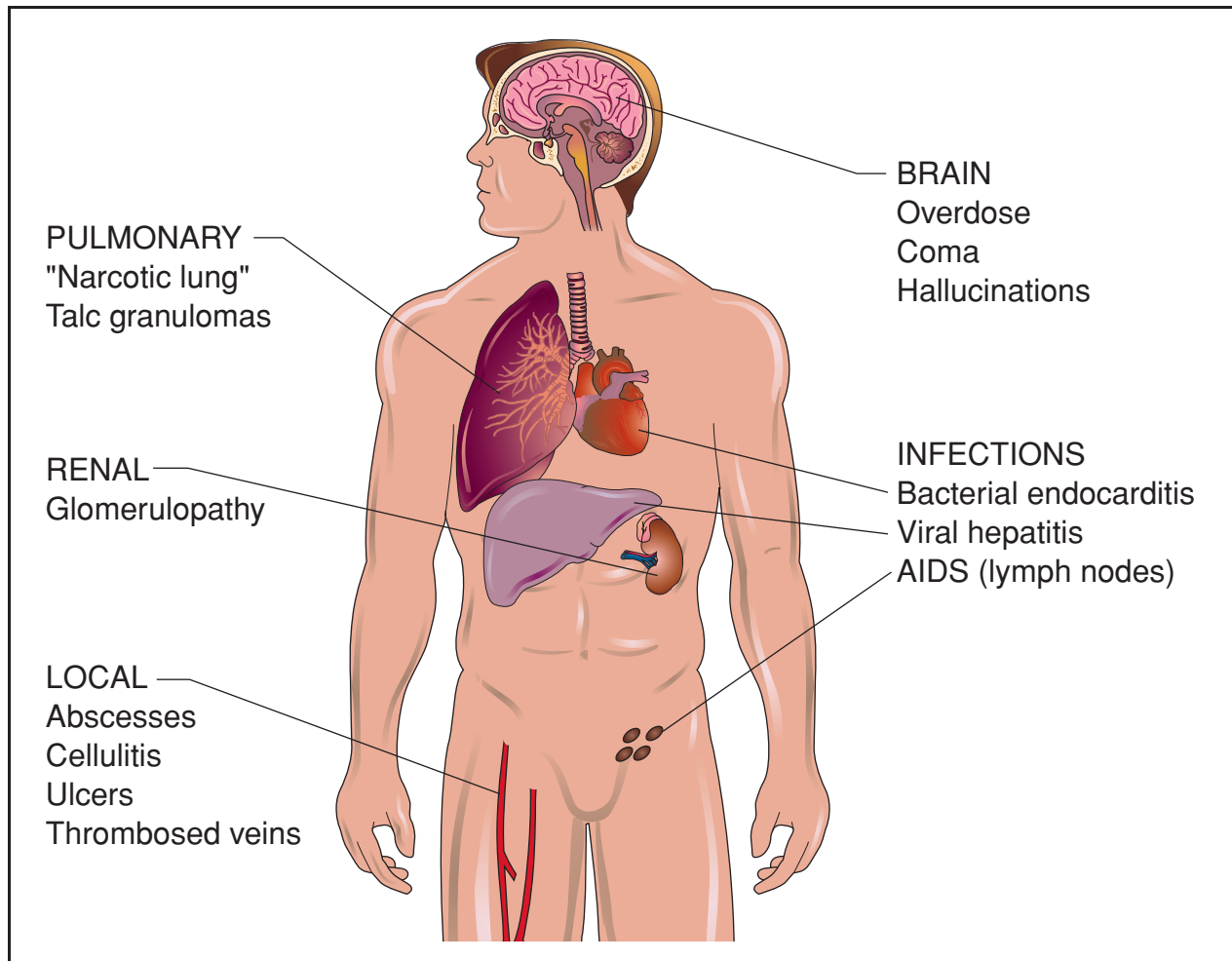
Other signs are related to the form in which the substance is used. For example, heroin, certain other opioid drugs, and certain forms of cocaine may be injected. A person using an injectable substance may have “track marks” (outwardly visible signs of the site of an injection, with possible redness and swelling of the vein in which the substance was injected). Furthermore, poor judgment brought on by substance use can result in the injections being made under dangerously unhygienic conditions. These unsanitary conditions and the use of shared needles are risk factors for major infections of the heart, as well as infection with HIV (the virus that causes **AIDS**), certain forms of hepatitis (a liver infection), and **tuberculosis**.

Cocaine is often taken as a powdery substance which is “snorted” through the nose. This can result in frequent nosebleeds, sores in the nose, and even erosion (an eating away) of the nasal septum (the structure that separates the two nostrils).

Overdosing on a substance is a frequent complication of substance abuse. **Drug overdose** can be purposeful (with **suicide** as a goal), or due to carelessness, the unpredictable strength of substances purchased from street dealers, mixing of more than one type of substance, or as a result of the increasing doses that a person must take to experience intoxicating effects. Substance overdose can be a life-threatening emergency, with the specific symptoms depending on the type of substance used. Substances with depressive effects may dangerously slow the breathing and heart rate, drop the body temperature, and result in a general unresponsiveness. Substances with stimulatory effects may dangerously increase the heart rate and blood pressure, produce abnormal heart rhythms, increase body temperature, induce seizures, and cause erratic behavior.

Diagnosis

Tools used in the diagnosis of substance dependence include screening questionnaires and patient histories, **physical examination**, and laboratory tests. A simple and popular screening tool is the CAGE questionnaire. CAGE refers to the first letters of each word that forms the basis of each of the four questions of the screening exam:



Substance abuse often causes a variety of medical abnormalities and conditions throughout the body, as shown in the illustration above. (Illustration by Electronic Illustrators Group.)

- Have you ever tried to Cut down on your substance use?
- Have you ever been Annoyed by people trying to talk to you about your substance use?
- Do you ever feel Guilty about your substance use?
- Do you ever need an Eye opener (use of the substance first thing in the morning) in order to start your day?

A "yes" answer to two or more of these questions is an indication that the individual should be referred for more thorough work-up for substance dependency or abuse.

In addition to CAGE, other screening questionnaires are available. Some are designed for particular population groups such as pregnant women, and others are designed to more thoroughly assess the severity of substance dependence. These

questionnaires, known by their acronyms, include AUDIT, HSS, HSQ, PRIME-MD, ACE, TWEAK, s-MAST, and SADD. There is some variability among questionnaires in terms of how accurately and comprehensively they can identify individuals as substance dependent.

Patient history, as taken through the direct interview, is important for identifying physical symptoms and psychiatric factors related to substance use. Family history of alcohol or other substance dependence is also useful for diagnosis.

A physical examination may reveal signs of substance abuse. These signs are specific to the substances used, as well as needle marks, tracks, or nasal erosion.

With the individual's permission, substance use can be detected through laboratory testing of his or her blood, urine, or hair. Laboratory testing, however,

may be limited by the sensitivity and specificity of the testing method, and by the time elapsed since the person last used the drug.

One of the most difficult aspects of diagnosis involves overcoming the patient's denial. Denial is a psychological state during which a person is unable to acknowledge the (usually negative) circumstances of a situation. In this case, denial leads a person to underestimate the degree of substance use and of the problems associated with substance use.

Treatment

According to the American Psychiatric Association, there are three goals for the treatment of people with substance use disorders: (1) the patient abstains from or reduces the use and effects of the substance; (2) the patient reduces the frequency and severity of relapses; and (3) the patient develops the psychological and emotional skills necessary to restore and maintain personal, occupational, and social functioning.

In general, before treatment can begin, many treatment centers require that the patient undergo **detoxification**. Detoxification is the process of weaning the patient from his or her regular substance use. Detoxification can be accomplished "cold turkey," by complete and immediate cessation of all substance use, or by slowly decreasing (tapering) the dose which a person is taking, to minimize the side effects of withdrawal. Some substances must be tapered because "cold turkey" methods of detoxification are potentially life threatening. In some cases, medications may be used to combat the unpleasant and threatening physical and psychological symptoms of withdrawal. For example, methadone is used to help patients adjust to the tapering of heroin use.

Treatment itself consists of three parts: (1) assessment; (2) formulation of a treatment plan; (3) psychiatric management. The first step in treatment is a comprehensive medical and psychiatric evaluation of the patient. This evaluation includes:

- a history of the patient's past and current substance use, and its cognitive, psychological, physiological, and behavioral effects
- a medical and psychiatric history and examination
- a history of psychiatric treatments and outcomes
- a family and social history
- screening of blood, breath, or urine for substances
- other laboratory tests to determine the presence of other conditions commonly found with substance use disorders

After the assessment is made, a treatment plan is formulated. Treatment plans vary according to the needs of the specific patient and can change for the same patient as he or she undergoes different phases of the disorder. Plans typically involve the following elements: (1) a strategy for the psychiatric management of the patient; (2) a strategy for reducing effects or use of substances, or for abstinence; (3) efforts to ensure compliance with the treatment program and to prevent relapse; (4) treatments for other conditions associated with substance use. Initial therapy and treatment setting (hospital, residential treatment, partial hospitalization, outpatient) decisions are made as part of the treatment plan, but because substance use disorders are considered a chronic condition requiring long-term care, these plans can and do change through the course of treatment.

The third step, psychiatric management of the patient, is the implementation of the treatment plan. Psychiatric management of the patient includes establishing a trusting relationship between clinician and patient; monitoring the patient's progress; managing the patient's relapses and withdrawal; diagnosing and treating associated psychiatric disorders; and helping the patient adhere to the treatment plan through therapy and the development of skills and social interactions that reinforce a drug-free lifestyle.

As part of the treatment process, patients typically undergo psychosocial therapy and, in some cases, pharmacologic treatment. Psychosocial therapeutic modalities include **cognitive-behavioral therapy**, behavioral therapy, individual psychodynamic or interpersonal therapy, **group therapy**, **family therapy**, and self-help groups. Pharmacologic treatment may include medications that ease withdrawal symptoms, reduce craving, interact negatively with substances of abuse to discourage drug-taking, or treat associated psychiatric disorders.

Alternative treatment

The efficacy of alternative treatments for substance use disorders remains for the most part ambiguous. One treatment that has been recently shown to have variable success is the use of **acupuncture** in treating substance dependence. In 2000, a randomized controlled trial of the effect of acupuncture on cocaine addiction reported that acupuncture significantly reduced the cocaine use of study participants. A 1999 meta-analysis (summary analysis of studies), however, reported that acupuncture had no statistically significant effect on **smoking** cessation.

There has been movement toward examining some touted treatments in more rigorous clinical trials. In particular, there has been some interest in *Pueraria lobata*, or kudzu, an herb that has reputedly been used in Chinese medicine to treat alcoholism. Preclinical trials of an herbal formula with kudzu have shown that increased consumption of the herbal formula is associated with decreased consumption of alcohol. Toxicity studies show few ill effects of the formula, and human trials are currently being undertaken to more fully evaluate the efficacy of this treatment.

The effectiveness of electroacupuncture (the practice of acupuncture accompanied by the application of low levels of electrical current at acupuncture points) in alleviating opiate withdrawal symptoms is also being examined. Preclinical trials suggest that electroacupuncture treatment given prior to the administration of naxolone seems to alleviate the withdrawal effects of naxolone.

Prognosis

Recovery from substance use is notoriously difficult, even with exceptional treatment resources. Although relapse rates are difficult to accurately obtain, the NIAAA cites evidence that 90% of alcohol dependent users experience at least one relapse within the 4 years after treatment. Relapse rates for heroin and nicotine users are believed to be similar. Certain pharmacological treatments, however, have been shown to reduce relapse rates.

Relapses are most likely to occur within the first 12 months of having discontinued substance use. Triggers for relapses can include any number of life stresses (problems on the job or in the marriage, loss of a relationship, death of a loved one, financial stresses), in addition to seemingly mundane exposure to a place or an acquaintance associated with previous substance use.

The development of adaptive life skills and ongoing drug-free social support are believed to be two important factors in avoiding relapse. The effect of the support group Alcoholics Anonymous has been intensively studied, and a 1996 meta-analysis noted that long-term sobriety appears to be positively related to Alcoholics Anonymous attendance and involvement. Support for family members in addition to support for the individual in recovery is also important. Because substance dependence has a serious impact on family functioning, and because family members may inadvertently maintain behaviors that initially led to the substance dependence, ongoing therapy and support for family members should not be neglected.

KEY TERMS

Addiction—The state of being both physically and psychologically dependent on a substance.

Dependence—A state in which a person requires a steady concentration of a particular substance to avoid experiencing withdrawal symptoms.

Detoxification—A process whereby an addict is withdrawn from a substance.

Intoxication—The desired mental, physical, or emotional state produced by a substance.

Street drug—A substance purchased from a drug dealer; may be a legal substance, sold illicitly (without a prescription, and not for medical use), or it may be a substance that is illegal to possess.

Tolerance—A phenomenon whereby a drug user becomes physically accustomed to a particular dose of a substance, and requires increasing dosages in order to obtain the same effects.

Withdrawal—Those side effects experienced by a person who has become physically dependent on a substance, upon decreasing the substance's dosage or discontinuing its use.

Prevention

Prevention is best aimed at teenagers and young adults aged 18–24 who are at very high risk for substance experimentation. Prevention programs should include an education component that outlines the risks and consequences of substance use and a training component that gives advice on how to resist peer pressure to use drugs.

Furthermore, prevention programs should work to identify and target children who are at relatively higher risk for substance abuse. This group includes victims of physical or sexual abuse, children of parents who have a history of substance abuse, and children with poor school performance and/or attention deficit disorder. These children may require more intensive intervention.

Resources

BOOKS

American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*. 4th ed., revised. Washington, DC: American Psychiatric Association, 2000.

PERIODICALS

- "Inhalant Abuse Becomes Focus of SAMHSA Guidelines, Prevention Efforts." *Alcoholism & Drug Abuse Weekly* March 22, 2004: 1–4.
- Schneider, Robert K., James L. Levenson, and Sidney H. Schnoll. "Update in Addiction Medicine." *Annals of Internal Medicine* 134 (March 6, 2001): 387–395.

Genevieve Pham-Kanter
Teresa G. Odle

Substance dependence see **Substance abuse and dependence**

Sucralfate see **Antilulcer drugs**

Sucrose intolerance see **Carbohydrate intolerance**

Sudden cardiac death

Definition

Sudden cardiac **death** (SCD) is an unexpected death due to heart problems, which occurs within one hour from the start of any cardiac-related symptoms. SCD is sometimes called cardiac arrest.

Description

When the heart suddenly stops beating effectively and breathing ceases, a person is said to have experienced sudden cardiac death.

SCD is not the same as actual death. In actual death, the brain also dies. The important difference is that sudden cardiac death is potentially reversible. If it is reversed quickly enough, the brain will not die.

Sudden cardiac death is also not the same as a **heart attack**. A heart attack (myocardial infarction) is the result of a blockage in an artery which feeds the heart, so the heart becomes starved for oxygen. The part that has been starved is damaged beyond repair, but the heart can still beat effectively.

Causes and symptoms

Sudden cardiac death is usually caused by **ventricular fibrillation** (the lower chamber of the heart quivers instead of pumping in an organized rhythm). Ventricular fibrillation almost never returns to normal by itself, so the condition requires immediate intervention. **Ventricular tachycardia** can also lead to sudden

KEY TERMS

Defibrillator—A device which delivers a controlled electric shock to the heart to return it to normal beating rhythm.

Ventricular fibrillation—When The lower chamber of the heart quivers instead of pumping in an organized way.

Ventricular tachycardia—A rapid heartbeat, usually over 100 beats per minute.

cardiac death. The risk for SCD is higher for anyone with heart disease.

When the heart stops beating effectively and the brain is being deprived of oxygenated blood, a medical emergency exists.

Diagnosis

Diagnosis of sudden cardiac death is made when there is a sudden loss of consciousness, breathing stops, and there is no effective heart beat.

Treatment

When sudden cardiac death occurs, the first priority is to establish the flow of oxygenated blood to the brain. The next priority is to restore normal rhythm to the heart. Forcing air into the mouth will get oxygen into the lungs. Compressing the chest simulates a pumping heart and will get some blood flow to the lungs, brain, and coronary arteries. This method is called **cardiopulmonary resuscitation (CPR)**. When trained help arrives, they will attempt to establish a normal heart beat by using a device called a defibrillator.

When If sudden cardiac death occurs outside the hospital setting, **cardiopulmonary resuscitation (CPR)** must begin within four to six minutes and advanced **life support** measures must begin within eight minutes, to avoid brain death. CPR requires no special medical skills and training is available for the ordinary person nationwide.

Prognosis

Sudden cardiac death is reversible in most people if treatment is begun quickly. However, of the people who are resuscitated, 40% will have another SCD within two years if they do not receive appropriate treatment for the underlying cause of the episode.

Prevention

When In order to prevent sudden cardiac death, underlying heart conditions must be addressed. Medications and implantable cardioverter-defibrillators may be used.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Dorothy Elinor Stonely

Sudden infant death syndrome

Definition

Sudden infant **death** syndrome (SIDS) is the unexplained death without warning of an apparently healthy infant, usually during sleep.

Description

Also known as crib death, SIDS has baffled physicians and parents for years. In the 1990s, advances have been made in preventing the occurrence of SIDS, which killed more than 4,800 babies in 1992 and 3,279 infants in 1995. Education programs aimed at encouraging parents and caregivers to place babies on their backs and sides when putting them to bed have helped contribute to a lower mortality rate from SIDS.

In the United States, SIDS strikes one or two infants in every thousand, making it the leading cause of death in newborns. It accounts for about 10% of deaths occurring during the first year of life. SIDS most commonly affects babies between the ages of two months and six months; it almost never strikes infants younger than two weeks of age or older than eight months. Most SIDS deaths occur between midnight and 8 A.M.

Causes and symptoms

Risk factors for SIDS

The exact causes of SIDS are still unknown, although studies have shown that many of the infants had recently been under a doctor's care for a cold or other illness of the upper respiratory tract. Most SIDS deaths occur during the winter and early spring, which are the peak times for respiratory infections. The most common risk factors for SIDS include:

Ten Leading Causes Of Infant Death (U.S.)

Congenital anomalies
Pre-term/Low birthweight
Sudden Infant Death Syndrome (SIDS)
Respiratory Distress Syndrome
Problems related to complications of pregnancy
Complications of placenta, cord, and membrane Accidents
Perinatal infections
Pneumonia/Influenza
Intrauterine hypoxia and birth asphyxia

- sleeping on the stomach (in the prone position)
- mother who smokes during **pregnancy**; smokers are as much as three times more likely than nonsmokers to have a SIDS baby
- the presence of passive smoke in the household
- male sex; the male/female ratio in SIDS deaths is 3:2;
- belonging to an economically deprived or minority family
- mother under 20 years of age at pregnancy
- mother who abuses drugs
- mother with little or no prenatal care
- prematurity or low weight at birth
- family history of SIDS

Most of these risk factors are associated with significantly higher rates of SIDS; however, none of them are exact enough to be useful in predicting which specific children may die from SIDS.

Theories about SIDS

MEDICAL DISORDERS. Currently, it is not known whether the immediate cause of death from SIDS is a heart problem or a sudden interruption of breathing. The most consistent **autopsy** findings are pinpoint hemorrhages inside the baby's chest and mild inflammation or congestion of the nose, throat, and airway. Some doctors have thought that the children stop breathing because their upper airway gets blocked. Others have suggested that the children have an abnormally high blood level of the chemicals that transmit nerve impulses to the brain, or that there is too much fetal hemoglobin in the blood. A third theory concerns the possibility that SIDS infants have an underlying abnormality in the central nervous system. This suggestion is based on the assumption that normal infants sense when their air supply is inadequate and wake up. Babies with an abnormal nervous

KEY TERMS

Congenital—Existing or present at the time of birth.

Crib death—Another name for SIDS.

Prone—Lying on the stomach with the face downward.

Supine—Lying on the back with the face upward.

system, however, do not have the same alarm mechanism in their brains. Other theories about the cause of death in SIDS include immune system disorders that cause changes in the baby's heart rate and breathing patterns during sleep, or a metabolic disorder that causes a buildup of fatty acids in the baby's system.

PHYSICAL SURROUNDINGS. A recent theory proposes that SIDS is connected to the child's rebreathing of stale air trapped in soft bedding. In addition to the infant's sleeping in the prone position, pillows, sheepskins, and other soft items may contribute to trapping air around the baby's mouth and nose, which causes the baby to breathe in too much carbon dioxide and not enough oxygen. Wrapping a baby too warmly has also been proposed as a factor.

Diagnosis

The diagnosis of SIDS is primarily a diagnosis of exclusion. This means that it is given only after other possible causes of the baby's death have been ruled out. Known risk factors aid in the diagnosis. Unlike the pattern in other diseases, however, the diagnosis of SIDS can only be given post-mortem. It is recommended that all infants who die in their sleep receive an autopsy to determine the cause. Autopsies indicate a definite explanation in about 20% of cases of sudden infant death. In addition, an autopsy can often put to rest any doubts the parents may have. Investigation of the location of the death is also useful in determining the child's sleeping position, bedding, room temperature, and similar factors.

Treatment

There is no treatment for SIDS, only identification of risk factors and preventive measures. The baby's parents may benefit from referral to counseling and support groups for parents of SIDS victims.

Prevention

SIDS appears to be at least partly preventable, which has been shown by a substantial decrease in

the case rate. The following are recommended as preventive measures:

- **Sleep position.** The United States Department of Health and Human Services initiated a "Back-to-Sleep" campaign in 1994 to educate the public about sleep position. Prior to that time, an estimated 70% of infants slept on their stomachs, since parents had been taught that a "back down" position contributed to **choking** during sleep. There are some conditions for which doctors will recommend the prone position, but for normal infants, side or back (supine) positions are better. When placing an infant on his or her side, the parent should pull the child's lower arm forward so that he or she is less likely to roll over onto the stomach. When babies are awake and being observed, they should be placed on their stomachs frequently to aid in the development of the muscles and skills involved in lifting the head. Once a baby can roll over to his or her stomach, he or she has developed to the point where the risk of SIDS is minimal.
- **Good prenatal care.** Proper prenatal care can help prevent the abnormalities that put children at higher risk for SIDS. Mothers who do not receive prenatal care are also more likely to have premature and low birth-weight babies. Expectant mothers should also be warned about the risks of **smoking**, alcohol intake, and drug use during pregnancy.
- **Proper bedding.** Studies have shown that soft bedding, such as beanbags, waterbeds and soft mattresses, contributes to SIDS. Babies should sleep on firm mattresses with no soft or fluffy materials underneath or around them— including quilts, pillows, thick comforters or lambskin. Soft stuffed toys should not be placed in the crib while babies sleep.
- **Room temperature.** Although babies should be kept warm, they do not need to be any warmer than is comfortable for the caregiver. An overheated baby is more likely to sleep deeply, perhaps making it more difficult to wake when short of breath. Room temperature and wrapping should keep the baby warm and comfortable but not overheated.
- **Diet.** Some studies indicate that breastfed babies are at lower risk for SIDS. It is thought that the mother's milk may provide additional immunity to the infections that can trigger sudden death in infants.
- **Bedsharing with parents.** Opinions differ on whether or not bedsharing of infant and mother increases or decreases the risk of SIDS. Bedsharing may encourage breastfeeding or alter sleep patterns, which could lower the risk of SIDS. On the other hand, some studies suggest that bedsharing increases the

risk of SIDS. In any case, mothers who choose to bring their babies to bed should observe the following cautions: Soft sleep surfaces, as well as quilts, blankets, comforters or pillows should not be placed under the baby. Parents who sleep with their infants should not smoke around the baby, or use alcohol or other drugs which might make them difficult to arouse. Parents should also be aware that adult beds are not built with the same safety features as infant cribs.

- Secondhand smoke. It is as important to keep the baby's environment smoke-free during infancy as it was when the mother was pregnant with the baby.
- Electronic monitoring. Electronic monitors are available for use in the home. These devices sound an alarm for the parents if the child stops breathing. There is no evidence, however, that these monitors prevent SIDS. In 1986, experts consulted by the National Institutes of Health (NIH) recommended monitors only for infants at risk. These infants include those who have had one or more episodes of breath stopping; premature infants with breathing difficulties; and babies with two or more older siblings that died of SIDS. Parents who use monitors should know how to use them properly and what to for the baby if the alarm goes off.
- Immunizations. There is no evidence that immunizations increase the risk of SIDS. In fact, babies who receive immunizations on schedule are less likely to die of SIDS.

Resources

ORGANIZATIONS

- Association of SIDS and Infant Mortality Programs. MN SID Center, Children's Hospitals and Clinics, 2525 Chicago Ave. So., Minneapolis, MN 55404. (612) 813-6285. <<http://www.asip1.org>>.
- National Institute of Child Health and Human Development. Bldg 31, Room 2A32, MSC 2425, 31 Center Drive, Bethesda, MD 20892-2425. (800) 505-2742. <<http://www.nichd.nih.gov/sids/sids.htm>>.
- National SIDS Resource Center. 2070 Chain Bridge Road, Suite 450, Vienna, VA 22181. (703) 821-8955. <<http://www.circsol.com/SIDS/>>.
- Sudden Infant Death Syndrome Alliance. 1314 Bedford Avenue, Suite 210, Baltimore, MD 21208. (800) 221-7437. <<http://www.sidsalliance.org>>.

Teresa Odle

Sugar diabetes see **Diabetes mellitus**

Sugar intolerance see **Carbohydrate intolerance**

Suicide

Definition

Suicide is defined as the intentional taking of one's own life. Prior to the late nineteenth century, suicide was legally defined as a criminal act in most Western countries. In the social climate of the early 2000s, however, suicidal behavior is most commonly regarded and responded to as a psychiatric emergency.

Description

Suicide is considered a major public health problem around the world as well as a personal tragedy. According to the National Institute of Mental Health (NIMH), suicide was the eleventh leading cause of **death** in the United States in 2000, and the third leading cause of death for people between the ages of 15 and 24. About 10.6 out of every 100,000 persons in the United States and Canada die by their own hands. There are five suicide victims for every three homicide deaths in North America as of the early 2000s. There are over 30,000 suicides per year in the United States, or about 86 per day; and each day about 1900 people attempt suicide.

The demographics of suicide vary considerably within Canada and the United States, due in part to differences among age groups and racial groups, and between men and women. Adult males are three to five times more likely to commit suicide than females, but females are more likely to attempt suicide. Most suicides occur in persons below the age of 40; however, elderly Caucasians are the sector of the population with the highest suicide rate. Americans over the age of 65 accounted for 18 percent of deaths by suicide in the United States in 2000. Geographical location is an additional factor; according to the Centers for Disease Control and Prevention (CDC), suicide rates in the United States are slightly higher than the national average in the western states, and somewhat lower than average in the East and the Midwest.

Race is also a factor in the demographics of suicide. Between 1979 and 1992, Native Americans had a suicide rate 1.5 times the national average, with young males between 15 and 24 accounting for 64% of Native American deaths by suicide. Asian American women have the highest suicide rate among all women over the age of 65. And between 1980 and 1996 the suicide rate more than doubled for black males between the ages of 15 and 19.

Causes & symptoms

Causes

Suicide is a complex act that represents the end result of a combination of factors in any individual. These factors include biological vulnerabilities, life history, occupation, present social circumstances, and the availability of means for committing suicide. While these factors do not “cause” suicide in the strict sense, some people are at greater risk of self-harm than others. Risk factors for suicide include:

- Male sex.
- Age over 75.
- A family history of suicide.
- A history of previous suicide attempts.
- A history of **abuse** in childhood.
- A local cluster of recent suicides or a local landmark associated with suicides. Examples of the latter include the Golden Gate Bridge in San Francisco; Sydney Harbor Bridge in Australia; St. Peter's Basilica in Rome; the Eiffel Tower in Paris; Prince Edward Viaduct in Toronto; and Mount Mihara, a volcano in Japan.
- Recent stressful events: separation or divorce, job loss, bankruptcy, upsetting medical diagnosis, death of spouse.
- Medical illness. Persons in treatment for such serious or incurable diseases as **AIDS**, Parkinson's disease, and certain types of **cancer** are at increased risk of suicide.
- Employment as a police officer, firefighter, physician, dentist, or member of another high-stress occupation.
- Presence of firearms in the house. Death by firearms is the most common method for women as well as men as of the early 2000s. In 2001, 55% of reported suicides in the United States were committed with guns.
- Alcohol or **substance abuse**. Mood-altering substances are a factor in suicide because they weaken a person's impulse control.
- Presence of a psychiatric illness. Over 90% of Americans who commit suicide have a significant mental illness. Major depression accounts for 60% (especially in the elderly), followed by **schizophrenia**, **alcoholism**, substance abuse, borderline personality disorder, Huntington's disease, and epilepsy. The lifetime mortality due to suicide in psychiatric patients is 15% for major depression; 20% for **bipolar disorder**; 18% for alcoholism; 10% for

schizophrenia; and 5–10% for borderline and certain other personality disorders.

Neurobiological factors may also influence a person's risk of suicide. Post-mortem studies of the brains of suicide victims indicate that the part of the brain associated with aggression and other impulsive behaviors (the frontal cortex) has a significantly lower level of serotonin, a neurotransmitter associated with **mood disorders**. Low serotonin levels are correlated with major depression. In addition, suicide victims have higher than normal levels of cortisol, a hormone produced in stressful situations, in the tissues of their central nervous system. Other research has indicated that abuse in childhood may have permanent effects on the level of serotonin in the brain, possibly “resetting” the level abnormally low. In addition, twin studies have suggested that there may be a genetic susceptibility to both suicidal ideation and suicide attempts which cannot be explained by inheritance of common psychiatric disorders.

Some psychiatrists propose psychodynamic explanations of suicide. According to one such theory, suicide is “murder in the 180th degree” that is, the suicidal person really wants to kill someone else but turns the anger against the self instead. Another version of this idea is that the suicidal person has incorporated the image of an abusive parent or other relative in their own psyche and then tries to eliminate the abuser by killing the self.

Diagnosis

When a person consults a doctor because they are thinking of committing suicide, or they are taken to a doctor's office or emergency room after a suicide attempt, the doctor will evaluate the patient's potential for acting on their thoughts or making another attempt. The physician's assessment will be based on several different sources of information:

- The patient's history, including a history of previous attempts or a family history of suicide.
- A clinical interview in which the physician will ask whether the patient is presently thinking of suicide; whether they have made actual plans to do so; whether they have thought about the means; and what they think their suicide will accomplish. These questions help in evaluating the seriousness of the patient's intentions.
- A suicide note, if any.
- Information from friends, relatives, or first responders who may have accompanied the patient.

- Short self-administered psychiatric tests that screen people for depression and suicidal ideation. The most commonly used screeners are the Beck Depression Inventory (BDI), the Depression Screening Questionnaire, and the Hamilton Depression Rating Scale.
- The doctor's own instinctive reaction to the patient's mood, appearance, vocal tone, and similar factors.

Treatment of attempted suicide

Suicide attempts range from well-planned attempts involving a highly lethal method (guns, certain types of poison, jumping from high places, throwing oneself in front of trains or subway cars) that fail by good fortune to impulsive or poorly planned attempts using a less lethal method (medication overdoses, cutting the wrists). Suicide attempts at the less lethal end of the spectrum are sometimes referred to as suicide gestures or pseudocide. These terms should not be taken to indicate that suicide gestures are only forms of attention-seeking; they should rather be understood as evidence of serious emotional and mental distress.

A suicide attempt of any kind is treated as a psychiatric emergency by the police and other rescue personnel. Treatment in a hospital emergency room includes a complete psychiatric evaluation; a **mental status examination**; blood or urine tests if alcohol or drug abuse is suspected; and a detailed assessment of the patient's personal circumstances (occupation, living situation, family or friends nearby, etc.). The patient will be kept under observation while decisions are made about the need for hospitalization.

A person who has attempted suicide can be legally hospitalized against his or her will if he or she seems to be a danger to the self or others. The doctor will base decisions about hospitalization on the severity of the patient's depression; the availability of friends, relatives, or other social support; and the presence of other suicide risk factors, including a history of previous suicide attempts, substance abuse, and **psychosis** (loss of contact with reality, often marked by **delusions** and **hallucinations**). If the attempt is judged to be a nonlethal suicide gesture, the patient may be released after the psychiatric assessment is completed. According to CDC figures, 132,353 Americans were hospitalized in 2002 following suicide attempts while 116,639 were released following emergency room treatment.

Related issues

Survivors of suicide

One group of people that is often overlooked in discussions of suicide is the friends and family left behind by the suicide. It is estimated that each person who kills him- or herself leaves six survivors to deal with the aftermath; thus there are at least 4.5 million survivors of suicide in the United States. In addition to the grief that ordinarily accompanies death, survivors of suicide often struggle with feelings of guilt and shame as well. They often benefit from group or individual psychotherapy in order to work through such issues as wondering whether they could have prevented the suicide or whether they are likely to commit suicide themselves. The American Foundation for Suicide Prevention (AFSP) has a number of online resources available for survivors of suicide.

Assisted suicide

One question that has been raised in developed countries as the average life expectancy increases is the legalization of assisted suicide for persons suffering from a painful terminal illness. Physician-assisted suicide has become a topic of concern since it was legalized in the Netherlands in 2001 and in the state of Oregon in 1997. It is important to distinguish between physician-assisted suicide and euthanasia, or "mercy killing." Assisted suicide, which is often called "self-deliverance" in Britain, refers to a person's bringing about his or her own death with the help of another person. Because the other person is often a physician, the act is often called "doctor-assisted suicide." Euthanasia strictly speaking means that the physician or other person is the one who performs the last act that causes death. For example, if a physician injects a patient with a lethal overdose of a pain-killing medication, he or she is performing euthanasia. If the physician leaves the patient with a loaded syringe and the patient injects himself or herself with it, the act is an assisted suicide. As of early 2005 assisted suicide is illegal everywhere in the United States except for Oregon, and euthanasia is illegal in all fifty states.

Media treatment of suicide

The Centers for Disease Control and Prevention (CDC) sponsored a national workshop in April 1994 that addressed the connection between sensationalized media treatments of suicide and the rising rate of suicide among American youth. The CDC and the American Association of Suicidology subsequently adopted a set of guidelines for media coverage of

suicide intended to reduce the risk of suicide by contagion.

The CDC guidelines point out that the following types of reporting may increase the risk of “copycat” suicides:

- Presenting oversimplified explanations of suicide, when in fact many factors usually contribute to a person’s decision to take their own life.
- Excessive or repetitive local news coverage.
- Sensationalizing the suicide by inclusion of morbid details or dramatic photographs.
- Giving “how-to” descriptions of the method of suicide.
- Describing suicide as an effective coping strategy or as a way to achieve certain goals.
- Glorifying the act of suicide or the person who commits suicide.

Alternative treatment

Some alternative treatments may help to prevent suicide by preventing or relieving depression. **Meditation** practice or religious faith and worship have been shown to lower a person’s risk of suicide. In addition, any activity that brings people together in groups and encourages them to form friendships helps to lower the risk of suicide, as people with strong social networks are less likely to give up on life.

Prognosis

The prognosis for a person who has attempted suicide is generally favorable, although further research needs to be done. A 1978 follow-up study of 515 people who had attempted suicide between 1937 and 1971 reported that 94% were either still alive or had died of natural causes. This finding has been taken to indicate that suicidal behavior is more likely to be a passing response to an acute crisis than a reflection of a permanent state of mind.

Prevention

One reason that suicide is such a tragedy is that most self-inflicted deaths are potentially preventable. Many suicidal people change their minds if they can be helped through their immediate crisis; Dr. Richard Seiden, a specialist in treating survivors of suicide attempts, puts the high-risk period at 90 days after the crisis. Some potential suicides change their minds during the actual attempt; for example, a number of people who survived jumping off the Golden Gate

Bridge told interviewers afterward that they regretted their action even as they were falling and that they were grateful they survived.

Brain research is another important aspect of suicide prevention. Since major depression is the single most common psychiatric diagnosis in suicidal people, earlier and more effective recognition of depression is a necessary preventive measure. Known biological markers for an increased risk of suicide can now be correlated with personality profiles linked to suicidal behavior under **stress** to help identify individuals at risk. In addition, brain imaging studies using **positron emission tomography (PET)** are presently in use to detect abnormal patterns of serotonin uptake in specific regions of the brain. Genetic studies are also yielding new information about inherited predispositions to suicide.

Another major preventive measure is education of clinicians, media people, and the general public. In 2002 the CDC, the National Institutes of Health (NIH), and several other government agencies joined together to form the National Strategy for Suicide Prevention, or NSSP. Education of the general public includes a growing number of medical and government websites posting information about suicide, publications available for downloading, lists of books for further reading, tips for identifying symptoms of depressed and suicidal thinking, and advice about helping friends or loved ones who may be at risk. Many of these websites also have direct connections to suicide hotlines.

The National Institute of Mental Health (NIMH) recommends the following action steps for anyone dealing with a suicidal person:

- Make sure that someone is with them at all times; do not leave them alone even for a short period of time.
- Persuade them to call their family doctor or the nearest hospital emergency room.
- Call 911 yourself.
- Keep the person away from firearms, drugs, or other potential means of suicide.

Resources

BOOKS

Alvarez, A. *The Savage God: A Study of Suicide*. New York: Random House, Inc., 1972. A now-classic study of suicide written for general readers. The author includes a historical overview of suicide along with accounts of his own suicide attempt and the suicide of his friend, the poet Sylvia Plath.

American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*, 4th edition, text revision.

KEY TERMS

Assisted suicide—A form of self-inflicted death in which a person voluntarily brings about his or her own death with the help of another, usually a physician, relative, or friend.

Cortisol—A hormone released by the cortex (outer portion) of the adrenal gland when a person is under stress. Cortisol levels are now considered a biological marker of suicide risk.

Euthanasia—The act of putting a person or animal to death painlessly or allowing them to die by withholding medical services, usually because of a painful and incurable disease. Mercy killing is another term for euthanasia.

Frontal cortex—The part of the human brain associated with aggressiveness and impulse control. Abnormalities in the frontal cortex are associated with an increased risk of suicide.

Psychodynamic—A type of explanation of human behavior that regards it as the outcome of interactions between conscious and unconscious factors.

Serotonin—A chemical that occurs in the blood and nervous tissue and functions to transmit signals across the gaps between neurons in the central nervous system. Abnormally low levels of serotonin are associated with depression and an increased risk of suicide.

Suicide gesture—Attempted suicide characterized by a low-lethality method, low level of intent or planning, and little physical damage. Pseudocide is another term for a suicide gesture.

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Soreff, Stephen, MD. “Suicide.” *eMedicine*, 3 September 2004. <<http://www.emedicine.com/med/topic3004.htm>>.

ORGANIZATIONS

American Academy of Child and Adolescent Psychiatry. 3615 Wisconsin Avenue, NW, Washington, DC 20016-3007. (202) 966-7300. Fax: (202) 966-2891. <<http://www.aacap.org>>.

American Association of Suicidology. Suite 408, 4201 Connecticut Avenue, NW, Washington, DC 20008. (202) 237-2280. Fax: (202) 237-2282. <<http://www.suicidology.org>>.

American Foundation for Suicide Prevention (AFSP). 120 Wall Street, 22nd Floor, New York, NY 10005. (888) 333-2377 or (212)

Centers for Disease Control and Prevention (CDC), National Center for Injury Prevention and Control (NCIPC). Mailstop K60, 4770 Buford Highway, Atlanta, GA 30341-3724. (770) 488-4362. Fax: (770) 488-4349. <<http://www.cdc.gov/ncipc.htm>>.

National Institute of Mental Health (NIMH). 6001 Executive Boulevard, Room 8184, MSC 9663, Bethesda, MD 20892-9663. (301) 443-4513 or (886) 615-NIMH. <www.nimh.nih.gov>.

OTHER

American Academy of Child and Adolescent Psychiatry (AACAP). *Teen Suicide*. AACAP Facts for Families #10. Washington, DC: AACAP, 2004.

Centers for Disease Control and Prevention, National Center for Injury Prevention and Control. “Suicide Contagion and the Reporting of Suicide: Recommendations from a National Workshop.” *Morbidity and Mortality Weekly Report* 43 (22 April 1994): 9-18. <<http://www.cdc.gov/mmwr/preview/mmwrhtml/00031539.htm>>.

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4594. Bethesda, MD: NIMH, 2003. <<http://www.nimh.nih.gov/publicat/NIMHharmsway.pdf>>. National Suicide Hotline: (800) 273-TALK (1-800-273-8255).

Rebecca Frey, PhD

Sulfacetamide *see* **Antibiotics, ophthalmic**

Sulfamethoxazole and trimethoprim *see* **Sulfonamides**

Sulfinpyrazone *see* **Gout drugs**

Sulfisoxazole *see* **Sulfonamides**

Sulfonamides

Definition

Sulfonamides are medicines that prevent the growth of bacteria in the body.

Purpose

Sulfonamides are used to treat many kinds of infections caused by bacteria and certain other microorganisms. Physicians may prescribe these drugs to treat urinary tract infections, ear infections, frequent or long-lasting **bronchitis**, bacterial **meningitis**, certain eye infections, *Pneumocystis carinii* **pneumonia**, **traveler's diarrhea**, and a number of other kinds of infections. These drugs will *not* work for colds, flu, and other infections caused by viruses.

Description

Sulfonamides, also called sulfa medicines, are available only with a physician's prescription. They are sold in tablet and liquid forms. Some commonly used sulfonamides are sulfisoxazole (Gantrisin) and the combination drug sulfamethoxazole and trimethoprim (Bactrim, Cotrim).

Recommended dosage

The recommended dosage depends on the type of sulfonamide, the strength of the medicine, and the medical problem for which it is being taken. Check with the physician who prescribed the drug or the pharmacist who filled the prescription for the correct dosage.

Always take sulfonamides exactly as directed. To make sure the infection clears up completely, take the medicine for as long as it has been prescribed. Do not stop taking the drug just because symptoms begin to improve. Symptoms may return if the drug is stopped too soon.

Sulfonamides work best when they are at constant levels in the blood. To help keep levels constant, take the medicine in doses spaced evenly through the day and night. Do not miss any doses. For best results, take the medicine with a full glass of water and drink several more glasses of water every day. This will help prevent some of the medicine's side effects.

Precautions

Symptoms should begin to improve within a few days of beginning to take this medicine. If they do not, or if they get worse, check with the physician who prescribed the medicine.

Although such side effects are rare, some people have had severe and life-threatening reactions to sulfonamides. These include sudden, severe liver damage, serious blood problems, breakdown of the outer layer of the skin, and a condition called Stevens-Johnson syndrome, in which people get blisters around the mouth, eyes, or anus. Call a physician immediately if any of these signs of a dangerous reaction occur:

- skin rash or reddish or purplish spots on the skin
- other skin problems, such as blistering or peeling
- fever
- sore throat
- cough
- shortness of breath
- joint **pain**
- pale skin
- yellow skin or eyes

This medicine may cause **dizziness**. Anyone who takes sulfonamides should not drive, use machines or do anything else that might be dangerous until they have found out how the drugs affect them.

Sulfonamides may cause blood problems that can interfere with healing and lead to additional infections. Avoid injuries while taking this medicine. Be especially careful not to injure the mouth when brushing or flossing the teeth or using a toothpick. Do not have dental work done until the blood is back to normal.

This medicine may increase sensitivity to sunlight. Even brief exposure to sun can cause a severe **sunburn** or a rash. While being treated with this medicine, avoid being in direct sunlight, especially between 10 A.M. and 3 P.M.; wear a hat and tightly woven clothing that covers the arms and legs; use a sunscreen with a skin protection factor (SPF) of at least 15; protect the lips with a sun block lipstick; and do not use tanning beds, tanning booths, or sunlamps.

Babies under 2 months should not be given sulfonamides unless their physician has ordered the medicine.

Older people may be especially sensitive to the effects of sulfonamides, increasing the chance of unwanted side effects, such as severe skin problems and blood problems. Patients who are taking water pills (**diuretics**) at the same time as sulfonamides may also be more likely to have these problems.

Special conditions

People with certain medical conditions or who are taking certain other medicines can have problems if they take sulfonamides. Before taking these drugs, be sure to let the physician know about any of these conditions:

ALLERGIES. Anyone who has had unusual reactions to sulfonamides, water pills (diuretics), diabetes medicines, or **glaucoma** medicine in the past should let his or her physician know before taking sulfonamides. The physician should also be told about any **allergies** to foods, dyes, preservatives, or other substances.

PREGNANCY. In studies of laboratory animals, some sulfonamides cause **birth defects**. The drugs' effects on human fetuses have not been studied. However, pregnant women are advised not to use this medicine around the time of labor and delivery, because it can cause side effects in the baby. Women who are pregnant or who may become pregnant should check with their physicians about the safety of using sulfonamides during **pregnancy**.

BREASTFEEDING. Sulfonamides pass into breast milk and may cause liver problems, anemia, and other problems in nursing babies whose mothers take the medicine. Because of those problems, women should not breastfeed when they are under treatment with this drug. Women who are breastfeeding and who need to take this medicine should check with their physicians to find out how long they need to stop breastfeeding.

OTHER MEDICAL CONDITIONS. Before using sulfonamides, people with any of these medical problems

should make sure their physicians are aware of their conditions:

- anemia or other blood problems
- kidney disease
- liver disease
- asthma or severe allergies
- alcohol **abuse**
- poor **nutrition**
- abnormal intestinal absorption
- porphyria
- folic acid deficiency
- deficiency of the enzyme glucose-6-phosphate dehydrogenase (G6PD)

USE OF CERTAIN MEDICINES. Taking sulfonamides with certain other drugs may affect the way the drugs work or may increase the chance of side effects.

Side effects

The most common side effects are mild **diarrhea**, **nausea**, **vomiting**, dizziness, **headache**, loss of appetite, and tiredness. These problems usually go away as the body adjusts to the drug and do not require medical treatment.

More serious side effects are not common, but may occur. If any of the following side effects occur, check with a physician immediately:

- itching or skin rash
- reddish or purplish spots on the skin
- other skin problems, such as redness, blistering, peeling
- severe, watery or bloody diarrhea
- muscle or joint aches
- fever
- sore throat
- cough
- shortness of breath
- unusual tiredness or weakness
- unusual bleeding or bruising
- pale skin
- yellow eyes or skin
- swallowing problems

Other rare side effects may occur. Anyone who has unusual symptoms while taking sulfonamides should get in touch with his or her physician.

KEY TERMS

Anemia—A lack of hemoglobin—the compound in blood that carries oxygen from the lungs throughout the body and brings waste carbon dioxide from the cells to the lungs, where it is released.

Bronchitis—Inflammation of the air passages of the lungs.

Fetus—A developing baby inside the womb.

Inflammation—Pain, redness, swelling, and heat that usually develop in response to injury or illness.

Meningitis—Inflammation of tissues that surround the brain and spinal cord.

***Pneumocystis carinii* pneumonia**—A lung infection that affects people with weakened immune systems, such as people with AIDS or people taking medicines that weaken the immune system.

Porphyria—A disorder in which porphyrins build up in the blood and urine.

Porphyrin—A type of pigment found in living things.

Urinary tract—The passage through which urine flows from the kidneys out of the body.

Interactions

Sulfonamides may interact with a large number of other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. Anyone who takes sulfonamides should let the physician know all other medicines he or she is taking. Among the drugs that may interact with sulfonamides are:

- acetaminophen (Tylenol)
- medicine for overactive thyroid
- male hormones (androgens)
- female hormones (estrogens)
- other medicines used to treat infections
- birth control pills
- medicines for diabetes such as glyburide (Micronase)
- anticoagulants such as warfarin (Coumadin)
- disulfiram (Antabuse), used to treat alcohol abuse
- amantadine (Symmetrel), used to treat flu and also Parkinson's disease
- water pills (diuretics) such as hydrochlorothiazide (HCTZ, HydroDIURIL)

- the anticancer drug methotrexate (Rheumatrex)
- antiseizure medicines such as valproic acid (Depakote, Depakene)

The list above does not include every drug that may interact with sulfonamides. Be sure to check with a physician or pharmacist before combining sulfonamides with any other prescription or nonprescription (over-the-counter) medicine.

Nancy Ross-Flanigan

Sumatriptan see **Antimigraine drugs**

Sunburn

Definition

Inflammation of the skin caused by overexposure to the sun.

Description

Sunburn is caused by exposure to the ultraviolet (UV) rays of the sun. There are two types of ultraviolet rays, UVA and UVB. UVA rays penetrate the skin more deeply and can cause melanoma in susceptible people. UVB rays, which don't penetrate as deeply, cause sunburn and wrinkling. Most UVB rays are absorbed by **sunscreens**, but only about half the UVA rays are absorbed.

Skin **cancer** from sun overexposure is a serious health problem in the United States, affecting almost a million Americans each year. One out of 87 will develop **malignant melanoma**, the most serious type of skin cancer, and 7,300 of them will die each year.

Fair-skinned people are most susceptible to sunburn, because their skin produces only small amounts of the protective pigment called melanin. People trying to get a tan too quickly in strong sunlight are also more vulnerable to sunburn. While they have a lower risk, even the darkest-skinned people can get skin cancer.

Repeated sun overexposure and burning can prematurely age the skin, causing yellowish, wrinkled skin. Overexposure can increase the risk of skin cancer, especially a serious burn in childhood.

Causes and symptoms

The ultraviolet rays in sunlight destroy cells in the outer layer of the skin, damaging tiny blood vessels



This person has a second-degree sunburn on the back of the neck. (Custom Medical Stock Photo. Reproduced by permission.)

underneath. When the skin is burned, the blood vessels dilate and leak fluid. Cells stop making protein. Their DNA is damaged by the ultraviolet rays. Repeated DNA damage can lead to cancer.

When the sun **burns** the skin, it triggers immune defenses which identify the burned skin as foreign. At the same time, the sun transforms a substance on the skin which interferes with this immune response. While this substance keeps the immune system from attacking a person's own skin, it also means that any malignant cells in the skin will be able to grow freely.

Sunburn causes skin to turn red and blister. Several days later, the dead skin cells peel off. In severe cases, the burn may occur with sunstroke (**vomiting**, **fever** and collapse).

Diagnosis

Visual inspection and a history of exposure to the sun.

Treatment

Aspirin can ease **pain** and inflammation. Tender skin should be protected against the sun until it has healed. In addition, apply:

- calamine lotion
- sunburn cream or spray
- cool tap water compress
- colloidal oatmeal (Aveeno) baths
- dusting powder to reduce chafing

People who are severely sunburned should see a doctor, who may prescribe corticosteroid cream to speed healing.

KEY TERMS

Malignant melanoma—The most deadly of the three types of skin cancer.

Sunscreen—Products which block the damaging rays of the sun. Good sunscreens contain either para-aminobenzoic acid (PABA) or benzophenone, or both. Sunscreen protection factors range from 2–45.

Alternative treatment

Over-the-counter preparations containing aloe (*Aloe barbadensis*) are an effective treatment for sunburn, easing pain and inflammation while also relieving dryness of the skin. A variety of topical herbal remedies applied as lotions, poltices, or compresses may also help relieve the effects of sunburn. Calendula (*Calendula officinalis*) is one of the most frequently recommended to reduce inflammation.

Prognosis

Moderately burned skin should heal within a week. While the skin will heal after a sunburn, the risk of skin cancer increases with exposure and subsequent burns. Even one bad burn in childhood carries an increased risk of skin cancer.

Prevention

Everyone from age six months on should use a water-resistant sunscreen with a sun protective factor (SPF) of at least 15. Apply at least an ounce 15–30 minutes before going outside. It should be reapplied every two hours (more often after swimming). Babies should be kept completely out of the sun for the first six months of life, because their skin is thinner than older children. Sunscreens have not been approved for infants.

In addition, people should:

- limit sun exposure to 15 minutes the first day, even if the weather is hazy, slowly increasing exposure daily
- reapply sunscreen every two hours (more often if sweating or swimming)
- reapply waterproof sunscreen after swimming more than 80 minutes, after toweling off, or after perspiring heavily
- avoid the sun between 10 A.M. and 3 P.M.

- use waterproof sunscreen on legs and feet, since the sun can burn even through water
- wear an opaque shirt in water, because reflected rays are intensified

If using a sunscreen under SPF 15, simply applying more of the same SPF won't prolong allowed time in the sun. Instead, patients should use a higher SPF in order to lengthen exposure safely. A billed cap protects 70% of the face; a wide-brimmed hat is better. People at very high risk for skin cancer can wear clothing that blocks almost all UV rays, but most people can simply wear white cotton summer-weight clothing with a tight weave.

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Carol A. Turkington

Sunscreens

Definition

Sunscreens are products applied to the skin to protect against the harmful effects of the sun's ultraviolet (UV) rays.

Purpose

Everyone needs a little sunshine. About 15 minutes of exposure a day helps the body make Vitamin D, which is important for healthy bones and teeth. But longer exposure may cause many problems, from wrinkles to skin **cancer**. One particularly deadly form of skin cancer, **malignant melanoma**, has been on the rise in recent decades, as tanning has become more popular. Over the same period, scientists have warned that the thin layer of ozone that protects life on Earth from the sun's ultraviolet (UV) radiation is being depleted. This allows more UV radiation to get through, adding to the risk of overexposure.

Sunscreens help protect against the sun's damaging effects. But just how much protection they provide is a matter of debate. The sun gives off two kinds of ultraviolet radiation, called UV-A and UV-B. For many years, experts thought that only UV-B was harmful. However, recent research suggests that UV-A may be just as dangerous as UV-B, although its

effects may take longer to show up. In particular, UV-A may have a role in causing melanoma. Most sunscreen products contain ingredients that provide adequate protection only against UV-B rays. Even those labeled as broad-spectrum sunscreens may offer only partial protection against UV-A radiation. Those containing the ingredient avobenzone give the most protection against UV-A rays.

Some medical experts are concerned that sunscreens give people a false sense of security, allowing them to stay in the sun longer than they should. Although sunscreens protect the skin from burning, they may not protect against other kinds of damage. A number of studies suggest that people who use sunscreens may actually increase their risk of melanoma because they spend too much time in the sun. This does not mean that people should stop using sunscreens. It means that they should not rely on sunscreens *alone* for protection. According to the American Academy of Dermatology, sunscreens should be one part of sun protection, along with wide-brimmed hats and tightly-woven clothing that covers the arms and legs.

Sunscreens are also recommended for patients with **rosacea** or other skin disorders that are aggravated by exposure to sunlight.

Description

Many brands of sunscreens are available, containing a variety of ingredients. The active ingredients work by absorbing, reflecting, or scattering some or all of the sun's rays. Most sunscreen products contain combinations of ingredients.

The U.S. Food and Drug Administration (FDA) has required sunscreen products to carry a sun protection factor (SPF) rating on their labels since 1999. This number tells the consumer how well the sunscreen protects against burning. The higher the number, the longer a person can stay in the sun without burning.

Sunscreens are usually grouped into two major categories, namely chemical absorbers and physical blockers. Chemical absorbers absorb high-intensity UV rays while physical blockers reflect or scatter them. Chemical absorber compounds include avobenzone, padimate O, octyl methoxycinnamate, octisalate, and octocrylene. Physical blocker compounds include titanium dioxide and zinc oxide. The chief drawback of the physical blockers is their tendency to leave a white film on the skin, causing many people to use less of the product than they should for full sun protection.

A plant-derived compound that shows promise as a sunscreen is nobiletin, a flavonoid extracted from *Citrus depressa* or flat orange, a small citrus fruit native to Taiwan and Okinawa. Topical application of nobiletin has been shown to be effective in preventing the swelling and reddening of the skin associated with overexposure to sunlight.

Sunscreen products are sold as lotions, creams, gels, oils, sprays, sticks, and lip balms, and can be bought without a physician's prescription.

Recommended dosage

One should be sure to read the instructions that come with the product. Some need to be applied as long as one or two hours before sun exposure. Others should be applied 15–30 minutes before exposure, and reapplied frequently during exposure.

People should apply sunscreen liberally to all exposed parts of the skin, including the hands, feet, nose, ears, neck, scalp (if the hair is thin or very short), and eyelids. They should take care not to get sunscreen in the eyes, as it can cause irritation. People should also use a lip balm containing sunscreen to protect the lips, and reapply sunscreen liberally every 1–2 hours—more frequently when perspiring heavily or after swimming.

Precautions

Sunscreen alone will not provide full protection from the sun. When possible, one should wear a hat, long pants, long-sleeved shirts or blouses, and sunglasses. Try to stay out of the sun between 10 A.M. and 2 P.M. (11 A.M. to 3 P.M. Daylight Saving Time), when the sun's rays are strongest. The sun can damage the skin even on cloudy days, so get in the habit of using a sunscreen every day. Be especially careful at high elevations or in areas with surfaces that reflect the sun's rays, such as sand, water, concrete, or snow.

Sunlamps, tanning beds, and tanning booths were once thought to be safer than the sun, because they give off mainly UV-A rays. However, UV-A rays are now known to cause serious skin damage and may increase the risk of melanoma. Health experts advise people not to use these tanning devices.

People with fair skin, blond, red or light-brown hair, and blue or light-colored eyes are at greatest risk for developing skin cancer. So are people with many large skin **moles**. These people should avoid exposure to the sun as much as possible. However, even dark-skinned people, including African Americans and

Hispanic Americans may suffer skin damage from the sun and should be careful about exposure.

Other groups of people who should minimize sun exposure are those who have had organ transplants or recent **plastic surgery**. Patients who have received organ transplants have a greatly increased risk of developing skin cancer, and the facial skin of people who have had face lifts or similar plastic surgery procedures sunburns more easily than intact skin.

Sunscreens should not be used on infants under 6 months of age because of the risk of side effects. Instead, children this young should be kept out of the sun. Children over 6 months should be protected with clothing and sunscreens of at least SPF 15, preferably lotions. Sunscreens containing alcohol should not be used on children because they may irritate the skin.

Older people who stay out of the sun and use sunscreens may not produce enough vitamin D in their bodies. They may need to increase the vitamin D in their **diets** by including foods such as fortified milk and salmon. A health care professional can help decide if this precaution is necessary.

Anyone who has had unusual reactions to any sunscreen ingredients in the past should check with a physician or pharmacist before using a sunscreen. The physician or pharmacist should also be told about any **allergies** to foods, dyes, preservatives, or other substances, especially the following:

- artificial sweeteners
- such anesthetics as benzocaine, procaine, or tetracaine
- diabetes medicine taken by mouth
- hair dyes
- sulfa medicines
- water pills
- cinnamon flavoring

People with skin conditions or diseases should check with their physicians before using a sunscreen. This is especially true of people with conditions that get worse with exposure to light.

Side effects

The most common side effects are drying or tightening of the skin. This problem does not need medical attention unless it does not improve.

KEY TERMS

Hair follicle—A tiny pit in the skin from which hair grows.

Melanoma—A rapidly spreading and deadly form of cancer that usually occurs on the skin.

Ozone—A gas found in the atmosphere. A layer of ozone about 15 mi (24 km) above Earth's surface helps protect living things from the damaging effects of the sun's ultraviolet rays.

Pus—Thick, whitish or yellowish fluid that forms in infected tissue.

Rosacea—A chronic skin disease characterized by persistent redness of the skin and periodic outbreaks of pustules, usually affecting the middle third of the face.

Ultraviolet rays—Invisible light rays with a wavelength shorter than that of visible light but longer than that of x rays.

Other side effects are rare, but possible. If any of the following symptoms occur, check with a physician as soon as possible:

- acne
- burning, **itching**, or stinging of the skin
- redness or swelling of the skin
- rash, with or without blisters that ooze and become crusted
- pain in hairy parts of body
- pus in hair follicles

Interactions

Anyone who is using a prescription or nonprescription (over-the-counter) drug that is applied to the skin should check with a physician before using a sunscreen.

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American Society of Plastic Surgeons (ASPS). 444 East Algonquin Road, Arlington Heights, IL 60005. (847) 228-9900. <<http://www.plasticsurgery.org>>.

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Sunstroke see **Heat disorders**

Superficial phlebitis see **Thrombophlebitis**

Superior vena cava syndrome

Definition

The superior vena cava is the major vein in the chest that carries blood from the upper part of the body in to the heart. A restriction of the blood flow (occlusion) through this vein can cause superior vena cava syndrome (SVCS).

Description

Superior vena cava syndrome is a partial occlusion of the superior vena cava. This leads to a lower than normal blood flow through this major vein.

SVCS is also called superior mediastinal syndrome and/or superior vena cava obstruction.

Causes and symptoms

More than 95% of all cases of SVCS are associated with cancers involving the upper chest. The cancers most commonly associated with SVCS are advanced lung cancers, which account for nearly 80% of all cases of SVCS, and lymphoma. Cancers that have spread (metastasized) to the chest, such as metastatic **breast cancer** to the chest and metastatic **testicular cancer** to the chest have also been shown to cause SVCS.

Other causes of SVCS include: the formation of a blood clot in the superior vena cava, enlargement of the thyroid gland, **tuberculosis**, and **sarcoidosis**.

The symptoms of SVCS include:

- change in voice
- confusion
- cough
- enlargement of the veins in the upper body, particularly those in the arms
- headache
- light-headedness
- shortness of breath
- swelling of the arms
- swelling of the face
- trouble swallowing

Diagnosis

SVCS should be considered in any **cancer** patient with swelling of the face and arms. This diagnosis can be confirmed by x ray, computerized tomography (CT) scan, or medical resonance imaging (MRI) of the chest that reveals a partial occlusion of the superior vena cava.

Treatment

Treatment of SVCS depends on the underlying cancer that is causing it. This treatment may include radiation, **chemotherapy**, or a combination of both. In some cases, surgical procedures may be performed to open (dilate) the vessel. These procedures are generally performed by a trained radiologist or vascular surgeon.

KEY TERMS

Metastasis—The spread of a cancer from one part of the body (where the cancer originated) to another part of the body.

Sarcoidosis—A disease of unknown origin in which there is chronic (recurrent) swelling in the lymph nodes and other tissues.

Superior vena cava—The major vein that carries blood from the upper body to the heart.

Thymoma—A tumor that originates in the thymus, a small gland just in front of the heart that produces hormones necessary for the development of certain components of the immune system.

Alternative treatment

Since treatment of SVCS is aimed at treating the underlying disorder that is causing SVCS, alternative treatments must also focus on treating these underlying causes. Alternative treatments for cancer include **acupuncture**, **aromatherapy**, herbal remedies, **hydrotherapy**, hypnosis, and massage, among many others.

Prognosis

The prognosis depends on the underlying cause of SVCS. In cases of SVCS caused by lung cancers, the prognosis is generally rather poor since SVCS does not generally occur until the later stages of these diseases.

Prevention

SVCS may be prevented by early medical intervention to halt and/or reverse the cancer which, in a later stage, would have lead to SVCS.

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Paul A. Johnson, Ed.M.

Supportive cancer therapy see **Cancer therapy, supportive**

Surfactant

Definition

Surfactant is a complex naturally occurring substance made of six lipids (fats) and four proteins that is produced in the lungs. It can also be manufactured synthetically.

Purpose

Surfactant reduces the surface tension of fluid in the lungs and helps make the small air sacs in the lungs (alveoli) more stable. This keeps them from collapsing when an individual exhales. In preparation for breathing air, fetuses begin making surfactant while still in the womb. Babies that are born very prematurely often lack adequate surfactant and must receive surfactant replacement therapy immediately after birth in order to breathe.

Precautions

Babies are considered premature if they are born before 37 weeks gestation. Fetuses begin to produce surfactant between weeks 24 and 28. By about 35 weeks, most babies have enough naturally produced surfactant to keep the alveoli from collapsing. Babies born before 35 weeks, especially those born very prematurely (before 30 weeks), are likely to need surfactant replacement therapy. Over half the babies born before 28 weeks gestation need this treatment, while about one-third born between 32 and 36 weeks need supplemental surfactant. Some very premature infants may also need to be placed on a mechanical ventilator.

Description

The lungs consist of spongy tissue filled with air spaces called alveoli. In the alveoli, oxygen is taken up by the blood and carbon dioxide, a waste product of cellular metabolism, is released and exhaled. For

efficient oxygen-carbon dioxide exchange to occur, the surface area of the alveoli must be as large as possible. Under normal conditions, when a person exhales, the alveoli would collapse into each other and form larger air sacs with less surface area. Surfactant prevents this collapse by reducing the surface tension of the fluids that line the lungs and helping to equalize the pressures between large and small air spaces.

Surface tension is a measure of the attraction molecules of a fluid have for each other. The attractive force pulls fluids into a shape with the smallest surface area. This is why a drop of water on a flat surface is rounded rather than flat. If the surface tension is lowered, the attraction among molecules of the fluid is decreased and the surface area of the fluid increases. For example, if a drop of detergent is added to a drop of water, the detergent reduces the surface tension and the drop of water flattens out.

In the lungs, surfactant reduces the surface tension and helps to maximize the surface area available for gas exchange. Without adequate surfactant, a baby works much harder to breathe, becomes exhausted, and does not get enough oxygen. Babies that do not have enough surfactant to breathe normally at birth are said to have infant **respiratory distress syndrome** (RDS) or hyaline membrane disease (HMD).

Babies with RDS are given replacement surfactant as soon as possible within the first six hours after birth. Manufactured surfactant is a white powder that is mixed with sterile water. It is given through a breathing tube (endotracheal tube) that is inserted in the baby's lungs. Multiple doses are usually required.

Surfactant replacement therapy continues until the baby's lungs have matured enough to make surfactant on their own. Some very premature babies are also put on mechanical respirators to help them breathe. Surfactant replacement therapy has reduced deaths due to respiratory distress by 50% since the early 1990s. This therapy is expensive, but it is normally covered by insurance.

Preparation

The administration of surfactant is often a neonatal emergency. The only way to prevent the need for surfactant replacement therapy is to prevent a premature birth. Mothers who are at known high risk to deliver prematurely are given drugs called **corticosteroids** toward the end of the **pregnancy** that stimulate the lungs of the fetus to mature and begin producing surfactant sooner. This helps reduce the need for

KEY TERMS

Alveolus (plural alveoli)—The terminal air sacs of the lungs where gas exchange occurs.

Hyaline membrane—A thin layer of cells that line the lung.

Surface tension—The attraction of molecules in a fluid for each other.

surfactant replacement therapy. Although babies of all races may be born prematurely, **prematurity** is more common if the mother is diabetic, is carrying multiple fetuses, or has delivered a previous premature baby. The decision to use surfactant replacement therapy is based on the condition of the baby, its blood oxygen level, and degree of respiratory distress.

Aftercare

Babies receiving surfactant therapy are normally cared for by a neonatologist, a pediatrician that specializes in newborn care. Premature newborns often have other health problems in addition to RDS. Aftercare varies depending on their other health risks.

Risks

Delivery of surfactant requires inserting a breathing tube into the baby's lungs. Complications of this therapy include air leaking into the area between the chest wall and the lungs and air leaking into the sac around the heart. Some infants also develop chronic lung disease.

Normal results

Normally surfactant replacement therapy keeps the infant alive until the lungs start producing their own surfactant.

Abnormal results

Surfactant replacement therapy is very effective if begun within six hours after birth. When it fails, **death** may result.

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Tish Davidson, A.M.

Surgical debridement see **Debridement**

Swallowing disorders

Definition

Swallowing disorders include a number of diseases and conditions that cause difficulty in passing food or liquid from the mouth to the stomach.

Description

Although normally swallowing is automatic, it is a complex process involving several phases and 29 muscles. Saliva helps soften food as it is chewed. The tongue helps move food to the back of the mouth, triggering a swallowing reflex that passes food through the pharynx. The epiglottis helps keep food from mistakenly going down the windpipe and into the esophagus, the canal that carries food to the stomach. Swallowing disorders can occur at any phase in the swallowing process. The medical term for difficult swallowing is dysphagia.

Each year, about 10 million people in the United States require medical evaluation for swallowing problems. Some experts say that about 10% of Americans develop symptoms of swallowing disorders in adulthood. Elderly people are the most likely to have problems with swallowing.

Causes and symptoms

Swallowing disorders often result from other conditions and diseases. For example, Parkinson's disease, **cerebral palsy**, **stroke**, **head injury**, and other central nervous system conditions can damage the

muscles and nerves involved in swallowing. Some people are born with abnormalities in the swallowing structures, such as infants with **cleft palate**.

Some cancers can lead to swallowing disorders. **Esophageal cancer** can cause narrowing and eventual blockage of the esophagus. Surgery and **radiation therapy for head and neck cancer** can restrict or weaken tongue motion, paralyze vocal cords, or cause muscle damage that affects swallowing. An inflamed esophagus, often resulting from gastroesophageal reflux disease (GERD), can cause painful or difficult swallowing. Infections of the esophagus also can inflame it and cause it to narrow. Swallowing difficulty may result from **aging**, though researchers are not certain why.

The most common symptoms people report are **choking** and the feeling that food feels stuck in the throat. Other symptoms include needing to swallow many times to clear food from the mouth and throat, a gurgly, wet sound to the voice after swallowing, having to clear the throat after eating, coughing, **pain** while swallowing, bringing food back up (regurgitation), food or acid backing up into the throat, unexpected weight loss, and not being able to swallow at all. Children also may gag during meals and may have excessive drooling or leaking of food or liquid from their mouths during meals. They may have difficulty breathing when eating or drinking, spit up frequently and lag behind in weight gain. They also may have recurring **pneumonia** or respiratory infections.

Diagnosis

A physician should perform a full head and neck examination based on the patient's symptoms. Speech-language pathologists may aid in the diagnosis. Physicians also might order a swallowing test to study how the patient swallows. The patient will be asked to drink a liquid with a contrast agent called barium that will show up on x rays of the throat and upper chest. The exam might be imaged with a technique called video fluoroscopy, which will take motion camera images in addition to still images. For this exam, the patient may be asked to swallow liquid, paste, and solids. A speech pathologist may work with the radiologist to perform this exam.

If the physician thinks the problem originates in the lower esophagus or has concerns about an abnormality in the esophagus, an endoscopy may be ordered. This test involves passing a thin, flexible instrument called an endoscope down the throat. The lighted endoscope helps the physician view the esophagus. Other tests may be used, including ultrasound.

KEY TERMS

Cleft palate—An opening or hole in the roof of the mouth that occurs at birth when the roof fails to fully develop in the infant.

Epiglottis—A thin layer of cartilage behind the tongue that helps block food from entering the windpipe.

Pharynx—The muscular cavity that leads from the mouth and nasal passages to the larynx and esophagus.

Treatment

Treatment will depend on the cause of the swallowing problem. Special exercises may help strengthen the muscles used for chewing and swallowing. Problems originating in the mouth may be treated with artificial saliva, improved hydration or better dental care. Esophageal problems will be treated depending on the cause. Patients with GERD will receive medications and instructions on how to better manage the disease. Esophageal **cancer** is a life-threatening disease that will involve coordinating care with an oncologist. Many patients will receive help with their disorders from speech pathologists. Special liquid **diets** may be ordered for patients who continue to have trouble chewing or swallowing. In severe cases, the patient may need a feeding tube that bypasses the part of the swallowing system that does not work.

Alternative treatment

Some herbs that may help improve swallowing include oil of peppermint and licorice. Valerian may be used as a tea. Homeopathic physicians may suggest some remedies aimed at improving bloating, **indigestion**, or **cough**. Alternative care should be sought from licensed practitioners and coordinated with physician care.

Prognosis

In many cases, these disorders can be corrected. If not treated, swallowing disorders can lead to serious complications, including **dehydration** and **malnutrition**. There also is a risk of food entering the airway (aspiration) as a person attempts to swallow, which can lead to aspiration pneumonia as the food particles enter the lungs.

Prevention

Many causes of swallowing disorders cannot be prevented. Slowly and fully chewing food helps. People with GERD should manage it to lower the risk of developing swallowing difficulties.

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Teresa G. Odle

Swan-Ganz catheterization see **Pulmonary artery catheterization**

Sweating, excessive see **Hyperhidrosis**

Swimmer's ear see **Otitis externa**

Swimming pool conjunctivitis see **Inclusion conjunctivitis**

Swollen glands see **Lymphadenitis**

occasionally in pregnant women. It is closely associated with **rheumatic fever** following a throat infection. The disorder is named for Thomas Sydenham (1624–1689), an English doctor who first described it in 1686. Other names for Sydenham's chorea include simple chorea, chorea minor, acute chorea, rheumatic chorea, juvenile chorea, and St. Vitus' dance. The English word "chorea" itself comes from the Greek word *choreia*, which means "dance." The disorder takes its name from the rapid involuntary jerking or twitching movements of the patient's face, limbs, and upper body.

Description

Sydenham's chorea is a disorder that occurs in children and is associated with rheumatic **fever**. Rheumatic fever is an acute infectious disease caused by certain types of streptococci bacteria. It usually starts with **strep throat** or **tonsillitis**. These types of streptococci are able to cause disease throughout the body. The most serious damage caused by rheumatic fever is to the valves in the heart. At one time, rheumatic fever was the most common cause of damaged heart valves, and it still is in most developing countries around the world. Rheumatic fever and rheumatic heart disease are still present in the industrialized countries, but the incidence has dropped substantially.

Both acute rheumatic fever and Sydenham's chorea are relatively uncommon disorders in the United States as of 2004. According to the Centers for Disease Control and Prevention (CDC), only 1%–3% of people with streptococcal throat infections develop acute rheumatic fever (ARF); thus the incidence of ARF in the United States is thought to be about 0.5 per 100,000 patients between 5 and 17 years of age.

With regard to age, the incidence of Sydenham's chorea is higher in childhood and adolescence than in adult life. It occurs more frequently in females than in males; the gender ratio is thought to be about 2 F: 1 M. Since the peak incidence of rheumatic fever in North America occurs in late winter and spring, Sydenham's chorea is more likely to occur in the summer and early fall. There is no evidence as of 2004 that the disorder selectively affects specific racial or ethnic groups.

Rheumatic fever may appear in several different forms. Sydenham's chorea is one of five major criteria for the diagnosis of rheumatic fever. There are also four minor criteria and two types of laboratory tests associated with the disease. The "Jones criteria" define the diagnosis. They require laboratory evidence of a streptococcal infection plus two or more of the

Sydenham's chorea

Definition

Sydenham's chorea is an acute but self-limited movement disorder that occurs most commonly in children between the ages of 5 and 15, and

criteria. The laboratory evidence may be identification of streptococci from a **sore throat** or antibodies to streptococcus in the blood. The most common criteria are arthritis and heart disease, occurring in half to three-quarters of the patients. Sydenham's chorea, characteristic nodules under the skin, and a specific type of skin rash occur only 10% of the time.

About 20% of patients diagnosed with Sydenham's chorea experience a recurrence of the disorder, usually within two years of the first episode. Most women who develop Sydenham's during **pregnancy** have a history of acute rheumatic fever in childhood or of using birth control pills containing estrogen.

Causes and symptoms

Sydenham's is caused by certain types of streptococci called Group A beta-hemolytic streptococci or GAS bacteria. In general, streptococci are spherical-shaped anaerobic bacteria that occur in pairs or chains. GAS bacteria belong to a subcategory known as pyogenic streptococci, which means that the infections they cause produce pus. These particular germs seem to be able to create an immune response that attacks the body's own tissues along with the germs. Those tissues are joints, heart valves, skin, and brain.

The initial throat infection that leads to Sydenham's chorea is typically followed by a symptom-free period of 1–5 weeks. The patient then develops an acute case of rheumatic fever (ARF), an inflammatory disease that affects multiple organ systems and tissues of the body. In most patients, ARF is characterized by fever, arthritis in one or more joints, and carditis, or inflammation of the heart. In about 20% of patients, however, Sydenham's chorea is the only indication of ARF. Sydenham's is considered a delayed complication of rheumatic fever; it may begin as late as 12 months after the initial sore throat, and it may start only after the patient's temperature and other physical signs have returned to normal. The average time interval between the pharyngitis and the first symptoms of Sydenham's, however, is eight or nine weeks.

It is difficult to describe a “typical” case of Sydenham's chorea because the symptoms vary in speed of onset as well as severity. Most patients have an acute onset of the disorder, but in others, the onset is insidious, which means that the symptoms develop slowly and gradually. In some cases, the child's physical symptoms are present for four to five weeks before they become severe enough for the parents to consult a doctor. In other cases, emotional or psychiatric

symptoms precede the clumsiness and involuntary muscular movements that characterize the disorder. The psychiatric symptoms that may develop in patients with Sydenham's chorea are one reason why it is sometimes categorized as a PANDAS disorder. PANDAS stands for Pediatric Autoimmune Neuropsychiatric Disorders Associated with **Streptococcal Infections**.

Behavioral or emotional disturbances that have been observed with Sydenham's include:

- frequent mood changes
- episodes of uncontrollable crying
- behavioral regression; that is, acting like much younger children
- mental confusion
- general irritability
- difficulty concentrating
- impulsive behavior

Some researchers think that children who have had Sydenham's are at increased risk of developing **obsessive-compulsive disorder (OCD)**. OCD is characterized by obsessions, which are unwanted recurrent thoughts, images, or impulses, and by compulsions, which are repetitive rituals, mental acts, or behaviors. Obsessions in children often take the form of fears of intruders or harm coming to a family member. Compulsions may include such acts as counting silently, washing the hands over and over, insisting on keeping items in a specific order, checking repeatedly to make sure a door is locked, and similar behaviors.

Diagnosis

Because rheumatic fever is such a damaging disease, a complete evaluation should be done whenever it is suspected. This includes cultures for streptococci, blood tests, and usually an electrocardiogram (heart-beat mapping to detect abnormalities).

The diagnosis of Sydenham's is also based on the doctor's observation of the patient's involuntary movements. Unlike tics, the movements associated with chorea are not repetitive; and unlike the behavior of hyperactive children, the movements are not intentional. The recent onset of the movements rules out a diagnosis of **cerebral palsy**. If the doctor suspects Sydenham's, he or she may ask the patient to stick out the tongue and keep it in that position, or to squeeze the doctor's hand. Many patients with Sydenham's cannot hold their mouth open and keep

the tongue out for more than a second or two. Another characteristic of Sydenham's is an inability to grip with a steady pressure; when the patient squeezes the doctor's hand, the strength of the grip will increase and decrease in an erratic fashion. This characteristic is sometimes called the "milking sign."

Treatment

Suspected streptococcal infections must be treated. All the other manifestations of rheumatic fever, including Sydenham's chorea and excluding heart valve damage, remit with the acute disease and do not require treatment. Sydenham's chorea generally lasts for several months.

Most patients with Sydenham's chorea recover after a period of bed rest and temporary limitation of normal activities. In most cases the symptoms disappear gradually rather than stopping abruptly.

Most doctors recommend ongoing treatment with penicillin to prevent a recurrence of rheumatic fever or Sydenham's chorea, although there is some disagreement as to whether this treatment should continue for 5 years after an acute attack or for the rest of the patient's life. The penicillin may be given orally or by injection. Patients who cannot take penicillin may be given erythromycin or sulfadiazine.

Prognosis

Sydenham's chorea usually clears up without complications when the rheumatic fever is treated. The heart valve damage associated with rheumatic fever may lead to heart trouble and require a surgical valve repair or replacement.

In most cases of Sydenham's, the patient recovers completely, although a recurrence is possible. In a very few cases—about 1.5% of patients diagnosed with Sydenham's—there may be increasing muscle stiffness and loss of muscle tone resulting in disability. This condition is occasionally referred to as paralytic chorea.

Prevention

All cases of strep throat in children should be treated with a full 10 days of **antibiotics** (penicillin or erythromycin). Treatment may best be delayed a day or two to allow the body to build up its own antibodies. In addition, for those who have had an episode of rheumatic fever or have damaged heart valves from any other cause, prophylactic antibiotics should be continued to prevent recurrence.

KEY TERMS

Arthralgia—Joint pain.

Chorea—A term that is used to refer to rapid, jerky, involuntary movements of the limbs or face that characterize several different disorders of the nervous system, including chorea of pregnancy and Huntington's chorea as well as Sydenham's chorea.

Electrocardiogram—Mapping the electrical activity of the heart.

Insidious—Developing in a stealthy or gradual manner. Sydenham's chorea may have an insidious onset.

PANDAS disorders—A group of childhood disorders associated with such streptococcal infections as scarlet fever and "strep throat." The acronym stands for Pediatric Autoimmune Neuropsychiatric Disorders Associated with Streptococcal Infections. Sydenham's chorea is considered a PANDAS disorder.

Pharyngitis—Inflammation of the throat, accompanied by dryness and pain. Pharyngitis caused by a streptococcal infection is the usual trigger of Sydenham's chorea.

Rheumatic fever—Chiefly childhood disease marked by fever, inflammation, joint pain, and Sydenham's chorea. It is often recurrent and can lead to heart valve damage.

St. Vitus' dance—Another name for Sydenham's chorea. St. Vitus was a fourth-century martyr who became the patron saint of dancers and actors during the Middle Ages. He was also invoked for protection against nervous disorders, epilepsy, and the disease that bears his name.

Streptococcus (plural, streptococci)—A genus of spherical-shaped anaerobic bacteria occurring in pairs or chains. Sydenham's chorea is considered a complication of a streptococcal throat infection.

Tonsillitis—Inflammation of the tonsils, which are in the back of the throat.

It is possible to eradicate dangerous GAS bacteria from a community by culturing everyone's throat and treating everyone who tests positive. This is worth doing wherever a case of rheumatic fever appears, but it is expensive and requires many resources.

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American Academy of Family Physicians (AAFP). 11400 Tomahawk Creek Parkway, Leawood, KS 66211-2672. (800) 274-2237 or (913) 906-6000. *lt;<http://www.aafp.org> > .

National Institute of Neurological Disorders and Stroke (NINDS). NIH Neurological Institute, P. O. Box 5801, Bethesda, MD 20824. (800) 352-9424 or (301) 496-5751. < <http://www.ninds.nih.gov> > .

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Sympathectomy

Definition

Sympathectomy is a surgical procedure that destroys nerves in the sympathetic nervous system. The procedure is done to increase blood flow and decrease long-term **pain** in certain diseases that cause narrowed blood vessels. It can also be used to decrease excessive sweating. This surgical procedure cuts or destroys the sympathetic ganglia, collections of nerve cell bodies in clusters along the thoracic or lumbar spinal cord.

Purpose

The autonomic nervous system that controls unwilling (involuntary) body functions, such as breathing, sweating, and blood pressure, are divided into the sympathetic and the parasympathetic nervous systems. The sympathetic nervous system speeds the heart rate, narrows (constricts) blood vessels, and raises blood pressure. Blood pressure is controlled by means of nerve cells that run through sheaths around the arteries. The sympathetic nervous system can be described as the "fight or flight" system because it allows us to respond to danger by fighting off an attacker or by running away. When danger threatens, the sympathetic nervous system increases heart and respiratory rate, increases blood flow to muscles, and decreases blood flow to other areas, such as skin, digestive tract, and limb veins. The net effect is an increase in blood pressure.

Sympathectomy is performed to relieve intermittent constricting of blood vessels (**ischemia**) when the fingers, toes, ears, or nose are exposed to cold (Raynaud's phenomenon). In Raynaud's phenomenon, the affected extremities turn white, then blue, and red as the blood supply is cut off. The color changes are accompanied by **numbness**, **tingling**, burning, and pain. Normal color and feeling are restored when heat is applied. The condition sometimes occurs without direct cause but it is more often caused by an underlying medical condition, such as **rheumatoid arthritis**. Sympathectomy is usually less effective

when Raynaud's is caused by an underlying medical condition. Narrowed blood vessels in the legs that cause painful cramping (claudication) are also treated with sympathectomy.

Sympathectomy may be helpful in treating **reflex sympathetic dystrophy** (RSD), a condition that sometimes develops after injury. In RSD, the affected limb is painful (causalgia) and swollen. The color, temperature, and texture of the skin change. These symptoms are related to prolonged and excessive activity of the sympathetic nervous system.

Because sweating is controlled by the sympathetic nervous system, sympathectomy is also effective in treating excessive sweating (**hyperhidrosis**) of the palms, armpits, or face.

Precautions

To determine whether sympathectomy is needed, a reversible block of the affected nerve cell (**ganglion**) should be done. A reversible ganglion block interrupts nerve impulses by means of steroid and anesthetic injected into it. If the block has a positive effect on pain and blood flow in the affected area, the sympathectomy will probably be helpful. The surgical procedure should be performed only if conservative treatment has not worked. Conservative treatment includes avoiding exposure to **stress** and cold, physical therapy, and medications.

Sympathectomy is most likely to be effective in relieving the pain of reflex sympathetic dystrophy if it is done soon after the injury occurs. However, increased benefit from early surgery should be balanced against time needed to promote spontaneous recovery and response to conservative treatment.

Description

Sympathectomy was traditionally done as an inpatient surgical procedure under **general anesthesia**. An incision was made on the mid-back, exposing the ganglia to be cut. Recent techniques are less invasive and may be done under **local anesthesia** and as outpatient surgery. If only one arm or leg is affected, it may be treated with a percutaneous radiofrequency technique. In this technique, the surgeon locates the ganglia by a combination of x ray and electrical stimulation. The ganglia are destroyed by applying radio waves through electrodes on the skin.

Sympathectomy for hyperhidrosis can be done by making a small incision under the armpit and introducing air into the chest cavity. The surgeon inserts a fiber optic tube (endoscope) that projects an image of

the operation on a video screen. The ganglia can then be cut with fine scissors attached to the endoscope. Laser beams can also be used to destroy the ganglia.

Preparation

As with any surgery, patients should discuss expected results and possible risks with their surgeons. They should tell their surgeons all medications they are taking and all their medical problems, and they should be in good general health. To improve general health, the patient may be asked to lose weight, give up **smoking** or alcohol, and get the proper sleep, diet, and **exercise**. Immediately before the surgery, patients will not be permitted to eat or drink, and the surgical site will be cleaned and scrubbed.

Aftercare

The surgeon will inform the patient about specific aftercare needed for the technique used. **Doppler ultrasonography**, a test using sound waves to measure blood flow, can help to determine whether sympathectomy has had a positive result.

Risks

Side effects of sympathectomy may include decreased blood pressure while standing, which may cause **fainting** spells. After sympathectomy in men, semen is sometimes ejaculated into the bladder, which may impair fertility. After a sympathectomy done by inserting an endoscope in the chest cavity, patients may experience chest pain with deep breathing. This problem usually disappears within two weeks. They may also experience **pneumothorax** (air in the chest cavity).

In 30% of cases, surgery for hyperhidrosis may cause increased sweating on the chest. In 2% of cases, this surgery causes increased sweating in other areas, including increased facial sweating while eating. Other complications occur less frequently. These complications include Horner's syndrome, a condition of the nervous system that causes the pupil of the eye to close, the eyelid to droop, and sweating to decrease on one side of the face. Other rare complications are nasal blockage and pain of the nerves supplying the skin between the ribs.

Normal results

Some studies report that sympathectomy relieves causalgia in as many as 75% of cases. The studies also show that it relieves hyperhidrosis in more than 90%

KEY TERMS

Causalgia—A severe burning sensation sometimes accompanied by redness and inflammation of the skin. Causalgia is caused by injury to a nerve outside the spinal cord.

Claudication—Cramping or pain in a leg caused by poor blood circulation. This condition is frequently caused by hardening of the arteries (atherosclerosis). Intermittent claudication occurs only at certain times, usually after exercise, and is relieved by rest.

Fiberoptics—In medicine, fiberoptics uses glass or plastic fibers to transmit light through a specially designed tube. The tube is inserted into organs or body cavities where it transmits a magnified image of the internal body structures.

Hyperhidrosis—Excessive sweating. Hyperhidrosis can be caused by heat, overactive thyroid glands, strong emotion, menopause, or infection.

Parasympathetic nervous system—The division of the autonomic (involuntary or unwilled) nervous system that slows heart rate, increases digestive and gland activity, and relaxes the sphincter muscles that close off body organs.

Percutaneous—Performed through the skin, from the Latin *per*, meaning through and *cutis*, meaning skin.

Pneumothorax—A collection of air or gas in the chest cavity that causes a lung to collapse. Pneumothorax may be caused by an open chest wound that admits air.

of cases. The less invasive procedures cause very little scarring. Most patients stay in the hospital for less than one day and return to work within the week.

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The American Institute for Hyperhidrosis Page. <<http://www.handsweat.com>>.

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Syncope see **Fainting**

Syndactyly see **Polydactyly and syndactyly**

Synergistic gangrene see **Flesh-eating disease**

Synovial fluid analysis see **Joint fluid analysis**

Synovial membrane biopsy see **Joint biopsy**

Syphilis

Definition

Syphilis is an infectious systemic disease that may be either congenital or acquired through sexual contact or contaminated needles.

Description

Syphilis has both acute and chronic forms that produce a wide variety of symptoms affecting most of the body's organ systems. The range of symptoms makes it easy to confuse syphilis with less serious diseases and ignore its early signs. Acquired syphilis has four stages (primary, secondary, latent, and tertiary) and can be spread by sexual contact during the first three of these four stages.

Syphilis, which is also called lues (from a Latin word meaning **plague**), has been a major public health problem since the sixteenth century. The disease was treated with mercury or other ineffective remedies until World War I, when effective treatments based on arsenic or bismuth were introduced. These were succeeded by **antibiotics** after World War II. At that time, the number of cases in the general population decreased, partly because of aggressive public health measures. This temporary decrease, combined with the greater amount of attention given to **AIDS** in recent years, leads some people to think that syphilis is no longer a serious problem. In actual fact, the number of cases of syphilis in the United States has risen since 1980. This increase affects both sexes, all races, all parts of the nation, and all age groups, including adults over 60. The number of women of childbearing age with syphilis is the highest that has been recorded since the 1940s. About 25,000 cases of infectious syphilis in adults are reported annually in the United States. It is estimated, however, that 400,000 people in the United States need treatment for syphilis every year, and that the annual worldwide total is 50 million persons.

In 1999, the Centers for Disease Control and Prevention (CDC) joined several other federal agencies in announcing the "National Plan to Eliminate Syphilis in the United States." Eliminating the disease was defined as the absence of transmission of the disease; that is, no transmission after 90 days following the report of an imported index case. The national goals for eliminating syphilis include bringing the annual number of reported cases in the United States below 1000, and increasing the number of syphilis-free counties to 90% by 2005. In November 2002, the CDC

released figures for 2000–2001, which indicate that the number of reported cases of primary and secondary syphilis rose slightly. This rise, however, occurred only among men who have sex with other men. The CDC also stated that the number of new cases of syphilis has actually declined among women as well as among non-Hispanic blacks.

The increased incidence of syphilis since the 1970s is associated with drug **abuse** as well as changes in sexual behavior. The connections between drug abuse and syphilis include needle sharing and exchanging sex for drugs. In addition, people using drugs are more likely to engage in risky sexual practices. As of 2002, the risk of contracting syphilis is particularly high among those who abuse crack **cocaine**.

With respect to changing patterns of conduct, a sharp increase in the number of people having sex with multiple partners makes it more difficult for public health doctors to trace the contacts of infected persons. Women are not necessarily protected by having sex only with other women; in the past few years, several cases have been reported of female-to-female transmission of syphilis through oral-genital contact. In addition, the incidence of syphilis among men who have sex with other men continues to rise. Several studies in Latin America as well as in the United States reported in late 2002 that unprotected sexual intercourse is on the increase among gay and bisexual men.

Changing patterns of sexual behavior have led to a striking increase in the number of cases of syphilis in eastern Europe since the collapse of the Soviet Union; Slovenia reported an 18-fold increase in reported cases of syphilis just between 1993 and 1994. Over half of the new cases were linked to a source of infection in another European country.

In general, high-risk groups for syphilis in the United States and Canada include:

- sexually active teenagers
- people infected with another sexually transmitted disease (STD), including AIDS
- sexually abused children
- women of childbearing age
- prostitutes of either sex and their customers
- prisoners
- persons who abuse drugs or alcohol

The chances of contracting syphilis from an infected person in the early stages of the disease during unprotected sex are between 30–50%.

Causes and symptoms

Syphilis is caused by a spirochete, *Treponema pallidum*. A spirochete is a thin spiral- or coil-shaped bacterium that enters the body through the mucous membranes or breaks in the skin. In 90% of cases, the spirochete is transmitted by sexual contact. Transmission by blood **transfusion** is possible but rare; not only because blood products are screened for the disease, but also because the spirochetes die within 24 hours in stored blood. Other methods of transmission are highly unlikely because *T. pallidum* is easily killed by heat and drying.

Primary syphilis

Primary syphilis is the stage of the organism's entry into the body. The first signs of infection are not always noticed. After an incubation period ranging between 10 and 90 days, the patient develops a chancre, which is a small blister-like sore about 0.5 in (13 mm) in size. Most chancres are on the genitals, but may also develop in or on the mouth or on the breasts. Rectal chancres are common in male homosexuals. Chancres in women are sometimes overlooked if they develop in the vagina or on the cervix. The chancres are not painful and disappear in three to six weeks even without treatment. They resemble the ulcers of **lymphogranuloma venereum**, herpes simplex virus, or skin tumors.

About 70% of patients with primary syphilis also develop swollen lymph nodes near the chancre. The nodes may have a firm or rubbery feel when the doctor touches them but are not usually painful.

Secondary syphilis

Syphilis enters its secondary stage between six to eight weeks and six months after the infection begins. Chancres may still be present but are usually healing. Secondary syphilis is a systemic infection marked by the eruption of skin **rashes** and ulcers in the mucous membranes. The skin rash may mimic a number of other skin disorders such as drug reactions, **rubella**, **ringworm**, mononucleosis, and **pityriasis rosea**. Characteristics that point to syphilis include:

- a coppery color
- absence of **pain** or **itching**
- occurrence on the palms of hands and soles of feet

The skin eruption may resolve in a few weeks or last as long as a year. The patient may also develop condylomata lata, which are weepy pinkish or grey areas of flattened skin in the moist areas of the body.

The skin rashes, mouth and genital ulcers, and condylomata lata are all highly infectious.

About 50% of patients with secondary syphilis develop swollen lymph nodes in the armpits, groin, and neck areas; about 10% develop inflammations of the eyes, kidney, liver, spleen, bones, joints, or the meninges (membranes covering the brain and spinal cord). They may also have a flulike general illness with a low **fever**, chills, loss of appetite, headaches, runny nose, **sore throat**, and aching joints.

Latent syphilis

Latent syphilis is a phase of the disease characterized by relative absence of external symptoms. The term latent does not mean that the disease is not progressing or that the patient cannot infect others. For example, pregnant women can transmit syphilis to their unborn children during the latency period.

The latent phase is sometimes divided into early latency (less than two years after infection) and late latency. During early latency, patients are at risk for spontaneous relapses marked by recurrence of the ulcers and skin rashes of secondary syphilis. In late latency, these recurrences are much less likely. Late latency may either resolve spontaneously or continue for the rest of the patient's life.

Tertiary syphilis

Untreated syphilis progresses to a third or tertiary stage in about 35–40% of patients. Patients with tertiary syphilis cannot infect others with the disease. It is thought that the symptoms of this stage are a delayed hypersensitivity reaction to the spirochetes. Some patients develop so-called benign late syphilis, which begins between three and 10 years after infection and is characterized by the development of gummas. Gummas are rubbery tumor-like growths that are most likely to involve the skin or long bones but may also develop in the eyes, mucous membranes, throat, liver, or stomach lining. Gummas are increasingly uncommon since the introduction of antibiotics for treating syphilis. Benign late syphilis is usually rapid in onset and responds well to treatment.

CARDIOVASCULAR SYPHILIS. Cardiovascular syphilis occurs in 10–15% of patients who have progressed to tertiary syphilis. It develops between 10 and 25 years after infection and often occurs together with neurosyphilis. Cardiovascular syphilis usually begins as an inflammation of the arteries leading from the heart and causes heart attacks, scarring of the aortic valves, congestive **heart failure**, or the formation of an **aortic aneurysm**.

NEUROSYPHILIS. About 8% of patients with untreated syphilis will develop symptoms in the central nervous system that include both physical and psychiatric symptoms. Neurosyphilis can appear at any time, from 5–35 years after the onset of primary syphilis. It affects men more frequently than women and Caucasians more frequently than African Americans.

Neurosyphilis is classified into four types:

- **Asymptomatic.** In this form of neurosyphilis, the patient's spinal fluid gives abnormal test results but there are no symptoms affecting the central nervous system.
- **Meningovascular.** This type of neurosyphilis is marked by changes in the blood vessels of the brain or inflammation of the meninges (the tissue layers covering the brain and spinal cord). The patient develops headaches, irritability, and visual problems. If the spinal cord is involved, the patient may experience weakness of the shoulder and upper arm muscles.
- **Tabes dorsalis.** Tabes dorsalis is a progressive degeneration of the spinal cord and nerve roots. Patients lose their sense of perception of one's body position and orientation in space (proprioception), resulting in difficulties walking and loss of muscle reflexes. They may also have shooting pains in the legs and periodic episodes of pain in the abdomen, throat, bladder, or rectum. Tabes dorsalis is sometimes called locomotor ataxia.
- **General paresis.** General paresis refers to the effects of neurosyphilis on the cortex of the brain. The patient has a slow but progressive loss of memory, ability to concentrate, and interest in self-care. Personality changes may include irresponsible behavior, depression, **delusions** of grandeur, or complete **psychosis**. General paresis is sometimes called **dementia** paralytica, and is most common in patients over 40.

Special populations

CONGENITAL SYPHILIS. Congenital syphilis has increased at a rate of 400–500% over the past decade, on the basis of criteria introduced by the Centers for Disease Control (CDC) in 1990. In 1994, over 2,200 cases of congenital syphilis were reported in the United States. The prognosis for early congenital syphilis is poor: about 54% of infected fetuses die before or shortly after birth. Those who survive may look normal at birth but show signs of infection between three and eight weeks later.

Infants with early congenital syphilis have systemic symptoms that resemble those of adults with secondary syphilis. There is a 40-60% chance that the child's central nervous system will be infected. These infants may have symptoms ranging from **jaundice**, enlargement of the spleen and liver, and anemia to skin rashes, condylomata lata, inflammation of the lungs, "snuffles" (a persistent runny nose), and swollen lymph nodes.

CHILDREN. Children who develop symptoms after the age of two years are said to have late congenital syphilis. The characteristic symptoms include facial deformities (saddle nose), Hutchinson's teeth (abnormal upper incisors), saber shins, dislocated joints, deafness, **mental retardation**, **paralysis**, and seizure disorders.

PREGNANT WOMEN. Syphilis can be transmitted from the mother to the fetus through the placenta at any time during **pregnancy**, or through the child's contact with syphilitic ulcers during the birth process. The chances of infection are related to the stage of the mother's disease. Almost all infants of mothers with untreated primary or secondary syphilis will be infected, whereas the infection rate drops to 40% if the mother is in the early latent stage and 6-14% if she has late latent syphilis.

Pregnancy does not affect the progression of syphilis in the mother; however, pregnant women should not be treated with **tetracyclines**.

HIV PATIENTS. Syphilis has been closely associated with HIV infection since the late 1980s. Syphilis sometimes mimics the symptoms of AIDS. Conversely, AIDS appears to increase the severity of syphilis in patients suffering from both diseases, and to speed up the development or appearance of neurosyphilis. Patients with HIV are also more likely to develop lues maligna, a skin disease that sometimes occurs in secondary syphilis. Lues maligna is characterized by areas of ulcerated and dying tissue. In addition, HIV patients have a higher rate of treatment failure with penicillin than patients without HIV.

Diagnosis

Patient history and physical diagnosis

The diagnosis of syphilis is often delayed because of the variety of early symptoms, the varying length of the incubation period, and the possibility of not noticing the initial chancre. Patients do not always connect their symptoms with recent sexual contact. They may go to a dermatologist when they develop the skin rash of secondary syphilis rather than to their primary care



This patient has secondary syphilis, evidenced by the appearance of lesions on the skin. (Custom Medical Stock Photo. Reproduced by permission.)

doctor. Women may be diagnosed in the course of a gynecological checkup. Because of the long-term risks of untreated syphilis, certain groups of people are now routinely screened for the disease:

- pregnant women
- sexual contacts or partners of patients diagnosed with syphilis
- children born to mothers with syphilis
- patients with HIV infection
- persons applying for marriage licenses

When the doctor takes the patient's history, he or she will ask about recent sexual contacts in order to determine whether the patient falls into a high-risk group. Other symptoms, such as skin rashes or swollen lymph nodes, will be noted with respect to the dates of the patient's sexual contacts. Definite diagnosis, however, depends on the results of laboratory blood tests.

Blood tests

There are several types of blood tests for syphilis presently used in the United States. Some are used in follow-up monitoring of patients as well as diagnosis.

NONTREPONEMAL ANTIGEN TESTS. Nontreponemal antigen tests are used as screeners. They measure the presence of reagin, which is an antibody formed in reaction to syphilis. In the venereal disease research

laboratory (VDRL) test, a sample of the patient's blood is mixed with cardiolipin and cholesterol. If the mixture forms clumps or masses of matter, the test is considered reactive or positive. The serum sample can be diluted several times to determine the concentration of reagin in the patient's blood.

The rapid plasma reagin (RPR) test works on the same principle as the VDRL. It is available as a kit. The patient's serum is mixed with cardiolipin on a plastic-coated card that can be examined with the naked eye.

Nontreponemal antigen tests require a doctor's interpretation and sometimes further testing. They can yield both false-negative and false-positive results. False-positive results can be caused by other infectious diseases, including mononucleosis, **malaria**, **leprosy**, **rheumatoid arthritis**, and lupus. HIV patients have a particularly high rate (4%, compared to 0.8% of HIV-negative patients) of false-positive results on reagin tests. False-negatives can occur when patients are tested too soon after exposure to syphilis; it takes about 14-21 days after infection for the blood to become reactive.

TREPONEMAL ANTIBODY TESTS. Treponemal antibody tests are used to rule out false-positive results on reagin tests. They measure the presence of antibodies that are specific for *T. pallidum*. The most commonly used tests are the microhemagglutination-*T. pallidum* (MHA-TP) and the fluorescent treponemal antibody absorption (FTA-ABS) tests. In the FTA-ABS, the patient's blood serum is mixed with a preparation that prevents interference from antibodies to other treponemal infections. The test serum is added to a slide containing *T. pallidum*. In a positive reaction, syphilitic antibodies in the blood coat the spirochetes on the slide. The slide is then stained with fluorescein, which causes the coated spirochetes to fluoresce when the slide is viewed under ultraviolet (UV) light. In the MHA-TP test, red blood cells from sheep are coated with *T. pallidum* antigen. The cells will clump if the patient's blood contains antibodies for syphilis.

Treponemal antibody tests are more expensive and more difficult to perform than nontreponemal tests. They are therefore used to confirm the diagnosis of syphilis rather than to screen large groups of people. These tests are, however, very specific and very sensitive; false-positive results are relatively unusual.

INVESTIGATIONAL BLOOD TESTS. Currently, ELISA, Western blot, and PCR testing are being studied as

additional diagnostic tests, particularly for congenital syphilis and neurosyphilis.

Other laboratory tests

MICROSCOPE STUDIES. The diagnosis of syphilis can also be confirmed by identifying spirochetes in samples of tissue or lymphatic fluid. Fresh samples can be made into slides and studied under darkfield illumination. A newer method involves preparing slides from dried fluid smears and staining them with fluorescein for viewing under UV light. This method is replacing darkfield examination because the slides can be mailed to professional laboratories.

SPINAL FLUID TESTS. Testing of cerebrospinal fluid (CSF) is an important part of patient monitoring as well as a diagnostic test. The VDRL and FTA-ABS tests can be performed on CSF as well as on blood. An abnormally high white cell count and elevated protein levels in the CSF, together with positive VDRL results, suggest a possible diagnosis of neurosyphilis. CSF testing is not used for routine screening. It is used most frequently for infants with congenital syphilis, HIV-positive patients, and patients of any age who are not responding to penicillin treatment.

Treatment

Medications

Syphilis is treated with antibiotics given either intramuscularly (benzathine penicillin G or ceftriaxone) or orally (doxycycline, minocycline, tetracycline, or azithromycin). Neurosyphilis is treated with a combination of aqueous crystalline penicillin G, benzathine penicillin G, or doxycycline. It is important to keep the levels of penicillin in the patient's tissues at sufficiently high levels over a period of days or weeks because the spirochetes have a relatively long reproduction time. Penicillin is more effective in treating the early stages of syphilis than the later stages.

In the fall of 2000, the CDC convened a group of medical advisors to discuss backup medications for treating syphilis. Although none of the newer drugs will displace penicillin as the primary drug, the doctors recommended azithromycin and ceftriaxone as medications that should have a larger role in the treatment of syphilis than they presently do.

Doctors do not usually prescribe separate medications for the skin rashes or ulcers of secondary syphilis. The patient is advised to keep them clean and dry, and to avoid exposing others to fluid or discharges from condylomata lata.

Pregnant women should be treated as early in pregnancy as possible. Infected fetuses can be cured if the mother is treated during the second and third trimesters of pregnancy. Infants with proven or suspected congenital syphilis are treated with either aqueous crystalline penicillin G or aqueous procaine penicillin G. Children who acquire syphilis after birth are treated with benzathine penicillin G.

Jarisch-Herxheimer reaction

The Jarisch-Herxheimer reaction, first described in 1895, is a reaction to penicillin treatment that may occur during the late primary, secondary, or early latent stages. The patient develops chills, fever, **head-ache**, and muscle pains within two to six hours after the penicillin is injected. The chancre or rash gets temporarily worse. The Jarisch-Herxheimer reaction, which lasts about a day, is thought to be an allergic reaction to toxins released when the penicillin kills massive numbers of spirochetes.

Alternative treatment

Antibiotics are essential for the treatment of syphilis. Recovery from the disease can be assisted by dietary changes, sleep, **exercise**, and **stress reduction**.

Homeopathy

Homeopathic practitioners are forbidden by law in the United States to claim that homeopathic treatment can cure syphilis. Given the high rate of syphilis in HIV-positive patients, however, some alternative practitioners who are treating AIDS patients with homeopathic remedies maintain that they are beneficial for syphilis as well. The remedies suggested most frequently are *Medorrhinum*, *Syphilinum*, *Mercurius vivus*, and *Aurum*. The historical link between homeopathy and syphilis is Hahnemann's theory of miasms. He thought that the syphilitic miasm was the second oldest cause of constitutional weakness in humans.

Prognosis

The prognosis is good for the early stages of syphilis if the patient is treated promptly and given sufficiently large doses of antibiotics. Treatment failures can occur and patients can be reinfected. There are no definite criteria for cure for patients with primary and secondary syphilis, although patients who are symptom-free and have had negative blood tests for two years after treatment are usually considered cured. Patients should be followed up with blood tests

at one, three, six, and 12 months after treatment, or until the results are negative. CSF should be examined after one year. Patients with recurrences during the latency period should be tested for reinfection.

The prognosis for patients with untreated syphilis is spontaneous remission for about 30%; lifelong latency for another 30%; and potentially fatal tertiary forms of the disease in 40%.

Prevention

Immunity

Patients with syphilis do not acquire lasting immunity against the disease. Currently, no effective vaccine for syphilis has been developed. Prevention depends on a combination of personal and public health measures.

Lifestyle choices

The only reliable methods for preventing transmission of syphilis are sexual abstinence or monogamous relationships between uninfected partners. Condoms offer some protection but protect only the covered parts of the body.

Public health measures

CONTACT TRACING. The law requires reporting of syphilis cases to public health agencies. Sexual contacts of patients diagnosed with syphilis are traced and tested for the disease. This includes all contacts for the past three months in cases of primary syphilis and for the past year in cases of secondary disease. Neither the patients nor their contacts should have sex with anyone until they have been tested and treated.

Because of the rising incidence of syphilis abroad, a growing number of public health physicians are recommending routine screening of immigrants, refugees, and international adoptees for syphilis as of late 2002.

All patients who test positive for syphilis should be tested for HIV infection at the time of diagnosis.

PRENATAL TESTING OF PREGNANT WOMEN. Pregnant women should be tested for syphilis at the time of their first visit for prenatal care, and again shortly before delivery. Proper treatment of secondary syphilis in the mother reduces the risk of congenital syphilis in the infant from 90% to less than 2%.

As of late 2005, many obstetricians and gynecologists are recommending routine screening of non-pregnant as well as pregnant women for syphilis. At present, only about half of obstetricians and

KEY TERMS

Chancere—The initial skin ulcer of primary syphilis, consisting of an open sore with a firm or hard base.

Condylomata lata—Highly infectious patches of weepy pink or gray skin that appear in the moist areas of the body during secondary syphilis.

Darkfield—A technique of microscope examination in which light is directed at an oblique angle through the slide so that organisms look bright against a dark background.

General paresis—A form of neurosyphilis in which the patient's personality, as well as his or her control of movement, is affected. The patient may develop convulsions or partial paralysis.

Gumma—A symptom that is sometimes seen in tertiary syphilis, characterized by a rubbery swelling or tumor that heals slowly and leaves a scar.

Index case—The first case of a contagious disease in a group or population that serves to call attention to the presence of the disease.

Jarisch-Herxheimer reaction—A temporary reaction to penicillin treatment for syphilis that includes fever, chills, and worsening of the skin rash or chancre.

Lues maligna—A skin disorder of secondary syphilis in which areas of ulcerated and dying tissue are formed. It occurs most frequently in HIV-positive patients.

Spirochete—A type of bacterium with a long, slender, coiled shape. Syphilis is caused by a spirochete.

Tabes dorsalis—A progressive deterioration of the spinal cord and spinal nerves associated with tertiary syphilis.

gynecologists in the United States screen nonpregnant women for chlamydia and **gonorrhea**, while fewer than a third screen them for syphilis.

EDUCATION AND INFORMATION. Patients diagnosed with syphilis should be given information about the disease and counseling regarding sexual behavior and the importance of completing antibiotic treatment. It is also important to inform the general public about the transmission and early symptoms of syphilis, and provide adequate health facilities for testing and treatment.

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Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rebecca J. Frey, PhD

Systemic antifungal drugs see **Antifungal drugs, systemic**

Systemic lupus erythematosus

Definition

Systemic lupus erythematosus (also called lupus or SLE) is a disease where a person's immune system attacks and injures the body's own organs and tissues. Almost every system of the body can be affected by SLE.

Description

The body's immune system is a network of cells and tissues responsible for clearing the body of invading foreign organisms, like bacteria, viruses, and fungi. Antibodies are special immune cells that recognize these foreign invaders, and begin a chain of events to destroy them. In an autoimmune disorder like SLE, a person's antibodies begin to recognize the body's own tissues as foreign. Cells and chemicals of the immune system damage the tissues of the body. The reaction that occurs in tissue is called inflammation. Inflammation includes swelling, redness, increased blood flow, and tissue destruction.

In SLE, some of the common antibodies that normally fight diseases are thought to be out of control. These include antinuclear antibodies and anti-DNA antibodies. Antinuclear antibodies are directed against the cell's central structure that contains genetic material (the nucleus). Anti-DNA antibodies are directed against the cell's genetic material. DNA is the chemical substance that makes up the chromosomes and genes.

SLE can occur in both males and females of all ages, but 90% of patients are women. The majority of these women are in their childbearing years. African Americans are more likely than Caucasians to develop SLE.



A close-up view of a woman's face with a lesion caused by systemic lupus erythematosus (SLE). One characteristic of this autoimmune disease is a butterfly rash present across the cheeks and nose. (Photograph by Dr. P. Marazzi, Custom Medical Stock Photo. Reproduced by permission.)

Occasionally, medications can cause a syndrome of symptoms very similar to SLE. This is called drug-induced lupus. Medications that may cause this syndrome include hydralazine (used for high blood pressure) and procainamide (used for abnormal heartbeats). Drug-induced lupus almost always disappears after the patient stops taking the medications that caused it.

Causes and symptoms

The cause of SLE is unknown. Because the vast majority of patients are women, some research is being done to determine what (if any) link the disease has to female hormones. SLE may have a genetic basis, although more than one gene is believed to be involved in the development of the disease. Because patients with the disease may suddenly have worse symptoms (called a flare) after exposure to things like sunlight, alfalfa sprouts, and certain medications, researchers suspect that some environmental factors may also be at work.

The severity of a patient's SLE varies over time. Patients may have periods with mild or no symptoms, followed by a flare. During a flare, symptoms increase in severity and new organ systems may become affected.

Many SLE patients have fevers, **fatigue**, muscle **pain**, weakness, decreased appetite, and weight loss. The spleen and lymph nodes are often swollen and enlarged. The development of other symptoms in SLE varies, depending on the organs affected.

- **Joints.** Joint pain and problems, including arthritis, are very common. About 90% of all SLE patients have these types of problems.
- **Skin.** A number of skin **rashes** may occur, including a red butterfly-shaped rash that spreads across the face. The “wings” of the butterfly appear across the cheekbones, and the “body” appears across the bridge of the nose. A discoid, or coin-shaped, rash causes red, scaly bumps on the cheeks, nose, scalp, ears, chest, back, and the tops of the arms and legs. The roof of the mouth may develop sore, irritated pits (ulcers). Hair loss is common. SLE patients tend to be very easily sunburned (photosensitive).
- **Lungs.** Inflammation of the tissues that cover the lungs and line the chest cavity causes pleuritis, with fluid accumulating in the lungs. The patient frequently experiences coughing and shortness of breath.
- **Heart and circulatory system.** Inflammation of the tissue surrounding the heart causes **pericarditis**; inflammation of the heart itself causes **myocarditis**. These heart problems may result in abnormal beats (**arrhythmias**), difficulty pumping the blood strongly enough (**heart failure**), or even sudden **death**. **Blood clots** often form in the blood vessels and may lead to complications.
- **Nervous system.** Headaches, seizures, changes in personality, and confused thinking (**psychosis**) may occur.
- **Kidneys.** The kidneys may suffer significant destruction, with serious life-threatening effects. They may become unable to adequately filter the blood, leading to kidney failure.
- **Gastrointestinal system.** Patients may experience **nausea**, **vomiting**, **diarrhea**, and abdominal pain. The lining of the abdomen may become inflamed (peritonitis).
- **Eyes.** The eyes may become red, sore, and dry. Inflammation of one of the nerves responsible for vision may cause vision problems, and blindness can result from inflammation of the blood vessels (**vasculitis**) that serve the retina.

Diagnosis

Diagnosis of SLE can be somewhat difficult. There are no definitive tests for diagnosing SLE.

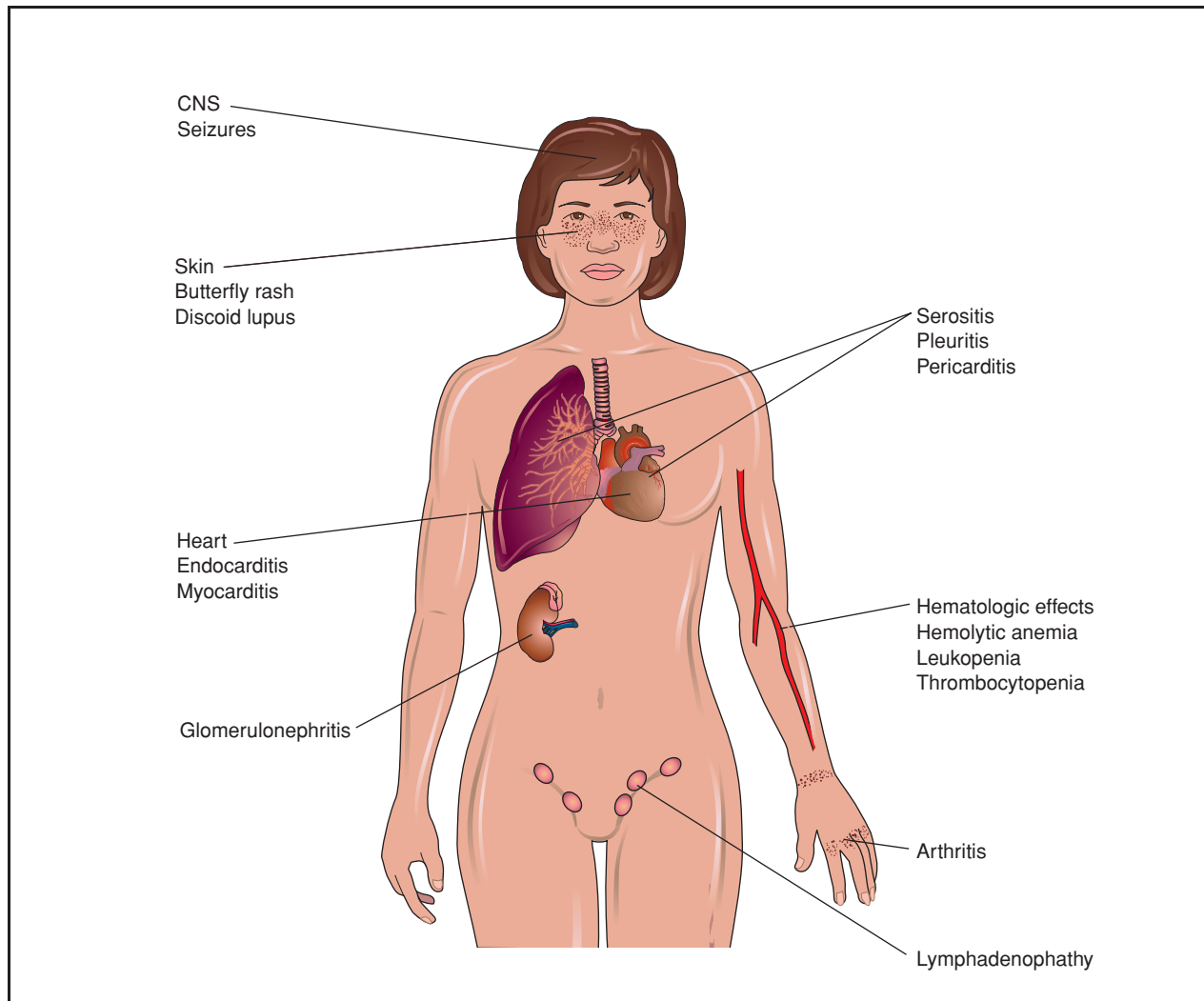
Many of the symptoms and laboratory test results of SLE patients are similar to those of patients with different diseases, including **rheumatoid arthritis**, **multiple sclerosis**, and various nervous system and blood disorders.

Laboratory tests that are helpful in diagnosing SLE include several tests for a variety of antibodies commonly elevated in SLE patients (including anti-nuclear antibodies, anti-DNA antibodies, etc.). SLE patients tend to have low numbers of red blood cells (anemia) and low numbers of certain types of white blood cells. The **erythrocyte sedimentation rate** (ESR), a measure of inflammation in the body, tends to be quite elevated. Samples of tissue (biopsies) from affected skin and kidneys show characteristics of the disease.

A test called the lupus erythematosus cell preparation (or LE prep) test is also performed. This test involves obtaining a sample of the patient's blood. Cells from the blood are damaged in the laboratory in order to harvest their nuclei. These damaged cells are then put together with the patient's blood serum, the liquid part of blood separated from the blood cells. Antinuclear antibodies within the patient's serum will clump together with the damaged nuclear material. A material called Wright's stain will cause these clumps to turn blue. These stained clumps are then reacted with some of the patient's white blood cells, which will essentially eat the clumps. LE cells are the white blood cells that contain the blue clumps. This test will be positive in about 70-80% of all patients with SLE.

The American Rheumatism Association developed a list of symptoms used to diagnose SLE. Research supports the idea that people who have at least four of the eleven criteria (not necessarily simultaneously) are extremely likely to have SLE. The criteria are:

- butterfly rash
- discoid rash
- photosensitivity
- mouth ulcers
- arthritis
- inflammation of the lining of the lungs or the lining around the heart
- kidney damage, as noted by the presence of protein or other abnormal substances called casts in the urine
- seizures or psychosis
- the presence of certain types of anemia and low counts of particular white blood cells



Systemic lupus erythematosus (SLE) is an autoimmune disease in which the individual's immune system attacks, injures, and destroys the body's own organs and tissues. Nearly every system of the body can be affected by SLE, as depicted in the illustration above. (Illustration by Electronic Illustrators Group.)

- the presence of certain immune cells, anti-DNA antibodies, or a falsely positive test for **syphilis**
- the presence of antinuclear antibodies

Treatment

Treatment depends on the organ systems affected by SLE and the severity of the disease. Some patients have a mild form of SLE. Their mild symptoms of inflammation can be treated with **nonsteroidal anti-inflammatory drugs** like ibuprofen (Motrin, Advil) and **aspirin**. Severe skin rashes and joint problems may respond to a group of medications usually used to treat **malaria**. More severely ill patients with potentially life-threatening complications (including

kidney disease, pericarditis, or nervous system complications) will require treatment with more potent drugs, including steroid medications. Because steroids have serious side effects, they are reserved for more severe cases of SLE. Drugs that decrease the activity of the immune system (called **immunosuppressant drugs**) may also be used for severely ill SLE patients. These include azathioprine and cyclophosphamide.

Other treatments for SLE try to help specific symptoms. Clotting disorders will require blood thinners. Psychotic disorders will require specific medications. Kidney failure may require the blood to be cleaned outside the body through a machine (dialysis) or even a **kidney transplantation**.

KEY TERMS

Autoimmune disorder—A disorder in which the body's antibodies mistake the body's own tissues for foreign invaders. The immune system then attacks and causes damage to these tissues.

Chromosomes—Spaghetti-like structures located within the nucleus (or central portion) of each cell. Chromosomes contain genes, structures that direct the growth and functioning of all the cells and systems in the body. Chromosomes are responsible for passing on hereditary traits from parents to child.

Immune system—The system of specialized organs, lymph nodes, and blood cells throughout the body that work together to prevent foreign organisms (bacteria, viruses, fungi, etc.) from invading the body.

Psychosis—Extremely disordered thinking with a poor sense of reality; may include hallucinations (seeing, hearing, or smelling things that are not really there).

Alternative treatment

A number of alternative treatments have been suggested to help reduce the symptoms of SLE. These include **acupuncture** and massage for relieving the pain of sore joints and muscles. **Stress** management is key for people with SLE and such techniques as **meditation**, hypnotherapy, and **yoga** may be helpful in promoting relaxation. Dietary suggestions include eating a whole foods diet with reduced amounts of red meat and dairy products in order to decrease pain and inflammation. **Food allergies** are believed either to contribute to SLE or to arise as a consequence of the digestive difficulties. Wheat, dairy products, and soy are the major offenders. An elimination/challenge diet can help identify the offending foods so that they can be avoided. Another dietary measure that may be beneficial is eating more fish that contain **omega-3 fatty acids**, like mackerel, sardines, and salmon. Because alfalfa sprouts have been associated with the onset of flares in SLE, they should be avoided. Supplements that have been suggested to improve the health of SLE patients include **vitamins B, C, and E**, as well as selenium, zinc, magnesium, and a complete trace mineral supplement. Vitamin A is believed to help improve discoid skin rashes. Botanical medicine can help the entire body through immune modulation and **detoxification**, as well as assisting

individual organs and systems. Homeopathy and flower essences can work deeply on the emotional level to help people with this difficult disease.

Prognosis

The prognosis for patients with SLE varies, depending on the organ systems most affected and the severity of inflammation. Some patients have long periods of time with mild or no symptoms. About 90-95% of patients are still living after 2 years with the disease. About 82-90% of patients are still living after 5 years with the disease. After 10 years, 71-80% of patients are still alive, and 63-75% are still alive after 20 years. The most likely causes of death during the first 10 years include infections and kidney failure. During years 11-20 of the disease, the most likely cause of death involves the development of abnormal blood clots.

Because SLE frequently affects women of child-bearing age, **pregnancy** is an important issue. For pregnant SLE patients, about 30% of the pregnancies end in **miscarriage**. About 25% of all babies born to mothers with SLE are premature. Most babies born to mothers with SLE are normal. However, a rare condition called neonatal lupus causes a baby of a mother with SLE to develop a skin rash, liver or blood problems, and a serious heart condition.

Prevention

There are no known ways to avoid developing SLE. However, it is possible for a patient who has been diagnosed with SLE to prevent flares of the disease. Recommendations for improving general health to avoid flares include decreasing sun exposure, getting sufficient sleep, eating a healthy diet, decreasing stress, and exercising regularly. It is important for a patient to try to identify the early signs of a flare (like **fever**, increased fatigue, rash, **headache**). Some people believe that noticing and responding to these warning signs will allow a patient with SLE to prevent a flare, or at least to decrease its severity.

Resources

ORGANIZATIONS

American College of Rheumatology. 1800 Century Place, Suite 250, Atlanta, GA 30345. (404) 633-3777. <<http://www.rheumatology.org>>.

Lupus Foundation of America. 1300 Piccard Dr., Suite 200, Rockville, MD 20850. (800) 558-0121. <<http://www.lupus.org>>.

Rosalyn Carson-DeWitt, MD

T

T-cell count see **Lymphocyte typing**

T-PA see **Thrombolytic therapy**

Taeniasis see **Tapeworm diseases**

Tagged red blood cell scan see **GI bleeding studies**

Tai chi

Definition

T'ai chi is a Chinese **exercise** system that uses slow, smooth body movements to achieve a state of relaxation of both body and mind.

Purpose

As a system of physical exercise used to improve and maintain health, t'ai chi can be helpful in achieving a state of physical and mental relaxation while also strengthening the cardiovascular and immune systems.

Precautions

As a very slow and gentle form of moving, tai chi has virtually no side effects. However, if a person has any doubts about the conditions of his or her joints, vertebrae, or heart, a physician should be consulted.

Description

Developed originally in China as a self-defense strategy, or martial art, tai chi—the “supreme ultimate fist”—is practiced in modern times primarily as a gentle exercise technique. Described as “meditation in motion,” tai chi consists of a standing person performing a series of postures or bodily movements in a

slow and graceful manner, with each movement flowing without pause to the next. According to Chinese legend, the technique was created by a Taoist monk who was inspired as he watched a crane and a snake do battle. Impressed by the ‘snake’s ability to subtly and swiftly avoid the bird’s thrusts, he devised a series of self-defense techniques that do not involve meeting the opponent’s force with force, but rather **stress** evading the blow; causing the opponent’s own momentum to work against him.

Tai chi is an ancient form of exercise, about 2,000 years old, that at one point had over 100 separate movements or postures. In current practice, there are two popular versions, of 18 and 37 movements respectively. The fact that in China 10 million people practice some type of t'ai chi daily suggests that it is the one of the most popular forms of exercise in the world. In the United States, t'ai chi is learned in classes in which students (or “players,” as they are called in China) wear loose, comfortable clothing and either go barefoot or wear only socks or soft shoes on the feet. In China, t'ai chi is almost always practiced outdoors at dawn, and ideally near trees. Unlike other martial arts, t'ai chi is not competitive. Classes usually begin with a few minutes of standing **meditation** to calm the mind and gather energy. Following warm-up exercises, students are taught the basics of a particular form or posture. Learning forms is not easy, and it takes some time to master what looks like a simple position. Properly done postures are done in a relaxed, artful, and linked way, with the circular and rhythmic movements of one position flowing seamlessly into the next.

While strict attention to body position is critical, proper breathing is considered to be equally important. Just as movements are slow and continuous and without strain, breathing should be effortless yet deep. Finally, both mental and physical balance is considered essential to t'ai chi. The experienced



Tai chi is a Chinese exercise system which uses slow, smooth body movements to achieve a state of relaxation. The posture above is part of the single whip sequence of Tai chi motions. (Illustration by Electronic Illustrators Group.)

practitioner of t'ai chi maintains perfect body balance throughout the exercise series. Altogether, the five essential qualities of t'ai chi are:

- Slowness. To develop awareness.
- Lightness. To make movements flow.
- Balance. To prevent body strain.
- Calmness. To maintain continuity.
- Clarity. To focus the mind.

T'ai chi has both physical and mental benefits. If done regularly, it improves muscle tone, flexibility, balance, and coordination. Many older people find that it boosts their energy, stamina, and agility, sharpens their reflexes, and gives an overall sense of well-being. The calming and meditative aspects of t'ai chi allow many to experience its ability to relieve stress. Some claim t'ai chi to be a healing therapy, and it is often used to support other treatments for chronic conditions; arthritis, **fibromyalgia**, and digestive disorders are just three examples. Like **yoga**, t'ai chi has several different styles to suit the individual. Also, it can eventually be done daily by oneself, and ultimately becomes a very personal endeavor. Most Westerners find it best to practice t'ai chi in the same place and at the same time of day; and those who enjoy it most are those who are not seeking major, dramatic breakthroughs, but rather who can take pleasure in small gains that accumulate over a long period of time.

KEY TERMS

Arthritis—Inflammation of the joints.

Cardiovascular—Relating to the heart and blood vessels.

Continuity—Uninterrupted and successive.

Fibromyalgia—A chronic disease syndrome characterized by fatigue, widespread muscular soreness, and pain at specific points on the body.

Meditation—An exercise of contemplation that induces a temporary feeling of relaxation.

Stamina—Staying power, endurance.

Yoga—A system of exercise aimed at promoting the control of the body and the mind.

Some research done in the United States focuses on the emotional and psychological benefits of t'ai chi. One recently discovered advantage of t'ai chi is its ability to hold people's interest longer than many other forms of exercise. One study in Oregon found that only 20% of people enrolled in a six-month t'ai chi program dropped out before the end, compared to an average of 55% for other forms of exercise. With regard to depression, a study of college students found that those who were taking t'ai chi classes had a lower rate of depression than students enrolled in other fitness programs.

Risks

T'ai chi is a safe exercise system for people of all ages and fitness levels. Done properly, without any over-stretching, t'ai chi should not leave a person feeling tired or sore.

Normal results

Besides its overall fitness benefits and **stress reduction** aspects, regular t'ai chi sessions are said to be especially helpful for seniors, as they lower their blood pressure. T'ai chi claims to benefit arthritis sufferers, those recovering from an injury or rehabilitating their hearts, and also improves balance, and therefore, reduces the risk of falling, especially important for the elderly. T'ai chi can result in a significant improvement in the quality of life for anyone. But, because of the low stress level of the exercises it is a particularly attractive form of exercise to seniors.

In addition to studying the cardiovascular and range-of-motion benefits of t'ai chi, researchers are

also investigating its positive effects on the immune system. A team of scientists in California reported in 2003 that t'ai chi boosts the resistance of older people to the **shingles** virus— a virus that is both more common and more severe in the elderly.

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- American Association of Oriental Medicine. 5530 Wisconsin Avenue, Suite 1210, Chevy Chase, MD 20815. (301) 941-1064. <www.aaom.org>.
- Canadian Taijiquan Federation. P.O. Box 421, Milton, Ontario L9T 4Z1. <www.canadiantaijiquanfederation.ca>.
- Northeastern Tai Chi Chuan Association. 163 West 23rd St., 5th Floor., New York, NY 10011 (212) 741-1922.

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Tailbone injuries see **Coccyx injuries**

Talipes see **Clubfoot**

Tamoxifen see **Anticancer drugs**

Tamponade see **Cardiac tamponade**

Tapeworm diseases

Definition

Tapeworms are a group of parasitic worms that live in the intestinal tracts of some animals. Several different species of tapeworms can infect humans. Tapeworm disease or cestodiasis occurs most commonly after eating raw or undercooked meat or fish that contains the immature form of the tapeworm.

Description

Tapeworm infections pose a serious public health problem in many less developed countries due to poor sanitation conditions. The disease is most common where livestock, such as cattle and pigs, are raised in areas where human feces are not disposed of in a sanitary manner. Another common source of human tapeworms are certain species of freshwater fish. Tapeworm infections tend to occur more frequently in areas of the world where the people regularly eat raw or undercooked beef, pork, or fish. Persons of all ages and both sexes are susceptible to tapeworm infection, but children are generally not exposed until they are old enough to begin eating meat or fish.

Tapeworm infections in humans are less common in industrialized regions of the world, although German public health experts reported in 2003 that the rate of these infections is higher in Europe than the official statistics indicate. Travel to areas where tapeworm infections are more common and immigration of people from these areas serve as new sources of the parasite. Infected persons are often unaware of the presence of adult tapeworms in their intestinal tract, as they may have no obvious symptoms of infection. Some tapeworms can live in an infected person for over 10 years if diagnosis is not made and treatment is not administered.

In addition to the typical infection caused by eating undercooked meat or fish, people may also be directly infected by ingesting tapeworm eggs shed by the adult worm. This type of tapeworm infection can lead to a condition referred to as cysticercosis, in which the

larvae continue to develop within tissues other than the intestinal tract. One of the most serious forms of this disease occurs when the tapeworm larvae infect the central nervous system, a disease referred to as neurocysticercosis. In contrast to a typical tapeworm infection, which may not be associated with symptoms, neurocysticercosis is a serious condition that may cause seizures and is potentially life-threatening.

Causes and symptoms

Several species of tapeworm can infect people. The two most common species are the pork tapeworm (*Taenia solium*) and the beef tapeworm (*Taenia saginata*). Improperly treated human sewage may be used to fertilize pastures or crops. Pigs or cattle become infected by grazing in contaminated pastures or drinking water contaminated with tapeworm eggs from human feces. The pea-sized larvae of these tapeworms are deposited in certain tissues of the body of infected pigs and cattle, including the muscles. The infection is then transmitted to people when raw or undercooked meat containing tapeworm larvae is consumed. The immature tapeworm develops into the adult form in the human intestine and may remain there for many years if not identified and treated.

The *Taenia* tapeworms attach to the intestinal walls but cause only mild inflammation at the site of attachment. As a result, most tapeworm carriers show no symptoms (asymptomatic) and usually become aware of the infection only after noticing tapeworm segments in their feces. Segments of the beef tapeworm may spontaneously pass through the anus causing a noticeable sensation. Mild gastrointestinal symptoms, such as **nausea** or abdominal **pain**, can occur in infected individuals. In rare cases where the tapeworm segments migrate into the appendix, pancreas, or bile duct, there may be a sudden onset of severe abdominal discomfort.

Cysticercosis is a potentially serious complication of *Taenia solium* infection in which the larvae develop outside the intestinal tract. This type of infection is less common and occurs following accidental consumption of tapeworm eggs released from the adult worm. These eggs initially are localized in the anal area, but they may also contaminate the fingers or other parts of the body. Infection can occur in the person harboring the adult tapeworm or in other people with whom that individual comes in contact. The tapeworm larvae may develop in various tissues throughout the body. The most serious clinical problems occur when the larvae develop in the central nervous system (neurocysticercosis), potentially causing seizures and other

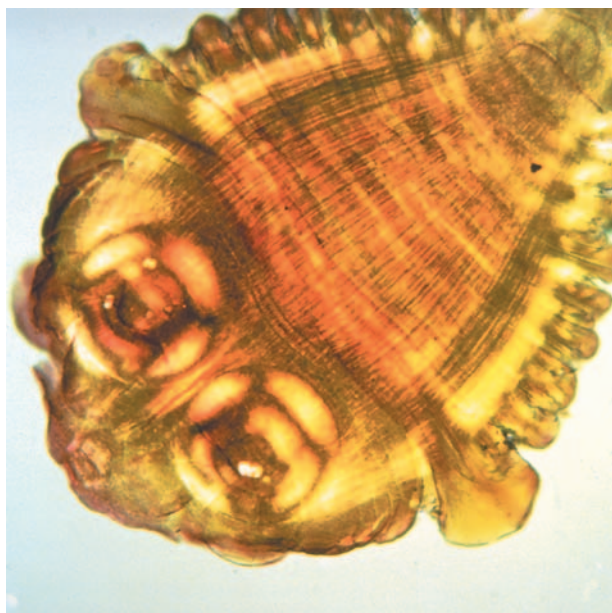
neurological problems. An important aspect of this type of infection is that poor hygiene on the part of the individuals harboring an adult tapeworm can lead to an infection in an individual who may never consume meat. This is a particular problem if infected individuals are employed as food handlers.

Another important tapeworm that may infect people is the fish tapeworm (*Diphyllobothrium latum*). This is a frequent human intestinal parasite in many areas where raw freshwater fish is consumed. Human infection with the fish tapeworm is referred to as diphyllobothriasis. Feces from infected hosts or raw sewage contaminates a fresh water source. Tapeworm larvae are initially ingested by freshwater crustaceans and then are eaten by fish. Human infection occurs when a person consumes raw fish contaminated with the tapeworm larvae. Adult tapeworms then develop in the human intestinal tract.

Most infections with the fish tapeworm are not associated with symptoms. The tapeworm causes little damage to the lining of the intestine. Infected individuals may report **diarrhea**, **fatigue**, weakness, or sensation of hunger more commonly than uninfected individuals. One problem unique to this tapeworm is that it may compete with the host for absorption of vitamin B12 from the small intestine, causing the person to become deficient in this vitamin and leading to a condition called **pernicious anemia**.

Two smaller species of tapeworms may also infect people. The dwarf tapeworm (*Hymenolepis nana*) is a common infection throughout the world that can be passed from one person to another. Transmission is usually the result of inadvertent ingestion of tapeworm eggs from feces eliminated by infected individuals. As a result, infection with this tapeworm is encountered most frequently in children, the developmentally disabled, and psychiatric patient populations. Abdominal pain that is not localized to any particular area is the most common complaint. Patients may experience loose bowel movements or diarrhea with mucus, but bloody diarrhea is rare.

Another small tapeworm capable of infecting people is the rodent tapeworm (*Hymenolepis diminuta*). Rats, mice, and other rodents are the usual hosts for the adult tapeworm (definitive host), but humans can become infected following accidental consumption of insects containing tapeworm larvae. Meal worms or grain beetles that infest cereal, flour, or dried fruit are the most likely source of infection. Most human infections are not associated with symptoms, although some individuals report headaches, anorexia, nausea, and diarrhea.



The head of an adult beef tapeworm. (Custom Medical Stock Photo. Reproduced by permission.)

Diagnosis

Identification of tapeworm segments or eggs in a stool sample is necessary for diagnosis of an adult tapeworm infection. In many cases, a tentative diagnosis may be made on the basis of a patient's description of short chains of tapeworm segments in their stool. Further evaluation is recommended to determine the actual species involved since infection with *Taenia solium* is potentially more serious due to the added risk of cysticercosis. Whenever possible, tapeworm segments should be carefully collected in water or salt solutions, using strict precautions to avoid contamination. Stool examination should be performed in a laboratory having experience in the diagnosis of intestinal parasites. It is recommended that at least three stool samples be collected on alternate days to increase the likelihood of being able to make an accurate diagnosis.

Although the general appearance of tapeworm segments from the two *Taenia* species is quite similar, trained laboratory personnel can detect distinct differences between the beef and pork tapeworms when samples are examined under a microscope. Tapeworm segments and eggs from the fish tapeworm and the dwarf tapeworm have characteristic appearances that allow accurate differentiation from the *Taenia* species of worms. Other diagnostic procedures may be necessary when cysticercosis is suspected. Blood samples from an infected individual are collected to look for

the presence of antibodies against the tapeworm larvae. In 2004, researchers isolated an antigen diagnostic for cysticercosis called GP50. In cases in which infection of the central nervous system is present, advanced imaging tests, such as **computed tomography scans** and **magnetic resonance imaging (MRI)**, may be necessary to determine the exact location of the tapeworm larvae within the body.

Ultrasound imaging is also being used as of 2003 to diagnose tapeworm infections. Because of the recent rise in immigration and travel, clinicians in developed countries are seeing more patients with tapeworms. The World Health Organization (WHO) has developed standardized sets of ultrasound images to help doctors recognize the signs of a parasitic infection. In some cases, ultrasound allows the doctor to see the tapeworm directly.

Treatment

Effective treatment of tapeworm infections involves administering compounds that are toxic to the adult worm. Many of the early treatments were also somewhat toxic to the patient, so treatment was often quite an ordeal. Newer medications are much more easily tolerated and are highly effective in eliminating the parasite from the body. It is important, however, to completely eliminate the head and neck regions of the tapeworm, as the entire worm can regenerate from these parts.

One treatment that has been in use since the early 1960s is niclosamide (Niclocide). This drug is poorly absorbed from the digestive tract and rapidly kills tapeworms upon exposure. It has been shown to be effective against *Taenia* species and the fish tapeworm, but treatment of the dwarf tapeworm (*Hymenolepis nana*) may require a more prolonged treatment schedule. Side effects reported with niclosamide are infrequent and typically mild. When present, side effects may include nausea, abdominal discomfort, **vomiting**, diarrhea, light-headedness, and skin rash. This medication should be taken in the morning on an empty stomach. The tablets are chewed thoroughly and swallowed with water. For young children, the tablets may be pulverized and mixed with water. Patients are allowed to eat two hours after treatment. Recommended dosage is 2 grams for adults and about half this for children.

Another oral medication that has been shown to be 95% effective in the treatment of tapeworm infections associated with both *Taenia* and *Diphyllobothrium latum* species is praziquantel (Biltricide). Side effects reported for praziquantel are mild and appear to be



The head of an adult pork tapeworm. (Custom Medical Stock Photo. Reproduced by permission.)

short-lived. They include nausea, abdominal pain, **itching**, sore joints, and muscle pain.

It is recommended that follow-up stool samples be examined at one month and three months after treatment has been completed. Treatment can be considered successful if no eggs are present in several stool samples. It should be noted that the tapeworm medications do not kill the tapeworm eggs when they kill the adult worm, so the potential for infection with eggs still exists as the dead worm segments are passed. Proper personal hygiene in individuals receiving treatment will greatly reduce this potential.

Cases of neurocysticercosis, where larvae have developed in the central nervous system, may also be treated with praziquantel or albendazole. If the patient is treated promptly, damage to the central nervous system will be minimized.

As of late 2003, researchers in developing countries are working on a vaccine for pigs to help control neurocysticercosis; however, the vaccine is not likely to be available for several years.

Prognosis

When confined to the intestinal tract, tapeworms cause minimal damage to their human host. Once the diagnosis of an intestinal tapeworm infection has been made, prognosis following treatment with niclosamide or praziquantel is good. The worms can be eliminated

KEY TERMS

Cestodiasis—Parasitic infection caused by the presence of adult tapeworms of the class Cestoda within the intestinal tract. Infection is caused by accidental consumption of tapeworm larvae.

Cysticercosis—Parasitic infection caused by the presence of immature tapeworm larvae (cysticerci) that have developed outside the intestinal tract. Infection is caused by accidental consumption of tapeworm eggs.

Diphyllobothriasis—Parasitic infection caused by the presence of tapeworms from the *Diphyllobothrium* genus, such as the fish tapeworm (*Diphyllobothrium latum*).

Hymenolepiasis—Parasitic infection caused by the presence of tapeworms from the *Hymenolepis* genus, such as the dwarf tapeworm (*Hymenolepis nana*) or the rodent tapeworm (*Hymenolepis diminuta*).

Neurocysticercosis—Parasitic infection caused by the presence of immature tapeworm larvae within the central nervous system.

Pernicious anemia—Type of anemia caused by a deficiency in vitamin B12.

Taeniasis—Parasitic infection caused by the presence of tapeworms from the *Taenia* genus, such as the pork tapeworm (*Taenia solium*) or the beef tapeworm (*Taenia saginata*).

from the intestines with oral treatment, and there are usually no residual side effects. Serious problems from tapeworm infections occur when tapeworm eggs are consumed and the larvae localize in tissues outside the digestive tract (cysticercosis). Prompt diagnosis and treatment of this condition is necessary to prevent permanent damage to the central nervous system and other internal organs. In fact, cysticercosis is becoming increasingly recognized as an important cause of severe neurologic disease in the United States. Untreated cases of cysticercosis have the rare potential to be life-threatening; in some cases, **liver transplantation** has been necessary to save the patient's life.

Prevention

The best way to prevent infection with tapeworms is to eliminate the exposure of livestock to the tapeworm eggs by properly disposing of human feces. The next best strategy is to thoroughly cook or freeze all meat and fish before it is eaten to prevent consumption of live

tapeworm larvae in infected samples. Larval cysts in pork and beef are killed by moderate temperatures of 150°F (65°C) or if frozen for at least 12 hours. Proper cooking of freshwater fish could also eliminate the possibility of human infection with the fish tapeworm. Freezing fresh fish for 24 hours will also kill the larval form.

People who raise sheep or horses should have these animals checked regularly by a veterinarian and dewormed if necessary.

The Centers for Disease Control and Prevention (CDC) recommends that people traveling abroad should wash their hands with soap and water before handling food; should wash and peel all raw vegetables and fruits before eating; and should drink only bottled or boiled water, or carbonated drinks in cans or bottles.

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ORGANIZATIONS

American Veterinary Medical Association (AVMA). 1931 North Meacham Road, Suite 100, Schaumburg, IL 60173-4360. <<http://www.avma.org>>.

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Tardive dyskinesia

Definition

Tardive dyskinesia is a mostly irreversible neurological disorder of involuntary movements caused by long-term use of antipsychotic or neuroleptic drugs.

Description

Antipsychotic or neuroleptic drugs are powerful tranquilizers generally prescribed for serious psychiatric disorders, as well as neurological and gastrointestinal disorders. Some common antipsychotics are: chlorpromazine HCl (Thorazine), thioridazine HCl (Mellaril), haloperidol (Haldol), perphenazine (Trilafon), thiothixene (Navane), trifluoperazine HCl (Stelazine), and fluphenazine HCl (Permitil, Prolixin).

When these drugs are used long term, tardive dyskinesia (TD) can result. About 20 percent of people taking **antipsychotic drugs** for more than one year become affected by TD. The prevalence of TD tends to be highest among elderly patients and among women.

Causes and symptoms

TD usually appears after years of antipsychotic drug use, and seems to be related to the total lifetime dose of medication. The symptoms include the following:

- tongue protrusion
- grimacing
- rapid eye blinking
- lip smacking, pursing, or puckering
- rapid movement of the arms or legs
- other involuntary movements of the head, face, neck and tongue muscles

Diagnosis

The diagnosis of TD is suspected upon observation of involuntary movements of the head, neck, face, and tongue in individuals who have a history of antipsychotic drug prescription.

Treatment

There is no standard treatment for TD. The primary approach is to discontinue or minimize the use of antipsychotic drugs while attempting to treat some of the symptoms. The treatment must be individualized to

KEY TERMS

Antipsychotics—Drugs used to treat psychotic conditions such as schizophrenia or psychosis. These medications are powerful tranquilizers that all have sedating and calming effects, but their major effect is to reduce psychotic thinking and behavior.

Neuroleptics—Any of a class of drugs used to treat psychotic conditions.

Psychosis—A condition where a person's ability to recognize reality and cope with everyday life is severely affected.

the patient, because discontinuation of the antipsychotic drug(s) may not be advisable, depending on the patient's condition. In some cases, substituting another drug for the antipsychotic drug may be beneficial.

Prognosis

Once TD appears in full-blown form, it can be permanent. With careful management, some symptoms may improve and even disappear with time. In less severe cases, some patients may recover from TD within three months of discontinuing the use of antipsychotic medication. Studies report that at least half of patients experience remission of major symptoms within 12 to 18 months following discontinuation of antipsychotic drugs. In some patients, however, decreasing the dose of the antipsychotic drug actually increases the symptoms of TD, while increasing the dose sometimes offers a temporary remission of the symptoms.

Prevention

TD can be prevented by early recognition and discontinuation of the antipsychotic medication if this is clinically possible. The use of antipsychotic drugs should in any case be kept to a minimum in all patients. Patients should be followed carefully to determine when the dose of the drug can be tapered off as the psychiatric condition improves. In all cases, the benefits of taking the antipsychotic medication should outweigh the risk of developing TD.

A study has shown that elderly institutionalized patients with **dementia** that were treated with risperidone had a low incidence of TD. Although further study is needed, this study shows that non-conventional neuroleptic drugs should be considered to avoid the risk of tardive dyskinesia, particularly in elderly patients.

Resources

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Tierney, Lawrence, et al. *Current Medical Diagnosis and Treatment*. Los Altos: Lange Medical Publications, 2001.

PERIODICALS

"Risperidone May Lower Incidence of TD." *Brown University geroPsych Report* August 2000: 2.

ORGANIZATIONS

National Institute for Mental Health. 6001 Executive Blvd., Room 8184, MSC 9663, Bethesda, MD 20892-9663. <<http://www.nimh.nih.gov>>.

Tardive Dyskinesia/Tardive Dystonia National Association. P.O. Box 45732, Seattle, WA 98145-0732. (206) 522-3166.

Kim A. Sharp, M.Ln.

Tarsorrhaphy

Definition

Tarsorrhaphy is a rare procedure in which the eyelids are partially sewn together to narrow the opening.

Purpose

The eye needs the a lid to protect it. It also needs tears and periodic blinking to cleanse it and keep it moist. There are many conditions that impair these functions and threaten the eye, specifically the cornea, with drying. Until they can be corrected, sewing the eyelids partially together helps protect the eye.

A partial list of the conditions that can require tarsorrhaphy includes:

- Paralysis or weakness of the eyelids so that they cannot close or blink adequately. Bell's palsy is a nerve condition that weakens the muscles of the face, including the eyelids. It is usually temporary. **Myasthenia gravis** also weakens facial muscles, but it is usually treatable. A **stroke** can also weaken eyelids so they do not close.
- Exophthalmos (the eyes sticking out of their sockets) occurs with Graves' disease of the thyroid and with tumors behind the eyes. If the eyes stick out too far, the lids cannot close over them.
- Enophthalmos is a condition in which the eye falls back into the socket so that the eyelid function is inadequate.

KEY TERMS

Cornea—The clear part of the front of the eye through which vision occurs.

Enophthalmos—A condition in which the eye falls back into the socket and inhibits proper eyelid function.

Exophthalmos—A condition in which the eyes stick out of their sockets and inhibit proper eyelid function.

Palpebral fissure—Eyelid opening.

Sjögren's syndrome—A connective tissue disease that hinders the production of tears and other body fluids.

- Several eye and corneal diseases cause swelling of the cornea and require temporary added protection until the condition resolves.
- Sjögren's syndrome reduces tear flow to the point where it can endanger the cornea.
- Dendritic ulcers of the cornea caused by viruses may need to be covered with the eyelid while they heal.

Precautions

The use of eye drops and contact lenses to moisten and protect the eyes must be considered first before tarsorrhaphy is performed.

Description

Stitches are carefully placed at the corners of the eyelid opening (called the palpebral fissure) to narrow it. This allows the eye better lubrication and less exposure to the air. Eyeball motion can then help bathe the cornea in tears when it rolls up under the lid. The outpatient procedure is done under local anesthetic.

Preparation

Tarsorrhaphy is a minor procedure done under **local anesthesia**. Special preparation is not necessary.

Aftercare

Eye drops or ointment may still be needed to preserve the cornea or treat accompanying disease.

Risks

Tarsorrhaphy carries few risks. If complications occur, they are usually minor eyelid swelling and superficial infection.

Resources

BOOKS

Sardegna, Jill Otis, and T. Paul. *The Encyclopedia of Blindness and Vision Impairment*. New York: Facts on File Inc., 1990.

J. Ricker Polsdorfer, MD

Tattoos see **Piercing and tattoos**

Tay-Sachs disease

Definition

Tay-Sachs disease is a genetic disorder caused by a missing enzyme that results in the accumulation of a fatty substance in the nervous system. This results in disability and **death**.

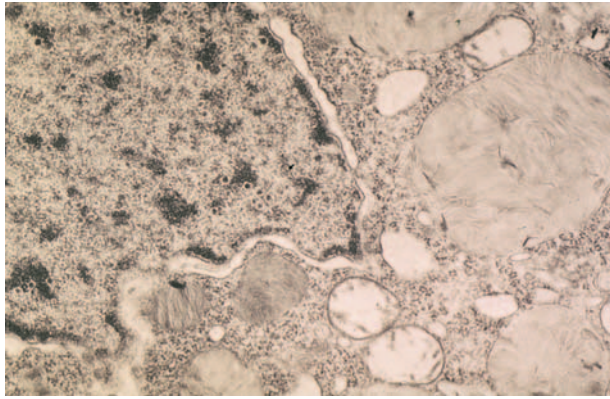
Description

Gangliosides are fatty substances necessary for the proper development of the brain and nerve cells (nervous system). Under normal conditions, gangliosides are continuously broken down, so that an appropriate balance is maintained. In Tay-Sachs disease, the enzyme necessary for removing excess gangliosides is missing. This allows gangliosides to accumulate throughout the brain, and is responsible for the disability associated with the disease.

Tay-Sachs disease is particularly common among Jewish people of Eastern European and Russian (Ashkenazi) origin. About one out of every 3,600 babies born to Ashkenazi Jewish couples will have the disease. Tay-Sachs is also more common among certain French-Canadian and Cajun French families.

Causes and symptoms

Tay-Sachs is caused by a defective gene. Genes are located on chromosomes, and serve to direct specific development/processes within the body. The genetic defect in Tay-Sachs disease results in the lack of an enzyme called hexosaminidase A. Without this enzyme, gangliosides cannot be degraded. They build up within the brain, interfering with nerve functioning.



Section of brain tissue from patient with Tay-Sachs disease.
(Custom Medical Stock Photo. Reproduced by permission.)

Because Tay-Sachs is a recessive disorder, only people who receive two defective genes (one from the mother and one from the father) will actually have the disease. People who have only one defective gene and one normal gene are called carriers. They carry the defective gene and thus the possibility of passing the gene and/or the disease onto their offspring.

When a carrier and a non-carrier have children, none of their children will actually have Tay-Sachs. It is likely that 50% of their children will be carriers themselves. When two carriers have children, their children have a 25% chance of having normal genes, a 50% chance of being carriers of the defective gene, and a 25% chance of having two defective genes. The two defective genes cause the disease itself.

Classic Tay-Sachs disease strikes infants around the age of six months. Up until this age, the baby will appear to be developing normally. When Tay-Sachs begins to show itself, the baby will stop interacting with other people and develop a staring gaze. Normal levels of noise will startle the baby to an abnormal degree. By about one year of age, the baby will have very weak, floppy muscles, and may be completely blind. The head will be quite large. Patients also present with loss of peripheral (side) vision, inability to breathe and swallow, and **paralysis** as the disorder progresses. Seizures become a problem between ages one and two, and the baby usually dies by about age four.

A few variations from this classical progression of Tay-Sachs disease are possible:

- Juvenile hexosaminidase A deficiency. Symptoms appear between ages two and five; the disease progresses more slowly, with death by about 15 years.
- Chronic hexosaminidase A deficiency. Symptoms may begin around age five, or may not occur until

KEY TERMS

Ganglioside—A fatty (lipid) substance found within the brain and nerve cells.

age 20–30. The disease is milder. Speech becomes slurred. The individual may have difficulty walking due to weakness, **muscle cramps**, and decreased coordination of movements. Some individuals develop mental illness. Many have changes in intellect, hearing, or vision.

Diagnosis

Examination of the eyes of a child with Tay-Sachs disease will reveal a characteristic cherry-red spot at the back of the eye (in an area called the retina). Tests to determine the presence and quantity of hexosaminidase A can be performed on the blood, specially treated skin cells, or white blood cells. A carrier will have about half of the normal level of hexosaminidase A present, while a patient with the disease will have none.

Treatment

There is no treatment for Tay-Sachs disease.

Prognosis

Sadly, the prognosis for a child with classic Tay-Sachs disease is certain death. Because the chronic form of Tay-Sachs has been discovered recently, prognosis for this type of the disease is not completely known.

Prevention

Prevention involves identifying carriers of the disease and providing them with appropriate information concerning the chance of their offspring having Tay-Sachs disease. When the levels of hexosaminidase A are half the normal level, a person is a carrier of the defective gene. Blood tests of carriers reveals reduction of hexosaminidase A.

When a woman is already pregnant, tests can be performed on either the cells of the baby (aminocentesis) or the placenta (**chorionic villus sampling**) to determine whether the baby will have Tay-Sachs disease.

Resources

ORGANIZATIONS

Late Onset Tay-Sachs Foundation. 1303 Paper Mill Road, Erdenheim, PA 19038. (800)672-2022.

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Avenue, White Plains, NY 10605. (888) 663-4637. resourcecenter@modimes.org. <http://www.modimes.org>.

Laith Farid Gulli, M.D.

TB see **Tuberculosis**

TCM see **Traditional Chinese medicine**

TE fistula see **Tracheoesophageal fistula**

Technetium heart scan

Definition

The technetium heart scan is a noninvasive nuclear scan that uses a radioactive isotope called technetium to evaluate blood flow after a **heart attack**.

Purpose

The technetium heart scan is used to evaluate the heart after a heart attack. It can confirm that a patient had a heart attack when the symptoms and **pain** usually associated with a heart attack were not present; identify the size and location of the heart attack; and provide information useful in determining the patient's post-heart attack prognosis. The scan is most useful when the electrocardiogram and cardiac enzyme studies do not provide definitive results—after heart surgery, for example, or when chest pain occurred more than 48 hours before the patient was examined. It is also used to evaluate the heart before and after heart surgery.

Precautions

Pregnant women and those who are breastfeeding should not be exposed to technetium.

Description

The technetium heart scan is a nuclear heart scan, which means that it involves the use of a radioactive isotope which targets the heart, and a radionuclide detector that traces the absorption of the radioactive isotope. The isotope is injected into a vein and

absorbed by healthy tissue at a known rate during a certain time period. The radionuclide detector, in this case a gamma scintillation camera, picks up the gamma rays emitted by the isotope.

The technetium heart scan uses technetium Tc-99m stannous pyrophosphate (usually called technetium), a mildly radioactive isotope that binds to calcium. After a heart attack, tiny calcium deposits appear on diseased heart valves and damaged heart tissue. These deposits appear within 12 hours of the heart attack. They are generally seen two to three days after the heart attack and are usually gone within one to two weeks. In some patients, they can be seen for several months.

After the technetium is injected into a blood vessel in the arm, it accumulates in heart tissue that has been damaged, leaving “hot spots” that can be detected by the scintillation camera. The technetium heart scan provides better image quality than commonly used radioactive agents such as thallium, because it has a shorter half-life and can thus be given in larger doses.

During the test, the patient lies motionless on the test table. Electrocardiogram electrodes are placed on the patient's body for continuous monitoring during the test. The test table is rotated so that different views of the heart can be scanned. The camera, which looks like an x-ray machine and is suspended above the table, moves back and forth over the patient. It displays a series of images of technetium's movement through the heart and records them on a computer for later analysis.

The test is usually performed at least 12 hours after a suspected heart attack, but it can also be done during triage of a patient who goes to a hospital emergency room with chest pain but does not appear to have had a heart attack. Recent clinical studies demonstrate that technetium heart scans are very accurate in detecting heart attacks while the patient is experiencing chest pain. They are far more accurate than electrocardiogram findings.

The technetium heart scan is usually performed in a hospital's nuclear medicine department but it can be done at the patient's bedside during a heart attack if the equipment is available. The scan is done two to three hours after the technetium is injected. Scans are usually done with the patient in several positions, with each scan taking 10 minutes. The entire test takes about 30 minutes to an hour. The scan is usually repeated over several weeks to determine if any further damage has been done to the heart. The test is also called technetium 99m pyrophosphate scintigraphy, hot-spot myocardial imaging, infarct avid imaging, or myocardial infarction scan.

KEY TERMS

Electrocardiogram—A test in which electronic sensors called electrodes are placed on the body to record the heart's electrical activities.

Noninvasive—A procedure that does not penetrate the body.

Radioactive isotope—One of two or more atoms with the same number of protons but a different number of neutrons with a nuclear composition. In nuclear scanning, radioactive isotopes are used as a diagnostic agent.

Technetium—A radioactive isotope frequently used in radionuclide scanning of the heart and other organs. It is produced during nuclear fission reactions.

The technetium heart scan is not dangerous. The technetium is completely gone from the body within a few days of the test. The scan itself exposes the patient to about the same amount of radiation as a **chest x ray**. The patient can resume normal activities immediately after the test.

Preparation

Two to three hours before the scan, technetium is injected into a vein in the patient's forearm.

Normal results

If the technetium heart scan is normal, no technetium will show up in the heart.

Abnormal results

In an abnormal technetium heart scan, hot spots reveal damage to the heart. The larger the hot spots, the poorer the patient's prognosis.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Texas Heart Institute. Heart Information Service. P.O. Box 20345, Houston, TX 77225-0345. <<http://www.tmc.edu/thi>>.

Lori De Milto

TEE see **Transesophageal echocardiography**

Teeth whitening

Definition

Teeth whitening is the process of using bleach or other materials to make teeth look whiter. The materials remove stains or other discoloration from the tooth surface.

Purpose

Teeth whitening is a cosmetic treatment done to improve the appearance of teeth. Teeth are whitened to remove the effects of coffee, cigarettes, and other substances that permanently stain or discolor teeth. Medications such as **antibiotics** like tetracycline may discolor teeth. Fluorosis, a condition caused by absorbing too much fluoride, could affect tooth color. Furthermore, **aging** also causes teeth to lose their bright color.

Precautions

Teeth whitening is not safe or effective for everyone, so a person should have a dental exam before starting treatment. The dentist can advise the patient about the most appropriate procedure. The oral health professional will also discuss the expected results of treatment. Patients may expect yellow or gray teeth to be replaced with a bright, white color. However, whitening may not work well on some stain colors.

The importance of a check-up

Whitening may not be recommended for people with gum disease, receding gums, or sensitive teeth. The dentist may advise against certain treatments if the enamel is worn. Enamel is the outer layer of the tooth. There is no enamel on an exposed tooth root, so the root cannot be whitened.

In addition, cavities must be filled before treatment begins. Otherwise, the patient could experience additional mouth sensitivity when treatment material comes into contact with decay or the tooth interior.

When whitening is not recommended

Women who are pregnant and nursing should avoid any whitening treatment except for toothpaste. Oral health care professionals advise that other treatments could contain levels of peroxide that are potentially dangerous to the child. Although no connections have been made between these treatments and harm to

the child's health, mothers are urged to take preventive action and delay whitening treatment.

Teenagers should not have their teeth bleached until they are between 14 and 16 years old. In a younger child, the nerve of the tooth called the pulp chamber has not fully developed. Whitening at this point could irritate the pulp and cause sensitivity.

People who are allergic to peroxide should not be treated with this whitening agent.

Cautions about tooth color

Treatments such as bleaching are most effective on yellowish stains, according to the American Dental Association (ADA). Teeth with brown stains may not bleach as well, and the treatment is even less effective on gray-stained teeth.

Furthermore, bleaching will not change the color of tooth-colored fillings, dentures, crowns, porcelain restorations, bonding, or other material used to restore or replace a tooth. If bleaching is done, the newly whitened teeth will stand out in contrast to fillings or other modifications.

Description

Teeth are whitened by the use of bleach or other material. The treatment may be done in the dental office, at home with guidance from a dentist, or at home with the use of over-the-counter products. Tools for self-treatment include bleaching trays, gels or strips that are applied to the teeth, and toothpaste.

Whitening treatments are cosmetic procedures, and are usually not covered by dental insurance.

Products used by oral health professionals such as those sold over the counter may have the ADA Seal of Acceptance. This endorsement indicates that products carrying the seal have met the American Dental Association's criteria for safety and effectiveness. Those standards are based on the patient following directions when the product is used.

The ADA evaluation program is voluntary. That means manufacturers are not required to submit products for review. As a result, the lack of a seal may not indicate that the product is unsafe. However, products on the Accepted list have the ADA endorsement, and the association may take positions on certain unevaluated procedures such as laser treatment.

Dental office treatment

The whitening treatment provided by dentists is known as chairside bleaching, in-office bleaching, or

power bleaching. The dentist first protects the patient's gums and tissue by applying a protective gel or a rubber shield. The dentist then applies a whitening solution on the teeth.

The whitening solution contains hydrogen peroxide, which is a bleaching agent that could change the tooth color. The bleach is used to remove surface (extrinsic) and deeper (intrinsic) stains. Professionally applied whiteners, those solutions used by dentists, usually contain hydrogen peroxide. This bleaching agent comes in concentrations ranging from 15% to 35%. As of March of 2005, all solutions with the ADA Seal had a 35% concentration of hydrogen peroxide.

After the gel is applied, a light may be shined on the teeth to accelerate the whitening agent. Some agents are enhanced by lasers. However, no treatments requiring lasers were on the ADA list of accepted products as of March 2005. Although lasers may be safe, the association had not seen published, peer-reviewed data on the safety and effectiveness of laser whitening.

Chairside bleaching treatment may last from thirty minutes to an hour, according to the ADA. In addition, patients may need to return for additional treatments. The cost of treatment for the whole mouth can cost from \$500 to \$1,000. Factors affecting cost include the patient's location and the number of treatments needed. At the high end of the range is laser treatment, which could cost \$1,000 or more.

Dentist-supervised treatment

Supervised treatment combines visits to the dentist with treatment at home. The procedure is also called tray bleaching or nightguard bleaching because the patient wears a tray on the teeth that protects the gums from the whitening solution.

For this treatment, the dentist takes an impression of the patient's teeth and makes a mouthpiece tray, or mouthguard, that will fit over the teeth. The dentist dispenses a whitening gel that the patient will place in the customized mouthguard.

The gel usually contains carbamide peroxide, which comes in concentrations of 10%, 16%, and 22%. Products with the ADA Seal have a 10% concentration. That amount is the equivalent of an approximately 3% concentration of hydrogen peroxide. The ADA endorsement applies only to home systems dispensed by dentists. The association's Seal reflects the importance of consulting with a dentist before undergoing treatment at home, according to the ADA.

The dentist will set up a schedule for wearing the mouthguard. Wearing times vary by product. A patient may wear the piece overnight for one to two weeks. For other systems, the patient wears the mouthguard for a set amount of time twice a day. This treatment usually lasts two weeks.

During supervised treatment, the dentist generally schedules appointments to monitor the patient's progress. In addition to checking the whitening process, the dentist may examine the fit of the mouthguard and look for signs of gum irritation.

Supervised home bleaching of the whole mouth costs from \$300 to \$600.

Over-the-counter (OTC) products

In-home treatments that can be purchased over-the-counter include products that use bleach in mouthguard trays as well as strips and gels. The bleaching agent is usually carbamide peroxide, which is not as strong as the hydrogen peroxide found in solutions that are used in chairside bleaching and supervised home treatment.

OTC treatments range in price from \$20 to \$150. Treatment lasts 14 days on average. Another treatment is the use of whitening toothpaste, a product that does not contain bleach.

TRAY TREATMENT. Mouthguard treatment kits can be bought in stores and over the Internet. The tray kits involve the use of a mouthguard and gel. While similar to dentist-supervised home treatment, the patient does not use a customized tray specifically for her or his mouth. Some kits have mouthguards that patients can mold to their teeth. However, the patient relies on the generic instructions provided by the manufacturer.

GELS AND WHITENING STRIPS. Gels are applied directly to the teeth. Whitening strips are thin, clear strips coated with a peroxide-based gel. The strips are applied to the teeth and worn for 30 minutes twice a day. Treatment time varies by product and generally lasts from five to 14 days.

WHITENING TOOTHPASTES. Whitening toothpastes do not contain bleach. Instead mild abrasives remove surface stains, but do not change tooth color. Products with the ADA Seal contain special chemicals or polishing agents that remove stains. A tube of whitening toothpaste costs about \$5.

Preparation

The ADA advises people to consult with a dentist before beginning any teeth whitening treatment. The

KEY TERMS

Enamel—The hard, white, outer layer of the tooth.

Fluoride—A compound believed to combat cavities in teeth.

Peroxide—A bleaching agent that is a compound consisting of two atoms of oxygen connected by a single bond.

dentist can review the patient's oral health history and discuss the appropriate treatment. If necessary, the dentist will fill cavities.

Aftercare

During supervised at-home treatment, the dentist may schedule appointments to check on the progress of whitening, side effects, and the tray fit.

After treatment is completed, people need to be aware that **smoking** will cause teeth to discolor. Beverages with **caffeine** should be consumed with a straw to reduce the effects of staining. Another preventive action is brushing the teeth after drinking or eating foods that cause stains.

Risks

Teeth-whitening may cause sensitivity to hot and cold food and beverages. This is a temporary side effect that usually ends when treatment is completed. Some patients also experience gum irritation if the tray does not fit properly.

Normal results

Dentists use a stronger bleaching agent than that found in commercial products, so in-office whitening treatment produces a more dramatic effect on teeth with yellow stains. Over-the-counter products with bleach provide some change in the tooth color, and whitening toothpaste works only on surface stains.

Bleaching does not leave teeth permanently white. Whitening can last from six months to a year. Sometimes teeth stay white even longer. However, smoking or consumption of food and beverages that stain can cause discoloration within one month.

Resources

PERIODICALS

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ORGANIZATION

Academy of General Dentistry. 211 East Chicago Avenue, Suite 900, Chicago, IL 60611-1999. 888-AGD-DENT (888-243-3368). < <http://www.agd.org> >.

American Dental Association. 211 East Chicago Avenue, Chicago, IL 60611-2678. 312-440-2500. < <http://www.ada.org> >.

American Dental Hygienists' Association. 444 North Michigan Avenue, Suite 3400, Chicago, IL 60611. 312-440-8927. < <http://www.adha.org> >.

OTHER

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Liz Swain

Template bleeding time see **Bleeding time**

Temporal arteritis

Definition

The term temporal arteritis literally means "inflammation of the temporal arteries." As implied by the name, these blood vessels run along the temples after they branch off from the carotid artery in the neck. They provide the blood supply to portions of the scalp, jaw muscles, and salivary glands. Inflammation of these arteries, probably resulting from an abnormal immune reaction, disrupts this blood supply, resulting in a variety of symptoms. They can range from relatively minor jaw **pain** or headache to major symptoms, including temporary or permanent blindness.

Temporal arteritis is also called giant cell arteritis or cranial arteritis. It is a rheumatic disease that affects large and medium-sized arteries throughout the body and can occur in a variety of patients. Although the temporal arteries are most commonly affected, other arteries throughout

the body may be affected. The disease seems to target arteries containing elastic tissue. Veins are rarely affected. Temporal arteritis is a type of **vasculitis**.

Description

Temporal arteritis almost always occurs in people over 50, and it becomes more common as people age. About 20 out of 100,000 people over the age of 50 suffer from temporal arteritis. Women are affected twice as often as men. Some authorities say that temporal arteritis is more common in Caucasians (especially Scandinavians) than in people of other races. Close relatives of patients with temporal arteritis may be more likely than others to get the disease.

Patients with temporal arteritis are diagnosed and overlap with a broader disorder called giant cell arteritis. This can affect parts of the body in addition to the scalp, eyes, and jaw. Sometimes the disease can cause restricted circulation to both arms or both legs, producing pain in the affected limbs. With other blood vessels involved, patients with advanced forms of the disease may experience strokes or transient ischemic attacks (TIA). These result in brief episodes of pain caused by decreased blood flow. Even heart attacks are occasionally caused by giant cell arteritis.

Causes and symptoms

This disease is one of a group of diseases in which the linings of large- or medium-sized blood vessels become inflamed. The elastic layer of these vessels is attacked by "giant" cells and chemicals produced by the immune system. This reaction reduces blood flow through the blood vessels, and the limited blood supply causes the symptoms.

The disease usually begins with "flu-like" symptoms, including a mild **fever** (100–101°F), general body discomfort, and a persistent, dull **headache**. The scalp may be tender to the touch over the affected blood vessels. Jaw muscles sometimes become painful when the patient chews.

As the disease progresses, more severe symptoms occur. These include blurred vision or temporary blindness that typically lasts ten minutes or less. Eventually, permanent loss of vision can occur. Transient ischemic attacks, strokes, and heart attacks may occur when the disease is far advanced.

Diagnosis

Doctors from a number of specialties develop experience in diagnosing and treating temporal arteritis. These include internists, who treat a broad range of diseases; rheumatologists, who focus on rheumatic diseases; geriatricians, who treat older people; ophthalmologists, who treat eye and vision disorders; neurologists, who treat headaches and problems of the optic nerve; and vascular surgeons, who treat blood vessel problems.

The doctor will generally take a medical history first. The patient can help the doctor tremendously by reviewing all symptoms—both major and minor—from the last two or three months. If possible, the patient should ask family or close friends for help in recalling his/her ailments from recent months. Then the doctor will conduct a complete **physical examination**. Often, he or she will detect a tender, swollen artery on the scalp.

The doctor will order blood tests as well. A standard and inexpensive test called the **erythrocyte sedimentation rate** (ESR or “sed” rate) is particularly helpful. Results from this test, which measures inflammation in the body, will almost always be higher than normal. Tests of the red blood cells may show mild anemia. Sometimes blood tests for liver function will also be abnormal.

The definitive diagnostic test is a temporal artery biopsy. A doctor will make one or more tiny incisions under **local anesthesia** to remove samples of the suspect artery. Under the microscope, a pathologist usually can identify the typical damage caused by temporal arteritis.

Treatment

The mainstay of treatment is a course of **corticosteroids** (steroid hormones that have an anti-inflammatory effect), usually prednisone. The initial prescription involves a fairly high dose of steroids (40–60 mg/day) which is gradually tapered down to a maintenance dose. Because of the high incidence of blindness in untreated cases, steroid therapy should be started immediately rather than waiting for biopsy results. Patients typically take this maintenance dose for periods of one to three years. Sometimes **nonsteroidal anti-inflammatory drugs** (NSAIDs) are prescribed for muscle aches or headaches, especially while steroid doses are being reduced.

Prognosis

The outlook for most patients with temporal arteritis is good, especially if the disorder is

KEY TERMS

Anemia—Lower than normal level of red blood cells, or of the oxygen-carrying chemical hemoglobin.

Biopsy—Removal and examination of a sample tissue from the body for diagnostic purposes.

Corticosteroids—A group of hormones, produced naturally by the adrenal gland and other organs. They are used to treat a wide variety of disorders, including many rheumatic disorders.

Erythrocyte sedimentation rate—The speed at which red blood cells sink in a tube of freshly drawn blood, which is a rough measure of clotting disorders or inflammation.

Prednisone—A corticosteroid often used to treat inflammation.

Rheumatic disease—A type of disease involving inflammation of muscles, joints, and other tissues.

Transient ischemic attack—A brief experience of stroke-like symptoms (for instance, numbness, paralysis, problems in speaking or understanding speech) that go away within hours, with no permanent damage. Also known as TIA.

Vasculitis—An inflammation of the blood vessels.

diagnosed early. Symptoms often diminish within a month once patients begin to take steroids. Although physicians do not talk about a “cure” for temporal arteritis, symptoms typically do not return after a full course of steroid treatment. Unfortunately, if the diagnosis is made late in the disease, lost vision may not return.

Prevention

There is no medically proven approach to prevention. The best way to prevent severe, permanent damage is to obtain expert medical advice if the patient or the family physician suspects this problem.

Resources

ORGANIZATIONS

National Headache Foundation. 428 W. St. James Place, Chicago, IL 60614. (800) 843-2256. <<http://www.headaches.org>>.

Richard H. Lampert

Temporomandibular joint disorders

Definition

Temporomandibular joint disorder (TMJ) is the name given to a group of symptoms that cause **pain** in the head, face, and jaw. The symptoms include headaches, soreness in the chewing muscles, and clicking or stiffness of the joints. They often have psychological as well as physical causes.

Description

TMJ disorder, which is also sometimes called TMJ syndrome, results from pressure on the facial nerves due to muscle tension or abnormalities of the bones in the area of the hinge joint between the lower jaw and the temporal bone. This hinge joint is called the temporomandibular joint. There are two temporomandibular joints, one on each side of the skull just in front of the ear. The name of the joint comes from the two bones that make it up. The temporal bone is the name of the section of the skull bones where the jaw bone (the mandible) is connected. The jaw bone is held in place by a combination of ligaments, tendons, and muscles. The temporomandibular joint also contains a piece of cartilage called a disc, which keeps the temporal bone and the jaw bone from rubbing against each other. The jaw pivots at the joint area in front of the ear. The pivoting motion of the jaw is complicated because it can move downward and from side to side as well as forward. Anything that causes a change in shape or functioning of the temporomandibular joint will cause pain and other symptoms.

Causes and symptoms

Causes

TMJ syndrome has several possible physical causes:

- **Muscle tension.** Muscle tightness in the temporomandibular joint usually results from overuse of muscles. This overuse in turn is often associated with psychological **stress** and clenching or grinding of the teeth (**bruxism**).
- **Injury.** A direct blow to the jaw or the side of the head can result in bone fracture, soft tissue bruising, or a dislocation of the temporomandibular joint itself.
- **Arthritis.** Both **osteoarthritis** and **rheumatoid arthritis** can cause TMJ.

- **Internal derangement.** Internal derangement is a condition in which the cartilage disk lies in front of its proper position. In most cases of internal derangement, the disc moves in and out of its correct location, making a clicking or popping noise as it moves. In a few cases, the disc is permanently out of position, and the patient's range of motion in the jaw is limited.
- **Hypermobility.** Hypermobility is a condition in which the ligaments that hold the jaw in place are too loose and the jaw tends to slip out of its socket.
- **Birth abnormalities.** These are the least frequent cause of TMJ but do occur in a minority of patients. In some cases, the top of the jawbone is too small; in others, the top of the jawbone outgrows the lower part.

Symptoms

The symptoms of TMJ depend in part on its cause. The most common symptoms are facial pain in front of the ears; headaches; sore jaw muscles; a clicking sound when chewing; a grating sensation when opening and closing the mouth; and temporary locking of the jaw. Some patients also report a sensation of buzzing or ringing in the ears. Usually, the temporomandibular joint itself is not painful. Most cases of TMJ are seen in women between 20-50 years of age.

Diagnosis

Dental examination and patient history

TMJ disorders are most frequently diagnosed by dentists. The dentist can often diagnose TMJ based on **physical examination** of the patient's face and jaw. The examination might include pressing on (palpating) the jaw muscles for soreness or asking the patient to open and close the jaw in order to check for misalignment of the teeth in the upper and lower jaw. This condition is called **malocclusion**. The dentist might also gently move the patient's jaw in order to check for loose ligaments.

Imaging studies

Imaging studies are not usually necessary to diagnose TMJ. In most cases, x rays and MRI scans of the temporomandibular joint will be normal. Consequently, these two tests are not commonly used to diagnose TMJ. If the dentist suspects that the patient has internal derangement of the disc, he or she can use a technique called **arthrography** to make the diagnosis. In an arthrogram, a special dye is injected into the joint, which is then x-rayed. Arthrography can be used to evaluate the movement of the jaw and the disc as well as size and shape, and to evaluate the effectiveness of treatment for TMJ.

KEY TERMS

Arthrography—An imaging technique that is sometimes used to evaluate TMJ associated with internal derangement.

Bruxism—Habitual clenching and grinding of the teeth, especially during sleep.

Electromyographic biofeedback—A method for relieving jaw tightness by monitoring the patient's attempts to relax the muscle while the patient watches a gauge. The patient gradually learns to control the degree of muscle relaxation.

Internal derangement—A condition in which the cartilage disc in the temporomandibular joint lies in front of its proper position.

Malocclusion—The misalignment of opposing teeth in the upper and lower jaws.

Mandible—The medical name for the lower jaw.

Osteoarthritis—A type of arthritis marked by chronic degeneration of the cartilage of the joints, leading to pain and sometimes loss of function.

Rheumatoid arthritis—A chronic autoimmune disorder marked by inflammation and deformity of the affected joints.

Temporal bones—The compound bones that form the left and right sides of the skull.

Transcutaneous electrical nerve stimulation—A method for relieving the muscle pain of TMJ by stimulating nerve endings that do not transmit pain. It is thought that this stimulation blocks impulses from nerve endings that do transmit pain.

Treatment

In many cases, the cause of pain in the TMJ area is temporary and disappears without treatment. About 80% of patients with TMJ will improve in six months without medications or physical treatments.

Medications

Patients with TMJ can be given **muscle relaxants** if their symptoms are related to muscle tension. Some patients may be given **aspirin** or **nonsteroidal anti-inflammatory drugs** (NSAIDs) for minor discomfort. If the TMJ is related to rheumatoid arthritis, it may be treated with **corticosteroids**, methotrexate (MTX, Rheumatrex) or gold sodium (Myochrysine).

Physical therapy and mechanical devices

Patients who have difficulty with bruxism are usually treated with splints. A plastic splint called a nightguard is given to the patient to place over the teeth before going to bed. Splints can also be used to treat some cases of internal derangement by holding the jaw forward and keeping the disc in place until the ligaments tighten. The splint is adjusted over a period of two to four months.

TMJ can also be treated with ultrasound, electromyographic **biofeedback**, stretching exercises, transcutaneous **electrical nerve stimulation**, stress management techniques, or friction massage.

Surgery

Surgery is ordinarily used only to treat TMJ caused by birth deformities or certain forms of internal derangement caused by misshapen discs.

Prognosis

The prognosis for recovery from TMJ is excellent for almost all patients. Most patients do not need any form of long-term treatment. Surgical procedures to treat TMJ are quite successful. In the case of patients with TMJ caused by arthritis or infectious diseases, the progression of the arthritis or the success of eliminating infectious agents determines whether TMJ can be eliminated.

Resources

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John T. Lohr, PhD

TEN see **Toxic epidermal necrolysis**

Tendinitis

Definition

Tendinitis is the inflammation of a tendon, a tough rope-like tissue that connects muscle to bone.

Description

Tendinitis usually occurs in individuals in middle or old age because it is often the result of overuse over

a long period of time. Tendinitis does occur in younger patients as a result of acute overuse.

tendons that commonly become inflamed include:

- tendons of the hand
- tendons of the upper arm that effect the shoulder
- achilles tendon and the tendon that runs across the top of the foot

Causes and symptoms

Sudden stretching or repeated overuse injures the connection between the tendon and its bone or muscle. The injury is largely mechanical, but when it appears, the body tries to heal it by initiating inflammation. Inflammation increases the blood supply, bringing nutrients to the damaged tissues along with immunogenic agents to combat infection. The result is swelling, tenderness, **pain**, heat, and redness if the inflammation is close to the skin.

Diagnosis

Some tendon injuries are superficial and easy to identify. These include “tennis elbow” (extensor tendinitis) over the outside of the elbow, and Achilles’ tendinitis just above the heel of the foot. There are several tendons in the shoulder that can be overused or stretched, and usually a shoulder will have more than one injury at a time. Tendinitis in the biceps, the infraspinatus, or the supraspinatus tendon may accompany a tear of the shoulder ligaments or an impingement of one bone or another. Careful pressure testing and movement of the parts is all that is necessary to identify the tendinitis.

Treatment

Rest, ice, compression, and elevation (RICE) will treat the acute condition. The best way to apply ice is in a bag with water. The water applies the cold directly to the skin. Chemical ice packs can get too cold and cause **frostbite**. Compression using an elastic wrap minimizes swelling and bleeding in an acute sprain. Splinting may help rest the limb. Pain and anti-inflammatory medications (**aspirin**, naproxen, ibuprofen) will help. Sometimes the inflammation lingers and requires additional treatment. Injections of cortisone-like medicine often relieve chronic tendinitis, but should be reserved for resistant cases since cortisone can occasionally cause problems of its own.

If tendinitis is persistent and unresponsive to non-surgical treatment, a surgery to remove the afflicted portion of tendon can be performed. Surgery is also

KEY TERMS

Biceps—The muscle in the front of the upper arm.

Infraspinatus—A muscle at the middle of the shoulder blade.

Supraspinatus—A muscle at the top of the shoulder blade.

conducted to remove calcium buildup that comes with persistent tendinitis.

Alternative treatment

An osteopathic soft-tissue treatment on the tendon may relieve pain and increase mobility. Increasing intake of antioxidant-rich foods and lowering intake of animal fats may help reduce the inflammation. **Acupuncture** has also been used to combat tendinitis. Hydrotherapies, such as whirlpool baths, help relax the surrounding muscles.

Prognosis

Generally, tendinitis will heal if the provoking activity is stopped.

Prevention

If given enough time, tendons will strengthen to meet the demands placed on them. They grow slowly because of their poor blood supply, so adequate time is required for good conditioning.

Resources

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J. Ricker Polsdorfer, MD

Tennis elbow

Definition

Tennis elbow is an inflammation of several structures of the elbow. These include muscles, tendons,

bursa, periosteum, and epicondyle (bony projections on the outside and inside of the elbow, where muscles of the forearm attach to the bone of the upper arm).

Description

The classic tennis elbow is caused by repeated forceful contractions of wrist muscles located on the outer forearm. The **stress**, created at a common muscle origin, causes microscopic tears leading to inflammation. This is a relatively small surface area located at the outer portion of the elbow (the lateral epicondyle). Medial tennis elbow, or medial epicondylitis, is caused by forceful, repetitive contractions from muscles located on the inside of the forearm. All of the forearm muscles are involved in tennis serves, when combined motions of the elbow and wrist are employed. This overuse injury is common between ages 20 and 40.

People at risk for tennis elbow are those in occupations that require strenuous or repetitive forearm movement. Such jobs include mechanics or carpentry. Sport activities that require individuals to twist the hand, wrist, and forearm, such as tennis, throwing a ball, bowling, golfing, and skiing, can cause tennis elbow. Individuals in poor physical condition who are exposed to repetitive wrist and forearm movements for long periods of time may be prone to tennis elbow. This condition is also called epicondylitis, lateral epicondylitis, medial epicondylitis, or golfer's elbow, where **pain** is present at the inside epicondyle.

Causes and symptoms

Tennis elbow pain originates from a partial tear of the tendon and the attached covering of the bone. It is caused by chronic stress on tissues attaching forearm muscles known as extensor muscles to the elbow area. Individuals experiencing tennis elbow may complain of pain and tenderness over either of the two epicondyles. This pain increases with gripping or rotation of the wrist and forearm. If the condition becomes longstanding and chronic, a decrease in grip strength can develop.

Diagnosis

Diagnosis of tennis elbow includes the individual observation and recall of symptoms, a thorough medical history, and **physical examination** by a physician. Diagnostic testing is usually not necessary unless there may be evidence of nerve involvement from underlying causes. X rays are usually always negative because

the condition is primarily soft tissue in nature, in contrast to a disorder of the bones. However, **magnetic resonance imaging** (MRI) has been shown to be helpful in diagnosing cases of early tennis elbow because it can detect evidence of swelling and tissue tears in the common extensor muscle group.

Treatment

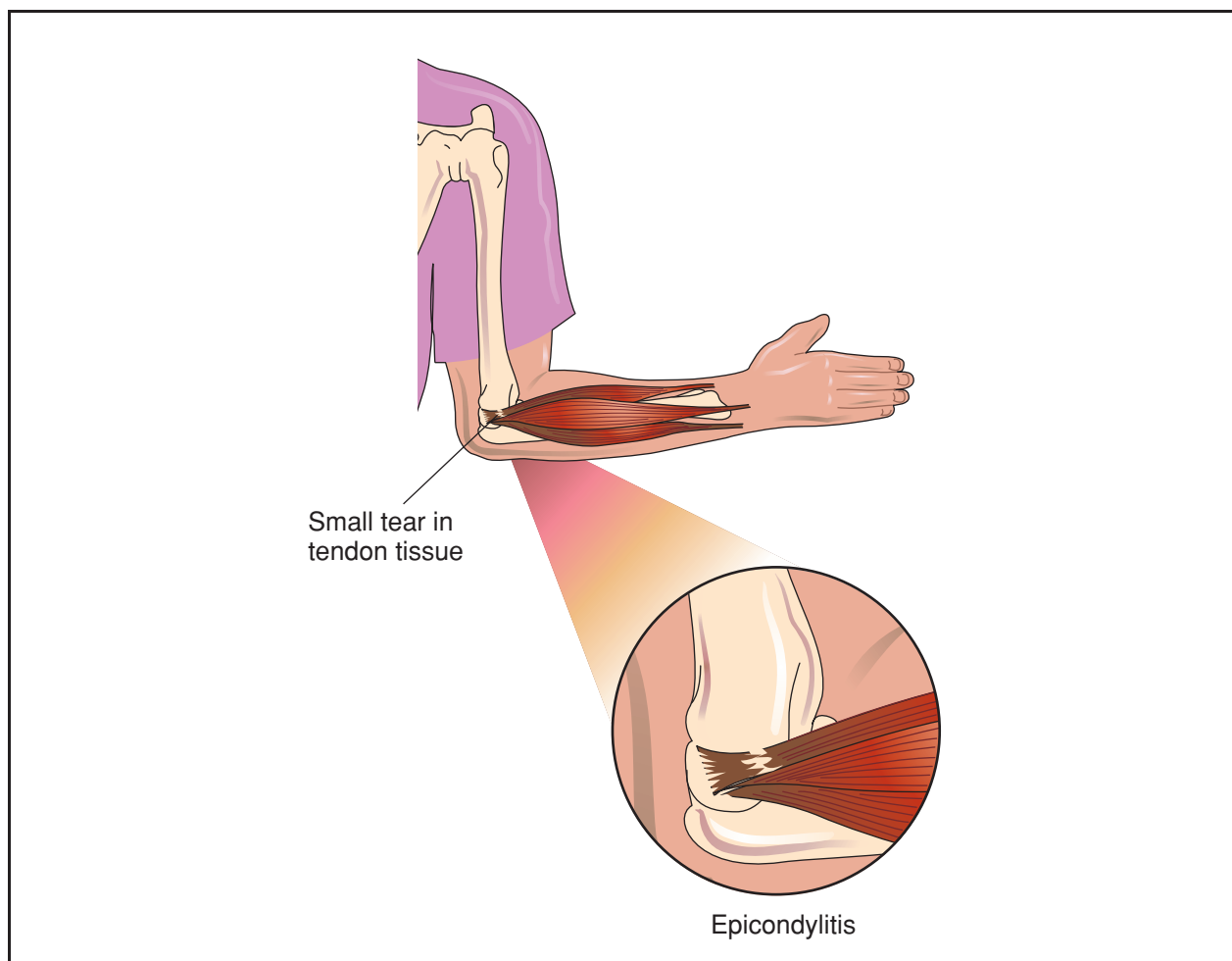
Conservative

Heat or ice is helpful in relieving tennis elbow pain. Once acute symptoms have subsided, **heat treatments** are used to increase blood circulation and promote healing. The physician may recommend physical therapy to apply diathermy or ultrasound to the inflamed site. These are two common modalities used to increase the temperature of the tissues in order to address both pain and inflammation. Occasionally, a tennis elbow splint or taping may be useful to help decrease stress on the elbow throughout daily activities. Exercises become very important to improve flexibility to all forearm muscles, and will aid in decreasing muscle and tendon tightness that has been creating excessive pull at the common attachment of the epicondyle.

The physician may also prescribe **nonsteroidal anti-inflammatory drugs** (NSAIDs) to reduce inflammation and pain. Injections of cortisone or anesthetics are often used if physical therapy is ineffective. Cortisone reduces inflammation, and anesthetics temporarily relieve pain. Physicians are cautious regarding an excessive number of injections as they have been found to weaken the tendon's integrity. In addition, a significant number of patients experience a temporary increase in pain following corticosteroid injections.

A newer method of treatment for tennis elbow is shock wave therapy, in which pulses of high-pressure sound are directed at the injured part of the tendon. The "shock" refers to the high pressure, which breaks down scar tissue and stimulates the regrowth of blood vessels in healthy tissue. Shock wave therapy sessions take about 20 minutes and have been reported to have a success rate of 80%. Shock wave therapy has very few side effects; one group of German physicians found that temporary reddening of the skin or small **bruises** were the most commonly reported side effects.

Botulinum toxin, or Botox, is also being tried as a treatment for tennis elbow as of late 2003. Although further research needs to be done, Botox appears to relieve pain in chronic tennis elbow by relaxing muscles that have gone into spasm from prolonged inflammation.



The classic tennis elbow is caused by repeated forceful contractions of wrist muscles located on the outer forearm. The stress created at a common muscle origin causes microscopic tears leading to inflammation. Persons who are most at risk of developing tennis elbow are those whose occupations requires strenuous or repetitive forearm movement. (Illustration by Electronic Illustrators Group.)

Surgery

If conservative methods of treatment fail, surgical release of the tendon at the epicondyle may be a necessary form of treatment. However, surgical intervention is relatively rare.

Alternative treatment

Massage therapy has been found to be beneficial if symptoms are mild. Massage techniques are based primarily on increasing circulation to promote efficient reduction of inflammation. Manipulation, **acupuncture**, and **acupressure** have been used as well. Contrast **hydrotherapy** (alternating hot and cold water or compresses, three minutes hot, 30 seconds cold, repeated three times, always ending with cold) applied to the elbow can help bring nutrient-rich blood

to the joint and carry away waste products. Botanical medicine and homeopathy may also be effective therapies for tennis elbow. For example, cayenne (*Capsicum frutescens*) ointment or prickly ash (*Zanthoxylum americanum*) oil applied topically may help to increase blood flow to the affected area and speed healing.

Prognosis

Tennis elbow is usually curable; however, if symptoms become chronic, it is not uncommon for treatment to continue for three to six months.

Prevention

Until symptoms of pain and inflammation subside, activities requiring repetitive wrist and forearm

KEY TERMS

Epicondyle—A projection on the surface of a bone; often an area for muscle and tendon attachment.

Epicondylitis—A painful and sometimes disabling inflammation of the muscle and surrounding tissues of the elbow caused by repeated stress and strain on the forearm near the lateral epicondyle of the humerus (arm bone).

Extensor muscles—A group of muscles in the forearm that serve to lift or extend the wrist and hand. Tennis elbow results from overuse and inflammation of the tendons that attach these muscles to the outside of the elbow.

Periosteum—A fibrous vascular membrane that covers bones.

Shock wave therapy—A method of treating tennis elbow and other musculoskeletal injuries that involves directing bursts of high-pressure sound waves at the affected area.

motion should be avoided. Once pain decreases to the point that return to activity can begin, the playing of sports, such as tennis, for long periods should not occur until excellent condition returns. Many times, choosing a different size or type of tennis racquet may help. Frequent rest periods are important despite what the wrist and forearm activity may be. Compliance with a stretching and strengthening program is very important in helping prevent recurring symptoms and exacerbation.

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American College of Occupational and Environmental Medicine (ACOEM). 1114 North Arlington Heights Road, Arlington Heights, IL 60004. (847) 818-1800. <www.acoem.org>.

American College of Sports Medicine. P.O. Box 1440, Indianapolis, IN 46206-1440. (317) 637-9200. <<http://www.acsm.org>>.

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TENS see **Electrical nerve stimulation**

Tensilon test

Definition

Tensilon is the trade name for edrophonium chloride. The Tensilon test is an injection of edrophonium chloride used to diagnosis **myasthenia gravis** (MG).

Purpose

Tensilon blocks the action of an enzyme, acetylcholinesterase, an important part of the system regulating neuromuscular transmission. To stimulate a muscle, a nerve cell (neuron) releases the chemical acetylcholine. To prevent prolonged muscle response

to a single nerve signal, acetylcholine is broken down by acetylcholinesterase after the muscle is stimulated.

In myasthenia gravis, there are too few receptors for acetylcholine on the muscle. The acetylcholine is broken down before it can fully stimulate this reduced number of receptors, and, as a result, the muscle is weak. By blocking the action of acetylcholinesterase, Tensilon prolongs the muscle stimulation, and temporarily improves strength. Increased strength following an injection of Tensilon strongly suggests a diagnosis of MG. The Tensilon test is most effective when easily observed weakness is present, and is less useful for vague or fluctuating complaints.

Precautions

The Tensilon test may cause heart rhythm abnormalities, especially in those patients with preexisting conditions.

Description

The Tensilon test involves the intravenous injection of a small amount of Tensilon. The needle is left in place. If no adverse reaction is observed within 30 seconds, an additional volume is injected. Results are apparent within one minute.

Preparation

Before the test, the patient must stop taking all drugs that can inhibit acetylcholinesterase. The referring physician can advise on specific drugs the patient is taking.

Aftercare

The effects of Tensilon subside quickly, and are completely gone after 30-60 minutes. No aftercare is needed.

Risks

Atrial fibrillation and bradycardia are possible in sensitive individuals. The administering physician must have appropriate resuscitative equipment available.

Normal results

In a patient without MG, the Tensilon test will not produce an obvious increase in a previously weak muscle. Some subjective feelings of increased strength are possible but not significant.

KEY TERMS

Acetylcholine—a molecule released by neurons at the neuromuscular junction that causes muscle contraction.

Abnormal results

An obvious increase in strength in weakened muscles strongly suggests the diagnosis of myasthenia gravis. The effect comes on very rapidly, and fades within minutes.

Resources

ORGANIZATIONS

Muscular Dystrophy Association. 3300 East Sunrise Drive, Tucson, AZ 85718. (800) 572-1717. <<http://www.mdaua.org>>.

Myasthenia Gravis Foundation of America. 222 S. Riverside Plaza, Suite 1540, Chicago, IL 60606. (800) 541-5454. <<http://www.med.unc.edu>>.

Richard Robinson

Tension headache

Definition

This most common type of **headache** is caused by severe muscle contractions triggered by **stress** or exertion. The American Council for Headache Education (ACHE) estimates that 95% of women and 90% of men in the United States and Canada have had at least one headache in the past twelve months. Tension headaches are considered a type of primary headache, which means that they are not caused by another medical condition or disorder.

Other names for tension headaches include muscle contraction headache, ordinary headache, psychomyogenic headache, and stress headache.

Description

While most American adults get a tension headache from time to time, women and people with more education are slightly more likely to suffer with them; the female: male ratio is reported to be 1.4: 1. People who are so anxious that they grind their teeth or hunch

their shoulders may find that the physical strain in their body can be experienced as **pain** and tension in the muscles of the neck and scalp, producing almost constant pain.

In terms of age groups, chronic tension headaches have been reported in children younger than six years, but most people report that their tension headaches start in adolescence or the early twenties.

Causes and symptoms

Tension headaches are caused by tightening in the muscles of the face, neck and scalp because of stress or poor posture. They can last for days or weeks and can cause pain of varying intensity. The tightening muscles cause more expansion and constriction of blood vessels, which can make head pain worse. Eyestrain caused by dealing with a large amount of paperwork or reading can cause a tension headache as well.

Many people report tension headache pain as a kind of steady ache (as opposed to a throb) that forms a tight band around the forehead, affecting both sides of the head. Tension headaches usually occur in the front of the head, although they also may appear at the top or the back of the skull.

Tension headaches often begin in late afternoon and can last for several hours; they can occur every day and last throughout most of the day, but most (82%) go away within a few hours. A tension headache that occurs on 15 or more days per month over a period of six months or longer is called a chronic tension headache. Unlike migraines, tension headaches don't cause **nausea and vomiting**, sensitivity to light, or any kind of aura before the headache begins.

It is possible for children as well as adults to have more than one type of headache. It is not unusual for patients with chronic tension headaches to suffer from migraine headaches as well.

Diagnosis

Diagnosis of tension headaches is made from a medical history, discussion of symptoms, and elimination of other types of headaches or underlying disorders.

Very few headaches are the sign of a serious underlying medical problem. However, sufferers should call a physician at once if they:

- have more than three headaches a week
- take painkillers almost every day
- need more than the recommended dose of painkiller

- have a stiff neck and/or **fever** in addition to headache
- are dizzy, unsteady, or have slurred speech, weakness, or numbness
- have confusion or drowsiness with the headache
- have headaches that began with a head injury
- have headaches triggered by bending, coughing or exertion
- have headaches that keep getting worse
- have severe **vomiting** with the headache
- have the first headache after age 50
- awaken with headache that gets better as the day goes on

Treatment

There are many different treatments for tension headaches, which respond well to both medication and massage. If these headaches become chronic, however, they are best treated by identifying the source of tension and stress and reducing or eliminating it.

Medication

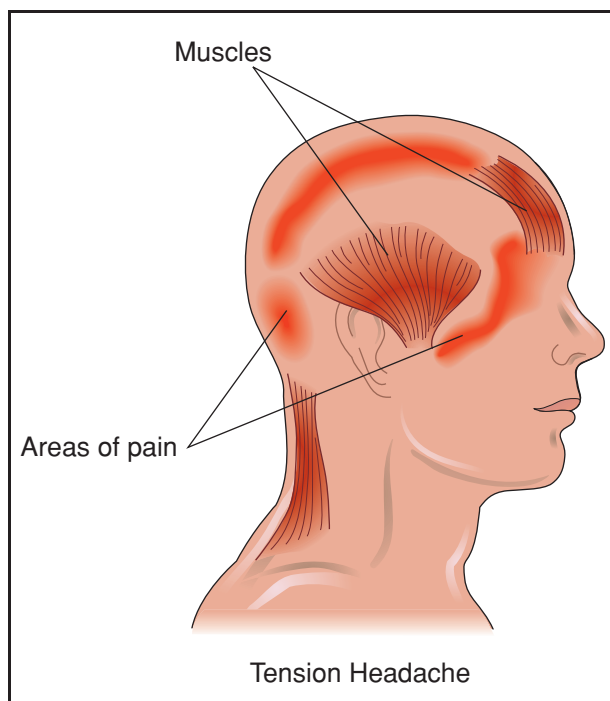
Tension headaches usually respond very well to such over-the-counter **analgesics** as **aspirin**, ibuprofen, or **acetaminophen**. However, some of these drugs (especially those that contain **caffeine**) may trigger rebound headaches if discontinued after they are taken for more than a few days.

More severe tension headaches may require combination medications, including a mild sedative such as butalbital; these should be used sparingly, though. Chronic tension headaches may respond to low-dose amitriptyline taken at night.

Massage

Massaging the tense muscle groups may help ease pain. Instead of directly massaging the temple, patients will get more relief from rubbing the neck and shoulders, because tension headaches can arise from tension in this area. In fact, relaxing the muscles of the neck can cut the intensity and duration of tension headaches at least in half.

To relax these muscles, the neck should be rotated from side to side as the shoulders shrug. Some people find that imagining a sense of warmth or heaviness in the neck muscles can help. Taking three very deep breaths at the first hint of tension can help prevent a headache.



Tension headache is the most common type of headache caused by severe muscle contractions triggered by stress or exertion. Tension headaches usually occur in the front of the head, although they may also appear at the top or the back of the skull, as shown in the illustration above. (Illustration by Electronic Illustrators Group.)

Other therapy

If tension headaches are a symptom of either depression or **anxiety**, the underlying problem should be treated with counseling, medication, or a combination of both.

Injections of botulinum toxin (Botox) have been reported to relieve tension headaches in some patients, but this treatment is still regarded as experimental as of 2004.

Alternative treatment

Eliminating the source of the tension as much as possible will help prevent tension headaches. **Acupuncture** may be helpful in treating some chronic tension headaches. Homeopathic remedies and botanical medicine can also help relieve tension headaches. Valerian (*Valeriana officinalis*), skullcap (*Scutellaria lateriflora*), and passionflower (*Passiflora incarnata*) are three herbal remedies that may be helpful. A tension headache can also be relieved by soaking the feet in hot water while an ice cold towel is wrapped around the neck.

KEY TERMS

Analgesic—A medication that relieves pain without causing loss of consciousness. Over-the-counter analgesics include aspirin and NSAIDs.

Primary headache—A headache that is not caused by another disease or medical condition. Tension headaches are a subtype of primary headache.

Rebound headache—A type of primary headache caused by overuse of pain relievers. It is also known as analgesic abuse headache.

Prognosis

Cutting down on stress and relying less on caffeine-containing medications can reduce the number of tension headaches for most people.

Prevention

Tension headaches can often be prevented by managing everyday stress and making some important lifestyle changes. Those who are prone to tension headaches should:

- take frequent “stress breaks”
- get regular exercise—even a brisk 15-minute walk can help prevent tension headaches
- get enough sleep
- release angry feelings

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ORGANIZATIONS

- American Headache Society (AHS). 19 Mantua Road, Mt. Royal, NJ 08061. (856) 423-0043. Fax: (856) 423-0082. < <http://www.ahsnet.org> >.
- National Headache Foundation. 428 W. St. James Place, Chicago, IL 60614. (800) 843-2256. < <http://www.headaches.org> >.
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Terazosin see **Alpha1-adrenergic blockers**

Testicular cancer

Definition

Testicular **cancer** is a disease in which cancer cells are discovered in one or both testicles. The testicles, also known as testes or gonads, are located in a pouch beneath the penis called the scrotum.

Description

The testicles make up one portion of the male reproductive system. Normally, they are each somewhat smaller than a golf ball in size and are contained within the scrotum. The testicles are a man's primary source of male hormones, particularly testosterone. They also produce sperm.

There are several types of cells contained in the testicles, and any of these may develop into one or more types of cancer. Over 90% of all testicular cancers begin in cells called germ cells. There are two main types of germ cell tumors in men: seminomas and nonseminomas. Seminomas make up about 40% of all testicular germ cell tumors. Nonseminomas make

up a group of cancers, which include **choriocarcinoma**, yolk sac tumors, embryonal carcinoma, and teratoma.

Although testicular cancer accounts for less than 2% of all cancers in men, it is the most commonly seen cancer in young men aged 15 to 35. It is also one of the most curable.

The American Cancer Society estimates that approximately 8,980 new cases of testicular cancer will be diagnosed in American men in 2004. In addition, about 360 men will die of the disease during that year. Although the incidence of testicular cancer is rising, having doubled since 1975, it is still rare. Scandinavian countries have the highest rate in the world. Germany and New Zealand also have high rates. The lowest incidences of testicular cancer are in Asia and Africa.

Causes and symptoms

The exact causes of testicular cancer are unknown. However, there is research showing that some men are more likely to acquire it than others. The risk for testicular cancer is much higher for boys born with one or both of their testicles located in the lower abdomen rather than in the scrotum. This condition is called cryptorchidism or undescended testicles. The lifetime risk of getting testicular cancer is four times higher for boys with cryptorchidism than the risk in the general population. This risk factor remains even if surgery is done to place the testicle back into the scrotum.

Boys born with **Down syndrome** are also at higher risk of developing testicular cancer, although the reasons for this increased risk are not yet fully understood as of 2004.

There are other risk factors as well. Men who have had abnormal development of their testicles are at increased risk, as are men with Klinefelter's syndrome (a disorder of the sex chromosomes). A family history of testicular cancer increases the possibility of getting the disease. Men infected with the human **immunodeficiency virus** (HIV), especially those with **AIDS**, have a higher incidence, as do infertile men. Certain testicular tumors appear more frequently among men who work in certain occupations, like miners, oil workers, and utility workers. There is no conclusive evidence that injuries to the testicles or environmental exposure to various chemicals cause the disease.

Testicular cancer usually shows no early symptoms. It is suspected when a mass or lump is felt in the testes, although a testicular mass does not always indicate cancer and is usually painless.

Symptoms:

- a lump in either testicle (usually pea-sized, but may be as large as a marble or an egg)
- any enlargement or significant shrinking of a testicle
- a sensation of heaviness in the scrotum
- a dull ache in the groin or lower abdomen
- any sudden collection of fluid in the scrotum
- tenderness or enlargement of the breasts
- pain or discomfort in a testicle or in the scrotum

Diagnosis

When a man exhibits symptoms that suggest a possibility of testicular cancer, several diagnostic steps will occur before a definitive diagnosis is made.

History and physical

The physician takes a personal and family medical history and a complete **physical examination** is performed. The doctor will examine the scrotum as well as the abdomen and other areas to check for additional masses.

Imaging studies

If a mass is found, the physician will likely have an ultrasound performed. Through the use of sound waves, ultrasounds can help visualize internal organs and may be useful in telling the difference between fluid-filled cysts and solid masses. If the tumor is solid, it is most likely cancerous.

Computed tomography as well as ultrasound may be used to diagnose malignant germ cell tumors in **undescended testes**.

Blood tests

Certain blood tests can be helpful in diagnosing some testicular tumors. **Tumor markers** are substances often found in higher-than-normal amounts in cancer patients. Some testicular cancers secrete high levels of certain proteins such as alpha-fetoprotein (AFP), human chorionic gonadotropin (HCG), and enzymes like lactate dehydrogenase (LDH). These markers may help find a tumor that is too small to be felt during a physical examination. In addition, these tests are also helpful in determining how much cancer is actually present, and in evaluating the response to treatment to make sure the tumor has not returned.

Surgery

If a suspicious growth is found, a surgeon will need to remove the tumor and send it to the laboratory for testing. A pathologist examines the testicular tissue microscopically to determine whether cancer cells are present. If cancer cells are found, the pathologist sends back a report describing the type and extent of the cancer. In almost all cases, the surgeon removes the entire affected testicle through an incision in the groin, though not through the scrotum. This procedure is called radical inguinal orchiectomy.

Once testicular cancer is determined, further tests are necessary to find out if the cancer has metastasized (spread) to other parts of the body, and to ascertain the stage or extent of the disease. This information helps the doctor plan appropriate treatment. These tests may include computed tomography (CT scan), **lymphangiography** (x rays of the lymph system), bone scans, and chest x rays.

Treatment

Staging

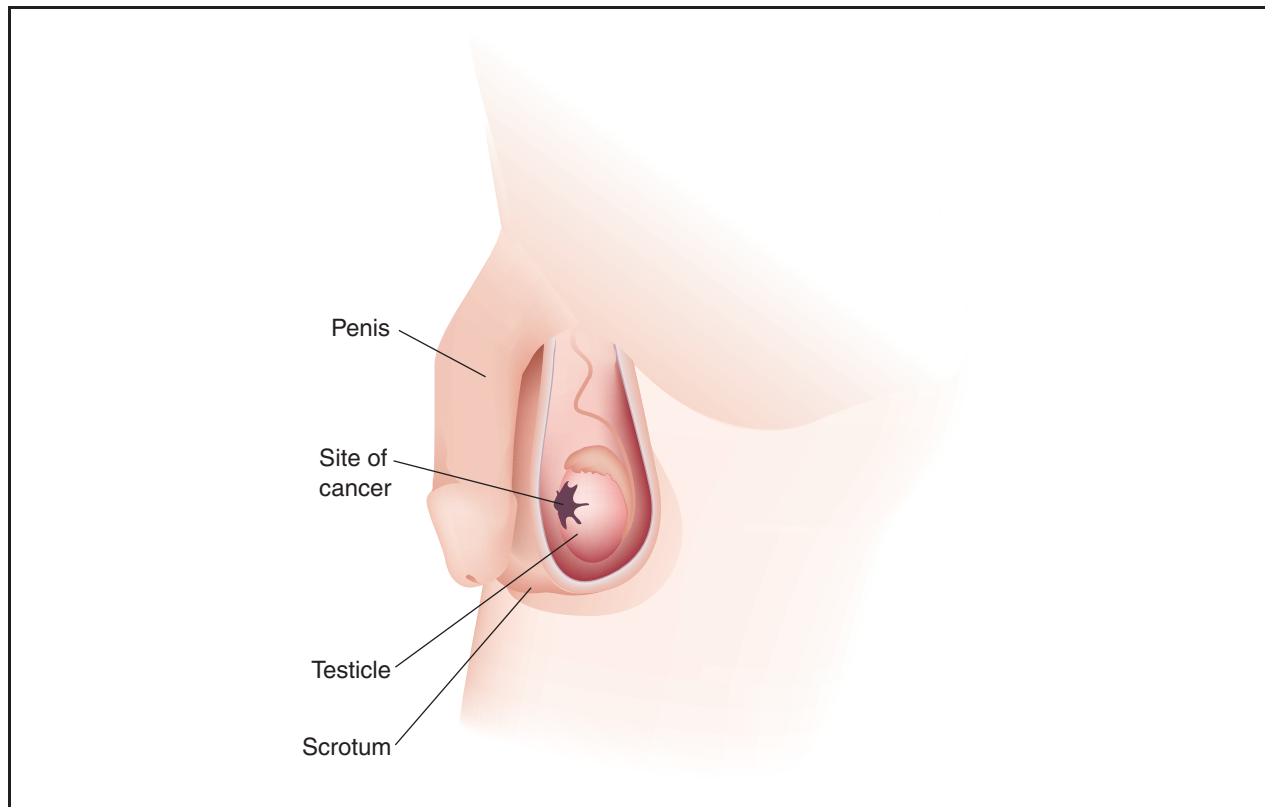
One method the cancer treatment team uses to describe the scope of a patient's cancer is the use of a staging system. Testicular cancer is classified using the TNM system. However, in order to simplify and summarize this information, the TNM description can be grouped according to stages.

Stages of testicular cancer:

- Stage I. This stage refers to a cancer found only in the testicle, with no spread to the lymph nodes or to distant organs.
- Stage II. This indicates that the cancer has spread to the lymph nodes in the abdomen, but not to lymph nodes in other parts of the body.
- Stage III. In this stage, the cancer has spread beyond the lymph nodes in the abdomen, and/or the cancer is in parts of the body far away from the testicles, such as the lungs or the liver.
- Recurrent. Recurrent disease indicates that the cancer has come back after it has already been treated. Testicular cancer can come back in the same testicle (if it was not surgically removed) or in some other body part.

Treatment

The treatment decisions for testicular cancer are dependent on the stage and cell type of the disease, as



A cancerous growth on the testicle. (Illustration by Argosy Inc.)

well as the patient's age and overall health. The four kinds of treatment most commonly used are surgery, **radiation therapy**, **chemotherapy**, and bone marrow or **stem cell transplantation**.

Surgery is normally the first line of treatment for testicular cancer and involves the removal of the affected testicle. This procedure is known as a radical inguinal orchiectomy. Depending on the type and stage of the cancer, some lymph nodes may also be removed at the same time, or possibly in a second operation. This procedure is called a retroperitoneal lymph node dissection, and can be a major operation. Some patients will experience temporary complications after surgery, including infections and bowel obstruction. If both of the testicles are taken out, a man will have no ability to produce sperm cells and will become infertile (unable to father a child). Surgery removing the lymph nodes may cause some damage to nearby nerves, which may interfere with the ability to ejaculate. Men undergoing surgery for testicular cancer may wish to discuss nerve-sparing surgery with their doctor, as well as sperm banking.

Radiation therapy for testicular cancer is delivered from a machine and is known as external beam

radiation. One potential problem with this type of radiation is that it can also destroy nearby healthy tissue as well as cancer cells. Other potential side effects include **nausea**, **diarrhea** and **fatigue**. A special device can be used to protect the unaffected testicle to preserve fertility.

Chemotherapy refers to the use of drugs in treating cancer. Since the drugs enter the bloodstream and circulate throughout the body, chemotherapy is considered a systemic treatment. The drugs primarily used in the treatment of testicular cancer are cisplatin, vinblastine, bleomycin, cyclophosphamide, etoposide, and ifosfamide. These drugs are given in various combinations, since the use of two or more drugs is considered more effective than using only one drug.

Since chemotherapy agents can affect normal as well as cancerous cells, several side effects are possible. These side effects include:

- **nausea and vomiting**
- changes in appetite
- hair loss (temporary)
- mouth sores

- increased risk of infections
- bleeding or bruising
- fatigue
- diarrhea or constipation

Several drugs are available to assist in treating these side effects, most of which will disappear after the treatment is completed. However, some of the chemotherapy agents used during treatment of testicular cancer may cause long-term side effects. These include **hearing loss**, nerve damage, and possible kidney or lung damage. Another potentially serious long-term complication is an increased risk of leukemia. This is a rare side effect, however, as it occurs in less than 1% of testicular cancer patients who receive chemotherapy. Chemotherapy may also interfere with sperm production. This may be permanent for some, but many will regain their fertility within a few years.

Studies are ongoing to determine whether high doses of chemotherapy combined with stem-cell transplantation will prove effective in treating some patients with advanced testicular cancer. In this treatment, blood-forming cells called stem cells are taken from the patient (either from the bone marrow or filtered out of the patient's blood). These cells are kept frozen while high-dose chemotherapy is administered. After receiving the chemotherapy, the patient is given the stem cells through an infusion. This treatment enables the use of extra large doses of chemotherapy that might increase the cure rate for some testicular cancers.

Preferred treatment plans by stage of disease

Stage I: Stage I seminomas are normally treated with a radical inguinal orchiectomy followed by radiation treatment aimed at the lymph nodes. More than 95% of Stage I seminomas are cured through this method. Another approach is to perform surgery only. Patients are then followed closely for several years with blood tests and imaging studies. If the cancer spreads later on, radiation or chemotherapy can still be used. Stage I non-seminomas are also highly curable with surgery, followed by one of three options. These options include the performance of a retroperitoneal lymph node dissection, two cycles of chemotherapy, or careful observation for several years.

Stage II: Stage II seminomas and non-seminomas are cured in 90% to 95% of the cases. For the purposes of treatment, stage II testicular cancers are classified as either bulky or nonbulky. Nonbulky seminomas (no lymph nodes can be felt in the

KEY TERMS

Cryptorchidism—Occurs when a boy is born with one or both testicles in the lower abdomen rather than the scrotum. Known also as undescended testicles, it is the primary risk factor for testicular cancer.

Metastatic testicular cancer—Testicular cancer that has spread to other parts of the body.

Radical inguinal orchiectomy—Surgical procedure performed to remove one or both testicles. It is done via a groin incision.

Testicles—Also called testes or gonads, they are part of the male reproductive system, and are located beneath the penis in the scrotum.

abdomen) are treated with an orchiectomy followed by radiation to the lymph nodes. Men with bulky seminomas have surgery, which may be followed by either radiation or a course of chemotherapy. Nonbulky Stage II non-seminomas are treated with surgery and lymph node removal, with possible chemotherapy. Men with bulky disease have surgery followed by chemotherapy.

Stage III: Stage III seminomas and non-seminomas are treated with surgery followed by chemotherapy. This produces a cure in about 70% of the cases. Those who are not cured may be eligible to participate in clinical trials of other chemotherapy agents.

Recurrent: Treatment of recurrent testicular cancer is dependent upon the initial stage and the treatment given. This might include further surgery and chemotherapy. Many men whose disease comes back after chemotherapy are treated with high-dose chemotherapy followed by bone marrow or stem cell transplantation.

As of 2004, there is growing evidence that men treated with cisplatin for testicular cancer are at increased risk of **coronary artery disease** ten years or longer after treatment. In addition, men who have had an orchiectomy followed by external beam radiation therapy have a significantly increased risk of dying from heart disease or a second cancer.

Alternative treatment

There are currently no scientifically proven alternative treatments known for testicular cancer. Nothing has been shown to be as successful as

conventional treatment. However, some patients may find certain alternative or complementary treatments supportive while undergoing surgery, chemotherapy or radiation. For example, **meditation** and relaxation exercises may prove effective in reducing nausea and **vomiting**. Some dietary modifications and **nutritional supplements** may be helpful in assisting with recovery after surgery. The testicular cancer patient considering alternative treatments should talk it over with members of the cancer care team. They may be able to offer additional information.

Prevention

The main risk factors associated with testicular cancer—cryptorchidism, family history of the disease, and being Caucasian—are unavoidable since they are present at birth. In addition, many men diagnosed with the disease have no known risk factors. Because of these reasons, it is not possible to prevent most incidences of testicular cancer.

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Rebecca J. Frey, PhD

Testicular scan see **Scrotal nuclear medicine scan**

Testicular self-examination

Definition

A testicular self-examination (TSE) is the procedure by which a man checks the appearance and consistency of his testes.

Purpose

Most testicular cancers are first noticed by the man himself. Men should do a TSE every month to find out if the testes contain any suspicious lumps or other irregularities, which could be signs of **cancer** or infection.

Precautions

None.

Description

A TSE should take place during a warm shower or bath, when the skin is warm, wet, and soapy. The man needs to step out of the tub so that he is in front of a mirror. The heat from the tub or shower will relax the scrotum (sac containing the testes) and the skin will be softer and thinner, making it easier to feel a lump. It is important that the exam be done very gently.

The man should stand facing his mirror and look for swelling on the scrotum. Using both hands, the scrotum should be gently lifted so that the area underneath can be checked.

KEY TERMS

Epididymis—A tube in the back of the testes that transports sperm.

Scrotum—The pouch containing the testes.

Testes—Egg-shaped male gonads located in the scrotum. Testes is the plural form of testis, which is a testicle.

Vas deferens—A tube that is a continuation of the epididymis. This tube transports sperm from the testis to the prostatic urethra.

The next step is examination by hand. The index and middle fingers should be placed under each testicle, with the thumbs on top. The testes should be examined one at a time. The man should roll each testicle between his fingers and thumbs. He should feel for lumps of any size (even as small as a pea) particularly on the front or side of each testicle. He should also look for soreness or irregularities. Next, the epididymis and vas deferens, located on the top and back of the testes, should be felt. This area feels like a cord, and should not be tender.

Normal results

It is normal for one testicle to be larger than the other is, and for them to hang at different levels; but the size should stay the same from one month to the next. The testes should be free from lumps, **pain**, irregularities and swelling.

Abnormal results

A TSE is considered abnormal if any swelling, tenderness, lumps, or irregularities are found. Hard, unmovable lumps are abnormal, even if they are painless. A lump could be a sign of an infection or a cancerous tumor. A change in testicle size from one month to the next is also abnormal. A feeling of heaviness in the scrotum is another abnormal sign. If any abnormality is found, a man is encouraged to check with his doctor as soon as possible because **testicular cancer** is highly curable if found early.

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Rhonda Cloos, R.N.

Testicular sonogram see **Scrotal ultrasound**

Testicular surgery

Definition

Testicular surgery is any surgical operation on the testicles.

Purpose

Testicular surgery is used primarily to correct developmental defects, treat infection or trauma, and treat **cancer** of the testes.

Precautions

Testicular surgery, a group of surgical operations performed on the testicles, is considered major surgery. In all cases, except when the testes are being removed, care must be taken not to damage any of the nerves and blood vessels supplying the testes and associated organs.

Description

Testicular surgery is commonly performed for the following reasons: to reposition **undescended testes** (orchiopexy); to correct **testicular torsion**; to treat **testicular cancer**, which may involve removal of the testicles (castration) or the testes (orchiectomy); to treat traumatic injuries of the testicles; and to correct **intersex states**.

Undescended testes

Undescended testes are testes that have not dropped into the scrotum. During the fetal stage of development, the testes are not in the scrotum, but in the body. As male children age, the testes descend from the body to the scrotum for proper maturation and function. Undescended testes must be treated with surgery. There are two types of undescended testes,

ectopic and cryptorchid. Ectopic testes are outside the normal route of descent. Cryptorchid testes are in the proper route of descent, but descent has been stopped before the testes reached the scrotum. The treatment for undescended testes is a surgical operation called orchiopexy, in which an incision allows the surgeon to reach the testes and pull them down into the scrotum. This operation is best done between the ages of one and two; otherwise, the testes are unlikely to mature normally. If the patient has one normal testis and one poorly developed testis, the undeveloped testis is usually removed.

Testicular torsion

Testicular torsion is a developmental defect in the tissues of the scrotum that allows the testes to rotate within the scrotum. This results in the blood vessels around other tubes in the scrotum to become wrapped around each other, resulting in blood supply to the testes being cut off. Torsion disease is seen in young boys. **Pain, nausea,** and scrotal swelling are the main symptoms. When torsion is suspected, immediate surgery is recommended. An incision is made in the scrotum, and the blood vessels and other tissues are untangled. During surgery, the testes are examined to determine their condition. If they have received enough blood to remain viable, the testes are surgically attached to scrotal tissue to prevent twisting from recurring. If the testes do not regain a healthy pink color after the blood vessels have been untangled, then it is best to remove the testes. The lack of a pink color indicates that the testes have been without blood for too long a time period, and are dead tissue. Unless removed, they will turn necrotic and cause further harm to the body. Usually, testicular torsion occurs in only one testis. However, because the other testicle has similar anatomy, it too is subject to torsion. During surgery, the other testicle is attached to scrotal tissue to prevent torsion from occurring.

Cancer

Carcinoma of the testes is the medical term for cancer of the testicles. For males between ages 20–35, carcinoma of the testes is the second most common cancer. It accounts for 1–2% of all cancers in all males. There are many kinds of cancer that can affect the testes. A mass of tissue that is suspected to be cancer should be removed surgically. It is recommended that a biopsy not be performed, but that the physician proceed directly to surgery. Biopsies have not proven to be better at diagnosing cancer of the testicles than exploratory surgery. If the presence of cancer is

confirmed during exploratory surgery, surgical excision of the cancer can be performed immediately.

The approach to the cancer during the operation depends on the location of the tissue mass. The two main approaches are through the scrotum and through the groin (inguinal region). The amount of tissue removed is variable and depends on the amount of cancerous tissue and the location. However, if a solid lesion is confirmed within a testis, a radical orchiectomy should be performed. A radical orchiectomy is a complete removal of one or both testes and associated lymphatic tissue. Other tumors allow partial removal of a testis. After surgery, the tumor is examined to determine the type of tumor for use as a guide in followup therapy.

Castration is the surgical removal of the testicles. Castration is performed as a cancer therapy, to reduce the amount of testosterone being produced, and as part of treatment for **prostate cancer**. In castration, an incision is made through one or both sides of the scrotum, depending on whether one or both testicles are being removed.

Trauma

Traumatic injuries to the testes may involve penetrating gunshot or knife **wounds**; explosions and other industrial accidents; and athletic injuries. Dislocation of the testes is a potential complication of blunt trauma to the abdomen, which commonly occurs in automobile or motorcycle accidents. A dislocated testis can usually be identified by either ultrasound or CT scans of the patient's pelvis.

Intersex states

Intersex states are a group of developmental diseases in which the patient has parts of both male and female genitalia. In testicular feminization syndrome, the patient appears to be a female and will have female genitalia but has internal testes. The internal testes are undescended. Genetic studies show that the person was to be a male. This form of intersex is also called male pseudohermaphroditism. There are a number of different causes of this condition. These patients produce the male hormone testosterone. Treatment consists of surgical removal of the internal testes, and the administration of the hormone estrogen, which produces female characteristics. Failure to remove the testes is associated with a higher rate of cancer in these patients.

Preparation

About one hour before receiving **general anesthesia**, the patient will get a shot that dries up internal

KEY TERMS

Biopsy—Removing tissue to test it for disease.

Lesion—An injury in the body tissue, such as a wound, sore, rash, or boil.

Orchiectomy—Surgical removal of one or both testes.

Orchiopexy—Surgical fixation of one or both testes.

Testes—The pair of male reproductive glands enclosed in the scrotum that produce the male sex hormone testosterone and the spermatozoa. The singular form is testis.

Testicles—The testes along with their enclosing structures.

fluids and makes him sleepy. Presurgical counseling is often recommended for patients whose reproductive abilities will be compromised by their surgeries.

Aftercare

A patient who has had a testicle removed should visit his physician once a month for the first year and every other month for the second year, with periodic followups thereafter.

Risks

Testicular surgery, like any major surgery, can have postoperative complications. These complications include internal bleeding and wound infection, as well as adverse reactions to anesthesia. About 10–15% of patients develop fertility problems.

As of 2004, there is growing evidence that men who have had an orchiectomy followed by external beam **radiation therapy** have a significantly increased risk of dying from heart disease or a second cancer.

Normal results

Undescended testes are pulled down into their correct position and mature normally. In testicular torsion, the affected testis either regains its healthy pink color and is attached to the surrounding tissue with sutures, or it is removed along with any dead tissue surrounding it. (So long as only one testis is removed, sexual function and fertility will not be affected.) Successful surgery for cancer results in the removal of malignant tissue.

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American Urological Association (AUA). 1000 Corporate Boulevard, Linthicum, MD 21090. (866) 746-4282 or (410) 689-3700. < <http://www.auanet.org> >.

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Testicular torsion

Definition

Testicular torsion is the twisting of a testis (testicle) on its connection.

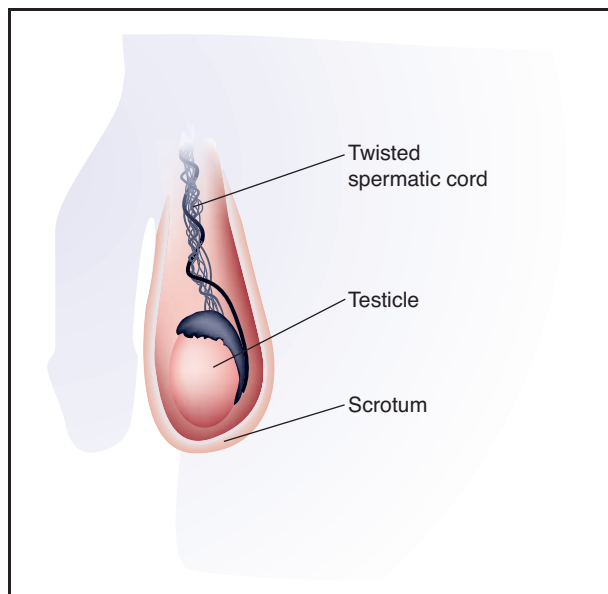
Description

The testes are suspended in the scrotum by a single bundle of tissues that also carries the blood supply to and from the testes. If the testicle rotates, the bundle kinks, and the blood supply is shut off. The resulting situation is an emergency because the testis will die within hours if the blood supply is not restored.

Causes and symptoms

Some testes hang in such a way that they twist more easily than others. Nearly all torsions happen to adolescent males—between the ages of 12 and 18—because their testes enlarge by a factor of five to six during **puberty**. A larger testis is more likely to twist. Torsion can also occur in a newborn.

Symptoms of testicular torsion are sudden severe **pain** in the scrotum, swelling, **nausea and vomiting**.



A rare condition, testicular torsion occurs when the spermatic cord is twisted and cuts off the blood supply to the testicle. (Illustration by Argosy Inc.)

Diagnosis

A nuclear scan of the scrotum may be performed. In this procedure, a tiny amount of radioactive fluid is injected into the blood and detected as it flows through the scrotum and testicles. Torsion is indicated if the radioactive fluid does not flow through the sore testis. Ultrasound scan accompanied by a contrast agent can also be used to diagnose testicular torsion.

Treatment

Surgery must be performed within 24 hours to ensure the health of the affected testis. During the procedure, the surgeon untwists the cord and secures the testis in place so that it cannot rotate again. The other testicle should also be secured to deter future testicular torsion. This procedure is called orchiopexy.

Prognosis

If the torsion is relieved within 24 hours, the testis will recover normal blood flow and function.

Prevention

Torsion of the unaffected testis is prevented by securing it during the surgery to correct the twisted testis.

KEY TERMS

Orchiopexy—The surgical securing of the testis to prevent torsion.

Scrotum—The bag of skin below the penis that contains the testes.

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J. Ricker Polsdorfer, MD

Testicular ultrasound see **Scrotal ultrasound**

Testicular x ray see **Scrotal nuclear medicine scan**

Testosterone test see **Sex hormones tests**

Tetanus

Definition

Tetanus is a rare but often fatal disease that affects the central nervous system by causing painful muscular contractions. It begins when tetanus bacteria enter the body, usually through a wound or cut exposed to contaminated soil. Tetanus is easily preventable through **vaccination**.

Description

Tetanus is rare in the United States, with nearly all cases occurring in adults who were not vaccinated as children. About 100 cases are reported each year; 63% of these occur in people over the age of 50. The number of tetanus cases in the United States has steadily decreased since the 1940s (500 to 600 cases per year); the number of reported cases has remained at approximately 50 to 100 cases per year since the mid-1970s. In 1999, however, the lowest number of annual cases to date was reported (33, or 0.02 per 100,000).

Tetanus causes convulsive **muscle spasms** and rigidity that can lead to respiratory **paralysis** and **death**. It is sometimes called “lockjaw” because one of the most common symptoms is a stiff jaw, unable to

be opened. Sometimes, tetanus affects only the part of the body where the infection began, but in almost all of reported cases, it spreads to the entire body. The incubation period from the time of the injury until the first symptoms appear ranges from two to 50 days. Symptoms usually occur within five to 10 days. When symptoms occur early, the chance of death is increased. Tetanus is not contagious.

Causes and symptoms

Tetanus is caused by a bacteria called *Clostridium tetani*, whose spores (the dormant form) are found in soil, street dust, and animal (or even human) feces. Tetanus spores germinate in the body, producing a highly poisonous neurotoxin in the blood, spreading to the nervous system. The infection is usually transmitted through deep puncture **wounds** or cuts or scratches that are not cleaned well. Between 1997 and 1999, approximately 64% of tetanus cases in the United States were associated with such wounds as punctures, lacerations, or abrasions. Many people associate tetanus with rusty nails and other dirty objects, but any wound can be a source. Less common ways of getting tetanus are animal scratches and bites, surgical wounds, dental work, and **therapeutic abortion**. About 18% of cases reported between 1997 and 1999 were a result of intravenous drug use. Cases have also been reported in people with no known wound or medical condition.

The first symptom of tetanus is often a stiff or “locked” jaw that prevents the patient from opening his/her mouth or swallowing. This is also called trismus and results in a facial expression called a sardonic smile (or risus sardonicus). Stiffness of the neck and other muscles throughout the body and uncontrollable spasms often follow. Sometimes these convulsions are severe enough to cause broken bones. The bacterial toxin (*tetanospasmin*) affects the nerve endings, causing a continuous stimulation of muscles. Other symptoms include irritability, restlessness, loss of appetite, and drooling. People with tetanus that is localized experience **pain** and **tingling** only at the wound site and spasms in nearby muscles.

In the underdeveloped world, neonatal tetanus accounts for about one-half of tetanus deaths and is related to infection of the umbilical stump in a baby born of an unimmunized mother. The Centers for Disease Control and Prevention (CDC) estimate that over 270,000 deaths occur annually worldwide as a result of neonatal tetanus. In contrast, only two cases of neonatal tetanus in the United States were reported to the CDC between 1989 and 1999. Mothers who



One characteristic of tetanus bacillus is the recurrent contracture of a muscle. Here, the patient's left hand is affected. (Custom Medical Stock Photo. Reproduced by permission.)

have been adequately immunized against tetanus protect their newborns by passing the antibody through the placenta.

Diagnosis

Tetanus is diagnosed by the clinical symptoms and a medical history that shows no tetanus immunization. Early diagnosis and treatment is crucial to recovery from tetanus.

Treatment

Tetanus is a life-threatening disease that requires immediate hospitalization, usually in an intensive care unit (ICU). Treatment can take several weeks and includes **antibiotics** to kill the bacteria and shots of antitoxin to neutralize the toxin. It also includes muscle-relaxing drugs to control muscle spasms or **barbiturates** for **sedation**. In severe cases, patients are placed on an artificial respirator. Recovery can take six weeks or more. After recovery, since the levels of circulating toxin are too low to stimulate natural antibody production, the patient must still be immunized against this disease to prevent reinfection.

Prognosis

Up to 30% of tetanus victims in the United States die. Early diagnosis and treatment improves the prognosis. Neonatal tetanus has a mortality rate of more than 90%.

Prevention

Pre-exposure vaccination

Tetanus is easily preventable through vaccination. All children should have a series of five doses of DTaP, a combined vaccine that offers protection against **diphtheria**, tetanus, and pertussis, before the age of seven, according to the Centers for Disease Control and Prevention's national immunization guidelines, the Advisory Committee on Immunization Practices, the Committee on Infectious Diseases of the American Academy of Pediatrics, and the American Academy of Family Physicians. Children will not be admitted to school without proof of this and other immunizations.

The DTaP (diphtheria, tetanus, acellular pertussis) vaccine should be given at ages two months, four months, six months, 15 to 18 months, and four to six years. DTaP is the preferred vaccine for children up to the age of seven in the United States; it has fewer side effects than DTP and can be used to complete a vaccination schedule begun with DTP. DTaP was first approved by the Food and Drug Administration (FDA) in September 1996. In December 1996, it was approved for use in infants. Between the ages of 11 and 13, children should have a booster for diphtheria and tetanus, called Td.

Adults should have a Td booster every 10 years. Statistics from the Centers for Disease Control and Prevention (CDC) show that fewer than half of Americans 60 years of age and older have antibodies against tetanus. The CDC suggests adults may be revaccinated at mid-decade birthdays (for example, 45, 55). Adults who have never been vaccinated against tetanus should get a series of three injections of Td over six to 12 months and then follow the 10-year booster shot schedule.

Side effects of the tetanus vaccine are minor: soreness, redness, or swelling at the site of the injection that appear anytime from a few hours to two days after the vaccination and go away in a day or two. Rare but serious side effects that require immediate treatment by a doctor are serious allergic reactions or deep, aching pain and muscle wasting in the upper arms. These symptoms could start from two days to four weeks after the shot and could continue for months.

In early 2001, a shortage of the tetanus vaccine became evident after the pharmaceutical company Wyeth-Ayerst Laboratories decided to stop production of the tetanus vaccine, leaving Aventis-Pasteur as the sole manufacturer of the vaccine. As a result, hospitals were provided with only a minimal amount of the drug on a weekly basis—enough to vaccinate patients with potentially infected wounds and other

KEY TERMS

Clostridium—A genus of deadly bacteria that are responsible for tetanus and other serious diseases, including botulism and gangrene from war wounds. Clostridia thrives without oxygen.

DTaP—Diphtheria and tetanus toxoids and acellular pertussis combination vaccine.

DTP—Diphtheria, tetanus, and whole-cell pertussis vaccine.

Td—Tetanus and diphtheria vaccine.

Toxin—A poisonous substance that flows through the body.

Wound—Any injury that breaks the skin, including cuts, scratches, and puncture wounds.

priority cases. Despite stepped-up production efforts on the part of the manufacturer, however, a spokesperson for Aventis-Pasteur predicted that the shortage would last until the end of 2001, as the vaccine takes 11 months to produce.

Post-exposure care

Keeping wounds and scratches clean is important in preventing infection. Since this organism grows only in the absence of oxygen, wounds must be adequately cleaned of dead tissue and foreign substances. Run cool water over the wound and wash it with a mild soap. Dry it with a clean cloth or sterile gauze. To help prevent infection, apply an antibiotic cream or ointment and cover the wound with a bandage. The longer a wound takes to heal, the greater the chance of infection. If the wound doesn't heal, or, it is red, warm, drains, or swells, consult a doctor.

Following a wound, to produce rapid levels of circulating antibody, a doctor may administer a specific antitoxin (human tetanus immune globulin, TIG) if the individual does not have an adequate history of immunization. The antitoxin is given at the same sitting as a dose of vaccine but at separate sites. Some individuals will report a history of significant allergy to "tetanus shots." In most cases, this occurred in the remote past and was probably due to the previous use of antitoxin derived from horse serum.

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Lori De Milto

Tetracyclines

Definition

Tetracyclines are medicines that kill certain infection-causing microorganisms.

Purpose

Tetracyclines are called “broad-spectrum” **antibiotics**, because they can be used to treat a wide variety of infections. Physicians may prescribe these drugs to treat eye infections, **pneumonia**, **gonorrhea**, **Rocky Mountain spotted fever**, urinary tract infections, certain bacteria that could be used in biological weapons, and other infections caused by bacteria. The medicine is also used to treat **acne**. The tetracyclines will *not* work for colds, flu, and other infections caused by viruses. Tetracyclines are generally a low-cost alternative among antibiotics.

Interestingly, a form of tetracycline has recently been used in prevention of **cancer** recurrence. Chemically modified tetracyclines such as COL-3 are derived from antibiotic tetracyclines, but because of their modifications, don’t act as antibiotics. Instead, they inhibit certain enzymes and processes that normally encourage cancer growth. By making cancer cells less aggressive, these drugs may show potential for long-term management of some cancers.

Description

Tetracyclines are available only with a physician’s prescription. They are sold in capsule, tablet, liquid, and injectable forms. Some commonly used medicines in this group are tetracycline (Achromycin V, Sumycin) and doxycycline (Doryx, Vibramycin).

Recommended dosage

The recommended dosage depends on the type of tetracycline, its strength, and the type and severity of

infection for which it is being taken. The physician who prescribed the drug or the pharmacist who filled the prescription know the correct dosage.

To make sure the infection clears up completely, patients are told to take the medicine for as long as it has been prescribed. They should not stop taking the drug just because symptoms begin to improve.

Tetracyclines work best when they are at constant levels in the blood. To help keep levels constant, the doses are spaced evenly through the day and night. No doses should be missed.

This medicine works best when taken on an empty stomach, with a full glass of water. The water will help prevent irritation of the stomach and esophagus (the tube-like structure that runs from the throat to the stomach). If the medicine still causes stomach upset, it may be necessary to take it with food. However, tetracyclines should *never* be taken with milk or milk products, as these may prevent the medicine from working properly. Milk or dairy products should be avoided within one to two hours of taking tetracyclines (except doxycycline and minocycline).

Precautions

Taking outdated tetracyclines can cause serious side effects. The medicine should not be taken if:

- its color, appearance, or taste have changed
- it has been stored in a warm or damp area
- the expiration date on its label has passed. Any such medicine should be flushed down the toilet; if there is any question about whether the medicine is still good, the patient should check with a physician or pharmacist

Antacids, calcium supplements, such as the salicylates Magan or Trilisate, magnesium-containing **laxatives**, or sodium bicarbonate (baking soda) should not be taken within one to two hours of taking tetracyclines.

Medicines that contain iron (including multivitamin and mineral supplements) should not be taken within 2-3 hours of taking tetracyclines.

Some people feel dizzy when taking these drugs. The medicine also may cause blurred vision. Because of these possible effects, anyone who takes these drugs should not drive, use machines or do anything else that might be dangerous until they have found out how the drugs affect them.

Birth control pills may not work properly while tetracyclines are being taken. To prevent **pregnancy**,

alternative methods of birth control should be used while taking tetracyclines.

This medicine may increase sensitivity to sunlight. Even brief exposure to sun can cause a severe **sunburn** or rash. While being treated with this medicine, direct sunlight should be avoided, especially between 10 A.M. and 3 P.M. A hat and tightly woven clothing that covers the arms and legs; a sunscreen with a skin protection factor (SPF) of at least 15; protecting the lips with a sun block lipstick; and avoiding use of tanning beds, tanning booths, or sunlamps are advised when on tetracyclines. The sensitivity to sunlight and sunlamps may continue for two weeks to several months after stopping the medicine.

Tetracyclines may permanently discolor the teeth of people who took the medicine in childhood. The drugs may also slow down the growth of children's bones. Tetracyclines are generally not given to infants or children under 8 years of age unless directed by the child's physician. The medical community continues to express concern about resistance to tetracyclines and other antibiotics. As patients continue to use, and often overuse, these medications, eventually, strains of bacteria develop resistance to the medicines. The use of tetracycline is much broader than it was 20 or 30 years ago and physicians will prescribe the medicine only when appropriate.

Special conditions

People with certain medical conditions or who are taking certain other medicines may have problems if they take tetracyclines. Before a patient takes these drugs, the physician should be advised of any of these conditions:

ALLERGIES. Anyone who has had unusual reactions to tetracyclines in the past should let his or her physician know before taking the drugs again. The physician also should be told about any **allergies** to foods, dyes, preservatives, or other substances.

PREGNANCY. Pregnant women should not take tetracyclines during the last half of pregnancy. These drugs can prevent the baby's bones and teeth from developing properly and can cause the baby's adult teeth to be permanently discolored. The medicine also can cause liver problems in pregnant women.

BREASTFEEDING. Women who are breastfeeding should not take tetracyclines. The drugs pass into breast milk and can affect the nursing baby's teeth and bones. They may also make the baby more sensitive to sunlight and may increase risk of fungal infections.

KEY TERMS

Gonorrhea—A sexually transmitted disease (STD) that causes infection in the genital organs and may cause disease in other parts of the body.

Microorganism—An organism that is too small to be seen with the naked eye.

Rocky Mountain spotted fever—An infectious disease that is caused by a microorganism and spread by ticks. High fever, muscle pain, and spots on the skin are among the symptoms.

Salicylates—A group of drugs that includes aspirin and related compounds. Salicylates are used to relieve pain, reduce inflammation, and lower fever.

OTHER MEDICAL CONDITIONS. Before using tetracyclines, people with any of these medical problems should make sure their physicians are aware of their conditions:

- diabetes
- liver disease
- kidney disease

USE OF CERTAIN MEDICINES. Taking tetracyclines with certain other drugs may affect the way the drugs work or may increase the chance of side effects.

Side effects

The most common side effects are stomach cramps or a burning sensation in the stomach, mild **diarrhea**, **nausea**, or **vomiting**. These problems usually go away as the body adjusts to the drug and do not require medical treatment. Less common side effects, such as sore mouth or tongue and **itching** of the rectal or genital areas also may occur and do not need medical attention unless they do not go away or they are bothersome.

Other rare side effects may occur. Anyone who has unusual symptoms during or after treatment with tetracyclines should get in touch with his or her physician.

Interactions

Tetracyclines may interact with other medicines. When this happens, the effects of one or both of the drugs may change or the risk of side effects may be greater. Anyone who takes tetracyclines should let a physician know all other medicines he or she is

taking. Among the drugs that may interact with tetracyclines are:

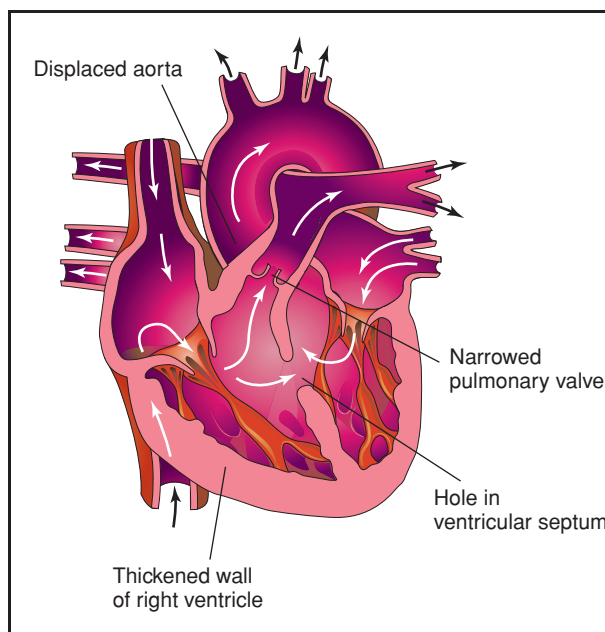
- antacids
- calcium supplements
- medicines that contain iron (including multivitamin and mineral supplements)
- laxatives that contain magnesium
- cholesterol-lowering drugs such as cholestyramine (Questran) and colestipol (Colestid)
- salicylates such as Magan and Trilisate
- penicillins
- birth control pills

Resources

PERIODICALS

- “Modified Tetracycline May Help Prevent Cancer Recurrence.” *Cancer Weekly* December 10, 2002: 2.
- Roberts, Marilyn C. “Tetracycline Therapy: Update.” *Clinical Infectious Diseases* February 15, 2003: 462–466.

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Tetralogy of Fallot is a common syndrome of congenital heart defects. This condition, present *in utero*, is caused by the narrowing of the pulmonary artery and a hole between the ventricles. When the baby is born and begins to breathe on its own, the baby turns cyanotic, or blue, due to the deoxygenated blood that bypasses the lungs because the narrowed pathway and the hole between the ventricles has remained open. (Illustration by Electronic Illustrators Group.)

Tetralogy of Fallot

Definition

Tetralogy of Fallot is a common syndrome of congenital heart defects.

Description

The heart is two pumps in one. The ventricle on the left side pumps blood full of oxygen through the body; the ventricle on the right side pumps the same blood through the pulmonary artery to the lungs to take up oxygen. The left ventricle operates at pressures about four times as high as the right ventricle. Blood is supposed to flow through one side, then the other.

Tetralogy of Fallot is a condition that is characterized by several congenital heart defects occurring at once. They include:

- **ventricular septal defect** (Abnormal passageway between the right and left ventricles)
- displaced aorta
- narrowed pulmonary valve
- thickened right ventricle wall

Each defect acts in combination with the others to create a malfunction of the heart. The problem starts very early in the uterus with a narrowed pulmonary valve and a hole between the ventricles. This is not particularly a problem for a fetus because hardly any blood flows through the lungs until birth. It is only after birth that the defects pose a problem. The blood that is supposed to start flowing through the lungs cannot easily get there because of the narrowed valve; however, the hole between the ventricles remains open. Because of the opening between ventricles, much of the blood that comes back to the heart needing oxygen is sent out without being properly oxygenated. In addition, the right heart has to pump at the same pressure as the left side. Several changes follow. First, the baby turns blue (cyanotic) because of the deoxygenated blood that bypasses the lungs. Deoxygenated blood is darker and appears blue through the skin. Second, the right side of the heart (ventricle) hypertrophies (gets more muscular) from the extra **exercise** demanded of it. Next, the low oxygen causes the blood to get thicker and clot more easily. Clots in the veins can now pass through the hole in the heart and directly enter the aorta, where they can do much more damage than in the lungs—such

KEY TERMS

Aorta—Main arterial trunk that moves blood from the heart to the arteries, which transport the blood throughout the body.

Cyanosis—Blue-colored skin due to oxygen-deficient blood.

Endocarditis—Inflammation of the lining of the heart.

Infarct—Death of tissue due to shutting off the blood supply.

Septicemia—Blood poisoning.

Systemic circulation—Through the body, as opposed to “pulmonary”—through the lungs.

Ventricles—The muscular chambers of the heart that do the pumping.

as causing infarcts in the brain. In addition, these anomalies make the lining of the heart more susceptible to infection—endocarditis— which can damage valves and lead to blood **poisoning** (septicemia).

Causes and symptoms

Tetralogy of Fallot is a congenital defect with unknown causes.

Babies with tetralogy of Fallot are blue at birth (**cyanosis**). Sometimes the blue color appears only when they cry. They also have detectable **heart murmurs**. Infants with mild forms can have surgery postponed until they are older. Infants with more severe symptoms often have attacks of worsened cyanosis. During attacks, they turn very blue, have **shortness of breath**, and can faint. This usually occurs during heightened activity, such as crying.

Diagnosis

A complete evaluation of the circulation is required, including testing the blood for its oxygen content; ultrasound; and x rays of the heart accompanied by a contrast agent to determine the amount of blood flowing in the wrong direction. A search for other **birth defects** is also necessary, because they tend to happen together.

Treatment

Correction of the defects are done through surgery. Surgery must be carefully timed with attention to the progression of the disease process, the size of the

infant, and the size of the various defects. There are temporary surgical procedures that can prolong the time before corrective surgery while the baby grows larger and stronger.

During surgery, the pulmonary valve is widened, the ventricular septal defect is closed, and any interim corrections removed.

Prognosis

Surgical correction has a high rate of success, returning the child to near-normal health.

Resources

BOOKS

Friedman, William F., and John S. Child. “Congenital Heart Disease in the Adult.” In *Harrison's Principles of Internal Medicine*, edited by Anthony S. Fauci, et al. New York: McGraw-Hill, 1997.

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Thalassemia

Definition

Thalassemia describes a group of inherited disorders characterized by reduced or absent amounts of hemoglobin, the oxygen-carrying protein inside the red blood cells. There are two basic groups of thalassemia disorders: alpha thalassemia and beta thalassemia. These conditions cause varying degrees of anemia, which can range from insignificant to life threatening.

Description

All types of thalassemias are considered *quantitative* diseases of hemoglobin, because the quantity of hemoglobin produced is reduced or absent. Usual adult hemoglobin is made up of three components: alpha globin, beta globin, and heme. Thalassemias are classified according to the globin that is affected, hence the names *alpha* and *beta* thalassemia. Although both classes of thalassemia affect the same protein, the alpha and beta thalassemias are distinct diseases that affect the body in different ways.

Beta thalassemia

Beta thalassemia may be the most best-known type of thalassemia and is also called Cooley's anemia.

It is caused by a change in the gene for the beta globin component of hemoglobin. Beta thalassemia causes variable anemia that can range from moderate to severe, depending in part on the exact genetic change underlying the disease. Beta thalassemia can be classified based on clinical symptoms. *Beta thalassemia major* usually causes severe anemia that can occur within months after birth. If left untreated, severe anemia can result in insufficient growth and development, as well as other common physical complications that can lead to a dramatically decreased life-expectancy. Fortunately, in developed countries beta thalassemia is usually identified by screening in the newborn period, before symptoms have developed. Children who are identified early can be started on ongoing blood **transfusion** therapy as needed. Although transfusion therapy prevents many of the complications of severe anemia, the body is unable to eliminate the excess iron contained in the transfused blood. Over time, the excess iron deposits in tissues and organs, resulting in damage and organ failure. Another medication must be administered to help the body eliminate the excess iron and prevent iron-overload complications. *Beta thalassemia intermedia* describes the disease in individuals who have moderate anemia that only requires blood transfusions intermittently, if at all.

Alpha thalassemia

Alpha thalassemia is the result of changes in the genes for the alpha globin component of hemoglobin. There are two main types of alpha thalassemia disease: hemoglobin H disease and alpha thalassemia major. The two diseases are quite different from beta thalassemia as well as from one another. Individuals with hemoglobin H disease can experience events of hemolytic anemia—anemia caused by the rapid breakdown of the red blood cells. These events are thought to be triggered by various environmental causes, such as infection and/or exposure to certain chemicals. Hemoglobin H disease is in most cases milder than beta thalassemia. It does not generally require transfusion therapy. *Alpha thalassemia major* is a very serious disease that results in severe anemia that begins even before birth. Most affected babies do not survive to be born or die shortly after birth.

The thalassemias are among the most common genetic diseases worldwide. Both alpha and beta thalassemia have been described in individuals of almost every ancestry, but the conditions are more common among certain ethnic groups. Unaffected carriers of all types of thalassemia traits do not experience health problems. In fact, the thalassemia trait is protective

against **malaria**, a disease caused by blood-borne parasites transmitted through mosquito bites. According to a widely accepted theory, most genetic changes—mutations—that cause thalassemia occurred multiple generations ago. Coincidentally, these mutations increased the likelihood that carriers would survive malaria infection. Survivors passed the mutation onto their offspring, and the trait became established throughout areas where malaria is common. As populations migrated, so did the thalassemia traits.

Beta thalassemia trait is seen most commonly in people with the following ancestry: Mediterranean (including North African, and particularly Italian and Greek), Middle Eastern, Indian, African, Chinese, and Southeast Asian (including Vietnamese, Laotian, Thai, Singaporean, Filipino, Cambodian, Malaysian, Burmese, and Indonesian). Alpha-thalassemia trait is seen with increased frequency in the same ethnic groups. However, there are different types of alpha thalassemia traits within these populations. The frequency of hemoglobin H disease and alpha thalassemia major depends on the type of alpha thalassemia trait. The populations in which alpha thalassemia diseases are most common include Southeast Asians and Chinese (particularly Southern Chinese).

It is difficult to obtain accurate prevalence figures for various types of thalassemia within different populations. This difficulty arises due to testing limitations in determining exact genetic diagnoses, as well as the fact that many studies have focused on small, biased hospital populations.

Two studies reflect prevalence figures that can be helpful counseling families and determining who to screen for beta thalassemia. Between the years of 1990 and 1996, the State of California screened more than 3.1 million infants born in the state for beta thalassemia. Approximately 1 in 114,000 infants had beta thalassemia major, with prevalence rates being highest among Asian Indians (about one in 4,000), Southeast Asians (about one in 10,000), and Middle Easterners (about one in 7,000). Another type of beta thalassemia disease, E/beta thalassemia, was represented in approximately one in 110,000 births, all of which occurred in families of Southeast Asian ancestry. Among Southeast Asians, the prevalence of E/beta thalassemia was approximately one in 2,600 births. This is in keeping with the observation that hemoglobin E trait carrier rates are relatively high within the Southeast Asian population: 16% in a study of 768 immigrants to California, and up to 25% in some specific Southeast Asian populations such as Cambodians. While these California studies address some of the limitations of earlier population

studies, the pattern observed in California is expected to be different in other areas of the United States and the world. For example, Italians are underrepresented in this population when compared to the population of the East Coast of the United States.

Determining prevalence figures for alpha thalassemia is even more difficult due to increased limitations in diagnostic testing. All types of alpha thalassemia disease are most common among people of Southeast Asian and Chinese descent, for reasons that become clearer with an understanding of the underlying genetics of alpha thalassemia. One study of 500 pregnant women in Northern Thailand estimated a frequency of one in 500 pregnancies affected by alpha thalassemia major, for example. Prevalence of alpha thalassemia disease is significantly lower in the United States primarily because of immigration patterns; although at least one state, California, has observed growing hemoglobin H disease incidence rates that are high enough to justify universal newborn screening for the condition.

Causes

Genetics

Humans normally make several types of the oxygen-carrying protein hemoglobin. An individual's stage in development determines whether he or she makes primarily embryonic, fetal, or adult hemoglobins. All types of hemoglobin are made of three components: heme, alpha (or alpha-like) globin, and beta (or beta-like) globin. All types of thalassemia are caused by changes in either the alpha- or beta-globin gene. These changes cause little or no globin to be produced. The thalassemias are, therefore, considered *quantitative* hemoglobin diseases. All types of thalassemias are recessively inherited, meaning that a genetic change must be inherited from both the mother and the father. The severity of the disease is influenced by the exact thalassemia mutations inherited, as well as other genetic and environmental factors. There are rare exceptions, notably with beta thalassemia, where globin gene mutations exhibit a dominant pattern of inheritance in which only one gene needs to be altered in order to see disease expression. Scientists continue to study the causes. For instance, a new mutation for alpha-thalassemia was discovered for the first time among Iranian patients in 2004.

BETA-THALASSEMIA. Most individuals have two normal copies of the beta globin gene, which is located on chromosome 11 and makes the beta globin component of normal adult hemoglobin, hemoglobin A. There are approximately 100 genetic mutations that

have been described that cause beta thalassemia, designated as either beta0 or beta+ mutations. No beta globin is produced with a beta0 mutation, and only a small fraction of the normal amount of beta globin is produced with a beta+ mutation.

When an individual has one normal beta globin gene and one with a beta thalassemia mutation, he or she is said to carry the beta thalassemia trait. Beta thalassemia trait, like other hemoglobin traits, is protective against malaria infection. Trait status is generally thought not to cause health problems, although some women with beta thalassemia trait may have an increased tendency toward anemia during **pregnancy**.

When two members of a couple carry the beta thalassemia trait, there is a 25% chance that each of their children will inherit beta thalassemia disease by inheriting two beta thalassemia mutations, one from each parent. The clinical severity of the beta thalassemia disease—whether an individual has beta thalassemia intermedia or beta thalassemia major—will depend largely on whether the mutations inherited are beta0 thalassemia or beta+ thalassemia mutations. Two beta0 mutations generally lead to beta thalassemia major, and two beta+ thalassemia mutations generally lead to beta thalassemia intermedia. Inheritance of one beta0 and one beta+ thalassemia mutation tends to be less predictable.

Although relatively uncommon, there are other thalassemia-like mutations that can affect the beta globin gene. Hemoglobin E is the result of a substitution of a single nucleotide. This change results in a structurally altered hemoglobin that is produced in decreased amounts. Therefore, hemoglobin E is unique in that it is both a quantitative (i.e. thalassemia-like) and qualitative trait. When co-inherited with a beta thalassemia trait, it causes a disease that is almost indistinguishable from beta thalassemia disease. Large deletions around and including the beta globin gene can lead to delta/beta thalassemia or hereditary persistence of fetal hemoglobin (HPFH). Interestingly, delta/beta thalassemia trait behaves very similarly to beta thalassemia trait in its clinical manifestations. However, HPFH trait does not tend to cause hemoglobin disease when co-inherited with a second thalassemia or other beta globin mutation.

ALPHA-THALASSEMIA. Most individuals have four normal copies of the alpha globin gene, two copies on each chromosome 16. These genes make the alpha globin component of normal adult hemoglobin, which is called hemoglobin A. Alpha globin is also a component of fetal hemoglobin and the other major adult hemoglobin called hemoglobin A2. Mutations

of the alpha globin genes are usually deletions of the gene, resulting in absent production of alpha globin. Since there are four genes (instead of the usual two) to consider when looking at alpha globin gene inheritance, there are several alpha globin types that are possible.

Absence of one alpha globin gene leads to a condition known as silent alpha thalassemia trait. This condition causes no health problems and can be detected only by special **genetic testing**. Alpha thalassemia trait occurs when two alpha globin genes are missing. This can occur in two ways. The genes may be deleted from the same chromosome, causing the ‘cis’ type of alpha thalassemia trait. Alternately, they may be deleted from different chromosomes, causing the ‘trans’ type of alpha thalassemia trait. In both instances, there are no associated health problems, although the trait status may be detected by more routine blood screening.

Hemoglobin H disease results from the deletion of three alpha globin genes, such that there is only one functioning gene. Typically, this can occur when one parent carries the silent alpha thalassemia trait, and the other parent carries the ‘cis’ type of the alpha thalassemia trait. In this situation, there is a 25% chance for hemoglobin H disease in each of such a couple’s children.

Hemoglobin H disease-like symptoms can also be a part of a unique condition called *alpha thalassemia mental retardation syndrome*. Alpha thalassemia **mental retardation** syndrome can be caused by a deletion of a significant amount of chromosome 16, affecting the alpha globin genes. This is usually not inherited, but rather occurs sporadically in the affected individual. Affected individuals have mild hemoglobin H disease, mild-to-moderate mental retardation, and characteristic facial features. This syndrome can also occur as a sex-linked form in which a mutation is inherited in a particular gene on the X-chromosome. This gene influences alpha globin production, as well as various other developmental processes. Individuals affected with this form of the syndrome tend to have more severe mental retardation, delayed development, nearly absent speech, characteristic facial features, and genital-urinary abnormalities. The remaining discussion will focus only on aspects of hemoglobin H disease.

Alpha thalassemia major results from the deletion of all four alpha globin genes, such that there are no functioning alpha globin genes. This can occur when both parents carry the ‘cis’ type of the alpha thalassemia trait. In this situation, there is a 25% chance for alpha thalassemia major in each of such a couple’s children.

Symptoms

Beta thalassemia

Beta thalassemia major is characterized by severe anemia that can begin months after birth. In the United States and other developed countries beta thalassemia is identified and treated early and effectively. Therefore, the following discussion of symptoms applies primarily to affected individuals in the past and unfortunately in some underdeveloped countries now. If untreated, beta thalassemia major can lead to severe lethargy, paleness, and delays in growth and development. The body attempts to compensate by producing more blood, which is made inside the bones in the marrow. However, this is ineffective without the needed genetic instructions to make enough functioning hemoglobin. Instead, obvious bone expansion and changes occur that cause characteristic facial and other changes in appearance, as well as increased risk of **fractures**. Severe anemia taxes other organs in the body—such as the heart, spleen, and liver—which must work harder than usual. This can lead to **heart failure**, as well as enlargement and other problems of the liver and spleen. When untreated, beta thalassemia major generally results in childhood **death**, usually due to heart failure. In 2004, the first known **heart attack** associated with beta thalassemia major was reported. Fortunately, in developed countries diagnosis is usually made early, often before symptoms have begun. This allows for treatment with blood transfusion therapy, which can prevent most of the complications of the severe anemia caused by beta thalassemia major. Individuals with beta thalassemia intermedia have a more moderate anemia that may only require treatment with transfusion intermittently, such as when infections occur and **stress** the body. As a person with beta thalassemia intermedia gets older, however, the need for blood transfusions may increase to the point that they are required on a regular basis. When this occurs their disease becomes more similar to beta thalassemia major. Other genetic and environmental factors can influence the course of the disease as well. For example, co-inheritance of one or two alpha thalassemia mutations can tend to ameliorate some of the symptoms of beta thalassemia disease, which result in part from an imbalance in the amount of alpha- and beta-globin present in the red blood cells.

Hemoglobin H disease

Absence of three alpha globin genes causes an imbalance of alpha and beta globin proteins in the red blood cells. The excess beta globin proteins tend to

come together to form hemoglobin H, which is unable to release oxygen to the tissues. In addition, hemoglobin H tends to precipitate out in the cells, causing damage to the red blood cell membrane. When affected individuals are exposed to certain drugs and chemicals known to make the membrane more fragile, the cells are thought to become vulnerable to breakdown in large numbers, a complication called **hemolytic anemia**. Fever and infection are also considered to be triggers of hemolytic anemia in hemoglobin H disease. This can result in **fatigue**, paleness, and a yellow discoloration of the skin and whites of eyes called **jaundice**. Usually, the anemia is mild enough not to require treatment. Severe anemia events may require blood transfusion, however, and are usually accompanied by such other symptoms as dark feces or urine and abdominal or back **pain**. These events are uncommon in hemoglobin H disease, although they occur more frequently in a more serious type of hemoglobin H disease called hemoglobin H/Constant Spring disease. Individuals affected with this type of hemoglobin H disease are also more likely to have enlargement of and other problems with the spleen.

Alpha thalassemia major

Because alpha globin is a necessary component of all major hemoglobins and some minor hemoglobins, absence of all functioning alpha globin genes leads to serious medical consequences that begin even before birth. Affected fetuses develop severe anemia as early as the first trimester of pregnancy. The placenta, heart, liver, spleen, and adrenal glands may all become enlarged. Fluid can begin collecting throughout the body as early as the start of the second trimester, causing damage to developing tissues and organs. Growth retardation is also common. Affected fetuses usually miscarry or die shortly after birth. In addition, women carrying affected fetuses are at increased risk of developing complications of pregnancy and delivery. Up to 80% of such women develop toxemia, a disturbance of metabolism that can potentially lead to convulsions and **coma**. Other maternal complications include premature delivery and increased rates of delivery by **cesarean section**, as well as hemorrhage after delivery.

Diagnosis

Thalassemia may be suspected if an individual shows signs that are suggestive of the disease. In all cases, however, laboratory diagnosis is essential to confirm the exact diagnosis and to allow for the provision of accurate **genetic counseling** about recurrence risks and testing options for parents and affected

individuals. Screening is likewise recommended to determine trait status for individuals of high-risk ethnic groups.

The following tests are used to screen for thalassemia disease and/or trait:

- complete blood count
- **hemoglobin electrophoresis** with quantitative hemoglobin A2 and hemoglobin F
- free erythrocyte-protoporphyrin (or ferritin or other studies of serum iron levels)

A *complete blood count* will identify low levels of hemoglobin, small red blood cells, and other red blood cell abnormalities that are characteristic of a thalassemia diagnosis. Since thalassemia trait can sometimes be difficult to distinguish from iron deficiency, tests to evaluate iron levels are important. A *hemoglobin electrophoresis* is a test that can help identify the types and quantities of hemoglobin made by an individual. This test uses an electric field applied across a slab of gel-like material. Hemoglobins migrate through this gel at various rates and to specific locations, depending on their size, shape, and electrical charge. *Isoelectric focusing* and *high-performance liquid chromatography (HPLC)* use similar principles to separate hemoglobins and can be used instead of or in various combinations with hemoglobin electrophoresis to determine the types and quantities of hemoglobin present. Hemoglobin electrophoresis results are usually within the normal range for all types of alpha thalassemia. However, hemoglobin A2 levels and sometimes hemoglobin F levels are elevated when beta thalassemia disease or trait is present. Hemoglobin electrophoresis can also detect structurally abnormal hemoglobins that may be co-inherited with a thalassemia trait to cause thalassemia disease (i.e., hemoglobin E) or other types of hemoglobin disease (i.e., sickle hemoglobin). Sometimes DNA testing is needed in addition to the above screening tests. This can be performed to help confirm the diagnosis and establish the exact genetic type of thalassemia.

Diagnosis of thalassemia can occur under various circumstances and at various ages. Several states offer thalassemia screening as part of the usual battery of blood tests done for newborns. This allows for early identification and treatment. Thalassemia can be identified before birth through the use of prenatal diagnosis. **Chorionic villus sampling (CVS)** can be offered as early as 10 weeks of pregnancy and involves removing a sample of the placenta made by the baby and testing the cells. CVS carries a risk of causing a **miscarriage** that is between 0.5%-1%. **Amniocentesis** is generally offered between 15 and 22 weeks of pregnancy, but can

sometimes be offered earlier. Two to three tablespoons of the fluid surrounding the baby is removed. This fluid contains fetal cells that can be tested. The risk of miscarriage associated with amniocentesis ranges from 0.33–0.5%. Pregnant woman and couples may choose prenatal testing in order to prepare for the birth of a baby that may have thalassemia. Alternately, knowing the diagnosis during pregnancy allows for the option of pregnancy termination. Preimplantation genetic diagnosis (PGD) is a relatively new technique that involves in-vitro fertilization followed by genetic testing of one cell from each developing embryo. Only the embryos unaffected by **sickle cell disease** are transferred back into the uterus. PGD is currently available on a research basis only and is relatively expensive.

Treatment

Beta Thalassemia

Individuals with beta thalassemia major receive regular blood transfusions, usually on a monthly basis. This helps prevent severe anemia and allows for more normal growth and development. Transfusion therapy does have limitations, however. Individuals can develop reactions to certain proteins in the blood—called a transfusion reaction. This can make locating appropriately matched donor blood more difficult. Although blood supplies in the United States are very safe, particularly relative to the past and to other areas of the world, there remains an increased risk of exposure to such blood-borne infections as hepatitis. Additionally, the body is not able to get rid of the excess iron that accompanies each transfusion. An additional medication called desferoxamine is administered, usually five nights per week over a period of several hours, using an automatic pump that can be used during sleep or taken anywhere the person goes. This medication is able to bind to the excess iron, which can then be eliminated through urine. If desferoxamine is not used regularly or is unavailable, iron overload can develop and cause tissue damage and organ damage and failure. The heart, liver, and endocrine organs are particularly vulnerable. Desferoxamine itself may rarely produce allergic or toxic side effects, including hearing damage. Signs of desferoxamine toxicity are screened for and generally develop in individuals who overuse the medication when body iron levels are sufficiently low. Overall, however, transfusion and desferoxamine therapy have increased the life expectancy of individuals with the most severe types of beta thalassemia major to the 4th or 5th decade. This can be expected to improve with time and increased developments in treatment, as well as for those with more mild forms of the disease.

New treatments offer additional options for some individuals with beta thalassemia major. There are various medications that target the production of red blood cells (i.e. erythropoietin) or fetal hemoglobin (i.e. hydroxyurea and butyrate). Their effectiveness in ameliorating the severity of beta thalassemia is currently being investigated. Another promising new treatment is **bone marrow transplantation**, in which the bone marrow of an affected individual is replaced with the bone marrow of an unaffected donor. If successful, this treatment can provide a cure. However, there is an approximately 10–15% chance the procedure could be unsuccessful (i.e. the thalassemia returns); result in complications (i.e. graft-versus-host disease); or result in death. The risk for specific individuals depends on current health status, age, and other factors. Because of the risks involved and the fact that beta thalassemia is a treatable condition, transplant physicians require a brother or sister donor who has an identically matched tissue type, called HLA type. HLA type refers to the unique set of proteins present on each individual's cells, which allows the immune system to recognize “self” from “foreign.” HLA type is genetically determined, so there is a 25% chance for two siblings to be a match. Transplant physicians and researchers are also investigating ways to improve the safety and effectiveness of bone marrow transplantation. Using newborn sibling umbilical cord blood—the blood from the placenta that is otherwise discarded after birth but contains cells that can go on to make bone marrow—seems to provide a safer and perhaps more effective source of donor cells. Donors and recipients may not have to be perfect HLA matches for a successful transplant using cord blood cells. Trials are also underway to determine the effectiveness of “partial transplants,” in which a safer transplant procedure is used to replace only a percentage of the affected individual's bone marrow. Other possible treatments on the horizon may include **gene therapy** techniques aimed at increasing the amount of normal hemoglobin the body is able to make.

Hemoglobin H disease

Hemoglobin H disease is a relatively mild form of thalassemia that may go unrecognized. It is not generally considered a condition that will reduce one's life expectancy. Education is an important part of managing the health of an individual with hemoglobin H disease. It is important to be able to recognize the signs of severe anemia that require medical attention. It is also important to be aware of the medications, chemicals, and other exposures to avoid due to the theoretical risk they pose of causing a severe anemia event.

KEY TERMS

Anemia—A blood condition in which the level of hemoglobin or the number of red blood cells falls below normal values. Common symptoms include paleness, fatigue, and shortness of breath.

Bilirubin—A yellow pigment that is the end result of hemoglobin breakdown. This pigment is metabolized in the liver and excreted from the body through the bile. Bloodstream levels are normally low; however, extensive red cell destruction leads to excessive bilirubin formation and jaundice.

Bone marrow—A spongy tissue located in the hollow centers of certain bones, such as the skull and hip bones. Bone marrow is the site of blood cell generation.

Bone marrow transplantation—A medical procedure used to treat some diseases that arise from defective blood cell formation in the bone marrow. Healthy bone marrow is extracted from a donor to replace the marrow in an ailing individual. Proteins on the surface of bone marrow cells must be identical or very closely matched between a donor and the recipient.

Desferoxamine—The primary drug used in iron chelation therapy. It aids in counteracting the life-threatening buildup of iron in the body associated with long-term blood transfusions.

Globin—One of the component protein molecules found in hemoglobin. Normal adult hemoglobin has a pair each of alpha-globin and beta-globin molecules.

Heme—The iron-containing molecule in hemoglobin that serves as the site for oxygen binding.

Hemoglobin—Protein-iron compound in the blood that carries oxygen to the cells and carries carbon dioxide away from the cells.

Hemoglobin A—Normal adult hemoglobin that contains a heme molecule, two alpha-globin molecules, and two beta-globin molecules.

Hemoglobin electrophoresis—A laboratory test that separates molecules based on their size, shape, or electrical charge.

Hepatomegaly—An abnormally large liver.

HLA type—Refers to the unique set of proteins called human leukocyte antigens. These proteins are present on each individual's cell and allow the immune system to recognize 'self' from 'foreign'. HLA type is particularly important in organ and tissue transplantation.

Hydroxyurea—A drug that has been shown to induce production of fetal hemoglobin. Fetal hemoglobin has a pair of gamma-globin molecules in place of the typical beta-globins of adult hemoglobin. Higher-than-normal levels of fetal hemoglobin can ameliorate some of the symptoms of thalassemia.

Iron overload—A side effect of frequent blood transfusions in which the body accumulates abnormally high levels of iron. Iron deposits can form in organs, particularly the heart, and cause life-threatening damage.

Jaundice—Yellowing of the skin or eyes due to excess of bilirubin in the blood.

Mutation—A permanent change in the genetic material that may alter a trait or characteristic of an individual, or manifest as disease, and can be transmitted to offspring.

Placenta—The organ responsible for oxygen and nutrition exchange between a pregnant mother and her developing baby.

Red blood cell—Hemoglobin-containing blood cells that transport oxygen from the lungs to tissues. In the tissues, the red blood cells exchange their oxygen for carbon dioxide, which is brought back to the lungs to be exhaled.

Screening—Process through which carriers of a trait may be identified within a population.

Splenomegaly—Enlargement of the spleen.

When severe anemia occurs, it is treated with blood transfusion therapy. For individuals with hemoglobin H disease, this is rarely required. For those with the hemoglobin H/Constant Spring form of the disease, the need for transfusions may be intermittent or ongoing, perhaps on a monthly basis and requiring desferoxamine treatment. Individuals with this more severe form of the disease may also have an increased

chance of requiring removal of an enlarged and/or overactive spleen.

Alpha thalassemia major

Because alpha thalassemia major is most often a condition that is fatal in the prenatal or newborn period, treatment has previously been focused on

identifying affected pregnancies in order to provide appropriate management to reduce potential maternal complications. Pregnancy termination provides one form of management. Increased prenatal surveillance and early treatment of maternal complications is an approach that is appropriate for mothers who wish to continue their pregnancy with the knowledge that the baby will most likely not survive. In recent years, there have been a handful of infants with this condition who have survived long-term. Most of these infants received experimental treatment including transfusions before birth, early delivery, and even bone marrow transplantation before birth, although the latter procedure has not yet been successful. For those infants that survive to delivery, there seems to be an increased risk of developmental problems and physical effects, particularly heart and genital malformations. Otherwise, their medical outlook is similar to a child with beta thalassemia major, with the important exception that ongoing, life-long blood transfusions begin right at birth.

Prognosis

As discussed above, the prognosis for individuals with the most serious types of thalassemia has improved drastically in the last several years following recent medical advances in transfusion, chemo-, and transplantation therapy. Advances continue and promise to improve the life expectancy and quality of life further for affected individuals.

Resources

PERIODICALS

"First Known Heart Attack Associated With Beta-thalassemia Major Reported." *Heart Disease Weekly* February 22, 2004: 10.

"Novel Alpha-thalassemia Mutations Identified." *Hematology Week* January 26, 2004: 19.

ORGANIZATIONS

Children's Blood Foundation. 333 East 38th St., Room 830, New York, NY 10016-2745. (212) 297-4336. cfc@nyh.med.cornell.edu.

Cooley's Anemia Foundation, Inc. 129-09 26th Ave. #203, Flushing, NY 11354. (800) 522-7222 or (718) 321-2873. <<http://www.thalassemia.org>>.

March of Dimes Birth Defects Foundation. 1275 Mamaroneck Ave., White Plains, NY 10605. (888) 663-4637. resourcecenter@modimes.org. <<http://www.modimes.org>>.

National Heart, Lung, and Blood Institute. PO Box 30105, Bethesda, MD 20824-0105. (301) 592-8573. nhlbiinfo@rover.nhlbi.nih.gov. <<http://www.nhlbi.nih.gov>>.

National Organization for Rare Disorders (NORD). PO Box 8923, New Fairfield, CT 06812-8923. (203) 746-6518 or (800) 999-6673. Fax: (203) 746-6481. <http://www.rarediseases.org>.

OTHER

Bojanowski J. "Alpha Thalassemia Major: The Possibility of Long-Term Survival." Pamphlet from the Northern California Comprehensive Thalassemia Center. (1999).

Children's Hospital Oakland, Northern California Comprehensive Thalassemia Center website. <<http://www.thalassemia.com>>.

Cooley's Anemia Foundation, Inc. website. <<http://www.thalassemia.org/gohome.html>>.

Joint Center for Sickle Cell and Thalassemic Disorders website. <<http://cancer.mgh.harvard.edu/medOnc/sickle.htm>>.

Jennifer Bojanowski, MS, CGC
Teresa G. Odle

Thallium heart scan

Definition

A thallium heart scan is a test using a special camera and a small amount of radioactive substance injected into the bloodstream to make an image of the blood flow to the heart.

Purpose

A thallium heart scan is used to evaluate the blood supply to the heart muscle. It can identify areas of the heart that may have a poor blood supply as a result of damage from a previous **heart attack** or blocked coronary arteries. While **exercise** testing has long been a standard examination in the diagnosis of **coronary artery disease**, in some cases, the thallium scan may be more sensitive and more specific in the information it provides. In other words, the test may be better able to detect a problem and to differentiate one condition from another. A thallium heart scan may more accurately detect ischemic heart disease. This type of scan is most likely to be helpful in cases in which the exercise test is inconclusive, the patient cannot exercise adequately, or a quantitative evaluation of blood flow is required. In addition to evaluating coronary artery disease, thallium scanning can help to evaluate blood flow following treatment of clogged arteries with **coronary artery bypass graft surgery** or **angioplasty**.

Precautions

Radioisotopes such as thallium 201 should not be administered during **pregnancy** because they may be harmful to the fetus.

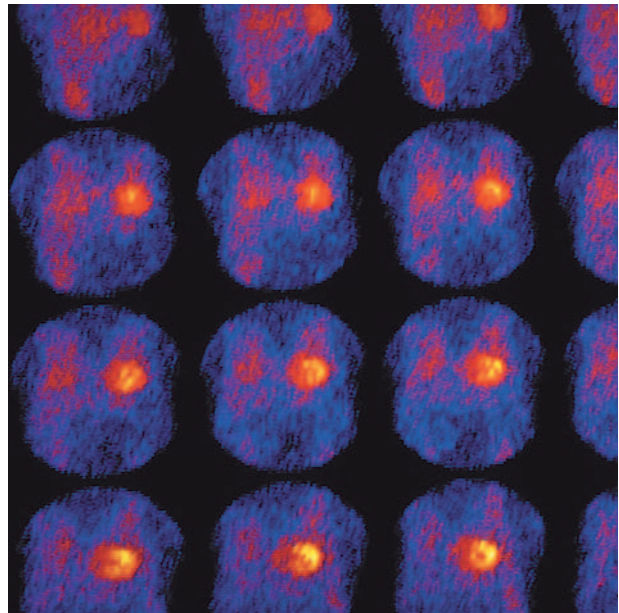
Description

The thallium scan is performed in conjunction with an exercise **stress test**. At the end of the **stress test** (once the patient has reached the highest level of exercise he or she can comfortably achieve), a small amount of the harmless radioisotope thallium 201 is injected into the patient's bloodstream through an intravenous (IV) line. The patient then lies down under a special camera called a gamma scintillation camera, which makes photographs from the gamma rays emitted by the thallium.

The thallium attaches itself to the red blood cells and is carried throughout the body in the bloodstream. It enters the heart muscle by way of the coronary arteries and collects in the cells of the heart muscle that come into contact with the blood. Since the thallium can reach only those areas of the heart with an adequate blood supply, no thallium will show up in poorly perfused areas of the heart (perfusion defects). These areas show up as “cold spots” on the thallium scan. The patient may then be given a second injection of thallium. Several hours later, the gamma scintillation camera takes more pictures in order to get an image of the heart when the patient is at rest.

Cold spots that appear at rest as well as during exercise often indicate areas where the heart tissue has been damaged (for example, as a result of a prior heart attack). Sometimes perfusion is adequate during rest but cold spots appear during exercise, when the heart has to work harder and has a greater demand for blood. This can indicate some blockage in the coronary arteries, producing a condition called **ischemia**. In ischemia, the heart temporarily does not get enough blood flow. People with perfusion defects, especially perfusion defects that appear only during exercise, have the greatest risk of such future cardiac events as heart attacks.

In recent years, there have been improvements in heart scanning. Many centers now use a single photon emission computed tomographic (SPECT) camera, which provides a clearer image. Some centers also use a type of radioactive chemical called sestamibi. Sestamibi is used along with a radioactive compound called technetium. While thallium may still be better for some uses, such as providing a



A thallium scan showing many images of a human heart with cold spots. (Custom Medical Stock Photo. Reproduced by permission.)

better image of the heart muscle itself, sestamibi may produce clearer images in overweight patients and is more useful in assessing how well the heart pumps blood.

If the patient is unable to exercise because of another medical condition, such as arthritis, he or she may be given a drug to mimic the effects of exercise on the heart. Some of these drugs include dipyridamole (Persantine), which dilates the coronary arteries; and dobutamine, which increases blood flow through the heart muscle.

Preparation

Patients should not drink alcoholic or caffeinated beverages, smoke tobacco, or ingest other nicotine products for 24 hours before the test. These substances can affect test results. Patients should also not eat anything for at least three hours before the test. They may also be instructed to stop taking certain medications during the test that may interfere with test results.

Aftercare

In some cases, another set of scans may be needed, and the patient may be given special instructions regarding eating and test preparation. Otherwise, the

KEY TERMS

Angioplasty—The reconstruction of damaged blood vessels.

Coronary bypass surgery—Surgery in which a section of blood vessel is used to bypass a blocked coronary artery and restore an adequate blood supply to the heart muscle.

Perfusion—The passage of fluid (such as blood) through a specific organ or area of the body (such as the heart).

Radioisotope—A radioactive form of a chemical element, which is used in medicine for therapeutic or diagnostic purposes.

patient is free to return to his or her normal daily activities.

Risks

Radioisotopes such as thallium 201 should not be administered during pregnancy because they may be harmful to the fetus.

Normal results

A normal thallium scan shows healthy blood flow through the coronary arteries and normal perfusion of the heart muscle, without cold spots, both at rest and during exercise.

Abnormal results

Cold spots on the scan, where no thallium shows up, indicate areas of the heart that are not getting an adequate supply of blood. Cold spots appearing both at rest and during exercise may indicate areas where the heart tissue has been damaged. However, “reversible” cold spots appearing only during exercise usually indicate some blockage of the coronary arteries.

Resources

ORGANIZATION

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-630 or (800) 242-8721. inquire@heart.org. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Robert Scott Dinsmoor

Thematic apperception test

Definition

The thematic apperception test (TAT) is a projective personality test that was designed at Harvard in the 1930s by Christiana D. Morgan and Henry A. Murray. Along with the MMPI and the Rorschach, the TAT is one of the most widely used **psychological tests**. A projective test is one in which a person's patterns of thought, attitudes, observational capacity, and emotional responses are evaluated on the basis of responses to ambiguous test materials. The TAT consists of 31 pictures that depict a variety of social and interpersonal situations. The subject is asked to tell a story about each picture to the examiner. Of the 31 pictures, 10 are gender-specific while 21 others can be used with adults of either sex and with children. As of 2001, the TAT is distributed by Harcourt Brace Educational Measurement.

Purpose

The original purpose of the TAT was to reveal the underlying dynamics of the subject's personality, such as internal conflicts, dominant drives and interests, motives, etc. The specific motives that the TAT assesses include the need for achievement, need for power, the need for intimacy, and problem-solving abilities. After World War II, however, the TAT was used by psychoanalysts and clinicians from other schools of thought to evaluate emotionally disturbed patients. Another shift took place in the 1970s, when the influence of the human potential movement led many psychologists to emphasize the usefulness of the TAT in assessment services—that is, using the test to help clients understand themselves better and stimulate their personal growth.

The TAT is widely used to research certain topics in psychology, such as dreams and fantasies, mate selection, the factors that motivate people's choice of occupations, and similar subjects. It is sometimes used in psychiatric evaluations to assess disordered thinking and in forensic examinations to evaluate crime suspects, even though it is not a diagnostic test. As mentioned earlier, the TAT can be used to help people understand their own personality in greater depth and build on that knowledge in making important life decisions. Lastly, it is sometimes used as a screener in psychological evaluations of candidates for high-stress occupations (law enforcement, the military, religious ministry, etc.).

KEY TERMS

Apperception—The process of understanding through linkage with previous experience.

Human potential movement—A movement in psychotherapy that began in the 1960s and emphasized maximizing the potential of each participant through such techniques as group therapy and sensitivity training.

Projective test—A type of psychological test that assesses a person's thinking patterns, observational ability, feelings, and attitudes on the basis of responses to ambiguous test materials. It is not intended to diagnose psychiatric disorders.

Precautions

The TAT has been criticized for its lack of a standardized method of administration as well as the lack of standard norms for interpretation. Studies of the interactions between examiners and test subjects have found that the race, sex, and social class of both participants influence both the stories that are told and the way the stories are interpreted by the examiner. Attempts have been made to design sets of TAT cards for African American and for elderly test subjects, but the results have not been encouraging. In addition, the 31 standard pictures have been criticized for being too gloomy or depressing, and therefore limiting the range of personality characteristics that the test can assess.

Description

There is no standardized procedure or set of cards for administering the TAT, except that it is a one-on-one test. It cannot be administered to groups. In one common method of administration, the examiner shows the subject only 10 of the 31 cards at each of two sessions. The sessions are not timed, but average about an hour in length.

Preparation

There is no specific preparation necessary before taking the TAT, although most examiners prefer to schedule sessions (if there is more than one) over two days.

Risks

The chief risks involved in taking the TAT are a bad “fit” between the examiner and the test subject, and misuse of the results.

Normal results

Since the TAT is used primarily for personality assessment rather than diagnosis of mental disorders, it does not yield a “score” in the usual sense.

Resources

BOOKS

Geiser, Lon, and Morris I. Stein. *Evocative Images: The Thematic Apperception Test and the Art of Projection*. Washington, DC: American Psychological Association, 1999.

Rebecca J. Frey, PhD

Therapeutic abortion see **Abortion, therapeutic**

Therapeutic baths

Definition

Bathing the skin in a variety of preparations in order to remove crusts, scales, and old medications or to relieve inflammation and **itching** is called a therapeutic bath.

Purpose

Baths or soaks (balneotherapy) are an easy way to treat a variety of skin disorders involving large areas of the skin. They relieve general aches and pains and can ease dry or oily, inflamed or itchy skin. Hot baths are relaxing and stimulating; cool baths can reduce inflammation.

Therapeutic baths are useful for itchy skin, **hives**, **sunburn**, chafing, poison ivy and oak, eczema, skin irritation, and dry skin. They may also help to relieve emotional tension and **stress**. Brief therapeutic baths may be useful in preventing pressure ulcers and other skin problems in the elderly.

Many family care physicians recommend warm-water therapeutic baths as a way to relieve labor pains during **childbirth** without administering drugs.

Precautions

The temperature of the water should be comfortable. The bath should not last longer than 20–30 minutes because of the tendency of these soaks to soften and wear away the skin.

KEY TERMS

Eczema—An inflammation of the skin that usually itches and sometimes forms scales or blisters.

A bath mat should be used, since medications may cause the floor of the tub to be slippery.

Eczema and other skin diseases can be treated with an ointment that contains a derivative of coal tar. Parts of the coal tar are volatile, so the bathroom should be well ventilated.

Description

The tub should be filled half–full with water at a comfortable temperature. The water should not be allowed to cool too much. If an emollient action is needed, the patient should apply a lubricating agent to the skin after the bath, since this increases hydration.

Different types of therapeutic baths are used for different conditions:

- colloidal oatmeal (oatmeal that has been ground into a fine powder, e.g. Aveeno) coats, soothes, stops itch and doesn't dry out the skin
- potassium permanganate—a dark purple salt—makes a good disinfectant
- bath oils are used as an emollient to ease itchy skin and eczema
- cornstarch is a soothing, drying bath for itchy skin
- sodium bicarbonate can be cooling for hot, dry skin conditions
- saline (salt) water baths are used to treat lesions scattered over the body

Preparation

Keep the room warm to minimize temperature fluctuations. This precaution is particularly important when bathing elderly patients.

Aftercare

After the bath, the skin should be blotted (not rubbed) carefully with a towel. The patient should wear loose, light clothing after the bath.

Resources

BOOKS

Pelletier, Dr. Kenneth R. *The Best Alternative Medicine, Part I: Naturopathic Medicine*. New York: Simon and Schuster, 2002.

PERIODICALS

Hunter, S., J. Anderson, D. Hanson, et al. "Clinical Trial of a Prevention and Treatment Protocol for Skin Breakdown in Two Nursing Homes." *Journal of Wound, Ostomy, and Continence Nursing* 30 (September 2003): 250–258.

Keegan, L. "Therapies to Reduce Stress and Anxiety." *Critical Care Nursing Clinics of North America* 15 (September 2003): 321–327.

Leeman, L., P. Fontaine, V. King, et al. "The Nature and Management of Labor Pain: Part I. Nonpharmacologic Pain Relief." *American Family Physician* 68 (September 15, 2003): 1109–1112.

ORGANIZATIONS

The American Association of Naturopathic Physicians. 8201 Greensboro Drive, Suite 300, McLean, Virginia 22102. (206) 298-0126. <<http://naturopathic.org>>.

Canadian Naturopathic Association/Association canadienne de naturopathie. 1255 Sheppard Avenue East at Leslie, North York, ON M2K 1E2. (800) 551-4381 or (416) 496-8633. <<http://www.naturopathicassoc.ca>>.

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Therapeutic drug monitoring see **Drug therapy monitoring**

Therapeutic massage see **Massage**

Therapeutic touch

Definition

Therapeutic touch, or TT, is a noninvasive method of healing that was derived from an ancient laying-on of hands technique. In TT, the practitioner alters the patient's energy field through an energy transfer that moves from the hands of the practitioner to the patient.

Origins

Therapeutic touch was developed in 1972 by Dora Kunz, a psychic healer, and **Dolores Krieger**, Ph.D., R.N, a nurse and professor of nursing at New York University. The year before, in 1971, when Krieger was working as a registered nurse in a hospital, she

became very frustrated when one of her patients, a 30-year-old female, lay dying from a gallbladder condition. In desperation, she tried what she was learning from Kunz. Within one treatment, the patient's condition began to shift and she lived, surprising the other hospital staff. Krieger and Kunz met during the study of Oskar Estebany, a world-renowned healer. They had invited Estebany to form a study for three years, observing his work with patients. In this study, Estebany practiced laying-on of hands healing on various patients. Using her psychic and intuitive abilities, Kunz would observe and assist in the healing, while Krieger recorded the activities of the healing session and created profiles of the patients.

As the study progressed, Kunz began teaching Krieger how to heal, based on her perceptions of Estebany's healing techniques. During her research of ancient healing methods, Krieger concluded that the energy transfer between the healer and the healee that takes place in a TT session is *prana*, an Eastern Indian concept representing energy, vitality, and vigor. Krieger then combined her research with Kunz's techniques to create TT.

TT was initially developed for persons in the health professions, but is currently taught worldwide to anyone who is interested in learning the technique. As of 1998, an estimated 100,000 people around the world have been trained in TT; 43,000 of those persons are health care professionals, many of whom use TT in conjunction with traditional medicine, as well as osteopathic, **chiropractic**, naturopathic, and homeopathic therapies. TT is taught in over 100 colleges, universities, and medical schools.

Benefits

The major effects of TT are relaxation, **pain** reduction, accelerated healing, and alleviation of psychosomatic symptoms. Studies have shown that TT has a beneficial effect on the blood as it has the ability to raise hemoglobin values. It also affects brain waves to induce a relaxed state. TT can induce the relaxation response often within five minutes.

Krieger has said that it is not individual illnesses that validate the effectiveness of TT, but rather, it is questioned which systems are most sensitive to TT. She and others have found that the most sensitive is the autonomic nervous system (ANS), which, for example, controls urination. The ANS is followed by dysfunctions of lymphatic and circulatory systems, and then finally musculoskeletal systems. In addition,

DOLORES KRIEGER (1935–)

Dolores Krieger, a prominent professor of nursing at the New York University Division of Nursing, conceived of therapeutic touch as a healing technique in the early 1970s and introduced the therapy in 1972. Therapeutic touch rarely consists of physical contact with the patient. The practitioner focuses positive energy through the hands, which are held or waved two to three inches away from the patient, and directs it towards the patient's energy field. Krieger developed the technique along with a colleague, Dora Van Gelder Kunz, who is believed to be clairvoyant. They initially taught the system to graduate students at the nursing school, and it evolved from that basis. Since the introduction of therapeutic touch, Krieger traveled the world in teaching the technique before she retired as professor emerita at the university. An estimated 70,000 nurses were trained by Krieger and Kunz.

In 1981 Dr. Krieger published *Foundations for Holistic Health Nursing Practices*. She later published a manual, *The Therapeutic Touch: How to Use Your Hands to Help or to Heal*, in 1992.

Krieger became embroiled in controversy over the potential benefits of therapeutic touch technique between 1996-98, when nine-year-old schoolgirl Emily Rosa challenged the validity of the therapy with a simple experiment. She gathered 21 practitioners and through a covered box held her hand over one of the practitioner's own to test whether they could sense her energy field. Only 44% of the time were the practitioners able to determine which of their hands that Rosa's was hovering over. Although Rosa contacted Krieger in 1997, Krieger refused to meet with her, refused to participate in Rosa's experiment, and disputed the relevancy of an elementary school student's observations. Krieger holds both an R.N. and a Ph.D. degree and dismissed the validity of the experiment due to the student's and practitioners' lack of experience.

Krieger continues to promote her technique; her latest book, *Living the Therapeutic Touch*, was published by in 1999.

the female endocrine system is more sensitive to TT than the corresponding male system. Thus, TT helps with **dysmenorrhea**, **amenorrhea**, problems with **contraception**, and the course of **pregnancy**.

TT is reported to have a positive effect on the immune system and thus accelerates the healing of **wounds**. Nurses use therapeutic touch in operating rooms to relax patients before surgery and in recovery rooms on postoperative patients to help speed the

healing process. TT is used in the treatment of terminally ill patients, such as those with **cancer** and autoimmune deficiency syndrome (**AIDS**), to relieve **anxiety** and **stress**, create peace of mind, and reduce pain.

Many nurses use TT in the nursery. The conditions of many premature babies who received TT reportedly improved rapidly. TT has been used to calm colicky infants, assist women in **childbirth**, and increase milk let-down in breast-feeding mothers.

Other claims of TT include relief of acute pain, **nausea**, **diarrhea**, tension and migraine headaches, **fever**, and joint and tissue swelling. TT has been used to treat thyroid imbalances, ulcers, psychosomatic illnesses, **premenstrual syndrome**, **Alzheimer's disease**, **stroke** and **coma**, **multiple sclerosis**, **measles**, infections, **asthma**, and bone and muscle injuries.

Therapeutic touch is performed in many different locations, including healing centers, delivery rooms, hospitals, hospice settings, accident scenes, homes, and schools.

Description

Therapeutic touch treats the whole person: relaxes the mind, heals the body, and soothes the spirit. The principle behind it is that it does not stop at the skin. The human body extends an energy field, or aura, several inches to several feet from the body. When illness occurs, it creates a disturbance or blockage in the vital energy field. The TT practitioner uses her/his hands to sense the blockage or disturbance. In a series of gentle strokes, the healer removes the disturbance and rebalances the energy to restore health.

The TT session generally lasts about 20–30 minutes. Although the name is therapeutic touch, there is generally no touching of the physical body, only the energetic body or field. It is usually performed on fully clothed patients who are either lying down on a flat surface or sitting up in a chair.

Each session consists of five steps. Before the session begins, the practitioner enters a state of quiet **meditation** in which he/she becomes centered and grounded in order to establish intent for the healing session and to garner the compassion necessary to heal.

The second step involves the assessment of the person's vital energy field. During this step, the practitioner places the palms of his/her hands 2–3 in (5–8 cm) from the patient's body and sweeps them over the energy field in slow, gentle strokes beginning at the head and moving toward the feet. The practitioner

might feel heat, coolness, heaviness, pressure, or a prickly or **tingling** sensation. These cues, as they are called, each signal a blockage or disturbance in the field.

To remove these blockages and restore balance to the body, the practitioner then performs a series of downward sweeping movements to clear away any energy congestion and smooth the energy field. This is known as the unruffling process and is generally performed from head to feet. To prevent any energy from clinging to him/her, the practitioner shakes his/her hands after each stroke.

During the next phase, the practitioner acts as a conduit to transfer energy to the patient. The energy used is not solely the energy of the practitioner. The practitioner relies on a universal source of energy so as not to deplete his/her own supply. In short, the healer acts as an energy support system until the patient's immune system is able to take over.

The practitioner then smoothes the field to balance the energy and create a symmetrical flow. When the session is over, it is recommended that the patient relax for 10–15 minutes in order for the energies to stabilize.

Side effects

The side effects reported occur when an excess of energy enters the body for an extended period of time creating restlessness, irritability, and hostility, or increasing anxiety and pain. **Burns** are sensitive to therapeutic touch, and it is recommended that TT be performed on burned tissue for short periods, generally two to three minutes at a time.

Research and general acceptance

Therapeutic touch is not generally accepted by Western medical professionals. Basic and anecdotal research has been performed on TT since its development in 1972, although little quantitative research has been carried out. It is based on a theory derived from formal research. It began as the basis of Dolores Krieger's postdoctoral research.

Dolores Krieger has performed extensive research on TT, including with pregnant women, and has noted that the following changes occur in a patient after short, consistent treatment: relaxation within the first five minutes of a session, a reduction of pain, and the acceleration of the healing process.

One study was created to determine the effect TT would have on wounds that resulted from a biopsy of the upper arm. Forty-four patients placed their injured arms through a hole in a door. Twenty-two of them received TT on their arms. The other half received no treatment. The wounds treated with TT healed more quickly than the wounds that received no treatment.

In 1998, a study was performed on 27 patients with **osteoarthritis** in at least one knee. For six weeks, the patients were treated with therapeutic touch, mock therapeutic touch, or standard care. According to *The Journal of Family Practice*, the journal who published the study, the results showed that the group who had received TT had “significantly decreased pain and improved function as compared with both the placebo and control groups.”

Therapeutic touch can be combined with a number of different therapies, including **acupressure**, massage, mental imagery, physical therapy, and **yoga**. When combined with massage and physiotherapy, TT may reduce tension headaches, back pain, stress-related problems, circulatory problems, and **constipation**. **Shiatsu** and TT may help **sinusitis**, digestive disorders, **muscle cramps**, menstrual difficulties, and **insomnia**. Yoga and TT may be beneficial in the treatment of **bronchitis**, asthma, blood pressure, **fatigue**, and anxiety.

TT is practiced in over 70 countries worldwide: by Egyptians and Israelis during fighting in the Gaza Strip; in South Africa to reduce racial strife; and in Poland, Thailand, and the former Soviet Union.

Training and certification

Therapeutic touch is taught at over 100 universities and nursing and medical schools around the United States and Canada. Although it was developed primarily for nurses, anyone can learn TT.

State laws vary regarding the practice of TT. In general, laypersons are allowed to practice TT within their families. Therapeutic touch is considered an extension of health care skills, so most health care professionals are covered under the state medical practice act.

Many hospitals have established policies allowing nurses and staff to perform TT on patients at no extra charge. The American Nurse's Association often holds workshops on TT at national conventions. Therapeutic touch classes are often held for the general public through community education, healing clinics, and holistic schools.

Resources

OTHER

The Nurse Healers Professional Associates International (NH-PAI), the Official Organization of Therapeutic Touch. 3760 S. Highland Dr. Salt Lake City, UT 84106. (801) 273-3399. nhpai@therapeutic-touch.org. <<http://www.therapeutic-touch.org>>.

Jennifer Wurges

Thiabendazole see **Antihelminthic drugs**

Thiamine deficiency see **Beriberi**

Thoracentesis

Definition

Also known as pleural fluid analysis, thoracentesis is a procedure that removes fluid or air from the chest through a needle or tube.

Purpose

The lungs are lined on the outside with two thin layers of tissue called pleura. The space between these two layers is called the pleural space. Normally, there is only a small amount of lubricating fluid in this space. Liquid and/or air accumulates in this space between the lungs and the ribs from many conditions. The liquid is called a **pleural effusion**; the air is called a **pneumothorax**. Most pleural effusions are complications emanating from metastatic malignancy (movement of **cancer** cells from one part of the body to another). Most malignant pleural effusions are detected and controlled by thoracentesis. Thoracentesis is also performed as a diagnostic measure. In these cases, only small amounts of material need to be withdrawn.

Symptoms of a pleural effusion include breathing difficulty, chest **pain**, **fever**, weight loss, **cough**, and **edema**. Removal of air is often an emergency procedure to prevent suffocation from pressure on the lungs. Negative air pressure within the chest cavity allows normal respiration. The accumulation of air or fluid within the pleural space can eliminate these normal conditions and disrupt breathing and the movement of air within the chest cavity. Fluid removal is performed to reduce the pressure in the pleural space and to analyze the liquid. In addition, thoracentesis

was traditionally used to remove blood from the chest cavity. This is rare now that the placement of a thoracostomy tube has proven to be a more effective and safer method.

Thoracentesis often provides immediate abatement of symptoms. However, fluid often begins to reaccumulate. A majority of patients will ultimately require additional therapy beyond a simple thoracentesis.

There are two types of liquid in the pleural space, one having more protein in it than the other. More watery liquids are called transudates; thicker fluids are called exudates. On the basis of this difference, the cause of the effusion can more easily be determined.

Transudates

Thin, watery fluid oozes into the chest either because back pressure from circulation squeezes it out or because the blood has lost some of its osmotic pressure.

- Heart failure creates back pressure in the veins as blood must wait to be pumped through the heart.
- A **pulmonary embolism** is a blood clot in the lung. It will create back pressure in the blood flow and also damage a part of the lung so that it leaks fluid.
- Cirrhosis is a sick, scarred liver that both fails to make enough protein for the blood and also restricts the flow of blood through it.
- Nephrosis is a collection of kidney disorders that change the osmotic pressure of blood and allow liquid to seep into body cavities.
- Myxedema is a disease caused by too little thyroid hormone.

Exudates

Thicker, more viscous fluid is usually due to greater damage to tissues, allowing blood proteins as well as water to seep out.

- Pneumonia, caused by viruses and by bacteria, damages lung tissue and can open the way for exudates to enter the pleural space.
- Tuberculosis can infect the pleura as well as the lungs and cause them to leak liquid.
- Cancers of many types settle in the lungs or the pleura and leak liquids from their surface.
- Depending upon its size and the amount of damage it has done, a pulmonary **embolism** can also produce an exudate.

- Several drugs can damage the lung linings as an unexpected side effect. None of these drugs is commonly used.
- An esophagus perforated by cancer, trauma, or other conditions can spill liquids and even food into the chest. The irritation creates an exudate in the pleural space.
- Pancreatic disease can cause massive fluid in the abdomen, which can then find its way into the chest.
- Pericarditis is an inflammation of the sac that contains the heart. It can ooze fluid from both sides—into the heart's space and into the chest.
- Radiation to treat cancer or from accidents with radioactive materials can damage the pleura and lead to exudates.
- A wide variety of autoimmune diseases attacks the pleura. Among these are **rheumatoid arthritis** and **systemic lupus erythematosus** (SLE).
- Many other rare conditions can also lead to exudates.

Blood

Blood in the chest (hemothorax) is infrequently seen outside of two conditions:

- major trauma can sever blood vessels in the chest, causing them to bleed into the pleural space
- cancers can ooze blood as well as fluid, they do not usually bleed massively

Chyle

Occasionally, the liquid that comes out of the chest is neither transparent nor bloody, but milky. This is due to a tear of the large lymphatic channel—the thoracic duct carrying lymph fluid from the intestines to the heart. It is milky because it is transporting fats absorbed in the process of digestion. The major causes of chylothorax are:

- injury from major trauma, such as an automobile accident
- cancers eroding into the thoracic duct

Air

Air in the pleural space is called pneumothorax. Air can enter the pleural space either directly through a hole between the ribs or from a hole in the lungs. Holes in the lungs are sometimes spontaneous, sometimes traumatic, and sometimes the result of disease opening a communication to the air in the lung.

Precautions

Care must be taken not to puncture the lung when inserting the needle. Thoracentesis should never be performed by inserting the needle through an area with an infection. An alternative site needs to be found in these cases. Patients who are on **anticoagulant drugs** should be carefully considered for the procedure.

Description

The usual place to tap the chest is below the armpit (axilla). Under sterile conditions and **local anesthesia**, a needle, a through-the-needle-catheter, or an over-the-needle catheter may be used to perform the procedure. Overall, the catheter techniques may be safer. Fluid or air is withdrawn. Fluid is sent to the laboratory for analysis. If the air or fluid continue to accumulate, a tube is left in place and attached to a one-way system so that it can drain without sucking air into the chest.

Preparation

The location of the fluid is pinpointed through x ray or ultrasound. Ultrasound is a more accurate method when the effusion is small. A sedative may be administered in some cases but is generally not recommended. Oxygen should be given to the patient.

Aftercare

As long as the tube is in the chest, the patient must lie still. After it is removed, x rays will determine if the effusion or air is reaccumulating—though some researchers and clinicians believe chest x rays do not need to be performed after routine thoracentesis.

Risks

Reaccumulation of fluid or air is a possible complication, as are hypovolemic **shock** (shock caused by a lack of circulating blood) and infection. Patients are at increased risk for poor outcomes if they have a recent history of anticoagulant use, have very small effusions, have significant amounts of fluid, have poor health leading into this condition, have positive airway pressure, and have **adhesions** in the pleural space. A pneumothorax can sometimes be caused by the thoracentesis procedure. The use of ultrasound to guide the procedure can reduce the risk of pneumothorax.

Thoracentesis can also result in hemothorax, or bleeding within the thorax. In addition, such internal

KEY TERMS

Axilla—Armpit.

Catheter—A tube that is moved through the body for removing or injecting fluids into body cavities.

Hypovolemic shock—Shock caused by a lack of circulating blood.

Osmotic pressure—The pressure in a liquid exerted by chemicals dissolved in it. It forces a balancing of water in proportion to the amount of dissolved chemicals in two compartments separated by a semi-permeable membrane.

Pleura—Two thin layers lining the lungs on the outside.

structures as the diaphragm, spleen, or liver, can be damaged by needle insertion. Repeat thoracenteses can increase the risk of developing hypoproteinemia (a decrease in the amount of protein in the blood).

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Mark A. Mitchell, M.D.

Thoracic aortic aneurysm see **Aortic aneurysm**

Thoracic empyema see **Empyema**

Thoracic outlet syndrome

Definition

Thoracic outlet syndromes are a group of disorders that cause **pain** and abnormal nerve sensations in the neck, shoulder, arm, and/or hand.

Description

The thoracic outlet is an area at the top of the rib cage, between the neck and the chest. Several anatomical structures pass through this area, including the esophagus, trachea, and nerves and blood vessels that lead to the arm and neck region. The area contains the first rib; collar bone (clavicle); the arteries beneath the collar bone (subclavian artery), which supply blood to the arms; a network of nerves leading to the arms (brachial plexus); and the top of the lungs.

Pain and other symptoms occur when the nerves or blood vessels in this area are compressed. The likelihood of blood vessels or nerves in the thoracic outlet being compressed increases with increased size of body tissues in this area or with decreased size of the thoracic outlet. The pain of thoracic outlet syndrome is sometimes confused with the pain of **angina** that indicates heart problems. The two conditions can be distinguished from each other because the pain of thoracic outlet syndrome does not appear or increase when walking, while the pain of angina does. Also, the pain of thoracic outlet syndrome usually increases if the affected arm is raised, which does not happen in cases of angina.

There are three types of thoracic outlet syndromes:

- True neurogenic thoracic outlet syndrome is caused by a compression of the nerves in the brachial plexus. Abnormal muscle or other tissue causes the problem.
- Arterial thoracic outlet syndrome is caused by compression of the major artery leading to the arm, usually by a rib.
- Disputed thoracic outlet syndrome describes patients who have chronic pain in the shoulders and arms and have no other disease or syndrome, but the underlying cause cannot be accurately determined.

Thoracic outlet syndrome is most common in women who are 35 to 55 years of age.

Causes and symptoms

Compression of blood vessels or nerves in the thoracic outlet causes pain and/or abnormal nerve sensations. Compression usually occurs at the location where the blood vessels and nerves pass out of the thoracic outlet into the arm.

There are several factors that contribute to a person developing thoracic outlet syndrome. Poor posture is a major cause and is easy to treat. A person's physical makeup also can cause thoracic outlet syndrome. For example, abnormalities of certain anatomical structures

can put pressure on blood vessels or nerves. Typical abnormalities that can cause problems are malformed ribs and too narrow an opening between the collar bone and the first rib.

The main symptom is pain in the affected area. The patient can also develop weakness in the arm and hands, **tingling** nerve sensations, and a condition called Raynaud's syndrome. In Raynaud's syndrome, exposure to cold causes small arteries in the fingers to contract, cutting off blood flow. This causes the fingers to turn pale. In very severe cases of blood vessel compression, **gangrene** can result. Gangrene is the death of tissue caused by the blood supply being completely cut off.

In the case of arterial thoracic outlet syndrome, the artery beneath the collar bone leading to the arm is compressed, causing the artery to increase in size. **Blood clots** (thrombi) may form in the blood vessel. When blood vessels are compressed, the hands, arms, and shoulders do not receive proper blood supply. They can swell and turn blue from a lack of blood.

In the case of true neurogenic thoracic outlet syndrome, the nerves most affected are those of the network of nerves supplying the chest, shoulder, arm, forearm, and hand (brachial plexus). When a nerve is affected in thoracic outlet syndrome it produces a tingling sensation (paresthesia). It can also cause weakness in the hand and reduced sensation in the palm and fingers.

Diagnosis

There are no specific diagnostic tests for thoracic outlet syndromes. The diagnosis is made by ruling out other diseases and by observing the patient. Two non-specific tests that can suggest the presence of thoracic outlet syndrome are the Adson test and the Allen test. In the Adson test, the patient takes a deep breath and tilts his or her head back and turns it to one side. The physician tests to see if the strength of the patient's pulse is reduced in the wrist on the arm on the opposite side of the head turn. In the Allen test, the arm in which the patient is experiencing symptoms is raised and rotated while the head is turned to the opposite side. The physician tests to see if the pulse strength at the wrist is reduced. If the strength of the pulse is reduced in either of these two tests it indicates compression of the subclavian artery.

Occasionally, examination with a stethoscope may reveal abnormal sounds in affected blood vessels. X rays can reveal constrictions in blood vessels if a special dye is injected into the blood stream to make the blood vessels visible (**angiography**).

KEY TERMS

Angina—A severe constricting pain in the chest, usually caused by a lack of oxygen to the heart.

Neurogenic—Caused by nerves; originating in the nerves.

Subclavian—Located beneath the collarbone (clavicle).

Certain tests are available to help with the diagnosis of nerve compression. These include the nerve conduction velocity test and somatosensory evoked potential test. In the nerve conduction velocity test, electrodes are placed at various locations on the skin along a nerve that is being tested. A mild electrical impulse is delivered through an electrode at one end of the nerve and the electrical activity is recorded by the other electrodes. The time it takes for the electrical impulse to travel down the nerve from the stimulating electrodes to the recording electrodes is used to calculate the nerve conduction velocity. This can be used to determine if any nerve damage exists.

In a somatosensory evoked potential test, electrodes are placed on the skin at the scalp, neck, shoulder, and wrist. A mild electrical impulse is delivered at the wrist, and a recording is made of the response by the brain and spinal cord. This test also can determine the presence of nerve damage.

Treatment

The main treatment for thoracic outlet syndrome is physical therapy. Exercises aimed at improving the posture of the affected person are also useful. In some cases, surgery can be performed to remove the cervical rib if this is causing the problem and physical therapy has failed to work. However, surgery is generally not used to treat thoracic outlet syndrome.

Prognosis

Treatment of true neurogenic and arterial thoracic outlet syndromes is usually successful. Treatment of disputed thoracic outlet syndrome is often unsuccessful. This may relate to the uncertainty of the underlying cause of the pain.

Resources

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Thoracic surgery

Definition

Thoracic surgery is the repair of organs located in the thorax, or chest. The thoracic cavity lies between the neck and the diaphragm, and contains the heart and lungs (cardiopulmonary system), the esophagus, trachea, pleura, mediastinum, chest wall, and diaphragm.

Purpose

Thoracic surgery repairs diseased or injured organs and tissues in the thoracic cavity. General thoracic surgery deals specifically with disorders of the lungs and esophagus. Cardiothoracic surgery also encompasses disorders of the heart and pericardium. Blunt chest trauma, reflux esophagitis, **esophageal cancer**, **lung transplantation**, **lung cancer**, and **emphysema** are just a few of the many clinical indications for thoracic surgery.

Precautions

Patients who have blood-clotting problems (coagulopathies), and who have had previous standard thoracic surgery may not be good candidates for video-assisted thoracic surgery (VATS). Because VATS requires the collapse of one lung, potential patients should have adequate respiratory function to maintain oxygenation during the procedure.

Description

Thoracic surgery is usually performed by a surgeon who specializes in either general thoracic surgery or cardiothoracic surgery. The patient is placed under **general anesthesia** and endotracheally intubated for the procedure. The procedure followed varies according to the purpose of the surgery. An incision that opens the chest (thoracotomy) is frequently performed

to give the surgeon access to the thoracic cavity. Commonly, the incision is made beginning on the back under the shoulder blade and extends in a curved arc under the arm to the front of the chest. The muscles are cut, and the ribs are spread with a retractor. The surgeon may also choose to open the chest through an incision down the breastbone, or sternum (sternotomy). Once the repair, replacement, or removal of the organ being operated on is complete, a chest tube is inserted between the ribs to drain the wound and re-expand the lung.

Video-assisted thoracic surgery (VATS) is a minimally invasive surgical technique that uses a thoracic endoscope (thoracoscope) to allow the surgeon to view the chest cavity. A lung is collapsed and 3-4 small incisions, or access ports, are made to facilitate insertion of the thoracoscope and the surgical instruments. During the procedure, the surgeon views the inside of the pleural space on a video monitor. The thoracoscope may be extracted and inserted through a different incision site as needed. When the surgical procedure is complete, the surgeon expands the lung and inserts a chest tube in one of the incision sites. The remaining incisions are sealed with adhesive.

The thoracic surgeon may also use a mediastinoscope or a bronchoscope to explore the thoracic cavity. **Mediastinoscopy** allows visualization of the mediastinum, the cavity located between the lungs. The bronchoscope enables the surgeon to view the larynx, trachea, and bronchi. These instruments may be used in a separate diagnostic procedure prior to thoracic surgery, or during the surgery itself.

Preparation

Except in the case of emergency procedures, candidates for general thoracic surgery should undergo a complete medical history and thorough **physical examination** prior to surgery. Particular attention is given to the respiratory system. The patient's **smoking** history will be questioned. If the patient is an active smoker, encouragement is always given for the patient to quit smoking prior to the surgery to facilitate recovery and reduce chances of complications.

Diagnostic tests used to evaluate the patient preoperatively may include, but are not limited to, x rays, MRI, CT scans, **blood gas analysis**, pulmonary function tests, **electrocardiography**, endoscopy, pulmonary **angiography**, and **sputum culture**.

Candidates for thoracic surgery should be fully educated by their physician or surgeon on what their surgery will involve, the possible risks and complications, and requirements for postoperative care.

Patients are instructed not to eat 10 to 12 hours prior to a thoracic surgery procedure. A sedative may be provided to relax the patient prior to surgery. An intravenous line (IV) is inserted into the patient's arm or neck to administer fluids and/or medication.

Aftercare

After surgery, the patient is taken to the recovery room, where vital signs are monitored; depending on the procedure performed, the breathing tube may be removed. The patient typically experiences moderate to severe **pain** following surgery. **Analgesics** or other pain medication are administered to keep the patient comfortable. Chest tubes are monitored closely for signs of fluid or air accumulation in the lungs that can lead to lung collapse. A urinary catheter will remain in the patient for 24 to 48 hours to drain urine from the bladder.

The hospital stay for thoracic surgery depends on the specific procedure performed. Patients who undergo a thoracotomy may be hospitalized a week or longer, while patients undergoing VATS typically have a shorter hospital stay of 2-3 days. During the recovery period, respiratory therapists and nurses work with the patient on deep breathing and coughing exercises to improve lung function.

Risks

Respiratory failure, hemorrhage, nerve injury, **heart attack**, **stroke**, **embolism**, and infection are all possible complications of general thoracic surgery. The chest tubes used for drainage after thoracic surgery may cause a build-up of fluid or the accumulation of air in the pleural space. Both of these conditions can lead to total lung collapse. Other specific complications may occur, depending on the procedure performed.

Normal results

Normal results of thoracic surgery are dependent on the type of procedure performed and the clinical purpose of the surgery.

KEY TERMS

Blood gas analysis—A blood test that measures the level of oxygen, carbon dioxide, and pH in arterial blood. A blood gas analysis can help a physician assess how well the lungs are functioning.

Electrocardiography—A cardiac test that measures the electrical activity of the heart.

Embolism—A blood clot, air bubble, or clot of foreign material that blocks the flow of blood in an artery. When blood supply to a tissue or organ is blocked by an embolism, infarction, or death of the tissue that the artery feeds, occurs. Without immediate and appropriate treatment, an embolism can be fatal.

Emphysema—A lung disease characterized by shortness of breath and a chronic cough. Emphysema is caused by the progressive stretching and rupture of alveoli, the air sacs in the lung that oxygenate the blood.

Endoscopy—The examination of organs and body cavities using a long, tubular optical instrument called an endoscope.

Intubation—Insertion of an endotracheal tube down the throat to facilitate airflow to the lung(s) during thoracic surgery.

Pericardium—The sac around the heart.

Pleural space—The space between the pleural membranes that surround the lungs and the chest cavity.

Pulmonary angiography—An x-ray study of the lungs, performed by insertion of a catheter into a vein, through the heart, and into the pulmonary artery. Pulmonary angiography is performed to evaluate blood circulation to the lungs. It is also considered the most accurate diagnostic test for detecting a pulmonary embolism.

Sputum culture—A laboratory analysis of the fluid produced from the lungs during coughing. A sputum culture can confirm the presence of pathogens in the respiratory system, and help to diagnose certain respiratory infections, including bronchitis, tuberculosis, and pneumonia.

Resources

ORGANIZATIONS

American Thoracic Society. 1740 Broadway, New York, NY 10019. (212) 315-8700. <<http://www.thoracic.org>>.

Paula Anne Ford-Martin

Thoracoscopy

Definition

Thoracoscopy is the insertion of an endoscope, a narrow-diameter tube with a viewing mirror or camera attachment, through a very small incision (cut) in the chest wall.

Purpose

Thoracoscopy makes it possible for a physician to examine the lungs or other structures in the chest cavity, without making a large incision. It is an alternative to thoracotomy (opening the chest cavity with a large incision). Many surgical procedures, especially taking tissue samples (biopsies), can also be accomplished with thoracoscopy. The procedure is done to:

- assess lung **cancer**
- take a biopsy for study
- determine the cause of fluid in the chest cavity
- introduce medications or other treatments directly into the lungs
- treat accumulated fluid, pus (**empyema**), or blood in the space around the lungs

For many patients, thoracoscopy replaces thoracotomy. It avoids many of the complications of open chest surgery and reduces **pain**, hospital stay, and recovery time.

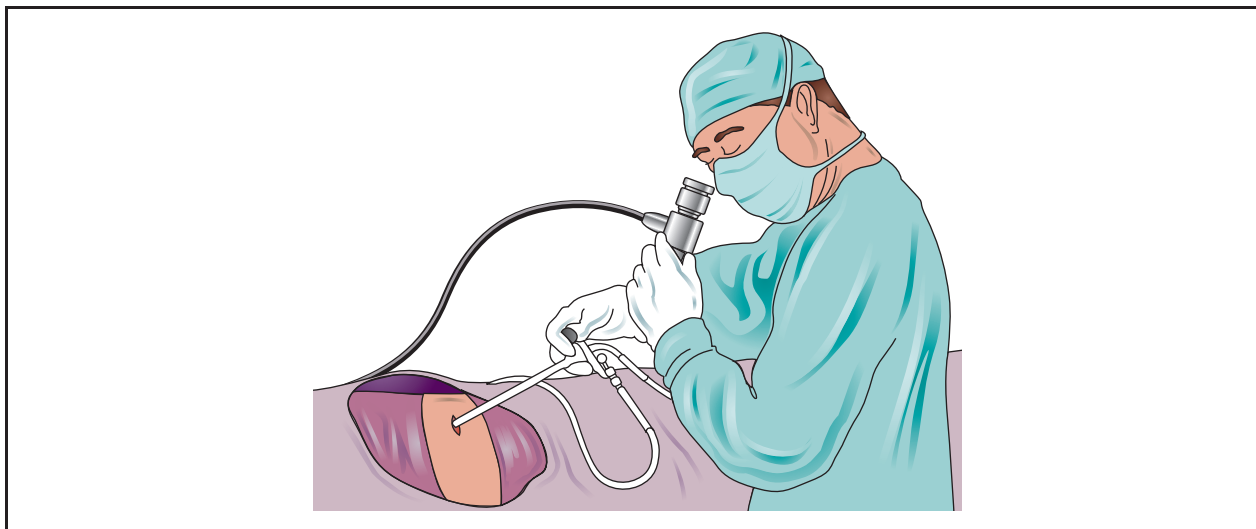
Precautions

Because one lung is partially deflated during thoracoscopy, the procedure cannot be done on patients whose lung function is so poor that they do not receive enough oxygen with only one lung. Patients who have had previous surgery that involved the chest cavity, or who have blood clotting problems, are not good candidates for this procedure.

Thoracoscopy gives physicians a good but limited view of the organs, such as lungs, in the chest cavity. Endoscope technology is being refined every day, as is what physicians can accomplish by inserting scopes and instruments through several small incisions instead of making one large cut.

Description

Thoracoscopy is most commonly performed in a hospital, and **general anesthesia** is used. Some of the



Thoracoscopy is a procedure in which a physician can view the chest cavity and the lungs by inserting an endoscope through the chest wall. Thoracoscopy is less evasive than surgical lung biopsy. (Illustration by Electronic Illustrators Group.)

procedures are moving toward outpatient services and **local anesthesia**. More specific names are sometimes applied to the procedure, depending on what the target site of the effort is. For example, if a physician intends to examine the lungs, the procedure is often called pleuroscopy. The procedure takes two to four hours.

The surgeon makes two or three small incisions in the chest wall, often between the ribs. By making the incisions between the ribs, the surgeon minimizes damage to muscle and nerves and the ribs themselves. A tube is inserted in the trachea and connected to a ventilator, which is a mechanical device that assists the patient with inhaling and exhaling.

The most common reason for a thoracoscopy is to examine a lung that has a tumor or a metastatic growth of cancer. The lung to be examined is deflated to create a space between the chest wall and the lung. The patient breathes with the other lung with the assistance of the ventilator.

A specialized endoscope, or narrow-diameter tube, with a video camera or mirrored attachment, is inserted through the chest wall. Instruments for taking necessary tissue samples are inserted through other small incisions. After tissue samples are taken, the lung is reinflated. All incisions except one are closed. The remaining open incision is used to insert a drainage tube. The tissue samples are sent to a laboratory for evaluation.

KEY TERMS

Endoscope—Instrument designed to allow direct visual inspection of body cavities, a sort of microscope in a long access tube.

Thoracotomy—Open chest surgery.

Trachea—Tube of cartilage that carries air into and out of the lungs.

Preparation

Prior to thoracoscopy, the patient will have several routine tests, such as blood, urine and **chest x ray**. Older patients must have an electrocardiogram (a trace record of the heart activity) because the anesthesia and the lung deflation put a big load on the heart muscle. The patient should not eat or drink from midnight the night before the thoracoscopy. The anesthesia used can cause **vomiting**, and, because anesthesia also causes the loss of the gag reflex, a person who vomits is in danger of moving food into the lungs, which can cause serious complications and **death**.

Aftercare

After the procedure, a chest tube will remain in one of the incisions for several days to drain fluid and release residual air from the chest cavity. Hospital

stays range from two to five days. Medications for pain are given as needed. After returning home, patients should do only light lifting for several weeks.

Risks

The main risks of thoracoscopy are those associated with the administration of general anesthesia. Sometimes excessive bleeding, or hemorrhage, occurs, necessitating a thoracotomy to stop it. Another risk comes when the drainage tube is removed, and the patient is vulnerable to lung collapse (**pneumothorax**).

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Tish Davidson, A.M.

Thoracotomy see **Lung surgery; Thoracic surgery**

Threadworm see **Enterobiasis**

Threadworm infection

Definition

Threadworm infection is an intestinal disease, which occasionally spreads to the skin, caused by a type of parasitic roundworm (helminth). In untreated patients, the disease has a high rate of reinfection caused by worms already present in the body. This type of disease recurrence is called autoinfection. Because of autoinfection, threadworms can remain inside humans for as long as 45 years after the initial infestation.

Description

Threadworm infection, which is also called strongyloidiasis, occurs in most countries of the world but is natural to (endemic in) tropical and subtropical climates. Strongyloidiasis is less common than other parasitic infections but may affect as much

as 25% of the population in some developing countries. In the United States, threadworm infection is most likely to be found among immigrants; returning travelers or military personnel; people who live in parts of Appalachia and the southeastern states; and persons in homes for the retarded and similar institutions.

Human beings are universally susceptible to threadworm infection, although adults and older children are at greater risk of infection than younger children. The disease does not confer immunity. In addition to humans, threadworms can infect dogs, cats, horses, pigs, rats, and monkeys.

Causes and symptoms

Threadworm infection is caused by *Strongyloides stercoralis*, a roundworm that lives in soil and can survive there for several generations. Mature threadworms may grow as long as 1–2 in (2.5–5 cm). The larvae have two stages in their life cycle: a rod-shaped (rhabdoid) first stage, which is not infective; and a threadlike (filariform) stage, in which the larvae can penetrate intact human skin and internal tissues.

The infection is most commonly transmitted when a person comes into contact—usually by walking barefoot—with soil containing *S. stercoralis* larvae in their filariform stage. The threadlike larvae penetrate the skin, enter the lymphatic system, and are carried by the blood to the lungs. Once in the lungs, the larvae burst out of the capillaries into the patient's main respiratory system. They migrate upwards—usually without symptoms—to the patient's throat, where they are swallowed and carried down into the digestive tract. The filariform larvae settle in the small intestine. They mature into adults that deposit eggs that hatch—usually in the intestines—into noninfectious rhabdoid larvae. The rhabdoid larvae then migrate into the patient's large intestine and are excreted in the feces. The time from initial penetration of the skin to excretion is 17–28 days. The rhabdoid larvae metamorphose into the infective filariform stage in the soil.

Threadworms are unique among human parasites in having both free-living and parasitic forms. In the free-living life cycle, some rhabdoid larvae develop into adult worms that live in contaminated soil and produce eggs that hatch into new rhabdoid larvae. The adult worms may live as long as five years.

The signs and symptoms of threadworm infection vary according to the stage of the disease as the larvae

migrate throughout the body. Patients who suffer from autoinfection may have chronic or intermittent symptoms for years after they are first infected.

Skin

The filariform larvae usually enter the body through the skin of the feet. There may be swelling, **itching**, and **hives** at the point of entry that may be confused with insect bites. Patients with chronic threadworm infection may also develop an itchy rash on their buttocks, thighs, or abdomen.

Digestive tract

Although some patients may notice only mild **diarrhea** and cramps, others may have **fever**, **nausea**, **vomiting**, general weakness, and blood or mucus in their stools. The **pain** may mimic a stomach ulcer.

Throat and lungs

When the larvae migrate to the lungs and air passages, the patient may have symptoms ranging from a simple dry **cough** to fever, difficulty breathing, and coughing up blood or pus.

Hyperinfection syndrome

Hyperinfection syndrome is a potentially fatal set of complications resulting from the spread of filariform larvae to the lungs and other organ systems. It can include inflammation of the heart tissue, stomach ulcers, perforation of the intestines, blood **poisoning**, **meningitis**, **shock**, and eventual **death**. Hyperinfection syndrome is most likely to occur in patients with immune disorders or **malnutrition**, or in those taking anti-inflammatory corticosteroid (anti-inflammatory) medications. It has been reported in only a few **AIDS** patients.

Autoinfection

Threadworm autoinfection in humans follows two patterns. In internal autoinfection, some rhabdoid larvae in the lower bowel develop into filariform larvae that enter the bloodstream from the intestines and migrate to the lungs. In external autoinfection, the skin around the patient's anus is infected by larvae in the feces.

Diagnosis

The doctor is likely to consider a diagnosis of threadworm infection when a patient has the symptoms described earlier and a history of travel or

military service in areas where the disease is endemic. A definite diagnosis is made by finding rhabdoid or filariform larvae in the patient's body fluids. The larvae may be found in fresh stool specimens or in mucus coughed up when the infection has reached the lungs. Because the larvae cannot be detected in the stools of 25% of infected patients, the string test is often performed to confirm the diagnosis. In this test, the patient swallows a weighted string which is withdrawn after four hours. The digestive juices absorbed by the string are then examined for the presence of threadworm larvae.

Doctors can also use blood tests and diagnostic imaging to support the diagnosis. Between 85% and 95% of patients with threadworm infections will have a measurable level of antibodies in their blood, even though these antibodies do not prevent the disease from spreading. In addition, patients with severe infections often have unusually high levels of white cells in their blood. X rays of the intestines or the chest often help in locating specific areas of inflamed or ulcerated tissue.

Treatment

Threadworm infections are treated with medications. The drugs most often given are ivermectin, thiabendazole (Mintezol), and albendazole. Ivermectin is generally preferred because it has fewer side effects than thiabendazole. These drugs, which are taken by mouth over a period of two to seven days, work by preventing the development of eggs and new larvae. Patients with severe infections should be given protein replacement, blood transfusions, and fluids to replace losses from nausea, vomiting, and diarrhea.

Patients who are taking **corticosteroids** should be carefully evaluated if they have symptoms of threadworm infection, because these medications encourage the development of hyperinfection syndrome.

Prognosis

The prognosis for complete recovery is good for most patients, except those with hyperinfection syndrome or severe protein loss.

Prevention

There is no effective immunization against threadworm infection. Prevention of the disease requires careful attention to personal and institutional hygiene in endemic areas, including

KEY TERMS

Antibody—A protein molecule produced by the immune system that is specific to a disease agent, such as threadworm larvae. The severity of a patient's infection can be measured from the level of antibody in the blood.

Autoinfection—An infection caused by a disease agent that is already present in the body.

Corticosteroid—A class of drugs based on hormones formed in the adrenal gland, used to reduce inflammation. They increase the likelihood of hyperinfection syndrome in patients with threadworm infection.

Endemic—Natural to or characteristic of a particular place, population, or climate. Threadworm infections are endemic in the tropics.

Filariform—Threadlike in appearance, like the infectious stage of the threadworm larva.

Helminth—A type of parasitic worm. Threadworms belong to a subcategory of helminths called nematodes, or roundworms.

Hyperinfection syndrome—A condition of massive infection in which threadworm larvae multiply rapidly and spread throughout the body. It is usually associated with damage to the immune system, the use of steroid medications, or malnutrition.

Rhabdoid—Rod- or wand-shaped, like the first stage of the threadworm larva.

String test—A test performed to diagnose threadworm infection. The patient is asked to swallow a weighted string that absorbs stomach juices, which can be analyzed for the presence of threadworm larvae.

handwashing after defecating and before handling food. Other precautions include wearing shoes when visiting countries with high rates of threadworm infection, and monitoring close contacts of patients for signs of infection.

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Rebecca J. Frey, PhD

Throat culture

Definition

A throat culture is a technique for identifying disease bacteria in material taken from the throat. Most throat cultures are done to rule out infections caused by beta-hemolytic streptococci, which cause **strep throat**. Hemolytic means that these streptococci destroy red blood cells.

Purpose

The primary purpose of a throat culture is identification of the specific organisms that cause strep throat. These organisms are Group A streptococci, specifically *Streptococcus pyogenes*. Since most sore throats are caused by viral infections rather than by *S. pyogenes*, a correct diagnosis is important to prevent unnecessary use of **antibiotics** and to begin treatment of strep infections as soon as possible. Group A **streptococcal infections** are potentially life-threatening, often involving other parts of the body in addition to the throat. Besides causing **sore throat** (pharyngitis), streptococci can also cause **scarlet fever**, **rheumatic fever**, **kidney disease**, or abscesses around the tonsils.

Throat cultures can also be used to identify other disease organisms that are present in the patient's throat; and to identify people who are carriers of the organisms that cause **meningitis** and **whooping cough**.

Besides their use in diagnosis, throat cultures are sometimes used to test antibiotics for their effectiveness in treating different infections.

Precautions

Throat cultures should be taken before the patient is given any antibiotic medications. In addition, the patient's immunization history should be checked to evaluate the possibility that diseases other than strep are causing the sore throat. The care provider should wash the hands carefully after taking the specimen to prevent the spread of any infectious organisms.

Description

A throat culture test should be done on anyone who has symptoms of a strep throat. These symptoms include a sore throat that may be accompanied by a **fever**, body aches, and loss of appetite. Age is a consideration, in that strep throat is more common in children than in adults. The tonsils and the back of the throat often appear red, swollen, and streaked with



This nurse is taking a throat culture from a patient for laboratory analysis. (Photograph by David Weinstein & Associates, Custom Medical Stock Photo. Reproduced by permission.)

pus. These symptoms usually appear one to three days after being exposed to group A strep. Because strep is highly contagious, family members and close contacts of patients diagnosed with strep throat should also have throat cultures performed if they show signs of the disease.

The specimen for throat culture is obtained by wiping the patient's throat with a cotton swab. The patient is asked to tilt the head back and open the mouth wide. With the tongue depressed and the patient saying "ah," the care provider wipes the back of the throat and the tonsils with a sterile swab. The swab is applied to any area that appears either very red or discharging pus. The swab is removed gently without touching the teeth, gums, or tongue. It is then placed in a sterile tube for immediate delivery to a laboratory. Obtaining the specimen takes less than 30 seconds. Laboratory results are usually available in two to three days. The swabbing procedure may cause gagging but is not painful. The doctor makes a note for the laboratory to indicate if any disease organisms other than strep are suspected, because some require special growth conditions in the laboratory.

S. pyogenes is cultured on a growth medium called blood agar. Agar is a gel that is made from the cell walls of red algae. Blood agar is made from

agar gel and sheep's blood. When the throat swab reaches the laboratory, it is wiped across a blood agar plate. The plate is allowed to incubate for 24–48 hours to allow the growth of bacteria. If the organism is a Group A hemolytic streptococcus, the area immediately around the bacterial colony will be cleared of red blood cells. Hemolytic streptococci dissolve (lyse) red blood cells, leaving a clear zone surrounding the colony.

Alternative procedures

So-called instant strep tests are now available to help diagnose strep throat. They can be used in the doctor's office and take about 10–30 minutes to perform. Instant tests detect an antigen associated with the streptococcus. These tests are relatively new and not available at all clinics. Their reliability has improved since they were first introduced. If an instant throat test is negative, however, a standard throat culture can be performed to verify the results.

Preparation

The patient does not need to avoid food or fluids before the test. Recent gargling or treatment with antibiotics, however, will affect the culture results. The laboratory should be notified if the patient has been recently taking antibiotic medications.

Aftercare

No specific aftercare is needed.

Risks

There is a minor risk for the health professional of exposure to the patient's illness.

Normal results

Normal results would include finding organisms that grow in healthy throat tissues. These organisms include non-hemolytic and alpha-hemolytic streptococci, some *Neisseria* species, staphylococci, **diphtheria** and hemophilus organisms, pneumococci, yeasts, and Gram-negative rods.

Abnormal results

In addition to *S. pyogenes*, other disease agents may be identified in the throat culture. Infectious agents that can be identified include *Candida albicans*, which can cause thrush; *Corynebacterium*

KEY TERMS

Agar—A gel made from red algae that is used to culture certain disease agents in the laboratory.

Antibiotic—A drug given to stop the growth of bacteria. Antibiotics are ineffective against viruses.

Antigen—A substance that interacts with an antibody and causes an immune reaction.

Carrier—A person harboring an infectious disease who may be immune to it but who can give it to others.

Diphtheria—A serious disease caused by a bacterium, *Corynebacterium diphtheriae*.

Hemolytic—Able to dissolve red blood cells. The bacteria that cause strep throat are hemolytic organisms.

Streptococcus—A category (genus) of sphere-shaped bacteria that occur in pairs or chains.

Thrush—A disease occurring in the mouth or throat that is caused by a yeast, *Candida albicans*.

Whooping cough—An infectious disease of the respiratory tract caused by a bacterium, *Bordetella pertussis*.

diphtheriae, which can cause diphtheria; and *Bordetella pertussis*, which can cause whooping cough. In addition, the appearance of a normal organism in very high numbers may also be regarded as an abnormal result.

Resources

ORGANIZATIONS

American Medical Association. 515 N. State St., Chicago, IL 60612. (312) 464-5000. <<http://www.ama-assn.org>>.

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Cindy L. A. Jones, PhD

Thromboangiitis obliterans see **Buerger's disease**

Thrombocyte count see **Platelet count**

Thrombocytopenia

Definition

Thrombocytopenia is an abnormal drop in the number of blood cells involved in forming **blood clots**. These cells are called platelets.

Description

Thrombocytopenia is a blood disease characterized by an abnormally low number of platelets in the bloodstream. The normal amount of platelets is usually between 150,000 and 450,000 cells per microliter of blood. A microliter is an amount equal to one one-millionth of a liter (a liter is almost equal to a quart). Platelet numbers are counted by having a blood sample collected and placing a measured amount of blood in a machine called a cell counter. When the platelet number drops below 150,000 cells per microliter of blood, this person is said to be thrombocytopenic.

Causes and symptoms

Abnormal reductions in the number of platelets are caused when abnormalities occur in any of the following three processes: decreased platelet production by the bone marrow; increased trapping of platelets by the spleen; or a more rapid than normal destruction of platelets. Persons with this condition easily bruise and can have episodes of excess bleeding (a hemorrhage).

Platelets come from megakaryocytes, which are produced in the material located within the center cavity of the bones (bone marrow). When abnormalities develop in the marrow, the marrow cells can lose their ability to produce platelets in correct amounts. The result is a lower than normal level of platelets in the blood. Drugs used in **cancer chemotherapy** can cause the marrow to malfunction in this way, as can the presence of tumor cells in the marrow itself.

Normally, the spleen holds about one-third of the body's platelets as part of this organ's function to recycle **aging** or damaged red blood cells (the cells that carry oxygen in the blood). When **liver disease** or cancer of the spleen is present, the spleen can enlarge, resulting in a greater number of platelets staying in the organ. This condition results in abnormally low numbers of platelets in the blood.

Platelets can break down in unusually high amounts in persons with abnormalities in their blood vessel walls; with blood clots; or with man-made

KEY TERMS

Gamma globulin—One of a group of proteins found in the blood that is involved in helping the body fight infections.

Stent—A man-made surgical device, usually tube-shaped, that is placed into a blood vessel to keep it from closing.

Transfusion—The transfer of blood from one person to another. Transfusions can be direct, in which blood is transferred from the donor to the recipient; or indirect, in which the blood is taken from the donor, stored in a container, and then given to the recipient.

replacement heart valves. Devices placed inside blood vessels to keep them from closing (stents) due to weakened walls or fat build-up can also cause platelets to break down. In addition, infections and other changes in the immune system can speed up the removal of platelets from the circulation.

Diagnosis

Thrombocytopenia is diagnosed by having a blood sample taken and counting the platelets present in the sample. However, accurately determining the medical reason for this condition is complex.

Once a low **platelet count** is verified, a careful evaluation of the function of the bone marrow and spleen are necessary. Improper functioning of either or both of these organs can cause thrombocytopenia. In addition, the causes for the abnormal spleen or marrow function must be investigated since different cancers, blood disorders, or liver disease can be the true cause for the drop in platelets found in the blood.

Treatment

If low platelet counts are caused by an enlarged spleen, removal of the spleen can help raise the platelet level, since the spleen is no longer there to capture the platelets. However, proper treatment for what causes the enlarged spleen is necessary as well.

Low platelet counts can indicate more serious conditions. If a dysfunctional immune system is found to be the cause for this condition, drugs like steroids or gamma globulin can be used to help maintain platelet levels in certain cases.

If low platelet levels are due to an abnormally low level of platelet production, transfusions of platelets can be given as well.

Prognosis

Thrombocytopenia can result in fatal bleeding, but it also can indicate various other, more serious, cancers and disorders that affect the blood cells. This condition requires thorough medical evaluation.

Prevention

There is no known way to prevent thrombocytopenia.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Dominic De Bellis, PhD

Thrombocytosis

Definition

Thrombocytosis is a blood disorder in which the body produces a surplus of platelets (thrombocytes).

Description

Thrombocytosis is an abnormally increased number of platelets in the blood. Platelets are blood cells that stick together, helping blood to clot. Thrombocytosis is a condition that may have many causes.

Thrombocytosis is classified as one of two types. Secondary thrombocytosis can be traced to another cause, such as inflammation, severe bleeding, iron deficiency, or some cancers. Primary thrombocytosis (or essential thrombocythemia) is a single disease entity, with unique clinical characteristics.

Causes and symptoms

The cause of essential thrombocytosis is unknown.

Secondary thrombocytosis may develop as a result of:

- acute hemorrhage or infection
- anemia
- arthritis and other chronic inflammations
- cancer
- exercise
- iron deficiency
- medication
- osteoporosis
- removal of the spleen (**splenectomy**)
- polycythemia vera (a disorder affecting other red blood cells, as well as platelets)
- stress
- surgery

Symptoms

Two of every three patients who have thrombocytosis do not have any symptoms of the disease at the time of diagnosis. Younger patients may remain symptom-free for years.

Enlargement of the spleen is detected in 60% of patients with thrombocytosis. The liver may also be enlarged. As many as half of all patients experience bleeding from the skin, gums, or nose; and 20–50% have some blockage of veins or arteries.

Other symptoms of thrombocytosis include:

- bloody stools
- bruising
- dizziness
- headache
- hemorrhage
- prolonged bleeding after having surgery or after having a tooth pulled
- redness or **tingling** of the hands and feet
- weakness. In rare instances, the lymph nodes become enlarged

The highest platelet counts usually produce the most severe symptoms. Younger patients (especially women) may not have symptoms, even though their platelet counts are very high.

Complications

Complications of thrombocytosis include **stroke**, **heart attack**, and formation of **blood clots** in the arms and legs.

A doctor should be notified whenever bleeding is unexplained or prolonged or the patient develops:

- chest or leg **pain**
- confusion
- numbness
- weakness

Diagnosis

The patient's symptoms suggest the presence of thrombocytosis. Blood tests confirm the diagnosis.

Bone marrow aspiration (removal of a tissue sample for microscopic examination) may also be performed.

Treatment

The key to treating secondary thrombocytosis is treating the underlying condition.

Any patient who has thrombocytosis should be encouraged not to smoke.

In young people who have no symptoms, this condition can remain stable for many years. These patients should be monitored by a physician, but may not require treatment.

Treatment for patients who do have symptoms focuses on controlling bleeding, preventing the formation of blood clots, and lowering platelet levels. Treatment for secondary thrombocytosis involves treating the condition or disease responsible for excess platelet production.

In 1997, the United States Food and Drug Administration (FDA) approved the use of anagrelide HCl (Agrylin) to reduce elevated platelet counts and decrease the risk of clot formation. Some patients have benefited from the use of hydroxyurea, an anti-cancer drug.

Low doses of **aspirin** may prevent clotting, but can cause serious hemorrhages.

If drug therapy does not bring platelet counts down to an acceptable level as rapidly as necessary, plateletpheresis may be performed. Usually combined with drug therapy and used primarily in medical emergencies, this procedure consists of:

- withdrawing blood from the patient's body
- removing platelets from the blood
- returning the platelet-depleted blood to the patient

Prognosis

Many patients with thrombocytosis remain free of complications for long periods. However, some patients may die as a result of blood clots or uncontrolled bleeding.

Prevention

There is no known way to prevent thrombocytosis.

Resources

OTHER

“Primary Thrombocythemia.” *The Merck Page*. June 3, 1998. <<http://www.merck.com>>.

Maureen Haggerty

Thromboembolism see **Embolism**

Thrombolytic therapy

Definition

Thrombolytic therapy is the use of drugs that dissolve **blood clots**.

Purpose

When a blood clot forms in a blood vessel, it may cut off or severely reduce blood flow to parts of the body that are served by that blood vessel. This can cause serious damage to those parts of the body. If the clot forms in an artery that supplies blood to the heart, for example, it can cause a **heart attack**. A clot that cuts off blood to the brain can cause a **stroke**. Thrombolytic therapy is used to dissolve blood clots that could cause serious, and possibly life-threatening, damage if they are not removed. Research suggests that when used to treat stroke, thrombolytic therapy can prevent or reverse **paralysis** and other problems that otherwise might result.

Thrombolytic therapy also is used to dissolve blood clots that form in tubes put into people's bodies for medical treatments, such as dialysis or **chemotherapy**.

Description

Thrombolytic therapy uses drugs called thrombolytic agents, such as alteplase (Activase), anistreplase

(Eminase), streptokinase (Streptase, Kabikinase), urokinase (Abbokinase), and tissue plasminogen activator (TPA) to dissolve clots. These drugs are given as injections, only under a physician's supervision.

Recommended dosage

The physician supervising thrombolytic therapy decides on the proper dose for each patient. He or she will take into account the type of drug, the purpose for which it is being used, and in some cases, the patient's weight.

Precautions

For thrombolytic therapy to be effective in treating stroke or heart attack, prompt medical attention is very important. The drugs must be given within a few hours of the beginning of a stroke or heart attack. However, this treatment is not right for every patient who has a heart attack or a stroke. Only a qualified medical professional can decide whether a thrombolytic agent should be used. To increase the chance of survival and reduce the risk of serious, permanent damage, anyone who has signs of a heart attack or stroke should get immediate medical help.

Thrombolytic therapy may cause bleeding. Usually this is not serious, but severe bleeding does occur in some people. This is especially likely in older people. To lower the risk of serious bleeding, people who are given this drug should move around as little as possible and should not try to get up on their own unless told to do so by a health care professional. Following all the instructions of the health care providers in charge is very important.

Thrombolytic therapy may be more likely to cause serious bleeding in people who have certain medical conditions or have recently had certain medical procedures. Before being given a thrombolytic agent, anyone with any of these problems or conditions should tell the physician in charge about it:

- blood disease or current or past bleeding problems in any part of the body
- heart or blood vessel disease
- stroke (recent or in the past)
- high blood pressure
- brain tumor or other brain disease
- stomach ulcer or colitis
- severe **liver disease**
- active **tuberculosis**

- recent falls, injuries, or blows to the body or head
- recent injections into a blood vessel
- recent surgery, including dental surgery
- tubes recently placed in the body for any reason
- recent delivery of a baby

In addition, anyone who has had a recent streptococcal (strep) infection should tell the physician in charge. Some thrombolytic agents may not work properly in people who have just had a strep infection, so the physician may want to use a different drug.

People who take certain medicines may be at greater risk for severe bleeding when they are given a thrombolytic agent.

Women who are pregnant should tell the physician in charge before being given a thrombolytic agent. There is a slight chance that a woman who is given thrombolytic therapy during the first five months of **pregnancy** will have a **miscarriage**. However, streptokinase and urokinase have both been used without problems in pregnant women.

After being treated with thrombolytic therapy, women who are breastfeeding should check with their physicians before starting to breastfeed again.

Side effects

Anyone who has **fever** or who notices bleeding or oozing from their gums, from cuts, or from the site where the thrombolytic agent was injected should immediately tell their health care provider.

People who are given thrombolytic therapy should also be alert to the signs of bleeding inside the body and should check with a physician immediately if any of the following symptoms occur:

- blood in the urine
- blood or black, tarry stools
- constipation
- coughing up blood
- vomiting blood or material that looks like coffee grounds
- nosebleeds
- unexpected or unusually heavy vaginal bleeding
- dizziness
- sudden, severe, or constant headaches
- **Pain** or swelling in the abdomen or stomach
- back pain or backache

KEY TERMS

Arteries—Blood vessels that carry blood away from the heart to the cells, tissues, and organs of the body.

Blood clot—A hard mass that forms when blood gels.

Chemotherapy—Treatment of an illness with chemical agents. The term is usually used to describe the treatment of cancer with drugs.

Dialysis—A process used in people whose kidneys are not working well. By way of a filtering machine, dialysis separates waste and other useless materials from the blood – a job the kidneys usually do.

Paralysis—Loss of the ability to move one or more parts of the body.

Stroke—A serious medical event in which blood flow to the brain is stopped. This may be because of a blood clot in an artery or because an artery has burst. Strokes may cause paralysis and changes in speech, memory, and behavior.

- severe or constant muscle pain or stiffness
- stiff, swollen, or painful joints

Other side effects of thrombolytic agents are possible. Anyone who has unusual symptoms during or after thrombolytic therapy should tell a health care professional.

Interactions

People who take certain medicines may be at greater risk for severe bleeding when they receive a thrombolytic agent. Anyone who is given a thrombolytic agent should tell the physician in charge about all other prescription or nonprescription (over-the-counter) medicines he or she is taking. Among the medicines that may increase the chance of bleeding are:

- aspirin and other medicines for pain and inflammation
- blood thinners (anticoagulants)
- antiseizure medicines, such as Depakote (divalproex) and Depakene (valproic acid)
- cephalosporins, such as cefamandole (Mandol), cefoperazone (Cefobid), and Cefotetan (Cefotan)

Also, anyone who has been treated with anistreplase or streptokinase within the past year should tell the physician in charge. These drugs may not work

properly if they are given again, so the physician may want to use a different thrombolytic agent.

Nancy Ross-Flanigan

Thrombophlebitis

Definition

Thrombophlebitis is the inflammation of a vein with blood clot formation inside the vein at the site of the inflammation. Thrombophlebitis also is known as phlebitis, phlebothrombosis, and venous thrombosis.

Description

There are two parts to thrombophlebitis, inflammation of a vein and blood clot formation. If the inflammation is minor, the disease usually is called venous thrombosis or phlebothrombosis. Thrombophlebitis can occur in both deep veins and superficial veins, but most often occurs in the superficial veins of the extremities (legs and arms). Most cases occur in the legs. When thrombophlebitis occurs in a superficial vein, one that is near the surface of the skin and is visible to the eye, the disease is called superficial thrombophlebitis. Any form of injury to a blood vessel can result in thrombophlebitis. In the case of superficial thrombophlebitis, the blood clot usually attaches firmly to the wall of the affected blood vein. Since superficial blood veins do not have muscles that massage the veins, **blood clots** in superficial veins tend to remain where they form and seldom break loose. When thrombophlebitis occurs in a deep vein, a vein that runs deep within muscle tissue, it is called deep venous thrombosis. Deep venous thrombosis presents the threat of producing blood clots that will break loose to form emboli. These can lodge in other tissues where they can block the blood supply, typically in the lungs. This results in tissue damage and can sometimes be serious or fatal, for example; **pulmonary embolism**.

Causes and symptoms

The main symptoms are tenderness and **pain** in the area of the affected vein. Redness and/or swelling also may be seen. In the case of deep venous thrombosis, there is more swelling than is caused by superficial thrombophlebitis, and the patient may experience

muscle stiffness in the affected area. There are many causes of thrombophlebitis.

The main causes can be grouped into three categories; injury to blood veins, increased blood clotting, and blood stasis. When blood veins are damaged, collagen in the blood vein wall is exposed. Platelets respond to collagen by initiating the clotting process. Damage to a vein can occur as a consequence of indwelling catheters, trauma, infection, **Buerger's disease**, or the injection of irritating substances. Increased tendency of the blood to clot can be caused by malignant tumors, genetic disorders, and **oral contraceptives**, though newer generation birth control pills carry a lower risk for many women. Stasis, in which the blood clots due to decreased blood flow in an area, can happen following surgery, as a consequence of **varicose veins**, as a complication of postpartum states, and following prolonged bed rest. In the case of prolonged bed rest, blood clots form because of inactivity, which allows blood to move sluggishly and stagnate (collect) in blood veins. This can lead to blood clots. These clots (also called emboli) sometimes are released when the patient stands up and resumes activity. This can present a problem if the emboli lodge in vital organs. In the case of postpartum patients, a **fever** developing four to 10 days after delivery may indicate thrombophlebitis. A 2004 study revealed that postmenopausal women taking hormone therapy combining estrogen and progestin had more than a twofold higher risk of venous thrombosis than non-hormone users.

Diagnosis

In superficial thrombophlebitis, the location of the clot sometimes can be seen by the unaided eye. Blood clots are hard and can usually be detected by a physician using palpation (touching or massage). Deep venous thrombosis requires specialized diagnostic procedures to detect the blood clot. Among the exams a physician may use are ultrasound and x ray, coupled with dye injection (venogram).

Treatment

Superficial thrombophlebitis usually resolves without treatment. If treatment of superficial thrombophlebitis is given, it usually is limited to the application of heat or anti-inflammatory drugs, like **aspirin** or ibuprofen, which also help to relieve the pain. It can take from several days to several weeks for the clot to resolve and the symptoms to completely disappear. A 2003 study showed that

KEY TERMS

Emboli, embolus—Emboli is the plural form of embolus. Embolus is any mass of air, blood clot, or foreign body that travels through the blood stream and is capable of lodging in smaller blood vessels where they obstruct the blood flow to that vessel.

Embolism—The obstruction of a blood vessel by a clot clot.

Phlebitis—Inflammation of a vein.

Thrombus—A blood clot that forms within a blood vessel or the heart.

low intensity therapy with warfarin, a common blood thinner, prevented recurrent venous thromboembolism in study subjects. A newer anticoagulant called ximelagatran also has been shown as equally or more effective than warfarin in preventing **deep vein thrombosis**.

Deep venous thrombosis is a serious condition, treated with **anticoagulant drugs** and by keeping the affected limb elevated. The primary objective in treating deep venous thrombosis is prevention of a pulmonary **embolism**. The patient usually is hospitalized during initial treatment. The prescribed anticoagulant drugs limit the ability of blood clots to grow and new clots to form. Sometimes, a drug that dissolves blood clots is administered. These drugs must be used with caution because, as the clot dissolves, it may release from the site where it formed and become an embolus. Surgery may be used if the affected vein is likely to present a long-term threat of producing blood clots that will release emboli. When superficial thrombophlebitis occurs in the groin, where the superficial veins join the deep veins, the threat of emboli is present. In this case, blood clots formed in the superficial veins can extend into the much larger deep vein where they break off and are released into the blood stream. The affected veins are either removed or tied off to prevent the release of the blood clots. Tying off superficial blood veins is an outpatient procedure that can be performed with **local anesthesia**. The patient is capable of immediately resuming normal activities.

Prognosis

Superficial thrombophlebitis seldom progresses to a serious medical complication, although non-lethal embolisms may be produced. Deep venous thrombosis

may lead to embolism, especially pulmonary embolism. This is a serious consequence of deep venous thrombosis, and sometimes is fatal.

Resources

PERIODICALS

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Elliott, William T. "Warfarin Effectively Prevents Venous Thromboembolism." *Clinical Cardiology Alert* April 2003: 1–2.

"Registral: The Combined Pill, Part One." *GP* September 8, 2003: 46.

"WPrevention of VTE With Ximelagatran." *Neurology Alert* February 2004: SS3–SS5.

John T. Lohr, PhD
Teresa G. Odle

Thrush see **Candidiasis**

Thymol see **Antiseptics**

Thymoma

Definition

Thymomas are the most common tumor of the thymus.

Description

The thymus is located in the upper chest just below the neck. It is a small organ that produces certain types of white blood cells before birth and during childhood. These white blood cells are called lymphocytes and are an important part of the body's immune system. Once released from the thymus, lymphocytes travel to lymph nodes where they help to fight infections. The thymus gland becomes smaller in adulthood and is gradually taken over by fat tissue.

Although rare, thymomas are the most common type of thymic **cancer**. They arise from thymic epithelial cells, which make up the covering of the thymus. Thymomas frequently contain lymphocytes, which are noncancerous. Thymomas are classified as either noninvasive (previously called "benign") or invasive (previously called "malignant"). Noninvasive thymomas are those in which the tumor is encapsulated and easy to remove. Invasive thymomas have spread to nearby

structures (such as the lungs) and are difficult to remove. Approximately 30% to 40% of thymomas are of the invasive type.

Thymoma affects men and women equally. It is usually diagnosed between the ages of 40 and 60 years. Thymomas are uncommon in children.

Causes and symptoms

The cause of thymoma is unknown. Cancer is caused when the normal mechanisms that control cell growth become disturbed, causing the cells to continually grow without stopping. This is caused by damage to the DNA in the cell.

Approximately 40% of the patients diagnosed with thymoma have no symptoms. The symptoms in the remaining 60% of patients are caused by pressure from the enlarged thymus on the windpipe (trachea) or blood vessels, or by paraneoplastic syndromes. Paraneoplastic syndromes are collections of symptoms in cancer patients that cannot be explained by the tumor. Seventy-one percent of thymomas are associated with paraneoplastic syndromes. The most common syndromes related to thymoma are pure red cell aplasia (having abnormally low levels of red blood cells), **myasthenia gravis** (a muscular disorder), and hypogammaglobulinemia (having abnormally low levels of antibodies). These conditions are autoimmune diseases—those in which the body mounts an attack against certain normal cells of the body.

Symptoms of thymoma may include:

- **shortness of breath**
- swelling of the face
- coughing
- chest pain
- muscle weakness (especially in the eyes, neck, and chest, causing problems with vision, swallowing, and breathing)
- weakness
- dizziness
- shortness of breath
- **fatigue**

Diagnosis

The physician will conduct a complete physical exam. He or she may be able to feel a fullness in the lower neck region. Routine blood tests may be performed. Imaging studies are necessary because the symptoms of thymoma can be caused by many other

diseases. Thymomas can be identified by **chest x ray**, **magnetic resonance imaging**, and computed tomography.

A biopsy may be performed, in which a small sample of the tumor is removed and examined under the microscope. However, because of the risk of “seeding” cancerous cells, biopsies are not routinely performed. There are a few different methods to biopsy a thymoma. For a **mediastinoscopy**, a wand-like lighted camera (endoscope) and special instruments are passed through a small cut in the lower neck. The surgeon can see the tumor on a monitor and can cut off small samples for microscopic analysis. Mediastinoscopy is performed under **general anesthesia**. Alternatively, a needle biopsy will be taken in which a long needle is passed through the skin and into the tumor. Fine needle biopsy uses a thin needle and larger-core needle biopsy uses a wider needle. Needle biopsies may be performed in conjunction with CT imaging.

Patients who are having difficulty breathing may have a **bronchoscopy** performed to examine the windpipe. An endoscope, in this case a bronchoscope, is inserted through the mouth and into the windpipe. The physician will look for tumors and may perform biopsies.

Treatment

Clinical staging

There is more than one type of staging system for thymoma; but the Masaoka system, a surgical staging system developed in 1981, is used most often. Thymoma is categorized into four stages (I, II, III, and IV), which may be further subdivided (A and B) based on the spread of cancerous tissue. The Masaoka staging system is as follows:

- Stage I. The thymoma lies completely within the thymus.
- Stage II. The thymoma has spread out of the thymus and invaded the outer layer of the lung (pleura) or nearby fatty tissue.
- Stage III. The thymoma has spread to other neighboring tissues of the upper chest, including the outer layer of the heart (pericardium), the lungs, or the heart's main blood vessels.
- Stage IVA. The thymoma has spread throughout the pericardium and/or the pleura.
- Stage IVB. The thymoma has spread to organs in other parts of the body.

In 1999, the World Health Organization (WHO) adopted a new classification system for thymic tumors. This system is a histologic classification, which means that it is based on the microscopic features of the cells that make up the tumor. The WHO classification system ranks thymomas into types A, AB, B1, B2, B3, and C, by increasing severity.

The treatment for thymoma depends on the stage of cancer and the patients overall health. Because thymomas are so rare, there are no defined treatment plans. Treatment options include surgery, **radiation therapy**, and/or **chemotherapy**. Surgical removal of the tumor is the preferred treatment. Surgery is often the only treatment required for stage I tumors. Treatment of thymoma often relieves the symptoms caused by paraneoplastic syndromes.

A treatment that is intended to aid the primary treatment is called adjuvant therapy. For instance, chemotherapy may be used along with surgery to treat thymoma. Stages II, III, and IV thymomas are often treated with surgery and some form of adjuvant therapy.

Surgery

Thymoma may be treated by surgically removing (resecting) the tumor and some of the nearby healthy tissue. Removal of the entire thymus gland is called a thymectomy. Surgery on the thymus is usually performed through the chest wall by splitting open the breast bone (sternum), a procedure called a median sternotomy. When complete removal of the tumor is impossible, the surgeon will remove as much of the tumor as possible (debulking surgery, subtotal resection). In these cases, If the tumor has spread, surgery may include removal of such other tissues as the pleura, pericardium, blood vessels of the heart, lung, and nerves.

Radiation therapy

Radiation therapy uses high-energy radiation from x rays and gamma rays to kill the cancer cells. Radiation given from a machine that is outside the body is called external radiation therapy. Radiation therapy is often used as adjuvant therapy following surgery to reduce the chance of cancer recurrence. Radiation may be used to kill cancer cells in cases in which the tumor was only partially removed. It may be used before surgery to shrink a large tumor. Radiation therapy is not very effective when used alone, although it may be used alone when the patient is too sick to withstand surgery.

The skin in the treated area may become red and dry and may take as long as a year to return to normal. Radiation to the chest may damage the lung, causing shortness of breath and other breathing problems. Also, the tube that goes between the mouth and stomach (esophagus) may be irritated by radiation, causing swallowing difficulties. Fatigue, upset stomach, **diarrhea**, and **nausea** are also common complaints of patients having radiation therapy. Most side effects go away about two to three weeks after radiation therapy has ended.

Chemotherapy

Chemotherapy uses **anticancer drugs** to kill the cancer cells. The drugs are given by mouth (orally) or intravenously. They enter the bloodstream and can travel to all parts of the body. Chemotherapy may be given before surgery to shrink a tumor, which is called neoadjuvant therapy. Thymoma cells are very sensitive to anticancer drugs, especially cisplatin, doxorubicin, and ifosfamide. Generally, a combination of drugs is given because it is more effective than a single drug in treating cancer. **Corticosteroids** are also used to treat thymoma.

The side effects of chemotherapy are significant, and include stomach upset, **vomiting**, appetite loss, hair loss (**alopecia**), mouth sores, and fatigue. Women may experience vaginal sores, menstrual cycle changes, and **premature menopause**. There is also an increased chance of infections.

Alternative treatment

Although alternative and complementary therapies are used by many cancer patients, very few controlled studies on the effectiveness of such therapies exist. Mind-body techniques such as prayer, **biofeedback**, visualization, **meditation**, and **yoga**, have not shown any effect in reducing cancer; but they can reduce **stress** and lessen some of the side effects of cancer treatments. Gerson, macrobiotic, orthomolecular, and Cancell therapies are ineffective treatments for cancer.

Clinical studies of hydrazine sulfate found that it had no effect on cancer and even worsened the health and well-being of the study subjects. One clinical study of the drug amygdalin (Laetrile) found that it had no effect on cancer. Laetrile can be toxic and has caused deaths. Shark cartilage has been studied as a cancer treatment and is presently being studied by the FDA in clinical trials. Although the results are mixed, clinical studies suggest that melatonin may increase the survival time and quality of life for cancer patients.

KEY TERMS

Adjuvant therapy—A treatment that is intended to aid the primary treatment. Adjuvant treatments for thymic cancer are radiation therapy and chemotherapy.

Invasive—A descriptive term for thymoma that has spread beyond the outer wall of the thymus.

Lymphocyte—A type of white blood cell that is found in the thymus.

Neoadjuvant therapy—Radiation therapy or chemotherapy used to shrink a tumor before surgical removal of the tumor.

Paraneoplastic syndrome—A set of symptoms that is associated with cancer but is not directly caused by the cancer.

Pleura—The outer covering of the lungs.

Selenium, in safe doses, may delay the progression of cancer. Laboratory and animal studies suggest that curcumin, the active ingredient of turmeric, has anticancer activity. Maitake mushrooms may boost the immune system, according to laboratory and animal studies. The results of laboratory studies suggest that mistletoe has anticancer properties; however, clinical studies have not been conducted.

Prognosis

The five-year survival rates for thymomas are 96% for stage I, 86% for stage II, 69% for stage III, and 50% for stage IV. Thorough (radical) surgery is associated with a longer survival rate. Almost 15% of thymoma patients develop a second cancer.

Thymomas rarely spread (metastasize) outside of the chest cavity. Metastasis is usually limited to the pleura. Invasive thymomas are prone to recurrence, even 10 to 15 years following surgery. The recurrence rates are drastically reduced and the five-year survival rates are drastically increased in patients who receive adjuvant radiation therapy.

Prevention

Because there are no known risk factors for the development of thymoma, there are no preventive measures. However, there may be an association between thymic cancer and exposure of the chest to radiation.

Resources

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Thomas, Charles, Wright, Cameron, and Patrick Loehrer. "Thymoma: State of the Art." *Journal of Clinical Oncology* 17 (July 1999): 2280-89.

ORGANIZATIONS

American Cancer Society. 1599 Clifton Road NE, Atlanta, GA 30329. (800) ACS-2345. <<http://www.cancer.org>>.

Cancer Research Institute, National Headquarters. 681 Fifth Ave., New York, NY 10022. (800) 992-2623. <<http://www.cancerresearch.org>>.

National Institutes of Health. National Cancer Institute. 9000 Rockville Pike, Bethesda, MD 20982. Cancer Information Service: (800) 4-CANCER. <<http://cancernet.nci.nih.gov>>.

Belinda Rowland, Ph.D.

Thymus tumor see **Thymoma**

Thyroid biopsy

Definition

The thyroid biopsy is a procedure in which a sample of thyroid tissue is withdrawn for laboratory examination. The sample can be withdrawn through a needle or a surgical incision may be made to obtain a piece of thyroid tissue.

Purpose

The test is generally performed when a lump or a nodule is detected in the thyroid. The test may also be ordered if the thyroid gland is enlarged and the cause is

not apparent. The biopsy is usually a test for **thyroid cancer**.

Precautions

A patient with a bleeding disorder should not have a biopsy unless the bleeding problem can be corrected by a **transfusion** of the cells that cause blood to clot (platelets).

Description

The thyroid is a butterfly-shaped gland located at the base of the neck. It produces thyroxine, a hormone that plays a very crucial role in regulating the metabolism of the body and controlling several vital functions, such as the heart beat, blood pressure, and body temperature. The thyroid also regulates childhood growth and development.

A thyroid biopsy is usually ordered when a painless lump or a nodule is detected, either by the patient or by a doctor during a routine **physical examination**. A biopsy is the only test that can accurately determine whether the lump is non-cancerous (benign) or cancerous (malignant). The biopsy can be performed in several ways.

The “fine needle aspiration” (FNA) can be done in the doctor’s office. An anesthetic is not usually given. The patient will be asked to lie on his or her back. A pillow will be placed under the shoulders and the neck will be extended. The biopsy site will be cleansed with a sterile antiseptic solution. A thin needle will be inserted into the thyroid, and a sample of thyroid cells and some fluid will be collected. The needle will be quickly withdrawn. Pressure will be applied at the biopsy site to stop the bleeding and a bandage may be used to cover the area. The test takes three to five minutes.

For “large needle biopsy,” a mild sedative may be given an hour before the procedure, to relax the patient. The patient will be asked to lie on his or her back, with the head tipped back and the neck extended. The biopsy site will be thoroughly cleansed and the physician will inject a local anesthetic. A small incision (about 1 inch) will be made in the skin. The biopsy needle will be inserted through the incision into the thyroid. A sample of tissue will be removed and the needle withdrawn. Pressure is applied at the biopsy site to stem the bleeding and a bandage applied. This test takes five to ten minutes.

The “open incisional biopsy” is done in an operating room by a surgeon. The patient will be given a general anesthetic. A sedative will be given an hour

before the procedure to relax the patient. An intravenous line will be placed in the arm for infusion of fluids or drugs. An endotracheal tube will be inserted through the mouth into the lungs for administering anesthetic gases. After the patient is anesthetized, a small incision is made in the neck. Either the whole thyroid or a part of it is removed. If only a portion is being removed, the surgeon may send a small piece of remaining tissue to the laboratory for immediate testing while the patient is still on the operating table. If the pathologist’s report comes back stating that **cancer** is present in the remaining tissue, the entire thyroid is removed. The incision is closed with stitches. The whole procedure may take about an hour.

Preparation

The doctor should be informed of any **allergies** to medications and every medication the patient is taking. If the patient is pregnant, the doctor should be told.

The patient will be asked to sign the necessary consent forms. If a needle biopsy is done, no special preparation is needed. If a large needle biopsy is being done, the doctor may order some tests to determine the clotting ability of the blood. If an open incisional biopsy is being done, a general anesthetic is required and the patient will be asked to refrain from eating or drinking anything 8-12 hours before the test.

Aftercare

The needle used in fine needle aspiration is so thin, the whole procedure feels like a quick injection. There is no **pain** or tenderness at the site after the test. In large needle biopsy, a stinging needle prick may be felt when the local anesthetic is injected. The site may be sore for a few hours and tender for a day or two after the test.

In the open incisional biopsy, the patient will feel nothing during the procedure, because of the effects of the anesthetic and the sedative. However, the anesthetic may cause the patient to feel drowsy for several hours after the procedure. The anesthetic may also cause the patient to experience some **fatigue**, and general aches and pains for a day or two after the procedure. The endotracheal tube may make the throat feel mildly sore. If there is swelling at the biopsy site or if the patient develops a **fever**, the doctor should be notified immediately.

Risks

No risks are associated with fine needle aspiration. Large needle biopsy may cause bleeding into the

KEY TERMS

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Endotracheal tube—A hollow tube that is inserted into the windpipe (trachea), leading to the lungs.

Pathologist—A doctor who specializes in the diagnosis of disease by studying cells and tissues under a microscope.

thyroid gland. There is a small risk that the anesthetic used in open surgical biopsy may cause a life-threatening reaction.

Normal results

The normal appearance and architecture of the thyroid cells indicate that no cancer cells are present in the thyroid tissue.

Abnormal results

Any abnormalities of the thyroid tissue cells may indicate cancer benign tumors, or some other thyroid disease. If cancer is suspected, the pathologist may do some more testing to identify the extent of the cancer, so that it can be treated appropriately.

Resources

ORGANIZATIONS

American Thyroid Association, Inc. Montefiore Medical Center, 111 East 210th St., Bronx, NY 10467. <<http://thyroid.org>>.

Thyroid Foundation of America, Inc. Ruth Sleeper Hall, RSL 350, 40 Parkman St., Boston, MA 02114-2698. (800) 832-8321.

Lata Cherath, PhD

Thyroid cancer

Definition

Thyroid **cancer** is a disease in which the cells of the thyroid gland become abnormal, grow uncontrollably, and form a mass of cells called a tumor.

Description

Thyroid cancer is grouped into four types based on how its cells appear under a microscope. The types are papillary, follicular, medullary and anaplastic thyroid cancers. They grow at different rates and can spread to other parts of the body if left untreated.

The thyroid is a hormone-producing butterfly-shaped gland located in the neck at the base of the throat. It has two lobes, the left and the right. The thyroid uses iodine, a mineral found in some foods, to make several of its hormones. **Thyroid hormones** regulate essential body processes such as heart rate, blood pressure, body temperature, metabolism; and affect the nervous system, muscles and other organs. These hormones also play an important role in regulating childhood growth and development.

Diseases of the thyroid gland affect millions of Americans. The most common diseases of the thyroid are either **hyperthyroidism** (Graves' disease) or **hypothyroidism**, an overactive or an underactive gland, respectively. Sometimes lumps or masses may develop in the thyroid, and although most (ninety-five percent) of these lumps or nodules are noncancerous (benign), all thyroid lumps should be taken seriously. The American Cancer Society estimates that the approximately 17,200 new cases of thyroid cancer that occur in the United States account for 1% of all cancers.

Women are three times more likely to develop thyroid cancer than men. Although the disease affects teenagers and young adults, most people that develop thyroid cancer are over 50 years of age.

Causes and symptoms

The exact cause of thyroid cancer is not known; but it is more common in whites than in African Americans. Radiation was used in the 1950s and 1960s to treat **acne** and to reduce swelling in infections of the tonsils, adenoids and lymph nodes. It has been proven that this exposure is a risk factor for thyroid cancer. In some areas of the world, **diets** are low in iodine. Papillary and follicular cancers occur more frequently in these areas. Iodine deficiency is not a large problem in the United States because iodine is added to table salt and other foods. Approximately 7% of thyroid cancer are caused by the alteration (mutation) of a gene called the RET gene, which can be inherited.

Symptoms are rare so the lump is not usually painful. The symptoms of thyroid nodules are:

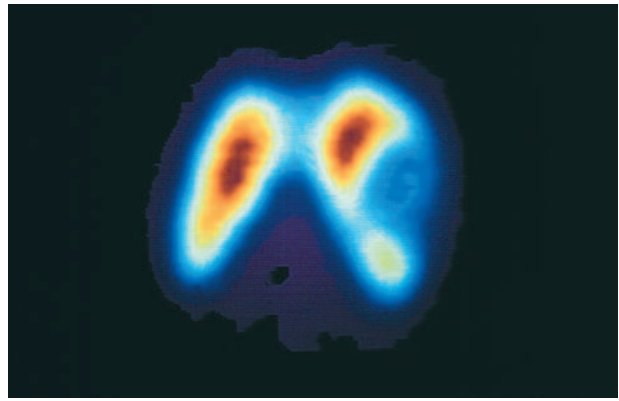
- a lump or nodule that can be felt in the neck is the most frequent sign of thyroid cancer
- the lymph nodes may be swollen and the voice may become hoarse because the tumor presses on the nerves leading to the voice box
- some patients experience a tight or full feeling in the neck and have difficulty breathing or swallowing

Diagnosis

Physicians use several tests to confirm the suspicion of thyroid cancer, to identify the size and location of the lump and to determine whether the lump is non-cancerous (benign) or cancerous (malignant). Blood tests such as the thyroid stimulating hormone (TSH) test check thyroid function. These are drawn by a technician with a needle and takes a few minutes. It take several days to be interpreted by a pathologist. Calcitonin is produced by the C cells (parafollicular cells) of the thyroid gland when the parafollicular cells of the thyroid become cancerous. Blood calcitonin levels are used to confirm the diagnosis of medullary thyroid cancer if it is suspected.

Computed tomography scan (CT scan) or an ultrasonography (ultrasound scan) are imaging tests used to produce a picture of the thyroid and usually last less than one hour. A radiologist usually interprets the results within 24 hours. In ultrasonography, high-frequency sound waves are bounced off the thyroid. The pattern of echoes that is produced by these waves is converted into a computerized image on a television screen. This test can determine whether the lumps found in the thyroid are benign fluid-filled cysts or solid malignant tumors.

A radioactive scan may take several hours and can be used to identify any abnormal areas in the thyroid by giving the patient a very small amount of radioactive iodine, which can either be swallowed or injected into the thyroid. Since the thyroid is the only gland in the body that absorbs iodine, the radioactive iodine accumulates there. An x-ray image can then be taken or an instrument called a “scanner” can be used to identify areas in the thyroid that do not absorb iodine normally. These abnormal spots are called “cold spots” and further tests are performed to check whether the cold spots are benign or malignant tumors. If a significant amount of radioactive iodine is concentrated in the nodule, then it is termed “hot” and is usually benign. Again a radiologist interprets the results within a day.



A gamma scan of the human thyroid gland revealing cancer.
(Custom Medical Stock Photo. Reproduced by permission.)

The most accurate diagnostic tool for thyroid cancer is a biopsy. In this process a sample of thyroid tissue is withdrawn and examined under a microscope by a pathologist. This usually takes a day or so. The tissue samples can be obtained either by drawing out a sample of tissue through a needle (needle biopsy) or by surgical removal of the nodule (surgical biopsy). A needle biopsy takes a few minutes and can be done by any trained physician, usually a radiologist. The surgical biopsy is done by a surgeon under **general anesthesia** with the help of an anesthesiologist and will take a few hours. If thyroid cancer is diagnosed, further tests may be done to learn about the stage of the disease and help doctors plan appropriate treatment.

Treatment

The aggressiveness of each type of thyroid cancer is different. Cancer staging considers the size of the tumor, whether it has grown into surrounding lymph nodes and whether it has spread to distant parts of the body (metastasized). Age and general health status are also taken into account. In patients less than 45 years old there are only two stages. I papillary or follicular type thyroid cancer, stage I refers to patients without evidence of cancer that has spread to the body. Stage II refers to patients with spread of cancer outside the thyroid gland. In patients over 45, patients with tumors smaller than one cm are classified as stage I, those with tumors not broken through the capsule (covering) of the thyroid belong to stage II, those with tumors outside the capsule or lymph node involvement are called stage III and those with spread outside the thyroid area are stage IV. In medullary-type thyroid cancer, stage I and IV are the same. Stage II

consists of patients with tumors greater than one cm and stage III comprises patients with lymph node involvement.

The papillary type (60–80% of all thyroid cancers) is a slow-growing cancer that develops in the hormone-producing cells (that contain iodine) and can be treated successfully. The follicular type (30–50% of thyroid cancers) also develops in the hormone-producing cells, has a good cure rate but may be difficult to control if the cancer invades blood vessels or grows into nearby structures in the neck. The medullary type (5–7% of all thyroid cancers) develops in the parafollicular cells (also known as the C cells) that produce calcitonin, a hormone that does not contain iodine. Medullary thyroid cancers are more difficult to control because they often spread to other parts of the body. The fourth type of thyroid cancer, anaplastic (2% of all thyroid cancers), is the fastest-growing and is usually fatal because the cancer cells rapidly spread to the different parts of the body.

More than 90% of patients who are treated for papillary or follicular cancer will live for 15 years or longer after the diagnosis of thyroid cancer. Eighty percent of patients with medullary thyroid cancer will live for at least 10 years after surgery. Only 3–17% of patients with anaplastic cancer survive for five years.

Like most cancers, cancer of the thyroid is best treated when it is found early by a primary physician. Treatment depends on the type of cancer and its stage. Four types of treatment are used: surgical removal, **radiation therapy**, hormone therapy and **chemotherapy**. Surgical removal is the usual treatment if the cancer has not spread to distant parts of the body.

The surgeon may remove the side or lobe of the thyroid where the cancer is found (lobectomy) or all of it (total **thyroidectomy**). If the adjoining lymph nodes are affected, they may also be removed during surgery. When the thyroid gland is removed and levels of thyroid hormones decrease, the pituitary gland starts to produce TSH that stimulates the thyroid cells to grow.

A radiation-oncologist uses radiation therapy with high-energy x-rays to kill cancer cells and shrink tumors. The radiation may come from a machine outside the body (external beam radiation), or the patient may be asked to swallow a drink containing radioactive iodine. Because the thyroid cells take up iodine, the radioactive iodine collects in any thyroid tissue remaining in the body and kills the cancer cells. A hematologist-oncologist uses chemotherapy either as a pill or an injection through a vein in the arm.

KEY TERMS

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Calcitonin—A hormone produced by the parafollicular cells (C cells) of the thyroid. The main function of the hormone is to regulate calcium levels in body serum.

Chemotherapy—Treatment of cancer with synthetic drugs that destroy the tumor either by inhibiting the growth of the cancerous cells or by killing them.

Hormone therapy—Treatment of cancer by inhibiting the production of hormones such as testosterone and estrogen.

Hyperthyroidism—A condition in which the thyroid is overactive due to overstimulation of the thyroid cells.

Hypothyroidism—A condition in which the thyroid gland is underactive.

Lobectomy—A surgical procedure that removes one lobe of the thyroid.

Radiation therapy—Treatment with high-energy radiation from x-ray machines, cobalt, radium, or other sources.

Total thyroidectomy—A surgical procedure that removes the entire thyroid gland.

Alternative treatment

Hormone therapy uses hormones after surgery to stop this growth and the formation of new cancerous thyroid cells. To prevent cancerous growth, the natural hormones that are produced by the thyroid are taken in the form of pills. Thus, their levels remain normal and inhibit the pituitary gland from making TSH. If the cancer has spread to other parts of the body and surgery is not possible, hormone treatment is aimed at killing or slowing the growth of cancer cells throughout the body.

A powerful phytochemical, lycopene, gives tomatoes their red color and appears to act as an antioxidant in the body, repairing damaged cells and scavenging free radicals, the molecules responsible for most types of degenerative diseases and **aging**. Antioxidants such as lycopene help inhibit DNA oxidation, which can lead to certain forms of cancer. Lycopene is a normal constituent of human blood and tissues, where it is

found in greater concentrations than beta-carotene or any other carotenoid. Tomatoes, including cooked or processed tomatoes, tomato juices, soups, sauces, paste and ketchup, contain more lycopene than any other food. Guava, rose hip, watermelon and grapefruit also contain lycopene.

Other antioxidants are: vitamin E (Dosage: 400 IU daily), vitamin C (Dosage: 1,000 to 4,000 mg daily), beta carotene (Dosage: 15 mg (25,000 IU) daily), lutein (Dosage: 6 to 20 mg daily), pycnogenol (Dosage: 25 to 50 mg daily), green tea (Dosage: 300 to 400 mg of green tea polyphenols daily), grape-seed extract (Dosage: 100 mg daily), alpha lipoic acid (Dosage: 50 to 200 mg daily), N-acetylcysteine (Dosage: 600 mg daily) and selenium (Dosage: 200 to 400 mcg daily). Pregnant women should consult a physician before taking any medication.

Prevention

Because most people with thyroid cancer have no known risk factor, it is not possible to completely prevent this disease. However, inherited cases of medullary thyroid cancer can be prevented and radiation to the neck is avoided. If a family member has had this disease, the rest of the family can be tested and treated early. The National Cancer Institute recommends that a doctor examine anyone who has received radiation to the head and neck during childhood at intervals of one or two years. The neck and the thyroid should be carefully examined for any lumps or enlargement of the nearby lymph nodes. Ultrasound may also be used to screen for the disease in people at risk for thyroid cancer.

Resources

ORGANIZATIONS

National Cancer Institute (National Institutes of Health).
9000 Rockville Pike, Bethesda, MD 20892. (800)
422-6237. <<http://www.nci.nih.gov>>.

Kulbir Rangi, DO

Thyroid drugs see **Thyroid hormones**

Thyroid function tests

Definition

Thyroid function tests are blood tests used to evaluate how effectively the thyroid gland is working. These tests include the thyroid-stimulating hormone

test (TSH), the thyroxine test (T4), the triiodothyronine test (T3), the thyroxine-binding globulin test (TBG), the triiodothyronine resin uptake test (T3RU), and the long-acting thyroid stimulator test (LATS).

Purpose

Thyroid function tests are used to:

- help diagnose an underactive thyroid (**hypothyroidism**) and an overactive thyroid (**hyperthyroidism**)
- evaluate thyroid gland activity
- monitor response to thyroid therapy

Precautions

Thyroid treatment must be stopped one month before blood is drawn for a thyroxine (T4) test.

Steroids, propranolol (Inderal), cholestyramine (Questran), and other medications that may influence thyroid activity are usually stopped before a triiodothyronine (T3) test.

Estrogens, anabolic steroids, phenytoin, and thyroid medications may be discontinued prior to a thyroxine-binding globulin (TBG) test. The laboratory analyzing the blood sample must be told if the patient cannot stop taking any of these medications. Some patients will be told to take these medications as usual so that the doctor can determine how they affect thyroxine-binding globulin.

Patients are asked not to take estrogens, androgens, phenytoin (Dilantin), salicylates, and thyroid medications before having a triiodothyronine resin uptake (T3RU) test.

Prior to taking a long-acting thyroid stimulant (LATS) test, the patient will probably be told to stop taking all drugs that could affect test results.

Description

Most doctors consider the sensitive thyroid-stimulating hormone (TSH) test to be the most accurate measure of thyroid activity. By measuring the level of TSH, doctors can determine even small problems in thyroid activity. Because this test is sensitive, abnormalities in thyroid function can be determined before a patient complains of symptoms.

TSH "tells" the thyroid gland to secrete the hormones thyroxine (T4) and triiodothyronine (T3). Before TSH tests were used, standard blood tests measured levels of T4 and T3 to determine if the thyroid

gland was working properly. The triiodothyrene (T3) test measures the amount of this hormone in the blood. T3 is normally present in very small amounts, but has a significant impact on metabolism. It is the active component of thyroid hormone.

The thyroxine-binding globulin (TBG) test measures blood levels of this substance, which is manufactured in the liver. TBG binds to T3 and T4, prevents the kidneys from flushing the hormones from the blood, and releases them when and where they are needed to regulate body functions.

The triiodothyronine resin uptake (T3RU) test measures blood T4 levels. Laboratory analysis of this test takes several days, and it is used less often than tests whose results are available more quickly.

The long-acting thyroid stimulator (LATS) test shows whether blood contains long-acting thyroid stimulator. Not normally present in blood, LATS causes the thyroid to produce and secrete abnormally high amounts of hormones.

It takes only minutes for a nurse or medical technician to collect the blood needed for these blood tests. A needle is inserted into a vein, usually in the forearm, and a small amount of blood is collected and sent to a laboratory for testing. The patient will usually feel minor discomfort from the “stick” of the needle.

Preparation

There is no need to make any changes in diet or activities. The patient may be asked to stop taking certain medications until after the test is performed.

Aftercare

Warm compresses can be used to relieve swelling or discomfort at the site of the puncture. With a doctor's approval, the patient may start taking medications stopped before the test.

Normal results

Not all laboratories measure or record thyroid hormone levels the same way. Each laboratory will provide a range of values that are considered normal for each test. Some acceptable ranges are listed below.

TSH

Normal TSH levels for adults are 0.5–5.0 mU/L.

T4

Normal T4 levels are:

- 10.1–2.0 ug/dl at birth
- 7.5–16.5 ug/dl at one to four months
- 5.5–14.5 ug/dl at four to 12 months
- 5.6–12.6 ug/dl at one to six years
- 4.9–11.7 ug/dl at 10 years
- 4–11 ug/dl at 10 years and older.

Levels of free T4 (thyroxine not attached to TBG) are higher in teenagers than in adults.

Normal T4 levels do not necessarily indicate normal thyroid function. T4 levels can register within normal ranges in a patient who:

- is pregnant
- has recently had contrast x rays
- has nephrosis or **cirrhosis**

T3

Normal T3 levels are:

- 90–170 ng/dl at birth
- 115–190 ng/dl at six to 12 years
- 110–230 ng/dl in adulthood

TBG

Normal TBG levels are:

- 1.5–3.4 mg/dl or 15–34 mg/L in adults
- 2.9–5.4 mg/dl or 29–54 mg/L in children.

T3RU

Between 25% and 35% of T3 should bind to or be absorbed by the resin added to the blood sample. The test indirectly measures the amount of thyroid binding globulin (TBG) and thyroid-binding prealbumin (TBPA) in the blood.

LATS

Long-acting thyroid stimulator is found in the blood of only 5% of healthy people.

Abnormal results

T4

Elevated T4 levels can be caused by:

- acute thyroiditis
- birth control pills

- clofibrate (Altromed-S)
- contrast x rays using iodine
- estrogen therapy
- heparin
- heroin
- hyperthyroidism
- **pregnancy**
- thyrotoxicosis
- toxic thyroid adenoma

Cirrhosis and severe non-thyroid disease can raise T4 levels slightly.

Reduced T4 levels can be caused by:

- anabolic steroids
- androgens
- antithyroid drugs
- cretinism
- hypothyroidism
- kidney failure
- lithium (Lithane, Lithonate)
- myxedema
- phenytoin
- propranolol

T3

Although T3 levels usually rise and fall when T4 levels do, T3 toxicosis causes T3 levels to rise while T4 levels remain normal. T3 toxicosis is a complication of:

- Graves' disease
- toxic adenoma
- toxic nodular **goiter**

T3 levels normally rise when a woman is pregnant or using birth-control pills. Elevated T3 levels can also occur in patients who use estrogen or **methadone** or who have:

- certain genetic disorders that do not involve thyroid malfunction
- hyperthyroidism
- thyroiditis
- t3 thyrotoxicosis
- toxic adenoma

Low T3 levels may be a symptom of:

- acute or chronic illness

- hypothyroidism
- kidney or **liver disease**
- starvation

Decreased T3 levels can also be caused by using:

- anabolic steroids
- androgens
- phenytoin
- propranolol
- reserpine (Serpasil)
- salicylates in high doses

TBG

TBG levels, normally high during pregnancy, are also high in newborns. Elevated TBG levels can also be symptoms of:

- acute hepatitis
- acute intermittent porphyria
- hypothyroidism
- inherited thyroid hormone abnormality

TBG levels can also become high by using:

- anabolic steroids
- birth control pills
- anti-thyroid agents
- clofibrate
- estrogen therapy
- phenytoin
- salicylates in high doses
- thiazides
- thyroid medications
- warfarin (Coumadin)

TBG levels can be raised or lowered by inherited liver disease whose cause is unknown.

Low TBG levels can be a symptom of:

- acromegaly
- acute hepatitis or other acute illness
- hyperthyroidism
- kidney disease
- malnutrition
- marked hypoproteinemia
- uncompensated acidosis

KEY TERMS

Acidosis—A condition in which blood and tissues are unusually acidic.

Acromegaly—A disorder in which growth hormone (a chemical released from the pituitary gland in the brain) causes increased growth in bone and soft tissue. Patients have enlarged hands, feet, noses, and ears, as well as a variety of other disturbances throughout the body.

Acute intermittent porphyria—An inherited disease affecting the liver and bone marrow. The liver overproduces a specific acid and the disease is characterized by attacks of high blood pressure, abdominal colic, psychosis, and nervous system disorders.

Anabolic steroids—Protein-building compounds used to treat certain anemias and cancers, strengthen bones, and stimulate weight gain and growth. Anabolic steroids are sometimes used to illegally enhance athletic performance.

Cholestyramine (Questran)—A drug used to bind with bile acids and prevent their reabsorption and to stimulate fat absorption.

Cirrhosis—Progressive disease of the liver, associated with failure in liver cell functioning and blood flow in the liver. Tissue and cells are damaged, the liver becomes fibrous, and jaundice can result.

Clofibrate (Altromed-S)—Medication used to lower levels of blood cholesterol and triglycerides.

Cretinism—Severe hypothyroidism that is present at birth and characterized by severe mental retardation.

Graves' disease—The most common form of hyperthyroidism, characterized by bulging eyes, rapid heart rate, and other symptoms.

Heparin—An organic acid that occurs naturally in the body and prevents blood clots. Heparin is also made synthetically and can be given as a treatment when required.

Hepatitis—Inflammation of the liver.

Hyperthyroidism—Overactive thyroid gland; symptoms include irritability/nervousness, muscle weakness, tremors, irregular menstrual periods, weight loss, sleep problems, thyroid enlargement, heat

sensitivity, and vision/eye problems. The most common type of this disorder is called Graves' disease.

Hypoproteinemia—Abnormally low levels of protein in the blood.

Hypothyroidism—Underactive thyroid gland; symptoms include fatigue, difficulty swallowing, mood swings, hoarse voice, sensitivity to cold, forgetfulness, and dry/coarse skin and hair.

Lithium (Lithane, Lithromate)—Medication prescribed to treat manic (excited) phases of bipolar disorder.

Myxedema—Hypothyroidism, characterized by thick, puffy features, an enlarged tongue, and lack of emotion.

Nephrosis—Any degenerative disease of the kidney (not to be confused with nephritis, an inflammation of the kidney due to bacteria).

Nodular goiter—An enlargement of the thyroid (goiter) caused when groups of cells collect to form nodules.

Phenytoin (Dilantin)—Anti-convulsive medication used to treat seizure disorders.

Propranolol (Inderal)—Medication commonly prescribed to treat high blood pressure; is a beta-adrenergic blocker and can also be used to treat irregular heartbeat, heart attack, migraine, and tremors.

Reserpine (Serpasil)—A drug prescribed for high blood pressure.

Salicylates—Aspirin and certain other nonsteroidal anti-inflammatory drugs (NSAIDs).

Thiazides—A group of drugs used to increase urine output.

Thyroid gland—A butterfly-shaped gland in front and to the sides of the upper part of the windpipe; influences body processes like growth, development, reproduction, and metabolism.

Thyroiditis—Inflammation of the thyroid gland.

Thyrotoxicosis—A condition resulting from high levels of thyroid hormones in the blood.

Toxic thyroid adenoma—Self-contained concentrations of thyroid tissue that may produce excessive amounts of thyroid hormone.

T3RU

A high degree of resin uptake and high thyroxine levels indicate hyperthyroidism. A low degree of resin uptake, coupled with low thyroxine levels, is a symptom of hypothyroidism.

Thyroxine and triiodothyronine resin uptake that are not both high or low may be a symptom of a thyroxine-binding abnormality.

LATS

Long-acting thyroid stimulator, not usually found in blood, is present in the blood of 80% of patients with Graves' disease. It is a symptom of this disease whether or not symptoms of hyperthyroidism are detected.

Resources**ORGANIZATIONS**

American Thyroid Association, Inc. Montefiore Medical Center, 111 E. 210th St., Bronx, NY 10467. <<http://www.thyroid.org>>.

Thyroid Foundation of America, Inc. Ruth Sleeper Hall, RSL350, 40 Parkman St., Boston, MA 02114-2698. (800) 832-8321. <<http://www.tfaeweb.org/pub/tfa>>.

Maureen Haggerty

Thyroid gland removal see **Thyroidectomy**

Thyroid hormones

Definition

Thyroid hormones are artificially made hormones that make up for a lack of natural hormones produced by the thyroid gland.

Purpose

The thyroid gland, a butterfly-shaped structure in the lower part of the neck, normally produces a hormone called thyroxine. This hormone controls the rate of metabolism – all the physical and chemical processes that occur in cells to allow growth and maintain body functions. When the thyroid gland does not produce enough thyroxine, body processes slow down. People with underactive thyroid glands feel unusually tired and may gain weight even though they eat less. They may also have trouble staying warm and may have other symptoms, such as dry skin, dry hair, and a puffy face. By making up for the

lack of natural thyroxine and bringing the rate of metabolism back to normal, artificially made thyroid hormone improves these symptoms.

Thyroid hormones also may be used to treat **goiter** (enlarged thyroid gland) and certain types of **thyroid cancer**.

Description

Thyroid hormones, also called thyroid drugs, are available only with a physician's prescription. They are sold in tablet form. A commonly used thyroid hormone is levothyroxine (Synthroid, Levoxyl, Levotheroid).

Recommended dosage

For adults and teenagers, the usual starting dose of levothyroxine tablets is 0.0125 mg (12.5 micrograms) to 0.05 mg (50 micrograms) per day. The physician who prescribes the medicine may gradually increase the dose over time.

For children, the dose depends on body weight and must be determined by a physician.

Taking thyroid hormones exactly as directed is very important. The physician who prescribes the medicine will figure out exactly how much of the medicine a patient needs. Taking too much or too little can make the thyroid gland overactive or underactive.

This medicine should be taken at the same time every day.

Precautions

People who take thyroid hormones because their thyroid glands do not produce enough natural hormone may need to take the medicine for the rest of their lives. Seeing a physician regularly while taking this medicine is important. The physician will make sure that the medicine is working and that the dosage is correct.

In patients with certain kinds of heart disease, this medicine may cause chest pains and **shortness of breath** during **exercise**. People who have this problem should be careful not to exert themselves too much.

Anyone who is taking thyroid hormones should be sure to tell the health care professional in charge before having any surgical or dental procedures or receiving emergency treatment.

This medicine is safe to take during **pregnancy**, but the dosage may need to be changed. Women who are pregnant should check with their physicians to make sure they are taking the proper dosage.

KEY TERMS

Adrenal glands—A pair of glands located next to the kidneys. The adrenal glands produce hormones that control many body functions.

Hormone—A chemical that is produced in one part of the body and then travels through the bloodstream to another part of the body where it has its effect.

Pituitary gland—A pea-sized gland at the base of the brain that produces many hormones that affect growth and body functions.

Anyone who has had unusual reactions to thyroid hormones in the past should let his or her physician know before taking the drugs again. The physician should also be told about any **allergies** to foods, dyes, preservatives, or other substances.

Before using thyroid hormones, people with any of these medical problems should make sure their physicians are aware of their conditions:

- heart disease
- high blood pressure
- hardening of the arteries
- diabetes
- history of overactive thyroid
- underactive adrenal gland
- underactive pituitary gland

Side effects

This medicine usually does not cause side effects if the dosage is right. Certain symptoms may be signs that the dose needs to be changed. Check with a physician if any of these symptoms occur:

- headache
- **fever**
- diarrhea
- vomiting
- changes in appetite
- weight loss
- changes in menstrual period
- tremors of the hands
- leg cramps

- increased sensitivity to heat
- sweating
- irritability
- nervousness
- sleep problems

Other side effects are possible. Anyone who has unusual symptoms while taking thyroid hormones should get in touch with his or her physician.

Interactions

Thyroid hormones may interact with other medicines. This may increase or decrease the effects of the thyroid medicine and may interfere with treatment. Anyone who takes thyroid hormones should not take any other prescription or nonprescription (over-the-counter) medicines without the approval of his or her physician. Among the drugs that may interact with thyroid hormones are:

- Medicine for colds, hay fever, and other allergies
- Medicine for **asthma** and other breathing problems
- Medicine for diabetes
- Blood thinners
- Amphetamines
- Diet pills (appetite suppressants)
- Cholesterol-lowering drugs such as cholestyramine (Questran) and colestipol (Colestid).

Nancy Ross-Flanigan

Thyroid nuclear medicine scan

Definition

A thyroid nuclear medicine scan is a diagnostic procedure to evaluate the thyroid gland, which is located in the front of the neck and controls the body's metabolism. A radioactive substance that concentrates in the thyroid is taken orally or injected into a vein (intravenously), or both. A special camera is used to take an image of the distribution of the radioactive substance in and around the thyroid gland. This is interpreted to evaluate thyroid function and to diagnose abnormalities.

Purpose

A thyroid scan may be ordered by a physician when the gland becomes abnormally large, especially if the enlargement is greater on one side, or when hard lumps (nodules) are felt. The scan can be helpful in determining whether the enlargement is caused by a diffuse increase in the total amount of thyroid tissue or by a nodule or nodules.

When other laboratory studies show an overactive thyroid (**hyperthyroidism**) or an underactive thyroid (**hypothyroidism**), a radioactive iodine uptake scan is often used to confirm the diagnosis. It is frequently done along with a thyroid scan.

Precautions

Women who are pregnant should not have this test.

Description

This test is performed in a radiology facility, either in an outpatient x ray center or a hospital department. Most often, the patient is given the radioactive substance in the form of a tasteless liquid or capsule. It may be injected into a vein (intravenously) in some instances. Images will be taken at a specified amount of time after this, depending on the radioisotope used. Most often, scanning is done 24 hours later, if the radioisotope is given orally. If it is given intravenously, the scan is performed approximately 20 minutes later.

For a thyroid scan, the patient is positioned lying down on his or her back, with the head tilted back. The radionuclide scanner, also called a gamma camera, is positioned above the thyroid area as it scans. This takes 30-60 minutes.

The uptake study may be done with the patient sitting upright in a chair or lying down. The procedure is otherwise the same as described for the thyroid scan. It takes approximately 15 minutes. There is no discomfort involved with either study.

A thyroid scan may also be referred to as a thyroid scintiscan. The name of the radioactive substance used may be incorporated and the study called a technetium thyroid scan or an iodine thyroid scan. The radioactive iodine uptake scan may be called by its initials, an RAIU test, or an iodine uptake test.

Preparation

Certain medications can interfere with iodine uptake. These include certain **cough** medicines, some

oral contraceptives, and thyroid medications. The patient is usually instructed to stop taking these medicines for a period of time before the test. This period may range from several days up to three to four weeks, depending on the amount of time the medicine takes to clear from the body.

Other nuclear medicine scans and x ray studies using contrast material performed within the past 60 days may affect this test. Therefore, patients should tell their doctors if they have had either of these types of studies before the thyroid scan is begun, to avoid inaccurate results.

Some institutions prefer that the patient have nothing to eat or drink after midnight on the day before the radioactive liquid or capsule is to be taken. A normal diet can usually be resumed two hours after the radioisotope is taken. Dentures, jewelry, and other metallic objects must be removed before the scanning is performed. No other physical preparation is needed.

The patient should understand that there is no danger of radiation exposure to themselves or others. Only very small amounts of radioisotope are used. The total amount of radiation absorbed is often less than the dose received from ordinary x rays. The scanner or camera does not emit any radiation, but detects and records it from the patient.

Aftercare

No **isolation** or special precautions are needed after a thyroid scan. The patient should check with his or her physician about restarting any medications that were stopped before the scan.

Risks

There are no risks with this procedure.

Normal results

A normal scan will show a thyroid of normal size, shape, and position. The amount of radionuclide uptake by the thyroid will be normal according to established laboratory figures. There will be no areas where radionuclide uptake is increased or decreased.

Abnormal results

An area of increased radionuclide uptake may be called a hot nodule or "hot spot." This means that a

KEY TERMS

Radioisotope—A radioactive or radiation-emitting form of an element.

Radionuclide—A substance that emits radiation as it disintegrates.

benign growth is overactive. Despite the name, hot nodules are unlikely to be caused by **cancer**.

An area of decreased radionuclide uptake may be called a cold nodule or “cold spot.” This indicates that this area of the thyroid gland is underactive. A variety of conditions, including cysts, nonfunctioning benign growths, localized inflammation, or cancer may produce a cold spot.

A thyroid nuclear medicine scan is rarely sufficient to establish a clear diagnosis. Frequently, the information revealed will need to be combined with data from other studies to determine the problem.

Resources

BOOKS

Goldsmith, Stanley J. “Endocrine System.” In *Nuclear Medicine*, edited by Donald R. Bernier, et al. St. Louis: Mosby, 1997.

Ellen S. Weber, MSN

Thyroid sonogram see **Thyroid ultrasound**

Thyroid storm see **Hyperthyroidism**

Thyroid ultrasound

Definition

Thyroid ultrasound is an imaging technique used for diagnosing suspected thyroid disease. It uses harmless, high-frequency sound waves to form an image. The sound waves are reflected by thyroid tissue to form a picture of internal structures. It is not invasive and involves no radiation.

Purpose

The thyroid gland is an organ located in front of the neck. It plays an important role in controlling the

body’s metabolism. Most thyroid ultrasounds are performed to evaluate a small lump (nodule) in the thyroid found during a **physical examination** or found by a radionuclide study (thyroid scan). The ultrasound can establish if the nodule is a cyst, which is an abnormal lump that contains fluid, or a solid mass. Cysts are almost always noncancerous (benign), although in some cases the fluid may be taken out for additional testing.

If there are several masses or nodules, this indicates the presence of enlargement of the thyroid gland (**goiter**). If there is only one mass, it may be cancerous and needs further evaluation. Specialized thyroid ultrasounds, such as color Doppler flow studies, can add valuable information. By showing an image of the blood circulation in the gland, this study can assess some ambiguous masses in greater detail, to further refine diagnosis. In some cases, a needle will be inserted to remove some tissue from the mass for evaluation in a laboratory (needle biopsy). Ultrasound is used during this procedure to help the physician guide the needle to the mass that needs to be evaluated.

Thyroid ultrasound can measure the size of the thyroid with great precision. Ultrasound studies may be done periodically to assess the response of the thyroid gland to medical therapy. An enlarged gland or a benign nodule should decrease in size when appropriate thyroid medication is taken.

Patients who have received therapeutic radiation to the head or neck may be monitored at regular intervals using thyroid ultrasound. The radiation puts these patients at higher risk for developing **thyroid cancer** or other abnormalities. In the early stages, these conditions may not cause symptoms or be apparent during a physical examination. They can, however, be detected by ultrasound.

Certain invasive medical procedures may be performed under ultrasound guidance. This is because ultrasound allows the physician to observe a needle as it enters body tissue below the skin. This is useful to direct the removal of fluid from a cyst (aspiration) or needle biopsy. Medications to treat recurrent cysts may be administered directly to the area using ultrasound guidance.

Precautions

Thyroid ultrasound is safe for people of all ages. It is the preferred procedure to evaluate suspected disease in pregnant women because no radiation is involved.

Description

The study may be done in an outpatient facility or in a hospital department. The patient lies on his or her back. A pillow or rolled towel is placed under the shoulders and upper back, allowing the head to tilt back (hyperextend). A gel that enhances sound transmission is spread over the thyroid area. The technologist then gently places a transducer, an instrument about the size of an electric shaver, against the skin. It is moved over the thyroid area. The images from reflected sound waves appear on a monitor screen. There is no discomfort involved with this study. The examination takes 15–30 minutes.

Preparation

Some facilities recommend limiting food and drink for one hour before the study to prevent discomfort. No other preparation is needed.

Aftercare

No special restrictions or procedures are needed after a thyroid ultrasound.

Risks

There are no risks with this procedure.

Normal results

A normal study would reveal a thyroid gland of normal size, shape, position, and uniform texture.

Abnormal results

A thyroid ultrasound may reveal cysts, solid masses that may or may not be cancerous, or an enlarged thyroid gland (goiter). In many cases, the ultrasound can establish a diagnosis. Sometimes the information revealed will need to be combined with data from other studies to determine the problem.

Resources

PERIODICALS

Rifat, Sami F., and Mack T. Ruffin. "Management of Thyroid Nodules." *American Family Physician* 50 (September 15, 1994): 785-791.

Ellen S. Weber, MSN

Thyroid x ray see **Thyroid nuclear medicine scan**

Thyroidectomy

Definition

Thyroidectomy is a surgical procedure in which all or part of the thyroid gland is removed. The thyroid gland is located in the forward part of the neck (anterior) just under the skin and in front of the Adam's apple.

Purpose

All or part of the thyroid gland may be removed to correct a variety of abnormalities of the gland. If the patient has a **goiter** (an enlargement of the thyroid gland, causing a swelling in the front of the neck), it may cause difficulties with swallowing or breathing. **Hyperthyroidism** (over-functioning of the thyroid gland) produces hypermetabolism (abnormally increased use of oxygen, nutrients, and other materials). If medication cannot adequately treat this condition, or if the patient is a child or pregnant, the thyroid gland must be removed. Both cancerous tumors and noncancerous tumors (frequently called nodules) can occur and they must be removed, in addition to some or all of the thyroid gland.

Precautions

There are definite risks associated with the procedure. Therefore, the thyroid gland should be removed only if there is a pressing reason or medical condition that requires it.

Description

Thyroidectomy is an operative procedure done most commonly by a general surgeon, or occasionally by an otolaryngologist, in the operating room of a hospital. The operation begins when an anesthesiologist puts the patient to sleep. The anesthesiologist injects drugs into the patient's veins and then places an airway tube in the windpipe to ventilate (provide air for) the patient. The surgeon makes an incision in the front of the neck where a tight-fitting necklace would rest. He locates and takes care not to injure the parathyroid glands and the recurrent laryngeal nerves, while freeing the thyroid gland from these surrounding structures. The blood supply to the portion of the thyroid gland that is to be removed is clamped off. Then all or part of the gland is removed. If **cancer** is present, all, or almost all, of the gland is removed. If other diseases or a nodule is present, the surgeon may remove only part of the gland. The total amount of

thyroid gland removed depends upon the thyroid disease being treated. A drain (a soft plastic tube that drains fluid out of the area) may be placed before the incision is closed. The incision is closed either with sutures (stitches) or metal clips. A dressing is placed over the incision and the drain, if one is used.

Patients generally stay in the hospital one to four days after completion of the operation.

Preparation

Before a thyroidectomy is performed, a variety of tests and studies are usually required to determine the nature of the thyroid disease. Laboratory analysis of blood determines the levels of active thyroid hormone circulating in the body. Sonograms and computed tomography scans (CT scans) help to determine the size of the thyroid gland and location of abnormalities. A thyroid nuclear medicine scan assesses the function of the gland. A needle biopsy of an abnormality or aspiration (removal by suction) of fluid from the thyroid gland may also be done to help determine the diagnosis.

If the diagnosis is hyperthyroidism, the patient may be asked to take antithyroid medication or iodides before the operation; or continued treatment with antithyroid drugs may be the treatment of choice. Otherwise, no other special procedure must be followed prior to the operation.

Aftercare

The incision requires little to no care after the dressing is removed. The area may be bathed gently with a mild soap. The sutures or the metal clips are removed three to seven days after the operation.

Risks

As with all operations, patients who are obese, smoke, or have poor **nutrition** are at greater risk for developing complications related to the general anesthetic itself.

Hoarseness or voice loss may develop if the recurrent laryngeal nerve was injured or destroyed during the operation. This is more apt to occur in patients who have large goiters or cancerous tumors.

Hypoparathyroidism (under-functioning of the parathyroid glands) can occur if the parathyroid glands are injured or removed at the time of the thyroidectomy.

Hypothyroidism (under-functioning of the thyroid gland) can occur if all or nearly all of the thyroid gland

KEY TERMS

Endocrinologist—A physician who specializes in treating patients who have diseases of the thyroid, parathyroid, adrenal glands, and/or the pancreas.

Hyperthyroidism—Abnormal over-functioning of the thyroid glands. Patients are hypermetabolic, lose weight, are nervous, have muscular weakness and fatigue, sweat more, and have increased urination and bowel movements. This is also called thyrotoxicosis.

Hypothyroidism—Abnormal under-functioning of the thyroid gland. Patients are hypometabolic, gain weight, and are sluggish.

Recurrent laryngeal nerve—A nerve which lies very near the parathyroid glands and serves the larynx or voice box.

is removed. This may be intentional when the diagnosis is cancer. If the patient's thyroid levels remain high, he may be required to take thyroid replacement for the rest of his life.

The neck and the area surrounding the thyroid gland have a rich supply of blood vessels. Bleeding in the area of the operation may occur and be difficult to control or stop. Rarely is a blood **transfusion** required, although a hematoma (collection of blood) may develop. If this occurs, it may be life-threatening. As the hematoma enlarges, it may obstruct the airway and cause the patient to stop breathing. If a hematoma does develop in the neck, it may require drainage to clear the airway.

Wound infections can occur. If they do, the incision is drained, and there are usually no serious consequences.

Normal results

Most patients are discharged from the hospital one to four days after the procedure. Most resume their normal activities two weeks after the operation. Patients who have cancer may require subsequent treatment by an oncologist or an endocrinologist.

Resources

OTHER

"Thyroid Gland Removal." *ThriveOnline*. <<http://thriveonline.oxygen.com>>.

Mary Jeanne Krob, MD, FACS

Thyroiditis

Definition

Thyroiditis is inflammation of the thyroid gland, a butterfly-shaped organ next to the windpipe.

Description

The thyroid is the largest gland in the neck. It produces, secretes, and stores thyroxine (T4), a hormone that influences the metabolism of just about every body process.

When the thyroid gland is functioning properly, hormone release is carefully regulated. When bacteria or viruses invade and inflame the gland, T4 surges into the bloodstream and raises hormone levels that then discourage the gland from creating more T4. Eventually the hormone stores are exhausted, the thyroid loses its ability to manufacture T4, and an underactive thyroid (**hypothyroidism**) results.

The major cause of hypothyroidism, thyroiditis affects about 12 million people in the United States. This condition is more common in women than in men and usually develops between ages 30–50.

Hashimoto's disease

The most common type of thyroiditis is Hashimoto's disease, a painless disorder also known as:

- autoimmune thyroiditis
- chronic lymphocytic thyroiditis
- lymphadenoid **goiter**
- Struma lymphomatosa

Hashimoto's disease can develop at any age, but is most common in middle-aged women. This immune system disorder runs in families, and affects about 5% of adults in the United States.

Hashimoto's disease slowly destroys thyroid tissue and robs the gland of its ability to change iodine into T4. The condition progresses so gradually that many people who have it do not realize anything is wrong until the enlarged gland forms a goiter, a swelling seen and felt in the front of the neck. This may not happen until weeks or even years after an individual develops Hashimoto's.

Subacute thyroiditis

Much less common than Hashimoto's disease, subacute thyroiditis is a painful inflammation that develops suddenly in a patient who has had a viral

infection, such as **mumps** or an upper respiratory illness. Pain radiates throughout the neck and patients feel ill and feverish. It may take as long as several months for normal thyroid function to resume.

Subacute thyroiditis is also called:

- DeQuervain's thyroiditis
- giant cell thyroiditis
- granulomatous thyroiditis
- subacute granulomatous thyroiditis

Silent thyroiditis

The least common of the three major types, silent thyroiditis is characterized by rigidity and slight enlargement of the thyroid gland. Postpartum thyroiditis, a form of silent thyroiditis, develops in 5–9% of all women who have recently given birth. Postpartum thyroiditis develops within a year of the baby's birth and disappears within six months.

Acute thyroiditis

Caused by acute infection, this rare disease is a medical emergency. A patient who has acute thyroiditis has a high **fever** and feels very ill. The neck is red, hot, and very tender.

Causes and symptoms

Hashimoto's disease

Hashimoto's disease develops when the immune system attacks the thyroid gland. It may be related to such hormone-related (endocrine system) disorders as:

- Addison's disease. This condition, caused by malfunction of the adrenal gland, is characterized by weakness, loss of weight and appetite, and increased sensitivity to cold.
- Diabetes mellitus. This metabolic disorder is caused by a lack of insulin production or by the body's inability to process insulin.
- Graves' disease. This disease is the most common form of **hyperthyroidism**.
- Vitiligo. This is a noncancerous skin disease characterized by unpigmented patches of skin.

Being female and having a family history of Hashimoto's thyroiditis increases the likelihood of developing the disease. Its symptoms include:

- constipation
- fatigue
- goiter or enlarged neck

- inability to tolerate cold temperatures
- weight gain, some patients' faces swell and their joints stiffen.

Subacute thyroiditis

Characterized by painful gland enlargement that is sometimes mistaken for a **sore throat** that may last for months, subacute thyroiditis often follows:

- influenza
- mumps
- upper respiratory infections
- viruses that produce cold symptoms and inflammation of the membrane that protects the brain (**meningitis**), inflammation of the sac that surrounds the heart (pericarditis), inflammation of the heart muscle (**myocarditis**), and other diseases

People who have subacute thyroiditis feel feverish, weak, and tired. The thyroid is sore to the touch. They may be nervous, sweat, and have trouble tolerating heat or swallowing. Symptoms of subacute thyroiditis also include:

- rapid heartbeat
- tremors
- weight loss.

Silent thyroiditis

The cause of silent thyroiditis is uncertain, but the condition is believed to be an immune-system disorder triggered by **childbirth**. Although silent thyroiditis is painless, the condition's other symptoms are similar to those of subacute thyroiditis. The thyroid gland enlarges only slightly, and the eyes do not bulge.

Diagnosis

Family physicians and endocrinologists usually base a diagnosis of thyroiditis on:

- blood levels of **thyroid hormones**, thyroid-stimulating hormone, and anti-thyroid antibodies
- personal and family medical history
- the appearance of a patient's thyroid gland

Thyroid antibodies present in 95% of patients with Hashimoto's thyroiditis make it possible to diagnose this disease without surgery or biopsy. A blood test that measures sedimentation rate, an indication of the extent of inflammation, is a useful tool for diagnosing subacute thyroiditis.

Treatment

Medical therapy for thyroiditis includes:

- **antibiotics** to fight infection
- high doses of **aspirin** to relieve inflammation
- hormones to suppress or replace thyroid function
- **pain** medications

Cortisone drugs are sometimes prescribed to reduce persistent inflammation. In rare instances, surgery can be used to drain infection or relieve pressure near the thyroid gland.

Hashimoto's disease

The goal of treatment for Hashimoto's disease is to prevent the thyroid gland from getting larger. Regular monitoring may be the only treatment indicated for patients whose gland is only slightly enlarged, and who show no signs of hormone deficiency. Levothyroxine (Synthroid) may be prescribed to correct hormone deficiency in a patient who has a large goiter.

Subacute thyroiditis

The goal of treatment for subacute thyroiditis is to relieve pain, reduce inflammation, and regulate hyperthyroidism. Bed rest and **beta blockers** (propranolol, naldolol) may be necessary until thyroid activity is controlled, and the patient may have to take:

- anti-inflammatory medication for several weeks
- high doses of aspirin
- other **analgesics**

If subacute thyroiditis continues for a long time, cortisone and thyroid hormone medication may be prescribed to relieve inflammation and allow the gland to rest. Glucocorticoids (prednisone) are prescribed for symptoms that do not respond to other treatment. The original problem often becomes more pronounced after these medications are discontinued.

Silent thyroiditis

Most patients who have silent thyroiditis don't need any treatment, but:

- bed rest and beta blockers are occasionally needed to regulate rapid heart beat
- inderal (propanolol) may be prescribed for brief periods of hyperthyroidism
- steroids may be prescribed for severe episodes of acute inflammation

KEY TERMS

Addison's disease—A disease that results from a deficiency in adrenocortical hormones.

Diabetes mellitus—A disorder of the pancreas. This chronic disorder of carbohydrate metabolism results in hyperglycemia and glycosuria.

Goiter—An abnormal enlargement of the thyroid gland.

Graves' disease—Also called hyperthyroidism, this disease results from overactivity of the thyroid gland.

Subacute—An abnormal condition present in a person who appears to be clinically well.

Vitiligo—A benign skin disease that results in irregular patches of skin that are totally lacking in color.

Acute thyroiditis

Acute thyroiditis requires emergency treatment with antibiotics and surgery.

Prognosis

Thyroiditis usually responds to treatment, and some patients recover normal thyroid function without treatment. Because permanent loss of thyroid function is a possibility and life-long thyroid replacement therapy may be necessary, regular medical monitoring should continue even after the patient has apparently recovered.

Hashimoto's disease

Some cases of Hashimoto's disease remain stable for years. Others slowly progress to hypothyroidism, which is treated with thyroid hormone replacement therapy.

Subacute thyroiditis

Most patients with subacute thyroiditis recover fully after no more than a few months. This condition occasionally recurs, but severe or long-term complications are rare.

Silent thyroiditis

Four of every five patients with silent thyroiditis recover completely within three months. The thyroid status of these patients should be evaluated within

12 months. Because silent thyroiditis recurs in 10% of patients within three years and may progress to hypothyroidism, medical monitoring should continue for three years after recovery appears complete.

Prevention

Flu shots or immunizations for **measles**, mumps, and rubella may help prevent conditions associated with subacute thyroiditis. There is no known way to prevent other forms of thyroiditis.

Resources

ORGANIZATIONS

Thyroid Foundation of America. 350 Ruth Sleeper Hall, Parkman St., Boston, MA 02114. (800) 232-8321. < <http://www.clark.net/pub/tfa> >.

Thyroid Society for Education and Research. 7515 South Main St., Suite 545, Houston, TX 77030. (800) 849-7643. < <http://the-thyroid-society.org/thyroid.html> >.

Maureen Haggerty

Thyrotoxicosis see **Hyperthyroidism**

Thyroxine-binding globulin test see **Thyroid function tests**

Thyroxine test see **Thyroid function tests**

TIA see **Transient ischemic attack**

Tic douloureux see **Trigeminal neuralgia**

Tick fever see **Relapsing fever**

Tilt table test

Definition

The tilt table test is a test in which a patient is positioned in a supine position and brought to a predetermined angle or angles from the horizontal position. Such positioning helps to determine the cause of any decrease in oxygen to the brain. Different types of drugs may also be used in the testing process.

Purpose

The purpose of the tilt table test is to help determine appropriate therapy for individuals with **fainting** (syncope) and presyncope of unexplained origin.

Precautions

Precautions are few with the tilt table test. However, when any drug is used with this test, the appropriate precautions for that particular drug should be observed. For example, when isoproterenol or similar drugs are used during the tilt table test, the taking of non-prescription drugs for **asthma**, **cough**, cold, or allergy; appetite suppressants; sleeping pills; or drugs containing **caffeine** should be made known to the physician prior to the test. Likewise, the physician should be informed of any **allergies** to any sympathomimetic drugs, including several of the diet pills on the market. The physician should be told of any serious heart-rhythm disorders.

Description

Syncope is described as a pathological brief loss of consciousness caused by a temporary deficiency of oxygen in the brain. Previous studies have shown the effectiveness of tilt table testing in establishing the diagnosis of neurocardiogenic syncope, and in dictating therapy in patients with syncope of unknown origin. Despite its usefulness, small numbers of patients and brief followup reports have limited the majority of studies. Sensitivity-enhancing techniques, such as the administration of isoproterenol, are applied in specific cases to children and young adults to compensate for the otherwise low sensitivity (20-30%) observed in that population.

Preparation

In order for a patient to make informed decisions about any diagnostic test or procedure, there are important questions that need to be asked prior to the procedure. The information gained will be helpful for that patient in determining benefits, risks, and cost of the procedure, and alternatives. The patient should understand the purpose of the tilt table test, and the diagnosis that the physician is trying to confirm or rule out. If the tilt table test is positive, the patient should ask questions about the frequency of false-positive results for that particular tilt table procedure, and should inquire about the next step in treatment.

Aftercare

After the procedure, the patient is asked to transfer from the supine position to a sitting position, and is observed for a short period of time. During this time and after several minutes in the sitting position, any symptoms of **dizziness** and vertigo are noted.

KEY TERMS

Sympathomimetic—Denoting a drug that mimics the effects of stimulation of organs and structures by the sympathetic nervous system. The sympathetic nervous system pertains to the part of the nervous system originating in the thoracic and lumbar regions of the spinal cord. In general, it inhibits or opposes the physiological effects of another aspect of the nervous system, as in tending to reduce digestive secretions, speed up the heart, and contract the blood vessels.

Syncope—A loss of consciousness over a short period of time, caused by a temporary lack of oxygen in the brain.

Vertigo—The sensation of dizziness.

When ready, the individual transfers from the sitting position to standing. After additional observation and taking of vital signs, the individual is allowed to go home.

Risks

Risks of the tilt table test are low, but do include significant changes in blood pressure while in the supine position, and any adverse reactions to any drugs administered during the tilt table test.

Normal results

Normal results of the tilt table test should help the physician in assessing what may or may not be the cause of the syncope.

Abnormal results

Abnormal results include any pathologic reactions to the position changes or sensitivity enhancing techniques, such as the administration of isoproterenol or other related drugs.

Resources

ORGANIZATIONS

American Medical Association. 515 N. State St., Chicago, IL 60612. (312) 464-5000. <<http://www.ama-assn.org>>.

Jeffrey P. Larson, RPT

Tinea cruris see **Ringworm**

Tinea pedis see **Athlete's foot**

Tingling see **Numbness and tingling**

Tinnitus

Definition

Tinnitus is hearing ringing, buzzing, or other sounds without an external cause. Patients may experience tinnitus in one or both ears or in the head.

Description

Tinnitus affects as many as 40 million adults in the United States. It is defined as either objective or subjective. In objective tinnitus, the doctor can hear the sounds, as well as the patient. Objective tinnitus is typically caused by tumors, turbulent blood flow through malformed vessels, or by rhythmic muscular spasms. Most cases of tinnitus are subjective, which means that only the patient can hear the sounds.

Causes and symptoms

Subjective tinnitus is frequently associated with hearing loss. About 90% of patients have sensorineural **hearing loss**; 5% suffer from conductive hearing loss; 5% have normal hearing. The causes of subjective tinnitus include:

- impacted ear wax
- ear infections
- hardening of the structures of the inner ear
- hearing loss related to age or excessive noise
- ototoxic medications, including **aspirin**, quinine, some diuretics, heavy metals, alcohol, and certain antibiotics
- meniere's syndrome
- head trauma
- systemic diseases, including **syphilis**, **hypertension**, anemia, or hypothyroidism
- tumors of the ear

Diagnosis

Diagnosis of tinnitus includes a **physical examination** of the patient's head and neck. The doctor will use

an otoscope to examine the ears for wax, infection, or structural changes. He or she will also use a stethoscope to listen to the blood vessels in the neck. Additional tests may include the following:

Tuning fork tests

The Rinne and Weber tests are commonly used to evaluate the type and severity of hearing loss. In the Weber test, the doctor holds a tuning fork against the patient's forehead or front teeth. If the hearing loss is sensorineural, the sound radiates to the ear with better hearing; if the hearing loss is conductive, the sound will be louder in the damaged ear. In the Rinne test, the tuning fork is placed alternately on the mastoid bone (behind the ear) and in front of the ear. In conductive hearing loss, bone conduction (BC) is greater than air conduction (AC). In sensorineural hearing loss, AC is greater than BC.

Diagnostic imaging

Magnetic resonance **angiography** or **venography** (MRA and MRV) can be used to evaluate malformations of the blood vessels. **Computed tomography scans** (CT scans) or magnetic resonance imaging scans (MRIs) can be used to locate tumors or abnormalities of the brain stem.

Blood tests

The doctor may order a complete **blood count** (CBC) with specific antibody tests to rule out syphilis or immune system disorders.

Treatment

Some cases of tinnitus can be treated by removal of the underlying cause. These include surgical treatment of impacted ear wax, tumors, head injuries, or malformed blood vessels; discontinuance of ototoxic medications; and antibiotic treatment of infections.

Subjective tinnitus, especially that associated with age-related hearing loss, can be treated with **hearing aids**, noise generators or other masking devices, bio-feedback, antidepressant medications, or lifestyle modifications (elimination of **smoking**, coffee, and aspirin).

Alternative treatment

A variety of alternative therapies may be helpful in the treatment of tinnitus. Dietary adjustments, including the elimination of coffee and other

KEY TERMS

Conductive hearing loss—Hearing loss caused by loss of function in the external or middle ear.

Meniere's syndrome—A disease of the inner ear, marked by recurrent episodes of loss of balance (vertigo) and roaring in the ears lasting several hours. Its cause is unknown.

Ototoxic—Damaging to the nerves controlling the senses of hearing and balance.

Sensorineural hearing loss—Hearing loss caused by damage to the nerves or parts of the inner ear governing the sense of hearing.

stimulants, may be useful, since stimulants can make tinnitus worse. In addition, reducing the amount of fat and cholesterol in the diet can help improve blood circulation to the ears. Nutritional supplementation with vitamin C, vitamin E, **B vitamins**, calcium, magnesium, potassium, and essential fatty acids is also recommended. Ginkgo (*Ginkgo biloba*) is often suggested, since it is believed to enhance circulation to the brain. **Acupuncture** treatments may help decrease the level of tinnitus sounds the patient hears, and constitutional homeopathic treatment may also be effective.

Prognosis

The prognosis depends on the cause of the tinnitus and the patient's emotional response. Most patients with subjective tinnitus do not find it seriously disturbing, but about 5% have strong negative feelings. These patients are frequently helped by instruction in relaxation techniques.

Resources

BOOKS

Jackler, Robert K., and Michael J. Kaplan. "Ear, Nose, & Throat." In *Current Medical Diagnosis and Treatment*, 1998, edited by Stephen McPhee, et al., 37th ed. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Tissue compatibility see **Tissue typing**

Tissue plasminogen activator see
Thrombolytic therapy

Tissue typing

Definition

Tissue typing is a group of procedures that determines the type of histocompatibility antigens on a person's cells or tissues. This procedure is typically used prior to transplantation of tissues or organs.

Purpose

Tissue typing is done prior to transplantation to ensure as close a match as possible between the donor and the recipient. If the histocompatibility antigens do not match well, there is a much greater chance that the recipient will reject the donated tissue.

Histocompatibility antigens are molecules on the surface of all cells in the body. The specific types of histocompatibility antigens present on a person's cells determine their identity and distinguish each person. They are a "fingerprint."

Each person has a unique set of histocompatibility antigens. If the antigens on tissue or organs from a donor do not match that of the recipient, a rejection response can occur. The recipient's immune system will detect the difference between the two sets of antigen and start a rejection response to kill the donated tissue. Except in the case of identical twins, no two people are identical in terms of their histocompatibility antigen types. However, the closer two tissues come to matching, the more likely the recipient will accept the donated tissue or organ.

Human Lymphocyte Antigens (HLA) is the name given to the most commonly used histocompatibility antigens. The antigens can be grouped into two classes: class I antigens are found on almost all cells, and class II antigens are normally found only on B lymphocytes, macrophages, monocytes, dendritic cells, and endothelial cells.

Description

Generally, typing is performed on blood cells because they are an easy sample to obtain. Blood is withdrawn from a vein in the forearm, and the cells are separated. There are a number of different techniques used to identify the antigens on the cells. Typically, specific antibodies react with the cells. Each antibody preparation is specific for one histocompatibility antigen. If the antigen is present, the antibody will bind to it. Laboratory instruments are used to detect antibody binding to the cells. Class II antigens are determined by the mixed lymphocyte reaction (MLR) or by a polymerase chain reaction (PCR). In the mixed lymphocyte

KEY TERMS

Antibody—A molecule produced by the body that is part of the immune response to attack antigens.

Antigen—A molecule that causes the body to produce an immunological response to attack the antigen.

Cornea—The transparent outer layer of the eye. It covers the iris and lens.

Lymphocyte—A class of white blood cells that are responsible for creating the immune response to antigens.

reaction, lymphocyte replication occurs if there is a mismatch, and is detected by a specific assay. The PCR test is a new DNA-based test that can detect the presence or absence of antigens by determining whether cells have the genes for the antigens.

One type of transplant does not require tissue typing. In the case of corneal transplants, tissue typing is not needed because cornea do not have their own blood supply. This greatly reduces the chance that immune cells will come in contact with the cornea and recognize it as foreign. For this reason, corneas can be transplant from any person, and there is little chance of rejection.

Normal results

Because each person has their own histocompatibility antigen “fingerprint,” there is no true normal result. Each fingerprint is unique.

Resources

BOOKS

Berkow, Robert, editor. *Merck Manual of Medical Information*. Whitehouse Station, NJ: Merck Research Laboratories, 1997.

John T. Lohr, PhD

TMJ see **Temporomandibular joint disorders**

Tobramycin see **Aminoglycosides; Antibiotics, ophthalmic**

Tocopherol deficiency see **Vitamin E deficiency**

Toenail removal see **Nail removal**

Tonsil removal see **Tonsillectomy and adenoidectomy**

Tonsillectomy and adenoidectomy

Definition

Tonsillectomy and adenoidectomy (T & A) are surgical procedures to remove the tonsils from the back of the mouth or adenoids from the back of the nasal cavity – both are part of the lymphatic system, which is responsible for fighting infection. These operations are often performed together and are usually done on children. T & As are the most common childhood operations.

Purpose

Tonsillectomy

Tonsils are removed (with or without the adenoids) when the child has any of the following conditions:

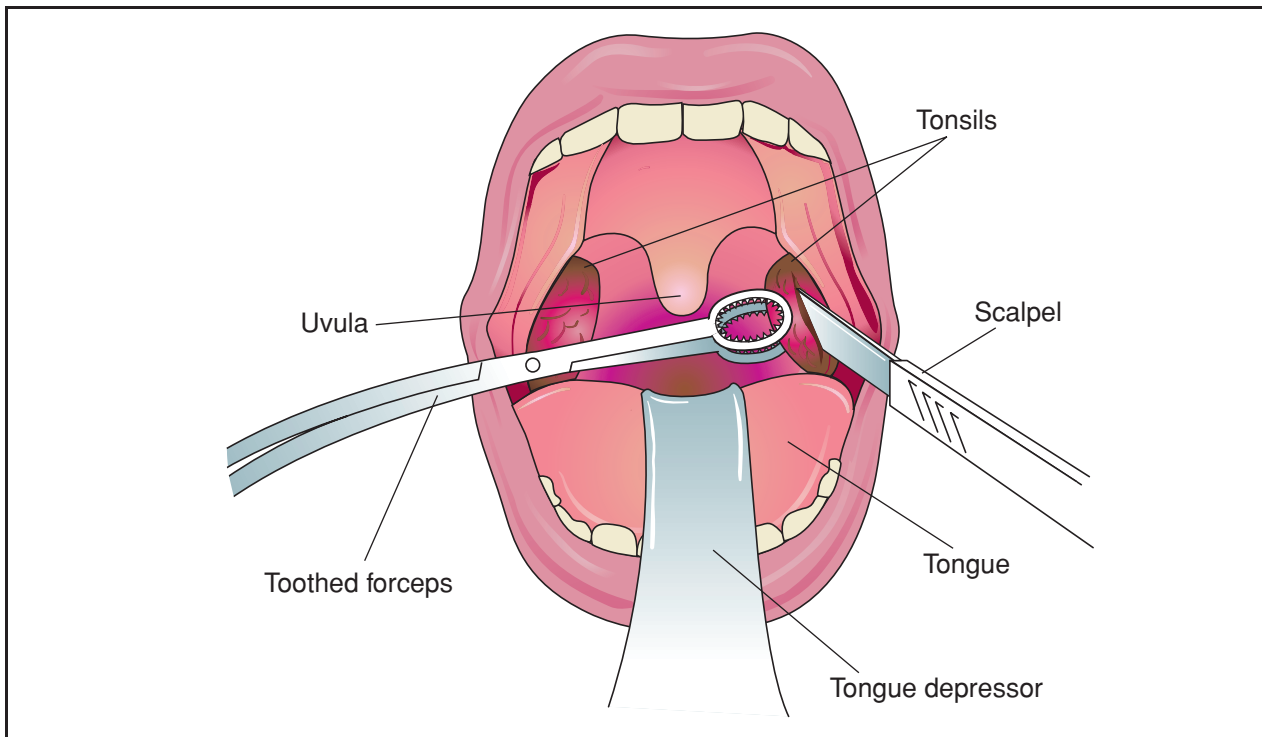
- obstruction
- sleep apnea. This is a condition in which the child snores loudly and stops breathing temporarily at intervals during sleep
- inability to swallow properly because of enlarged tonsils
- “hot potato” voice (breathy voice) and other speech abnormalities due to enlarged tonsils
- recurrent or persistent abscesses or throat infections

Doctors do not agree completely on the number of sore throats that make a tonsillectomy necessary. Most would agree that four cases of **strep throat** in any one year; six or more episodes of **tonsillitis** in one year; or five or more episodes of tonsillitis per year for two years indicate that the tonsils should be removed.

Adenoidectomy

Adenoids are removed (with or without the tonsils) when the child has any of the following conditions:

- alteration of facial growth because of enlarged adenoids
- upper airway obstruction



Tonsillectomy and adenoidectomy are surgical procedures performed to remove the tonsils or adenoids. The tonsils are removed in cases where they are a source of recurrent infection or have developed an abscess. Both operations are typically performed on children. The illustration above shows a tonsillectomy in progress. (Illustration by Electronic Illustrators Group.)

- development of an irregular bite (dental malocclusion)
- difficult speech or swallowing

Precautions

T & As are not performed as frequently today as they were in the past. One reason for a more conservative approach is that there is always some risk involved when a patient is put under **general anesthesia**.

In some cases, a T & A may need to be modified or postponed:

- children with cleft palates should not have the adenoids removed
- bleeding disorders; these must be brought under control before surgery
- acute tonsillitis; surgery should be postponed—usually for three to four weeks—until the infection is gone

Description

Tonsillectomies are hospital procedures. In adults, they may be performed under **local anesthesia**.

Children are usually placed under general anesthesia. The doctor depresses the tongue in order to see the throat and removes the tonsils with a scooplike instrument. The adenoids are usually removed through the nose.

Aftercare

Patients are turned on the side after the operation to prevent the possibility of blood being drawn into the lungs. The patient's vital signs are checked. After the patient is fully awake, he or she can drink water and other nonirritating liquids.

Adult patients are usually warned to expect some bleeding after the operation and a very **sore throat**. **Antibiotics** are given to prevent infection. Medications to relieve **pain** may also be given. For at least the first 24 hours, the patient is fed soft or pureed foods and fluids. If the adenoids alone were removed, the patient may be allowed solid food the day after surgery.

Patients are usually sent home the next day, with instructions to call the doctor if there is bleeding, an earache, or a **fever** that lasts longer than three days.

KEY TERMS

Abscess—A localized area of tissue destruction and pus formation.

Adenoids—Masses of lymphoid tissue that are found in the upper throat.

Sleep apnea—A condition marked by loud snoring during sleep and periodic episodes of suspended breathing.

Tonsils—Oval masses of lymphoid tissue on each side of the throat.

They are told to expect a white scab to form in the throat between five and 10 days after surgery.

Risks

About one in every fifteen thousand tonsillectomies ends in **death**, either from the anesthesia or from bleeding to death five to seven days after the operation. There is also a chance that children with previously normal speech will develop a nasal-sounding voice. In addition, children younger than five years may be badly emotionally upset by the hospital experience.

Normal results

Normal results include the correction of the condition for which the surgery was performed.

Resources

BOOKS

Berman, Stephen, and Ken Chan. "Ear, Nose, & Throat." In *Current Pediatric Diagnosis & Treatment*, edited by William W. Hay, Jr., et al. Stamford: Appleton & Lange, 1997.

Rebecca J. Frey, PhD

Tonsillitis

Definition

Tonsillitis is an infection and swelling of the tonsils, which are oval-shaped masses of lymph gland tissue located on both sides of the back of the throat.

Description

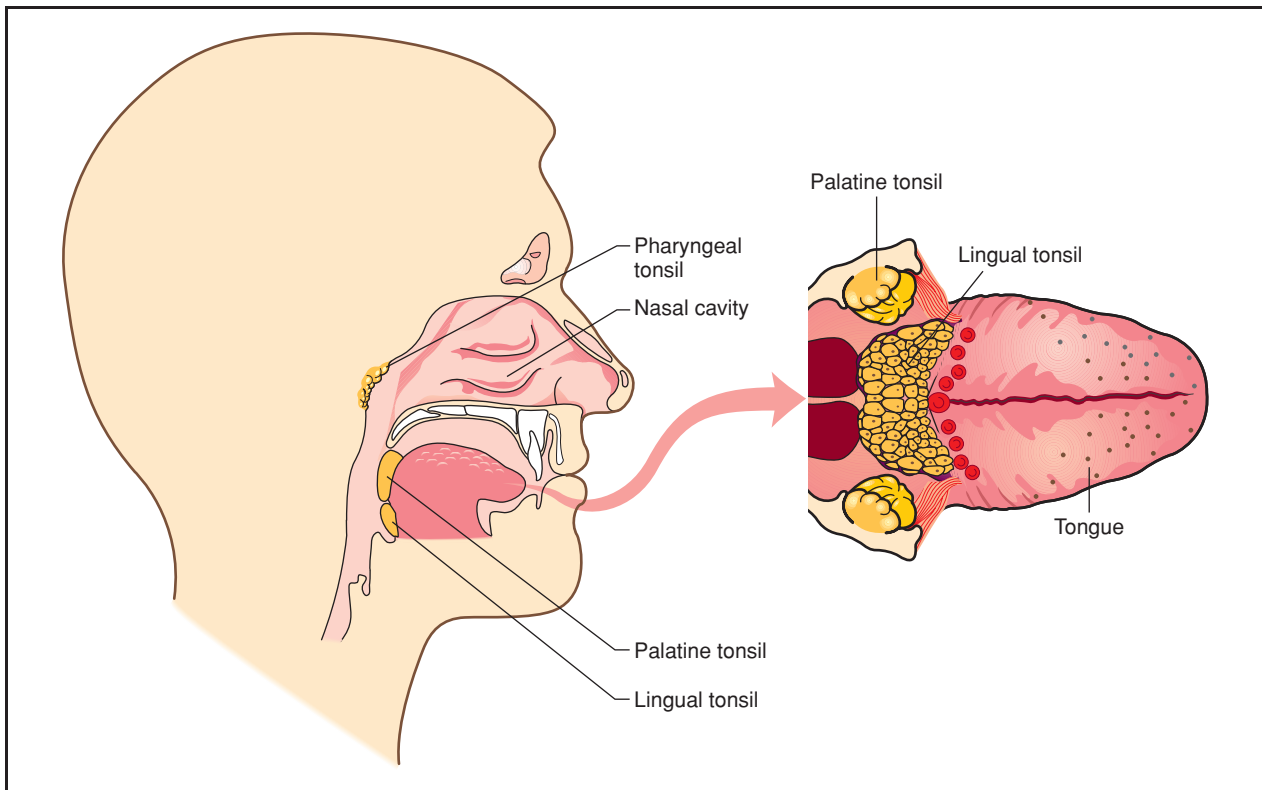
The tonsils normally help to prevent infections. They act like filters to trap bacteria and viruses entering the body through the mouth and sinuses. The tonsils also stimulate the immune system to produce antibodies to help fight off infections. Anyone of any age can have tonsillitis; however, it is most common in children between the ages of five and 10 years.

Causes and symptoms

Tonsillitis is caused by viruses or bacteria that cause the tonsils to swell and become inflamed. A mild or severe **sore throat** is one of the first symptoms of tonsillitis. Symptoms can also include **fever**, chills, tiredness, muscle aches, earache, **pain** or discomfort when swallowing, and swollen glands in the neck. Very young children may be fussy and stop eating. When a doctor or nurse looks into the mouth with a flashlight, the tonsils may appear swollen and red. Sometimes, the tonsils will have white or yellow spots or flecks or a thin coating. Symptoms usually last four to six days.

Diagnosis

The diagnosis of tonsillitis is made from the visible symptoms and a **physical examination** of the patient. The doctor will examine the eyes, ears, nose, and throat, looking at the tonsils for signs of swelling, redness, or a discharge. A careful examination of the throat is necessary to rule out **diphtheria** and other conditions that may cause a sore throat. Since most sore throats in children are caused by viruses rather than bacteria, the doctor may take a **throat culture** in order to test for the presence of streptococcal bacteria. A throat culture is performed by wiping a cotton swab across the tonsils and back of the throat, and sending the swab to a laboratory for culturing. *Streptococcus pyogenes*, the bacterium that causes strep throat, is the most common disease agent responsible for tonsillitis. Depending on what type of test is used for strep, the doctor may be able to determine within a few minutes if *S. pyogenes* is present. The quick tests for strep are not as reliable as a laboratory culture, which can take 24–48 hours. If the results of a quick test are positive, however, the doctor can prescribe **antibiotics** right away. If the quick test results are negative, the doctor can do a throat culture to verify the results and wait for the laboratory report before prescribing antibiotics. A blood test may also be done to rule out a more serious infection or condition, and to check the white blood cell count to see if the body is responding to the infection. In some cases, the doctor



The palatine, lingual, and pharyngeal tonsils. (Illustration by Hans & Cassady, Inc.)



An examination of this patient's mouth reveals acute tonsillitis. (Custom Medical Stock Photo. Reproduced by permission.)

may order blood tests for mononucleosis, since about a third of patients with mononucleosis develop **streptococcal infections** of the tonsils.

Treatment

Treatment of tonsillitis usually involves keeping the patient comfortable while the illness runs its course. This supportive care includes bed rest,

drinking extra fluids, gargling with warm salt water, and taking pain relievers—usually NSAIDs—to reduce fever. Frozen juice bars and cold fruit drinks can bring some temporary relief of sore throat pain; drinking warm tea or broth can be soothing. If the throat culture shows that *S. pyogenes* is present, penicillin or other antibiotics will be prescribed. An injection of benzathine or procaine penicillin may be most effective in treating the infection, but it is also painful. If an oral antibiotic is prescribed, it must be taken for the full course of treatment, usually 10-14 days. If the patient has several episodes of severe tonsillitis, the doctor may recommend a **tonsillectomy**, which is the surgical removal of the tonsils.

Alternative treatment

Strengthening the immune system is important whether tonsillitis is caused by bacteria or viruses. Naturopaths often recommend dietary supplements of vitamin C, bioflavonoids, and beta-carotenes—found naturally in fruits and vegetables—to ease inflammation and fight infection. A variety of herbal remedies also may be helpful in treating tonsillitis. Calendula (*Calendula officinalis*) and cleavers (*Galium aparine*) target the lymphatic system, while echinacea (*Echinacea*

KEY TERMS

Streptococcus pyogenes—A common bacterium that causes strep throat and can also cause tonsillitis.

Tonsillectomy—A surgical procedure to remove the tonsils if the patient has recurrent sore throats or throat infections, or if the tonsils have become so swollen that the patient has trouble breathing or swallowing.

Tonsils—Oval-shaped masses of glandular tissue located on both sides at the back of the throat. Tonsils act like filters to trap bacteria and viruses.

spp.) and astragalus (*Astragalus membranaceus*) stimulate the immune system. Goldenseal (*Hydrastis canadensis*), myrrh (*Commiphora molmol*), and bitter orange act as antibacterials. *Lomatium dissectum* and *Ligusticum porteri* have an antiviral action. Some of the homeopathic medicines that may be used to treat symptoms of tonsillitis include *Belladonna*, *Phytolacca*, *Mercurius*, *Lycopodium*, *Lachesis*, *Hepar sulphuris*, *Arsenicum*, or *Rhus toxicodendron*. As with any condition, the treatment and dosage should be appropriate for the particular symptoms and age of the patient.

Prognosis

Tonsillitis usually resolves within a few days with rest and supportive care. Treating the symptoms of sore throat and fever will make the patient more comfortable. If fever persists for more than 48 hours, however, or is higher than 102°F, the patient should be seen by a doctor. If antibiotics are prescribed to treat an infection, they should be taken as directed for the complete course of treatment, even if the patient starts to feel better in a few days. Prolonged symptoms may indicate that the patient has other upper respiratory infections, most commonly in the ears or sinuses. An **abscess** behind the tonsil (a peritonsillar abscess) may also occur. In rare cases, a persistent sore throat may point to more serious conditions, such as rheumatic fever or **pneumonia**.

Prevention

The bacteria and viruses that cause tonsillitis are easily spread from person to person. It is not unusual for an entire family or several students in the same classroom to come down with similar symptoms, especially if *S. pyogenes* is the cause. The risk of

transmission can be lowered by avoiding exposure to anyone who already has tonsillitis or a sore throat. Drinking glasses and eating utensils should not be shared and should be washed in hot, soapy water before reuse. Old toothbrushes should be replaced to prevent reinfection. People who are caring for someone with tonsillitis should wash their hands frequently, to prevent spreading the infection to others.

Resources

OTHER

“Tonsillitis.” *Kids Health Page*. <<http://KidsHealth.org/parent/common/tonsillitis.html>>.

Altha Roberts Edgren

Tooth decay

Definition

Tooth decay, which is also called dental cavities or dental caries, is the destruction of the outer surface (enamel) of a tooth. Decay results from the action of bacteria that live in plaque, which is a sticky, whitish film formed by a protein in saliva (mucin) and sugary substances in the mouth. The plaque bacteria sticking to tooth enamel use the sugar and starch from food particles in the mouth to produce acid.

Description

Tooth decay is a common health problem, second in prevalence only to the **common cold**. It has been estimated that 90% of people in the United States have at least one cavity, and that 75% of people had their first cavity by the age of five. Although anyone can have a problem with tooth decay, children and senior citizens are the two groups at highest risk. Other high-risk groups include people who eat a lot of starchy and sugary foods; people living in areas without a fluoridated water supply; and people who already have numerous dental restorations (fillings and crowns).

Baby bottle tooth decay

Baby bottle tooth decay is a dental problem that frequently develops in infants that are put to bed with a bottle containing a sweet liquid. Baby bottle tooth decay is also called nursing-bottle caries and bottle-mouth syndrome. Bottles containing such liquids as milk, formula, fruit juices, sweetened drink mixes, and

sugar water continuously bathe an infant's mouth with sugar during naps or at night. The bacteria in the mouth use this sugar to produce acid that destroys the child's teeth. The upper front teeth are typically the ones most severely damaged; the lower front teeth receive some protection from the tongue. Pacifiers dipped in sugar, honey, corn syrup, or other sweetened liquid also contribute to bottle-mouth syndrome. The first signs of damage are chalky white spots or lines across the teeth. As decay progresses, the damage to the child's teeth becomes obvious.

Causes and symptoms

Tooth decay requires the simultaneous presence of three factors: plaque bacteria, sugar, and a vulnerable tooth surface. Although several microorganisms found in the mouth can cause tooth decay, the primary disease agent appears to be *Streptococcus mutans*. The sugars used by the bacteria are simple sugars such as glucose, sucrose, and lactose. They are converted primarily into lactic acid. When this acid builds up on an unprotected tooth surface, it dissolves the **minerals** in the enamel, creating holes and weak spots (cavities). As the decay spreads inward into the middle layer (the dentin), the tooth becomes more sensitive to temperature and touch. When the decay reaches the center of the tooth (the pulp), the resulting inflammation (pulpitis) produces a **toothache**.

Diagnosis

Tooth decay develops at varying rates. It may be found during a routine six-month dental checkup before the patient is even aware of a problem. In other cases, the patient may experience common early symptoms, such as sensitivity to hot and cold liquids or localized discomfort after eating very sweet foods. The dentist or dental hygienist may suspect tooth decay if a dark spot or a pit is seen during a visual examination. Front teeth may be inspected for decay by shining a light from behind the tooth. This method is called transillumination. Areas of decay, especially between the teeth, will appear as noticeable shadows when teeth are transilluminated. X rays may be taken to confirm the presence and extent of the decay. The dentist then makes the final clinical diagnosis by probing the enamel with a sharp instrument.

Tooth decay in pits and fissures may be differentiated from dark shadows in the crevices of the chewing surfaces by a dye that selectively stains parts of the tooth that have lost mineral content. A dentist can also use this dye to tell whether all tooth decay has been removed from a cavity before placing a filling.

Diagnosis in children

Damage caused by baby bottle tooth decay is often not diagnosed until the child has a severe problem, because parents seldom bring infants and toddlers in for dental check-ups. Dentists want to initially examine primary teeth between 12 and 24 months. Children still drinking from a bottle anytime after their first birthday are likely to have tooth decay.

Treatment

To treat most cases of tooth decay in adults, the dentist removes all decayed tooth structure, shapes the sides of the cavity, and fills the cavity with an appropriate material, such as silver amalgam or composite resin. The filling is put in to restore and protect the tooth. If decay has attacked the pulp, the dentist or a specialist called an endodontist may perform **root canal treatment** and cover the tooth with a crown.

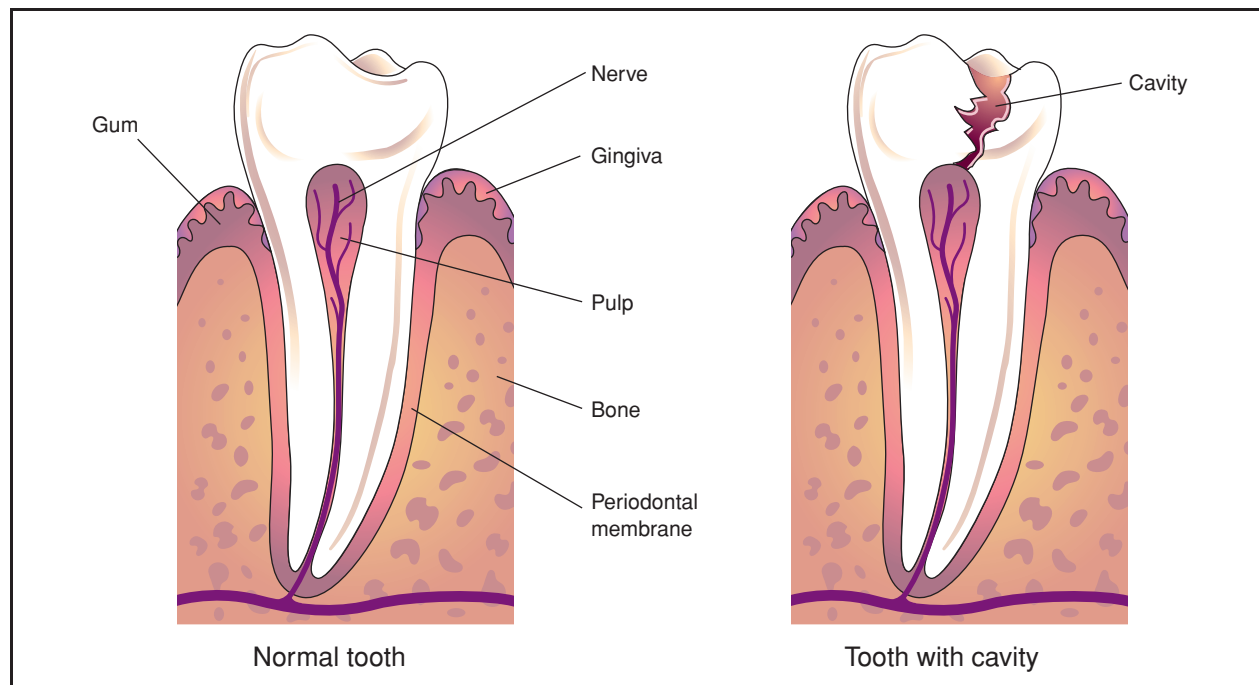
In cases of baby bottle tooth decay, the dentist must assess the extent of the damage before deciding on the treatment method. If the problem is caught early, the teeth involved can be treated with fluoride, followed by changes in the infant's feeding habits and better oral hygiene. Primary teeth with obvious decay in the enamel that has not yet progressed to the pulp need to be protected with stainless steel crowns. Fillings are not usually an option in small children because of the small size of their teeth and the concern of recurrent decay. When the decay has advanced to the pulp, pulling the tooth is often the treatment of choice. Unfortunately, loss of primary teeth at this age may hinder the young child's ability to eat and speak. It may also have bad effects on the alignment and spacing of the permanent teeth when they come in.

Prognosis

With timely diagnosis and treatment, the progression of tooth decay can be stopped without extended **pain**. If the pulp of the tooth is infected, the infection may be treated with **antibiotics** prior to root canal treatment or extraction. The longer decay goes untreated, however, the more destructive it becomes and the longer and more intensive the necessary treatment will be. In addition, a patient with two or more areas of tooth decay is at increased risk of developing additional cavities in the future.

Prevention

It is easier and less expensive to prevent tooth decay than to treat it. The four major prevention



Tooth decay is the destruction of the outer surface, or enamel, of a tooth. It is caused by acid buildup from plaque bacteria, which dissolves the minerals in the enamel and creates cavities. (Illustration by Electronic Illustrators Group.)

strategies include: proper **oral hygiene**; fluoride; sealants; and attention to diet.

Oral hygiene

GENERAL CARE OF THE MOUTH. The best way to prevent tooth decay is to brush the teeth at least twice a day, preferably after every meal and snack, and floss daily. Cavities develop most easily in spaces that are hard to clean. These areas include surface grooves, spaces between teeth, and the area below the gum line. Effective brushing cleans each outer tooth surface, inner tooth surface, and the horizontal chewing surfaces of the back teeth, as well as the tongue. Flossing once a day also helps prevent gum disease by removing food particles and plaque at and below the gum line, as well as between teeth. Patients should visit their dentist every six months for oral examination and professional cleaning.

MOUTH CARE IN OLDER ADULTS. Older adults who have lost teeth or had them removed still need to maintain a clean mouth. Bridges and dentures must be kept clean to prevent gum disease. Dentures should be relined and adjusted by a dentist whenever necessary to maintain proper fit. These adjustments help to keep the gums from becoming red, swollen, and tender.

MOUTH CARE IN CHILDREN. Parents can easily prevent baby bottle tooth decay by not allowing a child to fall asleep with a bottle containing sweetened liquids.

Bottles should be filled only with plain, unsweetened water. The child should be introduced to drinking from a cup around six months of age and weaned from bottles by twelve months. If an infant seems to need oral comfort between feedings, a pacifier specially designed for the mouth may be used. Pacifiers, however, should never be dipped in honey, corn syrup, or other sweet liquids. After the eruption of the first tooth, parents should begin routinely wiping the infant's teeth and gums with a moist piece of gauze or a soft cloth, especially right before bedtime. Parents may begin brushing a child's teeth with a small, soft toothbrush at about two years of age, when most of the primary teeth have come in. They should apply only a very small amount (the size of a pea) of toothpaste containing fluoride. Too much fluoride may cause spotting (fluorosis) of the tooth enamel. As the child grows, he or she will learn to handle the toothbrush, but parents should control the application of toothpaste and do the followup brushing until the child is about seven years old.

Fluoride application

Fluoride is a natural substance that slows the destruction of enamel and helps to repair minor tooth decay damage by remineralizing tooth structure. Toothpaste, mouthwash, fluoridated public drinking water, and vitamin supplements are all possible

KEY TERMS

Amalgam—A mixture (alloy) of silver and several other metals, used by dentists to make fillings for cavities.

Caries—The medical term for tooth decay.

Cavity—A hole or weak spot in the tooth surface caused by decay.

Dentin—The middle layer of a tooth, which makes up most of the tooth's mass.

Enamel—The hard, outermost surface of a tooth.

Fluoride—A chemical compound containing fluorine that is used to treat water or applied directly to teeth to prevent decay.

Mucin—A protein in saliva that combines with sugars in the mouth to form plaque.

Plaque—A thin, sticky, colorless film that forms on teeth. Plaque is composed of mucin, sugars from food, and bacteria that live in the plaque.

Pulp—The soft, innermost layer of a tooth containing blood vessels and nerves.

Sealant—A thin plastic substance that is painted over teeth as an anti-cavity measure to seal out food particles and acids produced by bacteria.

Transillumination—A technique of checking for tooth decay by shining a light behind the patient's teeth. Decayed areas show up as spots or shadows.

sources of fluoride. Children living in areas without fluoridated water should receive 0.5 mg/day of fluoride (0.25 mg/day if using a toothpaste containing fluoride) from three to five years of age, and 1 mg/day from six to 12 years.

While fluoride is important for protecting children's developing teeth, it is also of benefit to older adults with receding gums. It helps to protect their newly exposed tooth surfaces from decay. Older adults can be treated by a dentist with a fluoride solution that is painted onto selected portions of the teeth or poured into a fitted tray and held against all the teeth.

Sealants

Because fluoride is most beneficial on the smooth surfaces of teeth, sealants were developed to protect the irregular surfaces of teeth. A sealant is a thin plastic coating that is painted over the grooves of chewing surfaces to prevent food and plaque from being trapped there. Sealant treatment is painless

because no part of the tooth is removed, although the tooth surface is etched with acid so that the plastic will adhere to the rough surface. Sealants are usually clear or tooth-colored, making them less noticeable than silver fillings. They cost less than fillings and can last up to 10 years, although they should be checked for wear at every dental visit. Children should get sealants on their first permanent "6-year" molars, which come in between the ages of five and seven, and on the second permanent "12-year" molars, which come in between the ages of 11 and 14. Sealants should be applied to the teeth shortly after they erupt, before decay can set in. Although sealants have been used in the United States for about 25 years, one survey by the National Institute of Dental Research reported that fewer than 8% of American children have them.

Diet

The risk of tooth decay can be lowered by choosing foods wisely and eating less often. Foods high in sugar and starch, especially when eaten between meals, increase the risk of cavities. The bacteria in the mouth use sugar and starch to produce the acid that destroys the enamel. The damage increases with more frequent eating and longer periods of eating. For better dental health, people should eat a variety of foods, limit the number of snacks, avoid sticky and overly sweetened foods, and brush often after eating.

Drinking water is also beneficial for rinsing food particles from the mouth. Children can be taught to "swish and swallow" if they are unable to brush after lunch at school. Similarly, saliva stimulated during eating makes it more difficult for food and bacteria to stick to tooth surfaces. Saliva also appears to have a buffering effect on the acid produced by the plaque bacteria and to act as a remineralizing agent. Older patients should be made aware that some prescription medications may decrease salivary flow. Less saliva tends to increase the activity of plaque bacteria and encourage further tooth decay. Chewing sugarless gum increases salivation and thus helps to lower the risk of tooth decay.

Resources

ORGANIZATIONS

American Dental Association. 211 E. Chicago Ave., Chicago, IL 60611. (312) 440-2500. <<http://www.ada.org>>.

American Dental Hygienists' Association. 444 North Michigan Ave., Chicago, IL 60611. (800)847-6718.

National Institute of Dental Research. 31 Center Drive, MSC 2190, Building 31, Room 5B49, Bethesda, MD 20892-2190.

OTHER

Healthtouch Online Page. <<http://www.healthtouch.com>>.

Bethany Thivierge

Tooth extraction

Definition

Tooth extraction is the removal of a tooth from its socket in the bone.

Purpose

Extraction is performed for positional, structural, or economic reasons. Teeth are often removed because they are impacted. Teeth become impacted when they are prevented from growing into their normal position in the mouth by gum tissue, bone, or other teeth. Impaction is a common reason for the extraction of wisdom teeth. Extraction is the only known method that will prevent further problems. Teeth may also be extracted to make more room in the mouth prior to straightening the remaining teeth (orthodontic treatment), or because they are so badly positioned that straightening is impossible. Extraction may be used to remove teeth that are so badly decayed or broken that they cannot be restored. In addition, patients sometimes choose extraction as a less expensive alternative to filling or placing a crown on a severely decayed tooth.

Precautions

In some situations, tooth extractions may need to be postponed temporarily. These situations include:

- Infection that has progressed from the tooth into the bone. Infections may make anesthesia difficult. They can be treated with **antibiotics** before the tooth is extracted.
- The patient's use of drugs that thin the blood (anticoagulants). These medications include warfarin (Coumadin) and **aspirin**. The patient should stop using these medications for three days prior to extraction.
- Patients who have had any of the following procedures in the previous six months: heart valve replacement, open heart surgery, prosthetic **joint replacement**, or placement of a medical shunt.

These patients may be given antibiotics to reduce the risk of bacterial infection.

Description

Tooth extraction can be performed with **local anesthesia** if the tooth is exposed and appears to be easily removable in one piece. An instrument called an elevator is used to loosen (luxate) the tooth, widen the space in the bone, and break the tiny elastic fibers that attach the tooth to the bone. Once the tooth is dislocated from the bone, it can be lifted and removed with forceps.

If the extraction is likely to be difficult, the dentist may refer the patient to an oral surgeon. Oral surgeons are specialists who are trained to give nitrous oxide, an intravenous sedative, or a general anesthetic to relieve **pain**. Extracting an **impacted tooth** or a tooth with curved roots typically requires cutting through gum tissue to expose the tooth. It may also require removing portions of bone to free the tooth. Some teeth must be cut and removed in sections. The extraction site may or may not require one or more stitches to close the cut (incision).

Preparation

Before an extraction, the dentist will take the patient's medical history, noting **allergies** and prescription medications. A dental history is also taken, with particular attention to previous extractions and reactions to anesthetics. The dentist may then prescribe antibiotics or recommend stopping certain medications prior to the extraction. The tooth is x-rayed to determine its full shape and position, especially if it is impacted.

If the patient is going to have deep anesthesia, he or she should wear loose clothing with sleeves that are easily rolled up to allow for an intravenous line. The patient should not eat or drink anything for at least six hours before the procedure. Arrangements should be made for a friend or relative to drive the patient home after the surgery.

Aftercare

An important aspect of aftercare is encouraging a clot to form at the extraction site. The patient should put pressure on the area by biting gently on a roll or wad of gauze for several hours after surgery. Once the clot is formed, it should not be disturbed. The patient should not rinse, spit, drink with a straw, or smoke for at least 24 hours after the extraction and preferably



A close-up view inside a person's mouth following the extraction of the lower right molar. (Custom Medical Stock Photo. Reproduced by permission.)

longer. Vigorous **exercise** should not be done for the first three to five days.

For the first two days after the procedure, the patient should drink liquids without using a straw, and eat soft foods. Any chewing must be done on the side away from the extraction site. Hard or sticky foods should be avoided. The mouth may be gently cleaned with a toothbrush, but the extraction area should not be scrubbed.

Wrapped ice packs can be applied to reduce facial swelling. Swelling is a normal part of the healing process. It is most noticeable in the first 48–72 hours. As the swelling subsides, the patient may experience muscle stiffness. Moist heat and gentle exercise will restore jaw movement. The dentist may prescribe medications to relieve the postoperative pain.

Risks

Potential complications of tooth extraction include postoperative infection, temporary numbness from nerve irritation, jaw fracture, and jaw joint pain. An additional complication is called dry socket. When a blood clot does not properly form in the empty tooth socket, the bone beneath the socket is painfully exposed to air and food, and the extraction site heals more slowly.

Normal results

After an extraction, the wound usually closes in about two weeks. It takes three to six months for the bone and soft tissue to be restructured. Complications

KEY TERMS

Dry socket—A painful condition following tooth extraction in which a blood clot does not properly fill the empty socket. Dry socket leaves the underlying bone exposed to air and food.

Extraction site—The empty tooth socket following removal of the tooth.

Impacted tooth—A tooth that is growing against another tooth, bone, or soft tissue.

Luxate—To loosen or dislocate the tooth from the socket.

Nitrous oxide—A colorless, sweet-smelling gas used by dentists for mild anesthesia. It is sometimes called laughing gas because it makes some patients feel giddy or silly.

Oral surgeon—A dentist who specializes in surgical procedures of the mouth, including extractions.

Orthodontic treatment—The process of straightening teeth to correct their appearance and function.

such as infection or dry socket may prolong the healing time.

Resources

ORGANIZATIONS

American Association of Oral and Maxillofacial Surgeons.
9700 West Bryn Mawr Ave., Rosemont, IL 60018-5701.
(847) 678-6200. <<http://www.aaoms.org>>.

Bethany Thivierge

Tooth grinding see **Bruxism**

Tooth replacements and restorations

Definition

A tooth restoration is any artificial substance or structure that replaces missing teeth or part of a tooth in order to protect the mouth's ability to eat, chew, and speak. Restorations include fillings, inlays, crowns, bridges, partial and complete dentures, and dental implants.

Purpose

Restorations have somewhat different purposes depending on their extensiveness. Fillings, inlays, and crowns are intended to repair damage to individual teeth. They replace tooth structure lost by decay or injury, protect the part of the tooth that remains, and restore the tooth's shape and function. Bridges, dentures, and implants are intended to protect the shape and function of the mouth as a whole.

Precautions

Some patients are allergic to the medications used for **local anesthesia** in dental restorations. In addition, many people in the general population are afraid of dental work. Most dentists in present-day practice can help patients with this specific fear.

Description

Fillings

Fillings are restorations that are done to repair damage caused by **tooth decay** (dental caries). Tooth decay occurs when microorganisms in the mouth convert sugar from food to acid, which attacks the tooth. The acid forms cavities that start in the hard outer surface of the tooth (the enamel) and may extend inward to the pulp, which contains the tooth's nerves and blood vessels. Left untreated, tooth decay may lead to inflammation and infection that may cause **toothache** and perhaps more serious complications.

To stop the decay process, the dentist removes the decayed portion of the tooth using a high-speed drill or an air abrasion system, shapes the cavity walls, and replaces the tooth structure with a filling of silver amalgam, composite resin, or gold. The filling is placed in the cavity as a liquid or soft solid. It sets within a few minutes and continues to harden over the next several hours. Silver amalgam is commonly used to fill cavities on the biting surfaces of the back teeth, because it is strong enough to withstand the tremendous pressures exerted by grinding and chewing. Composite resin is typically used to fill cavities in front teeth and any other teeth that are visible when the patient smiles, because its color can be matched to the tooth surface. Gold as a filling material is far less common, but is being increasingly used. Although it is more expensive and less easily applied, it does not trigger the sensitivity reactions that some patients have to silver amalgam.

Inlays

An inlay resembles a filling in that it fills the space remaining after the decayed portion of a tooth has

been removed. The difference is that an inlay is shaped outside the patient's mouth and then cemented into place. After the decay is removed and the cavity walls are shaped, the dentist makes a wax pattern of the space. A mold is cast from the wax pattern. An inlay, usually of gold, is made from this mold and sealed into the tooth with dental cement.

Crowns

The crown of a tooth is the portion that is covered by enamel. A restorative crown replaces this outer part to protect the tooth. This protection becomes necessary when a tooth cracks or has its entire structure weakened by decay. As with a filling or inlay, the dentist first removes the decayed portion of the tooth. The tooth is then prepared for a crown. It may be tapered on the outside edges to a peg, reinforced with a cast metal core, or rebuilt with both a cast metal core and a post. A wax impression of the prepared tooth and the teeth next to it is made. The new crown is made to fit this mold. The crown may be made of gold or stainless steel alone, metal with a veneer of tooth-colored porcelain or resin, or of porcelain or resin alone. The finished crown is then placed over the prepared tooth, adjusted, and cemented into place.

Bridges

Bridges are a type of restoration that is done when one or more permanent teeth are lost or pulled. The resulting gap must be filled in to prevent the remaining teeth from shifting. If the other teeth shift, they will affect the patient's bite (occlusion), which sometimes produces **pain** in the jaw joint. As the teeth move and become crooked, they also become more difficult to keep clean. The risk of tooth decay and gum disease increases, increasing the likelihood that additional teeth will be lost. A bridge is inserted to prevent this risk. Bridges are nonremovable appliances of one or more artificial teeth (pontics) anchored by crowns on the adjacent teeth (abutment teeth). The abutment teeth carry the pressure when the patient chews food.

Partial dentures

A partial denture is similar to a bridge in that it fills a gap left by missing teeth with artificial teeth on a metal frame. A partial denture is removable, however. It attaches to a crown on the abutment tooth with a metal clasp or precision attachment. A partial denture is primarily used at the end of a row of natural teeth, where there is only one abutment tooth. The pressure exerted by chewing is shared by this abutment and the soft tissues of the gum ridge beneath the appliance.

Complete dentures

Complete dentures may be worn when all of the top or bottom teeth have been lost. A complete denture consists of artificial teeth mounted in a plastic base molded to fit the remaining oral anatomy. It may or may not be held in place with a denture adhesive.

Implants

Dental implants are a means of securing crowns, bridges, and dentures in the mouth. A hard plastic or metal fixture is implanted through the soft tissue into the bone. Over time, the bone grows around this fixture, firmly anchoring it. The exposed end of this fixture is covered with a crown and may serve as a stable abutment for a bridge or denture.

Preparation

Before a restoration is placed in the mouth, the dentist removes all traces of decay and shapes the remaining tooth structure for the restoration. Fillings are the only restoration created within the tooth itself—the others are made up in a laboratory using a model of the tooth structure. Thus, a filling may be placed in a single dental visit, while the other restorations usually take several appointments. Temporary crowns and dentures are put in place after the tooth is shaped until the permanent restoration is delivered by the laboratory.

Aftercare

Fillings

Fillings need time to harden for several hours after being placed, so the patient should chew food on the opposite side of the mouth for the first day.

Dentures

A partial or complete denture may take several weeks of getting used to. Inserting and removing the denture will take practice. Speaking clearly may be difficult at first—the patient may find it helpful to read out loud for practice. Eating may also feel awkward. The patient should begin by eating small pieces of soft foods. Very hard or sticky foods should be avoided.

Patients with dentures must work on good **oral hygiene**. Specialty brushes and floss threaders may be used to remove plaque and food from around crowns and bridges. Dentures should be removed and brushed

KEY TERMS

Abutment tooth—A crowned tooth that stabilizes a bridge or partial denture.

Bridge—An appliance of one or more artificial teeth anchored by crowns on the adjacent teeth.

Complete denture—A full set of upper or lower teeth, mounted in a plastic base. Dentures are also called false teeth.

Crown—A protective shell that fits over the tooth.

Dental caries—A disease of the teeth in which microorganisms convert sugar in the mouth to acid that erodes the tooth.

Enamel—The hard outermost surface of a tooth.

Filling—Dental material that occupies the space remaining within a tooth after the decayed portion has been removed.

Implant—A fixture with one end implanted into the bone and the other end covered with a crown, often to serve as a stable abutment for a bridge or denture.

Inlay—A filling that is made outside of the tooth and the cemented into place.

Occlusion—The way upper and lower teeth fit together during biting and chewing.

Partial denture—A removable bridge that usually clasps onto only one abutment.

Pontic—An artificial tooth.

Pulp—The soft innermost layer of a tooth that contains its blood vessels and nerves.

daily with a specially designed brush and a denture cleaner or other mild soap.

The patient should see the dentist for an adjustment if there is any discomfort or irritation resulting from a restoration. Otherwise, the patient should see the dentist at least twice a year for an oral examination.

Risks

Restoration procedures typically require local anesthesia. Some people may have allergic reactions to the medication. A very small number of people are allergic to one or more of the metals used in a dental restoration. In most cases, the dentist can use another material.

Normal results

A well-made restoration should feel comfortable and last a relatively long time with proper care. Artificial dental restorations only approximate the original tooth, however. A complete denture will never feel as comfortable or work as well as natural teeth. It is better, therefore, to prevent the need for restorative dental work than to replace teeth. Restorations are expensive, may require many appointments, and still need careful cleaning and attention.

Resources

ORGANIZATIONS

Academy of General Dentistry. Suite 1200, 211 East Chicago Ave., Chicago, IL 60611. (312) 440-4300. <<http://www.agd.org>>.

American Dental Association. 211 E. Chicago Ave., Chicago, IL 60611. (312) 440-2500. <<http://www.ada.org>>.

Bethany Thivierge

Toothache

Definition

A toothache is any **pain** or soreness within or around a tooth, indicating inflammation and possible infection.

Description

A toothache may feel like a sharp pain or a dull ache. The tooth may be sensitive to pressure, heat, cold, or sweets. In cases of severe pain, identifying the problem tooth is often difficult. Any patient with a toothache should see a dentist at once for diagnosis and treatment. Most toothaches get worse if not treated.

Causes and symptoms

Toothaches may result from any of a number of causes:

- **tooth decay** (dental caries)
- inflammation of the tooth pulp (pulpitis)
- abscesses
- gum disease, including periodontitis
- loose or broken filling

- cracked or **impacted tooth**
- exposed tooth root
- food wedged between teeth or trapped below the gum line
- tooth nerve irritated by clenching or grinding of teeth (bruxism)
- pressure from congested sinuses
- traumatic injury

Diagnosis

Diagnosis includes identifying the location of the toothache, as well as the cause. The dentist begins by asking the patient specific questions about the toothache, including the types of foods that make the pain worse, whether the tooth is sensitive to temperature or biting, and whether the pain is worse at night. The dentist then examines the patient's mouth for signs of swelling, redness, and obvious tooth damage. The presence of pus indicates an **abscess** or gum disease. The dentist may flush the sore area with warm water to dislodge any food particles and to test for sensitivity to heat. The dentist may then dry the area with gauze to determine sensitivity to touch and pressure. The dentist may probe tooth crevices and the edges of fillings with a sharp instrument, looking for areas of tooth decay. Finally, the dentist may take x rays, looking for evidence of decay between teeth, a cracked or impacted tooth, or a disorder of the underlying bone.

Treatment

Emergency self-care

Toothaches should always be professionally treated by a dentist. Some methods of self-treatment, however, may help manage the pain until professional care is available:

- rinsing with warm salt water
- using dental floss to remove any food particles
- taking **aspirin** or **acetaminophen** (Tylenol) to relieve pain; the drug should be swallowed—*never* placed directly on the aching tooth or gum
- applying a *cold* compress against the outside of the cheek; do not use heat, because it will tend to spread infection.
- using clove oil (*Syzygium aromaticum*) to numb the gums; the oil may be rubbed directly on the sore area or used to soak a small piece of cotton and applied to the sore tooth

KEY TERMS

Abscess—A hole in the tooth or gum tissue filled with pus as the result of infection.

Bruxism—Habitual clenching and grinding of the teeth as a result of stress. The behavior usually occurs during sleep.

Cavity—A hole or weak spot in the tooth surface caused by decay.

Dental caries—A disease of the teeth in which microorganisms convert sugar in the mouth to acid, which then erodes the tooth.

Enamel—The hard outermost surface of a tooth.

Endodontist—A dentist who specializes in diagnosing and treating diseases of the pulp and other inner parts of the tooth.

Impacted tooth—A tooth that is growing against another tooth, bone, or soft tissue.

Periodontitis—A gum disease that destroys the structures supporting the teeth, including bone.

Pulp—The soft innermost part of a tooth, containing blood vessels and nerves.

Pulpitis—Inflammation of the pulp of a tooth that involves the blood vessels and nerves.

Professional care

Treatment will depend on the underlying cause of the toothache. If the pain is due to tooth decay, the dentist will remove the decayed area and restore the tooth with a filling of silver amalgam or composite resin. Loose or broken fillings are removed, new decay cleaned out, and a new filling is placed. If the pulp of the tooth is damaged, root canal therapy is needed. The dentist or a specialist called an endodontist removes the decayed pulp, fills the space left behind with a soothing paste, and covers the tooth with a crown to protect and seal it. If the damage cannot be treated by these methods, or if the tooth is impacted, the tooth must be extracted.

Alternative treatment

Toothaches caused by infection or tooth decay must be treated by a dentist. Several alternative therapies may be helpful for pain relief until dental treatment is available. Clove oil (*Syzygium aromaticum*) may be rubbed on sensitive gums to numb them or added to a small cotton pellet that is then placed into

or over a hole in the tooth. The herb corydalis (*Corydalis yanhusuo*) may also help relieve toothache pain. Pain also may be reduced using **acupressure**, **acupuncture**, or **reiki**. Acupuncture should be done only by a licensed practitioner.

Prognosis

Prompt dental treatment provides a positive outcome for toothache. In the absence of active infection, fillings, root canal treatments, or extractions may be performed with minimal discomfort to the patient. When a toothache is left untreated, a severe infection may develop and spread to the sinuses or jawbone, and eventually cause blood **poisoning**.

Prevention

Maintaining proper **oral hygiene** is the key to preventing toothaches. The best way to prevent tooth decay is to brush at least twice a day, preferably after every meal and snack. Flossing once a day also helps prevent gum disease by removing food particles and bacteria at and below the gum line, as well as between teeth. People should visit their dentist at least every six months for oral examinations and professional cleaning.

Resources

ORGANIZATIONS

Academy of General Dentistry. Suite 1200, 211 East Chicago Ave., Chicago, IL 60611. (312) 440-4300. <<http://www.agd.org>>.

American Dental Association. 211 E. Chicago Ave., Chicago, IL 60611. (312) 440-2500. <<http://www.ada.org>>.

OTHER

Medical Source. Medical Alliances, Inc. 2121 Eisenhower Ave., Suite 603, Alexandria, VA 22314. (800) 463-6482 <<http://www.medsources.com>>.

Bethany Thivierge

Topical anesthesia

Definition

Topical anesthesia is a condition of temporary **numbness** caused by applying a substance directly to a surface of the body. Loss of feeling occurs in the specific areas touched by the anesthetic substance.

Purpose

Topical anesthesia typically either relieves existing **pain** from a body surface or prevents pain during medical examinations or procedures. Body surfaces include both skin and mucous membranes, which are the moist linings of areas such as the inside of the mouth or nose. Occasionally, use of a topical anesthetic may help clarify whether a patient's pain comes from a body surface, which the substance is able to touch, or from deeper structures beyond the reach of the substance. A mucous membrane, an area of skin, or areas just beneath the skin's surface temporarily lose feeling during topical anesthesia.

Precautions

Topical anesthetic medications must be selected carefully and used in proper amounts in order to prevent harmful reactions. For instance, some topical drugs should be used only on intact skin to avoid rapid absorption into the body. Therefore, it is important to choose the right type and dosage of anesthetic for each specific purpose.

Description

Most drugs in the early 2000s used to cause topical anesthesia have at least some chemical similarity to **cocaine**, which was recognized as a topical anesthetic in the early 1900s. These drugs work by blocking nerve impulses that carry pain messages to the brain. Commonly used, twenty-first-century drugs include benzocaine, lidocaine, prilocaine, and tetracaine. These drugs come in several application forms, such as cream, jelly, ointment, solution, and spray, are available in varying strengths, and may be used alone or in combinations.

One common use for topical anesthetics is to relieve pain from problems such as sores in the mouth, skin scrapes, and **hemorrhoids**. A person typically applies a cream or jelly to the affected area. Numbness begins within a few minutes, and the anesthetic effect may last an hour or more. Repeated applications are often necessary.

The other broad category of use for topical anesthetics is to prevent pain from medical examinations or procedures on areas such as the eye, nose, throat, urethra, rectum, or skin. For example, even a single drop of anesthetic may allow examination of a painfully irritated eye or removal of a speck of dirt from the eye surface. Careful inspection of a nostril or insertion of a urinary drainage tube into the urethra may be

KEY TERMS

Anesthetic—Not having sensation; related to the loss of sensation; or, a substance that produces loss of sensation.

EMLA—Eutectic mixture of local anesthetics, a drug combination for use on intact skin.

LET—A topical anesthetic mixture containing lidocaine, epinephrine, and tetracaine.

Mucous membranes—Moist linings of body surfaces such as the inside of the mouth or nose.

Numbness—Loss of feeling or sensation.

TAC—A topical anesthetic mixture containing tetracaine, epinephrine (adrenaline), and cocaine.

Topical—For use directly on a body surface.

difficult or impossible without the use of topical anesthetic spray or jelly beforehand.

Many uses of topical anesthetics involve application to the skin. Physicians and patients have long hoped for a way to numb an area of skin without having to use a needle to inject anesthetic. Original topical anesthetics were not effective in this way. A newer drug preparation called eutectic mixture of local anesthetics (EMLA, lidocaine 2.5%, and prilocaine 2.5%) can be used as a cream on intact skin. When applied for about 60 minutes, EMLA numbs the skin and penetrates as far as 5 mm below the surface. EMLA may help patients by decreasing the pain of needle pricks or simple skin procedures, such as repair of small lacerations. The main drawback to EMLA is that it takes so long for the anesthetic effect to begin, thus reducing its usefulness in settings such as the emergency rooms. Other drug combinations such as tetracaine, epinephrine (adrenaline), and cocaine (TAC), or lidocaine, epinephrine, and tetracaine (LET) have their own advantages and disadvantages.

The application of ice to a body area is a primitive method of producing topical anesthesia. Chemicals such as ethyl chloride can be sprayed onto intact skin to produce a momentary freezing and numbing effect.

Covering the ointment with plastic wrap increases the absorption of the ointment through the skin. The use of plastic wrap, or other occlusive dressings, is the standard means of applying EMLA cream. Other techniques include using gentle heat or electric current to help the drugs penetrate intact skin.

Risks

Overall, topical anesthesia tends to be very safe. Minor problems might include discoloration of the skin at the application site or an uncomfortable feeling of numbness that lasts longer than expected. Repeated use of a topical anesthetic on a damaged eye surface may interfere with the normal healing process. Rarely, too much of the drug may be absorbed into the body in a short time and cause a reaction such as seizure or rapid heartbeat. **Death** related to topical anesthesia is extremely rare.

Normal results

Successful use of a topical anesthetic produces temporary loss of sensation in the area where it is applied.

Resources

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Kenneth J. Berniker, M.D.
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Topical antibiotics see **Antibiotics, topical**

Topical antifungal drugs see **Antifungal drugs, topical**

TORCH test

Definition

The TORCH test, which is sometimes called the TORCH panel, belongs to a category of blood tests called infectious-disease antibody titer tests. This type of blood test measures the presence of antibodies (protein molecules produced by the human immune system in response to a specific disease agent) and their level of concentration in the blood. The name of the test comes from the initial letters of the five disease categories. The TORCH test measures the levels of an infant's antibodies against five groups of chronic

infections: *toxoplasmosis*, *other infections*, *rubella*, *cytomegalovirus (CMV)*, and *herpes simplex virus (HSV)*. The "other infections" usually include **syphilis**, **hepatitis B**, coxsackie virus, Epstein-Barr virus, varicella-zoster virus, and human parvovirus.

Since the TORCH test is a screening or first-level test, the pediatrician may order tests of other body fluids or tissues to confirm the diagnosis of a specific infection. In the case of **toxoplasmosis**, **rubella**, and syphilis, cerebrospinal fluid may be obtained from the infant through a spinal tap in order to confirm the diagnosis. In the case of CMV, the diagnosis is confirmed by culturing the virus in a sample of the infant's urine. In HSV infections, tissue culture is the best method to confirm the diagnosis.

Purpose

The five categories of organisms whose antibodies are measured by the TORCH test are grouped together because they can cause a cluster of symptomatic **birth defects** in newborns. This group of defects is sometimes called the TORCH syndrome. A newborn baby with these symptoms will be given a TORCH test to see if any of the five types of infection are involved.

The symptoms of the TORCH syndrome include:

- Small size in proportion to length of the mother's **pregnancy** at time of delivery. Infants who are smaller than would be expected (below the tenth percentile) are referred to as small-for-gestational-age, or SGA.
- Enlarged liver and spleen
- Low level of platelets in the blood
- Skin rash. The type of skin rash associated with the TORCH syndrome is usually reddish-purple or brown and is caused by the leakage of blood from broken capillaries into the baby's skin.
- Involvement of the central nervous system. These defects can include **encephalitis**, calcium deposits in the brain tissue, and seizures.
- Jaundice. The yellowish discoloration of the skin and whites of the eyes due to liver disease.

In addition to these symptoms, each of the TORCH infections has its own characteristic symptom cluster in newborns:

Toxoplasmosis

Toxoplasmosis is caused by *Toxoplasma gondii*, a parasite that the mother can acquire from handling

infected cats, drinking unpasteurized milk, or eating contaminated meat. The infection is carried to the infant through the mother's placenta, and can cause infections of the eyes or central nervous system. The organism can invade brain or muscle tissue and form tissue cysts. The later in pregnancy that the mother is infected, the higher the probability that the fetus will be infected. On the other hand, toxoplasmosis early in pregnancy is more likely to cause a **miscarriage** or serious birth defects. The incidence of toxoplasmosis in newborns is one in 1,000 live births.

Other (syphilis)

Syphilis is caused by a spirochete (spiral- or coil-shaped bacterium), *Treponema pallidum*. It is transmitted in the adult population by sexual intercourse. About 2-5% of children born to mothers diagnosed with syphilis will have the disease at birth. Syphilis was added to the TORCH panel because of a rapid increase in reported cases since 1990. It is also a potentially life-threatening infection for the fetus. Syphilis can cause early delivery, miscarriage, or **still-birth**. The mortality rate in infants infected with syphilis is about 54%.

Rubella

Rubella is a virus that has a seasonal pattern, with epidemics most likely in the spring. Between 0.1-2% of newborns will be infected with rubella. The rate of fetal infection varies according to the timing of the mother's infection during pregnancy. Birth defects, however, are most likely (85%) in infants infected during the first eight weeks of pregnancy. Infants born with rubella may already show signs of heart disease, retarded growth, **hearing loss**, blood disorders, vision problems, or **pneumonia**. They may also develop problems later in childhood, including **autism**, hearing loss, brain syndromes, immune system disorders, or thyroid disease.

Cytomegalovirus (CMV)

Cytomegalovirus belongs to the herpesvirus group of infections. It can be transmitted through body secretions, as well as by sexual contact; some newborns acquire CMV through the mother's breast milk. In adults, it produces symptoms resembling those of mononucleosis. About 1-2.2% of newborns in the United States are infected with CMV. Of this group, 10% will have measurable symptoms. The mortality rate for these symptomatic newborns is 20-30%. Surviving infants with CMV may suffer from hearing problems (15%) or

mental retardation (30%). Newborns that acquire CMV during the birth process or shortly after birth may develop pneumonia, hepatitis, or various blood disorders.

Herpes simplex virus (HSV)

Herpesvirus infections are among the most common viral infections in humans. They are spread by oral, as well as genital, contact. It is estimated that between 1 in 1,000 and 1 in 5,000 infants are born with HSV infections. About 80% of these infections are acquired during the birth process itself; the virus enters the infant through its eyes, skin, mouth, and upper respiratory tract. Of infants born with HSV infection, about 20% will have localized infections of the eyes, mouth, or skin. About 50% of infected infants will develop disease spread throughout the body (disseminated) within nine to 11 days after birth. Disseminated herpes infections attack the liver and adrenal glands, as well as other body organs. Without treatment, the mortality rate is 80%. Even with antiviral medication, the mortality rate is still 15-20%, with 40-55% of the survivors having long-term damage to the central nervous system. It is critical for the doctor to diagnose HSV infection in the newborn as soon as possible, for effective treatment.

Description

The TORCH panel requires a sample of the infant's blood. Samples from infants are usually obtained by the heelstick procedure when only a small quantity of blood is needed. The baby's foot is wrapped in a warm cloth for five minutes, to make the blood flow more easily. The foot is then wiped with an alcohol swab and a lancet is used to stick the baby's heel on one side. It is important to avoid the center of the heel, in order to prevent an inflammation of the bone.

Preparation

No special preparation, other than sterile technique, is required.

Risks

The only complications associated with the TORCH test are those resulting from the heelstick technique itself. These risks include scarring, infection of the bone, **cellulitis** (inflammation of cellular tissue), small lumpy calcium deposits, and inaccurate test results.

KEY TERMS

Antibody—A protein molecule produced by the immune system that is specific to a disease agent, such as CMV and the other organisms sought by the TORCH test. The antibody combines with the organism and disables it.

Perinatal—Referring to the period of time surrounding an infant's birth, from the last two months of pregnancy to the first 28 days of life. The TORCH panel tests for perinatal infections.

Small-for-gestational-age (SGA)—A term used to describe newborns who are below the 10th percentile in height or weight for their estimated gestational age. The gestational age is based upon the date of the mother's last menstrual period. SGA is one of the symptoms of TORCH syndrome.

Titer—The concentration of a substance in a given sample of blood or other tissue fluid.

Normal results

The normal result would be normal levels of immunoglobulin M (IgM) antibody in the infant's blood. IgM is one of five types of protein molecules found in blood that function as antibodies. IgM is a specific class of antibodies that seeks out virus particles. In contrast to adults, IgM is the most common type of immunoglobulin in newborn children. It is, therefore, the most useful indicator of the presence of a TORCH infection.

Abnormal results

The general abnormal, or positive, finding would be high levels of IgM antibody. The test can be refined further for antibodies specific to given disease agents. The TORCH screen, however, can produce both false-positive and false-negative findings. Doctors can measure IgM levels in the infant's cerebrospinal fluid, as well as in the blood, if they want to confirm the TORCH results.

Resources

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Rebecca J. Frey, PhD

Torticollis

Definition

Torticollis (cervical dystonia or spasmodic torticollis) is a type of movement disorder in which the muscles controlling the neck cause sustained twisting or frequent jerking.

Description

In torticollis, certain muscles controlling the neck undergo repetitive or sustained contraction, causing the neck to jerk or twist to the side. Cervical dystonia causes forward twisting, and is called antecollis. Backward twisting is known as retrocollis. The abnormal posture caused by torticollis is often debilitating, and is usually painful.

Torticollis most commonly begins between age 30–60, with females affected twice as often as males. According to the National Spasmodic Torticollis Association, torticollis affects 83,000 people in the United States. Dystonia tends to become more severe during the first months or years after onset, and may spread to other regions, especially the jaw, arm, or leg. Torticollis should not be confused with such other causes of abnormal neck posture as orthopedic or congenital problems.

Causes and symptoms

The nerve signals responsible for torticollis are thought to originate in the basal ganglia, a group of brain structures involved in movement control. The exact defect is unknown. Some cases of dystonia are due to the inheritance of a defective gene, whose function was unknown as of mid-1998. Other cases are correlated with neck or head trauma, such as from an automobile accident. Use of certain **antipsychotic drugs**, or neuroleptics, can induce dystonia.

There are three types of torticollis:

- tonic, in which the abnormal posture is sustained
- clonic, marked by jerky head movements.
- mixed, a combination of tonic and clonic movements

Symptoms usually begin gradually, and may be intermittent at first, worsening in times of stress. Symptoms usually progress over two to five years, and then remain steady. Symptoms may be relieved somewhat when lying down. Many people with torticollis can temporarily correct their head position by sensory tricks, as touching the chin or cheek on the

side opposite the turning. The reason for the effectiveness of this “geste antagoniste,” as it is called, is unknown.

Pain in the neck, back, or shoulder affects more than two-thirds of all people with torticollis. Pain may spread to the arm or hand.

Diagnosis

Diagnosis of torticollis is aided by an electrical study (**electromyography**) that can detect overactive muscles. Imaging studies, including x rays, may be done to rule out other causes of abnormal posture. A detailed medical history is needed to determine possible causes, including trauma.

Treatment

A variety of oral drugs are available to relax muscles, including baclofen. For a subgroup of patients, L-dopa provides effective relief. Denervation of the involved neck muscles may be performed with injection of alcohol or phenol on to the nerve.

Injection of botulinum toxin (BTX) is considered by many to be the treatment of choice. By preventing release of chemical messages from the nerve endings that stimulate the involved muscles, BTX partially paralyzes the muscles, therefore allowing more normal posture and range of motion. BTX treatment lasts several months, and may be repeated.

Physical therapy can help relieve secondary consequences of torticollis. Regular muscle stretching prevents contracture, or permanent muscle shortening. Pain and spasm may be temporarily lessened with application of heat or ice. **Stress** management techniques may help prevent worsening. An occupational therapist can suggest home or work modifications to reduce fatigue and improve function. Braces constructed to replace the patient's own sensory tricks may help reduce abnormal posture.

Alternative treatment

Biofeedback may be effective for some patients. Regular massage therapy can reduce additional pain in compensating areas of the body. Two energy-based therapies, **acupuncture** and homeopathic medicine, can work to rebalance the whole person, helping to correct the torticollis. Antispasmodic herbs may help to relax the muscles. In addition, herbs that can help balance the stimulus from the nervous system are often recommended.

Prognosis

Spontaneous remission is seen in up to 20% of patients, most often those patients with older onset and milder symptoms. Dystonia may spread to affect other regions of the body.

Prevention

There is no way known to prevent torticollis.

Resources

ORGANIZATIONS

National Spasmodic Torticollis Association. P.O. Box 5849, Orange, CA 92863-5849. (800) 487-8385. <<http://www.bluheronweb.com>>.

Worldwide Education and Awareness for Movement Disorders. One Gustave L. Levy Place, Box 1052, New York, NY 10029. (800) 437-6683. <<http://www.wemove.org>>.

Richard Robinson

Total parenteral nutrition

Definition

Total parenteral **nutrition** (TPN) is a way of supplying all the nutritional needs of the body by bypassing the digestive system and dripping nutrient solution directly into a vein.

Purpose

TPN is used when individuals cannot or should not get their nutrition through eating. TPN is used when the intestines are obstructed, when the small intestine is not absorbing nutrients properly, or a gastrointestinal **fistula** (abnormal connection) is present. It is also used when the bowels need to rest and not have any food passing through them. Bowel rest may be necessary in **Crohn's disease**, **pancreatitis**, **ulcerative colitis**, and with prolonged bouts of **diarrhea** in young children. TPN is also used for individuals with severe **burns**, multiple **fractures**, and in malnourished individuals to prepare them for major surgery, **chemotherapy**, or radiation treatment. Individuals with **AIDS** or widespread infection (**sepsis**) may also benefit from TPN.

Precautions

Individuals need to tell their doctor if they have any **allergies**, what medications they are taking, if they are diabetic, have had liver, kidney, heart, lung, or hormonal disorders, and if they are pregnant. All these factors can affect the type and amount of TPN required.

Description

TPN is normally given through a large central vein. A catheter is inserted into the vein in the chest area under **local anesthesia** and sterile conditions. Often the placement is done in an operating room to decrease the chance of infection. Several different types catheters are used based on the reason TPN is needed and the expected length of treatment. Catheters are made of silicone. Once the catheter is in place, a **chest x ray** is done to make sure the placement is correct.

Normally TPN is administered in a hospital, but under certain conditions and with proper patient and caregiver education, it may also be used at home for long-term therapy. TPN solution is mixed daily under sterile conditions. Maintaining sterility is essential for preventing infection. For this reason, the outside tubing leading from the bag of solution to the catheter is changed daily, and special dressings covering the catheter are changed every other day.

The contents of the TPN solution are determined based on the age, weight, height, and the medical condition of the individual. All solutions contain sugar (dextrose) for energy and protein (amino acids). Fats (lipids) may also be added to the solution. Electrolytes such as potassium, sodium, calcium, magnesium, chloride, and phosphate are also included, as these are essential to the normal functioning of the body. Trace elements such as zinc, copper, manganese and chromium are also needed. **Vitamins** can be included in the TPN solution, and insulin, a hormone that helps the body use sugar, may need to be added. The TPN catheter is used only for nutrients; medications are not added to the solution. Adults need approximately 2 liters of TPN solution daily, although this amount varies with the age, size, and health of the individual. Special solutions have been developed for individuals with reduced liver and kidney function. The solution is infused slowly at first to prevent fluid imbalances, then the rate is gradually increased. The infusion process takes several hours.

Successful TPN requires frequent, often daily monitoring of the individual's weight, glucose (blood

KEY TERMS

Catheter—A hollow tube that is inserted into veins or other passages in the body to insert or remove fluid.

Electrolyte—An ion such as potassium sodium, or chloride dissolved in fluid that helps to regulate metabolic activities of the cell.

Fistula—An abnormal passage that connects one organ to another or that connects the organ to the outside of the body.

sugar) level, **blood count**, blood gasses, fluid balance, urine output, waste products in the blood (plasma urea) and electrolytes. Liver and **kidney function tests** may also be performed. The contents of the solution are individualized based on the results of these tests.

Preparation

Preparation to insert the catheter involves creating a sterile environment. Other special preparations are not normally necessary.

Aftercare

During the time the catheter is in place, patients and caregivers must be alert to any signs of infection such as redness, swelling, **fever**, drainage, or **pain**.

Risks

TPN requires close monitoring. Two types of complications can develop. Infection, air in the lung cavity (**pneumothorax**) and blood clot formation (thrombosis) all can develop as a result of inserting the catheter into a vein. Metabolic and fluid imbalances can occur if the contents of the nutritional fluid are not properly balanced and monitored. The most common metabolic imbalance is **hypoglycemia**, or low blood sugar, caused by abruptly discontinuing a solution high in sugar.

Normal results

Ideally TPN will provide all the nutrients in the correct quantity to allow the body to function normally.

Abnormal results

Fluid and electrolyte imbalances may occur that can lead to major organ failure.

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Tish Davidson, A.M.

Total protein test see **Protein components test**

Tourette syndrome

Definition

Tourette syndrome (TS) is an inherited disorder of the nervous system, characterized by a variable expression of unwanted movements and noises (tics).

Description

The first references in the literature to what might today be classified as Tourette syndrome largely describe individuals who were wrongly believed to be possessed by the devil. In 1885 Gilles de la Tourette, a French neurologist, provided the first formal description of this syndrome, which he described as an inherited neurological condition characterized by motor and vocal tics.

Although vocal and motor tics are the hallmark of Tourette syndrome, such other symptoms as the expression of socially inappropriate comments or behaviors, obsessive compulsive disorder, attention deficit disorder, self-injuring behavior, depression, and **anxiety** also appear to be associated with Tourette syndrome. Most research suggests that

Tourette syndrome is an autosomal dominant disorder, although a gene responsible for Tourette syndrome has not yet been discovered.

Tourette syndrome is found in all populations and all ethnic groups, but is three to four times more common in males than females and is more common in children than adults. The exact frequency of Tourette syndrome is unknown, but estimates range from 0.05% to 3%.

Causes and symptoms

The cause of Tourette syndrome is unknown, although some studies suggest that the tics in Tourette syndrome are caused by an increased amount of a neurotransmitter called dopamine. A neurotransmitter is a chemical found in the brain that helps to transmit information from one brain cell to another. Other studies suggest that the defect in Tourette syndrome involves another neurotransmitter called serotonin; or involves other chemicals required for normal functioning of the brain.

Most studies suggest that Tourette syndrome is an autosomal dominant disorder with decreased penetrance, although this hypothesis has not been proven and may not be true in all families. An autosomal dominant disorder results from a change in one copy of a pair of genes. Individuals with an autosomal dominant disorder have a 50% chance of passing on the changed gene to their children. Decreased penetrance means that not all people who inherit the changed gene will develop symptoms. There is some evidence that females who inherit the Tourette syndrome gene have a 70% chance of exhibiting symptoms and males have a 99% chance of having symptoms. It has been suggested that other genetic and environmental factors may play a role in the development of symptoms in people who inherit the changed gene, but none have been discovered. Some researchers believe that Tourette syndrome has different causes in different individuals or is caused by changes in more than one gene, although these theories are less substantiated. Further research is needed to establish the cause of Tourette syndrome.

Motor and vocal tics

The principal symptoms of Tourette syndrome include simple and complex motor and vocal tics. Simple motor tics are characterized by brief muscle contractions of one or more limited muscle groups. An eye twitch is an example of a simple motor tic.

Complex motor tics tend to appear more complicated and purposeful than simple tics, and involve coordinated contractions of several muscle groups. Some examples of complex motor tics include the act of hitting oneself and jumping. Copropraxia, the involuntary display of unacceptable/obscene gestures; and echopraxia, the imitation of the movement of another individual, are other examples of complex motor tics.

Vocal tics are actually manifestations of motor tics that involve the muscles required for vocalization. Simple vocal tics include **stuttering**, stammering, abnormal emphasis of part of a word or phrase, and inarticulate noises such as throat clearing, grunts, and high-pitched sounds. Complex vocal tics typically involve the involuntary expression of words. Perhaps the most striking example of this is coprolalia, the involuntary expression of obscene words or phrases, which occurs in fewer than one-third of people with Tourette syndrome. The involuntary echoing of the last word, phrase, sentence or sound vocalized by oneself (phalilalia) or of another person or sound in the environment (echolalia) are also classified as complex tics.

The type, frequency, and severity of tics exhibited varies tremendously between individuals with Tourette syndrome. Tourette syndrome has a variable age of onset and tics can start anytime between infancy and age 18. Initial symptoms usually occur before the early teens; the mean age of onset for both males and females is approximately seven years of age. Most individuals with symptoms initially experience simple muscle tics involving the eyes and the head. These symptoms can progress to tics involving the upper torso, neck, arms, hands, and occasionally the legs and feet. Complex motor tics are usually the latest-onset muscle tics. Vocal tics usually have a later onset than motor tics. In some rare cases, people with Tourette syndrome suddenly present with multiple, severe, or bizarre symptoms.

Not only is there extreme variability in clinical symptoms between individuals with Tourette syndrome, but individuals commonly experience a variability in type, frequency, and severity of symptoms over the course of their lifetime. Adolescents with Tourette syndrome often experience unpredictable and variable symptoms, which may be related to fluctuating hormone levels and decreased compliance in taking medications. Adults often experience a decrease in symptoms or a complete end to symptoms.

A number of factors appear to affect the severity and frequency of tics. **Stress** appears to increase the frequency and severity of tics, while concentration on another part of the body that is not involved in a tic

can result in the temporary alleviation of symptoms. Relaxation, following attempts to suppress the occurrence of tics, may result in an increased frequency of tics. An increased frequency and severity of tics can also result from exposure to such drugs as steroids, **cocaine**, amphetamines, and **caffeine**. Hormonal changes such as those that occur prior to the menstrual cycle can also increase the severity of symptoms.

Other associated symptoms

People with Tourette syndrome are more likely to exhibit non-obscene, socially inappropriate behaviors such as expressing insulting or socially unacceptable comments or socially unacceptable actions. It is not known whether these symptoms stem from a more general dysfunction of impulse control that might be part of Tourette syndrome.

Tourette syndrome appears to also be associated with attention deficit disorder (ADD). ADD is a disorder characterized by a short attention span and impulsivity, and in some cases hyperactivity. Researchers have found that 21–90% of individuals with Tourette syndrome also exhibit symptoms of ADD, whereas 2–15% of the general population exhibit symptoms of ADD.

People with Tourette syndrome are also at higher risk for having symptoms of **obsessive-compulsive disorder** (OCD). OCD is a disorder characterized by persistent, intrusive, and senseless thoughts (obsessions) or compulsions to perform repetitive behaviors that interfere with normal functioning. A person with OCD, for example, may be obsessed with germs and may counteract this obsession with continual hand washing. Symptoms of OCD are present in 1.9–3% of the general population, whereas 28–50% of people with Tourette syndrome have symptoms of OCD.

Self-injurious behavior (SIB) is also seen more frequently in those with Tourette syndrome. Approximately 34–53% of individuals with Tourette syndrome exhibit some form of self-injuring behavior. The SIB is often related to OCD but can also occur in those with Tourette syndrome who do not have OCD.

Symptoms of anxiety and depression are also found more commonly in people with Tourette syndrome. It is not clear, however, whether these symptoms are symptoms of Tourette syndrome or occur as a result of having to deal with the symptoms of moderate to severe Tourette syndrome.

People with Tourette syndrome may also be at increased risk for having learning disabilities and

personality disorders; and may be more predisposed to such behaviors as aggression, antisocial behaviors, severe temper outbursts, and inappropriate sexual behavior. Further controlled studies need to be performed, however, to ascertain whether these behaviors are symptoms of Tourette syndrome.

Diagnosis

Tourette syndrome cannot be diagnosed through a blood test. The diagnosis is made through observation and interview of the patient and discussions with other family members. The diagnosis, of Tourette syndrome is complicated by a variety of factors. The extreme range of symptoms of this disorder makes it difficult to differentiate Tourette syndrome from other disorders with similar symptoms. Diagnosis is further complicated by the fact that some tics appear to be within the range of normal behavior. For example an individual who only exhibits such tics as throat clearing and sniffing may be misdiagnosed with a medical problem such as allergies. In addition, such bizarre and complex tics as coprolalia may be mistaken for psychotic or “bad” behavior. Diagnosis is also confounded by individuals who attempt to control tics in public and in front of health care professionals and deny the existence of symptoms. Although there is disagreement over what criteria should be used to diagnosis Tourette syndrome, one aid in the diagnosis is the DSMMD (DSM-IV). The DSM-IV outlines suggested diagnostic criteria for a variety of conditions including Tourette syndrome.

DSM-IV criteria

- presence of both motor and vocal tics at some time during the course of the illness
- the occurrence of multiple tics nearly every day through a period of more than one year, without a remission of tics for a period of greater than three consecutive months
- the symptoms cause distress or impairment in functioning
- age of onset of prior to 18 years of age
- the symptoms are not due to medications or drugs and are not related to another medical condition

Some physicians critique the DSM-IV criteria, arguing that they do not include the full range of behaviors and symptoms seen in Tourette syndrome. Others criticize the criteria since they limit the diagnosis to those who experience a significant impairment, which may not be true for individuals with milder symptoms. For this reason many physicians

use their clinical judgment as well as the DSM-IV criteria as a guide to diagnosing Tourette syndrome.

Treatment

There is no cure for Tourette syndrome. Treatment involves the control of symptoms through educational and psychological interventions and/or medications. The treatment and management of Tourette syndrome varies from patient to patient and should focus on the alleviation of the symptoms that are most bothersome to the patient or that cause the most interference with daily functioning.

Psychological and educational interventions

Psychological treatments such as counseling are not generally useful for the treatment of tics but can be beneficial in the treatment of such associated symptoms as obsessive-compulsive behavior and attention deficit disorder. Counseling may also help individuals to cope better with the symptoms of this disorder and to have more positive social interactions. Psychological interventions may also help people cope better with stressors that can normally be triggers for tics and negative behaviors. Relaxation therapies may, however, increase the occurrence of tics. The education of family members, teachers, and peers about Tourette syndrome can be helpful and may help to foster acceptance and prevent social isolation.

Medications

Many people with mild symptoms of Tourette syndrome never require medications. Those with severe symptoms may require medications for all or part of their lifetime. The most effective treatment of tics associated with Tourette syndrome involves the use of drugs such as haloperidol, pimozide, sulpiride, and tiapride, which decrease the amount of dopamine in the body. Unfortunately, the incidence of side effects, even at low dosages, is quite high. The short-term side effects can include **sedation**, dysphoria, weight gain, movement abnormalities, depression, and poor school performance. Long-term side effects can include phobias, memory difficulties, and personality changes. These drugs are therefore better candidates for short-term rather than long-term therapy.

Tourette syndrome can also be treated with such other drugs as clonidine, clonazepam, and risperidone, but the efficacy of these treatments is unknown. In many cases, treatment of such as associated conditions such as ADD and OCD is often more of a concern than the tics themselves. Clonidine used in

KEY TERMS

Attention deficit disorder (ADD)—Disorder characterized by a short attention span, impulsivity, and in some cases hyperactivity.

Autosomal dominant—A pattern of genetic inheritance in which only one abnormal gene is needed to display the trait or disease.

Coprolalia—The involuntary expression of obscene words or phrases.

Copropexia—The involuntary display of unacceptable/obscene gestures.

Decreased penetrance—Individuals who inherit a changed disease gene but do not develop symptoms.

Dysphoria—Feelings of anxiety, restlessness, and dissatisfaction.

Echolalia—Involuntary echoing of the last word, phrase, or sentence spoken by someone else or sound in the environment.

Echopraxia—The imitation of the movement of another individual.

Neurotransmitter—Chemical in the brain that transmits information from one nerve cell to another.

Obsessive compulsive disorder (OCD)—Disorder characterized by persistent, intrusive, and senseless thoughts (obsessions) or compulsions to perform repetitive behaviors that interfere with normal functioning.

Phalilalia—Involuntary echoing of the last word, phrase, sentence, or sound vocalized by oneself.

Tic—Brief and intermittent involuntary movement or sound.

conjunction with such stimulants as Ritalin may be useful for treating people with Tourette syndrome who also have symptoms of ADD. Stimulants should be used with caution in individuals with Tourette syndrome since they can sometimes increase the frequency and severity of tics. OCD symptoms in those with Tourette syndrome are often treated with such drugs as Prozac, Luvox, Paxil, and Zoloft.

In many cases the treatment of Tourette syndrome with medications can be discontinued after adolescence. Trials should be performed through the gradual tapering off of medications and should always be done under a doctor's supervision.

Prognosis

The prognosis for Tourette syndrome in individuals without associated psychological conditions is often quite good, and only approximately 10% of Tourette syndrome individuals experience severe tic symptoms. Approximately 30% of people with Tourette syndrome will experience a decrease in the frequency and severity of tics, and another 30–40% will experience a complete end of symptoms by late adolescence. The other 30–40% will continue to exhibit moderate to severe symptoms in adulthood. There does not appear to be a definite correlation between the type, frequency, and severity of symptoms and the eventual prognosis. Patients with severe tics may experience social difficulties and may isolate themselves from others in fear of shocking and embarrassing them. People with Tourette syndrome who have such other symptoms as obsessive compulsive disorder, attention deficit disorder, and self-injurious behavior usually have a poorer prognosis.

Resources

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ORGANIZATIONS

National Institute of Neurological Disorders and Stroke. 31 Center Drive, MSC 2540, Bldg. 31, Room 8806, Bethesda, MD 20814. (301) 496-5751 or (800) 352-9424. < <http://www.ninds.nih.gov> >.

National Tourette Syndrome Association, Inc. 42-40 Bell Blvd., Bayside, NY 11361-2820. (718) 224-2999. Fax: (718) 279-9596. tourette@ix.netcom.com.

Tourette Syndrome Foundation of Canada. 194 Jarvis Street, #206, Toronto, ONT M5B 2B7. Canada (800) 361-3120. tsfc.org@sympatico.ca. < <http://www.tourette.ca> >.

OTHER

"About Tourette Syndrome." Tourette Help. < <http://www.tourettehelp.com/pages/patient/about.html> >.

"Tourette's Disorder." Internet Mental Health. < <http://www.mentalhealth.com/fr20.html> >.

Lisa Maria Andres, MS, CGC

Toxic encephalopathy see **Delirium**

Toxic epidermal necrolysis

Definition

Toxic epidermal necrolysis is a rare condition that causes large portions of the epidermis, the skin's outermost layer, to detach from the layers of skin below. A reaction to a medication is the primary cause.

Description

Toxic epidermal necrolysis (TEN) begins with **fever**, **cough**, and other nonspecific symptoms, and is soon followed by purplish, bloody-looking lesions on the skin and mucous membranes. These early lesions, typically found on the head, neck, and upper chest, soon merge and blister. Sheets of epidermis then begin to detach from the skin layers below. In time, the entire surface of the skin may be involved, with detachment of 100% of the epidermis.

Causes and symptoms

The main cause of TEN is a severe drug reaction. Some investigators believe there may be additional infectious causes. A severe reaction in transplant patients, called **graft-vs.-host disease**, can also produce TEN. One study reported more than 100 different drugs as causes of TEN. The drugs most commonly implicated, however, include antibacterial **sulfonamides** such as sulfadiazine, **antibiotics** such as aminopenicillins and **cephalosporins**, and anticonvulsants like phenytoin. TEN is extremely rare. Researchers estimate that there are 0.2 cases per million users of aminopenicillins and 4.5 cases per million users of sulfonamides.

Exactly what leads to detachment of the epidermis remains unclear. People with TEN seem to have difficulty metabolizing the offending drug. Some researchers suggest that certain substances that should be cleared from the body instead get deposited on the outer shell of the epidermis, causing an immune response that leads the body to "reject" the skin.

Diagnosis

Diagnosis is made primarily on the appearance and spread of the **skin lesions**, and on a history that includes introduction of a new medication within the previous one to three weeks. A biopsy of the early lesions will confirm the diagnosis. Physicians will consider other potential diseases that cause similar symptoms before reaching a diagnosis of TEN. One is **erythema multiforme**, a recurrent skin disorder that produces lesions

KEY TERMS

Epidermis—The outermost layer of the skin.

Erythema multiforme—A recurrent skin disorder that produces lesions similar in appearance to TEN, but is not caused by a drug reaction and does not lead to sheet-like shedding of the skin.

Staphylococcal scalded skin syndrome—A disease caused by *Staphylococcus aureus*, in which large sheets of skin may peel away from the body. It most often affects infants, young children, and people with weakened immune systems.

Stevens-Johnson syndrome—A drug-induced skin disease that some experts believe is really a milder form of TEN.

similar in appearance to TEN. However, this disorder is not caused by a drug reaction and does not lead to sheet-like shedding of the skin. Another disease, Stevens-Johnson syndrome, is a drug-induced skin disease that some experts believe is really a milder form of TEN. **Staphylococcal scalded skin syndrome** (SSSS) also looks like TEN, but it is caused by a staphylococcal infection. Unlike TEN, which occurs rarely in children, SSSS primarily affects infants, young children, and adults with weakened immune systems.

Treatment

There is no specific treatment for TEN. Patients are typically treated in an intensive care unit or in a burn unit and receive treatment similar to that given to patients with major **burns**. With the loss of skin, severe **dehydration** is a major risk, so health care workers will attempt to replace fluids intravenously. Nutritional supplementation from a tube routed through the nose to the stomach may also be contemplated to promote the healing of the skin. Infection is a major risk, so some physicians "paint" the open lesions with topical **antiseptics**. Others use skin grafts taken from cadavers or cultured skin substitutes to cover large open areas until healing can occur. Some investigators believe system **corticosteroids** are useful in the treatment of TEN. But since these medications have also been implicated as a cause in some cases of TEN and are known to suppress the immune system, their use should be considered carefully.

Prognosis

About 25-30% of patients with TEN die. Elderly patients, those with extensive skin lesions, and those

with **AIDS** have the worst prognosis. Widespread systemic infection (**sepsis**) is the primary cause of **death**. Survivors, however, will be completely healed in three to four weeks.

Prevention

There is no prevention for TEN. No reliable test can indicate that a specific drug may cause TEN in a specific patient. Some researchers believe skin tests of potentially offending drugs may prove useful in the future.

Resources

ORGANIZATIONS

American Academy of Dermatology, 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

Richard H. Camer

Toxic shock syndrome

Definition

Toxic **shock** syndrome (TSS) is an uncommon, but potentially serious, illness that occurs when poisonous substances (toxins) produced by certain bacteria enter the bloodstream. The toxins cause a type of blood **poisoning** caused by staphylococcal, or less commonly streptococcal, infections in the lungs, throat, skin or bone, or from injuries. Women using super-absorbent tampons during menstruation were found to be most likely to get toxic shock syndrome.

Description

TSS first came to the attention of the public in the 1970s. Shortly after the introduction of a super-absorbent tampon, young women across the United States experienced an epidemic of serious but unexplained symptoms. Thousands went to emergency rooms with high **fever**, **vomiting**, peeling skin, low blood pressure, **diarrhea**, and a rash resembling **sunburn**. The only thing they had in common was that they all were menstruating at the time they felt sick, and all were using tampons, especially super-absorbent products.

At its height, the epidemic affected 15,000 people in the United States each year between 1980 and 1984; 15% of the women died. Since the offending products were taken off the market, the number of TSS cases

has declined sharply. As of 1998, only about 5,000 cases are diagnosed annually in the United States, 5% of which are fatal. The decline most likely is due to tampon manufacturers discontinuing the use of some synthetic materials, and the removal from the market of the brand of tampon associated with most cases of TSS. Today, most of these products are made with rayon and cotton.

In spite of TSS' association with menstruating women, the disease can affect anyone of either sex or any age or race. The infection may occur in children, men, and non-menstruating women who are weakened from surgery, injury, or disease, and who cannot fight off a staphylococcal infection. New mothers also are at higher risk for TSS.

Most cases reported in the recent past, however, still involve menstruating women under age 30. TSS still occurs in about 17 out of every 100,000 menstruating girls and women each year; more than half of these cases are related to tampons. Between five and 10% of patients with TSS die.

Streptococcal toxic shock syndrome (STSS)

A new type of toxic shock syndrome is caused by a different bacteria, called Group A streptococcus. This form of TSS is called streptococcal toxic shock syndrome, or STSS. Officially recognized in 1987, STSS is related to the strain of streptococcus nicknamed the "flesh-eating bacterium." STSS affects only one or two out of every 100,000 Americans. It almost never follows a simple **strep throat** infection.

Causes and symptoms

Transmission

TSS is caused by a strain of *Staphylococcus aureus* found in the nose, mouth, and occasionally the vagina. The bacteria produce a characteristic toxin. In large enough quantities, the toxin can enter the bloodstream, causing a potentially fatal infection.

While experts know the name of the bacterium, more than 10 years after the 1980s epidemic scientists still do not fully understand the link between TSS and tampons. Most medical researchers today suspect that the absorbent tampons introduce oxygen into the vagina, which is normally an oxygen-free area of the body. Oxygen triggers bacterial growth, and the more absorbent the tampon, the more bacteria it can harbor. Some experts believe that the reason TSS is linked to tampons in particular is that bacteria can contaminate and multiply in a tampon. If left in place

for a long time—as a woman could do with a super-absorbent product—the bacteria have a better chance of multiplying and producing a large amount of toxin. It also is possible that the tampons or the chemicals they contain may irritate the vaginal lining, enabling the toxin to enter the bloodstream.

These types of bacteria normally are present either on hands or in the vagina, and it takes an amount of bacteria only the size of a grain of sand to start an infection. Of the 15% of women who carry *Staphylococcus aureus*, only about 5% have the strain that produces the TSS toxin.

Symptoms

TSS. TSS begins suddenly, with a high fever of 102°F (38.9°C) or above, vomiting and watery diarrhea, **headache**, and sunburn-like rash, together with a **sore throat** and body aches. Blood pressure may plummet a day or two after the first symptoms appear. When the blood pressure drops, a woman may become disoriented or go into shock. Her kidneys may fail. After these developments, the skin on her hands and feet may peel.

STSS. STSS can occur after a streptococcal infection in the body, usually from an infected wound or even **chickenpox**. Within 48 hours, the patient's blood pressure drops. There also is fever, **dizziness**, breathing problems, and a weak, rapid pulse. The area around the wound may swell, the liver and kidneys can fail, and bleeding problems may occur.

Diagnosis

Any woman who is wearing a tampon and begins to experience the symptoms of toxic shock syndrome should remove the tampon right away and seek medical care.

The doctor will probably examine the vagina for signs of inflammation and rule out common **sexually transmitted diseases** with similar symptoms. A variety of blood tests, tests of vaginal secretions, and a physical examination are needed to identify this condition.

Treatment

TSS

In a menstruating woman, the vagina first is cleansed with an antiseptic solution to eliminate some of the bacteria that produce the toxin. TSS is

KEY TERMS

Shock—A condition in which the amount of blood circulating in the body is inadequate to meet the body's needs. Shock can be caused by certain diseases, serious injury, or blood loss.

Staphylococcus—A genus of bacteria that is commonly found on human skin and mucous membranes.

Streptococcus—A genus of sphere-shaped bacteria that can cause a wide variety of infections.

Toxin—A poisonous protein that is produced by some bacteria. A toxin is less complex than a poison.

treated with **antibiotics**, together with other drugs and fluids to lower fever and control blood pressure.

STSS

Antibiotics are used to treat STSS. Surgery may be needed to remove dead skin and muscle. Research continues on new therapy options to help improve treatment of STSS, including intravenous therapy with immunoglobulin. The therapy doesn't replace use of antibiotics but works along with it.

Prognosis

TSS lasts as long as three weeks, and has a tendency to recur. About a third of the women who are treated for TSS have it again within six months. In addition, TSS can affect the liver, kidneys, lungs, and other organs, depending on the severity of the infection. Untreated toxic shock syndrome can be fatal. A 2003 report noted that a quick diagnosis of TSS is critical to avoiding serious complications and **death**.

Prevention

TSS

Women who wear tampons should change them often and use different brands and types of pads and tampons. If a woman really prefers tampons, experts recommend using the lowest possible absorbency product made of cotton and rayon, and wearing it only during the day. In the past, it was difficult to compare absorbency rates for different products. Today, the Food and Drug Administration (FDA) requires standardized absorbency measurements on all tampon boxes. Above

all, women should wash their hands before inserting a tampon, and change the tampon every four to six hours.

Anyone who has had TSS even once should not use tampons again.

STSS

Doctors still are not sure how people can avoid STSS, but they advise patients to clean and bandage open **wounds** immediately. Anyone with a red, swollen, or tender wound, or a sudden fever should seek medical care.

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Carol A. Turkington
Teresa G. Odle

Toxocariasis see **Roundworm infections**

Toxoplasma gondii infection see

Toxoplasmosis

Toxoplasmosis

Definition

Toxoplasmosis is an infectious disease caused by the one-celled protozoan parasite *Toxoplasma gondii*. Although most individuals do not experience any symptoms, the disease can be very serious, and even fatal, in individuals with weakened immune systems.

Description

Toxoplasmosis is caused by a one-celled protozoan parasite known as *Toxoplasma gondii*. Cats, the primary carriers of the organism, become infected by eating rodents and birds infected with the organism. Once ingested, the organism reproduces in the intestines of cats, producing millions of eggs known as oocysts, which are excreted in cat feces daily for approximately two weeks. In the United States, it is estimated that approximately 30% of cats have been

infected by *T. gondii*. Oocysts are not capable of producing infection until approximately 24 hours after being excreted, but they remain infective in water or moist soil for approximately one year. When cattle, sheep, or other livestock forage through areas with contaminated cat feces, these animals become carriers of the disease. Fruits and vegetables can also become contaminated when irrigated with untreated water that has been contaminated with cat feces. In humans and other animals, the organisms produce thick-walled, dormant structures called cysts in the muscle and other tissues of the body.

Most humans contract toxoplasmosis by eating cyst-contaminated raw or undercooked meat, vegetables, or milk products. Humans can also become infected when they come into contact with the *T. gondii* eggs while cleaning a cat's litterbox, gardening, or playing in a sandbox, for instance. Once infected, an individual is immune to reinfection. The incubation period or period between infection and the start of the disease ranges from several days to months.

Anyone can be infected by *T. gondii*, but usually only those individuals with weakened immune systems (immunocompromised) develop symptoms of the disease. For them, toxoplasmosis can be severe, debilitating, and fatal. Immunocompromised individuals at risk include those with **AIDS**, **cancer**, or other chronic illnesses.

There is no person-to-person transmission, except from an infected mother to her child in the womb. Approximately six out of 1,000 women contract toxoplasmosis during **pregnancy**. Nearly half of these maternal infections are passed on to the fetus. Known as congenital toxoplasmosis, this form of the disease is acquired at birth by approximately 3,300 infants in the United States every year. The risk of fetal infection is estimated to be between one in 1,000 to one in 10,000. In children born with toxoplasmosis, symptoms may be severe and quickly fatal, or may not appear until several months or even years after birth.

Causes and symptoms

Healthy individuals do not usually display symptoms. When symptoms do occur, they are usually mild, resembling **infectious mononucleosis**, and include the following:

- enlarged lymph nodes
- muscle pains
- intermittent fever
- general sick feeling

The distinction is made between acquired toxoplasmosis, in which an individual becomes infected, and neonatal congenital toxoplasmosis, in which a fetus is born with the infection because the mother became infected during pregnancy. If a fetus becomes infected early in pregnancy, the disease can cause the fetus to spontaneously abort, be stillborn. If full-term, the infant may die in infancy or suffer from central nervous system lesions. If the mother becomes infected in the last three months of pregnancy, however, the prognosis is good and the baby may not even display any symptoms.

In adults, if the infection continues for an extended period of time, chronic toxoplasmosis can cause an inflammation of the eyes called retinochoroiditis, which can lead to blindness, severe yellowing of the skin and whites of the eyes (**jaundice**), easy bruising, and convulsions.

Adults with weakened immune systems have a high risk of developing cerebral toxoplasmosis, including inflammation of the brain (**encephalitis**), one-sided weakness or **numbness**, mood and personality changes, vision disturbances, **muscle spasms**, and severe headaches. If untreated, cerebral toxoplasmosis can lead to **coma** and **death**. This form of encephalitis is the second most common AIDS-related nervous system infection that takes advantage of a person's weakened immune system (opportunistic infection).

Diagnosis

A diagnosis of toxoplasmosis is made based on clinical signs and supporting laboratory results, including visualization of the protozoa in body tissue or isolation in animals and blood tests. Laboratory tests for toxoplasmosis are designed to detect increased amounts of a protein or antibody produced in response to infection with the toxoplasmosis organism. Antibody levels can be elevated for years, however, without active disease.

Treatment

Most individuals who contract toxoplasmosis do not require treatment because their immune systems are able to control the disease. Symptoms are not usually present. Mild symptoms may be relieved by taking over-the-counter medications, such as **acetaminophen** (Tylenol) and ibuprofen (Motrin, Advil). **Sore throat** lozenges and rest may also ease the symptoms.

Although the treatment of women infected with toxoplasmosis during pregnancy is controversial,

KEY TERMS

Cyst—The thick-walled dormant form of many organisms.

Immunocompromised—A state in which the immune system is suppressed or not functioning properly.

Oocyst—The egg form of the toxoplasmosis organism.

Protozoan—A single-celled, usually microscopic, organism.

most physicians feel that treatment is justified. Transmission of toxoplasmosis from the mother to the fetus may be prevented if the mother takes the antibiotic spiramycin. Later in a pregnancy, if the fetus has contracted the disease, treatment with the antibiotic pyrimethamine (Daraprim, Fansidar) or **sulfonamides** may be effective. Babies born with toxoplasmosis who show symptoms of the disease may be treated with pyrimethamine, the sulfa drug sulfadiazine (Microsulfon), and folinic acid (an active form of **folic acid**).

AIDS patients who have not been infected may be given a drug called TMP/SMX (Bactrim or Septra) to prevent toxoplasmosis infection. To treat cases of toxoplasmosis in immunocompromised AIDS patients, a combination of pyrimethamine and a sulfa-based drug, either sulfadiazine or clindamycin (Cleocin), have been used together and can be effective in treating this disease. Other antibiotic combinations and dosing schedules are still being investigated. Physicians have reported success in alleviating symptoms by using trimethoprim-sulfamethoxazole (Proloprim or Trimpex) or dapsone (DDS) plus pyrimethamine. These drugs can produce side effects, such as allergic reaction, **itching**, rashes, and **nausea** and patients must be monitored closely.

Prognosis

The prognosis is poor when congenital toxoplasmosis is acquired during the first three months of pregnancy. Afflicted children die in infancy or suffer damage to their central nervous systems that can result in physical and **mental retardation**. Infection later in pregnancy usually results in only mild symptoms, if any. The prognosis for acquired toxoplasmosis in adults with strong immune systems is excellent. The disease often disappears by itself after several weeks. However, the

prognosis for immunodeficient patients is not as positive. These patients often relapse when treatment is stopped. The disease can be fatal to all immunocompromised patients, especially AIDS patients, and particularly if not treated. As a result, immunocompromised patients are typically placed on anti-toxoplasmosis drugs for the rest of their lives.

Prevention

There are no drugs that can eliminate *T. gondii* cysts in animal or human tissues. Humans can reduce their risks of developing toxoplasmosis by practicing the following:

- freezing (to 10.4°F/−12°C) and cooking foods to an internal temperature of 152°F/67°C will kill the cyst
- practicing sanitary kitchen techniques, such as washing utensils and cutting boards that come into contact with raw meat
- keeping pregnant women and children away from household cats and cat litter
- disposing of cat feces daily, because the oocysts do not become infective until after 24 hours
- helping cats to remain free of infection by feeding them dry, canned, or boiled food and by discouraging hunting and scavenging
- washing hands after outdoor activities involving soil contact and wearing gloves when gardening

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Maury M. Breecher, PhD

Toxoplasmosis, Other, Rubella,
Cytomegalovirus, Herpes simplex test see
TORCH test

Trabeculectomy

Definition

Trabeculectomy is a surgical procedure that removes part of the trabeculum in the eye to relieve pressure caused by **glaucoma**.

Purpose

Glaucoma is a disease that injures the optic nerve, causing progressive loss of vision. Presently, glaucoma is a major cause of blindness in the United States. If caught early, glaucoma-related blindness is easily prevented. However, since it does not produce symptoms until late in its cycle, periodic tests for the disease are necessary.

Glaucoma is usually associated with an increase in the pressure inside the eye. This increase occurs in front of the iris in a fluid called the aqueous humor. Aqueous humor is supposed to exit through tiny channels between the iris and the cornea, in an area called the trabeculum. When the trabeculum is blocked, pressure from the build up of aqueous humor either increases rapidly with considerable **pain** and redness, or, as in most cases, the pressure builds slowly with no symptoms until much of the vision is lost. Trabeculectomy is the last treatment employed for either type of glaucoma. It is used only after medications and laser trabeculoplasty (less invasive procedure that uses a laser to open the blocked trabeculum) have failed to alleviate the pressure.

Description

A trabeculectomy involves removing a tiny piece of the eyeball right at the place where the cornea connects to the sclera (the white part), and creating a flap to allow fluid to escape the anterior chamber without deflating the eye. Along with that tiny piece of cornea and sclera comes a piece of the iris. The whole area is called the trabeculum. Fluid can then flow out onto the surface of the eye and be absorbed by the conjunctiva, the transparent membrane that lines the sclera and the eyelids. Sometimes, an additional piece is taken out of the iris so that anterior chamber fluid can also flow backward into the vitreous part of the eye. This procedure is called an iridectomy.

Preparation

The procedure and its benefits and possible complications are fully explained. Antiglaucoma drugs are prescribed before surgery. Added pressure on the eye caused from coughing or sneezing should be avoided.

Aftercare

Eye drops, and perhaps patching, will be needed until the eye is healed. The pressure inside the eye will still be monitored. Immediately following the procedure, the patient may experience blurred vision.

KEY TERMS

Cornea—Transparent film that covers the iris and pupil.

Iris—Colored part of the eye, which is suspended in aqueous humor and perforated by the pupil.

Sclera—White, outer coating of the eyeball.

Trabeculoplasty—Laser surgery that creates perforations in the trabeculum, to drain built up aqueous humor and relieve pressure.

Trabeculum—Tissue that is a drainage point for aqueous humor in the eye.

Risks

Infection and bleeding are risks of any surgery. Scarring can cause the drainage to stop. A third of patients with trabeculectomies will develop **cataracts**.

Resources

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Tracheoesophageal fistula

Definition

Tracheoesophageal **fistula** (TEF) is commonly a birth defect, with the trachea connected to the esophagus. In most cases, the esophagus is discontinuous, causing immediate feeding difficulties. TEFs may develop in adult life, secondary to the invasion of **cancer** in the area. In addition, TEFs may be deliberately constructed with surgery to aid talking in a patient who has the larynx removed (a **laryngectomy**).

Description

The trachea, or windpipe, carries air to the lungs. The esophagus carries food to the stomach. Sometimes during development, these two tubes do not separate completely, but remain connected by a short passage. When this happens, air enters the gastrointestinal system, causing the bowels to distend,

and mucus is breathed into the lungs causing aspiration **pneumonia** and breathing problems.

Most tracheoesophageal fistulas are diagnosed when a child is born. There are three types. In 85–90% of tracheoesophageal fistulas, the top part of the esophagus ends in a blind sac, and the lower part inserts into the trachea. In the second type, the upper part of the esophagus is connected directly to the trachea, while the lower part ends in a pouch. In a rare type of fistula called an H type, both the esophagus and trachea are complete, but they are connected. This is the most difficult type of tracheoesophageal fistula to diagnose, because both eating and breathing are possible.

Causes and symptoms

Tracheoesophageal fistulas arise as a developmental abnormality. At birth, the infant has difficulty swallowing. Eating produces severe coughing spells that interfere with breathing. Aspiration pneumonia can develop from fluid breathed into the lungs.

Small H type fistulas may go undiagnosed until later in life. Symptoms of an H type fistula include frequent pulmonary infections and bouts of abdominal bloating.

Diagnosis

Diagnosis that the esophagus is interrupted is confirmed by the inability to insert a nasogastric suction tube into the stomach. The exact type and location of the fistula can be determined using a radiopaque catheter, which allows pictures to be taken of the esophagus. X rays may show air in the bowels. Endoscopy often fails to locate the fistula if it is small.

Treatment

Babies with all but H type fistulas are unlikely to survive without surgical separation and repair of the trachea and the esophagus. Surgery cannot always be performed immediately because of **prematurity**, the presence of other birth defects, or complications from aspiration pneumonia. It is usually done at a hospital that has special facilities for treating seriously ill newborns.

While awaiting surgery, the infant's condition is stabilized. Preoperative care concentrates on avoiding aspiration pneumonia and includes:

- elevating the head to avoid reflux and aspiration of the stomach contents

KEY TERMS

Endoscopy—A procedure in which an instrument containing a camera and a light source is inserted into the gastrointestinal tract so that the physician can visually inspect the gastrointestinal system.

Gastrostomy tube—Stomach tube for feeding.

Laryngectomy—Surgical removal of the larynx to treat cancer.

- using a suction catheter to continuously removed mucus and saliva that could be inhaled
- when necessary, placement of a **gastrostomy** tube
- withholding feeding by mouth

When surgery is performed, the esophagus is reconnected to make it continuous and separate from the trachea. If the two ends of the esophagus are too far apart to be reattached, a piece of tissue from the large intestine is used to join the parts.

Prognosis

Infants who have tracheoesophageal fistula often have other **birth defects** that affect their recovery. Even when the esophagus is successfully separated and reattached, many infants have difficulty swallowing, because the contractility of the esophagus is impaired. Infants may also have problems with gastroesophageal reflux, in which the acidic contents of the stomach back up into the bottom of the esophagus and cause ulcers and scarring.

Prevention

Tracheoesophageal fistulas are not preventable birth defects.

Resources

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Tish Davidson, A.M.

Tracheostomy see **Tracheotomy**

Tracheotomy

Definition

A tracheotomy is a surgical procedure in which a cut or opening is made in the windpipe (trachea). The surgeon inserts a tube into the opening to bypass an obstruction, allow air to get to the lungs, or remove secretions. The term tracheostomy is sometimes used interchangeably with tracheotomy. Strictly speaking, however, tracheostomy usually refers to the opening itself while a tracheotomy is the actual operation.

Purpose

A tracheotomy is performed if enough air is not getting to the lungs, if the person cannot breathe without help, or is having problems with mucus and other secretions getting into the windpipe because of difficulty swallowing. There are many reasons why air cannot get to the lungs. The patient's windpipe may be blocked by a swelling; by a severe injury to the neck, nose or mouth; by a large foreign object; by **paralysis** of the throat muscles; or by a tumor. The patient may be in a **coma**, or need a ventilator to pump air into the lungs for a long period of time.

Precautions

Doctors perform emergency tracheotomies as last-resort procedures. They are done only if the patient's windpipe is obstructed and the situation is life-threatening.

Description

Emergency tracheotomy

There are two different procedures that are called tracheotomies. The first is done only in emergency situations and can be performed quite rapidly. The emergency room physician or surgeon makes a cut in a thin part of the voice box (larynx) called the cricothyroid membrane. A tube is inserted and connected to an oxygen bag. This emergency procedure is sometimes called a cricothyroidotomy.

Nonemergency tracheotomy

The second type of tracheotomy takes more time and is usually done in an operating room. The surgeon first makes a cut (incision) in the skin of the neck that lies over the trachea. This incision is in the lower part of the neck between the Adam's apple and top of the breastbone. The neck muscles are separated and

the thyroid gland, which overlies the trachea, is usually cut down the middle. The surgeon identifies the rings of cartilage that make up the trachea and cuts into the tough walls. A metal or plastic tube, called a tracheotomy tube, is inserted through the opening. This tube acts like a windpipe and allows the person to breathe. Oxygen or a mechanical ventilator may be hooked up to the tube to bring oxygen to the lungs. A dressing is placed around the opening. Tape or stitches (sutures) are used to hold the tube in place.

After a nonemergency tracheotomy, the patient usually stays in the hospital for three to five days, unless there is a complicating condition. It takes about two weeks to recover fully from the surgery.

Preparation

Emergency tracheotomy

In the emergency tracheotomy, there is no time to explain the procedure or the need for it to the patient. The patient is placed on his or her back with face upward (supine), with a rolled-up towel between the shoulders. This positioning of the patient makes it easier for the doctor to feel and see the structures in the throat. A local anesthetic is injected across the cricothyroid membrane.

Nonemergency tracheotomy

In a nonemergency tracheotomy, there is time for the doctor to discuss the surgery with the patient, to explain what will happen and why it is needed. The patient is then put under **general anesthesia**. The neck area and chest are then disinfected as preparation for the operation, and surgical drapes are placed over the area, setting up a sterile field.

Aftercare

Postoperative care

A **chest x ray** is often taken, especially in children, to check whether the tube has become displaced or if complications have occurred. The doctor may prescribe **antibiotics** to reduce the risk of infection. If the patient can breathe on their own, the room is humidified; otherwise, if the tracheotomy tube is to remain in place, the air entering the tube from a ventilator is humidified. During the hospital stay, the patient and his or her family members will learn how to care for the tracheotomy tube, including suctioning and clearing it. Secretions are removed by passing a smaller tube (catheter) into the tracheotomy tube.

It takes most patients several days to adjust to breathing through the tracheotomy tube. At first, it will be hard even to make sounds. If the tube allows some air to escape and pass over the vocal cords, then the patient may be able to speak by holding a finger over the tube. A patient on a ventilator will not be able to talk at all.

The tube will be removed if the tracheotomy is temporary. Then the wound will heal quickly and only a small scar may remain. If the tracheotomy is permanent, the hole stays open and, if it is no longer needed, it will be surgically closed.

Home care

After the patient is discharged, he or she will need help at home to manage the tracheotomy tube. Warm compresses can be used to relieve **pain** at the incision site. The patient is advised to keep the area dry. It is recommended that the patient wear a loose scarf over the opening when going outside. He or she should also avoid contact with water, food particles, and powdery substances that could enter the opening and cause serious breathing problems. The doctor may prescribe pain medication and antibiotics to minimize the risk of infections. If the tube is to be kept in place permanently, the patient can be referred to a speech therapist in order to learn to speak with the tube in place. The tracheotomy tube may be replaced four to 10 days after surgery.

Patients are encouraged to go about most of their normal activities once they leave the hospital. Vigorous activity is restricted for about six weeks. If the tracheotomy is permanent, further surgery may be needed to widen the opening, which narrows with time.

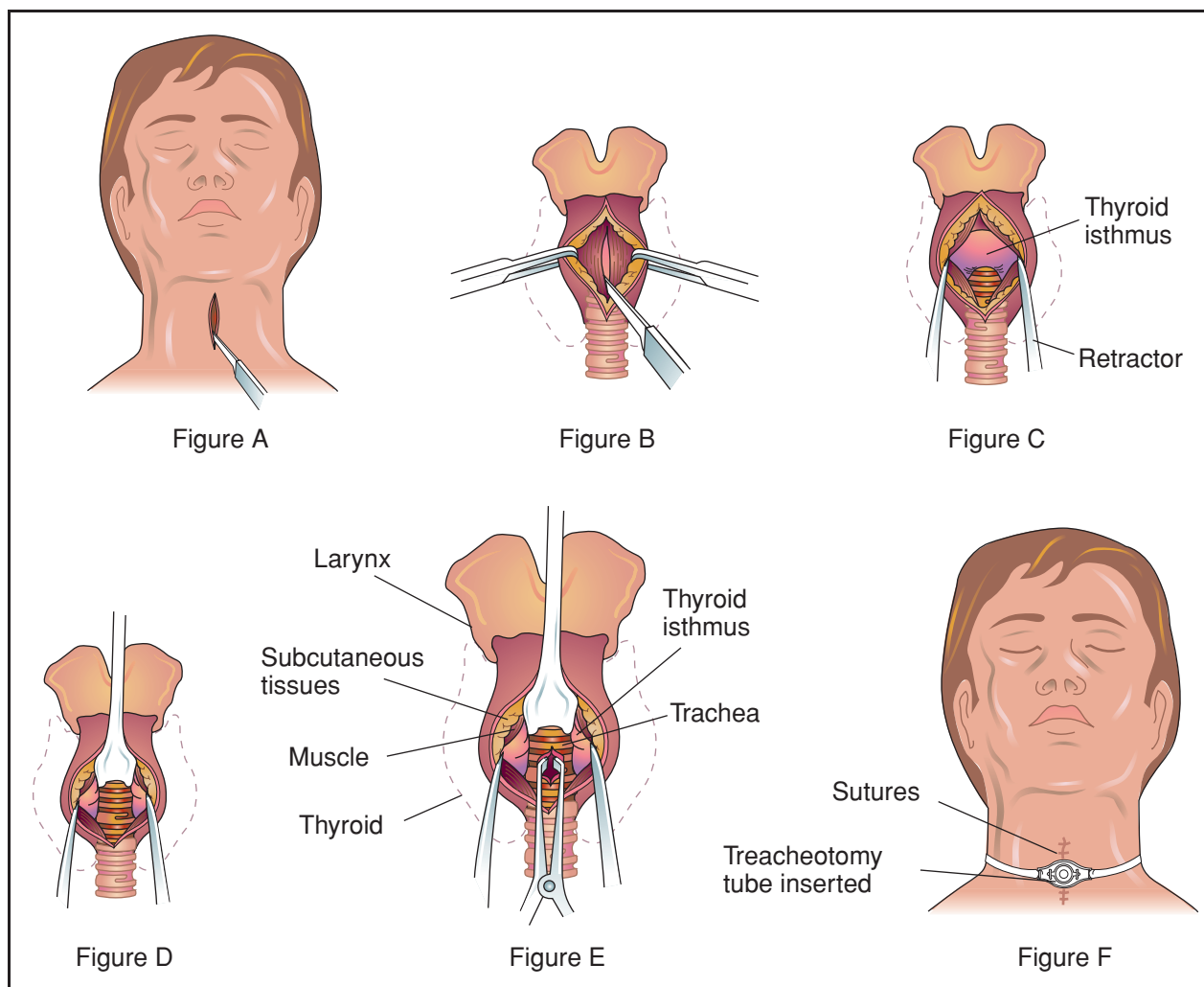
Risks

Immediate risks

There are several short-term risks associated with tracheotomies. Severe bleeding is one possible complication. The voice box or esophagus may be damaged during surgery. Air may become trapped in the surrounding tissues or the lung may collapse. The tracheotomy tube can be blocked by **blood clots**, mucus, or the pressure of the airway walls. Blockages can be prevented by suctioning, humidifying the air, and selecting the appropriate tracheotomy tube. Serious infections are rare.

Long-term risks

Over time, other complications may develop following a tracheotomy. The windpipe itself may



Tracheotomy is a surgical procedure in which an opening is made in the windpipe or trachea. As shown in the illustration above, the physician or surgeon will follow these steps in performing this procedure: Figure A: A vertical incision is made through the skin. Figure B: Another incision is made through the subcutaneous tissues and muscles of the neck. Figure C: The neck muscles are separated using retractors. Figure D: The thyroid isthmus is either cut or retracted. Figure E: The surgeon identifies the rings of cartilage that make up the trachea and cuts into the walls. Figure F: A metal or plastic tube is inserted into the opening and sutures are used to hold the tube in place. (Illustration by Electronic Illustrators Group.)

become damaged for a number of reasons, including pressure from the tube; bacteria that cause infections and form scar tissue; or friction from a tube that moves too much. Sometimes the opening does not close on its own after the tube is removed. This risk is higher in tracheotomies with tubes remaining in place for 16 weeks or longer. In these cases, the wound is surgically closed.

High-risk groups

The risks associated with tracheotomies are higher in the following groups of patients:

- children, especially newborns and infants
- smokers
- alcoholics
- obese adults
- persons over 60
- persons with chronic diseases or respiratory infections
- persons taking **muscle relaxants**, sleeping medications, tranquilizers, or cortisone

The overall risk of **death** from a tracheotomy is less than 5%.

KEY TERMS

Cartilage—A tough, fibrous connective tissue that forms various parts of the body, including the trachea and larynx.

Cricothyroidotomy—An emergency tracheotomy that consists of a cut through the cricothyroid membrane to open the patient's airway as fast as possible.

Larynx—A structure made of cartilage and muscle that connects the back of the throat with the trachea. The larynx contains the vocal cords.

Trachea—The tube that leads from the larynx or voice box to two major air passages that bring oxygen to each lung. The trachea is sometimes called the windpipe.

Ventilator—A machine that helps patients to breathe. It is sometimes called a respirator.

Normal results

Normal results include uncomplicated healing of the incision and successful maintenance of long-term tube placement.

Resources

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Trachoma

Definition

Trachoma, which is also called granular **conjunctivitis** or Egyptian ophthalmia, is a contagious, chronic inflammation of the mucous membranes of the eyes, caused by *Chlamydia trachomatis*. It is characterized by swelling of the eyelids, sensitivity to light, and eventual scarring of the conjunctivae and corneas of the eyes.

Description

Trachoma is a major cause of blindness in the world. It is found in the Far East, as well as countries with desert climates. In the United States, it is most common among certain Native Americans and in parts of Appalachia. The infection is highly contagious in its early stages. Blindness results from recurrent untreated infections.

The conjunctiva is the clear mucous membrane that lines the inside of the eyelid and covers the white part (sclera) of the eye. Conjunctivitis is an inflammation of the conjunctiva.

Causes and symptoms

Trachoma is caused by *C. trachomatis*, a parasitic organism closely related to bacteria. It is transmitted by insects, by hand-to-eye contact, or by the sharing of infected handkerchiefs or towels. The incubation period is about a week.

The early symptoms of trachoma include the development of follicles (small sacs) on the conjunctivae of the upper eyelids, **pain**, swollen eyelids, a discharge, tearing, and sensitivity to light. If the infection is not treated, the follicles develop into large yellow or gray pimples, and small blood vessels develop inside the cornea. In most cases, both eyes are infected.

Repeated infections eventually lead to contraction and turning-in of the eyelids, scarring of the corneas and conjunctivae, eventual blockage of the tear ducts, and blindness.

Diagnosis

Diagnosis is based on a combination of the patient's history (especially living or traveling in areas with high rates of trachoma) and examination of the eyes. The doctor will look for the presence of follicles or scarring. He or she will take a small sample of cells from the patient's conjunctivae and examine them, following a procedure called Giemsa staining, to confirm the diagnosis.

Treatment

Treatment of early-stage trachoma consists of four to six weeks of antibiotic treatment with tetracycline, erythromycin, or **sulfonamides**. **Antibiotics** should be given without waiting for laboratory test results. Treatment may combine oral medication with antibiotic ointment applied directly to the eyes. A single-dose treatment with azithromycin is an alternative method.



A close-up of a human eye with trachoma. Trachoma is caused by *Chlamydia trachomatis* and commonly results in blindness if left untreated. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Conjunctivitis—Inflammation of the conjunctivae, which are the mucous membranes covering the white part of the eyeball (sclera) and lining the inside of the eyelids.

Cornea—The transparent front part of the eye that allows light to enter.

Ophthalmia—Inflammation of the eye. Usually severe and affecting the conjunctiva. Trachoma is sometimes called Egyptian ophthalmia.

Tetracyclines should not be given to pregnant women or children below the age of seven years.

Patients with complications from untreated or repeated infections are treated surgically. Surgery can be used for **corneal transplantation** or to correct eyelid deformities.

Prognosis

The prognosis for full recovery is excellent if the patient is treated promptly. If the infection has progressed to the stage of follicle development, prevention of blindness depends on the severity of the follicles, the presence of additional bacterial infections, and the development of scarring.

Prevention

There are vaccines available that offer temporary protection against trachoma, but there is no

permanent immunization. Prevention depends upon good hygiene and public health measures:

- seek treatment immediately if a child shows signs of eye infection, and minimize his or her contact with other children
- teach children to wash hands carefully before touching their eyes
- protect children from flies or gnats that settle around the eyes
- if someone has trachoma (or any eye infection), do not share towels, pillowcases, etc; Wash items well
- if medications are prescribed, follow the doctor's instructions carefully

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Traction

Definition

Traction is the use of a pulling force to treat muscle and skeleton disorders.

Purpose

Traction is usually applied to the arms and legs, the neck, the backbone, or the pelvis. It is used to treat **fractures**, **dislocations**, and long-duration muscle spasms, and to prevent or correct deformities. Traction can either be short-term, as at an accident scene, or long-term, when it is used in a hospital setting.

Traction serves several purposes:

- it aligns the ends of a fracture by pulling the limb into a straight position
- it ends muscle spasm
- it relieves pain
- it takes the pressure off the bone ends by relaxing the muscle

There are two main types of traction: skin traction and skeletal traction. Within these types, many specialized forms of traction have been developed to address problems in particular parts of the body. The application of traction is an exacting technique that requires training and experience, since incorrectly applied traction can cause harm.

Positioning the extremity so that the angle of pull brings the ends of the fracture together is essential. Elaborate methods of weights, counterweights, and pulleys have been developed to provide the appropriate force while keeping the bones aligned and preventing muscle spasm. The patient's age, weight, and medical condition are all taken into account when deciding on the type and degree of traction.

Precautions

People who are suffering from skin disorders or who are allergic to tape should not undergo skin traction, because the application of traction will aggravate their condition. Likewise, circulatory disorders or **varicose veins** can be aggravated by skin traction. People with an inflammation of the bone (osteomyelitis) should not undergo skeletal traction.

Description

Skin traction

Skin traction uses five-to seven-pound weights attached to the skin to indirectly apply the necessary pulling force on the bone. If traction is temporary, or if only a light or discontinuous force is needed, then skin traction is the preferred treatment. Because the procedure is not invasive, it is usually performed in a hospital bed.

Weights are attached either through adhesive or nonadhesive tape, or with straps, boots, or cuffs. Care must be taken to keep the straps or tape loose enough to prevent swelling and allow good circulation to the part of the limb beyond the spot where the traction is applied. The amount of weight that can be applied through skin traction is limited because excessive weight will irritate the skin and cause it to slough off.

Specialized forms of skin traction have been developed to address specific problems. Dunlop's traction is used on children with certain fractures of the upper arm, when the arm must be kept in a flexed position to prevent problems with the circulation and nerves around the elbow. Pelvic traction is applied to the lower spine, with a belt around the waist. Buck's skin traction is used to treat **knee injuries** other than

fractures. The purpose of this traction is to stabilize the knee and reduce muscle spasm.

Skeletal traction

Skeletal traction is performed when more pulling force is needed than can be withstood by skin traction; or when the part of the body needing traction is positioned so that skin traction is impossible. Skeletal traction uses weights of 25-40 pounds.

Skeletal traction requires the placement of tongs, pins, or screws into the bone so that the weight is applied directly to the bone. This is an invasive procedure that is done in an operating room under general, regional, or **local anesthesia**.

Correct placement of the pins is essential to the success of the traction. The pin can be kept in place several months, and must be kept clean to prevent infection. Once the hardware is in place, pulleys and weights are attached to wires to provide the proper pull and alignment on the affected part.

Specialized forms of skeletal traction include cervical traction used for fractures of the neck vertebrae; overhead arm traction used for certain types of upper arm fractures; and tibia pin traction used for some fractures of the femur, hip, or pelvis.

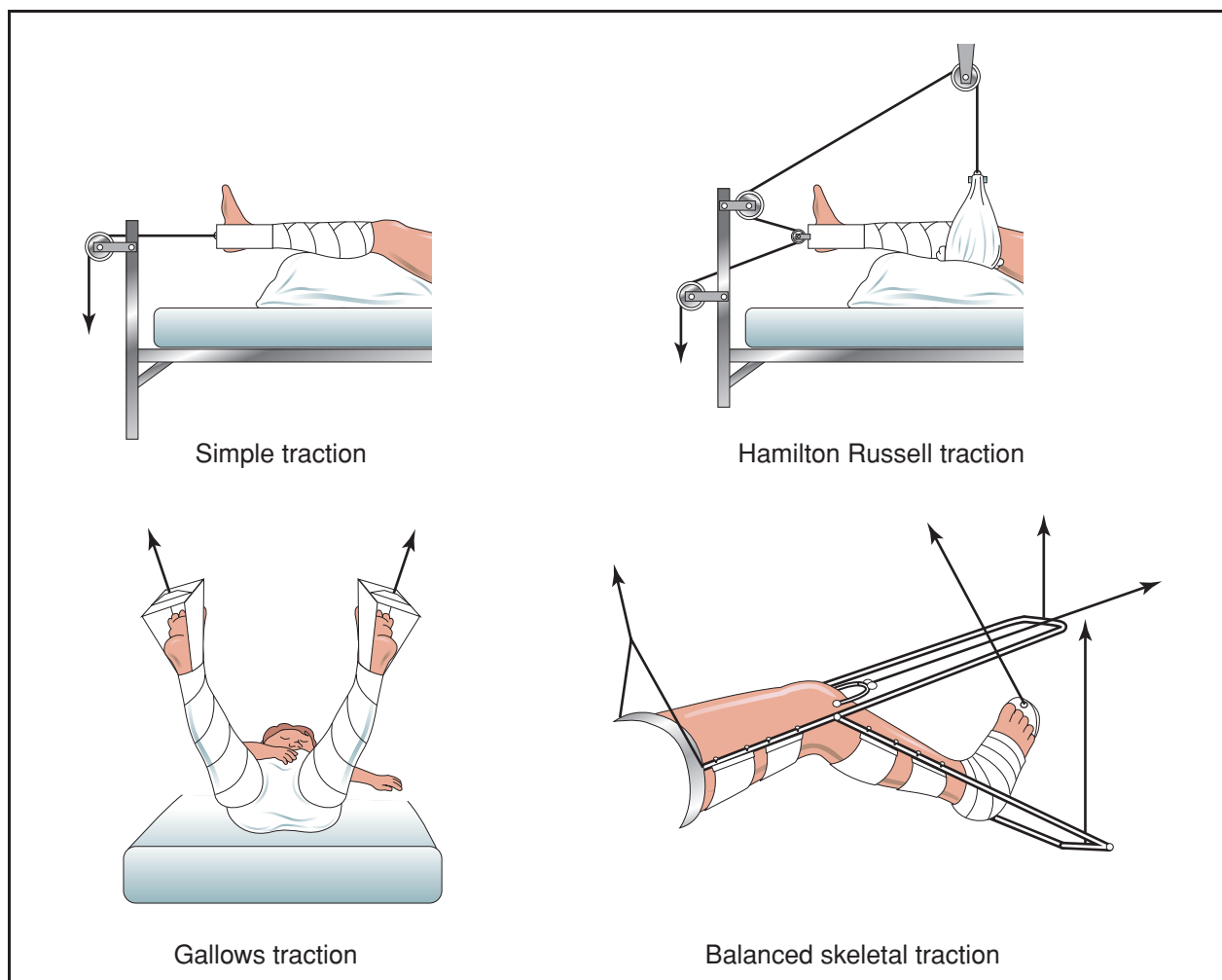
Preparation

X rays are done prior to the application of both forms of traction, and may be repeated during treatment to assure that the affected parts are staying in alignment and healing properly. Since the insertion of the anchoring devices in skeletal traction is a surgical procedure, standard preoperative blood and urine testing are done, and the patient may meet with an anesthesiologist to discuss any health conditions that might affect the administration of anesthesia.

Aftercare

Aftercare for skin traction involves making sure the limb stays aligned, and caring for the skin so that it does not become sore and irritated. The patient should also be alert to any swelling or **tingling** in the limb that would suggest that the limb has been wrapped too tightly.

Aftercare for skeletal traction is more complex. The patient is likely to be immobile for an extended period. Deep breathing exercises are taught so that respiratory function is maintained during this time of little activity. Patients are also encouraged to do range-of-motion exercises with the unaffected parts of the body. The patient is taught how to use a trapeze



Traction refers to the usage of a pulling force and special devices, such as a cast or splint, to treat muscle and skeletal disorders. It is used to treat fractures, dislocations, and long-duration muscle spasms, and to prevent or correct deformities. The illustration above features several commonly used forms of traction. (Illustration by Electronic Illustrators Group.)

(an overhead support bar) to shift on and off a bedpan, since it is not possible to get up to use the toilet. In serious injuries, traction may be continued for several months until healing is complete.

Risks

The main risks associated with skin traction are that the traction will be applied incorrectly and cause harm, or that the skin will become irritated. There are more risks associated with skeletal traction. Bone inflammation may occur in response to the introduction of foreign material into the body. Infection can occur at the pin sites. If caught early, infection can be treated with antibiotics, but if severe, it may require removal of the pin.

Both types of traction have complications associated with long periods of immobility. These include

the development of bed sores, reduced respiratory function, urinary problems, and circulatory problems. Occasionally, fractures fail to heal. Being confined to traction for a long period can take an emotional toll on the patient, also.

Normal results

When correctly applied, traction generally produces very good, if slow, results.

Resources

BOOKS

"Traction." In *Everything You Need to Know About Medical Treatments*. Springhouse, PA: Springhouse Corp., 1996.

Tish Davidson, A.M.

Traditional Chinese medicine

Definition

Traditional Chinese medicine (TCM) is an ancient and still very vital holistic system of health and healing, based on the notion of harmony and balance, and employing the ideas of moderation and prevention.

Purpose

TCM is a complete system of health care with its own unique theories of anatomy, health, and treatment. It emphasizes diet and prevention and using **acupuncture**, herbal medicine, massage, and **exercise**; and focuses on stimulating the body's natural curative powers.

Precautions

In situations of severe trauma, TCM should not be substituted for contemporary modern trauma practice; it is most useful as an adjunct to the healing regimen. TCM is not the first line of treatment for bacterial infection or **cancer**, but may usefully complement contemporary medical treatment for those conditions.

Description

In theory and practice, traditional Chinese medicine is completely different from Western medicine, both in terms of considering how the human body works and how illness occurs and should be treated. As a part of a continuing system that has been in use for thousands of years, it is still employed to treat over one-quarter of the world's population. Since the earliest Chinese physicians were also philosophers, their ways of viewing the world and human beings role in it affected their medicine. In TCM, both philosophically and medically, moderation in all things is advocated, as is living in harmony with nature and striving for balance in all things. Prevention is also a key goal of Chinese medicine, and much emphasis is placed on educating the patient to live responsibly. The Chinese physician also is more of an advisor than an authority; he or she believes in treating every patient differently, based on the notion that one does not treat the disease or condition but rather the individual patient. Thus two people with the same complaint may be treated entirely differently, if their constitutions and life situations are dissimilar. Disease is also considered to be evidence of the failure of preventive health care and a falling out of balance or harmony.

There is some confusion in the West about the fundamental philosophical principles upon which

traditional Chinese medicine is based – such as the concept of yin and yang, the notion of five elements (wood, fire, earth, metal and water), and the concept of *chi*—yet each can be explained in a way that is understandable to Westerners.

Yin and yang describe the interdependent relationship of opposing but complementary forces believed to be necessary for a healthy life. Basically, the goal is to maintain a balance of yin and yang in all things.

The five elements, or five phase theory, is also grounded in the notion of harmony and balance. The concept of *chi*, which means something like “life force” or “energy,” is perhaps most different from Western ideas. TCM asserts that *chi* is an invisible energy force that flows freely in a healthy person, but is weakened or blocked when a person is ill. Specifically, the illness is a result of the blockage, rather than the blockage being the result of the illness.

Besides these philosophical concepts that differ considerably from infection-based principles of medicine and health, the methods employed by traditional Chinese medicine are also quite different. If allopathic Western practitioners could be described as interventionist and dependent on synthetic pharmaceuticals, TCM methods are mostly natural and noninvasive. For example, where Western physicians might employ surgery and **chemotherapy** or radiation for a cancer patient, a TCM physician might use acupuncture and dietary changes. TCM believes in “curing the root” of a disease and not merely in treating its symptoms.

Another major difference is how the patient is regarded. In Western medicine, patients with similar complaints or diseases, usually will receive virtually the same treatment. In TCM however, the physician treats the patient and not the condition, believing that identical diseases can have entirely different causes. In terms of the principles upon which it is based and the methods used, traditional Chinese medicine, therefore, is considered by many in the West to be a radically different system of health care.

To some in the Western world, this very strangeness is the reason why it might be attractive. To others, tired of what they perceive as their physician's perfunctory, analytical, and sometimes cold manner, TCM offers a more humane, patient-oriented approach that encourages a high degree of practitioner-patient interaction and is not overly dependent on technology.

For example, during a consultation with a TCM practitioner, the patient will receive a considerable amount of time and attention. During the important

first visit, the practitioner will conduct four types of examinations, all extremely observational and all quite different from what patients usually experience.

First, the practitioner will ask many questions, going beyond the typical patient history to inquire about such particulars as eating and bowel habits or sleep patterns. Next, the physician looks at the patient, observing his or her complexion and eyes, while also examining the tongue very closely, believing that it is a barometer of the body's health and that different areas of the tongue can reflect the functioning of different body organs. After observing, they listen to the patient's voice or **cough** and then smell his or her breath, body odor, urine, and even bowel movements. Finally, the practitioner touches the patient, palpating his or her abdomen and feeling the wrist to take up to six different pulses. It is through these different pulses that the well-trained practitioner can diagnose any problem with the flow of the all-important *chi*. Altogether, this essentially observational examination will lead the physician to diagnose or decide the patient's problem. This diagnosis is very different from one in contemporary Western medicine. No blood or urine samples are tested in a laboratory. The key to this technique lies in the experience and skill of the practitioner.

After making a diagnosis, the physician will suggest a course of treatment from one or all of the available TCM methods. These fall into four main categories: herbal medicine, acupuncture, dietary therapy, and massage and exercise. A typical TCM prescription consists of a complex variety of many different herbal and mineral ingredients. Chinese herbal remedies are intended to assist the body's own systems so that eventually the patient can stop taking them and never becomes dependent on them. Herbal formulas are usually given as teas, which differ according to the patient.

Other common techniques used in a TCM prescription are as follows:

- Acupuncture is based on the notion that the body's vital energy force, *chi*, travels through known channels or "meridians." The acupuncturist inserts tiny, thin sterile needles at particular, selected points on the body to unblock or correct the flow of energy. These needles are hardly felt as they are inserted and are left in place for 15 to 20 minutes. Some patients report immediate improvement, others feel exhilarated, while some feel like sleeping. In some cases, patients say their condition worsens before it improves. No contemporary scientific explanation exists as to how or why acupuncture works.

KEY TERMS

Allopathic—Pertaining to conventional medical treatment of disease symptoms that uses substances or techniques to oppose or suppress the symptoms.

Anatomy—The science of the body structure of an organism and its parts.

Holistic—That which pertains to the entire person, including the mind, body, and spirit.

Palpate—To examine the body by touching or pressing with the fingers or the palm of the hand.

Pharmaceutical—Pertaining to drugs.

Therapeutic—Curative or healing.

Trauma—Injury or damage to the body.

- Moxibustion is a variation sometimes employed. Moxibustion is the slow burning on or over the body of special herbal "cones." These are placed on specific acupoints and provide penetrating, relaxing heat.
- Massage is often recommended, and a deep finger pressure technique known as **acupressure** is often used to promote the proper flow of *chi*.
- Diet is considered essential to good health, and what might be called "kitchen medicine" is just another aspect of herbalism. One example is a delicious DQ black bean soup that is traditionally eaten by women in China after **childbirth** and each menstrual cycle.
- Therapeutic exercises are sometimes prescribed as well. In both the exact and flowing movements of **tai chi**, and the breathing techniques of Qi Dong exercise is considered essential to relieving **stress** and promoting the smooth flow of *chi*.

As a system of total health care, TCM is prepared to deal with any physical or mental problem, condition, or disease. However, unlike Western medicine at its best, TCM is not able to render the kind of emergency crisis intervention that saves lives during physical traumas. Nonetheless, it works best at achieving its goal of practicing preventive medicine. It has proven effective in treating many types of aches and pains and in helping people with depression and **fatigue**, as well as circulation and digestive problems. Overall, its emphasis on good diet and exercise, as well as on individual responsibility and moderation in all things, suggest that it is grounded in fundamentally sound principles.

Risks

In the hands of a qualified practitioner, TCM is very safe. However, there is a small chance of not only getting an infection from acupuncture, but also that an existing infection could be spread to other parts of the body by increased blood flow and circulation.

Normal results

Traditional Chinese medicine seeks to harmonize and rebalance the entire human system rather than to treat just symptoms. Since proper internal balance is considered to be the key to human health, TCM strives to cure disease by restoring that balance and therefore allowing the body to repair itself. Its continuing medical goal is to detect and correct abnormalities before they cause permanent physical damage.

Resources

ORGANIZATIONS

American Academy of Medical Acupuncture. 2520 Milvia St., Berkeley, CA 94704. (415) 841-3220.

American Association of Acupuncture & Oriental Medicine. 4101 Lake Boone Trail, Suite 201, Raleigh, NC 27607. (919) 787-5181.

Leonard C. Bruno, PhD

Trager psychophysical integration

Definition

Trager psychophysical integration therapy, also known as the Tragerwork system of physical integration, is a combination of hands-on tissue mobilization, relaxation, and movement reeducation called Mentastics. The underlying principle of psychophysical integration is that clients learn to be lighter, easier, and freer by experiencing lightness, ease, and freedom of movement in their bodies.

The Trager method is a psychologically grounded physical approach to muscle relaxation, which is induced when a practitioner and patient achieve a state of mind called hook-up. Hook-up is described as a connection to a state of grace or a powerful and nourishing life force. It is the opposite of strain or effort.

MILTON TRAGER (1909–1997)

Milton Trager was a medical doctor and a somatic educator, specializing in body learning. He was a contemporary of F. Matthias Alexander, Moshe Feldenkrais, and Ida Rolf.

As a young man in the 1920s, he occupied himself with gymnastics and boxing. Through his intensely physical pursuits, he arrived at his self-taught body learning theories. The techniques that he nurtured emphasized body control over strength, prowess, and endurance. For example, in striving to leap as high as possible, Trager focused his concentration on landing as softly as possible. He obtained a degree in physical medicine before serving in the military during World War II.

Upon his return, Trager funded his medical school education with his GI benefits. He established a private practice and spent the ensuing 50 years refining his body learning techniques and assisting afflicted individuals in the process. When Trager's father was stricken with sciatic pain, Trager learned to relieve the spasms by hand. In time he learned to alleviate the symptoms of polio victims and others who suffered from muscle spasms.

Trager established the Trager Institute in the 1970s to propagate the techniques that he had developed. By the year 2000, an estimated 2,000 students and practitioners had embraced the Trager Approach.

Trager lived with his wife, Emily, in Southern California at the time of his death in January 1997.

Purpose

Psychophysical integration therapy has been helpful in relieving muscle discomfort in patients afflicted with **polio**, muscular dystrophy, Parkinson's disease, **multiple sclerosis**, post-stroke trauma, and psychiatric disturbances. The therapy is useful in alleviating such chronic conditions as back and leg **pain**. Athletes may benefit from this system to increase resilience to injuries and to improve their mental attitudes. In addition, the Trager Institute maintains that Tragerwork helps clients achieve greater mental clarity through the release of "deep-seated physical and mental patterns."

Description

Origins

Psychophysical integration therapy began with Dr. Milton Trager (1908–1997), who earned a medical degree in midlife after working out his approach to healing chronic pain. Trager was born with a spinal

KEY TERMS

Hook-up—A state of effortless connection with a life-enhancing force. Trager practitioners enter a state of hook-up before working with clients in order to focus on their needs. Trager himself described hook-up as a meditative process of “becoming one with the energy force that surrounds all living things.”

Mentastics—The active phase of Trager therapy. Mentastics are a form of movement reeducation in which clients learn to reexperience movement as pleasurable and positive.

Tablework—The passive phase of Trager therapy, in which the practitioner uses gentle and noninvasive movements to allow the client to relax deeply and experience physical movement as free and effortless.

deformity and overcame it through practicing a variety of athletic exercises. At the time that he discovered his approach to bodywork, he was training to become a boxer. His therapy came to public attention when Esalen Institute in California, the famous center of the human potential movement, invited him to give a demonstration of his technique during the mid-1970s. Trager abandoned his private medical practice in 1977 to devote full energy to the development and further understanding of psychophysical integration. The Trager Institute, which continues his work, was founded in 1980.

The Trager method consists of two parts, a passive aspect referred to as tablework and an active aspect called Mentastics, which is a self-care exercise program. Although the benefits of the Trager approach are said to be cumulative, practitioners and clients appear to be free to set their own schedules for a series of sessions. There is no minimum number of sessions that clients must agree to take.

Tablework

The tablework is performed on a comfortable padded table. Sessions last about 60–90 minutes. The practitioner moves the client in ways that he or she naturally moves, in such a way that he or she experiences how it feels to move effortlessly and freely on one's own. The movements resemble general mobilization techniques, and incorporate some manual, cervical, and lumbar **traction**. The goal of tablework is to

allow the client “slowly to give up muscular and mental control and sink into a very deep state of relaxation not unlike that experienced in hypnosis.”

Mentastics

Mentastics are free-flowing dance-like movements intended to increase the client's self-awareness, as well as providing tools to help the client move through and control chronic pain. The client is encouraged to “let go,” which means that they are asked to begin a movement, then release their muscle tension and allow the weight of the body part involved to complete the motion. By experiencing movement as something pleasurable and positive rather than painful or negative, clients begin to loosen up, learn new movements more easily, and even begin inventing their own. In the early stages of treatment, clients are advised to do Mentastic movements at home for 10–15-minute sessions, three times per day.

Preparations

Prior to a session of tablework, the client dresses for comfort, “with a minimum of swimwear or briefs,” according to the Trager Institute. The client is also covered with a drape. No oils or lotions are used.

The practitioner prepares for the session by clearing his or her mind of everything but the client, until he or she achieves a state of hook-up. This attitude of “relaxed meditative awareness” on the part of the practitioner is one of the unique features of Tragerwork. It is described as allowing the therapist “to connect deeply with the recipient in an unforced way and enables the practitioner to perceive the slightest responses from the [client's] body.”

Precautions

Because of the unusual sensitivity and heightened awareness that is associated with the practitioner's touch, pain should never result from tablework sessions. It is important for clients to alert the practitioner to any pain associated with either the tablework or the Mentastics program.

Although the movements used in Trager tablework are gentle and noninvasive, clients who have had recent injuries or surgery should wait to heal before undertaking a course of Tragerwork.

Side effects

The Trager method should not produce physical side effects when employed by a qualified practitioner.

It is possible that some clients may have emotional reactions associated with the release of physical patterns acquired as a response to trauma, but such reactions are unusual.

Research and general acceptance

Tragerwork, like other forms of bodywork, has gained increasing acceptance as a form of treatment since the 1980s. In 2000 there were 1,200 certified psychophysical integration practitioners in 15 countries worldwide. The therapy has been reported as a commonly employed treatment for mainstream athletes. In addition, the National Institutes of Health lists psychophysical therapy as a mind-body form of complementary alternative medicine.

Resources

ORGANIZATIONS

Florida Institute of Psychophysical Integration: Quantum Balance. 5837 Mariner Drive. Tampa, FL 33609-3411. (813) 186-2273. Fax: (813) 287-2870.

Trager Institute. 21 Locust Avenue. Mill Valley, CA 94941-2806. (415) 388-2688. Fax: (415) 399-2710. admin@trager.com. <<http://www.trager.com>>.

Gloria Cooksey

Tranquilizers see **Antianxiety drugs**

Transcranial Doppler ultrasonography

Definition

Transcranial **Doppler ultrasonography** is a noninvasive method of analyzing blood flow in the brain.

Purpose

The blood that flows through the brain distributes nutrients to the brain and removes wastes. This flow maintains the high rate of metabolism necessary for the brain to function. Restrictions in blood flow may occur from vessel narrowing (stenosis), clot formation (thrombosis), blockage (**embolism**), or blood vessel rupture (hemorrhage). Lack of sufficient blood flow (**ischemia**) threatens brain tissue and may cause a **stroke**.

The flow of blood through the arteries in the brain can be analyzed using transcranial Doppler ultrasonography (TCD). TCD is a form of ultrasound, in which high frequency sound waves bounce off or pass through body tissues. While most other types of ultrasonography create images of the tissue being studied, the results of TCD are audible sounds that the examiner listens to and records.

Doppler ultrasonography uses what is called the Doppler effect to measure the rate and direction of blood flow in the vessels. Just as a siren's pitch sounds higher when its source is moving toward you and lower as it moves away, so too will ultrasound waves change pitch, or frequency, as they bounce off the red blood cells moving in the blood. It is these pitch changes that produce the audible sounds during the exam.

Changes in frequency can be used to measure both the direction and the speed of blood flow. Faster blood flow causes a greater change in frequency. Combined with other tests, this information can be used to locate restrictions in the blood vessels in the brain, and to track changes in blood flow over time. In this way, TCD gives valuable information about the site of a stroke and the patient's progress after a stroke. TCD is also used to evaluate the contraction of blood vessels that can occur if a blood vessel ruptures.

Precautions

Ultrasonography procedures are safe, noninvasive, and painless. No special precautions are necessary.

Description

TCD is done with either one or two probes placed against the skin. The examiner spreads a clear gel on the areas of the head where the probe will be placed. Usually, the probes are placed on the temple, on the base of the skull at the back of the neck, and over the closed eyelid. In these places, there is the least amount of thick protective bone and the sound waves can penetrate the best. The examiner adjusts the probe position and orientation to direct the sound waves toward the blood vessels of interest. Finding the best approach may take some time. A compression test may be performed during the exam. In this test, the main artery in the neck (carotid artery) is briefly compressed, and changes in blood flow patterns are observed. A full TCD exam may last 30–45 minutes, and often longer in patients with disease.

Preparation

No special preparation is needed. The patient should remove contact lenses, and may wish to avoid the use of eye makeup, since the gel is likely to smear it.

Aftercare

The gel is washed off with soap and water. No other after care is needed.

Risks

TCD is noninvasive and has no risks. A compression test is occasionally, though very rarely, hazardous for a patient with narrowed arteries (**atherosclerosis**), since the increased pressure may dislodge a piece of the substance that causes the narrowing (plaque).

Normal results

TCD produces an audible sound that varies with the heartbeat. It also varies depending on the direction and rate of flow through the vessel being examined. Each of the vessels in the brain has a characteristic direction of flow, which can be detected by TCD. Flow rates are somewhat variable from person to person.

Abnormal results

Lack of flow indicates a vessel has been completely blocked (although absence of a signal may also be due to absorption of sound waves by bone). If blood flows in the wrong direction or alternates between normal and reverse flow, it may mean there is a blockage elsewhere. This happens because blood is rerouted due to abnormalities in pressure caused by the blockage.

If the speed of flow is increased, it may mean that blood is flowing through a restricted area that is just “upstream” from the probe. Intuitively, one might think that a restricted blood vessel would cause the speed of blood flow to slow down. However, the opposite is true. This is because the same amount of blood going through a narrower opening must go faster. Increased speed is also seen if a vessel is carrying rerouted blood.

Resources

BOOKS

Samuels, Martin, and Steven Feske, editors. *Office Practice of Neurology*. New York: Churchill Livingstone, 1996.

Richard Robinson

Transesophageal echocardiography

Definition

Transesophageal **echocardiography** is a diagnostic test using an ultrasound device that is passed into the esophagus of the patient to create a clear image of the heart muscle and other parts of the heart. A tube with a device called a transducer is passed down into the patient's throat and into the esophagus (the food tube that connects the mouth to the stomach). The transducer directs ultrasound waves into the heart, and the reflected sound waves picked up by the transducer are translated into an image of the heart.

Purpose

Since the esophagus is right next to the heart, transesophageal echocardiography provides a very clear picture of the heart. It can provide information on the size of the heart, its pumping strength, and the location and extent of any damage to its tissues. It can detect abnormal tissue growth around the heart valves. It is also good at detecting abnormalities in the pattern of blood flow, such as the backward flow of blood through partly closed heart valves, known as regurgitation or insufficiency. It is especially useful in cases in which conventional echocardiography (a test where the transducer is kept on the patient's chest) cannot offer a good image, such as when the patient is obese or has a thick chest wall. It is useful for monitoring heart function during cardiac surgery and detecting **blood clots** in the left atrium of the heart.

Precautions

Patients should avoid consuming alcohol for a day or so before the procedure, since alcohol may amplify the effects of the sedative used with the procedure.

Description

Echocardiography creates an image of the heart using ultra-high-frequency sound waves—sound waves that are much too high in frequency to be heard by the human ear. The technique is very similar to ultrasound scanning commonly used to visualize the fetus during pregnancy.

A transesophageal echocardiography examination generally lasts 30–60 minutes. The patient is given a mild sedative and the back of the throat is

sprayed with a local anesthetic, in order to suppress the gag reflex. Next, a special viewing tube called an endoscope, containing a tiny transducer, is passed through the mouth and into the esophagus. It is carefully moved until it is positioned directly next to the heart. Essentially a modified microphone, the transducer directs ultrasound waves into the heart, some of which get reflected (or “echoed”) back to the transducer. Different tissues and blood all reflect ultrasound waves differently. These sound waves can be translated into a meaningful image of the heart, which is displayed on a monitor or recorded on paper or tape. The transducer may be moved several times during the test to help doctors get a better view of the heart.

Preparation

The patient may be given a mild sedative before the procedure, and an anesthetic is sprayed into the back of the throat in order to suppress the gag reflex.

Aftercare

After the test, it is important to refrain from eating or drinking until the gag reflex has returned—otherwise, the patient may accidentally inhale some of the food or beverage. If a sedative has been given, patients should not drive or operate heavy machinery for at least 10-12 hours. They should avoid consuming alcohol for a day or so, since alcohol may amplify the effect of the sedative.

Risks

Transesophageal echocardiography may cause gagging and discomfort when the transducer is passed down into the throat. Patients may also experience sore throat for a few days after the test. In rare cases, the procedure may cause bleeding or perforation of the esophagus or an inflammatory condition known as infective **endocarditis**. The patient may have an adverse reaction to the sedative or local anesthetic.

Normal results

A normal transesophageal echocardiogram shows a normal heart structure and the normal flow of blood through the heart chambers and heart valves.

Abnormal results

A transesophageal echocardiogram may show a number of abnormalities in the structure and function of the heart, such as thickening of the wall of the heart muscle (especially the left ventricle). Other

KEY TERMS

Endoscope—An instrument used to see and examine the inside of a body cavity or organ.

Gag reflex—A normal reflex consisting of elevation of the palate, retraction of the tongue, and contraction of the throat muscles.

Regurgitation—Backward flow of blood through a partly closed valve.

Transducer—A device that converts electrical signals into ultrasound waves and ultrasound waves back into electrical impulses.

Ultrasound—Sound waves at a frequency of 20,000 kHz, often used for diagnostic imaging.

abnormalities can include blood leaking backward through the heart valves (regurgitation), or blood clots in the left atrium of the heart.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Robert Scott Dinsmoor

Transferrin test see **Iron tests**

Transfusion

Definition

Transfusion is the process of transferring whole blood or blood components from one person (donor) to another (recipient).

Purpose

Transfusions are given to restore lost blood, to improve clotting time, and to improve the ability of the blood to deliver oxygen to the body's tissues.

Precautions

For donors, the process of giving blood is very safe. Only sterile equipment is used and there is no chance of catching an infection from the equipment. There is a slight chance of infection at the puncture site if the skin is not properly washed before the collection needle is inserted. Some donors feel light-headed upon standing for the first time after donating. Occasionally, a donor will faint. Donors are advised to drink lots of liquids to replace the fluid lost with the donated blood. It is important to maintain the fluid volume of the blood so that the blood pressure will remain stable. Strenuous **exercise** should be avoided for the rest of the day. Most patients have very slight symptoms or no symptoms at all after donating blood.

For recipients, a number of precautions must be taken. The blood given by transfusion must be matched with the recipient's blood type. Incompatible blood types can cause a serious adverse reaction (transfusion reaction). Blood is introduced slowly by gravity flow directly into the veins (intravenous infusion) so that medical personnel can observe the patient for signs of adverse reactions. People who have received many transfusions can develop an immune response to some factors in foreign blood cells (see below). This immune reaction must be checked before giving new blood. Infectious diseases can also be transmitted through donated blood.

Description

Either whole blood or blood components can be used for transfusion. Whole blood is used exactly as it was received from the donor. Blood components are parts of whole blood, such as red blood cells (RBCs), plasma, platelets, clotting factors, immunoglobulins, and white blood cells. Whole blood is used only when needed or when components are not available. Most of the time whole blood is not used because the patient's medical condition can be treated with a blood component and too much whole blood can raise a recipient's blood pressure. High blood pressure can have medical side effects and should be avoided. Use of blood components is a more efficient way to use the blood supply, because blood that has been processed (fractionated) into components can be used to treat more than one person. Each year, more than 20 million units of blood or blood products are transfused in the United States.

Whole blood is generally used when a person has lost a lot of blood. Such blood loss can be caused by injury or surgical procedures. Whole blood is given to

help restore the blood volume, which is essential for maintaining blood pressure. Whole blood is also given to ensure that the body's tissues are receiving enough oxygen. Whole blood is occasionally given when a required blood component is unavailable in isolated form.

Red blood cells are the blood component most frequently used for transfusion. RBCs are the only cells in the body that transport oxygen. A transfusion of RBCs increases the amount of oxygen that can be carried to the tissues of the body. RBCs that have been separated from the liquid plasma (packed RBCs) are given to people who have anemia or who have lost a lot of blood. Platelets are another component frequently given by transfusion. Platelets are a key factor in blood clotting. The clear fluid that carries blood cells (plasma) also contains blood-clotting factors. The platelets and plasma clotting factors are extracted from donated blood and concentrated for use. These factors are used to treat people with such clotting disorders as hemophilia. Immunoglobulins, also called gamma globulin or immune serum, are collected from plasma for use in temporarily boosting the immune capability of a patient. White blood cells (WBCs) are another infection-fighting component of the blood. White blood cells are given by transfusion only rarely.

Blood donation

Each year in the United States, about 14,000,000 pints of blood are donated. Blood collection is strictly regulated by the U.S. Food and Drug Administration (FDA). The FDA has rules for the collection, processing, storage, and transportation of blood and blood products. In addition, the American Red Cross, United Blood Services, the American Association of Blood Banks, and most states have specific rules for the collection and processing of blood. The main purpose of regulation is to ensure the quality of blood and to prevent the transmission of infectious diseases through donated blood. Before blood and blood products are used, they are extensively tested for such infectious agents as hepatitis and **AIDS**.

DONORS. Blood donors are questioned about their general health, their lifestyle, and any medical conditions that might disqualify them as donors. These conditions include hepatitis, **AIDS**, cancer, heart disease, asthma, malaria, bleeding disorders, and high blood pressure. Screening prevents **blood donation** by people who could transmit diseases or by people whose medical condition would place them at risk if they donated blood.

The blood pressure, temperature, and pulse of donors are taken to ensure that they are physically able to donate blood. One pint (450 ml) of blood is usually donated, although it is possible to donate smaller volumes. The average man has 10-12 pints and the average woman 8-9 pints of blood. Within hours after donating, most people have replaced the fluid lost with the donated blood, bringing their blood volume back to normal. Replacing donated blood cells and platelets can take several weeks. People with low blood pressure or anemia, and pregnant women should not donate blood or should limit the amount of blood they donate. Generally, people are allowed to donate blood only once every two months. This delay ensures the health of the donor and discourages people from selling their blood. The practice of paying donors for blood has essentially stopped. Donors who sell blood tend to have a high risk for the transmission of infectious agents.

BLOOD COLLECTION. Blood is collected from the donor by inserting a large needle into a vein in the arm. Usually, one of the larger veins near the inside of the elbow is used. A tourniquet is placed on the upper arm to increase the pressure in the arm veins and make them swell and become more accessible. Once a suitable vein is identified, the area where the needle will be inserted is sterilized by washing with soap solution or an iodine-containing antiseptic. Sometimes both are used. The donor lies on a bed or cot during the procedure, which takes about 10 minutes. Generally, an 18-gauge needle is used. This needle can easily fit into the veins and yet is large enough that the blood flows easily. Blood will sometimes clot in a smaller needle and stop flowing. Blood is collected in sterile plastic bags that hold one pint (450 ml). The bags contain an anticoagulant to prevent clotting and preservatives to keep the blood cells alive. Properly handled and refrigerated, whole blood can last for 42 days. While emphasis has been on screening of blood donors and testing of blood products before they are released for transfusion, the emphasis is beginning to shift to future techniques designed to cleanse the blood and rid it of possible pathogens or reactants.

AUTOLOGOUS TRANSFUSION. Autologous transfusion is a procedure in which patients donate blood for their own use. Patients who are to undergo surgical procedures for which a blood transfusion might be required may elect to donate a supply of blood for the purpose ahead of time. The blood is stored at the hospital for the exclusive use of the patient. This procedure assures that the blood type is an exact match. It also assures that no infection will be transmitted through the blood transfusion.

DIRECTED DONATION. Directed donors are family or friends of the patient who needs a transfusion. Some people think that family and friends provide a safer source of blood than the general blood supply. Studies do not show that directed donor blood is any safer. Blood that is not used for the identified patient becomes part of the general blood supply.

APHERESIS. Apheresis is a special procedure in which only the necessary components of a donor's blood are collected. The remaining components are returned to the donor. A special blood-processing instrument is used in apheresis. It separates the blood into components, saves the desired component, and pumps all the other components back into the donor. Because donors give only part of their blood, they can donate more frequently. For example, people can give almost 10 times as many platelets by apheresis as they could give by donating whole blood.

BLOOD PROCESSING. A sample of the donor's blood is collected at the time of donation and tested for infectious diseases. Blood is not used until the results from these tests confirm that it is safe.

BLOOD TYPING. The donated blood is typed. There are major and minor blood types, also called blood groups. The major types are classified by the ABO system. This system groups blood by two substances, called antigen A and antigen B, in the red blood cells. The four ABO blood types are A, B, AB, and O. Type A blood has the A antigen, type B has the B antigen, type AB has both, and type O has neither. These four types of blood are further sorted by the Rh factor. The Rh, or rhesus factor, is also an antigen in the red blood cells. A person who has the Rh factor is Rh positive; a person who does not have the factor is Rh negative. If a person has red blood cells with both the B and the Rh antigens, that person is said to have a B positive (B+) blood type. Blood types determine what blood a patient can receive. Often, patients are limited to receiving only blood of the same ABO and Rh type as their own. An exception is type O, which can be transfused to people of other types. Those who are Rh negative can only receive Rh negative blood (Rh positive recipients also can safely receive Rh negative blood of a compatible ABO type.) This means that people with O negative blood are considered universal donors; their blood type always is in demand at blood centers.

Blood can be typed by several other minor antigens, such as Kell, Duffy, and Lewis. These minor antigens can become important when a patient has received many transfusions. These patients tend to build up an immune response to the minor blood

groups that do not match their own. Upon receiving a transfusion with a mismatched minor blood group, they may have an adverse reaction. A third group of antigens to which a patient can react are residues from the donor's plasma that have attached to the RBCs. To eliminate this problem, the RBCs are rinsed to remove plasma residues. These rinsed cells are called washed RBCs.

Blood components used in transfusion

Most blood collected from donors is broken down (fractionated) into components that are used to treat specific problems or diseases. Treating patients with blood components is the most efficient way to use the blood supply.

RED BLOOD CELLS. Red blood cells (RBCs) carry oxygen throughout the body. They obtain oxygen as they pass through the lungs and give up oxygen to the other tissues of the body as they are pumped through arteries and veins. When patients do not have enough RBCs to properly oxygenate their bodies, they can be given a transfusion with RBCs obtained from donors. RBCs are recovered from whole blood after donation. They are then typed, removed from the watery blood plasma to minimize the volume (packed), and stored. They are given to people who have anemia (including **thalassemia**), whose bone marrow does not make enough RBCs, or who have other conditions that decrease the number of RBCs in the blood. Occasionally, red blood cells from rare blood types are frozen. Once frozen, RBCs can survive for as long as 10 years. Packed RBCs are given in the same manner as whole blood.

PLASMA. Plasma is the liquid portion of blood. It contains many useful proteins, especially clotting factors and immunoglobulins. After they are processed, plasma or plasma factors (fractions) are usually frozen. Some plasma fractions are freeze-dried. These fractions include clotting factors I through XIII. Some people have an inherited disorder in which the body produces too little of the plasma clotting factors VIII (**hemophilia A**) or IX (**hemophilia B**). Transfusions of these clotting factors help people with hemophilia stop bleeding. Frozen plasma must be thawed before it is used and freeze-dried plasma must be mixed with liquid (reconstituted). In both cases, these blood fractions are usually small in volume and can be injected by syringe and needle.

PLATELETS. Platelets are small bodies in the blood that are essential for clotting. People who do not have enough platelets have bleeding problems. People who have lymphoma, leukemia, or thrombocytopenia, and

people who are receiving **cancer** therapy do not make enough platelets. Platelets have a very short shelf life; they must be used within five days of blood donation. Platelets are packed into bags. A platelet transfusion is given in the same manner as whole blood.

IMMUNOGLOBULINS. Immunoglobulins are the infection-fighting fraction of blood plasma. They are also known as gamma globulin, antibodies, and immune serum. This blood fraction is given to people who have difficulty fighting infections, especially people whose immune systems are depressed by diseases, such as AIDS. Immunoglobulins are also used to prevent **tetanus** after cuts, to treat animal bites when **rabies** infection is suspected, or to treat severe childhood diseases. Generally, the volume used is small, and the immunoglobulins can be injected.

WHITE BLOOD CELLS. White blood cells (WBCs) are another infection-fighting component of the blood. On rare occasions, white blood cells are given by transfusion to treat life-threatening infections. Such transfusions are given when the WBC count is very low or when WBCs are not functioning normally. Most of the time, however, **antibiotics** are used in these cases.

Preparation

A person receiving a transfusion is treated in much the same way as a blood donor. The site where the needle will be inserted is carefully washed with a soap-based solution, followed by an iodine-containing antiseptic. The skin is then dried and the transfusion needle inserted into the recipient's vein. During the early stages of a transfusion, the recipient is monitored closely to detect any adverse reactions. If no signs of adverse reaction are evident, the patient is monitored occasionally for the duration of the transfusion period. Upon completion of the transfusion, a compress bandage is placed over the needle-insertion site to prevent bleeding.

Aftercare

Recipients of blood transfusion are monitored during and after the transfusion for signs of adverse reaction.

Risks

Adverse reaction to mismatched blood (transfusion reaction) and transmission of infectious disease are the two major risks of blood transfusion. Transfusion reaction occurs when antibodies in the

KEY TERMS

ABO blood groups—A system in which human blood is classified by whether the red blood cells contain A or B antigens. Type A blood has the A antigen; type B has the B antigen, AB has both, and O has neither.

Antibody—A simple protein produced by the body to destroy bacteria, viruses, or other foreign bodies. Production of each antibody is triggered by a specific antigen.

Antigen—A substance that stimulates the immune system to manufacture antibodies (immunoglobulins). The function of antibodies is to fight off intruder cells, such as bacteria or viruses, in the body. Antigens stimulate the blood to fight other blood cells that have the wrong antigens. If a person with blood type A is given a transfusion with blood type B, the A antigens will fight the foreign blood cells as though they were an infection.

Immunoglobulin—An antibody.

Infusion—Introduction of a substance directly into a vein or tissue by gravity flow.

Injection—Forcing a fluid into the body by means of a needle and syringe.

Rh (rhesus) factor—An antigen present in the red blood cells of 85% of humans. A person with Rh factor is Rh positive (Rh+); a person without it is Rh negative (Rh-). The Rh factor was first identified in the blood of a rhesus monkey.

recipient's blood react to foreign blood cells introduced by the transfusion. The antibodies bind to the foreign cells and destroy them (hemolytic reaction). Transfusion reaction may also cause a hypersensitivity of the immune system that, in turn, may cause tissue damage within the patient's body. The patient may also have an allergic reaction to mismatched blood. The first symptoms of transfusion reaction are a feeling of general discomfort and **anxiety**. Breathing difficulties, flushing, a sense of pressure in the chest, and back **pain** may develop. Evidence of a hemolytic reaction can be seen in the urine, which will be colored from the waste of destroyed red blood cells. Severe hemolytic reactions are occasionally fatal. Reactions to mismatches of minor factors are milder. These symptoms include itchiness, **dizziness**, **fever**, **headache**, rash, and swelling. Sometimes the patient will experience breathing difficulties and **muscle spasms**. Most

adverse reactions from mismatched blood are not life-threatening. The infectious diseases most often acquired from blood transfusion in the United States are hepatitis and AIDS.

Patients who are given too much blood can develop high blood pressure, a concern for people who have heart disease. Very rarely, an air **embolism** is created when air is introduced into a patient's veins through the tubing used for intravenous infusion. The danger of embolism is greatest when infusion is begun or ended. Care must be taken to ensure that all air is bled out of the tubing before infusion begins, and that infusion is stopped before air can enter the patient's blood system.

Resources

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OTHER

United Blood Services. <<http://www.unitedbloodservices.org>>.

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Transhepatic biliary catheterization

Definition

Transhepatic biliary catheterization is a surgical procedure during which a catheter is inserted into the bile duct to relieve an obstruction.

Purpose

Bile is a fluid made in the liver and stored in the gall bladder. The function of bile is to break down fats during digestion. When fatty foods move into the intestine, bile is released from the gall bladder, travels through the bile duct, and enters the first part of the small intestine (duodenum).

If the bile duct is blocked, the skin becomes yellowish (jaundiced), the abdomen is painful, and a **fever** develops. The bile duct can be blocked by **gallstones**, surgical injury, infection in the duct, or by tissue growth due to **cancer**. Transhepatic biliary catheterization is performed to relieve bile duct blockage. The most common reason for this procedure is to relieve obstruction from the overgrowth of cancer cells. Obstruction due to gallstones is usually cleared by other means.

Precautions

Transhepatic biliary catheterization is done when cancer has progressed to the point where all the malignant cells cannot be removed by surgery. Patients who need transhepatic biliary catheterization often suffer from additional complications of their cancer. Because of the likelihood of bleeding from the liver, this procedure should not be done on patients who have blood clotting abnormalities.

Description

Transhepatic biliary catheterization is performed by inserting a needle through the skin, into the abdomen, through the liver, and into the bile duct. A wire attached to the needle then guides the catheter into place. The procedure can take several hours. The patient is given medication for **pain**.

The catheter can either reestablish bile flow into the duodenum or reroute the bile so it drains into a bag outside the body. The choice depends on the extent and position of the obstruction.

Preparation

The standard preoperative blood tests are performed. The patient should not eat or drink the day of the procedure.

Aftercare

The patient must stay in bed after the procedure for at least six hours, to reduce the risk of bleeding. A nurse checks vital signs and looks for indications of complication such as pain, cramping, or leakage around the catheter. The catheter is flushed periodically to keep it open. Patient and caregiver education on how to keep the catheter clean and irrigated is an important part of aftercare.

Risks

The most common complication of transhepatic biliary catheterization is bleeding as a result of

puncturing the liver. Infection may also result from this procedure. Sometimes the catheter itself becomes blocked and is ineffective.

Normal results

Transhepatic biliary catheterization is a treatment, not a cure. Successful treatment relieves the blocked bile duct, but does not change the underlying conditions that caused the blockage.

Resources

ORGANIZATIONS

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. < <http://www.nci.nih.gov> >.

OTHER

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Tish Davidson, A.M.

Transient ischemic attack

Definition

A transient ischemic attack, or TIA, is often described as a mini-stroke. Unlike a **stroke**, however, the symptoms can disappear within a few minutes. TIAs and strokes are both caused by a disruption of the blood flow to the brain. In TIAs and most strokes, this disruption is caused by a blood clot blocking one of the blood vessels leading to the brain. The blockage produces symptoms such as sudden weakness or **numbness** on one side of the body, sudden dimming or loss of vision, and difficulty speaking or understanding speech. If the symptoms are caused by a TIA, they last less than 24 hours and do not cause brain damage. Stroke-associated symptoms, on the other hand, do not go away and may cause brain damage or **death**. TIAs can serve as an early warning sign of stroke and require immediate medical attention.

Description

Strokes are the third leading cause of death in the United States and the leading cause of disability. Approximately 500,000-600,000 people have strokes each year, and more than 160,000 die as a result. About 85% of these strokes are classified as ischemic. In ischemic stroke, a blood vessel leading to the brain

becomes blocked and an area of the brain is deprived of oxygenated blood. (The other 15% of strokes are caused by bleeding from a blood vessel that has ruptured.) Without the blood supply, the cells in that area of the brain die. Since brain cells cannot grow back, the functions that are controlled by that brain area may be permanently lost.

Approximately 10% of strokes are preceded by one or more TIAs. The estimated annual number of TIAs is about 50,000; an exact count is difficult because TIAs are not always reported. They may be under-reported because they typically last less than an hour, perhaps only for a few minutes. Because they are so brief, TIAs may not seem important. However, an estimated one-third of all TIAs are followed by a stroke within five years. They are considered a medical emergency and prompt medical attention is very important.

Risk factors for strokes and TIAs are very similar. The risk of a TIA or stroke is higher among men, African Americans, people over age 65, and people with heart disease or diabetes. Smokers, people with high blood pressure, and people who are overweight also have a greater risk for TIAs and strokes.

Causes and symptoms

A TIA is caused by a temporary blockage of one of the arteries that leads into the brain. Small **blood clots**, called microemboli, are the immediate cause of the blockage. The blockage forms because of damage or disease within the circulatory system. Blood clots can form in blood vessels because of artery damage, heart disease, and other cardiovascular problems. For example, **atherosclerosis** is strongly associated with TIAs. Atherosclerosis is the build-up of fatty deposits or plaque at certain areas in the circulatory system. Clotting cells in blood, called platelets, tend to stick to atherosclerotic plaques or other damaged sites within blood vessels. Occasionally, a clot may grow large enough to block a blood vessel, or a piece of a clot may break off and circulate to other areas of the body. If a clot does not dissolve quickly enough, it can lodge in a blood vessel and block it. In TIAs, the microemboli dissolve within a short time.

Blood flows into the brain through two main pathways: the carotid arteries and the vertebrobasilar arteries. The carotid arteries are located on the front of the neck; the vertebrobasilar arteries are at the base of the skull at the back of the head. The symptoms produced by a TIA are determined by the arteries affected.

If a vertebrobasilar artery is blocked, common symptoms include double vision and **dizziness, nausea**

and vomiting, difficulty speaking, and problems understanding and using spoken words. There may also be a numbness around the mouth and a **tingling** sensation in the limbs. Blockage of a carotid artery produces complete loss of vision, dimmed or foggy vision, and **paralysis** or weakness on one side of the body. These symptoms also may be accompanied by language problems and speech difficulty.

With either type of blockage, the microemboli dissolve within hours and full function returns.

Diagnosis

The goal of diagnosis is to identify the precise cause of the TIA and to recommend treatment. Initial information that an individual can supply includes a medical history, what drugs are currently being taken and why, and a full description of the symptoms. Blood tests are ordered to screen blood counts — the numbers of specific blood cell types — and to measure sugar and lipid (fats, including cholesterol) levels. Based on this information and a **physical examination** that includes blood pressure, pulse, and respiration measurements, one or more of the following imaging tests are ordered.

A computed tomography scan (CT scan) or a **magnetic resonance imaging** (MRI) scan is usually the first imaging test. CT or MRI can rule out other problems, such as a tumor or **subdural hematoma**, which can mimic the symptoms of a TIA. A CT scan also can uncover aneurysms and arteriovenous malformation, both of which are blood vessel abnormalities that can cause bleeding in the brain.

Another useful imaging test is carotid ultrasonography, a noninvasive procedure that allows examination of the interior of the carotid artery. This examination can detect carotid stenosis, a condition in which the artery is abnormally narrow because of atherosclerosis. Ultrasonography is very reliable in identifying stenosis, but it does not give enough information to accurately assess the degree of stenosis. Because the treatment used depends on the degree of stenosis, treatment decisions cannot be based on ultrasonography. Another type of ultrasonography, called **transcranial doppler ultrasonography**, is used to detect stenosis of the blood vessels within the brain and in the vertebrobasilar arteries.

If stenosis is identified, a further test called cerebral arteriography may be done. This test is not done if the individual is in poor health, because it may be too risky. Arteriography involves injecting a special dye into the blood vessels that makes them visible on

x rays. This procedure also is used to find suspected problems with blood vessels in the brain. Because it is an invasive procedure, complications may arise. Typically, these complications are minor and temporary. In a very small percentage of people with cardiovascular disease, the procedure may cause serious complications, such as stroke.

Although TIAs affect the brain, the ultimate cause of the problem may be found in the heart. Heart disease or damage to the heart's blood vessels is assessed by **echocardiography**. Echocardiography is a type of ultrasonography and is a noninvasive procedure.

Treatment

Treatment is aimed at preventing further TIAs and especially at preventing a stroke. The particular therapy depends on the root cause of the TIA and is not begun until this cause is identified. If at all possible, drug therapy is the preferred method of treating TIAs. Surgical intervention may be required if an individual's situation is not likely to respond to medication or if medication has failed.

Aspirin often is chosen for drug therapy. It is sometimes called a blood thinner because it blocks the function of platelets, the sticky cells that trigger blood clotting. Since aspirin can cause gastrointestinal side effects, other drugs may be prescribed. These drugs include dipyridamole or ticlopidine hydrochloride (Ticlid). Dipyridamole, which works by relaxing the smooth muscles of the arteries, is not as effective as aspirin. Ticlopidine hydrochloride is an anti-platelet drug that is slightly more effective than aspirin, especially in women. However, it may cause **diarrhea** or lowered blood cell counts. Blood tests must, therefore, be done frequently when patients are taking ticlopidine.

If carotid arteriography reveals at least a 70% blockage of the carotid artery, surgical treatment usually is recommended. The particular surgical method is called carotid **endarterectomy**. In endarterectomy, the artery is opened and the material clogging it is removed. Another procedure, called **angioplasty**, has been suggested for treating carotid stenosis, but it is not widely used. This procedure is performed by threading a thin tube through the blood vessel to the site that is clogged. A balloon or a stent (a slender rod) is then passed through the tube to mechanically widen the narrowed area. This procedure is successfully used in other blood vessels in the body, but there is some worry that using it close to the brain may be too dangerous. Surgical treatment of blockage of the vertebralbasilar arteries is not usually recommended.

Treatment of TIAs also focuses on underlying problems. High blood pressure, heart disease, and high levels of blood lipids all require medical intervention. Condition-specific medications often are prescribed and lifestyle changes are strongly encouraged. These changes include giving up **smoking** or excess alcohol consumption, engaging in physical **exercise**, and eating sensibly.

Prognosis

One-third of TIAs are followed by stroke in the next five years; in the other two-thirds, the TIAs may either continue or disappear on their own. However, because of the risk of stroke-related disability and death, all TIAs should be treated as emergency medical situations. In 2003, researchers discovered a naturally occurring system in the bodies of those who have TIAs that shields them from some of the brain damage that might occur in a second stroke. The researchers said that the changes in the brain that seem to protect it are similar to those that occur in hibernating animals. They hope by studying the genetics of these changes, they can develop medications that will help protect the bodies of people who have not had TIAs. It still is important to protect against further strokes following a TIA, however.

Medical treatment significantly decreases the risk of stroke for people who experience one or more TIAs. Anti-platelet therapy with aspirin or ticlopidine may reduce risk as much as 31%. Carotid endarterectomy also substantially reduces stroke risk. The procedure itself carries some risk, but the complication rate is less than 5%. The risk of complication can be lowered by choosing to have the procedure done in a facility experienced with it and by a surgeon with a low complication rate.

Prevention

Treatment for TIAs is complemented by lifestyle changes. These practices also may prevent TIAs and strokes from ever occurring. Doctors and other health-care providers universally recommend that individuals stop smoking and consume alcohol in moderation. Regular health checkups can detect high blood pressure, heart disease, and other underlying problems. Adhering to treatment for these problems can help minimize TIA and stroke risks. A Danish study seemed to confirm in 2003 that eating more citrus fruit reduces the risk of TIA and stroke. Finally, maintaining a healthy weight and engaging in regular exercise as able are strongly recommended.

KEY TERMS

Angioplasty—A medical procedure in which a catheter, or thin tube, is threaded through blood vessels. The catheter is used to place a balloon or stent (a small metal rod) at an area of stenosis and expand it mechanically.

Arteriography—A medical test in which an x-ray visible dye is injected into blood vessels. This dye allows the blood vessels to be imaged with x rays.

Atherosclerosis—A build-up of fatty tissue called plaque inside arteries that can impede or block blood flow.

Carotid artery—One of the major blood vessels leading to the brain; it runs up the front of the neck.

Echocardiography—A type of ultrasonography that is used to create an image of the heart and its functioning.

Endarterectomy—A surgical procedure in which diseased tissue and atherosclerotic plaque are removed from the inside of an artery.

Ischemia—A condition in which blood flow is cut off or restricted from a particular area. The surrounding tissue, starved of oxygen and nutrients, dies.

Microemboli—Small blot clots in the bloodstream.

Platelets—Tiny cells in the blood that help form blood clots.

Stenosis—The narrowing of an opening or passage-way in the body. In arteries, stenosis is caused by a build-up of atherosclerotic plaque, disease, or other disorder.

Stroke—A condition in which blood flow to the brain has been blocked, thereby causing brain cells to die from lack of oxygen and nutrients; also called a “brain attack.”

Ultrasonography—A medical test in which sound waves are directed against internal structures in the body. As sound waves bounce off the internal structure, they create an image on a video screen.

Vertebrobasilar arteries—Major blood vessels that lead to the brain. They are located at the base of the skull at the back of the head.

Resources

PERIODICALS

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ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

National Stroke Association. 9707 E. Easter Lane, Englewood, Co. 80112. (800) 787-6537. <<http://www.stroke.org>>.

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Transplant reaction screening test
see Cytomegalovirus **antibody screening test**

Transplantation see **Corneal transplantation; Hair transplantation; Heart transplantation; Kidney transplantation; Lung transplantation; Pancreas transplantation**

Transposition of the great arteries

Definition

Transposition of the great arteries is a birth defect causing a fatal condition in which there is a reversal, or switch, in the truncal connections of the two main (great) blood vessels to the heart, the aorta and pulmonary artery.

Description

There are two great arteries, the pulmonary artery and the aorta. Normally, the pulmonary artery carries blood from the right ventricle to the lungs. The aorta carries blood from the left ventricle to the vessels of the rest of the body.

Normally, blood returning to the heart is depleted in oxygen. It goes first to the right atrium of the heart and then to the right ventricle where it is pumped to the lungs. While in the lungs, the

blood picks up more oxygen. After the lungs, the blood flows to the left atrium, then the left ventricle, which pumps the blood out through the aorta to the rest of the body, thereby supplying the body with oxygenated blood.

Transposition of the great arteries results in oxygen-depleted blood going to the body. The reason is that the connection of the two great arteries is reversed. In this case, the aorta is connected to the right ventricle. Blood returning to the heart goes to the right atrium and ventricle, which is normal. Then, when the right ventricle pumps the blood out, it goes into the aorta for distribution throughout the body. At the same time, blood in the lungs goes to the left atrium, the left ventricle, but then back to the lungs. This happens because the pulmonary artery is connected to the left ventricle. The result is that highly-oxygenated blood keeps recycling through the lungs, while oxygen-depleted blood recycles through the body without going through the lungs to reoxygenate.

This condition develops during the fetal stage and must be treated promptly after birth if the newborn is to survive. The newborn can survive for a few days because the foramen ovale, a small hole in the septum that separates the two atria, is open, allowing some oxygenated blood to escape and mix into the blood that is being pumped throughout the body. However, the foramen ovale normally closes within a few days after birth.

Causes and symptoms

Transposition of the great arteries is a birth defect that occurs during fetal development. There is no identifiable disease or cause. The main symptom is a “blue” baby appearance, caused by a general lack of oxygen in the body’s tissues.

Diagnosis

Diagnosis is made immediately after birth, when it is observed that the newborn is lacking oxygen. This is noted by the bluish color of the newborn, indicating **cyanosis**, a lack of oxygen. A definite diagnosis is made by x ray, **electrocardiography** (ECG), and **echocardiography**.

Treatment

The only treatment for this condition is prompt heart surgery shortly after birth. In surgery, the two great arteries are reconnected to their proper destination. This restores the normal blood flow pattern. The

coronary arteries are also reconnected, so that they can supply blood to the heart itself. A catheter may be used to maintain or enlarge the opening between the two atria until surgery can be performed.

Prognosis

Left untreated, this disease is fatal within the first weeks of life.

Prevention

Because there is no identifiable cause, there is no way to prevent this condition.

Resources

BOOKS

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John T. Lohr, PhD

Transsexualism see **Gender identity disorder**

Transurethral bladder resection

Definition

Transurethral bladder resection is a surgical procedure, performed under **sedation** or anesthesia, with a lighted tube inserted through the urethra (the small tube-like structure that allows urine to empty from the bladder), into the bladder. It plays both a diagnostic and therapeutic role in the treatment of bladder cancers.

Purpose

Transurethral resection is the initial form of treatment for bladder cancers. The procedure is performed to remove and examine bladder tissue and/or tumor. It may also serve to remove lesions and be the only treatment necessary for noninvasive tumors.

Description

For this procedure, a lighted tube (resectoscope) is inserted through the urethra, into the bladder. A clear solution is infused to maintain visibility, and the tumor or tissue to be examined is cut away using an electric current. Tumor and muscle fibers are biopsied

(a sample is cut out and examined, usually under a microscope) in order to evaluate the depth of tissue involvement, while avoiding perforation of the bladder wall. Every attempt is made to remove all visible tumor tissue, along with a small border of healthy tissue. The resected tissue is examined under the microscope for diagnostic purposes. An indwelling catheter may be inserted to ensure adequate drainage of the bladder postoperatively. At this time, interstitial radiation therapy may be initiated if necessary.

Preparation

Preoperative x rays with dye studies are helpful as a guide in determining the character and extent of tumor involved. As with any surgical procedure, the patient is asked to sign a consent form after the procedure is thoroughly explained.

Aftercare

As with any surgical procedure, blood pressure and pulse will be monitored. Urine is expected to be blood-tinged in the early postoperative period. Continuous bladder irrigation (rinsing) may be used for approximately 24 hours after surgery. Most operative sites should be completely healed in three months. The patient is followed closely for possible recurrence with visual examination, using a special viewing device (cystoscope) at regular intervals as the physician deems necessary.

Abnormal results

Complications of the procedure may include bleeding, which may require bladder irrigation postoperatively, during which time the patient's activity is limited to bedrest. Perforation of the bladder is another risk, in which case the urinary catheter is left in place for four to five days postoperatively. The patient is started on antibiotic therapy preventively. If the bladder is lacerated, accompanied by spillage of urine into the abdomen, an abdominal incision may be required.

Resources

ORGANIZATIONS

American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>>.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. <<http://www.nci.nih.gov>>.

Kathleen D. Wright, RN

Transvaginal ultrasound

Definition

Transvaginal ultrasound is a imaging technique used to create a picture of the genital tract in women. The hand-held device that produces the ultrasound waves is inserted directly into the vagina, close to the pelvic structures, thus often producing a clearer and less distorted image than obtained through transabdominal ultrasound technology, where the probe is located externally on the skin of the abdomen.

Purpose

Transvaginal ultrasound can used to evaluate problems or abnormalities of the female genital tract. It may provide more accurate information than transabdominal ultrasound for women who are obese, for women who are being evaluated or treated for **infertility**, or for women who have difficulty keeping a full bladder. However, it does provide a view of a smaller area than the transabdominal ultrasound. Types of conditions or abnormalities that can be examined include:

- the endometrium of women with infertility problems or who are experiencing abnormal bleeding
- sources of unexplained **pain**
- congenital malformations of the ovaries and uterus
- ovarian cysts and tumors
- pelvic infections, such as pelvic inflammatory disease
- bladder abnormalities
- a misplaced **IUD** (intrauterine device)
- other causes of infertility

Transvaginal ultrasound can also be used during **pregnancy**. Its capability of producing more complete images means that it is especially useful for identifying **ectopic pregnancy**, fetal heartbeat, and abnormalities of the uterus, placenta, and associated pelvic structures.

Transvaginal ultrasound is performed by an ultrasound technologist, who is supervised by a radiologist or a specialist such as an obstetrician.

Precautions

Studies have shown that ultrasound is not hazardous and has no harmful side effects.

Description

A ultrasound, or sonogram, is a procedure that utilizes reflected sound waves to produce a picture of organs and structures within the body. A transducer sends out high-pitched sounds, sounds that are above the range of human hearing, that are reflected back to the transducer. A computer is used to analyze the sound waves, transforming them into a picture (which is called a sonogram, echogram, or ultrasound scan) on a video monitor. These pictures can be saved as a permanent record of the test.

A transvaginal ultrasound is used for looking at organs and structures within the pelvic area that are solid and uniform, like the uterus, and ovaries, or for organs that are fluid-filled, like the bladder. Mineralized structures, like bones, or air-filled organs, such as the intestines, do not show up well on a sonogram and may disrupt the ultrasound beam so that deeper organs and structures cannot be seen clearly.

The transvaginal transducer that produces the ultrasound waves is shaped to fit within a woman's vagina. It is lubricated and covered with a sheath such as a **condom**. A woman should tell the health care provider if she is allergic to latex so that a latex-free cover can be applied to the transducer before it is inserted into the vagina.

Sometimes a woman may have both transvaginal and transabdominal ultrasound scans done to obtain a complete evaluation of the pelvic area.

Preparation

There is no special preparation required by a woman before a transvaginal ultrasound. She can continue to take medications prescribed by her health care provider, although she should not drink liquids for four hours before the test. A full bladder is not required for a transvaginal ultrasound, as is required for a transabdominal ultrasound. The test can be administered during any stage of the woman's menstrual cycle.

During the procedure, the woman lies on her back, with her hips slightly raised. The thin, lubricated tip of the transducer is inserted gently into the vagina. The transducer is moved and rotated to adjust the view of the pelvic structures displayed on the monitor. The woman must lie very still during the ultrasound scan. She may feel a mild discomfort from the pressure of the vaginal transducer. The procedure usually takes from 15 to 30 minutes.

Occasionally, **hysterosonography** is done on a non-pregnant woman to evaluate the inside of the uterus

(endometrial cavity) and fallopian tubes by filling the uterus with fluid during a transvaginal ultrasound.

Aftercare

There is a slight risk of infection associated with a transvaginal ultrasound, especially if a biopsy was done in conjunction with the procedure. Therefore the patient should be aware of symptoms of an infection and seek care from a health care professional as necessary.

Risks

Since no ionizing radiation is associated with transvaginal ultrasound, there has been no documented adverse effects on patients or their fetuses with the use of the procedure.

Normal results

In a normal transvaginal ultrasound, the pelvic structures or organs or the fetus are found to have no abnormalities.

Abnormal results

Abnormal ultrasound results can be due to a variety of conditions and diseases. In non-pregnant women, these abnormal results may include:

- cancerous tumors or non-cancerous growths of the uterus, ovaries, vagina, or other pelvic structures
- ovarian torsion
- infections such as PID
- congenital malformations.

In pregnant women, the following abnormalities may be diagnosed through transvaginal ultrasound:

- ectopic pregnancies
- multiple pregnancies
- fetal death
- short cervix length, which may indicate the risk of preterm labor
- placental abnormalities such as placental previa and placental abruption
- tumors of pregnancy, including gestational trophoblastic disease.

A biopsy is needed to determine whether a tumor that is found during the transvaginal ultrasound scan is cancerous or non-cancerous.

KEY TERMS

Ectopic pregnancy—Pregnancy that develops outside the uterus, usually in one of the fallopian tubes

Endometrium—The mucous membrane that makes up the inner layer of the uterine wall

Gestational trophoblastic disease—A rare, aggressive, malignant, often metastatic (spreading to other organs) cancer in women of childbearing age in which cancer cells grow in the tissues that are formed in the uterus after conception

Intrauterine device (IUD)—Contraceptive device consisting of a piece of bent plastic or metal that is inserted through the vagina into the uterus

Transabdominal ultrasound—A small handheld instrument called a transducer is passed back and forth over the pelvic area to provide images of the abdomen

Ovarian torsion—Twisting of the ovary due to the influence of another condition or disease, resulting in extreme lower abdominal pain

Pelvic inflammatory disease (PID)—Inflammation of the female genital tract, especially of the fallopian tubes, caused by any of several microorganisms, chiefly chlamydia and gonococci, and characterized by severe abdominal pain, high fever, vaginal discharge, and in some cases destruction of tissue that can result in sterility

Placental abruption—Premature separation of the placenta from the uterus

Placenta previa—A low-lying placenta that covers part or all of the inner opening of the cervix. This can result in heavy bleeding during labor and delivery, which can be dangerous for mother and baby.

Resources

BOOKS

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Smith, Norman C., and Smith, Pat M. *Obstetric Ultrasound Make Easy*. London: Churchill Livingstone, 2001.

PERIODICALS

Singh, K., and Malhotra, N. *Step-by-Step Ultrasound in Gynecology*. New York: McGraw Hill Companies, 2004.

Singh, K., and Malhotra, N. *Step-by-Step Ultrasound in Infertility*. New York: McGraw Hill Companies, 2004.

ORGANIZATION

The American Institute of Ultrasound in Medicine, 14750 Sweitzer Lane, Suite 100, Laurel, MD 20707.
Telephone: 301-498-4100. Web site: www.aium.org/

Judith L. Sims

Transverse myelitis

Definition

Transverse myelitis (TM) is an uncommon neurological syndrome caused by inflammation (a protective response which includes swelling, **pain**, heat, and redness) of the spinal cord, characterized by weakness, back pain, and bowel and bladder problems. It affects one to five persons per million.

Description

TM affects the entire thickness of the spinal cord, producing both sensory and movement problems. It is believed to be linked to the immune system, which may be prompted to attack the body's own spinal cord. Striking rapidly without warning, its effects can be devastating.

Causes and symptoms

Transverse myelitis has many different causes, often triggered by a variety of viral and bacterial infections (especially those associated with a rash such as **measles** or **chickenpox**). Once the infection subsides, the inflammation in the cord begins. About a third of patients experience a flu-like illness with **fever** about the time they develop symptoms of TM. Sometimes, there appears to be a direct invasion of, and injury to, the spinal cord by an infectious agent (such as herpes zoster or the **AIDS** virus).

TM can also accompany a variety of diseases that break down tissue that surrounds and insulates the nerves (demyelinating diseases), such as **multiple sclerosis** (MS).

Some toxic substances, such as carbon monoxide, lead, or arsenic, can cause a type of myelitis characterized by inflammation followed by hemorrhage or bleeding that destroys the entire circumference of the spinal cord. Other types of myelitis can be caused by poliovirus; herpes zoster; **rabies**, **smallpox** or **polio vaccination**; or parasitic and fungal infections.

Many experts believe that TM can occur without any apparent cause, probably as the result of an autoimmune process. This means that a person's immune system attacks the spinal cord, causing inflammation and tissue damage.

Regardless of the cause of the myelitis, onset of symptoms is sudden and rapid. Problems with movement and sensation appear within one or two days after inflammation begins. Symptoms include soft (flaccid) **paralysis** of the legs, with pain in the lower legs or back, followed by loss of feeling and sphincter (muscles which close an opening, as in the anus) control. The earliest symptom may be a girdle-like sensation around the trunk.

The extent of damage occurring will depend on how much of the spinal cord is affected, but TM rarely involves the arms. Severe spinal cord damage also can lead to **shock**.

Diagnosis

A doctor will suspect transverse myelitis in any patient with a rapid onset of paralysis. Medical history, **physical examination**, brain and spinal cord scans, myelogram, spinal tap, and blood tests are used to rule out other neurological causes of symptoms, such as a tumor. If none of these tests suggest a cause for the symptoms, the patient is presumed to have transverse myelitis.

Treatment

There is no effective treatment for transverse myelitis, but any underlying infection must be treated. After this, the focus of care shifts from diagnosis and treatment to learning how to live with the effects of the syndrome. Patients are helped to cope psychologically with new limitations, and are given physical **rehabilitation**.

Physical adaptations include learning to cope with bowel and bladder control, sexuality, inability to control muscles (spasticity), mobility, pain, and activities of daily living (such as dressing).

As nerve impulses from the spinal cord are often scrambled and misinterpreted by the brain as pain, painkillers are given to ease discomfort. Antidepressants or anticonvulsants may also help.

Prognosis

The prognosis depends on how much of the cord was damaged. Some people recover completely, while others have lasting problems and need help in learning

KEY TERMS

Demyelinating disorders—A group of diseases characterized by the breakdown of myelin, the fatty sheath surrounding and insulating nerve fibers. This breakdown interferes with nerve function, and can result in paralysis. Multiple sclerosis is a demyelinating disorder.

Myelogram—An x-ray examination of the brain and spinal cord with the aid of a contrast dye, to look for tumors or spinal cord injury.

how to cope with activities of daily living. People who develop spastic reflexes early in the course of the condition are more likely to recover than those who do not. If spinal cord tissue death (necrosis) occurs, the chance of a complete recovery is poor. Most recovery occurs within the first three months. A certain percentage of patients with TM will go on to develop multiple sclerosis.

Resources

ORGANIZATIONS

Transverse Myelitis Association. 1787 Sutter Parkway, Powell, OH 43065-8806. (614) 766-1806. <<http://www.myelitis.org>>.

Carol A. Turkington

Tranlycypromine see **Monoamine oxidase inhibitors**

Traumatic amputations

Definition

Traumatic **amputation** is the accidental severing of some or all of a body part. A complete amputation totally detaches a limb or appendage from the rest of the body. In a partial amputation, some soft tissue remains attached to the site.

Description

Trauma is the second leading cause of amputation in the United States. About 30,000 traumatic amputations occur in this country every year. Four of every

five traumatic amputation victims are male, and most of them are between the ages of 15–30.

Traumatic amputation most often affects limbs and appendages like the arms, ears, feet, fingers, hands, legs, and nose.

Causes and symptoms

Farm and factory workers have greater-than-average risks of suffering injuries that result in traumatic amputation. Automobile and motorcycle accidents and the use of lawnmowers, saws, and power tools are also common causes of traumatic amputation.

Blood loss may be massive or minimal, depending on the nature of the injury and the site of the amputation. Patients who lose little blood and have less severe injuries sometimes feel more **pain** than patients who bleed heavily and whose injuries are life-threatening.

Diagnosis

When the patient and the amputated part(s) reach the hospital, an Emergency Department physician will assess the probability that the severed tissue can be successfully reattached.

The Mangled Extremity Severity Score (MESS) assigns numerical values to such factors as body temperature, circulation, **numbness**, **paralysis**, tissue health, and the patient's age and general health. This is one of the diagnostic tools used to determine how successful reattachment surgery is apt to be. The total score is doubled if blood supply to the amputated part has been absent or diminished for more than six hours.

A general, emergency, or orthopedic surgeon makes the final determination about whether surgery should be performed. The surgeon also considers the patient's wishes and lifestyle. Additional concerns are how and to what extent the amputation will affect the patient's quality of life and ability to perform everyday activities.

Treatment

First aid or emergency care given immediately after the amputation has a critical impact on both the physicians' ability to salvage and reattach the severed part(s) and the patient's ability to regain feeling and function.

Muscle tissue dies quickly, but a well-preserved part can be successfully reattached as much as 24 hours after the amputation occurs. Tissue that has



This man's hand was surgically reattached following a traumatic amputation. (Photograph by Michael English, M.D., Custom Medical Stock Photo. Reproduced by permission.)

not been preserved will not survive for more than six hours.

Initial response

The most important steps to take when a traumatic amputation occurs are:

- Contact the nearest emergency services provider, clearly describe what has happened, and follow any instructions given.
- Make sure the victim can breathe; administer **CPR** if necessary.
- Control bleeding, using direct pressure but minimizing or avoiding contact with blood and other body fluids.
- Patients should not be moved if back, head, leg, or neck injuries are suspected or if motion causes pain. If none are found by the EMT, lie the victim flat, with the feet raised 12 inches above the surface.
- Cover the victim with a coat or blanket to prevent shock.

The injured site should be cleansed with a sterile solution and wrapped in a clean towel or other thick material that will protect the wound from further injury. Tissue that is still attached to the body should not be forced back into place. If it cannot be gently replaced, it should be held in its normal position and supported until additional care is available.

KEY TERMS

Phantom pain—Pain, tingling, itching, or numbness in the place where the amputated part used to be.

Saving the patient's life is always more important than recovering the amputated part(s). Transporting the patient to a hospital or emergency center should never be delayed until missing pieces are located.

Preserving tissue

No amputated body part is too small to be salvaged. Debris or other contaminating material should be removed, but the tissue should not be allowed to get wet.

An amputated body part should be wrapped in bandages, towels, or other clean, protective material and sealed in a plastic bag. Placing the sealed bag in a cooler or in a container that is inside a second container filled with cold water or ice will help prevent tissue deterioration.

Prognosis

Possible complications of traumatic amputation include:

- excessive bleeding
- infection
- muscle shortening
- pulmonary embolism.

Improved medical and surgical care and **rehabilitation** have improved the long-term outlook for these patients.

Phantom pain

About 80% of all amputees over the age of four experience **tingling**, **itching**, numbness, or pain in the place where the amputated part used to be. Phantom sensations may begin immediately after the amputation, or they may develop months or years later. They often occur after an injury to the site of the amputation.

These intermittent feelings may:

- occur frequently or only once in a while
- be mild or intense

- last for a few minutes or several hours
- help patients adjust more readily to an artificial limb (prosthesis).

Prevention

The best way to prevent traumatic amputation is to observe common-sense precautions like using seat belts and obeying speed limits and other traffic regulations. It is important to take special precautions when using potentially dangerous equipment and make sure machinery is turned off and disconnected before attempting to service or repair it. Appropriate protective clothing should be worn at all times.

Resources

ORGANIZATIONS

American Amputation Foundation, Inc. P.O. Box 250218, Hillcrest Station, Little Rock, AR 72225. (501) 666-2523.

Amputee Coalition of America. P.O. Box 2528, Knoxville, TN 37901-2528. (888) 267-5669. <<http://www.amputee-coalition.org>>.

Maureen Haggerty

Traveler's diarrhea

Definition

The occurrence of multiple loose bowel movements in someone traveling to an area outside their usual surroundings (usually from temperate industrialized regions to tropical areas), is known as Traveler's **diarrhea** (TD). The cause is almost always due to a bacterial or viral infection acquired through ingesting contaminated food or water.

Description

It is estimated that anywhere from 20–50% of the 12–20 million travelers going from temperate industrialized countries to the tropics will develop TD. Fortunately, most of these episodes are of short duration. Nevertheless, about 40% of those affected will need to rearrange their schedule, and 20% will be ill enough to remain in bed for some days.

The chance of winding up with TD is directly related to the area one is traveling to; only about 8% of individuals visiting an industrialized country are affected, whereas at least half of those traveling to

non-industrialized regions become ill. It is also clearly related to the number of potentially contaminated foods or beverages consumed. Attention to recommended guidelines regarding food safety and sanitation can greatly decrease the risk of infection.

Causes and symptoms

Bacterial infections are the most common cause of the illness. Viruses and occasional parasites also can be the cause. As for the bacteria involved, toxin producing types of *E. coli* (called enterotoxigenic) account for approximately 40–60% of cases, with *Campylobacter* and *Shigella* each reported in at least 10% of cases. In some studies, *Campylobacter* has accounted for almost half of the attacks, especially during cooler seasons of the year. The cause can vary depending on several factors, including the season and country visited. More than one organism can be found in 15–30% of cases, while none is identified in up to 40% of cases worldwide.

Rotaviruses and a parvovirus called Norwalk agent also are responsible for TD. *Giardia* is probably the most common parasite identified, though amoebas (*Entamoeba histolytica*), *Cryptosporidium*, and *Cyclospora* are being found with increasing frequency. Amazingly, research released in 2003 found that those living in undeveloped countries with poor sanitation and high incidence of diarrheal disease also have the lowest incidence of colorectal **cancer**. It appears that an enterotoxin secreted from multiple bacteria leads to **colon cancer** resistance.

Younger age groups, particularly students, are at greatest risk, probably because of where and what they eat. Individuals over 55 years of age, persons staying with relatives, or business travelers are at lower risk. Foods with the highest chance of transmitting disease are uncooked vegetables, unpeeled fruits, meat, and seafood. Tap water and even ice can be dangerous unless one is sure of the source.

Symptoms usually start within a few days after arrival, but can be delayed for as long as two weeks. Illness lasts an average of 3 to 5 days, but is sometimes longer. Cramping abdominal pain, lack of appetite, and diarrhea are the main complaints. In approximately 10% of patients, diarrhea turns bloody. Fever develops in about half of them. The presence of bloody bowel movements and **fever** usually indicates a more severe form of illness and makes *Shigella* a more likely cause. Medications that decrease the motility or contractions of the intestine, such as loperamide (Imodium) or diphenoxylate (Lomotil), should not be used when fever or bleeding occur.

Complications

Diarrhea varies from a few loose stools per day to 10 or more. **Dehydration** and changes in the normal blood pH (acid-base balance) are the main dangers associated with TD. Signs of dehydration can be hard to notice, but increasing thirst, dry mouth, weakness or lightheadedness (particularly if worsening while standing), or a darkening/decrease in urination are suggestive. Severe dehydration and changes in the body's chemistry can lead to kidney failure and become life-threatening.

Another potential complication is “toxic megacolon,” in which the colon gradually stretches and its wall thins to the point where it can tear. The presence of a hole in the intestine leads to peritonitis and is fatal unless quickly recognized and treated.

Other complications related to TD can involve the nervous system, skin, blood, or kidneys.

Diagnosis

The occurrence of diarrhea in an individual while traveling is very suggestive of TD. Although there are other possible causes, these are less likely. In most instances, the specific organism responsible for the symptoms does not need to be identified, and the majority of patients need only rest and treatment to avoid potential complications.

When patients develop fever or bloody diarrhea, the illness is more serious and a specific diagnosis is needed. In those cases, or when symptoms last longer than expected, stool samples are obtained to identify the organism.

For this purpose, laboratories can either try to grow (culture) the organism, or identify it with high-powered microscopes (electron microscopy) or with the use of special tests or stains. These can show parasites such as *Giardia*, *Amoeba*, *Cryptosporidium* and others in freshly obtained stool specimens. New techniques that involve identification of DNA (the characteristic material that controls reproduction and is unique for all individuals) of the various organisms, also can be used in special circumstances.

Treatment

The best treatment of TD is prevention; however, once disease occurs, therapy is aimed at preventing or reducing dehydration, and using **antibiotics** when needed. Fortunately, severe dehydration is unusual in patients with TD, but any fluid losses should be treated early with either fruit juices and “clear fluids” such as tea or broth, or with the recommended Oral

Rehydration Solutions (ORS) suggested by the World Health Organization (WHO). Persons traveling to known areas of infection should consult with their physician prior to departure and obtain appropriate instructions. For example, it may be advised to take along pre-prepared packets of ORS designed for easy mixing or commercial preparations such as Pedialyte, Ceralyte, Ricelyte, etc.

When nothing else is available, the following WHO recipe can be made up from household items and taken in small frequent sips;

- table salt: 1-3/4 teaspoon
- baking powder: 1 teaspoon
- orange juice: 1 cup
- water: 1 quart or liter

A debate has occurred in the medical community over the amount of salt (sodium) in the WHO preparations; some physicians feel that the content is too much for use by well-nourished persons in developed countries. Therefore these preparations should not be used for extended periods of time without consulting a physician.

Pepto-Bismol (bismuth subsalicylate preparation) is effective in both preventing and treating TD. For treatment once symptoms begin, the drug must be taken more frequently than when used for prevention. Bismuth subsalicylate preparation (1 oz of liquid or two 262.5-mg tablets every 30 minutes for eight doses) has been shown to decrease the number of bowel movements and shorten the length of illness. However, there is some concern about the large doses of bismuth in patients with **kidney disease**; therefore patients should check with physicians before starting this or any other therapy. Patients should be aware that bismuth can turn bowel movements black in color.

In 2003, a new drug called rifamixin (Normix) for treating traveler's diarrhea was nearing approval by the U.S. Food and Drug Administration (FDA). Medications designed to decrease intestinal motility and contractions such as loperamide (Imodium), diphenoxylate (Lomotil), or others are safest when used by those without fever or bloody bowel movements. The presence of either of these symptoms indicates a more severe form of colitis.

Antibiotics are usually not prescribed, because most cases of TD rapidly improve with minimal treatment. For patients in whom symptoms are especially severe (4 or more stools per day or the onset of bloody diarrhea or fever), or those with compromised immune systems, antibiotics are indicated. Individuals with less severe attacks can be treated with either antimotility medications or bismuth subsalicylate.

KEY TERMS

Oral Rehydration Solution (ORS)—A liquid preparation developed by the World Health Organization that can decrease fluid loss in persons with diarrhea. Originally developed to be prepared with materials available in the home, commercial preparations have recently come into use.

Choice of an antibiotic should ideally be tailored to the most likely organism and then adjusted according to results of stool cultures. Trimethoprim-sulfamethoxazole (Bactrim) or ciprofloxacin (Cipro) are the antibiotics most often prescribed, but others are also used. The type and duration of treatment continues to be revised, and it is therefore extremely important that patients check with a physician prior to beginning treatment. In many instances, an antibiotic can be combined with an anti-motility agent to provide the quickest relief.

Prognosis

Up to 1% of patients with TD will become sick enough to require hospitalization, and 3% will continue to experience diarrhea for at least one month. The majority of patients rapidly recover with minimal therapy. Some will suffer symptoms for even longer. The small number who continue to suffer symptoms will need careful evaluation to rule out the many causes of chronic diarrhea (such as lactase deficiency, irritable bowel syndrome, parasites, etc.). It is unusual for diarrhea caused by bacteria to last over two weeks; therefore, more prolonged diarrhea indicates a non-bacterial cause.

Prevention

The best means of prevention is avoiding foods, beverages, and food handling practices that lead to infection with the organisms that cause TD. Drinking bottled water and using bottled water for brushing teeth, eating fruits that the traveler peels on his or her own, and eating well-cooked, hot foods can help prevent illness.

One effective means to prevent TD is liquid Pepto-Bismol; this bismuth-containing compound has been shown to be very effective in reducing the incidence of TD. Tablets are now available, which are easier to carry. Two tablets four times a day is recommended, but use should not go beyond three weeks.

Antibiotics can also prevent TD, but their use is controversial, unless it is absolutely necessary to avoid

infection (such as someone on an important business trip or who has a weakened immune system). There is the tendency for bacteria to become resistant to these medications if used excessively; and these drugs have side effects that can be worse than the effects of TD. The benefits and risks of antibiotic treatment should be carefully weighed.

Resources

PERIODICALS

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Seale, J. Paul. "Travel Risks: Update on Traveler's Diarrhea and Other Common Problems." *Consultant* December 2002: 1778.

OTHER

Centers for Disease Control. <<http://www.cdc.gov/nccdphp/ddt/ddthome.htm>>.

David Kaminstein, MD
Teresa G. Odle

Tremors

Definition

Tremor is an unintentional (involuntary), rhythmic alternating movement that may affect the muscles of any part of the body. Tremor is caused by the rapid alternating contraction and relaxation of muscles and is a common symptom of diseases of the nervous system (neurologic disease).

Description

Occasional tremor is felt by almost everyone, usually as a result of fear or excitement. However, uncontrollable tremor or shaking is a common symptom of disorders that destroy nerve tissue, such as Parkinson's disease or multiple sclerosis. Tremor may also occur after **stroke** or head injury. Other tremors appear without any underlying illness.

Causes and symptoms

Tremor may be a symptom of an underlying disease, and it may be caused by drugs. It may also exist as the only symptom (essential tremor).

Underlying disease

Some types of tremor are signs of an underlying condition. About a million and a half Americans have Parkinson's disease, a disease that destroys nerve cells. Severe shaking is the most apparent symptom of Parkinson's disease. This coarse tremor features four to five muscle movements per second. The shaking is evident at rest but declines or disappears during movement.

Other disorders that cause tremor are **multiple sclerosis**, Wilson's disease, mercury poisoning, thyrotoxicosis, and liver encephalopathy.

A tremor that gets worse during body movement is called an "intention tremor." This type of tremor is a sign that something is amiss in the cerebellum, a region of the brain concerned chiefly with movement, balance and coordination.

Essential tremor

Many people have what is called "essential tremor," in which the tremor is the only symptom. This type of shaking affects between three and four million Americans.

The cause of essential tremor is not known, although it is an inherited problem in more than half of all cases. The genetic condition has an autosomal dominant inheritance pattern, which means that any child of an affected parent will have a 50% chance of developing the condition.

Essential tremor most often appears when the hands are being used, whereas a person with Parkinson's disease will most often have a tremor while walking or while the hands are resting. People with essential tremor will usually have shaking head and hands, but the tremor may involve other parts of the body. The shaking often begins in the dominant hand and may spread to the other hand, interfering with eating and writing. Some people also develop a quavering voice.

Essential tremor affects men and women equally. The shaking often appears at about age 45, although the disorder may actually begin in adolescence or early adulthood. Essential tremor that begins very late in life is sometimes called "senile tremor."

Drugs and tremor

Several different classes of drugs can cause tremor as a side effect. These drugs include amphetamines, **antidepressant drugs**, **antipsychotic drugs**, **caffeine**, and lithium. Tremor also may be a sign of withdrawal from alcohol or street drugs.

Diagnosis

Close attention to where and how the tremor appears can help provide a correct diagnosis of the cause of the shaking. The source of the tremor can be diagnosed when the underlying condition is found. Diagnostic techniques that make images of the brain, such as computed tomography scan (CT scan) or **magnetic resonance imaging** (MRI), may help form a diagnosis of multiple sclerosis or other tremor caused by disorders of the central nervous system. Blood tests can rule out such metabolic causes as thyroid disease. A family history can help determine whether the tremor is inherited.

Treatment

Neither tremor nor most of its underlying causes can be cured. Most people with essential tremor respond to drug treatment, which may include propranolol, primidone, or a benzodiazepine. People with Parkinson's disease may respond to levodopa or other **antiparkinson drugs**.

Research has shown that about 70% of patients treated with botulinum toxin A (Botox) have some improvement in tremor of the head, hand, and voice. Botulinum is derived from the bacterium *Clostridium botulinum*. This bacterium causes **botulism**, a form of **food poisoning**. It is poisonous because it weakens muscles. A very weak solution of the toxin is used in cases of tremor and **paralysis** to force the muscles to relax. However, some patients experience unpleasant side effects with this drug and cannot tolerate effective doses. For other patients, the drug becomes less effective over time. About half of patients don't get any relief of tremor from medications.

Tremor control therapy

Tremor control therapy is a type of treatment using mild electrical pulses to stimulate the brain. These pulses block the brain signals that trigger tremor. In this technique, the surgeon implants an electrode into a large oval area of gray matter within the brain that acts as a relay center for nerve impulses and is involved in generating movement (thalamus). The electrode is attached to an insulated wire that runs through the brain and exits the skull where it is attached to an extension wire. The extension is connected to a generator similar to a heart pacemaker. The generator is implanted under the skin in the chest, and the extension is tunneled under the skin from the skull to the generator. The patient can control his or

her tremor by turning the generator on with a hand-held magnet to deliver an electronic pulse to the brain.

Some patients experience complete relief with this technique, but for others it is of no benefit at all. About 5% of patients experience complications from the surgical procedure, including bleeding in the brain. The procedure causes some discomfort, because patients must be awake while the implant is placed. Batteries must be replaced by surgical procedure every three to five years.

Other surgical treatments

A patient with extremely disabling tremor may find relief with a surgical technique called thalamotomy, in which the surgeon destroys part of the thalamus. However, the procedure is complicated by **numbness**, balance problems, or speech problems in a significant number of cases.

Pallidotomy is another type of surgical procedure sometimes used to decrease tremors from Parkinson's disease. In this technique, the surgeon destroys part of a small structure within the brain called the globus pallidus internus. The globus is part of the basal ganglia, another part of the brain that helps control movement. This surgical technique also carries the risk of disabling permanent side effects.

Fetal tissue transplantation (also called a nigral implant) is a controversial experimental method to treat Parkinson's disease symptoms. This method implants fetal brain tissue into the patient's brain to replace malfunctioning nerves. Unresolved issues include how to harvest the fetal tissue and the moral implications behind using such tissue, the danger of tissue rejection, and how much tissue may be required. Although initial studies using this technique looked promising, there has been difficulty in consistently reproducing positive results.

Small amounts of alcohol may temporarily (sometimes dramatically) ease the shaking. Some experts recommend a small amount of alcohol (especially before dinner). The possible benefits, of course, must be weighed against the risks of alcohol **abuse**.

Prognosis

Essential tremor and the tremor caused by neurologic disease (including Parkinson's disease) slowly get worse and can interfere with a person's daily life. While the condition is not life-

KEY TERMS

Computed tomography (CT) scan—An imaging technique in which cross-sectional x rays of the body are compiled to create a three-dimensional image of the body's internal structures.

Essential tremor—An uncontrollable (involuntary) shaking of the hands, head, and face. Also called familial tremor because it is sometimes inherited; it can begin in the teens or in middle age. The exact cause is not known.

Fetal tissue transplantation—A method of treating Parkinson's and other neurological diseases by grafting brain cells from human fetuses onto the affected area of the human brain. Human adults cannot grow new brain cells but developing fetuses can. Grafting fetal tissue stimulates the growth of new brain cells in affected adult brains.

Intention tremor—A rhythmic purposeless shaking of the muscles that begins with purposeful (voluntary) movement. This tremor does not affect muscles that are resting.

Liver encephalopathy—A condition in which the brain is affected by a buildup of toxic substances that would normally be removed by the liver. The condition occurs when the liver is too severely damaged to cleanse the blood effectively.

Multiple sclerosis—A degenerative nervous system disorder in which the protective covering of the nerves in the brain are damaged, leading to tremor and paralysis.

Magnetic resonance imaging (MRI)—An imaging technique that uses a large circular magnet and radio waves to generate signals from atoms in the body. These signals are used to construct images of internal structures.

Pallidotomy—A surgical procedure that destroys a small part of a tiny structure within the brain called the globus pallidus internus. This structure is part of the basal ganglia, a part of the brain involved in the

control of willed (voluntary) movement of the muscles.

Parkinson's disease—A slowly progressive disease that destroys nerve cells. Parkinson's is characterized by shaking in resting muscles, a stooping posture, slurred speech, muscular stiffness, and weakness.

Thalamotomy—A surgical procedure that destroys part of a large oval area of gray matter within the brain that acts as a relay center for nerve impulses. The thalamus is an essential part of the nerve pathway that controls intentional movement. By destroying tissue at a particular spot on the thalamus, the surgeon can interrupt the nerve signals that cause tremor.

Thalamus—A large oval area of gray matter within the brain that relays nerve impulses from the basal ganglia to the cerebellum, both parts of the brain that control and regulate muscle movement.

Thyrotoxicosis—An excess of thyroid hormones in the blood, causing a variety of symptoms that include rapid heart beat, sweating, anxiety, and tremor.

Tremor control therapy—A method for controlling tremor by self-administered shocks to the part of the brain that controls intentional movement (thalamus). An electrode attached to an insulated lead wire is implanted in the brain; the battery power source is implanted under the skin of the chest, and an extension wire is tunneled under the skin to connect the battery to the lead. The patient turns on the power source to deliver the electrical impulse and interrupt the tremor.

Wilson's disease—An inborn defect of copper metabolism in which free copper may be deposited in a variety of areas of the body. Deposits in the brain can cause tremor and other symptoms of Parkinson's disease.

threatening, it can severely disrupt a person's everyday experiences.

Prevention

Essential tremor and tremor caused by a disease of the central nervous system cannot be prevented. Avoiding use of stimulant drugs such as caffeine and

amphetamines can prevent tremor that occurs as a side effect of drug use.

Resources

ORGANIZATIONS

American Academy of Neurology. 1080 Montreal Ave., St. Paul, MN 55116. (612) 695-1940. <<http://www.aan.com>>.

American Parkinson Disease Association. 60 Bay Street, Suite 401, Staten Island, NY 10301. (800) 223-2732. <<http://www.apdaparkinson.org>>.

International Tremor Foundation. 7046 West 105th St., Overland Park, KS 66212. (913) 341-3880.

National Parkinson Foundation. 1501 N.W. 9th Ave., Miami, FL 33136-1494. (800) 327-4545. <<http://www.parkinson.org>>.

Carol A. Turkington

Trench fever

Definition

Trench **fever** is a bacterial infection that causes repeated cycles of high fever.

Description

The term trench fever refers to the crowded conditions in which troops fought in during World War I and World War II. Because the causative bacteria are passed among humans through contact with body lice, overcrowding, and conditions which interfere with good hygiene (including regular washing of clothing) soldiers were predispose to this disease. Currently, homeless people in the United States are sometimes diagnosed with this illness. The bacteria are sometimes passed through the bite of an infected tick. This can cause the illness in people who participate in outdoor activity and encounter ticks in that particular area.

Causes and symptoms

Two different bacteria can cause trench fever: *Bartonella quintana* and *Bartonella henselae*. *B. quintana* is carried by body lice; *B. henselae* is carried by ticks.

Infection with *B. quintana* occurs when an infected louse defecates while feeding on a human. When the person scratches, the feces (which are full of bacteria) are rubbed into the tiny wound. Infection with *B. henselae* occurs when an infected tick bites a human, passing the bacteria along through the tiny bite wound.

Symptoms of trench fever begin about 2 weeks to a month after exposure to the bacteria. Sudden fever, loss of energy, **dizziness**, **headache**, weight loss, skin rash, severe muscle and bone **pain** can occur. Pain is particularly severe in the shins, leading to the

nickname “shin bone fever.” The fever can reach 105°F (40.5°C) and stays high for five to six days at a time. The temperature then drops, and stays down for several days, usually recurring in five- to six-day cycles. An individual may experience as many as eight cycles of fever with the illness.

Diagnosis

Diagnosis is usually made on the basis of the patient's symptoms, and on knowledge of the conditions in which the patient lives. A blood sample can be drawn and bacteria in the sample are allowed to grow. Identification is made by looking at the number of bacteria that may be present on a glass slide seen under the lens of a microscope. However, this technique can take up to four weeks, because this type of bacterium grows very slowly. By this time, the practitioner has often decided to treat the patient anyway.

Treatment

Erythromycin and azithromycin are both used to treat trench fever. Four weeks of treatment are usually necessary. Inadequate treatment often results in a relapse. In fact, relapses have been reported to occur as long as 10 years after the first episode.

Prognosis

Prognosis for patients with trench fever is excellent. Recovery may take a couple of months. Without treatment, there is always a risk of recurrence, even years after the original illness.

Prevention

Prevention involves good hygiene and decent living conditions. When this is impossible, insecticide dusting powders are available to apply to clothing. Avoidance of areas known to harbor ticks or the use of insect repellents is necessary to avoid the type of infection passed by ticks.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Treponema carateum infection see **Pinta**

Treponema pallidum infection see **Syphilis**

Tretinoin see **Antiacne drugs**

Trichinella spiralis infectio see **Trichinosis**

Trichiniasis see **Trichinosis**

Trichinosis

Definition

Trichinosis is a disease caused by a roundworm (nematode) called *Trichinella spiralis*. An individual worm of this species is called a trichina, from the Greek word meaning “hairlike.” Trichinae can be readily avoided by proper handling and cooking of certain meats, particularly pork products.

Description

The life cycle of *T. spiralis* includes several different stages. The adult trichina lives in the intestinal lining of such meat-eating animals as swine, bears, walrus, and rodents. After mating, the male worm dies while the female goes on to produce the offspring.

Roundworms have a stage of development called the embryonic stage, which in many species occurs after birth. In trichinae, however, this embryonic stage occurs within the uterus of the female, so the offspring that are ultimately discharged into the host’s intestinal lining are in the larval second stage of life. These larvae—about 1500 from each female worm—travel through the circulatory system to the heart, then through the blood vessels leading to striated muscle (the muscle of the skeletal system and the heart). Most larvae that cannot find suitable locations in striated muscle will die.

Those larvae that reach striated muscle will grow to a length of about one millimeter, coil themselves, and enclose themselves within a protective wall called a cyst. This process is referred to as encysting. The worms in the cysts can live for up to ten years in this form.

A pig that has been infected with *T. spiralis*, then, has thousands of cysts lying dormant within its muscles—the very muscles that humans look forward to consuming in the form of pork chops, ham, barbecued ribs, etc. When humans sit down to a delicious meal of undercooked, trichina-infected pig dinner, they are ingesting *T. spiralis* cysts. The cyst walls are broken down by the usual process of food digestion in the stomach, allowing the larvae to escape into the new

host’s intestines. There the larvae mature to become adult worms, capable of producing a new crop of larvae. When these new larvae hatch, they begin their migration throughout the human host’s bloodstream to his or her muscles, where they live for a short while before encysting.

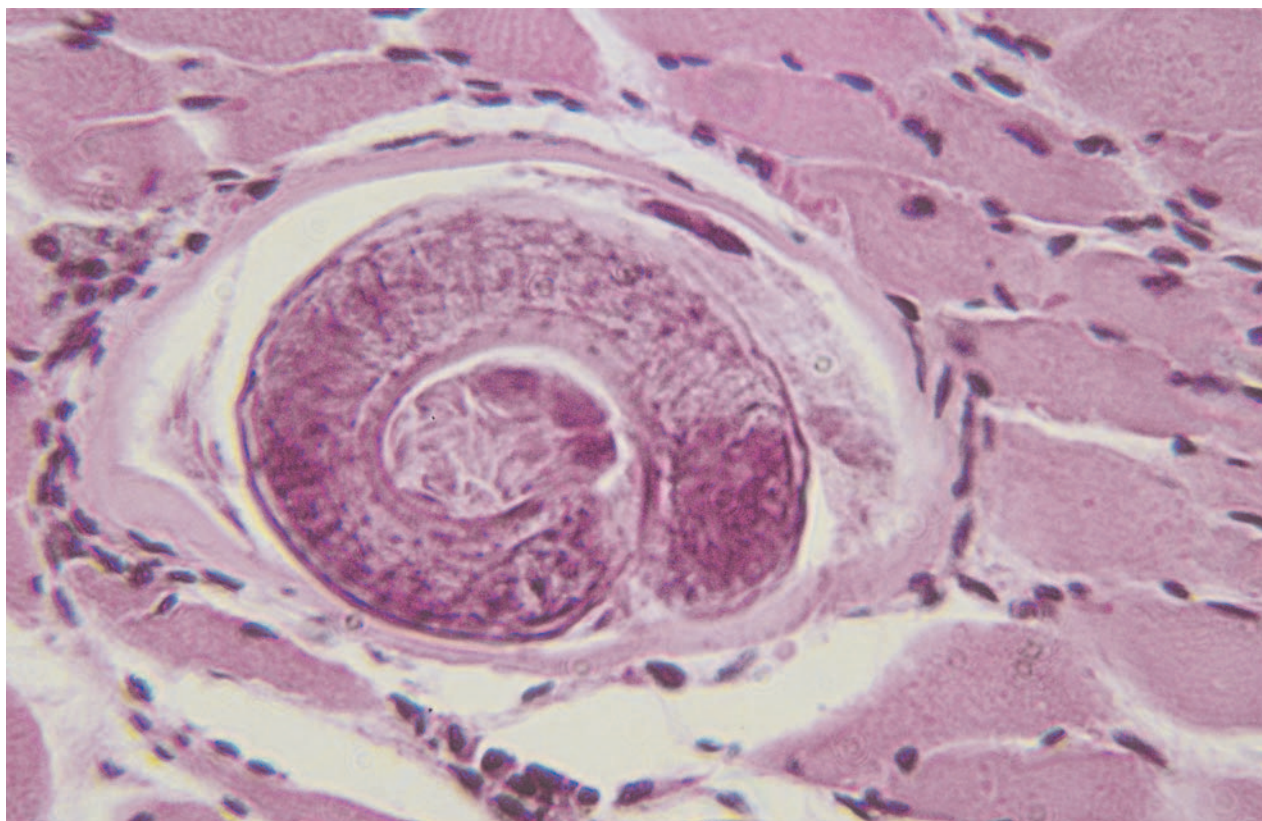
Causes and symptoms

Human hosts who eat meat infested with trichinae may experience symptoms in varying degrees. If the meat ingested has only a few cysts, then the human host’s load of parasites (worm burden) is said to be relatively small, and symptoms will be moderate. In fact, many trichinosis infections are subclinical, which means that the symptoms are so mild that the infection remains undiagnosed. In a host with a greater worm burden, the initial symptoms will be caused by the presence of the adult worms in the intestine. These symptoms usually include **fever**, **diarrhea**, abdominal **pain**, and perhaps **vomiting**. The symptoms begin about one to two days after eating the contaminated meat, and may last for a week or so.

When the larvae begin their migration through the blood vessels, the host will begin to experience symptoms that affect the whole body (systemic symptoms), such as fever; swelling of the face and the area around the eyes; rash; bleeding into the nail beds, retina, and whites of the eyes; and **cough**. In very severe cases of trichinosis, inflammation of the heart muscle (**myocarditis**), lungs (pneumonitis), or brain (encephalitis) may occur. These symptoms can lead to the few deaths caused by trichinosis.

The larvae begin to burrow into the host’s muscles and form cysts within two to three weeks of the initial infection. This encysting produces signs of muscle inflammation (myositis) including swelling of the affected muscle groups, pain, and weakness. The most frequently affected muscles are the muscles outside the eye (extraocular muscles) that control eye movements; the muscles of the jaw, neck, and upper arm (biceps muscle); the muscles of the lower back (lumbar region); and the diaphragm, which is the muscle that separates the abdominal and chest cavities and aids in breathing.

The symptoms of trichinosis are at their most severe at about three weeks after infection, and decrease very slowly in their severity. Recovery is extremely gradual, and symptoms may last for as long as three months. **Fatigue** and muscle pain (myalgia) may take several more months to subside.



A *Trichinella spiralis* cyst in striated muscle tissue. *T. spiralis* cysts can survive up to ten years in this form. (JLM Visuals. Reproduced by permission.)

Diagnosis

An initial diagnosis of trichinosis relies heavily on the presence of its classic symptoms—swelling around the eyes, muscle inflammation, fever, and high levels of a certain type of white blood cell (eosinophils)—coupled with the patient’s history. If the patient reports having eaten undercooked meat from an animal known to be a potential carrier of trichinosis, the doctor may order a muscle biopsy to confirm the diagnosis. By the third or fourth week of infection, muscle biopsies usually indicate the presence of larvae. Stool tests rarely reveal adult worms, although larvae can sometimes be found in blood or duodenal washings after the second week of infection. The blood test that is the most specific for trichinosis is the bentonite flocculation (BF) test.

T. spiralis can infect a number of different animal species used for food. The most common food culprit in the United States has been pork sausage, while outbreaks in Europe have been caused by wild boar and horse meat. Outbreaks of trichinosis in Asia and

Africa have been traced to dog meat, and outbreaks in Northern Canada have resulted from consumption of walrus and bear meat.

Treatment

Supportive care

Treatment of trichinosis is primarily aimed at decreasing the severity of the symptoms. Symptomatic relief includes bed rest and medications to relieve fever and muscle pain. The medications most commonly given are **aspirin** and **nonsteroidal anti-inflammatory drugs** (NSAIDs). Steroids such as prednisone (Deltasone, Meticorten) are reserved for the most severe cases of muscle inflammation, or for complicated cases that include myocarditis.

Anthelmintic medications

In addition to medications for pain relief, trichinosis can be treated with drugs that are called anti-worm medications or anthelmintics. Two related

KEY TERMS

Anthelmintic—A type of medication that is given to destroy or eliminate parasitic worms.

Cyst—In the life cycle of the round worm, a protective, walled-off capsule in which the larvae lie dormant.

Embryonic—In the life cycle of the round worm, a very early life stage occurring within the uterus of the female round worm.

Host—The animal within which a parasite lives, and from which the parasite receives its nutrition.

Inflammation—A reaction within the body to an invader (virus, bacteria, fungus, worm, etc.) or to tissue injury. The classic signs of inflammation include redness, heat, pain, and loss of function.

Larva—In the life cycle of the round worm, the second stage of life, sometimes considered the “adolescent” stage.

Nematode—A type of roundworm with a long, unsegmented body, usually parasitic on animals or plants.

Striated muscle—Also known as striped muscle; it includes muscles of the skeletal system and of the heart.

Trichina—An individual example of *Trichinella spiralis*.

anti-worm medications, mebendazole (Vermox) and thiabendazole (Mintezol), have been reported to be effective against intestinal larvae, but not against larvae encysted in the muscles. In particular, thiabendazole has worked best when given to patients who knew within 24 hours that they had eaten infested meat. Thiabendazole has, however, anti-inflammatory properties that can relieve some of the pain during the muscle stage of trichinosis.

Prognosis

The prognosis for recovery from trichinosis is generally good. Most people with the disease are unaware that they have even been infected. It is estimated that between 150,000 and 300,00 people in the United States become infected yearly, so that at any given time, 1.5 million people have *T. spiralis* infections. Most of these people have such light cases that trichinosis is never identified. Worm burden is measured in larvae per gram of muscle tissue;

people with 10 or fewer larvae per gram of muscle tissue usually have no significant symptoms. When the number climbs to 100 larvae per gram of muscle tissue, the symptoms become noticeable. People with over 1000 larvae per gram of muscle tissue are usually extremely ill, and often die. The mortality rate of trichinosis is about 1%.

Prevention

Prevention of trichinosis is relatively simple. Swine should be fed only grain or cooked garbage because uncooked garbage may contain contaminated pork scraps. Meat from animals prone to trichinosis infection should be cooked or smoked thoroughly until it is no longer pink. Freezing meat at an adequately low temperature (5°F/–15°C for three weeks) can kill most encysted larvae, except for species which infect such arctic mammals as walrus or bear.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Trichomonas vaginalis infection see
Trichomoniasis

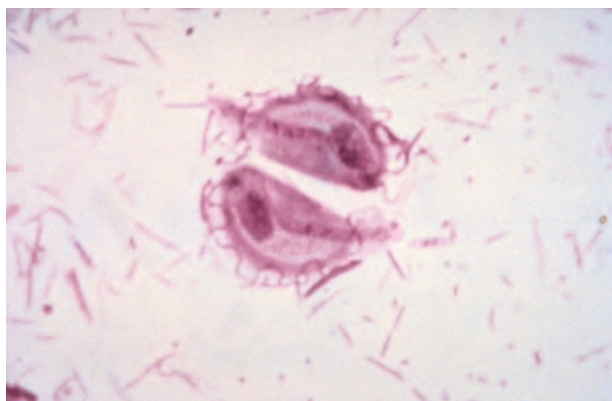
Trichomoniasis

Definition

Trichomoniasis refers to an infection of the genital and urinary tract. It is the most common sexually transmitted disease, affecting about 120 million women worldwide each year.

Description

Trichomoniasis is caused by a protozoan (the smallest, single-celled members of the animal kingdom). *Trichomonas vaginalis* is passed almost 100% of the time through sexual contact. Trichomoniasis is primarily an infection of women's vaginal and urinary tracts. A woman is most susceptible to infection just after having completed her menstrual period. Men may carry the organism unknowingly, since infection in men may cause mild or no symptoms.



A close up image of *Trichomonas vaginalis*, the parasite that causes vaginitis in humans. (Custom Medical Stock Photo. Reproduced by permission.)

Causes and symptoms

Because trichomoniasis is a sexually transmitted disease, it occurs more often in individuals who have multiple sexual partners. The protozoan is passed to an individual by contact within the body fluids of an infected sexual partner. It often occurs simultaneously with other sexually transmitted diseases, especially **gonorrhea**.

In women, the symptoms of trichomoniasis include an unpleasant vaginal odor, and a heavy, frothy, yellow discharge from the vagina. The genital area (vulva) is often very itchy, and there is frequently **pain** with urination or with sexual intercourse. The labia (lips) of the vagina, the vagina itself, and the cervix (the narrowed, lowest segment of the uterus which extends into the upper part of the vagina) will be bright red and irritated.

In men, there are usually no symptoms at all. Occasionally, a man will notice a small amount of yellowish discharge from his penis, usually first thing in the morning. There may be some inflammation of the urethra, or **urethritis**, which produces mild discomfort while urinating.

The use of **antibiotics** is a contributing factor to recurrent trichomoniasis in some women because antibiotics affect the balance of bacteria in the vagina, allowing such organisms as *T. vaginalis* to multiply more rapidly.

Diagnosis

Diagnosis is easily made by taking a sample of the discharge from the women's vagina, or from the opening of the man's penis. The sample is put on a slide, and viewed under a microscope. The protozoa, which are able to move about, are easily viewed.

KEY TERMS

Metronidazole—An anti-infective agent regarded as the best available drug for treating trichomoniasis. It is sold under the trade names Flagyl and MetroGel.

Protozoan—A one-celled organism belonging to the simplest phylum of the animal kingdom. Trichomoniasis is caused by a protozoan.

Urethritis—Inflammation of the urethra, which is the canal that carries urine from the bladder to the outside of the body.

Trichomoniasis tends to be underdiagnosed in men because of the relative mildness of symptoms in men and insufficiently sensitive diagnostic tests. The recent introduction of DNA amplification, however, indicates that the incidence of trichomoniasis in men is much higher than was previously thought.

Treatment

The usual treatment is a single large dose of metronidazole, or split doses over the course of a week. Sexual partners of an infected individual must all be treated, to prevent the infection being passed back and forth.

Women who are taking antibiotics for other illnesses should speak to their health care provider about the possible effects of the medication(s) on the balance of organisms in their vagina.

As of late 2003, the number of cases of metronidazole-resistant trichomoniasis appears to be increasing rapidly. Some success has been reported with the broad-spectrum anti-parasitic drug nitazoxanide, but further research needs to be done. A group of researchers in Thailand is currently investigating the effectiveness of a group of drugs known as bisquaternary quinolinium salt compounds in treating trichomoniasis.

Alternative treatment

Cure of trichomoniasis may be difficult to achieve with alternative treatments. Some practitioners suggest eliminating sweets and carbohydrates from the diet and supplementing with antioxidants, including **vitamins** A, C, and E, and zinc. Naturopaths may recommend treatment with two douches (a wash used inside the vagina), alternating one in the morning and one at

bedtime. One douche contains the herbs calendula (*Calendula officinalis*), goldenseal (*Hydrastis canadensis*), and **echinacea** (*Echinacea* spp.); the other douche contains plain yogurt. The herbal douche helps to kill the protozoa, while the yogurt reestablishes healthy flora in the vagina. Another herbal remedy that is sometimes used is tea tree oil. Acidifying the vagina by douching with boric acid or vinegar may also be useful.

Prognosis

Prognosis is excellent with appropriate treatment of the patient and all sexual partners. Without treatment, the infection can smolder on for a very long time, and can be passed to all sexual partners.

Prevention

All **sexually transmitted diseases** can be prevented by using adequate protection during sexual intercourse. Effective forms of protection include male and female condoms.

Resources

BOOKS

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Rosalyn Carson-DeWitt, MD
Rebecca J. Frey, PhD

Trichotillomania see **Alopecia**

Trichuriasis see **Roundworm infections**

Tricuspid incompetence see **Tricuspid valve insufficiency**

Tricuspid regurgitation see **Tricuspid valve insufficiency**

Tricuspid stenosis see **Tricuspid valve stenosis**

Tricuspid valve insufficiency

Definition

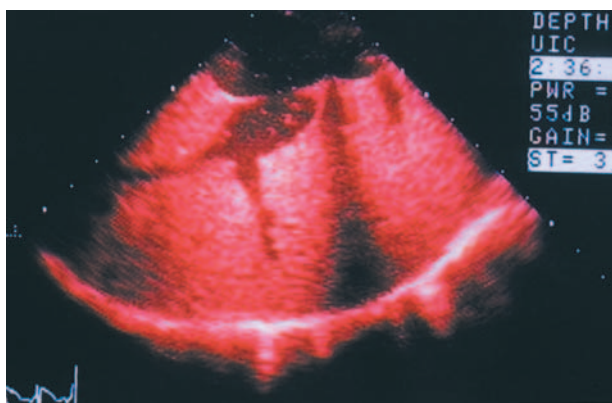
Tricuspid valve insufficiency occurs when a tricuspid valve does not close tightly enough to prevent leakage. This condition is also called tricuspid valve regurgitation and tricuspid incompetence.

Description

The tricuspid valve is located between the right atrium and the right ventricle of the heart. When the right ventricle contracts, it is supposed to pump blood forward into the lungs. If the tricuspid valve does not close tightly, some of that blood leaks back into the right atrium. When the atrium receives its usual quantity of blood from veins leading to the heart, plus the leaking blood, the pressure inside the atrium increases. This higher pressure creates resistance to the flow of blood in the veins that enter the atrium from the body. In addition, this increase in pressure causes the right atrium to enlarge over time. Congestion from fluid buildup occurs, particularly in the liver and legs.

Causes and symptoms

If a person has serious lung disease or a narrowing of the pulmonary valve, the right ventricle must pump harder to force the blood through the pulmonary valve. In order to pump harder, the right ventricle enlarges and the valve opening stretches, causing the valve to leak.



This echocardiogram of the heart shows tricuspid valve insufficiency. (Custom Medical Stock Photo. Reproduced by permission.)

Tricuspid valve insufficiency usually produces such vague symptoms, as general weakness and **fatigue**. As the conditions worsens, a person experiences pain in the upper right part of the abdomen, caused by a congested and enlarged liver. The legs may also swell (**edema**).

An enlarged right atrium can cause atrial fibrillation (the atria flutters, rather than pumping in a regular rhythm) and severe tricuspid regurgitation of blood, which can eventually lead to congestive **heart failure**.

Diagnosis

A leaky valve can be heard with a stethoscope; the sound is called a heart murmur. Additional support for diagnosing tricuspid valve insufficiency comes from a medical history, physical exam, and **chest x ray**. Further testing with **echocardiography**, to show an image of the leakage and its severity, is the most helpful diagnostic test for this condition.

Treatment

Tricuspid valve insufficiency itself usually does not require treatment, since a tiny leakage occurs in most normal people. In certain cases, however, if there is underlying pulmonary valve disease or lung disease, those conditions should be treated.

If irregular heart rhythms or heart failure are present, they are usually treated independently of the valve insufficiency.

Since a person with known tricuspid valve insufficiency is at risk for infections of the heart, **antibiotics**

KEY TERMS

Atrial fibrillation—A rapid, uncoordinated quivering of the upper chamber of the heart.

Atrium—The upper chamber of the heart.

Pulmonary valve—The valve at the opening from the right ventricle to the artery that leads to the lungs.

should be taken before and after oral or dental surgery, or urologic procedures.

Prognosis

Tricuspid valve insufficiency is not usually considered to be serious. If it is the result of other cardiopulmonary disease, the extent of those conditions effect the prognosis.

Prevention

In general, tricuspid valve insufficiency cannot be prevented.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave., Dallas, TX 75231-4596. (214) 373-6300 or (800) 242-8721. inquire@heart.org. <<http://www.americanheart.org>>.

Dorothy Elinor Stonely

Tricuspid valve stenosis

Definition

Tricuspid valve stenosis is a narrowing or stiffening of the opening in the valve. This stenosis causes increased resistance to blood flow through the valve.

Description

The tricuspid valve is located between the right atrium and the right ventricle of the heart. It is the largest of the four valves in the heart. When the tricuspid valve is narrowed or stiffened, it decreases the amount of blood that can flow through it. This

KEY TERMS

Rheumatic fever—An inflammatory illness that can follow strep throat, and could cause heart damage.

decrease raises the pressure in the right atrium and causes the atrium to enlarge. It also causes the right ventricle to shrink, and lowers the cardiac output.

Causes and symptoms

Tricuspid valve stenosis is most often the result of **rheumatic fever**. On rare occasions, it is caused by a tumor or disease of the connective tissue. The rarest cause is a birth defect.

A person with tricuspid valve stenosis may experience generalized weakness and **fatigue**. Many people have **palpitations** and can feel fluttering in their neck. Over time, there may be **pain** in the upper right abdomen, due to increased congestion and enlargement of the liver.

Diagnosis

The noise produced by blood trying to flow through a stenotic valve can be heard with a stethoscope, and is referred to as a murmur. An x ray of the chest will show the right atrium to be enlarged. Further support for this diagnosis is found on an echocardiogram of the heart, which will show an image of the stenotic valve and measure its severity.

Treatment

Tricuspid valve stenosis itself usually doesn't require treatment. However, if there is damage to other valves in the heart as well, then surgical repair or replacement must be considered.

Since a person with known tricuspid valve stenosis is at risk for infections of the heart, **antibiotics** should be taken before and after oral or dental surgery, or urologic procedures.

Prognosis

Mild tricuspid valve stenosis is not usually considered cause for surgery. The decision to repair or replace the tricuspid valve is often based on the health of the aortic and mitral valves, rather than on the severity of stenosis in the tricuspid valve.

Prevention

Rheumatic **fever**, the usual cause of tricuspid valve stenosis, has almost disappeared in North America and western Europe. Therefore, the number of people who acquired this condition in childhood will decline over time.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300 or (800) 242-8721. inquire @heart.org. <<http://www.americanheart.org>>.

Dorothy Elinor Stonely

Tricyclic antidepressants see
Antidepressants, tricyclic

Trigeminal neuralgia

Definition

Trigeminal **neuralgia** is a disorder of the trigeminal nerve (the fifth cranial nerve) that causes episodes of sharp, stabbing pain in the cheek, lips, gums, or chin on one side of the face.

Description

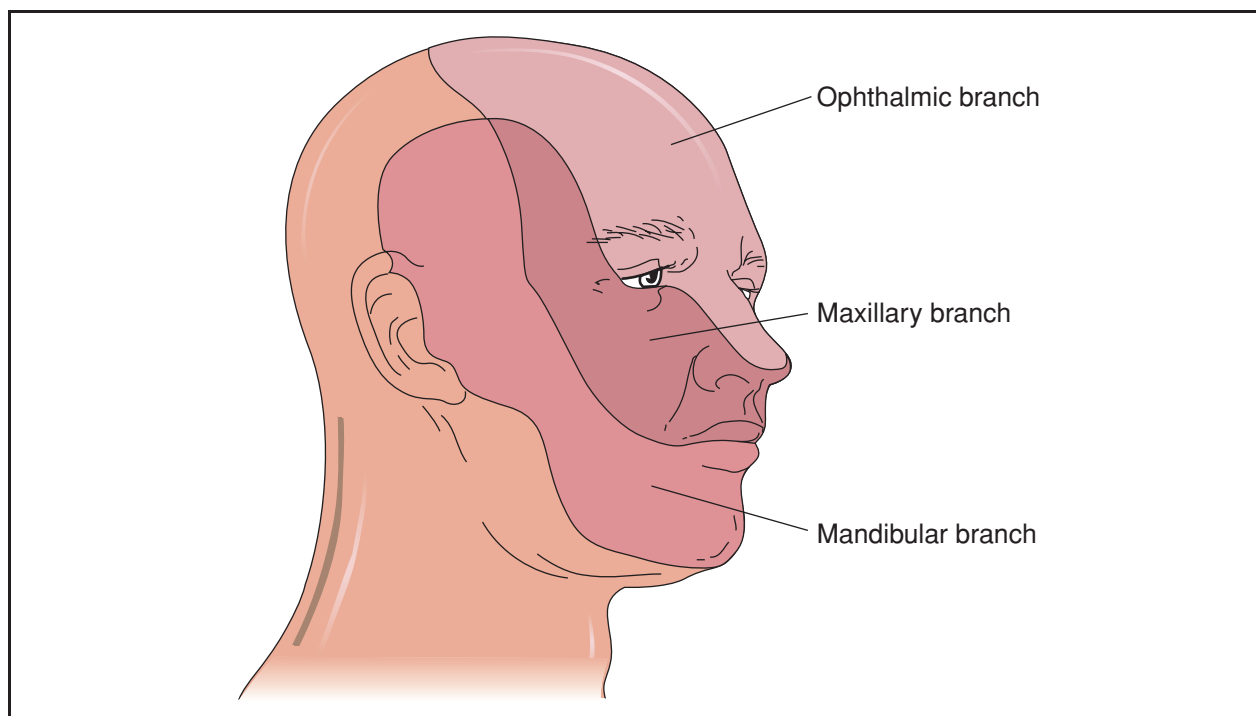
The trigeminal nerve, which is divided into three branches, is responsible for chewing, for producing saliva and tears, and for sending facial sensations to the brain. When this nerve breaks down for some reason, it can trigger brief but agonizing sizzles of **pain** on one side of the face.

This condition is unusual in those under age 50 and more often occurs after 70. Women are three times more likely to have the condition than are men. When trigeminal neuralgia does occur in younger people, it is often associated with multiple sclerosis.

The pain, while brief, is so severe that the sufferer often can't do anything else while the attack lasts. People with this pain often wince or twitch, which is where trigeminal neuralgia gets its French nickname *tic douloureux*, meaning "painful twitch."

Causes and symptoms

The origin of trigeminal neuralgia is not certain, but scientists believe it may be caused by degeneration,



Trigeminal neuralgia is a disorder of the trigeminal nerve (which is divided into three branches, as illustrated above) that causes episodes of sharp, stabbing pain in the cheek, lips, gums, or chin on one side of the face. The origin of this disorder is not certain, but scientists believe it may be caused by degeneration, pressure, or irritation of the trigeminal nerve. (Illustration by Electronic Illustrators Group.)

pressure, or irritation of the trigeminal nerve. Some doctors believe the pain may be triggered by pressure from a nearby abnormally-formed artery lying too close to the nerve.

Any part of the three branches of the trigeminal nerve may be affected. Neuralgia of the first branch leads to pain around the eyes and over the forehead; the second branch causes pain in the upper lip, nose and cheek; the third branch causes pain on the side of the tongue and lower lip.

The first episodes are usually fairly mild and brief, and it may be minutes, hours, or weeks before the next attack. However, attacks tend to occur in clumps that may last for weeks at a time. As the sufferer ages, the episodes become more frequent and painful, until the person begins to live in constant fear of the next one.

The momentary bursts of pain usually begin from the same spot on the face each time. The pain can be triggered by touching the area, washing, shaving, eating, drinking, or even talking. Even a cool breeze across the face can set off an attack. Pain is more severe at the ends of the affected nerve, especially over the lip, chin, nostrils, or teeth.

Diagnosis

Diagnosis is usually made by eliminating other problems that could cause similar pain in teeth, jaw, head, or sinuses. Because patients with the condition tend to avoid trigger points, avoiding chewing, shaving, touching or washing their faces can be a clue to diagnosis of trigeminal neuralgia.

Treatment

It is not easy to treat trigeminal neuralgia. Pain can be suppressed by a range of medicines, including the anti-epilepsy medicines carbamazepine (Tegretol) or phenytoin (Dilantin). These drugs slow down the nerve signals at certain nerve terminals, which eases the pain. However, these drugs cause a wide range of side effects, including **nausea**, **dizziness**, drowsiness, liver problems, and skin **allergies**. Some people develop resistance to the drugs or they can't tolerate the high dosage needed to control the discomfort. If the medicines are stopped, the pain usually returns.

If drug treatment fails, surgical treatment to block pain signals from the nerve may be effective. Radio-

KEY TERMS

Multiple sclerosis—A progressive disease of the central nervous system in which the coverings of nerves in the brain and spinal cord are destroyed.

frequency waves, gamma rays, or glycerol injections can deaden the nerve (and hence the pain). An operation that frees the nerve from whatever is compressing it (blood vessel or tumor) can permanently relieve pain, but this major neurosurgical procedure carries its own risks and complications. Alternatively, a new procedure seeks to place a cushioning sponge between the nerve and a pulsating artery wrapping around it to soothe the irritated nerve.

Prognosis

Although the pain is momentarily incapacitating, it's not life-threatening. As the person ages, the attacks can be expected to occur more and more frequently.

Prevention

While the condition itself can't be prevented, there are a number of things patients can do to avoid triggering attacks:

- wash with cotton pads and warm water over the face
- rinse the mouth with water after eating, if tooth-brushing triggers pain
- eat and drink food and beverages at room temperature
- chew on the unaffected side
- eat soft foods, if eating is becoming a problem

Resources

ORGANIZATIONS

Chronic Pain Outreach. 822 Wycliff Ct., Manassas, VA 22110. (703) 368-7357.

National Chronic Pain Outreach Association, Inc. P.O. Box 274, Millboro, VA 24460- 9606. (540) 997-5004.

National Institute of Neurological Disorders and Stroke. P.O. Box 5801, Bethesda, MD 20824. (800) 352-9424. <<http://www.ninds.nih.gov/index.htm>>.

Trigeminal Neuralgia Association. P.O. Box 785, Barnegat Light, NJ 08006. (609) 361-1014.

Carol A. Turkington

Trigger finger

Definition

Trigger finger is the popular name of stenosing tenosynovitis, a painful condition in which a finger or thumb locks when it is bent (flexed) or straightened (extended).

Description

Tendons are tough, fibrous cords that connect muscles to bones. Tendons must slide easily through their protective coverings (tendon sheaths). The finger and thumb bones have tendons that are responsible for bending and straightening the fingers. Problems start when a tendon sheath narrows (stenosis) and the outer covering of the tendon becomes inflamed (tenosynovitis). The tendon swells because of the constriction, sometimes forming a nodule, and is no longer able to move smoothly through its sheath. As a result, a finger may lock in an upward position as the person tries to straighten it. The condition usually happens in the ring and middle fingers and is more common in women, typically over age 30. In infants and small children, the condition generally occurs in the thumb.

Causes and symptoms

Trigger finger is often an overuse injury because of repetitive or frequent movement of the fingers. Trigger finger may happen because a person performs the same manipulation over and over on a job, from squeezing and gripping during a weekend of heavy pruning and gardening, or from such hobbies as playing a musical instrument or crocheting. Trigger finger may also result from trauma or accident. The symptoms of trigger finger are **pain** in the fingers and "popping" sensations. Sometimes the finger may lock down into the palm or lock out straight. Symptoms are usually worse in the morning and improve during the day.

Diagnosis

The diagnosis of trigger finger and thumb is obvious on physical examination. Often there is a click that can be felt as the nodule passes through the sheath. Most cases are uncomplicated although X rays are often taken to rule out other injuries or disease such as arthritis.

Treatment

Initial treatment for mild or infrequent symptoms of trigger finger include rest, avoiding or modifying those

KEY TERMS

Microcirculation—The passage of blood in the smallest blood vessels of the body, such as the capillaries in the hand and fingers.

Myofascial—The fibrous tissue that encloses and separates layers of muscles.

Nodule—A swelling or knob that may form on a tendon and make it difficult to slide smoothly through its sheath.

Stenosis—Narrowing of a passageway or opening in the body. In trigger finger it is the tendon sheath that narrows.

Synovial tendon sheath—Where the tendons cross joints, they are sheathed in thin membranes known as synovium, which provide lubrication to decrease friction.

Tendon sheath—A membrane covering a tendon.

Tenosynovitis—Inflammation of a tendon and its enveloping sheath, usually resulting from overuse injury.

activities that caused the inflammation, and the use of a nonsteroidal anti-inflammatory drug (NSAID) such as ibuprofen. This may relieve the swelling and inflammation that resulted in the constriction of the sheath and the restriction of the tendon. Injection of a steroid medication (cortisone) into the tendon sheath is the next option to treat trigger finger. Depending on the severity, there may be one more injection a week later. Two-thirds of patients improve after one injection. Some physicians will splint the finger in extension after the injection.

In severe cases that do not respond to injections and the finger or thumb remains in a locked position, surgery may be required to relieve the symptoms. A local anesthetic is used for the surgical procedure performed on an outpatient basis. An incision is made by a surgeon in the palm of the hand at the base of the affected finger or thumb to relieve the constriction of the tendon. Recovery may take up to four weeks. Sometimes physical therapy of the hand is required after surgery to regain good use.

Alternative treatment

Treatment should begin when a person starts having difficulty moving the fingers. If started early, non-invasive measures have a good chance for success. Alternative treatments include acupuncture to facilitate healing and microcirculation, pulsed ultrasound, and myofascial release work for the affected area.

Prognosis

At least half of cases can be cured non-surgically. The key to successful treatment is early intervention. A mistake people make is trying to work through the pain. Diabetics have a higher incidence of the condition and are sometimes left with a disability.

Prevention

Taking frequent breaks from a repetitive activity will do much to prevent the condition. Depending on the intensity, that may mean a 10-minute break every hour from the repetitive activity. The break should be spent stretching the hands and arms and generally moving around.

Resources

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Stroud, R. “Minimally Invasive Surgical Techniques of the Hand and Upper Extremities.” *Orthopedic Technology Review* September 2000: 18.

ORGANIZATIONS

American Society for Surgery of the Hand. 6300 N. River Rd., Suite 600, Rosemont, IL 60018. <<http://www.hand-surg.org>>.

OTHER

Jameson DC, CCSP, Timothy J.

“Explanation, Treatment, and Prevention of Trigger Finger.” *GuitarBase Articles*. <<http://www.gbase.com/articles/med/med4.html>>.

Ruthan Brodsky

Triglycerides test

Definition

Triglycerides test is a blood test to determine the amount of triglycerides, a form of fat, in the blood.

Purpose

The triglycerides test is one of the screening tests for excess lipids (fats) in the blood. It is usually part of an evaluation of risk factors for heart disease.

Description

Triglycerides are a form of fat that comes from foods. They can also be made and stored in the body and are used as an energy source. High levels of triglycerides in the blood can mean that there is too much fat in the diet. Hypertriglyceridemia (high levels of triglycerides) is associated with coronary heart disease, especially since elevated triglycerides levels are often associated with unhealthy low levels of hyper-density lipoproteins (the “good” cholesterol), which are necessary for good health.

Preparation

For triglycerides testing, blood is drawn from a vein in the arm. A vein at the inside of the elbow or on the back of the hand is usually selected. The area where the needle will be inserted is cleaned with antiseptic. A small needle is inserted through the skin and into the vein, allowing a small amount of blood to flow into a collection tube or syringe. Once the blood is collected, the needle is removed from the puncture site.

Before the blood test, the patient may be required to refrain from eating food from eight to–12 hours. Patients should not drink alcohol for 24 hours before the test. Some drugs may affect the test and the patient may be asked to cease taking certain medications before the test. Oral contraceptives, estrogen, and cholestyramine (a drug used to treat **high cholesterol**) can increase triglyceride levels. Ascorbic acid (vitamin C), asparaginase (an enzyme), and various drugs used to treat high blood lipids, can decrease blood triglyceride levels. These substances should not be taken prior to this test.

Aftercare

After the blood sample has been taken and the needle withdrawn from the puncture site, a cotton ball or gauze pad may be placed over the site and direct pressure applied to reduce bleeding. A piece of surgical tape or gauze adhesive bandage strip may be secured over the site to prevent further bleeding.

Risks

There is a very small risk that the puncture site may bleed excessively, a bruise or infection may develop at the site, or it may take several punctures to locate a vein. Some patients may feel faint or light-headed when blood is drawn.

Normal results

The normal range of triglycerides in the blood depends on the age and gender of the patient. Women naturally have higher levels of triglycerides than men. **Pregnancy** can also increase triglyceride levels. As people age and gain weight, triglyceride levels generally increase. For adults, a normal level is considered to be less than 200 mg/dl (milligrams per deciliter). Levels from 200–400 mg/dL are considered borderline high.

Abnormal results

Triglyceride levels ranging from 400-1000 mg/dL are considered high and levels greater than 1000 mg/dL are considered very high. High levels of triglycerides may indicate liver disease (cirrhosis), an underactive thyroid problem, uncontrolled diabetes, an infection of the pancreas (**pancreatitis**), **kidney disease**, or a diet too low in protein and too high in carbohydrates.

Extremely low triglycerides levels (less than 10 mg/dL) can also indicate a problem. Low levels may indicate **malnutrition** (not enough nutrients in the diet), malabsorption (inadequate absorption of nutrients in the intestinal tract), a diet too low in fat, or an overactive thyroid problem.

Resources

OTHER

“Triglyceride levels.” *HealthAnswers.com* <<http://www.healthanswers.co>>.

“Triglycerides.” *ThriveOnline*. <<http://thriveonline.oxygen.com>>.

Altha Roberts Edgren

Triiodothyronine test see **Thyroid function tests**

Triple screen

Definition

Triple screen is a blood test offered to pregnant women during the 16th-18th week of **pregnancy** to help identify if the unborn baby is at risk for **birth defects**.

Purpose

Researchers have found that women carrying a fetus with **Down syndrome** tend to have three substances in their blood in a distinctive abnormal pattern. Both alpha-fetoprotein (AFP), a protein produced by the fetus, and unconjugated estriol (uE3), a hormone manufactured by both the fetus and the placenta, are lower than normal, while human chorionic gonadotropin (hCG), a hormone produced in the placenta, is higher. The level of each substance is then divided by the median concentration of that substance for the given week of pregnancy to generate a multiple of the median value (MOM). These values, along with other characteristics of the mother, such as her age, are analyzed by a computer program to indicate the probability that the fetus has Down syndrome. Down syndrome is a condition that includes **mental retardation**, skeletal abnormalities such as upslanted eyes and **cleft palate**, and organ abnormalities such as heart disease and intestinal obstruction.

Guidelines developed from this research help healthcare workers to determine whether the risk for any one pregnancy is clearly different from that for the general population of the same age. For example, the average 25-year-old woman has one chance in a 1,000 of bearing a Down syndrome baby. After her blood is analyzed for the three hormones, her risk can be refigured. It might be considerably lower—or higher—than average.

The triple screen was first aimed toward detecting fetuses at high risk for Down syndrome, an abnormality that comes about when, at fertilization, the embryo receives three copies of chromosome 21 instead of the usual two. For unknown reasons, extra amounts of genetic material are usually associated with severe developmental defects. Although Down syndrome is the most common such disorder, it is by no means the only one. Chromosomes 13 or 18 can also turn up as a triple set (or trisomy). An abnormal pattern of the three hormones sometimes signals one of these trisomies or another, more rare chromosomal disorder.

Precautions

It is important that the correct age of the unborn baby be determined by last menstrual period dating and recorded for the risk calculation. Errors in determining the age of the fetus lead to errors when interpreting the test results. Since an AFP test is only a screening tool, an abnormal test result is not necessarily indicative of a birth defect. Accurate dating of the

fetus lowers the false-positive and false-negative rates associated with this screening test.

Description

Prior to 1964, when the association between low levels of AFP and an increased risk for Down syndrome was reported, risk assessment for chromosomal diseases was based upon maternal age. At age 35, the risk of carrying a Down syndrome pregnancy is approximately one in 270, and this was deemed sufficient to warrant **amniocentesis**, a more accurate but invasive test that involves using a needle to draw out a sample of the embryonic fluid that surrounds the fetus. However, three of four Down syndrome pregnancies occur in women under 35 years old. When AFP testing was used along with maternal age, the rate of detection of Down syndrome increased to about 45%, but this level of sensitivity did not justify the screening of younger women because of the risk of **miscarriage**. The inclusion of uE3 and hCG testing has improved the detection rate to approximately 65-80% of cases for all age groups. The triple screen test costs \$90-250 and is covered by most insurance plans.

Preparation

There is no specific physical preparation for this test. **Fasting** is not required.

Aftercare

After the blood sample is drawn, pressure should be applied to the puncture site until the bleeding stops to reduce bruising. A bandage or taped ball of cotton may be applied to the site. A warm pack may be applied to the site to relieve discomfort.

Risks

The complications associated with drawing blood are minimal, but may include bleeding from the puncture site, feeling faint or lightheaded after the blood is drawn, or blood accumulating under the puncture site (hematoma).

Normal results

Results are expressed as multiples of the median value used by the laboratory. Normal ranges expressed in concentration (for example, ng/mL) are dependent upon the age of the fetus, but MOMs are age-adjusted and do not change. These values are used to calculate risk. If the multiple of the median value is

KEY TERMS

Amniocentesis—A test performed to determine the health, sex, or genetic constitution of a fetus by taking a sample of amniotic fluid through a needle inserted into the womb of the mother.

Amniotic fluid—The fluid that surrounds a fetus while it is developing. It is the fluid that flows out in the “breaking of the water” before a baby is born.

Down syndrome—A genetic disorder characterized by a broad skull, blunt facial features, short stature, and learning difficulties. It is caused by the presence of an extra copy of chromosome 21.

Placenta—An organ that develops inside the uterus of pregnant women to supply food and oxygen to the fetus through the umbilical cord.

below 2.0 MOM or 2.5 MOM (depending on the laboratory), the fetus is considered to be at a lower risk for a genetic defect.

Abnormal results

If the multiple of the median value is above 2.0 MOM or 2.5 MOM (depending on the laboratory), the fetus is considered to be at a higher risk for a genetic defect. The MOM value for amniotic fluid is then used to calculate the exact probability the fetus is affected (1:100, for example). With respect to Down syndrome and trisomy 18, the MOM values are also used in the calculation of probability. The woman is considered to be “high risk” or “screen positive” for Down syndrome if the risk is greater than the standard risk for women who are 35 years old or older (one in 270). For trisomy 18, the cut-off is one in 150. In one study the triple marker screen test had a detection rate for Down syndrome of 67% and a false positive rate of 5%.

Resources

BOOKS

Health Education Board for Scotland. *A Parent's Guide to the Screening Test for Spina Bifida and Down Syndrome: A Blood Test You Can Chose to Have During Your Pregnancy*. Edinburgh, United Kingdom: Health Education Board for Scotland, 2002.

PERIODICALS

Graves, J. Christopher, et al. “Maternal Serum Triple Analyte Screening in Pregnancy.” *American Family Physician* (March 1, 2002): 915-921.

Pouliot, Janine S. “Triple Screen.” *Better Homes and Gardens* (October 2003): 256.

ORGANIZATION

National Down Syndrome Society. 666 Broadway, New York, NY 10012. (212) 460-9330 or (800) 221-4602. info@ndss.org. <http://www.ndss.org>.

Ken R. Wells

Triplets see **Multiple pregnancy**

Trisomy 18 see **Edwards' syndrome**

Trisomy G syndrome see **Down syndrome**

Trisomy 13 see **Patau's syndrome**

Trobofloxacin see **Fluoroquinolones**

Tropical spastic paraparesis

Definition

Tropical spastic paraparesis (TSP) is an incurable viral infection of the spinal cord that causes weakness in the legs. It is caused by the human T-cell lymphotropic virus-1 (HTLV-1) retrovirus.

Description

As the name implies, tropical spastic paraparesis usually occurs in tropical locales. Although isolated cases have been diagnosed in the southeastern United States and other places in the United States, TSP is most frequently found in:

- the Caribbean
- Japan
- the Seychelles Islands
- regions of South America
- western Africa

TSP usually affects adults between the ages of 30 and 40, and is far more common in women than in men.

The disease may remain undetected for years after infection is contracted. When the immune system's response to the virus causes nerve damage, the legs gradually lose strength and flexibility.

Causes and symptoms

TSP is caused by the HTLV-1 virus, which also causes leukemia. The virus can be spread through the

KEY TERMS

Retrovirus—A family of RNA viruses containing a reverse transcriptase enzyme which allows the viruses' genetic information to become part of the genetic information of the host cell upon replication.

Virus—A microorganism, smaller than bacteria, which can only replicate within the a cell of a living plant or animal. The virus provides the genetic code and the host cell provides the energy and raw materials for replication.

placenta, and also through blood transfusions, breast-feeding, contaminated needles, and sexual contact.

Symptoms may begin years after infection. In response to the infection, the body's immune response may injure nerve tissue, causing symptoms that include bladder abnormalities, leg **pain**, loss of feeling in the feet, **tingling** sensations, and unpleasant sensations when the skin is touched.

As many as 20% of patients with TSP may also experience:

- deafness
- double vision
- the tendency to incorrectly estimate the amount of motion necessary to accomplish a specific task (dysmetria)
- exaggerated reflexes
- facial paralysis
- tremor.

Diagnosis

Infectious disease specialists use blood tests and **magnetic resonance imaging** (MRI) of the spinal cord to diagnose this condition.

Treatment

While the disease is incurable, significant improvement has been reported in the condition of TSP patients treated with **corticosteroids**. These drugs are believed to alleviate symptoms by suppressing the immune system's response to the virus that causes them.

Plasmapheresis, a dialysis-like procedure in which symptom-producing antibodies are removed from the blood, also provides temporary relief.

Prognosis

As noted, TSP cannot be cured.

Prevention

The United States Food and Drug Administration (FDA) has approved screening procedures developed to detect HTLV-1 in donated blood and blood products designated for **transfusion**. These procedures, which can also be used to diagnose patients with TSP, are designed to prevent the spread of the disease.

Resources

OTHER

"Current Trends Licensure of Screening Tests for Antibody to Human T-Lymphotropic Virus Type I." Centers for Disease Control. May 27, 1998. <<http://www.cdc.gov/epo/mmwr/preview/mmwrhtml/00001311.htm>>.

Maureen Haggerty

Tropical sprue see **Malabsorption syndrome**

Troponins test

Definition

Troponins are specific proteins found in heart muscle. Troponin testing is done to diagnose heart attacks (myocardial infarctions).

Purpose

When heart muscle is damaged, as in a myocardial infarction (MI), troponins leak out of cells and into the bloodstream. Increased troponin levels indicate myocardial infarction or injury in a person with chest **pain** or pressure. Some MIs are silent, manifesting few if any symptoms.

If infarction is ruled out in a person with continuing or recurring chest pain (unstable angina), an increased troponin level indicates the person has heart muscle **ischemia** (a decreased supply of oxygenated blood to the body), and is at an increased risk for a future serious heart event.

Description

Although troponins also exist in other muscles, those in the heart are unique, and are measured separately in laboratory tests. Troponins in the heart are

called cardiac troponins. There are two main types of cardiac troponins; T and I. T is also referred to as cTnT, while I is also referred to as cTnI.

Both troponin T and I are cardiac markers used to diagnose myocardial infarctions. Cardiac markers are substances whose blood levels increase after a myocardial infarction. Others include CK (creatine kinase), myoglobin, and CK-MB (one of three CK isoenzymes).

Like all cardiac markers, troponins have a unique diagnostic window (the timeline during which the marker rises, peaks, and returns to normal). Troponin levels rise within four to six hours after the beginning of chest pain or heart damage, and stay elevated for at least one week. This long elevation allows detection of a myocardial infarction that occurred days earlier, but prevents detection of a second infarction if it occurred only days after the first.

Troponins I and T are considered superior cardiac markers for several reasons. The most significant is that cardiac troponins are the only markers specific for heart muscle. Other markers also increase following damage to other muscles. Troponin levels help predict the extent of heart muscle damage; higher levels are associated with increased damage, lower levels with less damage. Levels in a healthy person are negligible, so an increase is easily detected.

The main difference between troponins I and T is that cardiac troponin I tests measure only cardiac troponin; tests for cardiac troponin T may cross-react with troponin found in other muscles and give positive or increased results in the absence of heart damage.

Two types of tests for troponins T and I are available: a traditional quantitative test that provides an actual measurement of troponin, and a newer qualitative test that simply reports the result as positive or negative. The quantitative test takes 45–90 minutes, and helps distinguish between myocardial infarction and unstable **angina**. The qualitative test takes 15 minutes and is used in emergency rooms in which rapid patient care decisions can be made based on the presence or absence of troponins.

Preparation

Troponins tests require 5 mL of blood. Collection of the sample takes only a few minutes.

Aftercare

Discomfort or bruising may occur at the puncture site or the person may feel dizzy or faint. Pressure to the puncture site until the bleeding stops reduces

KEY TERMS

Angina—A temporary chest pain caused by the heart not receiving enough oxygen.

Cardiac marker—A substance in the blood whose level rises following a myocardial infarction.

Myocardial infarction—Commonly known as a heart attack, a myocardial infarction is an episode in which some of the heart's blood supply is severely cut off or restricted, causing the heart muscle to suffer and die from lack of oxygen.

bruising. Warm packs to the puncture site relieve discomfort.

Normal results

People without heart damage have troponin levels less than 0.5 ng/mL.

Abnormal results

Levels greater than 2.0 ng/mL indicate a person has had a significant myocardial injury, such as an infarction, and is at an increased risk for future serious heart events. Levels between 0.5 and 2.0 ng/mL indicate a diagnosis of unstable angina, other heart disorders, or **chronic kidney failure**.

Resources

BOOKS

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Nancy J. Nordenson

Trypanosoma cruzi infection see **Chagas' disease**

TSS see **Toxic shock syndrome**

Tsutsugamushi fever see **Scrub typhus**

Tubal ligation

Definition

Tubal ligation is a permanent voluntary form of birth control (**contraception**) in which a woman's

Fallopian tubes are surgically cut or blocked off to prevent **pregnancy**.

Purpose

Tubal ligation is performed in women who definitely want to prevent future pregnancies. It is frequently chosen by women who do not want more children, but who are still sexually active and potentially fertile, and want to be free of the limitations of other types of birth control. Women who should not become pregnant for health concerns or other reasons may also choose this birth control method. Tubal ligation is one of the leading methods of contraception, having been chosen by over 10 million women in the United States—about 15% of women of reproductive age. The typical tubal ligation patient is over age 30, is married, and has had two or three children.

Precautions

Tubal ligation should be postponed if the woman is unsure about her decision. While it is sometimes reversible, the procedure should be considered permanent and irreversible. Up to 10% of sterilized women regret having had the surgery, and about 1% seek treatment in attempts to restore fertility.

Description

Tubal ligation, or getting one's "tubes tied," refers to female sterilization, the surgery that ends a woman's ability to conceive. The operation is performed on the patient's Fallopian tubes. These tubes, which are about 10 cm long and 0.5 cm in diameter, are found on the upper outer sides of the uterus, and open into the uterus through small channels. It is within the Fallopian tube that fertilization, the joining of the egg and the sperm, takes place. During tubal ligation, the tubes are cut or blocked in order to close off the sperm's access to the egg.

Normally, tubal ligation takes about 20–30 minutes, and is performed under **general anesthesia**, spinal anesthesia, or **local anesthesia** with **sedation**. The surgery can be performed on either hospitalized patients within 24 hours after **childbirth** or on outpatients. The woman can usually leave the hospital the same day.

The most common surgical approaches to tubal ligation include **laparoscopy** and mini-laparotomy. In a laparoscopic tubal ligation, a long, thin telescope-like surgical instrument called a laparoscope is inserted into the pelvis through a small cut about 1 cm long near the navel. Carbon dioxide gas is pumped in to help move the abdominal wall to give the surgeon easier

access to the tubes. Often the surgical instruments are inserted through a second incision near the pubic-hair line. An instrument may be placed through the vagina to hold the uterus in place.

In a mini-laparotomy, a 3–4 cm incision is made just above the pubic bone or under the navel. A larger incision, or laparotomy, is rarely used today. Tubal ligation can also be performed at the time of a **cesarean section**.

Tubal ligation costs about \$2,000 when performed by a private physician, but is less expensive when performed at a family planning clinic. Most insurance plans cover treatment costs.

Tubal ligation is performed in several ways:

- **Electrocoagulation.** A heated needle connected to an electrical device is used to cauterize or burn the tubes. Electrocoagulation is the most common method of tubal ligation.
- **Falope ring.** In this technique, an applicator is inserted through an incision above the bladder and a plastic ring is placed around a loop of the tube.
- **Hulka clip.** The surgeon places a plastic clip across a tube held in place by a steel spring.
- **Silicone rubber bands.** A band placed over a tube forms a mechanical block to sperm.

Preparation

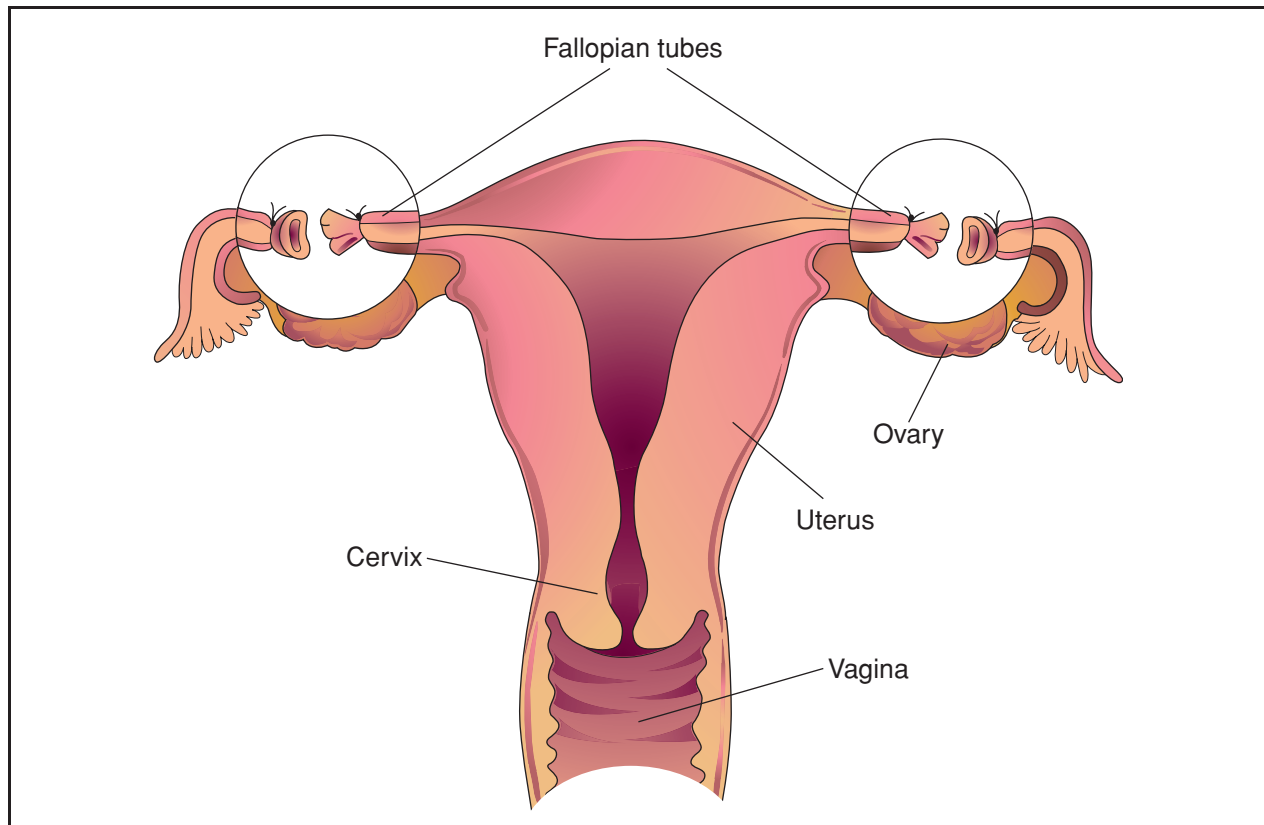
Preparation for tubal ligation includes patient education and counseling. Before surgery, it is important that the woman understand the permanent nature of tubal ligation, and the risks of anesthesia and surgery. Her medical history is reviewed, and a **physical examination** and laboratory testing are performed. The patient is not allowed to eat or drink for several hours before surgery.

Aftercare

After surgery, the patient is monitored for several hours before she is allowed to go home. She is instructed on care of the surgical wound, and what signs to watch for, such as **fever**, **nausea**, **vomiting**, faintness, or **pain**. These signs could indicate that complications have occurred.

Risks

While major complications are uncommon after tubal ligation, there are risks with any surgical procedure. Possible side effects include infection and bleeding. Rarely, **death** may occur as a complication of general anesthesia if a major blood vessel is cut. The



Tubal ligation is a permanent form of contraception in which a woman's Fallopian tubes are surgically cut, cauterized, tied, or blocked to prevent pregnancy. This procedure blocks the pathway sperm takes to fertilize an egg. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Contraception—The prevention of the union of the male's sperm with the female's egg.

Ectopic pregnancy—The implantation of a fertilized egg in a Fallopian tube instead of the uterus.

Electrocoagulation—The coagulation or destruction of tissue through the application of a high-frequency electrical current.

Female sterilization—The process of permanently ending a woman's ability to conceive by tying off or cutting apart the Fallopian tubes.

Laparoscopy—Abdominal surgery performed through a laparoscope, which is a thin telescopic instrument inserted through an incision near the navel.

death rate following tubal ligation is about four per 100,000 sterilizations.

After laparoscopy, the patient may experience pain in the shoulder area from the carbon dioxide used during surgery, but the technique is associated with less pain than mini-laparotomy, as well as a faster recovery period. Mini-laparotomy results in a higher incidence of pain, bleeding, bladder injury, and infection compared with laparoscopy. Patients normally feel better after three or four days of rest, and are able to resume sexual activity at that time.

Following tubal ligation, there is a low risk (less than 1%) of **ectopic pregnancy**. Ectopic pregnancy is a condition in which the fertilized egg implants in a place other than the uterus, usually in one of the Fallopian tubes. Ectopic pregnancies are more likely to happen in younger women, and in women whose tubes were ligated by electrocoagulation.

Normal results

After having her tubes ligated, a woman does not need to use any form of birth control to avoid pregnancy. Tubal ligation is almost 100% effective for the prevention of conception. The possibility for treatment failure is very low—fewer than one in 200 women (0.4%) will become pregnant during the first year after sterilization. Failure can happen if the cut ends of the tubes grow back together; if the tube was not completely cut or blocked off; if a plastic clip or rubber band is loose or comes off; or if the woman was already pregnant at the time of surgery.

Resources

ORGANIZATIONS

American College of Obstetricians and Gynecologists. 409
12th Street, S.W., P.O. Box 96920.

Planned Parenthood Federation of America, Inc. 810
Seventh Ave., New York, NY, 10019. (800) 669-0156.
< <http://www.plannedparenthood.org> >.

Mercedes McLaughlin

Tube compression of the esophagus and stomach

Definition

Tube compression of the esophagus and stomach is an emergency procedure used to stop bleeding from the upper digestive tract.

Purpose

Vomiting blood is both frightening and life-threatening. Among its causes are:

- bleeding from the nose and throat
- peptic ulcers
- stomach **cancer**
- esophageal cancer
- a tear in the esophagus caused by violent vomiting (**Mallory-Weiss syndrome**)
- breaking of blood vessels in the esophagus.

The most profuse bleeding comes from veins in the lower esophagus, just above the stomach, that have dilated to enormous dimensions as the result of **liver disease**. When the liver shrinks due to **cirrhosis** (scarring from chronic disease), its blood vessels shrink, forcing

blood from the intestines to find alternate routes back to the heart. The blood usually flows through tiny veins in the esophagus located just beneath the passageway where food passes downward and vomitus passes upward. Major causes for this rearrangement are alcoholic liver disease, chronic hepatitis, and **cholangitis**. Called esophageal varices, the affected veins can be easily damaged and bleed voraciously.

Description

One emergency method of stopping bleeding from esophageal varices is to tamponade it with a balloon. The Sengstaken-Blakemore tube is a complex rubber device with two balloons and three channels—one channel for each balloon and one that goes all the way through. The Minnesota tube has four channels, an extra one that opens above the first balloon. The bottom balloon is round; the upper balloon is long and narrow. The tube is passed through the nose or mouth into the stomach, where the bottom balloon is inflated. Then the tube is pulled back until the bottom balloon comes up against the narrow valve at the top of the stomach, when it can go no further. At this point, the upper balloon is inflated, putting pressure on a length of esophagus where the bleeding veins are located. The tube is then fixed so it cannot be dislodged. The third channel in the tube is used to aspirate (suck out) stomach contents to see if the bleeding has stopped. The fourth channel aspirates from the esophagus.

These tubes are a temporary measure. They stabilize the patient until bleeding has stopped, blood transfusions are received, and permanent repair is imminent.

Since the lower balloon effectively separates the esophagus from the stomach, it is possible to determine more accurately where the bleeding is located when it is in place.

This method of treating upper intestinal bleeding is being replaced by procedures that use a gastroscope, a flexible device that permits viewing and operating without surgery.

Preparation

The procedure is explained to the patient and family. A sedative may be given to prepare the patient for the procedure.

Aftercare

With the tube in place, the patient cannot eat and may have some difficulty breathing. The patient will be hospitalized until the tube can be removed.

KEY TERMS

Cholangitis—Inflammation of the system of tubes that drains bile from the liver into the intestines.

Chronic hepatitis—Long lasting inflammation of the liver due to viruses or other causes.

Peptic ulcers—Wounds in the stomach and duodenum caused by stomach acid and the bacterium *Helicobacter pylori*.

Risks

Major complications frequently occur, and **death** results about 3% of the time. Problems include damage to the esophagus and stomach and interference with the airway. Should the tube remain in place too long, there is danger of the pressure eroding the esophagus or the nose.

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J. Ricker Polsdorfer, MD

Tube feedings

Definition

Nutrients, either a special liquid formula or pureed food, are delivered to a patient through a tube directly into the gastrointestinal tract, usually into the stomach or small intestine.

Purpose

Tube feeding provides **nutrition** to patients who are unable or unwilling to eat food. Conditions where tube feeding is considered include **protein-energy malnutrition**, liver or kidney failure, **coma**, or in patients who cannot chew or swallow (dysphagia) due to stroke, **brain tumor**, or **head injury**. Patients who are receiving radiation therapy or **chemotherapy** treatments for **cancer** may also be candidates for tube feedings.

Precautions

Certain medications may interact with some formulas to inactivate the nutrients or change the way that the drug is absorbed.

Description

A flexible, narrow tube is inserted into some portion of the digestive tract and liquid formulas or liquefied foods are placed into the tube to meet the patient's nutritional needs. The feeding may be pumped into the tube or allowed to drip into the tube continuously or at scheduled feeding times.

A feeding tube can be inserted by a surgical or nonsurgical procedure in several positions along the gastrointestinal tract. The tube may be inserted into the nose and passed down the throat and through the esophagus. A nasogastric tube is inserted through the nose with the end of the tube reaching into the stomach. A nasoduodenal or nasojejunal tube is inserted through the nose and ends in either the duodenum or jejunum, both of which are portions of the small intestine. This type of tube placement is usually used for short-term feeding. Surgical placement of a feeding tube may be done if there will be a long-term need for feeding that bypasses the upper digestive tract. An esophagostomy creates an opening in the esophagus, a gastrostomy creates an opening into the stomach, and a jejunostomy creates an opening into the jejunum. The feeding tube is then inserted through the surgically created opening.

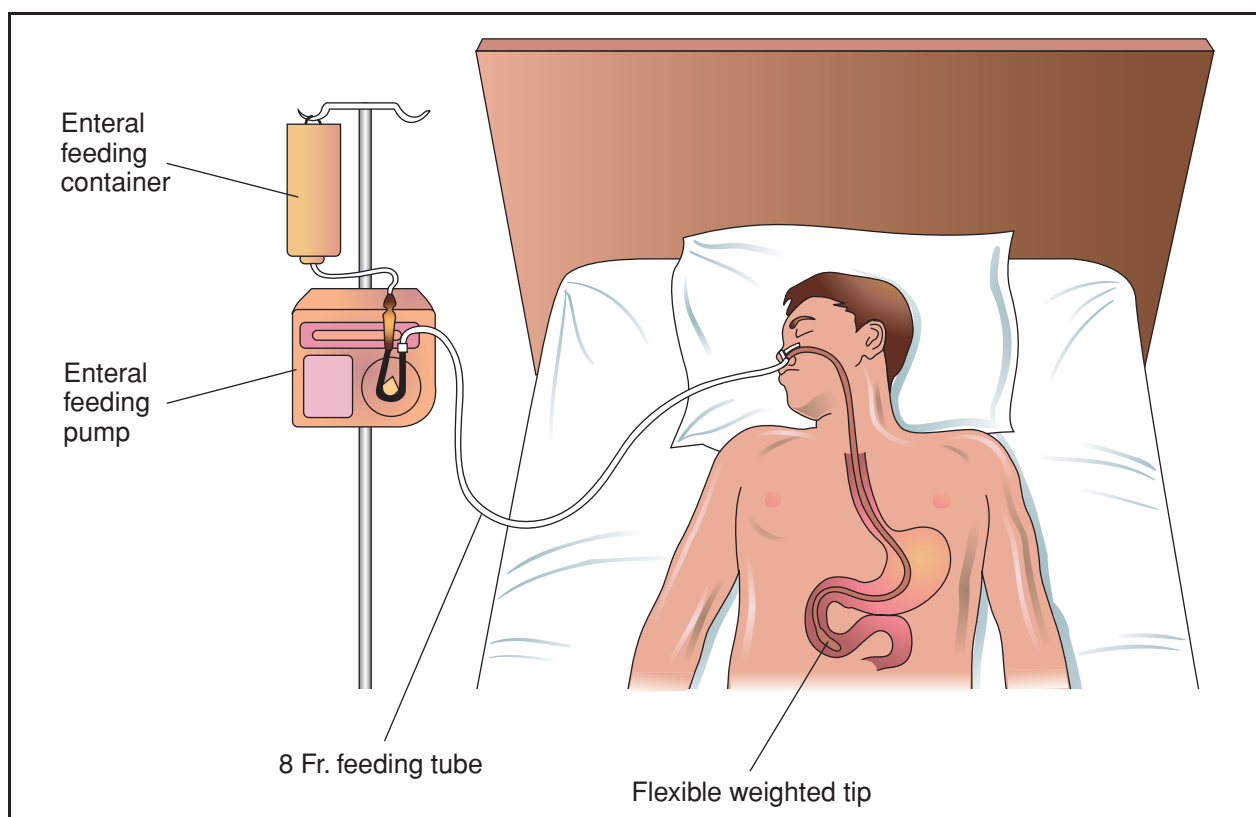
Tube feedings can be a mixture of regular foods that are blended with liquid to make a consistency that will pass through the tube. Nutritionally balanced liquid products are often more convenient to use and ensure a balance of proteins, fats, and carbohydrates along with **vitamins** and **minerals**. Specialized formulas are also available to meet almost any nutritional need. For example, patients with severe burns, protein-energy malnutrition, or slow wound healing may require formulas that are higher in protein. Patients with renal failure may require low-protein formulas with lower concentrations of minerals and vitamins.

Preparation

The reasons that tube feeding is necessary are discussed with the patient, as is the length of time that the feeding tube is expected to be in place. The specific procedure is also explained to the patient.

Aftercare

Patients with **ostomy** feeding tubes may have the tube positioned level with the surrounding skin. A cap



A feeding tube can be inserted by a surgical or nonsurgical procedure in several positions along the gastrointestinal tract to provide nutrition to patients who are unable or unwilling to eat food. The feeding may be pumped into the tube or allowed to drip into the tube continuously, or at scheduled feeding times. The illustration above features a nasojeunal tube which is inserted through the nose and ends in either the duodenum or jejunum. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Duodenum—The upper portion of the small intestine. It is approximately 10 in (25 cm) long and extends from the stomach to the jejunum.

Jejunum—The middle portion of the small intestine. It is approximately 8 ft (2.5 m) long and extends from the jejunum to the ileum.

or button can be placed over the opening so that it can be more comfortably concealed under clothing. The opening and surrounding tissue need to be cleaned and inspected regularly to prevent infection. For patients with a tube inserted through the nose, daily nasal hygiene is important and the mouth and lips should be kept moist. Good mouth care is necessary for any patient with a feeding tube.

Risks

Formula from the tube can back up in the esophagus and be breathed into the trachea and lungs, causing aspiration **pneumonia**. The placement of the tube should be checked frequently and the head of the bed elevated during and after feeding to prevent the solution from moving back up the digestive tract. Feeding tubes can also become clogged and should be flushed regularly with water. If the feeding formula is too concentrated or given too fast, the patient may experience **nausea**, **vomiting**, cramping, and bloating. The feeding may need to be diluted with liquid or the rate at which it is given decreased. **Diarrhea** or **constipation** can occur if the feeding is not the right composition or does not provide enough liquid. The tube itself can irritate the nasal passage, esophagus, or surrounding tissues.

Normal results

A patient may be able to return to a normal diet of solid foods after short-term supplementation with

formula through a feeding tube. In cases where long-term nutritional therapy is required, all of the patient's nutritional needs will have to be provided by the formula. The balance of fluids, calories, proteins, fats, vitamins, and minerals may need to be adjusted periodically.

Abnormal results

If formula feedings are not tolerated by the patient or are inadequate to meet his or her nutritional needs, the patient may need to receive nutrition through an intravenous line (parenteral nutrition). This type of therapy involves delivery of sterile nutrient solutions directly into the bloodstream through a needle inserted into a vein.

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Altha Roberts Edgren

Tuberculin skin test

Definition

Tuberculosis (TB) is an airborne infectious disease caused by the bacteria *Mycobacterium tuberculosis*. Besides culturing in the laboratory, the two most common types of tests to screen for exposure to this disease are the Mantoux PPD tuberculin skin test, which is generally considered more reliable, and the older TB tine test, which is now rarely used. These tests are designed to help identify individuals who may have been infected by the tuberculosis bacteria. A diagnosis of active, infectious tuberculosis is never made solely based on the results of a TB skin test, but requires further testing, including a sputum culture and a chest x-ray.

Purpose

Because TB is spread through the air, especially in poorly ventilated areas, it is more commonly found among people living in crowded conditions, such as jails, nursing homes, and homeless shelters. Often, a TB skin test will be given as part of a physical examination when a person is hiring a new employee,

particularly for those individuals seeking employment in the health care or food service professions.

People can be exposed to or infected with TB without showing any symptoms or necessarily developing the disease. Individuals with normally functioning immune systems generally prevent the spread of the bacteria by "walling off" or encysting the bacteria within the body. To be at risk for infection a person must have or had close contact with someone who has active tuberculosis (such as a friend or family member). Persons who are more at risk for developing the TB infection overtly include those with a weakened immune system (immunocompromised), either from a chronic disease, such as HIV infection; or as a result of a tissue or organ transplant or other medical treatment designed to suppress the immune system. In addition, persons who are heavy users of crack **cocaine** or alcohol are more susceptible to TB than those who do not **abuse** these substances. Symptoms of tuberculosis include a persistent **cough**, **fever**, weight loss, night sweats, **fatigue**, and loss of appetite.

Precautions

Although the test is generally considered safe, it is important to inform the person conducting the test if the patient may be pregnant, has had a positive TB test in the past, or has had tuberculosis in the past. People who have had a positive TB test in the past will probably always have a positive test and should not be tested again.

There are several situations when the TB test results might not be accurate. These include situations involving people who:

- have had vaccinations (such as those for measles, polio, **rubella** or **mumps**) within the last four weeks
- are taking steroids
- have severe malnutrition

Description

TB skin tests are usually given at a clinic, hospital, or doctor's office. Sometimes the tests are given at schools or workplaces and may be a pre-employment requirement. Many cities provide free TB skin tests and followup care. The Mantoux PPD tuberculin skin test involves injecting a very small amount of a substance called PPD tuberculin just under the top layer of the skin (intracutaneously). Tuberculin is a mixture of antigens obtained from the culture of *M. tuberculosis*. Antigens are foreign particles or proteins that stimulate the immune system to produce antibodies.

Two different tuberculin preparations are available, Old Tuberculin (OT) and Purified Protein Derivative (PPD). The latter is the preferred testing substance. The test is usually given on the inside of the forearm about halfway between the wrist and the elbow, where a small bubble will form as the tuberculin is injected. The skin test takes just a minute to administer.

After 48–72 hours, the test site will be examined by a trained person for evidence of swelling. People who have been exposed to tuberculosis will develop an immune response, causing a slight swelling at the injection site. If there is a lump or swelling, the health care provider will use a ruler to measure the size of the reaction. Some public health physicians recommend using a 72-hour waiting period as a general practice on the grounds that a 48-hour waiting period yields a higher percentage of false negative test results.

The other method of TB skin test is called the multiple puncture test or tine test because the small test instrument has several small tines that lightly prick the skin. The small points of the instrument are either coated with dried tuberculin or are used to puncture through a film of liquid tuberculin. The test is read by measuring the size of the largest papule. Because it is not possible to precisely control the amount of tuberculin used in the tine test, a positive test should be verified using the Mantoux test. For this reason, the tine test is not as widely used as the Mantoux test and is considered to be less reliable; as of 2000, it is no longer recommended for general use.

Preparation

There is no special preparation needed before a TB skin test. A brief personal history will be taken to determine whether the person has had tuberculosis or a TB test before, has been in close contact with anyone with TB, or has any significant risk factors. Directly before the test, the skin on the arm at the injection site is usually cleaned with an alcohol swab and allowed to air dry.

Aftercare

After having a TB skin test, it is extremely important to make sure that the patient keeps the appointment to have the test reaction read. The patient is instructed to keep the test site clean, uncovered, and to not scratch or rub the area. Should severe swelling, **itching**, or **pain** occur, or if the patient has trouble breathing, the clinic or health care provider should be contacted immediately.

Risks

The risk of an adverse reaction is very low. Occasionally, an individual who has been exposed to the TB bacteria will develop a large reaction in which the arm swells and is uncomfortable. This reaction should disappear in two weeks. A sore might develop where the injection was given, or a fever could occur, but these are extremely rare reactions.

It is possible that a person who has TB may receive a negative test result (called a “false negative”) or a person who does not have TB may receive a positive test result (called a “false positive”). If there is some doubt, the test may be repeated or the person may be given a diagnostic test using a **chest x ray** and/or sputum sample culture test to determine whether the disease is present and/or active in the lungs.

Normal results

In people who have not been exposed to TB, there will be little or no swelling at the test site after 48–72 hours. This is a negative test result. Negative test results can be interpreted to mean that the person has not been infected with the tuberculosis bacteria or that the person has been infected recently and not enough time has elapsed for the body to react to the skin test. Persons become sensitive between two and ten weeks after the initial infection. As a result, if the person has been in contact with someone with tuberculosis, the test should be repeated in three months. Also, because it may take longer than 72 hours for an elderly individual to develop a reaction, it may be useful to repeat the TB skin test after one week to adequately screen these individuals. Immunocompromised persons may be unable to react sufficiently to the Mantoux test, and either a chest x ray or sputum sample may be required.

A newer test that appears to be preferable to the tuberculin skin test in evaluating patients who are HIV-positive is the enzyme-linked immunospot (ELISPOT) assay. A group of researchers in the United Kingdom found that the ELISPOT assay was more accurate than the PPD test in detecting active as well as latent tuberculosis in HIV-positive patients.

Abnormal results

A reaction of 5 mm of induration (swelling) is considered positive for the following groups:

- household contacts of persons with active tuberculosis
- AIDS patients
- persons with old healed tuberculosis on chest x ray

KEY TERMS

Antibody—A specific protein produced by the immune system in response to a specific foreign protein or particle called an antigen.

Antigen—Any foreign particle or protein that causes an immune response.

Attenuated—Alive but weakened; an attenuated microorganism can no longer produce disease.

Cross-reaction—A positive reaction that occurs as a result of a person's exposure to other non-tuberculosis bacteria.

Immunocompromised—A state in which the immune system is suppressed or not functioning properly.

Induration—An abnormally hard spot or area on the skin. The tuberculin skin test produces an induration at the test site in persons who have been exposed to TB.

Intracutaneous—Into the skin, in this case directly under the top layer of skin.

Mantoux or PPD test—Other names for a tuberculin skin test. PPD stands for purified protein derivative.

Tuberculin—A mixture of antigens obtained from the cultured bacteria that cause tuberculosis, *Mycobacterium tuberculosis*.

- organ transplant recipients.
- persons receiving immunosuppressive medications

A reaction of 10 mm of induration is considered positive in individuals with one or more of the following risk factors which are either reasons to have a higher exposure to TB and/or a condition that increases the risk for progression to active TB:

- foreign-born immigrants from Asia, Africa, or Latin America
- injection drug users and persons who abuse alcohol
- residents and employees of such high-risk congregate settings as hospitals, homeless shelters, and jails
- medically under-served low income populations
- TB laboratory personnel
- children younger than four years of age or infants, children or adolescents exposed to adults in high risk categories
- residents of long-term care facilities

- individuals with certain medical conditions that increase the risk of developing tuberculosis; these medical conditions include being 10% or more below ideal body weight, silicosis, chronic renal failure, diabetes mellitus, high dose corticosteroid or other immunosuppressive therapy, some blood disorders like leukemia and lymphomas, and other cancer

Finally, a reaction of 15 mm of induration or greater is considered positive in those with no risk factors and are therefore at the lowest risk of developing TB.

A TB skin conversion is defined as an increase of 10 mm or greater of induration within a two year period, regardless of age.

A positive reaction to tuberculin may be the result of a previous natural infection with *M. tuberculosis*, infection with a variety of non-tuberculosis mycobacteria (cross-reaction), or tuberculosis **vaccination** with a live, but weakened (attenuated) mycobacterial strain. TB vaccination is not done in the US. Cross-reactions are positive reactions that occur as a result of a person's exposure to other non-tuberculosis bacteria. These tend to be smaller than those caused by *M. tuberculosis*. There is no reliable way of distinguishing whether a positive TB skin test is due to a previous vaccination against tuberculosis. Generally, however, positive results are not due to vaccination exposure because reactions in vaccinated people tend to be less than 10 mm, and an individual's sensitivity to tuberculin steadily declines after vaccination. If the skin test is interpreted as positive, a chest x ray will be performed to determine whether the person has active tuberculosis or whether the body has sufficiently handled the infection.

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ORGANIZATIONS

- American Lung Association. (800) LUNG-USA.
- Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.
- National Heart, Lung, and Blood Institute (NHLBI). P. O. Box 30105, Bethesda, MD 20824-0105. (301) 592-8573. <www.nhlbi.nih.gov>.
- National Tuberculosis Center. University of Medicine and Dentistry of New Jersey. Executive Office, Suite GB1, 65 Bergen Street, Newark, NJ 07107-3268. (800) 4TB-DOCS. <<http://www.umdj.edu/~ntbcweb/ntbchome.htm>>.

OTHER

- American Academy of Family Physicians. "Positive Skin Tests for Tuberculosis." <<http://www.aafp.org/healthinfo>>.
- "Diagnostic Standards and Classification of Tuberculosis." <<http://aepo-xdv-www.epo.cdc.gov/wonder/prevguid/p0000425/body006.htm>>.
- U.S. Department of Health and Human Services Public Health Service, Centers for Disease Control and Prevention. "Questions and Answers About TB." <<http://www.cdc.gov/nchstp/tb/faq.htm>>.

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Tuberculosis

Definition

Tuberculosis (TB) is a potentially fatal contagious disease that can affect almost any part of the body but is mainly an infection of the lungs. It is caused by a bacterial microorganism, the tubercle bacillus or *Mycobacterium tuberculosis*. Although TB can be treated, cured, and can be prevented if persons at risk take certain drugs, scientists have never come close to wiping it out. Few diseases have caused so much distressing illness for centuries and claimed so many lives.

Description

Overview

Tuberculosis was popularly known as consumption for a long time. Scientists know it as an infection caused by *M. tuberculosis*. In 1882, the microbiologist Robert Koch discovered the tubercle bacillus, at a time when one of every seven deaths in Europe was caused by TB. Because **antibiotics** were unknown, the only means of controlling the spread of infection was to isolate patients in private sanatoria or hospitals limited to patients with TB—a practice that continues to this day in many countries. The net effect of this pattern of treatment was to separate the study of tuberculosis from mainstream medicine. Entire organizations were set up to study not only the disease as it affected individual patients, but its impact on the society as a whole. At the turn of the twentieth century more than 80% of the population in the United States were infected before age 20, and tuberculosis was the single most common cause of **death**. By 1938 there were more than 700 TB hospitals in this country.

Tuberculosis spread much more widely in Europe when the industrial revolution began in the late nineteenth century. The disease became widespread somewhat later in the United States, because the movement of the population to large cities made overcrowded housing so common. When streptomycin, the first antibiotic effective against *M. tuberculosis*, was discovered in the early 1940s, the infection began to come under control. Although other more effective anti-tuberculosis drugs were developed in the following decades, the number of cases of TB in the United States began to rise again in the mid-1980s. This upsurge was in part again a result of overcrowding and unsanitary conditions in the poor areas of large cities, prisons, and homeless shelters. Infected visitors and immigrants to the United States have also contributed to the resurgence of TB. An additional factor is the **AIDS** epidemic. AIDS patients are much more likely to develop tuberculosis because of their weakened immune systems. There still are an estimated 8–10 million new cases of TB each year worldwide, causing roughly 3 million deaths.

High-risk populations

THE ELDERLY. Tuberculosis is more common in elderly persons. More than one-fourth of the nearly 23,000 cases of TB reported in the United States in 1995 developed in people above age 65. Many elderly patients developed the infection some years ago when

FLORENCE B. SEIBERT (1897–1991)



(Library of Congress.)

Florence Barbara Seibert was born on October 6, 1897, in Easton, Pennsylvania, the second of three children. She was the daughter of George Peter Seibert, a rug manufacturer and merchant, and Barbara (Memmert) Seibert. At the age of three she contracted polio. Despite her resultant handicaps, she completed high school, with the help of her highly supportive parents, and entered Goucher College

in Baltimore, where she studied chemistry and zoology. She graduated in 1918, then worked under the direction of one of her chemistry teachers, Jessie E. Minor, at the Chemistry Laboratory of the Hammersley Paper Mill in Garfield, New Jersey. She and her professor, having responded to the call for women to fill positions vacated by men fighting in World War I, coauthored scientific papers on the chemistry of cellulose and wood pulps.

A biochemist who received her Ph.D. from Yale University in 1923, Florence B. Seibert is best known for her research in the biochemistry of tuberculosis. She developed the protein substance used for the tuberculosis skin test. The substance was adopted as the standard in 1941 by the United States and a year later by the World Health Organization. In addition, in the early 1920s, Seibert discovered that the sudden fevers that sometimes occurred during intravenous injections were caused by bacteria in the distilled water that was used to make the protein solutions. She invented a distillation apparatus that prevented contamination. This research had great practical significance later when intravenous blood transfusions became widely used in surgery. Seibert authored or coauthored more than a hundred scientific papers. Her later research involved the study of bacteria associated with certain cancers. Her many honors include five honorary degrees, induction into the National Women's Hall of Fame in Seneca Falls, New York (1990), the Garvan Gold Medal of the American Chemical Society (1942), and the John Elliot Memorial Award of the American Association of Blood Banks (1962).

the disease was more widespread. There are additional reasons for the vulnerability of older people: those living in nursing homes and similar facilities are in close contact with others who may be infected. The **aging** process itself may weaken the body's immune system, which is then less able to ward off the tubercle bacillus. Finally, bacteria that have lain dormant for some time in elderly persons may be reactivated and cause illness.

RACIAL AND ETHNIC GROUPS. TB also is more common in blacks, who are more likely to live under conditions that promote infection. At the beginning of the new millennium, two-thirds of all cases of TB in the United States affect African Americans, Hispanics, Asians, and persons from the Pacific Islands. Another one-fourth of cases affect persons born outside the United States. As of 2002, the risk of TB is still increasing in all these groups.

As of late 2002, TB is a major health problem in certain specific immigrant communities, such as the

Vietnamese in southern California. One team of public health experts in North Carolina maintains that treatment for tuberculosis is the most pressing health care need of recent immigrants to the United States. In some cases, the vulnerability of immigrants to tuberculosis is increased by occupational exposure, as a recent outbreak of TB among Mexican poultry farm workers in Delaware indicates. Other public health experts are recommending tuberculosis screening at the primary care level of all new immigrants and refugees.

LIFESTYLE FACTORS. The high risk of TB in AIDS patients extends to those infected by human **immuno-deficiency** virus (HIV) who have not yet developed clinical signs of AIDS. Alcoholics and intravenous drug abusers are also at increased risk of contracting tuberculosis. Until the economic and social factors that influence the spread of tubercular infection are remedied, there is no real possibility of completely eliminating the disease.

Causes and symptoms

Transmission

Tuberculosis spreads by droplet infection. This type of transmission means that when a TB patient exhales, coughs, or sneezes, tiny droplets of fluid containing tubercle bacilli are released into the air. This mist, or aerosol as it is often called, can be taken into the nasal passages and lungs of a susceptible person nearby. Tuberculosis is not, however, highly contagious compared to some other infectious diseases. Only about one in three close contacts of a TB patient, and fewer than 15% of more remote contacts, are likely to become infected. As a rule, close, frequent, or prolonged contact is needed to spread the disease. Of course, if a severely infected patient emits huge numbers of bacilli, the chance of transmitting infection is much greater. Unlike many other infections, TB is not passed on by contact with a patient's clothing, bed linens, or dishes and cooking utensils. The most important exception is **pregnancy**. The fetus of an infected mother may contract TB by inhaling or swallowing the bacilli in the amniotic fluid.

Progression

Once inhaled, tubercle bacilli may reach the small breathing sacs in the lungs (the alveoli), where they are taken up by cells called macrophages. The bacilli multiply within these cells and then spread through the lymph vessels to nearby lymph nodes. Sometimes the bacilli move through blood vessels to distant organs. At this point they may either remain alive but inactive (quiescent), or they may cause active disease. Actual tissue damage is not caused directly by the tubercle bacillus, but by the reaction of the person's tissues to its presence. In a matter of weeks the host develops an immune response to the bacillus. Cells attack the bacilli, permit the initial damage to heal, and prevent future disease permanently.

Infection does not always mean disease; in fact, it usually does not. At least nine of ten patients who harbor *M. tuberculosis* do not develop symptoms or physical evidence of active disease, and their x-rays remain negative. They are not contagious; however, they do form a pool of infected patients who may get sick at a later date and then pass on TB to others. It is thought that more than 90% of cases of active tuberculosis come from this pool. In the United States this group numbers 10–15 million persons. Whether or not a particular infected person will become ill is impossible to predict with certainty. An estimated 5% of infected persons get sick within 12–24 months of

being infected. Another 5% heal initially but, after years or decades, develop active tuberculosis either in the lungs or elsewhere in the body. This form of the disease is called reactivation TB, or post-primary disease. On rare occasions a previously infected person gets sick again after a later exposure to the tubercle bacillus.

Pulmonary tuberculosis

Pulmonary tuberculosis is TB that affects the lungs. Its initial symptoms are easily confused with those of other diseases. An infected person may at first feel vaguely unwell or develop a **cough** blamed on **smoking** or a cold. A small amount of greenish or yellow sputum may be coughed up when the person gets up in the morning. In time, more sputum is produced that is streaked with blood. Persons with pulmonary TB do not run a high **fever**, but they often have a low-grade one. They may wake up in the night drenched with cold sweat when the fever breaks. The patient often loses interest in food and may lose weight. Chest **pain** is sometimes present. If the infection allows air to escape from the lungs into the chest cavity (**pneumothorax**) or if fluid collects in the pleural space (**pleural effusion**), the patient may have difficulty breathing. If a young adult develops a pleural effusion, the chance of tubercular infection being the cause is very high. The TB bacilli may travel from the lungs to lymph nodes in the sides and back of the neck. Infection in these areas can break through the skin and discharge pus. Before the development of effective antibiotics, many patients became chronically ill with increasingly severe lung symptoms. They lost a great deal of weight and developed a wasted appearance. This outcome is uncommon today—at least where modern treatment methods are available.

Extrapulmonary tuberculosis

Although the lungs are the major site of damage caused by tuberculosis, many other organs and tissues in the body may be affected. The usual progression is for the disease to spread from the lungs to locations outside the lungs (extrapulmonary sites). In some cases, however, the first sign of disease appears outside the lungs. The many tissues or organs that tuberculosis may affect include:

- **Bones.** TB is particularly likely to attack the spine and the ends of the long bones. Children are especially prone to spinal tuberculosis. If not treated, the spinal segments (vertebrae) may collapse and cause **paralysis** in one or both legs.

- **Kidneys.** Along with the bones, the kidneys are probably the commonest site of extrapulmonary TB. There may, however, be few symptoms even though part of a kidney is destroyed. TB may spread to the bladder. In men, it may spread to the prostate gland and nearby structures.
- **Female reproductive organs.** The ovaries in women may be infected; TB can spread from them to the peritoneum, which is the membrane lining the abdominal cavity.
- **Abdominal cavity.** Tuberculous **peritonitis** may cause pain ranging from the vague discomfort of stomach cramps to intense pain that may mimic the symptoms of appendicitis.
- **Joints.** Tubercular infection of joints causes a form of arthritis that most often affects the hips and knees. The wrist, hand, and elbow joints also may become painful and inflamed.
- **Meninges.** The meninges are tissues that cover the brain and the spinal cord. Infection of the meninges by the TB bacillus causes tuberculous **meningitis**, a condition that is most common in young children but is especially dangerous in the elderly. Patients develop headaches, become drowsy, and eventually comatose. Permanent brain damage is the rule unless prompt treatment is given. Some patients with tuberculous meningitis develop a tumor-like brain mass called a tuberculoma that can cause stroke-like symptoms.
- **Skin, intestines, adrenal glands, and blood vessels.** All these parts of the body can be infected by *M. tuberculosis*. Infection of the wall of the body's main artery (the aorta), can cause it to rupture with catastrophic results. Tuberculous **pericarditis** occurs when the membrane surrounding the heart (the pericardium) is infected and fills up with fluid that interferes with the heart's ability to pump blood.
- **Miliary tuberculosis.** Miliary TB is a life-threatening condition that occurs when large numbers of tubercle bacilli spread throughout the body. Huge numbers of tiny tubercular lesions develop that cause marked weakness and weight loss, severe anemia, and gradual wasting of the body.

Diseases similar to tuberculosis

There are many forms of mycobacteria other than *M. tuberculosis*, the tubercle bacillus. Some cause infections that may closely resemble tuberculosis, but they usually do so only when an infected person's immune system is defective. People who are HIV-positive are a prime example. The most common

mycobacteria that infect AIDS patients are a group known as *Mycobacterium avium* complex (MAC). People infected by MAC are not contagious, but they may develop a serious lung infection that is highly resistant to antibiotics. MAC infections typically start with the patient coughing up mucus. The infection progresses slowly, but eventually blood is brought up and the patient has trouble breathing. In AIDS patients, MAC disease can spread throughout the body, with anemia, **diarrhea**, and stomach pain as common features. Often these patients die unless their immune system can be strengthened. Other mycobacteria grow in swimming pools and may cause skin infection. Some of them infect **wounds** and artificial body parts such as a breast implant or mechanical heart valve.

Diagnosis

The diagnosis of TB is made on the basis of laboratory test results. The standard test for tuberculosis—which is the so-called tuberculin skin test—detects the presence of infection, not of active TB. Tuberculin is an extract prepared from cultures of *M. tuberculosis*. It contains substances belonging to the bacillus (antigens) to which an infected person has been sensitized. When tuberculin is injected into the skin of an infected person, the area around the injection becomes hard, swollen, and red within one to three days. Today skin tests utilize a substance called purified protein derivative (PPD) that has a standard chemical composition and is therefore is a good measure of the presence of tubercular infection. The PPD test is also called the Mantoux test. The Mantoux PPD skin test is not, however, 100% accurate; it can produce false positive as well as false negative results. What these terms mean is that some people who have a skin reaction are not infected (false positive) and that some who do not react are in fact infected (false negative). The PPD test is, however, useful as a screener. Anyone who has suspicious findings on a **chest x ray**, or any condition that makes TB more likely should have a PPD test. In addition, those in close contact with a TB patient and persons who come from a country where TB is common also should be tested, as should all healthcare personnel and those living in crowded conditions or institutions.

Because the symptoms of TB cover a wide range of severity and affected body parts, diagnosis on the basis of external symptoms is not always possible. Often, the first indication of TB is an abnormal chest x-ray or other test result rather than physical discomfort. On a chest x ray, evidence of the disease appears as numerous white, irregular areas against a dark

background, or as enlarged lymph nodes. The upper parts of the lungs are most often affected. A PPD test is always done to show whether the patient has been infected by the tubercle bacillus. To verify the test results, the physician obtains a sample of sputum or a tissue sample (biopsy) for culture. Three to five sputum samples should be taken early in the morning. If necessary, sputum for culture can be produced by spraying salt solution into the windpipe. Culturing *M. tuberculosis* is useful for diagnosis because the bacillus has certain distinctive characteristics. Unlike many other types of bacteria, mycobacteria can retain certain dyes even when exposed to acid. This so-called acid-fast property is characteristic of the tubercle bacillus.

Body fluids other than sputum can be used for culture. If TB has invaded the brain or spinal cord, culturing a sample of spinal fluid will make the diagnosis. If TB of the kidneys is suspected because of pus or blood in the urine, culture of the urine may reveal tubercular infection. Infection of the ovaries in women can be detected by placing a tube having a light on its end (a laparoscope) into the area. Samples also may be taken from the liver or bone marrow to detect the tubercle bacillus.

One important new advance in the diagnosis of TB is the use of molecular techniques to speed the diagnostic process as well as improve its accuracy. As of late 2005, four molecular techniques are increasingly used in laboratories around the world. They include polymerase chain reaction to detect mycobacterial DNA in patient specimens; nucleic acid probes to identify mycobacteria in culture; restriction fragment length polymorphism analysis to compare different strains of TB for epidemiological studies; and genetic-based susceptibility testing to identify drug-resistant strains of mycobacteria.

Treatment

Supportive care

In the past, treatment of TB was primarily supportive. Patients were kept in **isolation**, encouraged to rest, and fed well. If these measures failed the lung was collapsed surgically so that it could “rest” and heal. Today surgical procedures still are used when necessary, but contemporary medicine relies on drug therapy as the mainstay of home care. Given an effective combination of drugs, patients with TB can be treated at home as well as in a sanatorium. Treatment at home does not pose the risk of infecting other household members.

Drug therapy

Most patients with TB can recover if given appropriate medication for a sufficient length of time. Three principles govern modern drug treatment of TB:

- Lowering the number of bacilli as quickly as possible. This measure minimizes the risk of transmitting the disease. When sputum cultures become negative, this has been achieved. Conversely, if the sputum remains positive after five to six months, treatment has failed.
- Preventing the development of drug resistance. For this reason, at least two different drugs and sometimes three are always given at first. If drug resistance is suspected, at least two different drugs should be tried.
- Long-term treatment to prevent relapse.

Five drugs are most commonly used today to treat tuberculosis: isoniazid (INH, Laniazid, Nydrazid); rifampin (Rifadin, Rimactane); pyrazinamide (Tebrazid); streptomycin; and ethambutol (Myambutol). The first three drugs may be given in the same capsule to minimize the number of pills in the dosage. As of 1998, many patients are given INH and rifampin together for six months, with pyrazinamide added for the first two months. Hospitalization is rarely necessary because many patients are no longer infectious after about two weeks of combination treatment. Follow-up involves monitoring of side effects and monthly sputum tests. Of the five medications, INH is the most frequently used drug for both treatment and prevention.

Surgery

Surgical treatment of TB may be used if medications are ineffective. There are three surgical treatments for pulmonary TB: pneumothorax, in which air is introduced into the chest to collapse the lung; thoracoplasty, in which one or more ribs are removed; and removal of a diseased lung, in whole or in part. It is possible for patients to survive with one healthy lung. Spinal TB may result in a severe deformity that can be corrected surgically.

Prognosis

The prognosis for recovery from TB is good for most patients, if the disease is diagnosed early and given prompt treatment with appropriate medications on a long-term regimen. According to a 2002 Johns Hopkins study, most patients in the United States who die of TB are older—average age 62—and suffer from such underlying diseases as diabetes and kidney failure.

Modern surgical methods have a good outcome in most cases in which they are needed. Miliary

KEY TERMS

Bacillus Calmette-Guérin (BCG)—A vaccine made from a damaged bacillus akin to the tubercle bacillus, which may help prevent serious pulmonary TB and its complications.

Mantoux test—Another name for the PPD test.

Miliary tuberculosis—The form of TB in which the bacillus spreads through all body tissues and organs, producing many thousands of tiny tubercular lesions. Miliary TB is often fatal unless promptly treated.

Mycobacteria—A group of bacteria that includes *Mycobacterium tuberculosis*, the bacterium that causes tuberculosis, and other forms that cause related illnesses.

Pneumothorax—Air inside the chest cavity, which may cause the lung to collapse. Pneumothorax is both a complication of pulmonary tuberculosis and a means of treatment designed to allow an infected lung to rest and heal.

Pulmonary—Referring to the lungs.

Purified protein derivative (PPD)—An extract of tubercle bacilli that is injected into the skin to find out whether a person presently has or has ever had tuberculosis.

Resistance—A property of some bacteria that have been exposed to a particular antibiotic and have “learned” how to survive in its presence.

Sputum—Secretions produced in the infected lung and coughed up. A sign of illness, sputum is routinely used as a specimen for culturing the tubercle bacillus in the laboratory.

Tuberculoma—A tumor-like mass in the brain that sometimes develops as a complication of tuberculous meningitis.

tuberculosis is still fatal in many cases but is rarely seen today in developed countries. Even in cases in which the bacillus proves resistant to all of the commonly used medications for TB, other seldom-used drugs may be tried because the tubercle bacilli have not yet developed resistance to them.

Prevention

General measures

General measures such as avoidance of overcrowded and unsanitary conditions are also necessary

aspects of prevention. Hospital emergency rooms and similar locations can be treated with ultraviolet light, which has an antibacterial effect.

Vaccination

Vaccination is one major preventive measure against TB. A vaccine called BCG (Bacillus Calmette-Guérin, named after its French developers) is made from a weakened mycobacterium that infects cattle. Vaccination with BCG does not prevent infection by *M. tuberculosis* but it does strengthen the immune system of first-time TB patients. As a result, serious complications are less likely to develop. BCG is used more widely in developing countries than in the United States. The effectiveness of vaccination is still being studied; it is not clear whether the vaccine's effectiveness depends on the population in which it is used or on variations in its formulation.

Prophylactic use of isoniazid

INH can be given for the prevention as well as the treatment of TB. INH is effective when given daily over a period of six to 12 months to people in high-risk categories. INH appears to be most beneficial to persons under the age of 25. Because INH carries the risk of side-effects (liver inflammation, nerve damage, changes in mood and behavior), it is important to give it only to persons at special risk.

High-risk groups for whom isoniazid prevention may be justified include:

- close contacts of TB patients, including health care workers
- newly infected patients whose skin test has turned positive in the past two years
- anyone who is HIV-positive with a positive PPD skin test; Isoniazid may be given even if the PPD results are negative if there is a risk of exposure to active tuberculosis
- intravenous drug users, even if they are negative for HIV
- persons with positive PPD results and evidence of old disease on the chest x-ray who have never been treated for TB
- patients who have an illness or are taking a drug that can suppress the immune system
- persons with positive PPD results who have had intestinal surgery; have diabetes or **chronic kidney failure**; have any type of **cancer**; or are more than 10% below their ideal body weight

- people from countries with high rates of TB who have positive PPD results
- people from low-income groups with positive skin test results
- persons with a positive PPD reaction who belong to high-risk ethnic groups (African Americans, Hispanics, Native Americans, Asians, and Pacific Islanders)

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National Heart, Lung, and Blood Institute (NHLBI). P. O. Box 30105, Bethesda, MD 20824-0105. (301) 592-8573. <www.nhlbi.nih.gov>.

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Tubo-ovarian abscesses see **Pelvic inflammatory disease**

Tularemia

Definition

Tularemia is an illness caused by a bacterium. It results in **fever**, rash, and greatly enlarged lymph nodes.

Description

Tularemia infects a variety of wild animals, including rabbits, deer, squirrels, muskrat, and beaver. Humans can acquire the bacterium directly from contact with the blood or body fluids of these animals, from the bite of a tick or fly which has previously fed on the blood of an infected animal, or from contaminated food or water.

Tularemia occurs most often in the summer months. It is most likely to infect people who come into contact with infected animals, including hunters, furriers, butchers, laboratory workers, game wardens, and veterinarians. In the United States, the vast majority of cases of tularemia occur in the southeastern and Rocky Mountain states.

Causes and symptoms

Five types of illness may occur, depending on where/how the bacteria enter the body:

KEY TERMS

Conjunctiva—The lining of the eyelids and the surface of the white part of the eye.

Shock—A state in which drastically low blood pressure prevents adequate blood flow to the tissues and organs throughout the body.

- Ulceroglandular/glandular tularemia. Seventy-five to 85% of all cases are of this type. This type is contracted through the bite of an infected tick that has defecated bacteria-laden feces in the area of the bite wound. A tender red bump appears in the area of the original wound. Over a few weeks, the bump develops a punched-out center (ulcer). Nearby lymph nodes grow hugely swollen and very tender. The lymph nodes may drain a thick, pus-like material. Other symptoms include fever, chills, and weakness. In adults, the lymph nodes in the groin are most commonly affected; in children, the lymph nodes in the neck.
- Oculoglandular tularemia. This type accounts for only about 1% of all cases of tularemia. It occurs when a person's contaminated hand rubs his or her eye. The lining of the eyelids and the surface of the white of the eye (conjunctiva) becomes red and severely painful, with multiple small yellow bumps and pitted sores (ulcers). Lymph nodes around the ears, under the jaw, or in the neck may swell and become painful.
- Oropharyngeal and gastrointestinal tularemia. This type occurs when contaminated meat is undercooked and then eaten, or when water from a contaminated source is drunk. Poor hygiene after skinning and cleaning an animal obtained through hunting can also lead to the bacteria entering through the mouth. Sores in the mouth and throat, as well as abdominal **pain, nausea and vomiting**, ulcers in the intestine, intestinal bleeding, and **diarrhea** may all occur.
- Pulmonary tularemia. This rare type of tularemia occurs when a person inhales a spray of infected fluid, or when the bacteria reach the lungs through the blood circulation. A severe pneumonia follows.
- Typhoidal tularemia. This type of tularemia is particularly hard to diagnose, because it occurs without the usual skin manifestations or swelling of lymph glands. Symptoms include continuously high fever, terrible **headache**, and confusion. The illness may result in a severely low blood pressure, with signs of poor blood flow to the major organs (shock).

Diagnosis

Samples from the **skin lesions** can be prepared with special stains, to allow identification of the causative bacteria under the microscope. Other tests are available to demonstrate the presence of antibodies (special immune cells that the body produces in response to the presence of specific foreign invaders) which would be increasing over time in an infection with tularemia.

Treatment

Streptomycin (given as a shot in a muscle) and gentamicin (given as either a shot in a muscle or through a needle in the vein) are both used to treat tularemia. Other types of **antibiotics** have been tested, but have often resulted in relatively high rates of relapse (20%).

Prognosis

With treatment, **death** rates from tularemia are under 1%. Without treatment, however, the death rate may reach 30%. The **pneumonia** and typhoidal types have the worst prognosis without treatment.

Prevention

Prevention involves avoiding areas known to harbor ticks and flies, or the appropriate use of insect repellents. Hunters should wear gloves when skinning animals or preparing meat. Others (butchers, game wardens, veterinarians) who work with animals or carcasses should always wear gloves. A vaccine exists, but is usually only given to people at very high risk due to their profession or hobby (veterinarians, laboratory workers, butchers, hunters, game wardens).

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Tumor markers

Definition

Tumor markers are measurable biochemicals that are associated with a malignancy. They are either

produced by tumor cells (tumor-derived) or by the body in response to tumor cells (tumor-associated). They are typically substances that are released into the circulation and thus measured in the blood. There are a few exceptions to this, such as tissue-bound receptors that must be measured in a biopsy from the solid tumor or proteins that are secreted into the urine.

Purpose

Though tumor markers are rarely specific enough to be used alone to diagnose **cancer**, they do have a number of clinical uses. They can be used to stage cancer, to indicate a prognosis, to monitor treatment, or in follow-up to watch for cancer recurrence. Changes in some tumor markers have been sensitive enough to be used as targets in clinical trials. When used for diagnosis, tumor markers are used in conjunction with other clinical parameters such as biopsy and radiological findings. Although there are a multitude of tumor markers, very few of them have found their way into clinical practice because of their lack of specificity. However, some of these non-specific markers have found a place in monitoring cancer treatment rather than in diagnosis.

Description

As tumor cells grow and multiply, some of their substances can increase and leak into the bloodstream or other fluids. Depending upon the tumor marker, it can be measured in blood, urine, stool or tissue. Some widely used tumor markers include: AFP, beta-HCG, CA 15-3, CA 19-9, CA 27.29, CA 125, CEA, and PSA. Some tumor markers are associated with many types of cancer; others, with as few as one. Some tumor markers are always elevated in specific cancers; most are less predictable. However, no tumor marker is specific for cancer and most are found in low levels in healthy persons, or can be associated with non-neoplastic diseases as well as cancer. Also, no tumor marker test is free of false negatives or false positives.

Once cancer is diagnosed, tumor marker levels sometimes help to determine the extent of cancer. Higher levels can indicate more advanced cancer and a worse prognosis in some cases. The patient and their physician may use this information to choose between more or less aggressive treatments.

Monitoring cancer treatment is the most common use of tumor markers. As cancer is reduced, levels often decrease. Stable or increasing levels often indicate that the cancer is not responding to treatment.

The choice of tumor marker to use for monitoring is important. Only a marker elevated before treatment should be used to monitor a person during or after treatment. Timing of the tests is also important. Each tumor marker has a unique life span in the blood. To monitor a treatment's success, enough time must have passed for the initial marker to be cleared from the blood. Tests done too soon may be falsely elevated because the marker produced by the untreated cancer is still present.

Watching for cancer recurrence after treatment is another reason for tumor marker testing. Periodic testing can sometimes detect a recurrence often months earlier than could an ultrasound, x ray, or **physical examination**.

Tumor marker tests are performed in a lab using immunological techniques. A sample of blood or other tissue is mixed with a substance containing specific antibodies to each tumor marker. If that tumor marker is present, these very specific antibodies bind to the markers. Some type of label, often a radioactive substance, is then used to measure the amount of bound marker and antibody. From this measurement, the amount of tumor marker is calculated. The results are usually available within a few days.

Conclusions based on tumor marker tests are seldom based on one test result but on a series of test results, called serial measurements. A series of increasing or decreasing values is more significant than a single value.

Tumor marker testing is currently the object of much research and attention. Their use is directed by approval from the Food and Drug Administration (FDA) and guidelines established by organizations such as the American Society of Clinical Oncology and the American Cancer Society. Not all tumor receptor marker tests are widely available nor are they widely accepted.

Oncofetal antigens

There are two common oncofetal antigens, alpha-fetoprotein (AFP) and carcinoembryonic antigen (CEA). Carcinoembryonic antigen CA 72-4 is a more recently discovered oncofetal antigen just coming into usage. The oncofetal antigens are so named because they are normally produced during embryonic development and decrease soon after birth. Cancer cells tend to dedifferentiate, or revert to a more immature tissue and begin to produce fetal antigens again. Oncofetal antigens are very non-specific and expressed by a wide number of cancer types. However, they are

used both to monitor a patient's progress and their response to treatment over time.

ALPHA-FETOPROTEIN (AFP). Elevated AFP typically indicates a primary liver tumor or a germ cell tumor of the ovary or testicle. AFP is a glycoprotein produced in high amounts by fetal tissue and is elevated during **pregnancy**. It is most widely used as a marker for hepatocellular carcinoma and **testicular cancer** but is also associated with **ovarian cancer**. Seventy percent of people with **liver cancer** have increased AFP levels. In China, where liver cancer rates are high, AFP is used as a screening test for that disease. AFP levels indicate the extent of cancer, and serial measurements are used to monitor treatment response. Non-cancerous liver conditions such as **cirrhosis** and hepatitis have moderately increased levels of AFP.

CARCINOEMBRYONIC ANTIGEN (CEA). CEA is a glycoprotein most often associated with colorectal cancer, and used to monitor patients with this type of cancer. Its most popular use is in early detection of relapse in individuals already treated for colorectal cancer. After surgery, serial measurements indicate the surgery's success and are used to detect early signs of recurrence. It has recently been found to be useful when measured during surgery for colorectal cancer to help determine prognosis and who will benefit from adjuvant treatment.

CEA is measured in the blood plasma. It is very non-specific and can be increased in many types of cancer: gastrointestinal, colorectal, ovarian, bladder, cervical, stomach, kidney, lung, pancreatic, liver, prostate, thyroid, melanoma, lymphoma, and breast. People with noncancerous conditions, such as cirrhosis or peptic disease, or inflammatory intestinal conditions such as colitis or **diverticulitis**, may also have increased levels. CEA levels can be elevated in elderly patients and in those who smoke.

CANCER ANTIGEN 72-4 (CA 72-4). The more recently identified carcinoembryonic protein is CA 72-4. Although it is slightly elevated with most carcinomas, it is mostly associated with gastric carcinoma (**stomach cancer**). CA 72-4 is finding a role in the management of patients with gastric carcinoma.

Cancer antigen 15-3 (CA 15-3)

CA 15-3 is produced by cells in the breast and increased levels can be associated with **breast cancer**. Rarely increased in women with early breast cancer, it may be used to detect recurrence of cancer in women following treatment or **mastectomy** and to monitor treatment for women with advanced breast cancer.

However, adenocarcinomas of the ovary, lung, colon, and pancreas also express elevated CA 15-3 levels. Non-cancerous conditions sometimes associated with elevated CA 15-3 include benign breast or ovarian disease, **endometriosis**, **pelvic inflammatory disease**, and hepatitis. Pregnancy and **lactation** are also related to high CA 15-3 levels.

Cancer antigen 27-29 (CA 27-29)

CA 27-29, also called breast carcinoma-associated antigen, is used as a marker for breast cancer. Eighty percent of women with breast cancer have an increased CA 27-29 level. This marker may be used with other procedures and tumor marker levels such as CA 15-3 to check for recurrences of cancer in previously treated women. Serial measurements monitor treatment response and identify recurrence.

Levels of CA 27-29 may also be increased in cancers of the colon, stomach, kidney, lung, ovary, pancreas, uterus, and liver. Noncancerous conditions associated with elevated CA 27-29 include first trimester pregnancy, endometriosis, **ovarian cysts**, non-cancerous breast disease, **kidney disease**, and **liver disease**.

HER-2/neu

HER-2/neu is an oncogenic growth factor receptor also known as c-erbB-2. It is measured in the tissue from a biopsy either by immunological assays of the protein or polymerase chain reaction (PCR) to identify the DNA. The presence of HER-2/neu is generally associated with a poorer prognosis for breast cancer. It can also help to determine treatment options, since newer drugs can block this protein and decrease cancer growth. The most widely known of these drugs is trastuzumab (brand name Herceptin).

Estrogen receptor

Measurement of the estrogen receptor (ER) is used specifically to evaluate breast cancers. It gives an indication of prognosis and responsiveness to therapy. Tissue from a biopsy is used to measure the estrogen receptor. Most breast cancers in post-menopausal women are ER-positive, meaning that they require estrogen to grow. These ER positive breast cancers are less aggressive than ER negative breast cancers, which are found generally in pre-menopausal women.

Cancer antigen 125 (CA 125)

Although produced by a number of cell types, CA 125 is primarily produced by ovarian cancer cells. Eighty percent of women with ovarian cancer have

increased CA 125 levels. Although the test is not sensitive or specific enough to be used for screening, it contributes to a diagnosis when combined with an ultrasound and pelvic examination. Blood levels of CA 125 are used primarily to monitor the treatment of ovarian cancer. A falling CA 125 level usually indicates that cancer is responding to the treatment. After diagnosis and treatment, serial measurements help detect remaining or recurrent cancer. A negative or normal result, however, does not guarantee the absence of cancer.

Women may have increased CA 125 levels during menstruation and pregnancy. Increased levels are also found in pelvic inflammatory disease, endometriosis, **pancreatitis**, and liver disease. Elevated levels are also associated with non-ovarian cancers including cancers of the uterus, cervix, pancreas, liver, colon, breast, lung, or digestive tract.

Prostate specific antigen (PSA)

Prostate specific antigen (PSA) levels, along with the digital **rectal examination**, are used to screen for **prostate cancer**. PSA is a protein produced by the prostate gland and can be overproduced in prostate cancer. It is perhaps the best tumor marker in use because of its tissue specificity, meaning that it is produced only by the prostate. Men over the age of 50 years are advised to consider annual screening for prostate cancer. Men at high risk for prostate cancer, such as African Americans or those with a family history of the disease, should begin screening at age 40. Once a diagnosis of prostate cancer is made, PSA levels can help determine the stage of the cancer, monitor the response to treatment, and watch for recurrence.

Measurements of PSA following **prostatectomy** are useful in determining the success of surgery. Any PSA level following surgery would indicate residual prostate tissue, possibly from metastasis. PSA levels can also be used to detect a recurrence of prostate cancer. PSA is also increased in benign prostatic hyperplasia (BPH), an **enlarged prostate** condition common in older men.

PSA can be found in the serum in two states, bound and free. Measuring both PSA levels can provide more specificity to the test and reduce unnecessary biopsies. The percentage of free PSA is greater in BPH than prostate cancer. If the total PSA level is higher than 4.0 nanogram/milliliter (ng/mL) and the free PSA level is less than 25%, a **prostate biopsy** is indicated.

PSA levels may increase after ejaculation. Men are recommended to abstain from sexual intercourse or masturbation for 48 hours before the test. PSA levels may also increase after prostate manipulation following the digital rectal exam.

Prostatic acid phosphatase (PAP) originally found to be produced by the prostate and thought to be a marker for prostate cancer. It is now found to be elevated with testicular cancer, leukemia, non-Hodgkin's lymphoma and several noncancerous conditions.

Cancer antigen 19-9 (CA 19-9)

CA 19-9 has been identified in patients with digestive tract or intra-abdominal carcinomas such as colorectal cancer, pancreatic cancer, stomach cancer and **bile duct cancer**. In pancreatic cancer, higher levels are associated with more advanced disease. After diagnosis, levels help predict the success of surgery and monitor the course of the cancer. Not all people with pancreatic cancer have increased CA 19-9 levels. This antigen is related to the Lewis blood group and so only patients positive for the Lewis blood group antigen will test positive for CA 19-9. It is also increased in liver and gastrointestinal cancers and in noncancerous diseases, including pancreatitis, **gallstones** and **jaundice**.

Human chorionic gonadotropin (hCG)

Human chorionic gonadotropin is normally produced by the placenta during pregnancy. There are two protein subunits that make up HCG, beta and alpha. It is the beta subunit that is increased in women's serum during early pregnancy. It is also the beta subunit that is increased in some malignant tumors. Tumors that secrete beta-hCG are typically germ cell tumors such as teratocarcinomas. These are tumors found in the ovaries and testes that contain embryonal tissue. Rarely, these types of tumors are found in the pineal region of the brain where beta-hCG can serve as a marker. Levels of hCG rise with **choriocarcinoma** and with trophoblastic disease, a rare cancer that develops from an abnormally fertilized egg. Gestational trophoblastic tumors also secrete AFP and this test is often used in combination.

HCG is most often used to screen for cancer of the testis or ovary. Serial measurements monitor the progress and treatment of these cancers. This marker can be elevated in individuals who use **marijuana**.

Squamous cell carcinoma (SCC) Antigen

Squamous cell carcinoma (SCC) antigen was first identified in **cervical cancer**. It is a marker for squamous cell cancers, which can occur in the cervix, head

and neck, lung, and skin. Levels of SCC can be used as an aid to stage the carcinoma and to determine the response to treatment.

Bence-Jones protein

Patients with plasmacytomas such as myeloma overproduce monoclonal immunoglobulins, also called M proteins. The Bence-Jones protein refers to the immunoglobulin light chain, a portion of these immunoglobulins. The Bence-Jones protein is secreted into the urine where it can be measured. It was the first tumor marker identified.

Neuron-specific enolase (NSE)

NSE is a protein found mainly in neurons and neuroendocrine cells. It is elevated in tumors derived from these tissues, including **neuroblastoma** and **small cell lung cancer**. It can give information about the extent of the disease, the patient's prognosis and the patient's response to treatment. NSE can also be elevated in medullary thyroid cancers, carcinoid tumors, pancreatic endocrine tumors, and melanoma.

Hormone assays

Tumors of the endocrine glands oversecrete their corresponding hormones. By measuring particular hormones, clues can be obtained regarding certain cancers. For instance, breast cancer cells may secrete prolactin and estrogen. Medullary carcinoma can secrete calcitonin. Pheochromocytomas secrete catecholamines. Tumors of the pituitary gland may secrete growth hormone or cortisol. Carcinoid tumors secrete serotonin. Some tumors of the pancreas secrete insulin. Serial measurements can also monitor treatment for these tumors.

Enzymes

Several serum enzymes can be measured to help detect metastases in cancer patients. Tumors that metastasize to the liver cause increases in serum alkaline phosphatase, gamma-glutamyltransferase, and transaminases. Although these are not necessarily tumor markers, they indicate liver damage that may be caused by metastatic cancer. Tumors that metastasize to the bone sometimes secrete elevated alkaline phosphatase. Lactate dehydrogenase is an enzyme found throughout the body. Because of this it cannot be used as a marker for cancer. It can, however, be used to monitor the treatment of some types of cancer including germ cell tumors, testicular cancer, Ewing's sarcoma, non-Hodgkins lymphoma and some types of leukemia.

Precautions

There is not a good consensus in the medical community about the value of most tumor markers. Because they lack specificity and accuracy, their use is limited. False positives can cause emotional distress and fear. It is not yet determined if there is a savings of life or money with testing. Currently, much controversy surrounds the issue of mass screening for cancer using tumor markers.

Preparation

Tumor marker tests usually require 5-10 mL of blood. A healthcare worker ties a tourniquet on the patient's upper arm, locates a vein in the inner elbow region, and inserts a needle into that vein. Vacuum action draws the blood through the needle into an attached tube. Collection of the sample takes a few minutes and results are available within a few days.

Some markers, such as those for **bladder cancer**, **multiple myeloma**, and plasmacytomas, are measured in the urine. Typically this requires a 24-hour urine sample, which means that the individual must collect all of his or her urine for 24 hours. This is usually about 1.5 quarts or more. These results are then available within a few days.

Other tumor markers require tissue samples for analysis. These include receptor analysis such as estrogen receptor and Her-2/neu. Tissue samples are obtained by biopsy. This is usually done by inserting a needle through the skin and into the tumor. The area is typically numbed prior to the procedure. These results are also available within two to three days.

Aftercare

Discomfort or bruising may occur at the puncture site or the person may feel dizzy or faint. Pressure to the puncture site until the bleeding stops reduces bruising. Warm packs to the puncture site relieve discomfort. There is a rare chance of infection occurring especially after biopsy. Any sign of infections should be watched for such as **pain** and redness.

Normal results

- AFP: 99% of (nonpregnant) people have less than 15 ng/mL; 95% have less than 6 ng/mL. Serum AFP levels higher than 400 micrograms/L are associated with cancer or some other pathology.
- Beta-HCG: in males, less than 2.5 IU/L; in females, less than 5.0 IU/L; in postmenopausal females, less than 9.0 IU/L.

KEY TERMS

AFP (Alpha-fetoprotein)—A tumor marker associated with liver, testicular, and ovarian cancer.

Beta-HCG (Beta-human chorionic gonadotropin)—A tumor marker associated with testicular cancer and tumors, such as choriocarcinoma and molar pregnancies, that begin in placental cells called trophoblasts.

Biopsy—The process of taking a sample of tumor tissue through a needle.

CA 15-3 (Cancer antigen 15-3)—A tumor marker associated with breast cancer.

CA 19-9 (Cancer antigen 19-9)—A tumor marker associated with pancreatic cancer.

CA 27-29 (Breast carcinoma-associated antigen)—A tumor marker associated with breast cancer.

CA 125 (Cancer antigen 125)—A tumor marker associated with ovarian cancer.

CEA (Carcinoembryonic antigen)—A tumor marker associated with many cancers, especially liver, intestinal, and pancreatic.

Prognosis—The predicted outcome of a disease.

PSA (Prostate specific antigen)—A tumor marker associated with prostate cancer.

Sensitivity—A test's ability to detect all cases of a disease.

Serial measurements—A series of measurements looking for an increase or decrease over time.

Specificity—A test's ability to detect only the disease in question.

Tumor markers—Biochemicals produced by tumor cells or by the body in response to tumor cells. Their levels in the blood help evaluate people for certain kinds of cancer.

- CA 15-3: less than 40 U/mL.
- CA 19-9: less than 40 U/mL.
- CA 27.29: less than or equal to 38 U/mL.
- CA 125: less than 35 U/L.
- CEA: less than or equal to 5 ng/mL.
- PSA: less than 4 ng/mL; PSA levels increase with age. Age-specific values range from 2.0 micrograms/L at age 40 to 7.2 micrograms/L at age 80. Typically, levels below 4.0 micrograms/L rule out prostate cancer.

Abnormal results

The meaning of an increased tumor marker level depends on the specific marker, the person's medical history, and why the test was done. Knowledge of the patient's history and additional tests and physical examinations are needed to correctly interpret tumor marker test results.

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American Cancer Society. 1599 Clifton Rd., NE, Atlanta, GA 30329-4251. (800) 227-2345. <<http://www.cancer.org>>.

American Society of Clinical Oncology. 225 Reinekers Lane, Suite 650, Alexandria, VA 22314. (703) 299-0150. <<http://www.asco.org>>.

National Cancer Institute. Building 31, Room 10A31, 31 Center Drive, MSC 2580, Bethesda, MD 20892-2580. (800) 422-6237. <<http://www.nci.nih.gov>>.

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Nancy J. Nordenson

Tumor removal

Definition

Tumor removal is a surgical procedure to remove an abnormal growth.

Purpose

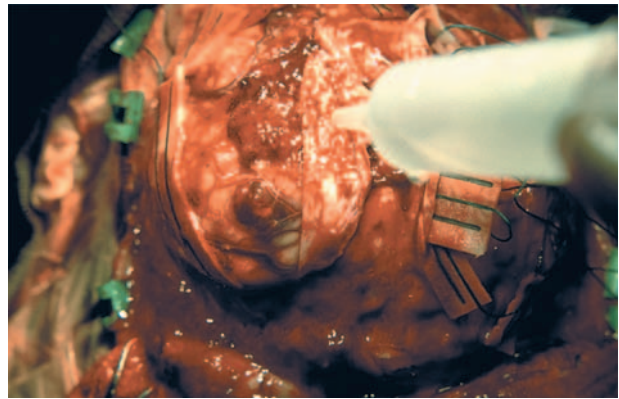
A tumor can be either benign, like a wart, or malignant, in which case it is a **cancer**. Benign tumors are well circumscribed and generally are easy to remove completely. In contrast, cancers pose some of the most difficult problems in all of surgery.

Currently 40% of all cancers are treated with surgery alone. In 55% of cases, surgery is combined with other treatments — usually **radiation therapy** or **chemotherapy**.

The doctor needs to decide if surgery should be done at all. Because cancers spread (metastasize) to normal tissues, sometimes at the other end of the body, the ability of surgery to cure must be addressed at the outset. As long as the cancer is localized, the initial presumption is that cure should be attempted by removing it as soon as possible.

Non-curative surgery may make other treatments more effective. “Debulking” a cancer—making it smaller—is thought to assist radiation and chemotherapy to get to the remaining pieces of the cancer and be more effective. Physicians apply standards and guidelines based on research to the specific situation of a given patient’s situation to decide the best approach. For example, reports released in 2003 said a study of patients with a common form of **brain tumor** called glioma had a more favorable outcome if their tumors were removed than if a biopsy were just performed.

Another important function surgery performs in cancer treatment is accurately assessing the nature and extent of the cancer. Most cancers cannot be adequately identified without a piece being placed under a microscope. This piece is obtained by surgery. Surgery also is the only way to determine exactly how far the tumor has spread. There are a few standard methods of comparing one cancer to another for the purposes of comparing treatments and estimating



A tumor inside the brain is being removed. (Photograph by Jennifer Watson-Holton, Custom Medical Stock Photo. Reproduced by permission.)

outcomes. These methods are called “staging.” The most universal method is the TNM system.

- “T” stands for “tumor” and reflects the size of the tumor.
- “N” represents the spread of the cancer to lymph nodes, largely determined by those nodes removed at surgery that contain cancer cells. Since cancers spread mostly through the lymph system and this is a useful measure of their ability to disperse.
- “M” refers to the metastases, how far they are from the original cancer and how often they have multiplied.

Other methods of staging include Duke’s method and similar systems, which add the degree of invasion of the cancer into the surrounding tissues to the above criteria.

Staging is particularly important with such lymphomas as **Hodgkin’s disease**. These cancers may appear in many places in the lymphatic system. Because they are very radiosensitive, radiation treatment often is curative if all the cancer is irradiated. Therefore, it must all be located. Surgery is a common, usually essential, method of performing this staging. If the disease is too widespread, the staging procedure will dictate chemotherapy instead of radiation.

Precautions

Curative cancer surgery demands special considerations. There is a danger of spreading or seeding the cancer during the process of removing it. Presuming the cancer cells can grow almost anywhere in the body to which they spread, the surgeon must not “spill” cells into the operating field or “knock them loose” into the blood stream. Special techniques called “block

MOSES JUDAH FOLKMAN (1933–)



(AP Wide World Photos. Reproduced by permission.)

Moses Judah Folkman was born in Cleveland, Ohio, on February 24, 1933. He was one of three children born to Bessie and Jerome Folkman. Because Folkman's father, a rabbi, would take his children with him when he visited

sick individuals in the hospital, his son Judah dreamed of becoming a surgeon. Young Folkman requested a microscope for his bah mitzvah present and upon receiving it, set up a laboratory in his parents' basement.

Folkman entered Ohio State University as a pre-med student, graduating in 1953 after only three years of study. He then went on to Harvard Medical School, where he helped create one of the first pacemakers ever produced and he received his medical degree in 1957. Folkman completed his internship and residency at Massachusetts General Hospital in Boston. In 1960, he married Paula Prial and the couple moved to Bethesda, Maryland where Folkman worked as a lieutenant in the U.S. Navy at the National Naval Medical Center. It was here, with the help of his colleague, Fredrich Becker, that Folkman would make his initial discovery dealing with angiogenesis. Folkman found that a tumor would only grow if it had blood supplied to it. Additionally, the tumor would promote the growth of new blood vessels.

Folkman also held positions as a professor and surgeon for many years, but in 1981, he retired and became director of the Children's Hospital Surgical Research Laboratories in Boston. Folkman's research has opened new doorways that may allow scientists to ultimately find a cure for cancer.

resection" and "no touch" are used. Block resection means taking the entire specimen out as a single piece. "No touch" means that only the normal tissue removed with specimen is handled; the cancer itself is never touched. This prevents "squeezing" cancer cells out into the circulation. Further, in this technique pains are taken to clamp off the blood supply first, preventing cells from leaving by that route later in the surgery.

Description

Diagnostic biopsies

There are four types of biopsy techniques. The physician will choose the appropriate technique depending on the tumor type, location, size, and other factors. Some types of biopsy are more invasive than others.

- Aspiration biopsy. A needle is inserted into the tumor and a sample is withdrawn.
- Needle biopsy. A special cutting needle is inserted into the core of the tumor and a core sample is cut out.

- Incisional biopsy. A portion of a large tumor is removed, usually before complete tumor removal.
- Excisional biopsy. A whole lesion is removed along with surrounding normal tissue.

Complete tumor removal

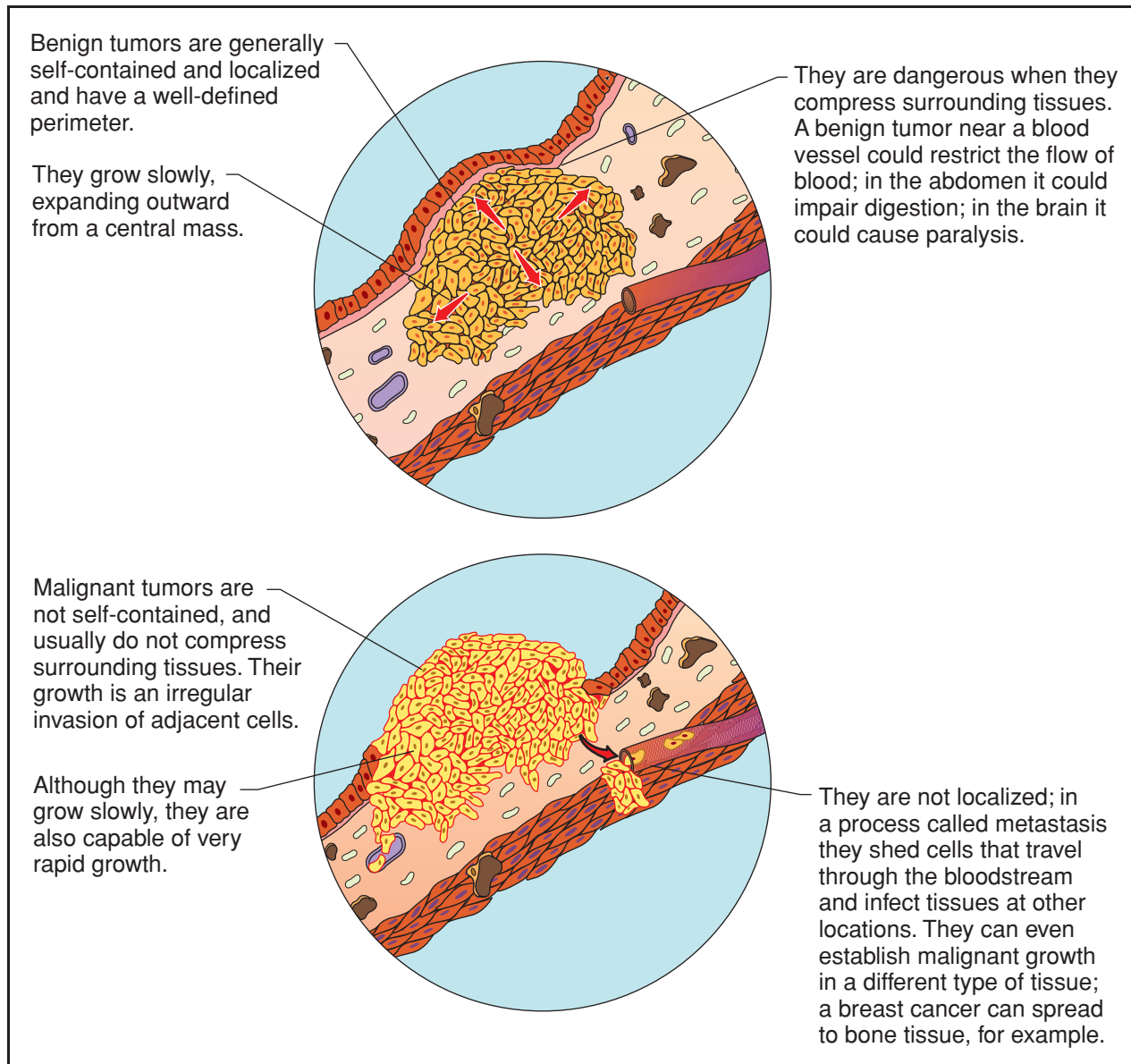
Once surgical removal has been decided, an oncologic surgeon will remove the tumor whole, taking with it a large section of the surrounding normal tissue. The healthy tissue is removed to minimize the risk of possible seeding.

Cytoreduction

When surgical removal of a tumor is unacceptable as a sole treatment, a portion of the tumor is removed to "debulk" the mass. Debulking aids radiation and chemotherapy treatments.

Aftercare

Retesting and periodical examinations are necessary to ensure that a tumor has not reformed after total removal.



A comparison of benign (top of illustration) and malignant tumor characteristics. (Illustration by Hans & Cassady, Inc.)

Risks

The possibility of metastasis and seeding are risks that have to be considered in consultation with an oncologist. The advancement in imaging techniques has led to some tumor staging by techniques like **positron emission tomography (PET)** scanning. A needle biopsy may still be required to confirm a diagnosis or staging decision, but in some cases, imaging can replace an initial need for biopsy.

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J. Ricker Polsdorfer, MD
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Turner syndrome

Definition

Turner syndrome is a chromosomal disorder affecting females wherein one of the two X-chromosomes is defective or completely absent.

Description

Chromosomes are structures in the nucleus of every cell in the human body. Chromosomes contain the genetic information necessary to direct the growth and normal functioning of all cells and systems of the body. A normal individual has a total of 46 chromosomes in each cell, two of which are responsible for determining gender. Normally, females have two X-chromosomes and males have one X and one Y-chromosome.

In Turner syndrome, an error occurring very early in development results in an abnormal number and arrangement of chromosomes. Most commonly, an individual with Turner syndrome will be born with 45 chromosomes in each cell rather than 46. The missing chromosome is an X-chromosome. The affected person is always female.

The prevalence of Turner syndrome is widely reported as being approximately one per 2,000 live female births, although researchers have reported prevalence rates that range from one in 3,125 to one in 5,000 live female births.

About 1% to 2% of all female conceptions have a missing X-chromosome. Of these, the majority (99%) spontaneously abort, usually during the first trimester of **pregnancy**. With ultrasound being used more frequently, researchers have realized that some pregnancies with a missing X-chromosome that progress into the second trimester are associated with nuchal cysts, severe **lymphedema**, or hydrops fetalis. These pregnancies are associated with a high frequency of fetal **death**.

Causes and symptoms

Turner's syndrome is a disorder associated with characteristic defects in the X-chromosome. The most common presentation is a female with a single X-chromosome and an absent X-chromosome. A Greek study from 1999 reported that the intact X-chromosome was as likely to come from the mother as from the father. This means that there is no parental pattern of responsibility for the missing or defective X-chromosome.

Another less common genetic pattern for Turner Syndrome (35%) is a mosaic. A Danish study reported that mosaicism has an effect on malformations that are associated with Turner syndrome. Research reported in 1997 noted that the karyotype can have a significant effect on the growth of children with Turner syndrome.

The exact location of the genes on the X-chromosome involved in Turner syndrome has not been determined as of 2001. At present, evidence exists that there is a locus for stature on the distal portion of the short arm; there are loci for normal ovarian function on both the short and long arms; and there are loci contributing to fetal viability on the long arm of X.

Turner syndrome is characterized by retarded growth that leads to a small stature and frequent **infertility**. Individuals with Turner syndrome report an increased incidence of **fractures** in childhood and osteoporotic fractures in adulthood. The incidence of **diabetes mellitus** (both insulin dependent and non-insulin dependent varieties) has been reported to be increased in Turner syndrome. Ischemic heart disease, **stroke** and **hypertension** are also more common.

Growth in children with Turner syndrome is characterized by a slight **intrauterine growth retardation**, relatively normal growth rates for the first several years of life, a progressive deceleration of growth later in childhood, and the lack of a pubertal growth spurt. Growth patterns of Chinese girls with Turner syndrome parallel those of Caucasians, although their ultimate height is still less than normal.

Contrary to earlier reports, most individuals with Turner syndrome are not mentally retarded. They may have some learning disabilities, particularly with regard to spatial perception, visual-motor coordination, and mathematics. As a result, the nonverbal IQ in Turner syndrome tends to be lower than the verbal IQ.

Cardiovascular malformations are well-recognized congenital anomalies in Turner syndrome. Dilation and dissection of the aorta are reported in approximately half of women with Turner syndrome. Because of the potential consequences of aortic dilation, some experts recommend screening all individuals with Turner syndrome. However, the specific timing for this screening remains controversial in 2001.

Juvenile arthritis, an autoimmune condition, has been recently (1998) associated with Turner syndrome. The prevalence seems to be at least six times greater than would be expected if the two conditions were only randomly associated. Women with Turner syndrome have an elevated prevalence rate of dental caries and such other periodontal conditions as gum disease and plaque.



A low hairline at the back of the neck is one of several characteristics of Turner syndrome. (Custom Medical Stock Photo. Reproduced by permission.)

Normal pubertal development and spontaneous menstrual periods do not occur in the majority of children with Turner syndrome. It is estimated that 3–8% of girls with a single X-chromosome and 12–21% of females with sex chromosome mosaicism may have normal pubertal development and spontaneous menstrual periods. A few pregnancies have been reported in women with Turner syndrome.

Diagnosis

Turner syndrome is diagnosed on the basis of genetic analysis of chromosomes. This can be done prior to birth. However, the predictive value of **amniocentesis** in diagnosing Turner syndrome varies from 21–67%. There is no significant relation between the mother's age and risk of Turner's syndrome.

Treatment

Because it is so dangerous, experts suggest screening for **aortic dissection**, although the specific timing for this screening is controversial. **Plastic surgery** to correct webbing of the neck should be considered at an early age (before entering school) for girls with Turner syndrome.

Most individuals with Turner syndrome require female hormone therapy to promote development of secondary sexual characteristics and menstruation. The time of beginning therapy varies with individuals.

KEY TERMS

Chromosome—A microscopic thread-like structure found within each cell of the body that consists of a complex of proteins and DNA. Humans have 46 chromosomes arranged into 23 pairs. Changes in either the total number of chromosomes or their shape and size (structure) may lead to physical or mental abnormalities.

Mosaic—A term referring to a genetic situation in which an individual's cells do not have the exact same composition of chromosomes. In Down syndrome, this may mean that some of the individual's cells have a normal 46 chromosomes, while other cells have an abnormal 47 chromosomes.

Ovary—The female reproductive organ that produces the reproductive cell (ovum) and female hormones.

Zygote—The cell formed by the uniting of egg and sperm.

Experts recommend that therapy begin when a woman expresses concern about her onset of **puberty**.

All women receiving long-term, exogenous female hormone therapy require periodic gynecological examinations, because those with Turner syndrome have an increased risk of developing neoplasms such as gonadoblastoma and dysgerminoma, which arise from their rudimentary streak gonads.

Prognosis

Most women with Turner syndrome can live relatively normal lives. The prognosis for a person with Turner syndrome is dependent on other conditions that may be present. Care must be taken to regularly monitor them for the health problems that are associated with Turner syndrome. For example, heart or kidney defects, **hearing loss**, or the development of inflammatory bowel disease may significantly impact the quality of life. Without these types of conditions, however, their life expectancy is normal. Support will be necessary to help an adolescent girl cope with body image issues and to help some women accept the fact that they will never be able to have children.

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American Academy of Pediatrics. 141 Northwest Point Blvd., Elk Grove Village, IL 60007-1098. (847) 434-4000. Fax: (847) 434-8000. <<http://www.aap.org/visit/contact.htm>>.

Endocrine Society. 4350 East West Highway, Suite 500, Bethesda, MD 20814-4410. (301) 941-0200. Fax: (301) 941-0259. endostaff@endo-society.org.

Human Growth Foundation. 997 Glen Cove Ave., Glen Head, NY 11545. (800) 451-6434. Fax: (516) 671-4055. <<http://www.hgfound.org>>.

Turner Syndrome Society of Canada. 7777 Keele St, Floor 2, Concord, ONT L4K 1Y7. Canada (800) 465-6744 or (416) 660-7766. Fax: (416) 660-7450.

Turner Syndrome Society of England. 2 Mayfield Ave., London, W41PW. UK 44 (0)181-994 7625. Fax: 44 (0)181-995 9075. <<http://www.exnet.com/staff/sys4/ts.html>> or <<http://www.tss.org.uk>>.

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University of Kansas Medical Center. <<http://www.kumc.edu/gec/support/turner.html>>.

L. Fleming Fallon, MD, PhD, DrPH

Twins see **Multiple pregnancy**

2,3-diphosphoglycerate test

Definition

2,3-diphosphoglycerate (2,3-DPG) is a substance made in the red blood cells. It controls the movement of oxygen from red blood cells to body tissues. 2,3-DPG testing is done to help investigate both a deficiency in red blood cells (anemia) and an unexplained increase of red blood cells, called erythrocytosis.

Purpose

Hemoglobin, the protein in the blood that carries oxygen, uses 2,3-DPG to control how much oxygen is released once the blood gets out into the tissues. The more 2,3-DPG in the cell, the more oxygen is delivered to body tissues. Conversely, the less 2,3-DPG in the cell, the less oxygen is delivered.

Increasing the amount of 2,3-DPG is the body's primary way of responding to a lack of oxygen. Anemia, obstructive lung disease, **cystic fibrosis**, and **congenital heart disease** are all accompanied by increases in 2,3-DPG. When more oxygen is required because of increased metabolism, such as in **hyperthyroidism**, more 2,3-DPG is produced.

Decreased 2,3-DPG results from an inherited lack of the red blood cell enzymes 2,3-DPG mutase and 2,3-DPG phosphatase. These enzymes are needed to make 2,3-DPG. Without 2,3-DPG to control the movement of oxygen to its tissues, the body responds by making more red blood cells, a condition called erythrocytosis. The outside membrane of the cell is weakened, causing it to have an irregular shape and burst, or hemolyze, easily. This condition is called nonspherocytic **hemolytic anemia**.

2,3-DPG levels are important in large blood transfusions, because stored blood quickly loses 2,3-DPG and its ability to deliver oxygen. After **transfusion**, the red cells rebuild the 2,3-DPG, but it takes about 24 hours to regain a normal level of 2,3-DPG and hemoglobin function.

Description

In the laboratory, a person's serum is mixed with a substance that will react with 2,3-DPG. The end product of this reaction is measured; and from that measurement, the amount of 2,3-DPG in the person's serum is determined. Results are usually available the next day.

Preparation

This test requires drawing 5-10 mL of blood. The patient should not **exercise** before having the blood drawn. Exercise increases the body's need for oxygen and could cause a temporary increase in levels of 2,3-DPG.

Aftercare

Discomfort or bruising may occur at the puncture site, or the person may feel dizzy or faint. Pressure to the puncture site until the bleeding stops will reduce bruising. Warm packs to the puncture site will relieve discomfort.

Normal results

Normal results will vary based on the laboratory and testing methods used.

KEY TERMS

Anemia—A reduction in the number of erythrocytes or red blood cells. Erythrocytes are necessary to form hemoglobin for transporting oxygen.

Erythrocytosis—Increased production of red blood cells.

Hemoglobin—A protein within the red blood cell that carries oxygen.

Nonspherocytic hemolytic anemia—Anemia caused by variably shaped red blood cells that burst, or hemolyze, easily.

Abnormal results

Decreased levels of 2,3-DPG are found in cases of erythrocytosis and nonspherocytic hemolytic anemia caused by 2,3-DPG mutase and 2,3-DPG phosphatase deficiencies. Lower levels are also commonly found after large blood transfusions.

Increased levels of 2,3-DPG are found in conditions in which the body needs more oxygen, such as anemia, obstructive lung disease, cystic fibrosis, congenital heart disease, and hyperthyroidism. High altitudes and participating in exercise sessions before the test can also give false high values.

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Nancy J. Nordenson

2,3-DPG see **2,3-diphosphoglycerate test**

Tylenol see **Acetaminophen**

Tympanic membrane perforation see **Perforated eardrum**

Tympanometry see **Audiometry**

Typhoid fever

Definition

Typhoid **fever** is a severe infection caused by a bacterium, *Salmonella typhi*. *S. typhi* is in the same

family of bacteria as the type spread by chicken and eggs, commonly known as salmonella **poisoning** or **food poisoning**. *S. typhi* bacteria do not have **vomiting** and **diarrhea** as the most prominent symptoms of their presence in humans. Instead, persistently high fever is the hallmark of *S. typhi* infection.

Description

S. typhi bacteria are passed into the stool and urine of infected patients. They may continue to be present in the stool of asymptomatic carriers, who are persons who have recovered from the symptoms of the disease but continue to carry the bacteria. This carrier state occurs in about 3% of all individuals recovered from typhoid fever.

Typhoid fever is passed from person to person through poor hygiene, such as incomplete or no hand washing after using the toilet. Persons who are carriers of the disease and who handle food can be the source of epidemic spread of typhoid. One such individual gave her name to the expression "Typhoid Mary," a name given to someone whom others avoid.

Typhoid fever is a particularly difficult problem in parts of the world with poor sanitation practices. There are about 16 million cases of typhoid reported around the world each year. In the United States, most patients who contract typhoid fever have recently returned from travel to another country where typhoid is much more common, including Mexico, Peru, Chile, India, and Pakistan. However, there have been reports in the early 2000s of typhoid outbreaks within the United States that were unrelated to recent travel. One such outbreak occurred in Queens, New York, and was traced to a worker in a local restaurant.

Causes and symptoms

S. typhi must be ingested to cause disease. Transmission often occurs when a person in the carrier state does not wash hands thoroughly (or not at all) after defecation and serves food to others. This pathway is sometimes called the fecal-oral route of disease transmission. In countries where open sewage is accessible to flies, the insects land on the sewage, pick up the bacteria, and then contaminate food to be eaten by humans.

After being swallowed, the *S. typhi* bacteria head down the digestive tract, where they are taken in by cells called mononuclear phagocytes. These phagocytes are cells of the immune system, whose job it is to engulf and kill invading bacteria and viruses. In the case of *S. typhi*, however, the bacteria are able to

MARY MALLON (1869–1938)

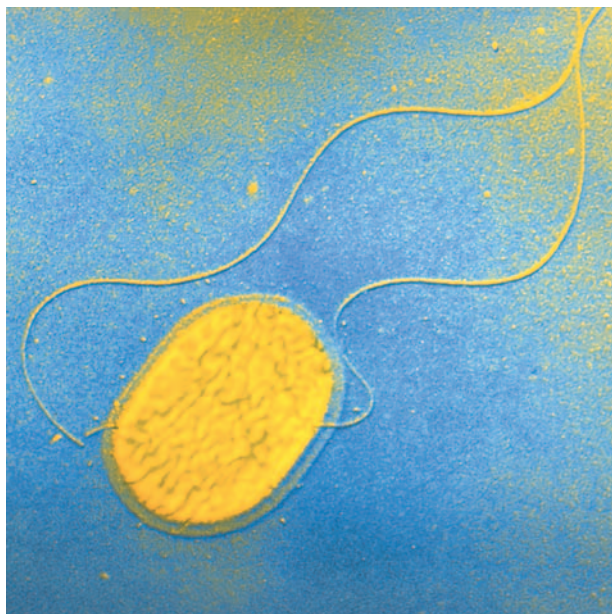


(Corbis. Reproduced by permission.)

Mary Mallon was born in Cookstown, Ireland, on September 23, 1869, to Catherine Igo and John Mallon. As a teenager, Mallon left her parents and immigrated to New York to live with an aunt and uncle. Until 1906, when George A. Soper began to study an outbreak of typhoid fever in Long Island, little was known about Mallon.

Soper was called to identify possible causes of an eruption of typhoid fever at a summer house in Oyster Bay. After examining the food and water in a futile attempt to discover contaminants, Soper decided that the disease was probably transmitted by a human carrier. He soon learned that the cook had disappeared and tracked Mallon to her new place of employment, expecting her cooperation in dealing with the matter. Soper eventually turned the case over to the New York City Department of Health. When Mallon was ultimately caught, she refused treatment and was held for three years as a threat to the public. In 1910, a judge granted her release with the stipulation that she not seek employment as a cook, since the disease was transmitted through food. Mallon agreed but, in 1915, an outbreak of typhoid at a hospital was, once again, linked to her. When Soper investigated this incident he learned that employees had nicknamed one of the cooks “Typhoid Mary.”

After Mallon was found, she was taken into custody, and spent the rest of her life at Riverside Hospital. Mallon died on November 11, 1938.



Transmission electron microscopy (TEM) scan of *Salmonella typhi*, the bacteria which causes typhoid fever in humans.
(Custom Medical Stock Photo. Reproduced by permission.)

survive ingestion by the phagocytes, and multiply within these cells. This period of time, during which the bacteria are multiplying within the phagocytes, is the 10 to 14-day incubation period of typhoid fever. When huge numbers

of bacteria fill an individual phagocyte, they spill out of the cell and into the bloodstream, where their presence begins to cause symptoms.

The presence of increasingly large numbers of bacteria in the bloodstream (**bacteremia**) is responsible for an increasingly high fever, which lasts throughout the four to eight weeks of the disease in untreated individuals. Other symptoms of typhoid fever include **constipation** (at first), extreme **fatigue**, **headache**, joint **pain**, and a rash across the abdomen known as rose spots.

The bacteria move from the bloodstream into certain tissues of the body, including the gallbladder and lymph tissue of the intestine (called Peyer's patches). The tissue's response to this invasion causes symptoms ranging from inflammation of the gallbladder (**cholecystitis**) to intestinal bleeding to actual perforation of the intestine. Perforation of the intestine refers to an actual hole occurring in the wall of the intestine, with leakage of intestinal contents into the abdominal cavity. This leakage causes severe irritation and inflammation of the lining of the abdominal cavity, which is called **peritonitis**. Peritonitis is a frequent cause of **death** from typhoid fever.

Other complications of typhoid fever include liver and spleen enlargement, sometimes so great that the spleen ruptures or bursts; anemia, or low red blood cell

KEY TERMS

Asymptomatic—A state in which a person experiences no symptoms of a disease.

Bacteremia—Bacteria in the blood.

Carrier—A person who has a particular disease agent present within his/her body, and can pass this agent on to others, but who displays no symptoms of infection.

Epidemic—A large number of cases of the same disease or infection all occurring within a short time period in a specific location.

Mononuclear phagocyte—A type of cell of the human immune system that ingests bacteria, viruses, and other foreign matter, thus removing potentially harmful substances from the bloodstream. These substances are usually then digested within the phagocyte.

Rose spots—A pinkish rash across the trunk or abdomen that is a classic sign of typhoid fever.

count due to blood loss from the intestinal bleeding; joint infections, which are especially common in patients with sickle cell anemia and immune system disorders; **pneumonia** caused by a bacterial infection— usually *Streptococcus pneumoniae*— which is able to take hold due to the patient's weakened state; heart infections; and **meningitis** and infections of the brain, which cause mental confusion and even **coma**. It may take a patient several months to recover fully from untreated typhoid fever.

Diagnosis

In some cases, the doctor may suspect the diagnosis if the patient has already developed the characteristic rose spots, or if he or she has a history of recent travel in areas with poor sanitation. The diagnosis, however, is confirmed by a **blood culture**. Samples of a patient's stool, urine, and bone marrow can also be used to grow *S. typhi* in a laboratory for identification under a microscope. Cultures are the most accurate method of diagnosis. Blood cultures usually become positive in the first week of illness in 80% of patients who have not taken **antibiotics**.

Treatment

Antibiotics are the treatment of choice for typhoid fever. As of the early 2000s, the most frequently used

drugs are ceftriaxone and cefoperazone. Ciprofloxacin is sometimes given as follow-up therapy.

Carriers of *S. typhi* must be treated even when they do not show any symptoms of the infection, because carriers are responsible for the majority of new cases of typhoid fever. Eliminating the carrier state is actually a fairly difficult task. It requires treatment with one or even two different medications over a period of four to six weeks. The antibiotics most commonly given are ampicillin (sometimes given together with probenecid) and amoxicillin. In the case of a carrier with **gallstones**, surgery may need to be performed to remove the gallbladder. This measure is necessary because typhoid bacteria are often housed in the gallbladder, where they may survive in spite of antibiotic treatment. In some patients, however, treatment with rifampin and trimethoprim-sulfamethoxazole is sufficient to eradicate the bacteria from the gallbladder without surgery.

Prognosis

The prognosis for recovery is good for most patients. In the era before effective antibiotics were discovered, about 12% of all typhoid fever patients died of the infection. Now, however, fewer than 1% of patients who receive prompt antibiotic treatment will die. The mortality rate is highest in the very young and very old, and in patients suffering from **malnutrition**. The most ominous signs are changes in a patient's state of consciousness, including stupor or coma.

Prevention

Hygienic sewage disposal systems in a community as well as proper personal hygiene are the most important factors in preventing typhoid fever. Immunizations are available for travelers who expect to visit countries where *S. typhi* is a known public health problem. Some of these immunizations provide only short-term protection (for a few months), while others may be effective for several years. Efforts are being made to develop vaccines that provide a longer period of protection with fewer side effects from the vaccine itself. The most commonly reported side effects are flu-like **muscle cramps** and abdominal pain. As of early 2004, these vaccines are also being studied as possible antibioterrorism agents.

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Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Typhus

Definition

Several different illnesses are called "typhus," all of them caused by one of the bacteria in the family Rickettsiae. Each illness occurs when the bacteria is passed to a human through contact with an infected insect.

Description

The four main types of typhus are:

- epidemic typhus
- Brill-Zinsser disease
- endemic or murine typhus
- scrub typhus

These diseases are all somewhat similar, although they vary in terms of severity. The specific type of *Rickettsia* that causes the disease also varies, as does the specific insect that can pass the bacteria along.

Epidemic typhus, which is sometimes called jail fever or louse-borne typhus, is caused by *Rickettsia prowazekii*, which is carried by body lice. When the lice feed on a human, they may simultaneously defecate.

When the person scratches the bite, the feces (which carries the bacteria) are scratched into the wound. Body lice are common in areas in which people live in overcrowded, dirty conditions, with few opportunities to wash themselves or their clothing. Because of this fact, this form of typhus occurs simultaneously in large numbers of individuals living within the same community; that is, in epidemics. This type of typhus occurs when cold weather, poverty, war, and other disasters result in close living conditions that encourage the maintenance of a population of lice living among humans. Some medical historians think that the Great **Plague** of Athens in 430 B.C. may have been epidemic typhus. Epidemic typhus is now found in the mountainous regions of Africa, South America, and Asia.

Brill-Zinsser disease is a reactivation of an earlier infection with epidemic typhus. It affects people years after they have completely recovered from epidemic typhus. When something causes a weakening of their immune system (like **aging**, surgery, illness), the bacteria can gain hold again, causing illness. This illness tends to be extremely mild.

Endemic typhus is carried by fleas. When a flea lands on a human, it may defecate as it feeds. When the person scratches the itchy spot where the flea was feeding, the bacteria-laden feces are scratched into the skin, thus causing infection. The causative bacteria is called *Rickettsia typhi*. Endemic typhus occurs most commonly in warm, coastal regions. In the United States, southern Texas and southern California have the largest number of cases.

Scrub typhus is caused by *Rickettsia tsutsugamushi*. This bacteria is carried by mites or chiggers. As the mites feed on humans, they deposit the bacteria. Scrub typhus occurs commonly in the southwest Pacific, southeast Asia, and Japan. It is a very common cause of illness in people living in or visiting these areas. It occurs more commonly during the wet season.

Causes and symptoms

The four types of typhus cause similar types of illnesses, though varying in severity.

Epidemic typhus causes fever, **headache**, weakness, and muscle aches. It also causes a rash composed of both spots and bumps. The rash starts on the back, chest, and abdomen, then spreads to the arms and legs. The worst types of complications involve swelling in the heart muscle or brain (**encephalitis**). Without treatment, this type of typhus can be fatal.

KEY TERMS

Antibody—Specialized cells of the immune system, which can recognize organisms that invade the body (such as bacteria, viruses, and fungi). The antibodies are then able to set off a complex chain of events designed to kill these foreign invaders.

Bioterrorism—The use of disease microorganisms to intimidate or terrorize a civilian population.

Endemic—Occurring naturally and consistently in a particular area.

Epidemic—A large cluster of cases all occurring at about the same time within a specific community or region.

Brill-Zinsser disease is quite mild, resulting in about a week-long fever, and a light rash similar to that of the original illness.

Endemic typhus causes about 12 days of high fever, with chills and headache. A light rash may occur.

Scrub typhus causes a wide variety of effects. The main symptoms include fever, headache, muscle aches and pains, **cough**, abdominal **pain**, **nausea and vomiting**, and **diarrhea**. Some patients experience only these symptoms. Some patients develop a rash, which can be flat or bumpy. The individual spots eventually develop crusty black scabs. Other patients go on to develop a more serious disease, in which encephalitis, **pneumonia**, and swelling of the liver and spleen (hepatosplenomegaly) occur.

Diagnosis

A number of tests exist that can determine the reactions of a patient's antibodies (immune cells in the blood) to the presence of certain viral and bacterial markers. When the antibodies react in a particular way, it suggests the presence of a rickettsial infection. Many tests require a fair amount of time for processing, so practitioners will frequently begin treatment without completing tests, simply on the basis of a patient's symptoms.

Treatment

The **antibiotics** tetracycline or chloramphenicol are used for treatment of each of the forms of typhus.

Prognosis

The prognosis depends on what types of complications an individual patient experiences. While children usually recover well from epidemic typhus, older adults may have as much as a 60% **death** rate without treatment. Brill-Zinsser, on the other hand, carries no threat of death. People usually recover uneventfully from endemic typhus, although the elderly, those with other medical problems, or people mistakenly treated with sulfa drugs may have a 1% death rate from the illness. Scrub typhus responds well to appropriate treatment, but untreated patients have a death rate of about 7%.

The relatively high death rate from untreated typhus is one reason why some researchers are concerned that its causative organisms might be used in the future as agents of bioterrorism.

Prevention

Prevention for each of these forms of typhus includes avoidance of the insects that carry the causative bacteria. Other preventive measures include good hygiene and the use of insect repellents.

Resources

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Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

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Tzanck preparation

Definition

Tzanck preparation is a rapid test done to diagnose infections caused by herpesviruses. Cells are examined under a microscope for signs of infection.

Purpose

Herpesviruses are responsible for several superficial infections. Varicella zoster virus causes **chickenpox** and **shingles**, herpes simplex type 1 causes the **common cold** sore or **fever** blister, and herpes simplex type 2 causes the sexually transmitted disease **genital herpes**. They are all characterized by blisters and ulcers.

Physicians usually can diagnose herpes infections simply by looking at the type of blisters and ulcers, and their distribution on the person's body. Sometimes laboratory evidence of herpes is needed to confirm the diagnosis. For example, herpes can be devastating to a newborn baby or a person with a weakened immune system. Treatment can begin once herpes is confirmed in a laboring mother's genital ulcers or in the skin blisters of an immunocompromised person. A lab tries to grow (culture) the virus that may be present in the blister. This lab test takes several days to complete, but the Tzanck preparation takes minutes.

Description

The Tzanck preparation is done by smearing cells taken from a fresh blister or ulcer onto a microscope slide. The cells are stained with a special stain, such as Wright's stain, and then examined under a microscope for characteristic changes caused by a herpesvirus. Herpes causes giant cells with multiple nuclei. The shape of each nucleus appears molded to fit together with those adjacent. The background of the cell looks like ground glass and contains small dark spots called inclusion bodies.

Tzanck preparation is also called a Tzanck smear, herpes stain for inclusion bodies, or inclusion bodies stain. Results are available the same or following day, often within minutes.

KEY TERMS

Herpes—A family of viruses including herpes simplex types 1 and 2, and herpes zoster (also called varicella zoster). Herpes viruses cause several infections, all characterized by blisters and ulcers, including chickenpox, shingles, genital herpes, and cold sores or fever blisters.

Preparation

A fresh blister is opened with a scalpel or sterile needle. The physician scrapes the base of the blister with the scalpel, gathers as much cellular material as possible, and gently spreads it on a microscope slide.

Normal results

A normal smear shows no evidence of a herpes infection. This test may also have false negatives. Studies have shown that the Tzanck preparation shows signs of infection in only 50–79% of people with a herpes infection. A negative Tzanck preparation may have to be confirmed by a herpes culture.

Abnormal results

A smear that shows evidence of herpes infection does not distinguish between the various infections caused by herpes virus. The physician uses the person's symptoms and other clinical findings to distinguish between these infections. In certain cases, the physician will follow a positive Tzanck smear with a culture for confirmation.

Resources

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Nancy J. Nordenson

Tzanck smear see **Tzanck preparation**

U

Ulcer surgery

Definition

Ulcer surgery is a procedure used to cure peptic ulcer disease when medications have failed.

Purpose

Ulcer surgery is used to relieve a present peptic ulcer disease and to prevent recurrence of it.

Surgery is usually required if the ulcer is in one of the following states:

- perforated and overflowed into the abdomen
- scarred or swelled so much that the bowel is obstructed
- acute bleeding
- defied all other types of treatment

The need for ulcer surgery has diminished greatly over the past 20–30 years due to the discovery of two new classes of drugs and the presence of the causal germ *Helicobacter pylori* in the stomach. The drugs are the H2 blockers such as cimetidine and ranitidine and the **proton pump inhibitors** such as omeprazole. These effectively arrest acid production. *H. pylori* can be eliminated from most patients with a combination of **antibiotics** and bismuth.

Precautions

There is a tumor of the pancreas that produces a hormone called gastrin. Gastrin causes ulcers by stimulating acid production. If this disease—Zollinger-Ellison syndrome—does not respond to medical treatment, either the tumor or the entire stomach must be removed.

Description

The two primary goals of ulcer surgery, elimination of the current problem and prevention of future problems bring with them a third problem—to perpetuate the normal function of the bowel. The vagus nerves relax the pylorus, allowing the stomach to empty. Cutting the vagus nerves, while reducing the stomach's acid production, also prevents stomach emptying. Therefore, the procedures described must guarantee stomach emptying along with their other goals.

Total gastrectomy

Removing the entire stomach is done only for resistant Zollinger-Ellison syndrome or extensive cancers.

Antrectomy

The lower half of the stomach makes most of the acid and gets all the peptic ulcers above the duodenum. Removing it leaves little place for ulcers to form and little acid to produce them.

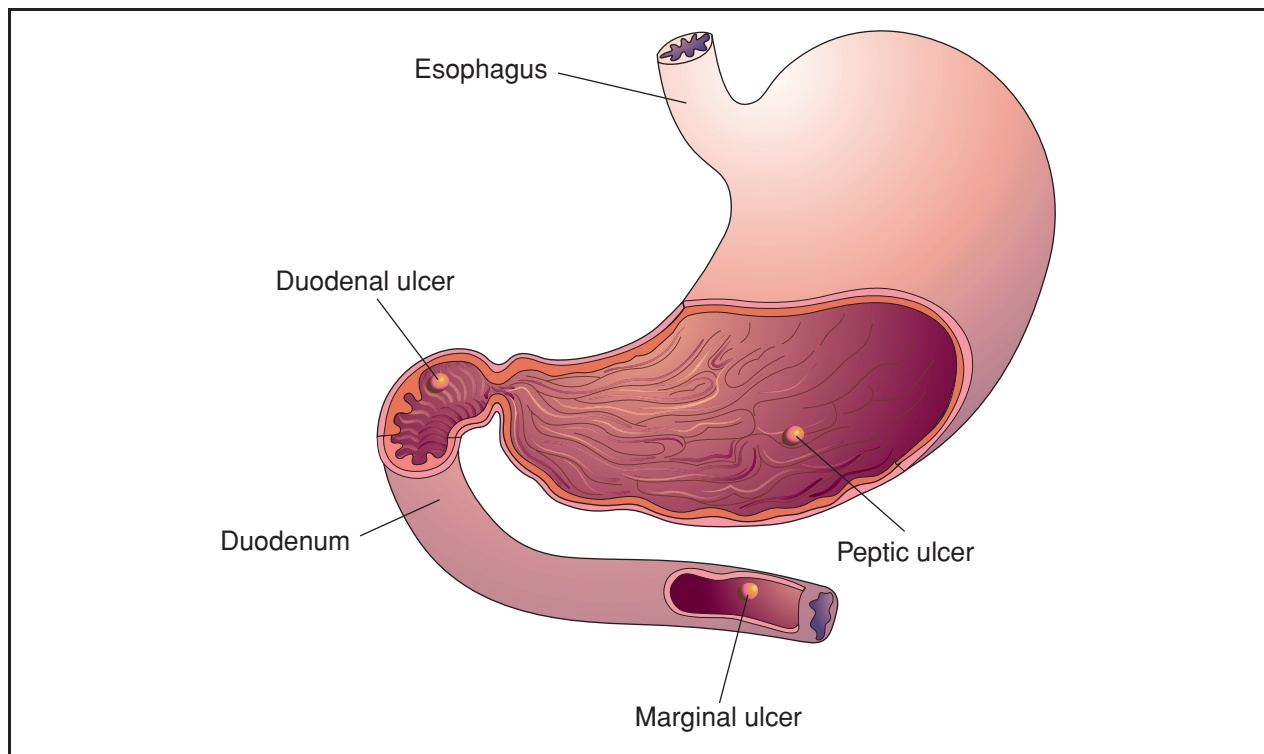
Vagotomy

Cutting the vagus nerves can be done in three ways:

- the main nerves can be cut completely as they enter the abdomen from the chest
- the branches that go to the stomach can be cut as they leave the main nerves
- the tiny branches that stimulate acid production can be cut on the surface of the stomach

Pyloroplasty

Opening up the valve at the outlet of the stomach guarantees that the stomach can empty, even without



Common sites of ulcers in the human stomach. The need for ulcer surgery has diminished over the past 20-30 years due to the discovery that *Helicobacter pylori*, an infectious bacterium, plays a major role in causing ulcers. *H. pylori* can be eliminated from most patients with a combination of antibiotics and bismuth. (Illustration by Electronic Illustrators Group.)

vagus nerve stimulation. **Pyloroplasty** is ordinarily done by cutting across the muscle that surrounds the outlet. It can also be done by passing a balloon down from the mouth and inflating it forcefully to stretch out the pylorus (opening from the stomach to the intestine).

Close perforation

For some patients all that can be done is to close the hole in the bowel and wait for the patient to recover before initiating corrective surgery.

Billroth I and II

After removing a piece of the stomach, the remainder must be reattached to the rest of the bowel. Simply joining the upper stomach back to the duodenum is called a Billroth I or gastroduodenostomy. It is sometimes better to attach the stomach with another piece of bowel (the jejunum), creating a “y” with the bile drainage and the duodenum forming the second branch of the “y.” This part of the procedure is called a gastrojejunostomy. A gastroenterostomy is a more general term for connecting the stomach with any piece of bowel.

A selective **vagotomy** can be done alone. A complete vagotomy requires either a pyloroplasty or antrectomy. An antrectomy must be reconnected with either a Billroth I or a Billroth II.

Some of these procedures are now being done through a laparoscope.

Risks

All of these procedures carry risks, generally in proportion to their benefits. The more extensive surgeries such as vagotomy and antrectomy with Billroth II reconnection have the highest success rate and the highest complication rate.

Complications include:

- Diarrhea after a meal
- Dumping syndrome occurring after a meal and characterized by sweating, abdominal **pain**, **vomiting**, lightheadedness, and diarrhea
- Hypoglycemia after a meal
- Alkaline reflux **gastritis** marked by abdominal pain, vomiting of bile, diminished appetite, and iron-deficiency anemia

KEY TERMS

Gastrin—A type of hormone that produces gastric juice.

Hypoglycemia—An abnormal decrease in blood sugar level.

Jejunum—Section of the small intestine.

Laparoscope—A pencil-thin telescope that allows surgery to be done through half-inch incisions.

Pylorus—The opening from the stomach to the intestine.

Vagus nerve—Cranial nerves that supply the internal organs (viscera).

Zollinger-Ellison syndrome—A syndrome marked by peptic ulcers and gastrinomas in the pancreas.

- Recurrence of an ulcer
- Malabsorption of necessary nutrients, especially iron, in patients who have had all or part of their stomachs removed

Resources

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J. Ricker Polsdorfer, MD

Ulcerative colitis

Definition

Ulcerative colitis is a form of inflammatory bowel disease (IBD). It causes swelling, ulcerations, and loss of function of the large intestine.

Description

The primary problem in IBD is inflammation, as the name suggests. Inflammation is a process that often occurs in order to fight off foreign invaders in the body, including viruses, bacteria, and fungi. In response to such organisms, the body's immune system begins to produce a variety of cells and chemicals intended to stop the invasion. These immune cells and

chemicals, however, also have direct effects on the body's tissues, resulting in heat, redness, swelling, and loss of function. No one knows what starts the cycle of inflammation in IBD, but the result is a swollen, boggy intestine.

In ulcerative colitis, the inflammation affects the lining of the rectum and large intestine. It is thought that the inflammation begins in the last segment of the large intestine, which empties into the rectum (sigmoid colon). This inflammation may spread through the entire large intestine, but only rarely affects the very last section of the small intestine (ileum). The rest of the small intestine remains normal.

Ulcerative colitis differs from **Crohn's disease**, which is a form of IBD that affects both the small and large intestines. The inflammation of ulcerative colitis occurs only in the lining of the intestine (unlike Crohn's disease which affects all of the layers of the intestinal wall). As the inflammation continues, the tissue of the intestine begins to slough off, leaving pits (ulcerations) which often become infected.

Like Crohn's disease, ulcerative colitis occurs in all age groups, with the most common age of diagnosis being 15–35 years of age. Men and women are affected equally. Whites are more frequently affected than other racial groups, and people of Jewish origin have 3–6 times greater likelihood of suffering from any IBD. IBD is familial; an IBD patient has a 20% chance of having other relatives who are fellow sufferers.

Causes and symptoms

No specific cause of ulcerative colitis has been identified. Although no organism (virus, bacteria, or fungi) has been found to set off the cycle of inflammation that occurs in ulcerative colitis, some researchers continue to suspect that some such organism is responsible for initiating the cycle. Other researchers are concentrating on identifying some change in the cells of the colon that would make the body's immune system accidentally begin treating those cells as foreign invaders. Other evidence for such a disorder of the immune system includes the high number of other immune disorders that tend to accompany ulcerative colitis.

The first symptoms of ulcerative colitis are abdominal cramping and **pain**, a sensation of urgent need to have a bowel movement (defecate), and blood and pus in the stools. Some patients experience diarrhea, fever, and weight loss. If the **diarrhea** continues, signs of severe fluid loss (**dehydration**) begin to appear, including low blood pressure, fast heart rate, and **dizziness**.



A specimen of a colon indicating ulcerative colitis. (Photo Researchers, Inc. Reproduced by permission.)

Severe complications of ulcerative colitis include perforation of the intestine (in which the wall of the intestine develops a hole), toxic dilation of the colon (in which the colon become quite large in diameter), and the development of **colon cancer**.

Intestinal perforation occurs when long-standing inflammation and ulceration of the intestine weakens the wall to such a degree that a hole occurs. This is a life-threatening complication, because the contents of the intestine (which under normal conditions contains a large number of bacteria) spill into the abdomen. The presence of bacteria in the abdomen can result in a massive infection called **peritonitis**.

Toxic dilation of the colon is thought to occur because the intestinal inflammation interferes with the normal function of the muscles of the intestine. This allows the intestine to become lax, and its diameter begins to increase. The enlarged diameter thins the walls further, increasing the risk of perforation and peritonitis. When the diameter of the intestine is quite large, and infection is present, the condition is referred to as “toxic megacolon.”

Patients with ulcerative colitis have a significant risk of developing colon **cancer**. This risk seems to begin around 10 years after diagnosis of ulcerative colitis. The risk becomes statistically greater every year:

- At 10 years, the risk of cancer is about 0.5–1%.
- At 15 years, the risk of cancer is about 12%.
- At 20 years, the risk of cancer is about 23%.
- At 24 years, the risk of cancer is about 42%.

The overall risk of developing cancer seems to be greatest for those patients with the largest extent of intestine involved in ulcerative colitis.

Patients with ulcerative colitis also have a high chance of experiencing other disorders, including inflammation of the joints (arthritis), inflammation of the vertebrae (spondylitis), ulcers in the mouth and on the skin, the development of painful, red bumps on the skin, inflammation of several areas of the eye, and various disorders of the liver and gallbladder.

Diagnosis

Diagnosis is first suspected based on the symptoms that a patient is experiencing. Examination of the stool will usually reveal the presence of blood and pus (white blood cells). Blood tests may show an increase in the number of white blood cells, which is an indication of inflammation occurring somewhere in the body. The blood test may also reveal anemia, particularly when a great deal of blood has been lost in the stool.

The most important method of diagnosis is endoscopy, during which a doctor passes a flexible tube with a tiny, fiberoptic camera device through the rectum and into the colon. The doctor can then examine the lining of the intestine for signs of inflammation and ulceration that might indicate ulcerative colitis. A tiny sample (biopsy) of the intestine will be removed through the endoscope, which will be examined under a microscope for evidence of ulcerative colitis. Because of the increased risk of cancer in patients with ulcerative colitis, endoscopic exam will need to be repeated frequently. Biopsies should be taken regularly, to closely monitor the intestine for the development of cancer or precancerous changes.

X-ray examination is helpful to determine the amount of intestine affected by the disease. However, x-ray examinations requiring the use of barium should be delayed until treatment has begun. Barium is a chalky solution that the patient drinks or is administered through the rectum and into the intestine (enema). The presence of barium in the intestine allows more detail to be seen on x-ray pictures. However, because of the risk of intestinal perforation in ulcerative colitis, most doctors begin treatment before stressing the wall of the intestine with the barium solution.

Treatment

Treatment for ulcerative colitis addresses the underlying inflammation, as well as the problems occurring due to continued diarrhea and blood loss.

Inflammation is treated with a drug called sulfasalazine. Sulfasalazine is made up of two parts. One

KEY TERMS

Endoscopy—A type of medical examination in which an instrument called an endoscope is passed into an area of the body (the bladder or intestine, for example). The endoscope usually has a fiberoptic camera, which allows a greatly magnified image to be projected onto a video screen, to be viewed by the operator. Many endoscopes also allow the operator to retrieve a small sample (biopsy) of the area being examined, in order to more closely view the tissue under a microscope.

Immune system—The system of the body that is responsible for producing various cells and chemicals that fight off infection by viruses, bacteria, fungi, and other foreign invaders. In autoimmune disease, these cells and chemicals are turned against the body itself.

Inflammation—The result of the body's attempts to fight off and wall off an area that is infected. Inflammation results in the classic signs of redness, heat, swelling, and loss of function.

part is related to the sulfa **antibiotics**; the other part is a form of the anti-inflammatory chemical salicylic acid (related to **aspirin**). Sulfasalazine is not well-absorbed from the intestine, so it stays mostly within the intestine, where it is broken down into its components. It is believed to be primarily the salicylic acid component that is active in treating ulcerative colitis, by fighting inflammation. For patients who do not respond to sulfasalazine, steroid medications (such as prednisone) are the next choice.

Depending on the degree of blood loss, a patient with ulcerative colitis may require blood transfusions and fluid replacement through a needle in the vein (intravenous or IV). Medications that can slow diarrhea must be used with great care, because they may actually cause the development of toxic megacolon.

A patient with toxic megacolon requires close monitoring and care in the hospital. He or she will usually be given steroid medications through an IV, and may be put on antibiotics. If these measures do not improve the situation, the patient will have to undergo surgery to remove the colon. This is done because the risk of **death** after perforation of toxic megacolon is greater than 50%.

Similarly, a patient with proven cancer of the colon, or even a patient who shows certain signs

thought to indicate a precancerous condition, will need his or her colon removed. Removal of the colon is called a colectomy. When a colectomy is performed, a piece of the small intestine (ileum) is pulled through an opening in the abdomen. This bit of intestine is fashioned surgically to allow a special bag to be placed over it, in order to catch the body's waste (feces) which no longer can be passed through the large intestine and out of the anus. This opening, which will remain for the duration of the patient's life, is called an ileostomy.

Prognosis

Remission refers to a disease becoming inactive for a period of time. The rate of remission of ulcerative colitis (after a first attack) is nearly 90%. Those individuals whose colitis is confined primarily to the left side of the large intestine have the best prognosis. Those individuals with extensive colitis, involving most or all of the large intestine, have a much poorer prognosis. Recent studies show that about 10% of these patients will have died by 10 years after diagnosis. About 20–25% of all ulcerative colitis patients will require colectomy. Unlike the case for patients with Crohn's disease, however, such radical surgery results in a cure of the disease.

Resources

ORGANIZATIONS

Crohn's and Colitis Foundation of America, Inc. 386 Park Avenue South, 17th Floor, New York, NY 10016-8804. (800) 932-2423.

Rosalyn Carson-DeWitt, MD

Ulcers (digestive)

Definition

In general, an ulcer is any eroded area of skin or a mucous membrane, marked by tissue disintegration. In common usage, however, ulcer usually is used to refer to disorders in the upper digestive tract. The terms ulcer, gastric ulcer, and peptic ulcer often are used loosely and interchangeably. Peptic ulcers can develop in the lower part of the esophagus, the stomach, the first part of the small intestine (the duodenum), and the second part of the small intestine (the jejunum).

Description

It is estimated that 2% of the adult population in the United States has active peptic ulcers, and that about 10% will develop ulcers at some point in their lives. There are about 500,000 new cases of peptic ulcer in the United States every year, with as many as 4 million recurrences. The male/female ratio for ulcers of the digestive tract is 3:1.

The most common forms of peptic ulcer are duodenal and gastric. About 80% of all ulcers in the digestive tract are duodenal ulcers. This type of ulcer may strike people in any age group but is most common in males between the ages of 20 and 45. The incidence of duodenal ulcers has dropped over the past 30 years. Gastric ulcers account for about 16% of peptic ulcers. They are most common in males between the ages of 55 and 70. The single most common cause of gastric ulcers is the use of **nonsteroidal anti-inflammatory drugs**, or NSAIDs. The widespread use of NSAIDs is thought to explain why the incidence of gastric ulcers in the United States is rising.

Causes and symptoms

Causes of peptic ulcers

There are three major causes of peptic ulcers: infection, certain types of medication, and disorders that cause oversecretion of stomach juices.

HELICOBACTER PYLORI INFECTION. *Helicobacter pylori* is a rod-shaped gram-negative bacterium that lives in the mucous tissues that line the digestive tract. Infection with *H. pylori* is the most common cause of duodenal ulcers. About 95% of patients with duodenal ulcers are infected with *H. pylori*, as opposed to only 70% of patients with gastric ulcers.

USE OF NONSTEROIDAL ANTI-INFLAMMATORY DRUGS (NSAIDS). Nonsteroidal anti-inflammatory drugs, or NSAIDs, are painkillers that many people use for headaches, sore muscles, arthritis, menstrual cramps, and similar complaints. Many NSAIDs are available without prescriptions. Common NSAIDs include aspirin, ibuprofen (Advil, Motrin), flurbiprofen (Ansaïd, Ocufen), ketoprofen (Orudis), and indomethacin (Indacin). Chronic NSAID users have 40 times the risk of developing a gastric ulcer as nonusers. Users also are three times more likely than nonusers to develop bleeding or fatal complications of ulcers. **Aspirin** is the NSAID that is most likely to cause ulcers.

MISCELLANEOUS SYNDROMES AND DISORDERS. Fewer than 5% of peptic ulcers are due to these disorders. They include Zollinger-Ellison

syndrome, a disorder in which small tumors, called gastrinomas, secrete a hormone (gastrin) that stimulates the production of digestive juices. Because of this excess secretion, these disorders are sometimes called hypersecretory syndromes.

OTHER RISK FACTORS. **Smoking** is an important risk factor that increases a patient's chance of developing an ulcer, decreases the body's response to therapy, and increases the chances of dying from ulcer complications. Blood type appears to be a predisposing factor for ulcer location; people with type A blood are more likely to have gastric ulcers, while those with type O are more likely to develop duodenal ulcers. The role of emotional **stress** in ulcer development is currently debated. Present research indicates that an individual's attitudes toward stress, rather than the amount of stress by itself, is a better predictor of vulnerability to peptic ulcers. Preferences for high-fat or spicy foods do not appear to be significant risk factors.

Symptoms

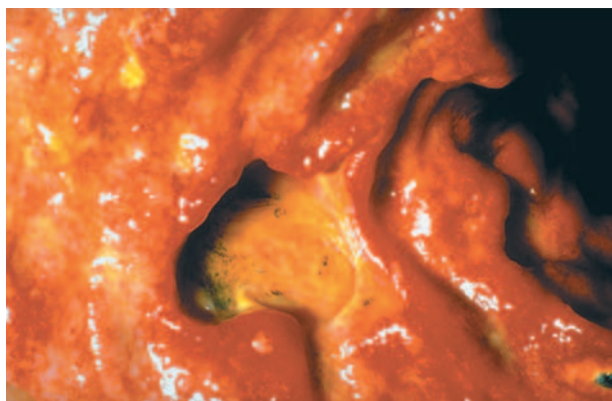
GASTRIC ULCERS. The symptoms of gastric ulcers include feelings of **indigestion** and **heartburn**, weight loss, and repeated episodes of gastrointestinal bleeding. Ulcer **pain** often is described as gnawing, dull, aching, or resembling hunger pangs. The patient may be nauseated and suffer loss of appetite. About 30% of patients with gastric ulcers are awakened by pain at night. Many patients have periods of chronic ulcer pain alternating with symptom-free periods that last for several weeks or months. This characteristic is called periodicity.

DUODENAL ULCERS. The symptoms of duodenal ulcers include heartburn, stomach pain relieved by eating or antacids, weight gain, and a burning sensation at the back of the throat. The patient is most likely to feel discomfort two to four hours after meals, or after having citrus juice, coffee, or aspirin. About 50% of patients with duodenal ulcers awake during the night with pain, usually between midnight and three a.m. A regular pattern of ulcer pain associated with certain periods of day or night or a time interval after meals is called rhythmicity.

Not all digestive ulcers produce symptoms; as many as 20% of ulcer patients have so-called painless or silent ulcers. Silent ulcers occur most frequently in the elderly and in chronic NSAID users.

Complications

Between 10–20% of peptic ulcer patients develop complications at some time during the course of their



A clinical photograph of a large duodenal ulcer after surgical removal. The ulcer is the prominent triangular crater at center. (Photo Researchers, Inc. Reproduced by permission.)

illness. All of these are potentially serious conditions. Complications are not always preceded by diagnosis of or treatment for ulcers; as many as 60% of patients with complications have not had prior symptoms.

HEMORRHAGE. Bleeding is the most common complication of ulcers. It may result in anemia, vomiting blood (hematemesis), or the passage of bright red blood through the rectum (melena). About half of all cases of bleeding from the upper digestive tract are caused by ulcers. The mortality rate from ulcer hemorrhage is 6–10%.

PERFORATION. About 5% of ulcer patients develop perforations, which are holes in the duodenal or gastric wall through which the stomach contents can leak out into the abdominal cavity. The incidence of perforation is rising because of the increased use of NSAIDs, particularly among the elderly. The signs of an ulcer perforation are severe pain, **fever**, and tenderness when the doctor touches the abdomen. Most cases of perforation require emergency surgery. The mortality rate is about 5%.

PENETRATION. Ulcer penetration is a complication in which the ulcer erodes through the intestinal wall without digestive fluid leaking into the abdomen. Instead, the ulcer penetrates into an adjoining organ, such as the pancreas or liver. The signs of penetration are more severe pain *without* rhythmicity or periodicity, and the spread of the pain to the lower back.

OBSTRUCTION. Obstruction of the stomach outlet occurs in about 2% of ulcer patients. It is caused by swelling or scar tissue formation that narrows the opening between the stomach and the duodenum (the pylorus). Over 90% of patients with obstruction have recurrent **vomiting** of partly digested or

undigested food; 20% are seriously dehydrated. These patients also usually feel full after eating only a little food, and may lose weight.

Diagnosis

Physical examination and patient history

The diagnosis of peptic ulcers should rarely be made on the basis of a physical examination alone. However, a 2003 report showed that many ulcer diagnoses made based solely on physical exams actually are only **dyspepsia**, or upper abdominal pain and discomfort not caused by ulcers. The only significant finding may be mild soreness in the area over the stomach when the doctor presses (palpates) it. The doctor is more likely to suspect an ulcer if the patient has one or more of the following risk factors:

- male sex
- age over 45
- recent weight loss, bleeding, recurrent vomiting, jaundice, back pain, or anemia
- history of using aspirin or other NSAIDs
- history of heavy smoking
- family history of ulcers or stomach cancer

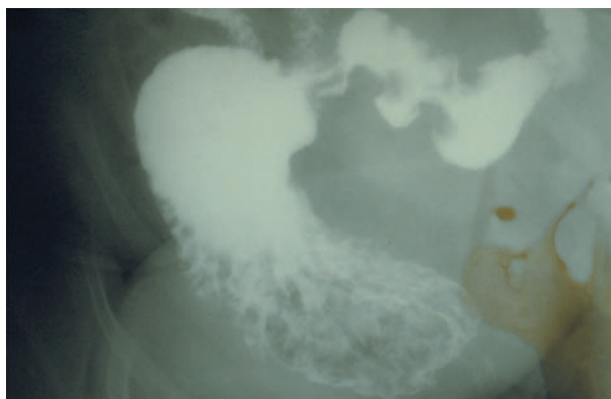
Endoscopy and imaging studies

An endoscopy is considered the best procedure for diagnosing digestive ulcers and for taking samples of stomach tissue for biopsies. An endoscope is a slender tube-shaped instrument that allows the doctor to view the tissues lining the stomach and duodenum. Duodenal ulcers are rarely malignant. If the ulcer is in the stomach, however, the doctor will take a tissue sample because 3–5% of gastric ulcers are malignant. Radiological studies are sometimes used instead of endoscopy because they are less expensive, more comfortable for the patient, and are 85% accurate in detecting malignancies.

Laboratory tests

BLOOD TESTS. Blood tests usually give normal results in ulcer patients without complications. They are useful, however, in evaluating anemia from a bleeding ulcer or a high white cell count from perforation or penetration. Serum gastrin levels can be used to screen for Zollinger-Ellison syndrome.

TESTS FOR *HELICOBACTER PYLORI*. It is important to test for *H. pylori* because almost all ulcer patients who are not taking NSAIDs are infected.



A barium x-ray image of a gastric ulcer. (Photograph by Bates, M.D., Custom Medical Stock Photo. Reproduced by permission.)

Noninvasive tests include blood tests for immune response and a breath test. In the breath test, the patient is given an oral dose of radiolabeled urea. If *H. pylori* is present, it will react with the urea and the patient will exhale radiolabeled carbon dioxide. Invasive tests for *H. pylori* include tissue biopsies and cultures performed from fluid obtained by endoscopy.

Treatment

Medications

Most drugs that are currently given to treat ulcers work either by lowering the rate of stomach acid secretion or by protecting the mucous tissues that line the digestive tract.

ANTISECRETORY DRUGS. Medications that lower the rate of stomach acid secretions fall into two major categories: proton pump inhibitors, which bind an enzyme that secretes stomach acid, and H₂ receptor antagonists, which work by reducing intracellular acid secretion. The **proton pump inhibitors** include omeprazole (Prilosec) and lansoprazole (Prevacid). The H₂ receptor antagonists include ranitidine (Zantac), cimetidine (Tagamet), famotidine (Pepcid), and nizatidine (Axid). Both types of drugs have few serious side effects and appear to be safe for long-term use.

PROTECTIVE DRUGS. The drugs that are currently used to protect the stomach tissues are sucralfate (Carafate), which forms a pastelike substance that clings to the mucous tissues and prevents further damage from stomach acid; and bismuth preparations. A third type of protective drug includes misoprostol (Cytotec), which is often given to patients with ulcers caused by NSAIDs.

KEY TERMS

Duodenum—The first of the three segments of the small intestine. The duodenum connects the stomach and the jejunum. Most peptic ulcers are in the duodenum.

Helicobacter pylori—A gram-negative rod-shaped bacterium that lives in the tissues of the stomach and causes inflammation of the stomach lining.

Zollinger-Ellison syndrome—A disorder characterized by the presence of tumors (gastrinomas) that secrete a hormone (gastrin), which stimulates the production of digestive juices.

Surgery

Surgical treatment of ulcers is generally used only for complications and suspected malignancies. The most common surgical procedures that are used are vagotomies, in which the connections of the vagus nerve to the stomach are cut in order to reduce acid secretion; and antrectomies, which involve the removal of a part of the stomach (the antrum).

Eradication of *Helicobacter pylori*

Most doctors presently recommend treatment to eliminate *H. pylori* in order to prevent ulcer recurrences. Without such treatment, ulcers recur at the rate of 80% per year. A 2003 report showed that eradication *H. pylori* alone usually prevents recurring bleeding ulcers. The usual regimen used to eliminate the bacterium is a combination of tetracycline, bismuth subsalicylate (Pepto-Bismol), and metronidazole (Metizol).

Alternative treatment

Alternative treatments can relieve symptoms and promote healing of ulcers. A primary goal of these treatments is to rebalance the stomach's hydrochloric acid output and to enhance the mucosal lining of the stomach.

Food allergies have been pointed to as a major cause of peptic (stomach) ulcers. An elimination/challenge diet can help identify the allergenic food(s) and continued elimination of these foods can assist in healing the ulcer. People with ulcers should not take aspirin. They also should stop smoking, since smoking irritates the mucosal lining of the stomach. **Antacids**

should be avoided by anyone with an ulcer, because they can cause a rebound effect of increasing gastric acid secretion, as well as deplete vital nutrients necessary for healing. Stress reduction is also important for ulcer sufferers.

Botanical medicine offers a variety of remedies that may be helpful in ulcer treatment. Deglycyrrhizinated licorice or DGL, in a chewable or powder form, can help heal the mucous membranes and increase mucous so that it mixes with saliva to protect the membranes. Raw cabbage juice, high in glutamic acid, is very effective in healing an ulcer (one quart per day in divided doses). Soothing herbs, such as plantain (*Plantago major*), marsh mallow (*Althaea officinalis*), and slippery elm (*Ulmus fulva*); astringent herbs, such as geranium (*Pelargonium odoratissimum*); and the antimicrobial herb goldenseal (*Hydrastis canadensis*) can all be effective. Nutritionists advise taking antioxidant nutrients, including **vitamins A, C, and E**, zinc, and selenium.

Prognosis

The prognosis for recovery from ulcers is good for most patients. Very few ulcers fail to respond to the medications that are currently used to treat them. Recurrences can be eliminated completely or cut to 5% by eradication of *H. pylori*. Most patients who develop complications recover without problems even when emergency surgery is necessary.

Prevention

Strategies for the prevention of ulcers or their recurrence include the following:

- eradication of *H. pylori* in patients already diagnosed with ulcers
- giving misoprostol to patients who must take NSAIDs
- avoiding unnecessary use of aspirin and NSAIDs
- giving up smoking
- cutting down on alcohol, tea, coffee, and sodas containing caffeine.

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American College of Gastroenterology. 4900-B South Thirty-First St., Arlington, VA 22206-1656. (703) 820-7400. <<http://www.acg.cgi.gi.org/acghome/html>>.
Digestive Health Initiative. 7910 Woodmont Ave., #914, Bethesda, MD 20814. (800) 668-5237. <<http://www.gastro.org/dhi.html>>.

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Ultrasonic lithotripsy see **Lithotripsy**

Ultraviolet light treatment

Definition

Ultraviolet light treatment uses a particular band of the nonvisible light spectrum to treat psoriasis and a variety of other skin diseases. It can be used alone or in combination with other medications applied directly to the skin or taken internally.

Purpose

Ultraviolet (UV) light treatment is used primarily in cases of severe **psoriasis** that have not responded to other medications or in cases affecting large portions of the body. Patients will typically receive a series of 3–5 weekly treatments for a month or more to bring their psoriasis symptoms into check. They may also receive periodic maintenance treatments to prevent recurrence of their psoriasis. Other skin conditions treated with UV light treatments are **vitiligo**, a condition in which people lose pigmentation in large patches of their skin, and **atopic dermatitis**, an allergy-related skin condition that produces itchy, reddish, and scaly patches of skin.

Precautions

Exposure to UV radiation is known to prematurely age the skin over time and increase the risk of skin **cancer**. These potential effects should be weighed against the potential benefits of the treatment. A history will be taken regarding sun exposure and burning, medications, such as **diuretics**, that may increase UV sensitivity exposure, and any history of skin cancers. Sometimes, UV light treatments are given in combination with photosensitizing agents, which maximize UV's effects on the skin. Patients who receive these agents, called psoralens, must take care to avoid exposure to sunlight, which also contains UV radiation. Exposure to UV radiation can also cause **cataracts** and

other eye damage, so the patient's eyes must be adequately shielded during the treatments.

Description

UV light treatment can employ one of two bands of the ultraviolet spectrum: ultraviolet A (UVA), and ultraviolet B (UVB). Patients receive full-body treatments in special light boxes; smaller areas of the skin are sometimes treated with hand-held devices.

UVB treatment

Psoriasis is the most common skin disease treated with UVB light treatment. Its mechanism of action remains unclear, but investigators speculate it may kill abnormal skin cells or alter immune system reactions in the skin. Most patients require 18–30 treatments before substantial improvement or complete clearing is seen. The intensity of the UV applied will vary depending on the patient's skin type. Fair-skinned patients will start with a relatively weaker dose; dark-skinned patients, a stronger dose. Physicians will first expose a small area of skin to UVB to determine the minimum erythema dose (MED), the minimum amount of UVB that produces redness 24 hours after exposure. Patients will be exposed for short times early in the treatment cycle, but these times will gradually increase over time.

The Goeckerman regimen, a treatment that combines UVB light with coal tar applied to the skin, is among the oldest and most frequently used treatments for patients with moderate to severe psoriasis. The coal tar is a photosensitizing agent, and, when it interacts with UVB, it appears to limit the abnormal turnover of skin cells characteristic of psoriasis. Although treatments with UVB and coal tar are highly effective, many patients dislike the smell. Some investigators believe that the use of petroleum jelly or other emollients is just as effective as the coal tar preparations.

In addition to their UVB treatments, many patients will receive such systemic agents as methotrexate, a drug used in severe case of psoriasis, and certain vitamin A derivatives called retinoids. A newer retinoid called bexarotene (Targretin), which was originally developed to treat **cutaneous T-cell lymphoma**, shows promise as a treatment for psoriasis in combination with UVB therapy.

Another new development in UV therapy is the use of a laser as the source of the UVB radiation. The type of laser that is used is known as a 308-nm excimer laser, which uses a specific mixture of gases to produce high-intensity, short pulses of UV light.

PUVA treatment

Psoralens are photosensitizing agents found in plants. They have been known since ancient Egypt but have only been available in a chemically synthesized form since the 1970s. Psoralens are taken systemically or can be applied directly to the skin. The psoralens allow a relatively lower dose of UVA to be used. When they are combined with exposure to UVA in PUVA, they are highly effective at clearing psoriasis. Like UVB light treatments, the reason remains unclear, though investigators speculate there may be similar effects on cell turnover and the skin's immune response.

Choosing the proper dose for PUVA is similar to the procedure followed with UVB. The physician can choose a dose based on the patient's skin type. Often, however, a small area of the patient's skin will be exposed to UVA after ingestion of a psoralen. The dose of UVA that produces uniform redness 72 hours later, called the minimum phototoxic dose (MPD), becomes the starting dose for treatment.

Some patients experience **nausea** and **itching** after ingesting the psoralen compound. For these patients "bath PUVA" may be a good option.

Preparation

No major preparation is required for UV light treatments. Areas of the skin that are especially sensitive to the effects of UV light, such as the groin, backside, or face, are shielded during the treatments. Areas not affected by psoriasis are also covered. Special goggles are worn to protect the eyes. Some physicians apply an emollient, such as petroleum jelly, to the skin or other topical agents, such as coal tar, to enhance the results. In PUVA treatments, the psoralen is usually taken one hour before the treatment.

Aftercare

No major aftercare is required following UV light treatments. Patients, however, must take great care to limit or eliminate other exposures to UV radiation, such as from sunlight or tanning beds, because of the increased risk of premature **aging** of the skin and the development of skin cancers. Patients should monitor their skin closely for any signs of precancerous or cancerous skin growths in the future.

Risks

People who receive UV light treatments are at higher risk of premature aging of the skin, and of developing skin cancer. These risks should be balanced

KEY TERMS

Erythema—The medical term for redness of the skin produced by congestion of the capillaries in the skin.

Goeckerman regimen—UVB light therapy combined with topical coal-tar preparations.

Minimum erythema dose—The minimum amount of UVB that produces redness 24 hours after exposure. It is the starting dose for UVB light treatments.

Minimum phototoxic dose—The dose of UVA that produces uniform redness 72 hours after ingesting a psoralen compound. It becomes the starting dose for PUVA treatment.

Psoralen—A family of photosensitizing chemicals that can be found in lemons, celery, and other plants. Chemically synthesized versions are used to augment the effects of UVA light treatments.

PUVA treatments—Treatments with the photosensitizers called psoralens and UVA.

Ultraviolet light—A portion of the light spectrum not visible to the eye. Two bands of the UV spectrum, UVA and UVB, are used to treat psoriasis and other skin diseases.

against the benefits of treatment. Patients must also take care to limit or eliminate their exposure to other sources of UV radiation, especially if they are taking a psoralen compound in addition to receiving the UV treatments.

Normal results

Psoriasis will normally show significant improvement to complete healing with three to five UVB treatments a week for about four to five weeks. PUVA treatments may require a bit longer to take effect, but because the overall dosage of UV is lower, they are thought by some investigators to be a safer alternative to UVB treatments.

Abnormal results

Modern light boxes carefully control the dosage of UV radiation and the exposure time. Overdose or overexposure is possible, however, and can lead to severe **burns**. It is important to choose a treatment provider who is experienced in the technique. It is also important to tell the physician about all medications being taken by the patient. Some medications, either alone or in combination with a psoralen, can provoke

an extreme reaction to UV radiation. One medication that has been shown to be helpful in treating burns caused by overexposure to UV radiation is a gel containing a platelet-activating antagonist factor, or PAF, known as WEB 2086.

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ORGANIZATIONS

- American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. <<http://www.aad.org>>.
- National Psoriasis Foundation. 6600 SW 92nd Ave., Suite 300, Portland OR 97223-7195. (503) 244-7404. <<http://www.psoriasis.org>>.

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Umbilical cord blood banking

Definition

Umbilical cord blood banking is the practice of preserving for future use fetal blood that remains in the umbilical cord at the time of birth.

Purpose

During **pregnancy**, a developing fetus is attached to its mother's placenta by the umbilical cord. At

birth, this cord is cut, and the placenta is expelled from the mother. Some fetal blood remains in the umbilical cord. Normally, this blood, along with the cord and the placenta, are simply disposed of. However, now there are ways to freeze the cord blood for future use.

Umbilical cord blood is special because it contains a lot of cells called hematopoietic stem cells. A stem cell is an unspecialized master cell that can develop into several different kinds of specialized cells. Hematopoietic stem cells normally develop into red blood cells that carry oxygen, white blood cells that fight infection, and platelets that help make blood clot. These hematopoietic stem cells are *not* the same as embryonic stem cells that are at the center political and ethical debate about their use in research. Their collection does not harm either the mother or the baby.

There are two other sources of hematopoietic stem cells—bone marrow and peripheral blood. Bone marrow is the spongy material inside bones that is the source for new blood cells produced throughout an individual's life. Peripheral blood is the blood that circulates through the body.

All cells have a protein profile that identifies them. When a foreign cell enters the body, immune system cells recognize the protein profile as foreign and attack the cell. The advantage of umbilical cord stem cells that they are more immature and have a less well developed protein profile than bone marrow or peripheral stem cells. As a result, they draw a weaker response from the body's immune system.

The protein profile of human cells is called the human leukocyte antigen (HLA) profile. Normally for transplanted cells such as bone marrow cells to be accepted, there must be a close to perfect match between the HLA profile of the donor and the recipient. Since there are over 10,000 HLA types, finding an exact match can be difficult. Because umbilical cord blood cells have a less well developed HLA profile, they do not have to match the recipient's HLA as closely to be accepted by the immune system of the transplant recipient.

Types of cord blood banks

There are two types of umbilical cord blood banks—private and public. Private blood cord banks can be for-profit or not-for-profit. They sell kits that can be used at any delivery to collect cord blood. They also charge a yearly fee to store the blood. The cord blood is available only to the child or members of the child's family from which it came. It is a perfect HLA match to child's blood. When a transplant is done

using cells that came from the individual who needs the transplant, it is called an autologous transplant. Autologous transplants have a high rate of success.

Public cord banks do not charge a fee to collect or store cord blood. Parents of newborns donate the cord blood, and it is then available to anyone who needs it and is an appropriate match. This type of transplant where the recipient is not the same as the donor is called an allogeneic transplant. Much of the success of allogeneic transplants depends on finding a good HLA match. Because of funding considerations, there are a limited number of public cord blood banks. To donate to a public blood cord bank, the mother must give birth in a facility associated with a public blood cord bank.

How cord blood is used

The first successful umbilical cord **stem cell transplantation** occurred in 1988, when a newborn girl donated cord blood to cure her brother of Fanconi's anemia, a genetic defect. Many other blood diseases can now be treated successfully with cord stem cell transplants. These same diseases can be treated with bone marrow transplant if an appropriately matched donor can be found. In addition, research is ongoing in the hope that cord blood stem cells can be induced into differentiating into other types of cells, such as nerve cells that can be used to treat other diseases.

Some diseases treated with blood stem cell transplant include:

- acute lymphatic leukemia
- acute myelogenous leukemia
- chronic myelogenous leukemia
- Hodgkin disease
- Non-Hodgkin lymphoma
- neuroblastoma
- severe combined immune deficiency
- anaplastic anemia
- thalassemia
- sickle cell anemia
- Fanconi's anemia

In addition to being more readily accepted by the donor's body, cord blood offers other advantages over **bone marrow transplantation**. It is relatively quick and easy to thaw the cord blood and infuse it, while bone marrow harvesting is an invasive procedure with risks to the donor. In addition, cord blood is less likely to be contaminated with viruses that may create complications for the recipient.

Considerations in deciding to bank cord blood

Although the list of diseases treated with umbilical cord stem cells is impressive and growing, controversy exists about whether parents should pay private cord cell banks to preserve their newborn's blood. Private cord cell banks sell their services as a type of "insurance." In case the child develops a treatable disease, the child's own stem cells will be a perfect match. However, if the disease is caused by a genetic defect, (e.g., sickle cell and other **anemias**) the child's own stem cells will be useless, because they will contain the same genetic defect. In addition, the quantity of cord blood that can be collected is only enough to treat a child. Although research is being done to try to multiply cord cells in the laboratory in quantities great enough to treat an adult, results have not been successful. Therefore, if an adult develops a treatable disease, even if his own cord blood was banked, the quantity will be insufficient to produce a cure. Also, cord blood storage is a new enough technology, that it is not clear how long the blood can remain frozen and still be effective.

In the most recent statement on cord blood banking released by the American Academy of Pediatrics in 1999, they concluded:

- There are no good estimates of how often a child will need a transplant of his own cells. Estimates range from 1 in 1,000 to 1 in 200,000
- Certain genetic disorders are not cured by autologous transplants
- Parents should be encouraged to donate their child's cord blood to free public cord blood banks for the widest possible use rather than paying private companies to store cord blood

The American College of Obstetricians and Gynecologists supports the position that parents should not be sold private blood cord banking without a realistic assessment of their need, nor should parents feel guilty if they are not eager or able to meet the annual expense associated with private storage. Although cord blood collection is safe, unless there is a family history of diseases for which hematopoietic stem cells are helpful, most pediatricians and obstetricians feel private cord blood storage is an unnecessary expense.

Precautions

Umbilical cord blood collection is usually not done in the case of multiple births.

KEY TERMS

Allogeneic—A transplant where the donated material comes from different (although often related) individual than the recipient

Autologous—A transplant where the material for the transplant comes from the individual who is also the recipient; thus, the transplant material is genetically identical to the donor's body.

Hematopoietic stem cell—A cell that can develop into any type of specialized blood cell.

Description

Cord blood collection is a simple procedure that takes between three and seven minutes. It is done immediately after the baby is delivered. Cord blood is drained into a standard blood bag, and then processed and frozen at very low temperatures.

Preparation

Parents who wish to bank their child's cord blood must plan ahead. Private storage requires ordering a kit in advance and coordinating with the health care providers doing the delivery to make sure the cord blood is collected properly. Mothers must test negative for infectious diseases such as HIV and hepatitis before delivery. Donation to a public cord blood bank is not always possible because of their limited number and location.

Aftercare

No special aftercare is required for mother or child, but the cord blood must remain frozen at low temperatures.

Risks

The main risk associated with umbilical cord blood collection is the possibility that it will become contaminated with bacteria during collection.

Parents who are interested in private cord blood storage should research the company with which they contract, just as they would in making any business decision. They need to know the costs, whether the annual fee can increase, and what will happen to the blood if the company goes out of business.

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Uncinariasis see **Hookworm disease**

Undescended testes

Definition

Also known as cryptorchidism, undescended testes is a congenital condition characterized by testicles that do not extend to the scrotum.

KEY TERMS

Cryptorchidism—Undescended testes.

Embryonic—Early stages of life in the womb.

Fetal—Refers to the fetus, also known in the first two months after conception as an embryo.

Orchiopexy—Surgical procedure that places the testicles in the scrotum.

Description

In the fetus, the testes are in the abdomen. As development progresses they migrate downward through the groin and into the scrotum. This event takes place late in fetal development, during the eighth month of gestation. Thirty percent of premature boys have testes that have not yet made the full descent. Only 3–4% of full-term baby boys have undescended testes, and half of those complete the journey by the age of three months. Eighty percent of all undescended testes cases naturally correct themselves during the first year of life. Undescended testes that are not corrected can lead to sterility and an increased risk of testicular cancer.

Causes and symptoms

The cause of undescended testes is presently unknown, however its symptoms are quite apparent. One or all of the testicles can be undescended, therefore the testicles appear to be either missing or lopsided.

Diagnosis

The newborn examination always checks for testes in the scrotum. If they are not found, a search will be conducted, but not necessarily right away. In most cases, the testes will drop into place later. If the testes are present at all, they can be anywhere within a couple inches of the appropriate spot. In 5% of cases, one testis is completely absent. In 10%, the condition occurs on both sides. Presence of undescended testes is indicated by measuring the amount of gonadotropin hormone in the blood.

Treatment

Once it is determined that the testes will not naturally descend, surgery becomes necessary. The procedure is called an orchiopexy and is relatively

simple once the testes are located. The surgery is usually performed when the boy is between one to two years old.

Prognosis

Undescended testes must be treated to eliminate the increased risk of **testicular cancer** and the possibility of sterility. Undescended testes are twice as likely to develop **cancer**. Ten percent of all testicular cancers are in undescended testes.

Resources

BOOKS

Rajfer, Jacob. "Congenital Anomalies of the Testes and Scrotum." In *Campbell's Urology*, edited by Patrick C. Walsh, et al. Philadelphia: W. B. Saunders Co., 1998.

J. Ricker Polsdorfer, MD

Undulant fever see **Brucellosis**

Unipolar depression see **Depressive disorders**

Upper GI exam

Definition

An upper GI examination is a fluoroscopic examination (a type of x-ray imaging) of the upper gastrointestinal tract, including the esophagus, stomach, and upper small intestine (duodenum).

Purpose

An upper GI series is frequently requested when a patient experiences unexplained symptoms of abdominal **pain**, difficulty in swallowing (dysphagia), regurgitation, **diarrhea**, or weight loss. It is used to help diagnose disorders and diseases of, or related to, the upper gastrointestinal tract, including cases of hiatal **hernia**, diverticuli, ulcers, tumors, obstruction, enteritis, gastroesophageal reflux disease, **Crohn's disease**, and pulmonary aspiration.

Precautions

Because of the risks of radiation exposure to the fetus, pregnant women are advised to avoid this procedure. Patients with an obstruction or perforation in their bowel should not ingest barium (a radioactive

substance used to show contrast in the images) for an upper GI, but may still be able to undergo the procedure if a water-soluble contrast medium is substituted for the barium.

Glucagon, a medication sometimes given prior to an upper GI procedure, may cause **nausea** and **dizziness**.

Description

An upper GI series takes place in a hospital or clinic setting and is performed by an x-ray technician and a radiologist. A radiologist typically is in attendance to oversee the procedure, and view and interpret the fluoroscopic pictures. Before the test begins, the patient is sometimes administered an injection of glucagon, a medication which slows stomach and bowel activity, to allow the radiologist to get a clearer picture of the gastrointestinal tract. In order to further improve the clarity of the upper GI pictures, the patient may be given a cup of baking soda crystals to swallow, which distend the stomach by producing gas.

Once these preparatory steps are complete, the patient stands against an upright x-ray table, and a fluoroscopic screen is placed in front of him. The patient will be asked to drink from a cup of flavored barium sulfate, a thick and chalky-tasting liquid that allows the radiologist to see the digestive tract, while the radiologist views the esophagus, stomach, and duodenum on the fluoroscopic screen. The patient will be asked to change positions frequently in order to coat the entire surface of the gastrointestinal tract with barium. The technician or radiologist may press on the patient's abdomen in order to spread the barium. The x-ray table will also be moved several times throughout the procedure. The radiologist will ask the patient to hold his breath periodically while exposures are being taken. The entire procedure takes approximately 30 minutes.

In some cases, in addition to the standard upper GI series, a doctor may request a detailed intestine, or small bowel, radiography and fluoroscopy series; it is also called a small bowel follow-through (SBFT). Once the preliminary upper GI series is complete, the patient will be escorted to a waiting area while the barium travels down through the rest of the small intestinal path. Every 15–30 minutes, the patient will return to the x-ray suite for additional x rays, or films. Once the barium has completed its trip down the small bowel tract, the test is completed. This procedure can take anywhere from one to four hours.

Esophageal radiography, also called a barium esophagram or a barium swallow, is a study of the esophagus only, and is usually performed as part of the upper GI series. It is commonly used to diagnose the cause of difficulty in swallowing (dysphagia) and for detecting hiatal hernia. A barium sulfate liquid, and sometimes pieces of food covered in barium, are given to the patient to drink and eat while a radiologist examines the swallowing mechanism on a fluoroscopic screen. The test takes approximately 30 minutes.

Preparation

Patients must not eat, drink, or smoke for eight hours prior to undergoing an upper GI examination. Longer dietary restrictions may be required, depending on the type and diagnostic purpose of the test. Patients undergoing a small bowel follow-through exam may be asked to take **laxatives** the day prior to the test. Upper GI patients are typically required to wear a hospital gown, or similar attire, and to remove all jewelry, so the camera has an unobstructed view of the abdomen.

Aftercare

No special aftercare treatment or regimen is required for an upper GI series. The patient may eat and drink as soon as the test is completed. The barium sulfate may make the patient's stool white for several days, and patients are encouraged to drink plenty of fluids in order to eliminate it from their system.

Risks

Because the upper GI series is an x-ray procedure, it does involve minor exposure to ionizing radiation. Unless the patient is pregnant, or multiple radiological or fluoroscopic studies are required, the small dose of radiation incurred during a single procedure poses little risk. However, multiple studies requiring fluoroscopic exposure that are conducted in a short time period have been known, on rare occasions, to cause skin death (necrosis) in some individuals. This risk can be minimized by careful monitoring and documentation of cumulative radiation doses administered to these patients.

Normal results

A normal upper GI series will show a healthy, functioning, and unobstructed digestive tract.

Abnormal results

Obstructions or inflammation, including ulcers of the esophagus, stomach, or small intestine; or irregularities in the swallowing mechanism are just a few of the possible abnormalities that may show up on an upper GI series.

Resources

PERIODICALS

Newman, J. "Radiographic and Endoscopic Evaluation of the Upper GI Tract." *Radiology Technology* 69, no. 3 (January-February 1998): 213-26.

Paula Anne Ford-Martin

Urea clearance by the kidneys see **Kidney function tests**

Ureteral stenting

Definition

Ureteral stents are thin catheters threaded into segments of the ureter that carry urine, produced by the kidney, either down into the bladder internally, or to an external collection system. Insertion is most often done through the skin (percutaneously); however, in the presence of kidney or ureteral stones, stenting is ideally done during **cystoscopy**.

Purpose

Ureteral stenting may be placed on a long-term basis (months to years) in order to bypass ureteral obstruction. Short-term stenting (weeks to months) may be used as an adjunct to open surgical procedures of the urinary tract to provide a mold around which healing can occur, or to divert the urinary flow away from areas of leakage. Following balloon dilation or incision of ureteral strictures, placement of stents maintains the functionality of the ureters. Stents may also be used in the presence of **kidney stones** to manipulate or prevent stone migration prior to treatment, or to make the ureters more easily identifiable during difficult surgical procedures. Ureteral stents may be used in those with active kidney infection or with markedly diseased, intolerant bladders (e.g., damage from **radiation therapy**, bladder invasion by adjacent neoplasm).

KEY TERMS

Cystoscopy—Examination or treatment of the interior of the urinary bladder by looking through a special instrument with reflected light.

Stricture—An abnormal narrowing of a tube or passageway.

Ureter—The tube-like passageway in the body that carries urine from the kidney to the bladder.

Preparation

The procedure should be thoroughly explained by a medical professional before it takes place. The patient will be asked to put on a hospital gown. If the procedure is performed with the aid of a cystoscope, the patient will assume a position that is typically used in a gynecological exam.

Aftercare

Stents must be periodically replaced to prevent **fractures** within the catheter wall, or buildup of encrustation. Stent replacement is recommended approximately every six months or more often in patients who form stones.

Normal results

Normally, a ureteral stent assures the patient of a free flow of urine. Postoperatively, urine flow will be monitored to ensure the stent has not been dislodged or obstructed.

Abnormal results

Serious complications of the procedure occur in approximately four percent of cases, with minor complications in another 10%. These may include:

- Bleeding. Usually minor and easily treated, occasionally requiring **transfusion**
- Catheter migration or dislodgement. May require readjustment with the fluoroscope in the Radiology Department
- Coiling of the stent within the ureter. May cause lower abdominal **pain** or flank pain on urination, urinary frequency, or blood in the urine
- Introduction or worsening of infection
- Penetration of adjacent organs (e.g., bowel, gallbladder, or lungs)

Resources

BOOKS

Schrier, Robert, and Carl Gottschalk. *Diseases of the Kidney*. Philadelphia: Little, Brown and Co., 1997.

Kathleen D. Wright, RN

Ureterostomy see **Urinary diversion surgery**

Urethra defects see **Hypospadias and epispadias**

Urethritis

Definition

Urethritis is an inflammation of the urethra that is usually caused by an infection.

Description

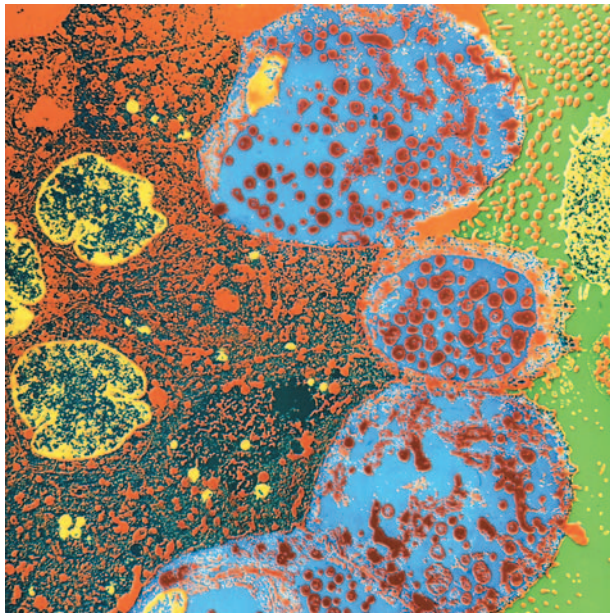
The urethra is the canal that moves urine from the bladder to the outside of the body. When this canal becomes infected, inflammation occurs due to the accumulation of white blood cells in the area. When this occurs, it is called urethritis. Besides the urethra, the urinary tract consists of the bladder, ureters, and kidneys. Inflammation can move up the urethra, causing **cystitis** in the bladder, or **nephritis** in the kidneys. Collectively, these inflammations are called urinary tract infections or UTIs.

Urinary tract infections are much more common in women than in men, probably due to anatomy. Infections are especially more common in older women, due to bladder problems.

Causes and symptoms

Uncomplicated urethritis usually results from infection by the bacteria *Escherichia coli*, commonly found in the bowel. Complicated urethritis can occur when other problems exist, such as **kidney stones**, malformations of the urinary tract, spinal cord injury, or a compromised immune system. People with diabetes tend to have more urinary tract infections, as well as hospitalized patients. Urinary tract infections can also be sexually transmitted. Some people seem to be susceptible to urinary tract infections, having them recurrently.

Frequently, a urinary tract infection has no symptoms. Common symptoms though, include pain and a



A false color transmission electron micrograph (TEM) scan of non-specific urethritis. (Photograph by Dr. R. Dourmashkin, Custom Medical Stock Photo. Reproduced by permission.)

burning sensation when urinating, frequent urination, or passing blood in the urine. Signs that the infection may be worsening include **fever** and chills, **nausea**, **vomiting**, and lower back **pain**.

Diagnosis

The diagnosis for a urinary tract infection is made by assessing the symptoms, feeling (palpating) the abdomen for tenderness, and a **urinalysis**. A urinalysis, or urine sample, is examined for both the presence of bacteria and white blood cells. After this, a **urine culture** to determine what bacteria is causing the infection may be done.

Treatment

Typical treatment for urinary tract infections is a course of **antibiotics**. In women who have recurrent urethritis, the diagnosis and treatment is often resolved over the phone. Additional drugs are sometimes given to relieve discomfort.

Alternative treatment

For those individuals who seem to be more susceptible to urinary tract infections, drinking lots of fluids at the first sign of an infection can ward it off by diluting the bacteria present and flushing the

system. Adding a teaspoon of baking soda to a glass of water and drinking it can change the pH of the urine, causing it to burn less. Also, cranberry juice contains a compound that can prevent bacteria from sticking to and thus growing in the urinary tract. Antimicrobial herbs, such as uva ursi (*Arctostaphylos uva-ursi*) and pipsissewa (*Chimaphila umbellata*), may be helpful. Other herbs, such as marsh mallow (*Althaea officinalis*), slippery elm (*Ulmus fulva*), comfrey (*Symphytum officinale*), plantain (*Plantago major*), and cornsilk, can soothe the urinary tract. *Lactobacillus acidophilus* and *L. bifidus* supplementation reintroduces normal flora into the urinary tract. **Acupuncture** and homeopathy can also be effective therapies for urethritis.

Prognosis

Given the appropriate antibiotic, urinary tract infections usually go away quickly. If not treated soon enough, however, urethritis can move up the urinary tract, infecting the bladder and possibly the kidneys, resulting in kidney damage. If the infection moves into the blood, additional complications can arise. Those who have previously had a urinary tract infection are more susceptible to additional urinary tract infections. Because of this, patients need to be aware of the symptoms so that a physician can be notified if the infection becomes recurrent.

Prevention

There are some steps that can be taken to keep the urinary tract healthy and prevent infection.

- drink plenty of fluids
- do not hold urine once the urge to urinate has occurred
- after a bowel movement, wipe from front to rear to keep bowel bacteria at a distance
- wear cotton underwear
- rinse soap off well in the shower
- urinate after sexual intercourse
- for post-menopausal women, estrogen replacement therapy can help prevent urinary tract infection

Resources

PERIODICALS

"Drink Away Urinary Tract Infections." *Prevention Magazine* January 1998: 135.

Cindy L. A. Jones, PhD

Uric acid tests

Definition

Uric acid tests are tests that are done to measure the levels of uric acid in blood serum or in urine.

Purpose

The uric acid tests are used to evaluate the blood levels of uric acid for **gout** and to assess uric acid levels in the urine for kidney stone formation. The urine test is used most often to monitor patients already diagnosed with **kidney stones**, but it can also be used to detect disorders that affect the body's production of uric acid and to help measure the level of kidney functioning.

Uric acid is a waste product that results from the breakdown of purine, a nucleic acid. (Nucleic acids are the building blocks of DNA.) Uric acid is made in the liver and excreted by the kidneys. If the liver produces too much uric acid or the kidneys excrete too little, the patient will have too much uric acid in the blood. This condition is called hyperuricemia. Supersaturated uric acid in the urine (uricosuria) can crystallize to form kidney stones that may block the tubes that lead from the kidneys to the bladder (the ureters).

Precautions

Blood test

Patients scheduled for a blood test for uric acid should be checked for the following medications: loop **diuretics** (Diamox, Bumex, Edecrin, or Lasix); ethambutol (Myambutol); vincristine (Oncovin); pyrazinamide (Tebrazid); thiazide diuretics (Naturetin, Hydrex, Diuril, Esidrix, HydroDiuril, Aquatensen, Renese, Diurese); **aspirin** (low doses); **acetaminophen** (Tylenol); ascorbic acid (vitamin C preparations); levodopa (Larodopa); or phenacetin. These drugs can affect test results.

Certain foods that are high in purine may increase the patient's levels of uric acid. These include kidneys, liver, sweetbreads, sardines, anchovies, and meat extracts.

Urine test

Patients should be checked for the following medications before the urine test: diuretics, aspirin, pyrazinamide (Tebrazid), phenylbutazone, probenecid (Benemid), and allopurinol (Lopurin). If the patient

needs to continue taking these medications, the laboratory should be notified.

The laboratory should also be notified if the patient has had recent x-ray tests requiring contrast dyes. These chemicals increase uric acid levels in urine and decrease them in blood.

Description

The uric acid blood test is performed on a sample of the patient's blood, withdrawn from a vein into a vacuum tube. The procedure, which is called a venipuncture, takes about five minutes. The urine test requires the patient to collect all urine voided over a 24-hour period, with the exception of the very first specimen. The patient keeps the specimen container on ice or in the refrigerator during the collection period.

Preparation

The uric acid test requires either a blood or urine sample. For the blood sample, the patient should be **fasting** (nothing to eat or drink) for at least eight hours before the test. The urine test for uric acid requires a 24-hour urine collection. The urine test does *not* require the patient to fast or cut down on fluids. Some laboratories encourage patients to drink plenty of fluids during the collection period.

Risks

Risks for the blood test are minimal, but may include slight bleeding from the puncture site, a small bruise or swelling in the area, or **fainting** or feeling lightheaded.

Normal results

Blood test

Reference values for blood uric acid vary from laboratory to laboratory but are generally found within the following range: Male: 2.1–8.5 mg/dL; female: 2.0–6.6 mg/dL. Values may be slightly higher in the elderly.

Urine test

Reference values for 24-hour urinary uric acid vary from laboratory to laboratory but are generally found within the following range: 250–750 mg/24 hours.

KEY TERMS

Fanconi's syndrome—A rare disorder caused by vitamin D deficiency or exposure to heavy metals.

Gout—A metabolic disorder characterized by sudden recurring attacks of arthritis caused by deposits of crystals that build up in the joints due to abnormally high uric acid blood levels. In gout, uric acid may be overproduced, underexcreted, or both.

Hyperuricemia—Excessively high levels of uric acid in the blood, often producing gout.

Purine—A white crystalline substance that is one of the building blocks of DNA. Uric acid is produced when purine is broken down in the body.

Uric acid—A compound resulting from the body's breakdown of purine. It is normally present in human urine only in small amounts.

Uricosuria—Increased levels of uric acid in the urine.

Wilson's disease—A rare hereditary disease marked by the buildup of copper in the liver and brain, causing loss of kidney function.

Abnormal results

The critical value for the blood test is a level of uric acid higher than 12 milligrams per deciliter (about 3.4 ounces).

Increased *production* of uric acid may result from eating foods that are high in purine. Increased uric acid levels due to overproduction may also be caused by gout, by a genetic disorder of purine metabolism, or by metastatic **cancer**, destruction of red blood cells, leukemia, or cancer **chemotherapy**.

Decreased *excretion* of uric acid is seen in chronic **kidney disease**, low thyroid, toxemia of **pregnancy**, and **alcoholism**. Patients with gout excrete less than half the uric acid in their blood as other persons. Only 10-15% of the total cases of hyperuricemia, however, are caused by gout.

Abnormally low uric acid levels may indicate that the patient is taking allopurinol or probenecid for treatment of gout; may be pregnant; or suffers from Wilson's disease or **Fanconi's syndrome**.

Resources

BOOKS

Pagana, Kathleen Deska. *Mosby's Manual of Diagnostic and Laboratory Tests*. St. Louis: Mosby, Inc., 1998.

Janis O. Flores

Urinalysis

Definition

Urinalysis is a diagnostic physical, chemical, and microscopic examination of a urine sample (specimen). Specimens can be obtained by normal emptying of the bladder (voiding) or by a hospital procedure called catheterization.

Purpose

Urinalyses are performed for several reasons:

- general evaluation of health
- diagnosis of metabolic or systemic diseases that affect kidney function
- diagnosis of endocrine disorders. Twenty-four hour urine studies are often ordered for these tests
- diagnosis of diseases or disorders of the kidneys or urinary tract
- monitoring of patients with diabetes
- testing for pregnancy
- screening for drug abuse

Precautions

Voided specimens

Urinalysis should not be performed while a woman is menstruating or having a vaginal discharge. A woman who must have a urinalysis while she has a vaginal discharge or is having her period should insert a fresh tampon before beginning the test. She should also hold a piece of clean material over the entrance to her vagina to avoid contaminating the specimen.

Patients do not have to fast or change their food intake before a urine test. They should, however, avoid intense athletic training or heavy physical work before the test because it may result in small amounts of blood in the urine.

The following drugs can affect urinalysis results. The patient may be asked to stop taking them until after the test:

- nitrofurantoin (Macrochantin, Furadantin). Nitrofurantoin is prescribed for infections of the urinary tract and other bacterial infections
- phenazopyridine (Pyridium). This medication is used to relieve burning and **pain** caused by urinary-tract infections

- rifampin (Rifadin). This medication is prescribed to treat **tuberculosis**, prevent the spread of meningitis, and treat other infections

Bladder catheterization

Bladder catheterization is sometimes used to collect urine samples from hospitalized patients. It should not, however, be used to collect specimens from males with acute inflammation of the prostate or from a patient of either sex with a fractured pelvis.

Description

Collecting a urine sample from emptying the bladder takes about two or three minutes. The sample can be collected at home as well as in a doctor's office. Urine specimens are usually collected early in the morning before breakfast. Urine collected eight hours after eating and at least six hours after the most recent urination is more likely to indicate abnormalities. Some people may be asked to void into a clean container before getting out of bed in the morning.

Specimen containers

The doctor or hospital will supply a sterile container for a specimen being collected for a colony count. A colony count is a test that detects bacteria in urine that has been cultured for 24–48 hours. It is used instead of a routine urinalysis when a patient's symptoms suggest a urinary tract infection. Nonsterile containers can be used for routine specimens that will not be tested immediately after being collected. An ordinary open-necked jar may be used after it and its lid have been soaked in very hot water for 15–20 minutes and then air-dried.

Laboratory procedures

STORAGE. Urine specimens should not remain unrefrigerated for longer than two hours. A urine specimen that cannot be delivered to a laboratory within two hours should be stored in a refrigerator. The reason for this precaution is that urine samples undergo chemical changes at room temperature. Blood cells begin to dissolve and the urine loses its acidity.

VISUAL EXAMINATION. A doctor, nurse, or laboratory technician will look at the specimen to see if the urine is red, cloudy, or looks unusual in any way. He or she will also note any unusual odor.

TESTING TECHNIQUES. Urine samples are tested with a variety of different instruments and techniques.

Some tests use dipsticks, which are thin strips of plastic that change color in the presence of specific substances. Dipsticks can be used to measure the acidity of the urine (its pH) or the presence of blood, protein, sugar, or substances produced during the breakdown of fatty acids (ketones). A urinometer is used to compare the density of the urine specimen with the density of plain water. This measurement is called specific gravity.

The urine specimen is also examined under a microscope to determine whether it contains blood cells, crystals, or small pieces of fibrous material (casts).

Preparation

Voided specimens

Most urine specimens from adults or older children are collected by the patient's voiding into a suitable container. Soaps and disinfectants may contaminate urine specimens and should not be used. The doctor or laboratory may supply a special antiseptic solution that won't irritate the skin. The method for collection varies somewhat according to age and sex.

WOMEN AND GIRLS. Before collecting a urine sample, a woman or girl should use a clean cotton ball moistened with lukewarm water to cleanse the external genital area. Gently separating the folded skin (labia) on either side of her vagina, she should move the cotton ball from the front of the area to the back. After repeating this process several times, using a fresh piece of cotton each time, she should dry the area with a clean towel.

To prevent menstrual blood, vaginal discharge, or germs from the external genitalia from contaminating the specimen, a woman or girl should release some urine before she begins to collect her sample. A urine specimen obtained this way is called a midstream clean catch.

MEN AND BOYS. A man or boy should use a piece of clean cotton, moistened with antiseptic, to cleanse the head of his penis and the passage through which urine leaves his body (the urethral meatus). He should draw back his foreskin if he has not been circumcised. He should move the cotton in a circular motion away from the urinary opening, using a fresh piece of cotton each time. After repeating this process several times, he should use a fresh piece of cotton to remove the antiseptic. After the area has been thoroughly cleansed, he should begin urinating and collect a small sample in a container without interrupting the stream of urine.

INFANTS. A parent, nurse, or doctor should cleanse the child's genitals and as much of the surrounding area as will fit into the sterile urine-collection bag provided by the hospital. When the area has been thoroughly cleansed, the bag should be attached to the child's genital area and left in place until the child has urinated. It is important to remember not to touch the inside of the bag and to remove it as soon as a specimen has been obtained.

Bladder catheterization

Bladder catheterization is a hospital procedure used to collect uncontaminated urine when the patient cannot void. A catheter is a thin flexible tube that the doctor inserts through the urethra into the bladder to allow urine to flow out. To minimize the risk of infecting the patient's bladder with bacteria, many doctors use a so-called Robinson catheter, which is a plain rubber or latex tube that is removed as soon as the specimen is collected.

Suprapubic bladder aspiration is a technique that is sometimes used to collect urine from infants younger than six months. The doctor withdraws urine from the bladder into a syringe through a needle inserted through the skin over the bladder. This technique is used only when the child cannot void because of an abnormal urethra or if he or she has a urinary tract infection that has not responded to treatment.

Aftercare

The patient may return to normal activities after collecting the sample and may start taking medications that were discontinued before the test.

Risks

There are no risks associated with voided specimens. The risk of bladder infection from catheterization with a Robinson catheter is about 3%.

Normal results

Contents and appearance

Normal urine is a clear straw-colored liquid. It has a slight odor. It contains some crystals, a small number of cells from the tissues that line the bladder, and transparent (hyaline) casts. Normal urine does *not* contain sugars, yeast cells, protein, ketones, bacteria, or parasitic organisms.

The time of day a urine sample is collected can make a difference in the appearance of the specimen.

Some foods and medicines, including red beets, asparagus, and penicillin, can affect the color or smell of urine. Although most color variations are harmless, they sometimes indicate the presence of serious disease. A doctor, nurse, or laboratory technician should be notified if the urine is red or cloudy or looks unusual in any way.

Acidity

The pH of normal urine is 4.5–8.0. Its specific gravity is 1.0005–1.035.

Abnormal results

Cloudiness

Urine may be cloudy (turbid) because it contains red or white blood cells, bacteria, fat, mucus, digestive fluid (chyle), or pus from a bladder or kidney infection.

Odor

Foul-smelling urine is a common symptom of urinary-tract infection. A fruity odor is associated with **diabetes mellitus**, **starvation** and **dehydration**, or ketone formation. Other distinctive odors are present in the urine of patients with maple syrup urine disease or **phenylketonuria** (PKU).

Specific gravity

The specific gravity of urine can be affected by a range of diseases and disorders. Low specific gravity (below 1.005) is associated with diabetes insipidus, nephrogenic **diabetes insipidus**, acute tubular necrosis, and inflammation of the upper urinary tract (**pyelonephritis**). In fixed specific gravity, the specific gravity of the urine remains at 1.010 no matter how much fluid the person drinks. This condition occurs in patients who have chronic inflammation of the small blood vessels in the kidneys (**glomerulonephritis**) and serious kidney damage. High specific gravity (above 1.035) occurs in patients who are in shock or who suffer from **nephrotic syndrome**, dehydration, acute glomerulonephritis, congestive **heart failure**, or liver failure.

pH

A pH factor greater than 7 (more alkaline) may result from Fanconi's syndrome, urinary tract infections, or metabolic or respiratory alkalosis. A pH factor below 7 (more acid) may be due to **fever**, PKU, the secretion of homogentisic acid in the urine (alkaptonuria), and acidosis.

Blood and tissue cells

Red blood cells in the urine can be due to vigorous exercise or exposure to toxic chemicals. Bloody urine can also be a sign of bleeding in the genitourinary tract as a result of systemic bleeding disorders, various kidney diseases, bacterial infections, parasitic infections including **malaria**, obstructions in the urinary tract, **scurvy**, subacute bacterial **endocarditis**, traumatic injuries, and tumors.

A high number of white blood cells in the urine is usually a symptom of urinary tract infection. A large number of cells from tissue lining (epithelial cells) can indicate damage to the small tubes that carry material into and out of the kidneys.

Casts

Casts are small fibrous objects that are formed when protein and other materials settle in the kidney tubules and collecting ducts. Casts are dislodged by normal urine flow. A large number of them in a urine specimen is a sign of **kidney disease**.

Crystals

There are several different chemicals in body fluids that can form crystals that appear in urine. Some of these appear in normal urine, such as calcium oxalate or uric acid crystals. A large number of calcium oxalate crystals, however, may be a sign of abnormally high levels of calcium in the blood (**hypercalcemia**). Other crystals, including tyrosine, leucine, and cholesterol, are abnormal. The presence of cystine crystals is a symptom of excessive urinary secretion of cystine (**cystinuria**). Cystine is an acid found in many proteins and normally reabsorbed by the kidney tubules.

Protein

Protein in the urine can be a symptom of kidney stones, inflammation of the kidneys, degenerative kidney disease, or multiple tumors.

Sugars

A high level of glucose and other sugars in the urine (glycosuria) is often a symptom of diabetes mellitus. Glycosuria can also be caused by advanced kidney disease, **Cushing's syndrome**, impaired tubular reabsorption, **shock**, a rare tumor of the adrenal gland (**pheochromocytoma**), or **cancer** of the pancreas.

Milk in the urine is normal if a woman is pregnant, has just given birth, or is breastfeeding. On the other hand, rare hereditary metabolic disorders are indicated when urine contains fruit sugar (fructose), milk sugar (galactose), or a simple sugar called pentose.

KEY TERMS

Acidosis—A condition of the blood in which bicarbonate levels are below normal.

Alkalosis—A condition of the blood and other body fluids in which bicarbonate levels are higher than normal.

Casts—Small fibrous objects formed from materials that collect in the kidney tubules and are washed out by normal urine flow.

Catheter—A thin flexible tube inserted through the urethra into the bladder to allow urine to flow out.

Clean catch specimen—A urine specimen that is collected from the middle of the urine stream after the first part of the flow has been voided.

Colony count—A measurement of the growth of bacteria in a urine sample that has been cultured for 24 to 48 hours.

Fanconi's syndrome—A rare disorder caused by vitamin D deficiency or exposure to heavy metals.

Ketones—Substances produced during the breakdown of fatty acids. They are produced in excessive amounts in diabetes and certain other abnormal conditions.

Nephrotic syndrome—A condition characterized by water retention, little or no protein in urine, and high blood cholesterol.

pH—A chemical symbol used to describe the acidity or alkalinity of a fluid, ranging from 0 (more acid) to 14 (more alkaline).

Urethra—The duct that carries urine from the bladder to the outside of the body.

Urinalysis (plural, urinalyses)—The diagnostic testing of a urine sample.

Voiding—Another word for emptying the bladder or urinating.

Ketones

The presence of abnormally high numbers of ketones in the urine (ketonuria) usually results from uncontrolled diabetes mellitus. Ketonuria can also be caused by prolonged **diarrhea** or **vomiting** that results in starvation.

Bilirubin

Bilirubin is an orange-yellow pigment found in bile, a fluid secreted by the liver. When it is found in urine, bilirubin may be a symptom of **liver disease**.

caused by the formation of fibrous tissue, medications that damage the liver, or obstructive **jaundice**.

Urobilinogen

Bacteria in the small intestine can convert bilirubin to urobilinogen, which is excreted in the feces, in bile, or in urine. An accumulation of urobilinogen in the urine may be a sign of severe infection, liver damage, or diseases that destroy red blood cells. Low levels of urobilinogen in the urine may be a result of antimicrobial therapy, inflammatory diseases, kidney disease, severe diarrhea, or blocked bile ducts.

Other findings

The presence of bacteria, parasites, or yeast cells in the urine may be a symptom of urinary tract infection or contamination of the external genitalia. Other factors that may affect urinalysis results include failure to collect a specimen during the day's first voiding; frequent urination; large dietary intake of vitamin C; and urine with a pH value lower than 6.

Resources

ORGANIZATIONS

American Association of Kidney Patients. 100 S. Ashley Dr., #280, Tampa, FL 33602. (800) 749-2257. <<http://www.aakp.org>>.

American Kidney Fund (AKF). Suite 1010, 6110 Executive Boulevard, Rockville, MD 20852. (800) 638-8299. <<http://216.248.130.102/Default.htm>>.

Maureen Haggerty

Urinary anti-infectives

Definition

Urinary anti-infectives are medicines used to treat or prevent infections of the urinary tract—the passage through which urine flows from the kidneys out of the body.

Purpose

Normally, no bacteria or other disease-causing organisms live in the bladder. Likewise, the urethra—the tube-like structure that carries urine from the bladder out of the body—usually has either no bacteria or not enough to cause problems. But the bladder, urethra, and other parts of the urinary tract may

become infected when disease-causing organisms invade from other body regions or from outside the body. Urinary anti-infectives are used to treat such infections or to prevent them in people who get them often.

Description

Commonly used urinary anti-infectives include methenamine (Urex, Hiprex, Mandelamine), nalidixic acid (NegGram) and nitrofurantoin (Macrobid, Furatoin, and other brands). These medicines are available only with a physician's prescription and come in capsule, tablet, granule, and liquid forms.

Recommended dosage

Methenamine

For adults and children 12 years and over, the usual dosage is 1 gram, taken either twice a day or four times a day, depending on the form in which the drug comes. For children aged 6–12 years, the dosage ranges from 500 mg taken 2–4 times a day to 1 gram taken twice a day, again depending on the form of the drug. For children under 6 years, a physician must determine the dose.

This medicine will not work properly unless the urine is acidic, with a pH reading of 5.5 or below. The physician who prescribes the medicine will explain how to test the urine's acidity. The physician also may suggest diet changes that will make the urine more acidic, such as eating more protein, drinking cranberry juice, eating plums and prunes, but avoiding most other fruits, and cutting down on milk and other dairy products. **Antacids** also should be avoided.

Nalidixic acid

The recommended dosage for adults and children 12 years and older is 1 gram every 6 hours. If the medicine is taken for more than one or two weeks, the dosage may be decreased to 500 mg every 6 hours. A physician must determine the correct dosage for children 3 months to 12 years old. Children under 3 months should not take this medicine because it causes bone problems in young animals and could have the same effect in young children.

Nitrofurantoin

CAPSULES, TABLETS, OR LIQUID. The usual dose for adults and teenagers is 50–100 mg every six hours.

EXTENDED-RELEASE CAPSULES. For adults and children 12 years and older, the usual dosage is 100 mg every 12 hours for seven days.

For all forms of nitrofurantoin, a physician must determine the correct dose for children one month and older, based on the child's body weight. Children under one month should not take this medicine.

Precautions

Methenamine

People with certain medical conditions may have problems if they take this medicine. For example, people with severe **liver disease** may have worsened symptoms of their disease. And people who are dehydrated or who have severe **kidney disease** may be more likely to have side effects that affect the kidneys.

Nalidixic acid

Some people feel drowsy, dizzy, or less alert than usual when using this drug. The medicine may also cause blurred vision or other vision changes. Because of these possible problems, anyone who takes nalidixic acid should not drive, operate machinery, or do anything else that might be dangerous until they have found out how the drugs affect them.

Nalidixic acid may increase sensitivity to sunlight. Even brief exposure to sun can cause a severe **sunburn** or a rash. While being treated with this medicine, avoid being in direct sunlight, especially between 10 a.m. and 3 p.m.; wear a hat and tightly woven clothing that covers the arms and legs; use a sunscreen with a skin protection factor (SPF) of at least 15; protect the lips with a sun block lipstick; and do not use tanning beds, tanning booths, or sunlamps.

Diabetic patients should be aware that this medicine may cause false results on some urine sugar tests. Check with a physician before making any changes in diet or diabetes medicine based on the results of a urine test.

In laboratory studies, nalidixic acid interferes with bone development in young animals. The drug's effects have not been studied in pregnant women, but because of its effects in animals, it is not recommended for use during **pregnancy**.

This medicine generally does not cause problems in nursing babies whose mothers take it. However, nursing babies with glucose-6-phosphate dehydrogenase (G6PD) deficiency (an inherited disorder that affects mainly black males) may have blood problems if their mothers take nalidixic acid.

People with certain medical conditions may be more likely to have particular side effects if they take

this medicine. For example, people with a history of seizures or severe hardening of the arteries in the brain may be more likely to have side effects that affect the nervous system. People with glucose-6-phosphate dehydrogenase (G6PD) deficiency are more likely to have side effects that affect the blood. Also, people with liver disease or severe kidney disease have an increased chance of having any of the drug's possible side effects.

Nitrofurantoin

Pregnant women should not take this medicine within two weeks of their delivery date and should not use it during labor and delivery, as this could cause problems in the baby.

Women who are breastfeeding should check with their physicians before using this medicine. It passes into breast milk and could cause problems in nursing babies whose mothers take it. This is especially true of babies with glucose-6-phosphate dehydrogenase (G6PD) deficiency. The medicine also should not be given directly to babies up to one month of age, as they are particularly sensitive to its effects.

Older people may be more likely to have side effects when taking nitrofurantoin, because they are more sensitive to the drug's effects.

Taking nitrofurantoin may cause problems for people with certain medical conditions. Side effects may be greater, for example, in people with lung disease or nerve damage. In people with kidney disease, the medicine may not work as well as it should, but may cause more side effects. Those with glucose-6-phosphate dehydrogenase (G6PD) deficiency who take nitrofurantoin may develop anemia.

Diabetic patients should be aware that this medicine may cause false results on some urine sugar tests. They should check with a physician before making any changes in diet or diabetes medicine based on the results of a urine test.

General precautions for all urinary anti-infectives

Symptoms should improve within a few days of starting to take a urinary anti-infective. If they do not, or if they become worse, check with a physician right away. Patients who need to take this medicine for long periods should see their physicians regularly, so that the physician can check their progress.

Anyone who has had unusual reactions to urinary anti-infectives in the past should let his or her physician know before taking the drugs again. The physician

should also be told about any allergies to foods, dyes, preservatives, or other substances. Patients taking nalidixic acid should tell their physicians if they have ever had reactions to related medicines such as cinoxacin (Cinobac), ciprofloxacin (Cipro), enoxacin (Penetrex), norfloxacin (Noroxin) or ofloxacin (Floxin), all of which are also used to treat or prevent infections. Anyone taking nitrofurantoin should let the physician know if he or she has had an unusual reaction to medicines such as furazolidone (Furoxone) or nitrofurazone (Furacin).

Side effects

Methenamine

Nausea and vomiting are not common but may occur. These side effects do not need medical attention unless they are severe. One side effect that should be brought to a physician's attention immediately is a skin rash.

Nalidixic acid

Some side effects are fairly minor and are likely to go away as the body adjusts to the drug. These include **dizziness**, drowsiness, **headache**, **nausea** or **vomiting**, stomach pain and **diarrhea**. Unless these problems continue or are bothersome, they do not need medical attention.

Other side effects, however, should have prompt medical attention. Anyone who has changes in vision, such as blurred vision, double vision, decreased vision, changes in color vision, halos around lights, or notices an excessive brightness of lights should check with a physician immediately.

Nitrofurantoin

This medicine may make the urine turn reddish-yellow to brown. This is nothing to worry about. Other possible side effects that do not need medical attention unless they are severe include pain in the stomach or abdomen, stomach upset, diarrhea, loss of appetite, and nausea or vomiting.

Anyone who has chest **pain**, breathing problems, fever, chills, or a **cough** while taking nitrofurantoin should check with a physician immediately.

General advice on side effects for all urinary anti-infectives

Other side effects are possible when taking any urinary anti-infective. Anyone who has unusual

KEY TERMS

Altitude sickness—A set of symptoms that people who normally live at low altitudes may have when they travel to high altitudes. The symptoms include nosebleed, nausea, and shortness of breath.

Anemia—A lack of hemoglobin—the compound in blood that carries oxygen from the lungs throughout the body and brings waste carbon dioxide from the cells to the lungs, where it is released.

Bacteria—Tiny, one-celled forms of life that cause many diseases and infections.

Glaucoma—A condition in which pressure in the eye is abnormally high. If not treated, glaucoma may lead to blindness.

Glucose-6-phosphate dehydrogenase (G6PD) deficiency—An inherited disorder in which the body lacks an enzyme that normally protects red blood cells from toxic chemicals. When people with this condition take certain drugs, their red blood cells break down, causing anemia. This may also happen when they have a fever or an infection. The condition usually occurs in males. About 10% of black males have it, as do a small percentage of people from the Mediterranean region.

Granule—A small grain or pellet. Medicines that come in granule form usually are mixed with liquids or sprinkled on food before they are taken.

Organism—An individual of some type of life form, such as a plant or an animal.

pH—A measure of how acidic or alkaline something is. The pH scale ranges from 0 to 14. Values below 7 are acidic; values above 7 are alkaline.

Seizure—A sudden attack, spasm, or convulsion.

symptoms while taking this type of medicine should get in touch with his or her physician.

Interactions

Methenamine

Certain medicines may make methenamine less effective. These include thiazide **diuretics** (water pills) and medicines that make the urine less acid, such as antacids, bicarbonate of soda, and the drugs acetazolamide (Diamox), dichlorphenamide (Daramide), and methazolamide (Neptazane), which are used to treat **glaucoma**, epilepsy, **altitude sickness**, and other conditions.

Nalidixic acid

People who are taking blood thinners (anticoagulants) may be more likely to have bleeding problems if they take this medicine.

Nitrofurantoin

Nitrofurantoin may interact with many other medicines. For example, taking nitrofurantoin with certain drugs that include methyldopa (Aldomet), **sulfonamides** (sulfa drugs), vitamin K, and diabetes medicines taken by mouth may increase the chance of side effects that affect the blood. General side effects are more likely in people who take nitrofurantoin with the **gout drugs** probenecid (Benemid) or sulfinpyrazone (Anturane). And the risk of side effects that involve the nervous system is higher in people who take nitrofurantoin with various drugs including lithium (Lithane), disulfiram (Antabuse), other anti-infectives, and the **cancer** drugs cisplatin (Platinol) and vincristine (Oncovin). Patients who have had a DPT (**diphtheria**, **tetanus**, and pertussis) vaccine within the last 30 days or who have one after taking nitrofurantoin are also more likely to have side effects that affect the nervous system. Because of the many possible interactions, anyone taking nitrofurantoin should be sure to check with a physician before combining it with any other medicine.

General advice about interactions

Not every drug that may interact with a urinary anti-infective is listed here. Be sure to check with a physician or pharmacist before combining a urinary anti-infective with any other prescription or nonprescription (over-the-counter) medicine.

Resources

PERIODICALS

Tucker, M. Susan. "Recurrent UTI: Who Should Treat Herself?" *Patient Care* 26 (July 15, 1992): 259.

Nancy Ross-Flanigan

Urinary antiseptics see **Urinary anti-infectives**

Urinary catheterization

Definition

Urinary catheterization is the insertion of a catheter into a patient's bladder. The catheter is used as a conduit to drain urine from the bladder into an attached bag or container.

Purpose

Urinary catheterization is employed in hospital and nursing home settings to maintain urine output in patients who are undergoing surgery, or who are confined to the bed and physically unable to use a bedpan. Critically ill patients who require strict monitoring of urinary output are also frequently catheterized.

Intermittent insertion of a urinary catheter is a treatment option for patients with certain types of **urinary incontinence**. Patients who are unable to completely empty the bladder during urination (urinary retention), or patients who have a bladder obstruction, may also require intermittent urinary catheterization. Disabled individuals with neurological disorders that cause **paralysis** or a loss of sensation in the perineal area may also use regular intermittent catheter insertion to void their bladders.

Precautions

Because urinary catheterization carries a risk of causing urinary tract infection (UTI), precautions should be used to keep the catheter clean and free of bacteria. Patients requiring intermittent catheterization should be well trained in the technique by a qualified health care professional.

Description

Intermittent catheterization is performed a minimum of four times a day by the patient or a care giver. The genital area near the urethral opening is wiped with an antiseptic agent, such as iodine. A lubricant may be used to facilitate the entry of the catheter into the urethra, and a topical local anesthetic may be applied to numb the urethral opening during the procedure. One end of the catheter is placed in a container, and the other end is inserted into and guided up the urethra until urine flow begins. When urine flow stops, the catheter may be moved or rotated, or the patient may change positions to ensure that all urine has emptied from the bladder. The catheter is then withdrawn, cleaned, and sterilized for the next use. Recommended cleaning practices vary, from the use of soap and water to submersion in boiling water or a disinfectant solution. Some patients prefer to use a new catheter with each insertion.

Nonintermittent catheterization, which is initiated in a hospital or nursing home setting, uses the same basic technique for insertion of the urinary tract catheter. The catheter is inserted by a nurse or other health care professional, and remains in the

patient until bladder function can be maintained independently. When the catheter is removed, patients will experience a pulling sensation and may feel some minor discomfort. If the catheter is required for an extended period of time, a long-term, indwelling catheter, such as a Foley catheter, is used. To prevent infection, it should be regularly exchanged for a new catheter every three to six weeks.

Use of indwelling catheters should be restricted to patients whose incontinence is caused by urinary tract obstruction that can not be treated, and for which alternative therapy is not feasible.

Preparation

If a patient wishes to perform intermittent catheterization himself, training in the technique by a qualified health care professional is required. Basic instruction in the anatomy, antiseptic techniques, catheter insertion, and proper catheter care should be provided. Patients learning chronic intermittent urinary catheterization may also benefit from an ultrasound examination to verify that they are completely emptying their bladder during the procedure.

Aftercare

Patients using intermittent catheterization as a treatment for incontinence will experience a period of adjustment as they try to establish a catheterization schedule that is adequate for their normal level of fluid intake.

Antibiotics may be prescribed as a preventative measure in long-term urinary catheterization patients who are at risk for urinary tract infection.

A patient with an indwelling catheter must be reassessed periodically to determine whether alternative treatment may be more effective in treating the problem.

Risks

Trauma to the urethra and/or bladder may result from incorrect insertion of the catheter. Repeated irritation to the urethra during catheter insertion may cause scarring and/or stricture, or narrowing, of the urethra. The catheter may introduce bacteria into the urethra and bladder, resulting in urinary tract infection. UTI can cause **fever** and inflammation of the bladder and urethra. Patients who practice intermittent catheterization can reduce their risks for UTI by using antiseptic techniques for insertion and catheter care.

KEY TERMS

Bladder obstruction—A blockage of the bladder caused by the presence of calculi (e.g., mineral deposits) or an anatomic abnormality.

Catheter—A long, thin, flexible tube.

Foley catheter—A two-channel catheter with a balloon on the bladder end of one channel. Once inflated, the balloon keeps the catheter securely in the bladder. The other channel of the catheter facilitates the flow of urine out of the bladder.

Local anesthetic—Medication applied topically to the skin or administered through an injection that deadens a specific part of the body and inhibits the sensation of pain.

Perineal area—The genital area between the vulva and anus in a woman, and between the scrotum and anus in a man.

Ultrasound examination—A diagnostic test that uses sound waves to generate a picture of an organ or organ system.

Urinary incontinence—The inability to control one's urine flow.

Normal results

When used correctly, catheterization facilitates complete voiding of the bladder.

Resources

PERIODICALS

Hunt, Gillian M., Pippa Oakeshott, and Robert Whitaker. "Intermittent Catheterization: Simple, Safe, and Effective but Underused." *British Medical Journal* 312, no. 7023 (Jan. 1996): 103-7.

Paula Anne Ford-Martin

Urinary diversion surgery

Definition

A urinary diversion involves removal of the urinary bladder and adjacent tissues and organs, and re-routing of the urinary stream. This may involve creation of an artificial opening in the abdomen called an **ostomy**.

Purpose

A urinary diversion is created as a means to treat **cancer** of the bladder, when conservative measures have been unsuccessful, or when there is recurrence of the disease invading the muscle wall. Congenital deformities or traumatic injury may also necessitate formation of a urinary diversion.

Description

Under **general anesthesia**, an incision is made in the abdomen. The ureters (tubes that carry urine away from the kidneys) are cut and tied. The bladder and surrounding tissues are cut free and removed. The ureters are then attached to a portion of the intestine. The most common types of urinary diversion are:

- **Ileal conduit.** Ureters are attached to a portion of the small intestine, the ileum, one end of which is brought through the abdominal wall as a conduit for the urine, creating a stoma
- **Ureterosigmoidostomy.** The ureters are attached to a portion of the large intestine, the sigmoid, which allows the urine to flow through the large intestine and out through the rectum
- **Cutaneous ureterostomy.** Bringing the detached ureters through the abdominal wall and attaching it to an opening in the skin

Following creation of an artificial opening to drain the urine, ureteral stents (tubes that go through the stoma and up into the ureters) are often inserted and left in place to allow urine to drain freely from the kidneys, without risk of blockage from swelling due to surgery. The muscles are replaced and sewn together. A transparent pouch is applied to the abdomen to collect urine, and attached to a bedside drainage bag. The incision is closed with sutures or clips (“staples”), which are usually removed about a week after surgery.

An alternative to a conventional urinary diversion is the continent urinary diversion. In this surgical procedure, a “false bladder” is constructed within the abdomen, using several lengths of small or large intestine. The ureters are sewn to this new reservoir for urine and nipple valves which are created at two sites; the abdominal wall for continence; and where the ureters are implanted, to prevent reflux of urine back to the kidneys. The patient is then taught to catheterize the reservoir to drain urine at regular intervals during the day. Although a continent diversion is not suitable for every patient who requires urinary diversion, it is an option to be considered.

Preparation

As with any surgical procedure, the patient will be required to sign a consent form after the procedure is explained thoroughly. Blood and urine studies, along with various x rays and an electrocardiogram (EKG), may be ordered as the doctor deems necessary. If creation of an ostomy is planned, the patient should visit an enterostomal therapist, who will mark an appropriate place on the abdomen for a stoma and offer preoperative education on ostomy management.

Eating or drinking is prohibited after midnight the night before the surgery. Oral anti-infectives, such as neomycin, erythromycin, or kanamycin sulfate, may be ordered to decrease bacteria in the intestine and help prevent postoperative infection. A nasogastric tube is inserted the day of surgery, or during surgery, to remove gastric secretions and prevent **nausea** and **vomiting**.

Aftercare

Postoperative care for the patient with a urinary diversion, as with those who have had any major surgery, involves monitoring of blood pressure, pulse, respirations, and temperature. Breathing tends to be shallow because of the effect of anesthesia, and the patient is reluctant to breathe deeply and experience **pain** that is caused by the abdominal incision. The patient is shown how to support the operative site during deep breathing and coughing, and is given pain medication as necessary. Fluid intake and output are measured, and the operative site is observed for color and amount of wound drainage. The nasogastric tube will remain in place, attached to low intermittent suction, until bowel activity resumes. Fluids and electrolytes are infused intravenously until the patient’s diet can gradually be resumed, beginning with liquids. The patient is usually able to move about in 8–24 hours after surgery, and is discharged from the hospital in 5–10 days.

If an ostomy has been placed, the patient and close family members will be educated on how to care for it. Determination of appropriate pouching supplies and a schedule of how often to change the pouch should be established. Regular assessment and meticulous care of the skin surrounding the stoma is important to maintain an adequate surface on which to apply the pouch. The pouch should be connected to a bedside drainage bag at night to prevent large volumes of urine from collecting in the pouch. Otherwise, the weight of the pouch could cause disruption of the pouch seal and leakage of urine onto the surrounding skin. Often, an enterostomal therapist

KEY TERMS

Ischemia—A compromise in blood supply to body tissues that causes tissue damage or death.

Ostomy—A surgically-created opening in the abdomen for elimination of waste products (urine or stool).

will visit the patient at home after discharge to help the new ostomy patient make the transition back to normal daily activities.

Risks

Potential complications of urinary diversion surgery include:

- excessive bleeding
- surgical wound infection
- thrombophlebitis (inflammation and blood clot to veins in the legs)
- pneumonia
- pulmonary **embolism** (blood clot or air bubble in the lungs' blood supply)

Normal results

Complete healing is expected without complications. The amount of time required for recovery from the surgery may vary depending of the patient's overall health status prior to surgery. The patient with a urinary diversion, without other medical complications, should be able to resume all daily activities once recovered from the surgery.

Abnormal results

The doctor should be made aware of any of the following problems after surgery:

- Increased pain, swelling, redness, drainage, or bleeding in the surgical area
- Headache, muscle aches, **dizziness**, or **fever**
- Increased abdominal pain or swelling, **constipation**, nausea, or vomiting

Stomal complications to be monitored include:

- Stomal tissue death (necrosis). This occurs because of inadequate blood supply, this is usually visible 12 to 24 hours after surgery and may require additional surgery

- Stoma flush or below the abdomen surface (retraction). Caused by insufficient stomal length, this may be managed by use of special pouching supplies. Elective revision of the stoma is also an option
- Narrowing at the opening of the stoma (stenosis). Often associated with infection around the stoma or scarring, mild stenosis can be removed under **local anesthesia**. Severe stenosis may require surgery for stomal revision
- Parastomal **hernia**. The bowel causes a bulge in the abdominal wall next to the stoma. This is usually due to placement of the stoma where the abdominal wall is weak, or an overly large opening in the abdominal wall. Use of an ostomy support belt and special pouching supplies may be adequate. If severe, the defect in the abdominal wall should be repaired and the stoma moved to another location

Resources

ORGANIZATIONS

United Ostomy Association, Inc. (UOA). 19772 MacArthur Blvd., Suite 200, Irvine, CA 92612-2405. (800) 826-0826. < <http://www.uoa.org> >.

Kathleen D. Wright, RN

Urinary incontinence

Definition

Urinary incontinence is unintentional loss of urine that is sufficient enough in frequency and amount to cause physical and/or emotional distress in the person experiencing it.

Description

Approximately 13 million Americans suffer from urinary incontinence. Women are affected by the disorder more frequently than are men; one in 10 women under age 65 suffers from urinary incontinence. A study published in late 2002 found that between 21% and 29% of adult women in the workforce reported at least one episode of urinary incontinence each month. Older Americans, too, are more prone to the condition. Twenty percent of Americans over age 65 are incontinent. In general, the condition is under-recognized and undertreated.

There are five major categories of urinary incontinence: overflow, **stress**, urge, functional, and reflex.

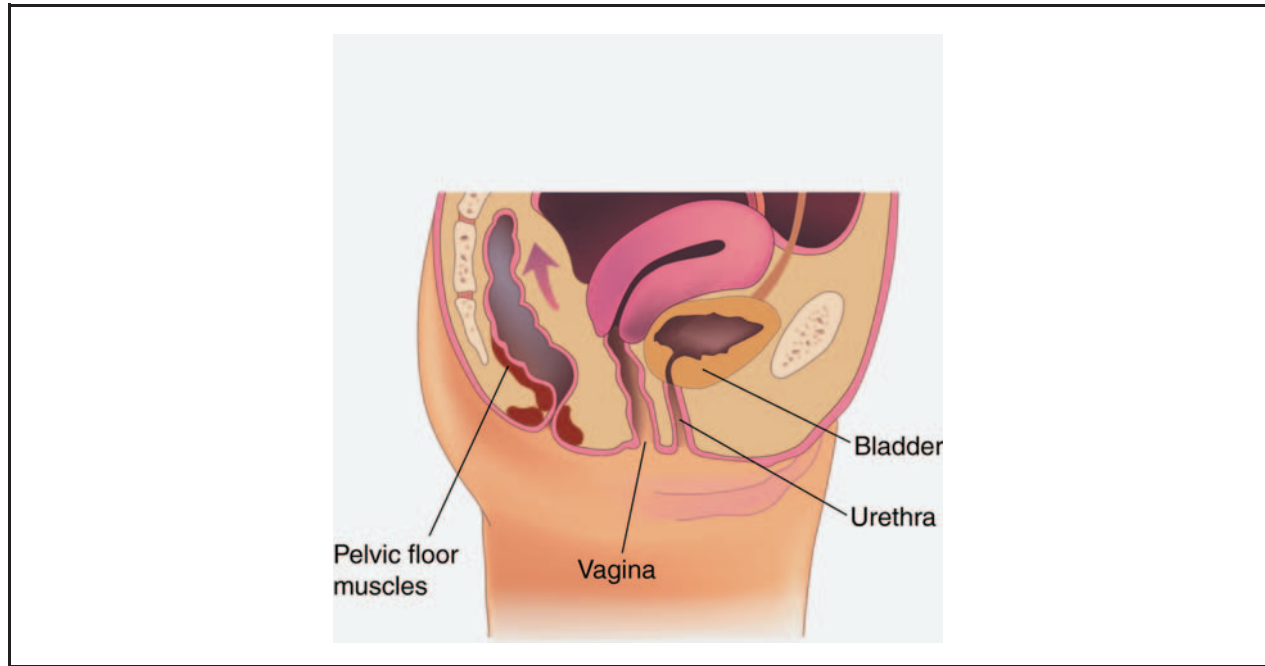
- **Overflow incontinence.** Overflow incontinence is caused by bladder dysfunction. Individuals with this type of incontinence have an obstruction to the bladder or urethra, or a bladder that doesn't contract properly. As a result, their bladders do not empty completely, and they have problems with frequent urine leakage.
- **Stress incontinence.** Stress incontinence occurs when an individual involuntarily loses urine after pressure is placed on the abdomen (i.e., during **exercise**, sexual activity, sneezing, coughing, laughing, or hugging).
- **Urge incontinence.** Urge incontinence occurs when an individual feels a sudden need to urinate, and cannot control the urge to do so. As a consequence, urine is involuntarily lost before the individual can get to the toilet.
- **Functional incontinence.** Individuals who have control over their own urination and have a fully functioning urinary tract, but cannot make it to the bathroom in time due to a physical or cognitive disability, are functionally incontinent. These individuals may suffer from arthritis, Parkinson's disease, **multiple sclerosis**, or **Alzheimer's disease**.
- **Reflex incontinence.** Individuals with reflex incontinence lose control of their bladder without warning. They typically suffer from neurological impairment.
- **Hysterectomy or other gynecological surgery.** Any surgery involving the urogenital tract runs the risk of damaging or weakening the pelvic muscles and causing incontinence.
- **Menopause.** The absence of estrogen in the postmenopausal woman can cause the bladder to drop, or prolapse.
- **Neurological conditions.** The nervous system sends signals to the bladder telling it when to start and stop emptying. When the nervous system is impaired, incontinence may result. Neurological conditions such as multiple sclerosis, **stroke**, spinal cord injuries, or a **brain tumor** may cause the bladder to contract involuntarily, expelling urine without warning, or to cease contractions completely, causing urinary retention.
- **Obesity.** Individuals who are overweight have undue pressure placed on their bladder and surrounding muscles.
- **Obstruction.** A blockage at the bladder outlet may permit only small amounts of urine to pass, resulting in urine retention and subsequent overflow incontinence. Tumors, calculi, and scar tissue can all block the flow of urine. A urethral stricture, or narrow urethra caused by scarring or inflammation, may also result in urine retention.

In some cases, an individual may develop short-term or *acute incontinence*. Acute incontinence may occur as a symptom or by-product of illness, as a side effect of medication, or as a result of dietary intake. The condition is typically easily resolved once the cause is determined and addressed.

Causes and symptoms

Urinary incontinence can be caused by a wide variety of physical conditions, including:

- **Childbirth.** Childbirth can weaken the pelvic muscles and cause the bladder to lose some support from surrounding muscles, resulting in stress incontinence.
- **Dysfunction of the bladder and/or the urinary sphincter.** In a continent individual, as the bladder contracts, the outlet that releases urine into the urethra (bladder sphincter) opens and urine exits the body. In individuals with overflow incontinence, bladder contractions and dilation of the sphincter do not occur at the same time.
- **Enlarged prostate.** In men, an enlarged prostate gland can obstruct the bladder, causing overflow incontinence.
- **Bladder irritants.** Substances in the urine that irritate the bladder may cause the bladder muscle to malfunction. The presence of a urinary tract infection and the ingestion of excess **caffeine** can act as irritants. Highly concentrated urine resulting from low fluid intake may also irritate the bladder.
- **Constipation.** Constipation can cause incontinence in some individuals. Stool that isn't passed presses against the bladder and urethra, triggering urine leakage.
- **Illness or disease.** Diabetes can greatly increase urine volume, making some individuals prone to incontinence. Other illnesses may temporarily impair the ability to recognize and control the urge to urinate, or to reach the toilet in time to do so.
- **Medications and alcohol.** Medications that sedate, such as tranquilizers and sleeping pills, can interfere with the proper functioning of the urethral nerves and bladder. Both sedatives and alcohol can also impair an individual's ability to recognize the need to urinate, and act on that need in a timely manner. Other medications such as **diuretics**, **muscle**



Strengthening the pelvic floor muscles by performing Kegel exercises helps to alleviate stress incontinence in women. Contract the pelvic floor muscles as if stopping an imaginary flow of urine. Hold for 10 seconds and repeat. (Illustration by Electronic Illustrators Group.)

relaxants, and blood pressure medication can also affect bladder function.

- **Surgery.** Men who undergo prostate surgery can suffer from temporary stress incontinence as a result of damage to the urethral outlet.

Diagnosis

Urinary incontinence may be diagnosed by a general practitioner, urologist, or gynecologist. If the patient is over age 65, a geriatrician may diagnose and treat the condition. A thorough medical history and **physical examination** is typically performed, along with specific diagnostic testing to determine the cause of the incontinence. Diagnostic testing may include x rays, ultrasound, urine tests, and a physical examination of the pelvis. It may also include a series of exams that measure bladder pressure and capacity and the urinary flow (urodynamic testing). The patient may also be asked to keep a diary to record urine output, frequency, and any episodes of incontinence over a period of several days or a week.

Treatment

There are numerous invasive and noninvasive treatment options for urinary incontinence:

- **Behavior modification therapy.** Behavior modification is a psychological approach to the treatment of urinary incontinence in which patients gradually increase the length of the time interval between voidings and “retrain” the bladder in other ways. It is reported to be highly effective in treating urge incontinence.
- **Biofeedback.** The use of sensors to monitor temperature and muscle contractions in the vagina to help incontinent patients learn to control their pelvic muscles.
- **Collagen injections.** Collagen injected in the tissue surrounding the urethra can provide urethral support for women suffering from stress incontinence.
- **External occlusive devices.** A new single-use disposable urethral cap is available without a prescription as of late 2002 for women suffering from stress urinary incontinence. The cap is noninvasive and appears to be quite effective in managing incontinence.
- **Inflatable urethral insert.** Sold under the trade name Reliance, this disposable incontinence balloon for women is inserted into the urethra and inflated to prevent urine leakage.
- **Intermittent urinary catheterization.** The periodic insertion of a catheter into a patient’s bladder to drain urine from the bladder into an attached bag or container.

KEY TERMS

Bladder neck—The place where the urethra and bladder join.

Bladder sphincter—The outlet that releases urine into the urethra.

Calculi (singular, calculus)—Mineral deposits that can form a blockage in the urinary system.

Occlusive—Closing off. One of the newest treatments for stress urinary incontinence in women is an external occlusive single-use cap that covers the urethral opening.

Perineal area—The genital area between the vulva and anus in a woman, and between the scrotum and anus in a man.

Sacral nerves—The five pairs of nerves that arise from the lowermost segments of the spinal cord and control bladder, bowel, and pelvic functions. Stimulation of the sacral nerves by an implanted device is a newer treatment for urinary incontinence.

- Medication. Estrogen **hormone replacement therapy** can help improve pelvic muscle tone in postmenopausal women. Other medications, including flurbiprofen, capsaicin and botulinum toxin, are sometimes prescribed to relax the bladder muscles or to tighten the urethral sphincter. As of late 2002, newer medications for the treatment of urinary incontinence were undergoing clinical trials. One of these drugs, duloxetine, differs from present medications in targeting the central nervous system's control of the urge to urinate rather than the smooth muscle of the bladder itself.
- Pelvic toning exercises. Exercises to tone the pelvic muscle can help alleviate stress incontinence in both men and women. These exercises involve tightening the muscles of the pelvic floor, and are also known as Kegel or PC muscle exercises.
- Perineal stimulation. Perineal stimulation is used to treat stress incontinence. The treatment uses a probe to deliver a painless electrical current to the perineal area muscles. The current tones the muscle by contracting it.
- Permanent catheterization. A permanent, or indwelling, catheter may be prescribed for chronic incontinence that doesn't respond to other treatments. A Foley catheter is usually used for urinary catheterization. One end is inserted through the urethra and into the bladder, and the external end is attached to a plastic reservoir bag that the patient may wear on the leg. A second alternative is a permanent catheter, called a suprapubic tube, surgically inserted into the bladder. The tube exits the body through the abdomen near the pubic bone, where it is attached to a drainage bag. As infection may result, this treatment should be reevaluated periodically, and the possibility of alternative treatment addressed.
- Sacral nerve stimulation (SNS). Also known as sacral neuromodulation, SNS is a procedure in which a surgeon implants a device that sends continuous stimulation to the sacral nerves that control the urinary sphincter. The FDA approved sacral nerve stimulation for the treatment of urinary urge incontinence in 1997 and for urinary frequency in 1999.
- Surgery. Bladder neck suspension surgery is used to correct female urinary stress incontinence. Surgical techniques such as the Marshall-Marchetti-Krantz and Burch procedures use sutures to raise and support the bladder neck and urethra. A sling procedure, which uses a strip of biocompatible material or the patient's own muscle or tissue as a supportive sling under the urethra and bladder neck, may also be used to treat stress incontinence. Bladder enlargement surgery may be recommended to treat incontinent men and women with unusually small bladders.
- Urinary sphincter implant. An artificial urinary sphincter may be used to treat incontinence in men and women with urinary sphincter impairment.
- Vaginal inserts. Devices constructed of silicone or other pliable materials that can be inserted into a woman's vagina to support the urethra.

Prognosis

Left untreated, incontinence can cause physical and emotional upheaval. Individuals with long-term incontinence suffer from urinary tract infections, and skin **rashes** and sores. Incontinence can also affect their self-esteem and cause depression and social withdrawal. They frequently stop participating in physical activities they once enjoyed because of the risk of embarrassing "accidents." However, with the wide variety of treatment options for incontinence available today, the prognosis for incontinent patients is promising. If incontinence cannot be stopped, it can be improved in the majority of cases.

Prevention

Women who are pregnant or who have gone through childbirth can reduce their risk for stress incontinence by strengthening their perineal area muscles with Kegel exercises. Men who have undergone prostate surgery may also benefit from pelvic muscle

exercises. Men and women should consult with their doctor before initiating any type of exercise program.

Resources

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- American Urological Association (AUA). 1120 North Charles Street, Baltimore, MD 21201. (410) 727-1100. <www.auanet.org>.
- Center for Biologics Evaluation and Research (CBER), U. S. Food and Drug Administration (FDA). 1401 Rockville Pike, Rockville, MD 20852-1448. (800) 835-4709 or (301) 827-1800. <www.fda.gov/cber>.
- National Association for Continence. P.O. Box 8310, Spartanburg, SC 29305-8310. (800) 252-3337. <<http://www.nafc.org>>.

National Kidney and Urologic Diseases Information Clearinghouse. 3 Information Way, Bethesda, MD 20892-3580. (800) 891-5390.

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Urinary tract infections see **Cystitis; Nongonococcal urethritis; Urethritis**

Urine concentration test see **Kidney function tests**

Urine culture

Definition

A urine culture is a diagnostic laboratory test performed to detect the presence of bacteria in the urine (bacteriuria).

Purpose

Culture of the urine is a method of diagnosis for urinary tract infection that determines the number of microorganisms present in a given quantity of urine.

Precautions

If delivery of the urine specimen to the laboratory within one hour of collection is not possible, it should be refrigerated. The health care provider should be informed of any **antibiotics** currently or recently taken.

Description

There are several different methods for collection of a urine sample. The most common is the midstream clean-catch technique. Hands should be washed before beginning. For females, the external genitalia (sex organs) are washed two or three times with a cleansing agent and rinsed with water. In males, the external head of the penis is similarly cleansed and rinsed. The patient is then instructed to begin to urinate, and the urine is collected midstream into a sterile container. In infants, a urinary collection bag (plastic bag with an adhesive seal on one end) is attached over the labia in girls or a boy's penis to collect the specimen.

Another method is the catheterized urine specimen in which a lubricated catheter (thin rubber tube) is inserted through the urethra (tube-like structure in

KEY TERMS

Bacteriuria—The presence of bacteria in the urine.

which urine is expelled from the bladder) into the bladder. This avoids contamination from the urethra or external genitalia. If the patient already has a urinary catheter in place, a urine specimen may be collected by clamping the tubing below the collection port and using a sterile needle and syringe to obtain the urine sample; urine cannot be taken from the drainage bag, as it is not fresh and has had an opportunity to grow bacteria at room temperature. On rare occasions, the health care provider may collect a urine sample by inserting a needle directly into the bladder (suprapubic tap) and draining the urine; this method is used only when a sample is needed quickly.

Negative culture results showing no bacterial growth are available after 24 hours. Positive results require 24–72 hours to complete identification of the number and type of bacteria found.

Preparation

Drinking a glass of water 15–20 minutes before the test is helpful if there is no urge to urinate. There are no other special preparations or aftercare required for the test.

Risks

There are no risks associated with the culture test itself. If insertion of a urinary catheter (thin rubber tube) is required to obtain the urine, there is a slight risk of introducing infection from the catheter.

Normal results

No growth of bacteria is considered the normal result, and this indicates absence of infection.

Abnormal results

Abnormal results, or a positive test, where bacteria are found in the specimen, may indicate a urinary tract infection. Contamination of the specimen from hair, external genitalia, or the rectum may cause a false-positive result. Identification of the number and type of bacteria, with consideration of the method used in obtaining the specimen, is significant in diagnosis.

Escherichia coli causes approximately 80% of infections in patients without catheters, abnormalities of the urinary tract, or calculi (stones). Other bacteria that account for a smaller portion of uncomplicated infections include *Proteus klebsiella* and *Enterobacter*.

Resources

ORGANIZATIONS

American Foundation for Urologic Disease. 300 West Pratt St., Suite 401, Baltimore, MD 21201.

Kathleen D. Wright, RN

Urine flow test

Definition

A urine flow test evaluates the speed of urination, or amount voided per second, and the total time of urination.

Purpose

A urine flow test is utilized to determine bladder function abnormalities, including a narrowed or obstructed urethra (the outflow passage from the bladder) and a weakened bladder muscle (detrusor).

Description

During a urine flow test, the patient urinates into a uroflowmeter, a funnel-shaped device that reads, measures, and computes the rate and amount of urine flow. The test takes approximately 10 minutes.

Preparation

The patient is prohibited from urinating at least two hours before the procedure.

Normal results

Average urine flow rates vary depending on age and gender.

Abnormal results

A urine flow test can indicate problems in bladder function, such as an obstruction, that will need further tests to diagnose.

KEY TERMS

Detrusor muscle—Bladder muscle.

Urethra—Passageway that carries urine from the bladder.

Resources

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J. Ricker Polsdorfer, MD

Urography see **Intravenous urography**

Urticaria see **Hives**

Uterine cancer see **Endometrial cancer**

Uterine fibroid embolization

Definition

Uterine fibroids are growths in the muscular tissue of the womb (uterus). Uterine fibroid embolization is non-surgical removal of the growths through the use of a tubular device called a catheter.

Purpose

It is estimated that between 20% and 40% of women over age 35 years have fibroids. These non-cancerous tumors of the uterus, also called myomas, fibromyomas, or leiomyomas, can be as tiny as a pea or grow to as large as a cantaloupe. They grow along the muscular wall of the uterus and are made of both muscle and fiber-like tissue. Although they are not cancerous, they can cause uncomfortable symptoms in women, particularly **pain** and a feeling of pressure in the area between the hip bones (the lower pelvic area). Some women experience a feeling that they constantly need to urinate. Others may feel pressure in the bowel, and experience bloating or **constipation**. Sometimes, fibroids cause heavy menstrual bleeding and lengthier menstrual periods.

In some cases, women need no treatment for fibroids, since the growths may cause no symptoms or may stop causing symptoms when a woman reaches **menopause**. However, when the fibroids cause heavy bleeding that can lead to anemia, pelvic pain, and

create pressure on other organs, the physician and patient may discuss treatment options. A woman still planning to have children may need to have fibroids removed to ensure they do not interfere with the ability of a fertilized egg to implant on her uterine lining. In rare cases, fibroids cause such severe and sharp pelvic pain that emergency treatment is required.

Although some medications may help ease the symptoms of fibroids or even slow their growth, they are not long-term solutions to the fibroid growths. Surgery also can remove the fibroids. **Myomectomy** is the most common surgery option still used to treat fibroids because it does not involve removing the uterus. In severe cases, women may have a **hysterectomy** and have their uterus removed.

The development of uterine fibroid embolization has offered women a non-surgical alternative to treating fibroids that are causing symptoms. It also is effective for treating multiple fibroids.

Precautions

Although uterine fibroid embolization has been shown in many studies to be effective, some physicians say it is a relatively new procedure that lacks long-term data on its effectiveness. Estimates show that at least 90% of women who have the procedure will have relief of symptoms and no return of fibroids. Some women have reportedly stopped having menstrual periods following the procedure. Since the long-term effects of uterine fibroid embolization on future pregnancies has not been studied extensively, patients still wanting to have children should discuss the latest research and fibroid treatment options with their physicians. Overall, however, the treatment is a safer and less invasive alternative than surgery.

Description

Uterine fibroid embolization usually is performed in a hospital, but sometimes in an outpatient imaging center. An interventional radiologist normally performs the procedure. Interventional radiologists are physicians who are specially trained to use x rays to see inside the body while guiding narrow tubes called catheters through blood vessels to diagnose or treat various diseases.

Uterine fibroid embolization also may be called uterine artery embolization. This is because the procedure injects tiny particles, called embolic agents, into the artery that supplies blood to the fibroid tumor, blocking off the tumor's blood supply. The radiologist reaches the uterine artery by first making a small nick

(about one-quarter of an inch) in the patient's skin in the groin area. From this incision, the radiologist can inset the catheter into the femoral artery and then guide the catheter toward the area of the uterus. Patients do not feel the catheter as it moves along the artery and a nurse or anesthesiologist will provide drugs for pain relief at the insertion site. The nurse or a technologist also will insert a Foley catheter into the patient's bladder. This keeps the bladder emptied so that the radiologist can have a better picture of the uterus and the anatomy that surrounds the uterus during the procedure.

The radiologist can view the progress of the catheter using a moving x-ray technique called fluoroscopy. When the catheter reaches the artery supplying blood to the fibroid tumor, the radiologist injects the embolic agents, which are only about the size of grains of sand. When they gather together, they form a clot and since no more blood can reach the fibroids, they eventually will shrink and disappear. If a woman has fibroids on both sides, the radiologist will guide the catheter around to repeat the procedure on the other side of the uterus. After removing the catheter, the medical team will clean the puncture area and cover it with a bandage. They also remove the Foley catheter. The entire procedure takes about one hour and should cost less than hysterectomy or other surgical treatments. Most insurance companies will pay for uterine fibroid embolization.

Preparation

When the embolization is scheduled, the hospital or facility should provide instructions regarding medications or food and drink to avoid prior to the procedure. Once a patient checks in, a nurse or technician may draw blood for routine tests and a nurse or radiologic technologist who specializes in vascular procedures will start an intravenous (IV) line through which the patient will receive **antibiotics** to prevent infection.

Aftercare

Although uterine fibroid embolization is an outpatient procedure, patients may stay in a recovery area for up to 23 hours. When patients are discharged, they will receive specific instructions about follow-up care and medication for relief of pain and swelling. Uterine fibroid embolization rarely requires stitches. Patients may be advised to use a stool softener while taking some of the aftercare medications to avoid constipation. They will be instructed to drink plenty of fluids for the first week following the procedure. Normally, the patient should avoid soaking in hot baths for the

KEY TERMS

Anemia—A condition in which the blood has a low number of red blood cells, low hemoglobin, or low total blood volume.

Artery—One of several tubular branches of muscular and elastic walled vessels that carry blood from the heart through the body.

Femoral artery—The chief artery of the thigh.

Groin—The fold that marks the meeting of the lower abdomen and the inner thigh.

first three days and should not drive a car for about three days following uterine fibroid embolization.

Strenuous activity, such as climbing stairs, squatting, or lifting heavy objects should be avoided for about one week. Patients also should not stand in one position for long periods of time in the first week following the procedure. It is usually best to avoid heavy **exercise** routines, athletic activity and sexual relations for about up to month. Women also may be advised to avoid using tampons for menstrual periods for up to three months following uterine fibroid embolization, substituting sanitary pads.

Risks

Most patients experience moderate to severe pain and cramping for several hours after the procedure and some will have **nausea** and **fever**. There is a risk of infection from uterine fibroid embolization, but the risk is small and the infection usually can be controlled by antibiotics. A small percentage (about one percent) of women may experience an injury to the uterus or an infection that may lead to a later hysterectomy. There also is a small risk that a woman will stop having periods after uterine fibroid embolization. Overall, these risks are relatively low compared to surgical removal of fibroids.

Normal results

About 90% of women who have the procedure experience significant or complete relief of heavy bleeding, pain, and other symptoms. Patients usually will follow up with their gynecologist about two weeks following the procedure. Recurrence of fibroids is rare, though long-term studies of 10 years or more have shown that about 10% of patients may need additional treatment after 10 years. Additional studies continue on the procedure's long-term effectiveness.

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- Society of Interventional Radiology. 10201 Lee Highway, Suite 500, Fairfax, VA 22030. 703-691-1805. <<http://www.sirweb.org>>.

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Uterine fibroids

Definition

Uterine fibroids (also called leiomyomas or myomas) are benign growths of the muscle inside the uterus. They are not cancerous, nor are they related to **cancer**. Fibroids can cause a wide variety of symptoms, including heavy menstrual bleeding and pressure on the pelvis.

Description

Uterine fibroids are extremely common. About 25% of women in their reproductive years have noticeable fibroids. There are probably many more women who have tiny fibroids that are undetected.

Fibroids develop between the ages of 30–50. They are never seen in women less than 20 years old. After **menopause**, if a woman does not take estrogen, fibroids shrink. It appears that African-American

women are much more likely to develop uterine fibroids.

Fibroids are divided into different types, depending on the location. Submucous fibroids are found in the uterine cavity; intramural fibroids grow on the wall of the uterus; and subserous fibroids are located on the outside of the uterus. Many fibroids are so large that they fit into more than one category. The symptoms caused by fibroids are often related to their location.

Causes and symptoms

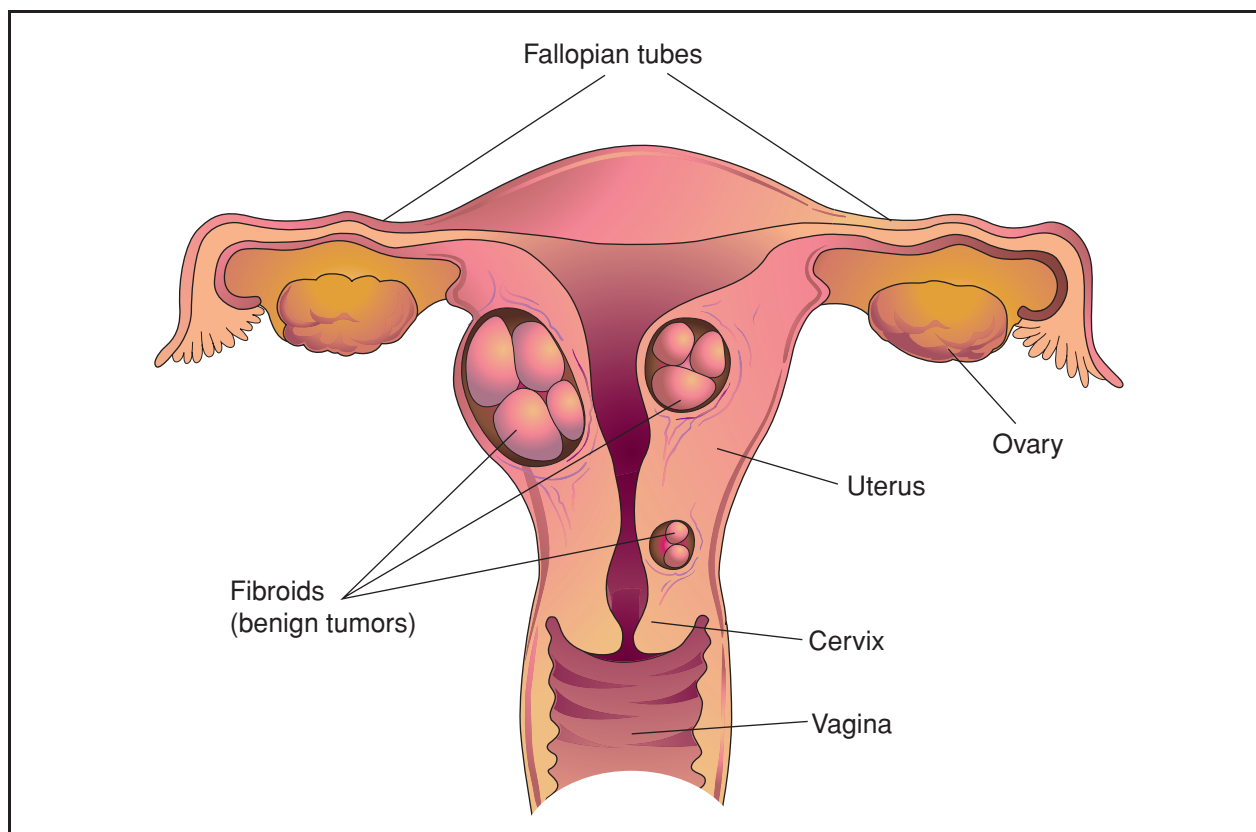
No one knows exactly what causes fibroids. However, the growth of fibroids appears to depend on the hormone estrogen. Fibroids often grow larger when estrogen levels are high, as in pregnancy. Medications that lower the estrogen level can cause the fibroids to shrink.

The signs and symptoms of fibroids include:

- **Heavy uterine bleeding.** This is the most common symptom, occurring in 30% of women who have fibroids. The excess bleeding usually happens during the menstrual period. Flow may be heavier, and periods may last longer. Women who have submucous or intramural fibroids are most likely to have heavy uterine bleeding.
- **Pelvic pressure and pain.** Large fibroids that press on nearby structures such as the bladder and bowel can cause pressure and pain. Larger fibroids tend to cause worse symptoms.
- **Infertility.** This is a rare symptom of fibroids. It probably accounts for less than 3% of infertility cases. Fibroids can cause infertility by compressing the uterine cavity. Submucous fibroids can fill the uterine cavity and interfere with implantation of the fertilized egg.
- **Miscarriage.** This is also an unusual symptom of fibroids, probably accounting for only a tiny fraction of the miscarriages that occur.
- **Pregnancy complications.** Fibroids can greatly increase in size during pregnancy, because of increased levels of estrogen. They can cause pain, and even lead to premature labor.

Diagnosis

A health care provider can usually feel fibroids during a routine pelvic examination. Ultrasound can be used to confirm the diagnosis, but this is not necessary.



Uterine fibroids are benign growths of uterine muscle and are very common. They are divided into three types, depending on the location. Submucous fibroids are found in the uterine cavity; intramural fibroids grow on the wall of the uterus; and subserous fibroids are located outside of the uterus. (Illustration by Electronic Illustrators Group.)

Treatment

Not all fibroids cause symptoms. Even fibroids that do cause symptoms may not require treatment. In the majority of cases, the symptoms are inconvenient and unpleasant, but do not result in health problems.

Occasionally, fibroids lead to such heavy menstrual bleeding that the woman becomes severely anemic. In these cases, treatment of the fibroids may be necessary. Very large fibroids are much harder to treat. Therefore, many doctors recommend treatment for moderately-sized fibroids, in the hopes of preventing them from growing into large fibroids that cause worse symptoms.

The following are possible treatment plans:

- **Observation.** This is the most common plan. Most women already have symptoms at the time their fibroids are discovered, but feel that they can tolerate their symptoms. Therefore, no active treatment is given, but the woman and her physician stay alert for signs that the condition might be getting worse.

- **Hysterectomy.** This involves surgical removal of the uterus, and it is the only real cure for fibroids. In fact, 25% of hysterectomies are performed because of symptomatic fibroids. By the time a woman has a hysterectomy for fibroids, she has usually endured several years of worsening symptoms. That's because fibroids tend to grow over time. A gynecologist can remove a fibroid uterus during either an abdominal or a vaginal hysterectomy. The choice depends on the size of the fibroids and other factors such as previous births and previous surgeries.
- **Myomectomy.** In this surgical procedure only the fibroids are removed; the uterus is repaired and left in place. This is the surgical procedure many women choose if they are not finished with childbearing. At first glance, it seems that this treatment is a middle ground between observation and hysterectomy. However, myomectomy is actually a difficult surgical procedure, more difficult than a hysterectomy. Myomectomy often causes significant blood loss, and blood transfusions may be required. In addition, some fibroids are so large, or buried so deeply within

KEY TERMS

Anemia—Low blood count.

GnRH antagonists—A group of medications that affect the reproductive hormones. These medications are used to treat fibroids, endometriosis, and infertility.

Hysterectomy—Removal of the uterus (with or without removal of the ovaries) by surgery. The surgery can be performed through an incision in the abdomen, or the uterus can be removed through the vagina.

Menopause—The end of the reproductive years, signaled by the end of menstrual periods. Also known as “the change.”

Osteoporosis—Brittle bones commonly found in elderly women.

the wall of the uterus, that it is not possible to save the uterus, and a hysterectomy must be done, even though it was not planned. There are exceptions to this, however. Sometimes, fibroids grow on a stalk (pedunculated fibroids), and these are easy to remove.

- Medical treatment. Since fibroids are dependent on estrogen for their growth, medical treatments that lower estrogen levels can cause fibroids to shrink. A group of medications known as GnRH antagonists can dramatically lower estrogen levels. Women who take these medications for three to six months find that their fibroids shrink in size by 50% or more. They usually experience dramatic relief of their symptoms of heavy bleeding and pelvic pain.

Unfortunately, GnRH antagonists cause unpleasant side effects in over 90% of women. The therapy is usually used for only three months, and should not be used for more than six months because the risk of developing brittle bones (**osteoporosis**) begins to rise. Once the treatment is stopped, the fibroids begin to grow back to their original size. Within six months, most of the old symptoms return. Therefore, GnRH agonists cannot be used as long-term solution. At the moment, treatment with GnRH antagonists is used mainly in preparation for surgery (myomectomy or hysterectomy). Shrinking the size of the fibroids makes surgery much easier, and reducing the heavy bleeding allows a woman to build up her blood count before surgery.

Fibroids can cause problems during pregnancy because they often grow in size. Large fibroids can

cause pain and lead to **premature labor**. Fibroids cannot be removed during pregnancy because of the risk of injury to the uterus and hemorrhage. GnRH antagonists cannot be used during pregnancy. Treatment is limited to pain medication and medication to prevent premature labor, if necessary.

Prognosis

Many women who have fibroids have no symptoms or have only minor symptoms of heavy menstrual bleeding or pelvic pressure. However, fibroids tend to grow over time, and gradually cause more symptoms. Many women ultimately decide to have some form of treatment. Currently, hysterectomy is the most popular form of treatment.

Prevention

Uterine fibroids cannot be prevented.

Resources

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Uterus x rays see **Hysterosalpingography**

Uveitis

Definition

Uveitis is an inflammation of the uveal tract, which lines the inside of the eye behind the cornea. Much of the uvea lies between the retina and tough, outer sclera. The uveal tract has three parts: the iris, the ciliary body, and the choroid. Uveitis is categorized according to the part of the uveal tract that is affected. Anterior uveitis is an inflammation of the front part of the uveal tract; it includes inflammation of the iris (iritis) and inflammation of the iris and the ciliary body (iridocyclitis). Posterior uveitis is an inflammation of the part of the uveal tract behind the lens of the eye. It includes inflammation of the choroid (choroiditis) and inflammation of the choroid and retina (chorioretinitis). Uveitis that affects the entire uveal tract is called panuveitis or diffuse uveitis.

Description

The uveal tract is made up of the iris, ciliary body, and choroid. The iris is the colored part of the eye. The ciliary body is inside the eye and produces a fluid called aqueous humor. Ciliary muscles aid in accommodation, the process of changing the shape of the lens in the eye to see things at various distances. The choroid lines the back of the eye and has many blood vessels. It helps nourish part of the retina. The choroid lies between the retina and outermost sclera.

Uveitis may either persist for a long time (chronic) or have a short-term duration (acute). Anterior uveitis is classified as either granulomatous or nongranulomatous. The distinction is based on the disease agents that were considered responsible for the condition. At one time, it was thought that granulomatous uveitis was caused by tuberculosis bacilli whereas nongranulomatous uveitis was thought to be caused by streptococci. The distinction is still used even though the causes of uveitis are now understood differently.

In most cases, uveitis affects only one eye, although posterior uveitis sometimes involves both eyes. About 60% of cases develop within the eye itself, but 40% are associated with systemic diseases or disorders ranging from **toxoplasmosis** to **syphilis**. Many of these are diseases of childhood and adolescence. Uveitis does not appear to run in families or to be associated with lifestyle choices, occupational history, geographical location, or environmental factors.

Uveitis is a serious condition that may develop rapidly and cause lasting damage to the eye. Patients who think they may have chronic uveitis should seek evaluation and treatment by an ophthalmologist (a physician who specializes in diseases of the eye) as soon as possible. If the patient has a sudden loss of vision and the eye looks inflamed, the patient should go *immediately* to the doctor for emergency treatment.

Causes and symptoms

The causes of uveitis are not fully understood, but they can be a result of trauma, allergy, or a response to a systemic or ocular disease. Uveitis may be a type of immune-response mechanism. In people with impaired immune systems, uveitis may be due to an infection.

Chronic uveitis is often associated with systemic disorders (e.g., Lyme disease, **sarcoidosis**, or juvenile **rheumatoid arthritis**).

Anterior uveitis

The so-called classic symptoms of anterior uveitis—severe **pain**; redness, particularly around the edge of the iris; and extreme sensitivity to light (photophobia)—occur mostly in acute uveitis. In anterior uveitis, the doctor will see a so-called “flare and cell” pattern when looking into the watery fluid (aqueous humor) between the cornea and the lens of the patient’s eye. The iris may adhere to the lens, thus increasing the intraocular pressure. There may be nodules on the iris. There may be tearing and the pupil may be constricted and nonreactive. In severe cases of anterior uveitis, there may be hypopyon (a small amount of pus or collection of white cells) visible when the doctor examines the eye.

GRANULOMATOUS UVEITIS. In granulomatous uveitis, there will be large yellowish-white cells visible on the back of the cornea, and possibly some small nodules on the iris. Granulomatous uveitis is usually less acute than the nongranulomatous form; the eye is only mildly inflamed and the patient’s vision is somewhat blurred.

Granulomatous uveitis can be produced by syphilis, toxoplasmosis, cytomegalovirus, sarcoidosis, **tuberculosis**, or Vogt-Koyonagi-Harada syndrome (VKH). VKH is marked by severe uveitis associated with hair loss, **hearing loss**, loss of pigment in the eyelashes and brows, and headaches. It occurs most commonly in Asians.

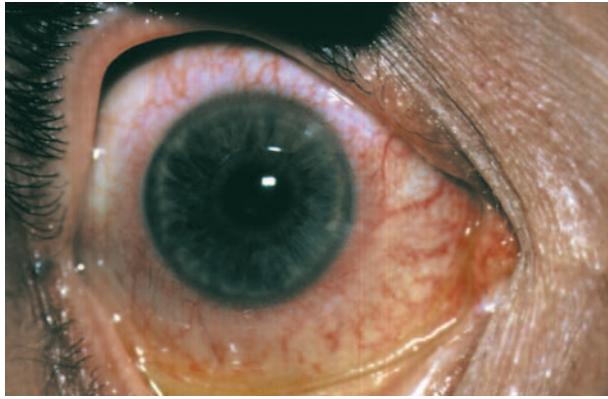
NONGRANULOMATOUS UVEITIS. In nongranulomatous uveitis, the cells visible on the cornea are smaller, and there are no masses on the iris. This type of anterior uveitis is, however, more painful. The eye is red and the patient experiences both photophobia and loss of vision.

Systemic diseases that can cause nongranulomatous uveitis include **ankylosing spondylitis**, Reiter’s syndrome, psoriasis, ulcerative colitis, Behcet’s syndrome, **Lyme disease**, and Crohn’s disease. Children—especially girls—with anterior uveitis should be screened for juvenile rheumatoid arthritis (JRA).

Posterior uveitis

The symptoms of posterior uveitis are sometimes subtle. The patient may notice blurred or hazy vision, or floating black spots before the eyes. There may be pain and photophobia. The iris may attach to the lens in the eye thus increasing intraocular pressure.

Posterior uveitis may be acute or chronic. It is more likely to involve both eyes. When the doctor



This person has acute iritis, or inflammation of the iris. Symptoms include pain in the eye or forehead and reddening of the margin of the iris. Treatment requires total rest of the eye (dark glasses and atropine drops which paralyze the muscles of accommodation), accompanied by application of corticosteroids. (Photo Researchers, Inc. Reproduced by permission.)

examines the eye, cells may be seen in the vitreous humor, which is the normally transparent gel that fills the eyeball behind the lens. There will be yellowish or dark areas of inflammation on the choroid and the retina. The blood vessels in the retina develop a sheath or covering of inflammatory tissue. In severe cases, the vitreous humor is so cloudy that the doctor cannot see the retina at the back of the eye.

PARS PLANITIS. Pars planitis is an inflammation of the pars plana, which is a part of the ciliary body. Pars planitis usually occurs in older children or young adults, and can develop into posterior uveitis.

The diseases that cause granulomatous uveitis may also cause posterior uveitis.

Diagnosis

The eye doctor will examine the patient's eyes with a slit lamp in order to rule out conjunctivitis and certain types of **glaucoma**. The slit lamp is an instrument that combines a binocular microscope with a special light. The slit lamp can shine a narrow beam of very bright light into the eye and allow the doctor to examine the front part of the eye in detail. The slit-lamp exam is not painful, however if the patient is sensitive to light there will be discomfort.

The absence of a discharge from the eye and the absence of infectious organisms in a laboratory smear usually rule out **conjunctivitis**. In addition, the size of the pupil is often small in uveitis whereas it is normal in conjunctivitis. In acute glaucoma, the patient has

KEY TERMS

Choroid—The part of the uveal tract behind the ciliary body. The choroid underlies and nourishes the retina and absorbs scattered light.

Ciliary body—The part of the uveal tract between the iris and the choroid.

Cornea—The transparent front part of the eye that covers the iris and pupil.

Flare and cell—A pattern revealed by slit-lamp examination that indicates uveitis. Flare and cell resembles light filtered through smoke.

Hypopyon—A small amount of pus or collection of white cells that is visible in the front of the eye in severe cases of anterior uveitis.

Iris—The circular membrane that forms the colored portion of the eye and expands or contracts around the pupil.

Photophobia—Extreme sensitivity to light. Photophobia is a major symptom of acute uveitis.

Pupil—The opening in the center of the iris that allows light to pass through to the retina.

Retina—The innermost membrane at the back of the eyeball on which images are projected by the lens.

Slit lamp—An instrument that combines a binocular microscope with special lights. It allows an eye doctor to examine the front portion of the eye.

Uveal tract—The pigmented membrane that lines the back of the retina of the eye and extends forward to include the iris. The uveal tract is sometimes called the uvea and has three parts: the iris, the choroid, and the ciliary body.

Vitreous humor—The clear gel-like substance that fills the eyeball behind the lens.

severe pain, the cornea of the eye is cloudy, and the pressure level of the fluid inside the eye is abnormally high; whereas in uveitis the pain is moderate, the cornea is clear, and the fluid pressure is normal or possibly lower or slightly above normal. The doctor may also use the slit lamp and another lens to examine the back of the eye to get a good look at the retina and choroid. Other instruments, such as a hand-held ophthalmoscope or a binocular indirect ophthalmoscope, can be used to examine the back of the eye. There should be no discomfort with these tests except if the patient is sensitive to the bright light.

Laboratory testing

Laboratory testing is used to rule out conjunctivitis in some patients. The doctor wipes the inside of the patient's eyelid with a swab in order to obtain a sample for testing. Although blood tests are not necessary to diagnose uveitis by itself, they are used to diagnose the cause if the doctor suspects that toxoplasmosis or another systemic disease is responsible for the uveitis.

Treatment

Uveitis is generally treated by an ophthalmologist because therapy requires topical and oral medications, however, some optometrists (O.D.) are state licensed to use therapeutic medications. Other doctors may be involved in treating the underlying disease, if the patient has one, and in monitoring the patient's responses to medications.

Anterior uveitis is treated with corticosteroid drops; in severe cases, the patient may be given steroid injections in the area of the eye or oral steroids. Atropine sulfate drops may be given to dilate the patient's pupil. Posterior uveitis is treated with systemic **corticosteroids**. It is usually not necessary to dilate the pupil.

Prolonged steroid use may increase intraocular pressure, thereby increasing the risk of glaucoma. Steroid use has also been connected to cataract formation. Patients should be monitored closely and frequently.

Prognosis

The prognosis depends upon the location of the uveitis, on whether it is chronic or acute, and on the promptness of treatment. The prognosis for untreated uveitis is poor. Untreated anterior uveitis usually progresses to posterior uveitis, resulting in **cataracts**, scar tissue, and eventual glaucoma. If treated promptly, anterior uveitis usually clears up in several days or weeks, but is likely to recur. Posterior uveitis usually results in some permanent loss or blurring of vision.

Prevention

Patients with anterior uveitis should be warned about the possibility of recurrence and instructed about its symptoms, especially inflammation of the iris. They should be advised to seek treatment at once at the first signs of recurrence.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424.

< <http://www.eyenet.org> >.

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. < <http://www.aoanet.org> >.

Rebecca J. Frey, PhD

V

Vaccination

Definition

Vaccination is the use of vaccines to prevent specific diseases.

Purpose

Many diseases that once caused widespread illness, disability, and **death** now can be prevented through the use of vaccines. Vaccines are medicines that contain weakened or dead bacteria or viruses. When a person takes a vaccine, his or her immune system responds by producing antibodies—substances that weaken or destroy disease-causing organisms. When the person is later exposed to live bacteria or viruses of the same kind that were in the vaccine, the antibodies prevent those organisms from making the person sick. Vaccines usually also stimulate the so-called cellular immune system as well. In other words, the person becomes immune to the disease the organisms normally cause. The process of building up immunity by taking a vaccine is called immunization.

Vaccines are used in several ways. Some, such as the rabies vaccine, are given only when a person is likely to have been exposed to the virus that causes the disease—through a dog bite, for example. Others are given to travelers planning to visit countries where certain diseases are common such as typhoid fever or **yellow fever**. Vaccines such as the **influenza** vaccine, or “flu shot,” are given mainly to specific groups of people—older adults and others who are at high risk of developing influenza or its complications. Then, there are vaccines that are given to almost everyone, such as the ones that prevent **diphtheria**, tetanus, polio and **measles**.

Children routinely have a series of vaccinations that begins at birth. Given according to a specific schedule, these vaccinations protect against hepatitis B,

diphtheria, **tetanus**, pertussis (**whooping cough**), measles, mumps, **rubella** (German measles), varicella (**chickenpox**), **polio**, pneumococcus and *Haemophilus influenzae* type b (Hib disease, a major cause of spinal meningitis) and, in some states, hepatitis A. This series of vaccinations is recommended by the American Academy of Family Physicians, the American Academy of Pediatrics, and the Centers for Disease Control and Prevention and is required in all states before children can enter school. All states will make exceptions for children who have medical conditions such as **cancer** that prevent them from having vaccinations, and some states also will make exceptions for children whose parents object for religious or other reasons.

Description

In addition to those discussed above, vaccines are available for preventing **anthrax**, **cholera**, **hepatitis A**, **Japanese encephalitis**, meningococcal **meningitis**, **plague**, pneumococcal infection (meningitis, **pneumonia**), **tuberculosis**, **typhoid fever**, and **yellow fever**. Most vaccines are given as injections, but a few are given by mouth.

Some vaccines are combined in one injection, such as the measles-mumps-rubella (MMR) or diphtheria-pertussis-tetanus (DPT) combinations.

Recommended dosage

The recommended dosage depends on the type of vaccine and may be different for different patients. The healthcare professional who gives the vaccine will decide on the proper dose.

A vaccination health record will help parents and health care providers keep track of a child's vaccinations. The record should be started when the child has his or her first vaccination and should be updated with each additional vaccination. While most physicians

ALBERT BRUCE SABIN (1906-1993)

Albert Bruce Sabin was born on August 26, 1906, in Bialystok, Russia, to Jacob and Tillie Sabin. In order to escape extreme poverty, the Sabins immigrated to the United States and settled in Paterson, New Jersey. Following his graduation from high school in 1923, Sabin was able to attend dentistry school at New York University due to his uncle's generous offer for financing. However, after reading Paul deKruif's *Microbe Hunters* he became intrigued by virology and the idea of curing epidemic diseases. After two years of dentistry school, Sabin decided to switch to medicine, earning his M.D. in 1931. Sabin completed his residency and internship in the United States and then went to London to conduct research.

Sabin returned to the United States in 1935 to resume his research of polio at the Rockefeller Institute. In 1953, Jonas Salk announced that he had created a dead-virus polio vaccine that was safe, but soon after its administration many people died. Sabin, however, wanted to create a live-virus vaccine, which he felt would be safer. Sabin diluted three strains of the polio virus and tested these on himself, his family, and other volunteers. These live-virus vaccines (given orally) proved safe and effective and soon became the vaccinations of choice around the world. Sabin's published works include *Viruses and Cancer: A Public Lecture in Conversational Style* (1965), *Behavior of Chimpanzee-Avirulent Poliomyelitis Viruses in Experimentally Infected Human Volunteers* (1955), and *Recent Advances in Our Knowledge of Dengue and Sand Fly Fever* (1955). Sabin died of congestive heart failure on March 3, 1993.

follow the recommended vaccination schedule, parents should understand that some flexibility is allowed. For example, vaccinations that are scheduled for age two months may be given anytime between six to 10 weeks. When possible, follow the schedule. However, slight departures will not prevent the child from developing immunity, as long as all the vaccinations are given at around the right times. The child's physician is the best person to decide when each vaccination should be given.

Anyone planning a trip to another country should check to find out what vaccinations are needed. Some vaccinations must be given as much as 12 weeks before the trip, so getting this information early is important. Many major hospitals and medical centers have travel clinics that can provide this information. The

Recommended Immunization Schedule

Age	Vaccine
0–2 months	Hepatitis B
1–4 months	Hepatitis B
2 months	DTP (diphtheria-tetanus-pertussis) Hib (Haemophilus influenzae type B) Polio
4 months	DTP Hib Polio
6 months	DTP Hib
6–18 months	Hepatitis B Polio
12–15 months	Hib MMR (measles-mumps-rubella)
12–18 months	Varicella (chickenpox)
15–18 months	DTP
4–6 years	DTP Polio MMR
11–12 years	Hepatitis B (if not already completed) MMR (if not already completed) Varicella (if not already completed)
11–16 years	DT (diphtheria-tetanus booster shot; and then booster shot every 10 to 15 years)

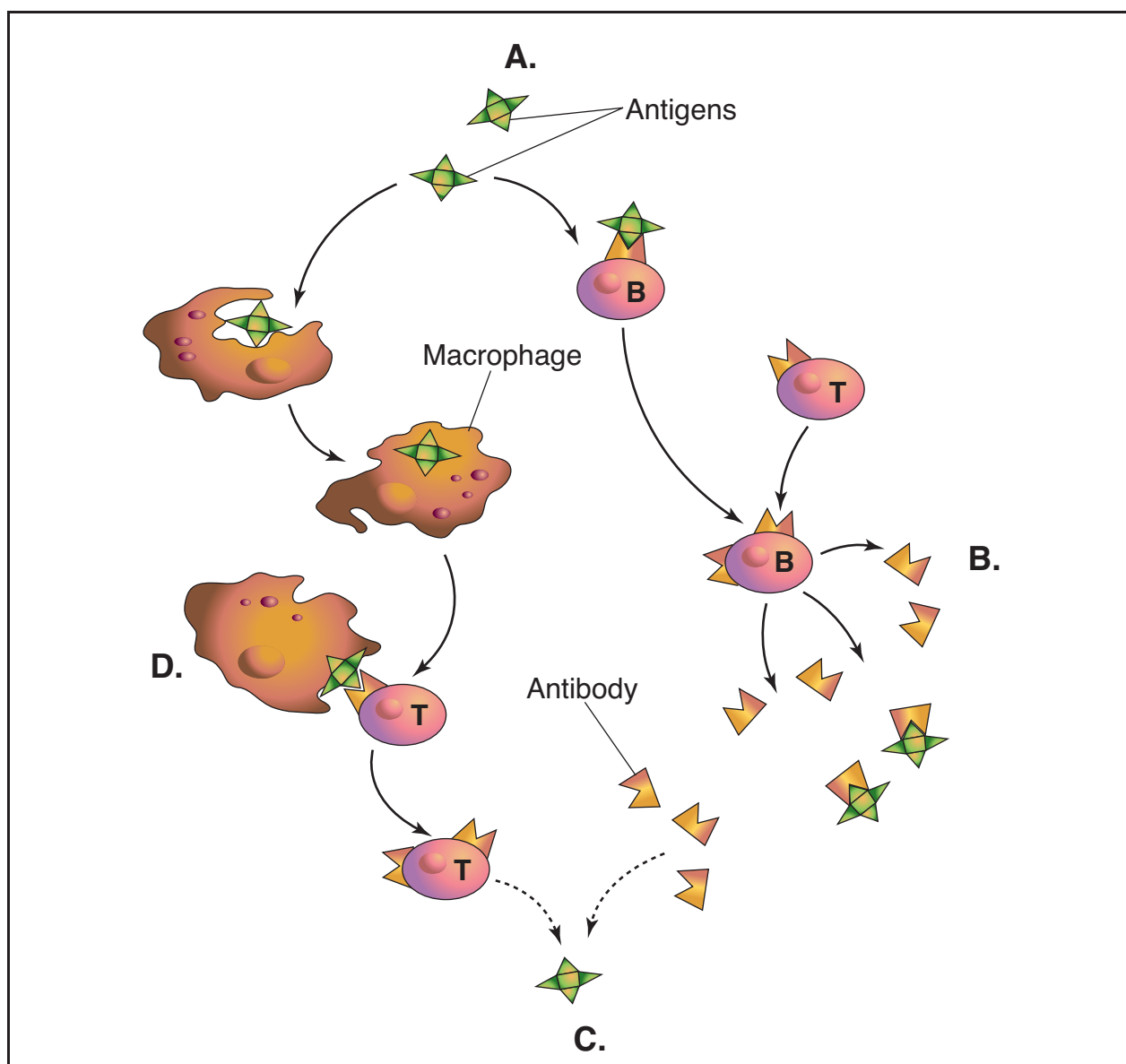
Traveler's Health Section of the Centers for Disease Control and Prevention also has information on vaccination requirements.

Precautions

Vaccines are not always effective, and there is no way to predict whether a vaccine will "take" in any particular person. To be most effective, vaccination programs depend on whole communities participating. The more people who are vaccinated, the lower everyone's risk of being exposed to a disease. Even people who do not develop immunity through vaccination are safer when their friends, neighbors, children, and coworkers are immunized.

Like most medical procedures, vaccination has risks as well as substantial benefits. Anyone who takes a vaccine should make that sure he or she is fully informed about both the benefits and the risks. Any questions or concerns should be discussed with a physician or other health care provider. The Centers for Disease Control and Prevention, located in Atlanta, Georgia, also is a good source of information.

Vaccines may cause problems for people with certain **allergies**. For example, people who are allergic to the **antibiotics** neomycin or polymyxin B should not take rubella vaccine, measles vaccine, **mumps** vaccine or the combined measles-mumps-rubella (MMR)



How vaccines work: **A.** Vaccines contain antigens (weakened or dead viruses, bacteria, and fungi that cause disease and infection). When introduced into the body, the antigens stimulate the immune system response by instructing B cells to produce antibodies, with assistance from T-cells. **B.** The antibodies are produced to fight the weakened or dead viruses in the vaccine. **C.** The antibodies “practice” on the weakened viruses, preparing the immune system to destroy real and stronger viruses in the future. **D.** When new antigens enter the body, white blood cells called macrophages engulf them, process the information contained in the antigens, and send it to the T-cells so that an immune system response can be mobilized. (Illustration by Electronic Illustrators Group.)

vaccine. Anyone who has had a severe allergic reaction to baker’s yeast should not take the **hepatitis B** vaccine. Patients who are allergic to antibiotics such as gentamicin sulfate, streptomycin sulfate or other **amino-glycosides** should check with their physicians before taking influenza vaccine, as some influenza vaccines contain small amounts of these drugs. Also, some vaccines, including those for influenza, measles and

mumps, are grown in the fluids of chick embryos and should not be taken by people who are allergic to eggs. In general, anyone who has had an unusual reactions to a vaccine in the past should let his or her physician know before taking the same kind of vaccine again. The physician also should be told about any allergies to foods, medicines, preservatives, or other substances.



An allergic reaction to a vaccination shot. (Photograph by Lester V. Bergman, Corbis Images. Reproduced by permission.)

People with certain other medical conditions should be cautious about taking vaccines. Influenza vaccine, for example, may reactivate Guillain-Barré syndrome (GBS) in people who have had it before. This vaccine also may worsen illnesses that involve the lungs, such as **bronchitis** or pneumonia. Vaccines that cause fever as a side effect may trigger seizures in people who have a history of seizures caused by fever.

Certain vaccines are not recommended for use during **pregnancy**, but some may be given to women at especially high risk of getting a specific disease such as polio. Vaccines also may be given to pregnant women to prevent medical problems in their babies. For example, vaccinating a pregnant woman with tetanus toxoid can prevent her baby from getting tetanus at birth.

Women should avoid becoming pregnant for three months after taking rubella vaccine, measles vaccine, mumps vaccine or the combined measles-mumps-rubella (MMR) as these vaccines could cause problems in the unborn baby.

Women who are breastfeeding should check with their physicians before taking any vaccine.

Side effects

Most side effects from vaccines are minor and easily treated. The most common are pain, redness, and swelling at the site of the injection. Some people

may also develop a fever or a rash. In rare cases, vaccines may cause severe allergic reactions, swelling of the brain, or seizures. Anyone who has an unusual reaction after receiving a vaccine should get in touch with a physician right away.

Interactions

Vaccines may interact with other medicines and medical treatments. When this happens, the effects of the vaccine or the other medicine may change or the risk of side effects may be greater. For example, **radiation therapy** and cancer drugs may reduce the effectiveness of many vaccines or may increase the chance of side effects. Anyone who takes a vaccine should let the physician know all other medicines he or she is taking and should ask whether the possible interactions could interfere with the effects of the vaccine or the other medicines.

Resources

OTHER

Centers for Disease Control National Immunization Program. <<http://www.cdc.gov/nip>>.

National Immunization Information Hotline. Centers for Disease Control and Prevention. (800) 232-2522.

Larry I. Lutwick, MD

Vaccines see **Vaccination**

Vaginal pain

Definition

Pain in the vaginal canal is usually associated with an underlying medical and/or psychological condition.

Description

Vaginal pain is experienced usually during vaginal manipulation or sexual intercourse. Approximately 50–85% of the causes are due to organic (medical) conditions. However, it is typical for the medical condition to be compounded by psychological issues such as depression and problems associated with sexual identity. The primary entity concerns dyspareunia, a vaginal pain experienced during sexual intercourse. The vagina has three physiological functions: an outflow duct for menstrual discharge, to receive the penis during sexual intercourse, and as the birthing canal. The overall prevalence for dyspareunia is 20% (15% of women and 57% of

KEY TERMS

Anthrax—An infectious disease caused by a type of bacterium. The disease can be passed from animals to people and usually is fatal. Symptoms include sores on the skin.

Antibody—A type of protein produced in the blood or in the body tissues that helps the body fight infection.

Bacteria—Tiny, one-celled forms of life that cause many diseases and infections.

Cholera—An infection of the small intestine caused by a type of bacterium. The disease is spread by drinking water or eating seafood or other foods that have been contaminated with the feces of infected people. It occurs in parts of Asia, Africa, Latin America, India, and the Middle East. Symptoms include watery diarrhea and exhaustion.

Encephalitis—Inflammation of the brain, usually caused by a virus. The inflammation may interfere with normal brain function and may cause seizures, sleepiness, confusion, personality changes, weakness in one or more parts of the body, and even coma.

Feces—The solid waste that is left after food is digested. Feces form in the intestines and pass out of the body through the anus. Also called stool.

Guillain-Barré syndrome (GBS)—A disease of the nerves with symptoms that include sudden numbness and weakness in the arms and legs, sometimes leading to paralysis. The disease is serious and requires medical treatment, but most people recover completely.

Immune system—The body's natural defenses against disease and infection.

Immunization—A process or procedure that protects the body against an infectious disease. A vaccination is a type of immunization.

Inflammation—Pain, redness, swelling, and heat that usually develop in response to injury or illness.

Meningitis—Inflammation of tissues that surround the brain and spinal cord.

Microorganism—An organism that is too small to be seen with the naked eye.

Organism—An individual of some type of life form, such as a plant, an animal, or a microorganism.

Plague—A highly infectious disease that can be fatal if not treated promptly. The bacteria that cause plague mainly infect rats, mice, squirrels, and other wild rodents. The disease is passed to people through fleas. Infected people can then spread the disease to other people.

Seizure—A sudden attack, spasm, or convulsion.

Tuberculosis—An infectious disease that usually affects the lungs, but may also affect other parts of the body. Symptoms include fever, weight loss, and coughing up blood.

Typhoid fever—An infectious disease caused by a type of bacterium. People with this disease have a lingering fever and feel depressed and exhausted. Diarrhea and rose-colored spots on the chest and abdomen are other symptoms. The disease is spread through poor sanitation.

Virus—A tiny, disease-causing particle that can reproduce only in living cells.

Yellow fever—An infectious disease caused by a virus. The disease, which is spread by mosquitoes, is most common in Central and South America and Central Africa. Symptoms include high fever, jaundice (yellow eyes and skin) and dark-colored vomit, a sign of internal bleeding. Yellow fever can be fatal.

men). A significant percentage of **breast cancer** and **hysterectomy** patients demonstrated **sexual dysfunction**.

Causes and symptoms

The causes can be categorized as organic, due to a medical condition and/or psychological difficulties. Medical conditions can include chronic diseases, minor ailments, breast **cancer**, and medications. Psychological cause can be related to physical or sexual **abuse**. **Pregnancy** and hormonal changes (decreased estrogen) have significant negative impact on sexual activity, desire, and satisfaction. Dyspareunia can be divided

into three types of pain: superficial, vaginal, and deep. Superficial pain is associated with attempted penetration. This is usually caused by changes in anatomy, irritative condition, or vaginismus. Vaginal pain is associated with friction, indicating a problem with lubrication and /or arousal disorders. Deep pain is related to thrusting and is indicative of pelvic disease or an inability for **pelvic relaxation**.

Diagnosis

The diagnosis must be pursued with diligence and in a comprehensive manner. A careful history and

KEY TERMS

Laparoscopic surgery—A surgical procedure to correct or diagnose an underlying disease.

physical examination is essential. Procedures that can be used include surgical investigation (**laparoscopy**) and treatment of the underlying cause(s).

Treatment

Treatment is directed at diagnosing the underlying condition, which can be medical and/or psychological cause(s). Treatment can include surgery, hormonal therapy (replacements), psychotherapy, and pain control protocols.

Prognosis

The prognosis depends on the primary cause. If treatment is aggressively pursued and patient compliance is satisfactory the overall outcome is favorable.

Prevention

There are no precise preventive measures since the condition can result from normal **aging** and/or progressively worsening psychological disease.

Resources

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Ryan, Kenneth J., et al, editors. *Kistner's Gynecology & Women's Health*. 7th ed. Mosby, Inc., 1999.

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Kathleen Berrisford, MSW, CSW

Vaginal warts see **Genital warts**

Vaginitis see **Vulvovaginitis**

Vagotomy

Definition

Vagotomy is the surgical cutting of the vagus nerve to reduce acid secretion in the stomach.

Purpose

The vagus nerve splits into branches that go to different parts of the stomach. Stimulation from these branches causes the stomach to produce acid. Too much stomach acid leads to ulcers that may eventually bleed and create an emergency situation.

Vagotomy is performed when acid production in the stomach can not be reduced by other means. It is used when ulcers in the stomach and duodenum do not respond to medication and changes in diet. It is an appropriate surgery when there are ulcer complications, such as obstruction of digestive flow, bleeding, or perforation. The frequency with which elective vagotomy is performed has decreased in the past 20 years as drugs have become increasingly effective in treating ulcers. However, the number of vagotomies performed in emergency situations has remained about the same.

Vagotomy is often performed in conjunction with other gastrointestinal surgery, such as partial removal of the stomach (antrectomy or subtotal **gastrectomy**). There are several types of vagotomies. Truncal vagotomy severs the trunk of the vagus nerve as it enters the abdomen. Parietal cell or proximal gastric vagotomy leaves the trunk intact, but severs the branches that go to different parts of the stomach.

Precautions

Patients who receive vagotomies are most often seen in emergency situations where bleeding and perforated ulcers make it necessary to act immediately. As with any major surgery, people who use alcohol excessively, smoke, are obese, and are very young or very old are at higher risks for complications.

Description

Vagotomy is performed under **general anesthesia** by a surgeon in a hospital. The surgeon makes an incision in the abdomen and locates the vagus nerve. Either the trunk or the branches leading to the stomach are cut. Then the abdominal muscles are sewn back together, and the skin is closed with sutures.

Often, other gastrointestinal surgery is performed at the same time as the vagotomy. Part of the stomach may be removed, for instance. Vagotomy causes a decrease in peristalsis and a change in the emptying patterns of the stomach. To ease this, a **pyloroplasty** is often performed. This procedure widens the outlet from the stomach to the small intestine.

KEY TERMS

Duodenum—The section of the small intestine immediately after the stomach.

Peristalsis—The rhythmic contractions that move material through the bowel.

Preparation

A gastroscopy and x rays of the gastrointestinal system are performed as diagnostic procedures to determine the position and condition of the ulcer. Standard preoperative blood and urine tests are done. The patient should discuss with the anesthesiologist any medications or conditions that might affect the administration of anesthesia.

Aftercare

Patients who have had a vagotomy stay in the hospital for about seven days. For the first three or four days, nasogastric suctioning is required. A tube is inserted through the nose and into the stomach. The stomach contents are then suctioned out. Patients eat a clear liquid diet until the gastrointestinal tract is functioning again. When patients return to a regular diet, spicy and acidic food should be avoided.

It takes about six weeks to fully recover from the surgery. The sutures that close the skin can be removed in seven to ten days. Patients are encouraged to move around soon after the operation to prevent the formation of deep vein **blood clots**. **Pain** medication, stool softeners, and **antibiotics** may be prescribed following the operation.

Risks

As with all surgery, excessive bleeding and infection are possible complications. In addition, the emptying patterns of the stomach are changed. This can lead to dumping syndrome and **diarrhea**. Dumping syndrome is a condition where shortly after eating, the patient experiences **palpitations**, sweating, **nausea**, cramps, **vomiting**, and diarrhea.

Normal results

Normal recovery is expected for most patients. In about 10% of those who have vagotomy without stomach removal, ulcers recur. Two to three percent of patients who have some portion of their stomach removed also have recurrent ulcers.

Resources

BOOKS

Way, Lawrence W., editor. "Stomach and Duodenum." In *Current Surgical Diagnosis and Treatment*. 10th ed. Stamford: Appleton & Lange, 1994.

Tish Davidson, A.M.

Valacyclovir see **Antiviral drugs**

Valley fever see **Coccidioidomycosis**

Valsalva maneuver

Definition

The Valsalva maneuver is performed by attempting to forcibly exhale while keeping the mouth and nose closed. It is used as a diagnostic tool to evaluate the condition of the heart and is sometimes done as a treatment to correct abnormal heart rhythms or relieve chest **pain**.

Purpose

The Valsalva maneuver is used with patients who have suspected heart abnormalities, often in conjunction with **echocardiography**. The maneuver is based on the fact that when a patient forcibly exhales against a closed nose and mouth while bearing down, as if having a bowel movement, specific changes occur in blood pressure and the rate and volume of blood returning to the heart.

Comparing the changes in a diseased heart to those expected in a normal heart gives clues to the type and location of heart damage. In addition, when a doctor listens to the chest with a stethoscope during the Valsalva maneuver, characteristic heart sounds are heard. Variations in these sounds can indicate the type of abnormality present in the heart. A 2004 study found that blood pressure response to the Valsalva maneuver could predict mortality in elderly patients with congestive **heart failure**. This could prove to be a new noninvasive way to help determine how long elderly patients with congestive heart failure are expected to live.

The Valsalva maneuver also corrects some rapid heartbeats originating in the atria. When the maneuver is done correctly, blood pressure rises. This forces the heart to respond by correcting its rhythm and beating more slowly. On rare occasions, the Valsalva

KEY TERMS

Atria—The heart has four chambers. The right and left atria are at the top of the heart and receive returning blood from the veins. The right and left ventricles are at the bottom of the heart and act as the body's main pumps.

Echocardiography—An ultrasound test that shows the size, shape, and movement of the heart.

maneuver can be used to diminish chest pain in patients with mild coronary disease.

Unrelated to any evaluation of the heart, the Valsalva maneuver also is taught to patients with **multiple sclerosis** who are unable to fully empty the bladder (flaccid bladder). It sometimes is used in sexual therapy to help men avoid **premature ejaculation**.

Precautions

The Valsalva maneuver should not be performed by patients who have severe **coronary artery disease**, have experienced recent **heart attack**, or have a moderate to severe reduction in blood volume.

Description

When performed formally, the patient is asked to blow against an aneroid pressure measuring device (manometer) and maintain a pressure of 40 millimeters of mercury (mm Hg) for 30 seconds. Or, less formally, the patient may be asked to bear down, as if having a bowel movement. During this 30 second period, a recording is made of the changes in blood pressure and murmurs of the heart.

Preparation

The patient may be connected to a heart monitor and echocardiograph or the physician may simply use a stethoscope to monitor the heart. Sometimes an indwelling needle is inserted for accurate pressure measurements, depending on whether the procedure is being done for corrective or diagnostic purposes.

Aftercare

When this procedure is done to regulate irregular heart rhythms, the patient usually remains on a heart monitor to evaluate heartbeat.

Risks

The patient may feel dizzy or faint during the procedure, but serious consequences are rare. There is a risk that the Valsalva maneuver can cause **blood clots** to detach, bleeding, and abnormal rhythms originating in the ventricle. It can also cause cardiac arrest. Consequently, the procedure is usually performed in a setting where emergency equipment is accessible.

Normal results

There are four characteristic changes or phases in a normal heart's response to the Valsalva maneuver. An abnormality in any of these phases indicates a cardiovascular abnormality.

Resources

PERIODICALS

Jancin, Bruce. "New Mortality Predictor Found for Heart Failure." *Family Practice News* March 15, 2004: 48–49.

Tish Davidson, A.M.
Teresa G. Odle

Valvular heart disease

Definition

Valvular heart disease refers to several disorders and diseases of the heart valves, which are the tissue flaps that regulate the flow of blood through the chambers of the heart.

Description

The human heart consists of four chambers—two upper chambers (the atria) and two lower chambers (the ventricles) that are responsible for pumping blood. The heart valves are like one-way doors, which open and close with each beat of the heart, controlling the blood flow from one chamber to the next. Each of these valves is made up of a few thin folds of tissue. When functioning correctly, they keep blood from flowing backward into a chamber when closed.

The four valves function in the following manner:

- The mitral valve is located between the left atrium and the left ventricle. It is the only valve with two flaps, or cusps.

- The tricuspid valve is located on the right side of the heart, between the right atrium and right ventricle. It is made up of three cusps, each a different size.
- The aortic valve is located on the left side of the heart and opens to allow blood to leave the heart from the left ventricle into the aorta, which is the main artery of the body. It closes to prevent blood from flowing back into the left ventricle.
- The pulmonary valve is situated on the right side of the heart, between the right ventricle and pulmonary artery. It allows blood to exit the heart and enter the lungs via the pulmonary artery. It closes to prevent blood from flowing back into the right ventricle.

Patients with valvular heart disease have a malfunction of one or more of these valves. There are several types of valvular heart diseases with distinct symptoms and treatments. These are:

- mitral valve prolapse (displacement)
- mitral valve insufficiency (regurgitation)
- mitral valve stenosis (narrowing)
- aortic valve insufficiency
- aortic valve stenosis
- tricuspid valve insufficiency
- tricuspid valve stenosis
- pulmonic stenosis
- pulmonic insufficiency

Certain types of heart disease can lead to one of the specific conditions listed above. These include **rheumatic fever** and infective inflammation of the heart (endocarditis). Multivalvular heart disease refers to a condition involving more than one of the heart valves.

Causes and symptoms

Problems with heart valves may occur as a result of infection, degeneration, or congenital abnormality. The most common infections are rheumatic **fever** and infective **endocarditis**.

Rheumatic fever

Rheumatic fever is a condition that results from the body's immune response to certain types of streptococcal bacteria. It occurs rarely. When it occurs, it is most often in children who have had **strep throat** that was not completely treated. The symptoms of rheumatic fever include heart inflammation, uncontrolled movement of face and limbs, arthritis that moves from joint to joint, shortness of breath, weakness, and either

lumps under the skin or raised red patches on the skin. The most common valvular heart diseases to result from rheumatic fever are mitral stenosis, tricuspid stenosis, aortic insufficiency, aortic stenosis, multivalvular involvement, and less commonly, pulmonic stenosis. Chronic rheumatic heart disease can result from one occurrence or from repeated attacks. It is not as common as it once was in the United States, but still occurs frequently in developing countries.

Infective endocarditis

Infective endocarditis is an infection and inflammation of the inner layer, or membrane, of heart tissue (endocardium). Most people with a healthy, normal heart are not at significant risk for contracting infective endocarditis. However, people who have had rheumatic fever, with its resulting scarring, or a congenital cause of heart malformation, may contract this disease when certain bacteria enter the bloodstream and become lodged in the heart. In particular, dental surgery or any surgery involving the mouth, bladder, prostate, or female pelvic organs increases risk for this infection. The disease also may occur in drug addicts who inject their veins using unsterilized needles, even if they have normal heart valves. Symptoms of infective endocarditis include fever, a new or changing heart murmur, and abnormal loss of appetite or **fatigue**.

The use of appetite suppressants

In 1997 the manufacturers of fen-phen, the abbreviation for a combination of the two weight-loss drugs fenfluramine and phentermine, withdrew the supplements from the market. Preliminary research showed that they caused heart valve problems in some patients. Physicians were concerned that this drug combination could affect the heart valves because the drugs alter metabolism of serotonin in the body. Serotonin is a natural substance found in the brain and intestines that can affect blood vessels. Until the issue could be studied more, physicians recommended that patients taper off the drugs, finally stopping them altogether. Eventually, the company reached a \$3.75 billion settlement of most lawsuits by former users of the diet drugs.

Other valvular heart disease

The mitral and aortic valves may also be affected by deposits of calcium in the heart that occurs with **aging**. This can lead to thickening and leakage of heart valves. Heart attacks also can damage the mitral valve structures. Additionally, certain connective tissue disorders can adversely affect the heart valves, for

example, Marfan's syndrome and myxomatous degeneration.

Diagnosis

Specific types of valvular heart disease are diagnosed using **electrocardiography** (EKG), echocardiography, certain x-ray studies, and/or **cardiac catheterization**. An EKG provides a record of electrical changes in the heart muscle during the heartbeat. Echocardiography uses sound waves to make images of the heart. These images can show if there are any abnormalities of the heart valves. Cardiac catheterization is a procedure in which a small tube (called a catheter) is inserted into an artery and passed into the heart. It is used to measure pressure in the heart and the amount of blood pumped by the heart.

Rheumatic fever

Rheumatic fever may be suspected when a recent throat infection has occurred and other major or minor symptoms appear, such as joint ache, abnormal EKG, or a blood test indicating heart inflammation. **Heart murmurs** may be detected from routine examination.

Infective endocarditis

A diagnosis of infective endocarditis can be obtained through patient history, EKG, ultrasound, or cardiac catheterization. Patients who have developed the disease rapidly may report fever, fatigue, night sweats, chills, and joint inflammation. Those whose disease has developed more slowly will show signs of rapid heart rate, an enlarged spleen, various skin colors or spots, and heart murmur. The physician may order blood tests to determine what is causing the infection.

Appetite suppressants

People with a history of using appetite suppressants may be sent for EKGs or further testing if any of the symptoms of valvular heart disease, such as swelling, considerable fatigue, or shortness of breath occur.

Treatment

The treatment of specific valvular heart diseases will vary, depending on the valve involved and the extent of damage or malfunction. Some patients will not require treatment and many will be treated with medication. Sometimes, patients need surgery. If

multivalvular disease is suspected or involved, different valves may be evaluated during surgery on one of the affected valves. Women with heart valve disease who want to become pregnant should receive a thorough check-up and see a cardiologist regularly throughout their **pregnancy**.

Rheumatic fever

Patients with rheumatic fever will be treated with **antibiotics** to eliminate streptococcal organisms that may still remain in the heart. Patients may receive antibiotics to prevent further infection, and inflammation may be treated with **aspirin** or cortisone-like drugs.

Infective endocarditis

Physicians will use the appropriate antibiotic or some combination of antibiotics to treat infective endocarditis, depending on the type of bacterium that caused the disease. Severe cases of this disease may be corrected by valve replacement surgery.

Appetite suppressants

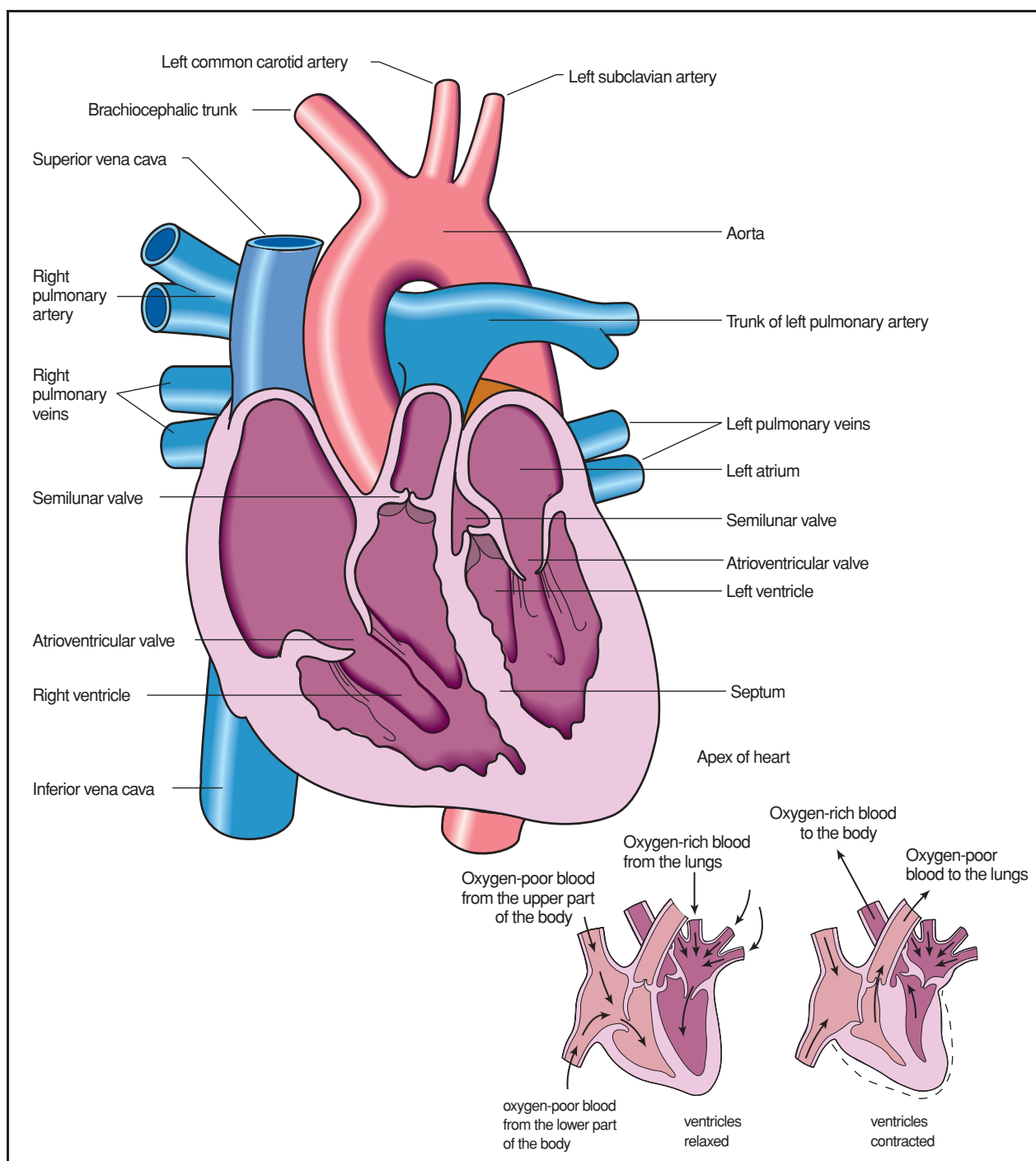
The role of appetite suppressants (fen-phen) in valvular heart disease has been under study. Many of the people who suffered complications of the diet drugs experienced regurgitation. However, some 2001 studies found that these valve problems often did not worsen or even improved after a period of time after the patients stopped taking fen-phen.

Prognosis

The prognosis for patients with valvular heart disease varies depending on the underlying cause, age and health of the patient, and the degree of valvular damage or involvement.

Rheumatic fever

Patients with rheumatic heart disease face a lifetime of caution over contact with the same bacterium that caused the disease. Since it can cause inflammation of one or more organs or joints, complications can occur. The inflammation of the heart may subside without side effects. Permanent scarring of one or more heart valves is a possibility and may require surgery to repair or replace damaged valves. In severe cases, rheumatic fever can lead to **death** from **heart failure**.



Anatomy of the human heart. The illustration at the bottom right shows how the right and left ventricles contract and relax when blood is pumped through the heart. (Illustration by Hans & Cassady, Inc.)

Infective endocarditis

The prognosis for patients with infective endocarditis depends on the underlying heart disease and resulting complications. If the disease further

damages heart valves, symptoms may occur for years after initial treatment. Sometimes, endocarditis can result in heart or renal failure. If untreated, it can be fatal.

KEY TERMS

Congenital—Used to describe a condition or defect present at birth.

Stenosis—An abnormal valve condition which is characterized by tightening or narrowing of the opening.

Streptococcal (*Streptococcus*)—*Streptococcus* is a bacterium that causes infection in people. Its most commonly known strain causes the infection strep throat.

Throat culture—A test for strep throat that involves swabbing the back of the throat and sending the swab to a laboratory, which will determine whether bacteria is present.

Appetite suppressants

Since it is believed that different valves may be affected, treatment for those taking these drugs most likely follows a similar course as that for the specific valvular disease.

Prevention

Certain measures can be taken to prevent some valvular disease. However, once valvular heart disease that results from congenital abnormality occurs, it may not be prevented. Steps can be taken to prevent further complications.

Rheumatic fever

The best prevention for rheumatic fever is prompt and thorough treatment of any suspected streptococcal infection, particularly strep throat in children. A physician should check any sore throat with fever that persists for more than 24 hours. The physician will probably order a **throat culture**. Completion of the antibiotic treatment even after symptoms diminish is important to be certain the infection is eliminated.

Infective endocarditis

Anyone who was born with a defective heart valve, those with artificial (prosthetic) valves, or those who have had a valve scarred by rheumatic fever, should use prescribed antibiotics by mouth before and after a dental procedure. These patients also may need to receive injected antibiotics prior to procedures involving the bladder, prostate, and pelvic organs.

Appetite suppressants

The drug associated with valvular heart disease, fen-phen, was removed from the market, so as long as people avoid obtaining this appetite suppression combination, they can prevent its negative effects.

Resources

PERIODICALS

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- Reimold, Sharon C., and John D. Rutherford. "Valvular Heart Disease in Pregnancy." *The New England Journal of Medicine* July 3, 2003: 52–58.
- "What Happened After Patients Stopped Taking Fen-Phen?" *Heart Disease Weekly* March 11, 2001: 8.

ORGANIZATIONS

- American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.
- National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Teresa G. Odle

Valvuloplasty see **Heart valve repair**

Varicella see **Chickenpox**

Varicocele removal see **Testicular surgery**

Varicose veins

Definition

Varicose veins are dilated, tortuous, elongated superficial veins that are usually seen in the legs.

Description

Varicose veins, also called varicosities, are seen most often in the legs, although they can be found in other parts of the body. Most often, they appear as lumpy, winding vessels just below the surface of the skin. There are three types of veins, superficial veins that are just beneath the surface of the skin, deep veins that are large blood vessels found deep inside muscles, and perforator veins that connect the superficial veins to the deep veins.

The superficial veins are the blood vessels most often affected by varicose veins and are the veins seen by eye when the varicose condition has developed.

The inside wall of veins have valves that open and close in response to the blood flow. When the left ventricle of the heart pushes blood out into the aorta, it produces the high pressure pulse of the heartbeat and pushes blood throughout the body. Between heartbeats, there is a period of low blood pressure. During the low pressure period, blood in the veins is affected by gravity and wants to flow downward. The valves in the veins prevent this from happening. Varicose veins start when one or more valves fail to close. The blood pressure in that section of vein increases, causing additional valves to fail. This allows blood to pool and stretch the veins, further weakening the walls of the veins. The walls of the affected veins lose their elasticity in response to increased blood pressure. As the vessels weaken, more and more valves are unable to close properly. The veins become larger and wider over time and begin to appear as lumpy, winding chains underneath the skin. Varicose veins can develop in the deep veins also. Varicose veins in the superficial veins are called primary varicosities, while varicose veins in the deep veins are called secondary varicosities.

Causes and symptoms

The predisposing causes of varicose veins are multiple, and lifestyle and hormonal factors play a role. Some families seem to have a higher incidence of varicose veins, indicating that there may be a genetic component to this disease. Varicose veins are progressive; as one section of the veins weakens, it causes increased pressure on adjacent sections of veins. These sections often develop varicosities. Varicose veins can appear following **pregnancy**, **thrombophlebitis**, congenital blood vessel weakness, or **obesity**, but is not limited to these conditions. **Edema** of the surrounding tissue, ankles, and calves, is not usually a complication of primary (superficial) varicose veins and, when seen, usually indicates that the deep veins may have varicosities or clots.

Varicose veins are a common problem; approximately 15% of the adult population in the United States have varicose veins. Women have a much higher incidence of this disease than men. The symptoms can include aching, **pain**, itchiness, or burning sensations, especially when standing. In some cases, with chronically bad veins, there may be a brownish discoloration of the skin or ulcers (open sores) near the ankles. A condition that is frequently associated with varicose veins is spider-burst veins. Spider-burst veins are very small veins that are enlarged. They may be



Varicose veins on a man's leg. (Custom Medical Stock Photo. Reproduced by permission.)

KEY TERMS

Congenital—Existing at or before birth; a condition that developed while the fetus was in utero or as a consequence of the birth process.

Edema—Swelling caused by a collection of fluid in a tissue or body cavity.

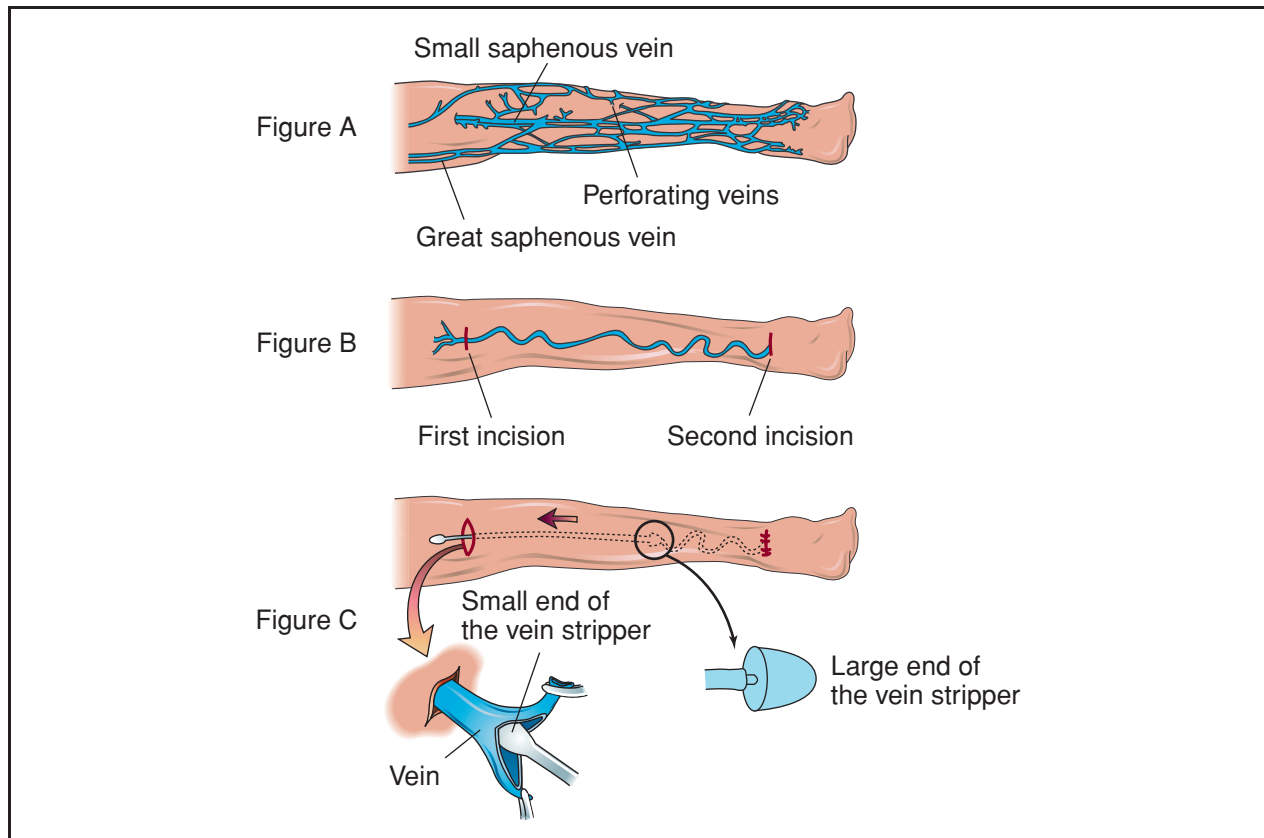
Hemorrhage—Bleeding from blood vessels.

Palpation—The process of examining a patient by touch.

caused by back-pressure from varicose veins, but can be caused by other factors. They are frequently associated with pregnancy and there may be hormonal factors associated with their development. They are primarily of cosmetic concern and do not present any medical concerns.

Diagnosis

Varicose veins can usually be seen. In cases where varicose veins are suspected, but can not be seen, a physician may frequently detect them by palpation (pressing with the fingers). X rays or ultrasound tests can detect varicose veins in the deep and perforator veins and rule out **blood clots** in the deep veins.



Varicose veins may be surgically removed from the body when they are causing pain and when hemorrhaging or recurrent thrombosis appear. Surgery involves making an incision through the skin at both ends of the section of vein being removed (figure B). A flexible wire is inserted through one end and extended to the other. The wire is then withdrawn, pulling the vein out with it (figure C). (Illustration by Electronic Illustrators Group.)

Treatment

There is no cure for varicose veins. Treatment falls into two classes; relief of symptoms and removal of the affected veins. Symptom relief includes such measures as wearing support stockings, which compress the veins and hold them in place. This keeps the veins from stretching and limits pain. Other measures are sitting down, using a footstool when sitting, avoiding standing for long periods of time, and raising the legs whenever possible. These measures work by reducing the blood pressure in leg veins. Prolonged standing allows the blood to collect under high pressure in the varicose veins. **Exercise** such as walking, biking, and swimming, is beneficial. When the legs are active, the leg muscles help pump the blood in the veins. This limits the amount of blood that collects in the varicose veins and reduces some of the symptoms. These measures reduce symptoms, but do not stop the disease.

Surgery is used to remove varicose veins from the body. It is recommended for varicose veins that are

causing pain or are very unsightly, and when hemorrhaging or recurrent thrombosis appear. Surgery involves making an incision through the skin at both ends of the section of vein being removed. A flexible wire is inserted through one end and extended to the other. The wire is then withdrawn, pulling the vein out with it. This is called “stripping” and is the most common method to remove superficial varicose veins. As long as the deeper veins are still functioning properly, a person can live without some of the superficial veins. Because of this, stripped varicose veins are not replaced.

Injection therapy is an alternate therapy used to seal varicose veins. This prevents blood from entering the sealed sections of the vein. The veins remain in the body, but no longer carry blood. This procedure can be performed on an out-patient basis and does not require anesthesia. It is frequently used if people develop more varicose veins after surgery to remove the larger varicose veins and to seal spider-burst veins for people concerned about cosmetic appearance.

Injection therapy is also called sclerotherapy. At one time, a method of injection therapy was used that did not have a good success rate. Veins did not seal properly and blood clots formed. Modern injection therapy is improved and has a much higher success rate.

Prognosis

Untreated varicose veins become increasingly large and more obvious with time. Surgical stripping of varicose veins is successful for most patients. Most do not develop new, large varicose veins following surgery. Surgery does not decrease a person's tendency to develop varicose veins. Varicose veins may develop in other locations after stripping.

Resources

BOOKS

Alexander, R. W., R. C. Schlant, and V. Fuster, editors. *The Heart*. 9th ed. New York: McGraw-Hill, 1998.

John T. Lohr, PhD

Variola see **Smallpox**

Vascular headache see **Migraine headache**

Vasculitis

Definition

Vasculitis refers to a varied group of disorders which all share a common underlying problem of inflammation of a blood vessel or blood vessels. The inflammation may affect any size blood vessel, anywhere in the body. It may affect either arteries and/or veins. The inflammation may be focal, meaning that it affects a single location within a vessel; or it may be widespread, with areas of inflammation scattered throughout a particular organ or tissue, or even affecting more than one organ system in the body.

Description

Inflammation is a process which occurs when the immune system of the body responds to either an injury or a foreign invader (virus, bacteria, or fungi). The immune system response involves sending a variety of cells and chemicals to the area in question. Inflammation causes blood vessels in the area to leak, causing swelling. The inflamed area becomes red, hot to the touch, and tender.

Antibodies are immune cells which recognize and bind to specific markers (called antigens) on other cells (including bacteria and viruses). These antibody-antigen complexes can then stimulate the immune system to send a variety of other cells and chemicals involved in inflammation to their specific location.

Some researchers believe that the damaging process of vasculitis is kicked off by such antibody-antigen complexes. These complexes are deposited along the walls of the blood vessels. The resulting inflow of immune cells and chemicals causes inflammation within the blood vessels.

The type of disease caused by vasculitis varies depending on a number of factors:

- the organ system or tissue in which the vasculitis occurs
- the specific type of inflammatory response provoked
- whether the affected vessels are veins (which bring blood to the heart) or arteries (which carry blood and oxygen from the heart to the organs and tissues)
- the degree to which blood flow within the affected vessel is reduced

Causes and symptoms

Some types of vasculitis appear to be due to a type of allergic response to a specific substance (for example, a drug). Other types of vasculitis have no identifiable initiating event. Furthermore, researchers have not been able to consistently identify antibody-antigen complexes in all of the types of diseases caused by vasculitis. The types of antigens responsible for the initial immune response have often gone unidentified as well. Furthermore, not all people with such complexes deposited along the blood vessels go on to develop vasculitis. Some researchers believe that, in addition to the presence of immune complexes, an individual must have some other characteristics which make him or her susceptible to vasculitis. Many questions have yet to be answered to totally explain the development these diseases.

Symptoms

Symptoms of vasculitis depend on the severity of the inflammation and the organ system or systems affected. Some types of vasculitis are so mild that the only symptoms noted are small reddish-purple dots (called petechiae) on the skin due to tiny amounts of blood seeping out of leaky blood vessels. In more widespread types of vasculitis, the patient may have general symptoms of illness, including **fever**, achy

muscles and joints, decreased appetite, weight loss, and loss of energy. The organ systems affected by vasculitis may include:

- The skin. **Rashes**, bumps under the skin, petechiae, larger reddish-purple circles (purpura), or bruising (ecchymoses) may appear. Areas of skin totally deprived of blood flow, and therefore of oxygen, may die, resulting in blackened areas of gangrene.
- The joints. In addition to joint **pain**, the joints themselves may become inflamed, resulting in arthritis.
- Brain and nervous system. Inflammation of the blood vessels in the brain can cause headaches, changes in personality, confusion, and seizures. If an area of the brain becomes totally deprived of oxygen, a **stroke** occurs. A stroke means that an area of brain tissue is either severely injured or completely dead from lack of oxygen. This may leave the individual with a permanent disability. If the vessels that lead to the eyes are affected, vision may become seriously disturbed. Nerves in the arms and legs may result in painful **tingling** sensations, loss of feeling, and weakness.
- Gastrointestinal system. Patients may have significant abdominal pain, **vomiting**, and diarrhea. If blood flow is completely cut off to an area of intestine, that part of the intestine will die off. The liver may be affected.
- Heart. This is an extremely serious type of vasculitis. The arteries of the heart (coronary arteries) may develop weakened areas, called aneurysms. The heart muscle itself may become inflamed and enlarged. With oxygen deprivation of the heart muscle, the individual may suffer a heart attack.
- Lungs. The patient may experience **shortness of breath** with chest pain, and may **cough** up blood. There may be **wheezing**.
- Kidney. Changes in the arteries of the kidney may result in high blood pressure. The kidneys may become increasingly unable to appropriately filter the blood, and kidney failure may occur.

Specific diseases

Multiple types of disease are associated with vasculitis. Many autoimmune diseases have vasculitis as one of their complications. These include systemic lupus erythematosus, **rheumatoid arthritis**, **scleroderma**, and **polymyositis**. Other types of diseases which have vasculitis as their major manifestations include:

- Polyarteritis nodosa. This is an extremely serious, systemic (affecting systems throughout the body) form of vasculitis. Small and medium arteries are



This person's legs are afflicted with leukocytoblastic vasculitis, a condition in which a blood or lymph vessel becomes inflamed. (Custom Medical Stock Photo. Reproduced by permission.)

involved, and the inflammation is so severe that the walls of the arteries may be destroyed. Any organ system, or multiple organ systems, may be affected. The most serious effects include kidney failure, complications involving the heart, gastrointestinal problems, and high blood pressure.

- Kawasaki's disease is an acute disease which primarily strikes young children. Fever and skin manifestations occur in all patients. While most patients recover completely, a few patients suffer from vasculitis in the heart. This is frequently fatal.
- Henoch-Schonlein purpura. While this frequently occurs in children, adults may also be affected. This disease tends to affect the skin, joints, gastrointestinal tract, and kidneys.
- **Serum sickness** occurs when an individual reacts to a component of a drug, for example penicillin.

KEY TERMS

Aneurysm—A weakened area in the wall of a blood vessel which causes an outpouching or bulge. Aneurysms may be fatal if these weak areas burst, resulting in uncontrollable bleeding.

Antibody—Specialized cells of the immune system which can recognize organisms that invade the body (such as bacteria, viruses, and fungi). The antibodies are then able to set off a complex chain of events designed to kill these foreign invaders.

Antigen—A special, identifying marker on the outside of cells.

Autoimmune disorder—A disorder in which the body's antibodies mistake the body's own tissues for foreign invaders. The immune system therefore attacks and causes damage to these tissues.

Immune system—The system of specialized organs, lymph nodes, and blood cells throughout the body which work together to prevent foreign invaders (bacteria, viruses, fungi, etc.) from taking hold and growing.

Inflammation—The body's response to tissue damage. Includes hotness, swelling, redness, and pain in the affected part.

Petechia—A tiny, purplish-red spot on the skin. Caused by the leakage of a bit of blood out of a vessel and under the skin.

Purpura—A large, purplish-red circle on the skin. Caused by the leakage of blood out of a vessel and under the skin.

Symptoms of this are often confined to the skin, although fevers, joint pain, and swelling of lymph nodes may also occur.

- **Temporal arteritis** (also called giant cell arteritis) tends to involve arteries which branch off the major artery that leads to the head, called the carotid. An artery which feeds tissues in the area of the temple (the temporal artery) is often affected. Severe headaches are the most classic symptom. Other symptoms include **fatigue**, loss of appetite and then weight, fever, heavy sweating, joint pain, and pain in the muscles of the neck, shoulders, and back. If the vasculitis includes arteries which supply the eye, serious visual disturbance or even blindness may result.

- Takayasu's arteritis affects the aorta (the very large main artery that exits the heart and receives all of the blood to be delivered throughout the body), and arteries which branch off of the aorta. Initial symptoms include fatigue, fever, sweating at night, joint pain, and loss of appetite and weight. Every organ may be affected by this disease. A common sign of this disease is the inability to feel the pulse in any of the usual locations (the pulse is the regular, rhythmic sensation one can feel with a finger over an artery, for example in the wrist, which represents the beating of the heart and the regular flow of blood).
- **Wegener's granulomatosis**: This disease exerts its most serious effects on the respiratory tract. The vasculitis produced by this disease includes the formation of fibrous, scarring nodules called granulomas. Symptoms include nose bleeds, ear infections, cough, shortness of breath, and chest pain. There may be bleeding in the lungs, and a patient may cough up blood. The kidneys, eyes, and skin are also frequently involved.

Diagnosis

Diagnosis of any type of vasculitis involves demonstrating the presence of a strong inflammatory process. Tests which reveal inflammation throughout the body include **erythrocyte sedimentation rate**, blood tests which may reveal anemia and increased white blood cells, and tests to demonstrate the presence of immune complexes and/or antibodies circulating in the blood. An x-ray procedure, called **angiography**, involves injecting dye into a major artery, and then taking x-ray pictures to examine the blood vessels, in order to demonstrate the presence of inflammation of the vessel walls. Tissue samples (biopsies) may be taken from affected organs to demonstrate inflammation.

Treatment

Even though there are many different types of vasculitis, with many different symptoms based on the organ system affected, treatments are essentially the same. They all involve trying to decrease the activity of the immune system. Steroid medications (like prednisone) are usually the first types of drugs used. Steroids work by interfering with the chemicals involved in the inflammatory process. More potent drugs for severe cases of vasculitis have more serious side effects. These include drugs like cyclophosphamide. Cyclophosphamide works by actually killing cells of the patient's immune system.

Prognosis

The prognosis for vasculitis is quite variable. Some mild forms of vasculitis, such as those brought on by reactions to medications, may resolve totally on their own and not even require treatment. Temporal arteritis, serum sickness, Henoch-Schonlein purpura, and Kawasaki's disease usually have excellent prognoses, although when Kawasaki's affects the heart, there is a high death rate. Other types of vasculitis were always fatal, prior to the availability of prednisone and cyclophosphamide, and continue to have high rates of fatal complications. These include polyarteritis nodosa and Wegener's granulomatosis.

Prevention

Because so little is known about what causes a particular individual to develop vasculitis, there are no known ways to prevent it.

Resources

ORGANIZATIONS

Lupus Foundation of America. 1300 Piccard Dr., Suite 200, Rockville, MD 20850. (800) 558-0121. <<http://www.lupus.org>>.

Wegener's Foundation, Inc. 3705 South George Mason Drive, Suite 1813 South, Falls Church, VA 22041. (703) 931-5852.

Rosalyn Carson-DeWitt, MD

Vasectomy

Definition

A vasectomy is a surgical procedure performed on males in which the vas deferens (tubes that carry sperm from the testicles to the seminal vesicles) are cut, tied, cauterized (burned or seared) or otherwise interrupted. The semen no longer contains sperm after the tubes are cut, so conception cannot occur. The testicles continue to produce sperm, but they die and are absorbed by the body.

Purpose

The purpose of this operation is to provide reliable **contraception**. Research indicates that the level of

effectiveness is 99.6%. Vasectomy is the most reliable method of contraception.

Description

Vasectomies are often performed in the doctor's office using a **local anesthesia**. The patient's scrotum area will be shaved and cleaned with an antiseptic solution to reduce the chance of infection. A small incision is made into the scrotum (the sac containing the testicles that produce the sperm). Each of the vas deferens (one from each testicle) is tied in two places with nonabsorbable (permanent) sutures and the tube is severed between the ties. The ends may be cauterized (burned or seared) to decrease the chance that they will leak or grow back together.

Sterility does not occur immediately after the procedure is finished. Men must use other methods of contraception until two consecutive semen analyses confirm that there are no sperm present in the semen. This will take four to six weeks or 15-20 ejaculations to clear all of the sperm from the tubes.

"No scalpel" vasectomies are gaining popularity. Instead of an incision, a small puncture is made into the scrotum. The vas deferens are cut and sealed in a manner similar to that described above. No stitches are necessary and the patient has less pain. Other advantages include less damage to the tissues, less bleeding, less risk of infection, and less discomfort after the procedure.

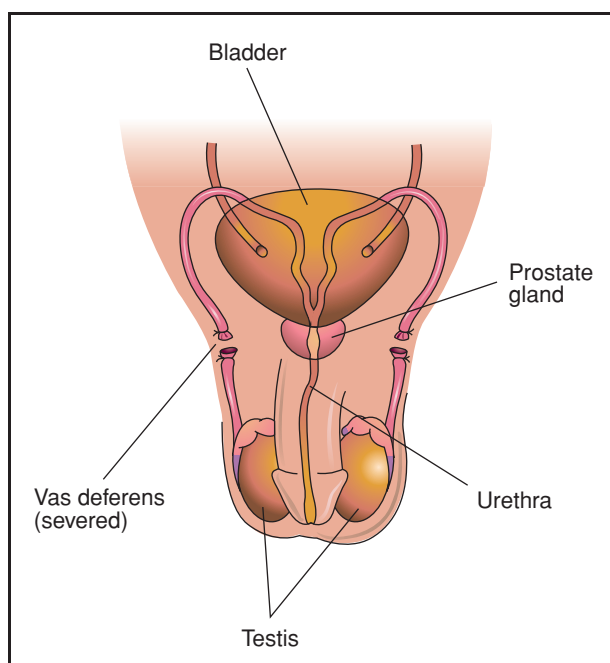
In some, cases vasectomies may be reversed. However, this procedure should be considered permanent as there is no guarantee of successful reversal.

Preparation

No special physical preparation is required. The physician will first assess the patient's general health in order to identify any potential problems that could occur. The doctor will then explain possible risks and side effects. The patient is asked to sign a consent form which indicates that he understands the information he has received, and gives the doctor permission to perform the operation.

Aftercare

Following the surgery, ice packs are often applied to scrotum to decrease **pain** and swelling. A dressing (or athletic supporter) which supports the scrotum can also reduce pain. Mild over-the-counter pain medication such as **aspirin** or **acetaminophen** (Tylenol) should be able to control any discomfort. Activities may be



Vasectomy is a surgical procedure performed on males in which the vas deferens (tubes that conduct sperm from the testicles to the penis) are cut, tied, cauterized, or otherwise interrupted. Although the testicles still produce sperm, the sperm die and are absorbed by the body. Men who have had vasectomies may continue to ejaculate the same amount of semen as before the procedure. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Ejaculation—The act of expelling the sperm through the penis during orgasm.

Epididymitis—Inflammation of the small tube that rests on top of the testicle and is part of the system that carries sperm from the testicle to the penis. The condition can be successfully treated with antibiotics if necessary.

Scrotum—The sac which contains the testicles.

Sperm granuloma—A collection of fluid that leaks from an improperly sealed or tied vas deferens. They usually disappear on their own, but can be drained if necessary.

Testicles—The two egg-shaped organs found in the scrotum that produce sperm.

Tubal ligation—A surgical procedure in which the fallopian tubes are tied in two places and cut between. This prevents eggs from moving from the ovary to the uterus.

restricted for one or two days, and sexual intercourse for three to four days.

Risks

There are very few risks associated with vasectomy other than infection, bruising, epididymitis (inflammation of the tube that carries the sperm from the testicle to the penis), and sperm granulomas (collection of fluid that leaks from a poorly sealed or tied vas deferens). These are easily treated if they do occur. Patients do not experience difficulty achieving an erection, maintaining an erection, or ejaculating. There is no decrease in the production of the male hormone (testosterone), and sex drive and ability are not altered. Vasectomy is safer and less expensive than tubal ligation (sterilization of a female by cutting the fallopian tube to prevent conception).

Normal results

Normally, vasectomies are 99% successful in preventing conception. As such, it is one of the most effective methods available to consumers.

Resources

ORGANIZATIONS

Planned Parenthood Federation of America, Inc.. 810 Seventh Ave., New York, NY 10019. (800) 669-0156. <<http://www.plannedparenthood.org>>.

Donald G. Barstow, RN

Vasodilators

Definition

Vasodilators are medicines that act directly on muscles in blood vessel walls to make blood vessels widen (dilate).

Purpose

Vasodilators are used to treat high blood pressure (**hypertension**). By widening the arteries, these drugs allow blood to flow through more easily, reducing blood pressure. Controlling high blood pressure is important because the condition puts a burden on the heart and the arteries, which can lead to permanent damage over time. If untreated, high blood pressure increases the risk of **heart attack**, **heart failure**, **stroke**,

or kidney failure. Vasodilators usually are prescribed with other types of blood pressure drugs and rarely are used alone.

Description

Examples of vasodilators are hydralazine (Apresoline) and **minoxidil** (Loniten). The vasodilator hydralazine also may be used to control high blood pressure in pregnant women or to bring down extremely high blood pressure in emergency situations. In the forms used for treating high blood pressure (tablets or injections), these drugs are available only with a physician's prescription. A liquid form of minoxidil, used to promote hair growth in people with certain kinds of baldness and is applied directly to the scalp, and is sold without a prescription.

Recommended dosage

The recommended dosage depends on the type of vasodilator. The physician who prescribed the drug or the pharmacist who filled the prescription can recommend the correct dosage. This medicine should only be used as directed.

Physicians usually prescribe vasodilators along with other blood pressure medicines. Taking each drug at the correct time is extremely important. Health care providers can offer suggestions on how to remember when to take each drug.

Precautions

Seeing a physician regularly while taking a vasodilator is important, especially during the first few months. The physician will check to make sure the medicine is working as it should and will watch for unwanted side effects. People who have high blood pressure often feel fine. But even when they feel well, patients should keep seeing their physicians and taking their medicine.

Vasodilators will not cure high blood pressure, but will help control the condition. To avoid the serious health problems that high blood pressure can cause, patients may have to take medicine for the rest of their lives. Furthermore, medicine alone may not be enough. People with high blood pressure also may need to avoid certain foods and keep their weight under control. The health care professional who is treating the condition can offer advice on what measures may be necessary.

Some people feel dizzy or have headaches while using this medicine. These problems are especially

likely to occur in older people, who are more sensitive than younger people to the medicine's effects. Anyone who takes these drugs should not drive, use machines, or do anything else that might be dangerous until they know how the drugs affect them.

Special conditions

People who have certain medical conditions or who are taking certain other medicines may have problems if they take vasodilators. In the past, patients with severe aortic stenosis could not use vasodilators because they could cause blood pressure to drop too low and result in severe complications. However, in 2003, a report announced that the vasodilator nitroprusside was safe in these patients and rapidly improved their cardiac function. Before a patient takes vasodilators, he or she should be sure to let the physician know about any of these conditions:

ALLERGIES. Anyone who has had an unusual reaction to a vasodilator in the past should let his or her physician know before taking this type of drug again. The physician also should be told about any allergies to foods, dyes, preservatives, or other substances.

PREGNANCY. Several problems—from excess hair growth to blood abnormalities—have been reported in babies whose mothers take certain vasodilators during **pregnancy**. In studies of laboratory animals, hydralazine caused **birth defects** in mice and rabbits, but not in rats. The effects of taking vasodilators during pregnancy have not been specifically studied in humans. Women who are pregnant or who may become pregnant should check with their physicians before using this medicine. Women who become pregnant while taking a vasodilator should tell their physicians right away.

BREASTFEEDING. Women who are breastfeeding their babies or who plan to breastfeed should check with their physicians before using this medicine.

OTHER MEDICAL CONDITIONS. Using a vasodilator to lower blood pressure may worsen the problems that result from heart disease, blood vessel disease, or a recent heart attack or stroke. This medicine also may make **angina** (chest **pain**) worse. Vasodilators may make pheochromocytomas (tumors of the adrenal medulla), more active. Before using a vasodilator, people with any of these medical problems should make sure their physicians are aware of their conditions.

People with **kidney disease** also should check with their physicians before using a vasodilator. Side effects may be greater in these people because their kidneys are slow to clear the medicine from the body.

USE OF CERTAIN MEDICINES. Taking vasodilators with certain other drugs may affect the way the drugs work or may increase the chance of side effects. Any other prescription or nonprescription (over-the-counter) medicine should not be taken along with a vasodilator unless it has been discussed with the physician who prescribed the vasodilator.

Side effects

Some side effects of vasodilators go away as the body adjusts to the drug and do not need medical attention unless they continue or interfere with normal activities. These include:

- headache
- nausea or **vomiting**
- diarrhea
- loss of appetite

In addition, minoxidil may cause a temporary increase in hair growth, especially on the face, arms, and back. Patients who are bothered by this should check with their physicians.

Other side effects of vasodilators should have medical attention. Rapid or irregular heartbeat are reasons to check with a physician immediately. And if any of the following problems occur, a physician should be consulted as soon as possible:

- chest pain
- muscle pain
- joint pain
- pain, **numbness**, **tingling** or weakness in the hands or feet
- swollen feet or lower legs
- swollen lymph nodes
- bloating
- **fever** and sore throat
- general discomfort or feeling of illness
- weakness
- blisters on skin; skin rash or **itching**; flushing or redness of the skin

KEY TERMS

Adrenal gland—One of a pair of organs located next to the kidneys. The adrenal glands produce hormones that control many body functions.

Adrenal medulla—The inner part of the adrenal gland. The adrenal medulla produces the hormones epinephrine (adrenaline), which stimulates the heart, tightens blood vessels, and relaxes some smooth muscles; and norepinephrine, which has similar effects.

Arteries—Blood vessels that carry blood away from the heart to the cells, tissues, and organs of the body.

Additional side effects are possible. Anyone who has unusual symptoms while taking a vasodilator should get in touch with his or her physician.

Interactions

Vasodilators may interact with other medicines. When this happens, the effects of one or both of the drugs may change or the chance of side effects may be greater. In addition, many prescription and nonprescription (over-the-counter) drugs may affect blood pressure. *No other medication should be taken without the approval of the physician who prescribed the vasodilator.* In particular, avoiding over-the-counter medicines for appetite control, colds, **cough**, sinus problems, **asthma**, hay fever and other **allergies**, should be avoided as these may increase blood pressure. At the other extreme, dangerously low blood pressure may result when drugs such as the blood pressure medicine guanethidine (Ismelin) or nitrates, used to treat chest pain, are combined with vasodilators.

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Nancy Ross-Flanigan
Teresa G. Odle

Vasodilatory see **Shock**

Vasopressin test see **Antidiuretic hormone (ADH) test**

Vasovagal faint see **Fainting**

Vegetarianism

Definition

Vegetarianism is the voluntary abstinence from eating meat. Vegetarians refrain from eating meat for various reasons, including religious, health, and ethical ones. Lacto-ovo vegetarians supplement their diet with dairy (lactose) products and eggs (ovo). Vegans (pronounced vee-guns) do not eat any animal-derived products at all.

Purpose

Vegetarianism is recommended as a dietary therapy for a variety of conditions, including heart disease, **high cholesterol**, type 2 diabetes, and **stroke**. Vegetarianism is a major dietary therapy in the alternative treatment of cancer. Other conditions treated with a dietary therapy of vegetarianism include **obesity**, **osteoporosis**, arthritis, **allergies**, **asthma**, environmental illness, hypertension, gout, **gallstones**, **hemorrhoids**, **kidney stones**, ulcers, colitis, premenstrual syndrome, anxiety, and depression. Vegetarians often report higher energy levels, better digestion, and mental clarity. Vegetarianism is an economical and easily implemented preventative practice as well.

Description

The term vegetarian was coined in 1847 by the founders of the Vegetarian Society of Great Britain, but vegetarianism has been around as long as people have created **diets**. Some of the world's oldest cultures advocate a vegetarian diet for health and religious purposes. In India, millions of Hindus are vegetarians because of their religious beliefs. One of the ancient mythological works of Hinduism, the *Mahabharata*, states that, "Those who desire to possess good memory, beauty, long life with perfect health, and physical, moral and spiritual strength, should abstain from animal foods." The **yoga** system of living and health is vegetarian, because its dietary practices are based on the belief that healthy food contains *prana*. Prana is the universal life energy, which yoga experts believe is abundant in fresh fruits, grains, nuts and vegetables, but absent in meat because meat has been killed. Yogis also believe that spiritual health is influenced by the practice of *ahimsa*, or not harming living beings. The principle of *ahimsa* (non-violence) appears in the Upanishads (Vedic literature) from c. 600–300 B.C. Taking of animal life or human life under any circumstances is sinful and results in rebirth as a lower organism. It became a fundamental element of Jainism,

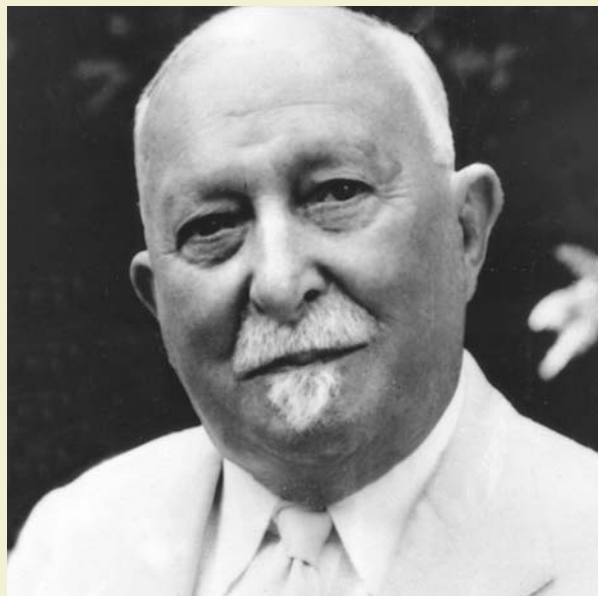
another religion of India. Some Buddhists in Japan and China are also vegetarian because of spiritual beliefs. In the Christian tradition, the Trappist Monks of the Catholic Church are vegetarian, and some vegetarians argue that there is evidence that Jesus and his early followers were vegetarian. Other traditional cultures, such as those in the Middle East and the Mediterranean regions, have evolved diets that frequently consist of vegetarian foods. The Mediterranean diet, which a Harvard study declared to be one of the world's healthiest, is primarily, although not strictly, vegetarian.

The list of famous vegetarians forms an illustrious group. The ancient Greek philosophers, including Socrates, Plato, and Pythagoras, advocated vegetarianism. In modern times, the word to describe someone who likes to feast on food and wine is "epicure," but it is little known that Epicurus, the ancient philosopher, was himself a diligent vegetarian. Other famous vegetarians include Leonardo da Vinci, Sir Isaac Newton, Leo Tolstoy, Ralph Waldo Emerson, and Henry Thoreau. This century's celebrated vegetarians include Gandhi, the physician Albert Schweitzer, writer George Bernard Shaw, musician Paul McCartney, and champion triathlete Dave Scott. Albert Einstein, although not a strict vegetarian himself, stated that a vegetarian diet would be an evolutionary step for the human race.

Vegetarianism in America received a lot of interest during the last half of the nineteenth century and the beginning of the twentieth century, during periods of experimentation with diets and health practices. Vegetarianism has also been a religious practice for some Americans, including the Seventh-day Adventists, whose lacto-ovo vegetarian diets have been studied for their health benefits. Vegetarianism has been steadily gaining acceptance as an alternative to the meat-and-potatoes bias of the traditional American diet. In 1997, Vegetarian Resource Group performed a Roper poll that showed that 13 million Americans, or 5% of the population, identified themselves as vegetarians.

Several factors contribute to the interest in vegetarianism in America. Outbreaks of food **poisoning** from meat products, as well as increased concern over the additives in meat such as hormones and **antibiotics**, have led some people and professionals to question meat's safety. There is also an increased awareness of the questionable treatment of farm animals in factory farming. But the growing health consciousness of Americans is probably the major reason for the surge in interest in vegetarianism. Nutrition experts have built up convincing evidence that there are major problems

DR. JOHN HARVEY KELLOGG (1852–1943)



(AP/Wide World Photos. Reproduced by permission.)

John Harvey Kellogg is known as the father of modern breakfast cereal. He was born in Tyrone Township, Michigan, on February 26, 1852, into a Seventh Day Adventist family. At age 12, he became an apprentice at the Review and Herald Press, a publishing company run by the church. He attended school in Battle Creek, Michigan.

He attended Bellevue Hospital Medical College in New York where he received his medical degree in 1875. In 1876, at the age of 24, Kellogg became an abdominal surgeon and superintendent of the Western Health Reform Institute, which he renamed the Battle Creek Sanitarium. There, he began applying his theories about natural living to his medical practice. Himself a vegetarian, he first advocated a diet high in whole grains, fruits, nuts, and legumes. He later included all types of vegetables in the diet. His controversial health regimen included morning calisthenics, open-air sleeping, cleansing enemas, chewing food hundreds of times before swallowing, and drinking plenty of water.

In the 1890s, Kellogg established a laboratory at the sanitarium to develop more nutritious foods. His brother, Will Keith Kellogg, joined in his research. In 1895 they developed a breakfast cereal of wheat flakes called Granose. The cereal quickly grew in popularity and was soon sold by mail order. This was followed by rice flakes and corn flakes. The brothers established the Sanitas Food Company. But philosophical differences led them to split into two companies. Will founded the W. K. Kellogg Company, which retained the rights to the cereal products. John set up the Battle Creek Food Company, which produced coffee substitutes and soymilk. John Kellogg also edited *Good Health Magazine*, which promoted vegetarianism, for 60 years. In 1904, he published a book, *The Miracle of Life*. He continued to promote his version of healthy living and radical techniques until his death in 1943.

with the conventional American diet, which is centered around meat products that are high in cholesterol and saturated fat and low in fiber. Heart disease, **cancer**, and diabetes, which cause 68% of all deaths in America, are all believed to be influenced by this diet. Nutritionists have repeatedly shown in studies that a healthy diet consists of plenty of fresh vegetables and fruits, complex carbohydrates such as whole grains, and foods that are high in fiber and low in cholesterol and saturated fat. Vegetarianism, a diet that fulfills all these criteria, has become part of many healthy lifestyles. In alternative medicine, vegetarianism is a cornerstone dietary therapy, used in **Ayurvedic medicine**, **detoxification** treatments, macrobiotics, the Ornish diet for heart disease, and in therapies for many chronic conditions.

Preparations

Some people, particularly those with severe or chronic conditions such as heart disease or cancer,

may be advised by a health practitioner to become vegetarian suddenly. For most people, nutritionists recommend that a vegetarian diet be adopted gradually, to allow people's bodies and lifestyles time to adjust to new eating habits and food intake.

Some nutritionists have designed transition diets to help people become vegetarian in stages. Many Americans eat meat products at nearly every meal, and the first stage of a transition diet is to substitute just a few meals a week with wholly vegetarian foods. Then, particular meat products can be slowly reduced and eliminated from the diet and replaced with vegetarian foods. Red meat can be reduced and then eliminated, followed by pork, poultry, and fish. For those wishing to become pure vegetarians or vegans, the final step would be to substitute eggs and dairy products with other nutrient-rich foods. Individuals should be willing to experiment with transition diets, and should have patience when learning how combine vegetarianism with social activities such as dining out.

The transition to vegetarianism can be smoother for those who make informed choices with dietary practices. Sound nutritional guidelines include decreasing the intake of fat, increasing fiber, and emphasizing fresh fruits, vegetables, legumes, and whole grains in the diet while avoiding processed foods and sugar. Everyone can improve their health by becoming familiar with recommended dietary and nutritional practices, such as reading labels and understanding basic nutritional concepts such as daily requirements for calories, protein, fat, and nutrients. Would-be vegetarians can experiment with meat substitutes, foods that are high in protein and essential nutrients. Thanks to the growing interest in vegetarianism, many meat substitutes are now readily available. Tofu and tempeh are products made from soybeans that are high in protein, calcium, and other nutrients. There are “veggie-burgers” that can be grilled like hamburgers, and vegetarian substitutes for turkey and sausage with surprisingly authentic textures and taste. There are many vegetarian cookbooks on the market as well.

A set of guidelines for North American vegetarian diets, updated for 2004, is available from the American Dietetic Association and the Dietitians of Canada. The new guidelines are intended to promote variety within vegetarian diets and to meet the needs of different stages in the life cycle as well as incorporate the most recent findings of medical research.

One remaining drawback to the widespread practice of vegetarianism is the unpleasant taste or smell of many vegetables. A number of phytonutrients have a bitter, astringent, or acrid taste that they impart to products made from vegetables that contain them. Some experts think that people tend to reject such strong-smelling or bitter-tasting vegetables as turnips, cabbage, brussels sprouts, or broccoli because humans have been programmed in the course of evolution to associate bitter taste with poisonous plants. It is increasingly recognized that the major barrier to dietary change for the sake of health is taste. One recommendation for improving the taste appeal of vegetarian diets is more frequent use of spices. In addition to pleasing the human palate, spices derived from plants have been shown to have chemoprotective effects, boosting the immune system, reducing inflammation, and fighting harmful bacteria and viruses.

Precautions

In general, a well-planned vegetarian diet is healthful and safe; in the summer of 2003, a position paper endorsed by the American Dietetic Association

and the Dietitians of Canada referred to vegetarian diets as “healthful, nutritionally adequate, and [able to] provide health benefits in the prevention and treatment of certain diseases.” However, vegetarians, and particularly vegans who eat no animal products, need to be aware of particular nutrients that may be lacking in non-animal diets. These are amino acids, vitamin B₁₂, vitamin D, calcium, iron, zinc, and essential fatty acids. Furthermore, pregnant women, growing children, and those with health conditions have higher requirements for these nutrients.

Vegetarians should be aware of getting *complete protein* in their diets. A complete protein contains all of the essential amino acids, which are the building blocks for protein essential to the diet because the body cannot make them. Meat and dairy products generally contain complete proteins, but most vegetarian foods such as grains and legumes contain incomplete proteins, lacking one or more of the essential amino acids. However, vegetarians can easily overcome this by combining particular foods in order to create complete proteins. For instance, beans are high in the amino acid lysine but low in tryptophan and methionine, but rice is low in lysine and high in tryptophan and methionine. Thus, combining rice and beans makes a complete protein. In general, combining legumes such as soy, lentils, beans, and peas with grains like rice, wheat, or oats forms complete proteins. Eating dairy products or nuts with grains also makes proteins complete. Oatmeal with milk on it is complete, as is peanut butter on whole wheat bread. Proteins do not necessarily need to be combined in the same meal, but generally within four hours.

Getting enough vitamin B₁₂ may be an issue for some vegetarians, particularly vegans, because meat and dairy products are the main sources. Vitamin supplements that contain vitamin B₁₂ are recommended. Spirulina, a nutritional supplement made from algae, is also a vegetarian source, as are fortified soy products and nutritional yeast.

Vitamin D can be obtained by **vitamins**, fortified foods, and sunshine. Calcium can be obtained in enriched tofu, seeds, nuts, legumes, dairy products, and dark green vegetables including broccoli, kale, spinach, and collard greens. Iron is found in raisins, figs, legumes, tofu, whole grains (particularly whole wheat), potatoes, and dark green leafy vegetables. Iron is absorbed more efficiently by the body when iron-containing foods are eaten with foods that contain vitamin C, such as fruits, tomatoes, and green vegetables. Zinc is abundant in nuts, pumpkin seeds, legumes, whole grains, and tofu. For vegetarians who

do not eat fish, getting enough omega-3 essential fatty acids may be an issue, and supplements such as flaxseed oil should be considered, as well as eating walnuts and canola oil.

Vegetarians do not necessarily have healthier diets. Some studies have shown that some vegetarians consume large amounts of cholesterol and saturated fat. Eggs and dairy products contain cholesterol and saturated fat, while nuts, oils, and avocados are vegetable sources of saturated fat. To reap the full benefits of a vegetarian diet, vegetarians should be conscious of cholesterol and saturated fat intake. Vegetarians may also consider buying organic foods, which are grown without the use of synthetic chemicals, as another health precaution.

Research and general acceptance

A vegetarian diet has many well-documented health benefits. It has been shown that vegetarians have a higher life expectancy, as much as several years, than those who eat a meat-centered diet. The U.S. Food and Drug Administration (FDA) has stated that data has shown vegetarians to have a strong or significant probability against contracting obesity, heart disease, lung cancer, **colon cancer**, **alcoholism**, **hypertension**, diabetes, gallstones, **gout**, kidney stones, and ulcers. However, the FDA also points out that vegetarians tend to have healthy lifestyle habits, so other factors may contribute to their increased health besides diet alone.

A vegetarian diet, as prescribed by Dr. Dean Ornish, has been shown to improve heart disease and reverse the effects of **atherosclerosis**, or hardening of the arteries. It should be noted that Dr. Ornish's diet was used in conjunction with **exercise**, **stress reduction**, and other holistic methods. The Ornish diet is lacto-ovo vegetarian, because it allows the use of egg whites and non-fat dairy products.

Vegetarians have a resource of statistics in their favor when it comes to presenting persuasive arguments in favor of their eating habits. Vegetarians claim that a vegetarian diet is a major step in improving the health of citizens and the environment. Americans eat over 200 lbs (91 kg) of meat per person per year. The incidence of heart disease, cancer diabetes, and other diseases has increased along with a dramatic increase in meat consumption during the past century. Many statistics show significantly smaller risks for vegetarians contracting certain conditions. The risks of women getting **breast cancer** and men contracting prostate cancer are nearly four times as high for frequent meat eaters as for those

KEY TERMS

Cholesterol—A steroid fat found in animal foods that is also produced in the body from saturated fat for several important functions. Excess cholesterol intake is linked to many diseases.

Complex carbohydrates—Complex carbohydrates are broken down by the body into simple sugars for energy, are found in grains, fruits and vegetables. They are generally recommended in the diet over refined sugar and honey, because they are a more steady source of energy and often contain fiber and nutrients as well.

Legume—Group of plant foods including beans, peas, and lentils, which are high in protein, fiber, and other nutrients.

Organic food—Food grown without the use of synthetic pesticides and fertilizers.

Saturated fat—Fat that is usually solid at room temperature, found mainly in meat and dairy products but also in vegetable sources such as some nuts, seeds, and avocados.

Tempeh—A fermented cake of soybeans and other grains; it is a staple food in Indonesia.

Tofu—A soft cheeselike food made from curdled soybean milk.

Unsaturated fat—Fat found in plant foods that is typically liquid (oil) at room temperature. They can be monounsaturated or polyunsaturated, depending on the chemical structure. Unsaturated fats are the most recommended dietary fats.

who eat meat sparingly or not at all. For heart attacks, American men have a 50% risk of having one, but the risk drops down to 15% for lacto-ovo vegetarians and to only 4% for vegans. For cancer, studies of populations around the world have implied that plant-based diets have lower associated risks for certain types of cancer.

Vegetarians claim other reasons for adopting a meat-free diet. One major concern is the amount of pesticides and synthetic additives such as hormones that show up in meat products. Chemicals tend to accumulate in the tissue of animals that are higher in the food chain, a process called *bioaccumulation*. Vegetarians, by not eating meat, can avoid the exposure to these accumulated toxins, many of which are known to influence the development of

cancer. One study showed that DDT, a cancer-causing pesticide, was present in significant levels in mother's milk for 99% of American women, but only 8% of vegetarian women had significant levels of the pesticide. Women who eat meat had 35 times higher levels of particular pesticides than vegetarian women. The synthetic hormones and antibiotics added to American cattle has led some European countries to ban American beef altogether. The widespread use of antibiotics in livestock has made many infectious agents more resistant to them, making some diseases harder to treat.

Vegetarians resort to ethical and environmental arguments as well when supporting their food choices. Much of U.S. agriculture is dedicated to producing meat, which is an expensive and resource-depleting practice. It has been estimated that 1.3 billion people could be fed with the grain that America uses to feed livestock, and **starvation** is a major problem in world health. Producing meat places a heavy burden on natural resources, as compared to growing grain and vegetables. One acre of land can grow approximately 40,000 lbs (18,000 kg) of potatoes or 250 lbs (113 kg) of beef, and it takes 50,000 gal (200,000 l) of water to produce 1 lb (0.45 kg) of California beef but only 25 gal (100 l) of water to produce 1 lb (0.45 kg) of wheat. Half of all water used in America is for livestock production. Vegetarians argue that the American consumption of beef may also be contributing to global warming, by the large amounts of fossil fuels used in its production. The South American rainforest is being cleared to support American's beef consumption, as the United States yearly imports 300 million lbs (136 million kg) of meat from Central and South America. The production of meat has been estimated as causing up to 85% of the loss of topsoil of America's farmlands. A German researcher in the field of **nutrition** ecology has summarized the environmental benefits of vegetarian diets: "Research shows that vegetarian diets are well suited to protect the environment, to reduce pollution, and to minimize global climate changes."

Despite the favorable statistics, vegetarianism does have its opponents. The meat industry in America is a powerful organization that has spent millions of dollars over decades advertising the benefits of eating meat. Vegetarians point out that life-long eating habits are difficult to change for many people, despite research showing that vegetarian diets can provide the same nutrients as meat-centered diets.

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Dietitians of Canada/Les diététistes du Canada. 480 University Avenue, Suite 604, Toronto, ON M5G IV2. (416) 596-0857. <<http://www.dietitians.ca/>>.

North American Vegetarian Society (NAVS). PO Box 72, Dolgeville, NY 13329. (518) 568-7970.

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Vegetarian Nutrition and Health Letter. 1707 Nichol Hall, Loma Linda, CA 92350. (888) 558-8703.

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Douglas Dupler, MA
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Vegetative state

Definition

A coma-like state characterized by open eyes and the appearance of wakefulness is defined as vegetative.

Description

The vegetative state is a chronic or long-term condition. This condition differs from a persistent vegetative state (PVS, a state of **coma** that lacks both awareness and wakefulness) since patients have awakened from coma, but still have not regained awareness. In the vegetative state patients can open their eyelids occasionally and demonstrate sleep-wake cycles. They also completely lack cognitive function. The vegetative state is also called coma vigil.

Causes and symptoms

The vegetative state can be caused by:

- cardiac arrest
- prolonged and profound **hypoglycemia** (an abnormal and severe decrease in blood sugar)
- carbon monoxide poisoning
- head injury
- brain hemorrhage
- compression of the brainstem
- tumors
- bilateral hemispheric demyelination (a loss of nerve cells)
- injury of the brain following infections (**meningitis** or **encephalitis**)
- neurodegenerative diseases
- anencephaly (an abnormality of the brain and skull)
- diffuse nerve cell injury

Patients in a vegetative state apparently have functioning of a special area in the brain called the reticular activating system (RAS) responsible for sleep-wake cycles. The connections that integrate more complex abilities such as awareness are interrupted. Patients in the vegetative state can open and close eyes spontaneously. They may appear to track or follow objects with their eyes. Patients may chew and swallow food placed in the mouth. The vegetative patient does not respond to sound, hunger, or **pain**. Patients cannot obey verbal commands and lack local motor responses. Additionally these patients cannot

KEY TERMS

Cognitive—The ability (or lack of) to think, learn, and memorize.

Hypermobility—Increased movement of joints.

talk in comprehensible terms and they may become noisy, restless, and hypermobile. These patients are in a state of arousal but completely lack awareness.

Diagnosis

Diagnosis of vegetative state depends on the primary cause of brain dysfunction. A comprehensive history and neurological examination, neuroimaging studies, and chemical analysis of the blood are essential. Additionally, special tests such as cerebrospinal fluid (CSF, the fluid that bathes and nourishes the brain and spinal cord) analysis and electroencephalography, (EEG analyzes the electrical activity within the brain) may be indicated to establish a diagnosis.

Treatment

Treatment is directed to presenting symptoms and patient needs. Patients require constant monitoring and assistance with feeding, hydration hygiene, assisted movement (to help prevent ulcers and **blood clots** in the legs), and elimination of waste products.

Alternative treatment

There is no known alternative treatment for vegetative patients.

Prognosis

The prognosis is generally poor and the condition can persist chronically.

Prevention

There is no known prevention since this state can occur as a result of unavoidable situations such as an accident, tumor, and bleeding or genetic abnormality.

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Laith Farid Gulli, M.D.

Velopharyngeal insufficiency

Definition

Velopharyngeal insufficiency is the improper closing of the velopharyngeal sphincter (soft palate muscle) during speech characterized by an acute nasal quality of the voice.

Description

At the back of the mouth is a circle of structures that include the tonsils, the tongue, and the palate. During speech, this apparatus must close off the nose for proper articulation of the explosive consonants “p,” “b,” “g,” “t,” and “d.” If it does not close, there is a snort sound produced through the nose. Improper function of this structure also produces a nasal tone to the voice.

Causes and symptoms

There are three main causes for this defect:

- Cleft palate is a congenital condition, producing a defect in the palate that allows air to escape upward during speech.
- If tonsil and adenoid surgery is done improperly, velopharyngeal insufficiency may result. The occurrence rate is approximately one in every 2,000-3,000 tonsillectomies.
- Nerve or muscle disease may paralyze the muscles that operate the velopharyngeal sphincter.

The primary symptom is the speech impediment. Some people develop a change in their speaking pattern or a series of facial grimaces to try to overcome the difficulty. If the condition is acute, regurgitation through the nose may occur.

Diagnosis

Examination of the velopharyngeal sphincter through ultrasound scans, fiber-optic nasopharyngoscopy, and videofluoroscopy will reveal the extent of velopharyngeal insufficiency. Speech and velopharyngeal sphincter movement are compared to make the diagnosis.

KEY TERMS

Adenoids—Lymph glands just above the tonsils and the palate.

Cleft palate—Congenital defect marked by a split in the roof of the mouth.

Nasopharyngoscopy—A diagnostic procedure that examines the nasal passageways and pharynx with an instrument outfitted with an optical system.

Pharynx—A canal located between the mouth cavity and the esophagus.

Tonsillectomy—Surgical removal of the tonsils.

Tonsils—Lymph glands in the throat, just behind the back teeth.

Treatment

Velopharyngeal insufficiency is treated with a combination of surgery and speech therapy. There are several surgical procedures that can be performed to correct the physical malfunction. They include:

- Pharyngeal flap procedure that moves the skin flap from the pharynx to the soft palate.
- Palatal push-back that separates the hard and soft palate in order to lengthen the soft palate.
- Pharyngoplasty that lengthens the soft palate by turning the pharyngeal skin flaps.
- Augmentation pharyngoplasty that inserts an implant into the pharyngeal wall to enlarge it, thus narrowing the velopharyngeal opening.
- Velopharyngeal sphincter reconstruction.

Prognosis

The combination of surgery to correct the insufficiency and speech therapy to retrain the voice successfully alleviate velopharyngeal insufficiency.

Resources

PERIODICALS

Huang M. H., S. T. Lee, and K. Rajendran. “Anatomic Basis of Cleft Palate and Velopharyngeal Surgery: Implications from a Fresh Cadaveric Study.” *Plastic & Reconstructive Surgery* 101 (Mar. 1998): 613-627.

J. Ricker Polsdorfer, MD

Vena cava filter

Definition

A vena cava filter is a device inserted into a major vein to prevent a blood clot from entering the lungs.

Purpose

The purpose of a vena cava filter is to prevent a blood clot from potentially traveling to the lungs. A thrombus clot traveling to the lungs is called a **pulmonary embolism** (PE). A thrombus in the deep venous system (the part of the circulation that brings blood back to the heart) represents a disorder of normal hemostasis (the normal clotting of blood).

Insertion of a vena cava filter is indicated for patients who:

- cannot receive medications that can dissolve the clot (anticoagulation therapy)
- have a thrombus in a deeply situated vein
- experience complication of anticoagulation therapy such as bleeding
- experience failure of anticoagulation therapy to prevent pulmonary **embolism**
- have an embolus in the lungs (pulmonary embolism) removed
- have a recurrent embolism while receiving adequate medications
- have significant bleeding complications during anticoagulation

Precautions

There are no significant precautions concerning insertion of a vena cava filter. The devices are usually effective and short-term complications are unusual

Description

Vena cava filters are usually inserted in to prevent PE caused by a thrombosis in a deep vein (DVT). Approximately 60% of patients who die in a hospital have evidence of PE during **autopsy**. The incidence (number of new cases) of DVT is highest for patients undergoing surgical repair of a fractured hip. However, DVT is common in both surgical and medical patients. DVT is found in 29–33% of patients in medical intensive care units (MICU) and in 27–40% of patients with a **heart attack** (myocardial infarction). Vena cava filters are placed to prevent thrombi from

KEY TERMS

Embolus—An embolus (or emboli the plural form) is a blood clot that has detached from its site of origin and travels to the lungs (pulmonary artery), where it can rupture the artery, causing death.

Pulmonary embolism—A traveling thrombus that has lodged in the pulmonary artery.

Thrombus—A thrombus (or thrombi the plural form) is a blood clot that can form in a deeply situated vein.

entering the lungs. There is currently a new type of filter called the Kim-Ray-Greenfield filter.

Preparation

Insertion of a vena cava filter is an invasive procedure. The patient is prepared for this procedure using standard surgical protocols. The VCF is commonly implanted in the jugular vein in the neck or the femoral vein in the groin. The procedure is generally well tolerated.

Aftercare

This depends on the patient's health status and recommendation's for continued care.

Risks

Many patients have died from PE even with a vena cava implantation. Use of a VCF is primarily indicated if there are contraindications for anticoagulation therapy. VCF can increase a patient's susceptibility for developing recurrent DVT.

Normal results

Patient progresses well and prevention of large emboli that can cause a PE is successful.

Abnormal results

The desired effect is not accomplished and the patient develops a PE resulting in **death**.

Resources

BOOKS

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- Isnard, R., and M. Komajda. "Thromboembolism in Heart Failure, Old Ideas and New Challenges." *European Journal of Heart Failure* June 2001.

ORGANIZATIONS

- American College of Angiology. 295 Northern Blvd., Ste. 104 Great Neck, NY 11021-4701.

Laith Farid Gulli, M.D.

Venereal diseases see **Sexually transmitted diseases**

Venography

Definition

Venography is an x-ray test that provides an image of the leg veins after a contrast dye is injected into a vein in the patient's foot.

Purpose

Venography is primarily performed to diagnose **deep vein thrombosis** (a condition that can lead to **pulmonary embolism**). It is the standard procedure used to detect this type of disorder. Venography also can be used to distinguish **blood clots** from obstructions in the veins, to evaluate congenital vein problems, to see how the deep leg vein valves are working, and to identify a vein for arterial bypass grafting.

Precautions

Venography usually is not performed in patients with kidney (renal) problems.

Description

Venography (also called phlebography, ascending contrast phlebography, or contrast venography) is an invasive diagnostic test that provides a constant image

of leg veins on a fluoroscope screen. Venography identifies the location, extent, and degree of attachment of the blood clots, and enables the condition of the deep leg veins to be assessed. It is especially useful when there is a strong suspicion of deep vein thrombosis, but non-invasive tests have failed to identify the disease.

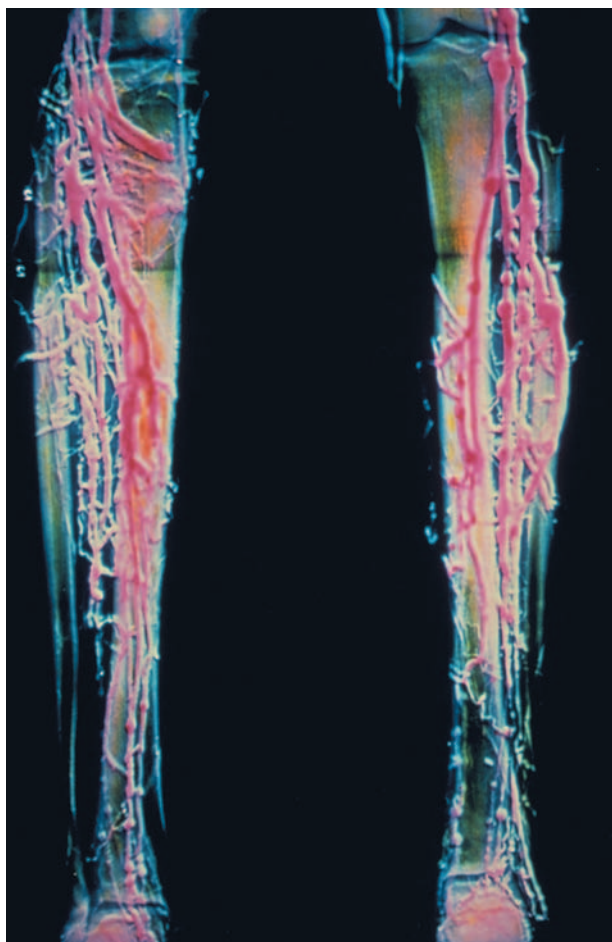
Venography is the most accurate test for detecting deep vein thrombosis. It is nearly 100% sensitive and specific in making this diagnosis (**pulmonary embolism** is diagnosed in other ways). Accuracy is crucial since deep vein thrombosis can lead to pulmonary embolism, a condition that can be fatal.

Venography is not used often, however, because it is painful, expensive, exposes the patient to a fairly high dose of radiation, and can cause complications. In about 5% of cases, there are technical problems in conducting the test. In addition, the test is less accurate in diagnosing problems below the knee. Venography takes between 30-45 minutes and can be done in a physician's office, outpatient center, or a hospital.

In 2003, a report said that computed tomography (CT) scanning could be used to diagnose pulmonary embolism and deep venous thrombosis in one examination. By combining CT **angiography** and CT venography, researchers could look for both conditions in one procedure with high-speed CT scanners. The procedure was quick and delivered a reduced radiation dose. However, it has not become accepted as a replacement for traditional venography and is only preferred if the patient is clinically stable and requires immediate diagnosis and treatment.

During the procedure, the patient lies on a tilting x-ray table. The area where the catheter will be inserted will be shaved, if necessary, and cleaned. Sometimes a local anesthetic is injected to numb the skin at the site of the insertion. A small incision may be required to make a point for insertion. The catheter is inserted and the contrast solution (or dye) is slowly injected. Injection of the dye causes a warm, flushing feeling in the leg that may spread through the body. The contrast solution also may cause slight **nausea**. About 18% of patients experience discomfort from the contrast solution.

In order to fill the deep venous system with dye, a tight band (or tourniquet) may be tied around the ankle of the foot the dye is injected into, or the lower extremities may be tilted. The patient is asked to keep the leg still. The doctor also observes the movement of the solution through the vein with a fluoroscope. At the same time, a series of x rays are taken. When the



A venographic image of a patient's legs with varicose veins.
(Custom Medical Stock Photo. Reproduced by permission.)

test is finished, fluid is injected to clear the dye from the veins, the catheter is removed, and a bandage is applied over the site of the injection.

Preparation

Fasting or drinking only clear liquids is necessary for four hours before the test. However, sometimes the test is done in an emergency even if the patient has eaten. The contrast solution contains iodine, to which some people are allergic. Patients who have **allergies** or hay **fever**, or have had a bad reaction to a contrast solution, should tell the doctor. A sedative, such as diazepam (Valium), may be prescribed to help the patient relax.

Aftercare

Patients should drink large amounts of fluids to flush the remaining contrast solution from their

KEY TERMS

Contrast solution—A liquid dye injected into the body that allows veins to be seen by x rays. Without the dye, the veins could not be seen on x rays.

Deep vein thrombosis—The development or presence of a blood clot in a vein deep within the leg. Deep vein thrombosis can lead to pulmonary embolism.

Invasive—A diagnostic test that invades healthy tissue; in the case of venography, through an incision in a healthy vein.

Pulmonary embolism—An obstruction of a blood vessel in the lungs, usually due to a blood clot, that blocks a pulmonary artery. Pulmonary embolism can be very serious and in some cases is fatal.

bodies. The area around the incision will be sore for a few days. If there is swelling, redness, **pain**, or fever, the doctor should be notified. Pain medication may be needed. In most cases, the patient can resume normal activities the next day.

Risks

Venography also can cause complications such as phlebitis, tissue damage, and the formation of deep vein thrombosis in a healthy leg. A rare side effect in up to 8% of cases is a severe allergic reaction to the dye. This usually happens within 30 minutes after injection of the dye and requires medical attention.

Normal results

Normal venography results show proper blood flow through the leg veins.

Abnormal results

Abnormal venography results show well-defined filling defects in veins. Findings include:

- blood clots
- consistent filling defects
- an abrupt end of a test dye column
- major deep veins that are unfilled
- dye flow that is diverted.

These results confirm a diagnosis of deep vein thrombosis.

Resources

PERIODICALS

Abella, H.A. "CT Dose Techniques Address PE, DVT — Indirect Venography and Collimation Changes Reduce Exposure." *Diagnostic Imaging* November 1, 2003.

Lori De Milto
Teresa G. Odle

Venous access

Definition

Venous access introduces a needle into a vein, usually for the purpose of withdrawing blood or administering medication.

Purpose

Venous access is necessary for fluid administration, medication administration, and obtaining blood for chemical analysis. Sites for access include veins located in the peripheral arms or legs, scalp, neck, and bone.

Venous access in children may pose special problems since finding appropriate veins and immobilization may be difficult but essential. For complicated procedures **sedation** may be indicated. Venous access can be performed during emergency situations, for outpatients, inpatients, and those who require long term chemotherapy.

Precautions

There are no major precautions for access during emergency procedures. The main concern during an emergency would be to secure a portal of entry to infuse potentially life saving medications and fluids. For all methods of access the main precautionary measures include attention to accurate procedures. Proper procedures are necessary to minimize the possibility of infection, **embolism**, phlebitis, or destruction of neighboring tissue.

Description

For peripheral venipuncture the common site is usually a vein in the arm (the antecubital fossa located on the opposite side of the elbow) or on the flat bony area of the hand (dorsum of the hand). Scalp veins are accessible in infants under one year of age. The selected vein should be long and straight for needle

accommodation. It should be identified by straightness, lack of pulsation (characteristic of an artery), and filling with blood from above (arteries fill from below). Internal jugular catheterization is performed in the neck using special bone and muscle landmarks. The external jugular vein can be cannulized by immobilizing tilted and rotating the head. The subclavian approach is a complicated procedure and emergency access can be performed if attempts for access a vein in other areas have failed. Intraosseous venous access is usually accomplished through a leg bone. Catheters implanted in the front of the chest (anterior chest wall) can accomplish long-term venous access. A large leg vein is preferably used and isolated by dissection. A catheter is inserted into the vein and they are tied together.

Preparation

For peripheral vein access in the arm, a tourniquet is applied a few inches over the puncture site. The skin over the puncture site is sterilized with an alcohol pad. The needle is inserted and either blood is drawn and the needle is removed, or a catheter is inserted to place an intravenous line. Scalp veins can be accessible by immobilizing the head, shaving the area from hair, and using a rubber band as a tourniquet. Internal jugular vein catheterization is accomplished by extending the patient's head over the edge of a table or cart and rotating away from the intended puncture site. Immobilizing the head and extending it 15–20 degrees over the edge of a bed or cart and rotating away from the puncture site can cannulize the external jugular vein. The subclavian vein access is a complicated procedure and requires sedation and special positioning (Trendelenburg). A towel should be place in the back of the area. The skin should be cleansed and the puncture site is anesthetized. For the femoral approach the leg is externally rotated. The artery should be felt and along with specific anatomical landmarks the vein can be localized. The skin should be cleaned and anesthetized. During venous cutdown a large vein near the anklebone is careful dissected away from underlying tissues. The area must be properly cleaned and anesthetize prior to making an incision. A catheter is inserted and secured in place with sutures.

Aftercare

For simple procedures such as peripheral venous access, applying simple pressure (to stop bleeding) and a bandage may be sufficient. For more complicated procedures, the primary cause for access should be treated as well as care to avoid or treat potential complications that may arise from access.

KEY TERMS

Cannula—Insertion of a tube.

Catheterization—The process of inserting a tubular instrument into a body cavity to permit passage of fluid.

Phlebitis—Inflammation of a vein.

Pneumothorax—The presence of air in the cavity that surrounds the lungs.

Risks

For access into a peripheral vein, care must be taken not to puncture both sides of the vein. After removal of the needle or catheter, a piece of cotton and pressure should be applied over the puncture site to prevent unwanted bleeding. Access with a scalp vein should be preformed with care to avoid hematoma formation (localized blood clot), accidental puncture of an artery, or infection. Access into the internal jugular vein in the neck can cause laceration of an artery or nerve. This procedure can also cause hematoma (blood clot) formation; damage to local nerves within the area, **pneumothorax**, or misplaced catheterization. Venous access into the external jugular vein can cause hematoma or placement outside the thorax. Subclavian vein access can cause air to enter a vein (resulting in an air embolus) or pneumothorax. Cannulation of the femoral vein in the groin area can cause infection or **thrombophlebitis**. Intraosseous venous access commonly performed in a leg bone can cause hematomas, infection or damage to bone marrow. This procedure should not be performed if the attempts in one leg is unsuccessful, the skin over the legs is diseased (from a burn or infection), or there is a broken leg bone or bone disease. Venous cutdown can cause infection, loss of the catheter in the vein, phlebitis, or nerve damage.

Laith Farid Gulli, M.D.
Bilal Nasser, M.Sc.

Venous insufficiency

Definition

Venous insufficiency is described as abnormal blood flow through veins that can cause local damage, damage to affected legs, or **death**.

Description

Syndromes related to venous insufficiency are caused by valve incompetence. Venous insufficiency is a chronic (long term) condition. The number of new and existing cases is dependent on age and gender. Some patients may have a positive family history. Usually older persons and females are more commonly affected. Deep situated and superficial veins can be affected. **Cancer** obstructing veins in the pelvis area can cause superficial venous insufficiency. Deep venous insufficiency is commonly caused by **thrombophlebitis**, causing obstruction of valves that regulate blood flow in veins. Small veins that have been occluded by a thrombosis may re-canalize (opening up new channels to re-direct blood flow). These re-canalized veins are inadequate and cannot correct the impairment of flow. However, larger veins may still remain occluded. When a thrombosis occurs the valves that regulate venous blood flow become thickened and incompetent, rendering them incapable of regulating back flow of blood. This valvular incompetence will cause an increase in the presence within veins (venous **hypertension**). Venous hypertension is responsible for most of the symptoms associated with venous disease. Superficial veins can become dilated causing **varicose veins** (veins that bulge and seem tortuous). Leg ulcers can be severe and are responsible for 100,000 cases of disability in the United States alone.

Causes and symptoms

The symptoms of chronic venous insufficiency can be subjective and objective. Subjective symptoms include throbbing, cramping, burning sensations, and leg **fatigue**. Patients can also develop chronic leg ulcers that may not heal. Varicose veins in the legs can bleed (since veins are delicate structures with thin walls) and cause death. Patients often develop fluid retention (**edema**) in the affected limb. Skin changes can occur and affected areas can become thin, shiny, discolored (blue-purple), and atrophic. The skin usually becomes thick and tough.

Diagnosis

There are several techniques used to diagnose venous disease. Electrical impedance plethysmography (IPG) provides a functional evaluation for out-flow obstruction ultrasound (a machine that transmits sound waves) studies can visualize the venous system in certain areas. Another technique called duplex scanning can measure velocity within a vein.

Treatment

Periodic elevation of legs and bed rest can help with leg swelling. Patients are advised to avoid

KEY TERMS

Atrophic—A wasting of cells and tissues.

Thrombophlebitis—Venous inflammation with formation of a thrombus.

Thrombus—A clot in the cardiovascular system (the system that circulates blood throughout the body).

prolonged periods of standing or sitting. Wearing compression stockings can also reduce swelling of the leg. Mild skin infections can be treated with compresses, steroids, and, if infection is present, with **antibiotics**. Ulcerations can be treated with compresses, possible surgery, special ointments, and a semi-rigid boot that helps improve blood flow. Varicose veins can be treated with elastic stockings. About 15–20% of patients require surgery, but only after careful evaluation and specialized testing confirms a beneficial value.

Prognosis

The prognosis is variable and depends on the progression of disease, extent of damage, and the presence of other diseases, which may affect the cardiovascular system.

Prevention

Persons who have a strong family history, evidence of disease, and/or those who stand on their legs many hours daily should discuss the option of elastic stocking with their primary clinician.

Resources

BOOKS

Goroll, Alan H., et al. *Primary Care Medicine*. 4th ed. Lippincott, Williams & Wilkins, 2000.

Townsend, Courtney M. *Sabiston Textbook of Surgery*. 16th ed. W. B. Saunders Company, 2001.

Laith Farid Gulli, M.D.

Ventricular aneurysm

Definition

Ventricular aneurysm is a complication of a **heart attack** (myocardial infarction). It is a ballooning of a

section of a blood vessel in the heart that first appears several days or weeks after an acute myocardial infarction.

Description

A myocardial infarction occurs when a section of the heart wall is deprived of blood and dies (undergoes necrosis, or tissue death, and scarring). The heart wall is mainly muscle. It has two ventricles, the right and left ventricles, which pump blood to and from the lungs, and to the body. When part of the heart muscle dies, pumping power from that part of the wall is lost. After a myocardial infarction, the part of the heart wall that did not die must continue pumping blood and compensate for the dead muscle.

Ventricular aneurysm is one of the complications that follow a myocardial infarction.

An aneurysm is the outward swelling, or ballooning, of a blood vessel at a weak spot in the wall of the blood vessel. In the case of ventricular aneurysm, the aneurysm occurs in the wall of the heart at the spot where the myocardial infarction occurred. A scar usually forms in the area of the dead muscle tissue, and may eventually calcify. Ventricular aneurysms generally do not rupture. The left ventricle is involved in most cases of ventricular aneurysm.

Causes and symptoms

The principle symptom of a ventricular aneurysm is cardiac insufficiency, a condition in which not enough blood is being pumped to the body. Ventricular aneurysm is usually found after a large infarction in the muscle wall of a ventricle. Ventricular aneurysm is seldom seen immediately after a myocardial infarction. It takes several days or weeks to several months to develop. Frequently, recurrent ventricular irregular heartbeats (**arrhythmias**) and low cardiac output result from the presence of a ventricular aneurysm. **Blood clots** (thrombi) may form on the inside wall of the aneurysm and produce systemic blood clots that get stuck in a blood vessel (embolisms), which could lead to **stroke** or an ischemic leg (a usually painful condition in which lack of blood circulation leads to reduced function).

Diagnosis

A number of signs may indicate ventricular aneurysm, including an abnormal precordial impulse in the heartbeat, persistent elevation of the S-T segment of an electrocardiogram, and a characteristic bulge seen

KEY TERMS

Arrhythmia—A disturbance in the beating pattern of the heart.

Myocardial infarction—Commonly known as a heart attack, a myocardial infarction occurs when a part of the heart muscle is deprived of blood and dies.

on the heart when x rayed. The bulge is typically seen when the heart contracts, driving blood to the aorta, in the systolic phase of the heartbeat. Echolocation (**echocardiography** or ultrasound) can confirm the presence of an aneurysm. **Cardiac catheterization** may be performed to determine the extent of the aneurysm and the status of the coronary arteries. Stethoscopic examination reveals abnormal heart sounds, especially those associated with a backflow of blood from the left ventricle to the left atrium in systole or contraction beat (mitral regurgitation). This heart murmur is caused by the heart muscles no longer being able to properly operate the mitral valve.

Treatment

Most cases of ventricular aneurysm are treated by close medical follow-up and limiting patient activity. Surgical removal of the aneurysm is an option when persistent left ventricular failure or arrhythmia occurs, and the aneurysm is large. **Vasodilators**, **diuretics**, and digoxin are used to treat **heart failure**. **Anticoagulant drugs** are used to prevent the formation of blood clots. **Antiarrhythmic drugs** are used to treat heart arrhythmias.

Prognosis

Ventricular aneurysm occurs more frequently than is commonly thought. Based on postmortem examination, ventricular aneurysm occurs in as many as 15% of myocardial infarction cases. Patients with a large ventricular aneurysm in the left ventricle have a reduced survival rate. Many patients have mild symptoms which are not life-threatening. The survival rate is dependent on the function of the left ventricle.

Resources

BOOKS

Alexander, R. W., R. C. Schlant, and V. Fuster, editors. *The Heart*. 9th ed. New York: McGraw-Hill, 1998.

John T. Lohr, PhD

Ventricular assist device

Definition

A ventricular assist device (VAD) is a mechanical pump used for temporary blood circulation support. It decreases the workload of the heart while maintaining adequate flow and blood pressure.

Purpose

A VAD is a temporary life-sustaining device. VADs can replace the left ventricle (LVAD), the right ventricle (RVAD), or both ventricles (BIVAD). They are used when the heart muscle is damaged and needs to rest in order to heal or when blood flow from the heart is inadequate. VADs can also be used as a bridge in patients awaiting heart transplantation or in patients who have rejected a transplanted heart.

Examples of patients who might be candidates for a VAD are those who:

- have suffered a massive heart attack
- cannot be weaned from heart-lung bypass after treatment with intravenous fluids, medications, and insertion of a balloon pump in the aorta
- have an infection in the heart wall that does not respond to conventional treatment
- are awaiting a heart transplant and are unresponsive to drug therapy and intravenous fluids
- are undergoing high-risk procedures to clear the blockages in a coronary artery

Although one in five people suffer left side ventricular failure, only a minority are candidates for VADs. To be considered for a VAD, patients must meet specific criteria concerning blood flow, blood pressure, and general health.

Precautions

Poor candidates for a VAD include those with:

- irreversible renal failure
- severe disease of the vascular system of the brain
- cancer that has spread (metastasized)
- severe liver disease
- blood clotting disorders
- severe lung disease
- infections that do not respond to antibiotics
- extreme youth or age

KEY TERMS

Coronary blood vessels—The arteries and veins that supply blood to the heart muscle.

Diaphragm—The muscle that separates the chest cavity from the abdominal cavity.

Ventricle—The heart has four chambers. The right and left ventricles are at the bottom of the heart and act as the body's main pumps.

Description

There are four types of VADs, each appropriate for a different condition. Surgery to install a VAD is performed under **general anesthesia** in a hospital operating room. An incision is made in the chest, then catheters are inserted into the heart and the correct artery. The surgeon sutures the catheters in place, then attaches tubing to connect the catheters to the pump. The pump stays outside the body. Once it is turned on, blood flows out of the diseased ventricle and into the pump, then is returned to the correct blood vessel leaving the heart.

Preparation

Before the operation the patient meets with an anesthesiologist to determine any special conditions that will affect the administration of anesthesia. Standard preoperative blood and urine studies are performed, and the heart is monitored both before and during the operation with an electrocardiograph.

Aftercare

The patient is monitored in intensive care, with follow-up blood, urine, and neurological studies. Blood thinning medications are given to prevent blood clotting.

Except for those patients awaiting a heart transplant, patients are slowly and gradually weaned from the VAD. Even when patients no longer need the VAD, they will require supportive drug therapy and/or a balloon pump inserted in the aorta.

Risks

VAD insertion carries risks of severe complications. Bleeding from surgery is common and occurs in as many as 30-50% of patients. Other complications include the development of blood clots, partial

paralysis of the diaphragm, **respiratory failure**, kidney failure, failure of the VAD, damage to the coronary blood vessels, **stroke**, and infection.

Sometimes when the left ventricle is supported, the right ventricle begins to need assistance. If VADs are inserted in both ventricles, the heart may become so dependent on their support that they cannot be removed.

Normal results

Because conditions for which VADs are used vary widely and because of the high risks associated with VAD insertion, the outcome of surgery cannot be predicted.

Resources

BOOKS

Zaret, Barry L., et al., editors. "Ventricular Assist Device." In *The Patient's Guide to Medical Tests*. Boston: Houghton Mifflin, 1997.

OTHER

"Ventricular Assist Devices." *Department of Biological and Agricultural Engineering*. New York State University. < <http://www.bae.ncsu.edu> >.

Tish Davidson, A.M.

Ventricular ectopic beats

Definition

A ventricular ectopic beat (VEB) is an extra heart-beat originating in the lower chamber of the heart. This beat, also called a premature ventricular contraction (PVC), occurs before the beat triggered by the heart's normal function.

Description

Ventricular ectopic beats are common and do not indicate a problem in people without heart disease. However, if a person has aortic stenosis, **heart failure**, or a previous **heart attack**, VEBs may be followed by **ventricular tachycardia** and fibrillation, which can lead to sudden **death**.

Causes and symptoms

Although the origin of a VEB is well documented, the exact cause or causes are not well understood.

KEY TERMS

Angioplasty—A surgical procedure which dilates a narrowed or blocked part of an artery.

Aortic stenosis—A stiffening of the artery which carries blood from the heart to the body.

Beta-blockers—A class of medication used to block the cellular response to chemicals normally present in the body.

Coronary artery—The vessel which brings blood to the muscle of the heart.

Fibrillation—Rapid, uncoordinated quivering of the heart.

Heart failure—A term used when the heart is unable to pump enough blood to supply the needs of the body.

Some physicians believe the beat is caused by a trigger of specific origin, while other physicians believe the beat is random. Occasional ventricular ectopic beats occur in healthy people. If there is no evidence of heart disease, there is little or no danger to the individual.

A single ventricular ectopic beat has very little effect on the pumping ability of the heart and usually does not cause any symptoms. If a symptom is felt, it is the feeling of a strong or skipped beat, often described as a thump, kick, or flip-flop. Sometimes, the sensation is referred to as a fullness in the neck.

Diagnosis

Ventricular ectopic beats are easily seen on an electrocardiogram.

Treatment

If a person is otherwise healthy, the only treatment needed is to decrease **stress** and limit the use of alcohol and **caffeine**. Cold medicines, available without prescription, sometimes contain drugs (e.g., **decongestants**) that stimulate the heart and should be used with caution.

If symptoms are uncomfortable, or the pattern of VEBs indicates a problem, the physician may prescribe drug therapy. Beta-blockers are quite safe and are usually tried first.

A person who has a history of heart attack or heart disease, and is experiencing frequent or complex VEBs, is at greater risk of sudden death. Drug therapy

with beta-blockers will be recommended. In addition, **angioplasty** or coronary artery bypass surgery may relieve any underlying coronary artery blockage and reduce the danger of sudden death.

Treatment with **antiarrhythmic drugs** can suppress VEBs, but they can also increase the risk of a fatal abnormal rhythm. Often, extensive electrophysiologic testing and risk evaluation will be done before this method of treatment is prescribed.

Prognosis

In healthy people, VEBs are inconsequential. If the person with heart disease is able to find an effective means of controlling ventricular ectopic beats, the outlook is good.

Prevention

Occasional ventricular ectopic beats in healthy people do not need to be prevented. People with a history of heart disease can usually control VEBs with medication.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Dorothy Elinor Stonely

Ventricular failure see **Heart failure**

Ventricular fibrillation

Definition

Ventricular fibrillation is a very rapid, uncoordinated, ineffective series of contractions throughout the lower chambers of the heart. Unless stopped, these chaotic impulses are fatal.

Description

When the ventricles begin to quiver and do not employ coordinated contractions, the heart is said to be fibrillating. In this condition the ventricles cannot pump blood from the heart. Ventricular fibrillation (V-fib) is the worst kind of abnormal heart rhythm, and is a form of cardiac arrest. It involves the pumping

of the lower chambers of the heart, while atrial fibrillation involves the upper chambers.

Causes and symptoms

Ventricular fibrillation often is associated with acute ischemic events (**ischemia** involves the deprivation of oxygenated blood to an area of tissue), and with chronic ischemic heart disease. It is frequently seen immediately following a **heart attack**. It also may develop during hypoxia, atrial fibrillation, or improper grounding of electrical devices. An extremely low level of potassium in the blood also can cause ventricular fibrillation.

The first, and usually the only, symptom of V-fib is sudden unconsciousness.

Diagnosis

When an individual suddenly collapses, the possibility of ventricular fibrillation should be considered immediately. A quick assessment usually shows no pulse or heartbeat. The diagnosis of ventricular fibrillation is confirmed with an electrocardiogram.

Treatment

Basic **life support** with standard **cardiopulmonary resuscitation (CPR)** must be started within a few minutes, followed as soon as possible with **cardioversion**. Cardioversion is an electric shock delivered to the heart to stop the fibrillating. Early **defibrillation** is the key to survival. If left untreated, irreversible brain damage, due to lack of oxygen to the brain, occurs after about five minutes. After the heart resumes its normal rhythm, medications are given to help maintain the rhythm.

Research continues into methods to deliver defibrillation as soon as possible to those experiencing ventricular fibrillation. One of the studies addressed in 2003 researched various clinical trials that implanted defibrillators into patients to prevent **sudden cardiac death**. The devices worked in many instances but more proof of their success was needed for widespread use.

Prognosis

Early and effective CPR may provide the time necessary for medical personnel to arrive with a defibrillator. If a defibrillator is able to promptly restore a normal rhythm, up to 25% of victims are able to leave the hospital without evidence of brain damage.

KEY TERMS

Atrial fibrillation—A condition in which the upper chambers of the heart quiver instead of contracting effectively

Cardiopulmonary resuscitation (CPR)—Using rescue breathing and chest compressions to help a person whose breathing and heartbeat have stopped

Cardioversion—An electrical shock delivered to the heart to restore a normal rhythm

Electrocardiogram—A visual representation of the heart beat

Heart failure—A term used when the heart is unable to pump enough blood to supply the needs of the body

Hypoxia—Insufficient oxygen in the cells of the body

Ischemic—Insufficient blood reaching the tissues

If ventricular fibrillation occurs in the hospital in conjunction with a heart attack, defibrillation has a 95% success rate. If shock and **heart failure** are present at the time, even with immediate defibrillation, only about 30% of those stricken are successfully restored to a normal heart rate.

Prevention

A healthy lifestyle to reduce the risk of heart diseases which lead to ventricular fibrillation is the best prevention. For people who have experienced an episode of V-fib, an internal cardioverter-defibrillator may prevent further episodes.

Resources

PERIODICALS

Ezekowitz, Justin A., et al. "Implantable Cardioverter Defibrillators in Primary and Secondary Prevention: A Systematic Review of Randomized, Controlled Trials." *Annals of Internal Medicine* January 2002: 445.

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Dorothy Elinor Stonely
Teresa G. Odle

Ventricular septal defect

Definition

A ventricular septal defect is a hole in the wall of the heart (septum) that separates the left lower chamber (left ventricle) from the right lower chamber (right ventricle). The hole allows blood to flow from the left ventricle to the right ventricle instead of entering the aorta for distribution throughout the body. Ventricular septal defect is one of a group of heart problems found in newborn babies that are collectively called congenital heart disease.

Description

The heart has four chambers. The two lower chambers are called ventricles and are responsible for pumping blood. The right ventricle pumps blood to the lungs and the left ventricle pumps blood throughout the body. If there is a hole in the septum that separates the two ventricles, blood from the left ventricle can enter the right ventricle. This blood recycles through the lungs before returning to the left ventricle. This results in less oxygenated blood reaching the body. If the hole is sufficiently large, the lack of oxygen being delivered to the body can cause severe problems, including **heart failure** and breathlessness. Approximately 0.7% of all babies have a congenital heart defect. Of these, 20% have a ventricular septal defect.

Causes and symptoms

Congenital heart defects are errors in the development of the heart structure. They occur early in the life of the embryo. There is no known cause of congenital heart defects. They can be associated with several diseases, such as German measles (**rubella**) and **Down syndrome**. Genetics does not seem to play a role in ventricular septal defect. People with a heart defect do not have an increased chance of passing it on to their children.

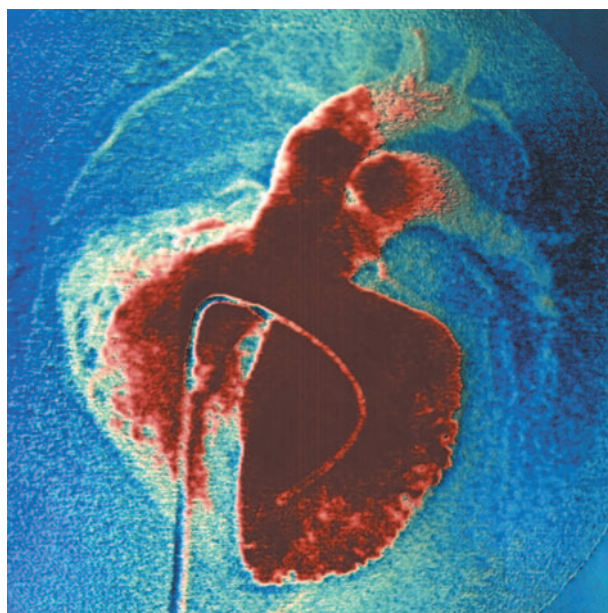
Symptoms result from a reduced amount of oxygen going to the body. Symptoms are proportional to the size of the defect. They may appear at any time in the life of the child. In cases where the hole in the septum is very small, few or no symptoms may appear and the child may develop normally. In cases where the ventricular septal defect is large, the newborn will show signs of heavy breathing, sweating, and feeding difficulties. Children with this defect tire easily. Ventricular septal defect can also result in stunted growth resulting from insufficient oxygen being delivered to the growing body. Children with ventricular

KEY TERMS

Echocardiogram—An image of the heart produced by an instrument that uses sound waves to create image the heart.

Electrocardiogram—A graph of the heart's beating action.

Endocarditis—An inflammation of the interior lining of the heart that is frequently caused by infectious agents.



An angiogram of a ventricular septal defect. This is a hole in the ventricular septum causing blood to flow from the left ventricle (right of image) to the right ventricle and to the lungs. The bent catheter at the center, which is used to take the angiogram, passes through the hole between the ventricles. (Photograph by Simon Fraser, Photo Researchers, Inc. Reproduced by permission.)

septal defect tend to suffer more frequent colds and **pneumonia**, and have a higher rate of inflammation and infection of the heart (**endocarditis**).

Diagnosis

The condition is first suspected based on observation of the child. The physician will listen to the heart with a stethoscope (auscultation) to detect a heart murmur. X rays, electrocardiogram (ECG), and **echocardiography** can all be used to evaluate ventricular septal defect.

Treatment

Most small holes close without treatment. Often, as the child grows, the hole closes or becomes smaller. If the hole is large or fails to close, the child is usually treated with drugs. Holes that persist and are causing problems in development are corrected by open heart surgery. Usually, surgery is performed after one year of age, but before the child enters school. This allows time for a trial of drug therapy, which could potentially eliminate the need for surgery. The operation is generally safe.

Prognosis

Children with small septal defects tend to develop normally and without any effect on their ability to participate in physical activities. Surgery allows children with larger defects to live nearly normal lives.

Resources

BOOKS

Alexander, R. W., R. C. Schlant, and V. Fuster, editors.
The Heart. 9th ed. New York: McGraw-Hill, 1998.

John T. Lohr, PhD

Ventricular shunt

Definition

Ventricular shunt is a surgical procedure in which a tube is placed in one of the fluid-filled chambers inside the brain (ventricles). The fluid around the brain and the spinal column is called the cerebrospinal fluid. When infection or disease causes an excess of this cerebrospinal fluid in the ventricles, the shunt is placed to drain it and thereby relieve excess pressure.

Purpose

Ventricular shunt relieves **hydrocephalus**, a condition in which the ventricles are enlarged. In hydrocephalus, pressure from the cerebrospinal fluid usually increases. It may be caused by tumor of the brain or of the membranes covering the brain (meninges), infection of or bleeding into the cerebrospinal fluid, or inborn malformations of the brain. Symptoms of hydrocephalus may include **headache**, personality disturbances and loss of intellectual abilities (**dementia**),

problems in walking, irritability, **vomiting**, abnormal eye movements, or a low level of consciousness.

Normal pressure hydrocephalus is associated with progressive dementia, problems in walking, and loss of bladder control (**urinary incontinence**). Even though the cerebrospinal fluid is not thought to be under increased pressure in this condition, it may also be treated by ventricular shunting.

Precautions

As with any surgical procedure, the surgeon must know about any medications or health problems that may increase the patient's risk. Because infections are both common and serious complications, **antibiotics** are often given before and after surgery.

Description

The ventricular shunt tube is placed to drain fluid from the ventricular system in the brain to the cavity of the abdomen or to the large vein in the neck (jugular vein). Therefore, surgical procedures must be done both in the brain and at the drainage site. The tubing contains valves to insure that fluid can only flow out of the brain and not back into it. The valve can be set at a desired pressure to allow cerebrospinal fluid to escape whenever the pressure level is exceeded.

A small reservoir may be attached to the tubing and placed under the scalp. This reservoir allows samples of cerebrospinal fluid to be removed with a syringe to check the pressure. Fluid from the reservoir can also be examined for bacteria, **cancer** cells, blood, or protein, depending on the cause of hydrocephalus. The reservoir may also be used to inject antibiotics for cerebrospinal fluid infection or **chemotherapy** medication for meningeal tumors.

Preparation

The diagnosis of hydrocephalus should be confirmed by diagnostic techniques that make images of the brain, such as computed tomography scan (CT scan) or **magnetic resonance imaging** (MRI), before the shunting procedure is performed. These techniques will also show any associated brain abnormalities. Cerebrospinal fluid should be examined if infection or tumor of the meninges is suspected. Patients with dementia or **mental retardation** should undergo neuropsychological testing to establish a baseline psychological profile before the shunting procedure.

Patients with normal pressure hydrocephalus may experience a temporary improvement in walking and

KEY TERMS

Cerebrospinal fluid—Fluid bathing the brain and spinal cord.

Computed tomography (CT) scan—An imaging technique in which cross-sectional x rays of the body are compiled to create a three-dimensional image of the body's internal structures.

Dementia—Progressive loss of mental abilities.

Magnetic resonance imaging (MRI)—An imaging technique that uses a large circular magnet and radio waves to generate signals from atoms in the body. These signals are used to construct images of internal structures.

mental abilities upon removal of a moderate amount of cerebrospinal fluid. This improvement may be an indication that shunting will improve their condition. However, patients who do not improve after temporary cerebrospinal fluid drainage may still benefit from ventricular shunt. When a case is in doubt, continuous monitoring of cerebrospinal fluid pressure (which in itself requires a surgical procedure) may indicate whether shunting is likely to be helpful.

Aftercare

To avoid infections at the shunt site, the area should be kept clean. Cerebrospinal fluid should be checked periodically by the doctor to be sure there is no infection or bleeding into the shunt. Cerebrospinal fluid pressure should be checked to be sure the shunt is operating properly. The eyes should be examined regularly because shunt failure may damage the nerve to the eyes (optic nerve). If not treated promptly, damage to the optic nerve causes irreversible loss of vision. Patients or caregivers should understand the life-threatening nature of shunt problems. All symptoms and signs of potential shunt failure or infection must be taken seriously.

Risks

Complications of shunting occur in 30% of cases, but only 5% are serious. Serious and long-term complications are bleeding under the outermost covering of the brain (**subdural hematoma**), infection, **stroke**, and shunt failure. Infection at the shunt site may cause a loss of intelligence. When shunts drain to the abdomen (ventriculoperitoneal shunts), fluid may accumulate in the abdomen or abdominal organs

may be injured. If cerebrospinal fluid pressure is lowered too much, patients may have severe headaches, often with **nausea and vomiting**, whenever they sit up or stand.

Normal results

Of patients with normal pressure hydrocephalus who are treated with shunting, 25-80% experience long-term improvement. Normal pressure hydrocephalus is more likely to improve when it is caused by infection of or bleeding into the cerebrospinal fluid than when it occurs without an underlying cause. Walking difficulties and bladder control are more likely to improve than dementia is.

After shunting, the ventricles get smaller within three or four days. This shrinkage occurs even when hydrocephalus has been present for a year or more. Clinically detectable signs of improvement occur within a few weeks. The cause of hydrocephalus, duration of hydrocephalus before shunting, and associated brain abnormalities affect the outcome.

Resources

ORGANIZATIONS

American Academy of Neurology. 1080 Montreal Ave., St. Paul, MN 55116. (612) 695-1940. <<http://www.aan.com>>.

Laurie Barclay, MD

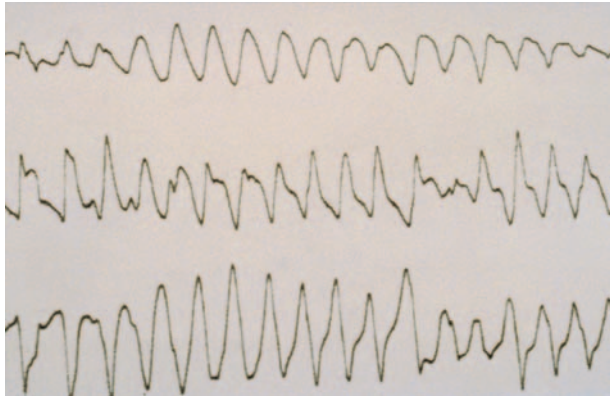
Ventricular tachycardia

Definition

Ventricular tachycardia (V-tach) is a rapid heart beat that originates in one of the lower chambers (the ventricles) of the heart. To be classified as tachycardia, the heart rate is usually at least 100 beats per minute.

Description

A rapid heart rate can originate in either the left or right ventricle. Ventricular tachycardia which lasts more than 30 seconds is referred to as sustained ventricular tachycardia. A period of three to five rapid beats is called a salvo, and six beats or more lasting less than 30 seconds is called nonsustained ventricular tachycardia. Rapid ventricular rhythms are more serious than rapid atrial rhythms because they make the heart extremely inefficient. They also tend to cause



An electrocardiographic image indicating a rapid heart beat. (Custom Medical Stock Photo. Reproduced by permission.)

more severe symptoms, and have a much greater tendency to result in **death**.

Although generally considered to be among the life-threatening abnormal rhythms, harmless forms of sustained V-tach do exist. These occur in people without any structural heart disease.

Causes and symptoms

Most ventricular tachycardias are associated with serious heart disease such as coronary artery blockage, **cardiomyopathy**, or **valvular heart disease**. V-tach is often triggered by an extra beat originating in either the right or left ventricle. It also occurs frequently in connection with a **heart attack**. V-tach commonly occurs within 24 hours of the start of the attack. It must be treated quickly to prevent fibrillation. After 48 to 72 hours of the heart attack, the risk of ventricular tachycardia is small. However, people who have suffered severe damage to the larger anterior wall of the heart have a second danger period, because V-tach often occurs during convalescence from this type of heart attack.

Sustained ventricular tachycardia prevents the ventricles from filling adequately so the heart can not pump normally. This results in loss of blood pressure, and can lead to a loss of consciousness and to **heart failure**.

The individual with V-tach almost always experiences palpitation, though some episodes cause no symptoms at all.

Diagnosis

Diagnosis is easily made with an electrocardiogram.

KEY TERMS

Atrial—Having to do with the upper chambers of the heart.

Cardiomyopathy—A disease of the heart muscle.

Cardioversion—A electrical shock delivered to the heart to restore a normal rhythm.

Coronary artery—The artery that supplies blood to the heart muscle itself.

Electrocardiogram—A visual representation of the heart beat.

Fibrillation—Rapid, uncoordinated, quivering of the heart.

Palpitations—Uncomfortable feeling of the heart beat in the chest.

Valvular—Having to do with the valves inside the heart.

Treatment

Any episode of ventricular tachycardia that causes symptoms needs to be treated. An episode that lasts more than 30 seconds, even without symptoms, also needs to be treated. Drug therapy can be given intravenously to suppress episodes of V-tach. If blood pressure falls below normal, a person will need electric **cardioversion** (“shock”) immediately.

Prognosis

With appropriate drug or surgical treatment, ventricular tachycardia can be controlled in most people.

Prevention

A person susceptible to sustained ventricular tachycardia often has a small abnormal area in the ventricles that is the source of the trigger event. This area can sometimes be surgically removed. If surgery is not an option, and drug therapy is not effective, a device called an automatic cardioverter-defibrillator may be implanted.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300. <<http://www.americanheart.org>>.

Dorothy Elinor Stonely

Verrucae see **Warts**

Vertigo see **Dizziness**

Vesicle see **Skin lesions**

Vesicoureteral reflux

Definition

Vesicoureteral reflux (VUR) refers to a condition in which urine flows from the bladder, back up the ureter, and back into the kidneys.

Description

The normal flow of urine begins in the collecting system of each kidney. Urine then flows out of each kidney and into a tube called the ureter. Each ureter leads into the bladder, where the urine collects until it is passed out of the body. Normally, urine should flow only in this direction. In vesicoureteral reflux, however, urine that has already collected in the bladder is able to flow backwards from the bladder, up the ureter, and back into the collecting system of the kidney. VUR may be present in either one or both ureters.

Vesicoureteral reflux causes damage to the kidneys in two ways:

- The kidney is not designed to withstand very much pressure. When VUR is present, back pressure of the urine on the kidney is significant. This can damage the kidney.
- The kidney is usually sterile, meaning that no bacteria are normally present within it. In VUR, bacteria that enter through the urinary tract may be carried back up the ureter with the urine. These bacteria can enter the kidney, causing severe infection.

Causes and symptoms

Most cases of VUR are due to a defect in the way the ureter is implanted into the bladder. The angle may be wrong, or the valve (which should allow urine only one-way entrance into the bladder) may be weak. Structural defects of the urinary system may also cause VUR. These include a situation in which two ureters leave a kidney, instead of the usual one (duplicated ureters), and in which the ureter is greatly enlarged at the end leading into the bladder (ureterocele).

VUR alone does not usually cause symptoms. Symptoms develop when an infection has set in. The usual symptoms of infection include frequent need to urinate, **pain** or burning with urination, and blood or pus in the urine. Occasionally, VUR is suspected when a child has a difficult time becoming toilet trained. In these cases, the bladder may become irritable and spasm, because it is never totally empty of urine. When the kidneys have been damaged, high blood pressure may develop.

Diagnosis

VUR is diagnosed by taking a series of x-ray pictures. These are taken after putting a small tube (catheter) into the bladder. The bladder is then filled with a dye solution which lights up on the x-ray picture. Pictures are taken immediately, followed by x rays taken while the patient is urinating. This will allow reflux to be demonstrated, and will reveal whether the level of reflux increases when pressure increases during urination. Reflux is then graded based on the height and effects of the VUR:

- Grade I. VUR enters just the portion of the ureter closest to the bladder. The ureter appears normal in size.
- Grade II. VUR enters the entire ureter, and goes up into the collecting system of the kidney. The ureter and the collecting system appear normal in size and structure.
- Grade III. VUR enters the entire ureter and kidney collecting system. Either the ureter or the collecting system are abnormal in size or shape.
- Grade IV. Similar to Grade III, but the ureter is greatly enlarged.
- Grade V. Similar to Grade IV, but the ureter is also abnormally twisted/curved, and the collecting system is greatly enlarged, with absence of the usual structural details.

Treatment

Treatment depends on the grade that is diagnosed. In grades I and II, the usual treatment involves long-term use of a small daily dose of **antibiotics** to prevent the development of infections. The urine is tested regularly to make sure that no infection occurs. The kidneys are evaluated regularly to make sure that they are growing normally and that no new scarring has occurred. Grade III VUR can be treated with antibiotics and careful monitoring. New infections, scarring, or stunting of kidney growth may result in a

KEY TERMS

Bladder—The muscular sac which receives urine from the kidneys, stores it, and ultimately works to remove it from the body during urination.

Reflux—A condition in which flow is backwards from normal.

Ureter—A muscular tube leading from the kidney to the bladder, down which the urine flows.

need for surgery. Grades IV and V are extremely likely to require surgery.

Surgery for VUR consists of reimplanting the ureters into the bladder at a more normal angle. This usually improves the functioning of the valve leading into the bladder. When structural defects of the urinary system are present, surgery will almost always be required to repair these defects.

Prognosis

Prognosis is dependent on the grade of VUR. About 80% of children with grades I and II VUR simply grow out of the problem. As they grow, the ureter lengthens, changing its angle of entry into the bladder. About 50% of children with grade III VUR will require surgery. Nearly all children with grades IV and V VUR will require surgery. In these cases, it is usually best to perform surgery at a relatively young age, in order to avoid damage and scarring to the kidneys.

Prevention

While there is no known method of preventing VUR, it is important to note that a high number of the siblings of children with VUR will also have VUR. Many of these siblings (about 36%) will have no symptoms, but will be discovered through routine examinations prompted by their brother's or sister's problems. It is important to identify these children, so that antibiotic treatment can be used to prevent the development of infection and kidney damage.

Resources

ORGANIZATIONS

American Foundation for Urologic Disease. 300 West Pratt St., Suite 401, Baltimore, MD 21201. (800) 242-2383.

Rosalyn Carson-DeWitt, MD

Vestibulitis see **Labyrinthitis**

Viagra see **Sildenafil citrate**

Vibrio cholera infection see **Cholera**

Vibrio parahaemolyticus infection see **Vibriosis**

Vibriosis

Definition

Vibriosis is a disease caused by an infection with bacteria of the *Vibrio* genus, most commonly *Vibrio parahaemolyticus* or *Vibrio vulnificus*. *Vibrio* bacteria cause **diarrhea**, skin infections, and/or blood infections. The diarrhea-causing *Vibrio parahaemolyticus* is a relatively harmless infection, but *Vibrio vulnificus* infection, though rare, can lead to blood **poisoning** and **death** in many cases.

Description

Vibriosis is a general term referring to an infection by any member of the large group of *Vibrio* bacteria. The bacteria that causes **cholera** is in this group. Alternate names include non-cholera *Vibrio* infection, *Vibrio parahaemolyticus* infection, and *Vibrio vulnificus* infection.

Vibrio parahaemolyticus and *Vibrio vulnificus* are found in salt water. Infection with either of these two bacteria primarily occurs through eating contaminated raw seafood. Raw oysters are the usual source, although other seafood can carry the bacteria.

Vibrio parahaemolyticus causes severe diarrhea. *Vibrio vulnificus* may cause diarrhea, but in persons with an underlying disease it may cause severe blood infections (septicemia or blood poisoning). Contact of a wound with seawater or contaminated seafood can lead to a *Vibrio vulnificus* skin infection.

Vibriosis is not very common in the United States. Most cases occur in coastal states between June and October. Between 1988 and 1991, there were only 21 reported cases of *Vibrio parahaemolyticus* infection in the United States. Between 1988 and 1995, there were over 300 reports of *Vibrio vulnificus* infection in the United States.

Causes and symptoms

Vibriosis is caused by eating seafood contaminated with *Vibrio parahaemolyticus* or *Vibrio vulnificus*.

These bacteria damage the inner wall of the intestine, which causes diarrhea and related symptoms. *Vibrio vulnificus* can get through the intestinal wall and into the bloodstream.

Persons at risk for severe, often fatal vibriosis include those with **liver disease (cirrhosis)**, excess iron (hemochromatosis), thalassemia (a blood disorder), **AIDS**, diabetes, or those who are immunosuppressed.

Symptoms of intestinal infection occur within two days of eating contaminated seafood. Symptoms last for two to 10 days and include watery diarrhea, abdominal cramps, nausea, vomiting, **headache**, and possibly **fever**. Symptoms of a blood infection develop one to two days after eating contaminated seafood, and include fever, chills, low blood pressure, and large fluid-filled blisters on the arms or legs. Similar blisters can also be produced by a *Vibrio vulnificus* skin infection.

Diagnosis

Vibriosis can be diagnosed and treated by an infectious disease specialist. It is diagnosed when *Vibrio* bacteria are grown from samples of stool, blood, or blister fluid. The symptoms and a recent history of eating raw seafood are very important clues for diagnosis.

Treatment

To counteract the fluid loss resulting from diarrhea, the patient will receive fluids either by mouth or intravenously. **Antibiotics** are not helpful in treating *Vibrio parahaemolyticus* diarrhea.

However, *Vibrio vulnificus* infections are treated with antibiotics such as tetracycline (Sumycin, Achromycin V), or doxycycline (Monodox) plus cef-tazidime (Ceftaz, Fortraz, Tazicef). One out of five patients with vibriosis requires hospitalization.

Prognosis

Most healthy persons completely recover from diarrhea caused by *Vibrio* bacteria. *Vibrio vulnificus* blood infection affects persons with underlying illness and is fatal in half of those cases. *Vibrio vulnificus* wound infections are fatal in one quarter of the cases.

Prevention

Contamination with *Vibrio* bacteria does not change the look, smell, or taste of the seafood. Vibriosis can be prevented by avoiding raw or

undercooked shellfish, keeping raw shellfish and its juices away from cooked foods, and avoiding contact of wounded skin with seawater or raw seafood.

Resources

OTHER

Center for Food Safety and Applied Nutrition. <<http://vm.cfsan.fda.gov>>.

Belinda Rowland, PhD

Viral diarrhea see **Rotavirus infections**

Viral meningitis see **Meningitis**

Visceral I see **Roundworm infections**

Vision training

Definition

Vision training, also known as vision therapy or orthoptics, consists of a variety of programs to enhance visual performance. It includes treatments for focusing, binocularity, and eye movement problems. Vision training is generally provided by an optometrist (O.D.).

Purpose

While visual acuity refers to how clearly each eye can see, vision training addresses how well the two eyes work together as a team. When looking at an object, the eyes must focus on the object (e.g., focusing for near or far objects). This involves the lens system of the eyes. The eyes must also work as a team and point at the same object so that the person does not see double. Aiming precisely at the same object will aid in depth perception (stereopsis) and seeing objects in three-dimensions (3D).

Although crossed eyes (**strabismus**) is an obvious condition, many defects in the coordination of eye movement are far less apparent. Even so, they can cause problems in reading, driving vehicles, and other complex tasks that require the integrated function of eyes and body. It is the goal of vision therapy to improve these subtle interactions using carefully devised exercises and devices.

The discipline, called “behavioral optometry,” involves a careful evaluation of visual function, concentrating on complex skills such as rapid reading, distance perception, peripheral field awareness,

accommodative facility, and the coordinated movement of each eye in relationship to the other. From that assessment the doctor goes on to design a course of exercises to correct the problems discovered. Like any other type of training, success requires practice and persistence until habits and reflexes can be retrained.

There are a number of different techniques and instruments used in vision therapy; the field is evolving rapidly in many directions. Some computerized exercises are being developed that promise better patient motivation. A device called the Dynavision apparatus, has produced positive results in retraining **stroke** victims to operate motor vehicles. And traditional forms of vision therapy have increased reading efficiency in an older age group (62 to 75 years).

Because the goal of vision training is to improve visual efficiency and visual processing, people having problems reading should consider a vision training evaluation. Children rubbing their eyes while reading, avoiding reading, or getting headaches while reading should be evaluated. Problems with sustaining focusing (accommodative insufficiency) or problems keeping words single (convergence or divergence problems) may be present. A full eye-health evaluation and vision training workup may reveal a problem. Vision training is also appropriate for people learning how to coordinate the eyes after surgery for strabismus. Vision training can also be used in lazy eye (**amblyopia**) and includes patching the eye and doing various exercises.

Dyslexia is a problem with following the flow of words when reading. Often the order of letters or words is reversed. It is a complex problem involving the way the brain processes the stream of information coming in from the eyes. While vision therapy is not a treatment for dyslexia or learning disabilities, there may be an underlying visual processing problem that may be present. Vision therapy can be part of a multidisciplinary approach to treating learning disabilities.

Sports vision deals with visual performance in sport-related activities. Protective eyewear is also a large consideration when participating in sports. Basketball, baseball, racquetball, and swimming (and other sports as well) can all cause injury to the eyes. Batting helmets with face shields, protective goggles with polycarbonate lenses, or something as simple as ultraviolet (UV) coatings on glasses to protect the eyes from the sun in outdoor sports such as golf can protect the eyes. Hitting a baseball or throwing a basketball into a hoop requires accurate fixation. Golfers need to see clearly and judge distance. Bifocals may need to be adjusted to allow for putting, driving, and reading the

score card. While many of these issues (e.g., UV coatings) can be addressed at a regular eye exam, sports vision may be able to help with more specific, individual problems.

Precautions

Behavioral optometry is a relatively new field of study. Results are mixed. Newer techniques, more refined evaluation methods, and newer pieces of apparatus are continuously being appraised. More study results are needed to define the scope and benefits of this discipline.

Description

Vision therapy is individually tailored to the subject and the discovered problems. It can be a lengthy process with many variations that requires repetition until eye muscles, coordination, reflexes, habits, and the way the brain handles visual input are all retrained. Each program will be individualized. The patient should be aware of the time involved for treatment. Treatment can be from several weeks to several months depending upon the condition. Some insurance plans may cover vision training.

Preparation

If vision therapy is recommended, the optometrist will discuss thoroughly what is expected and necessary for success. The patient must be prepared to perform some eye exercises at home.

Aftercare

Even after the treatment is successful, it may be necessary to continue the exercises to maintain the benefits. It may be necessary to repeat treatment in the future.

Risks

No risk is involved. The treatment is safe.

Normal results

A carefully and individually tailored program of vision therapy should result in a gradual improvement in whatever complex visual function is being addressed. This progress ought to be measurable by using the same tests that were used to diagnose it. If the patient had symptoms, such as headaches or double vision while reading, it should be alleviated.

KEY TERMS

Accommodation—The focusing of the lens of the eye.

Binocular—Both eyes accurately pointing to the same object.

Stereopsis—The visual perception of depth, or the ability to see three-dimensionally. For this to occur, the person must be binocular.

Abnormal results

Because the treatment is safe, the only abnormal result is failure. At the start of treatment, the optometrist should provide a reasonable estimate of what improvement to expect and how long it will take. Should this prove incorrect, either the treatment needs to be modified or the problem deemed untreatable by that method.

Resources

ORGANIZATIONS

American Optometric Association. 243 North Lindbergh Blvd., St. Louis, MO 63141. (314) 991-4100. <<http://www.aoanet.org>>.

Prevent Blindness America. 500 East Remington Road, Schaumburg, IL 60173. (800) 331-2020. <<http://www.preventblindness.org>>.

J. Ricker Polsdorfer, MD

Visual evoked potential study see **Evoked potential studies**

Visual impairment

Definition

Total blindness is the inability to tell light from dark, or the total inability to see. Visual impairment or low vision is a severe reduction in vision that cannot be corrected with standard glasses or contact lenses and reduces a person's ability to function at certain or all tasks. Legal blindness (which is actually a severe visual impairment) refers to a best-corrected central vision of 20/200 or worse in the better eye or a visual acuity of better than 20/200 but with a visual field no greater than 20° (e.g., side vision that is so reduced that it appears as if the person is looking through a tunnel).

Description

Vision is normally measured using a Snellen chart. A Snellen chart has letters of different sizes that are read, one eye at a time, from a distance of 20 ft. People with normal vision are able to read the 20 ft line at 20 ft—20/20 vision—or the 40 ft line at 40 ft, the 100 ft line at 100 ft, and so forth. If at 20 ft the smallest readable letter is larger, vision is designated as the distance from the chart over the size of the smallest letter that can be read.

Eye care professionals measure vision in many ways. Clarity (sharpness) of vision indicates how well a person's central visual status is. The diopter is the unit of measure for refractive errors such as nearsightedness, farsightedness, and **astigmatism** and indicates the strength of corrective lenses needed. People do not just see straight ahead; the entire area of vision is called the visual field. Some people have good vision (e.g., see clearly) but have areas of reduced or no vision (blind spots) in parts of their visual field. Others have good vision in the center but poor vision around the edges (peripheral visual field). People with very poor vision may be able only to count fingers at a given distance from their eyes. This distance becomes the measure of their ability to see.

The World Health Organization (WHO) defines impaired vision in five categories:

- Low vision 1 is a best corrected visual acuity of 20/70.
- Low vision 2 starts at 20/200.
- Blindness 3 is below 20/400.
- Blindness 4 is worse than 5/300
- Blindness 5 is no light perception at all.
- A visual field between 5° and 10° (compared with a normal visual field of about 120°) goes into category 3; less than 5° into category 4, even if the tiny spot of central vision is perfect.

Color blindness is the reduced ability to perceive certain colors, usually red and green. It is a hereditary defect and affects very few tasks. Contrast sensitivity describes the ability to distinguish one object from another. A person with reduced contrast sensitivity may have problems seeing things in the fog because of the decrease in contrast between the object and the fog.

According to the WHO there are over forty million people worldwide whose vision is category 3 or worse, 80% of whom live in developing countries. Half of the blind population in the United States is over 65 years of age.

Causes and symptoms

The leading causes of blindness include:

- **macular degeneration**
- **glaucoma**
- **cataracts**
- **diabetes mellitus**

Other possible causes include infections, injury, or **nutrition**.

Infections

Most infectious eye diseases have been eliminated in the industrialized nations by sanitation, medication, and public health measures. Viral infections are the main exception to this statement. Some infections that may lead to visual impairment include:

- **Herpes simplex keratitis**. A viral infection of the cornea. Repeated occurrences may lead to corneal scarring.
- **Trachoma**. This disease is responsible for six to nine million cases of blindness around the world, of the third of a billion who have the disease. Trachoma is caused by an incomplete bacterium, *Chlamydia trachomatis*, that is easily treated with standard **antibiotics**. It is transmitted directly from eye to eye, mostly by flies. The chlamydia gradually destroy the cornea.
- **Leprosy (Hansen's disease)**. This is another bacterial disease that has a high affinity for the eyes. It, too, can be effectively treated with medicines.
- **River blindness**. Much of the tropics of the Eastern Hemisphere are infested with *Onchocerca volvulus*, a worm that causes "river blindness." This worm is transmitted by fly bites and can be treated with a drug called ivermectin. Nevertheless, twenty-eight million people have the disease, and 40% of them are blind from it.

Other causes

Exposure of a pregnant woman to certain diseases (e.g., **rubella** or **toxoplasmosis**) can cause congenital eye problems. Injuries to the eyes can result in blindness. Very little blindness is due to disease in the brain or the optic nerves. **Multiple sclerosis** and similar nervous system diseases, brain tumors, diseases of the eye sockets, and head injuries are rare causes of blindness.

Nutrition

Vitamin A deficiency is a widespread cause of corneal degeneration in children in developing nations. As many as five million children develop

xerophthalmia from this deficiency each year. Five percent end up blind.

Diagnosis

A low vision exam is slightly different from a general exam. While a case history, visual status, and eye health evaluation are common to both exams, some things do differ. Eye charts other than a Snellen eye chart will be used. Testing distance will vary. A trial frame worn by the patient is usually used instead of the instrument containing the lenses the patient sits behind (phoropter). Because the low vision exam is slightly more goal oriented than a general exam, for example, what specifically is the patient having trouble with (reading, seeing street signs, etc.) different optical and nonoptical aids will generally be tried. Eye health is the last thing to be checked so that the lights necessary to examine the eyes will not interfere with the rest of the testing.

Treatment

There are many options for patients with visual impairment. There are optical and nonoptical aids. Optical aids include:

- **Telescopes**. May be used to read street signs.
- **Hand magnifiers**. May be used to read labels on things at the store.
- **Stand magnifiers**. May be used to read.
- **Prisms**. May be used to move the image onto a healthy part of the retina in some eye diseases.
- **Closed circuit television (CCTV)**. For large magnification (e.g., for reading).

Nonoptical aids can include large print books and magazines, check-writing guides, large print dials on the telephone, and more.

For those who are blind, there are enormous resources available to improve the quality of life. For the legally blind, financial assistance for help may be possible. Braille and audio books are increasingly available. Guide dogs provide well-trained eyes and independence. Orientation and mobility training is available. There are special schools for blind children and access to disability support through Social Security and private institutions.

Prognosis

The prognosis generally relates to the severity of the impairment and the ability of the aids to correct it.

KEY TERMS

Cornea—The clear dome-shaped structure that is part of the front of the eye. It lies in front of the colored part of the eye (iris).

Diabetic retinopathy—Retinal disease caused by the damage diabetes does to small blood vessels.

Phoropter—The instrument used to measure refractive status of the eyes. It contains many lenses which are then changed in front of the eyes while the patient is looking at an eye chart. This is when the doctor usually asks, “Which is better, one or two?”

Xerophthalmia—A drying of the cornea and conjunctiva.

A good low vision exam is important to be aware of the latest low vision aids.

Prevention

Regular eye exams are important to detect silent eye problems (e.g., glaucoma). Left untreated, glaucoma can result in blindness.

Corneal infections can be treated with effective antibiotics. When a cornea has become opaque beyond recovery it must be transplanted. Good hygiene (e.g., washing hands frequently) to prevent infection, proper use of contact lenses, and not sharing makeup are just some ways to guard against corneal infections.

Cataracts should be removed when they interfere with a person’s quality of life.

Primary prevention addresses the causes before they ever begin. Fly control can be accomplished by simple sanitation methods. Public health measures can reduce the incidence of many infectious diseases. Vitamin A supplementation (when appropriate) will eliminate xerophthalmia completely. It is possible that protecting the eyes against ultraviolet (UV) light will reduce the incidence of cataracts, macular degeneration, and some other eye diseases. UV coatings can be placed on regular glasses, sunglasses, and ski goggles. Patients should ask their eye care professional about UV coatings. Protective goggles should also be worn in certain situations (e.g., certain jobs, sports, even mowing the lawn).

Secondary prevention addresses treating established diseases before they cause irreversible eye

damage. Having general physical checkups can also detect systemic diseases such as diabetes or high blood pressure. Control of diabetes is very important in preserving sight.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street, P.O. Box 7424, San Francisco, CA 94120-7424. < <http://www.eyenet.org> >.

American Foundation for the Blind. 11 Penn Plaza, Suite 300, New York, NY 10001. (800) 232-5463.

Guide Dogs for the Blind. P.O. Box 1200, San Rafael, CA 94915. (415) 499-4000.

International Eye Foundation. 7801 Norfolk Ave., Bethesda, MD 20814. (301) 986-1830.

Lighthouse National Center for Education. 111 E. 59th Street. New York, NY 10022. (800) 334-5497. < <http://www.lighthouse.org> >.

National Association for the Visually Handicapped. 22 West 21st St., New York, NY 10010. (212) 889-3141.

National Center for Sight. (800) 221-3004.

National Children’s Eye Care Foundation. One Clinic Center, A3-108, Cleveland, OH 44195. (216) 444-0488.

National Eye Institute. 2020 Vision Place, Bethesda, MD 20892-3655. (301) 496-5248. < <http://www.nei.nih.gov> >.

National Federation of the Blind. 1800 Johnson St., Baltimore, MD 21230. (301) 569-9314.

Prevent Blindness America. 500 East Remington Road, Schaumburg, IL 60173. (800) 331-2020. < <http://www.preventblindness.org> >.

Research to Prevent Blindness. 598 Madison Ave., New York, NY 10022. (212) 363-3911.

J. Ricker Polsdorfer, MD

Visualization see **Guided imagery**

Vitamin A deficiency

Definition

Vitamin A deficiency exists when the chronic failure to eat sufficient amounts of vitamin A or beta-carotene results in levels of blood-serum vitamin A that are below a defined range. Beta-carotene is a form of pre-vitamin A, which is readily converted to vitamin A in the body. Night blindness is the first symptom of vitamin A deficiency. Prolonged and severe vitamin A deficiency can produce total and irreversible blindness.

Description

Vitamin A (called retinol in mammals) is a fat-soluble vitamin. The recommended dietary allowance (RDA) for vitamin A is 1.0 mg/day for the adult man and 0.8 mg/day for the adult woman. Since beta-carotene is converted to vitamin A in the body, the body's requirement for vitamin A can be supplied entirely by beta-carotene. Six mg of beta-carotene are considered to be the equivalent of 1 mg of vitamin A. The best sources of vitamin A are eggs, milk, butter, liver, and fish, such as herring, sardines, and tuna. Beef is a poor source of vitamin A. Plants do not contain vitamin A, but they do contain beta-carotene and other carotenoids. The best sources of beta-carotene are dark-green, orange, and yellow vegetables; spinach, carrots, oranges, and sweet potatoes are excellent examples. Cereals are poor sources of beta-carotene.

Vitamin A is used for two functions in the body. Used in the eye, it is a component of the eye's light-sensitive parts, containing rods and cones, that allow for night-vision or for seeing in dim-light circumstances. Vitamin A (retinol) occurs in the rods. Another form of Vitamin A, retinoic acid, is used in the body for regulating the development of various tissues, such as the cells of the skin, and the lining of the lungs and intestines. Vitamin A is important during embryological development, since, without vitamin A, the fertilized egg cannot develop into a fetus.

Causes and symptoms

Vitamin A deficiency occurs with the chronic consumption of **diets** that are deficient in both vitamin A and beta-carotene. When vitamin A deficiency exists in the developed world, it tends to happen in alcoholics or in people with diseases that affect the intestine's ability to absorb fat. Examples of such diseases are **celiac disease** (chronic nutritional disorder), **cystic fibrosis**, and **cholestasis** (bile-flow failure or interference). Vitamin A deficiency occurred in infants during the early 1900s in Denmark. The deficiency resulted when milk fat was made into butter for export, leaving the by-product (skimmed milk) for infant feeding. Vitamin A deficiency has taken place in infants in impoverished populations in India, where the only foods fed to the infants were low in beta-carotene. Vitamin A deficiency is also common in areas like Southeast Asia, where polished rice, which lacks the vitamin, is a major part of the diet.

KEY TERMS

Bitot's spots—Bitot's spots are superficial, foamy gray, triangular spots on the white of the eyeball.

Carotenoids—Carotenoids are yellow to deep-red pigments.

Conjunctiva—The conjunctiva is a clear layer of cells that covers the eye and directly contacts the atmosphere. The conjunctiva is about five-cells thick.

Cornea—The cornea is a clear layer of cells that covers the eye, just under the conjunctiva. The cornea is about 50-cells thick.

Fat-soluble vitamin—Fat-soluble vitamins can be dissolved in oil or in melted fat. Water-soluble vitamins can be dissolved in water or juice.

Keratomalacia—Keratomalacia is ulceration of the cornea.

Recommended Dietary Allowance (RDA)—The Recommended Dietary Allowances are quantities of nutrients in the diet that are required to maintain good health in people. RDAs are established by the Food and Nutrition Board of the National Academy of Sciences, and may be revised every few years. A separate RDA value exists for each nutrient. The RDA values refer to the amount of nutrient expected to maintain good health in people. The actual amounts of each nutrient required to maintain good health in specific individuals differ from person to person.

Xerophthalmia—Xerophthalmia is a dry, thickened, lusterless condition of the eyeball resulting from vitamin A deficiency.

The earliest symptom of vitamin A deficiency is night blindness. Prolonged deficiency results in drying of the conjunctiva (the mucous membrane that lines the inner surface of the eyelids and extends over the forepart of the eyeball). With continued vitamin A deficiency, the drying extends to the cornea (xerophthalmia). The cornea eventually shrivels up and becomes ulcerated (keratomalacia). Superficial, foamy gray triangular spots may appear in the white of the eye (Bitot's spots). Finally, inflammation and infection occur in the interior of the eye, resulting in total and irreversible blindness.

Diagnosis

Vitamin A status is measured by tests for retinol. Blood-serum retinol concentrations of 30-60 mg/dl are considered in the normal range. Levels that fall below this range indicate vitamin A deficiency. Night blindness is measured by a technique called electroretinography. Xerophthalmia, keratinomalacia, and Bitot's spots are diagnosed visually by trained medical personnel.

Treatment

Vitamin A deficiency can be prevented or treated by taking vitamin supplements or by getting injections of the vitamin. The specific doses given are oral retinyl palmitate (110 mg), retinyl acetate (66 mg), or injected retinyl palmitate (55 mg) administered on each of two successive days, and once a few weeks later if symptoms are not relieved.

Prognosis

The prognosis for correcting night blindness is excellent. Xerophthalmia can be corrected with vitamin A therapy. Ulcerations, tissue death, and total blindness, caused by severe vitamin A deficiency, cannot be treated with vitamin A.

Prevention

Vitamin A deficiency can be prevented by including foods rich in vitamin A or beta-carotene as a regular component of the diet; liver, meat, eggs, milk, and dairy products are examples. Foods rich in beta-carotene include red peppers, carrots, pumpkins, as well as those just mentioned. Margarine is rich in beta-carotene, because this chemical is used as a coloring agent in margarine production. In Africa, Indonesia, and the Philippines, vitamin A deficiency is prevented by public health programs that supply children with injections of the vitamin.

Resources

BOOKS

Brody, T. *Nutritional Biochemistry*. San Diego: Academic Press, Inc., 1998.

Tom Brody, PhD

Vitamin B₁ see **Beriberi**

Vitamin B₂ deficiency see **Riboflavin deficiency**

Vitamin B₆ deficiency

Definition

Vitamin B₆ is used by the body as a catalyst in reactions that involve amino acids. Vitamin B₆ deficiency is rare, since most foods eaten contain the vitamin.

Description

Vitamin B₆ is a water-soluble vitamin. The recommended dietary allowance (RDA) for vitamin B₆ is 2.0 mg/day for the adult man and 1.6 mg/day for the adult woman. Vitamin B₆ in the diet generally occurs as a form called pyridoxal phosphate. In this form, it cannot be absorbed by the body. During the process of digestion, the phosphate group is removed, and pyridoxal is produced. However, the body readily absorbs pyridoxal, and converts it back to the active form of the vitamin (pyridoxal phosphate).

Poultry, fish, liver, and eggs are good sources of vitamin B₆, comprising about 3-4 mg vitamin/kg food; meat and milk contain lesser amounts of the vitamin. The vitamin also occurs, at about half this level, in a variety of plant foods, including beans, broccoli, cabbage, and peas. Vitamin B₆ tends to be destroyed with prolonged cooking, with storage, or with exposure to light.

As mentioned, vitamin B₆ takes various forms. One of these forms, called pyridoxine, is relatively stable. For this reason, pyridoxine is the form of vitamin B₆ that is used in vitamin supplements, or when foods are fortified. Apples and other fruits are poor sources of the vitamin, containing only 0.2-0.6 mg vitamin/kg food.

Vitamin B₆, used mainly in the body for the processing of amino acids, performs this task along with certain enzymes. The enzyme that participates in this type of complex is aminotransferase. Several types of aminotransferase exist. With vitamin B₆ deficiency, while aminotransferase continues to occur in the various organs of the body, there is an abnormally low level of the active vitamin B₆/aminotransferase complex present. Thus, this vitamin deficiency results in the impairment of a variety of activities in the body. With supplement correction of the vitamin B₆ deficiency, the aminotransferase then readily forms the active complex, and normal metabolism is restored.

Vitamin B₆ converts certain amino acids (glutamic acid, aspartic acid, glycine) to energy. This allows the body to process all dietary protein, even when the

KEY TERMS

Amino acid—Amino acids are small molecules that are used as building blocks for all proteins. Some amino acids are also used in the body for the manufacture of hormones. There are about 20 nutritionally important amino acids, including glutamic acid, glycine, methionine, lysine, tryptophan, serine, and glycine.

Fat-soluble vitamins—Fat-soluble vitamins can be dissolved in oil or in melted fat.

Recommended Dietary Allowance (RDA)—The Recommended Dietary Allowances (RDAs) are quantities of nutrients in the diet that are required to maintain good health in people. RDAs are established by the Food and Nutrition Board of the National Academy of Sciences, and may be revised every few years. A separate RDA value exists for each nutrient. The RDA values refer to the amount of nutrient expected to maintain good health in people. The actual amounts of each nutrient required to maintain good health in specific individuals differ from person to person.

Water-soluble vitamins—Water-soluble vitamins can be dissolved in water or juice.

dietary protein is in excess of the body's needs. Vitamin B₆ also allows the body to synthesize certain amino acids. For example, if the diet is deficient or low in certain amino acids, such as glycine or serine, vitamin B₆ enables the body to make them from sugar. Vitamin B₆ is used also for the synthesis of certain hormones, such as adrenaline.

Causes and symptoms

Vitamin B₆ deficiency occurs rarely. When it does, it is usually associated with poor absorption of nutrients in the gastrointestinal tract (as in **alcoholism**, or with chronic **diarrhea**), the taking of certain drugs (as isoniazid, hydrolazine, penicillamine) that inactivate the vitamin, with genetic disorders that inhibit metabolism of the vitamin, or in cases of **starvation**.

The symptoms of vitamin B₆ deficiency in adults are only vaguely defined. These include nervousness, irritability, **insomnia**, muscle weakness, and difficulty in walking. Vitamin B₆ deficiency may produce fissures and cracking at the corners of the mouth. The deficiency occurred in infants fed early versions of commercial canned infant formula, when the vitamin

had been inadvertently omitted from the formula. This error resulted in infants failing to grow, in irritability, and in seizures.

Diagnosis

Vitamin B₆ status is measured by the transaminase stimulation test. This test requires extraction of red blood cells, and placement of the cells in two test tubes. Special chemicals (reagents) are added to both test tubes to allow for measurement of aminotransferase. This enzyme requires pyridoxal phosphate. A known quantity of pure pyridoxal phosphate is added to one of the test tubes. The activity level of the enzyme is measured, and compared, in both test tubes. If the added pyridoxal phosphate did not stimulate activity, the patient is considered not to be deficient in vitamin B₆. Neither is the patient considered deficient if only slight stimulation occurred. But if a stimulation of four-fold or more occurred, a vitamin B₆ deficiency is present.

Treatment

Vitamin B₆ deficiency can be prevented or treated with consumption of the recommended dietary allowance, as supplied by food or by vitamin supplements.

Prognosis

The prognosis for correcting vitamin B₆ deficiency is excellent.

Prevention

Vitamin B₆ deficiency is not a major concern for most people. The deficiency can be prevented with consumption of a mixed diet that includes poultry, fish, eggs, meat, vegetables, and grains.

Resources

BOOKS

Brody, T. *Nutritional Biochemistry*. San Diego: Academic Press, Inc., 1998.

Tom Brody, PhD

Vitamin B₁₂ deficiency anemia see **Pernicious anemia**

Vitamin C deficiency see **Scurvy**

Vitamin D deficiency

Definition

Vitamin D deficiency exists when the concentration of 25-hydroxy-vitamin D (25-OH-D) in the blood serum occurs at 12 ng/ml (nanograms/milliliter), or less. The normal concentration of 25-hydroxy-vitamin D in the blood serum is 25-50 ng/ml. When vitamin D deficiency continues for many months in growing children, the disease commonly referred to as **rickets** will occur. A prolonged deficiency of the vitamin in adults results in osteomalacia. Both diseases involve defects in bones.

Description

Vitamin D is a fat-soluble vitamin, meaning it is able to be dissolved in fat. While some vitamin D is supplied by the diet, most of it is made in the body. To make vitamin D, cholesterol, a sterol that is widely distributed in animal tissues and occurs in the yolk of eggs, as well as in various oils and fats, is necessary. Once cholesterol is available in the body, a slight alteration in the cholesterol molecule occurs, with one change taking place in the skin. This alteration requires the energy of sunlight (or ultraviolet light). Vitamin D deficiency, as well as rickets and osteomalacia, tends to occur in persons who do not get enough sunlight and who fail to eat foods that are rich in vitamin D.

Once consumed, or made in the body, vitamin D is further altered to produce a hormone called 1,25-dihydroxy-vitamin D (1,25-diOH-D). The conversion of vitamin D to 1,25-diOH-D does not occur in the skin, but in the liver and kidney. First, vitamin D is converted to 25-OH-D in the liver; it then enters the bloodstream, where it is taken-up by the kidneys. At this point, it is converted to 1,25-diOH-D. Therefore, the manufacture of 1,25-diOH-D requires the participation of various organs of the body—the liver, kidney, and skin.

The purpose of 1,25-diOH-D in the body is to keep the concentration of calcium at a constant level in the bloodstream. The maintenance of calcium at a constant level is absolutely required for human life to exist, since dissolved calcium is required for nerves and muscles to work. One of the ways in which 1,25-diOH-D accomplishes this mission is by stimulating the absorption of dietary calcium by the intestines.

The sequence of events that can lead to vitamin D deficiency, then to bone disease, is as follows: a lack of vitamin D in the body creates an inability to

manufacture 1,25-diOH-D, which results in decreased absorption of dietary calcium and increased loss of calcium in the feces. When this happens, the bones are affected. Vitamin D deficiency results in a lack of bone mineralization (calcification) in growing persons, or in an increased demineralization (decalcification) of bone in adults.

Causes and symptoms

Vitamin D deficiency can be caused by conditions that result in little exposure to sunlight. These conditions include: living in northern countries; having dark skin; being elderly or an infant, and having little chance to go outside; and covering one's face and body, such as for religious reasons. Many Arab women cover the entire body with black cloth, and wear a veil and black gloves when they go outside. These women may acquire vitamin D deficiency, even though they live in a sunny climate.

Most foods contain little or no vitamin D. As a result, sunshine is often a deciding factor in whether vitamin D deficiency occurs. Although fortified milk and fortified infant formula contain high levels of vitamin D, human breast milk is rather low in the vitamin. The term fortified means that **vitamins** are added to the food by the manufacturer.

To say that a food is high or low in vitamin D means how much of that food needs to be eaten in order to prevent vitamin deficiency and maintain good health. An exact meaning can be provided by comparing the Recommended Dietary Allowance of vitamin D with the amount of vitamin D supplied by a particular food per day. The Recommended Dietary Allowance, also referred to as RDA, is a recommendation based on data derived from different population groups and ages. The RDA for vitamin D for adults is 200 International Units (IU) per day, and can be supplied by eating approximately 1.5 kg of beef, 2.0 kg of corn oil, or 100 kg of cabbage. Few people, though, would want to eat a kilogram of beef in one day, and no human being is capable of eating 100 kg of cabbage in a day; therefore, these foods are poor sources of vitamin D. However, saltwater fish such as salmon, herring, and sardines are rich in vitamin D, supplied from the oils produced by these fish. The RDA can also be supplied by eating roughly 50 g of salmon or 2.0 g of cod liver oil, and since fortified milk contains 400 IU per quart, half a quart of milk provides the RDA. For comparison, human breast milk contains only 4 to 60 IU per quart.

No harm is likely to result from vitamin D deficiency that occurs for only a few days a year. If the

deficiency occurs for a period of many months or years, however, rickets or osteomalacia may develop. The symptoms of rickets include bowed legs and bowed arms. The bowed appearance is due to the softening of bones, and their bending if the bones are weight-bearing. Bone growth occurs through the creation of new cartilage, a soft substance at the ends of bones. When the mineral calcium phosphate is deposited onto the cartilage, a hard structure is created. In vitamin D deficiency, though, calcium is not available to create hardened bone, and the result is soft bone. Other symptoms of rickets include particular bony bumps on the ribs called rachitic rosary (beadlike prominences at the junction of the ribs with their cartilages) and knock-knees. Seizures may also occasionally occur in a child with rickets, because of reduced levels of dissolved calcium in the bloodstream.

Although osteomalacia is rare in the United States, symptoms of this disease include reduced bone strength, an increase in bone **fractures**, and sometimes bone **pain**, muscle weakness, and a waddling walk.

Diagnosis

Vitamin D deficiency is diagnosed by measuring the level of 25-hydroxy-vitamin D in the blood serum. The normal level or concentration of this form of the vitamin ranges from 25-50 ng/ml. Deficiency occurs when this level decreases to about 12 ng/ml or less. As mentioned previously, 25-OH-D is not the active form of the vitamin. It must be converted to 1,25-diOH-D in order to cause responses in various organs of the body. However, the levels of vitamin D, or of 1,25-dihydroxy-vitamin D in the blood, do not give a reliable picture of whether a person is deficient in the vitamin. For this reason, they are not measured when testing for vitamin D deficiency.

Rickets is diagnosed by x-ray examination of leg bones. A distinct pattern of irregularities, abnormalities, and a coarse appearance can be clearly seen with rickets. Osteomalacia is also diagnosed with x-ray examination. Measurements of blood plasma 25-OH-D, blood plasma calcium, and blood plasma parathyroid hormone must also be obtained for the diagnosis of these diseases. Parathyroid hormone and 1,25-diOH-D work together in the body to regulate the levels of calcium in the blood.

Treatment

Rickets heals promptly with 4,000 IU of oral vitamin D per day administered for approximately

one month. During this treatment, the doctor should monitor the levels of 25-OH-D in the plasma to make certain they are raised to a normal value. The bone abnormalities (visible by x ray) generally disappear gradually over a period of 3-9 months. Parents are instructed to take their infants outdoors for approximately 20 minutes per day with their faces exposed. Children should also be encouraged to play outside. Foods that are good sources of vitamin D include cod liver oil, egg yolks, butter, and oily fish. Some foods, including milk and breakfast cereals, are also fortified with synthetic vitamin D.

Osteomalacia is treated by eating 2,500 IU per day of vitamin D for about three months. Measurements of 25-OH-D, calcium, and parathyroid hormone should be obtained after the treatment period to make sure the therapy did, in fact, result in normal blood values.

Care must be taken in treating vitamin D deficiency, since high doses of vitamin D are toxic and can result in the permanent deposit of **minerals** in the heart, lungs, and kidneys. Symptoms of toxicity are **nausea**, **vomiting**, pain in joints, and lack of interest in eating food. In adults, vitamin D toxicity occurs with eating 50,000 IU or more per day. In infants, toxicity occurs with 1,000 IU per day. The continued intake of toxic doses results in **death**.

Rickets and osteomalacia are almost always treated with oral supplements of vitamin D, with the recommendation to acquire daily exposure to direct sunlight. An alternative to sunlight is the use of an ultraviolet (UV) lamp. When using UV lamps, the eyes must be covered to protect them against damage. Many types of sunglasses allow UV light to pass through, so only those that are opaque to UV light should be used. Attempts to acquire sunlight through glass windows fail to help the body make vitamin D. This is because UV light does not pass through window glass.

Rickets may also occur with calcium deficiency, even when a child is regularly exposed to sunshine. This type of rickets has been found in various parts of Africa. The bone deformities are similar to, or are the same as, those that occur in typical rickets; however, calcium deficiency rickets is treated by increasing the amount of calcium in the diet. No amount of vitamin D can cure the rickets of a child with a diet that is extremely low in calcium. For this reason, it is recommended that calcium be given in conjunction with vitamin D supplementation.

KEY TERMS

25-hydroxy-vitamin D—This is the form of vitamin D that is measured in order to assess vitamin D deficiency.

Cholesterol—A fat-soluble steroid alcohol (sterol) found in animal fats and oils, and in egg yolks. The human body needs cholesterol to produce vitamin D.

Fat-soluble vitamin—A vitamin that dissolves easily in fat or oil, but not in water. The fat-soluble vitamins are vitamins D, E, A, and K.

International unit (IU)—A measurement of biological activity in which one IU is equal to one mg (milligram).

Osteomalacia—Osteomalacia is a bone disease that occurs in adults and is caused by a prolonged period of vitamin D deficiency.

Rachitic rosary—Beadlike bumps present at the junction of the ribs with their cartilages—often seen in children with rickets.

Recommended Dietary Allowance (RDA)—The amount of nutrients, including vitamins, that should be supplied by foods on a daily basis to maintain normal health. Recommendations are based on data obtained from different population groups and ages.

Rickets—Rickets is a bone disease that occurs in infants and growing children and is caused by a prolonged period of vitamin D deficiency.

Prognosis

The prognoses for correcting vitamin D deficiency, rickets, and osteomalacia are excellent. Vitamin D treatment results in the return of bone mineralization to a normal rate, the correction of low plasma calcium levels, the prevention of seizures, and a recovery from bone pain. On the other hand, deformities such as bowed legs and the rachitic rosary persist throughout adult life.

Prevention

Food fortification has almost completely eliminated rickets in the United States. Vitamin D deficiency can be prevented by acquiring the RDA through drinking fortified milk and eating fortified cereals. For those who cannot drink milk, supplements of pills might be considered. In some older

people, a 400 IU supplement may not be enough to result in the normal absorption of calcium; therefore, daily doses of 10,000 IU per day may be needed. For infants who are fed only breast milk (and rarely exposed to sunshine), a daily supplement of 200-300 IU is recommended.

Rickets continues to be a problem in Africans and Asian Indians who migrate to Canada or Great Britain, especially where these immigrants do not drink fortified milk. Prevention of rickets in these populations is attempted through educational programs sponsored by the government.

Resources

PERIODICALS

Kinyamu, H., et al. "Serum Vitamin D Metabolites and Calcium Absorption in Normal Young and Elderly Free-Living Women and in Women Living in Nursing Homes." *American Journal of Clinical Nutrition* 65 (1997): 790-797.

Tom Brody, PhD

Vitamin E deficiency

Definition

Vitamin E deficiency is a very rare problem that results in damage to nerves. When vitamin E deficiency does occur, it strikes people with diseases that prevent the absorption of dietary fats and fat-soluble nutrients. Since vitamin E is a fat-soluble vitamin, it has some of the properties of fat.

Description

The recommended dietary allowance (RDA) for vitamin E is 10 mg/day for the adult man, 10 mg/day for the adult woman, and 3 mg/day for the infant. Vitamin E occurs in foods in a variety of related forms. The most potent and useful form of vitamin E is called alpha-tocopherol. The best sources of vitamin E are vegetable oils, such as corn oil, soy oil, and peanut oil. Animal fats, such as butter and lard, contain lower levels of the vitamin. Corn oil contains about 16 mg of alpha-tocopherol per 100 g oil. Wheat-germ oil contains 120 mg alpha-tocopherol per 100 g oil. Fish, eggs, and beef contain relatively low levels of the vitamin, with about 1 mg per 100 g food.

Vitamin E seems to have only one function in the body: the prevention of the natural and continual

process of deterioration of all body tissues. This deterioration is provoked by a number of causes; one of these is toxic oxygen. During the body's metabolism of atmospheric oxygen, toxic oxygen is produced continuously in the body by the formation of by-products. These toxic by-products include hydrogen peroxide, superoxide, and hypochlorite.

Hypochlorite is a natural product, produced by cells of the immune system. It is also the active component of bleach. Once formed, toxic oxygen can damage various parts of the body, such as the membranes which form the boundaries of every cell. Vitamin E serves the body in protecting membranes from toxic oxygen damage. In contrast, vitamin C serves to protect the aqueous, or watery, regions of the cell from toxic oxygen damage. The membranes that are most sensitive to toxic oxygen damage are the membranes of nerves; therefore, the main symptom of vitamin E deficiency is damage to the nervous system.

Causes and symptoms

As mentioned, when vitamin E deficiency occurs, it strikes people with diseases that prevent the absorption of dietary fats and fat-soluble nutrients. These diseases include **cystic fibrosis**, **pancreatitis**, and **cholestasis** (bile-flow obstruction). Bile salts, produced in the liver, are required for the absorption of fats. Cholestasis causes a decrease in the formation of bile salts and the consequent failure of the body to absorb dietary fats. For this reason, this disease may result in vitamin E deficiency. Premature infants may be at risk for vitamin E deficiency because they may be born with low tissue levels of the vitamin, and because they have a poorly developed capacity for absorbing dietary fats. Infants suffering from fat-malabsorption diseases can develop symptoms of vitamin E deficiency by age two. In adults, the onset of a fat-malabsorption disease can provoke vitamin E deficiency after a longer period, as an example, ten years.

Patients with colorectal **cancer** caused by the so-called Ki-ras mutation have also been shown to absorb less vitamin E from their diet than either normal control subjects or cancer patients without the mutation. The relationship between genetic mutations and dietary factors requires more intensive study.

Vitamin E deficiency in humans results in ataxia (poor muscle coordination with shaky movements), decreased sensation to vibration, lack of reflexes, and **paralysis** of eye muscles. One particularly severe

symptom of vitamin E deficiency is the inability to walk.

Another symptom of early vitamin E deficiency in children with cystic fibrosis is a decline in cognitive function, which results in difficulty with reading and falling behind in other intellectual skills during the elementary school years. Researchers have urged the introduction of neonatal screening in order to offset the potential effects of early vitamin E deficiency.

More recently, the suggestion has been made that vitamin E deficiency may be involved in the development of partial open-angle **glaucoma** (POAG), an eye disorder whose causes are not fully understood as of the early 2000s. The possibility that POAG is a vitamin-deficiency disorder, however, needs further research.

Diagnosis

Vitamin E status is measured by assessment of the content of alpha-tocopherol in the blood plasma, using a method called high-pressure liquid chromatography. Blood plasma levels of alpha-tocopherol that are 5.0 mg/l, or above, indicate normal vitamin E status; levels below 5.0 mg/l indicate vitamin E deficiency.

Treatment

Vitamin E deficiency that occurs with cholestatic **liver disease** or other malabsorption syndromes can be treated with weekly injections of 100 mg alpha-tocopherol that may continue for six months. Vitamin E deficiency in premature infants may require treatment for only a few weeks.

Prognosis

The prognosis for correcting the neurological symptoms of vitamin E deficiency is fair to excellent.

Prevention

The prevention of vitamin E deficiency should not be a concern for most people, since the vitamin is found in a wide variety of foods. Attention has been given to the theory that vitamin E serves to protect against cancer and **atherosclerosis**. The evidence that normal levels of vitamin E protect against atherosclerosis is fairly convincing. However, there is little or no proof that vitamin E intake, above and beyond the recommended daily allowance (RDA), can prevent cancer or atherosclerosis.

KEY TERMS

Fat-soluble vitamin—Fat-soluble vitamins can be dissolved in oil or in melted fat.

Recommended dietary allowance (RDA)—The quantity of a given nutrient in the diet that is required to maintain good health in people. RDAs are established by the Food and Nutrition Board of the National Academy of Sciences, and may be revised every few years. A separate RDA value exists for each nutrient. The RDA values refer to the amount of nutrient expected to maintain good health in people. The actual amounts of each nutrient required to maintain good health in specific individuals differ from person to person.

Toxic oxygen—Oxygen is required for life, as it is needed for energy production. When oxygen is used by the body, most of it is converted to water. However, a small fraction of the oxygen breathed is converted to toxic oxygen. The body uses several different processes for preventing and repairing toxic-oxygen damage. One of these processes involves vitamin E.

Water-soluble vitamins—Water-soluble vitamins can be dissolved in water or juice.

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Vitamin K deficiency

Definition

Vitamin K deficiency exists when chronic failure to eat sufficient amounts of vitamin K results in a tendency for spontaneous bleeding or in prolonged and excessive bleeding with trauma or injury. Vitamin K deficiency occurs also in newborn infants, as well as in people treated with certain **antibiotics**. The protein in the body most affected by vitamin K deficiency is a blood-clotting protein called prothrombin.

Description

Vitamin K is a fat-soluble vitamin. The recommended dietary allowance (RDA) for vitamin K is 80 mg/day for the adult man, 65 mg/day for the adult woman, and 5 mg/day for the newborn infant. The vitamin K present in plant foods is called phyloquinone; while the form of the vitamin present in animal foods is called menaquinone. Both of these **vitamins** are absorbed from the diet and converted to an active form called dihydrovitamin K.

Spinach, lettuce, broccoli, brussels sprouts, and cabbage are good sources of vitamin K, containing about 8 mg vitamin K/kg food. Cow milk is also a good source of the vitamin.

A portion of the body's vitamin K is supplied by bacteria living in the intestine rather than by dietary sources.

Vitamin K plays an important role in blood clotting. Without the vitamin, even a small cut would cause continuous bleeding in the body, and **death**. Blood clotting is a process that begins automatically when any injury produces a tear in a blood vessel. The process of blood clotting involves a collection of molecules, which circulate continuously through the bloodstream. When an injury occurs, these molecules rapidly assemble and form the blood clot. The clotting factors are proteins, and include proteins called Factor II, Factor VII, Factor IX, and Factor X. Factor II is also called prothrombin. These proteins require vitamin K for their synthesis in the body. The blood-clotting process also requires a dozen other proteins that do not need vitamin K for their synthesis.

Causes and symptoms

Newborns are especially prone to vitamin K deficiency. A nursing-mother's milk is low in the vitamin; breast milk can supply only about 20% of the infant's requirement. Infants are born with low levels of

KEY TERMS

Fat-soluble vitamin—Fat-soluble vitamins can be dissolved in oil or in melted fat.

Hemorrhage—Bleeding that continues for an abnormally long period of time.

Phylloquinone—An alternate name for vitamin K₁.

Prothrombin—Prothrombin is a blood-clotting protein. Injury to a blood vessel produces a signal which triggers the conversion of prothrombin to thrombin. Thrombin is a protein which plays a central role in provoking the assembly of other proteins to form the blood clot.

Recommended Dietary Allowance (RDA)—The Recommended Dietary Allowances (RDAs) are quantities of nutrients in the diet that are required to maintain good health in people. RDAs are established by the Food and Nutrition Board of the National Academy of Sciences, and may be revised every few years. A separate RDA value exists for each nutrient. The RDA values refer to the amount of nutrient expected to maintain good health in people. The actual amounts of each nutrient required to maintain good health in specific individuals differ from person to person.

Water-soluble vitamins—Water-soluble vitamins can be dissolved in water or juice.

vitamin K in their body; they do not have any vitamin K-producing bacteria in their intestines. Their digestive tracts are sterile. As a result, a form of vitamin K deficiency, called hemorrhagic disease of the newborn, may develop. This disease involves spontaneous bleeding beneath the skin or elsewhere in the infant's body, and occurs in about 1% of all infants. In rare cases, it causes death due to spontaneous bleeding in the brain.

Vitamin K deficiency in adults is rare. When it occurs, it is found in people with diseases that prevent the absorption of fat. These diseases include **cystic fibrosis**, **celiac disease**, and **cholestasis**. Vitamin K deficiency can exist in adults treated with antibiotics that kill the bacteria that normally live in the digestive tract. As mentioned, the intestine-bacteria supply part of our daily requirement of vitamin K. Vitamin K deficiency can result in bleeding gums, and in skin that is easily bruised.

Others who may benefit from supplemental vitamin K include those taking medications that interact with it or deplete the supply. It also appears to have

some effectiveness in preventing **osteoporosis**, but some studies done involved patients using a high dietary intake of the vitamin rather than supplements. In 2003, however, a group of Japanese researchers reported that supplemental doses of vitamin K₂ given together with vitamin D₃ appeared to reduce bone turnover and sustain bone density in postmenopausal women with mild osteoporosis.

Chronically low levels of vitamin K are correlated with higher risk of hip fracture in older men and women. A study done in 2003 reported that the current recommended dietary intake for vitamin K in adults may not be adequate for older women.

Diagnosis

Vitamin K status is measured by the **prothrombin time** test. The normal prothrombin time is about 13 seconds. With vitamin K deficiency, the prothrombin time can be several minutes. The test involves taking a sample of blood, placing it in a machine called a fibrometer, and measuring the time it takes for blood-clot formation. Blood-clotting problems can also be caused by a rare genetic disease called **hemophilia**. Hemophilia is not related to vitamin K deficiency. Once vitamin K deficiency is suspected, further tests must be used to distinguish it from possible hemophilia. Where a bleeding disorder can be corrected by vitamin K treatment, the diagnosis of vitamin K deficiency is proven to be correct.

Treatment

Vitamin K deficiency in newborn infants is treated and prevented with a single injection of phylloquinone (5 mg). Adults with vitamin K deficiency are treated with daily oral doses of 10 mg phylloquinone for one week.

Prognosis

The prognosis for correcting vitamin K deficiency, and associated blood-clotting problems, is excellent.

Prevention

Aside from newborns and young infants, vitamin K deficiency is not a concern for the general population. Vitamin K deficiency can be prevented by assuring that the diet contains such foods as spinach, cabbage, brussels sprouts, and eggs. Soybean oil, canola oil, and olive oil are good sources of the vitamin, while corn oil and peanut oil are very poor

sources. Elderly people, especially those living alone, should be checked for adequate intake of the vitamin.

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ORGANIZATIONS

American Academy of Pediatrics (AAP). 141 Northwest Point Boulevard, Elk Grove Village, IL 60007. (847) 434-4000. <<http://www.aap.org>>.

American Society for Clinical Nutrition. 9650 Rockville Pike, Bethesda, MD 20814. (301) 530-7110. <<http://www.faseb.org/ascn>>.

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Vitamin poisoning see **Vitamin toxicity**

Vitamin tests

Definition

Vitamin tests measure the levels of certain **vitamins** in an individual's blood. They are generally used to aid in the diagnosis of vitamin deficiencies or in detecting toxic amounts of a vitamin in a patient's system.

Purpose

Vitamins are components of food that are needed for growth, reproduction, and maintaining good health. The vitamins include vitamin D, vitamin E, vitamin A, and vitamin K, which are the fat-soluble vitamins, and folate, vitamin B₁₂, biotin, vitamin B₆,

niacin, thiamin, riboflavin, pantothenic acid, and ascorbic acid, which are the water-soluble vitamins. Vitamins are required in the diet in only tiny amounts, in contrast to the energy components of the diet, such as sugars, starches, and fats. However, not receiving sufficient quantities of a certain vitamin can be devastating, resulting in vitamin deficiency diseases such as **scurvy**, pellagra, or **rickets**. Conversely, consuming too much of a certain vitamin can be toxic to a person's system. Vitamin tests are used to assess the level of certain vitamins in an individual's blood so that doctors can more accurately diagnose vitamin deficiency diseases or vitamin overdoses and devise effective therapy. The vitamins that are most commonly measured by doctors are folate, vitamin B₁₂, vitamin K, vitamin D, and vitamin A.

Description

Most of the vitamin tests are conducted by acquiring a sample of blood, and then preparing plasma or serum from the blood sample. Each vitamin occurs at extremely small concentrations when compared to levels of most other molecules in the blood. Blood contains a great number of chemicals and molecules, and many of these tend to interfere with the vitamin tests. For this reason, a procedure that separates the vitamin from contaminating substances is usually performed immediately prior to conducting the actual test. Most laboratories use high pressure liquid chromatography (HPLC), also called high performance liquid chromatography, as this purification step. In HPLC, the sample is pumped at high pressure through a tube lined with an absorbent material, to which the different molecules cling at different rates. Following separation or purification by HPLC, the vitamin is detected by a color reaction or fluorescence reaction. In these reactions, the amount of color or fluorescence that is formed is proportional to the amount of vitamin in the sample, allowing the analyst to calculate the amount of vitamin present in the original sample. In the case of some vitamins, the purified vitamin is reacted with a special chemical (reagent) prior to detection.

Levels of some vitamins may be measured indirectly by a biological test that mimics the actual function of the vitamin in the body. Riboflavin status is often measured by a test in which the rate by which a certain enzyme converts one molecule into another indicates how much Vitamin B₂ is present in a person's blood. Vitamin K is often measured by a test that times how long it takes for a spontaneous blood clot to form in a prepared sample. Vitamin E status is often measured by placing the red blood cells in a test tube,

adding hydrogen peroxide, and assessing the resulting breakdown of the red blood cells. When a **vitamin E deficiency** exists, the red blood cells have a greater tendency to break.

Preparation

Most vitamin tests require no preparation; however, some may require that the patient fast for at least eight hours before giving a blood sample, or stop using some medications.

Normal results

The values that are considered to be normal for each vitamin can vary slightly. This variability can arise from different testing machines or from different types of chemistry that are used in conducting the vitamin assays. In interpreting data on plasma vitamin levels, it should also be noted that different normal ranges may exist for different age groups and genders. For example, the normal range for plasma vitamin B₆ for males is 7-52 nanograms per milliliter (ng/mL) for males and 2-26 ng/mL for females.

The normal ranges for levels of certain vitamins are as follows. Please note that, by convention, the units referring to the levels of each of the vitamins may differ from each other. The units picogram/milliliter (pg/mL), nanogram/milliliter (ng/mL), and micrograms per deciliter (micrograms/dL) refer to the weight of vitamin in the specified volume. The units nanomoles/liter (nmol/L) and micromoles/liter (M/L) refer to the concentration of vitamin in the specified volume.

- folate (**folic acid**): 3.1-18.0 ng/mL
- vitamin B₁₂: 200-1100 pg/mL
- thiamin: 9-44 nmol/L
- riboflavin: 6.2-39 nmol/L
- vitamin B₆: 7-52 ng/mL
- vitamin C (ascorbic acid): 28-84 M/L
- vitamin A: 28-94 micrograms/dL
- vitamin D (25-hydroxy-vitamin D): 25-50 ng/mL
- vitamin K: 80-1160 pg/mL

Abnormal results

In all cases, abnormal results fall below or above the normal concentration range. However, as noted above, values that are considered to be borderline or severely abnormal can differ according to the discretion of the medical laboratory or physician.

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Vitamin toxicity

Definition

Vitamin toxicity is a condition in which a person develops symptoms as side effects from taking massive doses of **vitamins**. Vitamins vary in the amounts that are required to cause toxicity and in the specific symptoms that result. Vitamin toxicity, which is also called hypervitaminosis or vitamin **poisoning**, is becoming more common in developed countries because of the popularity of vitamin supplements. Many people treat themselves for minor illnesses with large doses (mega-doses) of vitamins.

Description

Overview

Vitamins are organic molecules in food that are needed in small amounts for growth, reproduction, and the maintenance of good health. Some vitamins can be dissolved in oil or melted fat. These fat-soluble vitamins include vitamin D, vitamin E, vitamin A (retinol), and vitamin K. Other vitamins can be dissolved in water. These water-soluble vitamins include folate (**folic acid**), vitamin B₁₂, biotin, vitamin B₆, niacin, thiamin, riboflavin, pantothenic acid, and vitamin C (ascorbic acid). Taking too much of any vitamin can produce a toxic effect. Vitamin A and vitamin D are the most likely to produce hypervitaminosis in large doses, while riboflavin, pantothenic acid, biotin, and vitamin C appear to be the least likely to cause problems.

Vitamins in medical treatment

Vitamin supplements are used for the treatment of various diseases or for reducing the risk of certain diseases. For example, moderate supplements of folic acid appear to reduce the risk for certain **birth defects** (neural tube defects), and possibly reduce the risk of **cancer**. Therapy for diseases brings with it the risk for irreversible vitamin toxicity only in the case of vitamin D. This vitamin is toxic at levels which are only moderately greater than the recommended

dietary allowance (RDA). Niacin is commonly used as a drug for the treatment of heart disease. Niacin is far less toxic than vitamin D. Vitamin toxicity is not a risk with medically supervised therapy using any of the other vitamins.

Vitamin megadoses

With the exception of folic acid supplements, the practice of taking vitamin supplements by healthy individuals has little or no relation to good health. Most adults in the United States can obtain enough vitamins by eating a well-balanced diet. It has, however, become increasingly common for people to take vitamins at levels far greater than the RDA. These high levels are sometimes called vitamin megadoses. Megadoses are harmless for most vitamins. But in the cases of a few of the vitamins—specifically vitamin D, vitamin A, and vitamin B₆—megadoses can be harmful or fatal. Researchers have also started to look more closely at megadoses of vitamin C and of vitamin E, since indirect evidence suggests that these two vitamins may reduce the risks of cancer, heart disease, and **aging**. It is not yet clear whether megadoses of either of these vitamins has any influence on health. Some experts think that megadoses of vitamin C may protect people from cancer. On the other hand, other researchers have gathered indirect evidence that vitamin C megadoses may cause cancer.

Causes and symptoms

Fat-soluble vitamins

VITAMIN D. Vitamin D and vitamin A are the most toxic of the fat-soluble vitamins. The symptoms of vitamin D toxicity are **nausea**, **vomiting**, **pain** in the joints, and loss of appetite. The patient may experience **constipation** alternating with **diarrhea**, or have **tingling** sensations in the mouth. The toxic dose of vitamin D depends on its frequency. In infants, a single dose of 15 mg or greater may be toxic, but it is also the case that daily doses of 1.0 mg over a prolonged period may be toxic. In adults, a daily dose of 1.0-2.0 mg of vitamin D is toxic when consumed for a prolonged period. A single dose of about 50 mg or greater is toxic for adults. The immediate effect of an overdose of vitamin D is abdominal cramps, **nausea and vomiting**. Toxic doses of vitamin D taken over a prolonged period of time result in irreversible deposits of calcium crystals in the soft tissues of the body that may damage the heart, lungs, and kidneys.

VITAMIN A. Vitamin A toxicity can occur with long-term consumption of 20 mg of retinol or more

per day. The symptoms of vitamin A overdosing include accumulation of water in the brain (hydrocephalus), vomiting, tiredness, constipation, bone pain, and severe headaches. The skin may acquire a rough and dry appearance, with hair loss and brittle nails. Vitamin A toxicity is a special issue during pregnancy. Expectant mothers who take 10 mg vitamin A or more on a daily basis may have an infant with birth defects. These birth defects include abnormalities of the face, nervous system, heart, and thymus gland. It is possible to take in toxic levels of vitamin A by eating large quantities of certain foods. For example, about 30 grams of beef liver, 500 grams of eggs, or 2,500 grams of mackerel would supply 10 mg of retinol. The livers of polar bears and other arctic animals may contain especially high levels of vitamin A.

VITAMIN E. Megadoses of vitamin E may produce headaches, tiredness, double vision, and diarrhea in humans. Studies with animals fed large doses of vitamin E have revealed that this vitamin may interfere with the absorption of other fat-soluble vitamins. The term absorption means the transfer of the vitamin from the gut into the bloodstream. Thus, large doses of vitamin E consumed over many weeks or months might result in deficiencies of vitamin D, vitamin A, and vitamin K.

VITAMIN K. Prolonged consumption of megadoses of vitamin K (menadione) results in anemia, which is a reduced level of red blood cells in the bloodstream. When large doses of menadione are given to infants, they result in the deposit of pigments in the brain, nerve damage, the destruction of red blood cells (hemolysis), and **death**. A daily injection of 10 mg of menadione into an infant for three days can kill the child. This tragic fact was discovered during the early days of vitamin research, when newborn infants were injected with menadione to prevent a disease known as hemorrhagic disease of the newborn. Today a different form of vitamin K is used to protect infants against this disease.

Water-soluble vitamins

FOLATE. Folate occurs in various forms in food. There are over a dozen related forms of folate. The folate in oral vitamin supplements occurs in only one form, however—folic acid. Large doses of folic acid (20 grams/day) can result in eventual kidney damage. Folate is considered, however, to be relatively non-toxic, except in cases where folate supplementation can lead to **pernicious anemia**.

VITAMIN B₁₂. Vitamin B₁₂ is important in the treatment of pernicious anemia. Pernicious anemia is more common among middle-aged and older adults; it

is usually detected in patients between the ages of 40 and 80. The disease affects about 0.1% of all persons in the general population in the United States, and about 3% of the elderly population. Pernicious anemia is treated with large doses of vitamin B₁₂. Typically, 0.1 mg of the vitamin is injected each week until the symptoms of pernicious anemia disappear. The patient then takes oral doses of vitamin B₁₂ for the rest of his or her life. Although vitamin B₁₂ toxicity is not an issue for patients being treated for pernicious anemia, treatment of these patients with folic acid may cause problems. Specifically, pernicious anemia is often first detected because the patient feels weak or tired. If the anemia is not treated, the patient may suffer irreversible nerve damage. The problem with folic acid supplements is that the folic acid treatment prevents the anemia from developing, but allows the eventual nerve damage to occur.

VITAMIN B₆. Vitamin B₆ is clearly toxic at doses about 1000 times the RDA. Daily doses of 2-5 grams of one specific form of this vitamin can produce difficulty in walking and tingling sensations in the legs and soles of the feet. Continued megadoses of vitamin B₆ result in further unsteadiness, difficulty in handling small objects, and numbness in the hands. When the high doses are stopped, recovery begins after two months. Complete recovery may take two to three years.

VITAMIN C. The RDA for vitamin C in adults is 60 mg per day. Large doses of vitamin C are considered to be toxic in persons with a family history of or tendency to form **kidney stones** or gallbladder stones. Kidney and gallbladder stones usually consist of calcium oxalate. Oxalate occurs in high concentrations in foods such as cocoa, chocolate, rhubarb, and spinach. A fraction of the vitamin C in the body is normally broken down in the body to produce oxalate. A daily supplement of 3.0 grams of vitamin C has been found to double the level of oxalate that passes through the kidneys and is excreted into the urine.

NIACIN. The RDA for niacin is 15-19 mg per day in adults. Niacin comes in two forms, nicotinic acid and nicotinamide. Either form can satisfy the adult requirement for this vitamin. Nicotinic acid, however, is toxic at levels of 100 times the RDA. It can cause flushing of the skin, nausea, diarrhea, and liver damage. Flushing is an increase in blood passing through the veins in the skin, due to the dilation of arteries passing through deeper parts of the face or other parts of the body. In spite of the side effects, however, large doses of nicotinic acid are often used to lower blood cholesterol in order to prevent heart disease. Nicotinic acid results in a lowering of LDL-cholesterol ("bad cholesterol"), an increase in HDL-

cholesterol ("good cholesterol"), and a decrease in plasma triglycerides. Treatment involves daily doses of 1.5-4.0 grams of nicotinic acid per day. Flushing of the skin occurs as a side effect when nicotinic acid therapy is started, but may disappear with continued therapy.

Diagnosis

The diagnosis of vitamin toxicity is usually made on the basis of the patient's dietary or medical history. Questioning the patient about the use of vitamin supplements may shed light on some of his or her physical symptoms. With some vitamins, the doctor can confirm the diagnosis by ordering blood or urine tests for specific vitamins. When large amounts of the water-soluble vitamins are consumed, a large fraction of the vitamin is absorbed into the bloodstream and promptly excreted into the urine. The fat-soluble vitamins are more likely to be absorbed into the bloodstream and deposited in the fat and other tissues. In the cases of both water-soluble and fat-soluble vitamins, any vitamin not absorbed by the intestines is excreted in the feces. Megadoses of many of the vitamins produce diarrhea, because the non-absorbed nutrient draws water out of the body and into the gut, resulting in the loss of this water from the body.

Treatment

In all cases, treatment of vitamin toxicity requires discontinuing vitamin supplements. Vitamin D toxicity needs additional action to reduce the calcium levels in the bloodstream because it can cause abnormally high levels of plasma calcium (**hypercalcemia**). Severe hypercalcemia is a medical emergency and may be treated by infusing a solution of 0.9% sodium chloride into the patient's bloodstream. The infusion consists of two to three liters of salt water given over a period of one to two days.

Prognosis

The prognosis for reversing vitamin toxicity is excellent for most patients. Side effects usually go away as soon as overdoses are stopped. The exceptions are severe vitamin D toxicity, severe vitamin A toxicity, and severe vitamin B₆ toxicity. Too much vitamin D leads to deposits of calcium salts in the soft tissue of the body, which cannot be reversed. Birth defects due to vitamin A toxicity cannot be reversed. Damage to the nervous system caused by megadoses of vitamin B₆ can be reversed, but complete reversal may require a recovery period of over a year.

KEY TERMS

Absorption—The transfer of a vitamin from the digestive tract to the bloodstream.

Ascorbic acid—Another name for vitamin C.

Hypercalcemia—Hypercalcemia is a condition marked by abnormally high levels of calcium in the blood. It is an issue during vitamin D toxicity.

Hypervitaminosis—Another name for vitamin toxicity.

Megadose—A very large dose of a vitamin, taken by some people as a form of self-medication.

Menadione—A synthetic form of vitamin K. It is sometimes called vitamin K₃.

Recommended Dietary Allowance (RDA)—The recommended dietary allowances (RDAs) are the quantities of nutrients in the diet that are needed for good health. RDAs are established by the Food and Nutrition Board of the National Academy of Sciences and may be revised every few years.

Retinol—Another name for vitamin A.

Prevention

Vitamin toxicity can be prevented by minimizing the use of vitamin supplements. If vitamin D supplements are being used on a doctor's orders, vitamin toxicity can be prevented by monitoring the levels of plasma calcium. The development of hypercalcemia with vitamin D treatment indicates that the patient is at risk for vitamin D toxicity.

Resources

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Tom Brody, PhD

Vitamins

Definition

Vitamins are organic components in food that are needed in very small amounts for growth and for maintaining good health. The vitamins include vitamin D, vitamin E, vitamin A, and vitamin K, or the

fat-soluble vitamins, and folate (folic acid), vitamin B₁₂, biotin, vitamin B₆, niacin, thiamin, riboflavin, pantothenic acid, and vitamin C (ascorbic acid), or the water-soluble vitamins. Vitamins are required in the diet in only tiny amounts, in contrast to the energy components of the diet. The energy components of the diet are sugars, starches, fats, and oils, and these occur in relatively large amounts in the diet.

Most of the vitamins are closely associated with a corresponding vitamin deficiency disease. **Vitamin D deficiency** leads to diseases of the bones such as **osteoporosis** and **rickets**. **Vitamin E deficiency** occurs only rarely, and causes nerve damage. **Vitamin A deficiency** is common throughout the poorer parts of the world, and causes night blindness. Severe vitamin A deficiency can result in xerophthalmia, a disease which, if left untreated, results in total blindness. **Vitamin K deficiency** results in spontaneous bleeding. Mild or moderate folate deficiency is common throughout the world, and can result from the failure to eat green, leafy vegetables or fruits and fruit juices. Folate deficiency causes megaloblastic anemia, which is characterized by the presence of large abnormal cells called megaloblasts in the circulating blood. The symptoms of megaloblastic anemia are tiredness and weakness. Vitamin B₁₂ deficiency occurs with the failure to consume meat, milk or other dairy products. Vitamin B₁₂ deficiency causes megaloblastic anemia and, if severe enough, can result in irreversible nerve damage. Niacin deficiency results in **pellagra**. Pellagra involves skin **rashes** and scabs, **diarrhea**, and mental depression. Thiamin deficiency results in **beriberi**, a disease that can cause atrophy, weakness of the legs, nerve damage, and **heart failure**. Vitamin C deficiency results in scurvy, a disease that involves bleeding. Specific diseases uniquely associated with deficiencies in vitamin B₆, riboflavin, or pantothenic acid have not been found in humans, though persons who have been starving, or consuming poor **diets** for several months, might be expected to be deficient in most of the nutrients, including vitamin B₆, riboflavin, and pantothenic acid.

Some of the vitamins serve only one function in the body, while other vitamins serve a variety of unrelated functions. Therefore, some vitamin deficiencies tend to result in one type of defect, while other deficiencies result in a variety of problems.

Purpose

People are treated with vitamins for three reasons. The primary reason is to relieve a vitamin deficiency, when one has been detected. Chemical tests

Essential Vitamins	
Vitamin	What It Does For The Body
Vitamin A (Beta Carotene)	Promotes growth and repair of body tissues; reduces susceptibility to infections; aids in bone and teeth formation; maintains smooth skin
Vitamin B-1 (Thiamin)	Promotes growth and muscle tone; aids in the proper functioning of the muscles, heart, and nervous system; assists in digestion of carbohydrates
Vitamin B-2 (Riboflavin)	Maintains good vision and healthy skin, hair, and nails; assists in formation of antibodies and red blood cells; aids in carbohydrate, fat, and protein metabolism
Vitamin B-3 (Niacinamide)	Reduces cholesterol levels in the blood; maintains healthy skin, tongue, and digestive system; improves blood circulation; increases energy
Vitamin B-5	Fortifies white blood cells; helps the body's resistance to stress; builds cells
Vitamin B-6 (Pyridoxine)	Aids in the synthesis and breakdown of amino acids and the metabolism of fats and carbohydrates; supports the central nervous system; maintains healthy skin
Vitamin B-12 (Cobalamin)	Promotes growth in children; prevents anemia by regenerating red blood cells; aids in the metabolism of carbohydrates, fats, and proteins; maintains healthy nervous system
Biotin	Aids in the metabolism of proteins and fats; promotes healthy skin
Choline	Helps the liver eliminate toxins
Folic Acid (Folate, Folacin)	Promotes the growth and reproduction of body cells; aids in the formation of red blood cells and bone marrow
Vitamin C (Ascorbic Acid)	One of the major antioxidants; essential for healthy teeth, gums, and bones; helps to heal wounds, fractures, and scar tissue; builds resistance to infections; assists in the prevention and treatment of the common cold; prevents scurvy
Vitamin D	Improves the absorption of calcium and phosphorus (essential in the formation of healthy bones and teeth) maintains nervous system
Vitamin E	A major antioxidant; supplies oxygen to blood; provides nourishment to cells; prevents blood clots; slows cellular aging
Vitamin K (Menadione)	Prevents internal bleeding; reduces heavy menstrual flow

suitable for the detection of all vitamin deficiencies are available. The diagnosis of vitamin deficiency often is aided by visual tests, such as the examination of blood cells with a microscope, the x-ray examination of bones, or a visual examination of the eyes or skin.

A second reason for vitamin treatment is to prevent the development of an expected deficiency. Here, vitamins are administered even with no test for possible deficiency. One example is vitamin K treatment of newborn infants to prevent bleeding. Food supplementation is another form of vitamin treatment. The vitamin D added to foods serves the purpose of preventing the deficiency from occurring in persons who may not be exposed much to sunlight and who fail to

consume foods that are fortified with vitamin D, such as milk. Niacin supplementation prevents pellagra, a disease that occurs in people who rely heavily on corn as the main source of food, and who do not eat much meat or milk. In general, the American food supply is fortified with niacin.

A third reason for vitamin treatment is to reduce the risk for diseases that may occur even when vitamin deficiency cannot be detected by chemical tests. One example is folate deficiency. The risk for cardiovascular disease can be slightly reduced for a large fraction of the population by **folate acid** supplements. And the risk for certain **birth defects** can be sharply reduced if certain pregnant women use folate acid supplements.

Vitamin treatment is important during specific diseases where the body's normal processing of a vitamin is impaired. In these cases, high doses of the needed vitamin can force the body to process or utilize it in the normal manner. One example is pernicious anemia, a disease that tends to occur in middle age or old age, and impairs the absorption of vitamin B₁₂. Surveys have revealed that about 0.1% of the general population, and 2–3% of the elderly, may have the disease. If left untreated, pernicious anemia leads to nervous system damage. The disease can easily be treated with large oral daily doses of vitamin B₁₂ (hydroxocobalamin) or with monthly injections of the vitamin.

Vitamin supplements are widely available as over-the-counter products. But whether they work to prevent or curtail certain illnesses, particularly in people with a balanced diet, is a matter of debate and ongoing research. For example, vitamin C is not proven to prevent the **common cold**. Yet, millions of people take it for that reason. A physician or pharmacist can provide more information on the appropriate use of multivitamin supplements. Likewise, though vitamin supplements have been touted as a prevention for **cancer**, a 2004 report by the U.S. Preventive Services Task Force concluded that the evidence is inadequate to recommend supplementation of vitamins A, C, or E, multivitamins with folate acid, or antioxidant combinations to decrease the risk of cancer.

Precautions

Vitamin A and vitamin D can be toxic in high doses. Side effects range from **dizziness** to kidney failure. A physician or pharmacist can help with the correct use of a multivitamin supplement that contains these vitamins.

KEY TERMS

Genetic disease—A genetic disease is a disease that is passed from one generation to the next, but does not necessarily appear in each generation. An example of genetic disease is Down's syndrome.

Plasma—Blood consists of red and white cells, as well as other components, that float in a liquid. This liquid is called plasma.

Recommended dietary allowance (RDA)—The Recommended Dietary Allowances (RDAs) are quantities of nutrients of the diet that are required to maintain human health. RDAs are established by the Food and Nutrition Board of the National Academy of Sciences and may be revised every few years. A separate RDA value exists for each nutrient. The RDA values refer to the amount of nutrient expected to maintain health in the greatest number of people.

Serum—Serum is blood plasma with the blood clotting proteins removed. Serum is prepared by removing blood from the subject, allowing the blood naturally to form a blood clot, and then using a centrifuge to remove the red blood cells and the blood clot. The blood clot takes the form of an indistinct clump.

Vitamin status—Vitamin status refers to the state of vitamin sufficiency or deficiency of any person. For example, a test may reveal that a patient's folate status is sufficient, borderline, or severely inadequate.

Description

Vitamin treatment usually is done in three ways: by replacing a poor diet with one that supplies the recommended dietary allowance, by consuming oral supplements, or by injections. Injections are useful for people with diseases that prevent absorption of fat-soluble vitamins. Oral vitamin supplements are especially useful for people who otherwise cannot or will not consume food that is a good vitamin source, such as meat, milk, or other dairy products. For example, a vegetarian who will not consume meat may be encouraged to consume oral supplements of vitamin B₁₂.

Treatment of genetic diseases that impair the absorption or utilization of specific vitamins may

require megadoses of the vitamin throughout one's lifetime. Megadose means a level of about 10-1,000 times greater than the recommended daily allowance (RDA). **Pernicious anemia**, homocystinuria, and biotinidase deficiency are three examples of genetic diseases that are treated with megadoses of vitamins.

Preparation

The diagnosis of a vitamin deficiency usually involves a blood test. An overnight fast usually is recommended as preparation prior to withdrawal of the blood test so that vitamin-fortified foods do not affect the test results.

Aftercare

Response to vitamin treatment can be monitored by chemical tests, by an examination of red blood cells or white blood cells, or by physiological tests, depending on the exact vitamin deficiency.

Risks

Few risks are associated with supervised vitamin treatment. Risks depend on the vitamin and the reason why it was prescribed. Ask a physician or pharmacist about how and when to take vitamin supplements, particularly those that have not been prescribed by a physician.

Resources

PERIODICALS

"The Next Big Deficiency." *Chain Drug Review* February 16, 2004: 26.

Sadovsky, Richard. "Can Vitamins Prevent Cancer and Heart Disease?" *American Family Physician* February 1, 2004: 631.

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Vitiligo

Definition

Vitiligo is a condition in which a loss of cells that give color to the skin (melanocytes) results in smooth, white patches in the midst of normally pigmented skin.



Loss of pigmentation is one characteristic of vitiligo, as seen on this woman's hand. (Custom Medical Stock Photo. Reproduced by permission.)

Description

Vitiligo is a common, often inherited disorder characterized by areas of well-defined, milky white skin. People with vitiligo may have eye abnormalities and also have a higher incidence of thyroid disease, **diabetes mellitus**, and **pernicious anemia**. Vitiligo affects about 1-2% of the world's population. It is more easily observed in sun-exposed areas of the body and in darker skin types, but it affects any area of the body and all races. Vitiligo seems to affect men and women equally, although women more frequently seek treatment for the disorder.

Vitiligo may appear as one or two well-defined white patches or it may appear over large portions of the body. Typical sites for generalized vitiligo are areas surrounding body openings, bony areas, fingers, and toes. It can begin at any age but about 50% of the time it starts before the age of 20.

Causes and symptoms

Vitiligo is a disorder with complex causes. People with vitiligo seem to inherit a genetic predisposition for the disorder, and the appearance of disorder can be brought on by a variety of precipitating causes. Many people report that their vitiligo first appeared following a traumatic or stressful event, such as an accident, job loss, **death** of a family member, severe **sunburn**, or serious illness. There are at least three theories about the underlying mechanism of vitiligo. One theory says nerve endings in the skin release a chemical that is toxic to the melanocytes. A second theory states that the melanocytes simply self-destruct. The third explanation is that vitiligo is a type of autoimmune disease

KEY TERMS

Autoimmune disease—A condition in which something triggers the immune system to react against and attack the body's own tissues.

Autologous transplantation—A procedure wherein the person donates blood or tissue to themselves.

Iris—The colored part of the eye.

Pernicious anemia—A disease in which red blood cells are abnormally formed due to the body's inability to absorb vitamin B₁₂.

Retina—The innermost layer of the eye, it contains the rods and cones, specialized light-sensitive cells.

in which the immune system targets the body's own cells and tissues.

The primary symptom of vitiligo is the loss of skin color. Hair growing from the affected skin areas also lacks color. In addition, people with vitiligo may have pigment abnormalities of the retina or iris of the eyes. A minority of patients also may have inflammation of the retina or iris, but vision is not usually impaired.

Diagnosis

The diagnosis of vitiligo is usually made by observation. Progressive, white areas found at typical sites point to a diagnosis of vitiligo. If the diagnosis is not certain, the doctor will test for other conditions which can mimic vitiligo, such as chemical leukoderma or **systemic lupus erythematosus**. If the tests rule out other conditions, vitiligo is confirmed.

Treatment

Vitiligo cannot be cured, but it can be managed. Cosmetics can be used to improve the appearance of the white areas not covered by clothing. Sunscreens prevent burning of the affected areas and also prevent the normal skin around the patches from becoming darker. Skin creams and oral medications are available for severe cases, but they have side effects that may make them undesirable. Autologous transplantation of skin is an option for those who are severely affected. Bleaching or depigmentation of the normal skin is another option.

In addition to treating the skin, attention should be paid to the psychological well-being of the individual. Extreme cases of vitiligo can be unattractive and may affect a person's outlook and social interactions.

Prognosis

The condition is usually gradually progressive. Sometimes the patches grow rapidly over a short period, and then the condition remains stable for many years.

Prevention

No measures are currently known to prevent vitiligo.

Resources

ORGANIZATIONS

Frontier's International Vitiligo Foundation. 4 Rozina Court, Owings Mills, MD 21117. (301) 594-0958.

National Foundation for Vitiligo and Pigment Disorders. 9032 South Normandy Drive, Centerville, OH 45459. (513) 885-5739.

National Vitiligo Foundation. P.O. Box 6337, Tyler, TX 75703. (903) 531-9767.
< 73071.33@compuserve.com >.

Dorothy Elinor Stonely

Vitrectomy

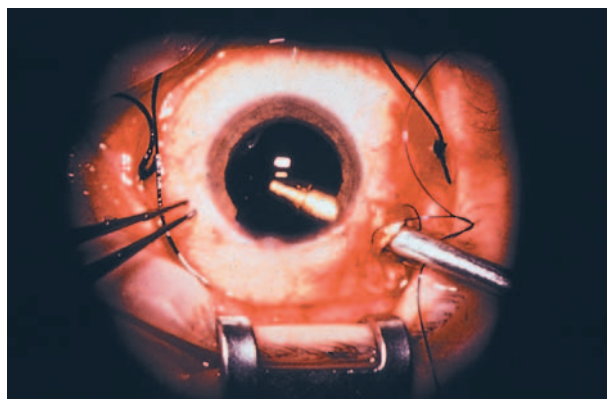
Definition

Vitrectomy is the surgical removal of the vitreous (transparent gel that fills the eye from the iris to the retina).

Purpose

The bulk of the contents of the eyeball is a clear jelly-like substance that is susceptible to several afflictions that impair vision by damaging its transparency.

- Infections
- Injuries
- Bleeding, particularly from diabetic retinopathy
- Blood vessels growing into the vitreous, again due to diabetes.



Vitrectomy is a surgical procedure in which the vitreous, the transparent gel that fills the eye from the iris to the retina, is removed. During this procedure, the surgeon penetrates the eyeball with a tiny instrument (shown above), which liquefies the vitreous and suctions it out of the eye. (Photo Researchers, Inc. Reproduced by permission.)

The retina is the light-sensitive membrane that receives images and transmits them to the brain. It covers the inside of the back of the eye. On occasion the retina will fall into the vitreous, a condition called **retinal detachment**. This may be due to disease in the vitreous that pulls the retina inward, small tears in the retina that allow liquid to seep behind it and push it forward, or injury to the eye that simply breaks the retina loose. It may be necessary to remove the vitreous in order to replace the retina and restore vision.

Description

Using instruments suited for microscopic surgery, the ophthalmologist (eye surgeon) penetrates the eyeball, aspirates the vitreous, and replaces it with saline. The saline replaces the vitreous at a constant pressure in order to keep the eye from collapsing. Once the saline is in place, both eyes are patched. The procedure takes two to three hours to complete.

Preparation

Because this is a major operation on the eye, the surgeon will perform a very extensive evaluation of both eyes. After looking inside with a variety of lenses, a CT, MRI, or ultrasound study may be needed. Immediately prior to the vitrectomy, the pupils will be dilated.

Aftercare

Eye drops and **antibiotics** are administered, and eye rest is advised until healing is completed.

KEY TERMS

Computed tomography (CT scan)—Computerized method of creating images of internal organs using x rays.

Diabetic retinopathy—Disease that damages the blood vessels in the back of the eye caused by diabetes mellitus.

Endophthalmitis—Inflammation of the eyeball.

Iatrogenic—Inadvertently caused by medical treatment.

Magnetic resonance imaging (MRI)—Computerized method of creating images of internal organs using magnetic fields.

Saline—A salt solution equivalent to that in the body—0.9% salt in water.

Risks

Risks associated with vitrectomy are retinal detachment, bleeding, iatrogenic (medically caused) **cataracts**, and endophthalmitis (inflammation of the eyeball).

Normal results

Vision is restored to useful levels in two-thirds of patients.

Resources

BOOKS

O'Malley, Conor. "Vitreous." In *General Ophthalmology*, edited by Daniel Vaughan, 13th ed. Stamford: Appleton & Lange, 1993.

J. Ricker Polsdorfer, MD

Vivax malaria see **Malaria**

Vocal cord nodules and polyps

Definition

Vocal cord nodules and polyps are noncancerous growths on the vocal cords that affect the voice.

Description

The vocal cords, located in the voice box in the middle of the neck, are two tough, fibrous bands that vibrate to produce sound. They are covered with a layer of tissue that is similar to skin. With use, this layer thickens. With heavy use, the thickening may localize, producing a nodule. Unlike skin, heavy usage over a short time may also produce polyps. A polyp is a soft, smooth lump containing mostly blood and blood vessels. A nodule is similar to a polyp, but tends to be firmer.

Causes and symptoms

Chronic infections caused by **allergies** and inhalation of irritants, such as cigarette smoke, may produce these lesions, but extensive use of the voice is the most common cause of vocal nodules and polyps. Nodules and polyps are more common in male children, female adolescents, and female adults. This may be due in part to the faster speed at which the cords vibrate to produce higher-pitched voices.

Voice alterations are most apparent in singers, who may notice the higher registers are the first to change. Hoarseness causes others to seek medical attention.

Diagnosis

The head and neck surgeon (otorhinolaryngologist) must see the vocal cords to diagnose these lesions. It is also important to confirm that there are not other problems instead of or in addition to these benign lumps. Other causes of hoarseness include throat cancers, **vocal cord paralysis**, and simple **laryngitis**. The cords can usually be seen using a mirror placed at the back of the tongue. More elaborate scopes, including a videostroboscope, allow better views while the cords are producing sounds.

A biopsy of a nodule or polyp will ensure they are not cancerous.

Treatment

Nodules usually only require voice therapy; less than 5% of nodules require surgery. Small polyps can be treated with voice therapy, but typically they are surgically removed.

Prognosis

Continued overuse of the voice will cause these lesions to regrow.

KEY TERMS

Laryngitis—Inflammation of the larynx (voice box).

Lesion—A wound or injury.

Otorhinolaryngologist—A physician specializing in ear, nose, and throat diseases. Also known as otolaryngologist.

Prevention

Careful use of the voice will prevent most vocal cord nodules and polyps. Avoiding inhaled irritants, may also prevent nodules and polyps from forming.

Resources

BOOKS

Ballenger, John Jacob. *Disorders of the Nose, Throat, Ear, Head, and Neck*. Philadelphia: Lea & Febiger, 1991.

J. Ricker Polsdorfer, MD

Vocal cord paralysis

Definition

Vocal cord **paralysis** is the inability to move the vocal cords and the resulting loss of vocal cord function.

Description

The vocal cords are a pair of tough, fibrous bands that lie across the air column in the middle of the voice box. They assist three functions: breathing, swallowing, and speaking. When vocal cords vibrate, they produce sound, allowing us to speak. Vocal cords temporarily stop breathing to aid coughing and for expelling **foreign objects**. During swallowing, the vocal cords shut the airway so that food is not inhaled. When vocal cords are paralyzed, all three functions are affected.

The relaxed position of the vocal cords is halfway open. There is one set of muscles that closes them all the way and one set that opens them. Each set of muscles is controlled by a different nerve. Each nerve comes from a different direction—one from above and one from below (the recurrent laryngeal nerve). Vocal

cords can either be partially paralyzed on one side or completely paralyzed on both sides.

Causes and symptoms

Vocal cord paralysis can result from injury, tumors, or surgery in the neck and upper chest. Brain tumors and **stroke** can also affect the nerves. Infectious diseases that damage nerves—like **whooping cough**, **tetanus** and polio—can also cause vocal cord paralysis. Vocal cord paralysis can also appear as a congenital defect. If congenital, the most frequent cause is a brain defect, which can often be effectively treated.

The most dangerous form of vocal cord paralysis is one that affects the opening function, controlled by the recurrent laryngeal nerve. If both vocal cords are paralyzed, breathing stops or becomes very labored. Fortunately, injury during trauma or surgery often involves only one side, but the congenital causes can damage both sides.

Vocal cord paralysis produces several symptoms.

- The voice is always affected; at best it is breathy and weak. At worst, it is not there at all. In infants, the cry can be weak. Older children will suppress laughing and coughing because it is hard to do.
- Swallowing may be hindered so that food ends up in the airway, causing violent coughing and often leading to pneumonia.
- Breathing is obstructed on inspiration, producing a condition known as **stridor**. Closing the airway while breathing in produces creaking noises in the throat and changes the shape of the chest. The breast bone is drawn inward, much more visibly in the flexible chest of a small child.

Diagnosis

The voice box must be observed during breathing to characterize the problem. A viewing instrument called a laryngoscope, either flexible or rigid, is passed through the nose or throat until the cords become visible. The motion of each cord can then be seen, and other problems in the area identified.

X rays, CT, or MRI scans of the skull may be done if a brain disorder is suspected.

Treatment

An adequate airway is immediately necessary, usually secured with an endotracheal tube in the windpipe. If a cure cannot be achieved, a permanent

KEY TERMS

Computed tomography (CT scan)—Computerized use of x rays to create images of internal organs.

Laryngoscope—A diagnostic instrument that is used to examine the interior of the larynx.

Magnetic resonance imaging (MRI)—Computerized use of magnetic fields and radio-frequency signals to create images of internal organs.

Recurrent laryngeal nerve—One of two offshoots of the vagus nerve that connect to the larynx. It is located below the larynx.

Stridor—A raspy sound that occurs during respiration when the airways are blocked.

Tracheostomy—Surgical opening in the neck to the trachea to aid respiration.

Voice box—The larynx.

breathing hole (tracheostomy) is cut in the neck. Brain problems that are relieved within 24 hours usually allow the cords to regain their function. Care must be taken to assure that swallowing takes place normally.

Alternative treatment

Vocal cord paralysis can be addressed with constitutional homeopathy. This will work with the whole person, not just the symptoms, to help bring about healing. Botanical medicine and deep tissue massage to the area can also bring some resolution, although it may not be long term.

Resources

BOOKS

Ballenger, John Jacob. *Disorders of the Nose, Throat, Ear, Head, and Neck*. Philadelphia: Lea & Febiger, 1991.

J. Ricker Polsdorfer, MD

Vocal cord polyps see **Vocal cord nodules and polyps**

Voiding cystourethrography see **Retrograde urethrography**

Volvulus see **Intestinal obstructions**

Vomiting see **Nausea and vomiting; Cyclic vomiting syndrome; Hyperemesis gravidarum**

Von Gierke's disease see **Glycogen storage diseases**

Von Recklinghausen disease see **Neurofibromatosis**

Von Willebrand disease

Definition

Von Willebrand disease is caused by a deficiency or an abnormality in a protein called von Willebrand factor and is characterized by prolonged bleeding.

Description

The Finnish physician Erik von Willebrand was the first to describe von Willebrand disease (VWD). In 1926 Dr. von Willebrand noticed that many male and female members of a large family from the Aland Islands had increased bruising (bleeding into the skin) and prolonged episodes of bleeding. The severity of the bleeding varied between family members and ranged from mild to severe and typically involved the mouth, nose, genital and urinary tracts, and occasionally the intestinal tract. Excessive bleeding during the menstrual period (menorrhagia) was also experienced by some of the women in this family. What differentiated this bleeding disorder from classical **hemophilia** was that it appeared not to be associated with muscle and joint bleeding and affected people of either sex rather than just men. Dr. von Willebrand named this disorder *hereditary pseudohemophilia*.

Pseudohemophilia, or von Willebrand disease (VWD) as it is now called, occurs when the body does not produce enough of a protein called von Willebrand factor (vWF) or produces abnormal vWF. vWF is involved in the process of blood clotting (coagulation). Blood clotting is necessary to heal an injury to a blood vessel. When a blood vessel is injured, vWF enables blood cells called platelets to bind to the injured area and form a temporary plug to seal the hole and stop the bleeding. vWF is secreted by platelets and by the cells that line the inner wall of the blood vessels (endothelial cells). The platelets release other chemicals called factors in response to a blood vessel injury that are involved in forming a strong permanent

clot. vWF binds to and stabilizes factor VIII, one of the factors involved in forming the permanent clot.

A deficiency or abnormality in vWF can interfere with the formation of the temporary platelet plug and also affect the normal survival of factor VIII, which can indirectly interfere with the production of the permanent clot. Individuals with VWD, therefore, have difficulty in forming blood clots and as a result they may bleed for longer periods of time. In most cases the bleeding is due to an obvious injury, although it can sometimes occur spontaneously.

VWD is classified into three basic types: type 1, type 2, and type 3. The definitions of each type are based on the amount and type of vWF that is produced. Type 1 is the most common and mildest form and results when the body produces slightly decreased amounts of typically normal vWF. Type 2 can be classified into four subtypes (IIA, IIB, IIM, and IIN) and results when the body produces an abnormal type of vWF. Type 3 is the rarest and most severe form and results when the body does not produce any detectable amount of vWF.

Approximately one out of 100 people are affected with VWD, making it the most common inherited bleeding disorder (hemophilia). VWD affects people of all ethnic backgrounds. Approximately 70–80% of people with VWD have type 1 and close to 20–30% have type 2. Type 3 is very rare and occurs in fewer than one percent of people with VWD.

Causes and symptoms

The genetics of VWD are complex and involve a gene that produces vWF and is found on the short arm of chromosome 12. Since humans inherit two of each type of chromosome, they inherit two vWF genes. There are different types of changes (mutations) in the vWF gene that can affect the production of vWF. Some types of changes can cause the vWF gene to produce decreased amounts of normal vWF, while other changes can cause the gene to produce abnormal vWF. Most of the gene changes are significant enough that a change in only one vWF gene is sufficient to cause VWD. Some gene mutations cause VWD only if both genes are changed, which often leads to more severe symptoms.

Type 1 VWD is called an autosomal dominant condition since it is caused by a change in only one vWF gene. Since type 1 VWD results in only a slight decrease in the amount of vWF produced, the symptoms are often mild and even non-existent in some patients. Most cases of Type 2 VWD are autosomal

dominant since they are caused by a change in only one vWF gene that results in the production of an abnormal protein. An autosomal dominant form of VWD can be inherited from either parent or can occur spontaneously in the embryo that is formed when the egg and sperm cells come together during fertilization.

Some cases of type 2 VWD and all cases of type 3 VWD are autosomal recessive since they are caused by changes in both vWF genes. A person with an autosomal recessive form of VWD has inherited a changed gene from his or her mother and a changed gene from his or her father. Parents who have a child with an autosomal recessive form of VWD are called carriers, since they each possess one changed vWF gene and one unchanged vWF gene. Many carriers for the autosomal recessive forms of type 2 VWD and type 3 VWD do not have any symptoms, although some people with type 3 VWD are born to parents who have type 1 VWD and may have symptoms. Each child born to parents who are both carriers for VWD has a 25% chance of having VWD, a 50% chance of being a carrier, and a 25% chance of being neither a carrier nor affected with VWD disease. A person with an autosomal dominant form of VWD has a 50% chance of passing the changed gene on to his or her children who may or may not have symptoms.

VWD is usually a relatively mild disorder characterized by easy bruising, recurrent nosebleeds, heavy menstrual periods, and extended bleeding after surgeries and invasive dental work. There is a great deal of variability in the severity of symptoms, which can range from clinically insignificant to life-threatening. Even people within the same family who are affected with the same type of VWD may exhibit different symptoms. An individual with VWD may exhibit a range of symptoms over the course of his or her lifetime and may experience an improvement in symptoms with age. The severity of the disease is partially related to the amount and type of vWF that the body produces, but is also influenced by other genetic and non-genetic factors.

Type 1

Type 1, the mildest form of VWD, is usually associated with easy bruising, recurrent nosebleeds, heavy menstrual periods, and prolonged bleeding after surgeries and invasive work. Many people with type 1 VWD do not have any noticeable symptoms or only have prolonged bleeding after surgery or significant trauma. The amount of vWF produced by the body increases during **pregnancy**, so prolonged bleeding during delivery is uncommon in people with type 1 VWD.

Type 2

People with type 2 VWD usually have symptoms from early childhood and symptoms may even be present at birth. They usually experience prolonged bleeding from cuts, easy bruising, nose bleeds, skin hematomas, and prolonged bleeding from the gums following teeth extraction and minor trauma. More than 50% of women with type 2 VWD experience heavy periods that may require a blood **transfusion**. Gastrointestinal bleeding is rare but can be life-threatening. Some women with type 2 VWD exhibit prolonged bleeding during delivery.

Type 3

Type 3 VWD can be quite severe and is associated with bruising and bleeding from the mouth, nose, intestinal, genital and urinary tracts. Type 3 is also associated with spontaneous bleeding into the muscles and joints, which can result in joint deformities. Some women with type 3 VWD experience prolonged bleeding during delivery.

Diagnosis

Diagnostic testing

Many people with VWD have mild symptoms or symptoms that can be confused with other bleeding disorders, making it difficult to diagnose VWD on the basis of clinical symptoms. VWD should be suspected in any person with a normal number of platelets in their blood and bleeding from such mucous membranes as the nose, gums and gastrointestinal tract. Testing for an individual with suspected VWD often includes the measurement of:

- how long it takes for the bleeding to stop after a tiny cut is made in the skin (the **bleeding time**)
- the amount of vWF (vWF antigen measurement)
- the activity of vWF (ristocetin co-factor activity)
- the amount of factor VIII (factor VIII antigen measurement)
- activity of factor VIII

Many doctors routinely screen women with menorrhagia for VWD, as heavy menstrual bleeding is the most common symptom of the disorder in females.

People with type 1 VWD usually have an increased bleeding time but they may have an intermittently normal bleeding time. They also have a decreased amount of vWF, and decreased vWF activity and usually have slightly decreased factor VIII

levels and activity. People with type 2 VWD have a prolonged bleeding time, decreased activity of vWF and may have decreased amounts of vWF and factor VIII, and may have decreased factor VIII activity. Type 3 individuals have undetectable amounts of vWF, negligible vWF activity, factor VIII levels of less than 5–10%, and significantly reduced factor VIII activity. The activity of vWF is reduced for all types of VWD, making it the most sensitive means of identifying all three types of VWD. Patients with borderline results should be tested two to three times over a three month period.

Once a patient is diagnosed with VWD, further testing such as vWF multimer analysis and ristocetin-induced platelet aggregation (RIPA) may need to be performed to determine the subtype. Multimer analysis evaluates the structure of the vWF, and RIPA measures how much ristocetin is required to cause the clumping of platelets in a blood sample. The vWF multimer analysis is able to differentiate people with a structurally normal vWF (type 1) from people with a structurally abnormal vWF (type 2) and is often able to identify the subtype of patients with type 2 VWD. People with type 1 VWD usually have normal to decreased RIPA concentrations. Depending on the subtype, patients with type 2 VWD either have increased or decreased RIPA. RIPA is usually absent and the multimer analysis shows undetectable vWF in people with type 3 VWD.

In some cases DNA testing can be a valuable adjunct to biochemical testing. The detection of gene alteration(s) can confirm a diagnosis and can determine the type and subtype of VWD. It can also help to facilitate prenatal testing and testing of other family members. Unfortunately, as of 2001, many people with VWD possess DNA changes that are not detectable through DNA testing. A person who has a mother, father, or sibling diagnosed with VWD should undergo biochemical testing for VWD. If the relative with VWD possesses a detectable gene change, then DNA testing should also be considered.

Prenatal testing

If one parent has been diagnosed with an autosomal dominant form of VWD or both parents are carriers for an autosomal recessive form of VWD, then prenatal testing can be considered. If the parent with an autosomal dominant form of VWD possesses a detectable gene change or both parents who are carriers for an autosomal recessive form of VWD possess detectable mutations, then DNA testing of their fetus would be available. DNA testing can be performed through amniocentesis or **chorionic villus sampling**.

If the DNA change in the parent(s) is unknown then prenatal testing can sometimes be performed through biochemical testing of blood obtained from the fetal umbilical cord, which is less accurate and is associated with a higher risk of pregnancy loss.

Treatment

VWD is most commonly treated by replacement of vWF through the administration of blood products that contain vWF or through treatment with desmopressin (DDAVP, 1-deamino-8-D-arginine vasopressin). DDAVP functions by increasing the amount of factor VIII and vWF in the bloodstream. Treatment with blood products or DDAVP may be started in response to uncontrollable bleeding or may be administered prior to procedures such as surgeries or dental work. The type of treatment chosen depends on the type of VWD and a patient's response to a preliminary treatment trial.

Treatment with desmopressin

DDAVP is the most common treatment for people with type 1 VWD. About 80% of people with type 1 VWD respond to DDAVP therapy. Treatment with DDAVP can also be used to treat some people with type 2 VWD. Patients with Type 2B VWD should not be treated with this medication since DDAVP can induce dangerous platelet clumping. Type 3 VWD should not be treated with DDAVP since this medication does not increase the level of vWF in type 3 patients. DDAVP should only be used in people who have been shown to be responsive through a pre-treatment trial transfusion with this medication.

DDAVP can be administered intravenously or through a nasal inhaler. DDAVP has relatively few side effects although some people may experience facial flushing, tingling sensations, and headaches after treatment with this medication. Often treatment with this medication is only required prior to invasive surgeries or dental procedures.

Treatment with blood products

Patients who are unable to tolerate or are unresponsive to drug-based treatments are treated with concentrated factor VIII obtained from blood products. Not all factor VIII concentrates can be used since some do not contain enough vWF. The concentrate is treated to kill most viruses, although caution should be used since not all types of viruses are destroyed. If the factor VIII concentrates are unable to manage a severe bleeding episode, then blood

products called cryoprecipitates, which contain concentrated amounts of vWF, or platelet concentrates should be considered. Caution should be used when treating with these blood products since they are not treated to kill viruses.

Other treatments and precautions

Medications called fibrinolytic inhibitors can be helpful in the control of intestinal, mouth, and nose bleeding. Estrogens such as are found in oral contraceptives increase the synthesis of vWF and can sometimes be used in the long-term treatment of women with mild to moderate VWD. Estrogens are also sometimes used prior to surgery in women with type 1 VWD. Some topical agents are available to treat nose and mouth bleeds.

Endometrial ablation, or the removal of the lining of the uterus by means of electrocautery or other thermal methods, is sometimes recommended as a treatment for menorrhagia associated with VWD. This procedure appears to be successful in lowering the amount of bleeding that these women experience during their menstrual periods.

Patients with VWD should avoid taking **aspirin**, ibuprofen, or other NSAIDs, which can increase their susceptibility to bleeding. They should also inform their dentist of their diagnosis, as many routine dental procedures can cause bleeding from the gums. People with severe forms of VWD should avoid activities that increase their risk of injury such as contact sports.

Patients with type 3 VWD living in the United States may wish to contact one of the 146 federally funded Hemophilia Treatment Centers (HTCs) for advice about prophylactic treatment and general follow-up.

Prognosis

The prognosis for VWD disease is generally fairly good and most individuals have a normal lifespan. The prognosis can depend, however on accurate diagnosis and appropriate medical treatment.

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KEY TERMS

Amniocentesis—A procedure performed at 16-18 weeks of pregnancy in which a needle is inserted through a woman's abdomen into her uterus to draw out a small sample of the amniotic fluid from around the baby. Either the fluid itself or cells from the fluid can be used for a variety of tests to obtain information about genetic disorders and other medical conditions in the fetus.

Autosomal dominant—A pattern of genetic inheritance where only one abnormal gene is needed to display the trait or disease.

Autosomal recessive—A pattern of genetic inheritance where two abnormal genes are needed to display the trait or disease.

Biochemical testing—Measuring the amount or activity of a particular enzyme or protein in a sample of blood or urine or other tissue from the body.

Carrier—A person who possesses a gene for an abnormal trait without showing signs of the disorder. The person may pass the abnormal gene on to offspring.

Chorionic villus sampling (CVS)—A procedure used for prenatal diagnosis at 10-12 weeks gestation. Under ultrasound guidance a needle is inserted either through the mother's vagina or abdominal wall and a sample of cells is collected from around the early embryo. These cells are then tested for chromosome abnormalities or other genetic diseases.

Chromosome—A microscopic thread-like structure found within each cell of the body and consists of a complex of proteins and DNA. Humans have 46 chromosomes arranged into 23 pairs. Changes in either the total number of chromosomes or their shape and size (structure) may lead to physical or mental abnormalities.

Deoxyribonucleic acid (DNA)—The genetic material in cells that holds the inherited instructions for growth, development, and cellular functioning.

Desmopressin (DDAVP)—A drug used in the treatment of von Willebrand's disease.

Diagnostic testing—Testing performed to determine if someone is affected with a particular disease.

DNA testing—Analysis of DNA (the genetic component of cells) in order to determine changes in genes that may indicate a specific disorder.

Endothelial cells—The cells lining the inner walls of the blood vessels.

Factor VIII—A protein involved in blood clotting that requires vWF for stability and long-term survival in the bloodstream.

Gene—A building block of inheritance, which contains the instructions for the production of a particular protein, and is made up of a molecular sequence found on a section of DNA. Each gene is found on a precise location on a chromosome.

Menorrhagia—Excessively heavy menstrual flow with cycles of normal length. It is also called hypermenorrhea.

Mutation—A permanent change in the genetic material that may alter a trait or characteristic of an individual, or manifest as disease, and can be transmitted to offspring.

Platelets—Small disc-shaped structures that circulate in the blood stream and participate in blood clotting.

Prenatal testing—Testing for a disease such as a genetic condition in an unborn baby.

Protein—Important building blocks of the body, composed of amino acids, involved in the formation of body structures and controlling the basic functions of the human body.

Skin hematoma—Blood from a broken blood vessel that has accumulated under the skin.

von Willebrand factor (vWF)—A protein found in the blood that is involved in the process of blood clotting.

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American Society of Hematology (ASH). 1900 M Street, NW, Suite 200, Washington, DC 20036. (202) 776-0544. Fax: (202) 776-0545. H <<http://www.hematology.org>>.

National Heart, Lung, and Blood Institute (NHLBI) Health Information Center, P. O. Box 30105, Bethesda, MD 20824-0105. (301) 592-8573. Fax: (240) 629-3246. <<http://www.nhlbi.nih.gov>>.

National Hemophilia Foundation. Soho Building, 110 Greene Street, Suite 406, New York, NY 10012. (212) 219-8180. <<http://www.hemophilia.org/home.htm>>.

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VSD *see* **Ventricular septal defect**

Vulvar cancer

Definition

Vulvar **cancer** refers to an abnormal, cancerous growth in the external female genitalia.

Description

Vulvar cancer is a rare disease that occurs mainly in elderly women. The vulva refers to the external female genitalia, which includes the labia, the opening of the vagina, the clitoris, and the space between the vagina and anus (perineum). There are two pairs of labia (a Latin term meaning lips). The labia meet to protect the openings of the vagina and the tube that connects to the bladder (urethra). The outer, most prominent folds of skin are called labia majora, and the smaller, inner skin folds are called labia minora. Vulvar cancer can affect any part of the female genitalia, but usually affects the labia.

Approximately 70% of vulvar cancers involve the labia (usually the labia majora), 15% to 20% involve the clitoris, and 15% to 20% involve the perineum. For approximately 5% of the cases, the cancer is present at more than one location. For approximately 10% of the cases, so much of the vulva is affected by cancer that the original location cannot be determined. Vulvar cancer can spread to nearby structures including the anus, vagina, and urethra.

Most vulvar cancers are squamous cell carcinomas. Squamous cells are the main cell type of the skin. Squamous cell carcinoma often begins at the edges of the labia majora or labia minora or the area around the vagina. This type of cancer is usually slow-growing and may begin with a precancerous condition referred to as vulvar intraepithelial neoplasia (VIN), or dysplasia. This means that precancerous cells are present in the surface layer of skin.

Other, less common types of vulvar cancer are melanoma, basal cell carcinoma, adenocarcinomas, Paget's disease of the vulva, and tumors of the connective tissue under the skin. Melanoma, a cancer that develops from the cells that produce the pigment that determines the skin's color, can occur anywhere on the skin, including the vulva. Melanoma is the second most common type of vulvar cancer, and accounts for 5% to 10% of the cases. Half of all vulvar melanomas involve the labia majora. Basal cell carcinoma, which is the most common type of cancer that occurs on parts of the skin exposed to the sun, very rarely occurs on the vulva. Adenocarcinomas develop from glands, including the glands at the opening of the vagina (Bartholin's glands) that produce a mucus-like lubricating fluid.

Vulvar cancer is most common in women over 50 years of age. The median age at diagnosis is 65 to 70 years old. Additional risk factors for vulvar cancer include having multiple sexual partners, **cervical cancer**, and the presence of chronic vaginal and vulvar inflammations. This type of cancer is often associated with **sexually transmitted diseases**.

Vulvar cancer is most common in women who are between the ages of 65 and 75 years. In the United States there are approximately 3,000 new cases of vulvar cancer diagnosed each year. Vulvar cancer accounts for only 1% of all cancers in women. Approximately 5% of all gynecologic cancers occur on the vulva. For unknown reasons, the incidence of vulvar cancer seems to be rising.

Causes and symptoms

Cancer is caused when the normal mechanisms that control cell growth become disturbed, causing

the cells to continually grow without stopping. This is usually the result of damage to the DNA in the cell. Although the cause of vulvar cancer is unknown, studies have identified several risk factors for vulvar cancer. These include:

- **Vulvar intraepithelial neoplasia (VIN).** This abnormal growth of the surface cells of the vulva can sometimes progress to cancer.
- **Infection with human papillomavirus (HPV).** This virus is sexually transmitted and can cause **genital warts**. Although HPV DNA can be detected in most cases of vulvar intraepithelial neoplasia, it is detected in fewer than half of all cases of vulvar cancer. Therefore, the link between HPV infection and vulvar cancer is unclear. As of 2001, it is theorized that two classes of vulvar cancer exist: one that is associated with HPV infection and one that is not.
- **Herpes simplex virus 2 (HSV2).** This sexually transmitted virus is also associated with increased risk for vulvar cancer.
- **Cigarette smoking.** Smoking in combination with infection by HPV or HSV2 was found to be a particularly strong risk factor for vulvar cancer.
- **Infection with human immunodeficiency virus (HIV).** This virus, which causes **AIDS**, decreases the body's immune ability, leaving it vulnerable to a variety of diseases, including vulvar cancer.
- **Chronic vulvar inflammation.** Long term irritation and inflammation of the vulva and vagina, which may be caused by poor hygiene, can increase the risk of vulvar cancer.
- **Abnormal Pap smears.** Women who have had abnormal Pap smears are at an increased risk of developing vulvar cancer.
- **Chronic immunosuppression.** Women who have had long-term suppression of their immune system caused by disease (such as certain cancers) or medication (such as those taken after organ transplantation) have an increased risk of developing vulvar cancer.

The hallmark symptom of vulvar cancer is **itching** (pruritus), which is experienced by 90% of the women afflicted by this cancer. The cancerous lesion is readily visible. Unfortunately, because of embarrassment or denial, it is not uncommon for women to delay medical assessment of vulvar abnormalities. Any abnormalities should be reported to a gynecologist.

If squamous cell vulvar cancer is present, it may appear as a raised red, pink, or white bump (nodule). It is often accompanied by **pain**, bleeding, vaginal

discharge, and painful urination. **Malignant melanoma** of the vulva usually appears as a pigmented, ulcerated growth. Other types of vulvar cancer may appear as a distinct mass of tissue, sore and scaly areas, or cauliflower-like growths that look like **warts**.

Diagnosis

A gynecological examination will be used to observe the suspected area. During this examination, the physician may use a special magnifying instrument called a colposcope to view the area better. Additionally, the area may be treated with a dilute solution of acetic acid, which causes some abnormal areas to turn white, making them easier to see. During this examination, if any area is suspected of being abnormal, a tissue sample (biopsy) will be taken. The biopsy can be performed in the doctor's office with the use of local anesthetic. A wedge-shaped piece of tissue, which contains the suspect lesion with some surrounding normal skin and the underlying skin layers and connective tissue, will be removed. Small lesions will be removed in their entirety (excisional biopsy). The diagnosis of cancer depends on a microscopic analysis of this tissue by a pathologist.

The diagnosis for vulvar cancer will determine how advanced the cancer is and how much it has spread. This is determined by the size of the tumor and how deep it has invaded the surrounding tissue and organs, such as the lymph nodes. It will also be determined if the cancer has metastasized, or spread to other organs. Tests used to determine the extent of the cancer include x ray and computed tomography scan (CT scan). Endoscopic examination of the bladder (**cystoscopy**) and/or rectum (proctoscopy) may be performed if it is suspected that the cancer has spread to these organs.

Treatment

Clinical staging

The International Federation of Gynecology and Obstetrics (FIGO) has adopted a surgical staging system for vulvar cancer. The stage of cancer is determined after surgery. The previous clinical staging system for vulvar cancer is no longer used. Vulvar cancer is categorized into five stages (0, I, II, III, and IV) which may be further subdivided (A and B) based on the depth or spread of cancerous tissue. The FIGO stages for vulvar cancer are:

- **Stage 0.** Vulvar intraepithelial neoplasia.
- **Stage I.** Cancer is confined to the vulva and perineum. The lesion is less than 2 cm (about 0.8 in) in size.

- Stage II. Cancer is confined to the vulva and perineum. The lesion is larger than 2 cm (larger than 0.8 in) in size.
- Stage III. Cancer has spread to the vagina, urethra, anus, and/or the lymph nodes in the groin (inguinofemoral).
- Stage IV. Cancer has spread to the bladder, bowel, pelvic bone, pelvic lymph nodes, and/or other parts of the body.

Treatments

Treatment for vulvar cancer will depend on its stage and the patient's general state of health. Surgery is the mainstay of treatment for most cases of vulvar cancer.

SURGERY. The primary treatment for stage I and stage II vulvar cancer is surgery to remove the cancerous lesion and possibly the inguinofemoral lymph nodes. Removal of the lesion may be done by laser, to burn off a minimal amount of tissue, or by scalpel (local excision), to remove more of the tissue. The choice will depend on the severity of the cancer. If a large area of the vulva is removed, it is called a vulvectomy. Radical vulvectomy removes the entire vulva. A vulvectomy may require skin grafts from other areas of the body to cover the wound and make an artificial vulva. Because of the significant morbidity and the psychosexual consequences of radical vulvectomy, there is a trend toward minimizing the extent of cancer excision. The specific inguinofemoral lymph node that would receive lymph fluid from the cancerous lesion, known as the sentinel node, may be exposed for examination (lymph node dissection) or removed (lymphadenectomy), especially in cases in which the cancerous lesion has invaded to a depth of more than 1 mm. Surgery may also be followed by **chemotherapy** and/or **radiation therapy** to kill additional cancer cells.

Surgical treatment of stage III and stage IV vulvar cancer is much more complex. Extensive surgery would be necessary to completely remove the cancerous tissue. Surgery would involve excision of pelvic organs (pelvic exenteration), radical vulvectomy, and lymphadenectomy. Because this extensive surgery comes with a substantial risk of complications, it may be possible to treat advanced vulvar cancer with minimal surgery by using radiation therapy and/or chemotherapy as additional treatment (adjuvant therapy).

An intraoperative technique that is used to identify the sentinel node in **breast cancer** and melanoma is being applied to vulvar cancer. This technique, called

lymphoscintigraphy, is performed during surgical treatment of vulvar cancer and allows the surgeon to immediately identify the sentinel node. A radioactive compound (technetium 99m sulfur colloid) is injected into the cancerous lesion approximately two hours prior to surgery. This injection causes little discomfort, so **local anesthesia** is not required. During surgery, a radioactivity detector is used to locate the sentinel node and any other nodes to which cancer has spread. Though still in the experimental stage, vulvar lymphoscintigraphy shows promise in reducing morbidity and hospital length of stay.

The most common complication of vulvectomy is the development of a tumor-like collection of clear liquid (wound seroma). Other surgical complications include urinary tract infection, wound infection, temporary nerve injury, fluid accumulation (**edema**) in the legs, **urinary incontinence**, falling or sinking of the genitals (genital prolapse), and **blood clots** (thrombus).

RADIATION THERAPY. Radiation therapy uses high-energy radiation from x rays and gamma rays to kill the cancer cells. The skin in the treated area may become red and dry and may take as long as a year to return to normal. **Fatigue**, upset stomach, **diarrhea**, and **nausea** are also common complaints of women having radiation therapy. Radiation therapy in the pelvic area may cause the vagina to become narrow as scar tissue forms. This phenomenon, known as vaginal stenosis, makes intercourse painful.

CHEMOTHERAPY. Chemotherapy uses **anticancer drugs** to kill the cancer cells. The drugs are given by mouth (orally) or intravenously. They enter the bloodstream and can travel to all parts of the body to kill cancer cells. Generally, a combination of drugs is given because it is more effective than a single drug in treating cancer. The side effects of chemotherapy are significant and include stomach upset, **vomiting**, appetite loss, hair loss, mouth or vaginal sores, fatigue, menstrual cycle changes, and **premature menopause**. There is also an increased chance of infections.

Alternative treatment

Although alternative and complementary therapies are used by many cancer patients, very few controlled studies on the effectiveness of such therapies exist. Mind-body techniques such as prayer, **biofeedback**, visualization, **meditation**, and **yoga** have not shown any effect in reducing cancer but can reduce **stress** and lessen some of the side effects of cancer treatments. Clinical studies of hydrazine sulfate found that it had no effect on cancer and even worsened the health and well-being of the study subjects.

KEY TERMS

Adjuvant therapy—A treatment that is intended to aid primary treatment. Adjuvant treatments for vulvar cancer are radiation therapy and chemotherapy.

Biopsy—Removal of a small piece of tissue for microscopic examination. This is done under local anesthesia and removed by either using a scalpel or a punch, which removes a small cylindrical portion of tissue.

Colposcope—An instrument used for examination of the vagina and cervix. Part of the instrument includes a magnifying lens for better visualization.

Metastasis—The movement of cancer cells from one area of the body to another. This occurs through the blood vessels or the lymph vessels.

Pelvic exenteration—Surgical removal of the organs of the true pelvis which includes the uterus, vagina, and cervix.

Sentinel lymph node—The first lymph node to receive lymph fluid from a tumor. If the sentinel node is cancer-free, then it is likely that the cancerous cells have not metastasized.

One clinical study of the drug amygdalin (Laetrile) found that it had no effect on cancer. Laetrile can be toxic and has caused **death**. Shark cartilage, although highly touted as an effective cancer treatment, is an improbable therapy that has not been the subject of clinical study.

The American Cancer Society has found that the “metabolic diets” pose serious risk to the patient. The effectiveness of the macrobiotic, Gerson, and Kelley **diets** and the Manner metabolic therapy has not been scientifically proven. The FDA was unable to substantiate the anticancer claims made about the popular Cancell treatment.

There is no evidence for the effectiveness of most over-the-counter herbal cancer remedies. However, some herbals have shown an anticancer effect. As shown in clinical studies, Polysaccharide krestin, from the mushroom *Coriolus versicolor*, has significant effectiveness against cancer. In a small study, the green alga *Chlorella pyrenoidosa* has been shown to have anticancer activity. In a few small studies, evening primrose oil has shown some benefit in the treatment of cancer.

Prognosis

Factors that are correlated with disease outcome include the diameter and depth of the cancerous lesion, involvement of local lymph nodes, cell type, HPV status, and age of the patient. Vulvar cancers that are HPV positive have a better prognosis than those that are HPV negative. The 5-year survival rate is 98% for stage I vulvar cancer and 87% for stage II vulvar cancer. The survival rate drops steadily as the number of affected lymph nodes increases. The survival rate is 75% for patients with one or two, 36% for those with three or four, and 24% for those with five or six involved lymph nodes. The previous statistics were obtained from studies of patients who received surgical treatment only and cannot be used to determine survival rates when adjuvant therapy is employed.

Vulvar cancer can spread locally to encompass the anus, vagina, and urethra. Because of the anatomy of the vulva, it is not uncommon for the cancer to spread to the local lymph nodes. Advanced stages of vulvar cancer can affect the pelvic bone. The lungs are the most common site for vulvar cancer metastasis. Metastasis through the blood (hematogenous spread) is uncommon.

Prevention

The risk of vulvar cancer can be decreased by avoiding risk factors, most of which involve lifestyle choices. Specifically, to reduce the risk of vulvar cancer, women should not smoke and should refrain from engaging in unsafe sexual behavior. Good hygiene of the genital area to prevent infection and inflammation may also reduce the risk of vulvar cancer.

Because vulvar cancer is highly curable in its early stages, women should consult a physician as soon as a vulvar abnormality is detected. Regular gynecological examinations are necessary to detect precancerous conditions that can be treated before the cancer becomes invasive. Because some vulvar cancer is a type of skin cancer, the American Cancer Society also recommends self-examination of the vulva using a mirror. If **moles** are present in the genital area, women should employ the ABCD rule:

- **Asymmetry.** A cancerous mole may have two halves of unequal size.
- **Border irregularity.** A cancerous mole may have ragged or notched edges.
- **Color.** A cancerous mole may have variations in color.
- **Diameter.** A cancerous mole may have a diameter wider than 6 mm (1/4 in).

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ORGANIZATIONS

- American Cancer Society. 1599 Clifton Rd. NE, Atlanta, GA 30329. (800) ACS-2345. <<http://www.cancer.org>>.
- Cancer Research Institute, National Headquarters. 681 Fifth Ave., New York, NY 10022. (800) 992-2623. <<http://www.cancerresearch.org>>.
- Gynecologic Cancer Foundation. 401 N. Michigan Ave., Chicago, IL 60611. (800) 444-4441 or (312) 644-6610. <<http://www.wcn.org/gcf>>.
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Vulvitis see **Vulvovaginitis**

Vulvodynia

Definition

Vulvodynia is chronic **pain** of the vulva, the external or visible region of the female genitalia, in the absence of vulvar or vaginal infection or skin disease.

Description

As with other types of chronic pain, vulvodynia can have a significant impact on the quality of life. The disease may interfere with daily activities including sitting, walking, physical **exercise**, and social interactions. Sexual intercourse may be painful or impossible. Vulvodynia can lead to exhaustion and depression. In extreme cases a woman may become bedridden.

Classification systems

Vulvodynia has not been well-defined, in part because it has not been well-studied. The symptoms vary greatly in severity and the causes of the condition are unknown. Thus vulvodynia is classified in different ways.

Most often vulvodynia is subdivided into two types:

- vulvar vestibulitis syndrome (VVS), also called vulval vestibulitis or vulvar dysesthesia, is a localized inflammation of the vestibule—the region immediately surrounding the opening of the vagina and the urethra.
- dysesthetic vulvodynia, also called generalized vulvodynia or generalized vulvar dysesthesia, usually is defined as vulvar burning that has no obvious cause and is not limited to the vestibule.

Some women appear to suffer from both types of vulvodynia simultaneously.

An older classification system is still in use. It distinguishes between vulvar vestibulitis and vulvodynia and subdivides vulvodynia into two major categories:

- organic vulvodynia, in which a cause can be identified
- essential vulvodynia, in which a cause cannot be identified and which usually affects older women, particularly women in their 60s.

Demographics

Females of any age can develop vulvodynia. A 2003 study suggested that up to 16% of women—or

14 million American women—experience chronic vulvar pain lasting three months or longer at some point in their lives. Approximately one-half of these women seek treatment. There is conflicting evidence concerning the prevalence of vulvodynia among different ethnic and racial groups. The 2003 study found that Hispanic women were much more likely to suffer from the condition. The study also found that women who experienced pain when first using a tampon were seven to eight times more likely to have vulvodynia at some point in their lives. Between 10% and 25% of women with interstitial cystitis—a bladder condition that causes frequent urination with burning—also have symptoms of vulvodynia.

The prevalence of VVS is unknown but some surveys have suggested that about 28% of women have a history of vestibular pain. It has been estimated that about 15% of women seeing private gynecologists and about 1% of women visiting genitourinary medical clinics suffer from VVS. Dysesthetic vulvodynia is more common in postmenopausal women and younger women who have had back injuries.

Causes and symptoms

Causes

Suggested causes of vulvodynia include:

- irritation or injury to the vulvar nerves
- hypersensitivity of nerves in the vulvar region
- an abnormal response of vulvar cells to environmental factors, including infection or injury
- irritation from oxalate in the urine
- a localized hypersensitivity to the yeast *Candida albicans*
- frequent yeast infections (candidiasis)
- genital **warts** caused by the human papillomavirus (HPV)
- genital rashes
- an allergic reaction to environmental irritants
- an autoimmune response to the body's own chemicals
- frequent use of antibiotics
- laser treatment or surgery on the external genitalia
- irritation or spasms in the pelvic floor muscles that support the bladder, uterus, and rectum.

A 2004 study found that women with vulvodynia were more likely than other women to have group D streptococcus (GDS) bacteria in their vaginas.

Preliminary evidence suggested that women with vulvodynia might have distinctive vaginal microflora—the various organisms that normally inhabit the vagina. Another 2004 study found that women with vulvodynia were more sensitive to pain in other parts of their bodies and may process pain differently than other women.

In a small subset of women VVS appears to be associated with chronic yeast infections. Some researchers have found an association between VVS and high levels of oxalate in the urine. It also has been suggested that genetic factors may increase a woman's susceptibility to chronic vestibular inflammation. In contrast, many researchers believe that dysesthetic vulvodynia is caused by nerve irritation or inflammation.

ORGANIC VULVODYNIA. Organic vulvodynia—in which a cause for the condition can be identified—includes contact or allergic **dermatitis** and chemical irritation commonly caused by:

- perfumes
- scented or deodorant soaps or laundry detergent
- shampoos
- fabric softener
- scented or dyed toilet paper
- scented menstrual pads
- vulvar wipes or deodorants—"feminine hygiene" products
- douches
- sexual lubricants
- spermicides
- some medications that are used to treat vulvar problems
- synthetic underwear
- excessive vaginal discharge
- urine or feces.

Other causes of organic vulvodynia include:

- chronic vulvar and vaginal yeast infections
- recurrent infection with herpes simplex virus
- other viral or bacterial infections
- nerve irritation
- injury from **childbirth** or vaginal/vulvar trauma.

Symptoms

GENERAL SYMPTOMS. Vulvodynia usually begins suddenly and may last for months or years.

Although the symptoms and their severity vary widely, a burning sensation within the female genitalia is the most common symptom. The pain may be intermittent or constant and localized or generalized throughout the pelvic region.

In addition to burning, sensations of vulvodynia often are described as:

- irritation
- stinging
- rawness
- aching
- stretching
- throbbing
- sharp, knife-like pain.

VULVAR VESTIBULITIS. VVS symptoms are confined to the vestibule and vary from mild to severe. VVS often begins suddenly, following an infection or trauma. Painful sexual intercourse may be the first symptom. The intense **itching** and painful burning can turn into chronic pain. Symptoms may occur daily or only with sexual intercourse.

In addition to painful burning, typical VVS symptoms include:

- an extremely tender, dry, or raw vestibule
- stinging or irritation of the vestibule
- a sensation of tight skin in the vestibule
- redness of the vestibule
- pain that is so severe as to preclude intercourse

The majority of women with VVS only have symptoms during or after:

- the vestibule is touched
- sexual intercourse
- tampon insertion
- a gynecological examination
- bicycle or horseback riding
- jogging
- wearing tight pants.

However in severe cases, sitting, walking, or no movement at all can cause pain.

DYSESTHETIC VULVODYNIA. Women with dysesthetic vulvodynia have constant burning throughout the genitalia or in different areas at different times. Symptoms do not necessarily occur in response to touch or pressure on the vulva, although activities

such as sexual intercourse or bicycle riding may increase the discomfort.

Areas commonly affected by dysesthetic vulvodynia include:

- hymen
- introitus—the membranes surrounding the vaginal opening
- small or minor vestibular glands around the vaginal opening
- vulvovaginal (Batholin's) glands located on each side of the vaginal opening
- paraurethral (Skene's) glands located on each side of the urethral opening
- vestibule
- labia minora or inner labia
- labia majora or outer labia

Some women also experience pain in the clitoris, the perineum, the mons pubis, the anus, the groin, and the inner thighs.

ORGANIC VULVODYNIA. Symptoms of organic vulvodynia caused by chronic yeast infection include vulvar itching and burning, especially just before menstruation. Symptoms caused by recurrent herpes simplex virus are intermittent, often reoccurring with **stress** and lasting for a few days to a week or more. Symptoms of vulvodynia caused by nerve irritation include pain that radiates from the vulva to the perineum, groin, and thighs, and may include lower back pain.

Diagnosis

Vulvodynia is characterized by pain or burning that lasts for more than three months with no apparent **skin lesions**. Although sometimes the vulvar tissue is inflamed, often it appears completely normal. Other conditions that can cause chronic vulvar pain may coexist with vulvodynia. Many women with vulvodynia are misdiagnosed for years. Women often are told that they have a chronic yeast infection or a psychological disorder. It is not uncommon for women to visit at least five doctors before vulvodynia is diagnosed.

To diagnose vulvodynia a healthcare provider may:

- take a medical history
- inquire about symptoms, sexual activity, diet, feminine hygiene, and medications
- examine the vulva and vaginal secretions for signs of infection or a skin disorder

- perform a pelvic examination
- take a vaginal culture to test for yeast or bacteria
- use a Q-tip to determine the location and severity of the pain
- use a magnifying glass to look for abnormalities
- perform a colposcopy—an examination with a special magnifying lens—to look for inflamed surface blood vessels
- perform a tissue biopsy—the removal of a small piece of tissue for microscopic examination—particularly if skin lesions are present.

Vulvar vestibulitis

Women with VVS are more likely to have yeast infections than the general population. A yeast infection must be treated before VVS can be diagnosed. VVS can be diagnosed by a “touch test” with a cotton-tipped applicator to reveal extreme sensitivity at various points on the vestibule. In about two-thirds of VVS cases these points have tiny red spots. Biopsies for VVS generally show signs of chronic inflammation.

Dysesthetic vulvodynia

Dysesthetic vulvodynia generally is diagnosed after ruling out other causes of vulvar pain including:

- yeast, bacterial, or viral infection
- contact or allergic dermatitis
- overuse of medications, particularly topical steroids
- vulvar dermatoses caused by one of numerous skin conditions and diseases
- cyclic vulvovaginitis—burning or itching that recurs at a particular stage of a woman’s menstrual cycle
- vulvovaginal atrophy due to estrogen deficiency in postmenopausal women.

Treatment

There is no cure for vulvodynia and, because there appear to be multiple causes for the disorder and because individual women have unique symptoms, no single treatment is appropriate for all women. A treatment may work for some women and not for others and some treatments may cause unacceptable side effects in some women. Women with vulvodynia should stop using all topical medications, soaps, douches, and other products that may cause irritation. Yeast infections may require long-term treatment.

Symptoms of vulvodynia may be treated with:

- topical anesthetics such as viscous or liquid 5% lidocaine ointment (Xylocaine) for temporary relief
- nerve blockers
- cortisone creams
- topical estrogen cream
- antihistamines, such as Atarax or Vistaril, that inhibit the release of histamines that can irritate mucous membranes
- physical and exercise therapy to strengthen and relax the pelvic-floor muscles—called pelvic floor therapy—particularly if lower back problems are present.

Vulvar vestibulitis

VVS often is treated first with a corticosteroid ointment, such as triamcinolone or desoximetasone, twice a week for one month. Corticosteroid treatment requires careful monitoring because it can cause irritation or thinning of the skin. If this treatment fails, VVS may be treated with the antidepressant amitriptyline (Elavil) or the anticonvulsant gabapentin (Neurontin). Initial dosages are low and can be increased steadily until symptoms are relieved or side effects become intolerable, up to a maximum daily dosage of 150 mg of amitriptyline or 3200 mg of gabapentin.

Other treatments for VVS include:

- interferon—an anti-viral drug—injected into the vulva, particularly in women who also have human papillomavirus
- vestibulectomy—surgery to remove the vestibule and hymen, usually as a last resort.
- Laser treatments the freeze the tissue have not proved successful and may worsen VVS.

Dysesthetic vulvodynia

The most common treatments for dysesthetic vulvodynia are medications that treat nerve irritation and pain throughout the body—tricyclic antidepressants such as amitriptyline, nortriptyline (Pamelor), desipramine (Norpramin), and imipramine (Tofranil), and anticonvulsants such as carbamazepine (Tegretol) or gabapentin.

Alternative treatment

Alternative treatments for vulvodynia include:

- biofeedback; sensitive detectors are available for use on the vulva

- acupuncture
- cognitive behavioral therapy
- aerobic exercise
- cool or lukewarm sitz baths for relieving irritation and burning
- douching with baking soda to relieve inflammation
- a compress of Aveeno—a powdered oatmeal bath treatment, made with two tablespoons of Aveeno to one quart of water—placed over the vulva three to four times per day.

Low-oxalate diet

Those who believe that vulvodynia may be caused by acidic, irritating oxalate crystals in the urine, recommend trying a low-carbohydrate, low-oxalate diet for three months to a year. The diet involves:

- eliminating all high-oxalate foods—those with more than 10 mg of oxalate per serving
- limiting moderate oxalate-containing foods (2–10 mg per serving) to three servings per week
- drinking 12–14 cups of water per day
- taking calcium citrate without Vitamin D, such as two Citracal tablets three times per day, approximately 45 minutes before eating.

Calcium citrate supplements decrease calcium oxalate formation in the urine, suppress oxalate secretion, and decrease the acidity of the urine.

Foods that are high in oxalates include:

- all beans
- wheat bran, wheat germ, and grits
- white corn
- all nuts and nut butters
- greens including chard, kale, escarole, collards, and spinach
- celery
- parsley and watercress
- green peppers
- beets
- tomato and vegetable soups
- rutabaga
- eggplant
- leeks
- okra
- sweet potatoes

- summer squash
- rhubarb
- berries
- purple grapes
- tangerines
- chocolate
- beer, beverage mixes, and tea

Prognosis

Vulvodynia sometimes disappears spontaneously. It often improves over time regardless of treatment. In some women treatment results in partial or complete relief of symptoms, although it may take weeks, months, or even years. Other women are never completely free of symptoms.

It has been reported that about 30% of VVS cases improve with corticosteroid treatment and another 60% improve with amitriptyline or other **tricyclic anti-depressants**. Vestibulectomy has the highest success rate of any VVS treatment.

Prevention

Numerous suggestions have been made for preventing vulvodynia and for relieving symptoms and preventing further irritation, including:

- using soft, white or unbleached, unscented toilet tissue
- using 100% cotton menstrual pads and tampons
- avoiding shampoo in the vulvar region
- avoiding bubble bath, feminine hygiene products, perfumed creams and soaps, petroleum jelly, and bath oils
- washing the vulva frequently with cool or lukewarm water
- using a bidet
- drying the perineum with a hair dryer on the cool setting
- urinating before the bladder is full
- rinsing the vulva with water from a squeeze bottle after urination
- eating fiber from whole grains, fruits, and vegetables, and psyllium products, such as Metamucil or Konsyl, to prevent constipation
- drinking eight glasses of water daily
- sitting on a foam rubber donut and standing periodically when sitting for long periods
- using relaxation techniques

KEY TERMS

Colposcopy—The use of a special microscope—a colposcope—to visualize the genitalia.

Contact dermatitis—Irritant dermatitis; direct skin contact with a substance that causes inflammation in some people.

Dysesthetic vulvodynia—Generalized vulvodynia; generalized vulvar dysesthesia; essential vulvodynia; chronic vulvar pain not limited to the vulva for which there is no apparent cause.

Hymen—A membrane that partially or completely covers the vaginal opening.

Interstitial cystitis—Chronic inflammation of the bladder; sometimes associated with vulvodynia.

Labia majora—Major lips; mounds of tissue forming the lateral boundaries of the vulva.

Labia minora—Minor lips; narrow folds of tissue between the labia majora, on either side of the urethral and vaginal openings.

Mons pubis—The fatty tissue over the area where the pubic bones meet.

Organic vulvodynia—Chronic vulvar pain for which a cause can be identified.

Oxalate—A salt of oxalic acid produced by the body's metabolism and excreted in the urine.

Perineum—The area between the external genitalia and the anus.

Tricyclic antidepressants—Medications used to treat mental depression and other conditions including chronic pain.

Vestibule—Vestibule of vulva; vestibule of vagina; the space between the labia minora containing the openings of the vagina and urethra.

Vestibulectomy—Surgical removal of the vestibule and hymen.

Vulva—Penum; the external female genitalia including the mons pubis, labia majora and minora, clitoris, vestibule, glands, and the vaginal opening.

Vulvar vestibulitis syndrome; VVS—Vulval vestibulitis; vulvar dysesthesia; inflammation of the vestibule.

Suggestions for clothing and laundry include:

- wearing all-white cotton underwear and washing new underwear before use
- wearing loose-fitting skirts and pants

- avoiding pantyhose and jeans
- promptly removing wet bathing suits and exercise clothing
- washing clothes in a dermatologically approved detergent, such as Purex or Clear, or baking soda soap
- double-rinsing underwear and other clothing that contacts the vulva
- avoiding fabric softener and dryer sheets.

Advice on physical activity includes:

- avoiding activities that put direct pressure on the vulva, such as bicycle and horseback riding
- limiting intense exercise that applies friction to the vulva
- applying a frozen gel pack wrapped in a towel after exercising to relieve symptoms
- performing stretching and relaxation exercises such as yoga
- avoiding hot tubs and highly-chlorinated swimming pools.

Suggestions for sexual intercourse include:

- applying a topical anesthetic, such as 5% lidocaine gel, to the vestibule 10–15 minutes before intercourse
- using pure almond oil, vegetable oil, or a water-soluble, glycerin-containing lubricant such as Astroglide, applied at the initiation of sexual activity
- avoiding lubricants containing preservatives or other chemicals
- avoiding contraceptive creams, spermicides, and devices that may irritate the vulva
- applying ice or a frozen blue gel pack after intercourse to prevent burning
- urinating and rinsing the vulva with cool water after intercourse.

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D.C. 20090-6920. 202-638-5577. <<http://www.acog.org>>.

Center for Vulvar Diseases, Department of Obstetrics and Gynecology, University of Michigan Health System.
1500 E. Medical Center Drive, Taubman Center,
Reception E, Box 0384, Ann Arbor, MI 48109-0384.
734-936-4000. <<http://www.med.umich.edu/obgyn/vulva/index.htm>>.

National Vulvodynia Association. P.O. Box 4491, Silver Spring, MD 20914-4491. 301-299-0775. <<http://www.nva.org>>.

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Vulvovaginitis

Definition

Inflammation of the vagina and vulva most often caused by a bacterial, fungal, or parasitic infection.

Description

Vulvovaginitis, vulvitis, and vaginitis are general terms that refer to the inflammation of the vagina and/or vulva (the external genital organs of a woman). These conditions can be caused by bacterial, fungal, or parasitic infections. Also, vulvovaginitis can be caused by low estrogen levels (called "atrophic vaginitis") or any type of allergic or irritation response from things such as spermicidal products, condoms, soaps, and bubble bath.

In general, vulvovaginitis causes vaginal discharge, irritation, and **itching**. One of the most common reasons why women visit their doctor is because of a change in vaginal discharge. It is completely normal for a woman to have a vaginal discharge, the amount and consistency of which varies during the course of the menstrual cycle. Each of the three most common types of vulvovaginitis will be described separately.

Bacterial vaginosis

Bacterial vaginosis is the most common cause of vaginitis during the childbearing years. Forty percent to 50% of vaginitis cases are caused by bacterial vaginosis. The occurrence of bacterial vaginosis is difficult to determine but studies have proposed that 10%–41% of women have had it at least once. The occurrence of bacterial vaginosis in the United States is highest among African-American women and women who have had multiple sexual partners and lowest among Asian women and women with no history of sexual contact with men. Bacterial vaginosis is not considered a sexually transmitted disease although it can be acquired by sexual intercourse.

Bacterial vaginosis is not caused by a particular organism but is a change in the balance of normal vaginal bacteria. Ninety percent of the bacteria found in a healthy vagina belong to the *Lactobacillus* family. For unknown reasons, there is a shift in the bacterial population that results in overgrowth of other bacteria. Patients suffering from bacterial vaginosis have very high numbers of such bacteria as *Gardnerella vaginalis*, *Mycoplasma hominis*, *Bacteroides* species, and *Mobiluncus* species. These

bacteria can be found at numbers 100 to 1000 times greater than found in the healthy vagina. In contrast, *Lactobacillus* bacteria are in very low numbers or completely absent from the vagina of women with bacterial vaginosis.

Candida vulvovaginitis

Candida vulvovaginitis also has been called “vulvovaginal candidiasis,” “candidal vaginitis,” “monilial infection,” or “vaginal yeast infection.” Twenty to 25% of the vaginitis cases are *Candida* vulvovaginitis. It has been estimated that about 75% of all women get a vaginal yeast infection at least once. In 80–90% of the cases, *Candida* vulvovaginitis is caused by an overgrowth of the yeast *Candida albicans*. The remaining cases are caused by other species of *Candida*. It is not known what causes the yeast overgrowth. However, **antibiotics** can inadvertently kill normal bacteria in the vagina and cause an overgrowth of *Candida*.

Candida vulvovaginitis is not considered a sexually transmitted disease because *Candida* species are commonly found in the healthy vagina. It is a rare disease in girls before **puberty** and celibate women. Vaginal yeast infections tend to occur more frequently in women who are pregnant, diabetic and not controlling their disease, taking birth control pills, or taking antibiotics. The **stress** of military deployment has also been found to be a factor in triggering vaginal yeast infections. Some women have four or more attacks per year, a condition called “recurrent vaginal candidiasis.”

Trichomoniasis

Trichomoniasis, which is sometimes called “trich,” accounts for 15–20% of the cases of vaginitis. It is estimated that two million to three million American women get trichomoniasis each year, and as many as 120 million women worldwide. Unlike the previous two causes of vulvovaginitis, trichomoniasis is a sexually transmitted disease. This means that the disease is passed from person to person only by sexual contact. Trichomoniasis occurs in both men and women and is caused by infection with the single-celled parasite *Trichomonas vaginalis*. Infection with *Trichomonas vaginalis* is frequently associated with other **sexually transmitted diseases** and assists the spread of the **AIDS** virus.

Causes and symptoms

Vulvovaginitis is most often caused by a bacterial, fungal, or parasitic infection as described above. Other

microorganisms may cause vulvovaginitis, or it may be caused by allergic reaction, irritation, injury, low estrogen levels, and certain diseases. Risk factors for bacterial vaginosis include using an intrauterine device (IUD), non-white race, prior **pregnancy**, first sexual activity at an early age, having multiple sexual partners, and having a history of sexually transmitted diseases. Persons at an increased risk for *Candida* vulvovaginitis include those who have had previous candida infections, frequent sexual intercourse, use birth control pills, have AIDS, are pregnant, are taking antibiotics or **corticosteroids**, are diabetic, use douches, use perfumed feminine hygiene sprays, wear tight clothing, or use vaginal sponges or an IUD.

The typical symptoms of vulvovaginitis are: vaginal discharge, itching, and irritation. Women may have few or no symptoms, while others may have pronounced symptoms. The main symptom of bacterial vaginosis is a fishy-smelling, thin, milky-white or gray vaginal discharge but itching and burning may also be present. The fishy smell is stronger after sexual intercourse. The symptoms of candida vulvovaginitis are itching, soreness, painful sexual intercourse, and a thick, curdy, white (like cottage cheese) vaginal discharge. Trichomoniasis symptoms are: painful urination, painful sexual intercourse, and a yellow-green to gray, foul smelling, sometimes frothy, vaginal discharge.

Diagnosis

Vulvovaginitis can be diagnosed and treated by a nurse practitioner or physician. Most insurance companies cover the costs of diagnosis and treatment. To diagnose vulvovaginitis, the doctor will examine the vagina (using a speculum to keep the vagina open) and take a sample of the vaginal discharge for tests and microscopic analysis. Laboratory culture results should be available in two to three days but the microscopic examination of the vaginal discharge may be immediately performed in the doctor's office. Diagnosis may be difficult because there are many different causes of vulvovaginitis. Women who think that they have vulvovaginitis should always visit their doctor to get an accurate diagnosis. Many women assume that they have a yeast infection and take over-the-counter medicines without first consulting their doctors.

There are four signs that indicate that a woman has bacterial vaginosis. These signs (known as Amsel's criteria) are: a thin, milky white discharge that clings to the walls of the vagina, presence of a fishy odor, a vaginal pH greater than 4.5, and the presence of clue

cells in the vagina. Clue cells are vaginal cells that are covered with small bacteria. A diagnosis of candida vulvovaginitis is made after finding a normal vaginal pH (4 to 4.5) and the presence of many yeast cells in the sample of vaginal discharge or growth of yeast on laboratory media. In the spring of 2004, the U.S. Food and Drug Administration (FDA) cleared a new quick test to diagnose trichomonas infection. The test provides results in 10 minutes. The company that manufactures the test also has a rapid result test to bacterial vaginosis. A trichomoniasis diagnosis is made when the parasites are found in the vaginal discharge either by microscopic examination or in laboratory cultures.

Trichomoniasis tends to be underdiagnosed in men because of the relative mildness of symptoms in men and insufficiently sensitive diagnostic tests. The recent introduction of DNA amplification, however, indicates that the incidence of trichomoniasis in men is much higher than was previously thought.

Treatment

Both bacterial vaginosis and trichomoniasis require prescription medication for treatment. Candida vulvovaginitis may be treated with either prescription or over-the-counter medicines. It is not advisable to take over-the-counter vaginal yeast infection medicines if one does not have a yeast infection. An Institute of Epidemiological Research survey of 390 gynecologists found that 44% of the women who were diagnosed with bacterial vaginosis had first treated themselves with over-the-counter yeast infection medications.

Bacterial vaginosis should be treated daily for one week with the antibiotics metronidazole (Flagyl, Protostat) or clindamycin (Cleocin) either as pills taken orally or in a gel or cream form put into the vagina. Trichomoniasis is treated with either a large, single dose of metronidazole or with a smaller dose taken twice daily for one week. Male sexual partners of women with trichomoniasis also must be treated.

Candida vulvovaginitis is most often treated by the application of medicated gels, creams, or suppositories applied directly to the vagina. The antifungal drugs used to treat candida vulvovaginitis include oral fluconazole (Diflucan), butoconazole (Femstat), clotrimazole (Gyne-lotrimin, Mycelex), miconazole (Monistat), and ticonazole (Vagistat). Most require only one or a few days of therapy to be effective. Women who have recurrent *Candida* infections may receive treatment for several weeks and then some form of a long-term preventative treatment.

KEY TERMS

Parasite—An animal or plant that can only survive by living inside or upon another animal or plant.

Vulva—The external genital organs of a woman, including the outer and inner lips, clitoris, and opening of the vagina.

Alternative treatment

One of the primary focuses of alternative treatment for vaginal conditions including vulvovaginitis is rebalancing the normal vaginal flora. To assist with this rebalancing, *Lactobacillus acidophilus* and *L. bifidus* are recommended, either taken internally or introduced directly into the vagina. Garlic (*Allium sativum*), both taken internally and inserted into the vagina (a peeled whole clove wrapped in gauze), may be helpful due to its antibacterial and antifungal actions. A variety of other herbs can be used as douches or in suppository form to help treat acute flare-ups of vaginal symptoms. For example a douche made by steeping 1–2 tsp. of calendula (*Calendula officinalis*) in boiling water (let the water cool before using) may help reduce inflammation. A boric acid douche can help to acidify the vaginal pH so that unwanted bacteria cannot survive and multiply. For atrophic vaginitis, especially in menopausal women, topical application of progesterone cream can help with symptoms caused by thinning of the tissue lining the vagina.

Dietary modification and nutritional supplementation may also be helpful in the treatment of vulvovaginitis. Antioxidant **vitamins**, including A, C, and E, as well as B complex vitamins, and vitamin D, are recommended. Foods to avoid include cheese, alcohol, chocolate, soy sauce, sugar, vinegar, fruits, and any fermented foods. Wearing cotton underwear and loose fitting clothes and avoiding panty hose can help keep the vagina cool and dry, thus helping to prevent some forms of vulvovaginitis. Cases of chronic vulvovaginitis should be addressed on systemic level by an alternative practitioner.

Prognosis

Vulvovaginitis is a disease with minor symptoms and most women respond well to medications. It is believed that certain vaginal infections, if left untreated, can lead to more serious conditions such as **pelvic inflammatory disease**, endometritis, postsurgical infections, and spread of the AIDS virus.

Prevention

Vaginal infections may be prevented by following these suggestions:

- Over-the-counter yeast infection treatments should not be taken unless the woman had been diagnosed with **candidiasis** before and recognizes the symptoms.
- Douching should be avoided because it may disturb the balance of organisms in the vagina and may spread them higher into the reproductive system.
- Thoroughly dry oneself after bathing and remove a wet bathing suit promptly.
- Avoid wearing tight clothing and wear cotton underwear.
- Clean diaphragms, cervical caps, and spermicide applicators after use. Use condoms to avoid sexually transmitted disease.
- After a bowel movement, wipe from front to back to avoid spreading intestinal bacteria to the vagina.

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Waldenström disease see **Waldenström's macroglobulinemia**

Waldenström's macroglobulinemia

Definition

Waldenström's macroglobulinemia is a rare, chronic **cancer** of the immune system that is characterized by hyperviscosity, or thickening, of the blood.

Description

Waldenström's (Waldenström, Waldenström's) macroglobulinemia (WM) is a lymphoma, or cancer of the lymphatic system. It was first identified in 1944, by the Swedish physician Jan Gosta Waldenström, in patients who had a thickening of the serum, or liquid part, of the blood. Their blood serum contained a great deal of a very large molecule called a globulin. Thus, the disorder is called macroglobulinemia.

Lymphomas are cancers that originate in tissues of the lymphatic system. All lymphomas other than **Hodgkin's disease**, including WM, are known collectively as non-Hodgkin's lymphomas. There are 13 major types of non-Hodgkin's lymphomas, and others that are very rare. Other names that are sometimes used for WM include: lymphoplasmacytic lymphoma, lymphoplasmacytic leukemia, macroglobulinemia of Waldenström, primary macroglobulinemia, Waldenström's syndrome, Waldenström's purpura, or hyperglobulinemic purpura. Purpura refers to purple spots on the skin, resulting from the frequent bleeding and bruising that can be a symptom of WM.

WM is classified as a low-grade or indolent form of lymphoma because it is a slow-growing cancer that produces fewer symptoms than other types of lymphomas. WM most often affects males over the age of 65. Frequently, this disease produces no symptoms and does not require treatment. It has not been studied as extensively as other types of lymphoma.

The lymphatic system

The lymphatic system is part of the body's immune system, for fighting disease, and part of the blood-producing system. It includes the lymph vessels and nodes, and the spleen, bone marrow, and thymus. The narrow lymphatic vessels carry lymphatic fluid from throughout the body. The lymph nodes are small, pea-shaped organs that filter the lymphatic fluid and trap foreign substances, including viruses, bacteria, and cancer cells. The spleen, in the upper left abdomen, removes old cells and debris from the blood. The bone marrow, the spongy tissue inside the bones, produces new blood cells.

B lymphocytes or B cells are white blood cells that recognize disease-causing organisms. They circulate throughout the body in the blood and lymphatic fluid. Each B lymphocyte recognizes a specific foreign substance, or antigen. When it encounters its specific antigen, the B cell begins to divide and multiply, producing large numbers of identical (monoclonal), mature plasma cells. These plasma cells produce large amounts of antibody that are specific for the antigen. Antibodies are large proteins called immunoglobulins (Igs) that bind to and remove the specific antigen.

A type of Ig, called IgM, is part of the early immune response. The IgM molecules form clusters

in the bloodstream. When these IgM clusters encounter their specific antigen, usually a bacterium, they cover it so that it can be destroyed by other immune system cells.

Plasma cell neoplasm

WM is a type of plasma cell neoplasm or B-cell lymphoma. These are lymphomas in which certain plasma cells become abnormal, or cancerous, and begin to grow uncontrollably. In WM, the cancerous plasma cells overproduce large amounts of identical (monoclonal) IgM antibody. This IgM also is called M protein, for monoclonal or myeloma protein.

Macroglobulinemia refers to the accumulation of this M protein in the serum of the blood. This large amount of M protein can cause the blood to thicken, causing hyperviscosity. The malignant plasma cells of some WM patients also produce and secrete partial immunoglobulins called light chains, or Bence-Jones proteins. The malignant plasma cells can invade various tissues, including the bone marrow, lymph nodes, and spleen, causing these tissues to swell.

WM accounts for about 1-2% of non-Hodgkin's lymphomas. It is estimated that it may affect about five out of every 100,000 people. It usually affects people over the age of 50, and most often develops after age 65. It is more common in men than in women. In the United States, WM is more common among Caucasians than among African Americans. The disease can run in families.

Causes and symptoms

The cause of WM is not known.

Many individuals with WM have no symptoms of the disease. This is known as asymptomatic macroglobulinemia. When symptoms of WM are present, they may vary greatly from one individual to the next.

Hyperviscosity syndrome

At least 50% of individuals with WM have hyperviscosity syndrome, an increased viscosity or thickening of the blood caused by the accumulation of IgM in the serum. Hyperviscosity can cause a slowing in the circulation through small blood vessels. This condition can lead to a variety of symptoms:

- fatigue
- weakness
- rash
- bruising

- nose bleeds
- gastrointestinal bleeding
- weight loss
- night sweats
- increased and recurrent infections
- poor blood circulation in the extremities

Poor blood circulation, or Raynaud's phenomenon, can affect any part of the body, but particularly the fingers, toes, nose, and ears.

Cold weather can cause additional circulatory problems, by further thickening the blood and slowing down circulation. In some cases, the excess blood protein may precipitate out of the blood in the cold, creating particles that can block small blood vessels. This is called cryoglobulinemia. The extremities may turn white, or a patchy red and white. The hands, feet, fingers, toes, ears, and nose may feel cold, numb, or painful.

Hyperviscosity may affect the brain and nervous system, leading to additional symptoms. These symptoms include:

- peripheral neuropathy, caused by changes in the nerves, leading to **pain** or **numbness** in the extremities
- dizziness
- headaches
- vision problems or loss of vision
- Mental confusion
- poor coordination
- temporary paralysis
- mental changes

Hyperviscosity can clog the tubules that form the filtering system of the kidneys, leading to kidney damage or kidney failure. Existing heart conditions can be aggravated by WM. In extreme cases, WM may result in **heart failure**. Late-stage WM also may lead to mental changes that can progress to **coma**.

Anemia

The accumulation of IgM in the blood causes an increase in the volume of the blood plasma. This effectively dilutes out the red blood cells and other blood components. The lowered concentration of red blood cells can lead to anemia and cause serious fatigue. Likewise, a deficiency in platelets (**thrombocytopenia**), which cause the blood to clot, can result in easy bleeding and bruising. As the cancer progresses, there may be abnormal bleeding from the gums, nose, mouth, and intestinal tract. There may be bluish

discoloration of the skin. In the later stages of the disease, leukopenia, a deficiency in white blood cells, also can develop.

Organ involvement

In 5-10% of WM cases, the IgM may be deposited in tissues. Thus, some individuals with WM have enlargement of the lymph nodes, the spleen, and/or the liver.

If Bence-Jones proteins are produced by the malignant plasma cells, they may be deposited in the kidneys. There they can plug up the tiny tubules that form the filtering system of the kidneys. This can lead to kidney damage and kidney failure.

Diagnosis

Since many individuals with WM have no symptoms, the initial diagnosis may result from blood tests that are performed for some other purpose. Blood cell counts may reveal low red blood cell and platelet levels. A **physical examination** may indicate enlargement of the lymph nodes, spleen, and/or liver. A retinal **eye examination** with an ophthalmoscope may show retinal veins that are enlarged or bleeding.

Blood and urine tests

Serum **protein electrophoresis** is used to measure proteins in the blood. In this laboratory procedure, serum proteins are separated in an electrical field, based on the size and electrical charge of the proteins. Serum **immunoelectrophoresis** uses a second antibody that reacts with IgM. A spike in the Ig fraction indicates a large amount of identical or monoclonal IgM in individuals with WM.

Normal serum contains 0.7-1.6 gm per deciliter (g/dl) of Ig, with no monoclonal Ig present. At serum IgM concentrations of 3-5 g/dl, symptoms of hyperviscosity often are present. However some individuals remain asymptomatic with IgM levels as high as 9 g/dl.

Urinalysis may indicate protein in the urine. A urine Bence-Jones protein test may indicate the presence of these small, partial Igs.

Bone marrow

Abnormal blood tests usually are followed by a **bone marrow biopsy**. In this procedure, a needle is inserted into a bone and a small amount of marrow is removed. Microscopic examination of the marrow may reveal elevated levels of lymphocytes and plasma

cells. However, less than 5% of patients with WM have lytic bone lesions, caused by cancerous plasma cells in the bone marrow that are destroying healthy cells. Bone lesions can be detected with x rays.

Treatment

Clinical staging, to define how far a cancer has spread through the body, is the common method for choosing a cancer treatment. However, there is no generally-accepted staging system for WM.

There also is no generally-accepted course of treatment for WM. Treatment may not be necessary for asymptomatic macroglobulinemia. However, if IgM serum levels are very high, treatment may be initiated even in the absence of symptoms. If symptoms are present, treatment is directed at relieving symptoms and retarding the disease's development. Of major concern is the prevention or alleviation of blood hyperviscosity. Therefore, the initial treatment depends on the viscosity of the blood at diagnosis.

Hyperviscosity

Plasmapheresis, or plasma exchange **transfusion**, is a procedure for thinning the blood. In this treatment, blood is removed and passed through a cell separator that removes the plasma, containing the IgM, from the red and white blood cells and platelets. The blood cells are transfused back into the patient, along with a plasma substitute or donated plasma. Plasmapheresis relieves many of the acute symptoms of WM. Individuals with WM may be given fluid to counter the effects of hyperviscous blood.

Low blood cell counts

Treatments for low blood cell levels include:

- the drug Procrit to treat anemia
- transfusions with packed red blood cells to treat anemia in later stages of the disease
- antibiotics to treat infections caused by a deficiency in white blood cells
- transfusions with blood platelets

Chemotherapy

Chemotherapy, the use of anti-cancer drugs, helps to slow the abnormal development of plasma cells, but does not cure WM. It can reduce the amount of IgM in the bone marrow. In particular, chemotherapy is used to treat severe hyperviscosity and anemia that are caused by WM.

Chlorambucil (Leukeran), possibly in combination with prednisone, is the typical chemotherapy choice for WM. This treatment is effective in 57% of cases. These drugs are taken by mouth. Prednisone is a corticosteroid that affects many body systems. It has anti-cancer and anti-inflammatory effects and is an immune system suppressant. Other drug combinations that are used to treat WM include cyclophosphamide (Cytosan), vincristine, and prednisone, with or without doxorubicin. Fludarabine, 2-chlorodeoxyadenosine, and **corticosteroids** also may be used.

side effects of chemotherapy may include:

- mouth sores
- nausea and indigestion
- hair loss
- increased appetite
- nervousness
- insomnia

These side effects disappear after the chemotherapy is discontinued.

The long-term management of WM usually is accomplished through a combination of plasmapheresis and chemotherapy.

Alternative and complementary therapies

Biological therapy or immunotherapy, with the potent, immune system protein interferon alpha, is used to relieve the symptoms of WM. Interferon alpha works by boosting the body's immune response. Interferon can cause flu-like symptoms, such as **fever**, chills, and fatigue. It also can cause digestive problems and may affect blood pressure.

The drug rituximab, an antibody that is active against antibody-producing cells, is effective in about 30% of individuals with WM. Rituximab is a monoclonal antibody produced in the laboratory. Monoclonal antibody treatment may cause an allergic reaction in some people.

Prognosis

There is no cure for WM. In general, patients go into partial or complete remission following initial treatments. However the disease is not cured and follow-up treatment may be necessary.

The prognosis for this cancer depends on an individual's age, general health, and genetic (hereditary) makeup. Males, individuals over age 60, and those with severe anemia have the lowest survival rates. The Revised European American Lymphoma

(REAL) classification system gives WM a good prognosis following treatment, with an average 5-year survival rate of 50-70%. However, many people with WM live much longer, some without developing any symptoms of the disease. About 16-23% of individuals with WM die of unrelated causes.

Prevention

There is no known prevention for WM.

Resources

BOOKS

Drum, David. *Making the Chemotherapy Decision*. 2nd ed. Los Angeles: Lowell House, 1996.

ORGANIZATIONS

Cure for Lymphoma Foundation. 215 Lexington Ave., New York, NY 10016. (212) 213-9595. (800)-CFL-6848. infocfl@cfl.org. <<http://www.cfl.org/home.html>>. An advocacy organization; education and support programs, research grants, information on clinical trials for Hodgkin's and non-Hodgkin's lymphomas.

International Waldenström's Macroglobulinemia Foundation. 2300 Bee Ridge Road, Sarasota, FL 34239-6226. (941) 927-IWMF. <<http://www.iwmf.com>>. Information, educational programs, support for patients and families, research support.

The Leukemia and Lymphoma Society. 600 Third Ave., New York, NY 10016. (800) 955-4572. (914) 949-5213. <<http://www.leukemia-lymphoma.org>>. Information, support, and guidance for patients and health care professionals.

The Lymphoma Research Foundation of America, Inc. 8800 Venice Boulevard, Suite 207, Los Angeles, CA 90034. (310) 204-7040. <<http://www.lymphoma.org>>. Research into treatments for lymphoma; educational and emotional support programs for patients and families.

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KEY TERMS

Anemia—Any condition in which the red blood cell count is below normal.

Antibody—Immunoglobulin produced by immune system cells that recognizes and binds to a specific foreign substance (antigen).

Antigen—Foreign substance that is recognized by a specific antibody.

Autosomal dominant—Genetic trait that is expressed when present on only one of a pair of non-sex-linked chromosomes.

B cell (B lymphocyte)—Type of white blood cell that produces antibodies.

Bence-Jones protein—Light chain of an immunoglobulin that may be overproduced in Waldenstrom's macroglobulinemia; it is excreted in the urine.

Biopsy—Removal of a small sample of tissue for examination under a microscope; used in the diagnosis of cancer.

Cryoglobulinemia—Condition in which protein in the blood forms particles in the cold, blocking blood vessels and leading to pain and numbness of the extremities.

Hyperviscosity—Thick, viscous blood, caused by the accumulation of large proteins, such as immunoglobulins, in the serum.

Immunoelectrophoresis—Use of an electrical field to separate proteins in a mixture (such as blood or urine), on the basis of the size and electrical charge

of the proteins; followed by the detection of an antigen (such as IgM), using a specific antibody.

Immunoglobulin (Ig)—Antibody such as IgM; large protein produced by B cells that recognizes and binds to a specific antigen.

Interferon alpha—Potent immune-defense protein; used as an anti-cancer drug.

Lymphatic system—The vessels, lymph nodes, and organs, including the bone marrow, spleen, and thymus, that produce and carry white blood cells to fight disease.

Lymphoma—Cancer that originates in lymphatic tissue.

M protein—Monoclonal or myeloma protein; IgM that is overproduced in Waldenstrom's macroglobulinemia and accumulates in the blood and urine.

Monoclonal—Identical cells or proteins; cells (clones) derived from a single, genetically-distinct cell, or proteins produced by these cells.

Plasma cell—Type of white blood cell that produces antibodies; derived from an antigen-specific B cell.

Plasmapheresis—Plasma exchange transfusion; the separation of serum from blood cells to treat hyperviscosity of the blood.

Platelet—Cell that is involved in blood clotting.

Stem cell—Undifferentiated cell that retains the ability to develop into any one of numerous cell types.

Waldenstroms.com. International Waldenstrom's Macroglobulinemia Foundation. June 28, 2001.
< <http://www.iwmf.com> >.

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Margaret Alic, PhD

Walleye see **Strabismus**

Warts

Definition

Warts are small, benign growths caused by a viral infection of the skin or mucous membrane. The virus

infects the surface layer. The viruses that cause warts are members of the human papilloma virus (HPV) family. Warts are not cancerous but some strains of HPV, usually not associated with warts, have been linked with **cancer** formation. Warts are contagious from person to person and from one area of the body to another on the same person.

Description

Particularly common among children, young adults, and women, warts are a problem for 7-10% of the population. There are close to 60 types of HPV that cause warts, each preferring a specific skin location. For instance, some types of HPV cause warts to grow on the skin, others cause them to grow inside the mouth, while still others cause them to grow on the

genital and rectal areas. However, most can be active anywhere on the body. The virus enters through the skin and produces new warts after an incubation period of one to eight months. Warts are usually skin-colored and feel rough to the touch, but they also can be dark, flat, and smooth.

Warts are passed from person to person, directly and indirectly. Some people are continually susceptible to warts, while others are more resistant to HPV and seldom get them. The virus takes hold more readily when the skin has been damaged in some way, which may explain why children who bite their nails tend to have warts located on their fingers. People who take a medication to suppress their immune system or are on long-term steroid use are also prone to a wart virus infection. This same is true for patients with AIDS.

Causes and symptoms

The more common types of warts include:

- common hand warts
- foot warts
- flat warts
- **genital warts**

Hand warts

Common hand warts grow around the nails, on the fingers, and on the backs of hands. They appear more frequently where skin is broken, such as in areas where fingernails are bitten or hangnails picked.

Foot warts

Foot warts are called plantar warts because the word plantar is the medical term for the sole of the foot, the area where the wart usually appears as a single lesion or as a cluster. Plantar warts, however, do not stick up above the surface like common warts. The ball of the foot, the heel and the plantar part of the toes are the most likely locations for the warts because the skin in those areas is subject to the most weight, pressure and irritation, making a small break or crack more likely.

Plantar warts are familiar to all ages groups, appearing frequently in children between the ages of 12-16. Adolescents often come into contact with a wart virus in a locker room, swimming pool area, or by walking barefooted on dirty surfaces. The blood vessels feeding them are the black dots that are visible on the wart. If left untreated, these warts can grow to an inch or more in circumference and spread into clusters of several warts. They are known to be very



Cluster of warts on finger. (Custom Medical Stock Photo. Reproduced by permission.)

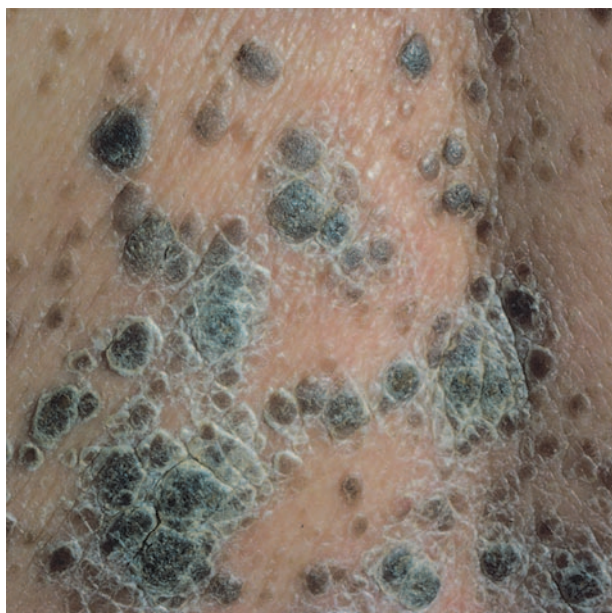
painful at times, the **pain** usually compared to the feeling of a permanent stone in the shoe particularly if the wart is on a pressure point of the foot. People with **diabetes mellitus** are prone to complications from plantar warts related to the development of sores or ulceration and the poor healing potential associated with diabetes.

Flat warts

Flat warts tend to grow in great numbers and are smaller and smoother than other warts. They can erupt anywhere, appearing more frequently on the legs of women, the faces of children, and on the areas of the face that are shaved by young adult males.

Genital warts

Genital warts, also called condyloma acuminata or venereal warts, are one of the most common causes of sexually transmitted disease (STD) in this country. According to the *Journal of the American Medical Association's* STD Information Center, they are contracted by sexual contact with an infected person who carries HPV and are more contagious than other warts. It is estimated that two-thirds of the people who have sexual contact with a partner with genital warts will develop the disease within three months of contact. As a result, about one million new cases of genital warts are diagnosed in the United States each year.



Seborrheic warts appearing on this patient's back. (Custom Medical Stock Photo. Reproduced by permission.)

Genital warts tend to be small flat bumps or they may be thin and tall. They are usually soft and not scaly like other warts. In women, genital warts appear on the genitalia, within the vagina, on the cervix, and around the anus or within the rectum. In men, genital warts usually appear on the tip of the penis but may also be found on the scrotum or around the anus. Genital warts can also develop in the mouth of a person who has had oral sexual contact with an infected person.

Diagnosis

Patients who notice warts in their genital area should see a doctor. The doctor may be able to diagnose the warts with a simple examination. If the warts are small, the doctor may put a vinegar-like liquid on the skin, which makes the warts turn white and easier to see, and then use a magnifying glass to look for them.

Treatment

Home/self treatment

Many of the nonprescription wart remedies available at drug stores will remove simple warts from hands and fingers. These medications may be lotions, ointments, or plasters and work by chemically removing the skin that was affected by the wart virus. The chemicals are strong, however, and should be used

with care since they can remove healthy as well as infected skin. These solutions should be avoided by diabetics and those with cardiovascular or other circulatory disorders whose skin may be insensitive and not appreciate irritation.

Flat warts are best treated with topical retinoids (retinoic acid) or a gel containing salicylic acid. The acid doesn't actually kill the wart virus, but waterlogs the skin so that the surface layer, with the virus, peels off. These products can take up to three months of treatment depending on the size and depth of the wart. Patches are also good to use. Rather than applying drops, a small pad is placed on the wart and left for 48 hours and then replaced with a new one. The patch usually contains a higher concentration of salicylic acid and may irritate the surrounding skin. If this occurs, patients should switch to a gel or stop medication for a period. To help the healing process for flat facial warts, men should shave with an electric shaver or temporarily grow a beard. Women with flat warts on areas that are shaved should use other methods to remove hair such as depilatory cream or wax.

Professional treatment

Physicians should be consulted if there are no signs of progress after a month of self treatment. Doctors have many ways of removing warts, including using stronger topically applied chemicals than those available in drugstores. Some of these solutions include podofilox, topical podophyllum, and trichloroacetic acid (TCA). Some burning and discomfort for one or more days following treatment can be expected. Although these chemicals are effective, they may not destroy all warts completely. A second method of removal is freezing or cryosurgery on the wart using liquid nitrogen. **Cryotherapy** is relatively inexpensive, does not require anesthesia, and usually does not result in scarring. Although temporarily uncomfortable, it provides an effective and safe way to deliver freezing temperatures to a particular area on the skin, and healing is usually quick. Physicians may also choose to burn the wart with liquid nitrogen or numb the skin and then scrape off the wart. Another removal process is electrocautery (electric burning), destroying the wart by burning it with an electric needle. Laser surgery is also becoming a more common option for removing warts.

Genital warts are the most difficult to treat. They can be removed, but the viral infection itself cannot be cured. Often, because the warts are so small, more than one treatment may be needed. The virus continues to live in the deeper skin, which is why warts often return after they have been removed. Strong

KEY TERMS

Condyloma acuminata—Another term for genital warts.

Cryotherapy—Freezing with liquid nitrogen for removal.

Endometritis—Inflammation of the endometrium or mucous membrane of the uterus.

Epidermis—The outer layer of human skin.

Human papilloma virus (HPV)—A family of viruses that causes hand warts, foot warts, flat warts and genital warts.

Retinoic acid—Vitamin A1 acid which is used topically to treat acne.

Salicylic acid—An agent prescribed in the treatment of hyperkeratotic skin conditions and fungal infections.

chemicals may be applied as well as surgical excision with or without electrocautery. This therapy requires a small operative procedure and a local anesthetic. Laser therapy, although more expensive, is often used for treating venereal warts that are more extensive. The use of lasers which vaporize the lesion can theoretically transmit the HPV. It is not at all clear, however, if this occurs.

There is no one recommended method for eliminating plantar warts. If detected early, cryotherapy is usually enough. However, they can be very resilient, requiring treatment over several months. Treatment ranges from the conservative approach of applying chemical solutions to the more aggressive option of surgery. Patients with diabetes or vascular disease are usually treated with the more conservative methods.

Alternative treatment

There are a variety of alternative approaches to the treatment of warts. The suggestions described below apply to common warts and plantar warts, not to genital or cervical warts. Since genital and cervical warts are transmitted sexually, they should be treated by a physician.

For the treatment of common or plantar warts, alternative practitioners may recommend these remedies.

- Apply a paste made of vitamin C powder to the wart for one to two weeks.

- Place a crushed or sliced garlic clove over the wart for seven consecutive nights while sleeping.
- Soak the wart in water, put cross-hatches over it with a sterile needle, and apply drops of thuja (*Thuja occidentalis*) tincture onto the wart. Repeat the cross-hatching and tincture application until the wart is saturated with the tincture. Repeat several times each day for one to two weeks. (A tincture is an herbal extract made with alcohol.)
- Tape a piece of banana peel, latex side down, over the wart and leave it on overnight. Repeat nightly for one to two weeks.

Because warts are caused by a virus, general immune system support can be effective in helping to keep warts from coming back after treatment or to keep them from multiplying or growing. Eating a well balanced diet high in sources of **vitamins A, C, and E** can help strengthen the immune system. Avoiding stress, which is believed to compromise the immune system, is also helpful.

Prognosis

Even though genital warts may be removed, the virus itself continues to live. The HPV can cause tissue changes in the cervix of women with cervical infection. The general recommendation for women who have a history of genital warts is to see their doctors every six months for Pap smears to monitor any changes that may occur.

For plantar warts, the treatment goal is to destroy the wart and its virus without causing much damage to healthy skin. It is not unusual for treatment to cause pain until the foot heals because of the weight put on the foot.

Prevention

Genital warts can be prevented by using condoms and avoiding unprotected sex. Barrier protection will not, however, prevent the spread of wart-causing HPV to uncovered areas such as the pubis and upper thighs. Plantar warts can be prevented by wearing shoes, changing shoes daily, keeping feet clean and dry, and not ignoring skin growths and changes in the skin.

Resources

ORGANIZATIONS

American Academy of Dermatology. 930 N. Meacham Road, P.O. Box 4014, Schaumburg, IL 60168-4014. (847) 330-0230. Fax: (847) 330-0050. <<http://www.aad.org>>.

American Academy of Family Physicians. 8880 Ward Parkway, Kansas City, MO 64114. (816) 333-9700. < <http://www.aafp.org> >.

American Podiatric Medical Association. 9312 Old Georgetown Road, Bethesda, MD 20814-1698. (301) 571-9200. < <http://www.apma.org> >.

Dermatology College of Medicine. The University of Iowa, 200 Hawkins Dr., Iowa City, IO 52242. (319) 356-2274. < <http://tray.dermatology.uiowa.edu> >.

Ruthan Brodsky

Water pills see **Diuretics**

Water therapy see **Hydrotherapy**

Waterhouse-Friderichsen syndrome see **Meningococemia**

Weber test see **Hearing tests with a tuning fork**

Wechsler intelligence test

Definition

The Wechsler Intelligence Scales are a series of standardized tests used to evaluate cognitive abilities and intellectual abilities in children and adults.

Purpose

The Wechsler Intelligence Scales for Children (regular, revised, and third edition) and Wechsler Preschool and Primary Scale of Intelligence are used as tools in school placement, in determining the presence of a learning disability or a developmental delay, in identifying giftedness, and in tracking intellectual development.

The Wechsler Adult Intelligence Scales (regular and revised) are used to determine vocational ability, to assess adult intellectual ability in the classroom, and to determine organic deficits. Both adult and children's Wechsler scales are often included in neuropsychological testing to assess the brain function of individuals with neurological impairments.

Precautions

Intelligence testing requires a clinically trained examiner. The Wechsler scales should be

administered, scored, and interpreted by a trained professional, preferably a psychologist or psychiatrist.

Description

All of the Wechsler scales are divided into six verbal and five performance subtests. The complete test takes 60-90 minutes to administer. Verbal and Performance IQs are scored based on the results of the testing, and then a composite Full Scale IQ score is computed. Although earlier editions of some of the Wechsler Scales are still available, the latest revisions are described below:

Wechsler Adult Intelligence Scale-Revised (WAIS-R)

The WAIS-R, the 1981 revision of the original Wechsler Adult Intelligence Scale, is designed for adults, age 16-74. The 11 subtests of the WAIS-R include information, digit span, vocabulary, arithmetic, comprehension, similarities, picture completion, picture arrangement, block design, object assembly, and digit symbol. An example of questions on the subtest of similarities might be: "Describe how the following pair of words are alike or the same—hamburger and pizza." A correct response would be "Both are things to eat."

Wechsler Intelligence Scale for Children, Third Edition (WISC-III)

The WISC-III subtests includes many of the same categories of subtests as the WAIS-R. In addition, there are two optional performance subtests: symbol search and mazes.

Wechsler Preschool and Primary Scale of Intelligence (WPPSI)

The WPPSI is designed for children age 4-6½ years. The test is divided into six verbal and five performance subtests. The eleven subtests are presented in the following order: information, animal house and animal house retest, vocabulary, picture completion, arithmetic, mazes, geometric design, similarities, block design, comprehension, and sentences.

The 1997 Medicare reimbursement rate for psychological and neuropsychological testing, including intelligence testing, is \$58.35 an hour. Billing time typically includes test administration, scoring and interpretation, and reporting. Many insurance plans cover all or a portion of diagnostic psychological testing.

KEY TERMS

Norms—Normative or mean score for a particular age group.

Representative sample—A random sample of people that adequately represents the test-taking population in age, gender, race, and socioeconomic standing.

Standard deviation—A measure of the distribution of scores around the average (mean). In a normal distribution, two standard deviations above and below the mean includes about 95% of all samples.

Standardization—The process of determining established norms and procedures for a test to act as a standard reference point for future test results.

Normal results

The Wechsler Intelligence Scales are standardized tests, meaning that as part of the test design, they were administered to a large representative sample of the target population, and norms were determined from the results. The scales have a mean, or average, standard score of 100 and a standard deviation of 15. The standard deviation indicates how far above or below the norm the subject's score is. For example, a ten-year-old is assessed with the WISC-III scale and achieves a full-scale IQ score of 85. The mean score of 100 is the average level at which all 10-year-olds in the representative sample performed. This child's score would be one standard deviation below that norm.

While the full-scale IQ scores provide a reference point for evaluation, they are only an average of a variety of skill areas. A trained psychologist will evaluate and interpret an individual's performance on the scale's subtests to discover their strengths and weaknesses and offer recommendations based upon these findings.

Resources

ORGANIZATIONS

American Psychological Association (APA). 750 First St. NE, Washington, DC 20002-4242. (202) 336-5700. <<http://www.apa.org>>.

Catholic University of America. Washington, DC 20064. (800) 464-3742. <<http://www.ericacae.net>>.

Paula Anne Ford-Martin

Wegener's granulomatosis

Definition

Wegener's granulomatosis is a very rare disease that affects many different organs and systems of the body. It mainly attacks the respiratory system (sinuses, nose, windpipe, and the lungs) and the kidneys. One of the main features of the disease is an inflammation of the blood vessels (**vasculitis**). The inflammation narrows the blood vessels and reduces the blood flow to the affected organs. This destroys tissues and damages vital organs.

Description

Wegener's granulomatosis (WG) is not a contagious disease, and there is no evidence to suggest that it is hereditary either. It is a very rare disease, affecting only 1 in every 30,000-50,000 people. About 500 new cases are diagnosed each year. The disease can occur at any age, however, it mostly affects individuals in their 30s and 40s. It affects males and females equally. Ninety seven percent of all patients are Caucasian, 2% are Black and 1% are of another race.

Causes and symptoms

No viral, bacterial, or other causative agent has yet been identified for WG. It is thought to be an autoimmune disease, meaning that the body's immune system attacks "itself," that is, the body's own tissues.

Whenever there is an infection in the body, proteins called antibodies, which are capable of attacking the infectious agent, are formed in the blood. In WG, the antibodies that are formed are directed against the white blood cells of the immune system. They are therefore called "auto-antibodies" (antibodies against one's own body cells). These auto-antibodies bind to the blood cells and forms clumps known as immune complexes. The complexes accumulate in the tissues and the blood vessels, leading to a tumor-like (granulomatous) inflammation of the blood vessels. This slows down the blood flow to the different organs and tissues, causing damage and resulting in the many symptoms of WG.

The symptoms of WG, and the severity of the symptoms, vary from patient to patient. One of the most common features is a chronic runny nose and other cold-like symptoms that do not respond to standard treatment. The cold symptoms gradually worsen and could lead to **sinusitis** (inflammation of the sinuses), middle ear infection (**otitis media**), **cough**,

coughing of blood, and inflammation of the lung (pleuritis and **pneumonia**). Other symptoms include **fever**, fatigue, loss of appetite, weight loss, joint **pain**, night sweats, change in urine color, and weakness.

Kidney (renal) disease is the most serious development of WG. Patients who do not have renal disease are said to have "Limited Wegener's."

Diagnosis

Early diagnosis is critical for the most effective treatment of the disease. However, there are no specific laboratory tests for WG. Blood tests are used to rule out other causes of the symptoms and to determine which organs are affected. The blood tests often show anemia (low red cell count) and high white blood cell counts. If the kidneys are involved, red blood cells are seen in the urine when viewed under a microscope. Also, blood tests aimed at measuring kidney function may show abnormalities.

Chest x rays are used to determine if the lungs are involved. Computed tomography scans (CT scans) of sinuses and lungs, and kidney biopsy, are also important tools used in diagnosing WG.

A specific type of antibody called anti-neutrophil cytoplasmic antibody (ANCA) is seen in the blood of about 90% of the patients with WG. The ANCA are a group of antibodies directed against the individual's own white blood cells (namely, the neutrophils). These anti-neutrophil cytoplasmic antibodies are also found in other inflammatory conditions and diseases (such as HIV infection). Though the ANCA test is useful, it cannot be used by itself to make a diagnosis of WG. However, the amount of ANCA in the blood can be measured and correlates well with the progression of the disease. When there is a relapse or a flare-up, the ANCA levels go up. Levels decrease when the disease is controlled by appropriate treatment.

Since there are no definitive laboratory tests for WG, and the initial symptoms of the disease are not very specific, it takes five-15 months, on an average, to make a diagnosis of WG.

Treatment

Cyclophosphamide (Cytoxan) which is an anti-cancer drug, and **corticosteroids**, such as prednisone, are used to treat WG. These are powerful drugs that suppress the immune system. However, they are also very toxic and can have serious side effects. The patient has to be watched carefully by the doctors and the dosage of the drugs has to be adjusted, if needed.

KEY TERMS

Auto-antibodies—An antibody that is produced in, and reacts with, an antigen in the same person or animal.

Autoimmune disease—Any disease which causes tissue injury due to an immunological reaction of antibodies against the patient's own tissues.

Granulomatous—Resembling a tumor made of granular material.

Immune complexes—Clusters or aggregates of antigen and antibody bound together.

Vasculitis—Inflammation of the walls of the blood vessels.

Since the patient's immune system is suppressed while on these drugs, he or she is at an increased risk for contracting infections. Vaccinations for flu and pneumonia are recommended.

Prognosis

In the past, approximately 80% of the patients with untreated WG died within a year of contracting the disease and 90% died within two years. Today, however, the prognosis has been dramatically improved. With appropriate treatment, patients can survive for much longer periods and lead relatively normal lives.

Approximately 50% of the patients with WG will have a relapse of the disease. This generally happens within two years of stopping the medication, but can occur at any point either during treatment or after stopping treatment. Therefore, it is extremely important that patients continue to see their doctors regularly even after stopping the medications.

Prevention

At present, there are no preventive measures known for Wegener's granulomatosis.

Resources

ORGANIZATIONS

National Organization for Rare Disorders. P.O. Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673. <<http://www.rarediseases.org>>.

Wegener's Foundation, Inc. 3705 South George Mason Drive, Suite 1813 South, Falls Church, VA 22041. (703) 931-5852.

Wegener's Granulomatosis Support Group, Inc. P.O. Box 28660, Kansas City, MO 64188-8660. (800) 277-9474. < <http://www.wgsg.org/wgsg> > .

Lata Cherath, PhD

Weight loss drugs

Definition

Weight loss drugs are medications that may help an obese person lose weight in combination with a low-calorie diet and physical activity.

Purpose

More than 60% of American adults are overweight or obese. Since the 1980s the number of overweight people has increased steadily and the number of obese people has almost doubled. Excess weight and physical inactivity account for more than 300,000 premature deaths in the United States each year, second only to smoking-related deaths.

Risks from Obesity

Overweight and obese people are at higher risk for developing the following problems:

- sleep apnea (interrupted breathing while asleep)
- osteoarthritis (wearing away of the joints)
- high blood pressure
- heart disease
- **stroke**
- diabetes
- gallbladder disease
- gout (joint **pain** caused by excess uric acid)
- many common types of cancer

Body mass index (BMI)

An overweight person has an excess of body weight but not necessarily an excess of body fat. **Obesity** refers to an excessively high proportion of body fat. The BMI is used to determine whether a person is overweight or obese. It defines body weight relative to height and, for most adults, correlates with total body fat.

BMI charts are widely available; however, an approximate BMI can be calculated by the following steps:

- multiplying body weight in pounds by 703
- dividing the result by height in inches
- dividing the result by height in inches again

With the exception of very muscular people, BMI is a measure of health:

- 18.5–24.9: healthy
- 25–29.9: overweight
- 30 or above: obese

Generally the higher the BMI the higher the risk of health problems; however, excess abdominal fat is a separate health risk. Men with waists of more than 40 in (102 cm) and women with waists of more than 35 in (89 cm) are at greater risk for health problems.

Prescription weight loss drugs are approved for use only by people with BMIs of 30 or above or those with BMIs of 27 and above who have other risk factors, including the following:

- high blood pressure
- type 2 diabetes
- dyslipidemia (abnormal amounts of fat in the blood)
- sleep apnea

Even short-term weight loss in some obese people has been found to reduce the following:

- blood pressure
- blood cholesterol
- triglycerides (fats)
- insulin resistance (the inability to utilize blood sugar)

Overweight or obese patients who have not lost an average of 1 lb (0.5 kg) per week after at least six months on a balanced low-calorie diet, including increased **exercise** and behavior therapy, may be considered candidates for weight loss drugs.

Description

Weight loss medications are prescribed only as a component of a weight loss program that includes a healthy low-calorie diet and regular physical activity. On average, people on prescription weight loss drugs lose about 5–10% of their original weight. They may lose in the range of 4.4–22 lb (2–10 kg) above what they might have lost following a diet and exercise program alone. Patients who lose weight initially on a drug tend to continue to respond. However, those

who fail to lose 4 lb (2 kg) within the first month on a weight loss medication are unlikely to respond to that particular drug even at an increased dosage. Over-the-counter weight loss drugs have been found to be ineffective, unsafe, or both.

Most weight loss occurs within the first six months of drug use, after which the weight levels off or increases. The majority of patients regain the lost weight after discontinuing the medication. Thus short-term use of weight loss drugs may not be beneficial. However, long-term use of weight loss drugs may help maintain the lower weight. Since obesity is a chronic disorder, treatment must be continued for years or for life. However, as of 2005, sibutramine (Merida) and orlistat (Xenical) were the only weight loss drugs approved by the U. S. Food and Drug Administration (FDA) for use for more than a few weeks.

Appetite suppressants (anorexiant)

Most FDA-approved weight loss drugs suppress appetite by affecting one or more neurotransmitters in the brain that control appetite and mood. Various appetite suppressants increase the secretion of dopamine, norepinephrine, or serotonin or inhibit the reuptake of these neurotransmitters into the nerve cell. Some drugs affect more than one neurotransmitter. These drugs may cause a person to feel less hungry or more satiated.

SIBUTRAMINE. Sibutramine was approved by the FDA in 1997 for long-term use. It reduces appetite by inhibiting the reuptake of norepinephrine, dopamine, and serotonin. One study found that patients taking sibutramine lost an average of 7–10 lb (3–5 kg) more over one year than those on a low-calorie diet alone.

AMPHETAMINES OR SYMPATHIMIMETIC DRUGS. Some amphetamines are FDA-approved for very short-term weight loss. Although most of these drugs do not cause weight loss over more than a few weeks, phentermine continues to be one of the two most commonly prescribed weight loss medications—along with sibutramine—in the United States.

Some amphetamine-type weight loss drugs, including methamphetamine (Desoxyn), were approved by the FDA in the mid-1990s based on very little data. Other FDA-approved weight loss drugs for short-term use include:

- benzphetamine (Didrex)
- diethylpropion (Tenuate, Tenuate Dospan, Tepanil Ten-Tab), approved in 1959

- mazindol (Mazanor, Sanorex)
- phendimetrazine (Adipost, Bontril PDM, Bontril Slow-Release, Melfiat, Obezine, Phendiet, Phendiet-105, PT 105, Plegine, Prelu-2, X-Trozone), approved in 1982
- phentermine (Adipex-P, Fastin, Ionamin, Obenix, Oby-trim, Phentercot, Phentride, Pro-Fast, Teramine, Zantryl), approved in 1959

Lipase inhibitors

In 1999 the FDA approved orlistat (the first of a new class of anti-obesity drugs called lipase inhibitors) for long-term use. Orlistat inhibits the pancreatic enzyme lipase that breaks down dietary fat, which in turn decreases the body's absorption of dietary fat by as much as 30%. The undigested fat is excreted in the stool.

Orlistat is prescribed for overweight or obese patients who also have **high cholesterol**, diabetes, high blood pressure, or heart disease

Off-label use

Physicians sometimes prescribe drugs for weight loss that are not FDA-approved for that purpose—so-called off-label use. The use of weight loss medications—other than sibutramine and orlistat—for more than a few weeks is also considered off-label use.

Off-label weight loss drugs include:

- antidepressants such as bupropion
- the **anticonvulsant drugs** topiramate and zonisamide
- the diabetes medication metformin

Antidepressants may result in moderate weight loss for up to six months after which the weight usually is regained. Metformin appears to reduce hunger and food intake and may promote small weight loss in the obese and those with type 2 diabetes.

Drugs withdrawn from the market

Although effective for long-term weight loss, fenfluramine (Pondimin and others), dexfenfluramine (Redux), and a fenfluramine/phentermine combination (fen/phen) were withdrawn from the market in 1997 after it was found that they caused potentially fatal complications.

In February of 2004 the FDA prohibited the sale of dietary supplements containing ephedrine alkaloids, including ephedra, Ma huang, Sida cordifolia, and pinellia, because of potential complications. These supplements were aggressively marketed for weight

control, as well as for energy and sports performance. Although they may effect modest short-term weight loss, these drugs raise blood pressure, **stress** the circulatory system, and have been linked to **heart attack** and stroke.

Recommended dosage

Dosages of appetite suppressants vary with the individual and a full dose may be more than necessary for some people. A low dose of subitramine often is prescribed for at least the first four weeks. Once-a-day appetite suppressors should be taken 10–14 hours before bedtime. When taking more than one dose per day, the last dose should be 4–6 hours before bedtime. Long-acting capsules or tablets should be swallowed whole, rather than chewed or cut up. Missed doses should be skipped.

The average doses of appetite suppressants are:

- sibutramin: a 10-mg capsule once a day, usually in the morning; a maximum of 15 mg per day
- benzphetamine: 25 to 50-mg tablets once a day in midmorning or mid-afternoon
- diethylpropion: 25-mg tablets, three times per day, one hour before meals; 75-mg extended-release tablet, once a day in midmorning
- mazindol: 1-mg tablet once a day, taken with food
- phendimetrazine: 17.5–35-mg tablets, two to three times per day, one hour before meals; 105-mg extended-release capsules, 30 to 60 minutes before the morning meal
- phentermine: 15–37.5-mg capsule or tablet, once a day, before breakfast or one to two hours after or smaller doses 30 minutes before meals; 15–30-mg oral resin capsule, once-per-day, before breakfast

The usual dose of orlistat is a 120-mg capsule taken three times a day, up to one hour after each main meal that contains fat. The meal should contain no more than 30% calories from fat. Orlistat should not be taken if a meal is missed, the meal does not contain fat, or it is more than one hour past the meal.

Precautions

With the exception of orlistat, all prescription weight loss medications are controlled substances. Very few studies have evaluated for their safety or effectiveness for periods of more than six months to two years. The use of any combination of weight loss drugs is not recommended. The risks of using appetite-suppressing drugs during **pregnancy** are unknown as

of 2005. Orlistat is approved for use in those aged 12 and above. No other weight loss drug is approved for use in children under age 16.

After beginning a weight loss drug, patients should return for follow-ups in two to four weeks, monthly for three months, and then every three months for the first year, to monitor weight, blood pressure, pulse, and side effects, and have laboratory testing.

Appetite suppressants

Appetite suppressants should not be taken within 14 days of taking a monoamine oxidase (MAO) inhibitor; extremely high blood pressure and seizures may develop suddenly. Taking subitramine along with a drug that has MAO-inhibitor activity can cause a life-threatening condition called serotonin syndrome.

Sibutramine should not be used by people with the following problems:

- poorly-controlled high blood pressure
- irregular heartbeat (arrhythmia)
- a history of stroke
- heart disease
- congestive heart failure

Appetite suppressants, including sibutramine, can be habit-forming. In late 2004 Meridia was singled out by critics as a drug that is only minimally effective and has been associated with several deaths.

Amphetamine-type weight loss drugs should not be use by anyone with the following problems:

- high blood pressure
- heart disease
- an overactive thyroid gland
- glaucoma

Amphetamine-type weight loss drugs can test positive on urine screenings for amphetamines. These drugs have a high potential for **abuse** and **addiction**.

Orlistat

Orlistat may decrease the absorption of fat-soluble **vitamins** and beta-carotene. Therefore, a daily multivitamin supplement, including vitamins A, E, K, and beta-carotene, should be taken at least two hours before or after orlistat or at bedtime. Foods containing more than 30% fat should be avoided while a person is using orlistat. Gallbladder problems may become worse with orlistat.

Other precautions

Various conditions may affect the choice of a weight loss drug, including the following:

- pregnancy or breastfeeding
- a history of drug or alcohol abuse or dependence
- a history of eating disorders
- a history of depression or manic-depressive disorder
- a family history of mental illness
- the use of MAO inhibitors or other antidepressants
- migraine headaches requiring medication
- glaucoma
- diabetes mellitus
- epilepsy
- thyroid disease
- kidney problems
- gallstones
- liver disease
- problems absorbing food (malabsorption syndrome)
- high blood pressure
- heart disease, a heart condition such as arrhythmia, blood vessel disease
- future surgery that will require general anesthesia

Some dietary supplements are marketed as weight loss products. They are not reviewed by the FDA and are not necessarily safe. They may interact with other drugs and may be dangerous for people with certain medical conditions. Herbal medications have unpredictable amounts of active ingredients and unpredictable and potentially harmful side effects.

Side effects

Many overweight people are generally healthy and the side effects of weight loss drugs may far outweigh the benefits. Side effects may be reduced by starting at the lowest effective dose, although any drug can cause an allergic reaction in any individual.

Appetite suppressants

Side effects of sibutramine and other appetite suppressants may include:

- increased blood pressure
- **dry mouth**
- headache
- abdominal cramps or pain

- constipation
- **nausea** or vomiting
- drowsiness
- insomnia
- irritability
- **anxiety**
- dizziness, lightheadedness, confusion
- poor coordination

Dry mouth can be relieved with sugarless candy or gum, ice, or a saliva substitute. However, dry mouth lasting more than two weeks can increase the risk of serious dental disease.

Less common side effects of appetite suppressants include:

- fast or irregular heart beat
- skin rash or hives
- fever
- sore throat
- sweating
- uncontrollable trembling or shaking
- diarrhea
- unusual bruising or bleeding
- depression

Although sibutramine side effects usually are mild and may improve with continued use, additional side effects may include:

- increased thirst
- changes in sense of taste
- heartburn
- stuffy or runny nose
- skin burning, **itching**, prickling, or tingling
- back pain
- painful menstruation
- weakness
- swelling in various parts of the body

Fast heart rate can be a symptom of sibutramine overdose. Other rare serious complications from sibutramine include:

- chest pain
- difficulty speaking, swallowing, or breathing
- muscle stiffness
- uncoordinated or abnormal movement
- fainting
- rapid mood shifts

- enlarged pupils, eye pain, vision changes
- confusion
- seizures

Other side effects of amphetamines can include:

- a false sense of well-being
- an unpleasant taste
- decreased alertness
- restlessness
- uncontrollable activity or excitement
- hallucinations
- increased need to urinate
- difficult or painful urination
- numbness, particularly on one side of the face or body
- blurred vision
- changes in libido or decreased sexual function

Higher dosages, more frequent doses, or longer-term use of amphetamines can lead to physical dependence or overdose, with symptoms that include:

- skin disease
- severe insomnia
- excessive excitability or irritability
- panic
- violent urges
- personality changes
- severe mental illness
- high or low blood pressure
- seizures
- coma

Symptoms of withdrawal may be lessened or prevented by gradually reducing the dosage.

Orlistat

Orlistat side effects usually are mild and can improve with continued use; they may include the following:

- intestinal discomfort or cramping
- flatulence
- frequent, urgent need to defecate
- inability to control bowel movements and leakage
- oily diarrhea
- flu- or cold-like symptoms
- irregular menstrual periods

Less common side effects of orlistate include:

- rectal pain
- back pain

- tooth or gum problems
- chest tightness
- wheezing
- anxiety

Orlistat side effects requiring immediate medical attention include:

- itching, rash, or hives
- fever
- frequent, difficult, or painful urination
- bloody or cloudy urine
- swelling of body parts
- chest pain
- difficulty breathing
- earache or hearing changes

Orlistat side effects can be worsened by eating meals with at least 30% of total calories from fat. Side effects usually disappear within two to three days after discontinuing the drug.

Interactions

Appetite-suppressants can interact negatively with numerous other drugs, including the following:

- other appetite suppressants and amphetamines
- caffeine
- cold and allergy medicines
- **asthma** medications
- selective serotonin reuptake inhibitors (SSRIs)
- tricyclic antidepressants
- medications used during surgical, dental, or emergency procedures
- cocaine and other street drugs

In addition to the above, subitramine is known to interact with numerous other drugs, including the following:

- pain relievers and muscle relaxants
- sulfa antibiotics
- antifungal medications
- sedatives, sleeping pills, tranquilizers
- high-blood-pressure medications
- migraine-headache medications
- medications for nausea, anxiety, mental illness, seizures
- blood thinners such as warfarin (Coumadin)

KEY TERMS

Amphetamines—Sympathomimetic amines; sometimes called speed; synthetic chemicals that stimulate the central nervous system.

Body mass index (BMI)—The correlation of body weight with height; usually correlates with total body fat.

Dopamine—A neurotransmitter and precursor of norepinephrine; found in high concentrations in the brain.

Ephedrine—A sympathomimetic amine from plants of the genus *Ephedra* or chemically synthesized; an asthma medication that previously was used in weight loss drugs.

Lipase—A fat-splitting enzyme found in pancreatic juice, blood, and many tissues.

Neurotransmitter—A substance that helps transmit impulses between two nerve cells or a nerve and muscle.

Norepinephrine—A hormone that constricts blood vessels.

Orlistat—A drug that inhibits lipase.

Phentermine—An amphetamine prescribed for appetite suppression.

Serotonin—5-Hydroxytryptamine; a chemical in various parts of the body that stimulates smooth muscle contraction, constricts blood vessels, and inhibits stomach secretions.

Sibutramine—An appetite-suppressing drug that may increase the activity of norepinephrine and serotonin in the brain.

• chemotherapies

Orlistat may interact with the following:

- other weight loss medications
- diabetes medications, including insulin
- blood thinners
- herbal products
- cyclosporine

Resources

BOOKS

Mitchell, Deborah R., and David Charles Dodson. *The Diet Pill Book: A Consumer's Guide to Prescription and Over-the-Counter Weight-Loss Pills and Supplements*. New York: St. Martin's Press, 2002.

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ORGANIZATIONS

- American Obesity Association*. 1250 24th Street, NW, Suite 300, Washington, DC 20037. (800)986-2373. <<http://www.obesity.org>>.
- Food and Drug Administration*. U.S. Department of Health and Human Services, Public Health Service, 5600 Fishers Lane, Rockville, MD 20857. <<http://www.fda.gov>>.
- Weight-control Information Network (WIN)*. National Institute of Diabetes and Digestive and Kidney Diseases. 1 Win Way, Bethesda, MD 20892-3665. (877)946-4627. <<http://www.niddk.nih.gov/health/nutrit/win.htm>>.

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Weil's disease see **Leptospirosis**

Wermer's syndrome see **Multiple endocrine neoplasia syndromes**

Wernicke-Korsakoff disease see **Alcohol-related neurologic disease**

Western equine encephalitis see **Arbovirus encephalitis**

Western herbalism see **Herbalism, western**

West Nile virus

Definition

West Nile virus is a mosquito-borne viral illness that can manifest with varying seriousness, ranging from no symptoms or mild flu-like symptoms to brain damage and **death**.

Description

West Nile virus (WNV) is a mosquito-borne flavivirus belonging to the **Japanese encephalitis** serocomplex, which includes St. Louis **encephalitis**, Murray Valley encephalitis, and Kunjin virus. Zoonotically maintained, infections occur generally between late summer and early fall in temperate areas, and throughout the year in southern climates. Although typical manifestation of WNV is asymptomatic, the virus can cross the blood-brain barrier and cause severe illness, **paralysis**, and even death in humans and animals.

The WNV was originally isolated in a feverish woman living in the West Nile District of Uganda during 1937. The virus was ecologically characterized in Egypt during the 1950s and later linked to severe human meningoencephalitis in elderly patients during a 1957 outbreak in Israel. Since 1937, subsequent outbreaks of WNV have been reported in Africa, Asia, Australia, Oceania, Western Europe, and the Middle East.

In the summer of 1999, the first North American cases of WNV occurred in the New York City area. It is still unknown how the WNV reached the continental United States, but it is suspected that the transport of infected birds or the international travel of infected humans may have been to blame. After its arrival in the New York area, the virus spread rapidly across the United States, as well as northward into Canada and southward into Mexico. In 2002, a severe outbreak of WNV in the United States killed 284 people and caused 2,944 cases of severe brain damage. During another outbreak a year, 9,858 cases of WNV infection were reported, of which over 2,900 were severe in nature and 262 of which resulted in fatality. As of January 2005, the CDC ArboNET has recorded

avian or animal WNV infections in every state except Alaska, Hawaii, and Washington. Additionally, the CDC ArboNET recorded human WNV infections in every state except Alaska, Delaware, Hawaii, Massachusetts, New Hampshire, Rhode Island, Vermont, Washington, and West Virginia. Experts believe that WNV is now firmly established in the Western Hemisphere.

Life Cycle and Transmission

Like most flaviviruses, the WNV is maintained in a natural host-vector-host cycle, where the primary vector is the mosquito. The zoonotic cycle begins with a reservoir host, which is most commonly of avian origin. When a mosquito feeds on the infected bird, the virus is passed to the insect along with the blood meal. The virus then multiplies rapidly within the mosquito's body and salivary glands over the next few days. When the insect feeds on another animal or human, the virus can be transmitted through the bite and cause serious illness.

Most mosquitoes can become infected with the WNV. However, female mosquitoes of the *Culex pipiens* species are of particular concern, as they live in suburban and urban areas, can survive through the winter, prefer to feed on birds, and frequently bite humans. The *Culex pipiens*, also known as the house mosquito, is also the most common vector for WNV transmission. *Culex restuans*, *Culex quinquefasciatus*, *Aedes Albopictus*, and *Aedes Vexans* are also common carriers of the WNV.

Common food sources for mosquitoes, birds represent the primary WNV reservoir species. American crows, in particular, are extremely susceptible to WNV, and have become the virus' primary host population. Indeed, an unusual crow die-off can be used as an excellent indicator for the regional presence of the WNV. The virus has also been identified in more than 250 bird species in the United States, including blue jays, ravens, magpies, sparrows, and starlings. Many in the scientific community believe that the rapid spread of WNV in North America may be due in part to the migratory nature of birds. Infected birds carry the virus with them as they travel in summer and winter, thus acting as reservoirs in their new nesting sites.

Most vertebrates, such as alligators, bats, chipmunks, skunks, squirrels, and rabbits, can also be infected with WNV. Horses, in particular, are commonly infected with WNV. Like humans, the majority of horses suffer either no or mild symptoms, but severe illness and death can and does occur. There are

relatively few cases of dogs and cats becoming infected with WNV. Animals of all species exhibiting **fever**, weakness, poor coordination, spasms, seizures, and/or personality changes may be infected with WNV.

There is no evidence of WNV transmission from person-to-person through touch, kissing, or other contact. However, there is evidence of WNV transplacental (mother-to-child) transmission, as well as viral transmission through breastfeeding. As such, pregnant mothers should be aware of the presence of WNV in their area and take appropriate precautions. The transmission of WNV has also been evidenced in blood-transfusions and organ transplants; although the current blood supply is now tested for the presence of the WNV. People that are immunocompromised (from disease or **chemotherapy**, for example) and people aged 50 and older represent the highest risk group for serious WNV infection.

Causes and symptoms

The exact mechanism of WNV-caused illnesses remains unclear. However, it is suspected that the virus enters the host's blood stream and multiplies. It can then develop to the point where it crosses the blood brain-barrier, which separates the blood from the central nervous system. When this occurs, the virus can infect the brain, spinal cord, and other vital systems, creating a potentially deadly inflammatory response.

Symptoms

The incubation period for WNV after infection typical ranges between 3 to 14 days. Eighty percent of infected persons will exhibit no clinically apparent symptoms whatsoever. Roughly 20% of infected persons will exhibit a series of mild flu-like symptoms, also known as West Nile Fever. These mild symptoms can persist for 3 to 6 days, possibly weeks, and include:

- eye pain
- fever
- **headache**
- loss of appetite
- lymphadenopathy (abnormal enlargement of the lymph nodes)
- malaise (nonspecific bodily discomfort)
- myalgia (nonspecific muscular pain/tenderness)
- nausea
- rash (on the neck, torso, and limbs)
- vomiting

In rare cases, approximately 1 in 150 cases (0.7%), WNV can cross the blood-brain barrier and develop into a severe neuroinvasive disease. Immunocompromised and elderly (> 50 years of age) patients are at an increased risk for developing more severe syndromes; a 20-fold increase in incidence among older patients. Symptoms indicating the possible presence of severe West Nile-related syndromes include:

- severe headache
- high fever
- acute muscle weakness
- neck stiffness
- convulsions and tremors
- disorientation and stupor
- paralysis
- coma

People exposed to WNV infection, especially the immunocompromised and elderly, should contact their health provider immediately if they develop a severe headache accompanied by high fever.

Typically, severe WNV syndromes manifest as one of three syndromes: West Nile encephalitis (inflammation of the brain); West Nile **meningitis** (inflammation of the meninges of the brain and spinal cord); or West Nile meningoencephalitis (inflammation of both the brain and the meninges). These three syndromes can cause severe brain damage and even death. Severe WNV disease carries a mortality rate ranging between 3% and 15%, with elderly patients suffering the highest mortality rate. The majority of these deaths are as a result of complication attributable to West Nile meningoencephalitis. Additionally, severe WNV disease can cause acute vision loss due to inflammatory disorders of the eye, such as chorioretinitis, **optic neuritis**, retinal **vasculitis**, **uveitis**, and vitritis. Less frequently, the patient can exhibit acute flaccid paralysis, similar to poliomyelitis (**polio**) or Guillain-Barré syndrome, caused by inflammation of the spinal cord and/or damage to the peripheral nerves. In some severe cases, this acute flaccid paralysis can disrupt muscles that control breathing and result in **respiratory failure**.

Diagnosis

A proper diagnosis of WNV infection depends heavily upon clinical presentation, laboratory testing, and patient history. Patients with a known

susceptibility to WNV (the elderly and immunocompromised) that exhibit symptoms during the late spring to early fall, or at any time in warmer climates, should be tested for WNV and other arboviral infections. Additionally, health providers should remain constantly aware of the local presence of WNV activity, such as reports of recent animal and/or human cases. Similarities of symptomology between and serological cross-reactivity of WNV and other flaviviruses, may lead to confusion and an incorrect diagnosis. Health providers must use thorough laboratory testing to differentiate WNV antibodies from those of other arboviruses.

Symptomatic WNV infection can be classified as either non-neuroinvasive or neuroinvasive, with each being identified according to certain criteria.

Non-neuroinvasive

The majority of WNV infections are asymptomatic. In approximately 20% of WNV cases, clinically recognizable symptoms can manifest. However, to be clinically classified as non-neuroinvasive West Nile disease, the following must be true:

- no neuroinvasive symptomology
- presence of fever without other recognizable cause
- four-fold or greater increase in serum antibody titer
- virus isolated from and or demonstrated in blood, tissue, cerebrospinal fluid (CSF), or other bodily fluid
- virus-specific immunoglobulin M (IgM antibodies demonstrated in CSF through antibody-capture methods

Neuroinvasive

In the rare cases (0.7%) of West Nile disease, the virus crosses the blood-brain barrier and manifests in severe and life-threatening symptomology. Clinical confirmation of neuroinvasive of neuroinvasive disease requires the presence of a fever and at least one of the following:

- acutely altered mental status, such as disorientation, stupor, or coma
- acute central or peripheral neurological difficulties, such as paralysis, nerve palsy, sensory deficits, and abnormal muscle function
- an increased white blood cell concentration in the CSF coupled with symptoms of meningitis, such as severe headache and neck pain

Treatment

Currently, there are no treatment modalities for WNV infection. Instead, supportive care is utilized to treat the varying symptoms and syndromes associated with the various West Nile diseases. Although milder symptoms can be treated at home, severe symptoms can require hospitalization. Treatment of severe symptoms may require the use of intravenous infusions, airway and respiratory management and support, and use of preventative measures against secondary infection.

Prognosis

The majority of WNV infections will manifest asymptotically. West Nile fever offers an excellent prognosis associated with quick recovery and no adverse side-effects. The majority of symptoms will resolve within a few days or weeks of manifestation.

However, the prognosis is not a positive for patients suffering the more severe syndromes attributable to WNV infection. Symptoms of West Nile encephalitis, West Nile meningitis, and West Nile meningoencephalitis can last for several weeks, as well as cause severe and permanent neurological damage. Inflammation can interfere with the brain and central nervous system and result in death, especially amongst the elderly population. Patients with West Nile poliomyelitis may suffer prolonged muscle weakness and loss of motor control. Long-term **rehabilitation** is typically required and a full recovery is not assured. If the poliomyelitis affects muscles used for breathing, death from respiratory failure may result.

Prevention

Although there is a vaccine used for horses and exotic birds in zoos, there is no WNV vaccine for humans at the current time. Several pharmaceutical companies, however, have WNV vaccines in development.

Prevention techniques of WNV typically coincide with avoidance measures against mosquito bites; the primary source of the virus. These include the use of insect repellent (with 5% to 20% DEET) on exposed body parts, wearing loose-fitting clothes over the limbs and torso while outdoors, using mosquito coils and/or citronella candles outdoors, and limiting outdoor activities during peak biting periods and/or in areas with high mosquito density. While camping outdoors, knockdown spray or bed netting with pyrethrum is suggested. Mosquito eradication programs have been instituted in most major cities. Public health

KEY TERMS

Flavivirus—An arbovirus that can cause potentially serious diseases, such as dengue, yellow fever, Japanese encephalitis, and West Nile fever.

Guillain-Barré—A disorder in which the body's immune system attacks part of the peripheral nervous system. Weakness, tingling, and abnormal sensations in the arms and upper body can progress until the muscles become totally disabled and the patient is effectively paralyzed.

Meninges—A series of membranous layers of connective tissue that protect the central nervous system (brain and spinal cord). Damage or infection to the meninges, such as in meningitis, can cause serious neurological damage and even death.

Zoonotic diseases—Diseases caused by infectious agents that can be transmitted between (or are shared by) animals and humans. This can include transmission through the bite of an insect, such as a mosquito.

authorities can utilize United States Environmental Protection Agency-approved “adulticides” in areas suspected of the presence of WNV.

The *Culex pipiens* mosquito is the primary vector of WNV transmission and is also commonly live and feed in urban areas. Special precautions should be taken to reduce exposure to these potentially infected insects. Screen doors and enclosed porches can help keep mosquitoes from coming into the house. It should be noted that studies have shown that mosquito control devices such as “bug zappers” and CO₂-baited traps do not significantly reduce the risk of being bitten.

Removing potential mosquito breeding areas from near the home and from the neighbourhood can further reduce the risk of bites. Any container which can collect half an inch of standing water can become a potential breeding site in as little as five days. Old tires, empty plant pots, and empty trashcans should be removed, while water sources like ponds or birdbaths should be cleaned regularly. Standing water on any property should be drained, such as from clogged eaves. Swimming pools and hot tubs should be properly covered and chlorinated to prevent mosquitoes breeding in them.

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ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd, Atlanta, GA 30333. (800)311-3435. <<http://www.cdc.gov>>.

Jason Fryer

Wheezing

Definition

Wheezing is a high-pitched whistling sound associated with labored breathing.

Description

Wheezing occurs when a child or adult tries to breathe deeply through air passages that are narrowed or filled with mucus as a result of:

- allergy
- infection
- illness
- irritation

Wheezing is most common when exhaling. It is sometimes accompanied by a mild sensation of tightness in the chest. **Anxiety** about not being able to breathe easily can cause muscle tension that makes matters worse.

Causes and symptoms

Wheezing is the symptom most associated with **asthma**. It can be caused by:

- exposure to allergens (food, pollen, and other substances, that cause a person to have an allergic reaction)
- fumes
- ice-cold drinks, or very cold air
- medication
- strenuous **exercise**
- weather changes
- foreign objects trapped in the airway
- cystic fibrosis, and other genetic disorders
- respiratory illnesses such as **pneumonia**, **bronchitis**, congestive **heart failure**, and emphysema

Diagnosis

A family physician, allergist, or pulmonary specialist takes a medical history that includes questions about **allergies**, or unexplained symptoms that may be the result of allergic reactions. If the pattern of the patient's symptoms suggests the presence of allergy, skin and blood tests are performed to identify the precise nature of the problem.

A **pulmonary function test** may be ordered to measure the amount of air moving through the patient's breathing passages. X rays are sometimes indicated for patients whose wheezing seems to be caused by chronic bronchitis or **emphysema**.

In 2004, researchers in Japan discovered a new method for diagnosing asthma in infants by testing for certain antibodies in their sputum (mucus that spits up from the bronchi).

Treatment

Mild wheezing may be relieved by drinking plenty of juice, water, weak tea, and broth. Ice-cold drinks should be avoided.

A vaporizer can help clear air passages. A steam tent, created by lowering the face toward a sink filled with hot water, placing a towel over the head and sink, and inhaling the steam, can do likewise.

Bronchodilators (medications that help widen narrowed airways) may be prescribed for patients whose wheezing is the result of asthma. Newer asthma medications taken daily can help prevent asthma attacks, as can avoiding asthma and allergy triggers.

Antibiotics are generally used to cure acute bronchitis and other respiratory infections. **Expectorants** (cough-producing medications) or certain

bronchodilators are prescribed to remove excess mucus from the breathing passages.

If wheezing is caused by an allergic reaction, antihistamines will probably be prescribed to neutralize body chemicals that react to the allergen.

Medical emergencies

Breathing problems can be life-threatening. Immediate medical attention is required whenever an individual:

- turns blue or gray and stops breathing
- becomes extremely short of breath, and is unable to speak
- coughs up bubbly-pink or white phlegm
- seems to be suffocating
- develops a **fever** of 101°F (38.3°C) or higher
- wheezes most of the time, and coughs up gray or greenish phlegm

Alternative treatment

Certain **yoga** positions (Bridge, Cobra, Pigeon, and Sphinx) may relieve wheezing by improving breathing control and reducing **stress**. Patients whose wheezing is related to asthma, chronic bronchitis, emphysema, or a severe allergic reaction may benefit from these techniques, but must continue to have their condition monitored by a conventional physician.

Prognosis

Mild wheezing caused by infection or acute illness usually disappears when the underlying cause is eliminated.

Some doctors believe that childhood respiratory infections may activate parts of the immune system that prevent asthma from developing.

Prevention

Stopping **smoking** can eliminate wheezing. So can reducing or preventing exposure to other substances that cause the problem.

Resources

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Maureen Haggerty
Teresa G. Odle

Whiplash

Definition

Whiplash is a sudden, moderate-to-severe strain affecting the bones, discs, muscles, nerves, or tendons of the neck.

Description

The neck is composed of seven small bones. Known as the cervical spine, these bones:

- support the head
- help maintain an unobstructed enclosure for the spinal cord
- influence the shape and structure of the spine
- affect posture and balance

About 1,000,000 whiplash injuries occur in the United States every year. Most are the result of motor vehicle accidents or collisions involving contact sports. When unexpected force jerks the head back, then forward the bones of the neck snap out of position and irritated nerves can interfere with flow of blood and transmission of nerve impulses. Pinched nerves can damage or destroy the function of body parts whose actions they govern.

Risk factors

Osteoarthritis of the spine increases the risk of whiplash injury. So do poor driving habits, driving in bad weather, or driving when tired, tense, or under the influence of alcohol or other drugs.

Causes and symptoms

Tension shortens and tightens muscles. **Fatigue** relaxes them. Either condition increases the likelihood that whiplash will occur and the probability that the injury will be severe.

Sometimes symptoms of whiplash appear right away. Sometimes they do not develop until hours,

days, or weeks after the injury occurs. Symptoms of whiplash include:

- **pain** or stiffness in the neck, jaw, shoulders, or arms
- dizziness
- headache
- loss of feeling in an arm or hand
- nausea and vomiting

Depression and vision problems are rare symptoms of this condition.

Diagnosis

Whiplash is difficult to diagnose because x rays and other imaging studies do not always reveal changes in bone structure. Organs affected by nerve damage or reduced blood supply may generate symptoms not clearly related to whiplash.

Diagnosis is based on observation of the patient’s symptoms, medical history, **physical examination**, and neurological studies to determine whether the spine has been injured.

Treatment

Medication, physical therapy, and supportive measures are used to treat whiplash. Chiropractors gently realign the spine to relax pinched nerves or improve blood flow. A patient whose symptoms are severe may wear a soft, padded collar (Thomas collar or cervical collar) until the pain diminishes.

When pressure on the root of the nerve causes loss of strength or sensation in a hand or arm, a cervical **traction** apparatus may be recommended.

Self-care

Inflammation and cramping can be alleviated by wrapping ice or an ice pack in a thin towel and applying it to the injured area for 10-20 minutes every hour. After the first 24 hours, painful muscle spasms can be prevented by alternating cold packs with heat treatments. Letting a warm shower run on the neck and shoulders for 10-20 minutes twice a day is recommended. Between showers, warm towels or a heat lamp should be used to warm and soothe the neck for 10-15 minutes several times a day.

Improving posture is important, and gentle massage can be beneficial. Sleeping without a pillow promotes healing, and a cervical collar or small rolled towel pinned under the chin can provide support and prevent muscle fatigue.

Alcohol should be avoided. A chiropractor, primary care physician, or orthopedic specialist should be notified whenever a painful neck injury occurs. Another situation requiring attention is if the face or arm weaken or become painful or numb following a neck injury.

Prognosis

With treatment, whiplash can usually be cured in one week to three months after injury occurs. If nerve roots are damaged, **numbness** and weakness may last until recovery is complete.

Prevention

Chiropractors can recommend diet and **exercise** techniques to reduce **stress** and tension. Careful, defensive driving, wearing seatbelts, and using padded automobile headrests can lessen the likelihood of whiplash.

Resources

OTHER

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Maureen Haggerty

Whipple's disease see **Malabsorption syndrome**

White blood cell count and differential

Definition

The white blood cell count and differential determine the number of white blood cells and the percentage of each type of white blood cell in a person's blood. These tests are included in general health examinations and help investigate a variety of illnesses, including infection, allergy, and leukemia.

Purpose

The white blood cell count provides a clue to the presence of illness. White cells protect the body by fighting infection and attacking foreign material. When extra white cells are needed, the bone marrow increases production.

There are five types of white cells, each with different functions: neutrophils, lymphocytes, monocytes, eosinophils, and basophils. The differential reveals if these cells are present in a normal distribution, or if one cell type is increased or decreased. This information helps diagnose specific types of illness.

Conditions or medications that weaken the immune system, such as **AIDS** or **chemotherapy**, cause a decrease in white cells. The white cell count detects dangerously low numbers of white cells.

Recovery from illness can be monitored by the white cell count. Counts continuing to rise or fall to abnormal levels indicate a worsening condition; counts returning to normal indicate improvement.

Description

Neutrophils increase in response to bacterial infection. They destroy bacteria by enveloping and digesting them, a process called phagocytosis. When many neutrophils are needed, they are released from the bone marrow as immature cells, called bands or stab cells.

Lymphocytes fight viral infections and some bacterial infections. Certain lymphocytes directly attack invading microorganisms; others produce antibodies that attack and destroy microorganisms and other foreign material. Large lymphocytes, called atypical lymphocytes, are seen during **infectious mononucleosis** and other illnesses.

Monocytes increase during severe infections, and other conditions. They remove debris and microorganisms by phagocytosis. Eosinophils and basophils increase in response to allergic reactions and parasitic infection.

White cell counts are usually done on an automated instrument. A sample of blood is mixed with a chemical to burst the red blood cells. The remaining white cells are counted by the instrument.

The differential is done by spreading a drop of blood on a microscope slide. The slide is stained with a special stain and examined under a microscope. One-hundred white cells are counted and identified as either neutrophils, bands, lymphocytes, monocytes, eosinophils or basophils. Any atypical or immature cells also are counted. Cells are identified by the shape and appearance of the nucleus, the color of cytoplasm (the background of the cell), and the presence and color of granules. The percentage of each cell type is reported. At the same time, red cells and platelets are examined for abnormalities in appearance. Some instruments perform an automated differential.

KEY TERMS

Band—Immature neutrophil.

Basophil—White blood cell that increases in response to parasitic infections and allergic reactions.

Differential—Blood test that determines the percentage of each type of white blood cell in a person's blood.

Eosinophil—White blood cell that increases in response to parasitic infections and allergic reactions.

Leukocytosis—A white count increased to over 10,000/ μ L.

Leukopenia—A white count decreased to less than 4,000/ μ L.

Lymphocyte—White blood cell that fights viral and some bacterial infections by direct attack or the production of antibodies.

Monocyte—White blood cell that increases during a variety of conditions including severe infections. It removes debris and microorganisms by phagocytosis.

Neutrophil—White blood cell that increases in response to bacterial infection. It removes and kills bacteria through phagocytosis.

Phagocytosis—A process by which a white blood cell envelopes and digests debris and microorganisms to remove them from the blood.

Both the white blood cell count (also called white count or leukocyte count) and the differential (also called diff) are covered by insurance. Results are available the same day.

Preparation

This test requires 7 mL of blood. A healthcare worker ties a tourniquet on the person's upper arm, locates a vein in the inner elbow region, and inserts a needle into that vein. Vacuum action draws the blood through the needle into an attached tube. Collection of the sample takes only a few minutes.

Aftercare

Discomfort or bruising may occur at the puncture site. Pressure to the puncture site until the bleeding

stops reduces bruising; warm packs relieve discomfort. The person may feel dizzy or faint.

Normal results

Total white cell count 5,000-10,000 u/L. Neutrophils 50-60%. Lymphocytes 20-40%. Monocytes 2-6%. Eosinophils 1-4%. Basophils 0.5-1%. Bands 0-3%.

Abnormal results

The white cell count and differential are interpreted according to a person's clinical condition and medical history. **Leukocytosis** (a white count increased to over 10,000/uL) is seen in bacterial infections, inflammation, leukemia, trauma, and stress. **Leukopenia** (a white count decreased to less than 4,000/ μ L) is seen in some viral infections or severe bacterial infections, and conditions that affect the bone marrow such as dietary deficiencies, chemotherapy, radiation therapy, and autoimmune diseases.

Nancy J. Nordenson

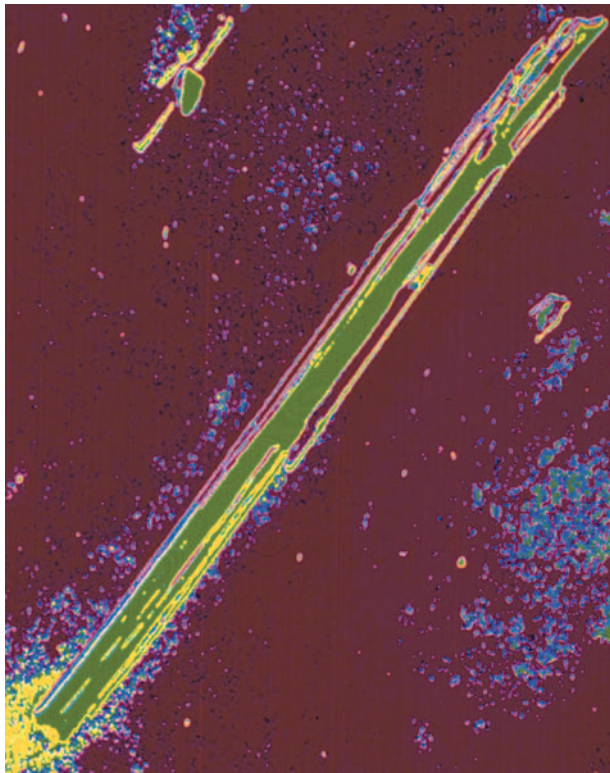
Whooping cough

Definition

Whooping **cough**, also known as pertussis, is a highly contagious disease which causes classic spasms (paroxysms) of uncontrollable coughing, followed by a sharp, high-pitched intake of air which creates the characteristic "whoop" of the disease's name.

Description

Whooping cough is caused by a bacteria called *Bordatella pertussis*. *B. pertussis* causes its most severe symptoms by attaching itself to those cells in the respiratory tract which have cilia. Cilia are small, hair-like projections that beat continuously, and serve to constantly sweep the respiratory tract clean of such debris as mucus, bacteria, viruses, and dead cells. When *B. pertussis* interferes with this normal, janitorial function, mucus and cellular debris accumulate and cause constant irritation to the respiratory tract, triggering coughing and increasing further mucus production.



A magnified image of a pertussis toxin crystal that causes whooping cough. (National Institutes of Health/Custom Medical Stock Photo. Reproduced by permission.)

Whooping cough is a disease which exists throughout the world. While people of any age can contract whooping cough, children under the age of two are at the highest risk for both the disease and for serious complications and **death**. Apparently, exposure to *B. pertussis* bacteria earlier in life gives a person some immunity against infection with it later on. Subsequent infections resemble the **common cold**.

Causes and symptoms

Whooping cough has four somewhat overlapping stages: incubation, catarrhal stage, paroxysmal stage, and convalescent stage.

An individual usually acquires *B. pertussis* by inhaling droplets infected with the bacteria coughed into the air by someone already suffering with the infection. Incubation is the symptomless period of seven to 14 days after breathing in the *B. pertussis* bacteria, and during which the bacteria multiply and penetrate the lining tissues of the entire respiratory tract.

The catarrhal stage is often mistaken for an exceedingly heavy cold. The patient has teary eyes, sneezing,

fatigue, poor appetite, and an extremely runny nose (rhinorrhea). This stage lasts about 10-14 days.

The paroxysmal stage, lasting two to four weeks, begins with the development of the characteristic whooping cough. Spasms of uncontrollable coughing, the “whooping” sound of the sharp inspiration of air, and **vomiting** are all hallmarks of this stage. The whoop is believed to occur due to inflammation and mucus which narrow the breathing tubes, causing the patient to struggle to get air into his/her lungs; the effort results in intense exhaustion. The paroxysms (spasms) can be induced by over activity, feeding, crying, or even overhearing someone else cough.

The mucus which is produced during the paroxysmal stage is thicker and more difficult to clear than the more watery mucus of the catarrhal stage, and the patient becomes increasingly exhausted attempting to clear the respiratory tract through coughing. Severely ill children may have great difficulty maintaining the normal level of oxygen in their systems, and may appear somewhat blue after a paroxysm of coughing, due to the low oxygen content of their blood. Such children may also suffer from swelling and degeneration of the brain (encephalopathy), which is believed to be caused both by lack of oxygen to the brain during paroxysms, and also by bleeding into the brain caused by increased pressure during coughing. Seizures may result from decreased oxygen to the brain. Some children have such greatly increased abdominal pressure during coughing that hernias result (hernias are the abnormal protrusion of a loop of intestine through a weak area of muscle). Another complicating factor during this phase is the development of **pneumonia** from infection with another bacterial agent; the bacteria takes hold due to the patient’s already-weakened condition.

If the patient survives the paroxysmal stage, recovery occurs gradually during the convalescent stage, usually taking about three to four weeks. However, spasms of coughing may continue to occur over a period of months, especially when a patient contracts a cold, or other respiratory infection.

Diagnosis

Diagnosis based just on the patient’s symptoms is not particularly accurate, as the catarrhal stage may appear to be a heavy cold, a case of the flu, or a simple **bronchitis**. Other viruses and tuberculosis infections can cause symptoms similar to those found during the paroxysmal stage. The presence of a pertussis-like cough along with an increase of certain specific white blood cells (lymphocytes) is suggestive of

KEY TERMS

Cilia—Tiny, hair-like projections from a cell. In the respiratory tract, cilia beat constantly in order to move mucus and debris up and out of the respiratory tree, in order to protect the lung from infection or irritation by foreign bodies.

Encephalopathy—Swelling and degeneration of the brain.

pertussis (whooping cough). However, cough can occur from other pertussis-like viruses. The most accurate method of diagnosis is to culture (grow on a laboratory plate) the organisms obtained from swabbing mucus out of the nasopharynx (the breathing tube continuous with the nose). *B. pertussis* can then be identified by examining the culture under a microscope.

Treatment

Treatment with the antibiotic erythromycin is helpful only at very early stages of whooping cough, during incubation and early in the catarrhal stage. After the cilia and the cells bearing those cilia, are damaged, the process cannot be reversed. Such a patient will experience the full progression of whooping cough symptoms; symptoms will only improve when the old, damaged lining cells of the respiratory tract are replaced over time with new, healthy, cilia-bearing cells. However, treatment with erythromycin is still recommended, to decrease the likelihood of *B. pertussis* spreading. In fact, all members of the household where a patient with whooping cough lives should be treated with erythromycin to prevent the spread of *B. pertussis* throughout the community. The only other treatment is supportive, and involves careful monitoring of fluids to prevent **dehydration**, rest in a quiet, dark room to decrease paroxysms, and suctioning of mucus.

Prognosis

Just under 1% of all cases of whooping cough cause death. Children who die of whooping cough usually have one or more of the following three conditions present:

- severe pneumonia, perhaps with accompanying encephalopathy
- extreme weight loss, weakness, and metabolic abnormalities due to persistent vomiting during paroxysms of coughing

- other pre-existing conditions, so that the patient is already in a relatively weak, vulnerable state (such conditions may include low birth weight babies, poor **nutrition**, infection with the **measles** virus, presence of other respiratory or gastrointestinal infections or diseases)

Prevention

The mainstay of prevention lies in programs similar to the mass immunization program in the United States which begins immunization inoculations when infants are two months old. The pertussis vaccine, most often given as one immunization together with **diphtheria** and **tetanus**, has greatly reduced the incidence of whooping cough. Unfortunately, there has been some concern about serious neurologic side effects from the vaccine itself. This concern led huge numbers of parents in England, Japan, and Sweden to avoid immunizing their children, which in turn has led to major epidemics of disease in those countries. However, several carefully constructed research studies have disproved the idea that the pertussis vaccine is the cause of neurologic damage. Furthermore, a newer formulation of the pertussis vaccine is available. Unlike the old whole cell pertussis vaccine, which is composed of the entire bacterial cell which has been deactivated (and therefore unable to cause infection), the newer acellular pertussis vaccine does not use a whole cell of the bacteria, but is made up of (between two and five) chemical components of the *B. pertussis* bacteria. The acellular pertussis vaccine appears to greatly reduce the risk of unpleasant reactions to the vaccine, including high fever and discomfort following **vaccination**.

Resources

PERIODICALS

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Rosalyn Carson-DeWitt, MD

Wilderness medicine

Definition

Wilderness medicine encompasses the prevention, diagnosis, and treatment of injuries and medical

conditions that may occur during activities in remote territories.

Purpose

Activities that may require wilderness medicine include backpacking, cross-country skiing, mountaineering, white water rafting, scuba diving, and exploration in undeveloped regions such as deserts or jungles. Wilderness medicine has evolved to deal with situations in which definitive medical care is hours or days away, and in which patients may require quick or extended attention. Wilderness medicine utilizes first aid techniques, but requires additional skills that take into account demanding environments, uncommon threats to health, hazardous or lengthy travel to medical facilities, and difficulties in obtaining food, water, and shelter.

Wilderness medicine uses techniques to assess and treat a variety of conditions and injuries, including:

- **wounds and burns**
- external and internal bleeding
- cardiac arrest
- head injuries
- spinal cord injuries
- **fractures and dislocations**
- **altitude sickness**
- problems from cold and heat
- allergic and anaphylactic reactions
- lightning strikes
- near drowning
- insect, reptile and animal bites
- poisoning
- emergency child birth

Precautions

In wilderness situations caregivers should follow the maxim of *first, do no harm*. Uninjured members of groups should not attempt rescues that place themselves in danger. People administering first aid or wilderness medicine should remain calm and organized at all times. Only those with experience should administer medications and medical procedures. Injured people should not be moved until they are fully evaluated, or unless environmental conditions are threatening and require immediate shelter.

People with certain medical conditions should avoid travel in the wilderness, which can make existing

conditions worse. These conditions include metastatic **cancer**, peptic ulcers, **coronary artery disease**, chronic obstructive pulmonary disease, clotting or bleeding disorders, high-altitude sickness, chronic **rheumatoid arthritis**, chronic severe back **pain**, and chronic knee and hip joint disease.

Description

The first stage of wilderness medicine begins with an assessment of the injury or condition. Primary assessment is used to quickly determine if a patient is in a life-threatening situation and to provide immediate emergency care. Secondary assessment is the thorough evaluation of a patient after life-threatening circumstances are relieved.

Primary assessment

A rule of thumb for the first steps of primary assessment, recommended by the Wilderness Medicine Institute, is the ABCDE procedure. It stands for Airway, Breathing, Circulation, Disability, and Exposure assessment. First, a patient's airway should be checked by close observation of whether or not air can move in and out and any obstructions to breathing should be alleviated. In unconscious people the tongue can often fall to the back of the throat and block breathing and the head should be tilted back and the lower jaw raised to alleviate the obstruction. If neck or spinal cord injuries are suspected, the head must be handled with extreme care to avoid further injury to the delicate spinal cord. In these cases, the lower jaw can be pulled forward to open the airway. If the neck is severely out of alignment due to an injury or fall, it may be gently realigned to free the airway.

After the airway is cleared and breathing is ensured, a patient's circulation is checked by noting the pulse of the carotid artery, on the neck, the pulse of the femoral artery in the front groin, and by listening to the heartbeat. If pulse is lacking, **cardiopulmonary resuscitation (CPR)** may be required, which requires chest compression and mouth-to-mouth breathing. Circulation checks include surveying a patient for bleeding. If severe bleeding is present, it should be stopped by direct pressure to the injured area, and by elevating the wound level to the heart if possible.

Disability assessment means checking for damage to the spinal cord, particularly in the cervical region of the neck. Assessment of exposure determines if environmental conditions, such as heat or cold, are immediate threats to a patient's life, which

may require actions such as seeking shelter or covering the patient with protective clothing.

Secondary assessment

During this stage, a thorough **physical examination** of the patient is made from head to toe to determine the extent of injuries or problems. Caretakers performing the assessment should write detailed notes in order to inform physicians or emergency workers later. Patients are thoroughly interviewed to determine the scope of problems and any previous medical issues that might be related. Patients should be spoken to calmly to determine their mental states and how well they respond to stimuli. Vital signs such as heart rate and respiration rate should be noted and monitored. The skin should be carefully observed for injuries, **boils**, **rashes**, and discoloration. Red or flushed skin may indicate **fever** or heat-related conditions, while pale or blotchy skin can point to **shock** or **hypothermia**. A bluish tint to the skin may mean a lack of oxygen. Contact lenses should be removed from patients in cold conditions, as they can freeze to the eyes. During secondary assessment the patient should be closely monitored over time until improvement is noted or further treatment decisions are made. At all times in wilderness injuries, shock must be watched for and immediately treated.

Shock

In wilderness situations shock should be suspected after traumatic injuries, significant loss of blood due to internal or external bleeding, extreme loss of fluids from **vomiting** or **diarrhea**, heart attacks, and spinal cord injuries. Shock is easiest to alleviate when it is treated early; when not treated properly, it can progress to unconsciousness and **death**. When the likelihood of shock occurs, patients should be continually monitored and supported.

Symptoms of shock begin with **anxiety** and restlessness, with increased heart rates and labored, shallow breathing. Shock victims tend to sweat profusely with cool and clammy skin. Thirst and **nausea** are also symptoms.

Shock is treated in the wilderness by maintaining an open airway for the patient to breathe, by treating any injuries such as bleeding wounds, by reducing pain if possible, and by replenishing fluids. Patients should be kept calm and warm and their feet should be elevated if possible to increase blood flow to the organs. If shock symptoms progress, plans should be quickly made to get help or evacuate the patient.

Evacuation

Evacuation of a patient may be a crucial decision in the wilderness, depending upon the severity of an injury or condition, the difficulty of moving the patient, the time considerations involved, and the availability of outside help. In general if a patient with severe symptoms is not improving despite care then evacuation becomes necessary. The Wilderness Medical Society lists symptoms that require postponing travel or evacuating patients:

- progressive deterioration with symptoms of **dizziness**, **fainting**, abnormally slow (bradycardia) or fast (tachycardia) heart rate, labored breathing, poor mental status, progressive weakness, constant vomiting or diarrhea, intolerance of oral fluids, or recurrent loss of consciousness due to head injuries
- debilitating pain
- inability to sustain pace due to medical problems
- passage of blood by mouth or rectum
- symptoms of serious high-altitude illness
- infections that get worse despite treatment
- chest pain that is not musculoskeletal in origin
- psychological status threatening the individual or group

If a patient cannot be moved without risk of further injury, then other members of a party, preferably two or more, should be sent to get outside help. When requesting outside assistance, the safety of incoming rescuers and time constraints should be weighed. Requests for outside help should be made in writing, and include an assessment of the patient and situation as well as a detailed location of the incident. In some regions, helicopter evacuation may be an option, and should be used if an injury is life-threatening.

During evacuation patients must be handled with extreme care, as well as insulated from heat, cold and further injuries. Larger wilderness expeditions may have special devices available for transporting injured members, while smaller parties may have to improvise transporting devices by using backpacks, ropes and other available materials.

Wounds and burns

In wilderness situations wound management strives to stop bleeding, prevent infection, and speed healing. Bleeding from wounds should be controlled by direct pressure. Wounds and burns should be cleaned gently and thoroughly, treated with antibiotic

ointment, and covered with bandages to avoid infections. Wounds that have high risks of infections, such as large cuts, open fractures, and animal bites, should be watched closely.

External and internal bleeding

External bleeding should be stopped by direct pressure, such as firmly applying a clean bandage or compress to an open wound. Secondary pressure may be applied to pressure points, such as the large arteries in the upper arm or groin, to slow bleeding. Tourniquets are recommended only in life-threatening situations, as they can cause complications and infections. Symptoms of internal bleeding include dizziness, fainting, rapid heartbeat, weak pulse, **shortness of breath**, thirst, loss of color, vomiting blood, blood in the feces or urine, and severe pain or swelling in the abdomen. If internal bleeding is suspected, medical help should be sought immediately. With all cases of significant blood loss, shock must be carefully considered.

Cardiac arrest

Cardiac arrest in the wilderness may require **CPR**, although CPR is less effective in remote regions that lack access to the **life support** technology that ambulances quickly supply. CPR should be administered to patients who have suffered near drowning, hypothermia, lightning strikes, and drug overdoses. CPR generally should not be administered in the wilderness if it endangers the rescuers, if the time of the cardiac arrest is unknown, if the patient appears to be dead or rigor mortis has set in, or if cardiac arrest was caused by severe trauma or lethal injuries.

Head injuries

Head injuries that do not cause loss of consciousness in the victim are rarely dangerous. Short-term loss of consciousness following head injuries is known as **concussion**, and these patients should be closely monitored for 24 hours, including waking them every three hours during sleep to check for mental alertness. For head injuries that cause prolonged unconsciousness, the airway and cervical spine must be protected. Severe brain injury is indicated by relapses into unconsciousness, bad headaches, bleeding from the ears, clear fluid draining from the nose, vomiting, persistent disorientation, personality changes, seizures, irregular heartbeat and breathing, and unequal or unreactive pupils. Severe head injuries must be treated by seeking immediate medical help or evacuation.

Spinal cord injuries

If spinal cord injuries are suspected, patients must be immobilized. Some expeditions or rescue teams may carry special splints or vests in their medical kits. If no such equipment is available, *spineboards* may be fashioned from available materials such as backpacks, poles, or ice axes to prevent unnecessary movement of the injured backbone.

Fractures and dislocations

Wilderness care for fractures recommends **immobilization** by using splints and slings. If manufactured splints and slings are not available in the medicine kit, they can be improvised by using natural materials, ski poles, ice axes, clothing, or parts of backpacks. In the case of dislocations, standard wilderness procedure is to splint, tape and stabilize the injury in the current position. However, if circulation or nerve function is impaired, or if the injured person is in extreme pain, relocation may be necessary by realigning the injured area. Relocation is most effective if it is done immediately following the injury, before stiffness or **muscle spasms** set in.

Altitude sickness

Symptoms of altitude sickness include **headache**, nausea, **fatigue**, vomiting, and bluish skin. Ataxia, or loss of muscular control and balance indicates more severe altitude sickness. Altitude sickness can occur at altitudes above 8,000 feet. The best prevention of the condition is allowing plenty of time for acclimatization at high altitudes, drinking plenty of fluids, and eating a diet rich in carbohydrates. **Aspirin** or **acetaminophen** may be taken, while the drug acetazolamide (Diamox) can relieve symptoms of mild acute mountain sickness (AMS). Other related conditions, which can cause death, are high altitude cerebral **edema** (HACE), which causes fluid accumulation on the brain, and high altitude **pulmonary edema** (HAPE), which causes fluid in the lungs. The main treatment for acute mountain sickness is to rapidly descend to lower altitudes. In some cases oxygen may be available to ease symptoms.

Problems from cold and heat

Frostbite is localized tissue damage from exposure to cold, and is remedied by the slow warming of exposed parts, preferably in heated water. Hypothermia is the condition resulting from lowered body core temperature, and is a common affliction in wilderness medicine. Mild hypothermia occurs when the body's core

temperature (measured rectally) falls from normal to 95°F (35°C) Fahrenheit. Moderate hypothermia gives temperatures between 90–95°F (32.2–35°C), while severe hypothermia occurs when a body's core temperature falls below 90°F (32.2°C). Symptoms include severe shivering, confusion, apathy, drowsiness, slurred speech, and impaired reflexes, and progresses to the point of unconsciousness.

Even cases of the mildest hypothermia must be cared for closely. Patients in whom hypothermia is suspected should be immediately warmed by gently removing wet clothing and providing dry clothing, blankets and shelter. They should be monitored for body temperature changes. Severe hypothermia cannot be remedied in the wilderness; victims must be immediately and gently evacuated. Warming severe hypothermia victims too quickly is dangerous. **Cardiopulmonary resuscitation** (CPR) may be initiated on victims of severe hypothermia who have cardiac arrest. In cases of near drowning, hypothermia must always be suspected.

Illness from heat includes heat exhaustion and the more severe heat **stroke**. Symptoms include confusion, rapid weak pulse, cramps, dizziness, nausea, diarrhea, headache, and high measured temperatures. Sweating may or may not occur, and the skin may be clammy and blotched. The principle treatment for heat illness in the wilderness is immediate cooling of the patient, by providing shade, fanning, sponging and immersion in cold water. Heat exhaustion will correct itself with enough rest and water. Heat stroke is life threatening and requires immediate cooling and rehydration with fluids, preferably intravenous ones. Prevention of heat illness includes proper conditioning, protective clothing, and avoiding **dehydration**.

Insect, reptile, and animal bites

Wilderness medicine must deal with an array of **bites and stings**, from bears, snakes, reptiles, spiders, scorpions, bees, fish and ticks. Prevention includes knowledge of the threats in the region being explored, as well as packing appropriate supplies such as bee sting kits for anaphylactic shock and snakebite kits for venomous attacks. The goal of treatment is to stop bleeding, prevent infection, and alleviate envenomation, or exposure to poison. The Sawyer Extractor is a suction tool used to remove snake venom, while the EpiPen and Ana-kit are available by prescription for anaphylactic shock due to stings and severe allergic reactions.

Preparation

Knowledge and sound planning can be the difference between success and disaster in the backcountry.

KEY TERMS

Anaphylactic shock—Severe allergic reaction characterized by airway constriction, tissue swelling, and lowered blood pressure.

Cardiac arrest—Heart failure or heart attack.

Dislocation—Displacement of bones at a joint.

Envenomation—Exposure to venom by bites or stings from insects, reptiles, and fish.

Wilderness—Large backcountry areas lacking roads, communication and other modern infrastructure.

Members of extended wilderness outings should undergo thorough examinations by their physicians and dentists prior to undertaking expeditions. People going on wilderness outings should begin in a state of sound physical fitness by undertaking appropriate conditioning programs, as well as becoming acclimated to special conditions such as altitude or extreme temperatures. Those with medications should be aware of potential side effects and complications, and inform other members of their group. At least two, and preferably all, members of wilderness expeditions should be familiar with first aid, wilderness medicine and rescue procedures. All members of wilderness outings should carry appropriate clothing, equipment, food, water, and first aid supplies. Trip itineraries should be recorded with park rangers or other official services. Means of communication with rescue facilities should be considered in advance in case emergencies arise.

Carrying adequate medical supplies is a crucial preparation for wilderness outings. These supplies will vary depending on the length of the trip and the region. Medical kits should contain basic first aid supplies such as bandages, dressings, pain relievers, water purification tablets, sunscreen, **antiseptics**, and ointments. Additional medical supplies include **antibiotics**, medications for gastrointestinal problems, **antihistamines** and emergency kits for **asthma** or allergic reactions, snake and insect bite kits, splints, and basic surgical supplies. Extended expeditions or those facing extreme conditions might include intravenous fluids, oxygen bottles for altitude problems, rescue gear and evacuation equipment, and specific medications for regional diseases and infections, such as **malaria**.

Immunizations are a very important preparation for those entering wilderness areas, particularly in

Third World countries. Immunizations should be planned as far in advance as possible, as some take several weeks to become effective and others cannot be given together. Some immunizations that may be required, depending on the region, include **tetanus**, poliovirus, **measles**, **mumps**, **rubella**, **cholera**, **yellow fever**, meningococcus, hepatitis, bubonic **plague**, **typhoid fever** and **rabies**. See Resources below for sources of specific immunization information.

Several organizations provide training and certification for various levels of wilderness medicine. The most basic levels of preparation are first aid and first responder certifications, followed by outdoor emergency care (OEC) training. More rigorous training provides the wilderness first responder (WFR), the wilderness emergency medical technician (WEMC), or the wilderness prehospital emergency care (WPHEC) certifications. The most advanced level of wilderness medical certification is search and rescue (SAR) emergency care, which provides expertise in a sophisticated array of rescue techniques and equipment.

Resources

BOOKS

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ORGANIZATIONS

International Association for Medical Assistance to Travelers (IAMAT). 417 Center St., Lewistown, NY 14092. (716) 754-4883.

U.S. Centers for Disease Control. 1600 Clifton Road NE, Atlanta, GA 30333. (404) 639-1610. <<http://www.cdc.gov>>. Publishes *Health Information for International Travel*.

Wilderness Medical Society. PO Box 2463, Indianapolis, IN 46204. (317) 631-1745.

Wilderness Medicine Institute. PO Box 9, 413 Main Street, Pitkin, CO 81241. (970) 641-3572. <<http://www.wildernessmed.com>>.

Douglas Dupler, MA

Wilms' tumor

Definition

Wilms' tumor is a cancerous tumor of the kidney that usually occurs in young children. It is named for Max Wilms, a German surgeon (1867–1918) and is also known as a nephroblastoma.

Description

When an unborn baby is developing, the kidneys are formed from primitive cells. Over time, these cells become more specialized. The cells mature and organize into the normal kidney structure. Sometimes, clumps of these cells remain in their original, primitive form. If these cells begin to multiply after birth, they may ultimately form a large mass of abnormal cells. This is known as a Wilms' tumor.

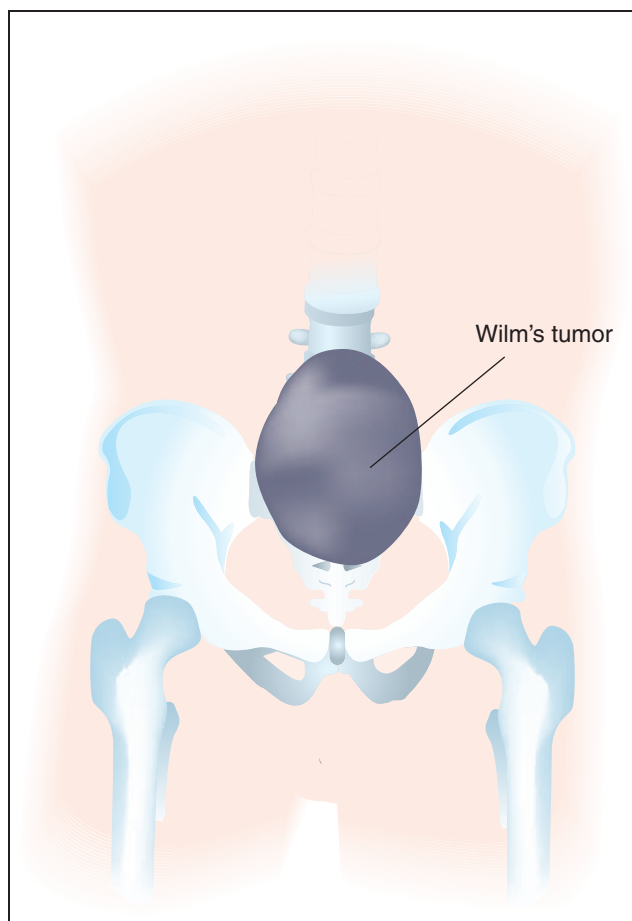
Wilms' tumor is a type of malignant tumor. This means that it is made up of cells that are significantly immature and abnormal. These cells are also capable of invading nearby structures within the kidney and traveling out of the kidney into other structures. Malignant cells can even travel through the body to invade other organ systems, most commonly the lungs and brain. These features of Wilms' tumor make it a type of **cancer** that, without treatment, would eventually cause **death**. However, advances in medicine during the last 20 years have made Wilms' tumor a very treatable form of cancer.

Wilms' tumor occurs almost exclusively in young children. The average patient is about three years old, although cases have been reported in infants younger than six months and adults in their early twenties. Females are only slightly more likely than males to develop Wilms' tumors. In the United States, Wilms' tumor occurs in 8.3 individuals per million in white children under the age of 15 years. The rate is higher among African-Americans and lower among Asian-Americans. Wilms' tumors are found more commonly in patients with other types of birth defects. These defects include:

- absence of the colored part (the iris) of the eye (aniridia)
- enlargement of one arm, one leg, or half of the face (hemihypertrophy)
- certain **birth defects** of the urinary system or genitals
- certain genetic syndromes (WAGR syndrome, Denys-Drash syndrome, and Beckwith-Wiedemann syndrome)

Causes and symptoms

The cause of Wilms' tumor is not completely understood. Because 15% of all patients with this type of tumor have other heritable defects, it seems clear that at least some cases of Wilms' tumor are due to an inherited alteration. A genetic defect known as WT1, the Wilms' tumor suppressor gene, has been identified in some patients on chromosome 11. It



Wilm's tumor. (Illustration by Argosy, Inc.)

appears that the tendency to develop a Wilms' tumor can run in families. In fact, about 1.5% of all children with a Wilms' tumor have family members who have also had a Wilms' tumor. The genetic mechanisms associated with the disease are unusually complex; it is thought as of 2004 that the tumor develops because the defective WT1 gene fails to stop its growth. Other genes that have been linked to Wilms' tumor are located on chromosomes 16q, 7p15, and 17q12.

Some patients with Wilms' tumor experience abdominal **pain**, **nausea**, **vomiting**, high blood pressure, or blood in the urine. However, the parents of many children with this type of tumor are the first to notice a firm, rounded mass in their child's abdomen. This discovery is often made while bathing or dressing the child and frequently occurs before any other symptoms appear. Rarely, a Wilms' tumor is diagnosed after there has been bleeding into the tumor, resulting in sudden swelling of the abdomen and a low red blood cell count (anemia).

About 4–5% of Wilms' tumor cases involve both kidneys during the initial evaluation. The tumor appears on either side equally. When pathologists look at these tumor cells under the microscope, they see great diversity in the types of cells. Some types of cells are associated with a more favorable outcome in the patient than others. In about 15% of cases, physicians find some degree of cancer spread (metastasis). The most common sites in the body where metastasis occurs are the liver and lungs.

Researchers have found evidence that certain types of lesions occur before the development of the Wilms' tumor. These lesions usually appear in the form of stromal, tubule, or blastemal cells.

Diagnosis

Children with Wilms' tumor generally first present to physicians with a swollen abdomen or with an obvious abdominal mass. The physician may also find that the child has **fever**, bloody urine, or abdominal pain. The physician will order a variety of tests before imaging is performed. These tests mostly involve blood analysis in the form of a white blood cell count, complete **blood count**, platelet count, and serum calcium evaluation. Liver and kidney function testing will also be performed as well as a **urinalysis**.

Initial diagnosis of Wilms' tumor is made by looking at the tumor using various imaging techniques. Ultrasound and **computed tomography scans** (CT scans) are helpful in diagnosing Wilms' tumor. Intravenous pyelography, where a dye injected into a vein helps show the structures of the kidney, can also be used in diagnosing this type of tumor. Final diagnosis, however, depends on obtaining a tissue sample from the mass (biopsy), and examining it under a microscope in order to verify that it has the characteristics of a Wilms' tumor. This biopsy is usually done during surgery to remove or decrease the size of the tumor. Other studies (chest x rays, CT scan of the lungs, **bone marrow biopsy**) may also be done in order to see if the tumor has spread to other locations.

Treatment

In the United States, treatment for Wilms' tumor almost always begins with surgery to remove or decrease the size of the kidney tumor. Except in patients who have tumors in both kidneys, this surgery usually will require complete removal of the affected kidney. During surgery, the surrounding lymph nodes, the area around the kidneys, and the entire abdomen will also be examined. While the tumor can spread to

these surrounding areas, it is less likely to do so compared to other types of cancer. In cases where the tumor affects both kidneys, surgeons will try to preserve the kidney with the smaller tumor by removing only a portion of the kidney, if possible. Additional biopsies of these areas may be done to see if the cancer has spread. The next treatment steps depend on whether/where the cancer has spread. Samples of the tumor are also examined under a microscope to determine particular characteristics of the cells making up the tumor.

Information about the tumor cell type and the spread of the tumor is used to decide the best kind of treatment for a particular patient. Treatment is usually a combination of surgery, medications used to kill cancer cells (**chemotherapy**), and x rays or other high-energy rays used to kill cancer cells (radiation therapy). These therapies are called adjuvant therapies, and this type of combination therapy has been shown to substantially improve outcome in patients with Wilms' tumor. It has long been known that Wilms' tumors respond to **radiation therapy**. Likewise, some types of chemotherapy have been found to be effective in treating Wilms' tumor. These effective drugs include dactinomycin, doxorubicin, vincristine, and cyclophosphamide. In rare cases, bone marrow transplantation may be used.

The National Wilms' Tumor Study Group (NWTSG) has developed a staging system to describe Wilms' tumors. All of the stages assume that surgical removal of the tumor has occurred. Stage I involves "favorable" Wilms' tumor cells and is usually treated successfully with combination chemotherapy involving dactinomycin and vincristine and without abdominal radiation therapy. Stage II tumors involving a favorable histology (cell characteristics) are usually treated with the same therapy as Stage I. Stage III tumors with favorable histology are usually treated with a combination chemotherapy with doxorubicin, dactinomycin, and vincristine along with radiation therapy to the abdomen. Stage IV disease with a favorable histology is generally treated with combination chemotherapy with dactinomycin, doxorubicin, and vincristine. These patients usually receive abdominal radiation therapy and lung radiation therapy if the tumor has spread to the lungs.

In the case of Stage II through IV tumors with unfavorable, or anaplastic, cells, then the previously-mentioned combination chemotherapy is used along with the drug cyclophosphamide. These patients also receive lung radiation therapy if the tumor has spread to the lungs. Another type of tumor cell can be present in Stages I through IV. This cell type is called clear cell sarcoma of the kidney. If this type of cell is

KEY TERMS

Biopsy—A procedure in which a small sample of tissue is removed, prepared, and examined with a microscope to determine the characteristics of the tissue's cells.

Blastemal—An immature material from which cells and tissues develop.

Cancer—A process where abnormal cells within the body begin to grow out of control, acquire the ability to invade nearby structures, and travel through the bloodstream in order to invade distant structures.

Malignant—Refers to cancer or cancer cells.

Sarcoma—A type of cancer that originates from connective tissue such as bone or muscle.

Stromal—A type of tissue that is associated with the support of an organ.

Tubule—Tissues and cells associated with the structures that connect the renal pelvis to the glomeruli.

present, then patients receive combination therapy with vincristine, doxorubicin, and dactinomycin. All of these patients receive abdominal radiation therapy and lung radiation therapy if the tumor has spread to the lungs.

As of 2004, there are significant differences between the treatment protocols of the NWTSG and its European counterpart, the Société Internationale d'Oncologie Pédiatrique (SIOP). Whereas American practice favors surgery followed by chemotherapy, European oncologists use preoperative chemotherapy and stage the tumor at the time of surgery rather than at the point of initial imaging studies.

Prognosis

The prognosis for patients with Wilms' tumor is quite good, compared to the prognosis for most types of cancer. One German study reported the overall five-year survival rate to be 89.5%. The patients who have the best prognosis are usually those who have a small-sized tumor, a favorable cell type, are young (especially under two years old), and have an early stage of cancer that has not spread. Modern treatments have been especially effective in the treatment of this cancer. Patients with the favorable type of cell have a long-term survival rate of 93%, whereas those

with anaplasia have a long-term survival rate of 43% and those with the sarcoma form have a survival rate of 36%.

Prevention

There are no known ways to prevent a Wilms' tumor, although it is important that children with birth defects associated with Wilms' tumor be carefully monitored.

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ORGANIZATIONS

- American Cancer Society. 1515 Clifton Rd. NE, Atlanta, GA 30329. (800) 227-2345. <<http://www.cancer.org>>.
- March of Dimes Birth Defects Foundation, National Office. 1275 Mamaroneck Ave., White Plains, NY 10605. <<http://www.modimes.org>>.
- National Cancer Institute (National Institutes of Health). 9000 Rockville Pike, Bethesda, MD 20892. (800) 422-6237. <<http://www.nci.nih.gov>>.

National Wilms Tumor Study Group (NWTSG). Fred Hutchinson Cancer Research Center, 1100 Fairview Avenue North, P. O. Box 19024, Seattle, WA 98109-1024. (800) 553-4878. Fax: (206) 667-6623. <<http://www.nwtsg.org>>.

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Rebecca J. Frey, PhD

Wilson disease

Definition

Wilson disease, or WD, is a rare inherited disorder that causes excess copper to accumulate in the body. It is also known as hepatolenticular degeneration. Steadily increasing amounts of copper circulating in the blood are deposited primarily in the brain, liver, kidneys, and the cornea of the eyes. WD is fatal if it is not recognized and treated. It is named for an American neurologist, Samuel A. K. Wilson, who first described it in 1912.

Description

Under normal conditions, copper that finds its way into the body through the diet is processed within the liver. This processed form of copper is then passed into the gallbladder, along with the other components of bile (a fluid produced by the liver, which enters the small intestine in order to help in digestive processes). When the gallbladder empties its contents into the first part of the small intestine (duodenum), the copper in the bile enters and passes through the intestine with the waste products of digestion. In healthy individuals, copper is then passed out of the body in stool.

In Wilson disease, copper does not pass from the liver into the bile, but rather begins to accumulate within the liver. As copper levels rise in the liver, the damaged organ begins to allow copper to flow into the bloodstream, where it circulates. Copper is then deposited throughout the body, building up primarily in the kidneys, the brain and nervous system, and the eyes. Wilson disease, then, is a disorder of copper poisoning occurring from birth.

Wilson disease affects approximately one in 30,000 to one in 100,000 individuals and can affect people from many different populations. Approximately one in 90 individuals are carriers of the gene for Wilson disease.

Causes and symptoms

Wilson disease is inherited in an autosomal recessive manner. Autosomal recessive refers to the pattern of inheritance where each parent carries a gene for the disease on one of his or her chromosome pairs. When each parent passes on the chromosome with the gene for Wilson disease, the child will be affected with the disease. Both males and females can be affected with Wilson disease. If an individual is a carrier of the Wilson disease gene they do not have any symptoms of this disease. In order to be affected, an individual must inherit two copies of the gene, one from each parent. Many cases of Wilson disease may not be inherited but occur as a spontaneous mutation in the gene.

The gene for Wilson disease is located on chromosome number 13. The name of the gene is called ATP7B and is thought to be involved in transporting copper. As of 2004, over 200 different mutations of this gene have been identified, making diagnosis by genetic testing difficult.

Symptoms typically present between the ages of three and 60, with age 17 considered to be the average age a diagnosis is made. About half of all patients experience their first symptoms in the liver. The illness causes swelling and tenderness of the liver, sometimes with **fever**, mimicking more common disorders, such as viral hepatitis and infectious mononucleosis. Abnormal levels of circulating liver enzymes reveal that the liver is being seriously damaged. This form of damage is referred to as fatty degeneration. Without medical intervention, the liver damage will progress to actual cirrhosis. An often-fatal manifestation of **liver disease** is called fulminant hepatitis. This extremely severe inflammation of the liver (hepatitis) results in jaundice, fluid leaking into the abdomen, low protein circulating in the blood, abnormalities of the blood clotting system, swelling of the brain, and anemia due to the abnormal destruction of red blood cells.

Neurological symptoms are the first to occur in half of all patients due to copper accumulation in the brain and nervous system. The average age of onset for neurological symptoms is 21. These symptoms include **tremors** of the hands, uncontrollable movements of the limbs, stiffness, drooling, difficulty swallowing,

difficulty talking, and headache. There is no change in patient's intelligence.

About one third of all patients with Wilson disease have a variety of psychiatric symptoms as the first signs of the disease. These symptoms include inability to cope, depression, irritability, increased anger, and inappropriate behavior. Often times patients have trouble completing tasks at work or in school.

Other symptoms that can affect patients with Wilson disease, and may occur before or after a diagnosis has been made include joint disorders, symptoms of arthritis and skeletal problems such as **osteoporosis**. Patients have occasionally been affected with kidney stones, abnormal handling of glucose in their body and women have menstrual cycle irregularities including stopping their regular cycle temporarily.

Diagnosis

The diagnosis of Wilson disease can be performed relatively easily through several different tests; however, because Wilson disease is so rare, diagnosis is often unfortunately delayed. The tests used to diagnose Wilson disease can be performed on patients who have and who have not already shown symptoms of the disease. It is extremely important to make a diagnosis as soon as possible since liver damage can occur before there are any signs of the disease.

An easy way to diagnose Wilson disease is to measure the amount of a glycoprotein found in the blood called ceruloplasmin. Low levels of ceruloplasmin can diagnose the disease in about 80% of affected patients. This procedure is not as effective for women taking birth control pills, pregnant women, or infants younger than six months of age.

A second test involving an **eye examination** to detect a characteristic ring of copper deposited in a membrane of the cornea (referred to as Kayser-Fleischer rings) is very easy to perform and is very useful in diagnosing patients who have already exhibited symptoms. This test is not as effective in persons without symptoms. This diagnostic test cannot be used by itself to make a diagnosis because some patients with liver disease but not Wilson disease will test positive.

A third test for diagnosing Wilson disease involves measuring the amount of copper in the liver. This can be accomplished by sampling a portion of the liver, called a biopsy. This is one of the most effective ways in which to diagnose Wilson disease, however the procedure itself is more difficult to perform than the others.

Other tests are also useful, for example measuring the amount of copper passed into the urine daily (high in Wilson disease). Another lab test measures the ability of a patient's ceruloplasmin to bind with a form of copper (decreased in Wilson disease). And finally, as discussed under genetic profile, some patients can be diagnosed through a DNA test to determine whether or not they carry two genes for Wilson disease. This test does not always prove to be useful in certain patients and is of most use when used to test the brothers and sisters of affected patients.

Molecular **genetic testing** is not particularly valuable in diagnosing WD because of the large number of possible gene mutations.

Treatment

Treatment involves life-long administration of either D-penicillamine (Cuprimine, Depen) or trientine hydrochloride (Syprine). Both of these drugs remove copper deposits throughout the body by binding to the copper which then leaves the body in the urine. This type of treatment is called **chelation therapy**. Zinc acetate (Galzin) and a low copper diet are other ways in which to treat Wilson disease.

Penicillamine has a number of serious side effects:

- joint pain
- neurological problems
- systemic lupus erythematosus
- decreased production of all blood elements
- interference with clotting
- allergic reactions

Careful monitoring is necessary. When patients have side effects from penicillamine, the dose can sometimes be lowered to an effective level that causes fewer difficulties. Alternatively, steroid medications may be required to reduce certain sensitivity reactions. Trientine has fewer potential side effects, but must still be carefully monitored.

Treatment with zinc acetate is also an effective way to remove excess copper from the body. Zinc is a metal that works to block copper absorption and bind copper in the intestinal cells until it is all released into the stool approximately one week later. The benefit of treatment with zinc is there are no toxic side effects however the zinc is a slower acting agent than the other drugs. It takes four to eight months for the zinc to be effective in reducing the overall amount of copper in the body.

Finally, patients with Wilson disease are encouraged to follow a diet low in copper, with an average copper intake of 1.0 mg/day. Foods to be avoided for the high levels of copper include liver and shellfish. Patients are also instructed to monitor their drinking water for excess levels of copper and drink distilled water instead.

Patients may be given a liver transplant in the event of liver failure as a complication of WD. **Liver transplantation** has been reported to have a relatively favorable outcome, in some cases decreasing the patient's neurologic symptoms.

Prognosis

Without treatment, Wilson disease is always fatal. With treatment, symptoms may continue to worsen for the first six to eight weeks. After this time, definite improvement should begin to be seen. However, it may take several years (two to five) of treatment to reach maximal benefit to the brain and liver. Even then, many patients are not returned to their original level of functioning. Patients with Wilson disease need to maintain some sort of anticopper treatment for the rest of their lives in order to prevent copper levels from rising in the body. Interruptions in treatment can result in a relapse of the disease which is not reversible, and can ultimately lead to **death**.

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KEY TERMS

Anemia—A blood condition in which the level of hemoglobin or the number of red blood cells falls below normal values. Common symptoms include paleness, fatigue, and shortness of breath.

Bile—A substance produced by the liver, and concentrated and stored in the gallbladder. Bile contains a number of different substances, including bile salts, cholesterol, and bilirubin.

Biopsy—The surgical removal and microscopic examination of living tissue for diagnostic purposes.

Cell—The smallest living units of the body which group together to form tissues and help the body perform specific functions.

Ceruloplasmin—A protein circulating in the bloodstream that binds with copper and transports it.

Chelation therapy—A method of removing copper or other heavy metals from the body by giving medications that bind to the metal and allow it to be excreted.

Chromosome—A microscopic thread-like structure found within each cell of the body and consists of a complex of proteins and DNA. Humans have 46 chromosomes arranged into 23 pairs. Changes in either the total number of chromosomes or their shape and size (structure) may lead to physical or mental abnormalities.

Cirrhosis—A chronic degenerative disease of the liver, in which normal cells are replaced by fibrous tissue. Cirrhosis is a major risk factor for the later development of liver cancer.

Deoxyribonucleic acid (DNA)—The genetic material in cells that holds the inherited instructions for growth, development, and cellular functioning.

Gallbladder—A small, pear-shaped organ in the upper right hand corner of the abdomen. It is connected by a series of ducts (tube-like channels) to the liver, pancreas, and duodenum (first part of the small intestine). The gallbladder receives bile from the liver, and concentrates and stores it. After a meal, bile is squeezed out of the gallbladder into the intestine, where it aids in digestion of food.

Gene—A building block of inheritance, which contains the instructions for the production of a particular protein, and is made up of a molecular sequence found on a section of DNA. Each gene is found on a precise location on a chromosome.

Glucose—One of the two simple sugars, together with galactose, that makes up the protein, lactose, found in milk. Glucose is the form of sugar that is usable by the body to generate energy.

Hepatitis—A viral disease characterized by inflammation of the liver cells (hepatocytes). People infected with hepatitis B or hepatitis C virus are at an increased risk for developing liver cancer.

Jaundice—Yellowing of the skin or eyes due to excess of bilirubin in the blood.

Toxic—Poisonous.

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ORGANIZATIONS

American Liver Foundation. 1425 Pompton Ave., Cedar Grove, NJ 07009. (800) 223-0179. <<http://www.liverfoundation.org>>.

National Organization for Rare Disorders (NORD). 55 Kenosia Avenue, P. O. Box 1968, Danbury, CT 06813-1968. (203) 744-0100. Fax: (203) 798-2291. <<http://www.rarediseases.org>>.

Wilson’s Disease Association. 4 Navaho Dr., Brookfield, CT 06804. (800) 399-0266.

OTHER

Wilson’s Disease Association. <<http://www.medhelp.org/wda/wil.htm>>.

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Wiskott-Aldrich syndrome

Definition

Wiskott-Aldrich syndrome (WAS) is a rare inherited disorder marked by a low level of blood platelets, eczema, recurrent infections, and a high risk of leukemia or lymph node tumors.

Description

WAS was named for the two physicians who first reported the disorder. In 1937, Dr. A. Wiskott, a physician working in Munich, described two affected boys of German ancestry who had repeated infections, a skin rash, and poor blood-clotting ability. Nearly twenty years later, Dr. R.A. Aldrich reported similar symptoms in members of an American family of Dutch ancestry.

WAS is inherited as an X-linked genetic disorder and will therefore only affect males. The gene responsible for WAS is located on the short arm of the X chromosome. Since males have only one X chromosome they only have one copy of the gene. If that copy carries the abnormal gene, they will have WAS. In contrast, females have two X chromosomes. They will have a normal copy of the gene on one chromosome even if an abnormal gene is on the other because the abnormal gene is very rare. The normal copy on one X chromosome is usually sufficient to prevent females from having WAS. However, women who have one abnormal copy of the WAS gene are designated as carriers. While they will not have WAS, they have a 50% risk of passing the gene to each of their sons who will have WAS. Carrier females also have a 50% risk of passing the defective copy of the gene to their daughters who also become carriers.

Researchers identified the gene for WAS in 1994 and pinpointed its location on the short arm of the X chromosome. As of 2000, over 100 different mutations have been found in the gene among WAS patients. The fact that there are many mutations many explain some of the variability of symptoms among boys with WAS. However, even within the same family, affected individuals with the identical WAS gene mutation may have different degrees of severity of the disease. The mild form, X-linked **thrombocytopenia**, is also caused by mutations in this same gene.

The WAS syndrome affects one in every 250,000 male children and occurs worldwide. In the year 2000, scientists estimated that about 500 Americans have WAS.

Causes and symptoms

The syndrome is caused by a defect (mutation) in a specific gene called the WAS gene that normally codes for the protein named Wiskott-Aldrich Syndrome Protein (WASP). This vital protein is a component of cells that are important in the body's defense against infection (lymphocytes). The same protein also functions in the cells that help prevent bleeding (platelets). A less severe form of the disease, X-linked thrombocytopenia affects mainly the platelets.

Increased susceptibility to infections, eczema, and excessive bleeding are the hallmarks of WAS, although the symptoms can vary significantly from one patient to another. The immune system of patients with WAS produces too few B and T cells. B cells are

the cells in the body that make antibodies. There are many types of T cells. Both B and T cells are needed to defend the body against infection. Because both types of cells are affected, WAS patients are subject to repeated infections from bacteria, fungi, and viruses. Ear infections, **meningitis**, and **pneumonia** are common in boys with WAS.

WAS patients also have thrombocytopenia, a decreased number of platelets. Platelets are the specialized blood cells that help to form **blood clots** and prevent uncontrolled bleeding. The platelets may also be smaller than normal. Some of the earliest symptoms of the syndrome are hemorrhage from **circumcision**, bloody **diarrhea**, and a tendency to bruise very easily.

Anemia and an enlarged spleen (splenomegaly) are seen in some patients. About 10% of patients develop malignancies, usually leukemia or tumors in the lymph nodes (non-Hodgkin's lymphoma).

Diagnosis

The diagnosis of WAS is usually suspected in male infants who have excessive bleeding, eczema, and frequent bacterial or viral infections. Special blood tests can then be ordered to confirm WAS. The blood of Wiskott-Aldrich patients will show a low platelet count and a weak immune (antibody) response. It is also possible to confirm the diagnosis by obtaining a small sample of the patient's blood and analyzing the DNA for a mutation in the WAS gene. Knowledge of the exact mutation combined with information about how much WAS protein the defective gene can produce may help predict how severe a form of the disease an individual will have.

Carrier Testing

If the specific WAS gene mutation is identified in an affected child, that child's mother can then be tested to confirm that she carries the gene. Other members of the mother's family may also want to consider testing to find out if they carry the same gene mutation. The first step in studying other family members is for a geneticist or genetic counselor to obtain a detailed family history and construct a pedigree (family tree) to determine which family members should be offered testing.

Prenatal Diagnosis

In families where there has been one child born with WAS, prenatal testing should be offered in subsequent pregnancies. There 50% chance with each

KEY TERMS

Amniocentesis—A procedure performed at 16-18 weeks of pregnancy in which a needle is inserted through a woman's abdomen into her uterus to draw out a small sample of the amniotic fluid from around the baby. Either the fluid itself or cells from the fluid can be used for a variety of tests to obtain information about genetic disorders and other medical conditions in the fetus.

Anemia—A blood condition in which the level of hemoglobin or the number of red blood cells falls below normal values. Common symptoms include paleness, fatigue, and shortness of breath.

Chorionic villus biopsy—A procedure used for prenatal diagnosis at 10-12 weeks gestation. Under ultrasound guidance a needle is inserted either through the mother's vagina or abdominal wall and a sample of cells is collected from around the early embryo. These cells are then tested for chromosome abnormalities or other genetic diseases.

Eczema—Inflammation of the skin with redness and other variable signs such as crusts, watery discharge, itching.

Gene—A building block of inheritance, which contains the instructions for the production of a particular protein, and is made up of a molecular sequence

found on a section of DNA. Each gene is found on a precise location on a chromosome.

Immune system—A major system of the body that produces specialized cells and substances that interact with and destroy foreign antigens that invade the body.

Mutation—A permanent change in the genetic material that may alter a trait or characteristic of an individual, or manifest as disease, and can be transmitted to offspring.

Platelets—Small disc-shaped structures that circulate in the blood stream and participate in blood clotting.

Prenatal diagnosis—The determination of whether a fetus possesses a disease or disorder while it is still in the womb.

Syndrome—A group of signs and symptoms that collectively characterize a disease or disorder.

Thrombocytopenia—A persistent decrease in the number of blood platelets usually associated with hemorrhaging.

X-linked—Located on the X chromosome, one of the sex chromosomes. X-linked genes follow a characteristic pattern of inheritance from one generation to the next.

subsequent **pregnancy** that the mother, who is a carrier, will transmit the abnormal copy of the gene to her baby. The key is to first identify the particular WAS gene mutation in the child with WAS. Then, early in a pregnancy, cells can be obtained from the developing fetus by **chorionic villus sampling** or amniocentesis, and checked for the same mutation. Women who carry the abnormal WAS gene and are considering prenatal diagnosis should discuss the risks and benefits of this type of testing with a geneticist or genetic counselor.

Treatment

Standard treatments for individuals with WAS include **antibiotics** for infections and platelet transfusions to limit bleeding. Immune globulin is given to strengthen the individual's immune system. Eczema can be treated with corticosteroid creams applied directly to the skin. The spleen is sometimes removed to reduce the risk of bleeding. In individuals with WAS, however, removal of the spleen also increases the risk of infection unless antibiotics are given to prevent

infections. About 50% of individuals with WAS are helped by treatment with transfer factor, which is a substance derived from the T cells of a healthy person. Transfer factor is given to improve both blood clotting and immune functions. Bone marrow transplantation has been successful in a number of cases. It has been most successful in boys under five years of age where the donor is a sibling whose tissue type closely matches that of the individual with WAS. As of 2000, attempts were also being made to treat individuals with WAS with umbilical cord blood from unrelated newborns in cases where the individual diagnosed with WAS has no matched sibling donor.

Prognosis

The prognosis for males diagnosed with Wiskott-Aldrich syndrome is poor. The average individual lives about four years; those who survive into adolescence often develop **cancer**. **Death** usually occurs from severe bleeding or overwhelming infection in the first few years of life.

Resources

BOOKS

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ORGANIZATIONS

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< <http://www.primaryimmune.org/inside.htm> >.

OTHER

"Entry 301000: Wiskott-Aldrich Syndrome." *OMIM—Online Mendelian Inheritance in Man*. < <http://www.ncbi.nlm.nih.gov/entrez/dispmim.cgi?id=301000> >.

NORD—National Organization for Rare Disorders, Inc.
< <http://www.rarediseases.org> >.

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Withdrawal syndromes

Definition

Withdrawal syndrome occurs in drug and alcohol addicted individuals who discontinue or reduce the use of their drug of choice. This process of eliminating drugs and alcohol from the body is known as **detoxification**. Anxiety, insomnia, **nausea**, perspiration, body aches, and **tremors** are just a few of the physical and psychological symptoms of drug and alcohol withdrawal that may occur during detoxification.

Description

Drugs and alcohol affect mood by altering brain chemistry, specifically the production of neurotransmitters. Neurotransmitters are chemicals in the central nervous system that enable nerve impulses to travel through the central nervous system and regulate thought processes, behavior, and emotion. Drugs that temporarily elevate neurotransmitter levels are called stimulants. Drugs that decrease neurotransmitter levels and depress the central nervous system are called depressants; they include opiates and sedative-hypnotic drugs such as alcohol and **barbiturates**. (There are exceptions: Benzodiazepine elevates the

level of an inhibitory neurotransmitter, GABA, therefore it serves as a tranquilizer.)

When drug or alcohol consumption becomes chronic, the body adjusts to the constant presence of the substance by changing its normal production of neurotransmitters. If drug and alcohol use suddenly stops, the body and central nervous system react to the absence of the substance with an array of symptoms known collectively as withdrawal syndrome.

Causes and symptoms

Acute withdrawal syndrome begins within hours of abstinence, and includes a full range of physical and psychological symptoms. More long-term, or subacute, withdrawal symptoms, such as intense drug craving, may occur weeks or months after detoxification has taken place.

Alcohol withdrawal

Alcohol withdrawal syndrome occurs in alcohol-dependent individuals who suddenly stop or dramatically reduce their alcohol intake. The onset of the syndrome is likely to occur within a week, but usually occurs within 24 hours of the individual's last drink, and is triggered when the central nervous system attempts to adjust to the sudden absence of ethyl alcohol in the body. Symptoms may include extreme **anxiety**, disorientation, **hallucinations**, **sleep disorders**, hand tremors, nausea, sweating, seizures, and racing pulse. **Delirium tremens** (DTs) are an extreme example of withdrawal. In the worst cases, untreated alcohol withdrawal syndrome can result in **death**. As many as two million Americans may experience symptoms of alcohol withdrawal conditions each year.

Barbiturate withdrawal

Barbiturates are prescribed as anticonvulsants, sedatives, and general anesthetics. They can also mimic some of the characteristics of alcohol intoxication (including euphoria, elation, and uninhibited behavior), which make them candidates for **abuse**. Commonly abused barbiturates include amobarbital (Amytal), pentobarbital (Nembutal), and secobarbital (Seconal). These drugs depress the respiratory and nervous system functions. Because abusers rapidly build up a tolerance to the effects of the drug, fatal overdose or **coma** can easily occur. Symptoms of withdrawal syndrome appear 12-20 hours after the last dose; they include anxiety, irritability, elevated heart and respiration rate, muscle **pain**, nausea, tremors, hallucinations, confusion, and seizures. Death is a

possibility if the condition is left untreated. Because barbiturates decrease REM (rapid eye movement) sleep, during which dreaming takes place, withdrawal often results in sleep disruptions such as nightmares, **insomnia**, or vivid dreaming.

Opiate withdrawal

Opiates are powerfully addictive analgesic drugs that deaden nerve pathways related to pain. Abusers of propoxyphene (Darvon), meperidine (Demerol), percocet (Oxycodone), heroin, morphine, and other powerfully addictive opiates quickly build up a tolerance to the drugs and need progressively larger doses to achieve the desired effect. Stopping or reducing the intake of the drug can cause severe withdrawal symptoms, which begin six to eight hours after the last dosage. Symptoms are flu-like, and include gastrointestinal distress, anxiety, nausea, insomnia, muscle pain, fevers, sweating, and runny nose and eyes.

Stimulant withdrawal

Use of stimulants, such as **cocaine**, crack, amphetamines, and methamphetamines, cause an increase in neurotransmitters in the central nervous system and produce feelings of alertness and increased energy. This initial “rush” is followed by a longer period of neurotransmitter loss, characterized by depression, lethargy, and a craving for more stimulants sometimes called a rebound effect. When a stimulant-dependent individual abstains from stimulant use, withdrawal symptoms, including depression, **fatigue**, insomnia, and loss of appetite, reflect this drop in neurotransmitter levels. Withdrawal typically takes one to two weeks.

Diagnosis

A detailed history of the patient’s drug or alcohol use taken before detoxification can be helpful in predicting the severity of withdrawal symptoms. Standardized clinical tests, such as the Clinical Institute Withdrawal Assessment for Alcohol, revised, (CIWA-Ar), are used to evaluate the severity of withdrawal symptoms throughout the detoxification procedure.

Treatment

Pharmacologic and medical management is often recommended for withdrawal syndrome. The physical condition of the patient is closely monitored throughout the detoxification procedure.

Alcohol withdrawal

Alcohol withdrawal syndrome can be treated at home or in a hospital or treatment setting. Inpatient treatment is recommended for patients who are at risk for serious withdrawal symptoms or re-intoxication if treated as an outpatient. Withdrawal symptoms are minimized through the administration of cross-tolerant sedatives. Long-acting **benzodiazepines**, such as diazepam (Valium), chlordiazepoxide (Librium), and lorazepam (Ativan), are the pharmacologic treatments of choice in managing the symptoms of alcohol withdrawal. Drug dosage is adjusted to offset the discomfort of withdrawal, without causing a euphoric effect, and is then gradually decreased as withdrawal symptoms lessen.

Barbiturate withdrawal

Because the risk for seizures and other severe complications is high, barbiturate withdrawal should be monitored in a hospital setting. Patients are given low doses of phenobarbital at a regular interval until mild intoxication is achieved. The dosage amount and frequency is then gradually decreased until withdrawal is complete.

Opiate withdrawal

Two basic treatment approaches are used for managing opiate withdrawal. The first involves treating the symptoms of the withdrawal with appropriate medication. Clonidine, an antihypertensive drug, is commonly prescribed to reduce muscle pain and cramping. Other symptom-specific drugs are administered on an as-needed basis.

The second treatment option is to replace the patient’s drug of choice with **methadone**, a long-acting, cross-tolerant opiate that does not normally produce a “high.” Doses of methadone are administered every four to six hours. The patient’s reaction is closely observed, and dosages are slowly decreased until withdrawal symptoms have disappeared, then dosages are discontinued. Methadone withdrawal can be completed within three weeks. It is important to note that methadone withdrawal treatment differs from a methadone maintenance program, in which patients who are unwilling to give up opiates are prescribed methadone as a legal, long-term substitute for their drug of choice.

Rapid opiate detoxification (ROD) is an emerging treatment option for opiate withdrawal. The ROD method is reported to be faster and to cause less physical discomfort than traditional forms of opiate detoxification. The treatment is typically performed in a hospital or private clinic setting. Naltrexone, an

KEY TERMS

Analgesics—Pain killing drugs that depress respiratory function. Opiates are analgesics.

Antagonist—A substance that tends to nullify the action of another.

Benzodiazepines—Sedatives used to treat anxiety, epilepsy, and alcohol withdrawal syndrome. Diazepam (Valium), alprazolam (Xanax), and chlor-diazepoxide (Librium) are all benzodiazepines.

Cross-tolerant—A drug that has the same pharmacological effect as another is considered cross-tolerant. Cross-tolerant drugs are often used in treating withdrawal syndromes.

Detoxification—The process of physically eliminating drugs and/or alcohol from the system of a substance-dependent individual.

Dysphoria—A depressed and anxious mood state.

Neurotransmitters—Chemicals in the brain that affect the nervous system and alter thinking patterns.

Opiates—Analgesic, pain killing drugs, such as heroin and morphine that depress the central nervous system.

opiate antagonist that blocks opiate receptors and reverses the effects of opiates, is administered to trigger the withdrawal response. Clonidine is given simultaneously to ease the symptoms of withdrawal. The patient is anesthetized throughout the three to four hour procedure, and withdrawal occurs while the patient sleeps. Vital signs are monitored closely and a ventilator may be employed.

In early 2004, a new single injection method for opiate **addiction** was being tested. It consisted of a slow-release form of buprenorphine and prevented symptoms for an entire month.

Stimulant withdrawal

Because of the depression and dysphoria (feeling of a psychological low) related to stimulant withdrawal, psychological and/or medical management is critical. Treatment may include a regimen of drugs that increase neurotransmitter production.

Prognosis

A closely observed, medically managed detoxification typically results in a safe and tolerable

withdrawal experience for the patient. Detoxification is only a short-term solution for obtaining abstinence. An addiction treatment and long-term recovery program is necessary to achieve long-term sobriety. Without such a treatment program, the likelihood of recurrence of abuse and, therefore, the recurrence of withdrawal syndrome is high.

Prevention

After detoxification, alcohol and drug dependent individuals are encouraged to maintain their abstinence through participation in **substance abuse** treatment or a twelve-step recovery program.

Resources

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"Single Injection May Relieve Opiate Withdrawal." *Alcoholism & Drug Abuse Weekly* January 19, 2004: 8.

ORGANIZATIONS

Alcoholics Anonymous. General Service Office. 475 Riverside Drive, New York, NY 10015. (212) 870-3400. < <http://www.alcoholics-anonymous.org> >.

National Clearinghouse for Alcohol and Drug Information. Center for Substance Abuse Prevention. P.O. Box 2345, Rockville, MD 20847-2345. (800) 729-6686. < <http://www.health.org> >.

National Council on Alcoholism and Drug Dependence. 12 West 21st St., New York, NY 10010. (800) 622-2255. < <http://www.ncadd.org> >.

National Institute on Alcohol Abuse and Alcoholism (NIAAA). 6000 Executive Boulevard, Willco Building, Bethesda, Maryland 20892-7003. < <http://www.niaaa.nih.gov> >.

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Teresa G. Odle

Wolff-Parkinson-White syndrome

Definition

Wolff-Parkinson-White syndrome is an abnormality in the electrical functioning of the heart which may cause rapid heart rates. The abnormality affects the electrical signal between the atria and ventricles.

Description

Blood is circulated through the heart and body by a muscular pump and valve system involving the atria and ventricles. The right atrium receives oxygen-lacking blood returning to the heart from the body. The blood is passed from the right atrium into the right ventricle, which contracts and sends blood out to the pulmonary artery. The pulmonary artery sends the blood into the lungs, where carbon dioxide is removed, and fresh oxygen is added. The left atrium receives blood with oxygen from the lungs and passes this arterial blood to the left ventricle, where it is emptied into the aorta, the main artery of the heart.

These functions are directed by electrical signals within the heart. In patients afflicted with Wolff-Parkinson-White syndrome, an abnormal pathway exists that causes additional electrical signals to pass between the atria and ventricles, possibly causing rapid heart rate.

Causes and symptoms

Congenital heart disease may contribute to this and other arrhythmias. Ebstein's anomaly, a congenital heart defect that involves displacement of the tricuspid valve, located on the right side of the heart, is one known cause of Wolff-Parkinson-White syndrome. This anomaly allows blood to flow via the small hole to the other side of the heart. Often, there is no known cause for Wolff-Parkinson-White syndrome. Many people with the syndrome have no symptoms. On the other hand, some people experience temporary rapid heartbeat due to certain drugs, **smoking**, and anxiety.

Diagnosis

Electrocardiography (ECG) is used to diagnose Wolff-Parkinson-White syndrome, and other cardiac **arrhythmias**. A trained physician, normally a cardiologist, can recognize patterns of electrical conduction. With this syndrome, the extra pathway will show a pattern different from those of normal conduction. If no irregular patterns show on the ECG, the patient may be sent home with a 24-hour heart monitor, called a Holter monitor, which will help detect intermittent occurrences. Other studies, such as the cardiac electrophysiologic study (EPS), may be ordered to pinpoint the location of the accessory pathway, and to determine a course of treatment.

Treatment

Various drugs may be used to treat Wolff-Parkinson-White syndrome, as well as other cardiac

KEY TERMS

Ablative—Used to describe a procedure involving removal of a tissue or body part, or destruction of its function.

Arrhythmia—Irregular heart beat.

Electrocardiograph (ECG)—A test of a patient's heartbeat that involves placing leads, or detectors, on the patient's chest to record electrical impulses in the heart. The test will produce a strip, or picture record of the heart's electrical function.

Tachycardia—Rapid heart rate, defined as more than 100 beats per minute.

arrhythmias. The purpose of these drugs is to slow the electrical signals and excitation of heart muscles. As some of these drugs may have side effects, including the rare production of new or more frequent arrhythmias, the patient should be carefully observed. Ablative therapies may be accomplished with radio-frequency or cardiac catheters to cut through the tissue which is causing the abnormal electrical signals.

At one time, only open heart surgery was used, but the procedure can be done now with local anesthesia in a special cardiac laboratory. In some cases, surgery may still be recommended to treat Wolff-Parkinson-White syndrome. Young people with this syndrome may be treated more successfully with surgery, rather than enduring a lifetime of drug treatments, or the possible threat of **sudden cardiac death**.

Alternative treatment

A provider may teach patients methods to help control heart rate. Relaxation techniques, acupuncture, botanical medicine, and homeopathy can all be helpful supportive therapies.

Prognosis

Most patients with this syndrome can lead normal lives, even with episodes of tachycardia. In many cases, the syndrome is secondary to the underlying congenital heart defect. However, Wolff-Parkinson-White syndrome can cause sudden cardiac arrest in certain instances.

Prevention

If the syndrome is not due to congenital heart disease, the patient may try avoiding behaviors

which lead to arrhythmia, such as elimination of **caffeine**, alcohol, **cocaine**, and smoking.

Resources

ORGANIZATIONS

American Heart Association. 7320 Greenville Ave. Dallas, TX 75231. (214) 373-6300 or (800) 242-8721. inquire@heart.org <<http://www.americanheart.org>>.

National Heart, Lung and Blood Institute. P.O. Box 30105, Bethesda, MD 20824-0105. (301) 251-1222. <<http://www.nhlbi.nih.gov>>.

Teresa Odle

Women's health

Definition

Women's health is the effect of gender on disease and health that encompasses a broad range of biological and psychosocial issues.

Description

Women's health is the concept that examines gender differences in health and disease states. The average life expectancy has almost doubled for women (79 years for women and 73 years for men), when compared with averages during the turn of the century. Because of the gender gap in lifespan, women comprise approximately two thirds of the population older than 65 and three fourths of the population aged 85 years and older. Currently the fastest growing group in the United States is persons aged 85 years and older. Because of gender life expectancy differences, it is estimated that at the beginning of the twenty-first century, women will outnumber men in the 85 years and older category by 3:1. The reasons for this variance are primarily due to physiological differences among men and women.

During different phases of a women's life cycle there are complex interactions that exist between sex hormones, physiological changes, and emotional issues. Physiological changes occur as early as embryonic development when hormones program structural differences between male and female brains. During reproductive years, sex hormones profoundly influence reproduction and development, which creates a spectrum of gender specific health issues. With advancing age and onset of **menopause**, women's risk factors for disease is comparably similar to men's.

Although the same disease may affect women as men, it is thought that biological mechanisms and psychosocial differences influence the clinical course of the disease (natural history) differently in women. The number of women working has doubled within the past 50 years. The effect of work **stress**, new environmental exposures and multiple roles is expected to have health and social impact.

The leading causes of **death** among women are cardiovascular disease, malignant **cancer**, cerebrovascular disease, chronic lung disease, pneumonia/influenza, and diabetes. Additionally, women can be prone to osteoporosis, alcohol **abuse**, psychological disorders, human immunodeficiency virus infection, and violence.

Heart Disease accounts for approximately a third of all deaths in women. About 250,000 women die annually of coronary heart disease or a one in three chance after age 40 years. The incidence of heart disease occurs about 10 years later in women than in men, since estrogens in premenopausal women has a protective effect. African American women are more prone to die from heart disease up to age 75. Beyond 75 years of age the propensity is reversed. Native American and Hispanic women have lower death rates from heart disease.

Malignant cancers are the most common cause of premature death among women. Breast cancer is the second leading cause of death in women and the most commonly diagnosed cancer. Lung cancer, secondary to cigarette **smoking** is the leading cause of cancer death among women.

Cerebrovascular disease, or stroked related deaths account for approximately 6% of all deaths in women and it is the third leading cause of mortality. The least common form of stroke, **subarachnoid hemorrhage**, is the more common cause in women.

The prevalence of cigarette smoking has increased greatly in women and this is correlated with pulmonary disease. Death rates for pulmonary disease including cancer and infectious causes of death are expected to rise for women.

Diabetes, a leading cause of death in women is more prevalent among Hispanic, African American, and Native American women. Past age 45, diabetes affects about one in six women.

Women can also develop:

- **osteoporosis**, or loss of the quantity of bone, common in postmenopausal women who have estrogen changes.

- alcohol abuse, characterized by repeated usage of alcohol despite negative consequences. These women frequently do not seek treatment because of fear of consequences (i.e., loss of child custody). This disease can also have adverse affects on fertility and in the developing fetus if the mother continues to consume alcohol (fetal alcohol syndrome).
- psychological disorders, such as depression and eating disorders.
- acquired **immunodeficiency** syndrome (**AIDS**), which represents the highest percent increase in death rates.
- violence, a leading cause of death, primarily caused by a perpetrator who is or was a partner.

Causes and symptoms

Cardiovascular disease can be caused by blockage of a blood vessel, high blood pressure, or a secondary complication to another disease. There may be an abnormal heart rhythm or cell death. Patients may complain of a broad spectrum of symptoms that may include **pain** chest discomfort, high blood pressure, or strain during physical exertion.

When attempting to define the cause and symptoms of cancer, it is important to assess the type of cancer and location. Additionally, if the tumor is localized (benign) or has spread to other areas (malignant), is vital for treatment planning and overall prognosis. In cases of breast cancer there may be a lump discovered during self-examination or **mammography** (special breast x rays).

Cerebrovascular disease may cause **tremors** (shaking), loss of balance and coordination, or functional and sensation loss of some parts of the body. Patients may have sudden transient strokes that could result in temporary loss of consciousness and amnesia of the incident. Patients may also develop chronic neurological states that causing memory loss and behavioral changes (**Alzheimer's disease**).

Patients with pulmonary (lung) cancer may develop shortness of breath, **fatigue**, weight loss, worsening **cough**, and coughing up bright red blood with sputum. Lung infections such as pneumonia may present with high **fever**, weakness, difficulty breathing, and abnormal breathe sounds heard with a stethoscope during **physical examination**.

Diabetes is a syndrome with disordered metabolism and high blood sugar due to an abnormality in the chemical that regulates sugar levels. It is characterized by an increased thirst, urination, and chronic skin infections.

Osteoporosis may cause the bones to be brittle and weak. It is usually not detected until bones start to break.

The alcohol abuser will continue to drink despite negative repercussions. The person may not seek treatment to evade legal and/or child custody problems. The patient may hide alcohol, or confine drinking to specific times. The disease progresses to where there may be permanent liver damage, memory blackouts and **malnutrition**.

Depression may manifest a loss of interest and desire. Patients may have difficulty getting out of bed. They may lack motivation to work or tend to daily activities.

Patients with AIDS may not have symptoms for years. When active disease occurs, patients will typically develop recurrent infections that are the usual cause of death.

Domestic violence is usually associated with a perpetrator who is in a relationship with the affected person. Abuse can be manifested by physical violence and/or homicide.

Diagnosis

Diagnosis can be accomplished with a history, physical examination, and specialized tests or procedures. For cardiovascular disease an electrocardiogram can determine the activity of the heart. Additional tests may include **echocardiography** (ultrasonic waves that generate an image), stress testing, and studies that require placing a catheter with a probe to examine the damage to heart tissue. Special tests with dyes may also be injected to enhance visualization. Cancer may be detected using specialized test called tumor makers and imaging studies such as MRI and CAT scans. Cerebrovascular disease can be detected with a complete neurological examination and specialized imaging technology. Diabetes is usually detected by a careful history presence of risk factors (**obesity**) and blood analysis of glucose levels. Osteoporosis can be evaluated with specialized bone densitometry. Alcohol abuse can be established by a bio-psycho-social assessment and standardized tests which screen for this disorder. Psychological evaluation (such as the **Minnesota Multiphasic Personality Inventory**, MMPI) can usually detect depression or eating disorders. AIDS can be established by a careful history, belonging to high-risk groups and Western blot analysis (examination of blood to detect the protein of human immunodeficiency virus). Violence can be established by physical signs of beating, such as cuts and **bruises**.

Treatment

Treatment depends on the extent of disease and the present health status of the patient. Additionally, in some cases treatment may be stopped at sometime, or it may altogether be refused. Treatment for cardiovascular disease may include surgical intervention and/or conservative medical treatment with medications. Diet, **exercise**, and weight reduction are important parameters for treatment planning. Appropriate referrals, counseling, and follow up are usually indicated. Treatment for cancer may include a combination of surgery, chemotherapy or radiation therapy. These treatment modalities may be given singly or in combination or at different times during disease progression. Cerebrovascular disease can be treated surgically and/or with medications that thin the blood. Symptomatic care may be indicated in addition to close monitoring if the patient develops disability and/or cognitive impairment. Diabetes can be treated by dietary modifications and medications, which treat abnormal levels of blood glucose (sugar). Osteoporosis can be treated with estrogen replacement and regular vitamin/mineral intake. Alcohol abuse may require long-term therapy, inpatient treatment and medications. Community centered support group meetings are also recommended as a form of treatment maintenance. To date there is no treatment for AIDS, other than medications, that offer symptomatic relief. Alcohol abuse, psychological disorders and violence require therapy, possible medication, and community centered support group meetings.

Alternative treatment

There are numerous studies which support intake of coenzyme Q₁₀ for cardiovascular health. Studies have shown that beta-carotene and vitamin E and C have no effect for cancer. Some studies indicate positive results for reproductive health using **acupuncture**. Some advocates proposed certain herbs may be beneficial during menopause. According to most medical literature, further research using scientific method is vital for general acceptance.

Prognosis

The prognosis depends on the extent of disease and the physical and emotional status of the patient. Prognosis is also related to tolerance of treatment, adverse drug effects, and complication during or after surgery, disease resurgence and patient compliance with treatment recommendations.

KEY TERMS

Electrocardiogram— An instrument that monitors heart rate and rhythm.

Prevention

One of the most reliable measures of prevention is education and training. The Council on Graduate Medical Education has provided funding for numerous centers to research women health issues. On more individual level preventive and personal habits are vital for good health. Most physicians believe that a baseline physical examination is a reliable comparative tool. Women should receive counseling for special issues concerning cigarette smoking, exercise, diet, primary disease prevention, safe sexual practices, alcohol abuse, psychological disorders, and violence. Additionally, knowledge of family history is important since many diseases have a strong propensity among first-degree relatives. Blood pressure should normally be measured every other year. Screening tests for breast, cervical, and colorectal cancer is recommended. Pap smears taken during routine pelvic examinations can screen for disease processes in the reproductive tract. Serum cholesterol monitoring and reduction are advised. Patients may require postmenopausal estrogen replacement therapy and vitamin/mineral supplements.

Resources

BOOKS

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PERIODICALS

Lautenbach, G. L., and M. Petri. "Women's Health: General Medical Care of the Patient with Rheumatic Disease." *Rheumatic Diseases Clinics of North America* 25 (August 1999).

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Wound culture

Definition

A wound culture is a laboratory test in which microorganisms from a wound are grown in a special

growth medium. It is done to find and identify the microorganism causing an infection in a wound or an **abscess**. If a microorganism is found, more testing is done to determine how to treat the infection.

Purpose

Wounds are injuries to body tissues caused by disease processes or events such as **burns**, punctures, and human or animal bites. Wounds or abscesses also occur within body tissues as a result of surgery or dental procedures. Wounds become infected when microorganisms from the outside environment, or from within the person's body, enter the open wound and multiply. A wound that is red, painful, swollen, and draining pus is probably infected. A **fever** following surgery indicates an infection at the site of surgery.

To enable healing and prevent the spread of infection to other body tissues, the infecting microorganisms must be killed. A wound culture discovers which type of microorganism is causing the infection and the best antibiotic with which to kill it. This is important as physicians have become less inclined to prescribe **antibiotics** until certain they are needed because of antibiotic resistance that has developed due to overuse of the drugs.

Description

A sample of material, such as pus or a portion of tissue, is taken from the wound, placed in a sterile container, and sent to the laboratory. In the laboratory, this material is spread over the surface of several different types of culture plates and placed in an incubator at body temperature for one to two days.

A Gram stain is done by staining the slide with purple and red stains, then examining it under a microscope. If many white blood cells and bacteria are seen, it is an early confirmation of infection. The color of stain retained by the bacteria (purple or red), their shape (such as round or rectangular), and their size provides valuable clues as to their identity, and helps the physician predict which antibiotics might work best even before the entire test is completed. Bacteria that stain purple are called gram-positive; those that stain red are called gram-negative.

Bacteria can be grouped into two categories: aerobes and anaerobes. Aerobes are bacteria that need oxygen to live; anaerobes live only where there is no oxygen. Deep wounds, closed-off from oxygen, are an ideal environment for an anaerobic infection to develop. Foul-smelling odor, gas, or **gangrene** at the infection site are signs of an infection caused by an

anaerobic bacteria. Routine cultures typically only look for aerobic bacteria. If the physician tells the laboratory to include a culture for anaerobes, a portion of the wound sample will be put on culture plates, or in a tube of culture broth, and incubated in a special chamber without oxygen.

Bacteria present in the wound sample will multiply and appear as visible colonies on the plates, or as cloudiness in the tube of broth. They are identified by the appearance of their colonies, the results of biochemical tests, and information from Gram staining part of the bacterial colony.

A sensitivity test, also called an antibiotic susceptibility test, is also done. The bacteria are tested against different antibiotics to determine which will treat the infection by killing the bacteria.

If the physician thinks the wound may be infected with a mold or yeast, a fungal culture is also done. The wound sample is spread on special culture plates that are treated to encourage the growth of mold and yeast. Different biochemical tests and stains are used to identify molds and yeast.

Other more unusual microorganisms, such as *Mycobacterium leprae*, may be the cause of a wound infection. The physician must notify the laboratory to culture specifically for these more unusual microorganisms.

The initial Gram stain result is available the same day, or in less than an hour if requested by the physician. An early report, known as a preliminary report, is usually available after one day. This report will tell if any microorganisms have yet been found, and, if so, their Gram stain appearance. For example, they may have the appearance of a gram-negative rod, or a gram-positive cocci (spherical shape). The final report, usually available in one to three days, includes complete identification, an estimate of the quantity of the microorganisms, plus a list of the antibiotics to which they are sensitive. Cultures for fungi and anaerobic bacteria may take two to three weeks.

Wound culture is also called soft tissue culture, abscess culture, or wound culture and sensitivity.

Preparation

A piece of the infected tissue is the best specimen. If this is not possible, the next best specimen is pus from the wound. Because many microorganisms normally live on skin and mucous membrane, the specimen must not be allowed to touch the area surrounding the wound.

KEY TERMS

Aerobe—Bacteria that require oxygen to live.

Anaerobe—Bacteria that live only where there is no oxygen.

Normal flora—The mixture of bacteria normally found at specific body sites.

The physician first cleans the surface of the wound using alcohol. Using a syringe, the physician suctions out (aspirates) as much pus as possible from the wound. Next, this is sent to the laboratory in a sterile container. If it is impossible to aspirate the pus, pus from within the wound can be collected on a swab.

The physician may choose to start the person on an antibiotic before the culture and sensitivity tests are completed. However, the specimen for culture should be collected before antibiotics are begun. Antibiotics in the person's system may prevent microorganisms present in the wound from growing in culture, and thus not be identifiable.

Normal results

A normal culture may be contaminated by a mixture of microorganisms normally found on a person's skin (normal flora).

It is not uncommon for the microorganism causing a wound infection to not grow in culture. This is particularly true if the specimen was collected with a swab rather than an aspirate or tissue biopsy.

Abnormal results

Streptococcus Group A, *Escherichia coli*, *Proteus*, *Klebsiella*, *Pseudomonas*, *Enterobacter*, Enterococci, *Staphylococcus aureus*, *Bacterioides*, and *Clostridium*, are common causes of wound infections. More than one microorganism may be the cause of the infection.

Resources

PERIODICALS

"Does Increased Use of Antibiotics Result in Increased Antibiotics Resistance?" *Clinical Infectious Diseases* July 1, 2004: 18–20.

ORGANIZATIONS

Wound Healing Society. 1550 South Coast Highway, Suite 201, Laguna Beach, CA 92651. (888) 434-4234.
< <http://wizard.pharm.wayne.edu/woundsoc/WHS.HTM> >.

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Wound flushing

Definition

Wound flushing is a method of cleaning a wound by applying pressurized water or antiseptic solutions to the tissues. It is also called irrigation.

Purpose

Wound flushing is used to help flush debris from a wound, lessening the risk of infection or treating an infection that already exists. If the wound is flushed with an antiseptic, it is more likely to heal correctly; flushing the wound can help prevent the surface from healing over a possibly infected area underneath.

Wound flushing is routinely performed by dentists or oral surgeons following tooth extractions, mouth trauma, or gum surgery, to prevent bacteria from the mouth entering the bloodstream as well as to cleanse the tooth socket.

Description

Wound flushing is usually done in a hospital or oral surgery center, though if it is performed at home, there is less chance of infection because of the higher risk of bacterial contamination in the hospital environment. Wound flushing is especially helpful in treating people with bites, lacerations, or crush injuries, which often become infected due to the presence of dead tissue and foreign debris, such as splinters or dirt. In a non-surgical situation, the procedure is usually performed by a nurse. An acute injury, such as a crushing wound or knife cut, the wound is flushed right before the injury is stitched closed. For people with chronic **wounds**, such as bed sores or abscesses, the wound may be flushed periodically to treat or prevent infection. During an operation, a surgeon uses an antibacterial solution to flush the surgical site just before stitching the wound closed. After surgery, the wounds may be flushed to treat or prevent infection.

KEY TERMS

Antiseptic—Chemicals applied to the skin to destroy bacteria and prevent infection.

Irrigation—The medical name for wound flushing.

Preparation

The nurse or doctor may inject the site with a local anesthetic before flushing the wound.

Aftercare

After the wound is flushed, the health care provider cleans the area around the wound to guard against infection. Packing to absorb excess fluids may be placed into the wound, followed by a sterile bandage.

Risks

Complications rarely occur, especially if the solution used to flush the wound is chosen carefully so as to avoid skin irritation; occasionally, however, serious infections are reported. In addition, damage to skin or internal organs has been reported from the use of hydrogen peroxide to flush wounds or irrigate the abdominal cavity after surgery.

Patients should call the doctor immediately if there is any sign of infection, such as **fever**, pus, or swelling.

Normal results

The wound will heal correctly, from the inside out, without infection.

Resources

BOOKS

Beers, Mark H., MD, and Robert Berkow, MD., editors. "Dentistry in Medicine." Section 9, Chapter 103 In *The Merck Manual of Diagnosis and Therapy*. Whitehouse Station, NJ: Merck Research Laboratories, 2002.

PERIODICALS

Watt, B. E., A. T. Proudfoot, and J. A. Vale. "Hydrogen Peroxide Poisoning." *Toxicological Reviews* 23 (January 2004): 51–57.

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Wounds

Definition

A wound occurs when the integrity of any tissue is compromised (e.g. skin breaks, muscle tears, **burns**, or bone **fractures**). A wound may be caused by an act, such as a gunshot, fall, or surgical procedure; by an infectious disease; or by an underlying condition.

Description

Types and causes of wounds are wide ranging, and health care professionals have several different ways of classifying them. They may be chronic, such as the skin ulcers caused by **diabetes mellitus**, or acute, such as a gunshot wound or animal bite. Wounds may also be referred to as open, in which the skin has been compromised and underlying tissues are exposed, or closed, in which the skin has not been compromised, but trauma to underlying structures has occurred (e.g. a bruised rib or cerebral contusion). Emergency personnel and first-aid workers generally place acute wounds in one of eight categories:

- **Abrasions.** Also called scrapes, they occur when the skin is rubbed away by friction against another rough surface (e.g. rope burns and skinned knees).
- **Avulsions.** Occur when an entire structure or part of it is forcibly pulled away, such as the loss of a permanent tooth or an ear lobe. Explosions, gunshots, and animal bites may cause avulsions.
- **Contusions.** Also called **bruises**, these are the result of a forceful trauma that injures an internal structure without breaking the skin. Blows to the chest, abdomen, or head with a blunt instrument (e.g. a football or a fist) can cause contusions.
- **Crush wounds.** Occur when a heavy object falls onto a person, splitting the skin and shattering or tearing underlying structures.
- **Cuts.** Slicing wounds made with a sharp instrument, leaving even edges. They may be as minimal as a paper cut or as significant as a surgical incision.
- **Lacerations.** Also called tears, these are separating wounds that produce ragged edges. They are produced by a tremendous force against the body, either from an internal source as in **childbirth**, or from an external source like a punch.
- **Missile wounds.** Also called velocity wounds, they are caused by an object entering the body at a high speed, typically a bullet.

- **Punctures.** Deep, narrow wounds produced by sharp objects such as nails, knives, and broken glass.

Causes and symptoms

Acute wounds have a wide range of causes. Often, they are the unintentional results of motor vehicle accidents, falls, mishandling of sharp objects, or sports-related injury. Wounds may also be an intentional result of violence involving assault with weapons, including fists, knives, or guns.

The general symptoms of a wound are localized **pain** and bleeding. Specific symptoms include:

- An abrasion usually appears as lines of scraped skin with tiny spots of bleeding.
- An avulsion has heavy, rapid bleeding and a noticeable absence of tissue.
- A contusion may appear as a bruise beneath the skin or may appear only on imaging tests; an internal wound may also generate symptoms such as weakness, perspiration, and pain.
- A crush wound may have irregular margins like a laceration; however, the wound will be deeper and trauma to muscle and bone may be apparent.
- A cut may have little or profuse bleeding depending on its depth and length; its even edges readily line up.
- A laceration too may have little or profuse bleeding; the tissue damage is generally greater and the wound's ragged edges do not readily line up.
- A missile entry wound may be accompanied by an exit wound, and bleeding may be profuse, depending on the nature of the injury.
- A puncture wound will be greater than its length, therefore there is usually little bleeding around the outside of the wound and more bleeding inside, causing discoloration.

Diagnosis

A diagnosis is made by visual examination and may be confirmed by a report of the causal events. Medical personnel will also assess the extent of the wound and the effect it has had on the patient's well being (e.g. profound blood loss, damage to the nervous system or skeletal system).

Treatment

Treatment of wounds involves stopping any bleeding, then cleaning and dressing the wound to prevent infection. Additional medical attention may



A close-up of a hard-contact gunshot wound with accompanying burn marks on the left and right sides of the wound. (Custom Medical Stock Photo. Reproduced by permission.)

be required if the effects of the wound have compromised the body's ability to function effectively.

Stopping the bleeding

Most bleeding may be stopped by direct pressure. Direct pressure is applied by placing a clean cloth or dressing over the wound and pressing the palm of the hand over the entire area. This limits local bleeding without disrupting a significant portion of the circulation. The cloth absorbs blood and allows clot formation; the clot should not be disturbed, so if blood soaks through the cloth, another cloth should be placed directly on top rather than replacing the original cloth.

If the wound is on an arm or leg that does not appear to have a broken bone, the wound should be elevated to a height above the person's heart while direct pressure is applied. Elevating the wound allows gravity to slow down the flow of blood to that area.

If severe bleeding cannot be stopped by direct pressure or with elevation, the next step is to apply pressure to the major artery supplying blood to the area of the wound. In the arm, pressure would be applied to the brachial artery by pressing the inside of the upper arm against the bone. In the leg, pressure would be applied to the femoral artery by pressing on the inner crease of the groin against the pelvic bone.

If the bleeding from an arm or leg is so extreme as to be life-threatening and if it cannot be stopped by any other means, a tourniquet may be required. However, in the process of limiting further blood loss, the tourniquet also drastically deprives the limb tissues of oxygen. As a result, the patient may live but the limb may die.



A defensive hand wound from a knife attack. (Photograph by D. Willoughby, Custom Medical Stock Photo. Reproduced by permission.)

In 2004, a new solution to stopping bleeding was reported. Called QuikClot, the FDA-approved substance is made up of synthetically made material called zeolite, which occurs naturally in volcanic rock. When used properly, it can be poured into a wound that will not stop bleeding and will slow blood loss. The trauma pack costs about \$50 and has shown particular promise in the battlefield and in wilderness situations.

Dressing the wound

Once the bleeding has been stopped, cleaning and dressing the wound is important for preventing infection. Although the flowing blood flushes debris from the wound, running water should also be used to rinse away dirt. Embedded particles such as wood splinters and glass splinters, if not too deep, may be removed with a needle or pair of tweezers that has been sterilized in rubbing alcohol or in the heat of a flame. Once the wound has been cleared of foreign material and washed, it should be gently blotted dry, with care not to disturb the blood clot. An antibiotic ointment may be applied. The wound should then be covered with a clean dressing and bandaged to hold the dressing in place.

Getting medical assistance

A person who has become impaled on a fixed object, such as a fence post or a stake in the ground, should only be moved by emergency medical personnel. **Foreign objects** embedded in the eye should only be removed by a doctor. Larger penetrating objects, such as a fishhook or an arrow, should only be removed by a doctor to prevent further damage as they exit.

Additional medical attention is necessary in several instances. Wounds which penetrate the

muscle beneath the skin should be cleaned and treated by a doctor. Such a wound may require stitches to keep it closed during healing. Some deep wounds which do not extend to the underlying muscle may only require butterfly bandages to keep them closed during healing. Wounds to the face and neck, even small ones, should always be examined and treated by a doctor to preserve sensory function and minimize scarring. Deep wounds to the hands and wrists should be examined for nerve and tendon damage. Puncture wounds may require a **tetanus** shot to prevent serious infection. Animal bites should always be examined and the possibility of **rabies** infection determined.

Infection

Wounds that develop signs of infection should also be brought to a doctor's attention. Signs of infection are swelling, redness, tenderness, throbbing pain, localized warmth, **fever**, swollen lymph glands, the presence of pus either in the wound or draining from it, and red streaks spreading away from the wound.

Emergency treatment

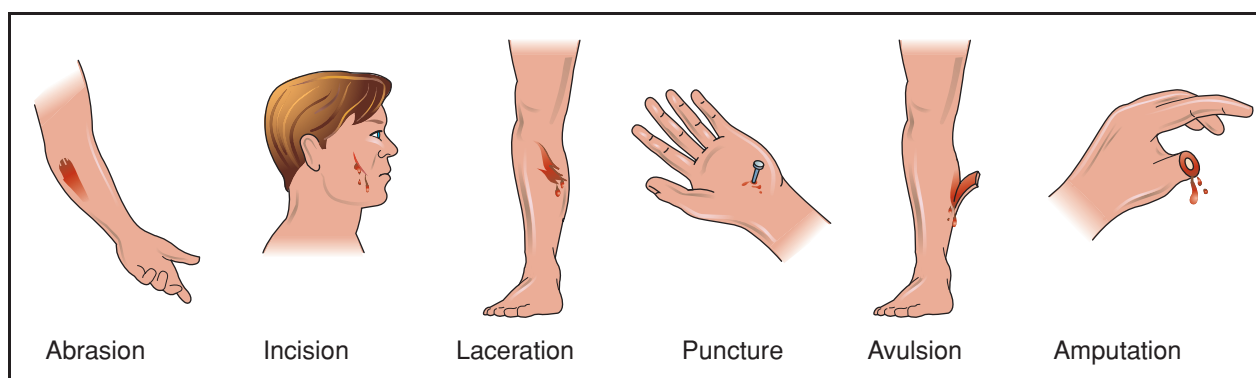
With even as little as one quart of blood lost, a person may lose consciousness and go into traumatic **shock**. Because this is life-threatening, emergency medical assistance should be called immediately. If the person stops breathing, artificial respiration (also called mouth-to-mouth resuscitation or rescue breathing) should be administered. In the absence of a pulse, **cardiopulmonary resuscitation (CPR)** must be performed. Once the person is breathing unassisted, the bleeding may be attended to.

In cases of severe blood loss, medical treatment may include the intravenous replacement of body fluids. This may be infusion with saline or plasma, or a **transfusion** of whole blood.

In some cases, clinicians have resorted to a Civil War-era treatment that does not sound appealing, but works well enough to receive FDA approval. Maggots can be placed on wounds that refuse to heal with high-tech medical methods. The maggots are dropped into the wound and covered with special mesh to keep them in place. They are removed in two to three days.

Alternative treatment

In addition to the conventional treatments described above, there are alternative therapies that



Examples of open wounds. (Illustration by Electronic Illustrators Group.)

KEY TERMS

Abrasion—Also called a scrape. The rubbing away of the skin surface by friction against another rough surface.

Avulsion—The forcible separation of a piece from the entire structure.

Butterfly bandage—A narrow strip of adhesive with wider flaring ends (shaped like butterfly wings) used to hold the edges of a wound together while it heals.

Cut—Separation of skin or other tissue made by a sharp edge, producing regular edges.

Laceration—Also called a tear. Separation of skin or other tissue by a tremendous force, producing irregular edges.

Plasma—The straw-colored fluid component of blood, without the other blood cells.

Puncture—An injury caused by a sharp, narrow object deeply penetrating the skin.

Tourniquet—A device used to control bleeding, consisting of a constricting band applied tightly around a limb above the wound. It should only be used if the bleeding is life-threatening and can not be controlled by other means.

Traumatic shock—A condition of depressed body functions as a reaction to injury with loss of body fluids or lack of oxygen. Signs of traumatic shock include weak and rapid pulse, shallow and rapid breathing, and pale, cool, clammy skin.

Whole blood—Blood that contains red blood cells, white blood cells, and platelets in plasma.

may help support the injured person. Homeopathy can be very effective in acute wound situations. *Ledum* (*Ledum palustre*) is recommended for puncture wounds (taken internally). *Calendula* (*Calendula officinalis*) is the primary homeopathic remedy for wounds. An antiseptic, it is used topically as a succus (juice), tea, or salve. Another naturally occurring antiseptic is tea tree oil (*Melaleuca* spp.), which can be mixed with water for cleaning wounds. Aloe (*Aloe barbadensis*) can be applied topically to soothe skin during healing. When wounds affect the nerves, especially in the arms and legs, St.-John's-wort (*Hypericum perforatum*) can be helpful when taken internally or applied topically. **Acupuncture** can help support the healing process by restoring the energy flow in the meridians that have been affected by the wound. In some cases, vitamin E taken orally or applied topically can speed healing and prevent scarring.

Prognosis

Without the complication of infection, most wounds heal well with time. Depending on the depth and size of the wound, it may or may not leave a visible scar.

Prevention

Most actions that result in wounds are preventable. Injuries from motor vehicle accidents may be reduced by wearing seat belts and placing children in size-appropriate car seats in the back seat. Sharp, jagged, or pointed objects or machinery parts should be used according to the manufacturer's instructions and only for their intended purpose. Firearms and explosives should be used only by adults with explicit training; they should

also be kept locked and away from children. Persons engaging in sports, games, and recreational activities should wear all proper protective equipment and follow safety rules.

Resources

PERIODICALS

“Maggots Make Medical Comeback for Healing Wounds.” *Health & Medicine Week* August 23, 2004: 287.

Thornton, Jim. “Stop the Bleeding! An Innovative Treatment for Wilderness Wounds” *Field & Stream* September 1, 2004: 48.

ORGANIZATIONS

American Red Cross. P.O. Box 37243, Washington, D.C. 20013. <<http://www.redcross.org>>.

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Wryneck see **Torticollis**



X-linked agammaglobulinemia

Definition

X-linked agammaglobulinemia (XLA) or Bruton's agammaglobulinemia is present at birth (congenital) and is characterized by low or completely absent levels of immunoglobulins in the bloodstream. Immunoglobulins are protein molecules in blood serum that function like antibodies. Without them, the body lacks a fully functioning immune system. Persons with XLA are vulnerable to repeated, potentially fatal bacterial infections.

Description

XLA occurs in one in 50,000 to one in 100,000 newborns. Almost all persons with the disorder are males. Although persons with XLA carry the genes to produce immunoglobulins, a genetic defect on the X chromosome prevents their formation. This defect is not associated with the immunoglobulins themselves, but rather with the B cells in the bloodstream that ordinarily secrete the immunoglobulins.

B cells are a type of white blood cell. They are the sole producers of immunoglobulins in the body. B cells are produced in the bone marrow and carried to the spleen, lymph nodes, and other organs as they mature. The maturation process depends on an enzyme called Bruton's agammaglobulinemia tyrosine kinase (Btk). If Btk is missing or defective, the B cells cannot mature and cannot produce immunoglobulins.

The gene for Btk is on the X chromosome. Certain changes (mutations) in this gene result in defective Btk. Since the gene is carried on the X chromosome, XLA individuals are almost always male. Females have two X chromosomes, which means they have two copies of the Btk gene, one of which is normal. Males have only one X chromosome.

Causes and symptoms

XLA is caused by a defect in the gene that codes for Btk. This defect leads to blocked maturation of B cells, the cells that produce immunoglobulins. Because other portions of the immune system are functional, people with XLA can fight off some types of infection, such as fungal and most viral infections. Immunoglobulins, however, are vital to combat bacterial infections. Infants with XLA usually do not show symptoms during the first six months of life because immunoglobulins from their mothers are circulating in their bloodstreams. As the mother's supply decreases, the baby becomes increasingly vulnerable to bacterial infections.

Common symptoms of immunoglobulin deficiency appear after the infant is six months old. They include frequent ear and sinus infections, **pneumonia**, and gastroenteritis. Certain viruses, such as hepatitis and **polio** viruses, can also pose a threat. Children with XLA grow slowly, have small tonsils and lymph nodes, and may develop chronic skin infections. Approximately 20% of these children develop arthritis, possibly as a result of joint infections.

Diagnosis

Frequent bacterial infections, a lack of mature B cells, and low-to-nonexistent levels of immunoglobulins point to a diagnosis of XLA. A sample of the infant's blood serum can be analyzed for the presence of immunoglobulins by a technique called **immunoelectrophoresis**. To make a definitive diagnosis, the child's X chromosome is analyzed for defects in the Btk gene. Similar analysis can be used for prenatal diagnosis or to detect carriers of the defective gene.

Treatment

Treatment of XLA consists of regular intravenous doses of commercially prepared gamma globulin (sold

KEY TERMS

Antibody—A molecule that is produced by the immune system in response to a protein, called an antigen, that is not recognized as belonging in the body.

B cell—A type of lymphocyte, or white blood cell, that is a key component of the body's immune system. Mature B cells produce immunoglobulins.

Bruton's agammaglobulinemia tyrosine kinase (Btk)—An enzyme vital for the maturation of B cells.

Carrier—A person who has a genetic defect but does not develop any symptoms or signs of the defect. The carrier's offspring may inherit the defect and develop the associated disorder.

Enzyme—A protein molecule that prompts rapid biochemical reactions.

Immunoglobulin—A protein molecule formed by mature B cells in response to foreign proteins in the body. There are five types of immunoglobulins, but the major one is gamma globulin or immunoglobulin G.

Mutation—A change in a gene that alters the function or other characteristics of the gene's product.

X chromosome—One of the two sex chromosomes (the other is Y) that determine a person's gender. Normal males have both an X and a Y chromosome, and normal females have two X chromosomes.

under the trade names Gamimune or Gammagard) to ward off infections. **Antibiotics** are used to treat infections as they occur. Children with XLA must be treated promptly for even minor cuts and scrapes, and taught to avoid crowds and people with active infections.

Prognosis

Prior to the era of gamma globulin and antibiotic treatment, approximately 90% of XLA individuals died before age 8. Early diagnosis and current therapy allows most individuals with XLA to reach adulthood and lead relatively normal lives. Infants who develop polio or persistent viral infections, however, have a poorer prognosis.

Prevention

Parents of a child with XLA should consider genetic counseling if they are planning to have more children.

Resources

ORGANIZATIONS

Immune Deficiency Foundation. 25 W. Chesapeake Ave., Suite 206, Towson, MD 21204. (800) 296-4433.
< <http://www.primaryimmune.org> >.

National Organization for Rare Disorders. P.O. Box 8923, New Fairfield, CT 06812-8923. (800) 999-6673.
< <http://www.rarediseases.org> >.

Julia Barrett

X rays of the eye's orbit see **X rays of the orbit**

X rays of the orbit

Definition

Orbital x rays are studies of the area and structures containing the eye. The orbit is the circle of thin bones that houses and protects the eye, even extending behind the eye and nearly wrapping around it. The orbit includes the eyebrow, the bridge of the nose and the cheekbone. X rays are a form of radiation (like light) that can penetrate body tissues.

Purpose

Orbital x ray, or orbital radiography, is often used to detect problems resulting from injury or trauma to the eye. The exam may also detect changes to the structure of the eye, which may indicate various diseases. An ophthalmologist may also order an orbital x ray if there is concern that foreign bodies may be present in the eye that cannot be detected with an instrument called an ophthalmoscope.

Precautions

Pregnant women and women who could possibly be pregnant should only receive orbital x rays when absolutely necessary. If the patient is in severe **pain** due to injury or trauma, a painkiller may be given to help ease discomfort during positioning of the head throughout the exam. No other precautions are necessary for orbital x rays.

Description

Each orbit is composed of a floor, a roof, a medial (in the center plane) and lateral (sides of the plane) walls. The orbital x ray involves several different views in order for the physician to clearly see various parts of

the eye without obstruction. In orbital x rays, images of the unaffected eye may also be taken to compare its shapes and structures to those of the affected eye. Views may include side view (lateral), back to front (posteroanterior), base view, views from both sides, and an image from the center to one outside edge (half-axial projection). Projections of the optical canal will also be included. For all of these views, the patient may be seated upright or asked to lie on a table in the x ray room.

The orbital x ray procedure should take about 15 minutes to complete. Following the procedure, the patient will usually be asked to wait until the films are developed to ensure they are high enough quality and that repeat x rays are not necessary. A physician may perform the x ray exam in his or her office, or refer the patient to an outpatient radiology facility or hospital radiology department. In the case of emergency, the exam may be performed in the emergency room or a nearby radiology area of the hospital.

Preparation

There are no special dietary preparations needed prior to an orbital x ray. As with any radiography procedure, the patient should remove any jewelry or metal objects, which may interfere with a clear image.

Aftercare

No aftercare is required following this diagnostic test.

Risks

Radiation exposure is low for this procedure and all certified radiology facilities follow strict personnel and equipment guidelines for radiation protection. Women of child bearing age and children should be offered protective shielding (lead aprons) to cover the genital and/or abdominal areas.

Normal results

Normal findings will show the bones of the orbit intact, and will show similarity between the orbit that is being studied and the unaffected orbit.

Abnormal results

Positive findings from an orbital x ray may show that there has been injury to the eye. Certain signs

KEY TERMS

Blowout fracture—A fracture or break in the orbit that is caused by sudden and violent impact to the area.

Malignancy—A malignancy is a tumor that is cancerous and growing.

Medial wall—The middle bone, or wall of the eye's orbit. It is generally thicker than the roof and floor walls.

Ophthalmologist—A physician who specializes in the workings, structures and care of the eyes.

Ophthalmoscope—An instrument routinely used by ophthalmologists to examine the interior of the eye. It consists of a small light, a mirror, and lenses of differing powers that magnify.

Radiography—Examination of any part of the body through the use of x rays. The process produces an image of shadows and contrasts on film.

X ray—A form of electromagnetic radiation with shorter wavelengths than normal light. X rays can penetrate most structures.

may indicate some disease that is affecting the orbital structures. Tiny **fractures** in the orbital bones can usually be detected on the radiograph. The floor bone, the medial wall and the ethmoid bone, which is a spongy bone that forms the upper part of the nasal cavity, are the most likely to break. In a blowout fracture (one involving the orbital floor), radiographic findings may include disruption to the orbital floor, an opaque look to the sinuses on the same side as the affected orbit (due to hemorrhage) or signs of sinus problems from the orbital root's interference. These indications can be seen in most typical orbital x ray views.

Since the physician examines both orbits side by side, indications of differences in size and shape of the various structures in the orbit may be apparent. The orbit may be enlarged, indicating irritation from an injury or foreign body. A number of growing tumors within the eye or brain area may also cause orbital enlargement. Destruction of the walls of the orbit may indicate a nearby infection or malignancy. Changes in density of the tiny orbit bones may also be a sign of bone disease or **cancer** spread to bone.

Children's orbits are more likely to be enlarged by a fast growing lesion, since their orbital bones have not fully developed.

Resources

ORGANIZATIONS

American Academy of Ophthalmology. 655 Beach Street,
P.O. Box 7424, San Francisco, CA 94120-7424.
< <http://www.eyenet.org> >.

National Eye Institute. 2020 Vision Place, Bethesda, MD
20892-3655. (301) 496-5248. < <http://www.nei.nih.gov> >.

OTHER

Scgeue Eye Institute. University of PennsylvaniaHealth
System. < <http://www.med.upenn.edu/opth> >.

Teresa Odle

Xerophthalmia see **Vitamin A deficiency**

Y

Yaws

Definition

Yaws is a chronic illness which first affects the skin, and then affects the bones.

Description

Yaws tends to strike children, particularly between the ages of two and five. It is common in areas where poverty and overcrowding interfere with good hygiene practices. The most common locations are in rural areas throughout Africa, Southeast Asia, and in locations bordering the equator in the Americas.

Causes and symptoms

Yaws is caused by a spiral-shaped bacterium (spirochete) called *Treponema pertenue*. This bacterium belongs to the same family as the bacterium that causes **syphilis**.

Yaws is passed among people by direct skin contact. It requires some kind of a scratched or insect bitten area in order for the bacteria to actually settle in and cause infection. An injured spot on the leg is the most common part of the body through which the bacteria enter. Young children, who are constantly bumping themselves in play, who wear little clothing, who do not wash their hands often, and who may frequently put their hands in their mouths, are particularly susceptible.

The first symptom of yaws occurs three to four weeks after acquiring the bacteria. The area where the bacteria originally entered the skin becomes a noticeable bump (papule). The papule grows larger and develops a punched-out center (ulcer), covered with a yellow crust. Lymph nodes in the area may become swollen and tender. This first papule may take as long

as six months to heal. Secondary soft, gummy growths then appear on the face, arms and legs, and buttocks. These soft, tumor-like masses may grow on the soles of the feet, causing the patient to walk in an odd and characteristic fashion on the sides of his or her feet (nicknamed “crab yaws”). More destructive tumors may then disrupt the bones of the face, the jaw, and the lower leg. Ulcers around the nose and on the face may be very mutilating.

Diagnosis

Samples taken from the first papules may be examined using a technique called dark-field microscopy. This often allows the spirochetes to be identified. They may also be identified in fluid withdrawn from swollen lymph nodes. Various tests can also be run on blood samples to determine if an individual is producing antibodies (special immune cells) which are specifically made in response to the presence of these spirochetes.

Treatment

A single penicillin injection in a muscle is sufficient to completely end the disease.

Prognosis

Without treatment, yaws is a terribly disfiguring chronic illness. With appropriate treatment, the progression of the disease can be completely halted.

Prevention

For a time, the World Health Organization (WHO) was working to totally eradicate yaws, just as **smallpox** was successfully eradicated. This has not occurred, however. WHO continues to work to identify and respond to outbreaks quickly, in an effort to at least slow the spread of yaws.

KEY TERMS

Papule—A raised bump on the skin.

Ulcer—A punched-out, irritated pit on the skin.

Resources

ORGANIZATIONS

Centers for Disease Control and Prevention. 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Rosalyn Carson-DeWitt, MD

Yellow fever

Definition

Yellow **fever**, which is also known as sylvatic fever and viral hemorrhagic fever or VHF, is a severe infectious disease caused by a type of virus called a flavivirus. This flavivirus can cause outbreaks of epidemic proportions throughout Africa and tropical America.

The first written evidence of such an epidemic occurred in the Yucatan in 1648. Since that time, much has been learned about the interesting transmission patterns of this devastating illness. It is thought that the disease originated in Africa and spread to the Americas in the 17th and 18th centuries through trading ships. The flavivirus that causes yellow fever was first identified in 1928 and the first vaccine against the disease was produced at the Rockefeller Institute in New York in 1937.

Description

In order to understand how yellow fever is passed, several terms need to be defined. The word “host” refers to an animal that can be infected with a particular disease. The term “vector” refers to an organism which can carry a particular disease-causing agent (such as a virus or bacteria) without actually developing the disease. The vector can then pass the virus or bacterium on to a new host.

Many of the common illnesses in the United States (including the **common cold**, many viral causes of **diarrhea**, and **influenza** or “flu”) are spread via direct passage of the causative virus between human

WILBUR AUGUSTUS SAWYER (1879–1951)

Wilbur Augustus Sawyer was born in Appleton, Wisconsin, on August 7, 1879, to Minnie Edmea (Birge) and Wesley Caleb Sawyer. The Sawyers moved to Oshkosh, Wisconsin and finally to Stockton, California in 1888. Sawyer spent two years at the University of California and then entered Harvard College where he earned his A.B. degree in 1902. In 1906, Sawyer graduated from Harvard Medical School and began a private practice, which lasted until he started his internship at Massachusetts General Hospital. Sawyer returned to California in 1908 in order to obtain a position at the University of California as a medical examiner. He then worked with California State Board of Health from 1910 until 1918. In 1911, Sawyer married Margaret Henderson. The couple had three children.

Sawyer's first publication (1913) dealt with his research of poliomyelitis. His discovery, in 1915, that examination of the individual's stool could lead to detection of the disease was later regarded as very significant. In 1918 and 1919, Sawyer worked to control venereal disease while employed by the Army Medical Corps. In 1926 and 1927, while director of the West African Yellow Fever Commission, Sawyer succeeded in isolating the yellow fever virus. He would ultimately return to the United States, where he and Wray Lloyd would devise an immunization against yellow fever (1931).

In 1944, Sawyer became director of health for the United Nations Relief and Rehabilitation Administration, a position he held for three years. He retired to Berkeley, California where he died on November 12, 1951.

beings. Yellow fever, however, cannot be passed directly from one infected human being to another. Instead, the virus responsible for yellow fever requires an intermediate vector, a mosquito, which carries the virus from one host to another.

The hosts of yellow fever include both humans and monkeys. The cycle of yellow fever transmission occurs as follows: an infected monkey is bitten by a tree-hole breeding mosquito. This mosquito acquires the virus, and can pass the virus on to any number of other monkeys that it may bite. This form of yellow fever is known as sylvatic yellow fever, and usually affects humans only incidentally. When a human is bitten by an infected mosquito, however, the human may acquire the virus. In the case of South American yellow fever, the infected human may return to the city, where an urban mosquito (*Aedes aegypti*) serves

as a viral vector, spreading the infection rapidly by biting humans. This form of the disease is known as urban yellow fever or epidemic yellow fever.

Yellow fever epidemics may also occur after flooding caused by earthquakes and other natural disasters. They result from a combination of new habitats available for the vectors of the disease and changes in human behavior (spending more time outdoors and neglecting sanitation precautions).

Cases of yellow fever are uncommon in the United States and Canada as of 2004. The last reported case in an American citizen concerned a man who contracted yellow fever in Brazil in 1996. The last epidemic in the United States occurred in New Orleans in 1905.

Symptoms

Once a mosquito has passed the yellow fever virus to a human, the chance of disease developing is about 5–20%. Infection may be fought off by the host's immune system, or may be so mild that it is never identified.

In human hosts who develop the disease yellow fever, there are five distinct stages through which the infection evolves. These have been termed the periods of incubation, invasion, remission, intoxication, and convalescence.

Yellow fever's incubation period (the amount of time between the introduction of the virus into the host and the development of symptoms) is three to six days. During this time, there are generally no symptoms identifiable to the host.

The period of invasion lasts two to five days, and begins with an abrupt onset of symptoms, including fever and chills, intense **headache** and lower backache, muscle aches, **nausea**, and extreme exhaustion. The patient's tongue shows a characteristic white, furry coating in the center, surrounded by a swollen, reddened margin. While most other infections that cause a high fever also cause an increased heart rate, yellow fever results in an unusual finding, called **Faget's sign**. This is the simultaneous occurrence of a high fever with a slowed heart rate. Throughout the period of invasion, there are still live viruses circulating in the patient's blood stream. Therefore, a mosquito can bite the ill patient, acquire the virus, and continue passing it on to others.

The next phase is called the period of remission. The fever falls, and symptoms decrease in severity for several hours to several days. In some patients, this signals the end of the disease; in other patients, this proves only to be the calm before the storm.

The period of intoxication represents the most severe and potentially fatal phase of the illness. During this time, lasting three to nine days, a type of degeneration of the internal organs (specifically the kidneys, liver, and heart) occurs. This fatty degeneration results in what is considered the classic triad of yellow fever symptoms: **jaundice**, black vomit, and the dumping of protein into the urine. Jaundice causes the whites of the patient's eyes and the patient's skin to take on a distinctive yellow color. This is due to liver damage, and the accumulation of a substance called bilirubin, which is normally processed by a healthy liver. The liver damage also results in a tendency toward bleeding; the patient's vomit appears black due to the presence of blood. Protein, which is normally kept out of the urine by healthy, intact kidneys, appears in the urine due to disruption of the kidney's healthy functioning.

Patients who survive the period of intoxication enter into a relatively short period of convalescence. They recover with no long term effects related to the yellow fever infection. Further, infection with the yellow fever virus results in lifelong immunity against repeated infection with the virus.

The course of yellow fever is complicated in some patients by secondary bacterial infections.

Diagnosis

Diagnosis of yellow fever depends on the examination of blood by various techniques in order to demonstrate either yellow fever viral antigens (the part of the virus that stimulates the patient's immune system to respond) or specific antibodies (specific cells produced by the patient's immune system which are directed against the yellow fever virus). The most rapid method of diagnosis as of 2004 is capture enzyme immunoassay. The diagnosis can be strongly suspected when **Faget's sign** is present. When the classic triad of symptoms is noted yellow fever is strongly suspected.

Treatment

There are no antiviral treatments available as of 2004 to combat the yellow fever virus, although researchers at the University of Texas are studying ribavirin (Virazole, Rebetol), a drug that is given by mouth to treat **hepatitis C**, as a potential treatment for liver damage caused by yellow fever. The only treatments for yellow fever are given to relieve its symptoms. Fevers and **pain** should be relieved with **acetaminophen**, not **aspirin** or ibuprofen, both of which could increase the already-present risk of bleeding. **Dehydration** (due to fluid loss both from fever and bleeding) needs to be

KEY TERMS

Epidemic—A situation in which a particular disease spreads rapidly through a population of people in a relatively short period of time.

Faget's sign—The simultaneous occurrence of a high fever with a slowed heart rate.

Host—The organism (such as a monkey or human) in which another organism (such as a virus or bacteria) is living.

Sylvatic—Pertaining to or living in the woods or forested areas. The form of yellow fever transmitted by mosquitoes to rainforest monkeys is called sylvatic yellow fever.

Vector—A carrier organism (such as a fly or mosquito) which serves to deliver a virus (or other agent of infection) to a host.

carefully avoided. This can be accomplished by increasing fluids. The risk of bleeding into the stomach can be decreased through the administration of **antacids** and other medications. Hemorrhage may require blood transfusions. Kidney failure may require dialysis (a process that allows the work of the kidneys in clearing the blood of potentially toxic substances to be taken over by a machine, outside of the body).

Prognosis

Five to ten percent of all diagnosed cases of yellow fever are fatal. Jaundice occurring during a yellow fever infection is an extremely grave predictor. Twenty to fifty percent of these patients die of the infection. **Death** may occur due to massive bleeding (hemorrhage), often following a lapse into a comatose state.

Prevention

A very safe and very effective yellow fever vaccine exists. The Arilvax vaccine is made from a live attenuated form of the yellow fever virus, strain 17D. In the United States, the vaccine is given only at Yellow Fever **Vaccination** Centers authorized by the U.S. Public Health Service. About 95% of vaccine recipients acquire long-term immunity to the yellow fever virus. Careful measures to decrease mosquito populations in both urban areas and jungle areas in which humans are working, along with programs to vaccinate all people living in such areas, are necessary to avoid massive yellow fever outbreaks.

Persons planning to travel to countries where yellow fever in endemic may obtain up-to-date information on yellow fever vaccination from the Centers for Disease Control and Prevention by telephone (404-332-4559) or fax (404-332-4265).

Resources

BOOKS

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Lo Re, V., III, and S. J. Gluckman. "Travel Immunizations." *American Family Physician* 70 (July 1, 2004): 89-99.

Reeder, G. David, MD, and Theodore E. Woodward, MD. "Yellow Fever." *eMedicine* February 25, 2002. <<http://www.emedicine.com/med/topic2432.htm>>.

Sbrana, E., S. Y. Xiao, H. Guzman, et al. "Efficacy of Post-Exposure Treatment of Yellow Fever with Ribavirin in a Hamster Model of the Disease." *American Journal of Tropical Medicine and Hygiene* 71 (September 2004): 306-312.

ORGANIZATIONS

Centers for Disease Control and Prevention (CDC). 1600 Clifton Rd., NE, Atlanta, GA 30333. (800) 311-3435, (404) 639-3311. <<http://www.cdc.gov>>.

Infectious Diseases Society of America (IDSA). 66 Canal Center Plaza, Suite 600, Alexandria, VA 22314. (703) 299-0200. Fax: (703) 299-0204. <<http://www.idsociety.org>>.

World Health Organization (WHO). <<http://www.who.int/en/>>.

OTHER

World Health Organization Fact Sheet. "Flooding and Communicable Diseases Fact Sheet: Risk Assessment and Preventive Measures." December 2004. <http://www.who.int/hac/techguidance/ems/flood_cds/en/index.html>.

Rosalyn Carson-DeWitt, MD
Rebecca J. Frey, PhD

Yersinia enterocolitica infection see
Yersinosis

Yersinia pestis see **Plague**

Yersinia pseudotuberculosis infection see
Yersinosis

Yersinosis

Definition

Yersinosis refers to infection by a genus of bacteria known as *Yersinia*. The two sub-types that are responsible for yersinosis are *Yersinia enterocolitica* and *Yersinia pseudotuberculosis*. The diseases produced by these organisms are called “zoonoses,” because the bacteria is passed to humans from animal sources.

The name *Yersinia* comes from Dr. Alexandre Yersin, who was the first person to grow a much more deadly type of *Yersinia* known as *Yersinia pestis*, the bacteria responsible for what is now known as bubonic **plague**. This article, however, will deal with the more common forms of *Yersinia*, namely *Y. enterocolitica* and *Y. pseudotuberculosis*.

Description

Yersinia are classified as gram-negative bacteria (bacteria that do not accept the color of a stain in a Gram stain test, which indicates the general chemical nature of the cell wall of the bacteria); they have a variety of appearances, and are therefore called pleomorphic. They belong to Enterobacteriaceae, the large group of organisms that inhabit the intestinal tract. There are many different subtypes of *Yersinia*.

They are found worldwide and have been isolated from soil, fresh water, contaminated foods, and many wild and domestic animals. For reasons not entirely clear, disease caused by these organisms occurs more frequently in areas of northern Europe, especially Scandinavia. Infection, particularly in children ages one through four years, is quite common, though often these infections produce few symptoms. Studies have shown that infection with these bacteria is almost as common as that with *Shigella* or *Campylobacter*.

Causes and symptoms

Animals are the most important sources of bacterial infection for humans. Whether from pets or undercooked meat (especially pork), these bacteria almost always enter the human body through the mouth (oral transmission). An incubation period of one to eleven days passes before signs of disease develop. Rare cases have been transmitted by way of contaminated blood transfusions.

Yersinia produces several different types of disease. The most common form is a short-lived inflammation of the intestine known as enterocolitis. Most

often the very end of the small intestine is involved, an area known as the terminal ileum. The result is gastroenteritis, with cramping abdominal **pain**, **fever**, and **diarrhea**. Diarrhea generally continues for two weeks or so, but can go on for many months. Up to 40% of patients also experience **nausea** and vomiting; and in one-third, inflammation of the intestine leads to bleeding.

In other patients, the same area of the intestine is involved, but instead of causing diarrhea, a syndrome resembling **appendicitis** occurs. In this syndrome, the lymph nodes surrounding the intestine are especially involved; this has led to the term mesenteric adenitis. Although this syndrome resolves without serious consequences, it is often difficult to differentiate from appendicitis, and leads to surgery in some instances. Ultrasound exam may be able to demonstrate a normal appendix and avoid surgery. Why some patients develop symptoms of **gastroenteritis**, and others only inflammation, pain, and fever, is unknown.

In some patients, *Yersinia* produces infection of areas other than the intestinal tract. These include:

- Inflammation of the throat (pharyngitis) and **tonsillitis**; this can be quite severe and even lead to **death**, particularly in adults.
- Septicemia, or infection of the blood stream, with spreading of infection to other organs such as bone, meninges, kidneys, and others. Individuals with decreased immunity due to liver disease, diabetes, **cancer**, and other diseases are at increased risk for this complication.

Different parts of the body may be affected (such as joints, eyes, and urinary system) by changes in the immune system caused by *Yersinia* infection. Arthritis, which is especially frequent in Scandinavia, occurs in up to 10% of *Yersinia* infections. About one week after typical intestinal symptoms, swelling and pain in multiple joints occurs. The knees and ankles are most often involved, and become inflamed over a period of two weeks. In two-thirds of those affected, symptoms gradually resolve over one to three months without need for treatment. Rarely does chronic joint disease develop.

Inflammation of the heart muscle, called myocarditis, sometimes occurs together with the arthritis. In about 15-20% of patients, the skin develops a red, raised area, usually located on the shins, called erythema nodosum. This appears within a few weeks of the intestinal symptoms and disappears over a month or so.

KEY TERMS

Mesenteric adenitis—Inflammation of the lymph nodes which serve the small intestine. Has symptoms similar to appendicitis.

Septicemia—Systemic disease associated with the presence of microorganisms or their toxins in the blood; blood poisoning.

Diagnosis

Identifying *Yersinia* as the cause of all or any of these symptoms is not an easy task. It is possible to grow the organism from stool cultures, but this is difficult to do unless special methods are used.

A change in antibody levels can also be used to determine the presence of infection. To be accurate, levels must be initially examined early in the illness. Therefore, it is most important for the possible diagnosis and examination to be thought of early.

Treatment

Since most of the symptoms caused by *Yersinia* are self limiting, specific antibiotic treatment is generally not needed. Patients with **dehydration** from gastroenteritis are given supportive therapy, including treatment aimed at replacing fluids.

Antibiotics are indicated, however, for those patients who develop more severe infections, such as invasion of the bloodstream (septicemia), or who develop infections at specific sites, such as bone. A variety of antibiotics have been used, but it is not clear which produces the best results.

No specific treatment is indicated for the joint, ocular, skin, or urinary symptoms that result from infection. As stated, these are not due to direct invasion by the bacteria, but are related to changes in immune reactions produced by the infection. However, treatment of those experiencing severe arthritic symptoms with NSAIDS (nonsteroidal anti-inflammatory drugs) or steroid injection at inflamed joints is used in selected cases.

Prognosis

As noted above, most of the time, *Yersinia* infection has an excellent outlook. However, when these bacteria invade the bloodstream or produce disease beyond the gastrointestinal tract, the outlook is less positive. This may be because more severe

infections occur in those with decreased immunity. Death rate from septicemia has been reported to be as high as 50%.

Prevention

Safe food handling procedures and food-preparation practices are by far the best means of avoiding infection. Undercooked food, especially pork or other animal products, should not be eaten.

Resources

OTHER

“Bad Bug Book.” *Center for Food Safety and Applied Nutrition*. <<http://vm.cfsan.fda.gov/~mow/intro.html>>.

David Kaminstein, MD

Yoga

Definition

The term *yoga* comes from a Sanskrit word which means yoke or union. Traditionally, yoga is a method joining the individual self with the Divine, Universal Spirit, or Cosmic Consciousness. Physical and mental exercises are designed to help achieve this goal, also called self-transcendence or enlightenment. On the physical level, yoga postures, called *asanas*, are designed to tone, strengthen, and align the body. These postures are performed to make the spine supple and healthy and to promote blood flow to all the organs, glands, and tissues, keeping all the bodily systems healthy. On the mental level, yoga uses breathing techniques (*pranayama*) and **meditation** (*dyana*) to quiet, clarify, and discipline the mind. However, experts are quick to point out that yoga is not a religion, but a way of living with health and peace of mind as its aims.

Purpose

Yoga has been used to alleviate problems associated with high blood pressure, **high cholesterol**, migraine headaches, asthma, shallow breathing, backaches, **constipation**, diabetes, **menopause**, **multiple sclerosis**, **varicose veins**, **carpal tunnel syndrome** and many chronic illnesses. It also has been studied and approved for its ability to promote relaxation and reduce **stress**.

PATANJALI (CIRCA SECOND CENTURY B.C.–CIRCA SECOND CENTURY B.C.)

There is little historical information available on Patanjali, who is credited with developing yoga, one of the six systems of Hindu philosophy. Several scholars suggest several persons may have developed yoga under the pseudonym of Patanjali. In any case, Patanjali existed around 150 B.C. in India. He developed yoga based on a loose set of doctrines and practices from the Upanishads, themselves a set of mystical writings. The Upanishads are part of the Aranyakas, philosophical concepts that are part of the Veda, the most ancient body of literature of Hinduism. Patanjali gave these combined philosophical and esoteric writings a common foundation in his *Yoga Sutra*, a set of 196 concise aphorisms (wise sayings) that form the principles of yoga. He also drew upon Samkhya, the oldest classic system of Hindu philosophy. Patanjali's yoga accepted Samkhya metaphysics and the concept of a supreme soul. He established an eight-stage discipline of self-control and meditation. The individual sutras (verses) lay out the entire tradition of meditation. They also describe the moral and physical disciplines needed for the soul to attain absolute freedom from the body and self.

As of late 2002, yoga is increasingly recommended for **dysmenorrhea**, **premenstrual syndrome**, and other disorders in premenopausal women, in Europe as well as in the United States.

Yoga can also provide the same benefits as any well-designed **exercise** program, increasing general health and stamina, reducing stress, and improving those conditions brought about by sedentary lifestyles. Yoga has the added advantage of being a low-impact activity that uses only gravity as resistance, which makes it an excellent physical therapy routine; certain yoga postures can be safely used to strengthen and balance all parts of the body. A study published in late 2002 summarized recent findings about the benefits of yoga for the cardiovascular and musculoskeletal systems. The review noted that yoga is still viewed as a “trendy” form of exercise rather than one with documented medical benefits.

Meditation has been much studied and approved for its benefits in reducing stress-related conditions. The landmark book, *The Relaxation Response*, by Harvard cardiologist Herbert Benson, showed that meditation and breathing techniques for relaxation could have the opposite effect of stress, reducing blood pressure and other indicators. Since then, much research has reiterated the benefits of meditation for **stress reduction** and

general health. Currently, the American Medical Association recommends meditation techniques as a first step before medication for borderline **hypertension** cases. Some 2002 studies indicate that yogic meditation by itself is effective in lowering serum cholesterol as well as blood pressure.

Modern psychological studies have shown that even slight facial expressions can cause changes in the involuntary nervous system; yoga utilizes the mind/body connection. That is, yoga practice contains the central ideas that physical posture and alignment can influence a person's mood and self-esteem, and also that the mind can be used to shape and heal the body. Yoga practitioners claim that the strengthening of mind/body awareness can bring eventual improvements in all facets of a person's life.

Description

Origins

Yoga originated in ancient India and is one of the longest surviving philosophical systems in the world. Some scholars have estimated that yoga is as old as 5,000 years; artifacts detailing yoga postures have been found in India from over 3000 B.C. Yoga masters (*yogis*) claim that it is a highly developed science of healthy living that has been tested and perfected for all these years. Yoga was first brought to America in the late 1800s when Swami Vivekananda, an Indian teacher and yogi, presented a lecture on meditation in Chicago. Yoga slowly began gaining followers, and flourished during the 1960s when there was a surge of interest in Eastern philosophy. There has since been a vast exchange of yoga knowledge in America, with many students going to India to study and many Indian experts coming here to teach, resulting in the establishment of a wide variety schools. Today, yoga is thriving, and it has become easy to find teachers and practitioners throughout America. A recent Roper poll, commissioned by *Yoga Journal*, found that 11 million Americans do yoga at least occasionally and 6 million perform it regularly. Yoga stretches are used by physical therapists and professional sports teams, and the benefits of yoga are being touted by movie stars and Fortune 500 executives. Many prestigious schools of medicine have studied and introduced yoga techniques as proven therapies for illness and stress. Some medical schools, like UCLA, even offer yoga classes as part of their physician training program.

Classical yoga is separated into eight limbs, each a part of the complete system for mental, physical and spiritual well-being. Four of the limbs deal with mental and physical exercises designed to bring the

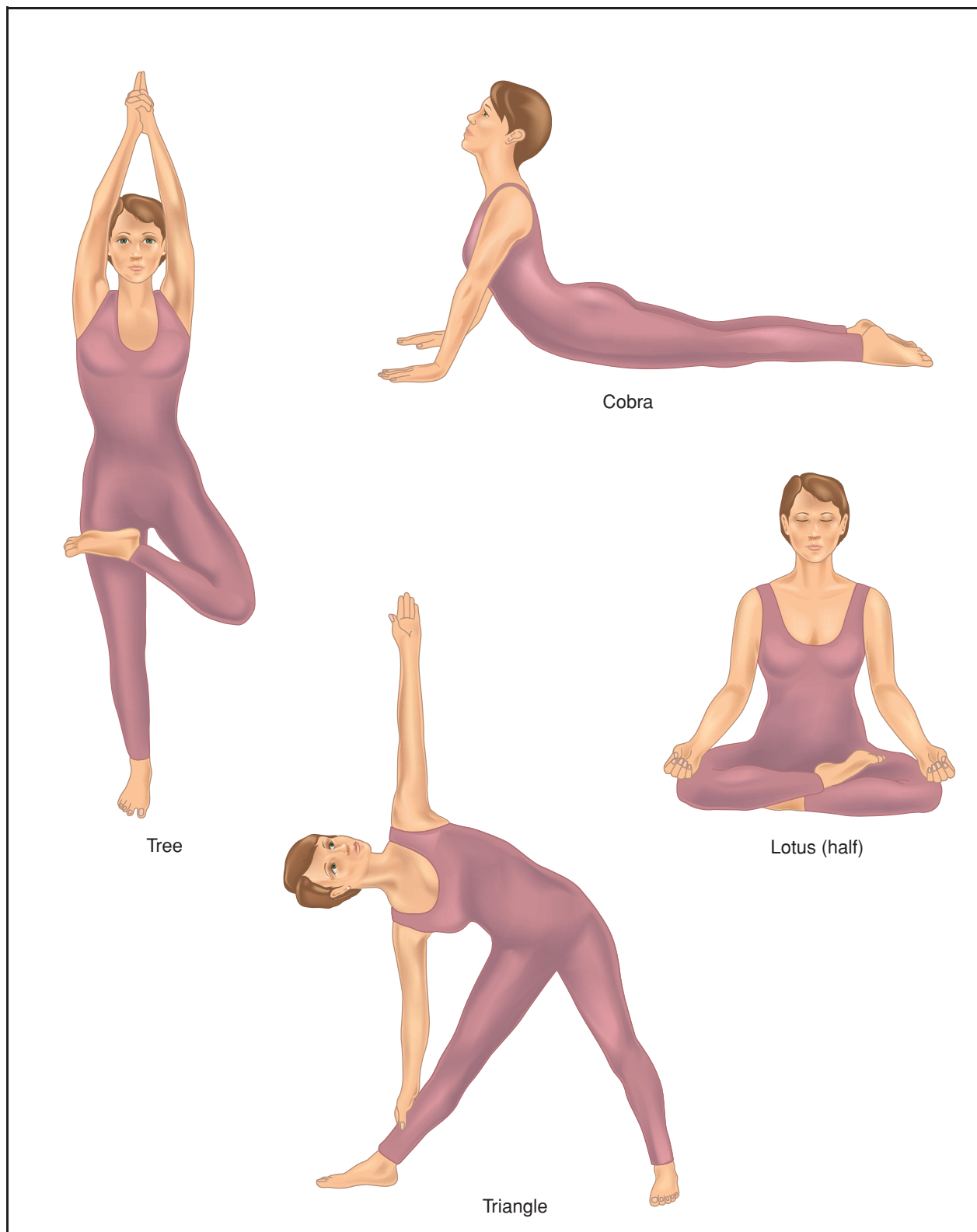
Yoga Positions

Name	Description
Abdominal massage	Kneel with arms folded. Bend torso toward ground and lower forehead to the floor. Slowly raise up, switch arms, and repeat.
Boat	Lying on stomach, raise head, torso, arms, and legs off the ground and stretch. Arms should be outstretched and pointing towards feet.
Bow	Lying on stomach, hold ankles from behind and slowly raise head, torso, and thighs off floor.
Bridge	Lying on back with knees bent and feet flat on floor, raise pelvis off floor and arch back. Arms should be stretched out on floor with hands grasped.
C	On hands and knees, move head and buttocks as far left as possible. Inhale as you return center and repeat on the right side.
Camel	While kneeling, arch back and bend head back toward feet. Hold heels with hands and exhale while in movement.
Cat	On hands and knees, arch back and exhale while in movement, rounding shoulders and back.
Child	Kneeling with arms to the side, roll torso to floor and rest forehead on the ground.
Cobra	Stretched out on floor with stomach down, place elbows parallel to shoulders and raise torso up. Arms should straighten with hands flat on floor.
Corpse	Lie on back with feet and arms outstretched. Breathe deeply.
Dog	On hands and knees, dip back and lift head and buttocks up. Exhale.
Downward Dog	On hands and knees form and inverted V by pushing pelvis up and pressing hands and heels to floor. Exhale while in movement.
Half Cobra	Stretched out on floor with stomach down, place elbows parallel to shoulders and raise torso up. Keep arms bent and only raise torso off the ground as far as the navel.
Half Locust	Lying on stomach with hands beneath the body, raise legs one at a time while tensing buttocks. Repeat with other leg.
Half Lotus	Sit with legs crossed (only one leg should be over the other) and knees touching the floor.
Half-Moon	Standing with feet together, hold hands above the head with arms outstretched. Exhale and stretch to the left. Inhale and return to center. Repeat on other side.
Hand and thumb squeeze	Make a fist around thumb and squeeze. Release slowly and repeat on other hand.
Head to knee	Sitting with right leg outstretched and the left leg bend toward the body with the left foot touching the right leg, stretch head to right knee. Repeat on other side.
Hero	On hands and knees, cross left knee in front of right knee while sitting back between the heels. Hold heels with hands.
Knee down twist	Lying on back with arms outstretched, place right foot on left knee and swivel right knee to the left side of floor. While in movement, turn head to left side. Repeat on opposite side.
Locust	Lying on stomach with hands under the body, squeeze buttocks and lift legs up and outward. Keep legs straight.
Mountain	Standing with feet together, inhale while raising arms straight above the head and clasp hands together. Exhale while lowering arms.
Pigeon	Kneeling, slide the left leg straight out from behind and inhale, stretching torso up. Release and repeat on other side.
Plow	Lying on back, inhale and raise legs over head while keeping hands flat on floor for support.

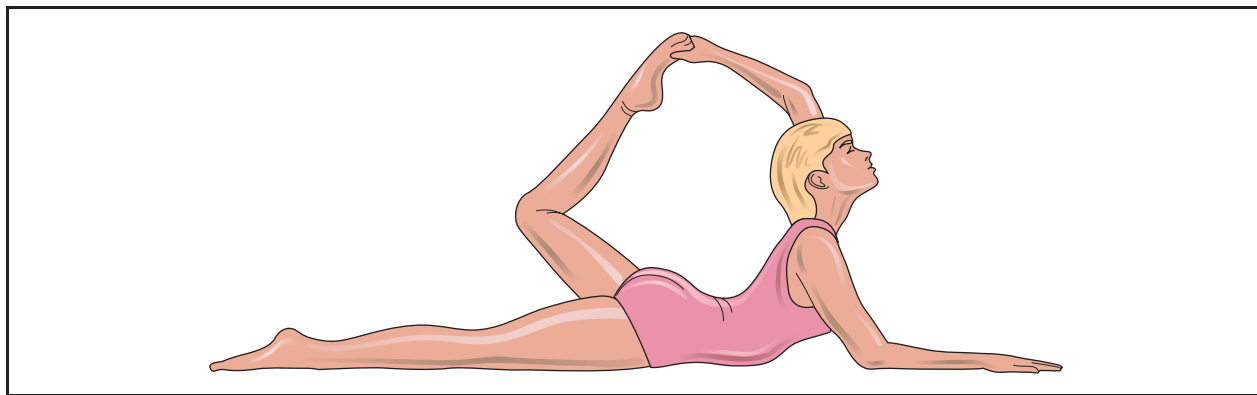
Yoga Positions (continued)

Name	Description
Posterior stretch	Sitting with legs outstretched and feet together, stretch head to toes.
Rag Doll	While standing, exhale and bend over toward toes, cupping elbows with hands. Breathe deeply.
Seated angle	Sitting with legs outstretched in a V shape, stretch arms to toes and head to floor.
Shoulder crunch	With back straight, slowly lift shoulder to ear and lower. Repeat on other side.
Shoulder stand	Lying on back, lift legs up and support back with hands. Slowly angle legs over head and then extend upward.
Sphinx	Lying on stomach with elbows parallel to shoulders and palms on the ground, push torso up and look upward.
Spider	Press fingertips together and move palms in and out.
Spinal twist	Sitting with right foot crossed over left leg and right leg held with left arm. Twist while supporting body with right hand on the floor. Repeat on other side.
Standing angle	Inhale and step into V position, stretching arms out and then down toward floor.
Standing yoga mudra	Standing with arms at sides, inhale and raise arms in front. Exhale and swing arms to back.
Tree	While standing, place one foot on the opposite thigh and outstretch arms above the head. Hold hands above with index fingers straight and the remaining fingers clasped.
Triangle	With arms parallel to floor and legs outstretched, turn one foot out and stretch to that side, keeping arms straight. Repeat on other side.
Upward Dog	Lying on stomach with hands down near the chest, lift torso off the floor while raising on toes. Hands should raise, but remain palms down. Arch back slightly.
Warrior I	Raise arms over head with palms together and lunge forward with one foot, keeping thigh parallel to the ground.
Warrior II	With arms straight out and parallel to the ground and legs in V, turn one foot out and lunge to the side, keeping hips straight.
Yoga Mudra	Sitting on heels, round torso to the ground with forehead to the floor while stretching arms overhead. Inhale while in movement and exhale while lowering arms.

mind in tune with the body. The other four deal with different stages of meditation. There are six major types of yoga, all with the same goals of health and harmony but with varying techniques: hatha, raja, karma, bhakti, jnana, and tantra yoga. Hatha yoga is the most commonly practiced branch of yoga in America, and it is a highly developed system of nearly 200 physical postures, movements and breathing techniques designed to tune the body to its optimal health. The yoga philosophy believes the breath to be the most important facet of health, as the breath is the largest source of *prana*, or life force, and **hatha yoga** utilizes *pranayama*, which literally means the science or control of breathing. Hatha yoga was originally developed as a system to make the body strong and healthy enough to enable mental awareness and spiritual enlightenment.



Demonstrations of the tree, triangle, cobra, and lotus poses. The tree and triangle are good for balance and coordination. Cobra stretches the pelvic and strengthens the back. Lotus is a meditative pose. (Illustration by Electronic Illustrators Group.)



Yoga is a system that benefits the body, mind, and spirit by teaching self-control through a series of postures and exercises as well as through breathing, relaxation, and meditation techniques. (Illustration by Electronic Illustrators Group.)

There are several different schools of hatha yoga in America; the two most prevalent ones are Iyengar and ashtanga yoga. Iyengar yoga was founded by B.K.S. Iyengar, who is widely considered as one of the great living innovators of yoga. Iyengar yoga puts strict emphasis on form and alignment, and uses traditional hatha yoga techniques in new manners and sequences. Iyengar yoga can be good for physical therapy because it allows the use of props like straps and blocks to make it easier for some people to get into the yoga postures. Ashtanga yoga can be a more vigorous routine, using a flowing and dance-like sequence of hatha postures to generate body heat, which purifies the body through sweating and deep breathing.

The other types of yoga show some of the remaining ideas which permeate yoga. Raja yoga strives to bring about mental clarity and discipline through meditation, simplicity, and non-attachment to worldly things and desires. Karma yoga emphasizes charity, service to others, non-aggression and non-harming as means to awareness and peace. Bhakti yoga is the path of devotion and love of God, or Universal Spirit. Jnana yoga is the practice and development of knowledge and wisdom. Finally, tantra yoga is the path of self-awareness through religious rituals, including awareness of sexuality as sacred and vital.

A typical hatha yoga routine consists of a sequence of physical poses, or asanas, and the sequence is designed to work all parts of the body, with particular emphasis on making the spine supple and healthy and increasing circulation. Hatha yoga asanas utilize three basic movements: forward bends, backward bends, and twisting motions. Each asana is named for a common thing it resembles, like the sun salutation, cobra, locust, plough, bow, eagle, tree, and

the head to knee pose, to name a few. Each pose has steps for entering and exiting it, and each posture requires proper form and alignment. A pose is held for some time, depending on its level of difficulty and one's strength and stamina, and the practitioner is also usually aware of when to inhale and exhale at certain points in each posture, as breathing properly is another fundamental aspect of yoga. Breathing should be deep and through the nose. Mental concentration in each position is also very important, which improves awareness, poise and posture. During a yoga routine there is often a position in which to perform meditation, if deep relaxation is one of the goals of the sequence.

Yoga routines can take anywhere from 20 minutes to two or more hours, with one hour being a good time investment to perform a sequence of postures and a meditation. Some yoga routines, depending on the teacher and school, can be as strenuous as the most difficult workout, and some routines merely stretch and align the body while the breath and heart rate are kept slow and steady. Yoga achieves its best results when it is practiced as a daily discipline, and yoga can be a life-long exercise routine, offering deeper and more challenging positions as a practitioner becomes more adept. The basic positions can increase a person's strength, flexibility and sense of well-being almost immediately, but it can take years to perfect and deepen them, which is an appealing and stimulating aspect of yoga for many.

Yoga is usually best learned from a yoga teacher or physical therapist, but yoga is simple enough that one can learn the basics from good books on the subject, which are plentiful. Yoga classes are generally inexpensive, averaging around 10 dollars per class, and students can learn basic postures in just a few

classes. Many YMCAs, colleges, and community health organizations offer beginning yoga classes as well, often for nominal fees. If yoga is part of a physical therapy program, its cost can be reimbursed by insurance.

Preparations

Yoga can be performed by those of any age and condition, although not all poses should be attempted by everyone. Yoga is also a very accessible form of exercise; all that is needed is a flat floor surface large enough to stretch out on, a mat or towel, and enough overhead space to fully raise the arms. It is a good activity for those who can't go to gyms, who don't like other forms of exercise, or have very busy schedules. Yoga should be done on an empty stomach, and teachers recommend waiting three or more hours after meals. Loose and comfortable clothing should be worn.

Precautions

People with injuries, medical conditions, or spinal problems should consult a doctor before beginning yoga. Those with medical conditions should find a yoga teacher who is familiar with their type of problem and who is willing to give them individual attention. Pregnant women can benefit from yoga, but should always be guided by an experienced teacher. Certain yoga positions should not be performed with a **fever**, or during menstruation.

Beginners should exercise care and concentration when performing yoga postures, and not try to stretch too much too quickly, as injury could result. Some advanced yoga postures, like the headstand and full lotus position, can be difficult and require strength, flexibility, and gradual preparation, so beginners should get the help of a teacher before attempting them.

Yoga is not a competitive sport; it does not matter how a person does in comparison with others, but how aware and disciplined one becomes with one's own body and limitations. Proper form and alignment should always be maintained during a stretch or posture, and the stretch or posture should be stopped when there is **pain**, **dizziness**, or **fatigue**. The mental component of yoga is just as important as the physical postures. Concentration and awareness of breath should not be neglected. Yoga should be done with an open, gentle, and non-critical mind; when one stretches into a yoga position, it can be thought of accepting and working on one's limits. Impatience,

KEY TERMS

Asana—A position or stance in yoga.

Dyana—The yoga term for meditation.

Hatha yoga—Form of yoga using postures, breathing methods and meditation.

Meditation—Technique of concentration for relaxing the mind and body.

Pranayama—Yoga breathing techniques.

Yogi (female, yogini)—A trained practitioner of yoga.

self-criticism and comparing oneself to others will not help in this process of self-knowledge. While performing the yoga of breathing (pranayama) and meditation (dyana), it is best to have an experienced teacher, as these powerful techniques can cause dizziness and discomfort when done improperly.

Side effects

Some people have reported injuries by performing yoga postures without proper form or concentration, or by attempting difficult positions without working up to them gradually or having appropriate supervision. Beginners sometimes report muscle soreness and fatigue after performing yoga, but these side effects diminish with practice.

Research and general acceptance

Although yoga originated in a culture very different from that of modern America, it has been accepted and its practice has spread relatively quickly. Many yogis are amazed at how rapidly yoga's popularity has spread in the United States and Canada, considering the legend that it was passed down secretly by handfuls of adherents for many centuries.

There can still be found some resistance to yoga, for active and busy Americans sometimes find it hard to believe that an exercise program that requires them to slow down, concentrate, and breathe deeply can be more effective than lifting weights or running. However, ongoing research in top medical schools is showing yoga's effectiveness for overall health and for specific problems, making it an increasingly acceptable health practice.

The growing acceptability of yoga as an alternative therapy for certain disorders or conditions is

reflected in the fact that the National Center for Complementary and Alternative Medicine (NCCAM) is conducting a series of clinical trials of ypga. As of the summer of 2004, NCCAM has five clinical trials in progress, evaluating yoga as a treatment for chronic **low back pain**; **insomnia**; depression in patients diagnosed with HIV infection; and **shortness of breath** in chronic obstructive pulmonary disease (COPD). The fifth clinical trial is an evaluation of yoga in improving attention span in **aging** and multiple sclerosis.

Resources

BOOKS

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- Pelletier, Kenneth R., MD. *The Best Alternative Medicine*, Chapter 10, "Ayurvedic Medicine and Yoga: From Buddha to the Millennium." New York: Simon & Schuster, 2002.

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- Vyas, R., and N. Dikshit. "Effect of Meditation on Respiratory System, Cardiovascular System and Lipid Profile." *Indian Journal of Physiology and Pharmacology* 46 (October 2002): 487-491.
- Yoga International Magazine*. R.R. 1 Box 407, Honesdale, PA 18431. <<http://www.yimag.com>>.
- Yoga Journal*. P.O. Box 469088, Escondido, CA 92046. <<http://www.yogajournal.com>>.

ORGANIZATIONS

- American Yoga Association. <<http://www.americanयोगassociation.org>>.
- International Association of Yoga Therapists (IAYT). 4150 Tivoli Ave., Los Angeles, CA 90066.
- National Center for Complementary and Alternative Medicine (NCCAM) Clearinghouse. P. O. Box 7923. Gaithersburg, MD 20898. (888) 644-6226. Fax: (866) 464-3616. <<http://nccam.nih.gov>>.
- Yoga Research and Education Center (YREC). 2400A County Center Drive, Santa Rosa, CA 95403. (707) 566-0000. <<http://www.yrec.org>>.

OTHER

- NCCAM Yoga Clinical Trials. <<http://nccam.nih.gov/clinicaltrials/yoga.htm>>.

Douglas Dupler, MA
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"Yuppie flu" see **Chronic fatigue syndrome**

Z

Zenker's diverticulum see **Esophageal pouches**

Zidovudine see **Antiretroviral drugs**

Zinc deficiency see **Mineral deficiency**

Zinc excess see **Mineral toxicity**

Zolpidem see **Anti-insomnia drugs**

Zoonosis

Definition

Zoonosis, also called *zoonotic disease* refers to diseases that can be passed from animals, whether wild or domesticated, to humans.

Description

Although many diseases are species specific, meaning that they can only occur in one animal species, many other diseases can be spread between different animal species. These are infectious diseases, caused by bacteria, viruses, or other disease causing organisms that can live as well in humans as in other animals.

There are different methods of transmission for different diseases. In some cases, zoonotic diseases are transferred by direct contact with infected animals, much as being near an infected human can cause the spread of an infectious disease. Other diseases are spread by drinking water that contains the eggs of parasites. The eggs enter the water supply from the feces of infected animals. Still others are spread by eating the flesh of infected animals. Tapeworms are spread this way. Other diseases are spread by insect vectors. An insect, such as a flea or

tick, feeds on an infected animal, then feeds on a human. In the process, the insect transfers the infecting organism.

The Centers for Disease Control (CDC) in Atlanta have said that most emerging diseases around the world are zoonotic. The director of the CDC has said that 11 of the last 12 emerging infections in the world with serious health consequences has probably arisen from animal sources. Wild animal trade occurs across countries and many people take in wild animals as domestic pets. However, pet shops and food markets are not properly testing for diseases and parasites that can cause harm to humans and other animals.

Some zoonotic diseases are well known, such as rats (**plague**), deer tick (**Lyme disease**). Others are not as well known. For example, elephants may develop **tuberculosis**, and spread it to humans.

Causes and symptoms

The following is a partial list of animals and the diseases that they may carry. Not all animal carriers are listed, nor are all the diseases that the various species may carry.

- Bats are important **rabies** carriers, and also carry several other viral diseases that can affect humans.
- Cats may carry the causative organisms for plague, **anthrax**, cowpox, tapeworm, and many bacterial infections.
- Dogs may carry plague, tapeworm, rabies, **Rocky Mountain Spotted Fever**, and Lyme disease.
- Horses may carry anthrax, rabies, and *Salmonella* infections.
- Cattle may carry the organisms that cause anthrax, European tick-borne **encephalitis**, rabies, tapeworm,

Salmonella infections and many bacterial and viral diseases.

- Pigs are best known for carrying tapeworm, but may also carry a large number of other infections including anthrax, **influenza**, and rabies.
- Sheep and goats may carry rabies, European tick-borne encephalitis, *Salmonella* infections, and many bacterial and viral diseases.
- Rabbits may carry plague and Q-Fever.
- Birds may carry *Campylobacteriosis*, *Chlamydia psittaci*, *Pasteurella multocida*, *Histoplasma capsulatum*, *Salmonellosis*, and others.

Zoonotic diseases may be spread in different ways. Tapeworms can often spread to humans when people eat the infected meat of cattle, and swine. Other diseases are transferred by insect vectors, often blood-feeding insects that carry the cause of the disease from one animal to another.

Diagnosis

Diagnosis of the disease is made in the usual manner, by identifying the infecting organism. Each disease has established symptoms and tests. Identifying the carrier may be easy, or may be more difficult when the cause is a fairly common infection. For example, tapeworms are usually species specific. Cattle, pigs, and fish all carry different species of tapeworms, although all can be transmitted to humans who eat undercooked meat containing live tapeworm eggs. Once the tapeworm has been identified, it is easy to tell which species the tapeworm came from.

Other zoonotic infections may be harder to identify. Sometimes the infection is fairly common among both humans and animals, and it is impossible to tell. Snakes may carry the bacteria *Escherichia coli* and *Proteus vulgaris*, but since these bacteria are already common among humans, it would be difficult to trace infections back to snakes.

Because of increased trade between nations, and changes in animal habitats, there are often new zoonotic diseases. These may be found in animals transported from one nation to another, bringing with them new diseases. In some cases, changes in the environment lead to changes in the migratory habits of animal species, bringing new infections.

Treatment

Treatment is the established treatment for the specific infection.

KEY TERMS

Anthrax—A disease of warm-blooded animals, particularly cattle and sheep, transmissible to humans. The disease causes severe skin and lung damage.

Bovine spongiform encephalopathy—A progressive, fatal disease of the nervous system of domestic animals. It is transmitted by eating infected food. Also known as Mad Cow disease.

Lyme disease—An acute disease which is usually marked by skin rash, fever, fatigue and chills. Left untreated, it may cause heart and nervous system damage.

Q-Fever—A disease that is marked by high fever, chills and muscle pain. It is seen in North America, Europe, and parts of Africa. It may be spread by drinking raw milk, or by tick bites.

Zoonotic—A disease which can be spread from animals to humans.

Prevention

Prevention of zoonotic infections may take different forms, depending on the nature of the carrier and the infection.

Some zoonotic infections can be avoided by immunizing the animals that carry the disease. Pets and other domestic animals should have rabies vaccinations, and wild animals are immunized with an oral vaccine that is encased in a suitable bait. In some places, the bait is dropped by airplane over the range of the potential rabies carrier. When the animal eats the bait, they also ingest the oral vaccine, thereby protecting them from rabies, and reducing the risk of spread of the disease. This method has been used to protect foxes, coyotes, and other wild animals.

Many zoonotic diseases that are passed by eating the meat of infected animals can be prevented by proper cooking of the infected meat. Tapeworm infestations can be prevented by cooking, and *Salmonella* infections from chickens and eggs can be prevented by being sure that both the meat and the eggs are fully cooked.

For other zoonotic diseases, programs are in place to eliminate the host, or the vector that spreads the disease. Plague is prevented by elimination of the rats—a common source of the infection—and of fleas that carry the disease from rats to humans. Efforts

around the world to control bovine spongiform encephalitis, better known as Mad Cow disease, have focused on the destruction of infected cattle to prevent spread of the disease. Regulations on the makeup of the cattle feed to ensure safety and prevent the disease have helped curb its spread.

Other means of prevention simply rely on care. People living in areas where Lyme disease is common are warned to take precautions against the bite of the deer tick, which transfers the disease. These precautions include not walking in tall grass, not walking bare legged, and wearing light-colored clothing so that the presence of the dark ticks can be readily seen.

Resources

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“When man’s best friend isn’t.” *Mother Earth News* June-July 2000.

“Zoonoses.” *Agricultural Research* February 2000.

“Zoonotic Diseases.” *Medical Laboratory Observer* March 2004: 12.

“The zoonotic threat: Curbing pet-to-people infections.” *Dog World* October 1999.

ORGANIZATIONS

American Association of Zoo Keepers (AAZK). Topeka Zoological Park 3601 SW 29th St., Ste. 133 Topeka, KS 66614-2054.

National Animal Disease Center Zoonotic Research Unit. 2300 Dayton Ave. PO Box 70 Ames, IA 50010.

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Zoonotic infections see **Zoonosis**

Zygomycosis see **Mucormycosis**

ORGANIZATIONS

The following is an alphabetical compilation of organizations listed in the *Resources* section of the main body entries. Although the list is comprehensive, it is by no means exhaustive. It is a starting point for further information, as well as other online and print sources. Many of the organizations listed provide information for multiple disorders and have links to additional related websites. E-mail addresses and web addresses listed were provided by the associations; Thomson Gale is not responsible for the accuracy of the addresses or the contents of the websites.

A

Academy of General Dentistry

Ste. 1200, 211 East Chicago Ave.
Chicago, IL 60611
(312) 440-4300
< <http://www.wagd.org> >

Academy for Guided Imagery

PO Box 2070
Mill Valley, CA 94942
(800) 726-2070

Achromatopsia Network

C/O Frances Futterman
PO Box 214
Berkeley, CA 94701-0214
< [http://www.wachromat.org/
how_to_join.html](http://www.wachromat.org/how_to_join.html) >

Acid Maltase Deficiency Association

PO Box 700248
San Antonio, TX 78270-0248
(210) 494-6144
< <http://www.amda-pompe.org> >

Acoustic Neuroma Association of Canada

Box 369 Edmonton, AB T5J 2J6
(800) 561-ANAC(2622)
(780) 428-3384
anac@compusmartab.ca
< <http://www.anac.ca> >

Action Against Allergy (AAA)

PO Box 278
Twickenham Middlesex, Greater
London TW1 4QQ
United Kingdom

Acupressure Institute

1533 Shattuck Ave
Berkeley, CA 94709

Advancement of Women's Health Research

1828 L St NW, Ste. 625
Washington, DC 20036

(202) 223-8224

< <http://www.womens-health.org> >

Aerospace Medical Association

320 S Henry St
Alexandria, VA 22314-3579
(703) 739-2240
< <http://www.asma.org> >

Agoraphobics Building Independent Lives

3805 Cutshaw Ave, Ste. 415, Dept W
Richmond, VA 23230
(804) 353-3964
< <http://www.anxietysupport.org> >

Agoraphobic Foundation of Canada

PO Box 132
Chomedey, Laval, Quebec H7W 4K2
Canada

Agoraphobics In Motion

605 W 11 Mile Rd
Royal Oak, MI 48067
(248) 547-0400

Al-Anon, Alanon Family Group, Inc

PO Box 862, Midtown Station
New York, NY 10018-0862
(800)356-9996
< <http://www.recovery.org/aa> >

Albert Ellis Institute

45 East 65th St
New York, NY 10021
(800) 323-4738
< <http://www.rebt.org> >

Alcoholics Anonymous (AA)

General Service Office 475
Riverside Dr
New York, NY 10015
(212) 870-3400
< <http://www.alcoholics-anonymous.org> >

Alexander Graham Bell Association for the Deaf

3417 Volta Place NW

Washington, DC 20007

(202) 337-5220

< <http://www.agbell.org> >

Alexander Technique International

1692 Massachusetts Ave, 3rd Floor
Cambridge, MA 02138
(888) 321-0856
Fax: (617) 497-2615
ati@ati-netcom
< <http://www.ati-net.com> >

Alliance of Genetic Support Groups

4301 Connecticut Ave NW, Ste. 404
Washington, DC 20008
(202)966-5557
< <http://www.geneticalliance.org> >

Alliance for Lung Cancer Advocacy, Support and Education

PO Box 849
Vancouver, WA 98666
(800) 298-2436
< <http://www.alcase.org> >

Alzheimer's Association

919 North Michigan Ave, Ste. 1000
Chicago, IL 60611
(800) 272-3900
< <http://www.alz.org> >

Ambiguous Genitalia Support Network

PO Box 313
Clements, CA 95227
(209) 727-0313

American Academy of Allergy Asthma and Immunology

611 East Wells St
Milwaukee, WI 53202
(800) 822-2762 < [http://
www.aaaai.org](http://www.aaaai.org) >

American Academy of Audiology

8201 Greensboro Dr, Ste. 300
McLean, VA 22102
(703) 610-9022
< <http://audiology.org> >

American Academy of Child and Adolescent Psychiatry (AACAP)

3615 Wisconsin Ave NW
Washington, DC 20016
(202) 966-7300
< <http://www.aacap.org> >

American Academy of Clinical Sexologists

1929 18th St NW, Ste. 1166
Washington, DC 20009
(202) 462-2122

American Academy of Cosmetic Surgery

401 N Michigan Ave
Chicago, IL 60611-4267
(313) 527-6713
< <http://www.cosmeticsurgeryonline.com> >

American Academy of Dermatology

930 N Meacham Road
PO Box 4014
Schaumburg, IL 60168-4014
(847) 330-0230
< <http://www.aad.org> >

American Academy of Emergency Medicine

611 East Wells St, Milwaukee, WI 53202
(800) 884-2236
Fax: (414) 276-3349
< <http://www.aaem.org/> >

American Academy of Environmental Medicine

PO Box CN 1001-8001
New Hope, PA 18938
(215) 862-4544

American Academy of Facial, Plastic, and Reconstructive Surgery

1110 Vermont Ave NW, Ste. 220
Washington, DC 20005
(800) 332-3223

American Academy of Family Physicians

11400 Tomahawk Creek Parkway
Leawood, KS 66211-2672
(913) 906-6000
< <http://www.aafp.org> >

American Academy of Husband-Coached Childbirth

PO Box 5224
Sherman Oaks, CA 91413
(800) 423-2397
(800) 422-4784 (in California)

American Academy of Medical Acupuncture

2520 Milvia St
Berkeley, CA 94704
(415) 841-3220

American Academy of Neurology

1080 Montreal Ave
St Paul, MN 55116
(612) 695-1940
< <http://www.aan.com> >

American Academy of Ophthalmology

655 Beach St
PO Box 7424
San Francisco, CA 94120-7424
< <http://www.eyenet.org> >

American Academy of Orthopaedic Surgeons

6300 N River Road
Rosemont, IL 60018
(847) 823-7186
< <http://www.aaos.org> >

American Academy of Otolaryngology-Head and Neck Surgery, Inc

One Prince St
Alexandria, VA 22314-3357
(703) 836-4444
< <http://www.entnet.org> >

American Academy of Pediatric Dentistry

211 East Chicago Ave, Ste.
700 Chicago, IL 60611-2616
(312) 337-2169
< <http://www.aapd.org> >

American Academy of Pediatrics

141 Northwest Point Blvd
Elk Grove Village, IL 60007-1098
(847) 434-4000
Fax: (847) 434-8000
< <http://www.aaporg/visit/contact.htm> >

American Academy of Wound Management

1255 23rd St NW
Washington, DC 20037
(202) 521-0368
< <http://www.aawm.org> >

American Allergy Association (AAA)

3104 E Camelback, Ste. 459
Phoenix, AZ 85016

American Amputation Foundation Inc

PO Box 250218
Hillcrest Station
Little Rock, AR 72225
(501) 666-2523

American Anorexia/Bulimia Association Inc

293 Central Park West, Ste. IR
New York, NY 10024
(212) 501-8351

American Art Therapy Association

1202 Allanson Rd
Mundelein, IL 60060-3808
(888) 290-0878
(847) 949-6064
Fax: (847) 566-4580
arttherapy@ntrnet
< <http://www.arttherapy.org> >

American Association of Acupuncture & Oriental Medicine

4101 Lake Boone Trail, Ste. 201
Raleigh, NC 27607
(919) 787-5181

American Association of Blood Banks

8101 Glenbrook Road
Bethesda, MD 20814
(301) 907-6977
< <http://www.aabb.org> >

American Association of Colleges of Osteopathic Medicine

5550 Friendship Blvd, Ste. 310
Chevy Chase, MD 20815-7231
(301) 968-4100
< <http://www.aacom.org> >

American Association for Chronic Fatigue Syndrome

7 Van Buren St
Albany, NY 12206
(518) 435-1765
< <http://weberuwashington.edu/~dedra/aacfs1.html> >

American Association of the Deaf-Blind

814 Thayer Ave, Ste. 302
Silver Spring, MD 20910
(301) 588-6545

American Association of Endodontists

211 East Chicago Ave, Ste. 1100
Chicago, IL 60611-2691
(800) 872-3636
< <http://www.aae.org> >

American Association of Kidney Patients (AAKP)

100 S Ashley Dr, Ste. 280
Tampa, FL 33602
(800) 749-2257
< <http://www.aakp.org> >

American Association of Nutritional Consultants

810 S Buffalo St

Warsaw, IN 46580
(888) 828-2262

**American Association for Marriage
and Family Therapy**

1133 15th St NW, Ste. 300
Washington, DC 20005-2710
(202) 452-0109
< <http://www.aamft.org> >

**American Association on Mental
Retardation (AAMR)**

444 North Capitol St NW, Ste. 846
Washington, DC 20001-1512
(800) 424-3688
< <http://www.aamr.org> >

**American Association of Naturopathic
Physicians**

601, VA lley St, Ste. 105
Seattle, WA 98109
(206) 298-0126
< <http://www.naturopathic.org> >

**American Association of Oral &
Maxillofacial Surgeons**

9700 W Bryn Mawr Ave
Rosemont, IL 60018
(847) 678-6200

**American Association of Oriental
Medicine**

433 Front St
Catasququa, PA 18032
(888) 500-7999
< <http://www.aaom.org> >

**American Association of
Orthodontists**

401 North Lindbergh Blvd
St Louis, MO 63141-7816
(314) 993-1700
< <http://www.aaortho.org> >

**American Association for Respiratory
Care**

11030 Ables Lane
Dallas, TX 75229
(972) 243-2272
< <http://www.aarc.org> >

**American Association of Sex
Educators Counselors and
Therapists**

PO Box 5488
Richmond, VA 23220
< <http://www.aasect.org> >

**American Association of Tissue
Banks**

1350 Beverly Road, Ste. 220-A
McLean, VA 22101
(703) 827-9582

American Behcet's Disease Association

PO Box 280240
Memphis, TN 38168-0240
< <http://www.behcets.com> >

American Board of Hypnotherapy

16842 Von Karman Ave, Ste. 476
Irvine, CA 92714
< <http://www.hypnosis.com/> >

American Botanical Council

PO Box 201660
Austin, TX 78720-1660

American Brain Tumor Association

2720 River Road, Ste. 146
Des Plaines, IL 60018-4110
(800) 886-2282
< <http://www.abta.org> >

American Burn Association

625 N Michigan Ave, Ste. 1530
Chicago, IL 60611
(800) 548-2876
< <http://www.ameriburn.org> >

**American Cancer Society (National
Headquarters)**

1599 Clifton Road NE
Atlanta, GA 30329
(800) 227-2345
< <http://www.cancer.org> >

American Celiac Society

58 Musano Court
West Orange, NJ 07052
(201) 325-8837

American Chiropractic Association

1701 Clarendon Blvd
Arlington, VA 22209
(800)986-4636
< <http://www.amerchiro.org/> >

American Chronic Pain Association

PO Box 850
Rocklin, CA 95677-0850
(916) 632-0922
< <http://memberstripod.com/~widdy/ACPA.html> >

**American College of Allergy Asthma &
Immunology**

85 West Algonquin Road, Ste. 550
Arlington Heights, IL 60005
(847) 427-1200

American College of Angiology

295 Northern Blvd, Ste. 104
Great Neck, NY 11021-4701

**American College of Emergency
Physicians**

PO Box 619911

Dallas, TX 75261-9911
(800) 798-1822
(972) 550-0911
Fax: (972) 580-2816
info@acep.org
< <http://www.acep.org/> >

American College of Gastroenterology

4900 B South 31st St
Arlington, VA 22206
(703) 820-7400
< http://www.acggi.org/ct_.html >

**American College of Hyperbaric
Medicine**

PO Box 25914-130
Houston, TX 77265
(713) 528-0657
< <http://www.hyperbaricmedicine.org> >

American College of Nuclear Medicine

PO Box 175
Landisville, PA 31906
(717) 898-6006

American College of Nutrition

722 Robert E Lee Dr
Wilmington, NC 20412-0927
(919) 152-1222

**American College of Obstetricians and
Gynecologists**

409 12th St SW
PO Box 96920
Washington, DC 20090-6920
< <http://www.acog.org> >

**American College of Osteopathic
Emergency Physicians**

142 E Ontario St, Ste. 550
Chicago, IL 60611
(312) 587-3709
(800) 521-3709
Fax: (312) 587-9951
< <http://www.acoep.org> >

American College of Radiology

1891 Preston White Dr
Reston, VA 22091
(800) 227-5463
< <http://www.acr.org> >

American College of Rheumatology

1800 Century Place, Ste. 250
Atlanta, GA 30345
(404) 633-3777
< <http://www.rheumatology.org> >

American College of Sports Medicine

401 W Michigan St
Indianapolis, IN 46202-3233
(317) 637-9200
Fax: (317) 634-7817

< <http://www.acsm.org/> >
mkeckhaver@acsmorg

American College of Surgeons
 633 North St Clair St
 Chicago, IL 60611-32311
 (312) 202-5000
 Fax: (312) 202-5001
postmaster@facsorg
 < <http://www.facs.org/> >

American Council for Headache Education (ACHE)
 19 Mantua Road
 Mt Royal, NJ 08061
 (800) 255-2243
 < <http://www.achenet.org> >

American Council on Transplantation
 PO Box 1709
 Alexandria, VA 22313
 (800) ACT-GIVE

American Dental Association
 211 E Chicago Ave
 Chicago, IL 60611
 (312) 440-2500
 < <http://www.ada.org> >

American Dental Hygienists' Association
 444 North Michigan Ave
 Chicago, IL 60611
 (800) 847-6718

American Diabetes Association
 1701 North Beauregard St
 Alexandria, VA 22311
 (800) 342-2383
 < <http://www.diabetes.org> >

American Dietetic Association
 216 W Jackson Blvd
 Chicago, IL 60606-6995
 (800) 745-0775
 < <http://www.eatright.org/cdr.html> >

American Epilepsy Society
 342 North Main St
 West Hartford, CT 06117-2507
 (860) 586-7505
 < <http://www.aesnet.org> >

American Foundation for the Blind
 11 Penn Plaza, Ste. 300
 New York, NY 10001
 (800) 232-5463

American Foundation for Homeopathy
 1508 S Garfield
 Alhambra, CA 91801

American Foundation for the Prevention of Venereal Disease Inc
 799 Broadway, Ste. 638
 New York, NY 10003
 (212) 759-2069

American Foundation for Urologic Disease
 1128 N Charles St
 Baltimore, MD 21201
 (401) 468-1800
 < <http://www.afud.org> >

American Gastroenterological Association (AGA)
 7910 Woodmont Ave 7th Floor
 Bethesda, MD 20814
 (310) 654-2055
aga001@aol.com
 < <http://www.gastro.org/index.html> >

American Hair Loss Council
 30 Grassy Plain Road
 Bethel, CT 06801
 (888) 873-9719
 < <http://www.ahlc.org/> >

American Hearing Research Foundation
 55 E Washington St, Ste. 2022
 Chicago, IL 60602
 (312) 726-9670
 < <http://www.american-hearing.org/> >

American Heart Association
 7272 Greenville Ave
 Dallas, TX 75231-4596
 (214) 373-6300
 (800) 242-8721
inquire@heartorg
 < <http://www.americanheart.org> >

American Hemochromatosis Society Inc.
 777 E Atlantic Ave PMB Z-363
 Delray Beach, FL 33483-5352
 (561) 266-9037
 (888) 655-4766
ahs@eminent
 < <http://www.americanhs.org> >

American Holistic Medical Association
 4101 Lake Boone Trail, Ste. 201
 Raleigh, NC 27607

American Humane Association Children's Division
 63 Inverness Dr East
 Englewood, CO 80112-5117
 (800) 227-4645
 < <http://www.americanhumane.org> >

American Institute for Cancer Research (AICR)
 1759 R St NW
 Washington, DC 20009
 (800) 843-8114
 < <http://www.aicr.org> >

American Institute of Nutrition
 9650 Rockville Pike
 Bethesda, MD 20814-3990
 (301) 530-7050

American Institute of Stress
 124 Park Ave
 Yonkers, NY 10703
 (914) 963-1200
 < <http://www.stress.org> >

American Institute of Ultrasound in Medicine
 14750 Sweitzer Lane, Ste. 100
 Laurel, MD 20707-5906
 (800) 638-5352
 < <http://www.aium.org> >

American Institute of Vedic Studies
 PO Box 8357
 Santa Fe NM 87504
 (505) 983-9385

American Kidney Foundation
 6110 Executive Blvd #1010
 Rockville, MD 20852
 (800) 638-8299

American Kidney Fund (AKF)
 Ste. 1010 6110 Executive Blvd
 Rockville, MD 20852
 (800) 638-8299
 < <http://www.arboncom/kidney> >

American Leprosy Missions
 1 ALM Way
 Greenville SC 29601
 (800)LEPROSY

American Liver Foundation
 1425 Pompton Ave
 Cedar Grove, NJ 07009
 (800) 223-0179
 < <http://www.liverfoundation.org> >

American Lung Association
 1740 Broadway
 New York, NY 10019
 (800) 586-4872
 (212) 315-8700
 < <http://www.lungusa.org> >

American Lyme Disease Foundation Inc
 Mill Pond Offices
 293 Route 100, Ste. 204
 Somers, NY 10589

(800) 876-LYME
< <http://www.w2com/docs2/d5/lyme.html> >

American Medical Association
515 N State St
Chicago, IL 60610
(312) 464-5000
< <http://www.ama-assn.org/> >

American Optometric Association
243 North Lindbergh Blvd
St Louis, MO 63141
(314) 991-4100
< <http://www.aoanet.org> >

American Oriental Bodywork Therapy Association
50 Maple Place
Manhasset, NY 11030

American Orthopedic Foot and Ankle Society
222 South Prospect
Park Ridge, IL 60068

American Orthopaedic Society for Sports Medicine
6300 N River Road, Ste. 200
Rosemont, IL 60018
(847) 292-4900
< <http://www.sportsmed.org> >

American Pain Society
4700 W Lake Ave
Glenview, IL 60025
(847) 375-4715
< <http://www.ampainsoc.org> >

American Parkinson Disease Association
60 Bay St, Ste. 401
Staten Island, NY 10301
(800) 223-2732
< <http://www.apdaparkinson.org> >

American Podiatric Medical Association
9312 Old Georgetown Road
Bethesda, MD 20814-1698
(301) 571-9200
< <http://www.apma.org> >

American Podiatry Association
20 Chevy Chase Circle NW
Washington, DC 20015

American Polarity Therapy Association
PO Box 19858
Boulder, CO 80308
(303) 545-2080
Fax: (303) 545-2161

American Physical Therapy Association
1111 North Fairfax St
Alexandria, VA 22314
(800) 999-2782
< <https://www.apta.org> >

American Porphyria Foundation
PO Box 22712
Houston, TX 77227
(713) 266-9617
< <http://www.enterprise.net/apf/> >

American Pseudo-Obstruction & Hirschsprung's Society
158 Pleasant St
North Andover, MA 01845
(978)685-4477

American Psychiatric Association
1400 K St NW
Washington, DC 20005
(888) 357-7924
< <http://www.psych.org> >

American Psychological Association (APA)
750 First St NE
Washington, DC 20002-4242
(202) 336-5700
< <http://www.apa.org> >

American Psychotherapy & Medical Hypnosis Association
210 S Sierra
Reno, NV 89501
< <http://membersxoom.com/Hypnosis/> >

American Red Cross
PO Box 37243 Washington,
DC 20013
< <http://www.redcross.org> >

American Skin Association Inc
150 E 58th St 3rd floor
New York, NY 10155-0002
(212) 688-6547

American Sleep Apnea Association
1424 K St NW, Ste. 302
Washington, DC 20005
(202) 293-3650
< <http://www.sleepapnea.org> >

American Sleep Disorders Association
1610 14th St NW, Ste. 300
Rochester, MN 55901
(507) 287-6006
< <http://www.asda.org> >

American Social Health Association
PO Box 13827 Research

Triangle Park, NC 27709
(800) 227-8922
< <http://www.ashastd.org> >

American Society of Addiction Medicine
4601 North Park Ave Arcade, Ste. 101
Chevy Chase, MD 20815
(301) 656-3920
< <http://www.asam.org> >

American Society of Cataract and Refractive Surgery
4000 Legato Road, Ste. 850
Fairfax, VA 22033-4055
(703) 591-2220
< <http://www.ascrs.org> >

American Society of Clinical Hypnosis
200 E Devon Ave
Des Plaines, IL 60018

American Society of Clinical Oncology
225 Reinekers Lane, Ste. 650
Alexandria, VA 22314
(703) 299-0150
< <http://www.asco.org> >

American Society of Clinical Pathologists
2100 West Harrison St
Chicago, IL 60612
(312) 738-1336
< <http://www.ascporg/index.asp> >

American Society of Colon and Rectal Surgeons
85 W Algonquin Road, Ste. 550
Arlington Heights, IL 60005
(847)290-9184

American Society for Colposcopy and Cervical Pathology
20 W Washington St, Ste. #1
Hagerstown, MD 21740
(800) 787-7227
< <http://www.asccp.org> >

American Society for Dermatologic Surgery
930 N Meacham Road
PO Box 4014
Schaumburg, IL 60168-4014
(847) 330-9830
< <http://www.asds-net.org> >

American Society of Extra-Corporeal Technology
11480 Sunset Hills Rd, Ste. 210E
Reston, VA 20190
(703) 435-8556
< <http://www.amsect.org> >

American Society for Gastrointestinal Endoscopy

13 Elm St Manchester, MA 01944
(508) 526-8330
< <http://www.asgeorg/doc/201> >

American Society of Hematology

1200 19th St NW, Ste. 300
Washington, DC 20036-2422
(202) 857-1118
< <http://www.hematology.org> >

American Society of Human Genetics

9650 Rockville Pike
Bethesda, MD 20814-3998
(301) 571-1825
< <http://www.faseb.org/genetics/ashg/ashgmenu.htm> >

American Society of Hypertension

515 Madison Ave, Ste. 1212
New York, NY 10022
(212) 644-0650
< <http://www.ash-us.org> >

American Society for Laser Medicine and Surgery

2404, Ste.wart Square
Wausau, WI 54401
(715) 845-9283
< <http://www.aslms.org> >

American Society of Microbiology

1752 N St NW
Washington, DC 20036
(202) 737-3600
< <http://www.asmtusa.org> >

American Society of Nuclear Cardiology

9111 Old Georgetown Road
Bethesda, MD 20814-1699
(301) 493-2360
Fax: (301) 493-2376
admin@asnrc.org
< <http://www.asnrc.org/> >

American Society of Ophthalmic Plastic and Reconstructive Surgery

1133 West Morse Blvd #201
Winter Park, FL 32789
(407) 647-8839
< <http://www.asopr.org> >

American Society of Plastic and Reconstructive Surgeons

44 E Algonquin Rd
Arlington Heights, IL 60005
(847) 228-9900
< <http://www.plasticsurgery.org> >

American Society for Prophylaxis in Obstetrics/LAMAZE (ASPO / LAMAZE)

1840 Wilson Blvd, Ste. 204
Arlington, VA 22201
(800) 368-4404

American Society of Radiologic Technologists

15000 Central Ave SE
Albuquerque NM 87123-3917
(505) 298-4500
< <http://www.asrt.org> >

American Society for Reproductive Medicine

1209 Montgomery Highway
Birmingham, AL 35216-2809
(205) 978-5000 < asrm@asrm.com >
< <http://www.asrm.com> >

American Society for Surgery of the Hand

6300 N River Rd, Ste. 600
Rosemont, IL 60018
< <http://www.hand-surg.org> >

American Speech-Language-Hearing Association

10801 Rockville Pike
Rockville, MD 20852
(800) 638-8255
< <http://www.asha.org> >

American Thoracic Society

1740 Broadway
New York, NY 10019
(212) 315-8700
< <http://www.thoracic.org> >

American Thyroid Association Inc

Montefiore Medical Center
111 E 210th St
Bronx, NY 10467
< <http://www.thyroid.org> >

American Tinnitus Association

PO Box 5
Portland, OR 97207
(503) 248-9985
tinnitus@ata.org

American Trauma Society

8903 Presidential Pkwy, Ste. 512
Upper Marlboro, MD 20227
(800) 556-7890
< <http://www.amtrauma.org> >

American Urological Association

1120 North Charles St
Baltimore, MD 21201-5559
(410) 727-1100
< http://www.auanet.org/index_hicfm >

Amputee Coalition of America

PO Box 2528
Knoxville, TN 37901-2528
(888) 267-5669
< <http://www.amputee-coalition.org> >

Amyloidosis Network International

7118 Cole Creek Dr
Houston, TX 77092-1421
(888) 1AMYLOID < <http://www.health.gov/nhic/Scripts/Entrycfm?HRCODE=HR2397> >

Anorexia Nervosa and Related Eating Disorders Inc

PO Box 5102
Eugene, OR 97405
(541) 344-1144

Anxiety Disorders Association of America

11900 Park Lawn Dr, Ste. 100
Rockville, MD 20852
(800) 545-7367
< <http://www.adaa.org> >

Aplastic Anemia Foundation of America

PO Box 613
Annapolis, MD 21404
(800)747-2820
< <http://www.aplastic.org> >

The Arc

900 Varnum St NE
Washington, DC 20017
(202) 636-2950
< <http://thearc.org> >

Arc of the United States (formerly Association for Retarded Citizens of the US)

500 East Border St, Ste. 300
Arlington, TX 76010
(817) 261-6003
< <http://thearc.org> >

Arteriovenous Malformation Support Group

168 Six Mile Canyon Road
Dayton, NV 89403
(702) 246-0682

Arthritis Foundation

1300 W Peachtree St
Atlanta, GA 30309
(800) 283-7800
< <http://www.arthritis.org> >

Association for the Advancement of Gestalt Therapy

400 East 58th St
New York, NY 10022
(212) 486-1581

< <http://www.aagt.org> >

Association for Applied Psychotherapy and Biofeedback

10200 W 44th Ave, Ste. 304
Wheat Ridge, CO 80033-2840
(303) 422-8436
< <http://www.aapb.org> >

Association of Birth Defect Children

3526 Emerywood Lane
Orlando, FL 32806
(305) 859-2821

Association for the Bladder Exstrophy Community

PO Box 1472
Wake Forest, NC 27588-1472
(919) 624-9447
< <http://www.bladderexstrophy.com/support.htm> >

Association for the Care of Children's Health (ACCH)

7910 Woodmont Ave, Ste. 300
Bethesda, MD 20814
(800) 808-2224

Association for the Cure of Cancer of the Prostate (CaPCure)

1250 Fourth St, Ste. 360
Santa Monica, CA 90401
(800) 757-CURE
< <http://www.capcure.org> >

Association for Glycogen Storage Disease

PO Box 896
Durant, IA 52747-9769
(319) 785-6038

Association for Neuro-Metabolic Disorders

5223 Brookfield Lane
Sylvania, OH 43560-1809
(419) 885-1497

Association of SIDS and Infant Mortality Programs

Minnesota SID Center Children's Hospitals and Clinics
2525 Chicago Ave S
Minneapolis, MN 55404
(612) 813-6285
< <http://www.asip1.org> >

Association for Spina Bifida and Hydrocephalus

42 Park Rd
Peterborough PE1 2UQ UK
0173 355 5988
Fax: 017 3355 5985
postmaster@asbahorg
< <http://www.asbahdemon.co.uk> >

Asthma and Allergy Foundation of America

1233 20th St NW, Ste. 402
Washington, DC 20036
(800) 727-8462
< <http://www.aafa.org> >

Aston Training Center

P O Box 3568 Incline Village, NV 89450
(775) 831-8228
Astonpat@aolcom
< <http://www.aston-patterning.com> >

Audiology Awareness Campaign

3008 Millwood Ave
Columbia SC 29205
(800) 445-8629

Auditory-Verbal International

2121 Eisenhower Ave, Ste. 402
Alexandria, VA 22314
(703) 739-1049
< avi@auditory-verbal.org >
< <http://www.auditory-verbal.org/contact.htm> >

Autism Network International

PO Box 448
Syracuse, NY 13210

Autism Research Institute

4182 Adams Ave
San Diego, CA 92116
(619) 281-7165

Autism Society of America

7910 Woodmont Ave, Ste. 300
Bethesda, MD 20814-3067
(800) 328-8476
< <http://www.autism-society.org> >

Ayurveda Holistic Center

Bayville
Long Island, NY
(516) 759-7731
mail@Ayurvedahccom
< <http://www.Ayurvedahc.com> >

Ayurvedic Institute

11311 Menaul NE
Albuquerque New Mexico 87112
(505) 291-9698
info@Ayurvedacom
< <http://www.Ayurveda.com> >

Ayurvedic and Naturopathic Medical Clinic

10025 NE 4th St
Bellevue, WA 98004
(206) 453-8022

B

Bastyr University of Natural Health Sciences

144 NE 54th St
Seattle, WA 98105
(206) 523-9585

Baylor College of Medicine

1 Baylor Plaza
Houston, TX 77030
(713) 798-4951
< <http://publicbcmcmc.edu> >

Beck Institute

GSB Building City Line and Belmont Aves, Ste. 700
Bala Cynwyd, PA 19004-1610
(610) 664-3020
< <http://www.beckinstitute.org> >

Behcet's Organization Worldwide Head Office

PO Box 27
Watchet Somerset TA23 OYJ
United Kingdom
< <http://www.behcets.org> >

Bell's Palsy Research Foundation

9121 E Tanque Verde, Ste. 105-286
Tucson, AZ 85749
(520) 749-4614

Beryllium Support Group

PO Box 2021
Broomfield, CO 80038-2021
(303) 412-7065
< <http://www.dimensional.com/~mhj> >

Better Hearing Institute

515 King St, Ste. 420
Alexandria, VA 22314
(703) 684-3391

Biofeedback Certification Institute of America

10200 W 44th Ave, Ste. 310
Wheat Ridge, CO 80033
(303) 420-2902

Bladder Health Council American Foundation for Urologic Disease

300 West Pratt St, Ste. 401
Baltimore, MD 21201
(800) 242-2383
(410) 727-2908

Brain Aneurysm Foundation Inc

66 Canal St
Boston, MA 02114
(617) 723-3870

< <http://neurosurgery.mgh.harvard.edu/baf> >

Brain Injury Association of America
105 North Alfred St
Alexandria, VA 22314
(800) 444-6443
< <http://www.biausa.org> >

Brain Tumor Information Services
Box 405 Room J341
University of Chicago Hospitals
5841 S Maryland Ave
Chicago, IL 60637
(312) 684-1400

British Coalition of Heritable Disorders of Connective Tissue
Rochester House
5 Aldershot Road
Fleet Hampshire GU13 9NG
United Kingdom
(012) 52-810472

**British Leprosy Relief Association
LEPRA**
Fairfax House
Causton Road
Colchester Essex CO1 1PU
United Kingdom

C

California Colon Hygienist Society
333 Miller Ave, Ste. 1
Mill, VA lley, CA 94941
(415) 383-7224

Canadian HIV/AIDS Clearinghouse
1565 Carling Ave, Ste. 400
Ottawa ON K1Z 8R1
Canada
(877) 999-7740
< http://www.clearinghousecpha.ca/clearinghouse_e.htm >

Canadian Society for Mucopolysaccharide and Related Diseases
PO Box 64714
Unionville ONT L3R-OM9
Canada
(905) 479-8701
(800) 667-1846
< <http://www.mpsociety.ca> >

Canadian MEN Society
PO Box 100
Meola Saskatchewan SOM 1X0
Canada
(306) 892-2080

Canadian 22q Group
320 Cote St Antoine West
Montreal Quebec H3Y 2J4
Canada

Cancer Care Inc
275 Seventh Ave
New York, NY 10001
(800) 813-HOPE
< <http://www.cancercare.org> >

Cancer Group Institute
1814 NE Miami Gardens Dr
North Miami Beach, FL 33179
(305) 651-5070
< <http://www.cancergroup.com/em19.html> >

Cancer Group Institute
17620 9th Ave NE
North Miami Beach Florida 33162
(305) 493-1980
< <http://www.cancergroup.com> >

Cancer Hope Network
Ste. A Two North Rd
Chester, NJ 07930
(877) HOPENET
< <http://www.cancerhopenetwork.org> >

Cancer Research Institute (National Headquarters)
681 Fifth Ave
New York, NY 10022
(800) 992-2623
< <http://www.cancerresearch.org> >

Cancer Prevention Coalition
2121 West Taylor St
Chicago, IL 60612
(312) 996-2297
< <http://www.preventcancer.com> >

Cancer Information Service National Cancer Institute
Building 31, Room 10A19
9000 Rockville Pike
Bethesda, MD 20892
(800)4-CANCER
< <http://www.nci.nih.gov/cancerinfo/index.html> >

CancerNet
National Cancer Institute 9000
Rockville Pike
Bldg 31, Rm 10A16
Bethesda, MD 20892
(800) 422-6237
< <http://www.icinc.nih.gov> >

Carcinoid Cancer Foundation Inc
1751 York Ave
New York, NY 10128

(212) 722-3132
< <http://www.carcinoid.org> >

Cardiac Arrhythmia Research and Education Foundation (CARE)
2082 Michelson Dr #301
Irvine, CA 92612
(800) 404-9500
< <http://www.longqt.com> >

Celiac Disease Foundation
13251 Ventura Blvd, Ste. 1
Studio City, CA 91604-1838
(818) 990-2354
< <http://www.cdf@celiac.org> >

Celiac Sprue Association/United State of America (CSA/USA)
PO Box 31700
Omaha NE 68131-0700
(402) 558-0600

Center for Cell and Gene Therapy
Baylor College of Medicine
1102 Bates St, Ste. 1100
Houston, TX 77030-2399
(713) 770-4663
< <http://www.bcm.tmc.edu/genetherapy> >

Center for Devices and Radiological Health
United States Food and Drug Administration
1901 Chapman Ave
Rockville, MD 20857
(301) 443-4109
< <http://www.fda.gov/cdrh> >

Center for Fertility and In Vitro Fertilization Loma Linda University
11370 Anderson St
Loma Linda, CA 92354
(909) 796-4851
< <http://www.llu.edu/llumc/fertility> >

Center for Holistic Urology
161 Fort Washington Ave
New York, NY 10032
(212) 305-0347
< <http://www.holisticurology.com> >

Center for Mind/Body Medicine
PO Box 1048
La Jolla, CA 92038
(619)794-2425

Center for Mindfulness
University of Massachusetts Medical Center
55 Lake Ave North
Worcester, MA 01655
(508) 856-2656
< <http://www.umassmed.edu/cfm> >

Center for Occupational and Environmental Medicine

7510 Northforest Dr
North Charleston SC 29420
(843) 572-1600
< <http://www.coem.com> >

Center for the Study of Anorexia and Bulimia

1 W 91st St
New York, NY 10024
(212) 595-3449

Center for Taste and Smell Disorders

University of Colorado Health Sciences Center
4200 E Ninth Ave
Denver, CO 80262
(303) 315-5660
< <http://www.hsccolorado.edu> >

Centers for Disease Control and Prevention

1600 Clifton Rd NE
Atlanta, GA 30333
(800) 311-3435
(404) 639-3311
< <http://www.cdc.gov> >

Central Institute for the Deaf

Washington University
St Louis MO
< <http://cidmacwustl.edu> >

Chalice of Repose Project at St Patrick Hospital

312 East Pine St
Missoula MT 59802
(406) 329-2810
Fax: (406) 329-5614
< <http://www.saintpatrick.org/chalice/> >

Charcot Marie Tooth Association (CMTA)

2700 Chestnut Parkway
Chester, PA 19013
(610) 499-9264
(800) 606-CMTA
Fax: (610) 499-9267
cmtassoc@aol.com
< www.charcot-marie-tooth.org >

Child Abuse Prevention Center of Utah

2955 Harrison Blvd #102
Ogden UT 84403
(888) 273-0071

Childbirth Education Foundation

PO Box 5
Richboro, PA 18954
(215) 357-2792

Children Living with Inherited Metabolic Diseases

The Quadrangle Crewe Hall Weston Rd
Crewe Cheshire CW1-6UR
United Kingdom
127 025 0221
Fax: 0870-7700-327
< <http://www.climb.org.uk> >

Children's Blood Foundation

333 East 38th St Room 830
New York, NY 10016-2745
(212) 297-4336
cfc@nyhmedcornell.edu

Children's Brittle Bone Foundation

7701 95th St
Pleasant Prairie, WI 53158
(847) 433-498
< <http://www.cbbf.org> >

Children's Gaucher Research Fund

PO Box 2123
Granite Bay, CA 95746-2123
(916) 797-3700
Fax: (916) 797-3707
< <http://www.childrensgaucher.org> >

Children's Health Information Network

1561 Clark Dr
Yardley, PA 19067
(215) 493-3068
< <http://www.tchin.org> >

Children's Organ Transplant Association Inc

2501 COTA Dr
Bloomington, IN 47403
(800) 366-2682
< <http://www.cota.org> >

Children's PKU Network (CPN)

3790 Via De La, VA 11e, Ste. 120
Del Mar, CA 92014
(800) 377-6677
< <http://www.pkunetwork.org/> >

Chinese National Chi Kung Institute

PO Box 31578
San Francisco, CA 94131
(800) 824-2433

Chromosome Deletion Outreach Inc

PO Box 724
Boca Raton, FL 33429-0724
(888) 236-6680

Chromosome 18 Registry & Research Society

6302 Fox Head
San Antonio, TX 78247

(210) 657-4968

< <http://www.chromosome18.org> >

Chronic Granulomatous Disease Association

2616 Monterey Road
San Marino, CA 91108-1646
(818) 441-4118

Chronic Pain Outreach

822 Wycliff Ct
Manassas, VA 22110
(703) 368-7357

CMT International

1 Springbank Dr
St Catherine's ONT L2S2K1
Canada
(905) 687-3630
< <http://www.cmtint.org> >

Cocaine Anonymous

6125 Washington Blvd, Ste. 202
Culver City, CA 90232
(800) 347-8998

Cochlear Implant Club International

5335 Wisconsin Ave NW, Ste. 440
Washington, DC 20015-2052
(202) 895-2781
< <http://www.cici.org> >

College of American Pathologists

325 Waukegan Road
Northfield, IL 60093
(800) 323-4040
< <http://www.cap.org> >

**College of Maharishi Ayur-Ved
Maharishi International University**

1000 4th St
Fairfield, IA 52557
(515) 472-7000

Coma Recovery Association Inc

570 Elmont Rd, Ste. 104
Elmont, NY 11003
(516) 355-0951

Compassionate Friends

PO Box 3696
Oak Brook, IL 60522
(877) 969-0010
< <http://www/compassionatefriends.org> >

**Congenital Heart Anomalies Support
Education and Resources
(CHASER)**

2112 North Wilkins Rd
Swanton, OH 43558
(419) 825-5575
< <http://www.csun.edu/~hfmth006/chaser> >

Congenital Heart Disease Information and Resources

1561 Clark Dr
Yardley, PA 19067
< <http://www.tchin.org> >

Congenital Nevus Support Group

1400 South Joyce St Number C-1201
Arlington, VA 22202
(703) 920-3249

Cooley's Anemia Foundation Inc

129-09 26th Ave #203
Flushing, NY 11354
(800) 522-7222
(718) 321-2873
< <http://www.thalassemia.org> >

Council for Homeopathic Certification

PO Box 157
Corte Madera, CA 94976

Cri du Chat Society

Dept of Human Genetics
Box 33, MCV Station
Richmond, VA 23298
(804) 786-9632

Crohn's and Colitis Foundation of America Inc

386 Park Ave South
17th Floor New York, NY 10016-8804
(800) 932-2423

Cystic Fibrosis Foundation

6931 Arlington Road
Bethesda, MD 20814
(800) 344-4823
< <http://www.cff.org> >

Cystinuria Support Network

21001 NE 36th St
Redmond, WA 98053
(425) 868-2996
< <http://www.cystinuria.com> >

D**Deafness Research Foundation**

1225 I St NW
No 500
Washington, DC 20005

Depression After Delivery (DAD)

PO Box 1282
Morrisville, PA 19067
(800) 944-4773

Dermatology College of Medicine

University of Iowa
200 Hawkins Dr

Iowa City, IA 52242

(319) 356-2274
< <http://traydermatologyuiowa.edu> >

Dermatomyositis and Polymyositis Support Group

146 Newtown Road
Southampton SO2 9HR
United Kingdom

Digestive Disease National Coalition

507 Capitol Court NE, Ste. 200
Washington, DC 20003
(202) 544-7497
< <http://www.ddnc.org> >

Digestive Health Initiative

7910 Woodmont Ave #914
Bethesda, MD 20814
(800) 668-5237
< <http://www.gastro.org/dhi.html> >

Divers Alert Network

The Peter B Bennett Center
6 West Colony Place
Durham, NC 27705
(800) 446-2671
< <http://www.diversalertnetwork.org> >

E**Ear Foundation**

1817 Patterson St
Nashville, TN 37203
(800) 545-4327
< <http://www.earfoundation.org> >

Eating Disorder Awareness & Prevention Inc

603, Ste.wart St, Ste. 803
Seattle, WA 98101
(206) 382-3587

Edward Bach Centre

Mount Vernon Bakers Lane
Sotwell Oxon OX10 OPX
United Kingdom
@bachcentre.com
< <http://www.bachcentre.com> >

Elhers-Danlos National Foundation

6399 Wilshire Blvd, Ste. 203
Los Angeles, CA 90048
(323) 651-3038
Fax: (323) 651-1366
< <http://www.ednf.org> >

Ehlers-Danlos Support Group - UK

PO Box 335
Farnham Surrey GU10 1XJ

United Kingdom 01252 690 940
< <http://www.atvndirect.co.uk> >

Emphysema Anonymous Inc

PO Box 3224
Seminole, FL 34642
(813) 391-9977

Endocrine Society

4350 East West Highway,
Ste. 500
Bethesda, MD 20814-4410
(301) 941-0200
Fax: (301) 941-0259
endostaff@endo-society.org

Endometriosis Association**International Headquarters**

8585 North 76th Place
Milwaukee, WI 53223
(800) 992-3636
< <http://EndometriosisAssn.org> >

Environmental Health Center

1025 Connecticut Ave NW
Washington, DC 20036
(202) 293-2270

Epilepsy Foundation of America

4351 Garden City Dr, Ste. 406
Landover, MD 20785-2267
(301) 459-3700
(800) 332-1000
< <http://www.epilepsyfoundation.org> >

Epilepsy International Service Group

1282 Wynnwood Dr
West Palm Beach, FL 33417
(407) 683-0044

ERIC Clearinghouse on Assessment and Evaluation

1131 Shriver Laboratory
Bldg 075
University of Maryland
College Park, MD 20742
(800) 464-3742
< <http://www.ericae.net> >

Extracorporeal Life Support Organization

1327 Jones Dr, Ste. 101
Ann Arbor MI 48105
(734) 998-6600
< <http://www.elsomedumich.edu> >

EyesOnThePrizeOrg

446 S Anaheim Hills Road #108
Anaheim Hills, CA 92807
< <http://www.eyesontheprize.org> >

F

Familial Polyposis Registry
Department of Colorectal Surgery
Cleveland Clinic Foundation
9500 Euclid Ave
Cleveland, OH 44195-5001
(216) 444-6470

Family Caregiver Alliance
425 Bush St, Ste. 500
San Francisco, CA 94108
(800) 445-8106
< <http://www.caregiver.org> >

Fasting Center International
32 West Anapurna St #360
Santa Barbara, CA 93101
< <http://www.fasting.com> >

Federal Drug Administration
5600 Fishers Lane
Rockville, MD 20857
(800) 532-4440
< <http://www.fda.gov> >

Federation for Children With Special Needs
1135 Tremont St, Ste. 420
Boston, MA 02120
(617) 236-7210
< <http://www.fcsn.org> >

Federation of Feminist Women's Health Centers
1469 Humboldt Rd, Ste. 200
Chico, CA 96928
(530) 891-1911

Feldenkrais Guild of North America
3611 SW Hood Ave, Ste. 100
Portland, OR 97201
(800) 775-2118
(503) 221-6612
Fax: (503) 221-6616
< <http://www.feldenkrais.com/> >

Female Sexual Medicine Center
UCLA Medical Center
924 Westwood Blvd, Ste. 520
Los Angeles, CA 90024
(310) 825-0025
< <http://www.newshe.com> >

Fetal Alcohol Syndrome Family Resource Institute
PO Box 2525
Lynnwood, WA 98036
(253) 531-2878
(800) 999-3429
< <http://www.fetalalcoholsyndrome.org> >

5p- Society
7108 Katella Ave. #502
Stanton, CA 90680
(888) 970-0777
< <http://www.fivepminus.org> >

Florida Institute of Psychophysical Integration: Quantum Balance
5837 Mariner Dr
Tampa, FL 33609-3411
(813) 186-2273
Fax: (813) 287-2870
DrJoy@JohnsonMail.com

Flower Essence Society
PO Box 459
Nevada City, CA 95959
(800) 736-9222 (US & Canada)
Fax: (530) 265-0584
mail@flowersociety.org
< <http://www.flowersociety.org> >

Food and Drug Administration
Office of Inquiry and Consumer Information
5600 Fisher Lane
Room 12-A-40
Rockville, MD 20857
(301) 827-4420
< <http://www.fda.gov/fdahomepage.html> >

Food and Nutrition Information Center
10301 Baltimore Blvd
Room 304
Beltsville, MD 20705-2351
< <http://www.nalusda.gov/fnic> >

Foundation Fighting Blindness
Executive Plaza I, Ste. 800
11350 McCormick Road
Hunt, VA lley, MD 21031-1014
(888) 394-3937
< <http://www.blindness.org> >

Foundation for Ichthyosis and Related Skin Types
650 N Cannon Ave, Ste. 17
Landsdale, PA 19446
(215) 631-1411
(800) 545-3286
Fax: (215) 631-1413
< <http://www.scalyskin.org> >

Frontier's International Vitiligo Foundation
4 Rozina Court
Owings Mills, MD 21117
(301) 594-0958

G

Gamblers Anonymous International Service Office
PO Box 17173
Los Angeles, CA 90017
(213) 386-8789
Fax: (213) 386-0030
< <http://www.gamblersanonymous.org/> >

Gay and Lesbian Medical Association
459 Fulton St, Ste. 107
San Francisco, CA 94102
(415) 225-4547
< <http://www.glma.org> >

Gluten Intolerance Group
PO Box 23053
Seattle, WA 98102-0353
(206) 325-6980

GriefNet
PO Box 3272
Ann Arbor MI 48106
< <http://rivendell.org> >

Guide Dogs for the Blind
PO Box 1200
San Rafael, CA 94915
(415) 499-4000

Guillain-Barré Syndrome Foundation International
PO Box 262
Wynnewood, PA 19096
(610) 667-0131
(610) 667-0131
< <http://www.webmast.com/gbs> >

Gynecologic Cancer Foundation
401 North Michigan Ave
Chicago, IL 60611
(800) 444-4441
< <http://www.wcn.org> >

H

Hairy Cell Leukemia Research Foundation
2345 County Farm Lane
Schaumburg, IL 60194
(800) 693-6173

HCF Nutrition Research Foundation Inc
PO Box 22124
Lexington, KY 40522
(606) 276-3119

Head Injury Hotline

PO Box 84151
Seattle, WA 98124
(206) 621- 8558
< <http://www.headinjury.com> >

Head Trauma Support Project Inc

2500 Marconi Ave, Ste. 203
Sacramento, CA 95821
(916) 482-5770

Health Services and Resources Administration

Division of Organ Transplantation
Room 11A-22
5600 Fishers Lane
Rockville, MD 20857

Hear Now

9745 E Hampden Ave, Ste. 300
Denver, CO 80231
(800) 648-HEAR
(202) 651-5258

Hearing Industries Association

1800 M St NW
Washington, DC 20036
(202) 651-5258

Hearing Loss Link

2600 W Peterson Ave, Ste. 202
Chicago, IL 60659
(312) 743-1032
(312) 743-1007 (TDD)

Heimlich Institute

PO Box 8858
Cincinnati, OH 45208
heimlich@iglou.com
< <http://www.heimlichinstitute.org/index.htm> >

Hellerwork

406 Berry St
Mt Shasta, CA 96067
(530) 926-2500
< <http://www.hellerwork.com> >

Hemochromatosis Foundation Inc

PO Box 8569
Albany, NY 12208-0569
(518) 489-0972
skleiner@shivahuntercuny.edu
< <http://www.hemochromatosis.org> >

Hepatitis B Foundation

101 Greenwood Ave, Ste. 570
Jenkintown, PA 19046
(215) 884-8786
info@hepb.org

Herb Research Foundation

1007 Pearl St, Ste. 200

Boulder, CO 80302
(303) 449-2265
< <http://www.herbs.org> >

Hermansky-Pudlak Syndrome Network Inc

One South Road
Oyster Bay, NY 11771-1905
(800) 789-9477
appell@theonramp.net

Hirshberg Foundation for Pancreatic Cancer Research

375 Homewood Rd
Los Angeles, CA 90049
(310) 472-6310
< <http://www.pancreatic.org> >

Histiocytosis Association of America

302 North Broadway
Pitman, NJ 08071
(800) 548-2758 (USA and Canada)
< <http://www.histio.org> >

Homeopathic Educational Services

2124B Kittredge St
Berkeley, CA 94704
(510) 649-0294
Fax: (510) 649-1955

Hospice Foundation of America

2001 S St NW, Ste. 300
Washington, DC 20009
(800) 854-3402
< <http://www.hospicefoundation.org> >

Hospicelink

Hospice Education Institute
190 Westbrook Rd
Essex, CT 06426-1510
(800) 331-1620
< <http://www.hospiceworld.com> >

Human Growth Foundation

997 Glen Cove Ave
Glen Head, NY 11545
(800) 451-6434
< <http://www.hgfound.org> >

Hydrocephalus Foundation Inc

(HyFI) 910 Rear Broadway
Saugus, MA 01906
(781) 942-1161
HyFI1@netscape.net
< <http://www.hydrocephalus.org> >

Hypoglycemia Association Inc

18008 New Hampshire Ave
PO Box 165
Ashton, MD 20861-0165

Hypospadias Association of America

4950 S Yosemite St

Box F2-156
Greenwood Village, CO 80111
hypospadiasasn@yahoo.com
< <http://www.hypospadias.net> >

IgA Nephropathy Support Network

964 Brown Ave
Huntington, VA lley, PA 19006
(215) 663-0536

Immune Deficiency Foundation

25 W Chesapeake Ave, Ste. 206
Towson, MD 21204
(800) 296-4433
< <http://www.primaryimmune.org> >

Impotence Institute of America

Impotents Anonymous
10400 Little Patuxent Parkway, Ste. 485
Columbia, MD 21044-3502
(800) 669-1603

INFOLEP Leprosy Information Services

Postbus 950051090 HA
Amsterdam Netherlands
< Infolep@antennanl >

Inherited High Cholesterol Foundation

410 Chipeta Way
Room 167
Salt Lake City UT 84104
(888) 244-2465

Insight Meditation Society

1230 Pleasant St
Barre, MA 01005
(978) 355-4378
Fax: (978) 355-6398
< <http://www.dharma.org> >

Institute for Families with Blind Children

PO Box 54700, Mail Stop 111
Los Angeles, CA 90054-0700
(213) 669-4649

Institute for Preventative Sports Medicine

PO Box 7032
Ann Arbor MI 48107
(313) 434-3390
< <http://www.ipsm.org> >

International Association of Enterostomal Therapy

27241 La Paz Road, Ste. 121
Laguna Niguel, CA 92656
(714) 476-0268

International Association of Infant Massage
PO Box 1045
Oak View, CA 93022

International Association of Laryngectomees (IAL)
7440 North Shadeland Ave, Ste. 100
Indianapolis, IN 46250
< <http://www.larynxlink.com/> >

International Association for Medical Assistance to Travelers (IAMAT)
417 Center St
Lewistown, NY 14092
(716) 754-4883

International Association of Parents and Professionals for Safe Alternatives in Childbirth
Rte 1, Box 646
Marble Hill, MO 63764
(314) 238-2010

International Association of Reiki Professionals
PO Box 481
Winchester, MA 01890
< <http://www.iarp.org> >

International Association of Yoga Therapists (IAYT)
4150 Tivoli Ave
Los Angeles, CA 90066

International Bio-Oxidative Medicine Foundation (IBOMF)
PO Box 891954
Oklahoma City OK 73109
(405) 634-7855
Fax: (405) 634-7320

International Childbirth Education Association
PO Box 20048
Minneapolis, MN 55420
(612) 854-8660

International Chi Kung/Qi Gong Directory
2730 29th St
Boulder, CO 80301
(303) 442-3131

International Cesarean Awareness Network
1304 Kingsdale Ave
Redondo Beach, CA 90278
(310) 542-6400

International Council for the Control of Iodine Deficiency Disorders
43 Circuit Road
Chester Hill, MA 02167

(207) 335-2221
< <http://www.tulane.edu/~icec/icciddhome.htm> >

International College of Applied Kinesiology
PO Box 905
Lawrence KS 66044-9005
(913) 542-1801

International Colour Vision Society: Forschungsstelle fuer Experimentelle Ophthalmologie
Roentgenweg 11
Tuebingen D-72076 Germany
< <http://orlaboptomunsw.edu.au/ICVS> >

International Council for Medical and Clinical Therapists
7361 McWhorter Place, Ste. 300
Annandale, VA 22003-5469
< <http://www.ultradept.com/ICMCT.htm> >

International DiGeorge/VCF Support Network
c/o Family Voices of New York
46 1/2 Clinton Ave
Cortland, NY 13045
(607) 753-1250

International Eye Foundation
7801 Norfolk Ave
Bethesda, MD 20814
(301) 986-1830

International Foundation for Functional Gastrointestinal Disorders
PO Box 17864
Milwaukee, WI 53217
(888) 964-2001
< <http://www.iffgd.org> >

International Foundation for Homeopathy
2366 Eastlake Ave East
#301
Seattle, WA 98102
(425) 776-4147

International Institute of Infant Massage
605 Bledsoe Rd NW
Albuquerque NM 87107
(505) 341-9381
Fax: (505) 341-9386
< <http://www.infantmassage.com> >

International Institute of Reflexology
PO Box 12642
St Petersburg, FL 33733-2642
(727) 343-4811

Fax: (727) 381-2807
ftflex@concentric.net

International Lactation Consultants Association
201 Brown Ave
Evanston, IL 60202
(708) 260-8874

International Lesch-Nyhan Disease Association
114 Winchester Way
Shamong, NJ 08088-9398
(215) 677-4206

International Medical and Dental Hypnotherapy Association
4110 Edgeland, Ste. 800
Royal Oak MI 48073-2285
< <http://www.infinityinst.com> >

International Myopia Prevention Association
RD No 5, Box 171
Ligonier, PA 15658
(412) 238-2101

International NLP Trainers Association Ltd
Coombe House Mill Road
Fareham Hampshire PO16 0TN
United Kingdom
(044) 01489 571171

International Ozone Association Ind Pan American Group
31 Strawberry Hill Ave
Stamford, CT 06902
(203) 348-3542
Fax: (203) 967-4845

International Polio Network
4207 Lindell Blvd, Ste. 110
St Louis, MO 63108-2915
(314) 534-0475

International Rett Syndrome Association
9121 Piscataway Road, Ste. 2B
Clinton, MD 20735
(800) 818-7388
< <http://www.rettsyndrome.org> >

International School of Shiatsu
10 South Clinton St
Doylestown, PA 18901

International Tremor Foundation
7046 West 105th St
Overland Park KS 66212
(913) 341-3880

Intersex Society
PO Box 31791
San Francisco, CA 94131

Intestinal Health Institute

4427 East Fifth St
Tucson, AZ 85711
(520) 325-9686
info@sheilas.com
<http://www.sheilas.com>

Irish Raynaud's and Scleroderma Society

PO Box 2958
Foxrock Dublin 18
Ireland
(01) 235 0900
irss@indigo.ie

Iron Disorders Institute Inc

PO Box 3021
Greenville SC 29602
(864) 241-0111
irondis@aol.com
<http://www.irondisorders.org>

Iron Overload Diseases Association Inc

433 Westwind Dr North
Palm Beach, FL 33408
(561) 840-8512
iod@ironoverload.org

J**Juvenile Diabetes Foundation**

120 Wall St 19th Floor
New York, NY 10005
(800) 533-2873
<http://www.jdf.org>

K**Kids with Heart**

1578 Careful Dr
Green Bay, WI 54304
(800) 538-5390
<http://www.execpc.com/~kdswhrt>

Klinefelter Syndrome and Associates Inc

PO Box 119
Roseville, CA 95678-0119
(916) 773-2999
(888) 999-9428
Fax: (916) 773-1449
ksinfo@geneticorg
<http://www.genetic.org/ks>

Klinefelter's Organization

PO Box 60
Orpington BR68ZQ
United Kingdom
<http://hometownaol.com/KSCUK/index.htm>

Komen Foundation

5005 LBJ Freeway, Ste. 250
Dallas, TX 75244
(972) 855-1600
<http://www.komen.org>

L**LaLeche League International**

1400 N Meacham Rd
Schaumburg, IL 60173-4048
(800) 525-3243
<http://www.lalecheleague.org>

Late Onset Tay-Sachs Foundation

1303 Paper Mill Road
Erdenheim, PA 19038
(800)672-2022

League for the Hard of Hearing

71 West 23rd St
New York, NY 10010-4162
(212) 741-7650
<http://www.lhh.org>

Learning Disabilities Association of America

4156 Library Road
Pittsburg, PA 15234
(412) 341-1515
<http://www.ldanatl.org>

Lesch-Nyhan Syndrome Registry

New York University School of Medicine
Department of Psychiatry
550 First Ave
New York, NY 10012
(212) 263-6458

Leukaemia Research Fund

43 Great Ormond St
London WC1N 3JJ
United Kingdom
(020) 7405-0101
<http://dspacedialpipex.com/lrf-//>

Leukemia Society of America Inc

600 Third Ave
New York, NY 10016
(800) 955 4572
<http://www.leukemia.org>

Lighthouse National Center for Education

111 E 59th St
New York, NY 10022
(800) 334-5497
<http://www.lighthouse.org>

Lighthouse National Center for Vision and Aging

111 E 59th St
New York, NY 10022
(800) 334-5497
<http://www.lighthouse.org>

Little People of America Inc

National Headquarters
PO Box 745
Lubbock, TX 79408
(806) 737-8186
(888) LPA-2001
lpadatabase@junocom
<http://www.lpaonline.org>

Lupus Foundation of America

1300 Piccard Dr, Ste. 200
Rockville, MD 20850
(800) 558-0121
<http://www.lupus.org>

Lyme Disease Network of, NJ Inc

43 Winton Road
East Brunswick, NJ 08816
<http://www.lymenet.org>

Lymphoma Research Foundation

8800 Venice Blvd, Ste. 207
Los Angeles, CA 90034
(310) 204 7040

M**March of Dimes Birth Defects Foundation**

1275 Mamaroneck Ave
White Plains, NY 10605
(888) 663-4637
resourcecenter@modimesorg
<http://www.modimes.org>

Massachusetts College of Emergency Physicians (MACEP)

P O Box 296
Swansea, MA 02777
(508) 643-0117
Fax: (508) 643-0141

Massachusetts General Hospital

Functional and, Ste.reotactic
Neurosurgery Cingulotomy Unit
Fruit St
Boston, MA 02114
(617) 726-2000
<http://neurosurgery.mgh.harvard.edu/cingulot.htm>

Ménière's Network

1817 Patterson St
Nashville, TN 37203

(800) 545-4327
< <http://www.earfoundation.org> >

Meningitis Foundation of America
7155 Shadeland Station, Ste. 190
Indianapolis, IN 46256-3922
(800) 668-1129
< <http://www.musa.org/welcome.htm> >

Metabolic Information Network
PO Box 670847
Dallas, TX 75367-0847
(214) 696-2188
(800) 945-2188

Micronutrient Initiative
(c/o International Development
Research Centre)
250 Albert St Ottawa Ontario
Canada K1G 3H9
(613) 236-6163, ext 2050
< <http://www.idrc.ca/mi/index.htm> >

Midlife Women's Network
5129 Logan Ave S
Minneapolis, MN 55419
(800) 886-4354

Midwest Heart Specialists
Physician Office Building
3825 Highland Ave Tower 2, Ste. 400
Downers Grove Illinois 60515
(630) 719-4799
< <http://www.midwestheart.com> >

Milne Institute Inc
PO Box 2716
Monterey, CA 93942-2716
(831) 649-1825
Fax: (831) 649-1826
< <http://www.milneinstitute.com> >
milneinst@aol.com

Mind-Body Medical Institute
Beth Israel Deaconess Medical Center
One Deaconess Road
Boston, MA 02215
(617) 632-9525
< <http://www.mindbodyharvard.edu> >

**Mine Safety and Health
Administration**
4015 Wilson Blvd
Arlington, VA 22203
(703) 235-1910
< <http://www.msha.gov> >

**Mommies Enduring Neonatal Death
(MEND)**
PO Box 1007
Coppell, TX 75067

(972) 459-2396
(888) 695-6363
< http://www.mend.org/home_index.asp >

**Multiple Myeloma Research
Foundation**
11 Forest St
New Canaan, CT 06840
(203) 972-1250
< <http://www.multiplemyeloma.org> >

Muscular Dystrophy Association
3300 East Sunrise Dr
Tucson, AZ 85718
(520) 529-2000
(800) 572-1717
< <http://www.mdausa.org> >

**Myasthenia Gravis Foundation of
America**
222 S Riverside Plaza, Ste. 1540
Chicago, IL 60606
(800) 541-5454
< <http://www.medunc.edu> >

Myelin Project Headquarters
2001 Pennsylvania Ave NW, Ste. 225
Washington, DC 20006-1850
(202) 452-8994
< <http://www.myelin.org> >

**Myelodysplastic Syndromes
Foundation**
464 Main St
PO Box 477
Crosswicks, NJ 08515
(800) MDS-0839
< <http://www.mds-foundation.org> >

**Myopia International Research
Foundation**
1265 Broadway
Room 608
New York, NY 10001
(212) 684-2777

Myositis Association of America
600-D University Blvd
Harrisonburg, VA 22801
(540) 433-7686
< <http://www.myositis.org> >

**Nar-Anon Family Group
Headquarters Inc**
PO Box 2562
Palos Verdes Peninsula, CA 90274
(310) 547-5800

Narcolepsy Network
PO Box 42460
Cincinnati, OH 45242
(973) 276-0115

National Adrenal Disease Foundation
505 Northern Blvd, Ste. 200
Great Neck, NY 11021
(516) 487-4992

**National AIDS Treatment Advocacy
Project**
580 Broadway, Ste. 403
New York, NY 10012
(888) 266-2827
< <http://www.natap.org> >

**National Alliance on Alcoholism and
Drug Dependence Inc**
12 West 21st St
New York, NY 10010
(212) 206-6770

**National Alliance of Breast Cancer
Organizations**
9 East 37th St
10th Floor
New York, NY 10016
(888) 806-2226
Fax: 212-689-1213
< <http://www.nabco.org/> >

**National Alliance for Breastfeeding
Advocacy**
254 Conant Rd
Weston, MA 02193
(617) 893-3553

**National Alliance of Methadone
Advocates (NAMA)**
435 Second Ave
New York, NY 10010
(212) 595-6262
< <http://www.methadone.org/> >

**National Alliance for the Mentally Ill
(NAMI)**
Colonial Place Three
2107 Wilson Blvd, Ste. 300
Arlington, VA 22201-3042
(800) 950-6264
< <http://www.nami.org> >

**National Alliance for Research on
Schizophrenia and Depression**
60 Cutter Mill Road, Ste. 200
Great Neck, NY 11021
(516) 829-0091
< <http://www.mhsource.com> >

**National Animal Disease
Center/Zoonotic Research Unit**
2300 Dayton Ave
PO Box 70
Ames, IA 50010

National Anxiety Foundation
3135 Custer Dr
Lexington, KY 40517
(606) 272-7166
< <http://www.lexington-on-line.com/naf.html> >

National Aphasia Association
156 5th Ave, Ste. 707
New York, NY 10010
(800) 922-4622
< <http://www.aphasia.org> >

National Association of Anorexia Nervosa and Associated Disorders
Box 7
Highland Park, IL 60035
(708) 831-3438

National Association of Cognitive-Behavioral Therapists
PO Box 2195
Weirton WV 26062
(800) 853-1135
< <http://www.nacbt.org> >

National Association for Continence
PO Box 8310
Spartanburg SC 29305-8310
(800) 252-3337
< <http://www.nafc.org> >

National Association for the Deaf
814 Thayer Ave
Silver Spring, MD 20910
(301) 587-1788
(301) 587-1789 (TDD)
< <http://www.nad.org> >

National Association of Holistic Aromatherapy
836 Hanley Industrial Court
St Louis, MO 63144
(888) ASK-NAHA
< <http://www.naha.org> >

National Association for Premenstrual Syndrome
7 Swift's Court High St
Seal Kent TN15 0EG
United Kingdom
+ 44 (0) 1732 760011
< <http://www.PMDD.org.uk> >

National Association for Proton Therapy
7910 Woodmont Ave, Ste. 1303
Bethesda, MD 20814
(301) 913-9360
< <http://www.proton-therapy.org/Default.htm> >

National Association for Pseudoxanthoma Elasticum
3500 East 12th Ave
Denver, CO 80206
(303) 355-3866
Fax: (303) 355-3859
Pxenape@eStcom
< <http://www.napxe.org> >

National Association for the Visually Handicapped
22 West 21st St
New York, NY 10010
(212) 889-3141

National Athletic Trainers' Association
2952, Ste.mmons Freeway
Dallas, TX 75247-6916
(800) 879-6282
(214) 637-6282
Fax: (214) 637-2206
< <http://www.nata.org/> >

National Attention Deficit Disorder Association (ADDA)
9930 Johnnycake Ridge Road, Ste. 3E
Mentor, OH 44060
(800) 487-2282
< <http://www.add.org> >

National Autism Hotline
c/o Autism Services Center
PO Box 507
605 Ninth St
Huntington, WV 25710
(304) 525-8014

National Birth Defects Prevention Network
Atlanta, GA (770) 488-3550
< <http://www.nbdpn.org> >

National Board for Hypnotherapy and Hypnotic Anaesthesiology
7841 West Ludlow Dr, Ste. A
Peoria, AZ 85381
< <http://www.nbha-medicine.com/index.html> >

National Breast Cancer Coalition
1707 L St NW, Ste. 1060
Washington, DC 20036
(800) 622-2838
Fax: 202-265-6854
< <http://www.natlbcc.org/> >

National Cancer Institute
Building 31, Room 10A31
31 Center Dr
MSC 2580
Bethesda, MD 20892-2580
(800) 422-6237
< <http://www.nci.nih.gov> >

National Center for Complementary and Alternative Medicine (National Institutes of Health)
PO Box 8218
Silver Spring, MD 20907-8218
(888) 644-6226
< <http://nccam.nih.gov> >

National Center for Homeopathy
801 North Fairfax St, Ste. 306
Alexandria, VA 22134
(703) 548-7790

National Center for Learning Disabilities (NCLD)
381 Park Ave South, Ste. 1401
New York, NY 10016
(410) 296-0232
< <http://www.nclld.org> >

National Center for Nutrition and Dietetics
American Dietetic Association
216 West Jackson Blvd, Ste. 800
Chicago, IL 60606-6995
(800) 366-1655

National Center for the Preservation of Medicinal Herbs
3350 Beech Grove Road
Rutland, OH 45775
(740) 742-4401

National Center on Shaken Baby Syndrome
2955 Harrison Blvd #102
Ogden UT 84403
(801) 627-3399
< <http://www.dontshake.com> >

National Center on Sleep Disorders Research
Two Rockledge Centre
6701 Rockledge Dr
Bethesda, MD 20892
(301) 435-0199

National Certification Board for Therapeutic Massage and Bodywork
8201 Greensboro Dr, Ste. 300
McLean, VA 22102

National Children's Eye Care Foundation
One Clinic Center A3-108
Cleveland, OH 44195
(216) 444-0488

National Cholesterol Education Program
NHLBI Information Center
PO Box 30105
Bethesda, MD 20824-0105
< <http://www.nhlbi.nih.gov> >

National Chronic Pain Outreach Association Inc
PO Box 274
Millboro, VA 24460
(540) 997-5004

National Clearinghouse for Alcohol and Drug Information
11426-28 Rockville Pike, Ste. 200
Rockville, MD 20852
(800) 729-6686
< <http://www.health.org/> >

National Clearinghouse on Child Abuse and Neglect Information
PO Box 1182
Washington, DC 20013-1182
(800) 394-3366
< <http://www.calib.com/nccanch> > <http://www.cansearch.org> >

National Committee to Prevent Child Abuse
200 S Michigan Ave 17th Floor
Chicago, IL 60604
(312) 663-3520
< <http://www.childabuse.org> >

National Congenital Port Wine Stain Foundation
123 East 63rd St
New York, NY 10021
(516) 867-5137

National Council on Alcoholism and Drug Dependence
12 West 21st St
New York, NY 10010
(800) 622-2255
< <http://www.ncadd.org> >

National Depressive and Manic-Depressive Association (NDMDA)
730 N Franklin St, Ste. 501
Chicago, IL 60610
(800) 826-3632
< <http://www.ndmda.org> >

National Digestive Diseases Information Clearinghouse
2 Information Way
Bethesda, MD 20892-3570
nddic@eriecom
< <http://www.niddk.nih.gov/Brochures/NDDIC.htm> >

National Easter Seal Society
230 W Monroe St, Ste. 1800
Chicago, IL 60606-4802
(312) 726-6200
(800) 221-6827
< <http://www.easter-seals.org> >

National Eating Disorders Organization (NEDO)
6655 South Yale Ave
Tulsa OK 74136
(918) 481-4044

National Enuresis Society
7777 Forest Lane, Ste. C-737
Dallas, TX 75230-2518
(800) 697-8080
< <http://www.pedsumn.edu/Centers/NES> >

National Eye Institute
2020 Vision Place
Bethesda, MD 20892-3655
(301) 496-5248
< <http://www.nei.nih.gov> >

National Familial Pancreas Tumor Registry
The Johns Hopkins Hospital
600 North Wolfe St
Baltimore, MD 21287-6417
(410) 377-7450

National Federation of the Blind
1800 Johnson St
Baltimore, MD 21230
(301) 569-9314

National Foundation for Vitiligo and Pigment Disorders
9032 South Normandy Dr
Centerville, OH 45459
(513) 885-5739

National Fragile X Foundation
PO Box 190488
San Francisco, CA 94119-0988
(800) 688-8765
(510) 763-6030
Fax: (510) 763-6223
natlfx@sprintmail.com
< <http://nfxf.org> >

National Gaucher Foundation
11140 Rockville Pike, Ste. 350
Rockville, MD 20852-3106
(800) 925-8885
< <http://www.gaucherdisease.org> >

National Guild of Hypnotists
PO Box 308
Merrimack NH
< <http://www.ngh.net> >

National Head Injury Foundation
333 Turnpike Rd
Southboro, MA 01722
(617) 485-9950

National Headache Foundation
428 W St James Place

Chicago, IL 60614
(800) 843-2256
< <http://www.headaches.org> >

National Hearing Aid Society
20361 Middlebelt
Livonia MI 48152
(800) 521-5247
(313) 478-2610

National Heart Lung and Blood Institute
PO Box 30105
Bethesda, MD 20824-0105
(301) 251-1222
< <http://www.nhlbi.nih.gov> >

National Hemophilia Foundation
116 West 32nd St
11th Floor
New York, NY 10001
(800) 42-HANDI
< <http://www.info@hemophilia.org> >

National Human Genome Research Institute
The National Institutes of Health
9000 Rockville Pike
Bethesda, MD 20892
(301) 496-2433
< <http://www.nhgri.nih.gov> >

National Hypoglycemia Association Inc
PO Box 120
Ridgewood, NJ 07451
(201) 670-1189

National Institute on Alcohol Abuse and Alcoholism (NIAAA)
6000 Executive Blvd
Bethesda Maryland 20892-7003
< <http://www.niaaa.nih.gov> >

National Institute of Allergy and Infectious Disease
Building 31 Room 7A-50
31 Center Dr
MSC 2520
Bethesda, MD 20892-2520
(301) 496-5717
< <http://www.niaid.nih.gov/default.htm> >

National Institute of Arthritis and Musculoskeletal and Skin Diseases
9000 Rockville Pike
Bldg 31, Rm 9A04
Bethesda, MD 20892

National Institute of Child Health and Human Development
Bldg 31, Room 2A32 MSC 2425

31 Center Dr
Bethesda, MD 20892-2425
(800) 505-2742
< <http://www.nichd.nih.gov/sids/sids.htm> >

National Institute on Deafness and Other Communication Disorders

National Institutes of Health
31 Center Dr
MSC 2320
Bethesda, MD 20892-2320
< <http://www.nidcd.nih.gov> >

National Institute of Dental Research

31 Center Dr
MSC 2190
Building 31, Room 5B49
Bethesda, MD 20892-2190

National Institute of Diabetes and Digestive and Kidney Diseases

31 Center Dr
USC2560 Building 31, Room 9A-04
Bethesda, MD 20892-2560
(301) 496-3583
< <http://www.Niddk.nih.gov> >

National Institute on Drug Abuse

PO Box 30652
Bethesda, MD 20824-0652
(888) 644-6432
< <http://www.drugabuse.gov> >

National Institute of General Medical Sciences

Division of Pharmacology Physiology and Biological Chemistry
45 Center Dr
MSC 6200
Bethesda, MD 20892-6200

National Institute of Mental Health

Mental Health Public Inquiries
5600 Fishers Lane
Room 15C-05
Rockville, MD 20857
(888) 826-9438
< <http://www.nimh.nih.gov> >

National Institute of Mental Health Panic Campaign

Rm 15C-05
5600 Fishers Lane
Rockville, MD 20857
(800) 647-2642
< <http://www.nimh.nih.gov> >

National Institute of Neurological Disorders and Stroke

PO Box 5801
Bethesda, MD 20824
(800) 352-9424
< <http://www.ninds.nih.gov/index.htm> >

National Jewish Center for Immunology and Respiratory Medicine

1400 Jackson St
Denver, CO 80206
(800) 222-5864
< <http://www.nationaljewish.org/main.html> >

National Kidney Foundation

30 East 33rd St
New York, NY 10016
(800) 622-9010
< <http://www.kidney.org> >

National Kidney and Urologic Diseases Information Clearinghouse

3 Information Way
Bethesda, MD 20892
(800) 891-5390
< <http://www.niddk.nih.gov/health/kidney/nkudic.htm> >

National Lead Information Center National Safety Council

1025 Connecticut Ave NW, Ste. 1200
Washington, DC 20036
(800) 532-3394
< <http://www.nsc.org/ehc/lead.htm> >

National Lymphedema Network (NLN)

2211 Post St, Ste. 404
San Francisco, CA 94115
(800) 541-3259
< <http://www.hooked.net> >

National Mental Health Association

1021 Prince St
Alexandria, VA 22314
(703) 684-7722
< <http://www.nmha.org> >

National MPS Society

102 Aspen Dr
Downingtown, PA 19335
(610) 942-0100
Fax: (610) 942-7188
info@mpssociety.org
< <http://www.mpssociety.org> >

National Necrotizing Fascitis Foundation

PO Box 145
Niantic, CT 06357
(616) 261-2538
< <http://www.nnfforg/> >

National Organization for Albinism and Hypopigmentation (NOAH)

1530 Locust St #29
Philadelphia, PA 19102-4415

(800) 473-2310
< <http://www.albinism.org> >

National Organization for Rare Disorders (NORD)

PO Box 8923
New Fairfield, CT 06812-8923
(203) 746-6518
(800) 999-6673
Fax: (203) 746-6481
< <http://www.rarediseases.org> >

National Osteoporosis Foundation

1150 17th St NW, Ste. 500
Washington, DC 20036-4603
< <http://www.nof.org> >

National Pancreas Foundation

PO Box 935
Wexford, PA 15090-0935
< <http://www.pancreasfoundation.org> >

National Parkinson Foundation

1501 NW Ninth Ave
Bob Hope Road
Miami, FL 33136
< <http://www.parkinson.org> >

National Prostate Cancer Coalition

1156 15th St NW
Washington, DC 20005
(202) 463-9455
< <http://www.4npcc.org> >

National Psoriasis Foundation

6600 SW 92nd Ave, Ste. 300
Portland, OR 97223
(800) 723-9166
< <http://www.psoriasis.org> >

National Respiratory Distress Syndrome Foundation

PO Box 723
ontgomeryville, PA 18936

National Safe Kids Campaign

1301 Pennsylvania Ave, Ste. 1000
Washington, DC 20004-1707
< <http://pedscmwustledu/All-Net/english/neurpage/protect/drown.htm> >

National Sleep Foundation

1522 K St NW, Ste. 500
Washington, DC 20005
< <http://www.sleepfoundation.org> >

National Society of Genetic Counselors

233 Canterbury Dr
Wallingford, PA 19086-6617
(610) 872-1192

< <http://www.nsgc.org/GeneticCounselingYouasp> >

National Society to Prevent Blindness
500 East Remington Rd
Schaumburg, IL 60173
(708) 843-2020
(800) 331-2020
< <http://www.preventblindness.org> >

National Stroke Association
9707 E Easter Lane
Englewood, CO 80112
(800) 787-6537
< <http://www.stroke.org> >

National Tay-Sachs and Allied Diseases Association
2001 Beacon St, Ste. 204
Brookline, MA 02146
(800) 906-8723
< <http://www.ntsad.org> >

National Tourette Syndrome Association Inc
42-40 Bell Blvd
Bayside, NY 11361-2820
(718) 224-2999
Fax: (718) 279-9596
tourette@ixnetcomcom

National, VA ginitis Association
117 South Cook St, Ste. 315
Barrington, IL 60010
(800) 909-8745
VagAssoc@aolcom
< <http://www.vaginalinfections.org> >

National Vitiligo Foundation
PO Box 6337
Tyler, TX 75703
(903) 531-9767
7307133@compuserve.com

National Women's Health Resource Center
120 Albany St, Ste. 820
New Brunswick, NJ 08901
(877) 986-9472
< <http://www.healthywomen.org> >

Neuropathy Association
60 E 42nd St, Ste. 942
New York, NY 10165
(212) 692-0662
< <http://www.neuropathy.org> >

North American Society of Homeopaths
10700 Old County Rd 15 #350
Minneapolis, MN 55441
(612) 593-9458

North American Vegetarian Society (NAVS)
PO Box 72
Dolgeville, NY 13329
(518) 568-7970

Northeastern T'ai Chi Chuan Association
163 West 23rd St 5th Floor
New York, NY 10011
(212) 741-1922

Northwest Center for Environmental Medicine
177 NE 102nd St
Portland, OR 97220
(503) 561-0966

Obsessive-Compulsive Anonymous
PO Box 215
New Hyde Park, NY 11040
(516) 741-4901
west24th@aol.com < <http://members@aol.com/west24th/index.html> >

Office on Smoking and Health
Centers for Disease Control and Prevention Mailstop K-50
4770 Buford Highway NE
Atlanta, GA 30341-3724
(800) 232-1311
< <http://www.cdcgov/tobacco/> >

Office of the Special Assistant for Gulf War Illnesses
5113 Leesburg Pike, Ste. 901
Falls Church, VA 22041
(703) 578-8518
< <http://www.gulfink.osd.mil> >

Optician Association of America
7023 Little River Turnpike, Ste. 207
Annandale, VA 22003
(703) 916-8856
< <http://www.opticians.org> >

Oral Health Education Foundation Inc
5865 Colonist Dr
PO Box 396
Fairburn, GA 30213
(770) 969-7400

Osteoporosis and Related Bone Diseases-National Resource Center
1150 17th S NW, Ste. 500
Washington, DC 20036
(800) 624-2663

Overeaters Anonymous World Service Office
6075 Zenith, CT NE
Rio Rancho NM 87124
(505) 891-2664
< <http://www.overeatersanonymous.org> >

Paget Foundation
200, VA rick St, Ste. 1004
New York, NY 10014-4810
(800)23-PAGET

Parents Families and Friends of Lesbians and Gays
1726 M St NW, Ste. 400
Washington, DC 20036
(202) 467-8180
< <http://www.pflag.org> >

Pediatric/Adolescent Gastroesophageal Reflux Association Inc
PO Box 1153
Germantown, MD 20875-1153
(301) 601-9541
< <http://www.reflux.org> >

Periodic Paralysis Association
5225 Canyon Crest Dr #71-351
Riverside, CA 92507
(909) 781-4401
< <http://www.periodicparalysis.org> >

Phoenix Project/Head Injury Hotline
Box 84151
Seattle, WA 98124
(206)621-8558
< <http://www.headinjury.com> >

Pilates Studio
2121 Broadway, Ste. 201
New York New York 10023-1786
(800)474-5283
(888) 474-5283
(212)875-0189
Fax: (212) 769-2368
< <http://www.pilates-studio.com> >

Planned Parenthood Federation of America Inc
810 Seventh Ave
New York, NY 10019
(800) 669-0156
< <http://www.plannedparenthood.org> >

Polycystic Kidney Disease Foundation
4901 Main St
Kansas City, MO 64112-2634

(800) PKD-CURE
< <http://www.pkdcure.org/home.htm> >

Polycystic Ovarian Syndrome Association
PO Box 80517
Portland, OR 97280
(877) 775-7267
info@pcosupport.org
< <http://www.pcosupport.org/> >

Postpartum Support International
927 North Kellogg Ave
Santa Barbara, CA 93111
(805) 967-7636

Prader-Willi Foundation
223 Main St
Port Washington, NY 11050
(800)253- 7993
< <http://www.prader-willi.org> >

Prader-Willi Syndrome Association(USA)
5700 Midnight Pass Rd
Sarasota, FL 34242
(800) 926-4797
< <http://www.pwsusa.org> >

Pregnancy and Infant Loss Support (SHARE)
St Joseph Health Center
300 First Capitol Dr
St Charles, MO 63301
(800) 821-6819
< <http://www.nationalshareoffice.com/index.html> >

Prevent Blindness America
500 East Remington Road
Schaumburg, IL 60173
(800) 331-2020
< <http://www.prevent-blindness.org> >

Project Inform
205 13th St #2001
San Francisco, CA 94103
(800) 822-7422
< <http://www.projinf.org> >

Prostate Health Council
American Foundation for Urologic Disease
1128 N Charles St
Baltimore, MD 21201-5559
(800) 828-7866
< <http://www.afud.org> >

Prostatitis Foundation Information Distribution Center
2029 Ireland Grove Park
Bloomington, IL 61704

(309) 664-6222
< <http://www.prostate.org> <

Pulmonary Fibrosis Foundation
1075 Santa Fe Dr
Denver, CO 80204
(720) 932-7850
< <http://pulmonaryfibrosis.org> >

Pulmonary Hypertension Association
PO Box 24733
Speedway, IN 46224-0733
(800) 748-7274
< <http://www.phassociation.org> >

PXE International Inc
23 Mountain St
Sharon, MA 02067
(781) 784-3817
Fax: (781) 784-6672
PXEInter@aol.com
< <http://www.pxe.org/> >

Q

Qigong Human Life Research Foundation
PO Box 5327
Cleveland, OH 44101
(216) 475-4712

R

Radiological Society of North America
820 Jorie Blvd
Oak Brook, IL 60523-2251
(630) 571-2670
< <http://www.rsna.org> >

Rainbows Down Under—A Trisomy 18 and Trisomy 13 Resource
SOFT Australia 198 Oak Rd
Kirrawee NSW 2232
Australia
02-9521-6039
< <http://membersoptushomecomau/karens> >

Rape Abuse and Incest National Network
635-B Pennsylvania Ave SE
Washington, DC 20003
(800) 656-HOPE

Raynaud's & Scleroderma Association (UK)
112 Crewe Road
Alsager Cheshire ST7 2JA
United Kingdom
(44) (0) 1270 872776

webmaster@raynaudsdaemon.co.uk
< <http://www.raynaudsdaemon.co.uk> >

Reflexology Association of America
4012 Rainbow St
KPMB#585
Las Vegas, NV 89103-2059

Rehabilitation International
25 East 21st St
New York, NY 10010
(212) 420-1500

Resolve
1310 Broadway
Somerville, MA 02144-1731
(617) 623-0744
< <http://www.resolve.org> >

Restless Legs Syndrome Foundation
1904 Banbury Road
Raleigh, NC 27608-4428
(919) 781-4428
< <http://www.rls.org> >

Retinoblastoma International
4650 Sunset Blvd
Mail Stop #88
Los Angeles, CA 90027
(323) 669-2299
info@retinoblastoma.net
< http://www.retinoblastoma.net/rbi/index_rbi.htm >

Retinoblastoma Society
Saint Bartholomew's Hospital
London EC1A 7BE
United Kingdom
020 7600 3309
Fax: 020 7600 8579
< <http://dsdialpipex.com/rbinfo> >

Rocky Mountain Institute of Yoga and Ayurveda
PO Box 1091
Boulder, CO 80306
(303)443-6923

Rolf Institute of Structural Integration
209 Canyon Blvd
PO Box 1868
Boulder, CO 80306-1868
(303) 449-5903
(800) 530-8875
< <http://www.rolf.org/> >

S

Schizophrenics Anonymous
15920 W Twelve Mile
Southfield MI 48076
(248) 477-1983

Scleroderma Foundation

12 Kent Way, Ste. 101
 Byfield, MA 01922
 (978) 463-5843
 (800) 722-HOPE
 Fax: (978) 463-5809
 < <http://www.scleroderma.org> >

Second Wind Lung Transplant Association Inc

9030 West Lakeview Court
 Crystal River, FL 34428
 (888) 222-2690
 < <http://www.arthouse.com/secondwind> >

Self Help for Hard of Hearing People Inc

7800 Wisconsin Ave
 Bethesda, MD 20814
 (301) 657-2248
 < <http://www.shhh.org> >

Sensory Integration International/The Ayres Clinic

1514 Cabrillo Ave
 Torrance, CA 90501-2817

Sexuality Information and Education Council of the US

130 W 42nd St, Ste. 350
 New York, NY 10036
 (212) 819-9770
 < <http://www.siecus.org> >

Shriners Hospitals for Children

International Shrine Headquarters
 2900 Rocky Point Dr
 Tampa, FL 33607-1460
 (813) 281-0300
 < <http://www.shrinershq.org> >

Shy-Drager Syndrome Support Group

2004 Howard Lane
 Austin, TX 78728
 (800) 288-5582
 < <http://www.shy-drager.com> >

Sickle Cell Disease Association of America Inc

200 Corporate Point, Ste. 495
 Culver City, CA 90230-8727
 (800) 421-8453
 Scdaa@sicklecelldiseaseorg
 < <http://sicklecelldisease.org/> >

Society for Light Treatment and Biological Rhythms

PO Box 591687
 174 Cook St
 San Francisco, CA 94159-1687
 < <http://www.websciences.org/sltbr> >

Society for Progressive Supranuclear Palsy Inc

Ste. #5065
 Johns Hopkins Outpatient Center
 601 N Caroline St
 Baltimore, MD 21287
 (800) 457-4777
 < <http://www.psp.org> >

Society for Mucopolysaccharide Diseases

46 Woodside Rd
 Amersham Buckinghamshire HP6 6AJ
 United Kingdom
 + 44 (01494) 434156
 < <http://www.mppsociety.co.uk> >

Society of Neuro-Lingistic Programming

PO Box 424
 Hopatcong, NJ 07843
 (201) 770-3600

Society of Nuclear Medicine

1850 Samuel Morse Dr
 Reston, VA 10016
 (703) 708-9000
 < <http://www.snm.org> >

Spina Bifida Association of America

4590 MacArthur Blvd NW, Ste. 250
 Washington, DC 20007-4226
 (800) 621-3141
 (202) 944-3285
 Fax: (202) 944-3295

Spine Center

1911 Arch St
 Philadelphia, PA 19103
 (215) 665-8300
 < <http://www.thespinecenter.com> >

SPOHNC Support for People with Oral and Head and Neck Cancer

PO Box 53
 Locust, VA lley, NY 11560-0053
 (800) 377-0928
 < <http://www.spoehnc.org> >

Sudden Arrhythmia Death Syndromes Foundation

540 Arapeen Dr, Ste. 207
 Salt Lake City UT 84108
 (800) STOP SAD
 < <http://www.sads.org> >
 < <http://www.ihc.com/research/longqt.html> >

Sudden Infant Death Syndrome Alliance

1314 Bedford Ave, Ste. 210
 Baltimore, MD 21208

(800) 221-7437
 < <http://www.sidsalliance.org> >

Support Organization for Trisomy 18 13 and Related Disorders (SOFT)

2982 South Union St
 Rochester, NY 14624
 (800) 716-7638
 < <http://www.trisomy.org> >

T

Tardive Dyskinesia/Tardive Dystonia National Association

PO Box 45732
 Seattle, WA 98145-0732
 (206) 522-3166

Texas Heart Institute Heart Information Service

PO Box 20345
 Houston, TX 77225-0345
 (800) 292-2221
 < <http://www.tmc.edu/thi/his.html> >

Thyroid Foundation of America Inc

Ruth Sleeper Hall RSL350
 40 Parkman St
 Boston, MA 02114-2698
 (800) 832-8321
 < <http://www.tfaeweb.org/pub/tfa> >

Thyroid Society for Education and Research

7515 South Main St, Ste. 545
 Houston, TX 77030
 (800) 849-7643
 < <http://the-thyroid-society.org/thyroid.html> >

Tourette Syndrome Foundation of Canada

194 Jarvis St #206
 Toronto ONT M5B 2B7
 Canada
 (800) 361-3120
 tsfcorg@sympatico.ca
 < <http://www.tourette.ca> >

Trager Institute

21 Locust Ave
 Mill, VA lley, CA 94941-2806
 (415) 388-2688
 Fax: (415) 399-2710
 admin@trager.com
 < <http://www.trager.com> >

Trans-Hyperboreau Institute of Science

PO Box 2344
 Sausalito, CA 94966
 (415) 331-0230

(800) 485-8095
Fax: (415) 331-0231

Transverse Myelitis Association
1787 Sutter Parkway
Powell, OH 43065-8806
(614) 766-1806
< <http://www.myelitis.org> >

Trichotillomania Learning Center Inc
1215 Mission St, Ste. 2
Santa Cruz, CA 95060
(831) 457-1004
Fax: (831) 426-4383
< <http://www.trich.org> >

Trigeminal Neuralgia/Tic Douloureux Association
PO Box 340
Barnegat Light, NJ 08006
(609) 361-1014

Turner Syndrome Society of England
2 Mayfield Ave
London W41PW UK
44 (0)181-994 7625
Fax: 44 (0)181-995 9075
< <http://www.exnet.com/staff/sys4/ts.html>
< <http://www.tss.org.uk> >

U

U.S. Department of Health and Human Services
200 Independence Ave SW
Washington, DC 20201
(877) 696-6775
< <http://www.hhs.gov> >

U.S. Department of Justice Drug Enforcement Administration
2401 Jefferson Davis Highway
Alexandria, VA 22301
(888) 644-6432
< <http://www.usdoj.gov/dea> >

U.S. Department of Justice Office for Victims of Crime
810 7th St NW
Washington, DC 20531

U.S. Food and Drug Administration (FDA) Center for Drug Evaluation and Research
Viagra Information
< <http://www.fda.gov/cder/consumerinfo/viagra/default.htm> >

U.S. National Library of Medicine
8600 Rockville Pike
Bethesda, MD 20894

(888) 346-3656
< <http://www.nlm.nih.gov> >

Undersea and Hyperbaric Medical Society
10531 Metropolitan Ave
Kensington, MD 20895
(301) 942-2980
< <http://www.uhms.org> >

United Cerebral Palsy Association Inc (UCP)
1660 L St NW, Ste. 700
Washington, DC 20036-5602
(202) 776-0406
(800) 872-5827
< <http://www.ucpa.org> >

United Network for Organ Sharing
1100 Boulders Parkway, Ste. 500
PO Box 13770
Richmond, VA 23225-8770
(804) 330-8500
< <http://www.unos.org> >

United Ostomy Association Inc (UOA)
19772 MacArthur Blvd, Ste. 200
Irvine, CA 92612-2405
(800) 826-0826
< <http://www.uoa.org> >

United Plant Savers
PO Box 98
East Barre VT 05649
(802) 479-9825
< <http://www.plantsavers.org> >

United States Renal Data System (USRDS)
The University of Michigan
315 W Huron, Ste. 240
Ann Arbor MI 48103
(734) 998-6611
< <http://www.medumichedu/usrds> >

University of California Los Angeles Harbor-UCLA Medical Center Research and Education Institute
1124 W Carson St B-4
South Torrance, CA 90502

University of Illinois Center for Narcolepsy Research
845 S Damen Ave
Chicago, IL 60612
(312) 996-5176

Upledger Institute
11211 Prosperity Farms Rd
Palm Beach Gardens, FL 33410
(800) 233-5880
Fax: (561) 622-4771
< <http://www.upledger.com> >

V

Vestibular Disorders Association
PO Box 4467
Portland, OR 97208-4467
(503) 229-7705
< <http://www.teleport.com/~veda> >

Veterans Administration
Persian Gulf Medical Information Helpline
400 South 18th St
St Louis, MO 63103-2271
(800) 749-8387

W

Wegener's Foundation Inc
3705 South George Mason Dr, Ste. 1813
South Falls Church, VA 22041
(703) 931-5852

Wegener's Granulomatosis Support Group Inc
PO Box 28660
Kansas City, MO 64188-8660
(800) 277-9474
< <http://www.wgsg.org/wgsg> >

Weight-Control Information Network
1 Win Way
Bethesda, MD 20896-3665
(301) 951-1120
< <http://www.navigatortufts.edu/special/win.html> >

Wellness Community
35 E Seventh St, Ste. 412 Cincinnati, OH 45202
(888) 793-9355
< <http://www.wellness-community.org> >

Wilderness Medical Society
PO Box 2463
Indianapolis, IN 46204
(317) 631-1745

Wilderness Medicine Institute
PO Box 9
413 Main St
Pitkin, CO 81241
(970) 641-3572
< <http://www.wildernessmed.com> >

Wilson's Disease Association
4 Navaho Dr
Brookfield, CT 06804
(800) 399-0266

Women's Cancer Network

c/o Gynecologic Cancer Foundation
401 N Michigan Ave
Chicago, IL 60611
(312) 644-6610
< <http://www.wcn.org> >

World Hypnosis Organization Inc

2521 W Montrose Ave
Chicago, IL 60618
< <http://www.worldhypnosis.org/about.html> >

Worldwide Education and**Awareness for Movement Disorders**

One Gustave L Levy Place
Box 1052
New York, NY 10029
(800) 437-6683
< <http://www.wemove.org> >

Wound Care Institute

1100 NE 163rd St, Ste. #101
North Miami Beach, FL 33162
(305) 919-9192
< <http://woundcare.org> >

Wound Healing Society

1550 South Coast Highway,
Ste. 201
Laguna Beach, CA 92651
(888) 434-4234
< <http://wizardpharmwayne.edu/woundsoc/WHS.htm> >

**Wright State University Aerospace
Medicine Program**

PO Box 92 Dayton, OH 45401-0927
(937) 276-8338
< <http://www.medwright.edu> >

Y**Y-ME National Organization for
Breast Cancer Information and
Support**

18220 Harwood Ave
Homewood, IL 60430
(800) 221-2141
(708) 799-8228

Z**Zain Hansen MPS Foundation**

23400 Henderson Rd
Covelo, CA 95420
(800) 767-3121

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